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NTA
UGC
NET/JRF/SET

PSYCHOLOGY

PAPER - II

As per updated syllabus from
JUNE 2019 onwards



NET

JRF

SET

PAPER 2

PSYCHOLOGY

Authors

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This document is created with the best intentions for NET/JRF/SET Psychology aspirants to help them during their preparation for the Examination.

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PREFACE

The key reason to compile this book is to provide a detailed knowledge about the NET/JRF/SET examination to the aspiring candidates and also to help them understand the scope associated with the exam. This book will serve as a complete package to help Psychology aspirants to crack NET/JRF/SET exam conducted by the National Testing Agency in India.

The book is a storehouse of extensive and detailed content covering the full latest syllabus of NET/JRF/SET prescribed by the NTA in 2019.

Key Features of the Book

As per the Latest syllabus of NET/JRF/SET exam changed in 2019. Comprehensive coverage of each topic.

Free Online Mock Test accessible via www.powerwithineducation.com

5000+ MCQs support in the form of Mock Tests accessible via the website.

Free complimentary Resources to boost your Strategy, and Preparation as per your learning style.

PowerWithin Education extends their humble gratitude to the authors, illustrators and editors who made this compilation of notes possible. We wish all our dear Psychology aspirants All the very Best for their exam preparations.

PowerWithin

ABOUT US

PowerWithin Education is a unit of PowerWithin (www.powerwithin.in). PW Education is a one-stop Hub for all Psychology Preparations for NET/JRF, SET, GRE, MA/MSc. Psychology, MPhil Clinical Psychology, PhD in Psychology and other examinations conducted at State and National Level in India. We offer online class for all Examinations in Psychology at our website and our android and iOS App (PW Education).

PW Education Services:

- Online Coaching for NET-JRF Psychology + Paper 1**
- Distance Mode for NET-JRF Psychology**
- Online Coaching for MA/MSc Psychology**
- Online Coaching for MPhil Clinical Psychology**
- Online Mock Tests**
- E-resources**

ABOUT POWERWITHIN

An organization with a vision to change the scope and meaning of “training workshops” in India. We provide need-based, experiential, and effective training sessions that help the participants tap into their dormant potential, ignite their passion, and find their inner power.

PW Services:

- Psychometric Assessments**
- Training Programs**
- Mental Health & Counseling Services**

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NET JRF Preparation: An Insight into the Examination

By: Amit Panwar

What is NET JRF Examination?

On behalf of University Grants Commission (UGC), the National Eligibility Test (NET) is conducted for determining the eligibility of Indian nationals for the Eligibility for Assistant Professor only or Junior Research Fellowship & Assistant Professor Both, in Indian Universities and Colleges.

The award of JRF and or Eligibility for Assistant Professor depends on the aggregate performance of the candidate in Paper-I (General Aptitude) and Paper-II (Psychology) of UGC-NET.

Who conducts it?

Till June 2018, the CBSE conducted the NET in 84 subjects at 91 selected Cities of spread across the country. Since Dec 2018 onwards, the UGC- NET is being conducted by the NTA (National Testing Agency).

When and How is it conducted?

UGC-NET is conducted Online twice a year, usually in June and December. The exam is for 3 hours in Online Mode for combined (Paper I and Paper II) without a break.

	Paper 1	Paper 2
Number of Questions	50	100
Negative Marking	0	0
Syllabus	General Aptitude (10 units)	Psychology (10 units)
Previous Year Questions	Available	Available
Official Website	https://ugcnet.nta.nic.in/	

Syllabus of the Exam

The NET-JRF exam has two Papers: Paper I and Paper II (the subject/specialization of your Master's degree).

Units	Paper 1: General Aptitude	Paper 2: Psychology
Unit 1	Teaching Aptitude	Emergence of Psychology
Unit 2	Research Aptitude	Research Methodology & Statistics
Unit 3	Comprehension	Psychological Testing
Unit 4	Communication	Biological Basis of Behavior
Unit 5	Mathematical Reasoning & Aptitude	Attention, Perception, Learning, Memory and Forgetting
Unit 6	Logical Reasoning	Thinking, Intelligence & Creativity
Unit 7	Data Interpretation	Personality, Motivation, Emotion, Stress and Coping

Unit 8	Information & Communication Technology	Social Psychology
Unit 9	People, Development & Environment	Human Development & Interventions
Unit 10	Higher Education System	Emerging Areas

What does it mean to Qualify NET Exam?

The candidates qualifying only for Assistant Professor (i.e. those who qualified NET Exam only) are not to be considered for the award of JRF. Candidates who qualify the test for eligibility for Assistant Professor are governed by the rules and regulations for recruitment of Assistant Professor of the concerned universities/colleges/state governments, as the case may be.

It means that if you have qualified NET then you are eligible for becoming the Assistant Professor but this does not mean that they will become Assistant Professor. The procedure, and other eligibility requirements posted by Universities would be different for different universities. Anywhere that you fulfill the basic requirements (majorly depends on the API or the Academic Performance Index) posted by a Colleges/University then you are eligible to appear for the Interview Process. This is why NET is the National Eligibility for the Assistant Professor or Lecturer in India.

*As per the guidelines by UGC, in 2021 and forth, NET Qualified + PhD degree is compulsory to become the Assistant Professor in India.

What does it mean to have JRF?

Anyone who qualifies JRF, means they have also qualified NET exam. JRF stands for Junior Research Fellowship, which is a scholarship amount given to those who qualify JRF exam as well as enroll themselves in a Higher Education Program (such as MPhil or PhD).

For the process to cash into this amount, you can contact your concerned Universities after enrolling PhD or MPhil Program. (Kindly confirm with your concerned hospital or institute for JRF as some MPhil Programs might now support JRF scholarship or only has their own Scholarship amounts.)

Other than Scholarship JRF also offer many benefits. Some of them includes:

Direct Admission into MPhil and PhD Programs.

Stipend of Rs. 31,000/- + HRA for initial 2 years and Rs. 35,000/- + HRA for next 3 years

Candidates with JRF are preferred over NET candidates in Research Projects by government agencies like UGC, NCERT, DRDO, IITs, etc. and also by private universities.

Difference between NET and JRF

	NET	JRF
Qualified the Exams means	Only Qualified for Assistant Professorship	Qualified for Assistant Professorship + JRF Scholarship Amount
	NET has no Scholarship amounts.	
Stipend	If enrolled into PhD, then Government Universities offer Rs. 8000/- to 10,000/- per month Scholarship under non-NET category (i.e. a Scholar who is not receiving any aid from anywhere else).	Rs. 31,000/- + HRA per month (for First two 2 years) Rs. 35,000/- + HRA per month (for next 3 years)

Other Benefits

- Direct Admission into almost in India every PhD Program
- Eligibility for some research Projects by NCERT, UGC, DRDO, etc.
- Additional benefit of Rs. 2,000/- to 3,000/- per month over non-NET qualifiers in Research Projects.

Direct Admission into almost every PhD and MPhil Program in India
 Eligibility for some research Projects by NCERT, UGC, DRDO, etc.
 Preferred in Research Projects over only NET qualified candidates

Scope and Opportunities in Psychology after NET-JRF?

- ✚ **Becoming an Assistant Professor using NET qualification.**
- ✚ **Taking Direct Admission into PhD and MPhil Programs using NET or JRF qualification.**
- ✚ **Stipend during MPhil and PhD for JRF qualified candidates. (NET has no stipend but other benefits or a non-NET fellowship for PhD and MPhil that can be availed via your concerned University/Institute.**
- ✚ **Eligible for Research Projects by many Government Agencies like UGC, NCERT, and DRDO, and private agencies using NET and JRF qualifications.**

What is SET and CSIR-NET exam? How are they different from NET-JRF Exam?

SET EXAM vs NET-JRF Exam

SET or the State Eligibility Exam, in a nutshell, is the NET exam conducted by the State under their jurisdictions. The SET exam is also conducted for determining the eligibility of Indian nationals for the Eligibility for Assistant Professor only in State Universities and Colleges in India.

In simpler terms, many a times, institutes or States in India prefer to conduct their own exams and not rely on NET Exam scores for Assistant Professorship, these exams are called SET exams. NET Exam is conducted only in English and Hindi languages whereas SET exams are conducted in English and other languages recognized by Indian States/ Government.

NET Exam is conducted twice a year whereas SET exam is conducted only once a year.

Syllabus of NET exam is given by NTA. Most of the SET exams follow the older syllabus of NET exam. The syllabus is similar or almost the same for NET exam with some topics added or subtracted by the State or the Institute.

NET Candidate can apply anywhere in India for Assistant Professorship but a SET candidate can only apply for the colleges/universities under that State jurisdiction.

As of now 15 states conduct their SET exams and other States only follow the NET exam scores for their selection criteria. These states are:

Andhra Pradesh State Eligibility Test (APSET)

Assam State Level Eligibility Test (Assam SLET)

Chhattisgarh State Eligibility Test (CG SET)

Gujarat State Eligibility Test (GSET)

Himachal Pradesh State Eligibility Test (HP SET)

Jammu & Kashmir State Eligibility Test (JK SET)

Kerala State Eligibility Test (Kerala SET)

Karnataka State Eligibility Test (KSET)

Madhya Pradesh State Eligibility Test (MP SET)

Maharashtra State Eligibility Test (MH SET)

Tamil Nadu State Eligibility Test (TNSET)

Telangana State Eligibility Test (TSSET)

Uttarakhand State Eligibility Test (USET)

West Bengal State Eligibility Test (WB SET)

Rajasthan State Eligibility Test (RSET)

CSIR UGC NET-JRF Exam or CSIR NET or CSIR UGC NET

It is similar to NTA UGC NET-JRF exam in every aspect other than the fact that it is for the Science and Technology field.

"It takes determination to see a dream come to pass. The question is not will you start, but will you finish." – Joel Osteen

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Preparing for the Exam:

Key to successfully crack an exam at times does not just depend on the how smart you are, or how much you studied but whether or not you are 'good at taking tests' or not. One of the most important keys here is to be improve at taking the tests. How can we improve there? Well, there is no one way to do it but let's experiment together on this using the following key points:

Start by first answering to yourself, "Who Am I"?

Focus on your Learning Style, how is it that you study? Do you make a timetable? Or Do you make one but unable to follow one? Do you make Notes or prefer studying from Visual or Presentational aids?

Your way of studying now, will help you to know what needs to be done to Ace at the exam. No exam is difficult, and never have I seen an easy one. You are unique, and so you need to study uniquely too.

Know Your Exam

This means knowing the Syllabus thoroughly. If you want to study in a relatively focused way then connect with the exam first. You have to learn to befriend the exam and not compete with it.

Then, learn the basic details about the exam: marking schemes, number of questions, duration of the exams, etc.

Once you've been through the basics, then move towards bringing together the Previous Year Questions. If possible, you can find the Previous Year Mock Tests and attempt the, first to know where you stand. Since, you are looking forward for the NET exam, that means either you had Psychology in your Bachelors and Masters or you had a different background in Bachelors but are doing Masters in Psychology. In both these cases, move forward with these Previous Year Mocks, they will help you when you will begin studying the Notes or your Study Material. These will help you to keep an eye for those concepts that you have now seen in the Previous Year papers. Also, this will help you to know which Units and what kind of topics were focused on in which year. This knowledge is a bliss when attempting the exam for the first time.

Anxiety and two sides of the Coin.

William James (1842-1910), an American Psychologist once advised his Harvard students that an

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"ounce of good nervous tone is an examination is worth many pounds of ... study".

By "good nervous tone", James was addressing the nervousness, or anxiety that an exam might provoke. Anxiety is natural and it is also natural to expect feeling less relaxed just before or during an exam. This same anxiety can also be caused during the preparation also or maybe while attempting Mock Tests, or maybe while trying to the understand the Exam itself.

Let me just come forth and tell you that Anxiety is natural because it is helpful. The "good nervous tone", alert our senses, gives sharpened perception, and provide readiness for action in an alarming situation.

Therefore, it can be helpful to know what good Anxiety can do and how it is beneficial for you. Additionally, learn to combat Anxiety effectively so that instead of hindering your preparation, it helps you to stay focused, stay creative, stay on track and in perfect sync to the duration of the Preparation.

Make a Plan

Sounds cliché, I know. But here, I am not asking you make an ideal time table and make your study plan accordingly. All I am asking you is to tackle the fear of initiating. Many a times we simply procrastinate by thinking that we will begin tomorrow and that tomorrow never comes.

I usually recommend my students to simply your syllabus by separating Units/Topics that needs more focus, and more study from Units that you are well

Here is a Strategic Planner for NET-JRF. Let's begin...

Strength Units			Challenging Units		
	Mock Test #1	Theory Checklist		Theory Checklist	Mock Test #1
Unit 1			Unit 1		
Unit 2			Unit 2		
Unit 3			Unit 3		
Unit 4			Unit 4		
Unit 5			Unit 5		
Unit 6			Unit 6		
Unit 7			Unit 7		
Unit 8			Unit 8		
Unit 9			Unit 9		
Unit 10			Unit 10		

versed with or have covered earlier to comes naturally to you.

This is a sample of my strategy that I provide with our Study Materials.

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Units that are your strength, start with their Mock Test first and then go to the theory. For Units that are challenging, begin with theory, then Take one Mock Test, then analyses, go back to the theory and re-attempt Mock Tests and post that attempt more and more Mock Tests of that Unit.

It is important to know that never try to score perfect in your Mocks. Mock Tests are a strategy tool that will help you to develop a certain state of mind for Test Taking, as well as help you to stay on track, and map you Progress. Grow by taking one step at a time.

Take your time

You are unique. You have your own pace, you own growth rate. Never ask too much around as to how other's preparation is going on or how much others

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have done. You have to crack the exam. Focus on yourself and Take your own Time. Remember, there is no secret ingredient. (KungFu Panda, is the movie I recommend to help you better understand this line)

Study not like any student but like a Psychology student

The NET-JRF Paper 2: Psychology syllabus that you would be preparing for will have most of the applied concepts that you also apply while preparing specially the ones you will learn in Unit 5 of the syllabus. Unit 8 Social Psychology is too interesting. So, be a Psychology student preparing for an exam that will also help you to revise, and learn so many beautiful concepts. So, trust Psychology and the exam shall be cracked.

For more Information and Preparation Tips, visit www.powerwithineducation.com

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CONTENT

PSYCHOLOGICAL THOUGHT IN SOME MAJOR EASTERN SYSTEMS: Bhagavad Gita, Buddhism, Sufism and Integral Yoga. Academic psychology in India: Pre-independence era; post-independence era; 1970s: The move to addressing social issues; 1980s: Indigenization; 1990s: Paradigmatic concerns, disciplinary identity crisis; 2000s: Emergence of Indian psychology in academia. Issues: The colonial encounter; Post colonialism and psychology; Lack of distinct disciplinary identity.

WESTERN: Greek heritage, medieval period and modern period. Structuralism, Functionalism, Psychoanalytical, Gestalt, Behaviorism, Humanistic-Existential, Transpersonal, Cognitive revolution, Multiculturalism. Four founding paths of academic psychology - Wundt, Freud, James, Dilthey. Issues: Crisis in psychology due to strict adherence to experimental-analytical paradigm (logical empiricism). Indic influences on modern psychology.

ESSENTIAL ASPECTS OF KNOWLEDGE PARADIGMS: Ontology, epistemology, and methodology. Paradigms of Western Psychology: Positivism, Post-Positivism, Critical perspective, Social Constructionism, Existential Phenomenology, and Co-operative Enquiry. Paradigmatic Controversies. Significant Indian paradigms on psychological knowledge: Yoga, Bhagavad Gita, Buddhism, Sufism, and Integral Yoga. Science and spirituality (avidya and vidya). The primacy of self-knowledge in Indian psychology.

Chapter 1

Unit 1: Emergence of Psychology

1. ACADEMIC PSYCHOLOGY IN INDIA

1990S (20TH CENTURY) PSYCHOLOGY IN INDIA:

A. PRE-INDEPENDENCE

Following the accession of India to the English empire after 1857 an education system modeled on the lines of Oxford and Cambridge was introduced in India. Formal education in Psychology began in the early part of the 20th century.

Owing its origin in the philosophical discourse of the 18 century thinkers, it was Wilhelm Wundt of the University of Leipzig, who first realized the importance of studying mental states under certain special experimental conditions. The closing years of the 19th century thus witnessed the appearance of a new branch of science, which came to be known as “**Experimental Psychology**”. Impressed by its promise and progress Sir **Asutosh Mukherjee**, the well-known jurist and renowned educationist, who was then the planning head of the University of Calcutta decided to include “Experimental Psychology” in the Post Graduate Course of studies of Calcutta University (Kolkata). Eminent philosopher **Dr. Brojendra Nath Seal**, King George V Professor of Mental and Moral Philosophy of the same University **drafted the syllabus in 1905 and established the first**

laboratory, for demonstrative purpose, on the request of Sir Asutosh Mukherjee. The honour of having the second Experimental Psychology Laboratory in Asia (after Japan) may be attributed to the Department of Psychology, Calcutta University.

Eleven years later this laboratory was upgraded as the first psychology department, the Department of Experimental Psychology. **Narendra Nath Sengupta**, who chaired this department, had his education at Harvard University with Hugo Munsterberg, a student of William Wundt. Laboratory research at Calcutta in the areas of **depth perception, psychophysics, and attention** inspired early work at other centers. Recognizing the scientific nature of research, **psychology was included as a separate section in the Indian Science Congress in 1923.** Due to this in India Psychology acquired the status of being a science along with biological sciences, faster than the status it has got in the west. The Indian Psychological Association was founded in 1924 and the Indian Journal of Psychology, the first psychology journal in India, appeared the very next year.

Currently the department has attained the status of being the **alma mater of Psychology in India.**

Before these experimental traditions could consolidate, Sengupta left Calcutta. He was succeeded by **Girindra Shekhar Bose**. Being a medical doctor and a psychiatrist who was in close contact with Sigmund Freud, Bose showed much enthusiasm to promote psychoanalysis. **In 1922 he founded the Indian Psychoanalytic Society, which two years later was affiliated with the International Psychoanalytic Society.** Bose received his Ph.D. from Calcutta on the 'concept of repression', the first Ph.D. from any Indian University in psychology. He established the Lumbini Park Mental Hospital in Calcutta in 1940, and in 1947 brought out a journal 'Samiksha'. The Department started an Applied Psychology Wing in 1938, when Jung, Meyers, and Spearman were invited to the Silver Jubilee Session of the Indian Science Congress.

(In 1924 down south, Mr. M.V.Gopalswamy was heading the department in Mysore and he was trained in 'Psychological Assessments').

B. POST-INDEPENDENCE

The first generation of psychologists after independence was people from the background of Philosophy. Separate Psychology Departments were largely started between 1940-1960.

Following the independence of India in 1947, there were many efforts from the part of government to integrate social science into science and technology research as part of the nation building programme. India's first Prime Minister Jawaharlal Nehru was a great promoter of psychology, encouraging students to go abroad to obtain their degrees under eminent psychology professors. He also invited and encouraged psychologists to research on certain issues in India.

Other departments of Psychology were soon established in Mysore and Patna. In

1946 Psychology was started in university of Patna under two separate wings Experimental Psychology and Department of Applied Psychology.

In **1956**, The University Grants Commission was set-up; its responsibility was to provide funds to various universities to start Psychology Departments. Due to this there were 32 departments in Psychology by 1960 (this has however increased now). Certain centers also provided advanced studies in Psychology. Soon another changing trend was that psychology was being offered in colleges offering Engineering, Agriculture and in management colleges. A lot of other centers such as NCERT, NIPCCD etc. started applied research in different fields of Psychology.

Training for **Clinical Psychologists** was provided in three major centres such as All India Institute of Mental Health in Bangalore (now NIMHANS), Hospital for Mental Disease, Ranchi and Mental Hospital in Calcutta. Around this time Clinical Psychologists also organized an all India Association of their own, and its first convention was held in October 1969, at the All India Institute of Mental Health Bangalore. The first time **Education Psychology was introduced at the MA level as a paper was in 1961** in the University of Calcutta.

The National Institute of Mental Health and Neurosciences, Bangalore is an institution of international repute for research in clinical psychology and training clinical psychologists. The Central Institute of Psychiatry, Ranchi is another highly reputed institution carrying out research in clinical psychology.

History of Clinical Psychology in NIMHANS; the department was started in the year 1954 as the department of Psychology and Human Relations. It is one of the oldest and largest departments in NIMHANS. It is involved in clinical services, human resource Development and research activities. The first training

programme was started in 1955 as Diploma in Medical Psychology and the first PhD was started in 1967. Most students were/are trained in Behavior Therapy, Rehabilitation, De-addiction, Adult and Child Psychiatry.

The Indian Association of Clinical Psychologists (IACP) was started in 1968.

The Indian Council of Social Science Research (ICCSR) established in 1968 monitored the status of research in Psychology in India.

According to **the Association of Indian Universities**, 51 of the 101 recognized universities were offering psychology by the end of 1975. **By 1995 the number of universities had risen to 219** and those offering psychology at various levels had risen to 70 (Prasadarao & Sudhir, 2001). The period after the mid-1990s has seen an even sharper growth in the popularity of psychology. Psychology has been introduced in the curriculum of senior secondary schools.

C. 1970S: THE MOVE TO ADDRESSING SOCIAL ISSUES

Initiatives towards socially relevant research India's Independence from the colonial rule in 1947 did bring changes in the content and concerns of psychological research. The National Government recognized the importance of social science teaching and research in attaining the objectives of national reconstruction and social development. There was much hope and expectation about the role that the social sciences could play in this endeavor. **Psychologists began to realize that they have a responsibility to engage in socially relevant research.** As an example, psychologists responded to the human tragedy of the partition of India. As an aftermath thousand were killed in Hindu-Muslim riots, followed by a massive influx of refugees from across the border. Many psychological studies of that

period evidenced this concern in the rioting behavior of the masses. Realizing the urgent need of research in this area, the Ministry of Education procured the services of Gardner Murphy through UNESCO in 1950 to develop a research programme to investigate the causes of communal violence. Many Indian psychologists collaborated on this project which culminated in the book edited by Murphy in 1953 entitled 'In the Minds of Men'. They continued working in this area in later years also.

Another area which emerged during this period but was not sustained in the later decades, is counseling and guidance. The guidance bureau at Patna inspired the establishment of similar bureaus in other states. U.P Psychological Bureau was among the first ones, established in 1947. This Bureau under the leadership of Sohan Lal, and thereafter of C.M. Bhatia and S.N. Mehrotra adapted many intelligence and aptitude tests in Hindi, and provided counseling services to the public. The Bihar Psychological Bureau under the guidance of Mohsin worked on similar lines. In Bombay, Parsi Panchayat Vocational Guidance Bureau provided services to students, as well as to referred cases. The Bureau also brought out the Journal of Vocational and Educational Guidance. This journal played an important role in furthering the guidance movement in this country. However, the social and political conditions were just not ripe for the expansion of this movement and not enough of a research base was built up to sustain this movement.

A new development after Independence was the **growth of psychology outside the university system. The Ahmedabad Textile Industries Research Association (ATIRA) was established in 1950, where Kamla Chowdhury** conducted large-scale surveys to study motivational problems in the textile industries. Erikson and McClelland were frequent visitors to this

Institute. In the clinical field, advanced training programmes were introduced at the All India Institute of Mental Health (now known as the National Institute of Mental Health and Neuro-Sciences) in 1955, followed by the Hospital for Mental Diseases in Ranchi (1962). Indian armed forces also showed interest in using psychological tests in personnel selection. For this, **the Psychological Research**

Wing of the Defense Science Organization was set up in 1949. The aim of this Wing was not only to help in the election of army personnel but also to do research on the whole range of defense related problems, such as motivation and morale of the armed forces, leadership, mental health, stress, rehabilitation of disabled war veterans and development of psychological tests. Later on this Wing was elevated as the Defense Institute of Psychological Research, employing a large number of psychologists.

An applied field that began to grow after Independence **was industrial psychology.** Rapid industrialization in the 1950s and 1960s created a need for better understanding of the labor-management relationship and organizational efficiency. Psychological research projects on job attitude, work incentive, absenteeism, and job satisfaction were quite popular till the 1970s. Ganguli's (1961) book, "Industrial Productivity and Motivation", is an example of the concerns of that period. With the launching of many self-employment schemes by the Central Government in the 1960s, the research focus shifted to developing training programmes for inculcating entrepreneurship. McClelland's theory of achievement motivation (1961) provided the basis for launching a unique experiment to impart training for achievement at the Small Industries Extension Training Institute, Hyderabad. The findings of this innovative training were not conclusive, and eventually Indian psychologists lost interest in such experimentation. 18 With the shift in

research from industrial to organizational behavior, the interest grew in studying all spheres of organized work activities.

Communal and caste conflicts being perennial problems in India, social prejudice, intergroup relations and socialization of the Indian child emerged as major research interest. Rath and Sircar (1960), and Anant (1970) studied caste stereotypes. Paranjpe's (1970) book, "Caste Prejudice and the Individual" was based on a survey of college students' prejudice against low caste members. The studies of attitude towards the Chinese (A. K. P. Sinha & Upadhyaya, 1960) are oft-quoted studies of change in attitude as a result of historical events (border dispute with China which led to war). A. K. Singh (1981) examined development of religious identity and prejudice in children, and more recently, Deridder and Tripathi (1992) studied social consequences of norm violation by different ethnic groups. The work in the field of socialization and child rearing practices in India has also been substantial. Kakar (1978) examined socialization and child rearing in light of the elaborate system of rituals and practices prevalent

D. 1990S: PARADIGMATIC CONCERNS, DISCIPLINARY IDENTITY CRISIS

Durganand Sinha, who may be considered as one of the architects of modern Indian psychology has observed in his prolific writings that early attempts at formulation of Indian Psychology were rejected by psychologists in India who were trained in the empirical tradition, because of such notions like rebirth, transmigration of souls, and supernatural powers. It was considered as "glib talk", "revivalism" and "uncritical worship of the past" and the term Indian Psychology acquired a "pejorative connotation". So what is "spiritual psychology" for western

academicians become “revivalism” for Indian academicians! **Reasons for such resistance and rejection of psychology grounded in traditional ideas is primarily due to the negative attitude that we have inherited from our colonial past on the one hand and the influence of the scientific worldview on the other** (Dalal, 2002; Paranjpe, 2002). It may be argued that a vast majority of Indian psychologists have shied away from Indian psychological perspectives because of this attitude and the religio-philosophical context in which those perspectives are embedded. But, paradoxically a majority of them share the same socio-cultural context with the rest of the Indian population and are guided by the same religio-philosophical perspectives, which have shaped the attitudes, emotions, motivations, morals, values, etc., of the Indian masses in their day-to-day living! **As Kiran Kumar (2008) observed, this situation has created some kind of a split in the personality – psychologist as a professional vs. psychologist as a person – and it has contributed for lack of creativity and originality in what one does, and draining of personal resources and energy resulting in „burn out“ among many. A psychologist in his/her role as a scientist conducts the professional activities with one set of assumptions and beliefs and as a person lives and acts with another set of assumptions and beliefs among fellow humans.**

The fact that psychologists in India lived in two worlds did not change much after Independence, but rather became more pronounced. In most universities, psychology departments were established by splitting the departments of philosophy. As a result, a large number of philosophy faculty moved to the newly formed psychology departments; many who opted for psychology were those who saw better career opportunities in the new departments. **This movement from philosophy to psychology was so pervasive that by the end of the 1960s a**

majority of chairmen in psychology departments had a philosophy background. In normal course this could have given a strong philosophical knowledge base to psychology, making it richer in terms of indigenous concepts and theories. But this did not happen. In their enthusiasm to establish a new identity as scientists, these faculty members with a background in philosophy completely dissociated themselves from their parent discipline. They were more fascinated by the idea of value-free and culturally-neutral experimental work. The areas they showed interest in were memory, psychophysics, perception, learning, pattern recognition, etc., which could be studied without bringing in the cultural context. These faculty members from philosophy had little training in research methodology. Their major quest, therefore, was to acquire methodological sophistication and mastery over statistical techniques. In the process they became ardent adherents of Western research methods.

E. CRISIS OF IDENTITY

In the 70s and 80s a number of review papers were published to take stock of the contributions made by Indian psychologists, and of the emerging trends. Interestingly, these appraisals of research publications brought home this realization that psychology in India is mostly a poor imitation of western research and does not lead to understanding Indian social reality. A crisis was perceptible in the discipline in the mid-70s, as many felt that Indian psychology was going nowhere. The following passages present a brief account of this crisis of direction and progress.

By the mid-1970s the enthusiasm with which the Western-educated Indian psychologists were conducting research was waning. Indian scholars were getting restive, as Western psychology was failing to throw light on Indian social issues.

Sixty years of Western psychology in India had not yielded any significant discoveries. **Nandy (1974) argued, "Indian psychology has become not merely imitative and subservient but also dull and replicative" (p. 5). K.G. Agrawal (1973) called psychology in India as that of "adoptology".** There was growing disillusionment about the applicability of Western theories and their mindless testing in India. The failure to resolve inner conflicts of cherishing Indian cultural values at the personal level and maintaining high standards of objectivity at the professional level was reflected in Methodologically sophisticated but socially irrelevant research. As a result, Indian psychologists were increasingly marginalized in the society. A strong need was felt to return to the cultural roots. D. Sinha (1977) urged that the scientific understanding of Indian social reality should benefit from its vast treasure of traditional psychological knowledge accumulated over centuries. He called for the development of an indigenous psychology with its own paradigms to understand developmental problems of the region. In short, psychology in India was seeking its own identity.

Indian psychologists were showing more interest in studying problems relevant to the country using Indian concepts and theories. Neki (1973), for example, suggested a teacher-pupil model in clinical counseling to break cultural and social barriers. **J.B.P. Sinha (1980) proposed a new leadership style - nurturant task-master,** which is more likely to succeed in Indian work organizations. **Kakar (1982, 1991) studied the role of traditional healers in maintaining mental health in traditional societies. Ramchandra Rao (1983) and Palsane, Bhavasari, Goswami, and Evans (1986) developed a concept of stress based on ancient scriptures. Pande and Naidu (1992) developed a measure to study the concept of detachment and its mental health consequences.**

However, this crisis of the discipline's identity was of concern for only a handful of psychologists. J.B.P. Sinha (1993) made a distinction between the bulk and the front of psychological research. Not much has changed in the mass produced bulk of the research which is still replicative and imitative. However, as a response to a long debate and changing global scenario, the front-runners have started taking indigenous psychology more seriously.

F. 1980S: INDIGENIZATION

A journey toward indigenous psychology the core of indigenization is the belief that all knowledge, including that of psychology, is rooted in the prevalent world-view of a community and is conditioned by historical and sociocultural factors. D. Sinha (1994) discussed in detail the indigenization of psychology in India. He refers to two facets of indigenization. The first is purely the product of culture, the concepts and categories which are culture-bound. The second is the product of the interaction of cultural variables with concepts, theories and methods introduced from outside. J. B. P. Sinha (2002) has referred to these two as endogenous and exogenous indigenization. He further expanded the domain by referring to two variants of each kind of indigenization.

Taking lead from the above discussion, it is argued in this paper that indigenous research has moved along two different streams. One considers indigenization as a gradual process. A majority of psychologists in India are trained in Western models, and for them it is not feasible to make any sudden shift towards indigenously developed theories and methods. It is but natural for them to continue using concepts and tests borrowed from the West, albeit with increasing sensitivity to the cultural context. The same trend is observable in

many other developing countries also (Marriott, 1992)

The Canadian psychologist, John Adair, **saw indigenization as a gradual process**, and in collaboration with his Indian colleagues, **developed a scale to measure the degree of indigenization**. Adair (1989) operationalized indigenization in terms of - **(1) cultural sensitivity of research in designing a study and discussing its findings, (2) citation from one's own culture, (3) problem orientation, development of culture-based concepts, theories and methods, (5) culturally anchored tests, and (6) cross-cultural comparisons.**

Viewing retrospectively, this line of thinking led to the development of three streams of research: **problem-oriented research, cross-cultural psychology and Indian psychology.**

However, as noted by D. Sinha (1997), the cross-cultural work only led to testing of Western theories on Indian samples. Very rarely studies originated from the needs of the Indian society or tested Indian concepts in other cultures. Indiscriminate search for cultural differences and similarities made such ventures superficial, without providing much understanding of the culture or the contemporary problems of Indian society. To a large extent cross-cultural psychology remained a methodological enterprise and culture remained a peripheral concern (Misra & Gergen, 1993). In recent years, **cross-cultural psychology itself is going through a crisis and the classical conception of culture**, that was the basis of most research in this area, is being seriously questioned (Miller, 1998).

Indian psychology has developed around the existential quest to overcome human suffering and in the process to raise the person to higher levels of awareness and achievement. Thus, **Indian psychology endeavors not only to provide an understanding of the nature of a person, the causes and consequences of his/her**

conduct, but also to explore the methods and means of transforming the person in pursuit of perfection in being, certainty in knowing and happiness in feeling. Indian psychology focuses on the inner-self which is accessible through **subjective methods of self-verification.** The beginning of Indian psychology can be traced in the writings of many eminent thinkers, like Vivekanand and Sri Aurobindo in the early part of the last Century. The monumental work of Jadunath Sinha (1934/1958; 1961) on Indian psychology can be considered as a landmark in formally establishing it as an independent discipline. The books of Ramachandra Rao (Development of psychological thought in India, 1962) Ragunath Safaya (Indian Psychology, 1975) gave it further impetus in the early years.

The scientific community is better prepared now than in the past to accept a psychology rooted in native wisdom and philosophical traditions. Psychologists in India are increasingly aware of the wide gap between their academic pursuits and the real-life problems of people. The replicative nature of research endeavours, antiquated and obsolete teaching programmes, and lack of applied orientation have devoid the discipline of any professional momentum. Indian psychologists want to break free from the theoretical and methodological constraints of the discipline to grapple with the real issues of development and to act as social catalysts in the change process. Indian psychologists are also aware that if they fail to take up the challenge, they are likely to be completely marginalized.

Four factors have contributed to this change of attitudes and perceptions of Indian psychology.

- a. **One** is the failure of Western psychology to deal with their own societal problems. It has increasingly been realized that positivistic psychology only

provides a fragmented and superficial understanding of human feelings and behavior. The growing problems of social and family violence, mental health, moral decay, etc., have brought out the gross limitations of mainstream western psychology.

Two, is the growing popularity of Yoga and other spiritual systems of India in the West. In the past few decades more and more persons with spiritual training and experiences from India have been visiting other countries and have acquired a large following. They have been responsible for disseminating the Indian spiritual tradition. Among them, Maharishi Mahesh Yogi and Swami Rama are some prominent gurus who have been able to draw the attention of academic psychologists. Sri Aurobindo's Integral Yoga and Integral Psychology attracted worldwide attention. Mahesh Yogi's Transcendental Meditation went through rigorous experimental testing at Harvard and many other universities. Swami Rama offered himself to extensive medical testing at the Menninger Foundation Laboratory, New York, where he demonstrated many yogic feats. These Indian masters were able to demonstrate convincingly the power of mind over body and have contributed significantly towards evoking interest in the Indian psychological perspectives.

Three, is the secular nature of Indian psychology. It is a gradual acknowledgement that Indian psychology of consciousness has much to offer in terms of self-

growth rather than as religious traditions. Interest in the study of altered state phenomena since 1960s and the founding of transpersonal psychology as a sub-discipline, encouraged psychologists involved in them to examine the indigenous perspectives available in many religious traditions of the east. It was recognized clearly that modern psychology has nothing to offer regarding the spiritual/transcendental dimension of human nature either to understand others or for personal growth (Tart, 1975). Unfortunately, most academic psychologists in India have not appreciated this fact, and consider Indian psychology as part of the revivalist movement.

Four, with India emerging as a major economic power in the world, there is a renewed interest in Indian values, philosophies and practices, as well as in the strength and resilience of Indian society. The very Indian culture and philosophy which was debunked for India's poverty and backwardness by many western Indologists (Max Muller, for example) is now seen behind India's success stories. Indian research is now taken more seriously by western psychologists, and as a consequence (ironically), by Indian psychologists themselves. The newly emerging psychology is rooted in traditional Indian thought and practices. As stated in the Pondicherry Manifesto (2002), "Rich in content, sophisticated in its methods and valuable in its applied aspects, Indian psychology is pregnant with possibilities for the birth of new

models in psychology that would have relevance not only for India but also to psychology in general.”

G. 2000S: EMERGENCE OF INDIAN PSYCHOLOGY IN ACADEMIA

Though psychology has traversed a long distance in India, lack of direction has cast doubts about the application of psychology in the context of a rapidly changing socio-economic scenario. Psychology in India has remained dissociated from its own vast storehouse of knowledge inherent in the Indian philosophical texts. These scriptures and texts provide immense possibilities of developing psychological theories of self and human development.

psychology was imported lock-stock-barrel from the West and was first implanted in 1916 in Calcutta University. The Western model of research and teaching provided the basis on which Indian research grew for a long period. For Indian psychologists trained in the western traditions, it has been a long journey to turn towards their own heritage and take Indian concepts and theories germane to understanding Indian social reality.

This chapter aims to examine the status of psychology in India as a scientific discipline, **identifying the factors responsible for its retarded growth**. It also discusses the parallel movement now underway to rediscover the knowledge rooted in scriptures and folk practices, and explores its relevance in the present times. **Three sets of arguments** are put forward to explain the retarded growth of academic psychology in India. **One**, which is more charitable, is the lack of a supportive intellectual climate. In a country where a vast population lives in a condition of subhuman poverty, and decisions about social developmental programmes are politically motivated, any scholarly pursuit is considered peripheral. The academic institutions plagued by a rising student

population, political manipulations and lack of funds have gradually become non-performers. There is no premium placed on excellence in teaching and research.

Adair, Pandey, Begam, Puhane, and Vora (1995) conducted a study on 64 Indian psychologists through a mailed questionnaire. The survey revealed three major impediments to research productivity: (a) lack of supportive intellectual climate, (b) poor professional support, and (c) inadequate research funding.

It, however, throws no light on 'why is psychology in India lagging behind other sisterly disciplines, such as sociology, anthropology and economics?' It brings forth the **second line of argument** that there are some inherent limitations in psychology as a scientific discipline. Its excessive conformity to empiricist-positivist methodologies and confining to micro-level problems have restricted the scope of its psycho-social inquiry.

The third set of arguments focus on the personal and professional background of Indian psychologists. Psychologists in the first half of this Century were a product of the colonial domination of the Indian society, greatly influenced by Western scholarly traditions. The first-generation psychologists after India's Independence were predominantly converts from the philosophy background. This combined with their elitist-urban background; fewer job openings and self-serving research orientation gave Indian psychologists an identity distinct from those who belonged to other sister disciplines.

We need a psychology which is a positive discipline conducive to self-growth and social harmony. Indian psychology has come a long way in the hundred years of its existence. Today, there are large number of teaching and research institutions offering wide range of courses in psychology. It is not known how many colleges and universities offer psychology courses and how many psychologists are

professionally active. There are no data which official agencies, like the University Grants Commission, Indian Council of Social Science Research, or Department of Science and Technology can furnish. One estimate (A. K. Jain, 2005) suggests that there are more than 15000 psychologists in India. In any case, India has the largest number of psychologists outside the Western block, and is considered a 'publication giant' among all developing countries (Gilgin & Gilgin, 1987). This rapid expansion of the discipline has aroused many hopes and expectations about its possible contribution to the success of nation-building projects. The problems of poverty, illiteracy, urban decay and disease control cannot be handled on the basis of sound economic planning only, but also require changes in the attitudes and beliefs of people, and their motivations for collective action.

In recent years there have been several good publications which enable a critical evaluation of the development of psychology in India. Some of these writings (Dalal, 1990, 1996, 2002; Misra

Gergen, 1993; D. Sinha, 1986, 1996; J.B.P. Sinha, 1993) have critically 4 evaluated the progress of psychology. The five surveys of research in psychology (Mitra, 1972; Pareek, 1980, 1981; J. Pandey, 1988, 2001, 2004) cover important research contributions since the beginning of the last century.

In the ancient Indian scriptures no rigid distinction among religion, philosophy, and psychology was maintained. The overriding consideration was **to help individuals in their pursuit of self-realization and liberation from the miseries of life.** In this **world-view**, the source of all **suffering was presumed to be within the person, and thus the emphasis was on exploring the 'world within', to alleviate the suffering.** The goal was to seek enduring harmony of spirit, mind and body for everlasting happiness. The yoga system evolved very

sophisticated mind-control techniques in this pursuit. In contemporary literature this broad field of inquiry is referred to as **"Indian Psychology"**.

Some of these common features which give Indian Psychology a distinct identity are briefly discussed here.

Indian psychology can be deemed as **universal**. It primarily deals with **the inner state** of a person, taking consciousness as the primary subject matter of study. Consciousness as a state of being is not an object but is conceived as undifferentiated subjectivity without any content. It can be studied only indirectly through its various manifestations. Buddhism refers to a stream of consciousness as the basis of the subjective feeling of continuity and identity which affects our all perceptions, thoughts, actions and emotions.

Human consciousness is considered hierarchical, the highest state being of pure consciousness, bliss and truth. Distortions in consciousness are due to active interference of mind and body which limits our awareness, obscures our knowledge and feelings and cause suffering. The goal of life is to attain this state of pure consciousness where the knowledge is direct, immediate and intuitive, and not mediated by sensory inputs. Yoga and meditation are the tools to attain this transcendental state of pure consciousness.

Indian psychology is **spiritual** in its orientation. Spiritual does not mean otherworldly, nor does it mean being religious or dogmatic. Spirituality hereby implies taking into consideration the whole range of human progression, without making a distinction between natural and supernatural.

It connotes the faith in the existence of higher powers and the possibility of relating to them to seek the higher order truth of life. Spirituality has opened up the possibilities of developing broader theories of human existence to understand paranormal powers, creativity and intuitive thinking, which many people exhibit.

H. ISSUES: THE COLONIAL ENCOUNTER; POST COLONIALISM AND PSYCHOLOGY; LACK OF DISTINCT DISCIPLINARY IDENTITY

Before independence, psychological research in India was greatly influenced by the British universities. Most of the Indian psychologists during this time were trained abroad and they followed the western theories of psychology. This psychology, transplanted to India as part of the total imperialist domination by the West, came as a ready-made intellectual package in the first decade of the century (Nandy, 1974). In doing so **it almost entirely challenged and to a greater**

Expansion of psychology departments within the university system continued in the 1970s. However, it became increasingly difficult to conduct research in the university departments. In a bureaucratic university set up, there was little organizational support for pursuing research projects. Again, due to mass entry of students in the higher education and chronic campus unrest for political reasons, the academic environment was on the decline. Heavy teaching, and no incentive for

RAPID BUT UNPLANNED EXPANSION: In the fifties and sixties the growth of psychology in Indian universities was phenomenal. In 1956 the University Grants Commission (UGC) was constituted and it provided funds to various universities to start psychology departments. As a result, the number of psychology departments increased to 32 by the end of 1960s. This rapid expansion of psychology, though impressive, was quite unplanned. In the absence of any definite educational policy of the government, these departments were often created as a part of

d. Indian psychology is based on veridical methods. It should be clear that as a human science of consciousness its methods ought to be different from the methods of physical sciences.

extent replaced the intellectual traditions and indigenous systems that had existed for thousands of years – systems that contained elaborate theories about human nature, actions, personality and their relationships with the world.

For instance, **mentally ill patients in India were historically treated using various approaches such as herbal or ayurvedic medicines, yoga and music** (PrasadaraoSudhir, 2001). But with the advent of scientific (Western) psychology all these traditional and effective treatments were replaced with ‘scientific’ techniques.

research had a telling effect on the research output of the 15 universities. As a result, research in psychology started growing outside the university system. Psychologists having research inclinations preferred to join various research institutes. A.N.S. Institute of Social Studies, Patna, Centre for the Study of Developing Societies, New Delhi, and National Institute of Community Development, Hyderabad are examples of some prominent institutes which became centres of research in psychology.

the general expansion of higher education and at times without any particular academic considerations.

No serious thinking preceded in establishing these departments in terms of their need-based specialization. Nevertheless, in the course of their growth many departments developed a distinct identity (R. E. Pandey, 1969). For example, departments became known for their research in the areas of rural and social psychology (Allahabad), test construction (Mysore), industrial psychology (Osmania), measurement and guidance (Patna), or verbal learning (Poona). Most departments developed around one dominant scholar, usually the person heading the department. His interest and training were decisive factors in the specialization of that department. Psychology during those days (this is more or less true even today) developed around personalities, and when those scholars departed, many centres collapsed, or showed shifts in specialization.

2. PSYCHOLOGICAL THOUGHT IN SOME MAJOR EASTERN SYSTEMS

A. BHAGAVAD GITA

ABOUT BHAGAVAD GITA

Bhagavad Gita is part of the great epic Mahabharata, a widely popular mythological story in Hindu philosophy; part of Bhishma Parva, Gita is almost in its entirety the dialogue between two individuals, Lord Krishna (considered as incarnation of Bhagawan Vishnu, Narayana) and Arjuna (The Pandava prince, Nara) in the battle field (war between the Pandavas and the Kauravas, the cousins, for control of the kingdom of Hasthinapura) of Kurukshetra. It has 18 yogas (chapters), with about 701 slokas (short poems), the first one being “Arjuna Vishada Yoga” (Sorrow of Arjuna) and the

last one “Moksha Sanyasa Yoga” (Nirvana and Renunciation).

Bhagavad Gita dates back to about 4000-5000 B.C. as part of mythology dictated by Sage Veda Vyasa and written by Lord Ganesha. **What is more important and relevant is NOT WHAT IT IS BUT WHAT TRANSPIRED in those 18 chapters of Bhagavad Gita; the process and content of the dialogue; its usefulness as a model of counseling and possible contemporary application value to current day psychological therapies.**

PSYCHOLOGY & BHAGAVAD GITA

Maharishi (1982) explains that the Bhagavad-Gita displays the practical application of the knowledge and experience of bringing the individual into contact with the field of cosmic intelligence. The Bhagavad-Gita describes what could be understood as a “**Vedic psychological session**” of approximately two hours, which is sufficient for Lord Krishna to raise Arjun, who has sought his guidance, from a state of suffering to the state of enlightenment. Lord Krishna is called “Lord” because he is understood in the Vedic tradition to most fully embody the totality of consciousness.

The Bhagavad-Gita addresses the cause and elimination of suffering, the levels of the mind, the mechanics of experiencing transcendental consciousness, and the development of higher states of consciousness, or enlightenment, in which life is lived at the highest level of bliss, effectiveness, and universality.

For a student of psychology Bhagavad Gita offers a valuable case study for lessons in psychotherapy – resolution of conflict and successful resumption of action from a state of acute anxiety and guilt laden depression that precipitated inaction. The therapy process involved in Bhagavad Gita in which Lord Krishna helped the grief-stricken Arjuna through dialogue and discussion.

Counseling/psychotherapy is essential and an integral component of psychiatric interventions in the management of a patient with psychological distress. The psychotherapeutic models available are developed and imported from the western literature. The applicability and usefulness of these models in the Indian context was discussed with some skepticism keeping in mind the varying cultural, religious, spiritual, societal attitudes (broadly described as eastern/oriental culture) by psychiatrists in the past. **The Guru-Chela concept, as a model in psychotherapy, popularized by Dr. Neki gets widely debated in the Indian context.** Eminent Indian psychiatrists and psychologists discussed and proposed Bhagavad Gita as a source and model to develop psychotherapeutic concepts suitable to Indian context.

The very first word in Bhagavad Gita is “Dharma” and the last word is “Mama”. “Mama Dharma” – My duties, responsibilities, rights, ethics, morals, attitude, action, activities and so on. Some commentators recommend Gita as an elaborate detailing of **MAMA DHARMA**.

The first chapter, Arjuna Vishada Yoga, narrates the expression of Arjuna's sorrow, anxiety, fear and guilt leading to a state of inaction after seeing his kith and kin (Gurus, cousins, uncles, nephews, friends...) lined up in the enemy camp in the battle field-Fighting this war, to win the kingdom, means killing all these people whom Arjuna respected and loved; a sin of commission from any angle. Overwhelmed by the acute state of sadness and guilt, Arjuna drops his weapons (Gandiva) and turns to Lord Krishna, his charioteer, for help and guidance.

***Sishyasthe Aham Sadhi Mam
Tvam Prapannam***

(I am your disciple; Guide me, And Help Me)

The one concept in Bhagavad Gita that received exceptional respect and applause from several great scholars is **the emphasis on KARMA (ACTION).**

Intelligent action (Gnana Karma) without performance anxiety and **without the greed for the fruits of the work (Nishkama Karma)** and never to have the choice of **nonperformance of duty (Akarma)** emerges as a key point in the teaching of Bhagavad Gita.

WHAT IS THE IMPORTANCE AND RELEVANCE OF BHAGAVAD GITA IN OUR PRESENT LIFE?

Upon deeper analysis, we can break the scene (Lord Krishna and Arjuna on the Chariot in the middle of the battleground) into 6 main components with certain psychological implications as explained in the highly acclaimed Kathopanishad, which are as follows:

THE CHARIOT– The chariot represents our physical body (Tanu).

KRISHNA AS THE CHARIOTEER– Lord Krishna represents our higher cognitive abilities, especially in the face of adversity i.e. the intellect (Buddhi).

ARJUNA– Arjuna represents the self.

HORSES– The horses represent our senses and limbs (Indriyagalu) by virtue of which we perceive and move ahead in our lives. These have no restraint and without a rein/lagaam can wander off in any direction.

THE REIN of the horses- The reins in the hands of Lord Krishna controlling the horses represents our mind (Manas).

THE BATTLEGROUND– the battleground represents our life (Jeevana). Our life is a continuing and continual process of daily battles fought-won and lost!

In our daily lives, our senses and limbs (Indriyagalu) are exposed to a variety of objects and experiences and impact our mind (Manas) in various ways. So how is it that our mind functions well enough to always make the best decision for us?

In the modern society battle field is the symbol of challenges of our life. We all have mountains of desires, dreams, duties and ambitions, obligations to family, society and self. Each of us has our own society, family and culture in which we embedded and through which our self, behavior, personality and behavior are conditioned. Our life is the perfect example of a battle between good and evil. Many types of question are come to our mind such as; who we are? What is the purpose and meaning of our life? What happen to us when we die? It is the existential question that is not directly come on the surface but run like a river beneath our busy life. Many time we come across a situation where we are afraid to do the right thing because of social pressure and conformity. We face conflict with a choice regarding what we want to do over what we should do. Bhagavad Gita is a story that tells us about the importance of doing the right thing, of conquering our own darkness, regardless of the cost or struggle.

SRI AUROBINDO's TAKE ON GITA

The argument of the Gita resolves itself into three great steps by which action rises out of the human into the divine plane. The first step is Karma yoga, the selfless sacrifice of works, and here the Gita's insistence is on action. The second is Jnanayoga, the self-realization and knowledge of the true nature of the self and the world; and here the insistence is on knowledge; but the sacrifice of works continues and the path of Works becomes one with but does not disappear into the path of Knowledge. The last step is Bhakti

yoga, adoration and seeking of the supreme Self as the Divine Being, and here the insistence is on devotion; but the knowledge is not subordinated, only raised, vitalized and fulfilled, and still the sacrifice of works continues; the double path becomes the triune way of knowledge, works and devotion. The Gita is not a book of practical ethics, but of the spiritual life which permits us to transcend the clash of all dharmas that the human mind can conceive, and to discover a new dharma, the law of divine action, **divyam karma**, by the attainment of divine freedom in which the nature of the individual transcends its limitations, the limitations of the nature subject to three gunas — tamas, rajas, and sattva, — and attains to the divine nature (*sādharmyam*).

PURUSHA & PRAKRITI: The intellectual exposition of the Sankhya begins with the statement of two ultimate principles of existence, — Purusha, the inactive being, and Prakriti, the active force of action. Purusha is a pure conscious Being, immobile, immutable and self-luminous. Prakriti is Energy and its process. The human soul evolves through at least three levels of Prakriti, its three gunas: **TAMAS, RAJAS, SATTVA**. The first level is the *tāmasic* or the darkened state of inertia and utter resistance to change. Here, the law of the masses, the rules of the herd like a subconscious beast or a half-conscious man, drive him. Next comes the *rājasic* or the egoic stage of kinesis and dynamic movement. Finally, there is the third or *sāttvic* stage wherein the individual learns to subordinate his ego and take from life only what is rightfully his.

B. SUFISM

Islam: A brief introduction

Islam is one of the largest and fastest growing world religions. It began with the prophetic vision of Mohammad approximately 1,400 years ago in the Arabian Peninsula and spread at an astonishing rate throughout the Old World. Islam teaches that there's only one God/ Ultimate Reality (Allah) and nothing is ultimately real but God. God is neither born nor gives birth, that is, it is not a sentient being.

Islam literally means 'peaceful surrender' in Arabic. This refers to the process of yielding of human will to the Divine Will, and ultimately the reconciliation and the unification of the two. The agency of human Will is called the nafs (self) which is the principle of embodiment of spirit and the immediate experience of the ordinary human being in the world; the nafs is of this material world. Nafs-al-ammara (the commanding self) is the human agency of organization, control, volition and action in the world.

SUFISM IN INDIA

Sufism is the esoteric or inner dimension of Islam, and not a branch of exoteric Islam. Scholars trace the root of the term **Sufi to suf, a woolen garment worn by some early Sufis**. Some argue that the term is related to the Arabic word 'saafi', meaning purity. Origins of **Sufism go back to the 8th century CE, when the first known Sufis lived**. Sufi orders began to be established by 12th and 13th centuries, and historically developed more or less as a matter of geographic location as Islam spread throughout the world.

The Naqshabandi is a major Sufi order, which became popular in Central Asia, Kurdistan and the Indian subcontinent.

The Chisti order was founded in India by Khawja Mu'in al-Din Hassan Chisti (1142–1236 CE), whose shrine is visited by millions of people every year. Sufism

was brought to Europe and America by the great Indian Sufi master, **Hazrat Inayat Khan**, in the 1910s and 1920s from the Chisti tradition, bringing the message of 'love, harmony, and beauty' to the West. Islam spread to India as Arabs began settling on the Indian west coast of Kerala as early as the eighth century, which is much earlier than the time of the invasion of India from the north by Mahmud Ghazni in 1001 CE. **The development of Sufism in India has a highly distinctive feature, in that Islamic mysticism has been influenced by Indian mysticism throughout the centuries.**

SUFI PHILOSOPHY

According to Sufi philosophy, **Reality is the universal will, the true knowledge, eternal light and supreme beauty, whose nature is self-manifestation, reflected in the mirror of the universe; The world in comparison with the reality is a mere illusion, or non-reality or not-being**. Among Sufis, as well as in the Indian schools of philosophy, some believed in the oneness of the existence. To them, multiplicity indicated a mode of unity. The phenomenal world is an outward manifestation of the one-real. The Real's essence is above human knowledge.

From the point of view of its attributes, it is a substance with two accidents, one as creator and the other as creature; one visible and the other invisible. In its essence, it is attribute less, nameless, indescribable, incomprehensible, but when covered with Avidya or descent from it absoluteness, names and attributes are formed. The sum of these names and attributes is the phenomenal world, which represents reality under the form of externality.

Man is the microcosm in whom divine attributes are manifested in most imperfect diminutive form. God is eternal beauty and the nature of beauty is self-manifestation and desire to be loved. Thus, the Sufis

base their doctrine on the principles of love and prefer the Course of love or the Indian Bhakti to other means of reaching God. They consider love to be the essence of all religions and the cause of creation and its Continuation. God is unknowable, but may be thought through some concrete comparison. Phenomenal diversity is the reflection of the supreme beauty. The attributes are identical with him in fact, though distinct in our thought. In His absolute beauty, He is called Jamal and in His phenomenal **Husn**.

Man possesses three natures, viz.

Sensual, which corresponds to the Indian **Tamas**.

Intellectual, somewhat like the Indian **Rajas**.

Spiritual or the Indian **Sat**.

He becomes virtuous or wicked according to the predominance of one of the three said qualities. His mind must receive gradual training for which a guide is absolutely necessary. **The selection and following of a spiritual guide is the most important duty of a Sufi.** A bad or imperfect guide may lead him to evil or leave him imperfect and bewildered. He must use all his intellectual ability and human endeavor to find out the true guide and once obtained, he must obey his direction.

In all brevity, the essence of Sufi thought is that the life is full of miseries and that man can get salvation from the miseries only by renouncing the worldly possessions. Different doctrines were propounded at different times to achieve renunciation.

Ibn Arabi believed that God and his existence are one: According to him, the apparent separation is due to ignorance. Thus, there is considerable variety in their conceptions of knowledge, God, self and the world. However, each sect of the Sufis declared their respective system and their doctrines to be the true knowledge and

they claimed that the realization of their knowledge leads to the goal.

The system has two aspects:

(1) Knowledge and (2) Practice.

According to the Sufis, the object of man's creation is the acquisition of the knowledge of God. Knowledge of God includes the knowledge of the Creator and the creation comprising the universe and the individuals and so on. The knowledge of the soul and the things of the world is an essential step towards the knowledge of God.

The Object Of man's creation, according to Sufism, is the acquisition Of knowledge Of God. The Sufis believe that the true knowledge of God can be gained only in intuition. The Sufis believe that the heart of man is a mirror in which he can see the vision of God. But the heart mirror is polluted with the dust of worldly desires. To see the vision of Reality it is essential that it should be cleansed of completely. The worldly desires are identified with nafs. It is not easy to kill the nays even for a moment. The theorists of the system have evolved a method of practice to kill it and to lead the novice to the realization of knowledge of God.

The practice is two-fold: **Firstly, to practice detachment from the worldliness and secondly to develop love of God through Muraqaba (Meditation) and Muhasaba (Self-examination) etc.** The practice is metaphorically called **Safr (Journey) to God** and the stages of spiritual attainment on it are described as Muqamat (gyations). The Salik (Practicant) realizes spiritual moods called Hal (States) at some of the stations and at the end.

PRINCIPAL TEACHINGS OF SUFISM

Absolute Being (God) is also Absolute Beauty.

Since beauty tends towards manifestation, Absolute Being developed the phenomenal world. Man should practice virtues like poverty, austerity, humility, fortitude and discipline. He should devote himself to the ways of inwardness like withdrawal, silence, solitariness, and self-examination. He should also keep in mind a constant awareness of God with faith, awe and desire. These virtues, inwardness and awareness will bring a sense of direct communication with God. It would be better if such slogans are raised which work as remainders of the mystic belief and aim and as aids to concentration on the quest for unification.

Man should follow these directions with sufficient perseverance as they will advance through the standard mystic stages of concentration, apprehension of everything, sudden and unpredictable illumination, blissful ecstasy, sense of union with the deity, sense of nothingness and sense of the nothingness beyond nothingness.

THE FOUR STAGES OF SUFISM

According to Ibn Al-Arabi, the great Andalusian Sufi master, Islam consists of **four dimensions** / stages, of practice and understanding in Sufism, **Shariah (religious Law), Tariqah (the mystical path), Haqiqah (Truth), and Marifah (Gnosis)**. Each is built upon the stages that go before.

SHARIAH, which is the basic foundation for the next three stages. The Shariah consists of teachings of Islam, basically the morality and ethics found in all religions. It provides guidance to us for living properly in this world. Trying to follow Sufism without following the Shariah is like trying to build a house on a foundation of sand. Without an ordered life built

on solid moral and ethical principles, Mysticism cannot flourish. In Arabic, Shariah means "road:" it is a clear track, a well-traveled route that anyone can follow. Second is the Tariqah, which refers to the practice of Sufism.

TARIQA, which literally means a path, like a trail in the forest or in the desert, which is not pre-established and requires the expert knowledge of a guide to traverse. When one enters the tarigha, he or she is initiated into a Sufi order with a help of a teacher (sheikh) who has experience and knowledge of the path. **Just as the Shariah refers to the external dimension of religion, the Tariqah refers to the inner practices of Sufism.** The guide you need in order to find your way is the Sheikh, or Sufi Teacher. The Shariah makes the outer day-to-day life clean and attractive. The Tariqah is designed to make the inner life clean and pure. Each of these supports the other.

HAQIQAH, or Truth: Haqiqah refers to the inner meaning of the practices and guidance found in the Shariah and Tariqah. It is the direct experience of the mystical states of Sufism, direct experience of the presence of God within. Without this experience, seekers follow blindly, attempting to imitate those who know, those who have attained the station of Haqiqah. The attainment of Haqiqah confirms and solidifies the practice of the first two stages. Before Haqiqah all practice is imitation. Without the deep inner understanding that comes from experience, one follows mechanically the teachings and practices of others.

MARIFAH, OR GNOSIS: Gnosis is superior wisdom or knowledge of Spiritual Truth. This is a deep level of inner knowing, beyond Haqiqah. More than momentary Spiritual experience, Marifah refers to an ongoing state of attunement with God and with Truth. It is the knowledge of Reality, attained by a very few. This is the station of the Messengers, the prophets, and the great sages and saints. The great Sufi sage Ibn 'Arabi explained these four levels as follows: At the level of the law (Shariah) there is "yours and mine." That is, the law guarantees individual rights and ethical relations between people. At the level of the Sufi path (Tariqah), "mine is yours and yours is mine." The dervishes are expected to treat one another as brothers and sisters-to open their homes, their hearts, and their purses to one another. At the level of Truth (Haqiqah), there is "no mine and no

yours." The advanced Sufis at this level realize that all things are from God, that they are really only caretakers and that they "possess" nothing. Those who realize Truth have gone beyond attachment to possessions and beyond attachment to externals in general, including fame and position. At the level of Gnosis (Marifah), there is "no me and no you," At this final level, the individual has realized that all is God, that nothing and no one is separate from God.

What is lawful at one level may not be lawful at another level of understanding. For example, the outer practice of fasting is required by religious law, but according to the Sufi path, one of the essential reasons for fasting is to develop self-discipline and to control the insatiable ego. If a person is proud of fasting, the fast is still technically lawful, but in terms of the Sufi path, the fast is a failure.

TYPES OF SELF IN SUFISM

The Commanding Self	The commanding self seeks to dominate and to control each individual. At this level there is unbridled selfishness and no sense of morality or compassion. Descriptions of this level are similar to descriptions of the id in psychoanalytic theory; it is closely linked to lust and aggression. These have been called the swine and the dogs of the self - the sensual traits are like swine, the ferocious ones like fierce dogs or wolves. This is the realm of physical and egoistic desires.
Regretful Self	At this level, wants and desires still dominate, but now the person repents from time to time and tries to follow higher impulses. At this second level, people do not yet have the ability to change their way of life in a significant way. However, as they see their faults more clearly, their regret and desire for change grow. At this level, people are like addicts who are beginning to understand the pain they have caused themselves and others.

Inspired Self	At the next level, the seeker begins to take genuine pleasure in prayer, meditation, and other spiritual activities. Only now does the individual taste the joys of spiritual experience. Now the seeker is truly motivated by ideals such as compassion, service, and moral values. This is the beginning of the real practice of Sufism. Before this stage, the best anyone can accomplish is superficial outer understanding and mechanical outer worship. Though one is not free from desires and ego, this new level of motivation and spiritual experience significantly reduces the power of these forces for the first time. What is essential here is to live in terms of higher values. Unless these new motivations become part of a way of life, they will wither and die away. Behaviors common to the inspired self-include gentleness, compassion, creative acts, and moral action. Overall, a person who is at the stage of the inspired self seems to be emotionally mature, respectable, and respected.
Contented Self	The seeker is now at peace. The struggles of the earlier stages are basically begins to let go, allowing the individual to come more closely in contact with the Divine. This level of self predisposes one to be liberal, grateful, trusting, and adoring. If one accepts difficulties with the same overall sense of security with which one accepts benefits, it may be said that one has attained the level of the contented self. Developmentally, this level marks a period of transition. The self can now begin to "disintegrate" and let go of all previous concern with self-boundaries and then begin to "reintegrate" as an aspect of the universal self.
Pleased Self	At this stage the individual is not only content with his or her lot, but pleased with even the difficulties and trials of life, realizing that these difficulties come from God. The state of the pleased self is very different from the way we usually experience the world, focused on seeking pleasure and avoiding pain.
Self-pleasing to God	Those who have reached the next stage realize that all power to act comes from God, that they can do nothing by themselves. They no longer fear anything or ask for anything. The Sufi sage Ibn 'Arabi described this level as the inner marriage or self and soul. The self-pleasing to God has achieved genuine inner unity and wholeness.
Pure / Serene / Secure Self	Those few who attain the final level have transcended the self entirely. There is no ego or no separate self is left, only union with God. At this stage, the individual has truly realized the truth, "There is no god but God." The Sufi now knows that there is nothing but God, that only the Divine exists, and that any sense of individuality or separateness is an illusion.

When an individual is ready and with God's grace, the journey of spiritual transformation begins and the nafs continues to transform first into the 'regretful self', before turning into the 'inspired self', and finally the 'serene/secure self'. The process of spiritual transformation involves seven classical (main) stages.

MAIN STAGES OF SUFISM

Islam has prescribed certain practices for Muslims for attaining perfection in life. These practices are Kalimah, Salat, Sawm, Haj and Zakat. In addition to these, there are certain duties for a disciplined life. The ritual practices in Islam are endless. The rituals of Sha'riat are not difficult to follow. But there is another method which is called Ma'rifat. It is called the path of Sufis. Ma'rifat is concerned mainly with immediate experience. This path is described as a journey by Sufis. It has seven stages which are called Maqamat. The path is known as "Path" of Tariqat. There are differences in Sufis in regard to "Stages" but the main stages are as follows: -

There are differences in Sufis in regard to "Stages" but the main stages are as follows: -

- Repentance (Tawbah)
- Abstinence (Witr)
- Renunciation (Zuhd)
- Poverty (Faqr)
- Patience (Sabr)
- Trust (Tawakkul)
- Satisfaction (Ridza)

These stages constitute the ascetic and ethical discipline of Sufi. These stages are by one's own efforts. Each of the stages is the result of the stage preceding it. The path of Sufi is not finished until he travels all the stages. In each stage he is to make himself perfect before advancing to the next. After completing all the stages, the seeker becomes Gnostic, A'rif

Repentance: It means, the sins are abandoned and it is resolved never to return to them. A novice should think of his sins with deep regret. Others should repent for forgetting God. They should forget their sins because the thought of the sins comes between them and God.

Abstinence: It means one should refrain himself from indulgence. In Sufi sense, it may be of three types: (a) Abstaining from acts which are doubtful (b) Abstaining from whatever one's consciousness does not allow (c) Abstaining from anything which diverts attention from God.

Renunciation: It means that one should give up voluntarily those things which give physical enjoyment. It has itself three stages. (a) Renouncing something which may bring name and fame. Investigating what to be given up and what not to be given up. (c) Not possessing nor wishing to possess any worldly thing.

Poverty: It means to be stripped of every wish that can turn one's thought from God. It may be of three types: (a) Neither to possess nor seek anything (b) Not to possess anything but not to decline if offered (c) Not to possess anything but to seek the help of friends in extreme need.

Patience: It means to keep mental equilibrium in adversity in distress and in trials from God. It may be of three types. (a) Patience in God. (b) Patience in and for God. (c) Patience in, for and with God. The

first type is of one who can endure distress at one time but loses patience at another time. The second type is of one who is not moved and who does not complain of his bad circumstances, the third type is of one whose equilibrium of mind is not lost in any adversity.

Trust in God: It means complete dependence on God in all affairs. It is of two types: (a) Renunciation of every personal initiative such as seeking food, taking medicine, etc.

Admission of some personal initiative such as working for the purpose of subsistence etc.

Satisfaction: In this stage the Sufi does everything to please God. The end Of the Sufi path is Fana followed by Baqa. Fana means passing away Of the individual self. Fana is a state of losing or forgetting self- consciousness in a state of ecstasy. In the highest stage of Fana, even the consciousness of attaining Fana disappears. This stage is known as Fana al-Fana which makes the beginning of the final stage. Baqa means passing from the phenomenal self to the real self.

ESSENTIALS OF SUFI PSYCHOLOGY: NAFS, HEART AND SOUL

Sufism offers a comprehensive approach to sacred psychology with the ultimate aim of self-knowledge and self-acceptance. The collective experience and knowledge from Sufi practice makes available to us today a thorough

understanding of who we are as human beings and what the purpose of our embodied existence here on earth is. It reveals to us the principal aspects of our nature and a way to transform our selves to become true instruments of manifestation of Divine love, Divine consciousness and Divine action.

The term **‘psycho-spiritual transformation’** highlights the importance of the psychological dimension in the overall process of spiritual transformation. It refers to transformation of consciousness by means of psychological insight. Insight into our emotional nature may help purify our emotions and emotional attachments.

Three principal aspects of the human being and their relationship comprise the triadic foundation of Sufi psychology: **the nafs (self), ghalb (heart) and ruh (soul).**

The spiritual destiny of each human being depends on whether the soul, or nafs, becomes the eventual winner of the inner battle (jihad) between the tendencies exercised by the nafs and the higher qualities of the soul. **If the nafs dominates, the heart becomes darkened and weakened and ineffective. If the forces of the soul take over the heart, it becomes transparent and illumined.**

NAFS

Nafs is simply the equivalent of the English word ‘self’. It has often been translated as ‘ego’ or ‘lower self’ both of which are somewhat inaccurate.

The nafs often craves worldly objects and ignores the wisdom of the heart which is the medium of transmission of the knowledge of the soul. The nafs needs spiritual guidance to awaken to its spiritual potential from the state of sleep-like ignorance (gheflat).

State of Consciousness	Self (<i>Nafs</i>)	Heart (<i>Ghalb</i>)	Soul (<i>Ruh</i>)
inorganic	mineral (<i>nafs al-jamaadi</i>)		mineral
vital+inorganic	vegetable (<i>nafs al-nabaati</i>)		vegetable
mental+vital+inorganic	animal (<i>nafs al-haywani</i>)		animal
rational+mental+vital+inorganic (intelligence/passions)	personal (<i>nafs al-ammarah</i>)	physical heart	human
heart/intuition/intellect	regretful (<i>nafs al-lawwamah</i>)	spiritual heart	individual
truth/inner consciousness	inspired (<i>nafs al-molhamah</i>)	inner heart	secret
unity/gnosis/innermost consciousness	secure/certain (<i>nafs al-motma-ennah</i>)	innermost heart	secret of secrets

Table 11.1. Seven planes of consciousness and the corresponding stations of the self, heart and soul

POINTS TO REMEMBER:

As the principle of embodiment, the human self is the crown of the evolutionary process, presiding over the animal self, vegetable self and the mineral self. *Nafs al-ammarah* undergoes a gradual process of transformation known as **fana**, which means ego-annihilation or loss of self-centered personality characteristics.

The term *samskāra* refers to mental impressions that are acquired through the mind and the senses associated with *nafs* and embodied existence. These impressions create numerous veils of illusion that keep human beings in the state of *gheflat* or sleep-like ignorance and unconsciousness.

In Sufi practice the goal is to open the seeker's heart, which is the gateway to the soul. The open heart first helps to transcend and then transform the *nafs*.

C. INTEGRAL YOGA

INTEGRAL PSYCHOLOGY:

A psychological framework for understanding the total human being is called integral psychology. It is a paradigm of psychology that emerges from a consciousness perspective.

Integral psychology is concerned with the study of the human psyche in its potential fullness. Accordingly, integral psychology is inspired by and founded upon four general postulates essential to an integral world view: **non-duality, multidimensionality, holism, and evolution**. - Sri Aurobindo was a yogi and a mystic. He has said that the materials of his spiritual philosophy were provided by **experiences obtained by practice of yoga**. According to Sri Aurobindo, **yoga has the same relation with the inner being and nature of man as science has with the forces of external nature like steam or electricity**. Yoga, he says, is scientific in that its methods are observation of and experiment with the

states, forces, functions of our subjective, that is, inner being and nature. Yoga is both science and art. It is a science because it knows by experience what man is inwardly and it is an art because it can apply that knowledge to change man's inner being and nature. **Yoga is known as a means of attaining spiritual liberation, *mukti* or *moksha*.**

INTEGRAL YOGA

Śrī Aurobindo is the founder of the system “Samagra Yoga” (Integral Yoga)

Integral yoga, also called **supramental yoga**, is the yoga-based philosophy and practice of Sri Aurobindo and *The Mother* (Mirra Alfassa).

ABOUT SRI AUROBINDO

He was born on 15th August 1872 at Calcutta. For the primary education he was taken along with his elder brothers to England. He returned to India in 1893 and joined the Baroda College as Professor of French and English. In 1906 he became the Principal of the Calcutta College and decided to work for the political and spiritual freedom of India. The pioneering work that he did was evidently a part of his larger work for the entire humanity and for the whole earth. In 1908 he was charged for sedition and kept in Alipur Jail. During the period he had numerous Yogic experiences and realized the assurance of the liberation of the country.

He authored various texts on Yoga and Spirituality. Some of them are *The Life Divine*, *The Synthesis of Yoga*, *The Ideal of Human Unity*, *The Human Cycle*, *Essays on Gita*, *The Supramental Manifestation upon Earth* and *The Savitri*.

Most yoga only develops a single aspect of the being, and has as their aim a state of liberation or transcendence. But the **aim of It is a synthesis of all the following yoga:**

integral yoga is the transformation of the entire being. Because of this, the various elements of one's make-up - Physical, Vital, Mental, Psychic, and Spiritual, and the means of their transformation, are described in great detail by Śrī Aurobindo, who in this way formulates an entire integral psychology.

Maharishi Aurobindo says **the meaning of the word Yoga is to join our eternal consciousness with our true self** – the divine within ourselves. He writes, all life is Yoga, while Yoga as a sadhana is a methodized effort towards self-perfection, which brings to expression the latent, hidden potentialities of being.

CENTRAL PURPOSE OF THE INTEGRAL YOGA

Transformation of our superficial, narrow and fragmentary human way of thinking, seeing, feeling and being into a deep and wide spiritual consciousness and an integrated inner and outer existence and of our ordinary human living into the divine way of life.

SÂDHANÂ OF THE INTEGRAL YOGA

The Sâdhanâ [practice] of the Integral Yoga does not proceed through any set mental teaching or prescribed forms of meditation, mantras or others, **but by aspiration, by a self-concentration inwards or upwards, by a self-opening to an Influence, to the Divine Power above us and its workings, to the Divine Presence in the heart and by the rejection of all that is foreign to these things.** It is only by faith, aspiration and surrender that this self-opening can come.

It is a synthesis of the various branches of Yoga. It is a scientific system for the harmonious development of every aspect of the individual.

Hatha Yoga	Path to physical practice of Yoga.	Includes asanas (postures), pranayama (breath control), mudras, bandhas, kriyas, yogic diet, and deep relaxation. These practices help to purify and strengthen the body and mind.
Raja Yoga	Path of concentration & meditation.	Balancing and controlling the mind utilizing the eight limbs of Yoga delineated in the Yoga Sutras of Patanjali. This leads ultimately to the state of samadhi, or super-consciousness.
Bhakti Yoga	Path of devotion.	Through prayer, kirtan, puja, and practices that help transcend the limited personality or ego, one attains union with the Divine.
Karma Yoga	Path of selfless service.	By engaging in one's duties without attachment to the fruits (or results) of the action, one becomes a pure channel of love and service.
Jnana Yoga	Path of wisdom.	With study, self-analysis, and awareness, one ceases to identify with the body and mind, and realizes the Oneness. The path of wisdom takes an analytical approach to the question: "Who am I?"
Japa Yoga	Path of Mantra Repetition.	The concentrated repetition of a mantra (a sound vibration representing an aspect of the Divine) leads to an awareness of and attunement to this cosmic vibration.

THE YOGA OF SELF-PERFECTION: PRACTICING INTEGRAL YOGA

The major topic in Sri Aurobindo's Integral Yoga is - **Triple transformation** which describes the process through which reality is transformed into the divinity.

TRIPLE TRANSFORMATION

PSYCHIC TRANSFORMATION: You can transform your consciousness by the opening of the psychic being within the heart. With the Divine supporting this psychic being takes charge of the sadhana and turns the whole being to the Truth, the Divine, with results in the mind, the vital and the physical consciousness.

SPIRITUAL TRANSFORMATION: This

leads to **realization of Cosmic Self**. We realize the one Self, Brahman, Divine, first above the body, life, mind and not only within the heart supporting them. It manifests as a Transcending Light, Knowledge, Power, Purity, Peace, Ananda of which we become aware and which descends into the being and progressively replaces the ordinary consciousness itself by its own movements. The final result of this recurring **ascent-descent-assimilation process** is known as **Liberation of the Soul**.

The limitations of the present being can be overcome by the Triple transformation, the process in which the lower nature is transformed into the divine nature.

PSYCHICISATION (PRONOUNCE: SI-KA-SI-ZATION).

Psychicisation is a turn inward, so that one realizes the psychic being, the psychic personality or Divine Soul, in the core of one's being. The Divine Soul serves as a spiritual Guide in the yoga, and enables one to transform the outer being.

SPIRITUALISATION

As a result of the Psychicisation, light, peace, and power descend into the body, transforming all of its parts, physical, vital, and mental. This is the Spiritual transformation, or Spiritualisation. It is equivalent to "enlightenment", as found in Vedanta and Buddhism.

INTERMEDIATE ZONE

Aurobindo asserted that spiritual aspirants may pass through an intermediate zone where experiences of force, inspiration, illumination, light, joy, expansion, power, and freedom from normal limits are possible. These can become associated with personal aspirations, ambitions, notions of spiritual fulfilment and yogic siddhi, and even be falsely interpreted as full spiritual realization. One can pass through this zone, and the associated spiritual dangers, without harm by perceiving its real nature, and seeing through the misleading experiences. Those who go astray in it may end in a spiritual disaster, or may remain stuck there and adopt some half-truth as the whole truth.

SUPRAMENTALISATION

Supramentalisation is the realization of the Super mind, or Supramental consciousness, and the resulting transformation of the entire being. Psychicisation and spiritualization serve as necessary prerequisites for the Supramentalisation of the entire being.

The supramental transformation is the final stage in the integral yoga, enabling the birth of a new individual, fully formed by the supramental power.

(FOR REFERENCE / EXTRA READING)

THREE TYPES OF BEING

Sri Aurobindo discerns three types of being, namely the Outer being, the Inner being, and the Psychic Being.

1. THE OUTER BEING

The Outer Being includes the physical, vital and mental levels of being, which characterizes our everyday consciousness and experience. It includes several levels of the subconscious: a mental subconscious, a vital subconscious, and a physical subconscious, down to the material Inconscient. Integral Yoga involves going beyond this surface consciousness to the larger life of the Inner Being, which is more open to spiritual realization.

2. THE INNER OR SUBLIMINAL BEING

The Inner Being, or Subliminal, includes the inner realms or aspects of the physical, vital and mental being. They have a larger, subtler, freer consciousness than that of the everyday consciousness. Its realization is essential for any higher spiritual realization. The Inner Being is also transitional between the surface or Outer Being and the Psychic or Inmost Being. By doing yoga practice (sadhana), the inner consciousness is being opened, and life turns away from the outward to the inward. The inner consciousness becomes more real than the outer consciousness, and becomes a peace, happiness and closeness to the Divine.

3. THE PSYCHIC BEING

The Psychic Being is Sri Aurobindo's term for the Personal Evolving Soul, the principle of Divine spirit in every individual. The Psychic is the "Innermost Being", the permanent being in us that stands behind and supports the physical, vital and mental principles. It "uses mind, life and body as its instruments," undergoing their fate yet also transcending them.

BUDDHISM

Buddhism begins with a man. In his later years, when India was a fire with his message and kings themselves were bowing before him, people came to him to ask who or what he was? Buddha's answer provided an identity for his entire message.

"Are you a god?" they asked. "An angel?" "NO." "A saint?" "Then what are you?" Buddha answered. "I am awake." His answer became his title, for this is what Buddha means. **The**

Sanskrit root budh denotes both to wake up and to know. Buddha, then, means the "Enlightened One," or the "Awakened One." While the rest of the world was wrapped in the womb of sleep, dreaming a dream known as the waking state of human life one of their numbers roused him. Buddhism begins with a man who shook off the doze, the dream-like vagaries of ordinary awareness. It begins with a man who woke up.

ABOUT SIDDHARTHA

The historical facts of his life are roughly these: He was born around 563 B.C. in what is now Nepal, near the Indian border. His full name was Siddhartha Gautama of the Sakyas. Siddhartha was his given name, Gautama his surname, and Sakya the name of the clan to which his family belonged. His father was a king, but as there were then many kingdoms in the subcontinent of India, it would be more accurate to think of him as a feudal lord. By the standards of the day his upbringing was luxurious. "I wore garments of silk and my attendants held a white umbrella over me My unguents were always from Banaras." He appears to have been exceptionally handsome, for there are numerous references to "the perfection of his visible body. At sixteen he married a neighboring princess, Yasodharā, who bore a son whom they called Rahula.

ABOUT BUDDHA'S DEATH:

After an arduous meal at the age of 80 and around the year 43 B.C., the Buddha died from dysentery after eating a meal of dried boar's flesh in the home Of Cunda the smith. Even on his death bed his mind moved toward others. In the midst of his pain, it occurred to him that Cunda might feel responsible for his death. "is request, therefore was that Cunda be informed that of all the meal he had eaten during his long life only two stood out as having blessed him exceptionally. One was the meal whose strength had enabled him to reach enlightenment under the Bo Tree and the Other One that was opening to him the final gates to nirvana.

• THE FOUR NOBLE TRUTHS

The Four Noble Truths are:

Dukkha

Dukkha-Samudaya, the arising or origin of dukkha,

Dukkha-Nirodha, the cessation of dukkha,

Dukkha-Magga, the way leading to the cessation of dukkha.

DUKKHA: THE FIRST NOBLE TRUTH

The First Noble Truth (Dukkha-ariyasacca) is generally translated by almost all scholars as 'The Noble Truth of Suffering', and it is interpreted to mean that life according to Buddhism is nothing but suffering and pain. First of all, Buddhism is neither pessimistic nor optimistic, but realistic. If anything at all, it is realistic, for it takes a realistic view of life and of the world. It looks at things objectively (yathabhutam). It does not falsely lull you into living in a fool's paradise, nor does it frighten and agonize you with all kinds of imaginary fears and sins. It tells you exactly and objectively what you are and what the world around you is, and shows you the way to perfect freedom, peace, tranquility and happiness.

The term dukkha as the First Noble Truth, which represents the Buddha's view of life and the world, has a deeper philosophical meaning and connotes enormously wider senses.

Fortunately the Buddha's teachings do not end with suffering; rather, they go on to tell us what we can do about it and how to end it.

SAMUDAYA: THE SECOND NOBLE TRUTH

Origin of suffering (Samudāya)

Our day-to-day troubles may seem to have easily identifiable causes: thirst, pain from an injury, sadness from the loss of a loved one. In the second of his Noble Truths, though, the Buddha claimed to have found the cause of all suffering - and it is much more deeply rooted than our immediate worries. The Buddha taught that the root of all suffering is desire, *tanhā*. This comes in three forms, which he described as the Three Roots of Evil, or the Three Fires, or the Three Poisons. These are the three ultimate causes of suffering: Greed and desire, represented in art by a rooster. Ignorance or delusion, represented by a pig. Hatred and destructive urges, represented by a snake. (*Tanhā* is a term in Pali, the language of the Buddhist scriptures that specifically means craving or misplaced desire. Buddhists recognize that there can be positive desires, such as desire for enlightenment and good wishes for others. A neutral term for such desires is *chanda*.)

- **NIRODHA: THE THIRD NOBLE TRUTH**

Cessation of suffering (Nirodha)

The Buddha taught that the way to extinguish desire, which causes suffering, is to liberate oneself from attachment. This is the third Noble Truth - the possibility of liberation.

Nirvana means extinguishing. Attaining nirvana - reaching enlightenment - means extinguishing the three fires of greed, delusion and hatred. Someone who reaches nirvana does not immediately disappear to a heavenly realm. Nirvana is better understood as a state of mind that humans can reach. It is a state of profound spiritual joy, without negative emotions and fears. Someone who has attained enlightenment is filled with compassion for all living things.

- **MAGGA: THE FOURTH NOBLE TRUTH**

Path to the cessation of suffering (Magga). The final Noble Truth is the Buddha's prescription for the end of suffering. This is a set of principles called the **Eightfold Path**.

THE EIGHTFOLD PATH IS ALSO CALLED THE MIDDLE WAY: It avoids both indulgence and severe asceticism, neither of which the Buddha had found helpful in his search for enlightenment.

The eight stages are not to be taken in order, but rather support and reinforce each other:

Right Understanding - Sammā ditthi

Accepting Buddhist teachings. (The Buddha never intended his followers to believe his teachings blindly, but to practice them and judge for themselves whether they were true.)

Right Intention - Sammā saṅkappa

A commitment to cultivate the right attitudes.

Right Speech - Sammā vācā

Speaking truthfully, avoiding slander, gossip and abusive speech.

Right Action - Sammā kammanta

Behaving peacefully and harmoniously; refraining from stealing, killing and overindulgence in sensual pleasure.

Right Livelihood - Sammā ājīva

Avoiding making a living in ways that cause harm, such as exploiting people or killing animals, or trading in intoxicants or weapons.

Right Effort - Sammā vāyāma

Cultivating positive states of mind; freeing one- self from evil and unwholesome states and preventing them arising in future.

Right Mindfulness - Sammā sati

Developing awareness of the body, sensations, feelings and states of mind.

Right Concentration - Sammā Samadhi

Developing the mental focus necessary for this awareness.

The eight stages can be grouped into Wisdom (right understanding and intention), Ethical Conduct (right speech, action and livelihood) and Meditation (right effort, mindfulness and concentration).

The Buddha described the Eightfold Path as a means to enlightenment, like a raft for crossing a river. Once one has reached the opposite shore, one no longer needs the raft and can leave it behind.

PRE-SCIENTIFIC PSYCHOLOGY ERA

(It means: Methodologically poor, Untestable (e.g. concept of denial) and Limited impact on scientific psychology)

(Greek Heritage, for e.g.)

Greek Heritage | The Basics

The ancient Greek philosophers gave us the basic categories of philosophy, beginning with metaphysics. Metaphysics is the part of philosophy that asks questions such as "What is the world made of?" and "What is the ultimate substance of all reality?" In fact, the ancient Greeks were among the first to suggest that there is a "true" reality (noumenon) under the "apparent" reality (phenomenon), an "unseen real" beneath the "unreal seen." The question is, what is this true reality? Is it matter and energy, i.e. something physical? This is called materialism. Or something more spiritual or mental, such as ideas or ideals? This is called idealism.

Materialism and idealism constitute the two extreme answers. Later, we will explore some other possibilities. A second aspect of philosophy is epistemology. Epistemology is the philosophy of knowledge: How do we know what is true or false, what is real or not? Can we know anything for certain, or is it ultimately hopeless?

Again, the Greeks outlined two opposing approaches to the problem of knowledge. One is called empiricism, which says that all knowledge comes through the senses. The other is called rationalism, which says that knowledge is a matter of reason, thought.

Socrates (469-399 BCE)	He, only, orated his teachings, Plato recorded his teachings in writing, Socrates work has been continued by Plato and was presented in Plato's published books. In Book VII of The Republic, Plato wrote the "Allegory of the Cave." This allegory represents the inherent capacity to learn through the concept of being, not becoming. Plato's allegory is written as a philosophical exchange between Socrates and his student. Socrates illustrates a cave wherein a line of prisoners with no possible range of movement face a cave wall.
Plato (428-348 BCE):	Socrates and his student Plato believed that the mind was separate from the body, that it continued to exist after death, and that ideas were innate. PLATO: Plato was an early proponent of individual differences. He recognized that we as individuals naturally differ in general aptitude, intelligence, and talent. He set forth a classification system to qualitatively rate people as being individuals of gold, silver, brass, or iron. In his work, The Republic, Plato describes a model for the perfect society in which a small collection of people with superior intellect, or the Guardians, governed under a philosopher king. Other members of this utopian society were given jobs based on their own personal strengths. Those who expressed great courage were posted as warriors, those who were endowed with a great appreciation of beauty and harmony would serve as artists and poets, and people who displayed little talent or ability would be delegated to the roles of servants and slaves. PLATO'S MIND BRAIN DUALISM: Plato might be given recognition as an early psychologist for his descriptions of the psyche. Plato stated that the psyche is an active entity which drives us and shapes our actions. Our most important task as human beings is the nurturing of our psyche by striving for good and gaining knowledge. For Plato, true knowledge is not gained through the senses, but rather through cognizance of the world of forms.

	Plato's position on the soul and his theory of the world of forms was an early view of mind brain dualism. The soul is in an intermediate position between the world of forms, and the physical world to which to body belongs. The body and soul, for Plato existed as two separate entities. Important Terms associated with Plato: - The allegory of the cave - World of Forms - Forms
Aristotle (384-322 BCE):	Aristotle suggested that the soul is not separable from the body and that knowledge (ideas) grow from experience. Aristotle {timeline} (384-322 BCE) was born in Macedonia. At 17 years old he left for Athens to study under Plato for 22 years. He mentored Alexander the Great. Aristotle founded his own school, the Lyceum in Athens (335 BCE). Unlike Plato, Aristotle did not completely agree that mental representations make us less aware of reality. In fact, he places them at the center of the cognitive psychology and developed the first theory in cognition. He refers to mental images as Phantasmata, and claims that they are necessary not just for recalling but even for thinking. In De Sensu et Sensili, translated as on sense and sensible, Aristotle distinguishes animals from plants based on the fact that they have perception. Perception according to Aristotle is the purpose of the soul. SENSATION AND PERCEPTION Aristotle's theory of sensation and perception was that there are special sensible and common sensible. Special sensible's are single pieces of information such as color determined by only one of our five senses. Common sensible's can be perceived by more than once sense such as movement. Common sense was what Aristotle argued, the five senses working together to form perceptions of something such as an apple. Aristotle felt the most basic sense was the sense of touch which developed to allow us to feel pleasure and pain.

	MEMORY In addition to how we sense objects, Aristotle covers how we imagine objects or recall them. Recollection or memory is when the senses are triggered that eventually causes a representation of the object being sensed in the mind. Furthermore, Aristotle introduced the idea that memory requires a process of recalling the image in the mind. He says that the extent that we recall an image depends on how we represent the object in our mind as well as how we associate it. INTELLIGENCE He believed that there are different types of intelligence or what he called “practical wisdom”. He believed that there were different ways to exercise these wisdoms including understanding (knowing), action (doing), and production (making). The different types of wisdoms paralleled Aristotle’s grouping of works of which were his theoretical works, his practical works, and his productive works. In relation to psychology today, Robert Sternberg (2003) created the Triarchic Theories of Intelligence which mirror Aristotle’s concepts of intelligence.
Hippocrates (450-380 BCE)	Considered by many to be the “father of medicine”. He developed the Theory of four Humors. One of the works he is credited with, “On the Sacred Disease,” is considered an important piece of work to the history of psychology, (Watson, 1968). Hippocrates who determined that the disease, epilepsy, was not “the intervention of a spirit who possessed the body of the sufferer,” but was a disease that had a nature and a cause like any other diseases, and could be treated as such. Hippocrates determined that all mental disorders belonged to one of three categories; mania, melancholia, and phrenitis. Hippocrates went on to express that he believed that this epilepsy was caused in the brain and more specifically was related to a “humoral congestion.” Theory of Four Humors aka (Theory of Disease) Blood, phlegm, yellow bile and black bile are what, according to Hippocrates theory, determine whether a person is healthy or has disease. When the four humors are in a state of balance, then a person is healthy but if there is any change in the amount of even one of the humors, then a person is considered to have a disease. The idea of therapy for curing disease, according to Hippocrates, dealt with the idea of “cure by opposites”. If one of the humors is in excess, then you cure the afflicted person by reducing the amount of that humor. Hippocrates believed that the brain was the center for all intelligence and activity and because of this, any mental illness that was

	discovered was due to brain pathology. He also believed that because so much of conscious life is connected to the brain, including emotions, all of our suffering comes from the brain when it is not healthy. If the brain is too hot, cold, moist, or dry then it is considered to be unhealthy and as a result disease occurs. He explains that the brain can be affected by phlegm and bile, most fevers are the thought to be caused by an imbalance with bile, too much bile can also cause overheating leading to fear, too much phlegm causes the body to be too cold, which can lead to feelings of anxiety and grief and can also result in memory loss.
Juan Huarte de San Juan (1529-88)	He lived and worked as a physician during the Spanish Renaissance era (Chilsom, 1910.) Examen de Ingenious par alas Ciencias, or The Trial of Men's Wits, is a source of information about Huarte's ideas on psychology. Huarte's writing was the first attempt at connecting psychology and physiology. The Trial of Men's Wits is considered the precursor to educational psychology, behaviorism, neuropsychology and psychiatry.
Rene Descartes (1596-1650)	René Descartes {timeline} is an extremely important figure to the history of psychology; he is frequently referred to as the "Father of Modern Philosophy". MIND-BODY DUALISM Descartes, like Plato, believed in soul (mind) body separation but speculated on how the immaterial mind and the physical body communicated. His definition of the mind was that it is "a thinking, non-extended thing while the body is an extended, non-thinking thing". With these two ideas he concluded that the mind is distinct from the body and can exist without it. This belief of the mind and body being two separate things is referred to as "mind-body dualism". Continental rationalism, mainly represented by René Descartes, insisted on the primacy of so-called innate ideas placed in the human mind at birth by God. These include mathematical principles, simple ideas, and the idea of God.
Francis Bacon (1561-1626)	Bacon was one of the founders of modern science, especially the experimental method.
John Locke (1632-1704)	Locke held that the mind is a tabula rasa or blank sheet at birth and experience writes on it.

	He theorized that the acquisition of human knowledge is a twofold process. The first is sensory experience, in which we learn about the world through our five senses. The second is reflection, in which we use our minds and emotions to build those sensory experiences into more complex ideas. As an example, one can experience an apple through sight and taste, and afterward, one can reflect that it is a good thing to eat, and conclude that mankind would benefit if we cultivated apples through farming. TABULA RASA: Theory that individuals are born without built-in mental content and that therefore all knowledge comes from experience or perception. Epistemological proponents of tabula rasa disagree with the doctrine of innatism (represented by: Rene Descartes) which holds that the mind is born already in possession of certain knowledge.
Christian Wolff (1679–1754) (Empirical vs. Rational Psychology)	He first popularized the term psychology to designate the study of mind. Wolff divided the discipline between empirical and rational psychology. The data of mind that resulted from observing ourselves and others constituted empirical psychology; rational psychology referred to the interpretation of the data of empirical psychology through the use of reason and logic. These psychologies were characterized as using knowledge acquired through experience (empirical psychology) or using knowledge that the mind possesses independent of experience (rational psychology) (Murray, 1988).
Immanuel Kant (1724–1804)	He denied the validity of any rational psychology because, he argued, rational mental processes must be activated by mental content derived from experience; therefore, the study of mind must be confined to questions appropriate to an empirical psychology (Leary, 1978). An empirical psychology of mental content could not, Kant contended, become a proper natural science because mental events cannot be quantified (i.e., measured or weighed), and thus its data are neither capable of being described mathematically nor subject to experimental manipulation. Finally, Kant asserted, the method of observing the mind - introspection - distorts the events observed by observing them. However, Kant suggested, psychology might improve its status as an empirical science by adopting the methods of anthropology to observe the activities of human beings in realistic settings.
<u>EXTRA REFERENCES FOR TIMELINE</u> (NOT IMPORTANT FOR EXAM but FOR PROVIDING MEANING TO THE TIMELINE) Jakob Friederich Fries (1773–1843) raised the status of introspection by arguing that it was not inherently more problematic than observing external phenomena; if introspection	

was unreliable, at least it was not any more so than any other kind of observation.

At the same time, **Johann Friederich Herbart (1776–1841)** offered a system of psychology that was both empirical and mathematical. If psychology needed to be mathematical to be a true science, Herbart proposed that numbers could be assigned to mental events of different intensities and a mathematical description of the relationship among them could be formulated. Herbart could assign numbers to describe experiences of different intensities, but he could not actually measure the subjective intensities in accord with an objective standard.

Eduard Friederich Beneke (1798–1854) argued that it was premature to apply mathematics to relationships among mental events absent more accurate empirical observations and reliable means of measurement; psychology could hope to become an experimental discipline by testing “**empirical results and theoretical hypotheses under controlled conditions and with the systematic variation of variables**”.

Scientific Psychology

School of psychology	Description	Important contributors
Structuralism	Uses the method of introspection to identify the basic elements or “structures” of psychological experience	Wilhelm Wundt, Edward B. Titchener
Functionalism	Attempts to understand why animals and humans have developed the particular psychological aspects that they currently possess	William James
Psychodynamic	Focuses on the role of our unconscious thoughts, feelings, and memories and our early childhood experiences in determining behavior	Sigmund Freud, Carl Jung, Alfred Adler, Erik Erickson
Behaviorism	Based on the premise that it is not possible to objectively study the mind, and therefore that psychologists should limit their attention to the study of behavior itself	John B. Watson, B. F. Skinner
Cognitive	The study of mental processes, including perception, thinking, memory, and judgments	Hermann Ebbinghaus, Sir Frederic Bartlett, Jean Piaget
Social-cultural	The study of how the social situations and the cultures in which people find themselves influence thinking and behavior	Fritz Heider, Leon Festinger, Stanley Schachter

Structuralism (1870s-1900)

Wilhelm Wundt—the father of psychology, set up the first psychology lab in 1879.

Wundt’s research in his laboratory in Leipzig focused on the nature of consciousness itself. Wundt and his students believed that it was possible to analyze the basic elements of the mind and to classify our conscious experiences scientifically. Wundt began the field known as structuralism, a school of psychology whose goal was to identify the basic elements or “structures” of psychological experience. Its goal was to create a “periodic table” of the “elements of sensations,”

similar to the periodic table of elements that had recently been created in chemistry. Structuralists used the method of introspection to attempt to create a map of the elements of consciousness. Introspection involves asking research participants to describe exactly what they experience as they work on mental tasks, such as viewing colors, reading a page in a book, or performing a math problem. A participant who is reading a book might report, for instance, that he saw some black and colored straight and curved marks on a white background. In other studies the structuralists used newly invented reaction time instruments to systematically assess not only what the participants were thinking but how long it took them to do so. Wundt discovered that it took people longer to report what sound they had just heard than to simply respond that they had heard the sound. These studies marked the first time researchers realized that there is a difference between the sensation of a stimulus and the perception of that stimulus, and the idea of using reaction times to study mental events has now become a mainstay of cognitive psychology.

Perhaps the best known of the structuralists was **Edward Bradford Titchener (1867–1927)**. Titchener was a student of Wundt who came to the United States in the late 1800s and founded a laboratory at Cornell University. In his research using introspection, Titchener and his students claimed to have identified more than 40,000 sensations, including those relating to vision, hearing, and taste.

An important aspect of the structuralist approach was that it was rigorous and scientific. The research marked the beginning of psychology as a science, because it demonstrated that mental events could be quantified. But the structuralists also discovered the limitations of introspection. Even highly trained research participants were often unable to report on their subjective experiences. When the participants were asked to do simple math problems, they could easily do them, but they could not easily answer how they did them. Thus the structuralists were the first to realize the importance of unconscious processes—that many important aspects of human psychology occur outside our conscious awareness, and that psychologists cannot expect research participants to be able to accurately report on all of their experiences.

Edward Titchener— student of Wundt, introduced experimental psychology to the USA.

Functionalism (1880s-current)	In contrast to Wundt, who attempted to understand the nature of consciousness, the goal of William James and the other members of the school of functionalism was to understand why animals and humans have developed the particular psychological aspects that they currently possess (Hunt, 1993). James and the other members of the functionalist school were influenced by Charles Darwin's (1809–1882) theory of natural selection, which proposed that the physical characteristics of animals and humans evolved because they were useful, or functional. The functionalists believed that Darwin's theory applied to psychological characteristics too. Just as some animals have developed strong muscles to allow them to run fast, the human brain, so functionalists thought, must have adapted to serve a particular function in human experience. Although functionalism no longer exists as a school of psychology, its basic principles have been absorbed into psychology and continue to influence it in many ways. The work of the functionalists has developed into the field of evolutionary psychology, a branch of psychology that applies the Darwinian theory of natural selection to human and animal behavior. Evolutionary psychology accepts the functionalists' basic assumption, namely that many human psychological systems, including memory, emotion, and personality, serve key adaptive functions.
Behaviorism (1915–current)	By 1920, Structuralism and Functionalism were replaced by Behaviorism, Gestalt Psychology & Psychoanalysis. Both structuralism and functionalism were essentially studies of the mind. The psychologists associated with the school of behaviorism, on the other hand, were reacting in part to the difficulties psychologists encountered when they tried to use introspection to understand behavior. Behaviorism is a school of psychology that is based on the premise that it is not possible to objectively study the mind, and therefore that psychologists should limit their attention to the study of behavior itself. Behaviorists believe that the human mind is a “black box” into which stimuli are sent and from which responses are received. They argue that there is no point in trying to determine what happens in the box because we can successfully predict behavior without knowing what happens inside the mind. Furthermore, behaviorists believe that it is possible to develop laws of learning that can explain all behaviors. The first behaviorist was the American psychologist John B. Watson (1878–1958). Watson was influenced in large part by the work of the Russian physiologist Ivan Pavlov (1849–1936), who had discovered that dogs would salivate at the sound of a tone that had previously been associated with the presentation of food. Watson and the other behaviorists began to use these ideas to explain how events that people and other organisms experienced in their environment (stimuli) could produce specific behaviors (responses). For instance, in Pavlov's research the stimulus (either the food or, after learning, the tone) would produce

the response of salivation in the dogs.

In his research Watson found that systematically exposing a child to fearful stimuli in the presence of objects that did not themselves elicit fear could lead the child to respond with a fearful behavior to the presence of the stimulus (Watson & Rayner, 1920; Beck, Levinson,

Irons, 2009). [11] In the best known of his studies, an 8-month-old boy named Little Albert was used as the subject. Here is a summary of the findings:

The boy was placed in the middle of a room; a white laboratory rat was placed near him and he was allowed to play with it. The child showed no fear of the rat. In later trials, the researchers made a loud sound behind Albert's back by striking a steel bar with a hammer whenever the baby touched the rat. The child cried when he heard the noise. After several such pairings of the two stimuli, the child was again shown the rat. Now, however, he cried and tried to move away from the rat.

In line with the behaviorist approach, the boy had learned to associate the white rat with the loud noise, resulting in crying. The most famous behaviorist was **Burrhus Frederick (B. F.) Skinner (1904– 1990)**, who expanded the principles of behaviorism and also brought them to the attention of the public at large. Skinner used the ideas of stimulus and response, along with the application of rewards or reinforcements, to train pigeons and other animals. And he used the general principles of behaviorism to develop theories about how best to teach children and how to create societies that were peaceful and productive. Skinner even developed a method for studying thoughts and feelings using the behaviorist approach.

The behaviorists made substantial contributions to psychology by identifying the principles of learning. Although the behaviorists were incorrect in their beliefs that it was not possible to measure thoughts and feelings, their ideas provided new ideas that helped further our understanding regarding the nature-nurture debate as well as the question of free will. The ideas of behaviorism are fundamental to psychology and have been developed to help us better understand the role of prior experiences in a variety of areas of psychology.

The methods and subject matter of physiology, especially sensory physiology, helped to provide the scientific basis for psychology.

SENSORY PHYSIOLOGY

Johannes Müller (1801–1858), the “Father of Physiology,” produced the classic systematic handbook (*Handbuch der Physiologie des Menschen*, 1833–1840) that set forth

what was then known about human physiology and offered observations and hypotheses for further research. Among the formulations that Müller provided in the Handbuch was the law of specific nerve energies, which stated that the mind is not directly aware of objects as such but can only be aware of the stimulation in the brain conveyed by sensory nerves.

The following Approaches would be understood using the following Case. The Case would be discussed from the lens of each therapy, thus mentioned.

Case I:

(The Case study has been taken from **Brief Interventions and Brief Therapies for Substance Abuse. Treatment Improvement Protocol (TIP) Series, No.**

34. Center for Substance Abuse Treatment.)

Sandra is a 38-year-old African-American woman who has abused a number of substances, including cocaine, heroin, alcohol, and marijuana over the past 15 years. She left high school and was a prostitute for 5 years. Later she found a job as a sales clerk at a home furnishings store. Sandra had two children in her early twenties, a daughter who is now 15, and a son, aged 18. Because of her substance abuse problems, they live with other relatives who agreed to raise them. Sandra has been in treatment repeatedly and has remained substance free for the last 5 years, with several minor relapses. She has been married for 2 years, to Steve, a carpenter; he is substance free and supports her attempts to stay away from substances.

Last month she became symptomatic with AIDS. She has been HIV-positive for 5 years but had not developed any illnesses related to the disease. Sandra has practiced safe sex with her husband who knew of her HIV status. Recently, after learning from the physician at her clinic about her HIV symptoms, she began to "shoot up," which led her back into treatment. Out of fear, she came to the treatment center and asked to see a counselor at the clinic one day after work. She is worried about her marriage and that her husband will be devastated by this news. She is afraid she is no longer strong enough to stay away from drugs since discovering the onset of AIDS. She is also concerned about her children and her job. Uncertain of how she will keep on living, she is also terrified of dying.

Gestalt Psychology

Gestalt-German word meaning “form” or “configuration”.

Although Max Wertheimer could be considered as the founding father of the Berlin school, his younger colleagues, Kurt Koffka and Wolfgang Köhler were just as important in its further development. Gestalt Psychologist’s primary interest was perception they believed that perceptual experiences depend on the patterns formed by stimulus and on the organization of experience.

It is a school of psychology founded in the 20th century that provided the foundation for the modern study of perception. Gestalt theory emphasizes that the whole of anything is greater than its parts. That is, the attributes of the whole are not deducible from analysis of the parts in isolation.

Gestalt Psychology is focused on the experience of contact that occurs in the here and now.

School of thought that emphasized the total experience of the individual and not just parts of the mind or behavior.

GESTALT PRINCIPLES

According to the gestalt psychologists, the way we form our perceptions are guided by certain principles or laws. These principles or laws determine what we see or make of things or situation.

Law of Proximity: Elements that are closer together will be perceived as coherent object.

Law of Similarity: Elements that look similar will be perceived as part of the same form. There seems to be a triangle in the square. We link similar elements together.

Law of Closure: We tend to fill the gaps or “close” the figures we perceive. We enclose a space by completing a contour and ignoring gaps in the figure.

Law of Good Continuation: Individuals have the tendency to continue contours whenever the elements of the pattern establish an implied direction. People tend to draw a good continuous line.

Law of Good Pragnanz: The stimulus will be organized into as good as figure as possible. In this example, good refers to symmetry, simplicity, and regularity. The figure is perceived as a square overlapping a triangle, not a combination of several complicated shapes. Based on our experiences with perception, we “expect” certain patterns and therefore perceive that expected pattern

{The German word “Prägnanz” is derived from the verb “prägen,” - to mint a coin. Hence, by describing the principle of Prägnanz as the tendency towards the formation of Gestalten which are as regular, simple, symmetric (“ausgezeichnet”, according to Wertheimer’s term). “Prägnanz” has often been translated as “goodness”}.

Law of Figure/Ground: We tend to pay attention and perceive things in the foreground first. A stimulus will be perceived as separate from its ground.

The idea of insight learning was first developed by Wolfgang Kohler in which he describes experiments with apes where the apes could use boxes and sticks as tools to solve problems.

INSIGHT LEARNING: In each of these problems, the important aspect of learning was not reinforcement, but the coordination of

thinking to create new organization (of material. Kohler referred to this behavior as insight or discovery learning. Kohler proposed the view that insight follows from the characteristics of objects under consideration. His theory suggested that learning could occur when an individual perceives the relationships of the elements before him and **reorganizes these elements and comes to a greater understanding or insight.** Learning could occur without reinforcement, and once it occurs, no review, training, or investigation is necessary.

RELEVANCE OF GESTALT IN LEARNING:

Gestalt is a theory of learning that focuses on the mind's perspective. It is useful as a behavioral tool as it enables the teacher to channel the pupil's energy into thinking of an item or subject as parts of a whole, e.g. a car, being metal, paint, wheels etc. By thinking of components and breaking down a situation it enables for a more psychological process to take place and over time will broaden a pupil's mind into thinking of the sum of the whole rather than just a complete thing of situation.

GESTALT THERAPY (BY FRITZ PERLS)

Fritz Perls coined the term 'Gestalt therapy' to identify the form of psychotherapy that he developed with his wife, Laura Perls, in the 1940s and 1950s.

Gestalt theory holds that the analysis of parts can never provide an understanding of the whole. In a therapeutic setting, this approach opposes the notion that human beings can be understood entirely through a rational, mechanistic, scientific process. The proponents of Gestalt therapy insist that the experiential world of a client can be understood only through that individual's direct experience and description. Gestalt therapists seek to help their clients gain awareness of themselves and the world. Discomfort arises from leaving elements and experiences of the psyche incomplete--primarily past relationships and intrapsychic conflicts that are unresolved, which Perls calls "unfinished business" (Perls, 1969). According to Gestalt theory

The organism should be seen as a whole (physical behavior is an important component, as is a client's mental and emotional life). Being in the "here and now" (i.e., being aware of present experience) is of primary importance. How is more important than why (i.e., causes are not as important as results). The individual's inner experience is central.

For Gestalt therapists the "power is in the present" (Polster and Polster, 1973). This means that the "now" is the only place where awareness, responsibility, and change can occur. Therefore, the process of therapy is to help the client make contact with the present moment.

Rather than seeking detailed intellectual analysis, the Gestalt therapist looks to create a "safe emergency" in the therapeutic encounter. Perls' invocation to "lose your mind and come to your senses" implies that a feeling-level, "here and now" experience is the optimal condition for therapeutic work. This may be accomplished in a fairly short amount of time by explicitly asking clients to pay attention (e.g., "What are you aware of now? How does your fear feel to you?"). The therapist may point out how the client could be avoiding the present moment through inauthentic "games" or ways of relating such as "talking about" feelings rather than experiencing them directly. Clients may be asked to exaggerate certain expressions (e.g., pounding a fist) or role-play certain internal dialogs (e.g., through an empty chair technique). These may all serve the goal of helping clients move into the immediacy of their experience rather than remaining distant from it through intellectualization or substance abuse.

The term contact in Gestalt refers to meeting oneself and what is other than oneself. Without appropriate contact and contact boundaries there is no real meeting of the world. Instead, one remains either engulfed by the world on one hand or, on the other hand, distant from the world and people.

RESPONSE TO THE CASE STUDY:

The Gestalt therapist begins with Sandra's current experience of the world, starting with awareness and attention. The therapist may simply help her become aware of basic sights, sounds, somatic reactions, feelings, and thoughts as well as what her attention drifts to. The immediate contact between therapist and client is a component of the "now" where these sensations are explored directly. The therapist might notice and ask about her style of eye contact, or her fidgeting body, or stream of thoughts (e.g., "What is it like to make eye contact now? What is the sensation in your body at this moment?").

Sandra may also identify certain issues such as substance abuse, relationship difficulties, and the threat of death from AIDS that seem to dominate her life. The therapist might invite her to name and explore the sensation that the thought of death, for example, brings; perhaps this involves a sense of a void, or feeling cold and dark, or a feeling of engulfment. She then may be asked to become these sensations--for example, the therapist may ask her to be "the void" and encourage her to speak as if she were that void. This may then open possibilities for a dialog with the void through acting out the

	opposite polarity: separateness and choice. This might involve using an empty chair technique in which the client would literally move into the chair of the "void," speak as if she were that, and then move into an opposite chair and respond in a dialog. A therapist could also explore her introjection through questions such as, "How is this void different or the same as from the feeling of alcohol or in relationships with your children or husband?" She might also use this same technique to dialog with family members, or certain aspects of herself. Sandra seems to have a great deal of "unfinished business" that involves unexpressed feelings (e.g., anger, longing, hurt). Experimentation with these sensations may begin to free her to express and meet these feelings more directly. All of this work encourages Sandra's experimentation with new ways of relating both during and outside of the session in order to move into the "here and now" and work toward the resolution of "unfinished business."
Psychoanalysis	A theory of personality and method of psychotherapy originated by Sigmund Freud. Concept of unconsciousness, thoughts, attitudes, impulses etc that we are unaware of. Repressed Unconscious wishes and desires expressed in Dreams, slips of tongue & physical mannerism. Method of Free Association. THE PRE-OEDIPAL STAGE Freud claimed that all human beings are born with certain instincts, i.e. with a natural tendency to satisfy their biologically determined needs for food, shelter and warmth. The satisfaction of these needs is both practical and a source of pleasure which Freud refers to as "sexual". Thus, when the infant, sucking at its mother's breast discovers the pleasure inherent in this activity, the first glimmers of sexuality are awakened. The child discovers an erotogenic zone which may be reactivated later in life through thumb sucking or kissing. Through this intimate interaction with the mother, upon whom the child is dependent, a sexual drive emerges. As this drive is separated out from its original function as a purely biological instinct, it achieves a relative autonomy. During the early stages of childhood development, other erotogenic zones emerge. The oral stage, associated with the drive to "incorporate" objects through the mouth, is followed by the anal stage during which the anus becomes an erotogenic zone as the child takes pleasure in defecation. This pleasure is characterized by Freud as "sadistic" because the child is

understood to be taking delight in expulsion and destruction. The anal stage is also associated with the desire for retention and possessive control (as in "granting or withholding" the faeces). The next stage the child enters is the phallic stage when the sexual drive is focused on the genitals. (Freud refers to this stage as "phallic" rather than "genital" because, he claims, only the male organ is recognized as significant.)

What is happening in this process -- though the stages overlap, and should not be seen as a strict sequence -- is a gradual organization of the libidinal drives, but one still centered on the child's own body. The drives themselves are extremely flexible, in no sense fixed like biological instinct: their objects are contingent and replaceable, and one sexual drive can substitute for another. What we can imagine in the early years of the child's life, then, is not a unified subject confronting and desiring a stable object, but a complex, shifting field of force in which the subject (the child itself) is caught up and dispersed, in which it has as yet no center of identity and in which the boundaries between itself and the external world are indeterminate. Within this field of libidinal force, objects and part-objects emerge and disappear again, shift places kaleidoscopically, and prominent among such objects is the child's body as the play of drives laps across it. One can speak of this as an 'auto eroticism', within which Freud sometimes includes the whole of infantile sexuality: the child takes erotic delight in its own body, but without as yet being able to view its body as a complete object. Auto-eroticism must thus be distinguished from what Freud will call 'narcissism', a state in which one's body or ego as a whole is 'cathected', or taken as an object of desire.

The child in this state is described by Freud as "anarchic, sadistic, aggressive, self-involved and remorselessly pleasure-seeking" - wholly within the grip of the pleasure principle. It is also ungendered. That is to say, even though it is riddled with sexual drives, it draws no distinction between the gender categories masculine and feminine.

THE OEDIPUS COMPLEX: GENDERING THE SUBJECT

At the center of Freud's theory of childhood development is the Oedipus Complex. According to Freud, a boy's close relation to his mother, as the primary love-object, leads to a desire for complete union with her. A girl, on the other hand, who is similarly attached to the mother and thus caught up in a "homosexual" desire, directs her libido (love, sexual energy broadly construed) toward her father (for reasons which we'll consider shortly). This produces a triadic relationship regardless of one's sex, with the parent of the same sex cast in the role of a rival for the affections of the parent of the opposite sex. The boy will eventually abandon his incestuous desire for his mother out of fear of being castrated by his father. (This fear arises when the boy comes to realize that females are "castrated" and

imagines that this may be his fate if he does not subordinate his desire for the mother.) Thus, the boy represses his incestuous desire, adjusts to the reality principle, and waits for the day when he will be the patriarch. In this way the boy identifies with his father and the symbolic role of manhood.

The girl's route through the Oedipal stage is far more problematic in Freud's view. "Realizing" that she is castrated and thus inferior, the girl turns away from her similarly castrated mother and attempts to "seduce" her father. When this fails, she returns to the mother and identifies with her feminine role. However, she still envies the penis that she will never have; so she unconsciously substitutes a desire to have her father's baby. (How she goes about giving up this desire is not made clear. Since she is already "castrated", fear of castration will not do the job.) Needless to say, Freud's theory shows little insight into femininity and the experience of women.

His claim that female sexuality is a "dark continent" says as much.

THE UNCONSCIOUS

As we say, the unconscious is that part of the mind that lies outside the somewhat vague and porous boundaries of consciousness, and is constructed in part by the repression of that which is too painful to remain in consciousness. (Not everything in the unconscious is repressed. However, repression is the ego's primary defense against disruption.) Freud distinguishes repression from sublimation - the rechanneling of drives that cannot be given an acceptable outlet. The unconscious also contains what Freud calls laws of transformation. These are the principles that govern the process of repression and sublimation. In general, we can say that the unconscious serves the theoretical function of making the relation between childhood experience and adult behavior intelligible.

EGO, ID AND SUPER-EGO

According to Freud, the ego is an aspect of the subject that emerges from the id - the biological, inherited, unconscious source of sexual drives, instincts, and irrational impulses. The ego develops out of the id's interaction with the external world. It is produced from the non-biological (social and familial) forces brought to bear on one's biological development and functions as an intermediary between the demands of the id and the external world. Thus, the ego can be thought of as a variable aspect of the subject constructed as a system of beliefs that organize one's dealings with the internal and external demands of life according to certain laws referred to by Freud as secondary process. It reconciles the biological, instinctual demands and drives (both unifying and destructive in nature) of the id (governed by primary process) with the socially determined constraints of the super-ego (internalized rules placing limits on the subject's satisfactions and pleasures) and the demands of reality. The healthy, mature ego translates the demands of both the id and the

	super-ego into terms which allow admission of them without destruction. Thus, constructive acceptance and transformation of the demands made by both the id and the superego are techniques of the ego and essential elements of mental health. Psychoanalytic therapy involves reliving repressed fantasies and fears both in feeling and in thought. This process involves a transference, i.e. a projection of the attitudes and emotions, originally directed towards the parents, onto the analyst. This is necessary for successful treatment. Access to these repressed fears is gained often through dream interpretation, where the manifest content in dreams is understood as a symbolic expression of the hidden or latent content. (Internal censorship demands that the wish be transformed, leading to a disguised or symbolic representation.) The sources of dream content results from – - lost memories - linguistic symbols - repressed experiences - "archaic" material inherited but not directly experienced. Dreams are "guardians of sleep", i.e. wish fulfillments that arise in response to inner conflicts and tensions whose function is to allow the subject to continue sleeping. Dream-Work is the production of dreams during sleep -- the translation of demands arising from the unconscious into symbolic objects of the preconscious and eventually the conscious mind of the subject. Dream Interpretation is the decoding of the symbols (manifest content) and the recovery of their latent content, i.e. the unconscious and, hence, hidden tensions and conflicts that give rise to the dreams in the first place. LIMITATION OF PSYCHOANALYSIS: 1. Freud's hypotheses are neither verifiable nor falsifiable. It is not clear what would count as evidence sufficient to confirm or refute theoretical claims. 2. The theory is based on an inadequate conceptualization of the experience of women. 3. The theory overemphasizes the role of sexuality in human psychological development and experience.
Humanistic-Existential	PHILOSOPHICAL ROOTS: Existential psychology is based on existential philosophy. Its philosophical roots can be traced to the works of Søren Kierkegaard (1813-1855), Friedrich Nietzsche (1844-1900), Karl Jaspers (1883-

1969), Edmund Husserl (1859-1938), and Martin Heidegger (1889-1976).

PSYCHOLOGICAL ROOTS:

Humanistic psychology is a psychological perspective which rose to prominence in the mid-20th century, drawing on the work of early pioneers like Carl Rogers and the philosophies of existentialism and phenomenology. It adopts a holistic approach to human existence through investigations of meaning, values, freedom, tragedy, personal responsibility, human potential, spirituality, and self-actualization.

HUMANISTIC PSYCHOLOGY:

Humanistic psychology, often referred to as the "**third force**" besides behaviorism and psychoanalysis, is concerned with human potential and the individual's unique personal experience. Humanistic psychologists generally do not deny the importance of many principles of behaviorism and psychoanalysis. They value the awareness of antecedents to behavior as well as the importance of childhood experiences and unconscious psychological processes. Humanistic psychologists would argue, however, that humans are more than the collection of behaviors or objects of unconscious forces. Therefore, humanistic psychology often is described as holistic in the sense that it tends to be inclusive and accepting of various theoretical traditions and therapeutic practices. The emphasis for many humanistic therapists is the primacy of establishing a therapeutic relationship that is collaborative, accepting, authentic, and honors the unique world in which the client lives. The humanistic approach is also holistic in that it assumes an interrelatedness between the client's psychological, biological, social, and spiritual dimensions. Humanistic psychology assumes that people have an innate capacity toward self-understanding and psychological health.

Some of the key proponents of this approach include Abraham Maslow, who popularized the concept of "self-actualization," **Carl Rogers, who formulated person-centered therapy, and Fritz Perls, whose Gestalt therapy focused on the wholeness of an individual's experience at any given moment.** Some of the essential characteristics of humanistic therapy are

Empathic understanding of the client's frame of reference and subjective experience

Respect for the client's cultural values and freedom to exercise choice

Exploration of problems through an authentic and collaborative approach to helping the client develop insight,

courage, and responsibility

Exploration of goals and expectations, including articulation of what the client wants to accomplish and hopes to gain from treatment

Clarification of the helping role by defining the therapist's role but respecting the self determination of the client

Assessment and enhancement of client motivation both collaboratively and authentically

Negotiation of a contract by formally or informally asking, "Where do we go from here?"

Demonstration of authenticity by setting a tone of genuine, authentic encounter

HUMANISTIC APPROACH TO THE CASE STUDY:

A client-centered therapist would engage in reflective listening, accepting the client and her past, and clarifying her current situation and feelings. As Sandra developed trust in the therapist, he would begin to emphasize her positive characteristics and her potential to make meaningful choices to become the person she wants to (and can) become. Another goal of therapy would be to help her develop sufficient insight so that she can make choices that reflect more closely the values and principles to which she aspires. For example, she may want to tell her husband about her symptoms and try to strengthen her marriage.

If Sandra began to feel guilt about her past as a prostitute, the therapist would demonstrate appreciation of her struggle to accept that aspect of herself, highlighting the fact that she did eventually choose to leave it. He may note that she did the best she could at that time and underscore her current commitment to choose a better life. Sandra would be supported and accepted, not criticized. She would be encouraged to express her fear of death and the effect this fear has on her. This might be the first time in her life that someone has been unconditionally accepting of her or focused on her strengths rather than her failings. She apparently has the ability to solve problems, which is reflected by her return to therapy and her insight about needing help. By being understood and accepted, her self-esteem and sense of hope would increase and her shame would decrease. She would feel supported in making critical choices in her life and more confident to resume her recovery.

EXISTENTIAL PSYCHOLOGY:

The existential approach to therapy emphasizes the following six propositions:

All persons have the capacity for self-awareness. As free beings, everyone must accept the responsibility that comes with freedom. Each person has a unique identity that can only be known through relationships with others. Each person must continually recreate himself. The meaning of life and of existence is never fixed; rather, it constantly changes. Anxiety is part of the human condition. Death is a basic human condition that gives significance to life.

The core question addressed in existential therapy is "How do I exist?" in the face of uncertainty, conflict, or death. An individual achieves authenticity through courage and is thus able to define and discover his own meaning in the present and the future. There are important choices to be made (e.g., to have true freedom and to take responsibility for one's life, one must face uncertainty and give up a false sense of security).

A core characteristic of the existential view is that an individual is a "being in the world" who has biological, social, and psychological needs. Being in the world involves the physical world, the world of relationships with others, and one's own relationship to self (May and Yalom, 1995, p. 265). The "authentic" individual values symbolization, imagination, and judgment and is able to use these tools to continually create personal meaning.

Existential therapy focuses on specific concerns rooted in the individual's existence. The contemporary existential psychotherapist, Irvin Yalom, identifies these concerns as death, isolation, freedom, and emptiness. Existential therapy focuses on the anxiety that occurs when a client confronts the conflict inherent in life. The role of the therapist is to help the client focus on personal responsibility for making decisions, and the therapist may integrate some humanistic approaches and techniques. Yalom, for example, perceives the therapist as a "fellow traveler" through life, and he uses empathy and support to elicit insight and choices. He strongly believes that because people exist in the presence of others, the relational context of group therapy is an effective approach (Yalom, 1980).

EXISTENTIAL APPROACH TO THE CASE STUDY:

An existential therapist may help Sandra understand that her diagnosis of AIDS forces her to confront the possibility of death and, consequently, face the responsibilities thrust upon her by life. The therapist could accomplish this by helping her understand that her life (like everyone else's) is finite. Therefore, she is challenged to forge meaning from her life and make difficult decisions about her relationships and ways of dealing (or not dealing) with choices about

	substance abuse. The focus in her therapy would be on choosing the life she wants to live. The therapist would assist her in dealing constructively with anxiety so that she can find meaning in the rest of her life. This could be accomplished by engaging her in the struggle to assume authorship of her choices. She may be encouraged to "play out" scenarios of choices she faces and acknowledge the accompanying fears and anxieties. She might be asked, "What keeps you from sharing your fears with your husband, and accepting the possibility of his support?" or "Imagine yourself expressing your love for your children and regret for the mistakes you have made." Thus, the therapist would help her understand that making difficult choices in the face of death is actually a way to find integrity, wholeness, and meaning. The teachings of the existential therapist, Yalom, can be a useful resource in dealing with issues related to death, since he has worked with terminally ill cancer patients for many years, helping them to use their crisis and their danger as an opportunity for change (Yalom, 1998). Yalom explains that although death is a primary source of anxiety for a client, incorporating death into life can enrich life and allow one to live more purposefully.
Transpersonal:	Transpersonal psychology emerged as a "fourth force" in psychology in the late 1960s and has strong roots in humanistic and existential psychologies, Jungian analysis, the East-West dialog, and ancient wisdom traditions. Transpersonal therapy may be thought of as a bridge between psychological and spiritual practice. A transpersonal approach emphasizes development of the individual beyond, but including, the ego. It acknowledges the human spiritual quest and recognizes the human striving for unity, ultimate truth, and profound freedom. It cultivates intuitive ways of knowing that complement rational and sensory modes. This approach also recognizes the potential for growth inherent in "peak" experiences and other shifts in consciousness. Although grounded in psychological theory, transpersonal practitioners also tend to incorporate perspectives from ancient wisdom traditions. The practice of transpersonal therapy is defined more by its orientation and scope rather than by a particular set of techniques or methods (Boorstein, 1980). Wittine suggests five postulates for a transpersonal psychotherapy (Wittine, 1989): Transpersonal psychotherapy is an approach to healing and growth that recognizes the centrality of the self in the therapeutic process. Transpersonal psychotherapy values wholeness of being and self-realization on all levels of the spectrum of identity (i.e., egoic, existential, transpersonal).

Transpersonal psychotherapy is a process of awakening from a limited personal identity to expanded universal knowledge of self. Transpersonal psychotherapy makes use of the healing restorative nature of subjective awareness and intuition in the process of awakening. In transpersonal psychotherapy, the therapeutic relationship is a vehicle for the process of awakening in both client and therapist.

Integrating insights and practices in everyday life is the goal of every therapy. Bringing the transpersonal dimension to the forefront may involve the following:

Exploration of "inner voices" including those of a higher self that provides guidance for growth of the individual (Rowan, 1993)

Refinement of intuition or nonrational knowing

Practice of creativity in "formal" (art) or informal (personal relationships) encounters

Meditation

Loving service

Cultivation of mindfulness

Use of dreams and imagery

These techniques may be taught and supported explicitly in the therapy session. At times, a therapist may directly cultivate shifts in consciousness (e.g., through meditation [Weil, 1972], or imaginal work [Johnson, 1987]), providing immediate insight and inspiration that may not be available through more conventional means (Hart, 1998). This may provide clients with a skill they can practice on their own; initiating such activity represents a potential for brief intervention.

Transpersonal therapy recognizes the need for basic psychological development to be integrated with spiritual growth (Nelson, 1994). Without such integration there is danger of "spiritual bypassing," where issues of basic psychological functioning are avoided in the name of spiritual development. In other words, the basic psychological work should be undertaken first.

RESPONSE TO THE CASE STUDY:

As the existentialists remind us, there is nothing like death to rivet our attention. A glimpse of death--for example, seeing the aftermath of a serious car crash--reminds the witness of how valuable life is, bringing up other issues as well. Sandra is now confronted with death due to AIDS. This opportunity to face death and life squarely provides a chance to reconsider and reprioritize her life. In fact, it could be argued that the best catalyst to brief therapy may be a death

	sentence precisely because it has the potential to wake up an individual. In many respects, helping the client wake from habitual, mechanical routines that are often based on ego protection and move toward an appreciation that the individual is not bound to or defined by a limited ego, is the goal of transpersonal therapy. This can be seen as a transformation of identity. Many inspiring instances of people facing death, including death through AIDS, have shown that emergent spirituality can change the quality and direction of existence very quickly. For treatment, the basic sharing of these experiences with a group of others in a similar predicament often quickly moves the client beyond isolation and a sense of self-separateness to connect intimately with others who understand her situation. This community may not only bring comfort and support but also a deep sense of communion with humanity. In this instance, breaking through the shell of isolation may enable Sandra to begin to make new connections with her family and with herself. A sense of interconnection, a central postulate and experience in the wisdom traditions, may replace her perceived isolation. Sandra may use this opportunity of facing possible death to begin to encounter and let go of such feelings as guilt, shame, disappointment, and anger that have kept her life less satisfying than it could be. Accessing the imaginal through art or dreams, for example, can provide a clear and symbolic expression of unresolved issues. The use of rituals or rites-of-passage inspired by the wisdom traditions can provide some catalyst for shifting her consciousness through forgiveness and release. The therapist may engage in a wide variety of methods (e.g., imagery, art, or dream work, meditation, rituals), but the heart of the work is in the simple and humane spirituality that is embodied by the therapist's loving presence along with the therapist's openness to explore the full range of human experience directly. For Sandra, this experience may be seen as an opportunity for practicing love and forgiveness, moving out from behind rigid self-separateness, facing fears, and transforming her self-definition.
Cognitive revolution:	Science is always influenced by the technology that surrounds it, and psychology is no exception. Thus it is no surprise that beginning in the 1960s, growing numbers of psychologists began to think about the brain and about human behavior in terms of the computer, which was being developed and becoming publicly available at that time. Cognitive psychology is a field of psychology that studies mental processes, including perception, thinking, memory, and judgment. These actions correspond well to the processes that computers perform. Although cognitive psychology began in earnest in the

1960s, earlier psychologists had also taken a cognitive orientation. Some of the important contributors to cognitive psychology include the German psychologist Hermann Ebbinghaus (1850–1909), who studied the ability of people to remember lists of words under different conditions, and the English psychologist Sir Frederic Bartlett (1886–1969), who studied the cognitive and social processes of remembering. Bartlett created short stories that were in some ways logical but also contained some very unusual and unexpected events. Bartlett discovered that people found it very difficult to recall the stories exactly, even after being allowed to study them repeatedly, and he hypothesized that the stories were difficult to remember because they did not fit the participants' expectations about how stories should go. The idea that our memory is influenced by what we already know was also a major idea behind the cognitive-developmental stage model of Swiss psychologist **Jean Piaget (1896–1980)**. Other important cognitive psychologists include **Donald E. Broadbent (1926–1993)**, **Daniel Kahneman (1934–)**, **George Miller (1920–)**, **Eleanor Rosch (1938–)**, and **Amos Tversky (1937–1996)**.

Cognitive psychology remains enormously influential today, and it has guided research in such varied fields as language, problem solving, memory, intelligence, education, human development, social psychology, and psychotherapy. The cognitive revolution has been given even more life over the past decade as the result of recent advances in our ability to see the brain in action using neuroimaging techniques. Neuroimaging is the use of various techniques to provide pictures of the structure and function of the living brain.

COGNITIVE REVOLUTION:

The cognitive revolution in psychology was a counter-revolution. The first revolution occurred much earlier when a group of experimental psychologists, influenced by Pavlov and other physiologists, proposed to redefine psychology as the science of behavior. Since the beginning of experimental psychology in the nineteenth century, there had been interest in the study of higher mental processes. Cognitive Revolution, as now it is termed, during the time (in the late 1950s) was a discontinuous process that rejected Behaviorism and accepted the model of Mind based on the computer. This marked the beginning of the counter-revolution called Cognitive revolution, and the view of mental processes that it spawned is called 'cognitive Psychology'.

The reference to the discontinuous process here means (in a way) the minds that were independently working without knowing that they

were building towards Cognitive Psychology. During this time, experimental psychologists were rethinking the definition of psychology, other important developments were occurring elsewhere. Norbert Wiener's cybernetics was gaining popularity, Marvin Minsky and John McCarthy were inventing artificial intelligence, and Alan Newell and Herb Simon were using computers to simulate cognitive processes. Finally, Chomsky was single-handedly redefining linguistics.

COMMUNICATIONS RESEARCH AND THE INFORMATION PROCESSING:

Approach during World War II, new concepts and theories were developed about signal processing and communication, and these ideas had a profound impact on psychologists active during the war years. One important work was Shannon's 1948 paper about Information Theory.

It proposed that information was communicated by sending a signal through a sequence of stages or transformations. This suggested that human perception and memory might be conceptualized in a similar way: sensory information enters the receptors, and then is fed into perceptual analyzers, whose outputs in turn are input to memory systems. This was the start of the 'information processing' approach—the idea that cognition could be understood as a flow of information within the organism, an idea that continues to dominate cognitive psychology. Perhaps the first major theoretical effort in information processing psychology was Donald Broadbent's Perception and Communication (Broadbent 1958). According to Broadbent's model, information output from the perceptual system encountered a filter, which passed only information to which people were attending. Although this notion of an all-or-none filter would prove too strong (Treisman, 1960) it offered a mechanistic account of selective attention, a concept that had been banished during behaviorism. Information that passed Broadbent's filter then moved on to a 'limited capacity decision channel,' a system that has some of the properties of short-term memory, and from there on to long-term memory. This last part of Broadbent's model—the transfer of information from short- to long-term memory—became the salient point of the dual-memory models developed in the 1970s. Another aspect of Information theory that attracted psychologist's interest was a quantitative measure of information in terms of 'bits'.

THE COMPUTER MODELING APPROACH

Technical developments during World War II also led to the development of digital computers. Questions soon arose about the comparability of computer and human intelligence (Turing 1950). By 1957, Alan Newell, J. C. Shaw, and Herb Simon had designed a computer program that could solve difficult logic problems, a

domain previously thought to be the unique province of humans. This work would also help establish the field of artificial intelligence.

THE GENERATIVE LINGUISTICS APPROACH

A third external influence that leads to the rise of modern cognitive psychology was the development of generative grammar in linguistics by Noam Chomsky.

THE APPROACH INTRINSIC TO PSYCHOLOGY:

At least one source of modern cognitive psychology came from within the field. This approach had its roots in Gestalt psychology, and maintained its focus on the higher mental processes. A signal event in this tradition was the 1956 book 'A Study of Thinking', by Bruner, Goodnow, and Austin (Bruner et al. 1956). The work investigated how people learn new concepts and categories, and it emphasized strategies of learning rather than just associative relations. The proposals fit perfectly with the information-processing approach—indeed, they were information processing proposals—and offered still another reason to break from behaviorism. This likened mind to a computer, and emphasized the representations and processes needed to give rise to activities ranging from pattern recognition, attention, categorization, memory, reasoning, decision making, problem solving, and language.

THE NEW PSYCHOLINGUISTICS

Beginning in the early 1960s there was great interest in determining the psychological reality of Chomsky's theories of language (these theories had been formulated with ideal listeners and speakers in mind). Some of these linguistically inspired experiments presented sentences in perception and memory paradigms, and showed that sentences deemed more syntactically complex by transformational grammar were harder to perceive or store (Miller 1962). Subtler experiments tried to show that syntactic units, like phrases, functioned as units in perception, STM, and LTM (Fodor et al. (1974) is the classic review). While many of these results are no longer seen as critical, this research effort created a new subfield of cognitive psychology, a psycholinguistics that demanded sophistication in modern linguistic theory. As psychologists became aware of related developments in linguistics and artificial intelligence, so researchers in the latter disciplines became aware of pertinent work in psychology. Thus, evolved the interdisciplinary movement called 'cognitive science'. In addition to psychology, AI, and linguistics, the fields of cultural anthropology and philosophy of mind also became involved. The movement eventuated in numerous interdisciplinary collaborations (e.g., Rumelhart et al. 1986), as well as in individual psychologists

	becoming more interdisciplinary.
Multiculturalism	Diversity is the central concern of almost all social, political and cultural theorists. The common argument of them is that thinking, perception, action and adjudication differs from culture to culture and society to society. Cultural diversity creates a climate in which different culture can engage in a mutually beneficial dialogue. Different artistic, literary, musical, moral and other traditions interrogate, challenge and probe each other, borrow and experiment with each other's idea and often throw up wholly new ideas and sensibilities that none of them could have generated of their own. Cultural Pluralism, Cultural Diversity and Multiculturalism are the terms very commonly used to describe societies of different cultures, religions, languages and races. In our day to day life these words are used interchangeably. The presence of many and different communities are common in these terms plural, diverse and multiculturalism. The term 'multiculturalism' has become very popular in the last few decades and, as often happens to popular words, it has ended up expressing quite different meanings. In fact, multicultural discussions refer to a wide variety of situations – integration policies and welfare state regimes, the legal and political accommodation of cultural diversity, the management of immigration and national borders, the recognition and respect of cultural/religious difference, living with 'difference' in daily contexts, the ideological representation of identities, cultures and 'the good society', to name only a few. These also involve different disciplines – philosophy, sociology, anthropology, social psychology, political science, pedagogy, and cultural and post-colonial studies – with their own distinct questions, perspectives and tool-kits. Multiculturalism is a body of thought in political philosophy about the proper way to respond to cultural diversity. Multiculturalism refers to situations in which people who hold 'different' habits, customs, traditions, languages and/or religions live alongside each other in the same social space, willing to maintain relevant aspects of their own difference and to have it publicly recognized. Usually the term involves a positive evaluation of cultural diversity and the institutional commitment to its preservation. Multiculturalism is closely associated with 'identity politics', 'the politics of difference' and 'the politics of recognition', all of which consider proper recognition of cultural diversity a necessary step towards revaluing disrespected identities and changing dominant pat-terns of representation and communication that marginalize certain groups (Song, 2010).

MULTICULTURALISM IN INDIA

The management of cultural diversity within state has become an increasingly prominent issue in recent times not only in developed countries but also in developing countries like India. Diversity has become the defining characteristic of our social and cultural worlds. We are now constantly confronted with a multitude of ways in which we can define ourselves, and categorize others. In fact, in itself India is a multi-ethnic and multi lingual society. The 2001 census of India reports 122 languages and 234 mother tongues.

Most of the major religions of the world such as Hinduism and Buddhism, originated in this land while others, such as Christianity and Islam, though came from outside has remained and grown in it for a thousand years and more. This great profusion of linguistic, religious and other customs and usages was associated with a multitude of castes, communities and tribes each of which was bearer of a particular sub-culture or even sub-sub-culture which is transmitted from generation to generation

(ONLY FOR REFERENCE)

MULTICULTURALISM IN COUNSELING:

Psychology is at a "pivotal period" in which the rules are being changed in ways that we are only beginning to appreciate (Mahoney & Pattetson, 1992). One clear indication that multiculturalism is introducing a permanent paradigm shift and not merely a passing fad in counselling is the strength of the multicultural controversies in the field of counselling. By "paradigm shift" I refer to changes in the underlying assumptions about psychology moving from a monocultural to a multicultural basis with profound implications for how psychology is applied in direct service. All change is painful and the more profound the change the more the pain. As the counselling profession responds to the requests and demands of many culturally different special interest groups, the rules have begun to change, causing controversy and confusion. Even when we recognize that the old rules are not working, it is not easy to replace them with new culturally different rules. The old rules of psychology focused on dissonance reduction. The new rules focus on tolerance of ambiguity.

Smith, Harre, and VanLangenhove (1995) contrast the new with the old paradigms. The new paradigms emphasize the following qualities: (a) understanding and description more than just measuring, (b) predicting consequences more than causation, (c) social significance more than statistical significance, (d) language and discourse more than numerical reductionism, (e) holistic

	perspectives more than atomistic trivia, (f) complex interacting particulars more than simplistic universals, and (g) subjectively derived meaning more than objectively imposed meaning. MULTICULTURALISM IN MENTAL HEALTH SERVICES https://www.apa.org/pubs/books/Foundations-of-Multicultural-Psychology-Intro-Sample.pdf
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WHEN TO USE BRIEF HUMANISTIC AND EXISTENTIAL THERAPIES

These approaches can be useful at all stages of recovery in creating a foundation of respect for clients and mutual acceptance of the significance of their experiences. There are, however, some therapeutic moments that lend themselves more readily to one or more specific approaches. The details of the specific approaches are laid out later in this chapter. Client-centered therapy, for example, can be used immediately to establish rapport and to clarify issues throughout the session. Existential therapy may be used most effectively when a client is able to access emotional experiences or when obstacles must be overcome to facilitate a client's entry into or continuation of recovery (e.g., to get someone who insists on remaining helpless to accept responsibility). Narrative therapy may be used to help the client conceptualize treatment as an opportunity to assume authorship and begin a "new chapter" in life. Gestalt approaches can also be used throughout therapy to facilitate a genuine encounter with the therapist and the client's own experience. Transpersonal therapy can enhance spiritual development by focusing on the

intangible aspects of human experience and awareness of unrealized spiritual capacity. These approaches increase self-awareness, which promotes self-esteem and allows for more client responsibility, thus giving the client a sense of control and the opportunity to make choices. All of these approaches can be used to support the goals of therapy for substance abuse disorders.

FOUR FOUNDING PATHS OF ACADEMIC PSYCHOLOGY:

WUNDT

Wilhelm Wundt (1832 –1920), the physiologist, psychologist, and philosopher, is generally acknowledged as the founder of experimental psychology, having established the first laboratory specifically devoted to a systematic program of research in psychology. The tributes of his contemporaries, his honors and obituaries indicate that Wundt was held in high esteem at that time.

Wundt's innovation was the attempt to project the experimental rigor of physiology into the domain of inner experience by supplementing these experiments with a purely psychological set of procedures. These procedures constituted Wundt's well-known method of 'Selbstbeobachtung', i.e., "introspection" or, better, "self-observation."

Wundt studied with Helmholtz and Bunsen, both famous. Many of Wundt's students became prominent too:

- G. Stanley Hall
- Cattell
- Kraepelin
- Munsterberg
- Kulpe
- Titchener

MEASUREMENT AND PSYCHOPHYSICS

Time was also Wundt's royal road towards a scientific model of the human mind. He relied on the skills of clock makers to construct the delicate instruments his methods demanded. Matthäus Hipp's chronoscope (1840) for example, an instrument that often featured in photographs of Wundt's laboratory, was theoretically capable of recording intervals of 1/1000th of a second. The Leipzig group used the chronoscope to measure 'reaction time', the time it took for a subject to respond to a stimulus.

INTROSPECTION

Wundt went on to develop another procedure: introspection. This technique involved the perception of a sensory event and then reporting on the phenomenological experience of the event. This procedure sounds simple. However in practice it was extremely tricky to get right. Firstly, of course, there were large variations in reaction times between subjects. Perceiving a sensory event and then reporting on the phenomenological experience of the event were not straightforward matters. Experimental subjects had to be knowledgeable about the purpose of the experiment and adopt the right 'mental set' to perceive stimuli correctly. They effectively had to know how to produce the kind of data Wundt wanted. As both experimenter and subject had to be experts on experimental protocol, their roles were therefore interchangeable. As a result, only a few experimenter-subjects needed to be trained in the art of introspection, because the aim was to discover the workings of the generalized human mind that everyone was assumed to share. According to Wundt, introspection was not about ruminating on one's own feelings or emotions, in the manner of a poet or philosopher. On the contrary, introspection was a highly controlled process — a systematic method used to study the mind by breaking up

conscious awareness into basic structures of thoughts, images and sensations.

In practice, introspective reports tended to consist of simple judgments of the size, intensity and duration of physical stimuli, occasionally supplemented by judgments of the simultaneity and succession of stimuli. Wundt's ambition to construct a theory of the human mind began with this simple technique of 'mental chronometry'.

PERCEPTION AND APPERCEPTION

Wundt knew that experimental methods were only useful for investigating the most elementary psychological processes. In his theory, he distinguished between perceptions on the one hand, and what he called apperception on the other. Stimuli outside awareness are merely perceived, he said, whereas stimuli receiving attention are apperceived. Apperception was therefore affected by 'higher level' psychological processes such as motives, innate tendencies, memory and so on. Higher mental processes such as emotion and language were not readily amenable to experimental investigation and could not be reliably studied using introspection and mental chronometry.

CULTURAL PSYCHOLOGY

Given the problems of investigating apperception, Wundt argued that there must therefore be an alternative to experimental psychology - a way to investigate social, cultural and historical phenomena beyond the confines of the laboratory. This would be a kind of social psychology based on the historical analysis of human cultural products, especially language, myth and customs. Experimental psychology rightly studied the lower perceptual processes of sensation and perception. But cultural psychology — or what Wundt called 'völkerpsychologie' — examined the higher processes of thinking, emotion and other products of collective activity. Two different laws, Wundt explained, governed the two

systems: the higher system was governed by the laws of apperception; the lower system by the laws of association. The higher processes presupposed the lower ones but they were not determined by them. A complete psychology must therefore embrace both quantitative and qualitative methods.

DEVELOPMENTS IN THE USA

While Wundt embarked on what would eventually become ten volumes of 'Völkerpsychologie', his students returned home to launch scientific psychology in America. The first American experimental psychology research laboratory was established by G. Stanley Hall at Johns Hopkins University in 1883. Four years later, James McKeen Cattell founded one at the University of Pennsylvania. They closely followed Wundt's practice of stocking their laboratories with impressive-looking 'brass instruments'. By 1913 there were ten times as many psychological laboratories in the USA as there were across the whole of Europe.

FREUD

Psychodynamic psychology is an approach to understanding human behavior that focuses on the role of unconscious thoughts, feelings, and memories. Freud developed his theories about behavior through extensive analysis of the patients that he treated in his private clinical practice. Freud believed that many of the problems that his patients experienced, including anxiety, depression, and sexual dysfunction, were the result of the effects of painful childhood experiences that the person could no longer remember. Freud's ideas were extended by other psychologists whom he influenced, including Carl Jung (1875–1961), Alfred Adler (1870–1937), Karen Horney (1855–1952), and Erik Erikson (1902–1994). These and others who follow the psychodynamic approach believe that it is possible to help the patient if the unconscious drives can be remembered,

particularly through a deep and thorough exploration of the person's early sexual experiences and current sexual desires. These explorations are revealed through talk therapy and dream analysis, in a process called psychoanalysis.

The first of the modern personality theories was developed by Sigmund Freud and is known as psychoanalytic theory. The psychiatric practice of this theory is called psychoanalysis. Freud's ideas were plentiful, profound, and often controversial. His theory about personality has had tremendous influence on societies around the world through many different disciplines. Not only psychology has been influenced and in formed by the ideas of Freud, but also literature, art, philosophy, cultural studies, film theory, and many other academic subjects. Freud's theory represents one of the major intellectual ideas of the modern world.

DEFENSE MECHANISM:

Psychoanalytic theory suggests that there are other ways in which our unconscious protects us besides by repression. These protective devices of the unconscious are known as defense mechanisms.

RATIONALIZATION: Sometimes our unconscious makes up a good-sounding reason to explain something we don't like. If we fail a test, we blame it on others. If our favorite candidate doesn't win the election, we say that it's for the best anyway. If we don't complete an assignment, we think the teacher was unfair to have given the assignment. "Sour grapes" is another example—if we don't get something we want, we find something wrong with it and convince ourselves we're better off without it. Rationalizing protects us

from the anxiety of seeing ourselves as deficient. This is a common defense mechanism because of the importance placed on giving good reasons for things. However, this is not rational, it is rationalizing. Being rational means being objective. In rationalization, our mind protects us with a reason that only sounds good; it is not objective, it just seems to be.

PROJECTION: In this case, when we have some thoughts or feelings that we consider to be wrong or upsetting, we project them onto other people instead of on our-selves. If I believe that a certain attitude or feeling that I have is terribly wrong, I will claim that others have it. A person who wants to use illegal drugs but who believes that it would make him a horrible person might expect everyone else to want to use illegal drugs. This defense mechanism deflects the anxiety away from us and onto others. Many of the predictions that a person makes about someone else are, in fact, true about the person making the prediction. Be careful what you say about others, it might be true about you! If a man says that he believes people lie on their resumes, perhaps it's an indication that he has an inclination to lie on his resume. Is he simply being objective? Or is he saying people lie because his unconscious knows he would act that way? That would be projection.

SUBLIMATION: We sublimate if we redirect or re-channel our undesirable emotions and thoughts

into a socially acceptable activity. If I am full of rage and horrible thoughts, I might vigorously wash my car. Many people sublimate by pouring their emotions into works of art. The famous painter Vincent van Gogh is the example that is most often given. His mental and emotional distress seems evident in the vivid colors, thick paint, and forceful brushstrokes of his paintings. We can imagine van Gogh's moods merely by looking at his paintings. Many famous composers and poets also are good examples of this defense mechanism. Their mental anguish is redirected into wonderful works of art. There is a long list of composers and poets who suffered from depression and bipolar disorder. That is a tragedy, but one that provided us with a world of music and literature. Through sublimation, unpleasant mental energy is redirected into acceptable work.

REACTION FORMATION:

Sometimes people's mental and emotional energy is so threatening that they adopt the reverse—the opposite—of what they really want. A person who believes that drinking alcohol is a terrible sin yet who has a desire to drink alcohol might be protected by reaction formation. In this case, the person's unconscious adopts a hatred of alcohol. The person might join groups that protest alcohol use and might attempt to pass laws against drinking alcohol. She becomes vociferous, wildly critical of alcohol. We might say, paraphrasing Shakespeare, that she

protests too much. If a man believes that being gay is a horrible thing yet feels attracted to other men, he might express a deep hatred of gays and attempt to harm them. In reaction formation, a person's unconscious takes on the beliefs that are opposite of the true desires, those repressed in the unconscious. This protects the conscious part of the mind from what the unconscious considers to be awful.

DISPLACEMENT:

Freud suggested this defense mechanism to explain how a person's unconscious wishes could appear in dreams but in disguise. A woman who is angry with her brother Tom might dream that she harms a noisy tomcat. Her conscious will not be aware of the connection between the names. Her anger is displaced onto a symbol of her brother. This defense mechanism is often used to explain behaviors outside of dreams; for instance, when a person's displeasure is directed toward some object other than the source of the displeasure (for example, if an employee displaces his anger toward his boss onto his wife, a subordinate, or his dog).

DENIAL:

This defense mechanism is a primitive form of repression. In this instance, a person simply denies things that produce anxiety. The term is often used today in referring to people who have obvious problems with alcohol, drugs, or relationships but refuse to accept that those problems exist.

REGRESSION: Under conditions of severe trauma or stress, a person might revert to developmentally earlier forms of behavior and thinking. This is known as regression. A person who is under significant stress, for example, might begin sucking his or her thumb. Freudian theory argues that regression provides a person with feelings of security and calm when under threatening conditions. Defense mechanisms protect us from anxiety and threats. In that sense, they are useful and good. However, they can go too far and take us into abnormality. When defense mechanisms become extreme, they cause more problems than they solve. A person might then develop symptoms of mental disturbance. Freud proposed a clinical therapy to deal with those instances, as noted above, a therapy known as psychoanalysis.

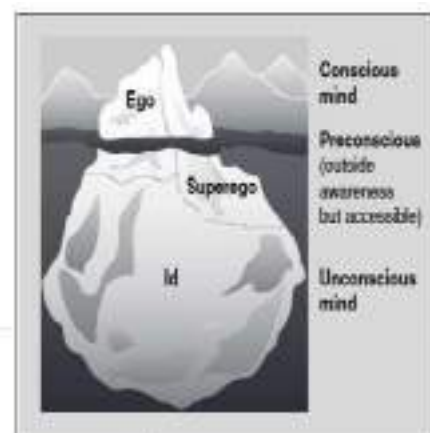
PERSONALITY STRUCTURES

Freud suggested an analogy about the mind. He said that the mind is like an iceberg in the ocean, floating 10% above the water and 90% below. The unconscious, Freud proposed, makes up the vast majority of our mind. In Freud's view, only about 10% of our behaviors are caused by conscious awareness—about 90% are produced by unconscious factors. According to psychoanalytic theory, most of what controls our behaviors, thoughts, and feelings is unknown to our aware minds. Normally, the unconscious guides us. Freud said that the mind could be divided into three abstract categories. These are the id, the ego, and the superego. Although these are known as structures, do not take the term literally.

Freud theorized that personality contains three structures—the id, ego, and superego—and that the mind is like an iceberg, the unconscious making up 90% while the conscious (like the tip of the iceberg floating above water) makes only 10% of the mind.

THE ID: Latin for the term “it,” this division of the mind includes our basic instincts, inborn dispositions, and animalistic urges. Freud said that the id is totally unconscious, that we are unaware of its workings. The id is not rational; it imagines, dreams, and invents things to get us what we want. Freud said that the id operates according to the pleasure principle—it aims toward pleasurable things and away from painful things. The id aims to satisfy our biological urges and drives. It includes feelings of hunger, thirst, sex, and other natural body desires aimed at deriving pleasure.

THE EGO: Greek and Latin for “I,” this personality structure



begins developing in childhood and can be interpreted as the “self.” The ego is partly conscious and partly unconscious. The ego operates according to the reality principle; that is, it attempts to help the id get what it wants by judging the difference between real and imaginary. If a person is hungry, the id might begin to imagine food and even dream about food. (The id is not rational.) The ego, however, will try to determine how to get some real food. The ego helps a person satisfy needs through reality.

THE SUPEREGO: This term means “above the ego,” and includes the moral ideas that a person learns within the family and society. The superego gives people feelings of pride when they do something correct (the ego ideal) and feelings of guilt when they do something, they consider to be morally wrong (the conscience). The superego, like the ego, is partly conscious and partly unconscious. The superego is a child’s moral barometer, and it creates feelings of pride and guilt according to the beliefs that have been learned within the family and the culture.

Freud theorized that healthy personality development requires a balance between the id and the superego. These two divisions of the mind are naturally at conflict with one another: The id attempts to satisfy animal, biological urges, while the super ego preaches patience and restraint. The struggle between these two is an example of intrapsychic conflict—conflict within the mind. According to psychoanalytic theory, defense mechanisms are automatic (unconscious) reactions to the fear that the id’s desires will overwhelm the ego. Freud believed that a healthy personality was one in which the id’s demands are met but also the

superego is satisfied in making the person feel proud and not overwhelmed by guilt. If the id is too strong, a person will be rude, overbearing, selfish, and animalistic. If the superego is too strong, a person is constantly worried, nervous, and full of guilt and anxiety and is always repressing the id’s desires. An overly strong id makes one a psychopath, lacking a conscience, or an ogre, selfishly meeting one’s needs without concern for others. An overly strong superego, on the other hand, makes one a worrier, a neurotic, so overwhelmed by guilt that it is difficult to get satisfaction. Sometimes it is said that the ego is the mediator between the id and the superego, but this is not what Freud said. The ego does not help to find compromise; the ego helps the id to satisfy its desires by focusing on what is real.

THE PSYCHOSEXUAL STAGES

Freud’s psychosexual stages are as follows:

ORAL: The first stage in Freud’s theory covers babies up to about the age of one and a half years. The driving force during this stage is interest and pleasure in activities involving the mouth (hence the term oral), such as sucking and biting. Adult oral personality traits that derive from the oral stage include anything to do with the mouth, such as smoking, overeating, or biting the nails, and anything that is babylike, such as being naïve (“swallowing” anything you are told) or being dependent on others.

ANAL: This stage centers on toilet training, beginning around the age of 18 months or two years and extending up to preschool, about age three. The term anal, of course, refers to the anus, the rear end (the opposite end of oral), and one of the jokes in psychology is that you can’t spell analysis without anal.

This joke makes light of the fact that Freud believed this stage to be crucial in planting the seeds for a number of adult personality traits. In the anal stage the child is being toilet trained and is learning to hold in and to let out at appropriate times. Therefore, Freud proposed that personality traits related to either holding in or letting out were formed during the anal stage. The following traits are known as anal-retentive (finding pleasure from holding in): neatness, orderliness, punctuality, cleanliness, compulsiveness, perfectionism, and stinginess. The following are called anal-expulsive (finding pleasure from letting out): being undisciplined, messy, disorderly, late, impulsive, and overly generous.

3. **PHALLIC:** This stage occurs approximately during the preschool years. The term phallic means any representation of the penis, which, according to Freud, is the main occupation of the unconscious during the childhood years of about three to six among both boys and girls. It is at this time, theoretically, that children become aware of whether or not they have a penis, and Freud believed that this causes a bit of anxiety in the unconscious parts of their minds. Boys, Freud reasoned, become protective of their penis and fear having it taken away. This is known as castration anxiety and might be manifested in a young boy's fear of knives, scissors, or being bitten by dogs. Girls, Freud thought, feel resentful that they do not have a penis and hence seek phallic things and activities that will provide them with feelings of power and possession. This is known as penis envy and might be

seen when preschool girls develop a deep fondness for horses, unicorns, and other strong, masculine things or long, pointed objects. Freud proposed an unconscious drama during this stage that he called his most important idea. It is called the Oedipus complex (some-times referred to as the Oedipal conflict). This unconscious process is named after the Greek story of Oedipus, the man who was raised by foster parents and grew up to unwittingly kill his biological father and marry his biological mother. Freud said that a similar drama occurs in the unconscious minds of preschool boys, who favor their mothers and fear their fathers (castration anxiety). The child resents the father for getting all of the mother's attentions. Many psychoanalysts suggest a similar conflict for preschool girls, referred to as the Electra complex; it is essentially the reverse of the situation for boys: love and desire for father, resentment for mother. According to psychoanalytic theory, these complexes become so severe and anxiety-producing that the child's unconscious must resolve them using a defense mechanism. The solution is for the child to begin to identify with the same-sexed parent. The child begins to internalize the personality of the same-sexed parent, thereby relieving the anxiety and vicariously winning the love of the opposite-sexed parent. For a little boy, being like daddy means no longer having to fear and resent him, and it also means getting mommy's love through daddy. For a little girl, it means winning daddy's love by being like mommy. This process is called identification with the aggressor; sometimes simply known as identification. The result is that

children begin to internalize the values, morals (the superego), traits, attitudes, and behaviors of their parents. In fact, in 1925, Freud concluded that he had been wrong about penis envy in young girls and theorized that the oedipal struggle for girls, as well as for boys, centered on love for the mother. As you can imagine, this remains a controversial idea among psychoanalysts.

LATENCY: After resolving the Oedipal conflict through identification (at about the age of six), children enter a stage during which sexual urges are dormant or resting. The term latent means that something is present or has potential without being active or evident. During this stage, sexual urges are taking a recess; they are at a minimum. From about the ages of 6 to 12, boys typically stick together and say that they do not like girls, or they act squeamish around girls. Similarly, girls during this stage are highly critical of boys, are shy around them, and avoid them. Apparently, the demands of the previous stage and the Oedipal drama were so overwhelming that the unconscious needs a bit of a rest.

GENITAL: This final of the psychosexual stages arises during adolescence when teenagers begin again to show sexual interests. This stage leads to adult affection and love. If all has gone well in the previous stages, Freud theorized, interest during adolescence is on heterosexual relationships. This is a time of exploring pleasure through more mature love and affection.

Stage	Approximate Ages	Main Features
1. Oral	Birth-1 1/2 or 2	Mouth, dependency
2. Anal	1 1/2-3	Toilet training, give and take
3. Phallic	3-6	Oedipus complex, identification, superego
4. Latency	6-12	Repression of sexuality
5. Genital	12-Adulthood	Development of normal sexuality

JAMES

William James (1842–1910), an American psychologist and philosopher who is referred to as the "Father of American psychology." He founded a psychology laboratory at Harvard University. James wrote and published extensively, and he was the first to offer a course in psychology in the United States. As a leading thinker during the latter half of the nineteenth century, his theories were influential in both psychology and philosophy.

Pragmatism and functionalism are two philosophies. William James used to further his understanding of the world around him. James posited the pragmatic theory of truth, and he used this philosophy to define and re-define, filtering answers to his questions through this theory. As a philosopher and psychologist, all his findings, theories, and inquiries were viewed through the lens of his pragmatic theory of truth.

To James, the truth was subjective, and he delved into all mental pursuits with an open mind. He was named the 14th most eminent psychologist of the 20th century.

WILLIAM JAMES' PRINCIPLES OF PSYCHOLOGY

In 1890, James wrote published his book about psychology, "The Principles of Psychology.

"The Principles of Psychology took twelve years to write, and the result consists of two volumes. During the last half of the nineteenth-century psychology was beginning to gain respectable ground in the United States and this book by William James helped to strengthen its hold.

The book, *The Principles of Psychology* was a comprehensive work covering the entire field of psychology as it was understood up until publication. Many individuals who work in the field of psychology today still find many of the concepts and theories in this book to be informative and interesting. Modern psychology has come a long way since the year 1890 and the publication of James' book, but his brilliance is still respected and discussed today.

Four main concepts are put forth by James in this book; stream of consciousness, emotion, habit, and will. Along with these four main concepts, James discusses theories and hypothesizes about centers in the brain receiving specific input from the physical senses. The concept of instinct is covered comparatively, and the evolution of brain function, particularly the cerebrum is also discussed.

The topic of experimental psychology is covered, and his experience with illusions both visual and auditory are explained. The illusion is explained as a physiological response; pathways in the brain are formed through repeated behaviors and use, this leads to illusion when similar stimuli occur and are guided through the same pathways. James goes on to explain that the mind becomes accustomed to recognizing something that is repeated and when stimuli are similar an assumption is made, this assumption is the "illusion."

The four main concepts of habit, stream of consciousness, emotion, and will make up the bulk of the work. Each of these concepts is complete with explanations and in some cases, empirical knowledge from James himself. These four concepts are a bit out sync with modern psychology, but they still have their merit.

PRAGMATISM AND FUNCTIONALISM

Pragmatism is a philosophy, and to William James, it was a philosophy of truth. James was a pragmatist, and he understood truth through the lens of pragmatism. Pragmatism means practical, and William James believed only the practical, those things that are beneficial and help to move us in the right direction, are worthwhile. Pragmatism as a philosophy of truth was something James believed in, to him the truth was arbitrary; it depends on belief.

The pragmatist approach was the approach James took whenever he was validating a theory of his own or others. The pragmatist movement, as a philosophical movement, was one James fell in with early in his career. It was an easy fit for James, he was a pragmatist at heart, and he scrutinized all the experience and information he accumulated, searching all the while for pragmatic answers to his inquiries.

William James founded the school of Functionalism. This school of thought in psychology was developed in direct response to the school of structuralism and the work of Wilhelm Wundt. Wundt criticized functionalism as nothing more than literature, and James criticized structuralism as "all school and no thought." When the criticisms faded, functionalism went on to influence the major schools of thought still in use today, such as the cognitive-behavioral and behavioral schools.

Functionalism focused on the human propensity for individualism and this heavily influenced how education was structured. James was influenced by his early physiology education and the work of Charles Darwin. Functionalism was built around a more systematic approach to understanding mental processes. William James developed functionalism to search for consciousness and behavior.

DILTHEY

Wilhelm Dilthey (1833–1911) was a German Philosopher, best known for the way he **distinguished between the natural and human sciences**. Whereas the main task of the natural sciences is to arrive at law-based causal explanations, the core task of the human sciences is the understanding of the organizational structures of human and historical life.

He promoted psychology as a “human science” (Geisteswissenschaft). His human scientific psychology took as **its subject matter experience in its totality**. Appropriate methods included description and analysis, with the goal of understanding. For Dilthey, **the mind was the medium through which meaning was formed; thus, the content, rather than the structure, of the mind was of most interest to him**. Understanding the meaning of human historical events requires being able to organize them in their proper contexts and to articulate the structural uniformities that can be found in this way. Dilthey’s reflections on the human sciences, historical contextualization and hermeneutics influenced many subsequent thinkers such as Husserl, Heidegger, Cassirer, Gadamer and Ricoeur.

Dilthey offered clear advice to psychology. First, he acknowledged that humans and human events both possess important physical properties. As an example, your brain is a physical object and its physical properties are going to be pretty important if you want to think. I’m sure you’d realized that (see ‘Aside’). This simple observation nonetheless creates a serious complication for psychology, because it means that our status as physical and material ‘objects’ has a massive effect on our capacity to be psychological. In other words, in order to fully grasp the psychological world, the discipline of psychology must also engage with aspects of the physical world. He wrote, explaining human actions are fundamentally different from explaining physical events. A woman

shooting a man is a physical event. However, understanding the event in human terms involves more than tracing the path of the bullet and showing how the bullet caused the man's death. We need to know why she shot the man, not just how she did so.

Dilthey defended his use of the term Geisteswissenschaft (literally, "science of the mind" or "spiritual knowledge") by pointing out that other terms such as "social science" and "cultural sciences" are equally one-sided and that the human mind or spirit is the central phenomenon from which all others are derived and analyzable.

Dilthey wanted psychology to remember that its primary subject matter was the subjective or psychological world itself, not just the physical properties that made this world possible. Dilthey suggested that if we want to find out ‘why’, priority must be given to the psychological world – the ways that people make sense of their experience and the meanings they attach to them. In other words, to understand why I am running, you will first need to understand my experience of the current situation and the meanings I assign to it, because only in this way can you find out my motives and reasons for acting. This task demands a social scientific approach. So, what should psychology do? If the psychological world was its proper subject matter, then surely psychology was a humanitarian discipline? For this reason, Dilthey felt a social scientific approach was preferable. But the psychological world can only exist through the physical world. If I don’t have a physical body I can’t see or hear, for example. Psychology couldn’t ignore this either. It needed to study the physical world as well and the natural scientific model dominated in this domain.

A lifelong concern was to establish a proper theoretical and methodological foundation for the "human sciences" (e.g. history, law, literary criticism), distinct

from, but equally "scientific" as, the "natural sciences" (e.g. physics, chemistry). He suggested that all human experience divides naturally into two parts: that of the surrounding natural world, in which "objective necessity" rules, and that of inner experience, characterized by "sovereignty of the will, responsibility for actions, a capacity to subject everything to thinking and to resist everything within the fortress of freedom of his/her own person".

This double-edged nature of psychology was (and remains) both a challenge and an opportunity. Psychology had the chance to bridge the divide between the natural and social sciences and it could do so by retaining a foot in both camps (Danziger, 1990). Nomothetic and idiographic knowledge, quantitative and qualitative methods, could all be embraced. It needed to study its subject matter from both perspectives. Wilhelm Wundt tried to support this vision (as did other early psychologists), but it was not a vision that psychology would ultimately sustain.

In his work *Ideas Concerning a Descriptive and Analytic Psychology*, he introduced a distinction between explanatory psychology and descriptive psychology. In his terminology, explanatory psychology is the study of psychological phenomena from a third-person point of view, which involves their subordination to a system of causality, while descriptive psychology is a discipline that attempts to explicate how different mental processes converge in the "structural nexus of consciousness."

In his later work (1910), he used the alternative term structural psychology for descriptive psychology.

WELTANSCHAUUNGEN OR WORLD VIEWS

In 1911, Dilthey developed a typology of the three basic Weltanschauung, or World-Views, which he considered to be "typical" (comparable to Max Weber's notion of

"ideal types") and conflicting ways of conceiving of humanity's relation to Nature. In **Naturalism**, represented by Epicureans of all times and places, humans see themselves as determined by nature. In **the Idealism of Freedom (or Subjective Idealism)**, represented by Friedrich Schiller and Immanuel Kant, humans are conscious of their separation from nature by their free will in **Objective Idealism**, represented by G. W. F. Hegel, Baruch Spinoza, and Giordano Bruno, humans are conscious of their harmony with nature. This approach influenced Karl Jaspers' *Psychology of Worldviews* as well as Rudolf Steiner's *Philosophy of Freedom*.

Issues: Crisis in psychology due to strict adherence to experimental analytical paradigm (logical empiricism). Indic influences on modern psychology.

All this part is covered in Identity crisis of the Discipline (in the above sections after 'Academic Psychology in India' content)

Essential aspects of knowledge paradigms: Ontology, epistemology, and methodology

ONTOLOGY

Ontology is the study of 'being' and is concerned with 'what is', i.e., the nature of existence and structure of reality as such (Crotty, 1998) or what it is possible to know about the world (Snape & Spencer, 2003). The SAGE Online Dictionary of Social Research Methods (2006) defines ontology as "a concept concerned with the existence of, and relationship between, different aspects of society such as social actors, cultural norms and social structures... Ontological issues are concerned with questions pertaining to the kinds of things that exist within society". For Richards, (2003), ontology is the assumptions we make about the kind and nature of reality and what exists. Snape and Spencer (2003) also define ontology as the nature of the world and what we can know about it.

EPISTEMOLOGY

Epistemology in general is the assumptions we make about the kind or the nature of knowledge (Richards, 2003) or how it is possible to find out about the world (Snape & Spencer, 200). For Crotty (1998), epistemology is a way of looking at the world and making sense of it.

It involves knowledge and, necessarily, it embodies a certain understanding of what that knowledge entails. He further explains that epistemology deals with the 'nature' of knowledge, its possibility (what knowledge is possible and can be attempted and what is not), its scope and legitimacy.

ONTOLOGICAL POSITIONS (nature of the world and existence)	
Position	Assumptions
Objectivism	• Reality exists <i>independently</i> of our beliefs or understanding • Reality can be observed <i>directly</i> and accurately • A clear distinction exists between our beliefs about the world and the way the world is • Only material or physical world is considered 'real' • Social phenomena and their meanings cannot change • Events have causes and determined by other circumstances • The casual links between events and their causes can eventually be uncovered by science • Life is defined in 'measurable' terms rather than inner experiences • Notions of choice, freedom, individuality and moral responsibility are excluded
Constructionism	• External reality exists but is only known through human mind and socially constructed meanings • There is no shared social reality, only a series of different individual constructions of it • Reality is subjective • There exist only estimate or approximate observations or views of reality • Social phenomena and their meanings are continually being accomplished by social actors • Social phenomena and their meanings are produced through social interaction and are in a constant state of revision • Life is defined in 'estimate' terms based on inner experiences of humans where choice, freedom and individual responsibility are appreciated

If knowledge, on one hand, is viewed as hard, objective and tangible, this demands of the researcher an observer role together with an allegiance to the methods of natural science such as testing, measuring, etc.

EPISTEMOLOGICAL POSITIONS (nature of knowledge and how it is acquired)	
Position	Assumptions
Positivism	• The world is independent of and unaffected by the researcher • Facts and values are distinct • Objective and value-free inquiry is possible • Disputes are resolved through observations • Methods of natural science are appropriate for the study of social phenomena • Knowledge is produced through the senses based on careful observation • Only phenomena (and hence knowledge) confirmed by the senses can be genuinely regarded as knowledge • Knowledge is seen as hard, tangible and objective • Knowledge is arrived at through gathering of facts • Social world is approached through the <i>explanation</i> of human behaviour
Interpretivism	• The researcher and the social world impact on each other • Facts and values are not distinct • Objective and value-free inquiry is not possible since findings are inevitably influenced by the researchers' perspectives and values • Methods of natural science are not appropriate for the study of social phenomena for the social world is not governed by law-like regularities but mediated through meaning and human agency • Knowledge is produced by exploring and understanding the social world of the people being studied • Knowledge is seen as personal, subjective and unique • The researcher understands the social world using both his/her as well as the participants' understanding • Social world is approached through the <i>understanding</i> of human behaviour

Table 1. Key philosophical research positions and their underlying assumptions.

If knowledge, on the other hand, is viewed as personal, subjective and unique, then this imposes on the researcher a rejection of the methods used by natural science and a greater involvement with their subjects.

THE THEORETICAL PERSPECTIVE

The theoretical perspective is defined as “the theoretical stance informing the methodology and thus providing a context for the process and grounding its logic and criteria.” (Crotty, 2003:7). For e.g., Let’s say the ontology of any given study is

mainly concerned with the human world of meanings and interpretations and the epistemological stance is mainly constructionist in nature, thus, it is logically sequenced that interpretivism is the theoretical perspective underpinning this study.

METHODOLOGY

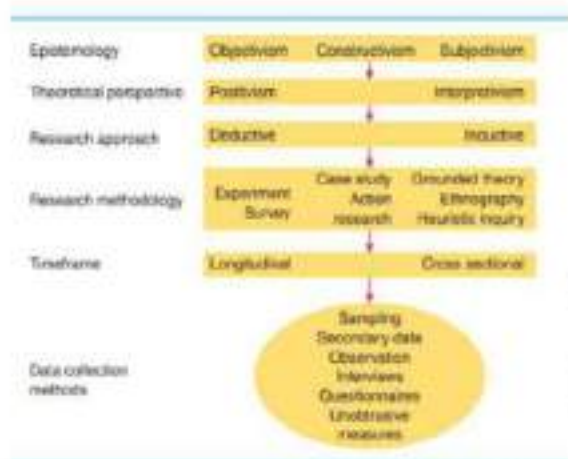
Methodology is “the strategy, plan of action, process or design lying behind the choice and use of particular methods and linking the choice and use of the methods to the desired outcomes.” Crotty, (2003:3).

It aims to describe, evaluate and justify the use of particular methods.

METHOD VS METHODOLOGY

Methods are ‘the techniques or procedures used to gather and collect data related to some research question or hypothesis’. (Crotty, 2003: 3).

For e.g., in a Research, a Case Study Methodology was used to design the given research. Therefore, the use of two methods – Interview Method and Reaction Papers Method, is justified.



Issue	Positivism	Post positivism	Critical Theory et al.	Constructionism
Ontology	Naïve realism - 'real' reality but apprehensible	critical realism - 'real' reality but only apprehensible and probably locally apprehensible	historical realism - critical reality shaped by social, political, cultural, economic, ethnic and gender values crystallised over time	relativism - local and specific constructed realities
Epistemology	deduct/objectivist findings true	modified deduct/objectivist, critical realism/community, findings probably true	transactional subjectivist, value mediated findings	transactional/subjectivist, created findings
Methodology	experimental manipulation, verification of hypotheses, strictly quantitative methods	modified experimental manipulation, critical analysis of hypotheses, may include qualitative methods	dialogic/dialectical	hermeneutic/dialectical
Axiology	propositional knowing about the world is an end in itself, is intrinsically valuable	propositional knowing about the world is an end in itself, is intrinsically valuable	propositional, transactional knowing is instrumentally valuable as a means to social emancipation, which is an end in itself, is intrinsically valuable	propositional, transactional knowing is instrumentally valuable as a means to social emancipation, which is an end in itself, is intrinsically valuable

Paradigms of Western Psychology: Positivism, Post-Positivism, Critical perspective,

Social Constructionism, Existential Phenomenology, and Co-operative Enquiry

POSITIVISM

The word 'Positivism' was coined by the French philosopher Auguste Comte. For Comte, positivism was the modern scientific outlook that was in the process of replacing previously dominant supernatural ways of thinking about the world. However, a more comprehensive version of positivism for 20th century social science was, 'Logical Positivism' or 'Logical Empiricism'

This became a prime or only model for inquiry and knowledge. The basic belief system of positivism is rooted in a realist ontology, i.e. the belief that there exists a reality 'out there', driven by immutable natural laws. Once committed to a realist ontology, the positivist is constrained to practice an objectivist epistemology. The positivists make use of manipulative or experimental methodology that allows the control and make use of manipulative or experimental methodology that allows the control and manipulation of variables to observe a cause-and-effect relationship. Therefore, the goal of inquiry is to form cause-and-effect relationship and thus values, beliefs, emotions, or anything "subjective" is prevented.

Hence, the basic belief system of logical positivism is-

Ontology: REALISM

Epistemology: DUALIST & OBJECTIVIST



Ontology:	<i>Critical realist—reality exists but can never be fully apprehended. It is driven by natural laws that can be only incompletely understood.</i>
Epistemology:	<i>Modified objectivist—objectivity remains a regulatory ideal, but it can only be approximated, with special emphasis placed on external guardians such as the critical tradition and the critical community.</i>
Methodology:	<i>Modified experimental/manipulative—emphasize critical multiplism. Redress imbalances by doing inquiry in more natural settings, using more qualitative methods, depending more on grounded theory, and reintroducing discovery into the inquiry process.</i>

Hence, the basic belief system of logical positivism is-

Ontology: REALISM

Epistemology: DUALIST & OBJECTIVIST

Methodology: EXPERIMENTAL & MANIPULATIVE

- Etic/Emic dilemma: Inability of general data to individual cases
- Received view of natural science followed blindly by positivists

Human science should study web of meanings that people live in.

DISENCHANTMENT WITH POSITIVISM

- Shattering of the 'objectivist' ideal of positivism
- Pluralization of life worlds
- Research conducted in the real world
- Knowledge needs to be 'locally', 'temporally' & 'situationally' relevant
- "Sensitizing Concepts" required to approach social contexts
- Recognition of central role of language & discourse
- Concern with process & individual rather than statistics & variables
- Human face' is lost in statistical manipulations
- Knowledge production is relative to frames of reference
- Truth is not absolute and is decided by human judgment

POSITIVISM: Context stripping, Exclusion of meaning, Theory ladenness

- Crisis of representation
- Exclusion of 'discovery' dimension in positivism

POST-POSITIVISM

CRITICAL PERSPECTIVE

It believes that nature cannot be seen as it "really is" or how it "really works" except through a value window. Thus, findings are value mediated. Reality is assumed to be constantly evolving over time & is 'apprehend able'. The reality is constantly shaped by social, political, cultural, economic, gender, and ethnic forces.

This critical form of research is a meta-process of investigation, which questions currently held values and assumptions and challenges conventional social structures. It invites both researchers and participants to discard what they term 'false consciousness' in order to develop new ways of understanding as a guide to effective action, confronting unjust social systems.

In a Marxist sense, the critical inquiry perspective is not content to interpret the world but also seeks to change it.

The aim of 'critical' paradigm is to smash myth & empower people to change society and thus Advocacy and Activism becomes its key concepts. At methodological level, critical theorists (ideologists) seem more consistent. The aim of the inquiry is to transform the world by raising the consciousness of participants so that they are energized and facilitated towards transformation. This is achieved by taking a DIALOGIC approach that seeks to eliminate false consciousness. In a way they critique conditions and imply a plan of change and provide vision of future to facilitate transformation. The believed criterion for progress is that "emancipation" should occur. The dialogue between the researcher and the participant must be dialectical in nature to transform ignorance.

Ontology: HISTORICAL REALISM

Epistemology: TRANSACTIONAL & SUBJECTIVIST

Methodology: DIALOGIC & DIALECTICAL

EXISTENTIAL PHENOMENOLOGY

Phenomenology holds that any attempt to understand social reality has to be grounded in people's experiences of that social reality. Hence, phenomenology insists that we must lay aside our prevailing understanding of phenomena and revisit our immediate experience of them in order that new meanings may emerge. Current understandings have to be 'bracketed' to the best of our ability to allow phenomena to 'speak for themselves', unadulterated by our preconceptions. The result will be new meaning, fuller meaning or renewed meaning. Attempts are made to avoid ways in which the prejudices of researchers bias the data. The key is gaining the subjective experience of the subject, sometimes by trying to put oneself in the place of the subject. Hence, phenomenology becomes an exploration, via personal experience, of prevailing cultural understandings. Value is ascribed not only to the interpretations of researchers, but also to the subjects of the research themselves.

SOCIAL CONSTRUCTIVISM

<i>Ontology:</i>	<i>Relativist—realities exist in the form of multiple mental constructions, socially and experientially based, local and specific, dependent for their form and content on the persons who hold them.</i>
<i>Epistemology:</i>	<i>Subjectivist—inquirer and inquired into are fused into a single (monistic) entity. Findings are literally the creation of the process of interaction between the two.</i>
<i>Methodology:</i>	<i>Hermeneutic, dialectic—individual constructions are elicited and refined hermeneutically, and compared and contrasted dialectically, with the aim of generating one (or a few) constructions on which there is substantial consensus.</i>

Phenomenological research

Study of the 'lifeworld' human experience
 Exploring the personal construction of the world
 Studying individuals
 Between 5 and 15 'participants'
 Use of in-depth, unstructured interviews
 Unit of analysis: meaning unit
 Reliability: confirmation by participants

BACKGROUND:

The German philosopher Friedrich Nietzsche (1844-1900) emphasized the sense of individuality and isolation to human existence by arguing that we have to identify our own meanings and direction without reference to God as our 'guiding-light' (Nietzsche, 1882). Thus, an engagement with the struggles and imperfections of life directly facilitates the process of establishing our own values and sense of self-empowerment (Deurzen-Smith, 1997). Here, Nietzsche introduced the key existential themes of choice, freedom, responsibility and courage (Deurzen-Smith, 1995). The German philosopher Martin Heidegger (1889-1976) (most often regarded as the founder of Existentialism) interpreted the term 'Dasein' (roughly translated as 'Being-in-the-world') to capture that the essence of human being is always in the world and always with others. In turn, this demands 'Authenticity', i.e. a full recognition of our ultimate 'being-towards-death' and other 'being' limitations whilst being true to our own sense of values and sense of direction (Heidegger, 1927). By contrast, 'Inauthenticity' is seen as a defense against the anxiety of this realization and is marked by a resignation to convention, conformity and duty, i.e. by doing what people imagine is expected of them (Deurzen-Smith, 1988).

Like Nietzsche, the French philosopher Jean-Paul Sartre (1905-1980) emphasized choice and freedom in human existence; we choose what we become through our actions which in turn, determines our possibilities for the future (Sartre, 1946). Sartre (1943) also contends that as humans we have no fixed essence but only and primarily our existence; we exist first and in the process of our existence have to create ourselves. However, in order to escape from the anxiety of this uncertainty of our nature we act at being something definite and 'fixed' (essentially a nothing pretending to be a something) through our *mauvaise foi* – 'bad faith' (or self -deception) (Sartre, 1943). The philosophy of Phenomenology, founded by the German philosopher Edmund Husserl (1859-1938) refers to the study of phenomena (the appearance of things) as they present to our consciousness (Spinelli, 1989). Thus, the 'reality' that each of us experiences is the result of an inseparable interaction between the raw matter of the physical world (e.g. things, people, places etc.) and our consciousness of it. This interactive process is known as 'intentionality', i.e. that our consciousness is always of something and involves an interpretative act from which our sense of reality is experienced and shaped (Spinelli, 1996). As such, our 'reality' is a phenomenal one open to a multiplicity of interpretations and meanings (Spinelli, 1989). From this perspective, there is no 'objective' and 'true' reality – we can only ever have an interpreted reality and one which is entirely ours' alone (Spinelli, 1996). Further still, this openness to our interpretations of experience directly invokes meaninglessness, i.e. that there is no ultimate 'truth' or meaning to our experiences (and therefore to life itself) other than what/how we interpret it; it is our construction based upon our personal values and beliefs.

Existential-phenomenological psychotherapy is therefore a philosophical approach to therapy (rather than a

technique per se). It concerns itself with how the 'givens' of human existence (death, anxiety, meaninglessness, isolation, choice, freedom and responsibility) are negotiated (or not) through the interpretations and meanings given to experience – to self, others and the world and how in turn, this shapes our sense of reality. As such, it regards that all peoples' forms of human dilemma, tragedy and suffering (from the normative to the extraordinary) are fundamentally problems about their 'Being-in-the-world'. Thus, all clients' difficulties are seen to reflect their attempts to avoid, resist or deny the angst and uncertainty that authentic living demands (Spinelli, 1989).

The basic framework through which the existential-phenomenological approach aims to understand and clarify clients' difficulties is through reference to the four realms of existence. **These are as follows:**

The Umwelt – the physical, biological dimension, our embodied existence;

The Mitwelt – the social/public dimension involving relating and interacting with others;

The Eigenwelt – the personal/psychological dimension, comprising our sense of identity;

The Uberwelt – the spiritual dimension referring to our sense of values, ideal and purpose (Deurzen-Smith, 1988).

Although people inhabit all four dimensions of existence at once, the salience and significance of each may become more/less apparent as the client's difficulties are explored and clarified in increasing depth and understanding. In order to facilitate the exploration of existence in these domains, there are three basic principles (or 'rules') of the **existential-phenomenological method which are as follows:**

Epoch – refers to a 'bracketing (or setting aside) of our initial assumptions, biases and prejudices within the therapeutic encounter in order to fully engage with the client and their material in a spirit of relative naivete.

Description – to facilitate a detailed description of the client's events, experiences and beliefs so that meanings within meanings may be identified and the implicit made explicit;

Horizontalization – each descriptive element of an event or experience is regarded as having equal value/significance so that an initial hierarchy of assumptions is avoided as the client's story unfolds (Spinelli, 1989).

CO-OPERATIVE ENQUIRY

Co-operative inquiry is one articulation of action research. The original proposal for **experiential inquiry** was put forward by **John Heron** in 1971. This **developed into practice of co-operative inquiry**. In co-operative inquiry, all those involved in the research endeavor are both co-researchers, whose thinking and decision-making contributes to generating ideas, designing and managing the project, and drawing conclusions from the experience; and also co-subjects, participating in the activity that is being researched.

PARTICIPATIVE WORLDVIEW

A participatory worldview is articulated based on a **subjective-objective ontology**; an extended **epistemology** of experiential, presentational, propositional, and practical ways of knowing; a **methodology** based on cooperative relations between coresearchers; and an **axiology** that affirms the primary value of practical knowing in the service of human flourishing.

Co-operative inquiry can be seen as cycling through four phases of reflection and action, drawing on a **fourfold “extended” epistemology**:

Experiential knowing is through direct face-to-face encounter with a person, place or thing; it is knowing through empathy and resonance, that kind of in-depth knowing which is almost impossible to put into words.

Presentational knowing grows out of experiential knowing, and provides the first form of expression through story, drawing, sculpture, movement, dance, drawing on aesthetic imagery.

Propositional knowing draws on concepts and ideas, and,

Practical knowing consummates the other forms of knowing in action in the world.

In **Phase I**, a group of co-researchers come together to explore an agreed area of human activity. In this first phase, they agree on the focus of their inquiry and together develop a set of questions or propositions they wish to explore. They agree to undertake some action, some practice, which will contribute to this exploration, and agree to a set of procedures by which they will observe and record their own and each other's experience. Phase I is primarily in the mode of propositional knowing, although it will also contain important elements of presentational knowing as group members use their imagination in story, fantasy, and graphics to help them articulate their interests and to focus on their purpose in the inquiry. Once the focal idea—what the inquiry is about—is agreed upon, Phase I will conclude with planning a method for exploring the idea in action, and with devising ways of gathering and recording data from this experience.

In **Phase II**, the co-researchers now also become co-subjects: they engage in the actions agreed, observing and recording the process and outcomes of their own and each other's experience. In particular, they are careful to notice the subtleties of experience, to hold lightly the propositional frame from which they started so that they are able to notice how practice does and does not conform to their original ideas. This phase involves primarily practical knowledge: knowing how (and how not) to engage in appropriate action, to bracket off the starting idea, and to exercise relevant discrimination.

In **Phase III** is, in some ways, the touchstone of the inquiry method. It is a stage in which the co-subjects become fully immersed in and engaged with their experience. They may develop a degree of openness to what is going on so free of pre-conceptions that they see it in a new way. They may deepen into the experience so that superficial understandings are elaborated and developed. Or they may be led away from the original ideas and proposals into new fields, unpredicted action and creative insights. It is also possible that they may get so involved in what they are doing that they lose the awareness that they are part of an inquiry group: there may be a practical crisis, they may become enthralled, they may simply forget. Phase III involves mainly experiential knowing, although it will be richer if new experience is expressed, when recorded, in creative presentational form through graphics, color, sound, movement, drama, story, poetry, and so on. In Phase IV, after an agreed period engaged in phases II and III, the co-researchers reassemble to consider their original propositions and questions in the light of their experience. As a result, they may modify, develop or reframe them; or reject them and pose new questions. They may choose, for the next cycle of action, to focus on the same or on different aspects of the overall inquiry. The group may also

choose to amend or develop its inquiry procedures—forms of action, ways of gathering data—in the light of experience.

Phase IV is primarily the stage of propositional knowing, although presentational forms of knowing will form an important bridge with the experiential and practical phases. In a more complete inquiry, the cycle will be repeated several times. The group itself may become more cohesive and self-critical, more skilled in its work and in the practices of inquiry. Ideally the inquiry is finished when the initial (and emergent) questions and concerns have been thoroughly addressed in practice, when there is a new congruence between the four kinds of knowing. It is, of course, rare for a group to complete an inquiry so fully. It should be noted that the actual process is not as straightforward as the model suggests: there are usually mini-cycles within major cycles; some cycles will emphasize one phase more than others; and some practitioners have advocated a more emergent process of inquiry, which is less structured into phases; nevertheless, the discipline of the research cycle is fundamental.

SIGNIFICANT INDIAN PARADIGMS ON PSYCHOLOGICAL KNOWLEDGE: YOGA, BHAGAVAD GITA, BUDDHISM, SUFISM, AND INTEGRAL YOGA.

From September 29 to October 1, 2002, more than 150 Indian psychologists met in Pondicherry at the National Conference on Yoga and Indian Approaches to Psychology. These psychologists issued a declaration that has become known as the **Manifesto on Indian Psychology**, which was published in *Psychological Studies*, the journal of the Indian National Academy of Psychology. The Manifesto affirmed that "Rich in content, sophisticated in its methods and valuable in its applied aspects, Indian psychology is pregnant with possibilities for the birth of

new models in psychology that would have relevance not only to India but also to psychology in general.... By Indian psychology we mean a distinct psychological tradition that is rooted in Indian ethos and thought, including the variety of psychological practices that exist in the country". The Manifesto also recommended eight "necessary steps for responsibly promoting psychology in India" that ranged from preparing resource materials to offering student fellowships, conducting seminars, offering courses, generating a website, and appointing a committee for follow-up action to ensure the implementation of the recommendations.

Rao and Paranjpe (2016) stated that Indian psychology, "subscribes to methodological pluralism. Without rejecting objectivity, control, and simplicity of experimental exploration, Indian psychology recognizes the need to go beyond experimentation to usefully capture the richness and manifold complexity of human experience to embrace and accommodate phenomenological methods as coequals."

INDIAN TRADITION: ONTOLOGY

Two major planes (satta) of reality:

TRANSCENDENTAL REALITY (PARMARTHIKA)

is considered to be non-changing and universal. it can be experienced in this life under certain conditions like samadhi. can be approximately termed as the spiritual realm.

EMPIRICAL REALITY (VYAVHARIKA)

is that which is apprehended through the sense modalities. can be approximately understood as the material realm (which includes all physio-psycho-social aspects of mundane existence).

EPISTEMOLOGY:

(a) PARMARTHIKA – Knower and known are one and the same (ahm brahma asmi).

(b) VYAVAHARIKA – Subject and object dichotomy is upheld (knowledge of object is within the reach of knower).

METHODOLOGY:**(A) EXPERIENTIAL; (B) EMPIRICAL**

(a) PARMARTHIKA– This involves sadhana (spiritual praxis) in the supervision of a rsi or drsta (seer). Ultimate authority in India is in practice held to belong not to ancient books nor their learned expositions, but to those who have personal experience of spiritual truth. It is a living vision that transforms the inner life, faculties and powers of the person who attains it.

Sadhana also denotes ‘making’, and thus implicates transformation of self. It involves detached truth seeking (jijnasa), sensitiveness (aksipartrakalpata), earnestness (samvega), maturity and wisdom (viveka), and realization of the constraints and limitations emanating from egoism and acquisitiveness. In general, three ways (marga) of spiritual realization are in practice; knowledge, action and devotion (jnana, karma and bhakti).

(b) VYAVAHARIKA– The major means of knowledge include the following:

pratyaksa (perception)
 anumana (inference)
 upamana (comparison)
 sabda (verbal testimony)
 arthapatti (postulation) and
 anupalabdhi (non-cognition)

SCIENCE AND SPIRITUALITY **(AVIDYA AND VIDYA)**

Vedic concept of knowledge is something entirely different from the scientific concept of knowledge. The ancients were aware of the more mundane type of knowledge, but they were less exclusively interested in it than modern science. **Avidya, or "ignorance"**, as they called it somewhat disdainfully, denotes all knowledge that is not knowledge of the Absolute and the word is specifically used for knowledge of the world, in other words, for science. According to Sri Aurobindo's interpretation of the Isha Upanishad, both **vidya (Knowledge of the One)** and **avidya (knowledge of the multiplicity) are needed for a complete understanding of reality.**

Vidya: The higher knowledge of the Self (Atman) and the Supreme Self. spiritual knowledge arises from the contact with the Self in a deeper state of self-absorption, at the end of a long spiritual journey and self-purification. It arises in a state in which the mind and the senses are withdrawn and fully resting. Spiritual knowledge is transcendental knowledge which arises in a person who is self-aware without the involvement of his mind or its faculties.

Avidya: The lower knowledge of the rituals, sacrifices, obligatory duties, occupational knowledge. This knowledge is considered inferior, because it does not liberate us from the cycle of births and deaths. Instead, it leads to more attachment, karma, egoism, worldliness and more involvement with things and Nature. All learned knowledge is accumulated knowledge and as long as the mind is clogged with worldly knowledge, peace and equanimity are not possible.

From the lower knowledge comes the discipline and the ability to practice the higher and realize the highest. They also affirm that knowledge of Brahman is the highest knowledge, because it is permanent, unchanging and indivisible, by knowing which there is no further knowing.

It is important to know that Brahman is not an object of knowledge, something that we need to know or gain or possess or remember but the goal of our spiritual effort to achieve liberation.

THE PRIMACY OF SELF-KNOWLEDGE IN INDIAN PSYCHOLOGY

REFERENCE:

<https://ipi.org.in/texts/matthijs/mc-tok-ppb.php>

A consistent view of orthodox Hindu philosophy is that the Self is an independent, incorporeal substance. The Yogic ontology, a subscriber to this theme, is inherently dualistic. It endorses two ultimate realities- Purusa or pure consciousness, which is non-corporeal and thus Nirguna and the psycho-physical substantiality called Prakriti, which is unconscious. Prakriti endorses three attributes or Guna and is thus Saguna. It is in the unique interaction of the higher Self or Pursha, and the Prakriti that the world and the human beings are created. Prakriti and its components namely Buddhi (intelligence), Ahamkara (ego-sense), Manas (mind) and Indriya (senses) borrow consciousness of the Self (Purusa). However, in course of interaction with the material world each of these components mistakes itself to be the absolute reality or the Self.

The four "knowledge realms" are as follows:

- **OBJECTIVE KNOWLEDGE.** This is the knowledge we have of the physical and socio-economic world

around us. It is sense-based and (supposed to be) guided by reason and "common sense". There are two varieties of it. The naive variety is simply whatever ordinary people know about the world outside of themselves. The expert variety is science. These two don't differ in principle, but they differ considerably in their actual processes and results. Science is more rigorous, specialized and cumulative; the senses are extended by instruments that have been constructed with the help of knowledge of this same type; the reason is extended in the form of mathematics. Modernity is the scene of an almost incredible collective growth of this type of knowledge.

- **SUBJECTIVE KNOWLEDGE.** Subjective knowledge is the knowledge we have of what is happening inside ourselves. The word "subjective" has nowadays largely negative connotations, and I use it here only for the naive variety of what we know about our own nature and our own self-existence. Within the realm of subjective knowledge one can distinguish three types: introspection which is a naive attempt at being "objective" about oneself (knowledge of type three), experiential knowledge which deals with processes we intimately identify with (knowledge of type two), and a superficial, ego-based awareness of one's own identity. All three are limited in scope and "subjective knowledge" has access only to a tiny fraction of what we are and what happens inside ourselves.

knowledge realm	usage	knowledge mode & type (acc. to Sri Aurobindo)	known reality		
objective knowledge	A. ordinary, sense-based knowing	naïve <i>separative, indirect knowledge</i> (4)	physical world	O W C	
	B. objective science	expert <i>separative, indirect knowledge</i> (4)			
subjective knowledge	C. Introspection	naïve <i>separative, direct knowledge</i> (3)	outer nature		Y O G A
	D. superficial experience	naïve <i>knowledge by intimate, direct contact</i> (2)			
	E. superficial awareness of own existence	naïve <i>knowledge by identity</i> (1)	surface self		
Inner knowledge	F. witness consciousness (<i>sakshibhava</i>); <i>purusha</i> -based self-observation	expert <i>separative, direct knowledge</i> (3)	inner nature	Y O G A	
	G. consciousness directly touching other consciousness	expert <i>knowledge by intimate, direct contact</i> (2)			
Self-knowledge	H. truth-consciousness, gnosis, intuition	expert <i>knowledge by identity</i> (1)	true Self, Real-Ideas		

INNER KNOWLEDGE. This consists of the sophisticated, expert variety of two types of knowledge of which subjective knowledge uses the naïve variety. Expert knowledge of Sri Aurobindo's type three is the pure, detached witness consciousness that allows genuinely "objective" knowledge of whatever happens in one's own inner nature. The expert variety of type two, knowledge by intimate direct contact, allows one's consciousness to touch directly the consciousness in others and even in things so that one can know these by an intimate, unmediated direct contact.

SELF-KNOWLEDGE. This is the expert variety of knowledge by identity (type one) and it leads us directly to who we are in the very essence of our being. The little of real self-knowledge that reaches our surface consciousness may never attain to that level of perfection, but in itself this type of knowledge is intrinsically true and perfect. It is the secret origin of whatever there is of real truth in all

other types of knowledge. According to the Indian tradition, a perfect knowledge of oneself automatically gives in principle the possibility of perfect knowledge of everything else.

As mentioned before, the realms of objective and subjective knowledge (as defined here) are the only ones that can be accessed fully in the **ordinary waking consciousness (or OWC)**.

Power Within

CONTENTS

RESEARCH: Meaning, Purpose, and Dimensions. Research problems, Variables and Operational Definitions, Hypothesis, Sampling. Ethics in conducting and reporting research

PARADIGMS OF RESEARCH: Quantitative, Qualitative, Mixed methods approach
Methods of research: Observation, Survey [Interview, Questionnaires], Experimental, Quasi-experimental, Field studies, Cross-Cultural Studies, Phenomenology, Grounded theory, Focus groups, Narratives, Case studies, Ethnography

STATISTICS IN PSYCHOLOGY: Measures of Central Tendency and Dispersion. Normal Probability Curve. Parametric [t-test] and Non-parametric tests [Sign Test, Wilcoxon Signed rank test, Mann-Whitney test, Kruskal-Wallis test, Friedman]. Power analysis. Effect size.

CORRELATIONAL ANALYSIS: Correlation [Product Moment, Rank Order], Partial correlation, multiple correlation. Special Correlation Methods: Biserial, Point biserial, tetrachoric, phi coefficient. Regression: Simple linear regression, Multiple regression. Factor analysis: Assumptions, Methods, Rotation and Interpretation.

EXPERIMENTAL DESIGNS: ANOVA [One-way, Factorial], Randomized Block Designs, Repeated Measures Design, Latin Square, Cohort studies, Time series, MANOVA, ANCOVA. Single-subject designs.

UNIT 2

UNIT 2: RESEARCH METHODOLOGY AND STATISTICS

RESEARCH: MEANING, PURPOSE, AND DIMENSIONS

Kerlinger- “Scientific research is a systematic, controlled, empirical and critical investigation of hypothetical

proposition about the presumed relations among natural phenomena.”

Aims of Research:

- **Explorative/Formulative Research**
Studies: Achieving new insights.
- Hypothesis Testing Research**

Point of Departure is driven by Research problem/ question.	Point of departure is driven by Specific tasks such as data collection, or sampling, etc.
Focuses on the logic of research: What evidence is required to address the question adequately?	Focuses on the individual steps in the research process.

RESEARCH PROBLEMS, VARIABLES AND OPERATIONAL DEFINITIONS, HYPOTHESIS, SAMPLING.

THE RESEARCH PROBLEM

Formulating a research problem determines almost every step that follows.

Sources of research problems: 4 P's

People
Problems
Programs
Phenomena

There must be some objectives to be attained to. (Researcher's Hypothesis)

There must be an alternative means to the objective. (Alternative Hypothesis)

IDENTIFYING VARIABLES

Variables are the properties that take on different values and is measurable.

TYPES OF VARIABLES:

1. **Independent Variables** vs. **Dependent Variables**

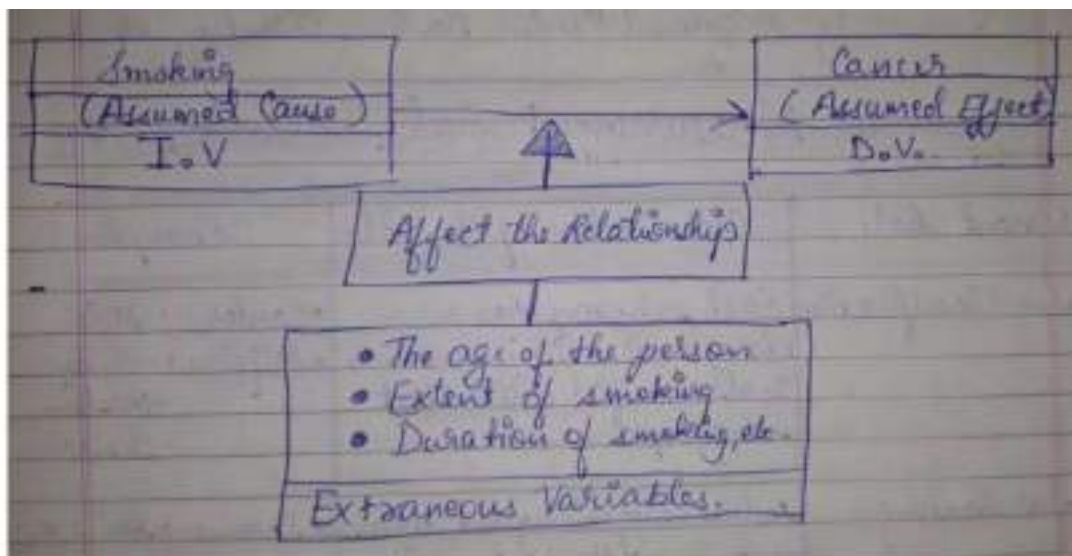
Independent variables	Dependent variables
Independent variables are manipulated by the experimenter	Dependent Variable is observed and recorded by the experimenter.
	It depends on the behavior of the participant, which in turn, is supposed to depend on the independent variable.

Control Variables/ Extraneous variables

A control variable is a potential independent variable that is held constant during an experiment.

It does not vary as it held under control.

E.g., Temperature, Time of the day, etc.

**2. Quantitative vs Qualitative Variables**

Quantitative are numbers (Interval/ Ratio)

Qualitative are attitude (good or bad). They can be compared but not measured (Nominal/Ordinal).

3. Active vs Attribute Variables

Active Variables are those that can be manipulated, changed or controlled experimentally.

Attribute variables are those that cannot be manipulated, or controlled, rather reflect the characteristics of the study population. They are, thus, the pre-existing qualities of the population.

E.g. level of motivation, Gender, Age etc.

4. Continuous vs Discrete variables

The Continuous variables exist between a range, say from 30 to 40, the value can be 30.1 to 39.9

The discrete Variables are the absolute values and can be divided into 3 parts-

Constant: When a variable can have only one value. E.g., Tree, Taxi, etc

Dichotomous: Can have two values. E.g., Yes/No, Good/Bad, etc

Polytomous: Can have many values. E.g., Religion

Controlling Extraneous Variables

1. Elimination: Noise can be eliminated by using sound-proof situations or settings.

2. Constancy: By holding the extraneous values constant in all situations. E.g., Choosing subjects of same sex.

Balancing: Participants are made equal in all aspects in both controlled and experimental groups.

Counter Balancing: Used to control variables occurring as a

result of practice or fatigue together called as order effect. E.g., AB BA design treatment.

Randomization: Each member of the population, having an equal chance to be selected. This technique is applied where the extraneous variable are known, but these effects can't be controlled by known technique.

ISG: INDUCTIVE IS SPECIFIC TO GENERAL

AIM OF A HYPOTHESIS

TESTING IS TO FIND-

Difference

Relationship (Statistically significant- Implies difference in result did not occur by chance)

Interaction

CONSTRUCTING HYPOTHESIS

Characteristics of Hypothesis

It should be simple, specific and conceptually clear.

It should be capable of verification.

It should be related to the existing body of knowledge.

It should be operationalized i.e. could be expressed in terms of measurement.

Difficulties in Formulating a Hypothesis:

The absence of knowledge of a theoretical framework.

When the investigator lacks the ability to utilize the knowledge of the theoretical framework.

When the investigator is not aware of the important scientific research techniques.

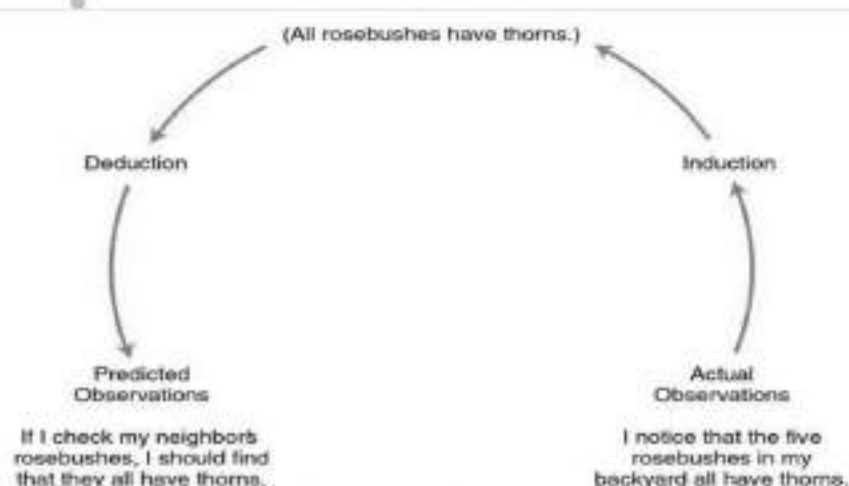
HYPOTHESIS:

A hypothesis (plural hypotheses) is a precise, testable statement of what the researchers predict will be the outcome of the study.

This usually involves proposing a possible relationship between two variables: the independent variable (what the researcher changes) and the dependent variable (what the research measures).

In research, there is a convention that the hypothesis is written in two forms, the null hypothesis, and the alternative hypothesis (called the experimental hypothesis when the method of investigation is an experiment).

TYPES OF RESEARCH



HYPOTHESES

Alternative Hypothesis

The alternative hypothesis states that there is a relationship between the two variables being studied (one variable has an effect on the other). It states that the results are not due to chance and that they are significant in terms of supporting the theory being investigated.

Null Hypothesis

The null hypothesis states that there is no relationship between the two variables being studied (one variable does not affect the other). It states results are due to chance and are not significant in terms of supporting the idea being investigated.

Nondirectional Hypothesis

Two-tailed non-directional hypothesis predicts that the independent variable will have an effect on the dependent variable, but the direction of the effect is not specified. E.g., there will be a difference in how many numbers are correctly recalled by children and adults.

Research Question: Is there a (statistically) significant difference between males and females with respect to math achievement? **H₀:** There is no (statistically) significant difference between males and females with respect to math achievement. **H_A:** There is a (statistically) significant difference between males and females with respect to math achievement.

Directional Hypothesis

A one-tailed directional hypothesis predicts the nature of the effect of the independent variable on the dependent variable. E.g., adults will correctly recall more words than children.

Research Question: Do males score significantly higher than females with respect to math achievement?

H₀: Males do not score significantly

higher than females with respect to math achievement.

H_A: Males score significantly higher than females with respect to math achievement.

There are other types of Hypothesis too which you can refer to for knowledge, not necessary to go into detail for NET. Above defined types are sufficient. They are:

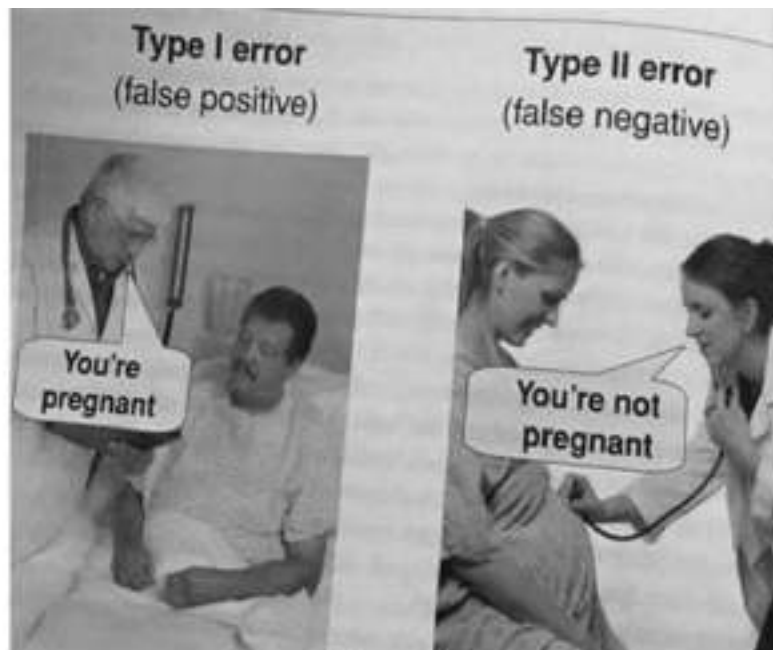
1. Research Hypothesis vs Null vs Alternate Hypothesis
2. Substantive vs Statistical Hypothesis
3. Simple vs Complex Hypothesis
4. Null vs Positive vs Negative Hypothesis
5. Universal vs Existential Hypothesis

TYPE I AND II ERRORS (IMPORTANT TO UNDERSTAND THE DIFFERENCE)

Type I error: reject a null hypothesis that is really true (with tests of difference this means that you say there was a difference between the groups when there really was not a difference). The probability of making a Type I error is the alpha level you choose. If you set your probability (alpha level) at $p < 0.05$, then there is a 5% chance that you will make a Type I error. You can reduce the chance of making a Type I error by setting a smaller alpha level ($p < .01$). The problem with this is that as you lower the chance of making a Type I error, you increase the chance of making a Type II error.

Type II error: fail to reject a null hypothesis that is false (with tests of differences this means that you say there

inferring patterns of behaviors within specific populations. We cannot study entire populations because of feasibility and cost constraints, and hence, we must select a representative sample from the



was no difference between the groups when there really was one)

SAMPLING

Sampling is the statistical process of selecting a subset (called a “sample”) of a population of interest for purposes of making observations and statistical inferences about that population. Social science research is generally about

population of interest for observation and analysis. It is extremely important to choose a sample that is truly representative of the population so that the inferences derived from the sample can be generalized back to the population of interest. Sampling techniques can be grouped into two broad categories: **probability (random) sampling and non-probability**

The Sampling Process



sampling. Probability sampling is ideal if generalizability of results is important for your study, but there may be unique circumstances where non-probability sampling can also be justified.

1. PROBABILITY SAMPLING

Probability sampling is a technique in which every unit in the population has a chance (non-zero probability) of being selected in the sample, and this chance can be accurately determined. Sample statistics thus produced, such as sample mean or standard deviation, are unbiased estimates of population parameters, as long as the sampled units are weighted according to their probability of selection. All probability sampling has two attributes in common: (1) every unit in the population has a known non-zero probability of being sampled, and (2) the sampling procedure involves random selection at some point. The different types of probability sampling techniques include:

Simple random sampling: In this technique, all possible subsets of a population (more accurately, of a sampling frame) are given an equal probability of being selected. The probability of selecting any set of n units out of a total of N units in a sampling frame is $\frac{1}{N C n}$. Simple random sampling involves randomly selecting respondents from a sampling frame, but with large sampling frames, usually a table of random numbers or a computerized random number generator is used. For instance, if you wish to select 200 firms to survey from a list of 1000 firms, if this list is entered into a spreadsheet like Excel, you can use **Excel's RAND()** function to generate random numbers for each of the 1000 clients on that list. Next, you sort the list in increasing order of their corresponding random number, and select the first 200 clients on that sorted list. This is the simplest of all probability sampling techniques; however, the simplicity is also

the strength of this technique. Because the sampling frame is not subdivided or partitioned, the sample is unbiased and the inferences are most generalizable amongst all probability sampling techniques.

Systematic sampling: In this technique, the sampling frame is ordered according to some criteria and elements are selected at regular intervals through that ordered list. Systematic sampling involves a random start and then proceeds with the selection of every ' k 'th element from that point onwards, where $k = N / n$, **where k is the ratio of sampling frame size N and the desired sample size n , and is formally called the sampling ratio.** It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first k elements on the list. In our previous example of selecting 200 firms from a list of 1000 firms, you can sort the 1000 firms in increasing (or decreasing) order of their size (i.e., employee count or annual revenues), randomly select one of the first five firms on the sorted list, and then select every fifth firm on the list. This process will ensure that there is no overrepresentation of large or small firms in your sample, but rather that firms of all sizes are generally uniformly represented, as it is in your sampling frame. In other words, the sample is representative of the population, at least on the basis of the sorting criterion.

Stratified sampling: In stratified sampling, the sampling frame is divided into homogeneous and non-overlapping subgroups (called "strata"), and a simple random sample is drawn within each subgroup. In the previous example of selecting 200 firms from a list of 1000 firms, you can start by categorizing the firms based on their size as large (more than 500 employees), medium (between 50 and 500 employees), and small (less than 50 employees). You can then randomly select 67 firms from each subgroup to

make up your sample of 200 firms. However, since there are many more small firms in a sampling frame than large firms, having an equal number of small, medium, and large firms will make the sample less representative of the population (i.e., biased in favor of large firms that are fewer in number in the target population). This is called non-proportional stratified sampling because the proportion of sample within each subgroup does not reflect the proportions in the sampling frame (or the population of interest), and the smaller subgroup (large-sized firms) is over-sampled. An alternative technique will be to select subgroup samples in proportion to their size in the population. For instance, if there are 100 large firms, 300 mid-sized firms, and 600 small firms, you can sample 20 firms from the “large” group, 60 from the “medium” group and 120 from the “small” group. In this case, the proportional distribution of firms in the population is retained in the sample, and hence this technique is called proportional stratified sampling. Note that the non-proportional approach is particularly effective in representing small subgroups, such as large-sized firms, and is not necessarily less representative of the population compared to the proportional approach, as long as the findings of the non-proportional approach is weighted in accordance to a subgroup’s proportion in the overall population.

Cluster sampling: If you have a population dispersed over a wide geographic region, it may not be feasible to conduct a simple random sampling of the entire population. In such case, it may be reasonable to divide the population into “clusters” (usually along geographic boundaries), randomly sample a few clusters, and measure all units within that cluster. For instance, if you wish to sample city governments in the state of New York, rather than travel all over the state to interview key city officials (as you may have to do with a simple random sample),

you can cluster these governments based on their counties, randomly select a set of three counties, and then interview officials from every official in those counties. However, depending on between- cluster differences, the variability of sample estimates in a cluster sample will generally be higher than that of a simple random sample, and hence the results are less generalizable to the population than those obtained from simple random samples.

EXTRA FOR REFERENCES

Matched-pairs sampling: Sometimes, researchers may want to compare two subgroups within one population based on a specific criterion. For instance, why are some firms consistently more profitable than other firms? To conduct such a study, you would have to categorize a sampling frame of firms into “high profitable” firms and “low profitable firms” based on gross margins, earnings per share, or some other measure of profitability. You would then select a simple random sample of firms in one subgroup, and match each firm in this group with a firm in the second subgroup, based on its size, industry segment, and/or other matching criteria. Now, you have two matched samples of high-profitability and low-profitability firms that you can study in greater detail. Such matched-pairs sampling technique is often an ideal way of understanding bipolar differences between different subgroups within a given population.

Multi-stage sampling: The probability sampling techniques described previously are all examples of single-stage sampling techniques. Depending on your sampling needs, you may combine these single-stage techniques to conduct multi-stage sampling. For instance, you can stratify a list of businesses based on firm size, and then conduct systematic sampling within each stratum. This is a two-stage combination of stratified and systematic sampling. Likewise, you can start with a

cluster of school districts in the state of New York, and within each cluster, select a simple random sample of schools; within each school, select a simple random sample of grade levels; and within each grade level, select a simple random sample of students for study. In this case, you have a four-stage sampling process consisting of cluster and simple random sampling.

NONPROBABILITY SAMPLING

It is a sampling technique in which some units of the population have zero chance of selection or where the probability of selection cannot be accurately determined. Typically, units are selected based on certain non-random criteria, such as quota or convenience. Because selection is non-random, nonprobability sampling does not allow the estimation of sampling errors, and may be subjected to a sampling bias. Therefore, information from a sample cannot be generalized back to the population. Types of non-probability sampling techniques include:

Convenience sampling: Also called accidental or opportunity sampling, this is a technique in which a sample is drawn from that part of the population that is close to hand, readily available, or convenient. For instance, if you stand outside a shopping center and hand out questionnaire surveys to people or interview them as they walk in, the sample of respondents you will obtain will be a convenience sample. This is a non-probability sample because you are systematically excluding all people who shop at other shopping centers. The opinions that you would get from your chosen sample may reflect the unique characteristics of this shopping center such as the nature of its stores (e.g., high end-stores will attract a more affluent demographic), the demographic profile of

its patrons, or its location (e.g., a shopping center close to a university will attract primarily university students with unique purchase habits), and therefore may not be representative of the opinions of the shopper population at large. Hence, the scientific generalizability of such observations will be very limited. Other examples of convenience sampling are sampling students registered in a certain class or sampling patients arriving at a certain medical clinic. This type of sampling is most useful for pilot testing, where the goal is instrument testing or measurement validation rather than obtaining generalizable inferences.

Quota sampling: In this technique, the population is segmented into mutually-exclusive subgroups (just as in stratified sampling), and then a non-random set of observations is chosen from each subgroup to meet a predefined quota. In proportional quota sampling, the proportion of respondents in each subgroup should match that of the population. For instance, if the American population consists of 70% Caucasians, 15% Hispanic-Americans, and 13% African-Americans, and you wish to understand their voting preferences in a sample of 98 people, you can stand outside a shopping center and ask people their voting preferences. But you will have to stop asking Hispanic-looking people when you have 15 responses from that subgroup (or African-Americans when you have 13 responses) even as you continue sampling other ethnic groups, so that the ethnic composition of your sample matches that of the general American population. Non-proportional quota sampling is less restrictive in that you don't have to achieve a proportional representation, but perhaps meet a minimum size in each subgroup. In this case, you may decide to have 50 respondents from each of the three ethnic subgroups (Caucasians, Hispanic-Americans, and African-Americans), and stop when your quota for each subgroup is

reached. Neither type of quota sampling will be representative of the American population, since depending on whether your study was conducted in a shopping center in New York or Kansas, your results may be entirely different. The non-proportional technique is even less representative of the population but may be useful in that it allows capturing the opinions of small and underrepresented groups through oversampling.

Expert sampling: This is a technique where respondents are chosen in a non-random manner based on their expertise on the phenomenon being studied. For instance, in order to understand the impacts of a new governmental policy such as the Sarbanes-Oxley Act, you can sample an group of corporate accountants who are familiar with this act. The advantage of this approach is that since experts tend to be more familiar with the subject matter than non-experts, opinions from a sample of experts are more credible than a sample that includes both experts and non-experts, although the findings are still not generalizable to the overall population at large.

Snowball sampling: In snowball sampling, you start by identifying a few respondents that match the criteria for inclusion in your study, and then ask them to recommend others they know who also meet your selection criteria. For instance, if you wish to survey computer network administrators and you know of only one or two such people, you can start with them and ask them to recommend others who also do network administration. Although this method hardly leads to representative samples, it may sometimes be the only way to reach hard-to-reach populations or when no sampling frame is available.

EXTRA FOR REFERENCES:

Statistics of Sampling

In the preceding sections, we introduced terms such as population parameter, sample statistic, and sampling bias. In this section, we will try to understand what these terms mean and how they are related to each other.

When you measure a certain observation from a given unit, such as a person's response to a Likert-scaled item, that observation is called a response (see Figure 8.2). In other words, a response is a measurement value provided by a sampled unit. Each respondent will give you different responses to different items in an instrument. Responses from different respondents to the same item or observation can be graphed into a frequency distribution based on their frequency of occurrences. For a large number of responses in a sample, this frequency distribution tends to resemble a bell-shaped curve called a normal distribution, which can be used to estimate overall characteristics of the entire sample, such as sample mean (average of all observations in a sample) or standard deviation (variability or spread of observations in a sample). These sample estimates are called sample statistics (a "statistic" is a value that is estimated from observed data). Populations also have means and standard deviations that could be obtained if we could sample the entire population. However, since the entire population can never be sampled, population characteristics are always unknown, and are called population parameters (and not "statistic" because they are not statistically estimated from data). Sample statistics may differ from population parameters if the sample is not perfectly representative of the population; the difference between the two is called sampling error. Theoretically, if we could gradually increase the sample size so that the sample approaches closer and closer to the population, then sampling error will decrease and a sample statistic will

increasingly approximate the corresponding population parameter.

ETHICS IN CONDUCTING AND REPORTING RESEARCH

ETHICAL DILEMMA DURING DIFFERENT STAGES OF THE RESEARCH

A research project follows a developmental trajectory. Each stage has a different purpose that, to a certain degree, shapes the respective roles of the participants and the researcher. The developmental nature of the research process leads to changes in power relations, which pose specific ethical issues to the researcher.

Three major stages in qualitative research are-

- Initial stage of subject/participant recruitment
- Data collection
- Data analysis and production of the report

INITIAL STAGE OF SUBJECT/PARTICIPANT RECRUITMENT

At this stage, ethical dilemmas involve questions about strategically obscuring some of the research goals to persuade the participants to take part in the study. This raises concerns about the participants feeling an obligation to participate. This is also the case when researchers tend to reduce the ethical principle of informed consent to a formal regulatory requirement by IRBs or other overseeing bodies.

After the data are transferred into the realm of research, they become research

property, to be used under conditions that commit the researcher to certain ethical guidelines.

These conditions are as follows:

Participants must fully understand (at the level known to the researcher at that point) the meaning of the study and truly volunteer to participate in it. Researchers must not distort the meaning of the participants' voices. Researchers must protect the anonymity of the participants. Researchers have an obligation to participants' beneficence—an obligation to provide benefits for the participant and to balance such benefits against risks. Researchers have an obligation to non-maleficence that requires doing no harm.

One of the major factors associated with observational studies is ethics. While observation is generally seen as the least intrusive data collection method, it can also be an abuse of an individual's privacy

It was emphasized that we should ensure that participation is voluntary, no harm comes to the respondents, the research is of some benefit to the participants, and that their privacy will be protected. One way of addressing all these concerns is through the informed consent process whereby verbal or written statements are shared with research participants to inform them about the topic of the study, its potential benefit or harm, and the specific steps taken to guard their privacy.

The difficulties inherent in qualitative research can be alleviated by awareness and use of well-established ethical principles, specifically autonomy, beneficence, and justice.

1. AUTONOMY

In a qualitative research study this principle is honoured by informed consent, which means making a reasonable balance between over-informing and under-informing. It also means that participants exercise their rights as autonomous persons to voluntarily accept or refuse to participate in the study.

2. BENEFICENCE

A second ethical principle closely linked with research is beneficence—doing good for others and preventing harm. Beneficence in some situations may be taken to the extreme as paternalism. A paternalistic approach indicates the denial of autonomy and freedom of choice. For example, the researcher may want to study the problem of violence among elderly women but may decide not to include them because they may be too vulnerable. In this case, the researcher is not giving elderly women the opportunity to decide for themselves and for their experiences to be heard.

If researchers are maintaining the principle of beneficence, overseeing the potential consequences of revealing participants' identities is a moral obligation. The use of pseudonyms is recommended.

3. JUSTICE

The principle of justice refers to equal share and fairness. One of the crucial and distinctive features of this principle is avoiding exploitation and abuse of participants. The researcher's understanding and application of the principle of justice in qualitative research studies is demonstrated by recognizing vulnerability of the participants and their contributions to the study.

PARADIGMS OF RESEARCH: QUANTITATIVE, QUALITATIVE, MIXED METHODS APPROACH

WHAT IS QUALITATIVE RESEARCH?

“Qualitative approach emphasizes the qualities of entities, processes and meanings that are not experimentally examined or measured in terms of quantity, amount, intensity or frequency. (Denzin & Lincoln, 2000)

The root term from which the phrase has been historically derived is from Latin, *QUALITAS* refers to a primary focus on qualities, the features, of entities- to distinction in kind- while the contrasting term *QUANTITAS* refers to the primary focus on differences in amount.

There is no single definition for qualitative research, as no single definition could encapsulate the vastness that Qualitative research offers. One such attempt for a definition is:

“Qualitative Research is a form of social inquiry that tends to adopt a flexible and data-driven research design, to use relatively unstructured data, to emphasize the essential role of subjectivity in the research process, to study a small number of naturally occurring cases in detail, and to use verbal rather than statistical form of analysis.”

Elaborating-

A flexible and data-driven research design: Qualitative research place more emphasis on generating and developing descriptions and explanations than upon testing pre-defined hypothesis. It means that a flexible research design is laid out at the start of the research and then ‘implemented’. The same is also reflected at the stage of analysing data, where the task is to generate categories rather than to place data into pre-determined ones.

Using relatively unstructured data: There is a little pressure to engage in formal counting, ranking, or

measurement. As in case of interviews, qualitative research typically involves a relatively unstructured approach where the aim is to invite informants to talk at length about matters that are broadly relevant to the research, with the interviewer following up to encourage more elaboration, detail, or exemplification where necessary.

- **Subjectivity:** The data and inferences made from them are always shaped by the social and personal characteristics of the researcher. It is impossible to eliminate this effect and also that this may facilitate insight as well as may lead to error.

The study of 'natural' setting:

Most Qualitative work investigates what goes on in the ordinary settings in which people live and work, and/or uses interviews that are designed to approximate to ordinary conversations in key respects.

A small number of cases studies: Qualitative inquiry often involves investigation of a small number of naturally occurring cases, perhaps just one as there is a need for in-depth examination of each case, in order of document complexity.

Verbal rather than statistical analysis of data: The predominant mode of analysis is verbal description and interpretation, supported by illustrative examples. Many approaches are very similar to that employed by historians, who produce narrative accounts of the events leading to some outcome that they are interested in explaining.

FEATURES OF QUALITATIVE RESEARCH-

It allows us to study what normally happens in the 'real' world, rather than what happens under experimental conditions. Thus allows a direct interaction with the people being studied in their context.

It allows us to observe what happens rather than to rely solely upon respondent's accounts in formal interviews and questionnaires.

It focuses on the need to allow people to speak in their own terms in interviews if we are able to understand their distinctive perspectives.

It contrasts the concern that the kind of variable analysis employed by quantitative researchers ignores the complex, contingent and context-sensitive character of social life, and the extent to which actions and outcomes are produced by people interpreting situations in diverse ways, and acting on the basis of these interpretations, rather than passively responding to the external causes.

NATURE OF QUANTITATIVE AND QUALITATIVE RESEARCH

Quantitative research derives from the natural sciences and is based on the premise that phenomena can be explained by objective and factual measures that help to keep data free from researcher bias. It makes epistemological assumptions that reality is unitary and objective and can only be 'discovered' through transcending the perspective of the individual, and, therefore, that phenomena must be explained through the analysis of data obtained by objective forms of

measurement. In that it attempts to measure the cause-effect relationships between discrete and pre-selected variables by controlling and manipulating data through experimental or quasi-experimental techniques (Leininger, 1985), the perceived knowledge outcome of quantitative inquiry is singular truth.

It is with the above epistemological assumptions that qualitative research takes issue (Wildemuth, 1993). Interpretative approaches such as ethnography, phenomenology and symbolic interactionism attempt to understand the nature of social reality through people's narrated accounts of their subjectively constructed processes and meanings, as opposed to the measurement of quantity, frequency and distribution across a given population. Qualitative research, therefore, is an approach that yields findings by means other than quantifiable statistical procedures (Strauss & Corbin, 1990). That is, it tends to generate verbal data as opposed to numerical (Knafl & Howard, 1984; Polgar & Thomas, 1995) and reflects an epistemology where phenomena have "multiple, socially defined realities" (McMurray, 1994, p. 18). The qualitative tradition assumes that the complexity of phenomena necessitates researcher's utilizing methods that will bring them closer to information sources, interacting with participants, interrogating

data, drawing upon their own experiences, following-up leads and checking out hunches. These data are usually generated in naturalistic settings and the qualitative researcher avoids controlling or manipulating situations, events or actions.

QUALITATIVE V/S QUANTITATIVE RESEARCH:

The word qualitative implies an emphasis on the qualities of entities and on processes and meanings. Qualitative researchers stress the socially constructed nature of reality, the intimate relationship between the researcher and what is studied, the situational constraints that shape inquiry. Such researchers emphasize the value-laden nature of enquiry. They seek answers to questions that stress how social experience is created and given meaning.

In contrast, Quantitative studies emphasize the measurement and analysis of causal relationships between variables, not processes. Proponents of such studies claim that their work is done from within a value-free framework.

Sr . No.	DIFFERENCES	QUALITATIVE	QUANTITATIVE

QUANTITATIVE

- Logical Positivism
- Natural science world view
- Hypo-deductive
- Particularistic
- Objective/ Outsider centered
- Outcome oriented
- Attempt to control variables
- Goal: find facts & causes
- Static reality
- Verification oriented
- Confirmatory
- Inquiry is value free/neutral

QUALITATIVE

- Interpretive/Phenomenological
- Naturalistic world view
- Inductive
- Holistic
- Subjective/ Insider centered
- Process oriented
- Relative lack of control
- Goal: understand actor's view
- Dynamic reality
- Discovery oriented
- Exploratory
- Inquiry is value bound

1.	Root Term (The root term from which the phrase has been historically derived)	Latin, 'QUALITAS' Latin refers to a primary focus on qualities, the features, of entities- to distinction in kind.	'QUANTITAS' The primary focus On differences In amount.
2.	Definition	Qualitative research methods focus on discovering and understanding the experiences, perspectives, and thoughts of participants—that is, qualitative research explores meaning, purpose or reality.	Quantitative research Methods To maximize objectivity, replicability, And generalizability Of findings, and Are typically interested in prediction.
3.	Paradigms (Philosophical Assumptions)	Interpretive perspective. “Reality is in the form of multiple, intangible mental constructions” i.e. multiple & subjective reality where researcher interacts with that being researched. Ontology: RELATIVIST Epistemology: SUBJECTIVE Methodology: HERMENEUTICAL & DIALECTICAL.	Positivist perspective i.e. objective reality. “Reality is assumed to ‘exist’ out there” Ontology: REALISM Epistemology: OBJECTIVE Methodology: EXPERIMENTAL & MANIPULATIVE.
4.	Method/Types of Research	Phenomenology, case study, ethnography, grounded theory, cultural studies.	Experimental, quasi-experimental, descriptive, correlational.

5.	Purpose/Goal of Research	Understanding actor's view, insight, contextualization and interpretation. The goal is to "better understand human behavior and experience... grasp the processes by which people construct meaning and to describe what those meaning are.	Generalizability, explanation, Prediction, Cause and effect. The goal Is "collecting 'facts' of human behavior, which When accumulated Will provide verification And elaboration on A theory that Will allow scientists To state causes And predict human behavior.
6.	Research Question / Hypothesis	Question is evolving, general and flexible. There are two general positions on question/design matters depending on the researcher's view. First, is an Inductive process where the researcher relies on what is observed in the field to develop a grounded theory to see what emerges. Second could be one of Preparation where researcher reviews the literature prior to entering the field. Here, the researcher "enters the field with an open mind, not an empty head".	Hypothesis Is informed guess or prediction. The Researcher uses the theories, results, And findings of Other studies in order To form a Hypothesis to test. This type Of inquiry Usually produces A research Design that is structured, formal, And specific, outlining a detailed plan of operation.

	Those Being Researched	Usually a small number of non-representative cases. Those being researched in a qualitative study are selected using purposeful sampling. Particular participants are chosen for a qualitative study because they are believed to facilitate the expansion of the developing theory.	A subset of the population, a sample, is usually randomly selected. The random selection is to ensure that the characteristics of the subjects in the study appear in the same proportion as they exist in the total population
8.	Those Conducting the Research	Interpretive and emic (other's points of view); personal involvement. The interpretive researcher's role is involved, trusting, intense and close to the participants.	Positivistic or Etic (outsider's point of view); objective, neutral, detached and impartial. Researcher has a neutral role Where he or she remains detached, uninvolved, And distant Bringing objectivity to The research.
9.	Data	Participant observation, direct observation, interview, focus group, discussions, case studies, visual data, life histories, document analysis. Most data come from fieldwork where the researcher spends time in the setting under study. The researcher makes first-hand observations of activities and interactions,	Questionnaires, surveys, tests, etc. in the form of numbers and statistics. Quantifiable coding with counts and measures And operationalized variables. Preconceived concepts And theories are used to determine What data will Be collected

		sometimes engaging personally in those activities as a “participant observer”.	
10.	Data Analysis	Inductive process: grounded theory, narrative analysis, thematic analysis, discourse analysis, content analysis, conversation analysis. Data analysis is an ongoing, inductive process where data are sorted, sifted through, read and reread. With some methods, codes are assigned to certain themes and patterns that emerge. Categories are formed and restructured until the relationships seem appropriately represented, and the story and interpretation can be written.	Deductive process, statistical Procedures. Statistical inference procedures are used to generalize findings from a sample to a defined population.
11.	Role of Values	Research can be subjective and value bound (i.e. the researcher’s own values). Values are personally relative and need to be understood.	Research can be conducted objectively and value free. Values are emotive and therefore outside the scientific inquiry.

ASSUMPTIONS OF QUALITATIVE RESEARCH

When researchers undertake a qualitative study, they are in effect agreeing to its underlying philosophical assumptions, while bringing to the study their own world views that end up shaping the direction of their research. They are beliefs about ontology (the nature of reality), epistemology (what counts as knowledge and how knowledge claims are justified), axiology (the role of values in research), and methodology (the process of research).

Creswell describes the following four philosophical assumptions:

ONTOLOGICAL (*What is the nature of the 'knowable'? or, what is the nature of 'reality'?*):

The ontological issue relates to the nature of reality and its characteristics. When researchers conduct qualitative research, they are embracing the idea of multiple realities and report on these multiple realities by exploring multiple forms of evidence from different individual's perspectives and experiences.

EPISTEMOLOGICAL:

It poses the following questions: What is the nature of relationship between the knower and what is known? How do we know what we know? What counts as knowledge? With the epistemological assumption, conducting a qualitative study means that researchers try to get as close as possible to the participants being studied. Therefore, subjective evidence is assembled based on individual views. This is how knowledge is known- through the subjective experiences of people. It becomes important, then, to conduct studies in the "field," where the participants live and work-these are important contexts for understanding what the participants are saying. The longer researchers stay

in the "field" or get to know the participants, the more they "know what they know" from firsthand information.

AXIOLOGICAL (The role of values in research):

In a qualitative study, the inquirers admit the value-laden nature of the study and actively report their values and biases as well as the value-laden nature of information gathered from the field. In a way, they "position themselves" in a study.

METHODOLOGY (How should the inquirer go about finding out knowledge?):

The procedures of qualitative research, or its methodology are characterized as inductive, emerging, and shaped by the researcher's experience in collecting and analyzing the data. The logic that the qualitative researcher follows is inductive, from the ground up, rather than handed down entirely from a theory or from the perspectives of the inquirer. Sometimes the research questions change in the middle of the study to reflect better the types of questions needed to understand the research problem. In response, the data collection strategy, planned before the study, needs to be modified to accompany the new questions. During the data analysis, the researcher follows a path of analyzing the data to develop an increasingly detailed knowledge of the topic being studied.

**[OUT OF SYLLABUS
(Subjectivity, Positionality, &
Triangulation) BUT IMPORTANT
TO UNDERSTAND
TERMINOLOGY ESPECIALLY
FOR PhD EXAMS (DU majorly)
AND TO UNDERSTAND**

CERTAIN OPTIONS IN THE QUESTIONS]

SUBJECTIVITY

The manner in which people make sense of their experiences is referred to as subjectivity by quantitative researchers, who recognize that people construct meanings in relation to their environment and previous experiences. Munhall (2001, p. 73) postulates that the objective of quantitative research is to "disclose subjectivity" through exploring and collecting data that describes the experience being researched.

In direct contrast to quantitative research traditions, which view objectivity as a goal or, at a minimum, as an aspiration, qualitative researchers acknowledge that the very nature of the data we gather and the analytic processes in which we engage are grounded in subjectivity. Depending on the underlying paradigm, we may work to limit, control, or manage subjectivity—or we may embrace it and use it as data.

Approaches to subjectivity have been referred to as **“bracketing,” “monitoring of self”, or being “rigorously subjective”**. Bracketing to describe the process of becoming aware of one’s implicit assumptions and predispositions and setting them aside to avoid having them unduly influence the research. In the absence of an articulated perspective on subjectivity, researchers leave themselves open to questions regarding whose perceptions are really being described in the findings. Qualitative researchers address a number of important issues to accomplish the goal of managing subjectivity, including making their implicit assumptions and biases overt to themselves and others, reflexivity, and representation.

Embedded in the conversation about subjectivity is the concept of representation. Referring to the “crisis of representation,” Denzin and Lincoln (2000b) described a growing concern with reflexivity. In particular, the crisis of representation deals with questions about whose reality is represented in the research. To conclude, all research is subject to researcher bias; qualitative and quantitative perspectives have their own ways of approaching subjectivity and are very much influenced by the paradigms guiding the research.

POSITIONALITY

A researcher’s knowledge is therefore always partial, because his/her positionality (perspective shaped by his/her unique mix of race, class, gender, nationality, sexuality and other identifiers), as well as location in time and space will influence how the world is viewed and interpreted. Drawing upon metaphors of spatiality, it can be therefore argued that knowledge is never pure but is situated in the complex and sometimes contradictory social locations of producers and audiences.

Interpretivists/constructivists and ideological/critical theorists are more likely to embrace the positioning of the researcher as co-creator of meaning, as integral to the interpretation of the data.

Researchers must especially take account of their own position in relation to the research participants and research setting. In particular, the reconstructing of insider/outsider status in terms of one’s positionality in respect of education, class, race, gender, culture and other factors, offer us better tools for understanding the dynamics of researching within and across one’s culture.

REFLEXIVITY

‘Re-flexivity’ involves looking again, turning your gaze to the self.

Reflexivity refers to active acknowledgement by the researcher that her/his own actions and decisions will inevitably impact upon the meaning and context of the experience under investigation.

Thus, it becomes a way for the researchers to inform their audiences about their perspectives as well as to manage their subjectivities. Researcher’s reflexivity provides an opportunity for the researcher to understand how his or her own experiences and understandings of the world affect the research process. Reflexivity can function both as a control for researcher bias and as a tool for gaining new depth in research. It is especially during the interview that expanded reflexivity may be seen in action.

Qualitative researchers have typically employed reflexivity as a means of controlling the effects of researcher bias and its influence on the research process. For example, practitioners of phenomenology may engage in **bracketing** – a process where researchers attempt to recognize and set aside their personal beliefs at the beginning of their study, while remaining cognizant of them throughout the entire research project.

In order to deal with biases and assumptions that come from their own life experiences or in interactions with research participants, which are often emotion-laden, qualitative researchers attempt to approach their endeavour reflexively.

HOW TO CARRY OUT REFLEXIVITY OR SELF-REFLECTION?

One of the most valuable is for the researcher to keep a self-reflective journal from the inception to the completion of the investigation. In it, the investigator keeps an ongoing record of her or his experiences, reactions, and emerging awareness of any assumptions or biases that come to the fore.

Another reflexive strategy is consulting with a research team or peer debriefers, who serve as a mirror, reflecting the investigator’s responses to the research process.

Engaging in **bracketing** – a process where researchers attempt to recognize and set aside their personal beliefs at the beginning of their study, while remaining cognizant of them throughout the entire research project.

Reflexivity may also be demonstrated by use of the ‘first person’ when the researcher describes the aspects of the research in which (s)he has had personal involvement. It provides sufficient information for the reader to identify the foundations upon which the findings of the study have been grounded.

To conclude, Reflexivity is often considered only as a concept of qualitative validity. At its most basic level, this may include raising researcher awareness of how their presence affects the research process and participants, as well as how the participants affect the researcher. Applied in this manner, reflexivity is the process of analysing how various elements affect and transform the research.

- Maintaining reflexive journal/memo taking

Personal
Reflexivity



Functional
Reflexivity



Disciplinary
Reflexivity

- Knowing who you are
- Your individuality as a researcher
- Influence of personal interests/biases on research

- Continuous examination of the practice/process of research to reveal assumptions, values & biases

- Reflecting on larger issues that include research methodology & questioning psychology itself

TRIANGULATION

Triangulation is the process by which several methods (data sources, theories or researchers) are used in the study of one

which the data were collected or measured. By linking different methods, the researcher intends that each method

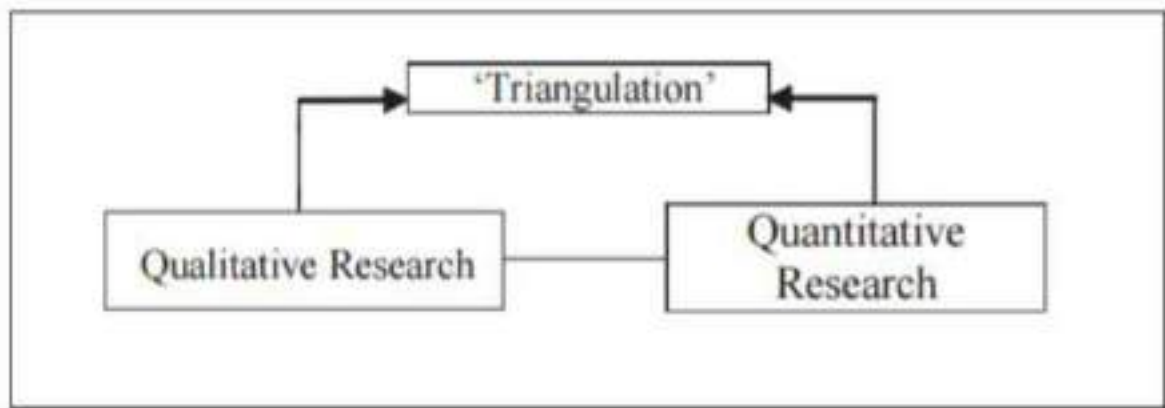


Figure 1: Basic 'Triangulation' Research Model

phenomena. The concept has its origin in ancient Greek mathematics; in modern times, it is employed in topographic surveying as a checking system.

'Triangulation' is a process of verification that increases validity by incorporating several viewpoints and methods. In the social sciences, it refers to the combination of two or more theories, data sources, methods or investigators in one study of a single phenomenon to converge on a single construct, and can be employed in both quantitative (validation) and qualitative (inquiry) studies.

When researchers employ triangulation, multiple measures are used to ensure that any data variance is not due to the way in enhances the other, since all the information that is collected potentially offers to be contextually richer than if it were seen from only one vantage point.

Denzin (1989) differentiates between four different types of triangulation:

Triangulation of Data

Triangulation of
Investigators/Observers

Triangulation of Theories

Triangulation of Methodologies

DATA TRIANGULATION:

Using one data origin may sometimes not be ideal. Collecting information from more than one source can extend and enhance the research process. Banister and colleagues suggest that more than one viewpoint, site, or source, increases diversity, thus leading to increased understanding of the research topic.

The authors propose it can be helpful to look at data collected at different times, or stages, of fieldwork, in order to re-evaluate (“research”) the material. This might mean checking if anything has been overlooked or given too much emphasis, during the research process. The use of triangulation can be very helpful when verification of data is needed.

The approach supports research being a reflexive, organic process, enriched by researchers’ increasing depth of knowledge as they investigate the area. This is linked to the role of reflexivity in qualitative research, considered by many to be an essential component in qualitative inquiry (Banister et al., 2011). Other factors which may be considered include whether flaws were found in the research design, how the research study might be improved or refined, what further research might be needed etc. Some researchers advocate keeping a journal or diary recording these reflexions during the actual research process.

INVESTIGATOR TRIANGULATION:

It means that more than one researcher is involved in the research. Investigator triangulation has the potential to enrich the research process, however, it also flags up the possibility of difficulties in using this method.

The authors suggest that- “it cannot be assumed that those around the table will have an equally shared degree of responsibility and contribution. If not, then once again fairness is challenged and ultimately more problems are created than solved.”

THEORY TRIANGULATION:

It involves the use of different theoretical perspectives in the study of one problem. Theoretical triangulation explores, and is informed by, more than one theory or theoretical framework. It acknowledges, and allows for, the broad range of theories, complexity and diversity of the real world and how different theories may be accounted for in research.

METHODOLOGICAL TRIANGULATION:

It involves using more than one research method or data collection technique. Here the researcher chooses the method of inquiry according to the question being researched e.g. by observing behaviours (an observational approach) or exploring how participants feel e.g. using interviews. Triangulated techniques are helpful for cross-checking and used to provide confirmation and completeness, which brings 'balance' between two or more different types of research. The purpose is to increase the credibility and validity of the results.

Although triangulation is generally considered helpful when using qualitative methods, it can just as equally be applied to quantitative or mixed methods research. It is a pragmatic and strategic approach, whether applied to qualitative or quantitative research (Denzin & Lincoln, 1998). It may be viewed as providing a way of expanding the research perspective and becomes another means of strengthening research findings.

'TRIANGULATION' AS AN EFFECTIVE TOOL OF RESEARCH METHODOLOGY

By combining multiple observers, theories, methods, and data, researchers can hope to overcome the weakness or intrinsic biases and the problems that come from single method, single-observer, and single-theory

studies. There are basically three types of biasness: firstly, the measurement bias is caused by the way in which data are collected; secondly, sampling bias causes as all the population under study are not covered; and thirdly, procedural bias occurs when participants are put under some kind of pressure to provide information.

RIGOR IN INTERPRETIVE RESEARCH / QUALITATIVE RESEARCH

While positivist research employs a “reductionist” approach by simplifying social reality into parsimonious theories and laws, interpretive research attempts to interpret social reality through the subjective viewpoints of the embedded participants within the context where the reality is situated. These interpretations are heavily contextualized, and are naturally less generalizable to other contexts. However, because interpretive analysis is subjective and sensitive to the experiences and insight of the embedded researcher, it is often considered less rigorous by many positivist (functionalist) researchers. Because interpretive research is based on different set of ontological and epistemological assumptions about social phenomenon than positivist research, the positivist notions of rigor, such as reliability, internal validity, and generalizability, do not apply in a similar manner. However, Lincoln and Guba (1985) [16] provide an alternative set of criteria that can be used to judge the rigor of interpretive research.

Dependability. Interpretive research can be viewed as dependable or authentic if two researchers assessing the same phenomenon using the same set of evidence independently arrive at the same conclusions or the same researcher observing the same or a similar phenomenon at different times arrives at similar conclusions. This concept is similar to that of reliability in positivist

research, with agreement between two independent researchers being similar to the notion of inter-rater reliability, and agreement between two observations of the same phenomenon by the same researcher akin to test-retest reliability. To ensure dependability, interpretive researchers must provide adequate details about their phenomenon of interest and the social context in which it is embedded so as to allow readers to independently authenticate their interpretive inferences.

Credibility. Interpretive research can be considered credible if readers find its inferences to be believable. This concept is akin to that of internal validity in functionalistic research. The credibility of interpretive research can be improved by providing evidence of the researcher’s extended engagement in the field, by demonstrating data triangulation across subjects or data collection techniques, and by maintaining meticulous data management and analytic procedures, such as verbatim transcription of interviews, accurate records of contacts and interviews, and clear notes on theoretical and methodological decisions, that can allow an independent audit of data collection and analysis if needed.

Confirmability. Confirmability refers to the extent to which the findings reported in interpretive research can be independently confirmed by others (typically, participants). This is similar to the notion of objectivity in functionalistic research. Since interpretive research rejects the notion of an objective reality, confirmability is demonstrated in terms of “inter-subjectivity”, i.e., if the study’s participants agree with the inferences derived by the researcher. For instance, if a study’s participants generally agree with the inferences drawn by a researcher about a phenomenon of interest (based on a review of the research paper or report), then the findings can be viewed as confirmable.

Transferability. Transferability in interpretive research refers to the extent to which the findings can be generalized to other settings. This idea is similar to that of external validity in functionalistic research. The researcher must provide rich, detailed descriptions of the research context (“thick description”) and thoroughly describe the structures, assumptions, and processes revealed from the data so that readers can independently assess whether and to what extent are the reported findings transferable to other settings.

METHODS OF RESEARCH:

OBSERVATION

Observation is defined as the systematic viewing, which is intentional and planned. Maybe done in a natural setting or in a lab. It is preferred as the supreme technique for nonverbal behavior.

Collecting facts that are in direct knowledge of the investigator, i.e. a deliberate study through the eyes.

Observation is the perception with the purpose, aka, ‘Regulated Perception’.

PROCEDURE OF OBSERVATION:

1. By Mechanical & Electronic Device

This include audio or visual recording of the scenario, etc.

2. Checklist & Schedules

This instrument tends to objectify the observation. After each listed item, a space is provided for the observer to write in few descriptive words. Some checklists are so designed that they enable the researcher to

arrive at a score that will facilitate comparative analysis.

3. Time Sampling

An observation procedure, in which the researcher records whether or not certain behavior occurs during a sample of short time intervals.

The observer, thus, records the frequency of observable form of occurrence during the time intervals that are systematically spaced.

4. Event Sampling

An observation procedure, wherein researcher records all instances of a particular event or behavior during specific time-period ignoring all other behavior.

5. Specimen Sampling

An observational procedure wherein researcher record the description of the subject’s entire scheme of behavior for a specific period.

TYPES OF OBSERVATION

Non-participant Observation Naturalistic	Participant Observation
Observing events that occur naturally. Belief: Artificial probes or manipulation might destroy all or basic character of the event being studied. Key Requirement: To avoid intrusion.	The participant have interface with the subjects. They are of two types. ▪ Undisguised Participant Observation: The individual being observed know that they are being observed about the social aspect of their behavior. ▪ Disguised Participant Observation: The observer disguise himself to the extent of being accepted as an insider. Allows the observer to watch the activities from a close quarter.
ADVANTAGES: Can be used where experimental method perspectives cannot be used. ▪ Original behavior can be observed.	ADVANTAGES: Allows better understanding of the natives ▪ Being a member, there is access to the private or the tabooed data. ▪ Any kind of reliance on any kind of informant is reduced and thus a higher objectivity is obtained.
LIMITATION: ▪ Cause-effect relationship Cannot be established. ▪ Replication opportunities are absent. Observer bias can produce subjectivity in the findings.	LIMITATION: ▪ Mental and Physical strain. ▪ Expectancy effect might bias the observation. ▪ May lead to emotional binding with the group under study and may dilute researcher's objectivity.
Primary problem unique to observation is that it is simply descriptive in nature and does not allow us to assess the relationship among events.	

SURVEY [INTERVIEW, QUESTIONNAIRES]

QUESTIONNAIRES

A notable feature: There is usually only

one opportunity to collect data from each informant. Therefore, questions needs to be clear, comprehensive and effective. The steps in developing questions is very

similar to developing questions for a semi-structured interview.

Closed Questions

Also called, MCQ's. They are straight-forward, quick to answer and lie within intellectual range of the majority of population.

The responses are easily turned into quantitative data.

Types of Closed Questions:

- a. **Checklist:** Presents a range of possible answers to tick as many as may apply.
- b. **Questions:** Answering the question by placing in rank-order in terms of preferences, frequency, etc.
- c. **Graded Response Questions:** Expresses degree of magnitude. Such as asking for degree of agreement.

Open-ended questions: Inviting a more precise, and personal response.

The MCQ's though are relatively inexpensive to analyze, poses a major drawback of fixed alternative questions is **putting answers in people's mouth**. In such situations, open-ended questions allow a free response from the respondent.

It is always advisable to conduct a 'pilot study' for testing the questionnaires due to their limitations in true representation. Pilot Survey is the replica and rehearsal of the main survey. This allows bringing in light the weakness of the questionnaires and also of the survey techniques.

INTERVIEWS

Interviews are different from questionnaires as they involve social interaction. Unlike questionnaires researchers need training in how to interview (which costs money).

Researchers can ask different types of questions which in turn generate different

types of data. For example, closed questions provide people with a fixed set of responses, whereas open questions allow people to express what they think in their own words.

Sometimes researchers use an interview schedule. This is a set of prepared questions designed to be asked exactly as worded. Interviews schedules have a standardised format which means the same questions are asked to each interviewee in the same order.

Quite often interviews will be recorded by the researcher and the data written up as a transcript (a written account of interview questions and answers) which can be analyzed at a later date.

The interviewer must ensure that they take special care when interviewing vulnerable groups, such as the children. For example, children have a limited attention span and for this reason lengthy interviews should be avoided.

Also the language the interviewer uses should be appropriate to the vocabulary of the group of people being studied. For example, the researcher must change the language of questions to match the social background of respondents' age / educational level / social class / ethnicity etc.

TYPES OF INTERVIEWS:

STRUCTURED INTERVIEW

This is also known as a formal interview. The questions are asked in a set / standardized order and the interviewer will not deviate from the interview schedule or probe beyond the answers received (so they are not flexible). These are based on structured, closed-ended questions.

Strengths Structured interviews are easy to replicate as a fixed set of closed questions are used, which are easy to quantify – this means it is easy to test for reliability.

Structured interviews are fairly quick to conduct which means that many interviews can take place within a short amount of time. This means a large sample can be obtained resulting in the findings being representative and having the ability to be generalized to a large population.

Limitations

Structure interviews are not flexible. This means new questions cannot be asked impromptu (i.e. during the interview) as an interview schedule must be followed.

The answers from structured interviews lack detail as only closed questions are asked which generates quantitative data. This means a research will won't know why a person behaves in a certain way.

UNSTRUCTURED INTERVIEW

These are sometimes referred to as '**discovery interviews**' & are more like a '**guided conversation**' than a strict structured interview. They are sometimes called informal interviews.

An interview schedule might not be used, and even if one is used, they will contain open-ended questions that can be asked in any order. Some questions might be added missed as the Interview progresses.

Strengths

Unstructured interviews are more flexible as questions can be adapted and changed depending on the respondents' answers. The interview can deviate from the interview schedule.

Unstructured interviews generate qualitative data through the use of open questions. This allows the respondent to talk in some depth, choosing their own words. This helps the researcher develop a real sense of a person's understanding of a situation.

They also have increased validity because it gives the interviewer the opportunity to probe for a deeper understanding, ask for clarification & allow the interviewee to steer the direction of the interview etc.

Limitations

It can be time consuming to conduct an unstructured interview and analyze the qualitative data (using methods such as thematic analysis).

Employing and training interviewers is expensive, and not as cheap as collecting data via questionnaires. For example, certain skills may be needed by the interviewer. These include the ability to establish rapport & knowing when to probe.

GROUP INTERVIEW

This refers to interviews where a dozen or so respondents are interviewed together – also known as a 'focus group'. This role of the interviewer is to make sure the group interact with each other and do not drift off topic.

A researcher must be highly skilled to conduct a group interview. For example, certain skills may be needed by the interviewer including the ability to establish rapport and knowing when to probe.

Strengths

Group interviews generate qualitative data through the use of open questions. This allows the respondents to talk in some depth, choosing their own words. This helps the researcher develop a real sense of a person's understanding of a situation.

They also have increased validity because some participants may feel more comfortable being with others as they are used to talking in groups in real life (i.e. it's more natural).

Limitations

The researcher must ensure that they keep all the interviewees details

confidential and respect their privacy. This is difficult when using a group interview. For example, the researcher cannot guarantee that the other people in the group will keep information private. Group interviews are less reliable as they use open questions and may deviate from the interview schedule making them difficult to repeat. Group interviews may sometimes lack validity as participants may lie to impress the other group members. They may conform to peer pressure and give false answers.

THE INTERVIEWER EFFECT

Because an interview is a social interaction the appearance or behavior of the interviewer may influence the answers of the respondent. This is a problem as it can bias the results of the study and make them invalid.

For example, the gender, ethnicity, body language, age, and social status of the interviewer can all create an interviewer effect.

For example, if a researcher was investigating sexism amongst males, would a female interview be more preferable than a male? It is possible that if a female interviewer was used male participants may lie (i.e. pretend they are not sexist) to impress the interviewer, thus creating an interviewer effect.

DESIGN OF INTERVIEWS

First you must choose whether to use a structured or non-structured interview.

Next, you must consider who will be the interviewer, and this will depend on what type of person is being interviewed. There are a number of variables to consider:

Gender and age: This can have a big effect on respondent's answer, particularly on person issues.

Personal characteristics: Some people are easier to get on with than others. Also, the accent and appearance (e.g.

clothing) of the interviewer can have an effect on the rapport between the interviewer and interviewee.

Ethnicity: People have difficulty interviewing people from a different ethnic group.

EXPERIMENTAL (IMPORTANT FOR PAPER 1 – RESEARCH APTITUDE & PAPER II)

Experimental research, often considered to be the “gold standard” in research designs, is one of the most rigorous of all research designs. In this design, one or more independent variables are manipulated by the researcher (as treatments), subjects are randomly assigned to different treatment levels (random assignment), and the results of the treatments on outcomes (dependent variables) are observed. The unique strength of experimental research is its internal validity (causality) due to its ability to link cause and effect through treatment manipulation, while controlling for the spurious effect of extraneous variable. Experimental research is best suited for explanatory research (rather than for descriptive or exploratory research), where the goal of the study is to examine cause-effect relationships. It also works well for research that involves a relatively limited and well-defined set of independent variables that can either be manipulated or controlled. Experimental research can be conducted in laboratory or field settings.

Experimental research can be grouped into two broad categories: true experimental designs and quasi-experimental designs.

Both designs require treatment manipulation, but while true experiments also require random assignment, quasi-experiments do not. Sometimes, we also refer to non-experimental research, which is not really a research design, but an all-inclusive term that includes all types of research that do not employ treatment manipulation or random assignment, such

as survey research, observational research, and correlational studies.

An experiment is an investigation in which a hypothesis is scientifically tested. In an experiment, an independent variable (the cause) is manipulated and the dependent variable (the effect) is measured; any extraneous variables are controlled.

An advantage is that experiments should be objective. The views and opinions of the researcher should not affect the results of a study. This is good as it makes the data more valid, and less bias.

1. LAB EXPERIMENT

A laboratory experiment is an experiment conducted under highly controlled conditions (not necessarily a laboratory), where accurate measurements are possible.

The researcher decides where the experiment will take place, at what time, with which participants, in what circumstances and using a standardized procedure. Participants are randomly allocated to each independent variable group.

Laboratory experiments, conducted in laboratory (artificial) settings, tend to be high in internal validity, but this comes at the cost of low external validity (generalizability), because the artificial (laboratory) setting in which the study is conducted may not reflect the real world.

An example is **Milgram's experiment on obedience** or **Loftus and Palmer's car crash study**.

Strengths:

It is easier to replicate (i.e. copy) a laboratory experiment. This is because a standardized procedure is used.

They allow for precise control of extraneous and independent variables. This allows a cause and effect relationship to be established.

Limitation:

The artificiality of the setting may produce unnatural behavior that does not reflect real life, i.e. low ecological validity. This means it would not be possible to generalize the findings to a real life setting.

Demand characteristics or experimenter effects may bias the results and become confounding variables.

2. FIELD EXPERIMENT

Field experiments are done in the everyday (i.e. real life) environment of the participants. The experimenter still manipulates the independent variable, but in a real-life setting (so cannot really control extraneous variables).

Field experiments, conducted in field settings such as in a real organization, and high in both internal and external validity. But such experiments are relatively rare, because of the difficulties associated with manipulating treatments and controlling for extraneous effects in a field setting.

An example is **Holfing's hospital study on obedience**.

Strength:

Behavior in a field experiment is more likely to reflect real life because of its natural setting, i.e. higher ecological validity than a lab experiment. There is less likelihood of demand characteristics affecting the results, as participants may not know they are being studied. This occurs when the study is covert.

Limitation:

There is less control over extraneous variables that might bias the results. This makes it difficult for another researcher to replicate the study in exactly the same way.

3. NATURAL EXPERIMENT

Natural experiments are conducted in the everyday (i.e. real life) environment of the participants, but here the experimenter has no control over the IV as it occurs naturally in real life.

For example, Hodges and Tizard's attachment research (1989) compared the long term development of children who have been adopted, fostered or returned to their mothers with a control group of children who had spent all their lives in their biological families.

Strength:

Behavior in a natural experiment is more likely to reflect real life because of its natural setting, i.e. very high ecological validity.

There is less likelihood of demand characteristics affecting the results, as participants may not know they are being studied.

Can be used in situations in which it would be ethically unacceptable to manipulate the independent variable, e.g. researching stress.

Limitation:

They may be more expensive and time consuming than lab experiments. There is no control over extraneous variables that might bias the results. This makes it difficult for another researcher to replicate the study in exactly the same way.

KEY TERMINOLOGIES:

Ecological validity: The degree to which an investigation represents real-life experiences.

Experimenter effects: These are the ways that the experimenter can accidentally influence the participant through their appearance or behavior.

Demand characteristics: The clues in an experiment that lead the participants to think they know what the researcher is looking for (e.g. experimenter's body language).

Independent variable (IV): Variable the experimenter manipulates (i.e. changes) – assumed to have a direct effect on the dependent variable.

Dependent variable (DV): Variable the experimenter measures. This is the outcome (i.e. result) of a study.

Extraneous variables (EV): All variables, which are not the independent variable, but could affect the results (DV) of the experiment. EVs should be controlled where possible.

Confounding variables: Variable(s) that have affected the results (DV), apart from the IV. A confounding variable could be an extraneous variable that has not been controlled.

Random Allocation: Randomly allocating participants to independent variable conditions means that all participants should have an equal chance of taking part in each condition. The principle of random allocation is to avoid bias in the way the experiment is carried out and to limit the effects of participant variables.

Order effects: Changes in participants' performance due to their repeating the same or similar test more than once. Examples of order effects include:

- (i) practice effect: an improvement in performance on a task due to repetition, for example, because of familiarity with the task;
- (ii) fatigue effect: a decrease in performance of a task due to repetition, for example, because of boredom or tiredness.

QUASI-EXPERIMENTAL

The prefix quasi means “resembling.” Thus, quasi-experimental research is research that resembles experimental research but is not true experimental research. Although the independent variable is manipulated, participants are not randomly assigned to conditions or orders of conditions (Cook & Campbell, 1979). Because the independent variable is manipulated before the dependent variable is measured, quasi-experimental research

eliminates the directionality problem. But because participants are not randomly assigned—making it likely that there are other differences between conditions—quasi-experimental research does not eliminate the problem of confounding variables. In terms of internal validity, therefore, quasi-experiments are generally somewhere between correlational studies and true experiments.

Quasi-experiments are most likely to be conducted in field settings in which random assignment is difficult or impossible. They are often conducted to evaluate the effectiveness of a treatment—perhaps a type of psychotherapy or an educational intervention. There are many different kinds of quasi-experiments, but we will discuss just a few of the most common ones here.

ADVANTAGES:

Quasi-experimental research may be more feasible because it often does not have the time and logistical constraints associated with many true experimental designs.

True experimental designs are sometimes impractical or impossible because the research can only effectively be carried out in natural settings. Experimental research can create artificial situations that do not always represent real-life situations. This is largely due to fact that all other variables are tightly controlled which may not create a fully realistic situation. For this reason, external validity is increased quasi-experimental research.

Reactions of test subjects are more likely to be genuine because it not an artificial research environment. It can be very useful in identifying general trends from the results,

especially in social science disciplines.

It reduces the difficulty and ethical concerns that may surround the pre-selection and random assignment of test subjects. For example, if examining the effects of cigarette smoking by pregnant women on the fetus, it would be unethical to randomly assign pregnant women to groups.

Matching procedures may be used to help create a reasonable control group, making generalization more feasible. For example, if one group of migraine sufferers received a new treatment and a second group did not receive the treatment, the difference in the pain levels may be attributed to the treatment if the control group is an appropriate comparison group.

The results generated can often be used to reinforce the findings of case studies by conducting research that may lend itself to statistical analysis.

DISADVANTAGES:

The lack of random assignment into test groups leads to non-equivalent test groups which can limit the generalizability of the results to a larger population. Beside of the lack of randomization and the reduced internal validity, conclusions about causality are less definitive in quasi-experimental designs.

Statistical analyses may not be meaningful due to the lack of randomization and the threats to internal validity.

Pre-existing factors and other influences are not considered because variables are less controlled in quasi-experimental research. For example, when examining the impact of smoking by pregnant mothers, there may be other factors such as diet,

education, overall health, and access to health care in general that may be playing a role in the outcome. If other variables are not controlled, the researcher can be assured that the treatment was the sole factor causing the outcome.

Human error also plays a key role in the validity of any project as discussed in previous modules.

The research must adhere to ethical standards in order to be valid. These will be discussed in the next module of this series.

FIELD STUDIES

Field research is defined as a qualitative method of data collection that aims to observe, interact and understand people while they are in a natural environment. For example, nature conservationists observe behavior of animals in their natural surroundings and the way they react to certain scenarios. In the same way, social scientists conducting field research may conduct interviews or observe people from a distance to understand how they behave in a social environment and how they react to situations around them.

Thus, it involves collecting data outside of an experimental or lab setting. Since, this type of data collection is most often done in natural settings or environments and can be done in a variety of ways for various disciplines, therefore, Field studies are known to be expensive and timely; however, the amount and diversity of the data collected can be invaluable.

Field studies collect original or unconventional data via face-to-face interviews, surveys, or direct observation. This research technique is usually treated as an initial form of research because the data collected is specific only to the purpose for which it was gathered. Therefore, it is not applicable to the general public.

Field research encompasses a diverse range of social research methods including direct observation, limited participation, analysis of documents and other information, informal interviews, surveys etc. Although field research is generally characterized as qualitative research, it often involves multiple aspects of quantitative research in it.

Field research typically begins in a specific setting although the end objective of the study is to observe and analyze the specific behavior of a subject in that setting. The cause and effect of a certain behavior, though, is tough to analyze due to presence of multiple variables in a natural environment. Most of the data collection is based not entirely on cause and effect but mostly on correlation. While field research looks for correlation, the small sample size makes it difficult to establish a causal relationship between two or more variables.



Figure: 5 distinct methods of Field Research

METHOD

Field studies should be carefully planned and prepared in order to ensure that the data collected is accurate, valid, and

collected efficiently. The equipment needed will depend on the type of study being conducted. The process first starts with clearly stating the problem and defining the area of study. From there, a hypothesis, or a theory of explanation, is set forth to explain any occurrences expected for the specified group or phenomena. This is why, before a field study is conducted, it is important to identify the data/phenomena to observe.

Once the hypothesis has been established, the data can be classified and scaled so that it will be easy to know how to categorize information. Observations are classified because not all field observations will be needed; therefore, the observer can know what to look for and what to disregard. Observations are also scaled to give the observer a way to rank the importance or significance of what has been observed. Once field observations are concluded, this data will be analyzed and processed in order to resolve the problem initially presented or to accept or reject the hypothesis that was presented.

EXAMPLES OF FIELD RESEARCH

Decipher social metrics in a slum

Purely by using observational methods and in-depth interviews, researchers can be part of a community to understand the social metrics and social hierarchy of a slum. This study can also understand the financial independence and day-to-day operational nuances of a slum. The analysis of this data can provide an insight into how different a slum is from structured societies.

Understand the impact of sports on a child's development

This method of field research takes multiple years to conduct and the sample size can be very large. The data

analysis of this research provides insights into how the kids of different geographical locations and backgrounds respond to sports and the impact of sports on their all-round development.

Study animal migration patterns

Field research is used extensively to study flora and fauna. A major use case is scientists monitoring and studying animal migration patterns with the change of seasons. Field research helps collect data across years and that helps draw conclusions about how to safely expedite the safe passage of animals.

ADVANTAGES OF FIELD RESEARCH

It is conducted in a real-world and natural environment where there is no tampering of variables and the environment is not doctored.

Due to the study being conducted in a comfortable environment, data can be collected even about ancillary topics.

The researcher gains a deep understanding into the research subjects due to the proximity to them and hence the research is extensive, thorough and accurate.

DISADVANTAGES OF FIELD RESEARCH

The studies are expensive and time-consuming and can take years to complete.

It is very difficult for the researcher to distance themselves from a bias in the research study.

The notes have to be exactly what the researcher says but the nomenclature is very tough to follow.

It is an interpretive method and this is subjective and entirely dependent on the ability of the researcher.

In this method, it is impossible to control external variables and this constantly alters the nature of the research.

CROSS-CULTURAL STUDIES

Cross-cultural research is a scientific method of comparative research which focuses on systematic comparisons that compares culture to culture and explicitly aims to answer questions about the incidence, distributions, and causes of cultural variation and complex problems across a wide domain, usually worldwide.

It commonly involves comparison of some cultural trait (or relationships between traits) across a sample of societies. What is most important to keep in mind is that cultures change over time, so most cross-cultural comparisons need to focus on particular time frames (and sometimes particular place foci) for each culture.

To compare culture is not to deny their individual uniqueness. Cross-culturalists do not deny the uniqueness of culture; uniqueness and similarity are always present, simultaneously. Cross-cultural research deals with what is general, what is true for some or even all human cultures.

Results of Cross-cultural research are generalizable to all types of society from a small hamlet with population in the hundreds or a few thousands to urban societies with population in the hundreds of millions. It helps researchers to make general (cross-cultural) statements about the similarities and differences of cultures

and to identify what may be universal and variable about human cultures, as well as to discover reasons why the variation exists. Cross-cultural psychologists generally collect primary data, they do not make use of secondary data derived from national surveys on issues such as crime rate, gross national product, etc. however, their comparison tend to be limited often only to two-cultures (the investigator's own and one other).

PERSPECTIVES IN CROSS-CULTURAL PSYCHOLOGY

Basically, there are two types of perspectives in cross-cultural psychology: the Absolutist Approach and the Relativist approach. The absolutists view in cross-cultural psychology is that psychological phenomena are basically the same in all cultures while the relativists view in cross-cultural psychology is that psychological phenomena should be studied only from "within" a culture where these phenomena occur.

TYPES OF CULTURES: SUBCULTURES OR POPULATIONS

Culture is a matrix of behaviours, beliefs, practices and values that typifies a particular group of people. It deals with a variety of things that influence all aspects of behaviour; emphasizes group influence rather than individual variability and essential part of the meaning-formation process.

Human communities have a variety of practices, beliefs, social roles, norms, expressions, forms of organization and conflicts (economic, political, legal, religious, expressive and artistic) that exhibit various sorts of internal coherence as well as cleavages within communities. These coherences and cleavages bear many close connections to the different historical experiences, physical and social environments in which people live. They include configurations of elements

and characteristic ways of interrelating that are shared with neighbouring and interacting groups, and shared among dispersed groups that have common historical experiences and similarities, including common origin, common membership in historical civilizations, and languages that are mutually understood or that derive common families. Lines of cleavage, conflict, and marginality, of course, are part of cultural phenomena. Elements and relationships that individuals or communities have in common are shared in a variety of ways. Some, such as the more intensive patterns of interaction that derive from common residence, joint experience, and discourse in a common language or system of signs, are relatively well bounded. Other patterns of sharing or similarity derive from processes of dispersal: migration, Diasporas, the trajectory of lives lived through spatial movements, social mobility, careers, and distinctive histories.

Interactions are by no means limited to localities, but to the trajectories of inhabitants who move through and between localities. Cultures consist of shared constructions that emerge out of social interactions of sets of individuals who inhabit overlapping social and physical spaces. Coherence may be viewed as an emergent property, but may be present or absent to varying degrees and along varying dimensions or trajectories.

PHENOMENOLOGY

Phenomenology is a research method that emphasizes the study of conscious experiences as a way of understanding the reality around us. It is based on the ideas of German philosopher Edmund Husserl in the early 20th century who believed that human experience is the source of all knowledge. Phenomenology is concerned with the systematic reflection and analysis

of phenomena associated with conscious experiences, such as human judgment, perceptions, and actions, with the goal of appreciating and describing social reality from the diverse subjective perspectives of the participants involved, and (2) understanding the symbolic meanings (“deep structure”) underlying these subjective experiences. Phenomenological inquiry requires that researchers eliminate any prior assumptions and personal biases, empathize with the participant’s situation, and tune into existential dimensions of that situation, so that they can fully understand the deep structures that drives the conscious thinking, feeling, and behavior of the studied participants.

Some researchers view phenomenology as a philosophy rather than as a research method. In response to this criticism, Giorgi and Giorgi (2003) developed an existential phenomenological research method to guide studies in this area. This method, illustrated in Figure 10.2, can be grouped into data collection and data analysis phases. In the data collection phase, participants embedded in a social phenomenon are interviewed to capture their subjective experiences and perspectives regarding the phenomenon under investigation.

Examples of questions that may be asked include “can you describe a typical day” or “can you describe that particular incident in more detail?” These interviews are recorded and transcribed for further analysis. During data analysis, the researcher reads the transcripts to:

Get a sense of the whole, and establish “units of significance” that can faithfully represent participants’ subjective experiences.

Examples of such units of significance are concepts such as “felt space” and “felt time,” which are then used to document participants’ psychological experiences. For instance, did participants feel safe, free, trapped, or joyous when experiencing phenomenon (“felt-space”)? Did they feel that their experience was pressured, slow, or discontinuous (“felt-time”)?

Phenomenological analysis should consider the participants’ temporal landscape (i.e., their sense of past, present, and future), and the researcher must transpose herself in an imaginary sense in the participant’s situation (i.e., temporarily live the participant’s life). The participants’ lived experience is described in form of a narrative or using emergent themes. The analysis then delves into these themes to identify multiple layers of meaning while retaining the fragility and ambiguity of subjects’ lived experiences.

GROUNDING THEORY

How can you analyze a vast set qualitative data acquired through participant observation, in-depth interviews, focus groups, narratives of audio/video recordings, or secondary documents? One of these techniques for analyzing text data is grounded theory – an inductive technique of interpreting recorded data about a social phenomenon to build theories about that phenomenon.

The technique was developed by Glaser and Strauss (1967) in their method of constant comparative analysis of grounded theory research, and further refined by Strauss and Corbin (1990) to further illustrate specific coding techniques – a process of classifying and categorizing text data segments into a set of codes (concepts), categories (constructs), and relationships. The interpretations are “grounded in” (or based on) observed empirical data, hence the name. To ensure that the theory is based solely on observed evidence, the grounded theory approach requires that researchers suspend any preexisting theoretical expectations or biases before data analysis, and let the data dictate the formulation of the theory.

Strauss and Corbin (1998) describe three coding techniques for analyzing text data:

**Open
Axial, and
Selective.**

OPEN CODING:

It is a process aimed at identifying concepts or key ideas that are hidden within textual data, which are potentially related to the phenomenon of interest. The researcher examines the raw textual data line by line to identify discrete events, incidents, ideas, actions, perceptions, and interactions of relevance that are coded as concepts (hence called **in-vivo codes**). Each concept is linked to specific portions of the text (coding unit) for later validation. Some concepts may be simple, clear, and unambiguous while others may be complex, ambiguous, and viewed differently by different participants. The coding unit may vary with the concepts being extracted. Simple concepts such as “organizational size” may include just a few words of text, while complex ones such as “organizational mission” may span several pages. Concepts can be named using the researcher’s own naming

convention or standardized labels taken from the research literature. Once a basic set of concepts are identified, these concepts can then be used to code the remainder of the data, while simultaneously looking for new concepts and refining old concepts. While coding, it is important to identify the recognizable characteristics of each concept, such as its size, color, or level (e.g., high or low), so that similar concepts can be grouped together later. This coding technique is called “open” because the researcher is open to and actively seeking new concepts relevant to the phenomenon of interest.

Next, similar concepts are grouped into higher order categories. While concepts may be context-specific, categories tend to be broad and generalizable, and ultimately evolve into constructs in a grounded theory. Categories are needed to reduce the amount of concepts the researcher must work with and to build a “big picture” of the issues salient to understanding a social phenomenon. Categorization can be done in phases, by combining concepts into subcategories, and then subcategories into higher order categories. Constructs from the existing literature can be used to name these categories, particularly if the goal of the research is to extend current theories. However, caution must be taken while using existing constructs, as such constructs may bring with them commonly held beliefs and biases. For each category, its characteristics (or properties) and dimensions of each characteristic should be identified. The dimension represents a value of a characteristic along a continuum. For example, a “communication media” category may have a characteristic called “speed”, which can be dimensionalized as fast, medium, or slow. Such categorization helps differentiate between different kinds of communication media and enables researchers identify patterns in the data, such as which communication media is used for which types of tasks.

AXIAL CODING:

The second phase of grounded theory is axial coding, where the categories and subcategories are assembled into causal relationships or hypotheses that can tentatively explain the phenomenon of interest. Although distinct from open coding, axial coding can be performed simultaneously with open coding. The relationships between categories may be clearly evident in the data or may be subtler and more implicit. In the latter instance, researchers may use a coding scheme (often called a “coding paradigm”, but different from the paradigms discussed in Chapter 3) to understand which categories represent conditions (the circumstances in which the phenomenon is embedded), actions/interactions (the responses of individuals to events under these conditions), and consequences (the outcomes of actions/ interactions). As conditions, actions/interactions, and consequences are identified, theoretical propositions start to emerge, and researchers can start explaining why a phenomenon occurs, under what conditions, and with what consequences.

SELECTIVE CODING:

The third and final phase of grounded theory is selective coding, which involves identifying a central category or a core variable and systematically and logically relating this central category to other categories. The central category can evolve from existing categories or can be a higher order category that subsumes previously coded categories. New data is selectively sampled to validate the central category and its relationships to other categories (i.e., the tentative theory). Selective coding limits the range of analysis, and makes it move fast. At the same time, the coder must watch out for other categories that may emerge from the new data that may be related to the

phenomenon of interest (open coding), which may lead to further refinement of the initial theory. Hence, open, axial, and selective coding may proceed simultaneously. **Coding of new data and theory refinement continues until theoretical saturation is reached, i.e., when additional data does not yield any marginal change in the core categories or the relationships.**

The “constant comparison” process implies continuous rearrangement, aggregation, and refinement of categories, relationships, and interpretations based on increasing depth of understanding, and an iterative interplay of four stages of activities: (1) comparing incidents/texts assigned to each category (to validate the category), (2) integrating categories and their properties, (3) delimiting the theory (focusing on the core concepts and ignoring less relevant concepts), and (4) writing theory (using techniques like **memoing, storylining, and diagramming**, discussed later in this section). Having a central category does not necessarily mean that all other categories can be integrated nicely around it. In order to identify key categories that are conditions, action/interactions, and consequences of the core category, Strauss and Corbin (1990) recommend **several integration techniques, such as storylining, memoing, or concept mapping.**

In **storylining**, categories and relationships are used to explicate and/or refine a story of the observed phenomenon.

Memos are theorized write-ups of ideas about substantive concepts and their theoretically coded relationships as they evolve during ground theory analysis, and are important tools to keep track of and refine ideas that develop during the analysis. Memoing is the process of using these memos to discover patterns and relationships between categories using

two-by-two tables, diagrams, or figures, or other illustrative displays.

Concept mapping is a graphical representation of concepts and relationships between those concepts (e.g., using boxes and arrows). The major concepts are typically laid out on one or more sheets of paper, blackboards, or using graphical software programs, linked to each other using arrows, and readjusted to best fit the observed data.

After a grounded theory is generated, it must be refined for internal consistency and logic. Researchers must ensure that the central construct has the stated characteristics and dimensions, and if not, the data analysis may be repeated. Researcher must then ensure that the characteristics and dimensions of all categories show variation. For example, if behavior frequency is one such category, then the data must provide evidence of both frequent performers and infrequent performers of the focal behavior. Finally, the theory must be validated by comparing it with raw data. If the theory contradicts with observed evidence, the coding process may be repeated to reconcile such contradictions or unexplained variations.

EXTRA FOR REFERENCE:

Standard software programs, such as **ATLAS.ti.5, NVivo, and QDA Miner**, can be used to automate coding processes in qualitative research methods. These programs can quickly and efficiently organize, search, sort, and process large volumes of text data using user-defined rules. To guide such automated analysis, a coding schema should be created, specifying the keywords or codes to search for in the text, based on an initial manual examination of sample text data. The schema can be organized in a hierarchical manner to organize codes into higher-order codes or constructs. The coding schema should be validated using a different

sample of texts for accuracy and adequacy. However, if the coding schema is biased or incorrect, the resulting analysis of the entire population of text may be flawed and non-interpretable. However, software programs cannot decipher the meaning behind the certain words or phrases or the context within which these words or phrases are used (such as those in sarcasms or metaphors), which may lead to significant misinterpretation in large scale qualitative analysis.

FOCUS GROUPS

Focus group research is a type of research that involves bringing in a small group of subjects (typically 6 to 10 people) at one location, and having them discuss a phenomenon of interest for a period of 1.5 to 2 hours. The discussion is moderated and led by a trained facilitator, who sets the agenda and poses an initial set of questions for participants, makes sure that ideas and experiences of all participants are represented, and attempts to build a holistic understanding of the problem situation based on participants' comments and experiences.

Internal validity cannot be established due to lack of controls and the findings may not be generalized to other settings because of small sample size. Hence, focus groups are not generally used for explanatory or descriptive research, but are more suited for exploratory research.

NARRATIVES

UNDERSTANDING NARRATIVE

In Latin, the noun *narratio* means a narrative or a story, and the verb *narrare* to tell or narrate (Heikkinen, 2002). A narrative is a story that tells a sequence of events that is significant for the narrator or

her or his audience. To repeat, when narratives are looked on within the framework of sociocultural theory, we have to remember the interlinking between the individual and her or his context. As individuals are telling their stories, they are not isolated and independent of their context. On the contrary, it is important to remember that the individual in question is irreducibly connected to her or his social, cultural and institutional setting (Wertsch, 1991). Narratives, therefore, capture both the individual and the context.

Like all of us, teachers tell and retell their stories of experience both for themselves and for others in different social settings, at different times and for different addressees. This means that the perspective on their experiences constantly changes form as they gain new experiences and engage in dialogues with other people (Heikkinen, 2002). Stories cannot be viewed simply as abstract structures isolated from their cultural context. They must be seen as rooted in society and as experienced and performed by individuals in cultural settings (Bruner, 1984). Human knowledge and personal identities are therefore continually constructed and revised. Experience of the world, like each person's perception of her-orhimself, is a continuously developing narrative that is constantly forming and changing form. Here human knowledge is regarded as a plurality of small narratives, local and personal in nature, that are always under construction (Heikkinen, 2002). There is no single, dominant, or static reality but, rather, a number of realities that are constructed in the process of interactions and dialogues. Human knowledge of the world is thus relative. It is dependent on the individual's past and present experiences, her or his values, the people the stories are being told to (the addressees), and when and where

they are being told (Bakhtin, 1986). Bruner (1984) maintained that you always create or hear about a narrative in terms of your life experiences and background.

UNDERSTANDING NARRATIVE RESEARCH:

Narrative research is a term that subsumes a group of approaches that in turn rely on the written or spoken words or visual representation of individuals. These approaches typically focus on the lives of individuals as told through their own stories. The emphasis in such approaches is on the story, typically both what and how is narrated.

Narrative research can be considered both a research method in itself but also the phenomenon under study.

Narrative methods can be considered "real world measures" that are appropriate when "real life problems" are investigated. In a basic linear approach, they encompass the study of the experiences of a single individual embracing stories of the life and exploring the learned significance of those individual experiences. However, in most cases one will be creating an aggregate of narratives each bearing on the others.

While some types of qualitative analysis have a standard set of procedures, narrative research is questionable in this regard. **One of the weaknesses of studying narratives** is that the text is by its own nature linguistically subjective. i.e. difficult to quantitatively access in an objective manner since it is subjective i.e. personally meaningful.

A number of data collection methods can be used, as the researcher and the research subjects work together in this collaborative dialogic relationship. Data can be in the form of field notes; journal records; interview transcripts; one's own and other's observations; storytelling; letter

writing; autobiographical writing; documents such as school and class plans, newsletters, and other texts, such as rules and principles; and pictures. To this list, one should add audio and video recordings, as these are also useful data in narrative research.

Dedicated research software like ATLAS.ti help the researcher to catalog, penetrate and analyze the data generated in a given research project. All forms of narrative research benefit extensively from the special capabilities of a dedicated data analysis tool like ATLAS.ti.

As stated above, the narrative approach is situated within the qualitative or interpretive research method (Gudmundsdottir, 1997, 2001). A qualitative approach to the field of investigation means that researchers study things in their natural settings, attempting to make sense of and interpret phenomena in terms of the meaning people bring to them.

THE NARRATIVE RESEARCH PROCESS

One of the main characteristics of narrative research is the collaboration process between the researcher and her or his research subjects. Within this approach, the research subject is regarded as a collaborator rather than an informant guided by the agenda of the researcher (Altork, 1998). This is the reason why I prefer research subject rather than the more traditional term informant. Other terms I just as easily could have preferred are collaborator or participant. In narrative research, stories of experience are shaped through discussions with the research subject in a dialogue. A number of data collection methods can be used, as the researcher and the research subjects work together in this collaborative dialogic relationship. Data can be in the form of

field notes; journal records; interview transcripts; one's own and other's observations; storytelling; letter writing; autobiographical writing; documents such as school and class plans, newsletters, and other texts, such as rules and principles; and pictures (Connelly & Clandinin, 1990). To this list, I would add video recordings, as these are also useful data in narrative research. Although other qualitative research approaches, such as case studies, biographies, phenomenological studies, grounded theory studies, and ethnographic studies, are described in detail, the literature on narrative research appears to be rather vague about concrete inquiry procedures. In this section I will focus on three recurrent issues in discussions on the narrative research approach. First is the relationship between the researcher and her or his research subjects. The second is how a narrative is developed from an experienced and orally told story into a written text, and the third is the hermeneutic or interpretive nature of narrative research.

EXTRA FOR REFERENCES:

BRUNER (1986): 2 DIFFERENT WAYS OF KNOWING/THINKING

‘Paradigmatic mode of thought’
...draws on reasoned analysis, logical proof, and empirical observation - used to explain ‘cause and effect’, to predict and control reality, and to create unambiguous objective ‘truth’ that can be proven or disproved.

‘Narrative knowing’ Narrative knowledge - created and constructed through stories of lived experiences, and the meanings created. Helps make sense of the ambiguity and complexity of human lives.

Narrative inquiry/analysis is a means by which we systematically gather, analyse, and represent people's stories as told by them, which challenges traditional and modernist views of truth, reality, knowledge and personhood.

Anything that is communicated from someone to another, is a narrative. A table, a list, a schedule, etc. are not a narrative. Stories... everywhere is a narrative, anything that can be described and communicated is a narrative, be it fairytale/ folktale, history, present, a painting, news...

Narrative vs a story: A story has a moralistic overtone.

Analysis is a detailed examination. Analysis is the process of breaking a complex topic or substance into smaller parts in order to gain a better understanding of it.

This focuses on people's narratives/stories either about themselves or a set of events. Instead of looking for themes that emerge from an account, it concentrates on the sequential unfolding of someone's story so there is an emphasis on emplotment and characters. It is time-consuming and usually includes a very small number of cases

Narrative inquiry is an umbrella term that captures personal and human dimensions of experience over time, and takes account of the relationship between individual experience and cultural context (Clandinin and Connelly 2000).

Narrative is referred to as the relating of experiences or events with temporal and spatial orientation.

Narratives can help construct individual or group identity, persuade, rationalize, make an argument, teach a lesson, remember, mobilize, offer perspective, entertain, cope

with or make sense of disturbing events/misfortune.

Narrative as a Genre

Early narrative forms, reaching back as far as 1500 BCE, reflect forms of recorded historical experience in epic formats and are argued to be instrumental in the creation of communal (tribal) education. In the course of sociogenesis, the epic form is joined and partly replaced by folk tales, fables, travelogues--all already foreshadowing the rise of the romantic fiction and the novel.

The new and innovative narrative techniques put to use in these genres--in concert with the development of the print culture--gave rise to the writing (and reading) of letters, confessions and memoirs. This in turn fed readers' interest in personal histories, the biography, life history, and auto-biographies--all making use of temporal sequences of lived events for a systematic and self-reflective quest of the (authentic) self.

Narrative has emerged as a new but central formatting device for the organization of self and (modern) identity. It successfully fed the commonly shared belief that who we are, or who we think we are, is realized in the stories we tell about ourselves; everyone not only has a story but also has a right to tell their story (Bamberg, 2010)

Narrative as method

The term narrative as method implies a general approach that views the individual within their social environments as actively conferring meaning onto objects in the world, including others and selves; the way this happens in everyday situations, as well as in interviews or surveys, is necessarily subjective and interpretive. If narrative is elevated into "the primary form by which human experience is made meaningful"

(Polkinghorne, 1988, p. 1), then it makes sense to argue that the stories we tell are such because they reflect the stories we are (McAdams, 1993)

Narrative Methods

In recent years, a number of qualitative, in-depth interviewing techniques have been designed to elicit explicitly narrative accounts--some open-ended and unstructured, others semi-structured and guided; the free association narrative interview method (Hollway & Jefferson, 2008), the biographic-narrative interpretive method--an interview technique that leads into personal experience, lived situations and life-histories, or narrative oriented inquiry, to name a few.

CASE STUDIES

Case studies are in-depth investigations of a single person, group, event or community. Typically, data are gathered from a variety of sources and by using several different methods (e.g. observations & interviews).

The case study research method originated in clinical medicine (the case history, i.e. the patient's personal history). In psychology, case studies are often confined to the study of a particular individual.

The information is mainly biographical and relates to events in the individual's past (i.e. retrospective), as well as to significant events which are currently occurring in his or her everyday life.

The case study is not itself a research method, but researchers select methods of data collection and analysis that will generate material suitable for case studies.

CONDUCTING A CASE STUDY:

The procedure used in a case study means that the researcher provides a description of the behavior. This comes from

interviews and other sources, such as observation. The client also reports detail of events from his or her point of view. The researcher then writes up the information from both sources above as the case study, and interprets the information. The research may also continue for an extended period of time, so processes and developments can be studied as they happen.

Amongst the sources of data the psychologist is likely to turn to when carrying out a case study are observations of a person's daily routine, unstructured interviews with the participant herself (and with people who know her), diaries, personal notes (e.g. letters, photographs, notes) or official document (e.g. case notes, clinical notes, appraisal reports).

The case study method often involves simply observing what happens to, or reconstructing 'the case history' of a single participant or group of individuals (such as a school class or a specific social group), i.e. the idiographic approach.

The interview is also an extremely effective procedure for obtaining information about an individual, and it may be used to collect comments from the person's friends, parents, employer, workmates and others who have a good knowledge of the person, as well as to obtain facts from the person him or herself. Most of this information is likely to be qualitative (i.e. verbal description rather than measurement) but the psychologist might collect numerical data as well.

ANALYZING CASE STUDY DATA:

The data collected can be analyzed using different theories (e.g. grounded theory, interpretative phenomenological analysis, text interpretation, e.g. thematic coding). All the approaches mentioned here use preconceived categories in the analysis and

they are ideographic in their approach, i.e. they focus on the individual case without reference to a comparison group.

Interpreting the information means the researcher decides what to include or leave out. A good case study should always make clear which information is the factual description and which is an inference or the opinion of the researcher.

STRENGTHS OF CASE STUDIES

Provides detailed (rich qualitative) information.
Provides insight for further research.

Permitting investigation of otherwise impractical (or unethical) situations.

Case studies allow a researcher to investigate a topic in far more detail than might be possible if they were trying to deal with a large number of research participants (nomothetic approach) with the aim of ‘averaging’. Because of their in-depth, multi-sided approach case studies often shed light on aspects of human thinking and behavior that would be unethical or impractical to study in other ways. Research which only looks into the measurable aspects of human behavior is not likely to give us insights into the subjective dimension to experience which is so important to psychoanalytic and humanistic psychologists.

Case studies are often used in exploratory research. They can help us generate new ideas (that might be tested by other methods). They are an important way of illustrating theories and can help show how different aspects of a person's life are related to each other. The method is therefore important for psychologists who

adopt a holistic point of view (i.e. humanistic psychologists).

LIMITATIONS OF CASE STUDIES

Can't generalize the results to the wider population.

Researchers' own subjective feeling may influence the case study (researcher bias).

Difficult to replicate.

Time-consuming.

Because a case study deals with only one person/event/group we can never be sure whether the conclusions drawn from this particular case apply elsewhere.

The results of the study are not generalizable because we can never know whether the case, we have investigated is representative of the wider body of "similar" instances. Because they are based on the analysis of qualitative (i.e. descriptive) data a lot depends on the interpretation the psychologist places on the information she has acquired. This means that there is a lot of scope for observer bias and it could be that the subjective opinions of the psychologist intrude in the assessment of what the data means.

ETHNOGRAPHY

The ethnographic research method, derived largely from the field of anthropology, emphasizes studying a phenomenon within the context of its culture. The researcher must be deeply immersed in the social culture over an extended period of time (usually 8 months to 2 years) and should engage, observe, and record the daily life of the studied culture and its social participants within their natural setting. The primary mode of data collection is participant observation, and data analysis involves a “sense-making” approach. In addition, the researcher must take extensive field notes,

and narrate her experience in descriptive detail so that readers may experience the same culture as the researcher. In this method, the researcher has two roles: rely on her unique knowledge and engagement to generate insights (theory), and convince the scientific community of the trans-situational nature of the studied phenomenon.

The advantages of this approach are its sensitiveness to the context, the rich and nuanced understanding it generates, and minimal respondent bias. However, this is also an extremely time and resource-intensive approach, and findings are specific to a given culture and less generalizable to other cultures.

The classic example of ethnographic research is Jane Goodall's study of primate behaviors, where she lived with chimpanzees in their natural habitat at Gombe National Park in Tanzania, observed their behaviors, interacted with them, and shared their lives. During that process, she learnt and chronicled how chimpanzees seek food and shelter, how they socialize with each other, their communication patterns, their mating behaviors, and so forth. A more contemporary example of ethnographic research is Myra Bluebond-Langer's (1996) [14] study of decision making in families with children suffering from life-threatening illnesses, and the physical, psychological, environmental, ethical, legal, and cultural issues that influence such decision-making. The researcher followed the experiences of approximately 80 children with incurable illnesses and their families for a period of over two years. Data collection involved participant observation and formal/informal conversations with children, their parents and relatives, and health care providers to document their lived experience.

EXTRA FOR REFERENCES: (For someone who wants to understand the Qualitative Research Methods)

Content Analysis

Content analysis is the systematic analysis of the content of a text (e.g., who says what, to whom, why, and to what extent and with what effect) in a quantitative or qualitative manner. Content analysis typically conducted as follows. First, when there are many texts to analyze (e.g., newspaper stories, financial reports, blog postings, online reviews, etc.), the researcher begins by sampling a selected set of texts from the population of texts for analysis. This process is not random, but instead, texts that have more pertinent content should be chosen selectively. Second, the researcher identifies and applies rules to divide each text into segments or "chunks" that can be treated as separate units of analysis. This process is called unitizing. For example, assumptions, effects, enablers, and barriers in texts may constitute such units. Third, the researcher constructs and applies one or more concepts to each unitized text segment in a process called coding. For coding purposes, a coding scheme is used based on the themes the researcher is searching for or uncovers as she classifies the text. Finally, the coded data is analyzed, often both quantitatively and qualitatively, to determine which themes occur most frequently, in what contexts, and how they are related to each other.

A simple type of content analysis is sentiment analysis – a technique used to capture people's opinion or attitude toward an object, person, or phenomenon. Reading online messages about a political candidate posted on an online forum and classifying each message as positive, negative, or neutral is an example of such an analysis. In this case, each message represents one unit of analysis. This analysis will help identify whether the

sample as a whole is positively or negatively disposed or neutral towards that candidate. Examining the content of online reviews in a similar manner is another example. Though this analysis can be done manually, for very large data sets (millions of text records), natural language processing and text analytics-based software programs are available to automate the coding process, and maintain a record of how people sentiments fluctuate with time.

A frequent criticism of content analysis is that it lacks a set of systematic procedures that would allow the analysis to be replicated by other researchers. Schilling (2006) [20] addressed this criticism by organizing different content analytic procedures into a spiral model. This model consists of five levels or phases in interpreting text: (1) convert recorded tapes into raw text data or transcripts for content analysis, (2) convert raw data into condensed protocols, (3) convert condensed protocols into a preliminary category system, (4) use the preliminary category system to generate coded protocols, and (5) analyze coded protocols to generate interpretations about the phenomenon of interest.

Content analysis has several limitations. First, the coding process is restricted to the information available in text form. For instance, if a researcher is interested in studying people's views on capital punishment, but no such archive of text documents is available, then the analysis cannot be done. Second, sampling must be done carefully to avoid sampling bias. For instance, if your population is the published research literature on a given topic, then you have systematically omitted unpublished research or the most recent work that is yet to be published.

Hermeneutic Analysis

Hermeneutic analysis is a special type of content analysis where the researcher tries to “interpret” the subjective meaning of a given text within its socio-historic context. Unlike grounded theory or content analysis, which ignores the context and meaning of text documents during the coding process, hermeneutic analysis is a truly interpretive technique for analyzing qualitative data. This method assumes that written texts narrate an author's experience within a socio-historic context, and should be interpreted as such within that context. Therefore, the researcher continually iterates between singular interpretation of the text (the part) and a holistic understanding of the context (the whole) to develop a fuller understanding of the phenomenon in its situated context, which German philosopher Martin Heidegger called the hermeneutic circle. The word hermeneutic (singular) refers to one particular method or strand of interpretation.

More generally, hermeneutics is the study of interpretation and the theory and practice of interpretation. Derived from religious studies and linguistics, traditional hermeneutics, such as biblical hermeneutics, refers to the interpretation of written texts, especially in the areas of literature, religion and law (such as the Bible). In the 20th century, Heidegger suggested that a more direct, non-mediated, and authentic way of understanding social reality is to experience it, rather than simply observe it, and proposed philosophical hermeneutics, where the focus shifted from interpretation to existential understanding. Heidegger argued that texts are the means by which readers can not only read about an author's experience, but also relive the author's experiences. Contemporary or modern hermeneutics, developed by Heidegger's students such as Hans-Georg Gadamer, further examined the limits of written texts for communicating social experiences, and

went on to propose a framework of the interpretive process, encompassing all forms of communication, including written, verbal, and non-verbal, and exploring issues that restrict the communicative ability of written texts, such as presuppositions, language structures (e.g., grammar, syntax, etc.), and semiotics (the study of written signs such as symbolism, metaphor, analogy, and sarcasm). The term hermeneutics is sometimes used interchangeably and inaccurately with exegesis, which refers to the interpretation or critical explanation of written text only and especially religious texts.

**STATISTICS IN
PSYCHOLOGY: MEASURES
OF CENTRAL TENDENCY AND
DISPERSION. NORMAL
PROBABILITY CURVE.
PARAMETRIC [T-TEST] AND
NON-PARAMETRIC TESTS
[SIGN TEST, WILCOXON
SIGNED RANK TEST, MANN-
WHITNEY TEST, KRUSKAL-
WALLIS TEST, FRIEDMAN].
POWER ANALYSIS. EFFECT
SIZE.**

**MEASURES OF CENTRAL
TENDENCY AND DISPERSION:**

Univariate analysis, or analysis of a single variable, refers to a set of statistical techniques that can describe the general properties of one variable. Univariate statistics include: (1) frequency distribution, (2) central tendency, and (3) dispersion. The frequency distribution of a variable is a summary of the frequency (or percentages) of individual values or ranges of values for that variable. For instance, we can measure how many times a sample of respondents attend religious services (as a measure of their “religiosity”) using a categorical scale: never, once per year, several times per year, about once a month, several times per month, several times per week, and an optional category for “did not answer.” If we count the number (or percentage) of observations within each category (except “did not answer” which is really a missing value rather than a category), and display it in the form of a table as shown in Figure 14.1, what we have is a frequency distribution. This distribution can also be depicted in the form of a bar chart, as shown on the right panel of Figure 14.1, with the horizontal axis representing each category of that variable and the vertical axis representing the frequency or percentage of observations within each category.

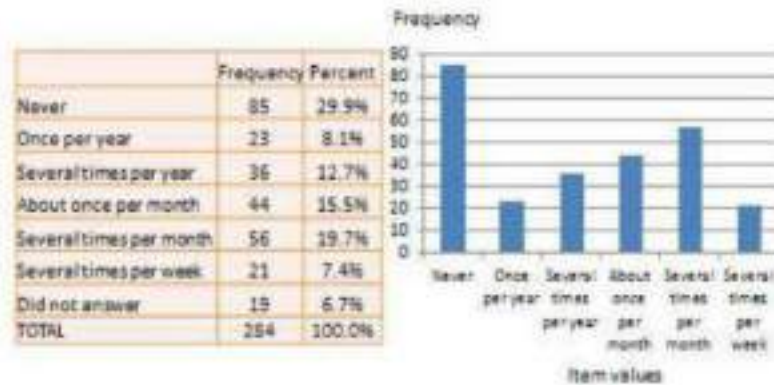


Figure 14.1 Frequency distribution of religiosity.

With very large samples where observations are independent and random, the frequency distribution tends to follow a plot that looked like a bell-shaped curve (a smoothed bar chart of the frequency distribution) similar to that shown in Figure 14.2, where most observations are clustered toward the center of the range of values, and fewer and fewer observations toward the extreme ends of the range. Such a curve is called a normal distribution.

Central tendency is an estimate of the center of a distribution of values. There are three major estimates of central tendency: mean, median, and mode. The arithmetic mean (often simply called the “mean”) is the simple average of all values in a given distribution. Consider a set of eight test scores: 15, 22, 21, 18, 36, 15, 25, 15. The arithmetic mean of these values is $(15 + 20$

$21+20+36+15+25+15)/8=20.875$. Other types of means include geometric mean (n th root of the product of n numbers in a distribution) and harmonic mean (the reciprocal of the arithmetic means of the reciprocal of each value in a distribution), but these means are not very popular for statistical analysis of social research data. The second measure of central tendency, the

median, is the middle value within a range of values in a distribution. This is computed by sorting all values in a distribution in increasing order and selecting the middle value. In case there are two middle values (if there is an even number of values in a distribution), the average of the two middle values represent the median. In the above example, the sorted values are: 15, 15, 15, 18, 22, 21, 25, 36. The two middle values are 18 and 22, and hence the median is $(18 + 22)/2 = 20$.

Lastly, the mode is the most frequently occurring value in a distribution of values. In the previous example, the most frequently occurring value is 15, which is the mode of the above set of test scores. Note that any value that is estimated from a sample, such as mean, median, mode, or any of the later estimates are called a statistic.

Dispersion refers to the way values are spread around the central tendency, for example, how tightly or how widely are the values clustered around the mean. Two common measures of dispersion are the range and standard deviation. The range is the difference between the highest and lowest values in a distribution. The range in our previous example is $36-15 = 21$.

The range is particularly sensitive to the presence of outliers. For instance, if the highest value in the above distribution was 85 and the other values remained the same, the range would be $85 - 15 = 70$. Standard deviation, the second measure of

dispersion, corrects for such outliers by using a formula that takes into account how close or how far each value from the distribution mean:

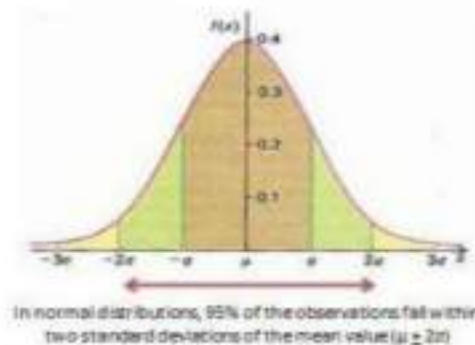
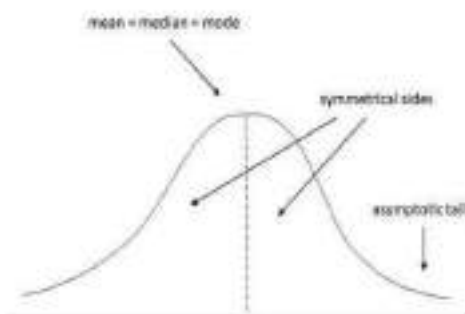


Figure 14.2. Normal distribution.

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n - 1}}$$

where σ is the standard deviation, x_i is the i th observation (or value), μ is the arithmetic mean, n is the total number of observations, and Σ means summation across all observations. The square of the standard deviation is called the variance of a distribution. In a normally distributed frequency distribution, it is seen that 68% of the observations lie within one standard deviation of the mean ($\mu + 1 \sigma$), 95% of the observations lie within two standard deviations ($\mu + 2 \sigma$), and 99.7% of the observations lie within three standard deviations ($\mu + 3 \sigma$), as shown in Figure 14.2.



NORMAL CURVE

PROBABILITY

A normal distribution has a bell-shaped curve and is symmetrical around its center, so the right side of the center is a mirror image of the left side. Most of the continuous data values in a normal distribution tend to cluster around the mean, and the further a value is from the mean, the less likely it is to occur. The tails are asymptotic, which means that they approach but never quite meet the horizon (i.e. x-axis).

For a perfectly normal distribution the mean, median and mode will be the same value, visually represented by the peak of the curve.

The normal distribution is often called the bell curve because the graph of its probability density looks like a bell. It is

Also known as called Gaussian distribution, after the German mathematician Carl Gauss who first described it

WHY IS THE NORMAL DISTRIBUTION IMPORTANT?

The bell-shaped curve is a common feature of nature and psychology

The normal distribution is the most important probability distribution in statistics because many continuous data in nature and psychology displays this bell-shaped curve when compiled and graphed.

For example, if we randomly sampled 100 individuals we would expect to see a normal distribution frequency curve for many continuous variables, such as IQ, height, weight and blood pressure.

Parametric significance tests require a normal distribution of the samples' data points

The most powerful (parametric) statistical tests used by psychologists require data to be normally distributed. If the data does not resemble a bell curve researchers may have to use a less powerful type of statistical test, called non-parametric statistics.

Converting the raw scores of a normal distribution to z-scores

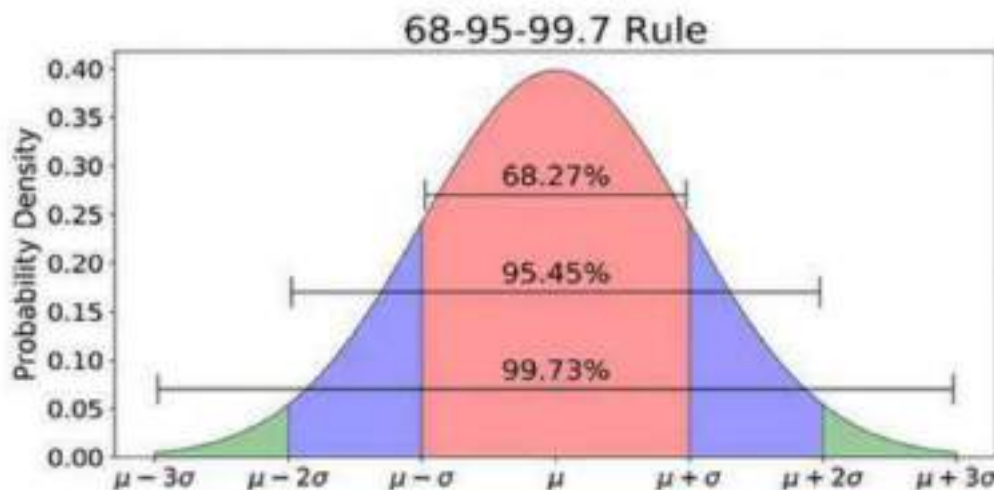
We can standardized the values (raw scores) of a normal distribution by converting them into z-scores.

This procedure allows researchers to determine the proportion of the values that fall within a specified number of standard deviations from the mean (i.e. calculate the empirical rule).

WHAT IS THE EMPIRICAL RULE FORMULA?

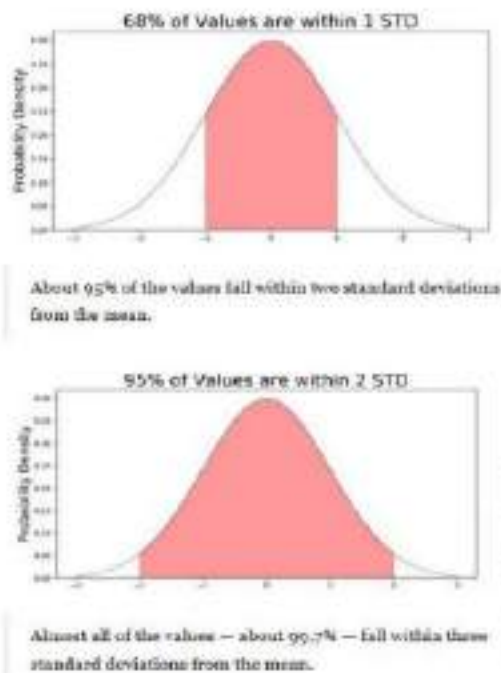
values that fall within certain distances from the mean. The empirical rule is often referred to as the three-sigma rule or the 68-95-99.7 rule.

The empirical rule in statistics allows researchers to determine the proportion of



If the data values in a normal distribution are converted to z-scores in a standard

normal distribution the empirical rule describes the percentage of the data that fall within specific numbers of standard deviations (σ) from the mean (μ) for bell-shaped curves. For example, in a normal distribution, 68% of the observations fall within ± 1 standard deviation from the mean.

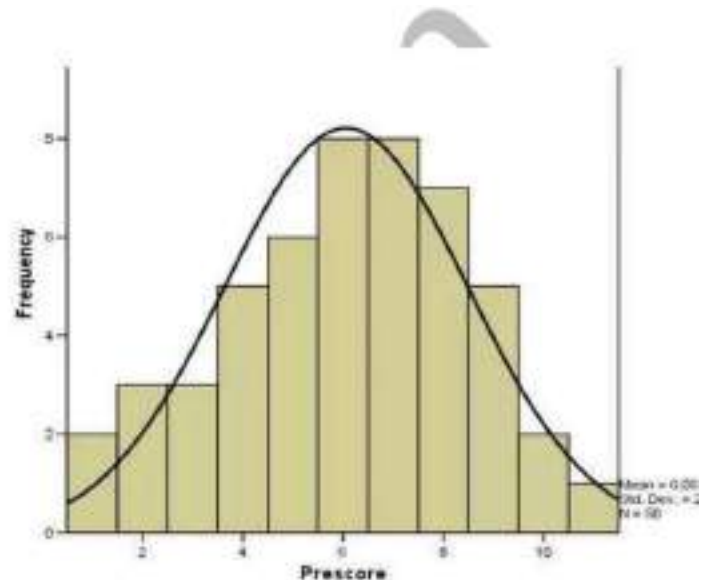


HOW CAN I CHECK IF MY DATA FOLLOWS A NORMAL DISTRIBUTION?

Statistical software (such as SPSS) can be used to check if your dataset is normally distributed by calculating the three measures of central tendency. If the mean, median and mode are very similar values there is a good chance that the data follows

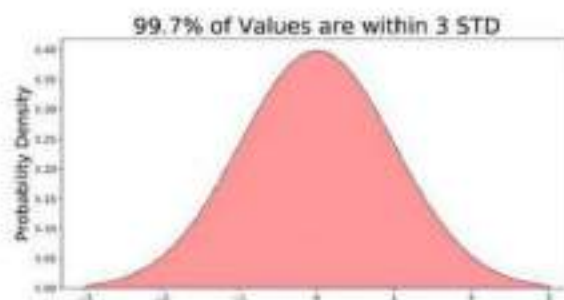
bell-shaped distribution (SPSS command here).

It is also advisable to a frequency graph too, so you can check the visual shape of your data (If your chart is a histogram, you can add a distribution curve using SPSS: From the menus choose: Elements > Show Distribution Curve).



Normal distributions become more apparent (i.e. perfect) the finer the level of measurement and the larger the sample from a population.

You can also calculate coefficients which tell us about the size of the distribution



tails in relation to the bump in the middle of the bell curve. For example, **Kolmogorov Smirnov and Shapiro-Wilk tests can be calculated using SPSS.**

These tests compare your data to a normal distribution and provide a p-value, which

if significant ($p < .05$) indicates your data is different to a normal distribution (thus, on this occasion we do not want a significant result and need a p-value higher than 0.05).

Tests of Normality				
Course		Kolmogorov-Smirnov ^a		
		Statistic	df	Sig.
Time	Beginner	.177	10	.200
	Intermediate	.166	10	.200 [*]
	Advanced	.151	10	.200 [*]

a. Lilliefors Significance Correction

*. This is a lower bound of the true significance.

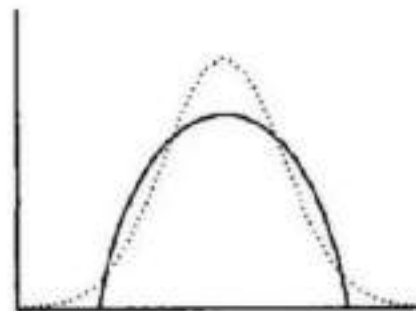
Kurtosis is a statistical measure used to describe the degree to which scores cluster in the tails or the peak of a frequency distribution. The peak is the tallest part of the distribution, and the tails are the ends of the distribution.

There are three types of kurtosis: **mesokurtic**, **leptokurtic**, and **platykurtic**.

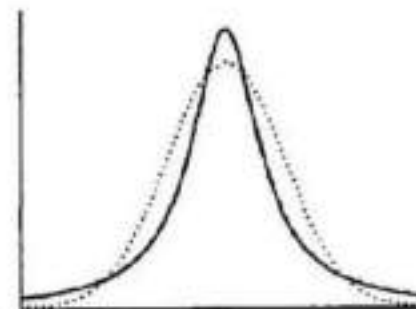
Mesokurtic: Distributions that are moderate in breadth and curves with a medium peaked height.

Leptokurtic: More values in the distribution tails and more values close to the mean (i.e. sharply peaked with heavy tails)

Platykurtic: Fewer values in the tails and fewer values close to the mean (i.e. the curve has a flat peak and has more dispersed scores with lighter tails).



**Negative kurtosis
(platykurtic)**



**Positive kurtosis
(leptokurtic)**

WHAT DOES IT MEAN WHEN KURTOSIS IS ZERO?

When kurtosis is equal to 0, the distribution is mesokurtic. This means the kurtosis is the same as the normal distribution, it is mesokurtic (medium peak). The kurtosis of a mesokurtic distribution is neither high nor low, rather it is considered to be a baseline for the two other classifications.

PARAMETRIC [T-TEST]

The t test is one type of inferential statistics. It is used to determine whether there is a significant difference between the means of two groups. With all inferential statistics, we assume the dependent variable fits a normal distribution. When we assume a normal distribution exists, we can identify the probability of a particular outcome. We specify the level of probability (alpha level, level of significance, p) we are

willing to accept before we collect data (p .05 is a common value that is used). After we collect data, we calculate a test statistic with a formula. We compare our test statistic with a critical value found on a table to see if our results fall within the acceptable level of probability. Modern computer programs calculate the test statistic for us and also provide the exact probability of obtaining that test statistic with the number of subjects we have.

STUDENT'S TEST (T TEST):

When the difference between two population averages is being investigated, a t test is used. In other words, **a t test is used when we wish to compare two means** (the scores must be measured on an interval or ratio measurement scale). We would use a t test if we wished to compare the reading achievement of boys and girls. With a t test, we have one independent variable and one dependent variable. The independent variable (gender in this case) can only have two levels (male and female). The dependent variable would be reading achievement. **If the independent had more than two levels, then we would use a one-way analysis of variance (ANOVA).**

The test statistic that a t test produces is a t-value. **Conceptually, t-values are an extension of z-scores.** In a way, the t-value represents how many standard units the means of the two groups are apart.

With a t test, the researcher wants to state with some degree of confidence that the obtained difference between the means of the sample groups is too great to be a chance event and that some difference also exists in the population from which the sample was drawn. In other words, the difference that we might find between the boys' and girls' reading achievement in our sample might have occurred by chance, or it might exist in the population.

If our t test produces a t-value that results in a probability of .01, we say that the likelihood of getting the difference we found by chance would be 1 in 100 times. We could say that it is unlikely that our results occurred by chance and the difference we found in the sample probably exists in the populations from which it was drawn.

Five factors contribute to whether the difference between two groups' means can be considered significant:

How large is the difference between the means of the two groups? Other factors being equal, the greater the difference between the two means, the greater the likelihood that a statistically significant mean difference exists. If the means of the two groups are far apart, we can be fairly confident that there is a real difference between them.

How much overlap is there between the groups? This is a function of the variation within the groups. Other factors being equal, the smaller the variances of the two groups under consideration, the greater the likelihood that a statistically significant mean difference exists. We can be more confident that two groups differ when the scores within each group are close together.

How many subjects are in the two samples? The size of the sample is extremely important in determining the significance of the difference between means. With increased sample size, means tend to become more stable representations of group performance. If the difference we find remains constant as we collect more and more data, we become more confident that we can trust the difference we are finding.

What alpha level is being used to test the mean difference (how confident do you want to be about your statement that there is a mean difference). A larger alpha level requires less difference between the means. It is much harder to find differences between groups when you are only willing to have your results occur by chance 1 out of a 100 times ($p < .01$) as compared to 5 out of 100 times ($p < .05$).

Is a directional (one-tailed) or non-directional (two-tailed) hypothesis being tested? Other factors being equal, smaller mean differences result in statistical significance with a directional hypothesis. For our purposes we will use non-directional (two-tailed) hypotheses.

ASSUMPTIONS UNDERLYING THE TEST

The samples have been randomly drawn from their respective populations

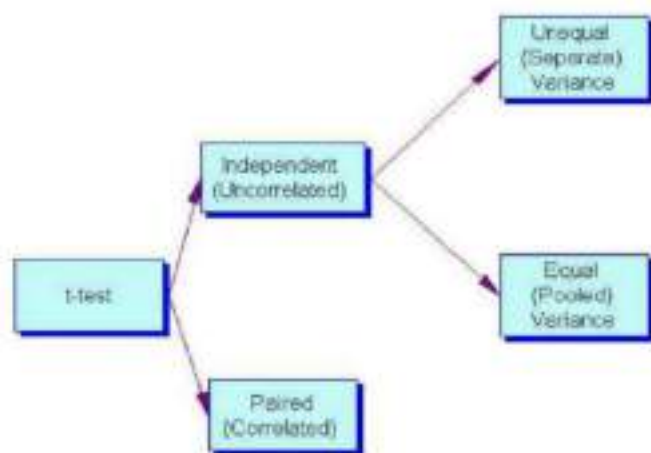
The scores in the population are normally distributed

The scores in the populations have the same variance ($s_1 = s_2$) Note: We use a different calculation for the standard error if they are not.

Pair-difference t test (a.k.a. t-test for dependent groups, correlated t test) $df = n$ (number of pairs) - 1

EXTRA for Reference

Three Types of t tests



This is concerned with the difference between the average scores of a single sample of individuals who are assessed at two different times (such as before treatment and after treatment). It can also compare average scores of samples of individuals who are paired in some way (such as siblings, mothers, daughters, persons who are matched in terms of a particular characteristics).

T-test for Independent Samples (with two options)

This is concerned with the difference between the averages of two populations. Basically, the procedure compares the averages of two samples that were selected independently of each other, and asks whether those sample averages differ enough to believe that the populations from which they were selected also have different averages. An example would be comparing math achievement scores

of an experimental group with a control group.

Equal Variance (Pooled-variance t-test) $df=n$ (total of both groups)

-2 Note: Used when both samples have the same number of subject or when $s_1=s_2$ (Levene or F-max tests have $p > .05$).

Unequal Variance (Separate-variance t test) df depends on a formula, but a rough estimate are one less than the smallest group.

The basic idea for calculating a t-test is to find the difference between the means of the two groups and divide it by the **STANDARD ERROR (OF THE DIFFERENCE)** — which is the standard deviation of the distribution of differences.

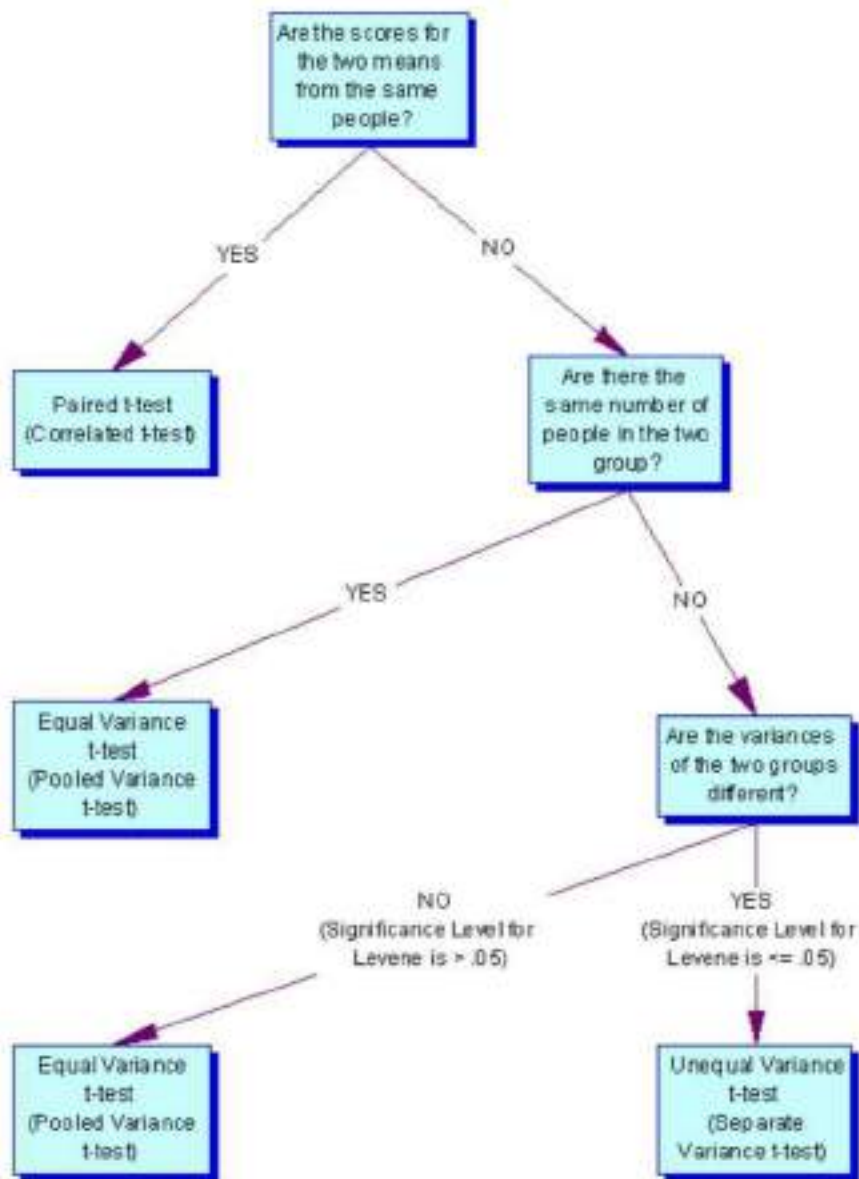
For information: A **CONFIDENCE INTERVAL** for a two-tailed t-test is calculated by multiplying the **CRITICAL VALUE** times the **STANDARD ERROR** and adding and subtracting that to and from the difference of the two means.

EFFECT SIZE is used to calculate practical difference. If you have several thousand subjects, it is very easy to find a statistically significant difference. Whether that difference is practical or meaningful is another questions. This is where effect size becomes important. With studies involving group differences, effect size is the difference of the two means divided by the standard deviation of the control group (or the average standard deviation of both groups if you do not have a control group). Generally, effect size is only important if you have statistical significance. An effect size of .2 is considered small, .5 is considered medium, and .8 is considered large.

FYI: William Sealy Gosset (1905) first published a t-test. He worked at the Guinness Brewery in Dublin and published under the name Student. The test was called **Student Test** (later shortened to t test).

How do I decide which type of t test to use?

How do I decide which type of t test to use?



NON-PARAMETRIC TESTS
[SIGN TEST, WILCOXON SIGNED RANK TEST, MANN-

WHITNEY TEST, KRUSKAL-WALLIS TEST, FRIEDMAN].

A non-parametric test (sometimes called a distribution free test) does not assume anything about the underlying distribution (for example, that the data comes from a normal distribution). That's compared to parametric test, which makes assumptions about a population's parameters (for example, the mean or standard deviation); When the word "non parametric" is used

in stats, it doesn't quite mean that you know nothing about the population. It usually means that you know the population data does not have a normal distribution.

For example, one assumption for the one-way ANOVA is that the data comes from a normal distribution. If your data isn't normally distributed, you can't run an ANOVA, but you can run the nonparametric alternative—the Kruskal-Wallis test.

If at all possible, you should use parametric tests, as they tend to be more accurate. Parametric tests have greater statistical power, which means they are likely to find a true significant effect. Use nonparametric tests only if you have to (i.e. you know that assumptions like normality are being violated). Nonparametric tests can perform well with non-normal continuous data if you have a sufficiently large sample size (generally 15-20 items in each group).

TYPES OF NONPARAMETRIC TESTS

When the word “parametric” is used in stats, it usually means tests like ANOVA or a t test. Those tests both assume that the population data has a normal distribution. Non-parametric do not assume that the data is normally distributed. The only non-parametric test you are likely to come across in elementary stats is the chi-square test. However, there are several others. For example: the Kruskal Willis test is the non-parametric alternative to the One-way ANOVA and the Mann Whitney is the non-parametric alternative to the two-sample t test.

THE MAIN NON-PARAMETRIC TESTS ARE:

1-SAMPLE SIGN TEST.

Use this test to estimate the median of a population and compare it to a reference value or target value.

The sign test is a statistical test to compare the sizes of two groups. It is a non-parametric or “distribution free” test, which means the test doesn't assume the data comes from a particular distribution, like the normal distribution. The sign test is an alternative to a one sample t test or a paired t test. It can also be used for ordered (ranked) categorical data.

The null hypothesis for the sign test is that the difference between medians is zero. For a one sample sign test, where the median for a single sample is analyzed.

HOW TO CALCULATE A PAIRED/MATCHED SAMPLE SIGN TEST: (FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/sign-test/>

1-SAMPLE WILCOXON SIGNED RANK TEST.

With this test, you also estimate the population median and compare it to a reference/target value. However, the test assumes your data comes from a symmetric distribution (like the Cauchy distribution or uniform distribution).

The Wilcoxon signed rank test (also called the Wilcoxon signed rank sum test) is a non-parametric test. When the word “non-parametric” is used in stats, it doesn't quite mean that you know nothing about the population. It usually means that you know the population data does not have a normal distribution. The Wilcoxon signed rank test should be used if the differences between pairs of data are non-normally distributed.

Two slightly different versions of the test exist:

The Wilcoxon signed rank test compares your sample median against a hypothetical median.

The Wilcoxon matched-pairs signed rank test computes the difference between each set of matched pairs, then follows the same procedure as the signed rank test to compare the sample against some median.

The term “Wilcoxon” is often used for either test. This usually isn’t confusing, as it should be obvious if the data is matched, or not matched.

The null hypothesis for this test is that the medians of two samples are equal. It is generally used:

- As a non-parametric alternative to the one-sample t test or paired t test.
- For ordered (ranked) categorical variables without a numerical scale.

HOW TO RUN THE WILCOXON SIGNED RANK TEST: (FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/wilcoxon-signed-rank-test/>

FRIEDMAN TEST

This test is used to test for differences between groups with ordinal dependent variables. It can also be used for continuous data if the one-way ANOVA with repeated measures is inappropriate (i.e. some assumption has been violated).

Friedman’s test is a non-parametric test for finding differences in treatments across multiple attempts. Nonparametric means the test doesn’t assume your data comes

from a particular distribution (like the normal distribution). Basically, it’s used in place of the ANOVA test when you don’t know the distribution of your data.

Friedman’s test is an extension of the sign test, used when there are multiple treatments. In fact, if there are only two treatments the two tests are identical.

Your data should meet the following requirements:

Data should be ordinal (e.g. the Likert scale) or continuous,

- Data comes from a single group, measured on at least three different occasions,

The sample was created with a random sampling method,

- Blocks are mutually independent (i.e. all of the pairs are independent — one doesn’t affect the other),
- Observations are ranked within blocks with no ties.

The null hypothesis for the test is that the treatments all have identical effects, or that the samples differ in some way. For example, they have different centers, spreads, or shapes. The alternate hypothesis is that the treatments do have different effects.

RUNNING THE TEST: (FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/friedmans-test/>

**GOODMAN
GAMMA:**

KRUSKA’S

A test of association for ranked variables.

RUNNING THE TEST: (FOR REFERENCE)

KRUSKAL-WALLIS TEST.

for the calculations, rather than the data. Use this test instead of a one-way ANOVA points themselves. To find out if two or more medians are different. Ranks of the data points are used

Rank	Os in Order
1	0.1
2	0.2
3	0.3
4	0.8
5	1.5
6	1.5
7	1.8
8	2.0
9	2.1
10	2.3
11	2.6
12	12.4

The Kruskal Wallis H test uses ranks instead of actual data.

The Kruskal Wallis test is the non parametric alternative to the One Way ANOVA. Non parametric means that the test doesn't assume your data comes from a particular distribution. The H test is used when the assumptions for ANOVA aren't met (like the assumption of normality). It is sometimes called the one-way ANOVA on ranks, as the ranks of the data values are used in the test rather than the actual data points.

The test determines whether the medians of two or more groups are different. Like most statistical tests, you calculate a test statistic and compare it to a distribution cut-off point. The test statistic used in this test is called the H statistic. The hypotheses for the test are:

- H_0 : population medians are equal.
- H_1 : population medians are not equal.

The Kruskal Wallis test will tell you if there is a significant difference between groups. However, it won't tell you which groups are different. For that, you'll need

to run a Post Hoc test.

Examples

1. You want to find out how test anxiety affects actual test scores. The independent variable "test anxiety" has three levels: no anxiety, low-medium anxiety and high anxiety. The dependent variable is the exam score, rated from 0 to 100%.
2. You want to find out how socioeconomic status affects attitude towards sales tax increases. Your independent variable is "socioeconomic status" with three levels: working class, middle class and wealthy. The dependent variable is measured on a 5-point Likert scale from strongly agree to strongly disagree.

Assumptions for the Kruskal Wallis Test

Your variables should have:

One independent variable with two or more levels (independent groups). The test is more commonly used when you have three or more levels. For two levels, consider using the Mann Whitney U Test instead. Ordinal scale, Ratio Scale Interval scale dependent variables.

- Your observations should be independent. In other words, there should be no relationship between the members in each group or between groups. For more information on this point, see: Assumption of Independence.
- All groups should have the same shape distributions. Most software (i.e. SPSS, Minitab) Part of the test.

RUNNING THE TEST:(FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/kruskal-wallis/>

THE MANN-KENDALL TREND TEST:

Looks for trends in time-series data.

MANN-WHITNEY TEST.

Use this test to compare differences between two independent groups when dependent variables are either ordinal or continuous. The Mann-Whitney U test is the nonparametric equivalent of the two sample t-test. While the t-test makes an assumption about the distribution of a population (i.e. that the sample came from t-distributed population), the Mann Whitney U Test makes no such assumption.

RUNNING THE TEST: (FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/mann-whitney-u-test/>

MOOD'S MEDIAN TEST.

Use this test instead of the sign test when you have two independent samples.

Mood's median test is used to compare the medians for two samples to find out if they are different. For example, you might want to compare the median number of positive calls to a hotline vs. the median number of negative comment calls to find out if you're getting significantly more negative comments than positive comments (or vice versa).

This test is the nonparametric alternative to a one way ANOVA; Nonparametric means that you don't have to know what distribution your sample came from (i.e. a

normal distribution) before running the test. That said, your samples should have been drawn from distributions with the same shape. This test has very low statistical power for samples drawn from normal distributions or short-tailed distributions.

Use this test instead of the sign test when you have two independent samples.

The null hypothesis for this test is that the medians are the same for both groups. The alternate hypothesis for the test is that the medians are different for both groups.

RUNNING THE TEST: (FOR REFERENCE)

<https://www.statisticshowto.datasciencecentral.com/moods-median-test/>

SPEARMAN RANK CORRELATION.

Use when you want to find a correlation between two sets of data. The Spearman rank correlation coefficient, r_s , is the nonparametric version of the Pearson correlation coefficient. Your data must be ordinal, interval or ratio. Spearman's returns a value from -1 to 1, where:

+1 = a perfect positive correlation between ranks
-1 = a perfect negative correlation between ranks
0 = no correlation between ranks.

RUNNING THE TEST: (FOR REFERENCE)

The following table lists the nonparametric tests and their parametric alternatives.

<https://www.statisticshowto.datasciencecentral.com/spearman-rank-correlation-definition-calculate/>

NONPARAMETRIC TEST	PARAMETRIC ALTERNATIVE
1-sample sign test	One-sample Z-test, One sample t-test
1-sample Wilcoxon Signed Rank test	One sample Z-test, One sample t-test
Friedman test	Two-way ANOVA
Kruskal-Wallis test	One-way ANOVA
Mann-Whitney test	Independent samples t-test
Mood's Median test	One-way ANOVA
Spearman Rank Correlation	Correlation Coefficient

Compared to parametric tests, nonparametric tests have several advantages, including:

- More statistical power when assumptions for the parametric tests have been violated. When assumptions haven't been violated, they can be almost as powerful.
 - Fewer assumptions (i.e. the assumption of normality doesn't apply). Small sample sizes are acceptable.
- They can be used for all data types, including nominal variables, interval variables, or data that has

outliers or that has been measured imprecisely. However, they do have their disadvantages. The most notable ones are:

Less powerful than parametric tests if assumptions haven't been violated.
 More labor-intensive to calculate by hand (for computer calculations, this isn't an issue).
 Critical value tables for many tests aren't included in many computer software packages. This is compared to tables for parametric tests (like the z-table or t-table) which usually are included.

Key Difference Between Parametric And Non-parametric

Properties	Parametric	Non-parametric
Assumptions	Yes	No
Value for central tendency	Mean value	Median value
Correlation	Pearson	Spearman
Probabilistic distribution	Normal	Arbitrary
Population knowledge	Requires	Does not require
Used for	Interval data	Nominal data
Applicability	Variables	Attributes & Variables
Examples	t-test, z-test, etc.	Kruskal-Wallis, Mann-Whitney

POWER

Power is defined as the probability of correctly rejecting a false null hypothesis. In terms of our example, it is the probability that given there is a difference between the population means of the new method and the standard method, the sample means will be significantly different. The probability of failing to reject a false null hypothesis is often referred to as β . Therefore, power can be defined as:

$$\text{power} = 1 - \beta.$$

It is very important to consider power while designing an experiment. You should avoid spending a lot of time and/or money on an experiment that has little chance of finding a significant effect.

Question 1 out of 2.

Power is:

- ☐ The probability that the null hypothesis is true.
- ☐ The probability that the null hypothesis is false.
- ☐ The probability a false null hypothesis will be rejected.
- ☐ The probability a true null hypothesis will be rejected.

1 It is the probability of correctly rejecting a false null hypothesis.

2 With low power, the null hypothesis is unlikely to be rejected. When the null hypothesis is not rejected, the experiment is inconclusive.

FACTORS AFFECTING POWER

Several factors affect the power of a statistical test. Some of the factors are under the control of the experimenter, whereas others are not.

Larger the sample size, the higher the power. Since sample size is typically under an experimenter's control, increasing sample size is one way to increase power.

Question 2 out of 2.

If the power of an experiment is low then

- ☐ The experiment will likely be inconclusive.
- ☐ Any significant findings obtained are suspect.
- ☐ The results are skewed.

However, it is sometimes difficult and/or expensive to use a large sample size.

Power is higher when the standard deviation is small. Larger the effect size, the more likely it is that an experiment would find a significant effect.

There is a trade-off between the significance level and power: the more stringent (lower) the significance level, the lower the power. Power is higher with a one-tailed test than with a two-tailed test as long as the hypothesized direction is correct. A one-tailed test at the 0.05 level has the same power as a two-tailed test at the 0.10 level. A one-tailed test, in effect, raises the significance level.

Question 2 out of 3.

Which of the following increase power?

- ☐ Increasing the standard deviation
- ☐ Increasing the sample size
- ☐ Increasing the significance level
- ☐ Increasing the size of the difference between means

Question 3 out of 3.

Which of the following decreases the probability of a type I error?

- ☐ Increasing the standard deviation
- ☐ Increasing the sample size
- ☐ Decreasing the significance level

2 All except (A)

3 Only decreasing the significance level.

The others have no effect.

EFFECT SIZE

Effect size is a quantitative measure of the magnitude of the experimenter effect. The larger the effect size the stronger the relationship between two variables.

It is used to calculate practical difference. If you have several thousand subjects, it is very easy to find a statistically significant difference. Whether that difference is practical or meaningful is another question. This is where effect size becomes important. With studies involving group differences, effect size is the difference of the two means divided by the standard deviation of the control group (or the average standard deviation of both groups if you do not have a control group). Generally, effect size is only important if you have statistical significance. An effect size of .2 is considered small, .5 is considered medium, and .8 is considered large. Thus, Effect sizes either measure the sizes of associations between variables or the sizes of differences between group means.

COHEN'S D

It is an appropriate effect size for the comparison between two means. It can be used, for example, to accompany the reporting of t-test and ANOVA results. It is also widely used in meta-analysis.

To calculate the standardized mean difference between two groups, subtract the mean of one group from the other ($M1 - M2$) and divide the result by the standard deviation (SD) of the population from which the groups were sampled.

$$\text{Effect Size} = \frac{[\text{Mean of experimental group}] - [\text{Mean of control group}]}{\text{Standard Deviation}}$$

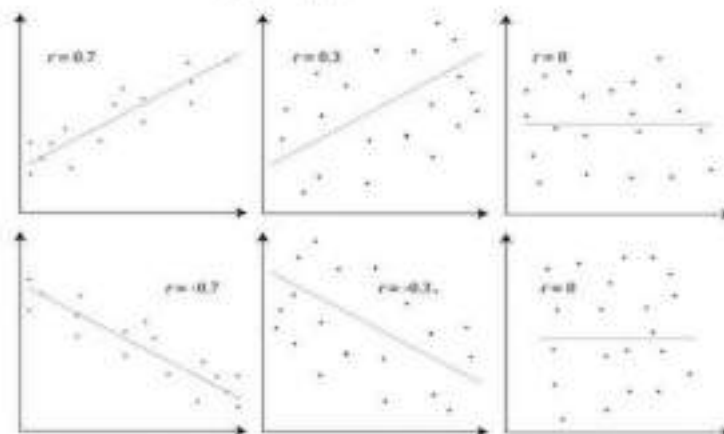
A d of 1 indicates the two groups differ by 1 standard deviation, a d of 2 indicates they differ by 2 standard deviations, and so on. Standard deviations are equivalent to **z-scores** (1 standard deviation = 1 z-score).

Relative size	Effect size	% of control group below the mean of experimental group
	0.0	50%
Small	0.2	58%
Medium	0.5	69%
Large	0.8	79%
	1.4	92%

Cohen suggested that $d=0.2$ be considered a 'small' effect size, 0.5 represents a 'medium' effect size and 0.8 a 'large' effect size. This means that if two groups' means don't differ by 0.2 standard deviations or more, the difference is trivial, even if it is statistically significant.

PEARSON R CORRELATION

This parameter of effect size summarizes the strength of the bivariate relationship. The value of the effect size of Pearson r correlation varies between -1 (a perfect negative correlation) to +1 (a perfect positive correlation).



According to Cohen (1988, 1992), the effect size is low if the value of r varies around 0.1, medium if r varies around 0.3, and large if r varies more than 0.5.

**CORRELATIONAL ANALYSIS:
PARTIAL CORRELATION,
MULTIPLE CORRELATION.
PARTIAL CORRELATION,
MULTIPLE CORRELATION**

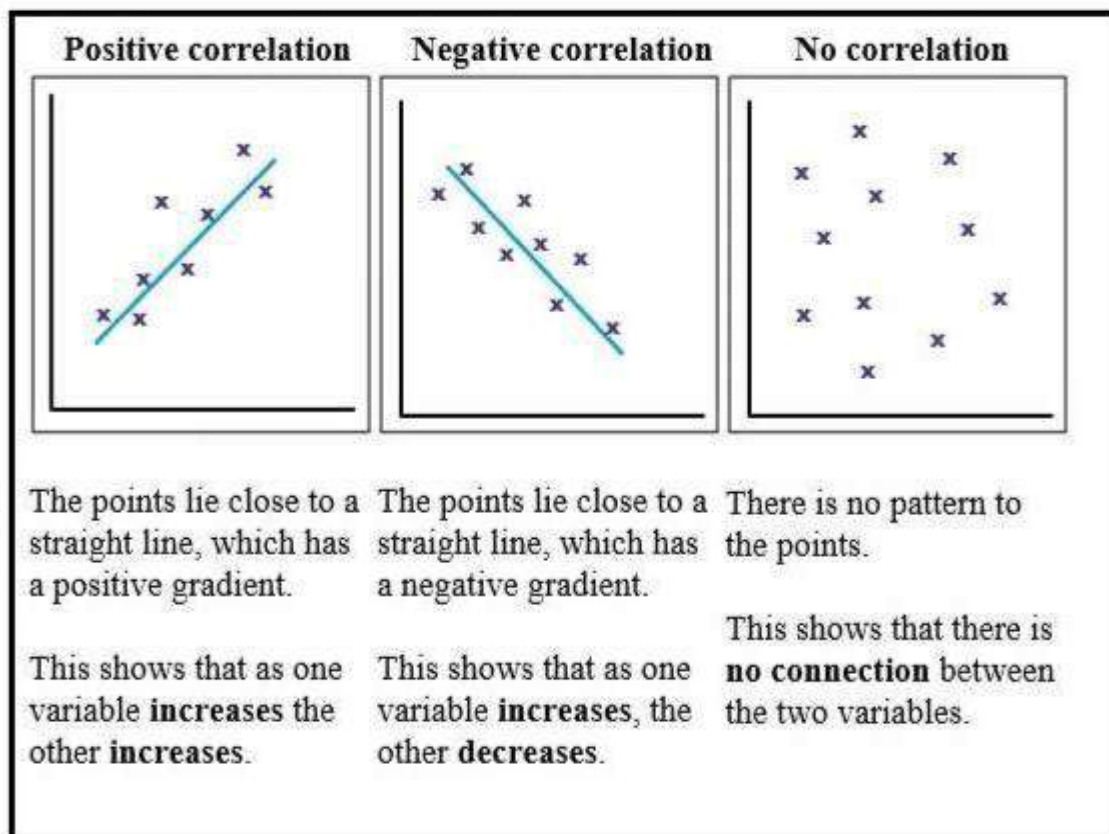
Correlation means association - more precisely it is a measure of the extent to which two variables are related. There are three possible results of a correlational study: a positive correlation, a negative correlation, and no correlation.

A positive correlation is a relationship between two variables in which both variables either increase or decrease at the same time. An example would be height and weight. Taller people tend to be heavier.

A negative correlation is a relationship between two variables in which an increase in one variable is associated with a decrease in the other. An example would be height above sea level and temperature. As you climb the mountain (increase in height) it gets colder (decrease in temperature).

A zero correlation exists when there is no relationship between two variables. For example, there is no relationship between the amount of tea drunk and level of intelligence.

A correlation can be expressed visually. This is done by drawing a scattergram - that is one can plot the figures for one variable against the figures for the other on a graph.



Some uses of Correlations

Prediction

- If there is a relationship between two variables, we can make predictions about one from another.

Validity

- Concurrent validity (correlation between a new measure and an established measure).

Reliability

- Test-retest reliability (are measures consistent).
- Inter-rater reliability (are observers consistent).

Theory verification

- Predictive validity.

that maybe unethical or impractical to test experimentally. For example, it would be unethical to conduct an experiment on whether smoking causes lung cancer.

Correlation allows the researcher to clearly and easily see if there is a relationship between variables. This can then be displayed in a graphical form.

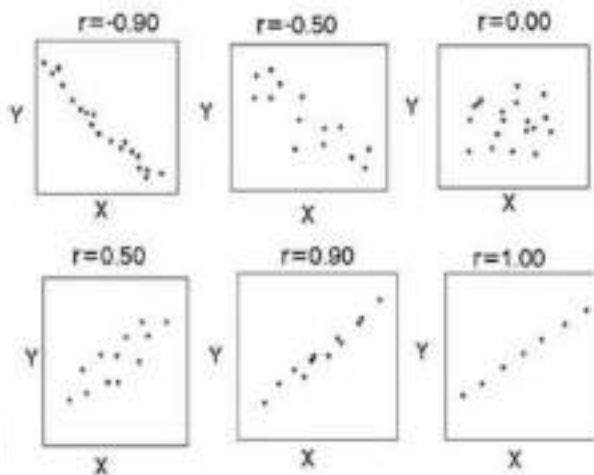
LIMITATIONS OF CORRELATIONS

Correlation is not and cannot be taken to imply causation. Even if there is a very strong association between two variables, we cannot assume that one causes the other. For example, suppose we found a positive correlation between watching violence on T.V. and violent behavior in adolescence. It could be that the cause of both these is a third (extraneous) variable - say for example, growing up in a violent home - and that both the watching of T.V. and the violent behavior are the outcome of this.

Correlation does not allow us to go beyond the data that is given. For example, suppose it was found that there was an association between time spent on homework (1/2 hour to 3 hours) and number of G.C.S.E. passes (1 to 6). It would not be legitimate to infer from this that spending 6 hours on homework would be likely to generate 12 G.C.S.E. passes.

CORRELATION COEFFICIENTS

Instead of drawing a scattergram a correlation can be expressed numerically as a coefficient, ranging from -1 to +1. When working with continuous variables, the correlation coefficient to use is Pearson's r .



The correlation coefficient (r) indicates the extent to which the pairs of numbers for these two variables lie on a straight line. Values over zero indicate a positive correlation, while values under zero indicate a negative correlation.

STRENGTHS OF CORRELATIONS

Correlation allows the researcher to investigate naturally occurring variables

SPECIAL CORRELATION METHODS: BISERIAL, POINT BISERIAL, TETRACHORIC, PHI COEFFICIENT.

<https://statistics.laerd.com/features-tests.php>

WHY SO MANY CORRELATION COEFFICIENTS?

The Pearson product moment correlation coefficient required quantitative (interval

or ratio) data for both x and y whereas the Spearman rho correlation coefficient applied to ranked (ordinal) data for both x and y. It is often the case that the data variables are not at the same level of measurement, or that the data might instead of being quantitative be categorical (nominal or ordinal). In addition to correlation coefficients based on the product moment and thus related to the Pearson product

moment correlation coefficient, there are coefficients which are instead measures of association which are also in common use.

For the purposes of correlation coefficients, we can generally lump the interval and ratio scales together as just quantitative. In addition, the regression of x on y is closely related to the regression of y on x, and the same coefficient applies.

Variable Y/X	Quantitative X (Interval/Ratio)	Ordinal X	Nominal X
Quantitative Y (Interval/Ratio)	Pearson's r	-	-
Ordinal Y	Biserial (Artificial Dichotomous- Continuous)	Tetrachoric r (Both variables are dichotomous + Continuous + Normally distributed)	-
Nominal Y	Point-biserial (Dichotomous- Dichotomous)	Rank Biserial	Phi Coefficient

POINT-BISERIAL CORRELATION:

Like all correlation analyses the **Point-Biserial Correlation** measures the strength of association or co-occurrence between two variables.

A **point-biserial correlation** is used to measure the strength and direction of the association that exists between one continuous variable and one dichotomous variable. It is a special case of the Pearson's product-moment correlation, which is applied when you have two continuous variables, whereas in this case one of the variables is measured on a dichotomous scale.

The point-biserial correlation coefficient, referred to as r_{pb} , is a special case of Pearson in which one variable is quantitative and the other variable is dichotomous and nominal. The calculations simplify since typically the

values 1 (presence) and 0 (absence) are used for the dichotomous variable.

For example, you could use a point-biserial correlation to determine whether there is an association between salaries, measured in US dollars, and gender (i.e., your continuous variable would be "salary" and your dichotomous variable would be "gender", which has two categories: "males" and "females"). Alternately, you could use a point-biserial correlation to determine whether there is an association between cholesterol concentration, measured in mmol/L, and smoking status (i.e., your continuous variable would be "cholesterol concentration", a marker of heart disease, and your dichotomous variable would be "smoking status", which has two categories: "smoker" and "non-smoker"). When you choose to analyse your data using a point-biserial correlation, part of the process involves checking to make sure that the data you want to analyse can actually be analysed using a point-biserial

correlation. You need to do this because it is only appropriate to use a point-biserial correlation if your data "passes" five assumptions that are required for a point-biserial correlation to give you a valid result.

Assumption #1: One of your two variables should be measured on a continuous scale. Examples of continuous variables include revision time (measured in hours), intelligence (measured using IQ score), exam performance (measured from 0 to 100), weight (measured in kg), and so forth.

Assumption #2: Your other variable should be dichotomous. Examples of dichotomous variables include gender (two groups: male or female), employment status (two groups:

employed or unemployed), smoker (two groups: yes or no), and so forth.

Assumption #3: There should be no outliers for the continuous variable for each category of the dichotomous variable. You can test for outliers using boxplots.

Assumption #4: Your continuous variable should be approximately normally distributed for each category of the dichotomous variable. You can test this using the Shapiro-Wilk test of normality.

Assumption #5: Your continuous variable should have equal variances for each category of the dichotomous variable. You can test this using Levene's test of equality of variances.

Correlations

		gender	engagement
gender	Pearson Correlation	1	-.358 [*]
	Sig. (2-tailed)		.023
	N	40	40
engagement	Pearson Correlation	-.358 [*]	1
	Sig. (2-tailed)	.023	
	N	40	40

*. Correlation is significant at the 0.05 level (2-tailed).

Note: The Correlations table actually states that the "Pearson C" is simply a special case of Pearson's product-moment correlation, in this case one of the variables is measured on a dichotomous scale. As long as you as discussed earlier, a point-biserial correlation will be run automatically.

EXTRA FOR REFERENCE:

The point-biserial correlation coefficient, which is simply the Pearson's correlation coefficient when one of the samples is dichotomous.

The Point-Biserial Correlation Coefficient is a correlation measure of the strength of association between a continuous-level

variable (ratio or interval data) and a binary variable. Binary variables are variables of nominal scale with only two values. They are also called dichotomous variables or dummy variables in Regression Analysis. Binary variables are commonly used to express the existence of a certain characteristic (e.g., reacted or did not react in a chemistry sample) or the membership in a group of observed specimen (e.g., male or female). If needed for the analysis, binary variables can also be created artificially by grouping cases or recoding variables. However, it is not advised to artificially create a binary variable from ordinal or continuous-level (ratio or scale) data because ordinal and

continuous-level data contain more variance information than nominal data and thus make any correlation analysis more reliable.

BISERIAL CORRELATION:

Biserial correlation (rbis) is a correlational index that estimates the strength of a relationship between an artificially dichotomous variable (X) and a true continuous variable (Y). Both variables are assumed to be normally distributed in their underlying populations.

The **biserial correlation coefficient** is also a correlation coefficient where one of the samples is measured as dichotomous, but where that sample is really normally distributed. In such cases, the point-biserial correlation generally under-reports the true value of the association. The biserial correlation coefficient provides a better estimate in this case.

The phi-coefficient and the tetrachoric correlation coefficient are two measures of association for dichotomous variables.

PHI COEFFICIENT

There is another special case of correlation called "phi" (or ϕ , the Greek letter f). Phi represents the correlation between two dichotomous variables. As with the point-biserial, computing the Pearson correlation for two dichotomous variables is the same as the phi.

Similar to the t-test/correlation equivalence, the relationship between two dichotomous variables is the same as the difference between two groups when the dependent variable is dichotomous. The appropriate test to compare group differences with a dichotomous outcome is the chi-square statistic. And, we can also

show that the test of the phi coefficient is equivalent to the chi-square test.

Remember, one of the ways chi-square is interpreted is as a test of independence. The test of independence refers to whether or not two variables are related. If two variables are related, they are correlated.

So, when we conduct a chi-square test, and we want to have a rough estimate of how strongly related the two variables are, we can examine phi. Squaring phi will give you the approximate amount of shared variance between the two variables, as does r-square.

If both variables instead are nominal and dichotomous, the Pearson simplifies even further. First, perhaps, we need to introduce contingency tables. A contingency table is a two dimensional table containing frequencies by category. For this situation it will be two by two since each variable can only take on two values, but each dimension will exceed two when the associated variable is not dichotomous. In addition, column and row headings and totals are frequently appended so that the contingency table ends up being $n + 2$ by $m + 2$, where n and m are the number of values each variable can take on. The label and total row and column typically are outside the gridded portion of the table, however.

EXTRA FOR REFERENCES

Cramer's V

Cramer's V is used to examine the association between two categorical variables when there is more than a 2 X 2 contingency (e.g., 2 X 3). In these more complicated designs, phi is not appropriate, but Cramer's statistic is. Cramer's V represents the association or correlation between two variables. I've also seen this statistic referred to as "Cramers Phi".

TETRACHORIC CORRELATION COEFFICIENT

The tetrachoric correlation coefficient, r_{tet} , is used when both variables are dichotomous, like the phi, but we need also to be able to assume both variables really are continuous and normally distributed. Thus it is applied to ordinal vs. ordinal data which has this characteristic.

Ranks are discrete so in this manner it differs from the Spearman. The tetrachoric correlation is used in a variety of contexts, one important one being in Item Response Theory (IRT) analyses of test scores, a second in the conversion of co-morbidity statistics to correlation coefficients.

The formula involves the cosine trigonometric function and can be applied to a 2x2 matrix or contingency table:

$$r_{tet} = \cos(180/(1 + \sqrt{(BC/AD)}))$$

Assumptions for the Test

The two main assumptions are:

The underlying variables come from a normal distribution. With only two variables, this is impossible to test. You should, therefore, have a good theoretical reason for using this particular type of correlation; in other words, you might know that the type of data you are dealing with tends to follow a normal distribution most of the time. Rating errors should also follow a normal distribution.

There is a latent continuous scale underneath your binary data. In other words, the trait you are measuring should be continuous and not discrete.

The tetrachoric correlation coefficient r_{tet} (sometimes written as r^* or r_t) tells you how strong (or weak) the association is between ratings for two raters. A “0” indicates no agreement and a “1” represents a perfect agreement. Most correlations will fall somewhere in between; what constitutes an acceptable

level of agreement largely depends on what type of data you’re dealing with. For example, medical ratings between medical professionals will require a higher level of agreement than most non-medical situations. In general, an agreement over about .7 is usually considered “strong enough.”

REGRESSION: SIMPLE LINEAR REGRESSION, MULTIPLE REGRESSION.

Simple & Multiple Regression Videos are accessible Online on PowerWithin YouTube Channel.

REGRESSION

Statisticians are often called upon to develop methods to predict one variable from other variables. For example, one might want to predict college grade point average from high school grade point average. Or, one might want to predict income from the number of years of education.

In simple linear regression, **we predict scores on one variable from the scores on a second variable.** The variable we are

predicting is called the **criterion variable and is referred to as Y**. The variable we are basing our predictions on is called the **predictor variable and is referred to as X**. When there is only one predictor variable, the prediction method is called simple regression. In simple linear regression, the topic of this section, the predictions of Y when plotted as a function of X form a straight line.

Linear regression consists of finding the best-fitting straight line through the points. The best-fitting line is called a regression line. The black diagonal line in Figure 2 is

the regression line and consists of the predicted score on Y for each possible value of X. The vertical lines from the points to the regression line represent the errors of prediction. As you can see, the red point is very near the regression line; its error of prediction is small. By contrast, the yellow point is much higher than the regression line and therefore its error of prediction is large.

The error of prediction for a point is the value of the point minus the predicted value (the value on the line).

Table 2. Example data.

X	Y	Y'	Y-Y'	(Y-Y') ²
1.00	1.00	1.210	-0.210	0.044
2.00	2.00	1.635	0.365	0.133
3.00	1.30	2.060	-0.760	0.578
4.00	3.75	2.485	1.265	1.600
5.00	2.25	2.910	-0.660	0.436

By far, the most commonly-used criterion for the best-fitting line is the line that minimizes the sum of the squared errors of prediction. That is the criterion that was used to find the line in Figure 2. The last

column in Table 2 shows the squared errors of prediction. The sum of the squared errors of prediction shown in Table 2 is lower than it would be for any other regression line.

The example data in Table 1 are plotted in Figure 1. You can see that there is a positive relationship between X and Y. If you were going to predict Y from X, the higher the value of X, the higher your prediction of Y.

Table 1. Example data.

X	Y
1.00	1.00
2.00	2.00
3.00	1.30
4.00	3.75
5.00	2.25

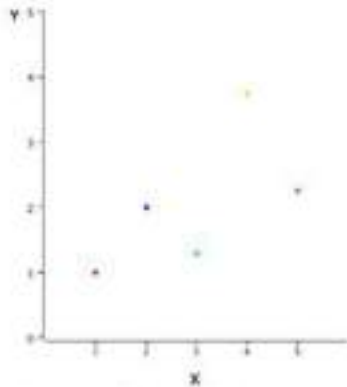


Figure 1. A scatter plot of the example data.

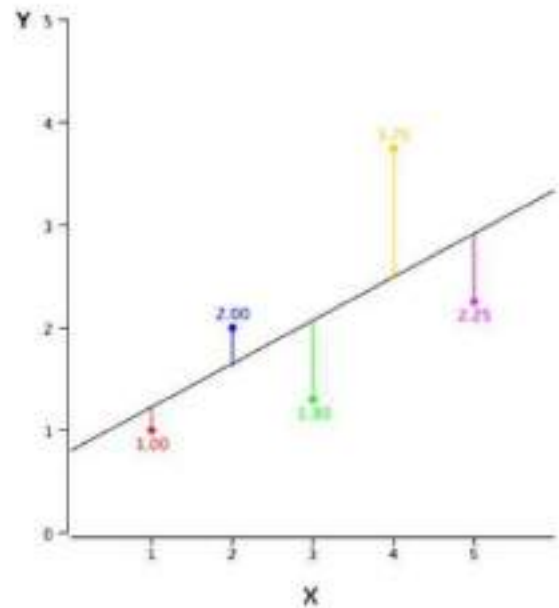


Figure 2. A scatter plot of the example data. The black line consists of the predictions, the points are the actual data, and the vertical lines between the points and the black line represent errors of prediction.

The formula for a regressior

$$Y' = bX + A$$

Question 1 out of 7.

The formula for a regression equation is $Y' = 3X - 2$. What would be the predicted Y score for a person scoring 4 on X?

Question 2 out of 7.

Suppose it is possible to predict a person's score on Test B from the person's score on Test A. The regression equation is: $B' = 2.3A + 9.5$. What is a person's predicted score on Test B assuming this person got a 40 on Test A?

Question 3 out of 7.

Suppose a person got a score of 32.5 on Test A and a score of 95.25 on Test B. Using the same regression equation as in the previous problem ($B' = 2.3A + 9.5$), what is the error of prediction for this person?

Question 4 out of 7.

What is the most common criterion used to determine the best-fitting line?

- ☐ The line that goes through the most points
- ☐ The line that has the same number of points above it as below it
- ☐ The line that minimizes the sum of squared errors of prediction

1 Plug $X = 4$ into the equation to find $Y' = 3(4) - 2 = 10$

2 Plug $A = 40$ into the equation to find $B' = 2.3(40) + 9.5 = 101.5$

3 Predicted value, $B' = 2.3(32.5) + 9.5 = 84.25$; Error of prediction $= B - B' = 95.25 - 84.25 = 11$

4 The most common criterion used to determine the best-fitting line is the line that minimizes the sum of squared errors of prediction. This line does not need to go through any of the actual data points, and it can have a different number of points above it and below it.

ASSUMPTIONS

Although no assumptions were needed to determine the best-fitting straight line, assumptions are made in the calculation of inferential statistics. Naturally, these assumptions refer to the population, not the sample.

Linearity: The relationship between the two variables is linear.

Homoscedasticity: The variance around the regression line is the same for all values of X . A clear violation of this assumption is shown in Figure 1. Notice that the predictions for students with high high-school GPAs are very good, whereas the predictions for students with low high-school GPAs are not very good. In other words, the points for students with

high high-school GPAs are close to the regression line, whereas the points for low high-school GPA students are not.

The errors of prediction are distributed normally. This means that the deviations from the regression line are normally distributed. It does not mean that X or Y is normally distributed.

KEY TERMINOLOGIES:

INFLUENCE

It is possible for a single observation to have a great influence on the results of a regression analysis. It is therefore important to be alert to the possibility of influential observations and to take them into consideration when interpreting the results.

The **influence** of an observation can be thought of in terms of how much the predicted scores for other observations would differ if the observation in question were not included.

A common rule of thumb is that an observation with a value of Cook's D over 1.0 has too much influence. As with all rules of thumb, this rule should be applied judiciously and not thoughtlessly. An observation's influence is a function of two factors: (1) how much the observation's value on the predictor variable differs from the mean of the predictor variable and (2) the difference between the predicted score for the observation and its actual score. The former factor is called the

observation's leverage. The latter factor is called the observation's distance.

LEVERAGE

The leverage of an observation is based on how much the observation's value on the predictor variable differs from the mean of the predictor variable. The greater an observation's leverage, the more potential it has to be an influential observation.

DISTANCE

The distance of an observation is based on the error of prediction for the observation: The greater the error of prediction, the greater the distance. The most commonly used measure of distance is the studentized residual. **The studentized residual** for an observation is closely related to the error of prediction for that observation divided by the standard deviation of the errors of prediction.

EXAMPLE

Table 1 shows the leverage, studentized residual, and influence for each of the five observations in a small dataset.

Table 1. Example Data.

ID	X	Y	h	R	D
A	1	2	0.39	-1.02	0.40
B	2	3	0.27	-0.56	0.06
C	3	5	0.21	0.89	0.11
D	4	6	0.20	1.22	0.19
E	8	7	0.73	-1.68	8.86

h is the leverage, R is the studentized residual, and D is Cook's measure of influence.

Observation A has fairly high leverage, a relatively high residual, and moderately high influence.

Observation B has small leverage and a relatively small residual. It has very little influence.

Observation C has small leverage and a relatively high residual. The influence is relatively low.

Observation D has the lowest leverage and the second highest residual. Although its residual is much higher than Observation A, its influence is much less because of its low leverage.

Observation E has by far the largest leverage and the largest residual. This combination of high leverage and high residual makes this observation extremely influential.

Question 1 out of 2.

Which of the following statements are true about studentized residuals?

- ☐ They are the deviations of the observed values from the fitted line.
- ☐ They're closely related to the errors of prediction of the observation.
- ☐ Large studentized residuals will lead to large influence on the regression model in all circumstances.

1 The Studentized residual measures the distance which is based on the error of the prediction for the observation. It is the combination of an observation's leverage and distance that determines its influence.

(HSGPA) and their total SAT score (verbal + math). The values of b (b_1 and b_2) are sometimes called "regression coefficients" and sometimes called "regression weights." These two terms are synonymous. The multiple correlation (R) is equal to the

Question 1 out of 4.

In a regression line, the _____ the standard error of the estimate is, the more accurate the predictions are.

- ☐ larger
- ☐ smaller
- ☐ The standard error of the estimate is not related to the accuracy of the predictions.

1 The standard error of the estimate is a measure of the accuracy of predictions. The regression line is the line that minimizes the sum of squared deviations of prediction (also called the sum of squares error), and the standard error of the estimate is the square root of the average squared deviation.

correlation between the predicted scores and the actual scores.

INTERPRETATION OF REGRESSION COEFFICIENTS

A regression coefficient in multiple regression is the slope of the linear relationship between the criterion variable and the part of a predictor variable that is independent of all other predictor variables.

MULTIPLE LINEAR REGRESSION

In simple linear regression, a criterion variable is predicted from one predictor variable. In multiple regression, the criterion is predicted by two or more variables. For example, in the SAT case study, you might want to predict a student's university grade point average on the basis of their High-School GPA

The sum of squares uniquely attributable to a variable is computed by comparing two regression models: **the complete model and a reduced model**. The complete model is the multiple regression with all the predictor variables included (HSGPA and SAT in this example). A reduced model is a model that leaves out one of the predictor variables.

FACTOR ASSUMPTIONS, ROTATION AND INTERPRETATION.

**Video available online on
PowerWithin YouTube Channel.**

Detailed and curated PPT on Factor Analysis by Raunaq Chawla (Team of Psychology Aaramse):

https://drive.google.com/open?id=1rXCnA7D1ogKi6z_YXchBalxlCEYMG7u

If you want to understand FA using a practical example, use the link:
<https://stats.idre.ucla.edu/spss/seminars/introduction-to-factor-analysis/a-practical-introduction-to-factor-analysis/>

Factor analysis is a way to condense the data in many variables into a just a few variables. For this reason, it is also sometimes called “dimension reduction” or “data reduction.” You can reduce the “dimensions” of your data into one or more “super-variables.” The most common technique is known as Principal Component Analysis (PCA).

It is, therefore, a procedure used to determine the extent to which **shared variance (the intercorrelation between measures)** exists between variables or items within the item pool for a developing measure. It is a means of determining to what degree individual items are measuring a something in common, such as a factor.

It is a systematic method for examining the meaning of the test studying its correlation with other variables. In the process of factor analysis the number of variables or categories is reduced from the number of original tests to a relatively small number of factors or common traits. Major purpose is to simplify the

ANALYSIS: METHODS,

description of behaviour by reducing the number of categories. After identifying the factors, each test can be characterised in terms of major factors determining its scores, weight or load of each factor and the correlation of each factor with the test. So, correlation of each factor with the test is known as factorial validity of the test. One of the objectives of factor analysis is to analyse the common variance to determine the number and types of common variances which are responsible for the correlations between variables. The values of the square roots of the common variance are known as factor loading. Each factor loading represents the amount of correlation of the sub-traits with that factor. Several methods are available, but principle component analysis is used most commonly.

TYPES OF FACTORING:

There are different types of methods used to extract the factor from the data set:

Principal component analysis: This is the most common method used by researchers. PCA starts extracting the maximum variance and puts them into the first factor. After that, it removes that variance explained by the first factors and then starts extracting maximum variance for the second factor. This process goes to the last factor.

Common factor analysis: The second most preferred method by researchers, it extracts the common variance and puts them into factors. This method does not include the unique variance of all variables. This method is used in SEM.

Image factoring: This method is based on correlation matrix. OLS Regression method is used to predict the factor in image factoring.

Maximum likelihood method: This method also works on correlation metric but it uses maximum likelihood method to factor.

FACTOR LOADING:

Factor loading is basically the correlation coefficient for the variable and factor. Factor loading shows the variance explained by the variable on that particular factor. In the SEM approach, as a rule of thumb, 0.7 or higher factor loading represents that the factor extracts sufficient variance from that variable.

EIGENVALUES:

Eigenvalues is also called characteristic roots. Eigenvalues shows variance explained by that particular factor out of the total variance. From the commonality column, we can know how much variance is explained by the first factor out of the total variance. For example, if our first factor explains 68% variance out of the total, this means that 32% variance will be explained by the other factor.

FACTOR SCORE:

The factor score is also called the component score. This score is of all row and columns, which can be used as an index of all variables and can be used for further analysis. We can standardize this score by multiplying a common term. With this factor score, whatever analysis we will do, we will assume that all variables will behave as factor scores and will move.

CRITERIA FOR DETERMINING THE NUMBER OF FACTORS:

According to the Kaiser Criterion, Eigenvalues is a good criteria for determining a factor. If Eigenvalues is greater than one, we should consider that a factor and if Eigenvalues is less than one, then we should not consider that a factor. According to the variance extraction rule, it should be more than 0.7. If variance is less than 0.7, then we should not consider that a factor.

ROTATION METHOD:

(Refer PPT)

Rotation method makes it more reliable to understand the output. Eigenvalues do not affect the rotation method, but the rotation method affects the Eigenvalues or percentage of variance extracted. There are a number of rotation methods available: (1) No rotation method, (2) Varimax rotation method, (3) Quartimax rotation method, (4) Direct oblimin rotation method, and (5) Promax rotation method. Each of these can be easily selected in SPSS, and we can compare our variance explained by those particular methods.

Assumptions:

No outlier: Assume that there are no outliers in data.

Adequate sample size: The case must be greater than the factor.

- No perfect multicollinearity: Factor analysis is an interdependency technique. There should not be perfect multicollinearity between the variables.

Homoscedasticity: Since factor analysis is a linear function of measured variables, it does not require homoscedasticity between the variables.

- Linearity: Factor analysis is also based on linearity assumption. Non-linear variables can also be used. After transfer, however, it changes into linear variable.

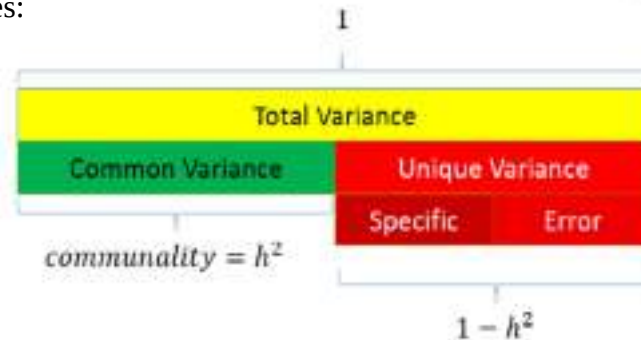
Interval Data: Interval data are assumed.

Key concepts and terms:

Exploratory factor analysis: Assumes that any indicator or variable may be associated with any factor. This is the most common factor analysis used by

researchers and it is not based on any prior theory.

Confirmatory factor analysis (CFA): Used to determine the factor and factor loading of measured variables, and to confirm what is expected on the basis of pre-established theory. CFA assumes that each factor is associated with a specified subset of measured variables. It commonly uses two approaches:



Common variance is the amount of variance that is shared among a set of items. Items that are highly correlated will share a lot of variance.

- Communality (also called h^2) is a definition of common variance that ranges between 0 and 1. Values closer to 1 suggest that extracted factors explain more of the variance of an individual item.

Unique variance is any portion of variance that's not common. There are two types:

- Specific variance: is variance that is specific to a particular item (e.g., Item 4 "All computers hate me" may have variance that is attributable to anxiety about computers in addition to anxiety about SPSS).
- Error variance: comes from errors of measurement and basically anything unexplained by common or specific

PARTITIONING THE VARIANCE IN FACTOR ANALYSIS

Since the goal of factor analysis is to model the interrelationships among items, we focus primarily on the variance and covariance rather than the mean. Factor analysis assumes that variance can be partitioned into two types of variance, common and unique:

variance (e.g., the person got a call from her babysitter that her two-year old son ate her favorite lipstick).

EXPERIMENTAL DESIGNS:
ANOVA [ONE-WAY, FACTORIAL], RANDOMIZED BLOCK DESIGNS, REPEATED MEASURES DESIGN, LATIN SQUARE, COHORT STUDIES, TIME SERIES, MANOVA, ANCOVA. SINGLE-SUBJECT DESIGNS.

EXPERIMENTAL DESIGNS

ANOVA [One-way, Factorial]

Two-group Experimental Designs /
One-way ANOVA Design

The simplest true experimental designs are **two group designs involving one treatment group and one control group**, and are ideally suited for **testing the**

effects of a single independent variable that can be manipulated as a treatment.

The two basic two-group designs are the pretest-posttest control group design and the posttest-only control group design, while variations may include covariance designs. These designs are often depicted using a standardized design notation, where R represents random assignment of subjects to groups, X represents the treatment administered to the treatment group, and O represents pretest or posttest observations of the dependent variable (with different subscripts to distinguish

between pretest and posttest observations of treatment and control groups).

PRETEST-POSTTEST CONTROL GROUP DESIGN: In this design, subjects are randomly assigned to treatment and control groups, subjected to an initial (pretest) measurement of the dependent variables of interest, the treatment group is administered a treatment (representing the independent variable of interest), and the dependent variables measured again (posttest).

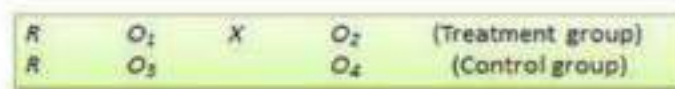


Figure 10.1. Pretest-posttest control group design

The effect E of the experimental treatment in the pretest posttest design is measured as the difference in the posttest and pretest scores between the treatment and control groups:

$$E = (O_2 - O_1) - (O_4 - O_3)$$

Statistical analysis of this design involves a simple analysis of variance (ANOVA) between the treatment and control groups. The pretest posttest design handles several threats to internal validity, such as maturation, testing, and regression, since these threats can be

expected to influence both treatment and control groups in a similar (random) manner. The selection threat is controlled via random assignment. However, additional threats to internal validity may exist. For instance, mortality can be a problem if there are differential dropout rates between the two groups, and the pretest measurement may bias the posttest measurement (especially if the pretest introduces unusual topics or content).

Posttest-only control group design: This design is a simpler version of the pretest-posttest design where pretest measurements are omitted.

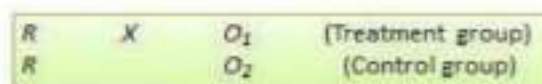


Figure 10.2. Posttest only control group design

The treatment effect is measured simply as the difference in the posttest scores between the two groups:
 $E = (O_1 - O_2)$

The appropriate statistical analysis of this design is also a two-group analysis of variance (ANOVA). The simplicity of this design makes it more attractive than the pretest-posttest design in terms of internal validity. This design controls for maturation, testing, regression, selection, and pretest-posttest interaction, though the mortality threat may continue to exist.

EXTRA FOR REFERENCES:

ANOVA TEST details (if anyone wants to know about it)

<https://www.statisticshowto.datasciencecentral.com/probability-and-statistics/hypothesis-testing/anova/#OneWayANOVA>

Because the pretest measure is not a measurement of the dependent variable, but rather a covariate, the treatment effect is measured as the difference in the posttest scores between the treatment and control groups as: $E = (O_1 - O_2)$

Due to the presence of covariates, the right statistical analysis of this design is a two-group analysis of covariance (ANCOVA). This design has all the advantages of post-

Covariance designs: Sometimes, measures of dependent variables may be influenced by extraneous variables called covariates. Covariates are those variables that are not of central interest to an experimental study, but should nevertheless be controlled in an experimental design in order to eliminate their potential effect on the dependent variable and therefore allow for a more accurate detection of the effects of the independent variables of interest. The experimental designs discussed earlier did not control for such covariates. A covariance design (also called a concomitant variable design) is a special type of pretest posttest control group design where the pretest measure is essentially a measurement of the covariates of interest rather than that of the dependent variables. The design notation is shown in Figure 10.3, where C represents the covariates:

R	C	X	O ₁	(Treatment group)
R	C		O ₂	(Control group)

Figure 10.3. Covariance design

test only design, but with internal validity due to the controlling of covariates. Covariance designs can also be extended to pretest-posttest control group design.

FACTORIAL DESIGNS

Two-group designs are inadequate if your research requires manipulation of two or more independent variables (treatments). In such cases, you would need four or higher-group designs. Such designs, quite popular in experimental research, are

commonly called factorial designs. Each independent variable in this design is called a factor, and each sub-division of a factor is called a level. Factorial designs enable the researcher to examine not only the individual effect of each treatment on the dependent variables (called main effects), but also their joint effect (called interaction effects).

The most basic factorial design is a 2 x 2 factorial design, which consists of two treatments, each with two levels (such as high/low or present/absent). For instance, let's say that you want to compare the learning outcomes of two different types of instructional techniques (in-class and online instruction), and you also want to examine whether these effects

vary with the time of instruction (1.5 or 3 hours per week). In this case, you have two factors: instructional type and instructional time; each with two levels (in-class and online for instructional type, and 1.5 and 3 hours/week for instructional time), as shown in Figure 8.1. If you wish to add a third level of instructional time (say 6 hours/week), then the second factor will consist of three levels and you will have a 2 x 3 factorial design. On the other hand, if you wish to add a third factor such as group work (present versus absent), you will have a 2 x 2 x 2 factorial design. In this notation, each number represents a factor, and the value of each factor represents the number of levels in that factor.

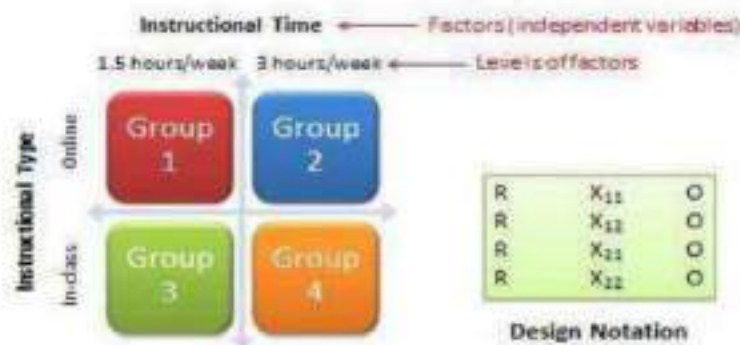


Figure 10.4. 2 x 2 factorial design

Factorial designs can also be depicted using a design notation, such as that shown on the right panel of Figure 10.4. R represents random assignment of subjects to treatment groups, X represents the treatment groups themselves (the subscripts of X represents the level of each factor), and O represent observations of the dependent variable. Notice that the 2 x 2 factorial design will have four treatment groups, corresponding to the four combinations of the two levels of each factor. Correspondingly, the 2 x 3 design will have six treatment groups, and the 2 x 2 x 2 design will have eight treatment groups. As a rule of thumb, each cell in a

factorial design should have a minimum sample size of 20 (this estimate is derived from Cohen's power calculations based on medium effect sizes). So, a 2 x 2 x 2 factorial design requires a minimum total sample size of 160 subjects, with at least 20 subjects in each cell. As you can see, the cost of data collection can increase substantially with more levels or factors in your factorial design. Sometimes, due to resource constraints, some cells in such factorial designs may not receive any treatment at all, which are called incomplete factorial designs. Such incomplete designs hurt our ability to draw inferences about the incomplete factors.

In a factorial design, a main effect is said to exist if the dependent variable shows a significant difference between multiple levels of one factor, at all levels of other factors. No change in the dependent variable across factor levels is the null case (baseline), from which main effects are evaluated. In the above example, you may see a main effect of instructional type, instructional time, or both on learning outcomes. An interaction effect exists when the effect of differences in one factor depends upon the level of a second factor. In our example, if the effect of instructional type on learning outcomes is greater for 3 hours/week of instructional time than for 1.5 hours/week, then we can say that there is an interaction effect between instructional type and instructional time on learning outcomes. Note that the presence of interaction effects dominates and make main effects irrelevant, and it is not meaningful to interpret main effects if interaction effects are significant.

HYBRID EXPERIMENTAL DESIGNS

Hybrid designs are those that are formed by combining features of more established designs. Three such hybrid designs are **randomized blocks design**, Solomon four-group design (not in syllabus), and switched replications design (not in syllabus).

RANDOMIZED BLOCK DESIGNS

This is a variation of the posttest-only or pretest-posttest control group design where the subject population can be grouped into relatively homogeneous subgroups (called blocks) within which the experiment is replicated. For instance, if you want to replicate the same posttest-only design among university students and full-time working professionals (two

homogeneous blocks), subjects in both blocks are randomly split between treatment group (receiving the same treatment) or control group (see Figure 10.5). The purpose of this design is to reduce the “noise” or variance in data that may be attributable to differences between the blocks so that the actual effect of interest can be detected more accurately.

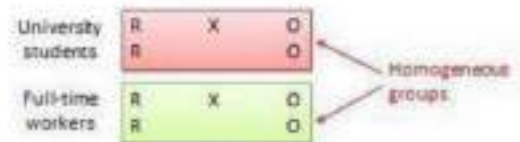


Figure 10.5. Randomized blocks design

REPEATED MEASURES DESIGN

Repeated measures designs, also known as a within-subjects designs, can seem like oddball experiments. They don't fit our impression of a typical experiment in several key ways. When we think of an experiment, we often think of a design that has a clear distinction between the treatment and control groups. Each subject is in one, and only one, of these non-overlapping groups. Subjects who are in a treatment group are exposed to only one type of treatment. This is the common independent groups experimental design.

Repeated Measures Design doesn't follow these rules.

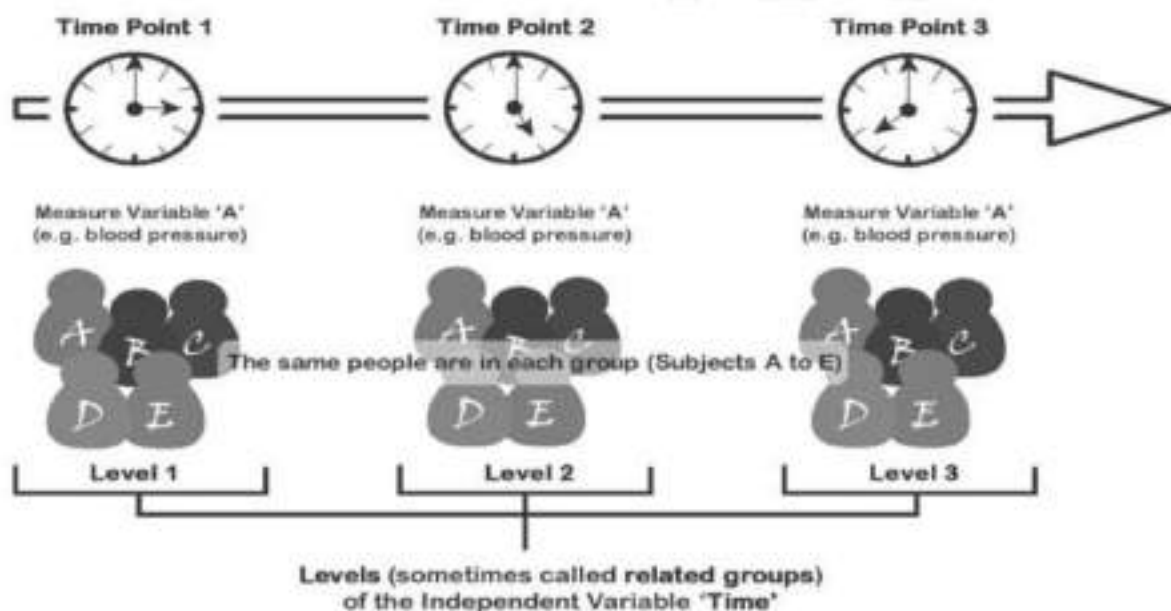
WHEN TO USE A REPEATED MEASURES ANOVA?

ANOVA is an acronym for Analysis of Variance. It's called Repeated Measures because the same group of study participants is being measured over and over again.

For example, you could be studying the glucose levels of the patients at 1 month, 6 months, and 1 year after receiving nutritional counseling.

We can analyse data using a repeated measures ANOVA for two types of study design. Studies that investigate either (1) changes in mean scores over three or more time points, or (2) differences in mean scores under three or more different conditions. For example, for (1), you might be investigating the effect of a 6-

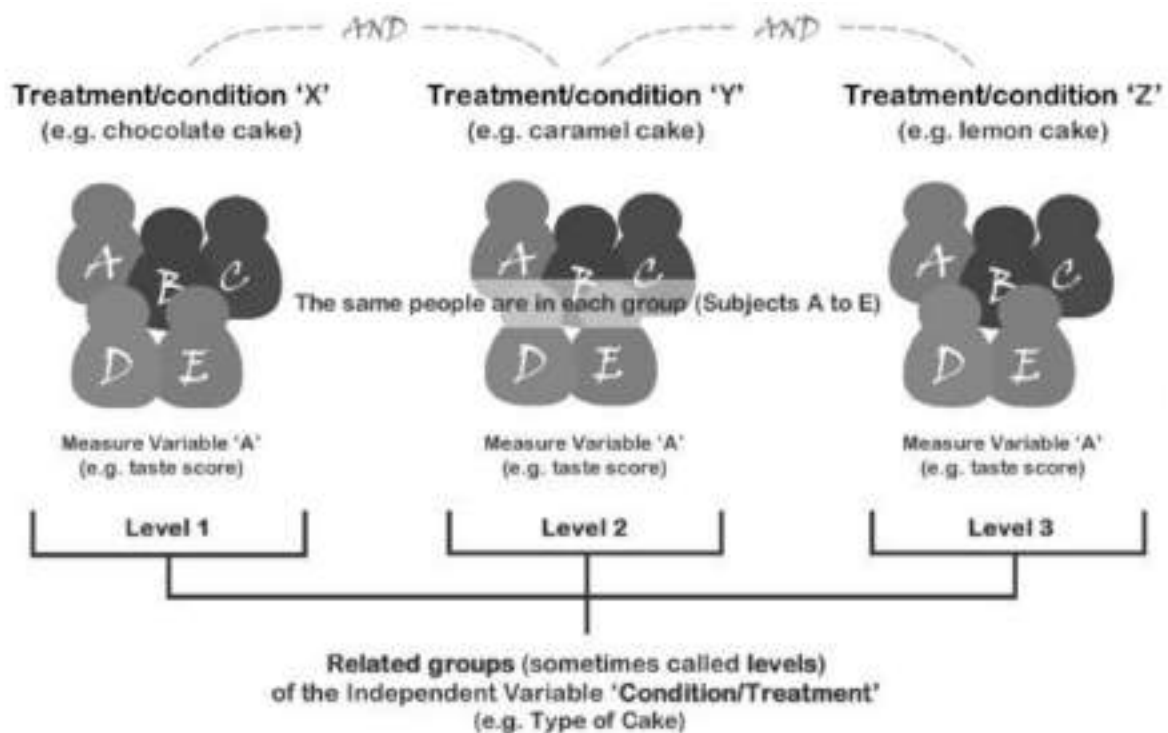
month exercise training programme on blood pressure and want to measure blood pressure at 3 separate time points (pre-, midway and post-exercise intervention), which would allow you to develop a time-course for any exercise effect. For (2), you might get the same subjects to eat different types of cake (chocolate, caramel and lemon) and rate each one for taste, rather than having different people taste each different cake. The important point with these two study designs is that the same people are being measured more than once on the same dependent variable (i.e., why it is called repeated measures).



In repeated measures ANOVA, the independent variable has categories called levels or related groups. Where measurements are repeated over time, such as when measuring changes in blood pressure due to an exercise-training programme, the independent variable is time. Each level (or related group) is a specific time point. Hence, for the exercise-training study, there would be three time points and each time-point is a

level of the independent variable (a schematic of a time-course repeated measures design is shown above).

Where measurements are made under different conditions, the conditions are the levels (or related groups) of the independent variable (e.g., type of cake is the independent variable with chocolate, caramel, and lemon cake as the levels of the independent variable). A schematic of different-conditions repeated measures design is shown below.



HYPOTHESIS FOR REPEATED MEASURE DESIGN

For our exercise-training example, the null hypothesis (H_0) is that mean blood pressure is the same at all time points (pre-3 months, and 6 months). The alternative hypothesis is that mean blood pressure is significantly different at one or more time points. A repeated measures ANOVA will not inform you where the differences between groups lie as it is an omnibus statistical test. The same would be true if you were investigating different conditions or treatments rather than time points, as used in this example. If your repeated measures ANOVA is statistically significant, you can run post hoc tests that can highlight exactly where these differences occur.

ADVANTAGES AND DISADVANTAGES

ADVANTAGES INCLUDE:

Fewer patients are needed overall. In a traditional experiment, you would need x patients per experimental group, plus a control; If you had three experimental conditions (e.g. nutrition, exercise, drugs) plus a control, you would need $4x$ patients. In a repeated measures design, you would only need one group of patients of size x , because all patients are given all conditions.

Fewer patients are needed to detect an effect size.

Higher statistical power: controlling for variability between subjects is built in.

DISADVANTAGES INCLUDE:

Order effects: the possibility that the position of the treatment in the order of treatments matters.
Randomization or

- counterbalancing can correct for this.
- Carryover Effects: administration of one part of the experiment can have trickle down effects to Subsequent parts.
- Subjects can drop out before the second or third part of the experiment, resulting in sample sizes that are too small to yield meaningful results.

Experimental design refers to how participants are allocated to the different conditions (or IV levels) in an experiment. There are three types:

- 1. Independent measures / between groups:** Different participants are used in each condition of the independent variable.
- 2. Repeated measures /within groups:** The same participants take part in each condition of the independent variable.
- 3. Matched pairs:** Each condition uses different participants, but they are matched in terms of important characteristics, e.g., gender, age, intelligence, etc.

LATIN SQUARE

Reference Article:
http://compneurosci.com/wiki/images/9/98/Latin_square_Method.pdf

A Latin square design is a method of placing treatments so that they appear in a balanced fashion within a square block or field. Treatments appear once in each row and column.

In other words, a Latin square design is the arrangement of t treatments, each one repeated t times, in such a way that each treatment appears exactly one time in each

row and each column in the design. We denote by Roman characters the treatments. Therefore, the design is called a Latin square design. This kind of design is used to reduce systematic error due to rows (treatments) and columns.

Replicates are also included in this design.

Treatments are assigned at random within rows and columns, with each treatment once per row and once per column.

There are equal numbers of rows, columns, and treatments.

Useful where the experimenter desires to control variation in two different directions

The Latin square design, perhaps, represents the most popular alternative design when two (or more) blocking factors need to be controlled for. A Latin square design is actually an extreme example of an incomplete block design,

with any combination of levels involving the two blocking factors assigned to one treatment only, rather than to all!

Following is an example of a four treatment Latin Square. The experimental layout is as follows:

Rows	Columns			
	Col1	Col2	Col3	Col4
Row 1	A	B	C	D
Row 2	B	C	D	A
Row 3	C	D	A	B
Row 4	D	A	B	C

In the above table, the four treatments are represented by the four letters: A, B, C, and D. The letters are arranged so that each letter occurs only once within each row and each column. Notice that a simple random design would require $4 \times 4 \times 4 = 64$ experimental units. This Latin Square needs only 16 experimental units—a reduction of 75%!

PROS AND CONS OF LATIN SQUARE DESIGN

The advantages of Latin square designs are:

They handle the case when we have several nuisance factors and we either cannot combine them into a single factor or we wish to keep them separate.

They allow experiments with a relatively small number of runs.

The disadvantages are:

The number of levels of each blocking variable must equal the number of levels of the treatment factor.

The Latin square model assumes that there are no interactions between the blocking variables or

between the treatment variable and the blocking variable.

COHORT STUDIES

Cohort studies are a type of medical research used to investigate the causes of disease and to establish links between risk factors and health outcomes.

The word **cohort** means a **group of people**. These types of studies look at groups of people. They can be forward-looking (prospective) or backward-looking (retrospective).

Prospective studies are planned in advance and carried out over a future period of time. **Retrospective cohort studies** look at data that already exist and try to identify risk factors for particular conditions. Interpretations are limited because the researchers cannot go back and gather missing data.

These long-term studies are sometimes called longitudinal studies.

potential long term consequences of the

Fast facts on cohort studies

- Cohort studies typically observe large groups of individuals, recording their exposure to certain risk factors to find clues as to the possible causes of disease.
- They can be prospective studies and gather data going forward, or retrospective cohort studies, which look at data already collected.
- The Nurses' Health Study is one example of a large cohort study, and it has produced many important links between lifestyle choices and health by following hundreds of thousands of women across North America.
- Such research can also help identify social factors that influence health.

The cohort study design is the best available scientific method for measuring the effects of a suspected risk factor.

In a prospective cohort study, researchers raise a question and form a hypothesis about what might cause a disease. Then they observe a group of people, known as the cohort, over a period of time. This may take several years. They collect data that may be relevant to the disease. In this way, they aim to detect any changes in health linked to the possible risk factors they have identified.

EXAMPLES

Some cohort studies have been very large and continued for a long time, producing a good deal of data that serves researchers in different fields.

NURSES HEALTH STUDY

One famous example of a cohort study is the Nurses' Health Study, a large, long-running analysis of women's health, originally set up in 1976 to investigate the

use of oral contraceptives.

This study recruited its second generation cohort for the Nurses' Health Study II in 1989, and its third-generation cohort of nurses from across the United States and Canada in 2010.

The nurses in the first NHS were married women aged 30 to 55 years. The NHS II and III aimed to look at a more diverse cohort including women aged between 20 and 46 years.

Numerous and important insights into health and wellbeing have already been gained by researchers using data from the Nurses' Health Study, which is run by the Harvard School of Public Health, and the Brigham and Women's Hospital, both based in Boston, MA.

The following headlines are from news stories published recently by MNT. They report on some of the findings from this huge study of hundreds of thousands of women:

Nuts may protect against heart disease

Weight gain in early adulthood linked to health risks later in life
Colon cancer: Could proinflammatory diets raise risk?

Because the Nurses' Health Study asks participants about their lifestyle choices, it has yielded much information about the harms and benefits of various factors, including specific types of food in the diet.

Cohort studies are also good at finding relationships between health and environmental factors such as chemicals in the air, water and food. These are issues that the World Health Organization (WHO) helps researchers to investigate with large-scale cohort studies.

Pooling data from different studies can increase the sample size, and this can make the results more reliable, especially for rare conditions such as some types of cancer.

FRAMINGHAM HEART STUDY

Another example is the Framingham Heart Study, which recruited over 5,209 male and female participants in 1948 from around the area of Framingham, MA. It has continued to serve as a source of data for cardiovascular risk factors.

A second cohort was recruited in 1971 and a third in 2002. The study has made important contributions to the understanding of heart health. The researchers are now looking into how genetic factors may affect cardiovascular risk.

LIMITATIONS

Cohort studies are graded as the most robust form of medical research after

experiments such as randomized controlled trials, but they are not always the best form of observational work.

Cohort studies do have some limitations:

They are less suited to finding clues about rare diseases. A case-control study identifies cases of disease first and then analyzes exposure to risk factors, whereas cohort studies follow exposure data and watch for any emerging cases of disease.

They are typically unsuitable for identifying the causes of a sudden outbreak of disease. A case-control study can give quicker results.

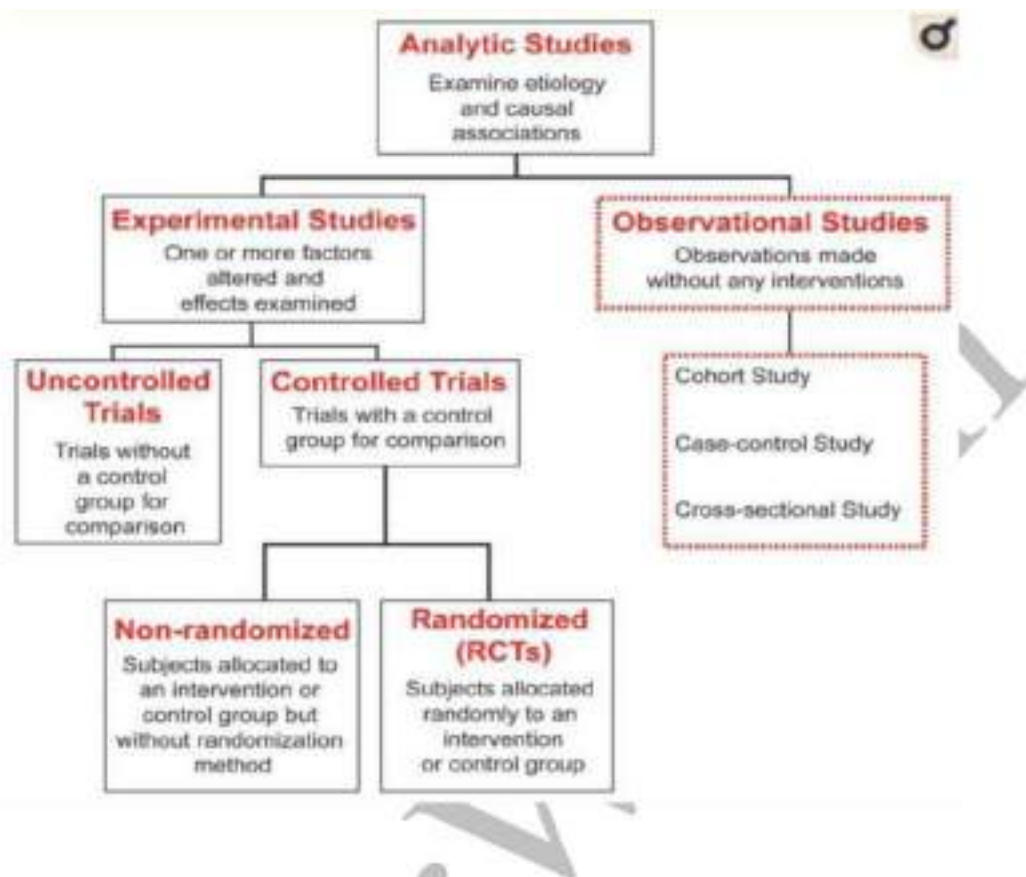
They are expensive to run and usually take many years, often decades, to produce results.

They can only offer clues about the causes of disease, rather than definitive proof of links between risk factors and health.

This is true of any observational medical research.

Participants may leave the cohort, perhaps move away, lose touch, or die from a cause that is not being studied. This can bias the results.

(The Below Image is FOR REFERENCE)



TIME SERIES

Time series analysis is a technique for analyzing time series data, or variables that continually changes with time. Examples of applications include forecasting stock market fluctuations and urban crime rates. This technique is popular in econometrics, mathematical finance, and signal processing. Special techniques are used to correct for auto-correlation, or correlation within values of the same variable across time.

Time-series analysis (TSA), thus, is a statistical methodology appropriate for longitudinal research designs that involve single subjects or research units that are measured repeatedly at regular intervals over time. TSA can be viewed as the exemplar of all longitudinal designs. TSA can provide an understanding of the underlying naturalistic process and the pattern of change over time, or it can evaluate the effects of either a planned or unplanned intervention.

MEASURING BEHAVIOR AS A TIME SERIES

According to Daniel T. Kaplan and Leon Glass (1995), there are two critical features of a time series that differentiate it from cross-sectional data-collection procedures:

Repeated measurements of a given behavior are taken across time at equally spaced intervals. Taking multiple measurements is essential for understanding how any given behavior unfolds over time, and doing so at equal intervals affords a clear investigation of how the dynamics of that behavior manifest at distinct time scales.

The temporal ordering of measurements is preserved. Doing so is the only way to fully examine the dynamics governing a particular process. If we expect that a given

stimulus will influence the development of a behavior in a particular way, utilizing summary statistics will completely ignore the temporal ordering of the data and likely occlude one's view of important behavioral dynamics.

Linear computations such as mean and variance merely describe global properties of a data set and thus may fail to capture meaningful patterns that only can be identified by looking at the sequential dependency between time points. Consequently, time-series techniques provide a valuable approach in studying psychological processes, which are, by their very nature, fundamentally embedded in time.

This design involves successive observations throughout a programmed intervention and assesses the characteristics of the change process. It is truly the mainstay of the proposed design package because it serves several simultaneous functions. First, it is descriptive. The descriptive function of the time series is particularly important when the intervention extends over a

considerable time period. The time series is the only design to furnish a continuous record of fluctuations in the experimental variables over the entire course of the program. Such record keeping should constitute an integral part of the experimental program; problems of reactivity (Webb, Camp-bell, Schwartz, & Sechrest, 1966) are avoided by incorporating the measurement operations as a natural part of the environment to which one wishes to generalize. Second, the time-series design functions as an heuristic device. When coupled with a carefully kept historical log of potentially relevant non-experimental events, the time series is an invaluable source of post hoc hypotheses regarding observed, but unplanned, changes in program variables. Moreover, where treatment programs require practical administrative decisions, the time series serves as a source of hypotheses regarding the most promising decisions, and later as a feedback source regarding the consequences and effectiveness of such decisions.

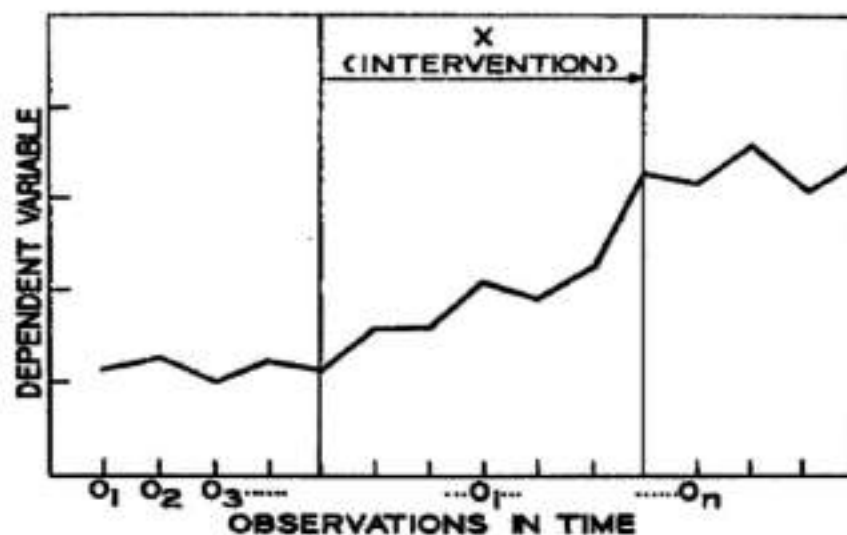


FIG. 1. A time series with an extended intervention.

ANALYZING TIME-SERIES DATA

(If needed details can be read from the following link:

<https://www.psychologicalscience.org/observer/time-series-methods-in-experimental-research>)

Once you've collected a series of behavioral measurements on your variable(s) of interest, there are a variety of ways to explore and quantify the observed dynamics.

MANOVA

Multivariate analysis of variance (MANOVA) is simply an ANOVA with several dependent variables.

MANOVA designs are appropriate when multiple dependent variables are included in the analysis. The dependent variables should represent continuous measures (i.e., interval or ratio data). Dependent variables should be moderately correlated. If there is no correlation at all, MANOVA offers no improvement over an analysis of variance (ANOVA); if the variables are highly correlated, the same variable may be measured more than once. In many MANOVA situations, multiple independent variables, called factors, with multiple levels are included. The independent variables should be categorical (qualitative). Unlike ANOVA procedures that analyze differences across two or more groups on one dependent variable, MANOVA procedures analyze differences across two or more groups on two or more dependent variables.

For example, we may conduct a study where we try two different textbooks, and we are interested in the students' improvements in math and physics. In that case, improvements in math and physics

are the two dependent variables, and our hypothesis is that both together are affected by the difference in textbooks. A multivariate analysis of variance (MANOVA) could be used to test this hypothesis. Instead of a univariate F value, we would obtain a multivariate F value (Wilks' λ) based on a comparison of the error variance/covariance matrix and the effect variance/covariance matrix. Although we only mention Wilks' λ here, there are other statistics that may be used, including **Hotelling's trace and Pillai's criterion**. The "covariance" here is included because the two measures are probably correlated and we must take this correlation into account when performing the significance test. Testing the multiple dependent variables is accomplished by creating new dependent variables that maximize group differences. These artificial dependent variables are linear combinations of the measured dependent variables.

RESEARCH QUESTIONS

The main objective in using MANOVA is to determine if the response variables (student improvement in the example mentioned above), are altered by the observer's manipulation of the independent variables.

Therefore, there are several types of research questions that may be answered by using MANOVA:

What are the main effects of the independent variables?

What are the interactions among the independent variables?

What is the importance of the dependent variables?

What is the strength of association between dependent variables?

What are the effects of covariates? How may they be utilized?

RESULTS

If the overall multivariate test is significant, we conclude that the respective effect (e.g., textbook) is significant. However, our next question would of course be whether only math skills improved, only physics skills improved, or both. In fact, after obtaining a significant multivariate test for a particular main effect or interaction, customarily one would examine the univariate F tests for each variable to interpret the respective effect. In other words, one would identify the specific dependent variables that contributed to the significant overall effect.

MANOVA is useful in experimental situations where at least some of the independent variables are manipulated. It has several advantages over ANOVA. First, by measuring several dependent variables in a single experiment, there is a better chance of discovering which factor is truly important. Second, it can protect against Type I errors that might occur if multiple ANOVA's were conducted independently. Additionally, it can reveal differences not discovered by ANOVA tests.

However, there are several cautions as well. It is a substantially more complicated design than ANOVA, and therefore there can be some ambiguity about which independent variable affects each dependent variable. Thus, the observer must make many potentially subjective assumptions. Moreover, one degree of freedom is lost for each dependent variable that is added. The gain of power obtained from decreased SS error may be offset by the loss in these degrees of freedom. Finally, the dependent variables should be largely uncorrelated. If the dependent variables are highly correlated, there is little advantage in including more than one in the test given the resultant loss in degrees of freedom.

Under these circumstances, use of a single ANOVA test would be preferable.

ASSUMPTIONS

Normal Distribution: - The dependent variable should be normally distributed within groups. Overall, the F test is robust to non-normality, if the non-normality is caused by skewness rather than by outliers. Tests for outliers should be run before performing a MANOVA, and outliers should be transformed or removed.

Linearity - MANOVA assumes that there are linear relationships among all pairs of dependent variables, all pairs of covariates, and all dependent variable-covariate pairs in each cell. Therefore, when the relationship deviates from linearity, the power of the analysis will be compromised.

Homogeneity of Variances: - Homogeneity of variances assumes that the dependent variables exhibit equal levels of variance across the range of predictor variables. Remember that the error variance is computed (SS error) by adding up the sums of squares within each group. If the variances in the two groups are different from each other, then adding the two together is not appropriate, and will not yield an estimate of the common within-group variance. Homoscedasticity can be examined graphically or by means of a number of statistical tests.

Homogeneity of Variances and Covariances: In multivariate designs, with multiple dependent measures, the homogeneity of variances assumption described earlier also applies. However, since there are multiple dependent variables, it is also required that their intercorrelations (covariances) are homogeneous across the cells of the design. There are various specific tests of this assumption.

LIMITATIONS

Outliers - Like ANOVA, MANOVA is extremely sensitive to outliers. Outliers may produce either a Type I or Type II error and give no indication as to which type of error is occurring in the analysis. There are several programs available to test for univariate and multivariate outliers.

Multicollinearity and Singularity - When there is high correlation between dependent variables, one dependent variable becomes a near-linear combination of the other dependent variables. Under such circumstances, it would become statistically redundant and suspect to include both combinations.

ANCOVA

ANCOVA is short for Analysis of Covariance. The analysis of covariance is a combination of an ANOVA and a regression analysis.

In basic terms, the ANCOVA examines the influence of an independent variable on a dependent variable while removing the effect of the covariate factor. ANCOVA first conducts a regression of the independent variable (i.e., the covariate) on the dependent variable. The residuals (the unexplained variance in the regression model) are then subject to an ANOVA. Thus, the ANCOVA tests whether the independent variable still influences the dependent variable after the influence of the covariate(s) has been removed. The One-Way ANCOVA can include more than one covariate, and SPSS handles up to ten. The ANCOVA model has more than one covariate it is possible to calculate the one-way ANCOVA using contrasts just like in the ANOVA to identify the influence of each covariate.

The purpose of including covariates in ANOVA is two-fold:

To reduce within-group error variance: In ANOVA we assess the effect of an experiment by comparing the amount of variability in the data that the experiment can explain, against the variability that it cannot explain. If we can explain some of this 'unexplained' variance (SSR) in terms of covariates, then we reduce the error variance, allowing us to more accurately assess the effect of the experimental manipulation (SSM).

Elimination of Confounds: In any experiment, there may be unmeasured variables that confound the results (i.e. a variable that varies systematically with the experimental manipulation). If any variables are known to influence the dependent variable being measured, then ANCOVA is ideally suited to remove the bias of these variables. Once a possible confounding variable has been identified, it can be measured and entered into the analysis as a covariate. Or, we can say the following (Interpretation or different statements to mean the same thing). The ANCOVA is most useful in that it explains an ANOVA's within-group variance, and controls confounding factors.

Firstly, as explained in the chapter on the ANOVA, the analysis of variance splits the total variance of the dependent variable into:

1. Variance explained by the independent variable (also called between groups variance)

Unexplained variance (also called within group variance)

The ANCOVA looks at the unexplained variance and tries to explain some of it with the covariate(s). Thus, it increases the power of the ANOVA by explaining more variability in the model.

The One-Way ANCOVA needs at least three variables. These variables are:

The independent variable, which groups the cases into two or more groups. The independent variable has to be at least of nominal scale.

The dependent variable, which is influenced by the independent variable. It has to be of continuous-level scale (interval or ratio data). Also, it needs to be homoscedastic and multivariate normal.

The covariate, or variable that moderates the impact of the independent on the dependent variable. The covariate needs to be a continuous-level variable (interval or ratio data). The covariate is sometimes also called confounding factor, or concomitant variable. The ANCOVA covariate is often a pre-test value or a baseline.

ASSUMPTIONS IN ANCOVA

ANCOVA has the same assumptions as any linear model except that there are two important additional considerations:

independence of the covariate and treatment effect, and homogeneity of regression slopes. The first one basically means that the covariate should not be different across the groups in the analysis (in other words, if you did an ANOVA or t-test using the groups as the independent variable and the covariate as the outcome, this analysis should be non-significant).

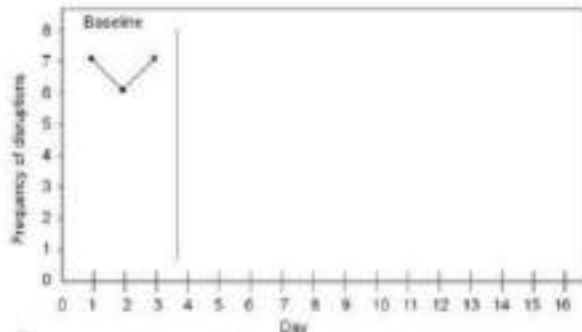
SINGLE-SUBJECT DESIGNS:

Single subject research (also known as single case experiments, single system designs) is popular in the fields of special education and counseling. This research design is useful when the researcher is attempting to change the behavior of an individual or a small group of individuals and wishes to document that change. Unlike true experiments where the researcher randomly assigns participants to a control and treatment group, in single subject research the participant serves as both the control and treatment group. The researcher uses line graphs to show the effects of a particular intervention or treatment. An important factor of single subject research is that only one variable is changed at a time. Single subject research designs are “weak” when it comes to external validity.

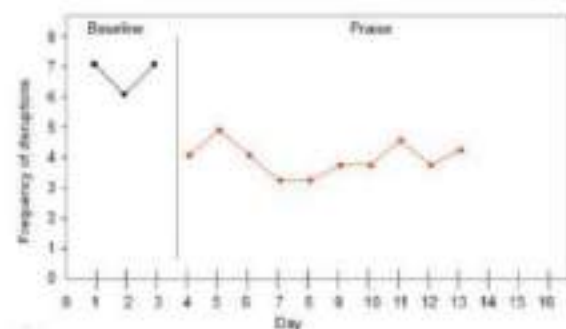
Single-subject research usually involves collecting data on one subject at a time. Single-subject researchers generally use line graphs to illustrate the effect of their intervention.

Suppose a researcher wished to investigate the effect of praise on reducing disruptive behavior over many days. First, she would need to establish a baseline of how frequently the disruptions occurred. She would measure how many disruptions occurred each day for several days. In the example below, the target student was disruptive seven times on the first day, six times on the second day, and seven times

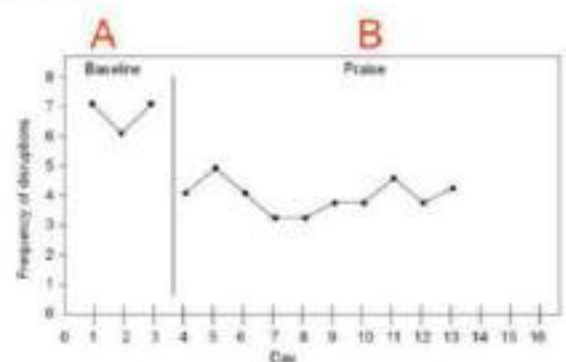
on the third day. Note how the sequence of time is depicted on the x-axis (horizontal axis) and the dependent variable (outcome variable) is depicted on the y-axis (vertical axis).



Once a baseline of behavior has been established (when a consistent pattern emerges with at least three data points), the intervention begins. The researcher continues to plot the frequency of behavior while implementing the intervention of praise.



In this example, we can see that the frequency of disruptions decreased once praise began. The design in this example is known as an A-B design. The baseline period is referred to as A and the intervention period is identified as B.



Sometimes an individual's behavior is so severe that the researcher cannot wait to establish a baseline and must begin with an intervention. In this case, a B-A-B design is used. The intervention is implemented immediately (before establishing a baseline). This is followed by a measurement without the intervention and then a repeat of the intervention.

Generally, in single subject research we count the number of times something occurs in a given time period and see if it occurs more or less often in that time period after implementing an intervention.

Extra for References:

Single subject designs are thought to be a direct result of the research of B.F. Skinner who applied the techniques of operant conditioning to subjects and measured the outcomes at various points in time. Because of this, single subject designs are often considered the design of choice when measuring behavioral change or when performing behavioral modification. Rather than comparing groups of subjects, this design relies on the comparison of treatment effects on a single subject or group of single subjects.

UNIT 2 IS COMPLETE

*

(EXTRA FOR REFERENCES – OUT OF SYLLABUS)

(Read further only if you have enough time for Preparation ahead for other topics)

Quasi-Experimental Designs

Quasi-experimental designs are almost identical to true experimental designs, but lacking one key ingredient: random assignment. For instance, one entire class section or one organization is used as the treatment group, while another section of the same class or a different organization in the same industry is used as the control group. This lack of random assignment potentially results in groups that are non-equivalent, such as one group possessing greater mastery of a certain content than the other group, say by virtue of having a better teacher in a previous semester, which introduces the possibility of selection bias. Quasi-experimental designs are therefore inferior to true experimental designs in interval validity.

Many true experimental designs can be converted to quasi-experimental designs by omitting random assignment. For instance, the quasi-equivalent version of pretest-posttest control group design is called **nonequivalent groups design (NEGD)**, with random assignment R replaced by non-equivalent (non-random) assignment N.

EXTRA FOR REFERENCES:

Idiographic or Nomothetic:

Explanations can be idiographic or nomothetic. Idiographic explanations are those that explain a single situation or event in idiosyncratic detail. For example, you did poorly on an exam because: (1) you forgot that you had an exam on that day, (2) you arrived late to the exam due to a traffic jam, (3) you panicked midway through the exam, (4) you had to work late the previous evening and could not study for the exam, or even (5) your dog ate your text book. The explanations may be detailed, accurate, and valid, but they may not apply to other similar situations, even involving the same person, and are hence not generalizable. In contrast, nomothetic

explanations seek to explain a class of situations or events rather than a specific situation or event. For example, students who do poorly in exams do so because they did not spend adequate time preparing for exams or that they suffer from nervousness, attention-deficit, or some

other medical disorder. Because nomothetic explanations are designed to be generalizable across situations, events, or people, they tend to be less precise, less complete, and less detailed. However, they explain economically, using only a few explanatory variables. Because theories are also intended to serve as generalized explanations for patterns of events, behaviors, or phenomena, theoretical explanations are generally nomothetic in nature.

Theory

While understanding theories, it is also important to understand what theory is not. Theory is not data, facts, typologies, taxonomies, or empirical findings. A collection of facts is not a theory, just as a pile of stones is not a house. Likewise, a collection of constructs (e.g., a typology of constructs) is not a theory, because theories must go well beyond constructs to include propositions, explanations, and boundary conditions. Data, facts, and findings operate at the empirical or observational level, while theories operate at a conceptual level and are based on logic rather than observations.

Building Blocks of a Theory

David Whetten (1989) suggests that there are four building blocks of a theory: constructs, propositions, logic, and boundary conditions/assumptions. Constructs capture the “what” of theories (i.e., what concepts are important for explaining a phenomenon), propositions capture the “how” (i.e., how are these concepts related to each other), logic represents the “why” (i.e., why are these concepts related), and boundary conditions/assumptions examines the “who, when, and where” (i.e., under what

circumstances will these concepts and relationships work).

Attributes of a Good Theory

Theories are simplified and often partial explanations of complex social reality. As such, there can be good explanations or poor explanations, and consequently, there can be good theories or poor theories. How can we evaluate the “goodness” of a given theory? Different criteria have been proposed by different researchers, the more important of which are listed below:

Logical consistency: Are the theoretical constructs, propositions, boundary conditions, and assumptions logically consistent with each other? If some of these “building blocks” of a theory are inconsistent with each other (e.g., a theory assumes rationality, but some constructs represent non-rational concepts), then the theory is a poor theory.

Explanatory power: How much does a given theory explain (or predict) reality? Good theories obviously explain the target phenomenon better than rival theories, as often measured by variance explained (R-square) value in regression equations.

Falsifiability: British philosopher Karl Popper stated in the 1940's that for theories to be valid, they must be falsifiable. Falsifiability ensures that the theory is potentially disprovable, if empirical data does not match with theoretical propositions, which allows for their empirical testing by researchers. In other words, theories cannot be theories unless they can be empirically testable. Tautological statements, such as “a day with high temperatures is a hot day” are not empirically testable because a hot day is defined (and measured) as a day with high temperatures, and hence, such statements cannot be viewed as a theoretical proposition. Falsifiability requires presence of rival explanations it ensures that the constructs are adequately measurable, and so forth. However, note that saying that a theory is falsifiable is not the same as saying that a theory should be

falsified. If a theory is indeed falsified based on empirical evidence, then it was probably a poor theory to begin with!

Parsimony: Parsimony examines how much of a phenomenon is explained with how few variables. The concept is attributed to 14th century English logician Father William of Ockham (and hence called “Ockham’s razor” or “Occam’s razor”), which states that among competing explanations that sufficiently explain the observed evidence, the simplest theory (i.e., one that uses the smallest number of variables or makes the fewest assumptions) is the best. Explanation of a complex social phenomenon can always be increased by adding more and more constructs. However, such approach defeats the purpose of having a theory, which are intended to be “simplified” and generalizable explanations of reality. Parsimony relates to the degrees of freedom in a given theory. Parsimonious theories have higher degrees of freedom, which allow them to be more easily generalized to other contexts, settings, and populations.

Key Terminologies:

Factors and Levels

The section on variables defined an independent variable as a variable manipulated by the experimenter. In the case study “Smiles and Leniency,” the effect of different types of smiles on the leniency shown to a person was investigated. Four different types of smiles (neutral, false, felt, and miserable) were shown. In this experiment, “Type of Smile” is the independent variable. In describing an ANOVA design, the term factor is a synonym of independent variable. Therefore, “Type of Smile” is the factor in this experiment. Since four types of smiles were compared, the factor “Type of Smile” has four levels.

An ANOVA conducted on a design in which there is only one factor is called a

one-way ANOVA. If an experiment has two factors, then the ANOVA is called a two-way ANOVA. For example, suppose an experiment on the effects of age and gender on reading speed were conducted using three age groups (8 years, 10 years, and 12 years) and the two genders (male and female). The factors would be age and gender. Age would have three levels and gender would have two levels.

Between- and Within-Subjects Factors

In the "Smiles and Leniency" study, the four levels of the factor "Type of Smile" were represented by four separate groups of subjects. When different subjects are used for the levels of a factor, the factor is called a between-subjects factor or a between-subjects variable. The term "between subjects" reflects the fact that comparisons are between different groups of subjects.

In the "ADHD Treatment" study, every subject was tested with each of four dosage levels (0, 0.15, 0.30, 0.60 mg/kg) of a drug. Therefore there was only one group of subjects, and comparisons were not between different groups of subjects but between conditions within the same subjects. When the same subjects are used for the levels of a factor, the factor is called a within-subjects factor or a within-subjects variable. Within-subjects variables are sometimes referred to as repeated-measures variables since there are repeated measurements of the same subjects.

Multi-Factor Designs

It is common for designs to have more than one factor. For example, consider a hypothetical study of the effects of age and gender on reading speed in which males and females from the age levels of 8 years, 10 years, and 12 years are tested. There would be a total of six different groups as shown in Table 1.

Table 1. Gender x Age Design.

Group	Gender	Age
1	Female	8
2	Female	10
3	Female	12
4	Male	8
5	Male	10
6	Male	12

This design has two factors: age and gender. Age has three levels and gender has two levels. When all combinations of the levels are included (as they are here), the design is called a factorial design. A concise way of describing this design is as a Gender (2) x Age (3) factorial design where the numbers in parentheses indicate the number of levels. Complex designs frequently have more than two factors and may have combinations of between- and within-subjects factors.

Question 1 out of 4.

Fifty subjects are each tested in both a control condition and an experimental condition. This is an example of:

- ☐ a between-subjects design
- ☐ a within-subjects design

Question 2 out of 4.

The times it took each of 20 subjects to name a set of colored squares and to read a set of color names were recorded. This is an example of:

- ☐ a between-subjects design
- ☐ a within-subjects design

Question 3 out of 4.

Subjects are randomly assigned to either a drug condition or a placebo condition. This is an example of:

- ☐ a between-subjects design
- ☐ a within-subjects design

Question 4 out of 4.

In a Gender (2) x Treatment (6) factorial between-subjects design, the total number of separate groups is

ANSWERS

1 (B): This is a within-subjects design since each subject was tested in each condition.

(B)

(A): This is a between-subjects design because each subject was tested in either one condition (drug) or another (placebo).

A total of 12. All combinations of Gender and Treatment would be in the design.

Experimental design

It refers to the manner in which the experiment was set up. Experimental design includes the way the treatments were administered to subjects, how subjects were grouped for analysis, how the treatments and grouping were combined.

In ANOVA there is a single dependent variable or score. In Psychology the dependent measure is usually some measure of behavior. If more than one measure of behavior is taken, multivariate analysis of variance, or MANOVA, may be the appropriate analysis. Because the ANOVA model breaks the score into component parts, or effects, which sum to the total score, the one must assume the interval property of measurement for this variable. Since in real life the interval property is never really met, one must be satisfied that at least an approximation of an interval scale exists for the dependent variable. To the extent that this assumption

is unwarranted, the ANOVA hypothesis testing procedure will not work.

In ANOVA there is at least one independent variable or factor. There are different kinds of factors; treatment, trial, blocking, and group. All factors, however, have some finite number of different levels. Each level is the same in either some quality or quantity. The only restriction on the number of levels is that there are fewer levels than scores, although in practice one seldom sees more than ten levels in a factor unless the data set is very large. It is not necessary that the independent variables or factors be measured on an interval scale. If the factors are measured on an (approximate) interval scale, then some flexibility in analysis is gained. The continued popularity of ANOVA can partially be explained by the lack of the necessity of the interval assumption for the factors.

Assumptions made by ANOVA:

The populations have the same variance. This assumption is called the assumption of homogeneity of variance.

The populations are normally distributed. Each value is sampled independently from each other value. This assumption requires that each subject provide only one value. If a subject provides two scores, then the values are not independent. The analysis of data with two scores per subject is shown in the section on within-subjects ANOVA later

in this chapter. These assumptions are the same as for a t test of differences between groups except that they apply to two or more groups, not just to two groups.

Kinds of Factors

Treatment Factors

Treatments will be defined as quantitatively or qualitatively different levels of experience. For example, in an experiment on the effects of caffeine, the treatment levels might be exposure to different amounts of caffeine, from none to .0375 milligrams. In a very simple experiment there are two levels of treatment, none, called the control condition, and some, called the experimental condition.

Treatment factors are usually the main focus of the experiment. A treatment factor is characterized by the following two attributes (Lee, 1975):

An investigator could assign any of his experimental subjects to any one of the levels of the factor.

The different levels of the factor consist of explicitly distinguishable stimuli or situations in the environment of the subject.

In the solitaire example, practice time would be a treatment factor if the experimenter controlled the amount of time that the subject practiced. If subject's came to the experiment having already practiced a given amount, then the experimenter could not arbitrarily or randomly assign that subject to a given practice level. In that case the factor would no longer be considered a treatment factor.

In an experiment where subjects are run in groups, it sometimes is valuable to treat each group as a separate level of a factor. There might be, for example, an obnoxious subject who affects the scores of all other subjects in that group. In this case the

second attribute would not hold and the factor would be called a group factor.

Group Factors

As described above, a group factor is one in which the subjects are arbitrarily assigned to a given group which differs from other groups only in that different subjects are assigned to it. If each group had some type of distinguishing feature, other than the subjects assigned to it, then it would no longer be considered as a group factor. If a group factor exists in an experimental design, it will be symbolized by G.

Trials Factors

If each subject is scored more than once under the same condition and the separate scores are included in the analysis, then a trials factor exists. If the different scores for a subject were found under different levels of a treatment, then the factor would be called a treatment factor rather than a trials factor. Trials factors will be denoted by T.

Trials factors are useful in examining practice or fatigue effects. Any change in scores over time may be attributed to having previously experienced similar conditions.

Blocking Factors

If subjects are grouped according to some pre-existing subject similarity, then that grouping is called a blocking factor. The experimenter has no choice but to assign the subject to one or the other of the levels of a blocking factor. For example, gender (sex) is often used as a blocking factor. A subject enters the experiment as either a male or female and the experimenter may not arbitrarily (randomly) assign that individual to one gender or the other.

Because the experimenter has no control over the assignment of subjects to a blocking factor, causal inference is made much more difficult. For example, if in the solitaire experiment, the practice factor was based on a pre-existing condition, then any differences between the groups may

be due either to practice or to the fact that some subjects liked to play solitaire, were better at the game and thus practiced more. Since the subjects are self-selected, it is not possible to attribute the differences between groups to practice, enjoyment of the game, natural skill in playing the game, or some other reason. It is possible, however, to say that the groups differed.

Even though causal inference is not possible, blocking factor can be useful. A factor that accounts for differences in the scores adds power to the experiment. That is, a blocking factor that explains some of the differences between scores may make it more likely to find treatment effects. For example, if males and females performed significantly different in the solitaire experiment, it might be useful to include sex as a blocking factor because differences due to gender would be included in the error variance otherwise.

In other cases blocking factors are interesting in their own right. It may be interesting to know that freshmen, sophomores, juniors, and seniors differ in attitude toward university authority, even though causal inferences may not be made.

In some cases the pre-existing condition is quantitative, as in an IQ score or weight. In these cases it is possible to use a median split where the scores above the median are placed in one group and the scores below the median are placed in another. Variations of this procedure divide the scores into three, four, or more approximately equal sized groups. Such procedures are not recommended as there are better ways of handling such data (Edwards, 1985).

Unit Factors

The unit factor is the entity from which a score is taken. In experimental psychology, the unit factor is usually a subject (human or animal), although classrooms, dormitories, or other units may serve the same function. In this text, the unit factor will be designated as S, with

the understanding that it might be some other type of unit than subject.

Error Factors

The error factor, designated as E, is not a factor in the sense of the previous factors and is not included in the experimental design. It is necessary for future theoretical development.

Relationships Between Factors

The following two relationships between factors describe a large number of useful designs. Not all possible experimental designs fit neatly into categories described by the following two relationships, but most do.

Crossed

When two factors are crossed, each level of each factor appears with each level of the other factor. An "X" indicates a crossing relationship.

For example, consider two factors, A and B, where A is gender (a_1 = Females, a_2 = Males) and B is practice (b_1 = none, b_2 = one hour, b_3 = five hours, and b_4 = twenty hours). If gender was crossed with practice, $A \times B$, then both males and females would participate in all four levels of practice. There would be eight groups of subjects including: ab_{11} , females who had no practice, ab_{12} , females who had one hour of practice, and so forth to ab_{24} , males who practiced twenty hours. An additional factor may be added to the design, say handedness (C), where c_1 = right handed and c_2 = left handed. If the design of the experiment was $A \times B \times C$, then there would be sixteen groups, including abc_{231} , left-handed males who practiced five hours.

If subjects (S) are crossed with treatments (A), $S \times A$, each subject sees each level of the treatment conditions. In a very simple experiment such as the effects of caffeine on alertness (A), each subject would be exposed to both a caffeine condition (a_1) and a no caffeine condition (a_2). For

example, using the members of a statistics class as subjects, the experiment might be conducted as follows. On the first day of the experiment the class is divided in half with one half of the class getting coffee with caffeine and the other half getting coffee without caffeine. A measure of alertness is taken for each individual, such as the number of yawns during the class period. On the second day the conditions are reversed, that is, the individuals who received coffee with caffeine are now given coffee without and vice-versa.

The distinguishing feature of crossing subjects with treatments is that each subject will have more than one score. This feature is sometimes used in referring to this class of designs as repeated measures designs. The effect also occurs within each subject, thus these designs are sometimes referred to as within subjects designs.

Crossing subjects with treatments has two advantages. One, they generally require fewer subjects, because each subject is used a number of times in the experiment. Two, they are more likely to result in a significant effect, given the effects are real. This is because the effects of individual differences between subjects is partitioned out of the error term.

Crossing subjects with treatments also has disadvantages. One, the experimenter must be concerned about carry-over effects. For example, individuals not used to caffeine may still feel the effects of caffeine on the second day, when they did not receive the drug. Two, the first measurements taken may influence the second. For example, if the measurement of interest was score on a statistics test, taking the test once may influence performance the second time the test is taken. Three, the assumptions necessary when more than two treatment levels are employed in a crossing subjects with treatments may be restrictive.

When a factor is a blocking factor, it is not possible to cross that factor with subjects.

It is difficult to find subjects for a $S \times A$ design where A is gender. I generally will take points off if a student attempts such a design.

Nested

Factor B is said to be nested within factor A if each meaningful level of factor B occurs in conjunction with only one level of A. This relationship is symbolized as B(A), and is read as "B nested within A". Note that B(A) is considerably different from A(B). In the latter, each meaningful level of A would occur in one and only one level of B. These types of designs are also designated as hierarchical designs in some textbooks.

A B(A) design occurs, for example, when the first three levels of factor B (b1, b3, and b3) appear only under level a1 of factor A and the next three levels of B (b4, b5, and b6) appear only under level a2 of factor A. Depending upon the labeling scheme, b4, b5, and b6 may also be called b1, b3, and b3, respectively. It is understood by the design designation that the b1 occurring under a1 is different from the b1 occurring under a2.

Nested or hierarchical designs can appear because many aspects of society are organized hierarchically. For example within the university, classes (sections) are nested within courses, courses are nested within departments, departments within colleges, and colleges within the university.

In experimental research it is also possible to nest treatment conditions within other treatment conditions. For example, suppose a researcher was interested in the effect of diet on health in hamsters. One factor (A) might be a high cholesterol (a1) or low cholesterol (a2) diet. A second factor (B) might be type of food, peanut butter (b1), cheese (b2), red meat (b3), chicken (b4), fish (b5), or vegetables(b6). Because type of food may be categorized as being either high or low in cholesterol, a B(A) experimental design would result. Chicken, fish, and vegetables would be

reabeled as b1 ,b3, and b3, respectively, but it would be clear from the experimental design specification that peanut butter and chicken, cheese and fish, and red meat and vegetables, were qualitatively different, even though they all share the same label.

While any factor may possibly be nested within any other factor, the critical nesting relationship is with respect to subjects. If S is nested within some combination of other factors, then each subjects appear under one, and only one, combination of factors within which they are nested. These effects are often called the Between Subjects effects. If S is crossed with come combination of other factors, then each subject see all combinations of factors with which they are crossed. These effects are referred to as Within Subjects effects.

As mentioned earlier subjects are necessarily nested within blocking factors. Subjects are necessarily nested within the

effects of gender and current religious preference, for example.

Treatment factors, however, may be nested or crossed with subjects. The effect of caffeine on alertness could be studied by dividing the subjects into two groups, with one receiving a beverage with caffeine and one group not. This design would nest subjects with caffeine and be specified as S(A), or simply A, as the S is often dropped when the design is completely between subjects.

If subjects appeared under both caffeine conditions, receiving caffeine on one day and no caffeine on the other, then subjects would be crossed with caffeine. The design would be specified as S X A. In this case the S would remain in the design.

REFERENCE ARTICLE

[:http://onlinestatbook.com/2/analysis_of_variance/anova_designs.html](http://onlinestatbook.com/2/analysis_of_variance/anova_designs.html)

Question 1 out of 20.

Unlike t tests, an ANOVA uses both differences between group means and differences within groups to determine whether or not the differences are significant.

- ☐ True
- ☐ False

Question 2 out of 20.

The "Smiles and Leniency" study uses a between-subjects design. The four types of smiles (false, felt, miserable, and neutral) are the four levels of one factor.

- ☐ True
- ☐ False

Question 3 out of 20.

If an experiment seeks to investigate the acquisition of skill over multiple sessions of practice, which of the following best describes the comparison of the subjects?

- ☐ Within-subjects
- ☐ Between-subjects
- ☐ Cannot be determined with the given information

Question 9 out of 20.

Which best describes the assumption of homogeneity of variance?

- ☐ The populations are both normally distributed to the same degree.
- ☐ The between and within population variances are approximately the same.
- ☐ The variances in the populations are equal.

Question 10 out of 20.

When performing a one-factor ANOVA (between-subjects), it is important that each subject only provide a single value. If a subject were to provide more than one value, the independence of each value would be lost and the test provided by an ANOVA would not be valid.

- ☐ True
- ☐ False

Question 13 out of 20.

Why can't an F ratio be below 0?

- ☐ Neither MSB nor MSE can ever be a negative value.
- ☐ MSB is never less than 1.
- ☐ MSE is never less than 1.

Question 14 out of 20.

Consider an experiment in which there are 7 groups and within each group there are 15 participants. What is the degrees of freedom for the numerator (between)?

Question 15 out of 20.

Consider an experiment in which there are 7 groups and within each group there are 15 participants. What is the degrees of freedom for the denominator (within)?

Question 16 out of 20.

The F distribution has a:

- ☐ positive skew
- ☐ no skew
- ☐ negative skew

Question 17 out of 20.

An independent-groups t test with 12 degrees of freedom was conducted and the value of t was 2.5. What would the F be in a one-factor ANOVA?

Question 18 out of 20.

If the sum of squares total were 100 and the sum of squares condition were 80, what would the sum of squares error be?

Question 19 out of 20.

If the sum of squares total were 100 and the sum of squares condition were 80 in an experiment with 3 groups and 8 subjects per group, what would the F ratio be?

Question 20 out of 20.

If a t test of the difference between means of two independent groups found a t of 2.5, what would be the value of F in a one-way ANOVA?

Answers

1 False, both t tests and ANOVAs use both. In a t test, the difference between means is in the numerator and the denominator is based on differences within groups. In an ANOVA, the variance of the group means (multiplied by n) is the numerator. The denominator is based on differences within groups.

True**A****C**

True. When a subject provides more than one data point, the values are not independent, thus violating one of the assumptions of between-subjects ANOVA.

13 F is defined as MSB/MSE . Since both MSB and MSE are variances and negative variance is impossible, an F score can never be negative.

14 $k-1 = 7-1 = 6$

$$\begin{aligned} df_{\text{numerator}} &= k-1 \\ df_{\text{denominator}} &= N-k \end{aligned}$$

15 $N-k = 105-7 = 98$

16 The F distribution has a long tail to the right which means it has a positive skew.

F equals $t^2 = 6.25$.

Sum of squares total equals sum of squares condition + sum of squares error.

Divide sums of squares by degrees of freedom to get mean squares. Then divide MSB by MSE to get F which equals 42.

$F = t(\text{square})$

Statistical Tests

Differences between groups

Independent-samples t-test
 Paired-samples t-test
 One-way ANOVA
 Repeated measures ANOVA
 Two-way ANOVA
 Factorial (three-way) ANOVA
 Within-within-subjects ANOVA
 Three-way repeated measures ANOVA
 Three-way mixed ANOVA
 Mixed ANOVA
 Hotelling's T^2
 One-way MANOVA
 Two-way MANOVA
 One-way MANCOVA
 One-way ANCOVA
 Two-way ANCOVA
 Mann-Whitney U test
 Wilcoxon signed-rank test
 Kruskal-Wallis H test
 Jonckheere-Terpstra test
 Friedman test
 McNemar's test
 Cochran's Q test
 Sign test
 Test of two proportions
 Chi-square test of homogeneity (2 x C)
 Chi-square test of homogeneity (R x 2)

One sample

One-sample t-test
 Chi-square goodness-of-fit

Survival analysis

Kaplan-Meier

Reliability

Cronbach's Alpha
 Cohen's kappa
 Fleiss' kappa
 Weighted kappa
 Kendall's W

Predictin

Linear reg
 Multiple r
 Hierarchi
 Logistic r
 Ordinal re

Associati

Pearson's
 Point-bise
 Pearson's
 Spearman
 Kendall's
 Goodman
 Somers' c
 Mantel-H
 Cochran-
 Chi-squar
 Chi-squar
 Fisher's e
 Relative r
 Odds rati
 Goodman
 Loglinear

Data red

Principal

Assumpt

Testing fo
 Transform

Moderati

Dichotom

Power Within

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Psychological Tests vs. Assessments vs. Measurements vs. Surveys

Types of tests Test construction: Item writing, item analysis

Test standardization: Reliability, validity and Norms

Areas of testing: Intelligence, creativity, neuropsychological tests, aptitude, Personality assessment, interest inventories

Attitude scales – Semantic differential, Staples, Likert scale. Computer-based psychological testing

Applications of psychological testing in various settings: Clinical, Organizational and business, Education, Counseling, Military. Career guidance.

UNIT

3

UNIT 3: PSYCHOLOGICAL TESTING

PSYCHOLOGICAL TESTING

Psychological testing is best defined as “The process of administering, scoring, and interpreting psychological tests.” (Maloney & Ward, 1976, p. 9).

Although all psychological tests require that you perform some behavior to measure personal attributes, traits, or characteristics or to predict outcomes, these tests can differ in various ways. For example, they can

differ in terms of the behavior they require you to perform, what they measure, their content, how they are administered and formatted, how they are scored and interpreted, and their psychometric quality (psychometrics is the quantitative and technical aspect of mental measurement).

For clinical psychologists, assessment is second only to psychotherapy in terms of its professional importance

CHARACTERISTICS:

(TABLE)

THE THREE DEFINING CHARACTERISTICS OF PSYCHOLOGICAL TESTS

All psychological tests have three common characteristics:

- First, a good test should representatively sample the behaviors thought to measure an attribute or predict an outcome. This ensures that the test measures what it says it measures.
- Second, the behavior samples should be obtained under standardized conditions. That

is, a test must be administered exactly the same way to all individuals so that we can be confident that a person's score accurately reflects the attribute being measured or the outcome being predicted.

- Third, there must be rules for scoring so that all examiners will score the test in the same way.

ASSUMPTIONS:

There are many assumptions that must be made when using psychological tests. The following are what we consider the most important assumptions:

Psychological tests measure what they say they measure, and any inferences that are drawn about test takers based on their test scores are appropriate.

An individual's behavior, and therefore test scores, will remain unchanged over time.

Individuals understand test items similarly.

Individuals can report about themselves accurately.

Individuals will report their thoughts and feelings honestly.

The test scores an individual receives is equal to his or her true ability plus some error, and this error may be attributable to the test itself the examiner, the examinee, or the environment.

HISTORY OF TESTING:

Psychological Tests: From Ancient China to the 20th Century

2200 BCE: Xia Dynasty

Some scholars believe that the use of psychological tests dates back approximately 4,000 years to 2200 BCE, when the Chinese emperor Yushun examined officials every third year to determine whether they were suitable to continue in office (Duflois, 1970; Martin, 1870). However, modern scholars of ancient China say that there is little archaeological evidence to support these claims. Reliable writing systems were developed by the Chinese somewhere between 1766 and 1122 BCE (Shang dynasty; Bowman, 1989). Nowhere in these writings were there any hints suggesting that leaders were examined as just described. Even in 1115 BCE, with the advent of more elaborate writing systems, there were no inscriptions or writings to suggest the existence of such an examination process (Martin, 1870).



200–100 BCE: Late Qin, Early Han Dynasty

Most modern scholars of ancient China agree that royal examinations began around 200 to 100 BCE, in the late Qin (Ch'in) or early Han dynasty (Eberhard, 1977; Franke, 1960; Pirazzoli-t'Serstevens, 1982; Rodzinski, 1979). Hacker (1978) believes that the first written examinations in world history began in 165 BCE, when the emperor administered written examinations to all nominees. Pirazzoli-t'Serstevens also believes that this was the beginning of all examination systems. Eberhard, on the other hand, admits that there may have been some assessment procedures before 165 BCE for selecting officials, who were probably tested more for literacy than for knowledge.

618–907 CE: Tang Dynasty

Such examination systems seem to have been discontinued until the Tang dynasty, when their use increased significantly (Bowman, 1989).

1368–1644: Ming Dynasty

During the Ming dynasty, the examinations became more formal. There were different levels of examinations (municipal, county, provincial, and national), and the results of examinations became associated with granting formal titles, similar to today's university degrees. On passing each level of examination, people received more titles and increasingly more power in the civil service (Bowman, 1989). These examinations were distressful, and this distress became a part of Chinese culture and also a part of folk stories and the literature (poems, comedies, and tragedies). Nonetheless, this examination system seemed to work well. Today, many scholars believe that this examination system kept talented men in the national government (Kracke, 1963) and kept members of the national government from becoming nobility because of their descent.

Seeing the value of these examinations for making important decisions, European governments, and eventually the governments of the United Kingdom, the United States, Canada, and other countries, adopted the use of such examination systems.

1791: France and Britain

France initially began using this kind of examination system in 1791. However, soon after, Napoleon temporarily abolished them. The system adopted by France served as a model for a British system started in 1833 to select trainees for the Indian civil service—the beginning of the British civil service.

1860s: United States

Due to the success of the British system, Senator Charles Sumner and Representative Thomas Jenckes proposed to Congress in 1860 that the United States use a similar system. Jenckes's report, *Civil Service in the United States*, described the British and Chinese systems in detail. This report laid the foundation for the establishment of the Civil Service Act Health and Psychosocial Instruments (HAPI), passed in January 1883.

20th Century: Western Europe and the United States

In 1879, Wilhelm Wundt introduced the first psychological laboratory, in Leipzig, Germany. At this time, psychology was the study of the similarities among people. For example, physiological psychologists studied how the brain and the nervous system function, and experimental psychologists conducted research to discover how people learn and remember. Strongly influenced by James McKeen Cattell, an American researcher in Wundt's laboratory, psychologists turned their attention to exploring individual differences. Cattell and others realized that learning about the differences among people was just as important as learning about the similarities among people. They believed that developing formal psychological tests to measure individual differences could help solve many social problems, such as who should be placed in remedial programs, who should be sent to battlefields, and who should be hired for particular jobs. At this time, scientists were particularly interested in finding a quantitative way of measuring general intelligence.

During the early 20th century, serious research efforts began on the use and usefulness of various testing procedures. Research conducted by scholars in the United States and Germany eventually led to Alfred Binet's research on intelligence in children.

Some scholars believe that the use of psychological tests can be traced to 2200 BCE in ancient China. Most scholars agree that serious research efforts on the use and usefulness of psychological tests did not begin until the 20th century with the advent of intelligence testing. Intelligence Tests Alfred Binet and the Binet–Simon Scale Late in the 19th century, Alfred Binet founded the first experimental psychology research laboratory in France. In his lab, Binet attempted to develop experimental techniques to measure intelligence and reasoning ability. He believed that intelligence was a complex characteristic that could be determined by evaluating a person's reasoning, judgment, and problem-solving abilities. Binet tried a variety of tasks to measure reasoning, judgment, and problem solving on his own children as well as on other children in the French school system. Binet was successful in measuring

intelligence, and in 1905 he and Théodore Simon published the **first test of mental ability, the Binet–Simon Scale**. Parisian school officials used this scale to decide which children, no matter how hard they tried, were unable to profit from regular school programs (Binet & Simon, 1905).

Lewis Terman and the Stanford Binet Binet's work influenced psychologists across the globe. Psychological testing became a popular method of evaluation, and the Binet–Simon Scale was adapted for use in many countries. In 1916, Lewis Terman, an American psychologist, produced **the Stanford–Binet Intelligence Scales**, an adaptation of Binet's original test. This test, developed for use with Americans ages 3 years to adulthood, was used for many years. A revised edition of the Stanford–Binet remains one of the most widely used intelligence tests today.

PSYCHOLOGICAL ASSESSMENTS V/S PSYCHOLOGICAL TESTS

Psychological assessments and psychological tests both are methods of collecting important information about people, and both are also used to help understand and predict behavior (Kline, 2000, Maloney & Ward, 1976). Assessment, however, is a broader concept than psychological testing. Psychological assessment involves multiple methods, such as personal history interviews, behavioral observations, and psychological tests, for gathering information about an individual. Psychological assessment involves both an objective component and a subjective component (Matarazzo, 1990), and psychological tests are only one tool in the assessment process. For example, a clinical psychologist may conduct a psychological assessment of a patient and, as a part of this assessment, may administer a psychological test such as the MMPI.

PSYCHOLOGICAL TESTS V/S MEASUREMENTS

Although the meanings overlap, psychological test and measurement are not synonyms. Measurement, broadly defined, is the assignment of numbers according to specific rules. Psychological tests require

PSYCHOLOGICAL TESTS V/S SURVEYS

Surveys, like psychological tests (and psychological assessments), are used to collect important information from individuals. Surveys differ from psychological tests in two important ways. First, psychological tests focus on individual outcomes, and surveys focus on group outcomes. Psychological tests provide important information about individual differences and help individuals and institutions make important decisions about

test takers to answer questions or perform tasks to measure personal attributes. The concept of a psychological test is represented by the lighter circle in the figure. With psychological tests, test takers' answers to questions or their performance on some task is not initially expressed in physical units of any kind; instead, scores are derived according to some predetermined method. In some cases, the end result of a psychological test is not a derived score at all, but rather a verbal description of an individual. For example, there are some personality tests that, although they have rules for scoring or summarizing information, do not produce overall scores. Instead, these tests yield profiles. The MBTI is an example of such a test. Psychological tests can be considered psychological measurements when a sample of behavior can be expressed as a numerical score.

Although most psychological tests are measurements, not all psychological tests, strictly defined, meet the definition of a measurement. Throughout the remainder of this text, we follow the common practice of referring to all psychological tests as measurements because most of them are, but keep in mind the distinctions we have drawn in this section.

individuals. For example, a psychological test may suggest that a child is unusually intelligent and therefore should be placed in a gifted or honors program. Surveys, on the other hand, provide important information about groups and help us make important decisions about groups. For example, an organizational survey may suggest that employees are displeased with a company benefits program and that a new benefits program is needed.

Second, the results of a psychological test are often reported in terms of an overall

derived score or scaled scores. Results of surveys, on the other hand, are often reported at the question level by providing the percentage of respondents who selected each answer alternative. Of course, in some

cases, surveys focus on individual outcomes and are constructed using scales. In such cases, the survey approximates a psychological test.

TYPES OF TESTS

There are tens of thousands of commercially available psychological tests, and professionals refer to these tests in various ways. Tests are classified based on multitude of perspectives considered.

MAXIMAL PERFORMANCE, BEHAVIOR OBSERVATION, OR SELF-REPORT

Most psychological tests can be defined as being tests of maximal performance, behavioral observation tests, or self-report tests.

Tests of maximal performance require test takers to perform a particular well-defined task such as making a right-hand turn, arranging blocks from smallest to largest, tracing a pattern, or completing mathematical problems. Test takers try to do their best because their scores are determined by their success in completing the task. Intelligence tests, tests of specific abilities (for example, mechanical ability), driving tests (road and written), and classroom tests all are good examples of tests of maximal performance.

Behavior observation tests involve observing people's behavior and how

people typically respond in a particular context. Unlike with tests of maximal performance, many times people do not know that their behavior is being observed and there is no single defined task for the individual to perform. Many restaurants use this technique to assess food servers' competence in dealing with customers. Sometimes managers hire trained observers to visit their restaurant disguised as a typical customer. In exchange for a free meal or some predetermined compensation, observers agree to record specific behaviors performed by a food server. For example, observers may document whether a food server greeted them in a friendly manner. Other examples of behavior observations include documenting job performance for performance appraisals or clinical interviews.

Self-report tests require test takers to report or describe their feelings, beliefs, opinions, or mental states. Many personality inventories, such as the Hogan Personality Inventory (HPI), are self-report tests. The HPI, a test used primarily for personnel selection and individualized assessment, asks test takers to indicate whether each of more than 200 statements about themselves is true or false.

STANDARDIZED OR NON-STANDARDIZED

Standardized tests are those that have been administered to a large group of individuals who are similar to the group for whom the test has been designed. For example, if a test is designed to measure the writing ability of high school students, the test would be administered to a large group of high school students. This group is called the **standardization sample**—people who are tested to obtain data to establish a frame of reference for interpreting individual test scores. These data, called norms, indicate the average performance of a group and the distribution of scores above and below this average. For example, if you took the SAT, the interpretation of your score included comparing it with the SAT standardization sample to determine whether your score was high or low in comparison with others and whether you scored above average, average, or below average. In addition, standardized tests always have specific directions for administration and scoring.

Non-standardized tests do not have standardization samples and are more common than standardized tests. Non-standardized tests are usually constructed by a teacher or trainer in a less formal manner for a single administration. For example, in many cases, the exams you take in your college courses are non-standardized tests.

OBJECTIVE OR PROJECTIVE

Sometimes people make a distinction between objective and projective tests.

Objective tests are structured and require test takers to respond to structured true/false questions, multiple-choice questions, or rating scales. What the test taker must do is clear, for example, answer true or false, circle the correct multiple-choice answer, or circle the correct item on the rating scale. The GRE, Stanford-Binet Intelligence Scales, General Aptitude Test Battery, and most classroom tests are examples of objective tests. Another example of an objective test is the NEO Personality Inventory, an objective self-report instrument designed to identify what makes individuals unique in their thinking, feeling, and interaction with others. Although there are two forms of the inventory, both measure five broad personality dimensions: neuroticism, extroversion, openness, agreeableness, and conscientiousness. Test takers are asked to indicate whether they strongly disagree, disagree, are neutral, agree, or strongly agree with each of 240 statements. These statements are about their thoughts, feelings, and goals. P

Projective tests are unstructured. They require test takers to respond to unstructured or ambiguous stimuli such as incomplete sentences, inkblots, and abstract pictures. The role of the test taker is less clear than with a standardized test. People who use projective tests believe that test takers project themselves into the task they are asked to perform and that their responses are based on what they believe the stimuli mean and, on the feelings, they experience while responding. These tests tend to elicit highly personal concerns. They are often used to detect unconscious thoughts or personality characteristics, and they may be used to identify the need for psychological

counseling. The TAT is an example of a projective test.

4. DIMENSION MEASURED

Psychological tests are often discussed in terms of the dimensions they measure. For example, sometimes we distinguish among achievement tests, aptitude tests, intelligence tests, personality tests, and interest inventories. We refer to these as dimensions because they are broader than a single attribute or trait level. Often these types of tests measure various personal attributes or traits.

ACHIEVEMENT TESTS

Achievement tests measure a person's previous learning in a specific academic area (for example, computer programming, German, trigonometry, psychology). A test that requires you to list the three characteristics of psychological tests would be considered an achievement test. Achievement tests are also referred to as tests of knowledge. Achievement tests are used primarily in educational settings to determine how much students have learned or what they can do at a particular point in time. Many elementary schools and high schools rely on achievement tests to compare what students know at the beginning of the year with what they know at the end of the year, to assign grades, to identify students with special educational needs, and to measure students' progress.

APTITUDE TESTS

Achievement tests measure a test taker's knowledge in a specific area at a specific point in time. Aptitude tests assess a test taker's potential for learning or ability to

perform in a new job or situation. Aptitude tests measure the product of cumulative life experiences—or what one has acquired over time. They help determine what "maximum" can be expected from a person. Schools, businesses, and government agencies often use aptitude tests to predict how well someone will perform or to estimate the extent to which an individual will profit from a specified course of training. Vocational guidance counseling may involve aptitude testing to help clarify the test taker's career goals. If a person's score is similar to scores of others already working in a given occupation, the test will predict success in that field.

INTELLIGENCE TESTS

Intelligence tests, like aptitude tests, assess the test taker's ability to cope with the environment, but at a broader level. Intelligence tests are often used to screen individuals for specific programs (for example, gifted programs, honors programs) or programs for the mentally challenged. Intelligence tests are typically used in educational and clinical settings.

INTEREST INVENTORIES

Interest inventories assess a person's interests in educational programs for job settings and provide information for making career decisions. Because these tests are often used to predict satisfaction in a particular academic area or employment setting, they are administered primarily to students by counselors in high schools and colleges. Interest inventories are not intended to predict success; rather, they are intended only to offer a framework for narrowing career possibilities.

PERSONALITY TESTS

Personality tests measure human character or disposition. The first personality tests were designed to assess and predict clinical disorders. These tests remain useful today for determining who needs counseling and who will benefit from treatment programs. Newer personality tests measure "normal" personality traits. For example, the Myers—Briggs Type Indicator (MBTI) is often used by industrial/organizational psychologists to increase employees' understanding of individual differences and to promote better communication between members of work teams. Career counselors also use the MBTI to help students select majors and careers consistent with their personalities. Personality tests can be either objective or projective. The MBTI is an example of an objective personality test. Projective personality tests, such as the TAT, serve the same purpose as some objective personality tests, but they require test takers to respond to unstructured or ambiguous stimuli.

SUBJECT TESTS

Many popular psychological testing reference books also classify tests by subject. For example, the Seventeenth Mental Measurements Yearbook (Geisinger, Spies, Carlson, & Plake, 2007) classifies

thousands of tests into 19 major subject categories:

- Achievement
- Behavior assessment
- Developmental
- Education
- English
- Fine arts
- Foreign languages
- Intelligence
- Mathematics
- Miscellaneous (for example, courtship and marriage, driving and safety education, etiquette)
- Multi-apptitude batteries
- Neuropsychological
- Personality
- Reading
- Science
- Sensorimotor

Reference books such as the Mental Measurements Year book often indicate whether a test is (a) a test of maximal performance, a behavior observation test, or self-report test; (b) standardized or non-standardized; and (c) objective or projective.

TEST CONSTRUCTION: ITEM WRITING, ITEM ANALYSIS

ITEM WRITING:

Item writing is essentially a creative art. There are no set rules to guide and guarantee writing of good items. A lot depends upon the item writer's intuition, imagination, experience, practice and ingenuity. However, there are some essential prerequisites which must be met if the item writer wants to write good and appropriate

items. These requirements are briefly discussed as follows;

Command on subject matter:

The item writer must have a thorough knowledge and complete mastery of the subject matter. In other words, he must be fully acquainted with all facts, principles, misconceptions, fallacies in a particular field so that he may be able to write good and appropriate items.

Fully aware of the population:

The item writer must be fully aware of those persons for whom the test is meant. He must also be aware of the intelligence level of those persons so that he may manipulate the difficulty level of the items for proper adjustment with their ability level. He must also be able to avoid irrelevant clues to correct responses.

Familiarity with different types of items:

The item writer must be familiar with different types of items along with their advantages and disadvantages. He must also be aware of the characteristics of good items and the common probable errors in writing items.

Command on language:

The item writer must have a large vocabulary. He must know the different

meanings of a word so that confusion in writing the items may be avoided. He must be able to convey the meaning of the items in the simplest possible language.

Expert opinion:

After writing down the items, they must be submitted to a group of subject experts for their criticism or suggestions, which must then be duly modified.

Cultivate a rich source of ideas:

The item writer must also cultivate a rich source of ideas for items. This is because ideas are not produced in mind automatically but rather require certain factors or stimuli. The common source of such factors are textbooks, journals, discussions, questions for interview, coarse outlines and other instructional materials.

CHARACTERISTICS OF A GOOD ITEM

An item must have the following characteristics;

Clarity:

An item should be phrased in such a manner that there is no ambiguity regarding its meaning for both the item writer as well as the examinees who take the test.

Moderately difficult:

The item should not be too easy or too difficult.

Discriminating power:

It must have discriminating power, that is, it must clearly distinguish between those who possess the trait and those who do not.

To the point:

It should not be concerned with the trivial aspects of the subject matter, that is, it must only measure the significant aspects of knowledge or understanding.

Not encourage guesswork:

As far as possible, it should not encourage guesswork by the subjects.

Clear in reading:

It should not present any difficulty in reading.

Independent for its meaning:

ITEM ANALYSIS:

Item analysis uses statistics and expert judgment to evaluate tests based on the quality of individual items, item sets, and entire sets of items, as well as the relationship of each item to other items. It “investigates the performance of items considered individually either in relation to some external criterion or in relation to the remaining items on the test”. It uses this information to improve item and test quality. Item analysis concepts are similar

ITEM DIFFICULTY:

Item difficulty is the percentage of people who answer an item correctly. It is the relative frequency with which examinees choose the correct response. It has an index ranging from a low of 0 to a high of +1.00.

Higher difficulty indexes indicate easier items. An item answered correctly by 75% of the examinees has an item difficulty level

It should not be such that its meaning is dependent upon another item and/or it can be answered by referring to another item. for norm-referenced and criterion-referenced tests, but they differ in specific, significant ways.

The item analysis is one of the most important steps in the process of test construction. It gives us two kinds of information i.e. The idea about difficulty index of the item and index of validity; index of validity is also known as internal consistency or discriminative power of the statement.

of .75. An item answered correctly by 35% of the examinees has an item difficulty level of .35. item difficulty is a characteristic of the item and the sample that takes the test. For example, a vocabulary question that asks for synonyms for English nouns will be easy for American graduate students in English literature, but difficult for elementary children.

For example, assume that 25 people were tested and 23 of them answered the first item correctly.

$$\text{Difficulty} = \frac{23}{25} \times 100 = .904$$

ITEM DISCRIMINATION

It compares the number of high scorers and low scorers who answer an item correctly. It is the extent to which items discriminate among trainees in the high and low groups. The higher the discrimination index, the better the item because high values indicate that the item discriminates in favor of the up-per group which should answer more items correctly. If more low scorers answer an item correctly, it will have a negative

value and is probably flawed. A negative discrimination index occurs for items that are too hard or poorly written, which makes it difficult to select the correct answer. On these items poor students may guess correctly, while good students, suspecting that a question is too easy, may answer in-correctly by reading too much into the question. Good items have a discrimination index of .40 and higher; reasonably good items from .30 to .39; marginal items from .20 to .29, and poor items less than .20 (Ebel & Frisbie, 1986).

DISTRACTORS

Analyzing the distractors (i.e., incorrect alternatives) is useful in determining the relative usefulness of the decoys in each item. Items should be modified if students consistently fail to select certain multiple-choice alternatives. The alternatives are probably totally implausible and therefore of little use as decoys in multiple choice

items. A discrimination index or discrimination coefficient should be obtained for each option in order to determine each distractor's usefulness (Millman & Greene, 1993). Whereas the discrimination value of the correct answer should be positive, the discrimination values for the distractors should be lower and, preferably, negative.

TEST STANDARDIZATION: RELIABILITY, VALIDITY AND NORMS

RELIABILITY:

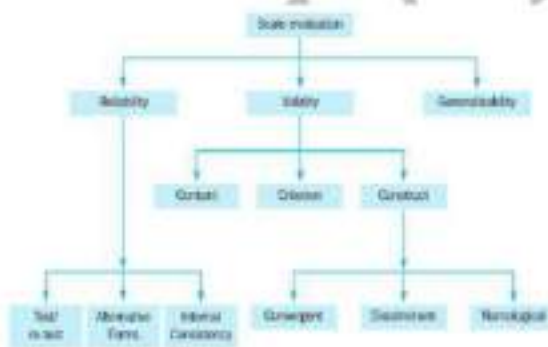
Reliability refers mainly to stability, internal consistency and equivalence of a measure. Reliability is one of the most essential characteristic of a test. If a test gives same result on different occasions, it is said to be reliable. So Reliability

means consistency of the test result, internal consistency and consistency of results over a period of time.

According to Anastasi and Ubrina (1982) "Reliability refers to the consistency of scores obtained by the same persons when they are re-examined with the same test on different occasions, or with different sets of equivalent items, or under other variable examining conditions."

Types of reliability	Definition	Example	Statistical tests
Stability	Consistency of repetitions, that is, how stable the measure is throughout time. ^{15,17}	If an individual concluded a research and repeats it in a few days, similar results are expected.	Test-retest (Intraclass correlation coefficient (ICC))
Internal consistency	It evaluates if the domains of an instrument measure the same characteristic, that is, the average correlation between all the construct items. ²¹	In an instrument that assesses satisfaction at work, all the items of a certain domain must measure such construct, not a different one.	Cronbach's alpha (continuous variables) Kuder-Richardson (dichotomous variables)
Equivalence	It is the concordance degree between two or more raters concerning the scores of an instrument.	Two qualified raters fill in the same instrument are supposed to obtain the same score.	Inter-observer reliability (Kappa)

METHODS OF ESTIMATING RELIABILITY:



There are different methods of establishing the reliability of the test. They are:

➤ INTER-RATER OR INTER-OBSERVER RELIABILITY

Measures the degree to which different observers give consistent estimates of the same persons. Often with essay and other constructed-response tests, two (or more) raters score individuals' performance. Each rater would provide a performance score for each individual (e.g., a total score summed over open-ended test items). Assuming "parallel raters" are created by selection and

training, akin to parallel tests, the correlation between the scores assigned by Rater 1 and by Rater 2 to persons can be used to determine interrater reliability – the consistency among raters (in the sense of relative or deviation scores assigned)

➤ STRATIFIED COEFFICIENT ALPHA

A reliability estimates that takes account this stratification is stratified coefficient alpha.

Tests may contain different types of items that can be categorized. For example, reading achievement tests may include both

vocabulary and reading comprehension items. Or, a battery of tests might be categorized into subsets as is done with the Wechsler intelligence test with both verbal (e.g., information, comprehension, arithmetic) and performance (e.g., block design, picture arrangement, matrix reasoning) categories. When components (items or subtests) fall within categories or strata, we might view the composite as a result of stratified random sampling of subtests or items. When we have such stratification, we would expect that items or subtests within a stratum would correlate more highly with each other than with items or subtests in other strata. When the stratification holds up and correlations within strata are higher than those between

strata, coefficient alpha will be smaller than stratified alpha.

TEST - RE-TEST RELIABILITY

Measures the consistency of measurement on two separate occasions. This type of Reliability is estimated by the Pearson product - moment coefficient of correlations between two administrations of the same inventory. Estimation is based on the correlation between scores of two or more administrations of the same inventory. In designing a reliability study to produce two sets of observations, one might give the same test on two occasions, separated (say) by two weeks.

ADVANTAGES:

Self-correlation or test-retest method, for estimating reliability coefficient is generally used. It is worthy to use in different situations conveniently. A test of an adequate length can be used after an interval of many days between successive testing.

DISADVANTAGES:

If the test is repeated immediately, many subjects will recall their first answers and spend their time on new material, thus tending to increase their scores—sometimes by a good deal.

Besides immediate memory effects, practice and the confidence induced by familiarity with the material will almost certainly affect scores when the test is taken for a second time. Index of reliability so obtained is less accurate.

If the interval between tests is rather long (more than six months) growth factor and maturity will affect the scores and tends to lower down the reliability index.

If the test is repeated immediately or after a little time gap, there may be the possibility of carry-over effect/transfer effect/memory/practice effect.

On repeating the same test, on the same group second time, makes the students disinterested and thus they do not like to take part wholeheartedly.

Sometimes, uniformity is not maintained which also affects the test scores.

Chances of discussing a few questions after the first administration, which may increase the scores at second administration affecting reliability.

PARALLEL-FORMS RELIABILITY

Estimating reliability by means of the equivalent form method involves the use of two different but equivalent forms of the test. Parallel form reliability is also known as Alternative form reliability or Equivalent form reliability or Comparable form reliability.

In this method two parallel or equivalent forms of a test are used. By parallel forms we mean that the forms are equivalent so far as the content, objectives, format, difficulty level and discriminating value of items, length of the test etc. are concerned.

Parallel tests have equal mean scores, variances and inter co-relations among items. That is, two parallel forms must be homogeneous or similar in all respects, but not a duplication of test items. Let the two forms be Form A and Form B.

The reliability coefficient may be looked upon as the coefficient correlation between the scores on two equivalent forms of test. The two equivalent forms are to be possibly similar in content, degree, mental processes tested, and difficulty level and in other aspects.

One form of the test is administered on the students and on finishing immediately another form of test is supplied to the same group. The scores, thus obtained are correlated which gives the estimate of reliability. Thus, the reliability found is called coefficient of equivalence.

Gulliksen 1950: has defined parallel tests as tests having equal means, equal variance and equal inter co-relations.

Guilford: The alternative form method indicates both equivalence of content and stability of performance.

ADVANTAGES:

This procedure has certain advantages over the test-retest method:

Here the same test is not repeated.

Memory, practice, carryover effects and recall factors are minimized and they do not effect the scores.

The reliability coefficient obtained by this method is a measure of both temporal stability and consistency of response to different item samples or test forms. Thus, this method combines two types of reliability.

Useful for the reliability of achievement tests.

This method is one of the appropriate methods of determining the reliability of educational and psychological tests.

LIMITATIONS:

It is difficult to have two parallel forms of a test. In certain situations (i.e. in Rorschach) it is almost impossible.

When the tests are not exactly equal in terms of content difficulty, length, the comparison between two set of scores obtained from these tests may lead to erroneous decisions.

Practice and carryover factors cannot be completely controlled.

Moreover, administering two forms simultaneously creates boredom. That is

why people prefer such methods in which only one administration of the test is required.

5. The testing conditions while administering the Form B may not be the same. Besides, the testees may not be in a similar physical, mental or emotional state at both the times of administration.



INTERNAL CONSISTENCY RELIABILITY

Measures the consistency of results across items within a test.

SPLIT - HALF RELIABILITY

Split-half method is an improvement over the earlier two methods, and it involves both the characteristics of stability and equivalence. The above discussed two methods of estimating reliability sometimes seems difficult.

It may not be possible to use the same test twice and to get an equivalent forms of test. Hence, to overcome these difficulties and to reduce memory effect as well as to economize the test, it is desirable to estimate reliability through a single administration of the test.

In this method the test is administered once on the sample and it is the most appropriate method for homogeneous tests. This method provides the internal consistency of a test scores.

Test scores of second form of the test are generally high.

Although difficult, carefully and cautiously constructed parallel forms would give us reasonably a satisfactory measure of reliability. For well-made standardized tests, the parallel form method is usually the most satisfactory way of determining the reliability.

All the items of the test are generally arranged in increasing order of difficulty and administered once on sample. After administering the test it is divided into two comparable or similar or equal parts or halves.

The scores are arranged or are made in two sets obtained from odd numbers of items and even numbers of items separately. As for example a test of 100 items is administered.

The scores of individual based on 50 items of odd numbers like 1, 3, 5, 99 and scores based on even numbers 2, 4, 6... 10 are separately arranged. In part 'A' odd number items are assigned and part 'B' will consist of even number of items.

After obtaining two scores on odd and even numbers of test items, co-efficient of correlation is calculated. It is really a correlation between two equivalent halves of scores obtained in one sitting. To estimate reliability, Spearman-Brown Prophecy formula is used.

The Spearman-Brown formula is given by:

$$r_{11} = \frac{2r_{11/22}}{1 + r_{11/22}}$$

in which r_{11} = the reliability of the whole test.

$r_{11/22}$ = the coefficient of correlation between two half tests.

Split-half can also be calculated using:

Rulon - Guttman's formulas: In this method, the variance of the differences between each person's scores on the two half-tests and the variance of total scores are considered.

Flanagan Formula: This formula is very close to Rulon's formula. In this formula the variance of two halves are added instead of difference between two halves.

ADVANTAGES:

Here we are not repeating the test or using the parallel form of it and thus the testee is not tested twice. As such, the carry over effect or practice effect is not there.

In this method, the fluctuations of individual's ability, because of environmental or physical conditions is minimized.

Because of single administration of test, day-to-day functions and problems do not interfere.

Difficulty of constructing parallel forms of test is eliminated.

coefficient of correlation in each case may be different.

This method cannot be used for estimating reliability of speed tests.

As the test is administered once, the chance errors may affect the scores on the two halves in the same way and thus tending to make the reliability coefficient too high.

This method cannot be used in power tests and heterogeneous tests.

In spite of all these limitations, the split-half method is considered as the best of all the methods of measuring test reliability, as the data for determining reliability are obtained upon occasion and thus reduces the time, labor and difficulties involved in case of second or repeated administration.

LIMITATIONS:

1. A test can be divided into two equal halves in a number of ways and the

• METHOD OF RATIONAL EQUIVALENCE:

This method is also known as "Kuder-Richardson Reliability" or '**Inter-Item**

Consistency'. It is a method based on single administration. It is based on consistency of responses to all items. The most common way for finding inter-item consistency is through the formula developed by Kuder and Richardson (1937). This method enables

to compute the inter-correlation of the items of the test and correlation of each item with all the items of the test. J. Cronbach called it as coefficient of internal consistency.

In this method, it is assumed that all items have same or equal difficulty value,

correlation between the items are equal, all the items measure essentially the same ability and the test is homogeneous in nature. Like split-half method this method also provides a measure of internal consistency.

The most popular formula is Kuder-Richardson i.e. KR-21 which is given below:

KR-21 :

$$r = \frac{n}{n-1} \left[1 - \frac{\sum pq}{\sigma_x^2} \right]$$

Where: n = Number of items in the test

$$p = \frac{\text{Number of persons answering item correctly}}{\text{Number of persons taking the test}}$$

$$q = \frac{\text{Number of persons answering item wrongly}}{\text{Number of persons taking the test}}$$

Σ = Summation sign indicating the pq is summated over all items

σ_x^2 = Variance of the total test.

r = Reliability of a test

$$q = 1 - p$$

$$p = 1 - q$$

An example will help us to calculate p and q .

EXAMPLE

60 students appeared a test and out of them 40 students have given correct response to a particular item of the test.

$$p = 40/60 = 2/3$$

This means $\frac{2}{3}$ portion of students have given correct response to one particular item of the test. In which 20 students have given incorrect response to that item.

$$\text{Thus } q = 20/60 \text{ or } 1 - 40/60$$

For each item we are to find out the value of p and q then pq is summated over all items to get $\sum pq$. Multiply p and q for each item and sum for all items. This gives $\sum pq$.

ADVANTAGES:

This coefficient provides some indications of how internally consistent or homogeneous the items of the tests are.

Rational equivalence is superior to the split-half technique in certain theoretical aspects, but the actual difference in reliability coefficients found by the two methods is often negligible.

Split-half method simply measures the equivalence but rational equivalence method measures both equivalence and homogeneity. Economical method as the test is administered once.

It neither requires administration of two equivalent forms of tests nor it requires to split the tests into two equal halves.

LIMITATIONS:

The coefficient obtained by this method is generally somewhat lesser than the coefficients obtained by other methods.

If the items of the tests are not highly homogeneous, this method will yield lower reliability coefficient.

Kuder-Richardson and split-half method are not appropriate for speed test. Different KR formula yield different reliability index.

CRONBACH'S ALPHA

This method is commonly used as a measure of internal consistency or reliability of a test. This was developed by Lee Cronbach in 1951. As an extension of the Kuder-Richardson formula (KR20). This method uses the variance of scores of odd, even and total items to work out the reliability. The coefficient alpha, or Cronbach's alpha, is the average of all possible split-half coefficients resulting from different ways of splitting the scale items.⁴⁷ This coefficient varies from 0 to 1, and a value of 0.6 or less generally indicates

Unsatisfactory internal consistency reliability. An important property of coefficient alpha is that its value tends to increase with an increase in the number of scale items. Therefore, coefficient alpha may be artificially, and inappropriately, inflated by including several redundant scale items. Another coefficient that can be employed in conjunction with coefficient alpha is coefficient beta.

Coefficient beta assists in determining whether the averaging process used in calculating coefficient alpha is masking any inconsistent items.

Reliability: Internal consistency

- Kuder-Richardson Formula 20 (KR-20)
 - First published in 1937
 - Designed for dichotomous (1/0, right/wrong) items
 - Values range from 0 to 1 (closer to 1 = higher reliability)
- Cronbach's Alpha
 - Cronbach published in 1951
 - Designed for dichotomous and non-dichotomous (continuous, 1 to 5) items
 - Generally values range from 0 to 1 (closer to 1 = higher reliability)
- Used in the Test Analysis Report and RMS

FACTORS INFLUENCING THE RELIABILITY OF TEST SCORES

(A) Intrinsic Factors:

The principal intrinsic factors (i.e. those factors which lie within the test itself) which affect the reliability are:

(i) Length of the Test:

Reliability has a definite relation with the length of the test. The more the number of items the test contains, the greater will be its

reliability and vice-versa. Logically, the more sample of items we take of a given area of knowledge, skill and the like, the more reliable the test will be. However, it is difficult to ensure the maximum length of the test to ensure an appropriate value of reliability. The length of the tests in such case should not give rise to fatigue effects in the testees, etc. Thus, it is advisable to use longer tests rather than shorter tests. Shorter tests are less reliable.

The number of times a test should be lengthened to get a desirable level of reliability is given by the formula:

$$n = \frac{r_{dx}(1 - r_1)}{r_1(1 - r_{dx})}$$

Where r_{dx} = the desired reliability

r_1 = the obtained reliability and

n = number of times a test is to be lengthened.

When a test has a reliability of 0.8, the number of items the test has to be lengthened to get a reliability of 0.95 is estimated in the following way:

Here r_{dx} = the desired reliability = .95

r_1 = the obtained reliability = .8

$$\text{Thus, } n = \frac{.95(1 - .8)}{.8(1 - .95)} = \frac{.95 \times .2}{.8 \times .05} = \frac{.19}{.04} = 4.75.$$

Hence the test is to be lengthened 4.75 times. However, while lengthening the test one should see that the items added to increase the length of the test must satisfy the conditions such as equal range of difficulty, desired discrimination power and comparability with other test items.

(II) HOMOGENEITY OF ITEMS:

Homogeneity of items has two aspects: item reliability and the homogeneity of traits measured from one item to another. If the items measure different functions and the inter-correlations of items are 'zero' or near to it, then the reliability is 'zero' or very low and vice-versa.

(III) DIFFICULTY VALUE OF ITEMS:

The difficulty level and clarity of expression of a test item also affect the reliability of test scores. If the test items are too easy or too difficult for the group members it will tend to produce scores of low reliability. Because

both the tests have a restricted spread of scores.

(IV) DISCRIMINATIVE VALUE:

When items can discriminate well between superior and inferior, the item total-correlation is high, the reliability is also likely to be high and vice-versa.

(V) TEST INSTRUCTIONS:

Clear and concise instructions increase reliability. Complicated and ambiguous directions give rise to difficulties in understanding the questions and the nature

of the response expected from the testee ultimately leading to low reliability.

(VI) ITEM SELECTION:

If there are too many interdependent items in a test, the reliability is found to be low.

(B) Extrinsic Factors:

The important extrinsic factors (i.e. the factors which remain outside the test itself) influencing the reliability are:

GROUP VARIABILITY: When the group of pupils being tested is homogeneous in ability, the reliability of the test scores is likely to be lowered and vice-versa.

(II) GUESSING AND CHANCE ERRORS: Guessing in test gives rise to increased error variance and as such reduces reliability. For example, in two-alternative response options there is a 50% chance of answering the items correctly in terms of guessing.

(VII) RELIABILITY OF THE SCORER:

The reliability of the scorer also influences reliability of the test. If he is moody, fluctuating type, the scores will vary from one situation to another. Mistake in him give rises to mistake in the score and thus leads to reliability.

(III) ENVIRONMENTAL CONDITIONS: As far as practicable, testing environment should be uniform. Arrangement should be such that light, sound, and other comforts should be equal to all testees, otherwise it will affect the reliability of the test scores.

(IV) MOMENTARY FLUCTUATIONS: Momentary fluctuations may raise or lower the reliability of the test scores. Broken pencil, momentary distraction by sudden sound of a train running outside, anxiety regarding non-completion of home-work, mistake in giving the answer and knowing no way to change it are the factors which may affect the reliability of test score.

Types of validity	Definition	Example	Statistical tests
Content validity	It is the degree in which a test includes all the necessary items to represent the concept to be measured. ¹⁷	An instrument that assesses the satisfaction at work must include not only work satisfaction, but other variables related to it, such as, salary, promotions, relationship with co-workers, among others.	- Qualitative approach (experts committee) - Quantitative approach (content validity index (IVC))
Criterion validity	It is assessed when a result can be compared to a 'gold standard'.		
Concurrent validity	It can be evaluated using both the target-test and the 'gold standard', at the same time,	In an investigation on depression, a new tool is used, and with it, a supposedly 'gold standard' question: Do you frequently feel sad or depressed? ¹⁸	Correlation tests
Predictive validity	First the target-test is applied, and then, the 'gold standard'. ¹⁸	Results on blood pressure and cholesterol levels are based on its predictive validity to project the risk of cardiovascular diseases. ¹⁸	Correlation tests
Construct validity	Is the extent in which a set of variables represent the construct that was projected to be measured. ¹⁴		
Known-groups technique	Different groups of individuals fill in the research instrument and then the groups' results are compared. ¹⁹	A test that assesses quality of life can be applied to a group of patients with chronic diseases and to a group of healthy youngsters. Differences in the scores on quality of life between these groups are expected. ¹⁹	Hypothesis testing
Convergent validity	It is obtained through the correlation between the instrument and another instrument that assesses a similar construct, expecting high correlation results between them. ²⁰	When administering two instruments that assess satisfaction at work, researchers expect to obtain strong correlation between them.	Correlation tests
Discriminant validity	It tests the hypothesis that the target-measurement is not improperly related to different constructs, that is, with variables from which it should differ. ²⁰	An instrument that assesses the motivation to work should present low correlation with an instrument that measures self-efficiency. ²¹	Correlation tests
Structural or factorial validity	It assesses if one measure captures the hypothetical dimension of a construct. ¹⁹	Researchers intend to assess if some characteristics of the work environment – such as autonomy and feedback – are predictors of professional satisfaction.	Factorial analysis and structural equation modeling
Cross-cultural validity	Measures in which the evidences support the inference that the original instrument and another one, culturally adapted are equivalent. ²⁶	A tool that assesses the satisfaction at work and that has been translated and adapted into another cultural context, must have a similar performance to the one of the original version. ²¹	- Independent translators and back-translators - Experts committee - Pre-test ²¹

ERRORS IN RELIABILITY:

At a time, scores are not consistent because some other factors also affect reliability e.g.

Noise
Health
Time

There is always a chance of 5% error in reliability which is acceptable.

TYPES OF ERRORS

Random error
Systematic error

RANDOM ERROR

Random error exists in every measurement and is often major source of uncertainty. These errors have no particular assignable cause. These errors can never be totally eliminated or corrected. These are caused by many uncontrollable variables that are inevitable part of every analysis made by human being. These variables are impossible to identified, even if we identify some they cannot be measured because most of them are so small.

TYPES OF VALIDITY

Face validity estimates whether a test measure what it claims to measure. It is the extent to which a test seems relevant, important, and interesting. It is the least rigorous measure of validity. It pertains to the fact that whether test looks valid or not. For this, the proposed instrument is to be sent to the experts in the related area and are requested to

SYSTEMATIC ERROR

Systematic error is caused due to instruments, machines, and measuring tools. It is not due to individuals.

VALIDITY:

Validity is the extent to which a test measure what it is supposed to measure. It is the most critical dimension of test development. Simply stated, validity is what a test measure and how well it does this.

In 1954, the American Psychological Association (APA) defined four categories of validity including content, predictive, concurrent, and construct. In 1966, the association combined predictive and concurrent validity into a single grouping called criterion validity (American Psychological Association, 1966) which remains the current classification (American Educational Research Association, American Psychological Association, & National Council on Measurement and Education, 1985). These aspects of validity are often mistakenly considered as three types of validity rather than **a concept about how a score can be interpreted.**

judge whether it actually measures for what it is constructed.

Content validity is the degree to which a test matches a curriculum and accurately measures the specific training objectives on which a program is based. Typically, it uses expert judgment of qualified experts to determine if a test is accurate, appropriate, and fair.

Criterion-related validity measures how well a test compares with an

external criterion. It reflects whether a scale performs as expected in relation to other selected variables (criterion variables) as meaningful criteria. If, for example, a scale is designed to measure loyalty in customers, criterion validity might be determined by comparing the results generated by this scale with results generated by observing the extent of repeat purchasing. Based on the time period involved, criterion validity can take two forms, concurrent validity and predictive validity. It includes:

- a. **Predictive validity** is the correlation between a predictor and a criterion obtained at a later time (e.g., test score on a specific competence and caseworker performance of a job-related tasks). It is concerned with how well a scale can forecast a future criterion. To assess predictive validity, the researcher collects data on the scale at one point in time and data on the criterion variables at a future time. For example, attitudes towards how loyal customers feel to a particular brand could be used to predict future repeat purchases of that brand. The predicted and actual purchases are compared to assess the predictive validity of the attitudinal scale.
- b. **Concurrent validity** is the correlation between a predictor and a criterion at the same point in time (e.g., performance on a cognitive test related to training and scores on a Civil Service examination). It is assessed when the data on the scale being evaluated (e.g. loyalty scale) and

the criterion variables (e.g. repeat purchasing) are collected at the same time. The scale being developed and the alternative means of encapsulating the criterion variables would be administered simultaneously and the results compared.

4. Construct validity addresses the question of what construct or characteristic the scale is, in fact, measuring. When assessing construct validity, the researcher attempts to answer theoretical questions about why the scale works and what deductions can be made concerning the underlying theory. Thus, construct validity requires a sound theory of the nature of the construct being measured and how it relates to other constructs. Construct validity is the most sophisticated and difficult type of validity to establish. Construct validity includes convergent, discriminant and nomological validity.

Convergent validity is the extent to which the scale correlates positively with other measurements of the same construct. It is not necessary that all these measurements be obtained by using conventional scaling techniques.

Discriminant validity is the extent to which a measure does not correlate with other constructs from which it is supposed to differ. It involves demonstrating a lack of correlation among differing constructs.

Nomological validity is the extent to which the scale correlates in theoretically predicted ways with measures of different but related constructs. A theoretical model is formulated that leads to further deductions, tests and inferences.

An example of construct validity can be evaluated in the following example. A researcher seeks to provide evidence of construct validity in a multi-item scale, designed to measure the concept of 'self-image'. These findings would be sought:

High correlations with other scales designed to measure self-concepts and with reported classifications by friends (convergent validity)

Low correlations with unrelated constructs of brand loyalty and variety-seeking (discriminant validity)

Brands that are congruent with the individual's self-concept are more preferred, as postulated by the theory (nomological validity)

A high level of reliability.

This type of validity is concerned with what qualities does a test measure. It is evaluated by demonstrating that certain explanatory constructs account for some degree of performance on the test. Construct validity can be estimated by following two methods.

a. Internal Consistency

Test falling under personality domain are validated by this method. The essential character of this

HERE ARE SOME FACTORS WHICH AFFECT INTERNAL VALIDITY:

INTERNAL VALIDITY is affected by flaws within the study itself such as not controlling some of the major variables (a design problem), or problems with the research instrument (a data collection problem).

Subject variability
Size of subject population
Time given for the data collection or experimental treatment

method is the total score on the test itself. It verifies that a particular item or section measures the same characteristic individually that the test as a whole measures. It can be determined in two ways. (a) By comparing the performance of upper criterion group with that of the lower criterion group. Biserual correlation between 'Pass-Fail' on each item and total test score is computed and items yielding significant item test correlations are retained. (b) By correlating sub-test scores with total score. Any subtest having low correlation is estimated. Internal consistency correlation are essential as they help to characterize the behavior domain or trait sampled by the test.

b. Factorial Validity

Factor analysis is a refined statistical technique for analyzing interrelationships of behavior data. The factorial validity of a test is the correlation between the test and the factor common to the test.

History
Attrition
Maturation
Instrument/task sensitivity

HERE ARE SEVEN IMPORTANT FACTORS AFFECT EXTERNAL VALIDITY:

EXTERNAL VALIDITY is the extent to which you can generalize your findings to a larger group or other contexts. If your research lacks external validity, the findings cannot be applied to contexts other than the

one in which you carried out your research. For example, if the subjects are all males from one ethnic group, your findings might not apply to females or other ethnic groups. Or, if you conducted your research in a highly controlled laboratory environment, your findings may not faithfully represent what might happen in the real world.

Population characteristics (subjects)

Interaction of subject selection and research

Descriptive explicitness of the independent variable

The effect of the research environment

Researcher or experimenter effects

Data collection methodology

The effect of time

RELIABILITY & VALIDITY

If a test is heterogeneous, it has low reliability and high validity.

On the other hand, maximum validity requires items differing in difficulty and low inter-correlation among items.

The validity of a test may not be higher than the reliability index.

Validity is the proportion of common factor variance.

A valid test is always reliable. If a test truthfully measures what it purports to measure is both valid and reliable.

A reliable test may not be Valid. A test maybe reliable but poor on validity.

NORMS:

A **norm** is the average or typical score on a particular test obtained by a set/group of defined individuals. Norms of the test are based on the distribution of scores obtained by the people of the standardization group. To develop norms, a test is administered to a large sample of individuals, and the **distribution of scores obtained by those individuals represent the norms of the test.**

In the development and application of test norms, considerable attention should be given to the standardization sample:

GENERALIZABILITY

Generalizability refers to the extent to which one can generalize from the observations at hand to a universe of generalizations. The set of all conditions of measurement over which the investigator wishes to generalize is the universe of generalization. These conditions may include items, interviewers, and situations of observation. A researcher may wish to generalize a scale developed for use in personal interviews to other modes of data collection, such as mail and telephone interviews. Likewise, one may wish to generalize from a sample of items to the universe of items, from a sample of times of measurement to the universe of times of measurement, from a sample of observers to a universe of observers, and so on.

The sample should be large enough to provide stable values

Sample must be representative of the population under consideration

For example, the mean intelligence test score for a group of 10-year olds. The raw score, that is, the actual number of units or points, obtained by an individual on a test does not in itself have much, if any, significance. One test may yield a score of 43 and cannot be directly compared with a score of 43 on another test.

The scores on a psychological test provide a relative, rather than an absolute, measure of

a test taker's performance. Scores on psychological tests are most commonly interpreted by reference to **norms** which represent the test performance of the standardization sample

Norms serve a **dual** purpose: -

They indicate an individual's relative position in the normative sample,

TYPES AND METHODS OF CALCULATING NORMS:

1. QUANTITATIVE NORMS:

They allow us to evaluate an individual's performance in terms of the performance of most nearly comparable standardization sample. They have a uniform and a clearly defined quantitative meaning. The examples

and thus permit an evaluation of his performance in reference to others in the normative sample.

They provide comparable measures which permit a direct comparison of performance on different tests. Norms only provide a frame of reference to interpret scores on a test.

of these types of norms are percentiles, deciles, standard scores and some variations of standard scores

Percentile norms

Deciles

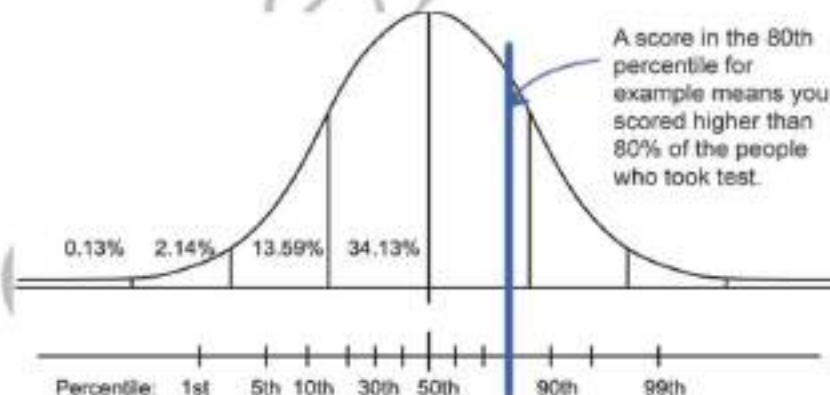
Standard score

T-score and z-score

Stanine

A. PERCENTILE NORMS

A percentile rank is a type of converted score that expresses a student's score relative to their group in percentile points.



This indicates the percentage of students tested who made scores equal to or lower than the specified score. If the percentile norms is to be made meaningful, it should be a sample which is made homogenous with respect to gender, age and other factors.

Easy to calculate.

Easy to understand and interpret.

No assumption about the characteristics of populations (not telling about how much have one scored but how better has one scored).

Advantages of Percentile Ranks:

B. DECILE

Deciles are points which divide the scale of measurement into 10 equal parts. Range of deciles is from Decile 1 - Decile 9.

There are Deciles scores and deciles ranks. Deciles score 1 indicates that 10 % of cases lie below you therefore Deciles rank 1 indicates that you are among the lowest 10 % of the group.

Deciles are based on same principles as of percentile instead of 1/100 it uses 1/10 parts of the group.

Deciles score 1 = PR 10. Decile 1 (lowest) and Decile 9 (highest).

C. STANDARD SCORES

These are derived scores, which designates an individual's position with respect to the total range and distribution of scores. Standard scores indicate, in terms of standard deviation as to how far a particular score is from the mean of the distribution. Mean is taken as zero point and therefore stand scores can be minus or plus. As they have fixed mean and standard deviation and thus there are different types of standard

scores like z scores, T scores, Stanine scores etc.

Advantages:

- Allow comparison of same person on different tests.
- Have equal units of measurement.

TRANSFORMATION

Raw scores can be converted into standard scores by two methods-

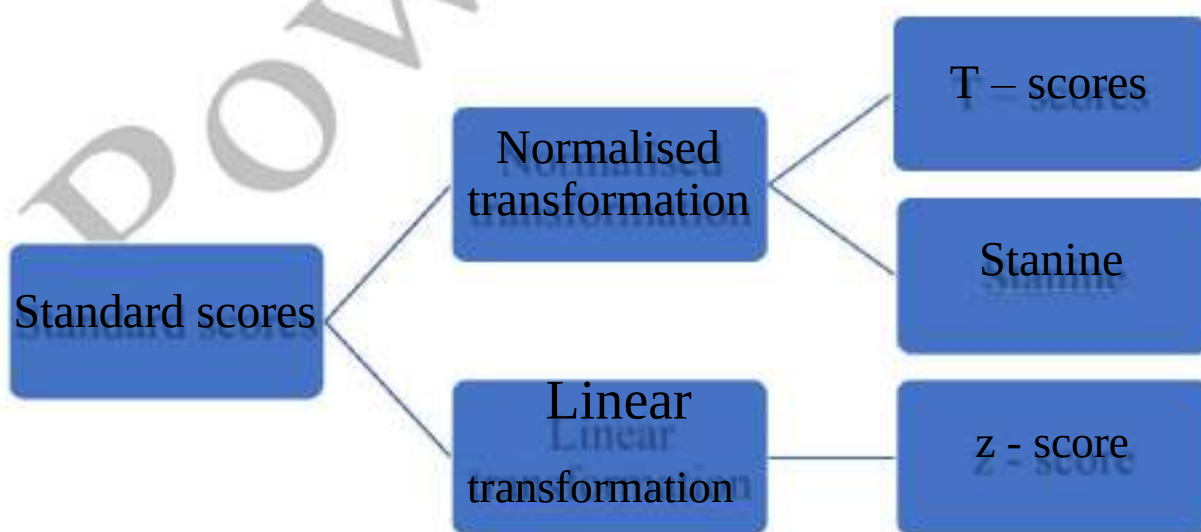
- i. Linear Transformation
- ii. Normative Transformation

For Linear Transformation:

► All characteristics of original distribution of raw score are retained without any change in distribution.

Example: z-scores (0 mean, 1 SD).

They can be compared only if both the distributions are approximately same. If one is skewed and other is normally distributed, then they cannot be compared.



For Normative Transformation:

Here, the raw scores of skewed distributions are adjusted to produce a normal frequency distribution and converted to a standard base.

Example: T-score and Stanine scores.

T-SCORE: Developed by McCall (1922)

T - Scores are standard scores which are based on: Mean of 50, Standard deviation of 10. Range 20-80.

STANINE: Developed by US Air Force during World War II.

It is an abbreviation for “standard nine”.

Scores are expressed in digits ranging 1 to 9. 1 (lowest) and 9 (highest).

Characteristics: Mean = 5, SD = 1.96

2. QUALITATIVE NORM:

Developmental norms are developed for psychological constructs which develop; for example, intelligence, reasoning, vocabulary, mathematical ability, etc. They are supplemented by percentiles or standard scores. They have considerable appeal for descriptive purposes, and for certain research purpose.

Age norms

Grade norms

Gender Norms

A. AGE NORMS:

An age norm relates the level of performance to the age of the people taken the test. It is the **median** score on a test obtained by persons of a given chronological age.

While age norms tend to **emphasize the average level** at a given age, it is important to remember that there is considerable variability within the same age. It implies that some children at one age will perform on a test will perform similarly to children at other ages.

B. GRADE NORMS:

A grade norm is the **median** score on a test obtained by students of a given grade level. They are most popular when reporting **achievement levels** of school children, and are useful for teachers to understand as to how well the students are progressing at a grade level.

SHORTCOMINGS:

Also, age and grade norms assume that the growth in psycho-motor, cognitive, and affective domains is uniform over the entire range of age or grade. But that is not the case.

Therefore, these norms can be misinterpreted to be implying that the rate of increase in tested abilities is constant from year to year.

CONCEPTS OF SCALING AND MEASUREMENT:

Four primary scales of measurement:
nominal, ordinal, interval and ratio.

Scaling Techniques:

Comparative scaling techniques, consisting of paired comparison, rank order, constant sum and Q-sort scaling

Non-comparative techniques are composed of continuous and itemized rating scales.

Popular itemized rating scales – the Likert, semantic differential and Stapel scales

Measurement means assigning numbers or other symbols to characteristics of objects according to certain pre-specified rules. We measure not the object but some characteristic of it. Thus, we do not measure consumers, only their perceptions, attitudes, preferences or other relevant characteristics. In marketing research, numbers are usually assigned for one of two reasons. First, numbers permit statistical analysis of the

resulting data. Second, numbers facilitate universal communication of measurement rules and results. The most important aspect of measurement is the specification of rules for assigning numbers to the characteristics. The assignment process must be isomorphic i.e., there must be one-to-one correspondence between the numbers and the characteristics being measured.

Scaling may be considered an extension of measurement. Scaling involves creating a continuum upon which measured objects are located. To illustrate, consider a scale for locating consumers according to the characteristic 'attitude towards banks'. Each respondent is assigned a number indicating an unfavorable attitude (measured as 1), a neutral attitude (measured as 2) or a favorable attitude (measured as 3). Measurement is the actual assignment of 1, 2 or 3 to each respondent. Scaling is the process of placing the respondents on a continuum with respect to their attitude towards banks. In our example, scaling is the process by which respondents would be classified as having an unfavorable, neutral or positive attitude.

Scales of measurement: nominal, ordinal, interval and ratio.

Table 1.3 Scales of Measurement and their Properties			
Type of Scale	Magnitude	Property	
		Equal Intervals	Absolute Zero
Nominal	No	No	No
Ordinal	Yes	No	No
Ratio	Yes	Yes	No
Interval	Yes	Yes	Yes

Although a larger number of scales exist and can be created for measuring attributes of people, objects, events, and so on, all scales belong to one of the four basic types:

1. Nominal,

Ordinal,
Interval and
Ratio.

These scales are actually four hierarchies of measurement procedures, the lowest in the

hierarchy being the Nominal Scale measurement and highest Ratio Scale measurement. That is why the expression 'levels of measurement' has been used by some scholars for scales of measurement.

1. NOMINAL SCALE

The word “*nominal*” is derived from the Latin word for “*name*”. Nominal measurements are also called categorical measurements as they reflect qualitative differences rather than quantitative ones. Common examples include categories such as yes/no, pass/fail, male/female or Conservative/Liberal/ Labor. When setting up a categorical measurement system the only requirements are those of **mutual exclusivity** and **exhaustiveness**. Mutual exclusivity means that each observation (person, case, score) cannot fall into more than one category; one cannot, for example, both pass and fail a test at the same time. Exhaustiveness simply means that your category system should have enough categories for all the observations. For biological sex there should be no observations (in this case people) who are neither male nor female.

This scale, some valid operations are equivalence and set membership. Therefore, nominal measures offer names or labels for certain characteristics and the variables assessed on a nominal scale are called categorical variables.

2. ORDINAL SCALE

An ordinal measurement describe order, but not relative size or degree of difference between the items measured. In this scale type, the numbers are assigned to the order, such as first, second, third, and so on. It is the next level of measurement in terms of complexity.

For example, we might determine that runners in a race finished in a particular order, and this order would provide us with useful information. Also, a lecturer might be asked to rank order their students in terms of general ability at statistics. They could put each student into one of five categories: excellent, good, average, poor, appallingly bad. Anjali might fall into the ‘excellent’ category and Shivani into the ‘good’ category. Anjali is better at statistics than Shivani, but what we do not know is just how much better Anjali is than Shivani. The rankings reflect more or less of something but not how much more or less.

An ordinal scale does not convey precise quantitative information. With an ordinal scale, we know the rank order, but we do not have any idea of the distance or interval between the rankings. As before, the assumptions of mutual exclusivity and exhaustiveness apply and cases are still assigned to categories. The big difference is that now the categories themselves can be rank-ordered with reference to some external criterion. Thus, A Likert Scale is a type of ordinal scale and may also use names with an order such as: "bad", "medium", and "good"; or "very satisfied", "satisfied", "neutral", "unsatisfied", "very unsatisfied."

3. INTERVAL SCALE

In an Interval Scale, we can specify both the order of events and the distance between events. The distance between any two intervals on this type of scale is equal throughout the scale. The central shortcoming of an interval scale is its lack of an absolute zero point. This type of scale often has an arbitrary zero point, sometimes called an *anchor point*.

As, there is no fixed zero point with these types of scales, thus, the result of no fixed zero is best demonstrated through an example. Let's say we want to measure individuals' "need for achievement" with a particular scale. With an interval level of measurement, the scores can be interpreted to mean that someone with a score of 15 is as different from someone with a score of 20, as someone with a score of 20 compared to someone with a score of 25. However, there is no universally accepted level of "zero need for achievement." Therefore, we cannot conclude that someone with a score of 20 has half the need for achievement as does a person with a score of 40. In order to make that claim, we would need to have an absolute zero point on the scale.

Though very few Interval Scales have been developed in the areas of social sciences, we can achieve interval measurement of attributes such as (a) interest, attitude, personality, motivation, and so on, (b) status, reading interests, and so on and (c) recall, relevance or usefulness of information for this purpose. We can design numerical rating scales beginning from an arbitrary zero point representing the total absence of an attribute or quality being

measured, and increasing the value in successive equal units on the scale up to the desired limit.

4. RATIO SCALE

Ratio scale measurement differs from interval measurement only in that it implies the existence of a potential absolute zero value. Good examples of ratio scales are length, time and number of correct answers on a test. It is possible to have zero (no) length, for something to take no time, or for someone to get no answers correct on a test.

An important corollary of having an absolute zero is that, for example, if person A is six feet tall and person B is three feet tall, it is true to say that person A is twice as tall as person B. The ratio of scores to one another now carries some sensible meaning which was not the case for the interval scale. Thus, a Ratio Scale has all the characteristics of Nominal, Ordinal and Interval Scales and, in addition, an absolute or natural zero point representing the absence of magnitude of a variable/ attribute.

Scale	Basic characteristics	Common examples	Marketing example	Permissible statistics	
				Descriptive	Inferential
Nominal	Numbers identify and classify objects	Student registration numbers, numbers on football players' shirts	Gender classification, bank types	Percentages, mode	Chi-square, binomial test
Ordinal	Numbers indicate the relative positions of the objects but not the magnitude of differences between them	Rankings of the top 4 teams in the football World Cup	Ranking of service quality delivered by a number of banks, Rank order of favourite television programmes	Percentile, median	Rank-order correlation, Friedman ANOVA
Interval	Differences between objects can be compared; zero point is arbitrary	Temperature (Fahrenheit, Celsius)	Attitudes, opinions, index numbers	Range, mean, standard deviation	Product-moment correlations, t-tests, ANOVA, regression, factor analysis
Ratio	Zero point is fixed; ratios of scale values can be computed	Length, weight	Age, income, costs, sales, market shares	Geometric mean, harmonic mean	Coefficient of variation

PROBLEMS IN PSYCHOLOGICAL MEASUREMENTS

1. INDIRECTNESS OF MEASUREMENT

Various psychological attributes are accessible to research and measurement only indirectly.

For example, if a researcher is interested in measuring the personality dimensions of a subject, then it is something that is not directly available for measurement as physical quantities- like length- are visible and

concretely available for observation and assessment. The only way to measure it is to assess the person on a set of overt or covert responses (for example, by administering a psychological test) related to his personality or other psychological attributes of interest.

2. LACK OF ABSOLUTE ZERO

Absolute zero, in case of psychological measurement, means a situation where the property being measured does not exist. The absolute zero is available in case of physical quantities, like length, but is very difficult to decide in the case of psychological attributes.

WE MEASURE A SAMPLE OF BEHAVIOUR NOT THE COMPLETE BEHAVIOUR

In psychological measurements, a complete set of behavioral dimensions is not possible and we take only a carefully chosen sample of behavioral dimensions to assess the attributes in question.

Although the sample is chosen only after fulfilling the various psychometric criteria, like randomness, representativeness, and so on, it is always questionable to reach at a conclusion about an aspect of behavior, only by measuring a small, though representative part of it.

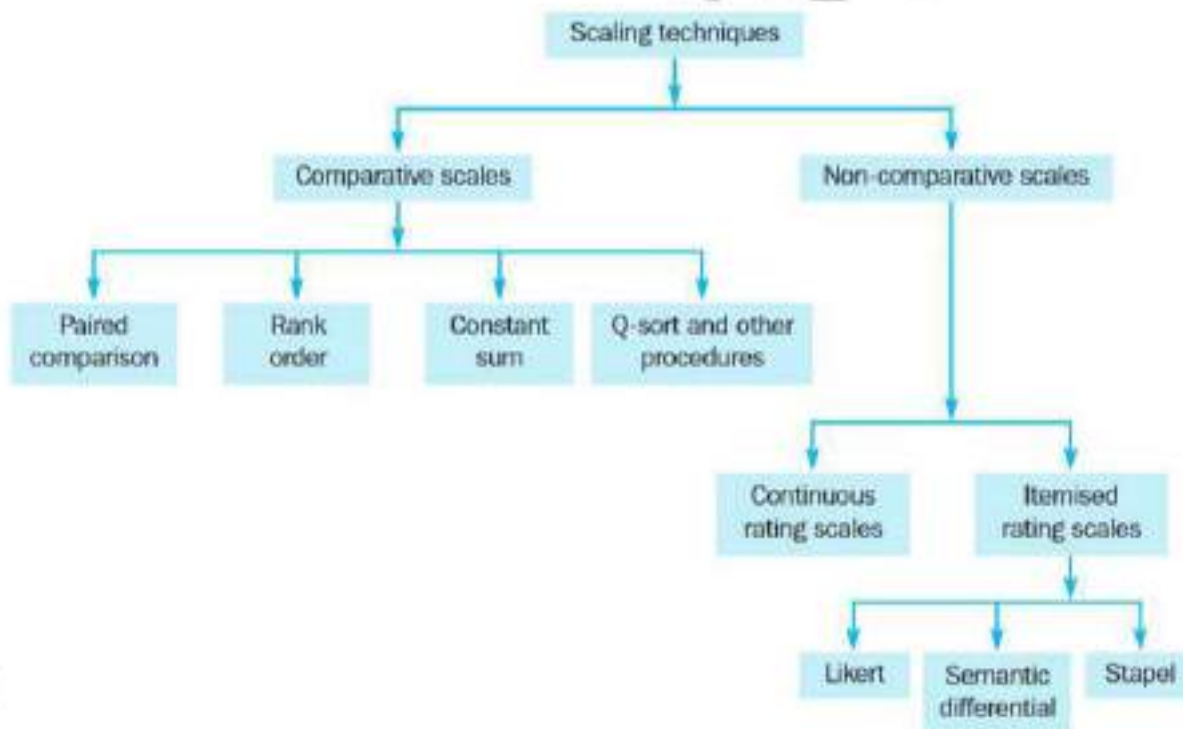
4. UNCERTAINTY AND DESIRABILITY INVOLVED IN HUMAN RESPONSES

Test subjects often give uncertain and desirable responses which generally negates the entire purpose of the psychological measurement. Uncertainty may arise either due to the negligence on part of the researcher, or carelessness on part of the subject(s), or due to uncontrolled extraneous variables.

5. VARIABILITY OF HUMAN ATTRIBUTES OVER TIME

Various human attributes, like intelligence, personality, attitude, and so on, are likely to vary over a period of time, and sometimes even hours are sufficient to provide scope for such variations. Psychological attributes are highly dynamic and they continuously undergo organization and reorganization. To capture these fluctuating attributes in terms of exact numbers is really an uphill task for any researcher.

SCALING TECHNIQUES:



COMPARATIVE TECHNIQUES

The scaling techniques commonly employed in marketing research can be classified into comparative and non-comparative scales.

Comparative scales involve the direct comparison of stimulus objects. For

SCALING

example, respondents may be asked whether they prefer Coke or Pepsi. Comparative scale data must be interpreted in relative terms and have only ordinal or rank order proper-ties. For this reason, comparative scaling is also referred to as non-metric scaling. **Comparative scales include**

paired comparisons, rank order, constant sum scales, Q-sort and other procedures.

The major benefit of comparative scaling is that small differences between stimulus objects can be detected. As they compare the stimulus objects, respondents are forced to choose between them. In addition, respondents approach the rating task from the same known reference points. Consequently, comparative scales are easily understood and can be applied easily. Other advantages of these scales are that they involve fewer theoretical assumptions, and they also tend to reduce halo or carryover effects from one judgement to another.⁶ The major disadvantages of comparative scales include the ordinal nature of the data and the inability to generalize beyond the stimulus objects scaled.

• PAIRED COMPARISON SCALING TECHNIQUE

As its name implies, in paired comparison scaling a respondent is presented with two objects and asked to select one according to some criterion. **The data obtained are ordinal in nature.** A respondent may state that he or she prefers Belgian chocolate to Swiss, likes Kellogg's cereals better than supermarket home brands, or likes Adidas more than Nike. **Paired comparison scales are frequently used when the stimulus objects are physical products.** Coca-Cola is reported to have conducted more than 190,000 paired comparisons before introducing New Coke. Paired comparison scaling is the most widely used comparative scaling technique.

In general, **with 'n' brands, $[n(n-1)/2]$ paired comparisons** include all possible pairings of objects.

Under the assumption of transitivity, it is possible to convert paired comparison data to a rank order. **Transitivity of preference** implies that if brand A is preferred to B, and brand B is preferred to C, then brand A is preferred to C.

Paired comparison scaling is useful when the number of brands is limited, since it requires direct comparison and overt choice. With a large number of brands, however, the number of comparisons becomes unwieldy. Other disadvantages are that violations of the assumption of transitivity may occur, and the order in which the objects are presented may bias the results.¹³ Paired comparisons bear little resemblance to the market place situation, which involves selection from multiple alternatives. Also respondents may prefer one object over certain others, but they may not like it in an absolute sense.

RANK ORDER SCALING

After paired comparisons, the most popular comparative scaling technique is rank order scaling. In rank order scaling respondents are presented with several objects simultaneously and asked to order or rank them according to some criterion. For example, respondents may be asked to rank brands of cars according to overall preference. These rankings are typically obtained by asking the respondents to assign a rank of 1 to the most preferred brand, 2 to the second most preferred, and so on, until a rank of 'n' is assigned to the least preferred brand. Like paired comparison, this approach is also comparative in nature, and it is possible that the respondent may dislike the brand ranked 1 in an absolute sense. Furthermore, rank order scaling also results in ordinal data.

Rank order scaling is commonly used to measure attributes of products and services as well as preferences for brands. Rank order data are frequently obtained from respondents in conjoint analysis, since rank order scaling forces the respondent to discriminate among the stimulus objects.

Moreover, compared with paired comparisons, this type of scaling process more closely resembles the shopping environment. It also takes less time and eliminates intransitive responses.

If there are 'n' stimulus objects, only (n– 1)/2 scaling decisions need be made in rank order scaling.

However, in paired comparison scaling, $[n(n-1)/2]$ decisions would be required. Another advantage is that most respondents easily understand the instructions for ranking. The major disadvantage is that this technique produces only ordinal data. Finally, under the assumption of transitivity, rank order data can be converted to equivalent paired comparison data, and vice versa.

CONSTANT SUM SCALING

In constant sum scaling, respondents allocate a constant sum of units, such as points or euros, among a set of stimulus objects with respect to some criterion. The respondents may be asked to allocate 100 points to attributes of bottled beers in a way that reflects the importance they attach to each attribute. If an attribute is unimportant, the respondent assigns it zero points. If an attribute is twice as important as some other attribute, it receives twice as many points. The sum of all the points is 100. Hence the name of the scale. The attributes are scaled by counting the points assigned to each one by all the respondents and dividing by the number of respondents. These results are presented for three groups, or segments, of respondents.

CONSTANT SUM SCALE DATA ARE SOMETIMES TREATED AS METRIC.

Although this may be appropriate in the limited context of the stimuli scaled, these results are not generalizable to other stimuli not included in the study. Hence, strictly speaking, the constant sum should be considered an ordinal scale because of its comparative nature and the resulting lack of generalizability.

The main advantage of the constant sum scale is that it allows for fine discrimination among stimulus objects without requiring too much time. It has two primary disadvantages, however. Respondents may allocate more or fewer units than those specified. For example, a respondent may allocate 108 or 94 points. The researcher must modify such data in some way or eliminate this respondent from analysis. Another potential problem is rounding error if too few units are used. On the other hand, the use of a large number of units may be too taxing on the respondent and cause confusion and fatigue.

• Q-SORT AND OTHER PROCEDURES

Q-sort scaling was developed to discriminate among a relatively large number of objects quickly. This technique uses a rank order procedure in which objects are sorted into piles based on similarity with respect to some criterion. For example, respondents are given 100 attitude statements on individual cards and asked to place them into 11 piles, ranging from 'most highly agreed with' to 'least highly agreed with'. The number of objects to be sorted should not be less than 60 nor more than 140; a reasonable range is 60 to 90 objects. The number of objects to be placed in each pile is pre-specified, often to result in a

roughly normal distribution of objects over the whole set. Another comparative scaling technique is magnitude estimation. In this technique, numbers are assigned to objects such that ratios between the assigned numbers reflect ratios on the specified criterion. For example, respondents may be asked to indicate whether they agree or disagree with each of a series of statements measuring attitude towards banks. Then they assign a number between 0 to 100 to each statement to indicate the intensity of their agreement or disagreement. Providing this type of number imposes a cognitive burden on the respondents.

Another particularly useful procedure (that could be viewed as a very structured combination of observation and depth interviewing) for measuring cognitive responses or thought processes consists of **verbal protocols**. Respondents are asked to 'think out loud' and verbalize anything going through their heads while making a decision or performing a task. The researcher says 'If you think anything, say it aloud, no matter how trivial the thought may be.' Even with such an explicit instruction, the respondent may be silent. At these times, the researcher will say 'Remember to say

aloud everything you are thinking.' Everything that the respondent says is tape recorded. This record of the respondent's verbalized thought processes is referred to as a protocol. Protocols have been used to measure consumers' cognitive responses in actual shopping trips as well as in simulated shopping environments. An interviewer accompanies the respondent and holds a microphone into which the respondent talks. Protocols, thus collected, have been used to determine the attributes and cues used in making purchase decisions, product usage behavior, and the impact of the shopping environment on consumer decisions. Protocol analysis has also been employed to measure consumer response to advertising. Immediately after seeing an ad, the respondent is asked to list all the thoughts that came to mind while watching the ad. The respondent is given a limited amount of time to list the thoughts so as to minimize the probability of collecting thoughts generated after, rather than during, the message. After the protocol has been collected, the individual's thoughts or cognitive responses can be coded into three categories as illustrated:

Category	Definition	Example
Support argument	Support the claim made by the message	'Diet Coke tastes great'
Counter-argument	Refute the claim made by the message	'Diet Coke has an aftertaste'
Source derogation	Negative opinion about the source of the message	'Coca-Cola is not an honest company'

Protocols are, typically, incomplete. The respondent has many thoughts that she or he

cannot or will not verbalize. The researcher must take the incomplete record and infer

from it a measure of the underlying cognitive response.

NON-COMPARATIVE SCALING TECHNIQUES

In non-comparative scales, also referred to as monadic or metric scales, each object is scaled independently of the others in the stimulus set. The resulting data are generally assumed to be interval or ratio scaled. For

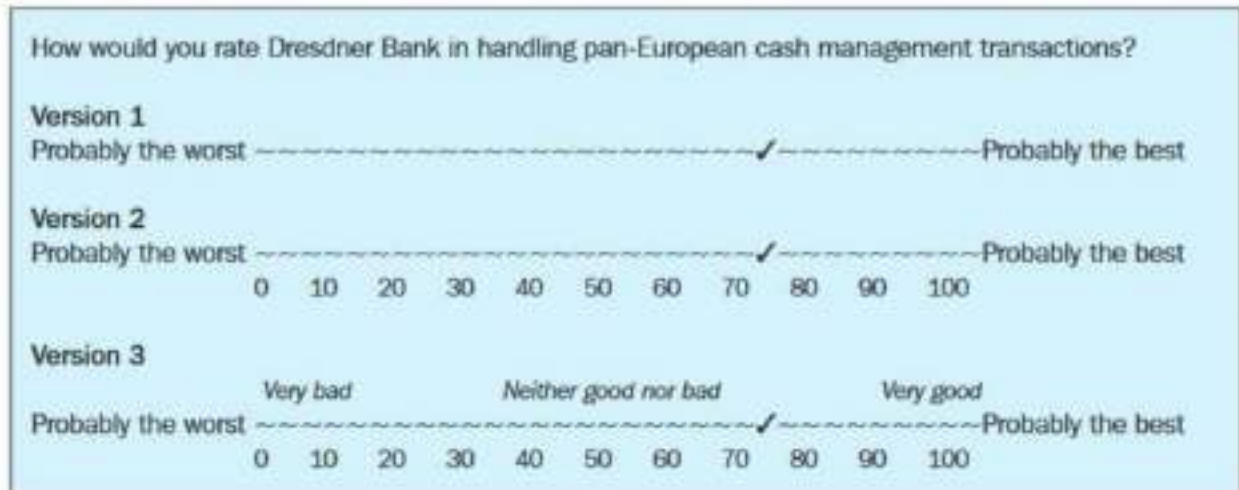
example, respondents may be asked to evaluate Coke on a 1 to 6 preference scale (1 = not at all preferred, 6 = greatly preferred). Similar evaluations would be obtained for Pepsi and Virgin Cola. As can be seen in Figure, non-comparative scales can be continuous rating or itemized rating scales. The itemized rating scales can be further classified as Likert, semantic differential or Stapel scales. Non-comparative scaling is the most widely used scaling technique in marketing research.

Scale	Basic characteristics	Examples	Advantages	Disadvantages
Continuous rating scale	Place a mark on a continuous line	Reaction to TV commercials	Easy to construct	Scoring can be cumbersome unless computerised
Itemised rating scales				
Likert scale	Degree of agreement on a 1 (strongly disagree) to 5 (strongly agree) scale	Measurement of attitudes	Easy to construct, administer and understand	More time-consuming
Semantic differential scale	Seven-point scale with bipolar labels	Brand product and company images	Versatile	Controversy as to whether the data are interval
Stapel scale	Unipolar 10-point scale, -5 to +5, without a neutral point (zero)	Measurement of attitudes and images	Easy to construct, administered over phone	Confusing and difficult to apply

CONTINUOUS RATING SCALE

It is also referred to as a **graphic rating scale**, respondents rate the objects by placing a mark at the appropriate position on a line that runs from one extreme of the criterion variable to the other. Thus, the respondents are not restricted to selecting from marks previously set by the researcher.

The form of the continuous scale may vary considerably. For example, the line may be vertical or horizontal; scale points, in the form of numbers or brief descriptions, may be provided; and if provided, the scale points may be few or many. Three versions of a continuous rating scale.



Once the respondent has provided the ratings, the researcher divides the line into as many categories as desired and assigns scores based on the categories into which the ratings fall.

The advantage of continuous scales is that they are easy to construct; however, scoring is cumbersome and unreliable. Moreover, continuous scales provide little new information. Hence, their use in marketing research has been limited. Recently, however, with the increased popularity of computer-assisted personal interviewing and other technologies, their use has become more frequent.

ITEMISED RATING SCALES

In an itemized rating scale, respondents are provided with a scale that has a number or brief description associated with each category. The categories are ordered in terms of scale position; and the respondents are required to select the specified category that best describes the object being rated. Itemized rating scales are widely used in marketing research and form the basic components of more complex scales, such as multi-item rating scales. We first describe the commonly used itemized rating scales – the Likert, semantic differential and Stapel scales.

LIKERT SCALE

Named after its developer, Rensis Likert, the Likert scale is a widely used rating scale that requires the respondents to indicate a degree of agreement or disagreement with each of a series of statements about the stimulus objects. Typically, each scale item has five response categories, ranging from ‘strongly disagree’ to ‘strongly agree’.

To conduct the analysis, each statement is assigned a numerical score, ranging either from –2 to +2 or from 1 to 5. The analysis can be conducted on an item-by-item basis (profile analysis), or a total (summed) score can be calculated for each respondent by summing across items. The Likert scale is also referred to as a summated scale.

The Likert scale has several advantages. It is easy to construct and administer, and respondents readily understand how to use the scale, making it suitable for Internet surveys, mail, telephone or personal interviews. The major disadvantage of the Likert scale is that it takes longer to complete than other itemized rating scales because respondents have to read and fully reflect upon each statement.

• SEMANTIC DIFFERENTIAL SCALE

The semantic differential is a seven-point rating scale with end points associated with bipolar labels that have semantic meaning.

In a typical application, respondents rate objects on a number of itemized, seven-point rating scales bounded at each end by one of two bipolar adjectives, such as 'cold' and 'warm'.

Instructions
What does Dresdner Bank mean to you? The following descriptive scales, bounded at each end by bipolar adjectives, summarises characteristics of the bank. Please mark **X** the blank that best indicates what the Dresdner Bank means to you.

Form
Dresdner Bank is:

Powerful	:	:	:	:	:	:	:	:	:	X	:	:	:	:	:	:	:	:	Weak
Unreliable	:	:	:	:	:	:	:	:	:	X	:	:	:	:	:	:	:	:	Reliable
Modern	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	X	Old-fashioned
Hi-tech	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	X	Low-tech
Careful	:	:	:	:	:	:	:	:	:	X	:	:	:	:	:	:	:	:	Careless

Individual items on a semantic differential scale may be scored either on a -3 to +3 or on a 1 to 7 scale. The resulting data are commonly analyzed through profile analysis. In profile analysis, means or median values on each rating scale are calculated and compared by plotting or statistical analysis. This helps determine the overall differences and similarities among the objects. To assess differences across segments of respondents, the researcher can compare mean responses of different segments. Although the mean is most often used as a summary statistic, there is some controversy as to whether the data obtained should be treated as an interval scale. On the other hand, in cases when the researcher requires an overall comparison of objects, such as to determine bank preference, the individual item scores are summed to arrive at a total score. Its versatility makes the semantic differential a popular rating scale in marketing research. It has been widely used in comparing brand, product and company images. It has also been used to develop advertising and promotion strategies and in new product development studies.

several modifications of the basic scale have been proposed.

STAPEL SCALE

The Stapel scale, named after its developer, Jan Stapel, is a unipolar rating scale with 10 categories numbered from -5 to +5, without a neutral point (zero).³⁰ This scale is usually presented vertically. Respondents are asked to indicate by selecting an appropriate numerical response category how accurately or inaccurately each term describes the object. The higher the number, the more accurately the term describes the object, as shown in Figure. In this example, Dresdner is perceived as not having high-quality products and having somewhat poor service. The data obtained by using a Stapel scale can be analyzed in the same way as semantic differential data. The Stapel scale produces results similar to the semantic differential. The Stapel scale's advantages are that it does not require a pre-test of the adjectives or phrases to ensure true bipolarity and that it can be administered over the telephone. Some researchers, however, believe the

Stapel scale is confusing and difficult to apply. Of the three itemized rating scales considered,

the Stapel scale issued least. Nonetheless, this scale merits more attention than it has received.

Instructions

Please evaluate how accurately each word or phrase describes Dresdner Bank. Select a positive number for the phrases you think describe the bank accurately. The more accurately you think the phrase describes the bank, the larger the plus number you should choose. You should select a minus number for the phrases you think do not describe the bank accurately. The less accurately you think the phrase describes the bank, the larger the negative number you should choose. You can select any number from +5 for phrases you think are very accurate, to -5 for phrases you think are very inaccurate.

Form

Dresdner Bank

+5	+5
+4	+4
+3	+3
+2	+2 ✗
+1	+1
<i>High quality</i>	<i>Poor service</i>
-1	-1
-2	-2
-3	-3
-4 ✗	-4
-5	-5

ITEMISED RATING SCALE DECISIONS

(REFERENCED CONTENT – OUT OF SYLLABUS BUT IS USEFUL)

As is evident from the discussion so far, non-comparative itemized rating scales can take many different forms. The researcher must make six major decisions when constructing any of these scales:

- a. The number of scale categories to use
- b. Balanced versus unbalanced scale
- c. Odd or even number of categories
- d. Forced versus non-forced choice
- e. The nature and degree of the verbal description
- f. The physical form of the scale.

1. Number of categories	Although there is no single, optimal number, traditional guidelines suggest that there should be between five and nine categories.
2. Balanced versus unbalanced	In general, the scale should be balanced to obtain objective data.
3. Odd or even number of categories	If a neutral or indifferent scale response is possible from at least some of the respondents, an odd number of categories should be used.
4. Forced versus unforced	In situations where the respondents are expected to have no opinion, the accuracy of the data may be improved by a non-forced scale.
5. Verbal description	An argument can be made for labelling all or many scale categories. The category descriptions should be located as close to the response categories as possible.
6. Physical form	A number of options should be tried and the best one selected.

BALANCED VERSUS UNBALANCED SCALE

In a balanced scale, the number of favorable and unfavorable categories is equal; in an unbalanced scale, the categories are unequal. Examples of balanced and unbalanced scales are given in Figure. In general, in order to obtain objective data, the scale should be

balanced. If the distribution of responses is likely to be skewed, however, either positively or negatively, an unbalanced scale with more categories in the direction of skewness may be appropriate. If an unbalanced scale is used, the nature and degree of imbalance in the scale should be taken into account in data analysis.

Balanced scale

Clinique moisturiser for men is:

Extremely good	
Very good	✓
Good	
Bad	
Very Bad	
Extremely bad	

Unbalanced scale

Clinique moisturiser for men is:

Extremely good	
Very good	✓
Good	
Somewhat good	
Bad	
Very bad	

FORCED VERSUS NON-FORCED CHOICE

On forced rating scales the respondents are forced to express an opinion because a 'no opinion' option is not provided. In such a case, respondents without an opinion may

mark the middle scale position. If a sufficient proportion of the respondents do not have opinions on the topic, marking the middle position will distort measures of central tendency and variance. In situations where the respondents are expected to have no opinion, as opposed to simply being reluctant to disclose it, the accuracy of data

may be improved by a non-forced scale that includes a 'no opinion' category.

COMPUTER-BASED PSYCHOLOGICAL TESTING

Computers have had a large impact on how tests can be administered. An important contribution of the computer is called computer adaptive testing (CAT). CAT is being widely adopted in tests of achievement and aptitude. In a CAT situation, often a small set—perhaps two or three—of midlevel difficulty locator items are given to all test takers. Depending on whether the individual passes or fails the items, subsequent items are presented that better match the test taker's ability level. So, if I fail all of the first locator items, the next item I get will be easier than the locator items. If I pass them all, I will be presented with a harder item. If I pass one and fail the others, I will be presented with an item at about the same difficulty as the locator items. Thus, whereas traditional tests present all test takers with the same items, in CAT,

each test taker gets, in essence, a different test, one that is customized to his or her own ability level. One valuable outcome of the CAT process is that a computer adaptive test is often shorter than a traditional test because fewer items are needed to determine test taker ability level. Thus, computer adaptive tests also take less time to administer. The ability level of the test taker is often very precisely determined using CAT because more items are presented at the test taker's individual capability level. In a more traditional testing situation, the test taker is presented with items that will be too easy and too difficult. Developing such tests takes tremendous amounts of resources. Item characteristics must be determined, and each item must be highly reliable to be included in the test item bank. To carry out the item analyses properly, thousands of cases are needed to have stable item parameter estimates. Therefore, it is not surprising that the large testing corporations are the ones most likely to develop CAT capacity in test delivery.

<i>Publisher</i>	<i>Website</i>	<i>Popular Published Tests</i>
Educational Testing Service	www.ets.org	• Advanced Placement (AP) Program Tests • Graduate Management Admission Test (GMAT) • Graduate Record Examinations (GRE) • Scholastic Assessment Test (SAT) • Test of English as a Foreign Language (TOEFL)
Pearson	www.pearsonassessments.com	• BarOn Emotional Quotient Inventory • Bayley Scales of Infant and Toddler Development—III • Bender Visual-Motor Gestalt Test—II • Watson–Glaser Critical Thinking Appraisal
Hogan Assessment Systems	www.hoganassessments.com	• Hogan Personality Inventory (HPI) • Hogan Development Survey (HDS) • Hogan Business Reasoning Inventory (HBRI) • Motives, Values, Preferences Inventory (MVPI)
IPAT	www.ipat.com	• 16 Personality Factors (16PF)
PAR	www3.parinc.com	• Self-Directed Search • NEO Personality Inventory • Personality Assessment Inventory • Slason Intelligence Test—Revised for Children and Adults
Psytech International	www.psytech.co.uk	• Occupational Interest Profile • Clerical Test Battery • Values and Motives Inventory
PSI	www.psionline.com	• Customer Service Battery • Firefighter Selection Test • Police Selection Test
Hogrefe	www.testagency.com	• Rorschach Inkblot Test • Trauma Symptom Inventory (TSI) • WPQ Emotional Intelligence Questionnaire
University of Minnesota Press Test Division	www.upress.umn.edu/tests/default.html	• Minnesota Multiphasic Personality Inventory (MMPI)
Wonderlic	www.wonderlic.com	• Wonderlic Personnel Test

AREAS OF TESTING: INTELLIGENCE, CREATIVITY, NEUROPSYCHOLOGICAL TESTS, APTITUDE, PERSONALITY ASSESSMENT, INTEREST INVENTORIES

LIST OF INDIAN TESTS:

http://www.ncert.nic.in/announcements/other/announcements/pdf_files/nlept_catalogue.pdf

The Mental Measurements Yearbook (MMY) is one of the most utilized sources of information about existing tests. The first edition was written in 1938 by Oscar Buros. Over the next 40 years, Buros published seven additional such yearbooks. These set

the standard for evaluation and critical assessment of tests and, in 1981, the Oscar K. Buros Center was established to continue the tradition of publishing the MMY. The most recent edition (Fifteenth Mental Measurements Yearbook) was edited by Plake, Impara, and Spies (2003). In addition, a Web site is available where reviews of specific tests can be accessed for a price (www.unl.edu/buros).

The fastest way to locate a test and/or obtain test reviews is through the World Wide Web. Test locator services can be accessed via www.unl.edu/buros, and www.ets.org, PsycINFO, PsycLIT, and ERIC searches pull up tests used in all types of research. The periodically updated Directory of

Unpublished Experimental Measures (Goldman & Mitchell, 2002) provides another resource. The experimental measures in this source are not commercially available but have been used by researchers and are often available at low or no cost.

SOME INTELLIGENCE TESTS

The term intelligence has caused much controversy in the literature. However, it is generally agreed that tests purporting to measure this construct assess general mental ability. An assumption of all general intelligence tests is that some individuals are better able to benefit and learn from

experience than are others and that these individuals have higher levels of general mental ability. The reviews are confined to two major commercially available intelligence tests. Strong theoretical frameworks have guided the development of these tests and given rise to the administration and scoring procedures used.

STANFORD-BINET INTELLIGENCE TESTS

Alfred Binet was a psychologist who was contracted by the French government to develop a tool to identify mentally retarded children in the Paris public school system. He believed that intelligence was manifested by performance on tasks. He also emphasized reasoning, judgment, and understanding in the development of test items. Because the context in which he was working was the school system, most of the items he developed for the test were academic in nature. The term items is used loosely insofar as the items he created were tasks.

Binet and his colleague, Theodore Simon, created 30 tasks initially and, by 1908, had revised and added to the tasks so that they numbered 58. They arranged these from least to most difficult and grouped them into age-level tasks with the levels ranging from

3 to 13 years. A final version of the scale was created in 1911, with 54 tasks arranged in increasing order of difficulty for ages 3 to adult. Based on the number of tasks the test takers could complete, a “mental age” could be calculated for them. For example, tasks such as “shows right hand and left ear” is a task that is age-appropriate for a 7-year-old. Another example is that, by age 15, the respondent is expected to be able to repeat seven digits correctly in order. In 1916, Lewis Terman at Stanford University translated and revised the 1911 version of the Simon-Binet tests; this version was called the Stanford-Binet Intelligence Scale. Like the previous tests, tasks were specific to an age level. The mental age of test takers was determined by how many of the tasks they could complete. The intelligence quotient (IQ) was calculated by taking the mental age (MA) and dividing it by the individual’s chronological age (CA) and then multiplying by 100.

WECHSLER INTELLIGENCE SCALES:

The Wechsler-Bellevue Intelligence Scale (Wechsler, 1939) was a response by David Wechsler to what he believed were several shortcomings of the Binet approach to assessing intelligence. Wechsler argued that the Binet tasks and items were inappropriate for use with adults, (b) the point system developed for scoring the Wechsler-Bellevue Intelligence Scale was superior to age-scaling, (c) performance was an important aspect of intelligence, and (d) speeded assessment penalized older individuals. For all these reasons, Wechsler embarked on the development of a new intelligence assessment tool specifically for use with adults.

The first edition (Wechsler-Bellevue Form and the second (Form II), published in 1947, were not standardized on a proper sample and thus were revised in 1955 and became known as the Wechsler Adult Intelligence Scale—or WAIS (Wechsler, 1955, 1958). It was revised again in 1981 (WAIS-R; Wechsler, 1981) to be used for testing adult intelligence for those aged 16 years to 74 years. The current version, revised in 1997 (WAIS-III; Tulskey, Zhu, & Ledbetter, 1997) was normed on 2,450

CREATIVITY

One of the most consensual definitions of creativity describes it as the capacity to produce new meaningful ideas from current existing concepts, emphasizing the variety, quantity and relevance of the results obtained (Guilford, 1987). It involves the

individuals ranging in age from 16 years to 89 years and stratified based on age, sex, educational level, and geographic region. The items were updated to more appropriate content for the era. Wechsler believed that intelligence was manifested by the capability of acting purposefully and adaptively to the environment. Although individual elements that made up intelligence could be assessed separately, the elements were interrelated. The sum of the capability on these elements would represent general intelligence. The WAIS-III scales are grouped into two major categories: verbal and performance. There are seven verbal subtests (e.g., vocabulary, arithmetic, comprehension) and seven performance subtests (e.g., picture completion, block design, object assembly), although only 11 are used in scoring (one fewer for the verbal and two fewer for the performance dimensions). The WAIS-III is an individually administered test but, like the group-administered tests, allows the test taker to attempt as many items as possible and earn points for each one completed correctly. After a certain number of items for each test is failed, the testing is completed. It takes about 75 minutes to administer all the tests.

notion of a process which combines different cognitive operations in order to generate innovative productions. This process includes two main concepts, such as previous experience and the skills to integrate recent and preceding perceptions. Originality (linked to innovation), fluency

(regarded to a big amount of ideas), flexibility (associated to constant change in the ideas which were generated), elaboration (connected to beautifying basic ideas), remote associations, interest, and cognitive styles, can be distinguished within the group of skills required to integrate former and current perceptions. All these cognitive operations get their expression through different means, such as graphic,

verbal, musical, dancing, etc. Moreover, the skills to handle these ways of expression represent another aspect which determines a person's degree of creativity. Due to the fact that in Western culture verbal and graphic creativity are socially valued over other forms of expression, both them are the most analyzed in education contexts.

Approaches of Measuring Creativity

Approach	Focus	Conception of Creativity	Instruments	Advantages	Weaknesses
Process	Creative processes or skills associated with creativity	Domain-general	WKCT (Wallach & Kogan, 1965); TTCT (Torrance, 1966, 2008); SOI (Guilford, 1967); CAP (Williams, 1980)	Widespread utility High reliability Standardized criteria for interpreting scores	Limited scope of measurement conflicting evidence for validity Bias due to scoring and sample size
Person	Personality traits or creative achievements	Domain-general or domain-specific	HDYT (Davis & Subkoviak, 1975); CPS (Gough, 1979); HCAY (Raudsepp, 1981); CBI (Hacevar, 1979c); CAQ (Carson et al., 2005); BICB (Batey, 2007)	Ease of use High reliability Standardized criteria for interpreting scores	Limited scope of measurement Low validity of self-reports Bias due to self-reporting Neglect of differences in creative personality across domains Low sensitivity to training Skewed scores
Product	Creative products	Domain-specific	CAT (Amabile, 1982)	Similar to evaluating creativity in real-life High reliability High validity	Limited scope of measurement Difficulty in selecting judges Bias due to judges Expensive and time-consuming Lack of standardized criteria
Press	Work environment or climate	Domain-general	SSSI (Siegel & Kaemmerer, 1978); CUCEI (Fraser et al., 1996); WES (Moos, 1986); WEI (Amabile & Gryskiewicz, 1989); KEYS (Amabile et al., 1996); TCI (Anderson & West, 1998); SOQ (Isaksen et al.,	Explore whether a work environment is supportive or inhibitive of creativity Evaluate the environmental improvement attempts and corrective actions	Limited scope of measurement Lack of research-based evidence Debate about "climate" meaning and measurement level Individual differences in conception of climate

NEUROPSYCHOLOGICAL TESTS

A neuropsychological evaluation typically includes an interview with a clinical neuropsychologist and assessment with a series of neuropsychometric tests. Neuropsychometric tests measure a wide range of abilities to determine an individual's strengths and weaknesses. The test results are compared to standardized norms and estimates of the person's pre-injury abilities. A neuropsychological evaluation is often completed soon after an individual has a traumatic brain injury or if an individual seems to be having difficulties related to an undiagnosed brain injury from the past. From the perspective of people with brain injury and their significant others, it is important to know what a neuropsychological evaluation is and how the results can be beneficial.

Neuropsychological tests are specifically designed tasks used to measure a psychological function known to be linked to a particular brain structure or pathway. They usually involve the systematic administration of clearly defined procedures in a formal environment.

LONDON OF TOWER TEST:

It is a test used in applied clinical neuropsychology for the assessment of executive functioning specifically to detect deficits in planning, which may occur due to a variety of medical and neuropsychiatric conditions. It is related to the classic problem-solving puzzle known as the Tower of Hanoi. The test was developed by the psychologist Tim Shallice and consists of

PACED AUDITORY SERIAL ADDITION TEST (PASAT):

Neuropsychological tests are typically administered to a single person working with an examiner in a quiet office environment, free from distractions. As such, it can be argued that neuropsychological tests at times offer an estimate of a person's peak level of cognitive performance. Neuropsychological tests are a core component of the process of conducting neuropsychological assessment.

Most neuropsychological tests in current use are based on traditional psychometric theory. In this model, a person's raw score on a test is compared to a large general population normative sample, that should ideally be drawn from a comparable population to the person being examined. Normative studies frequently provide data stratified by age, level of education, and/or ethnicity, where such factors have been shown by research to affect performance on a particular test. This allows for a person's performance to be compared to a suitable control group, and thus provide a fair assessment of their current cognitive functioning.

two boards with pegs and several beads with different colors. The examiner (usually a clinical psychologist or a neuropsychologist) uses the beads and the boards to present the examinee with problem-solving tasks.



It is a neuropsychological test used to assess capacity and rate of information processing and sustained and divided attention. Originally the test was known as the Paced Auditory Serial Addition Task (PASAT). The subjects are given in the version used as part of the Multiple Sclerosis Functional Composite a number every 3 seconds and are asked to add the number they just heard with the number they heard before. This is a challenging task that involves working memory, attention and arithmetic capabilities. Versions with numbers presented every 2 seconds are also available. The original version presented the numbers every 2.4 seconds with 0.4 decrements for subsequent trials. The PASAT was originally developed for use in evaluating patients with head injury. The advantage in this population was supposed to be minimal practice effects. This test has been widely used in other conditions besides traumatic brain injury.

One common use is for diagnosis of executive impairment. The performance of the examinee is compared to representative samples of individuals of the same age to derive hypotheses about the person's executive cognitive ability, especially as it may relate to brain damage. A certain degree of controversy surrounds the test's construct validity.

It has become widely used in the testing of patients with multiple sclerosis as patients with this disease frequently have an impaired performance on this test. The PASAT was included in the Multiple Sclerosis Functional Composite as a cognitive measure.

THE MINI-MENTAL STATE EXAMINATION (MMSE):

Also called Folstein test is a 30-point questionnaire that is used extensively in clinical and research settings to measure cognitive impairment. It is commonly used in medicine and allied health to screen for dementia. It is also used to estimate the severity and progression of cognitive impairment and to follow the course of cognitive changes in an individual over time; thus making it an effective way to document an individual's response to treatment. The MMSE's purpose has been not, on its own, to provide a diagnosis for any particular nosological entity.

Administration of the test takes between 5 and 10 minutes and examines functions including registration (repeating named prompts), attention and calculation, recall, language, ability to follow simple commands and orientation

Advantages to the MMSE include requiring no specialized equipment or training for administration, and has both validity and reliability for the diagnosis and longitudinal assessment of Alzheimer's disease. Due to its short administration period and ease of use, it is useful for cognitive assessment in the clinician's office space or at the bedside. Disadvantages to the utilization of the MMSE is that it is affected by demographic factors; age and education exert the greatest effect. The most frequently noted disadvantage of the MMSE relates to its lack of sensitivity to mild cognitive impairment and its failure to adequately discriminate patients with mild Alzheimer's disease from normal patients.

PERSONALITY ASSESSMENT:

Personality is usually defined as relatively stable patterns of responding to environmental conditions that vary from individual to individual. Structured personality tests attempt to evaluate various personality traits by asking respondents to indicate the extent to which an item or statement characterizes them. The interest in measuring aspects of personality first became an issue for society during World War I, when suitability for being a soldier became an important question. Because of the huge numbers of recruits that needed to be assessed, individual psychiatric screening was not possible. Instead, self-report structured personality inventories were developed that primarily assessed emotional well-being and adjustment to the military. Over the next 20 years, vast improvements in structured personality testing were developed. Various dimensions of personality were theoretically proposed, giving rise to multiscale inventories. Recognizing the problems inherent in self-reports, items were deliberately designed to assess faking good, faking bad, defensiveness, lying, and so forth.

CALIFORNIA PSYCHOLOGICAL INVENTORY:

The California Psychological Inventory (CPI) is one of the most widely used structured personality inventories. It assesses 20 attributes of normal personality. Some examples of these attributes are dominance, independence, well-being, and intellectual efficiency. The 480-item

instrument was first published by Harrison Gough in 1968 and was designed to assess normal personality in adolescents and adults. The scale was revised in 1987 (Gough, 1987) and reduced to 462 items. Norms for this version are based on more than 6,000 male and 7,000 female test takers. The revised inventory added several new scales, and some of the remaining items were updated. A third revision (Gough & Bradley, 1996) reduced by 28 the number of items. Norms for this version were based on 3,000 males and 3,000 females.

The CPI was intended to assess interpersonal behavior and social interaction. About half of the items on the original version were taken directly from the Minnesota Multiphasic Personality Inventory (MMPI), which was designed to differentiate individuals based on various psychiatric disorders. The CPI was developed emphasizing the empirical approach. Groups of individuals who had been identified as having a specific personality trait (e.g., high dominant or low dominant) would take the inventory. The items that the two groups scored significantly differently on were used to assess, for instance, the dominance trait. Thirteen of the 20 scales were developed this way. One of the criticisms of the CPI is that some of the criterion groups used in establishing the scales were identified by their friends as being high or low on the trait (for example, empathy or independent). Four of the scales (Social Presence, Self-Acceptance, Self-Control, and Flexibility) were developed by selecting items that theoretically were designed to measure the construct. Then these scales were refined by

assessing the intercorrelations among the

items.

NEO PERSONALITY INVENTORY:

The NEO Personality Inventory (Revised) (NEO-PI-R; Costa & McCrae, 1985, 1992) was designed to measure five primary dimensions of personality (called the Big Five) in normal adults ranging from 20 years to 80 years of age. The five personality constructs assessed are Neuroticism (N), Extraversion (E), Openness to Experience (O), Agreeableness (A), and Conscientiousness (C). There are six facets that underlie each of the major constructs. N indicates the degree to which a person is anxious and insecure versus calm and self-confident; the six facets are anxiety, hostility, depression, self-consciousness, impulsiveness, and vulnerability. E indicates the degree to which a person is sociable and assertive versus quiet and reserved; the six facets are warmth, gregariousness, assertiveness, activity, excitement-seeking, and positive emotions. O indicates the degree to which a person is imaginative and curious versus concrete and narrow-minded; the six facets are fantasy, aesthetics, feeling, actions, ideas, and values. A indicates the degree to which a person is warm and cooperative versus unpleasant and disagreeable; the six facets are trust, modesty, compliance, altruism, straightforwardness, and tender-mindedness. C indicates the degree to which a person is persevering and responsible; the six facets are competence, self-discipline, achievement striving, dutifulness, order, and deliberation. The full scale is 240 items (30 for each facet) with three additional validity-check items, and it takes about 30 minutes to

complete. However, a short version of 60 items, the NEO-FFI (NEO Five Factor Inventory), assesses only the five major constructs and takes about 15 minutes to complete. Items on the scale are rated on a five-point scale with the ends anchored with strongly disagree and strongly agree. Forms allow for self-report (Form S) or observer-report (such as spouse, colleague, friend; Form R). The response sheets can be hand- or machine-scored.

INTEREST INVENTORIES:

One of the oldest and most common uses to which psychological tests are put is to assist individuals in making vocational and career choices. High school and college counselors who see individuals who are making initial career choices use these instruments. As career progression is more and more punctuated by change, middle-aged individuals are also seeking assistance with career choices. Interest inventories do just that—they assess interests people have in various types of careers. They do not, however, purport to assess skills and abilities in those areas. Career guidance instruments provide a somewhat more fulsome picture of an individual's strengths, abilities, and interests.

STRONG INVENTORIES:

An early career interest inventory appeared in 1927 with the Strong Vocational Interest Blank (SVIB; Strong, 1927). The theoretical basis for the SVIB was that different professional groups showed consistent

differences in the things they liked versus things they did not like to do (e.g., work with people, perform repetitive tasks, engage in artistic work, etc.). Not only did this apply to work-related tasks but also to school subjects, hobbies, books, entertainment, and so forth. For years, Strong set out to differentiate men in one profession from men in other professions based on their likes and dislikes. Thus, the SVIB is primarily an empirically based scale. Items on the SVIB are activities and are responded to by the test taker with “like,” “dislike,” or “indifferent.” Strong used items for each scale (e.g., the dentistry scale) where the professional group answered statistically differently from “men in general.” When someone takes the SVIB, the interests that are most like certain professions form the basis of the career counseling process. In 1933, Strong created a form of the SVIB that included professions that were also populated by women. The

SELF-DIRECTED SEARCH:

While most interest inventories were developed to be administered, scored, and interpreted by a professional, John Holland developed the Self-Directed Search (SDS; 1971, 1979) to be self-administered, self-scored, and self-interpreted. The SDS is based on Holland’s theory of career choice. He defined personality types and work environments to be one of six: Realistic (R), Investigative (I), Artistic (A), Social (S), Enterprising (E), and Conventional (C). These six types are placed on a hexagon in the following order: R, I, A, S, E, C. Holland hypothesized that the types adjacent to one another (e.g., R and I, R and C) were

1966 version of the SVIB included 399 items that mapped onto 54 occupations for men.

A more recent version of the SVIB, called the Strong-Campbell Interest Inventory (SCII) was published in 1974 (Campbell, 1974). The men’s and women’s forms were merged into a single form and a theoretical basis for the interests was proposed. This theoretical linkage was based on the work of John Holland (1959, 1971), whose Self-Directed Search will be discussed next. The CSII was revised in 1977 (Campbell, 1977), in 1981 (Campbell & Hansen, 1981), and again in 1985 (Hansen & Campbell, 1985). The 1981 version attempted to use more up-to-date samples for norming purposes. The 1985 version attempted to expand the number and types of occupations represented in the inventory. This included adding nonprofessional vocations into the inventory.

more similar than those across from one another (e.g., R versus S, E versus I). The purpose of the SDS was to give the test taker an idea of his or her personality type and then to match that with the type of work environment that would be most congruent. This approach required Holland not only to define personality but also to define work environments.

Those who complete the SDS indicate their career aspirations, occupational interests, skills, and abilities in the six different areas by responding to a series of questions. Test takers then score their own sheets and calculate their scores on the six different areas. Let’s say that, based on my responses,

I score highest on the “E for enterprising” scale. In addition, I note that my next two highest scales are “A for artistic” and “S for Social.” Then, going to the occupational finder, I can locate occupations that have been defined as being “EAS.” These

occupations, then, would most likely be compatible with my aspirations, interests, and skills. Several revisions of the SDS occurred between 1971 and 1985. The 1994 revision (SDS Form R) was to make the SDS more useful for counselors and clients.

APPLICATIONS OF PSYCHOLOGICAL TESTING IN VARIOUS SETTINGS: CLINICAL, ORGANIZATIONAL AND BUSINESS, EDUCATION, COUNSELING, MILITARY; CAREER GUIDANCE.

APPLICATION OF PSYCHOLOGICAL TESTING AND ASSESSMENT IN THE MILITARY

The utility of psychological testing and assessment in the military is a “continuum”. It starts during selection for recruitment into the military and spans till retirement from the active service in the military. In between these periods, psychological testing and assessment are utilized in promotion, placement, special military mission, diagnosis and treatment for officers with psychopathology. However, testing and assessment are commonly applied in military selection, training and placement more than any other area.

Most military selection are large scaled and are influenced by societal situation such as special mission, peace or conflict periods. Devriendt (2003) noted that the nature of psychological tests and assessment utilized in the military are influenced by the targeted competencies or abilities a recruit or cadet must possess to do well in the military context and these competencies include

The ability to overpower people physically and psychologically.

The ability to live with the permanent threat to be killed, injured or taken prisoner.

The ability to obey unconditionally in response to orders from above.

Also, the main purposes of utilizing psychological testing and assessment in the military include the following:

- To classify prospective applicants according to their mental ability so that further placement could be easy.
- To provide the personnel officers with mental rating for each soldier which will help in building and organization of the military arena.
- To assist officers in charge of military human resource development with classification, grading, training, and ultimate assignment of soldiers.
- To Assist in discovering soldiers of superior mental ability who should be selected for officers’ training camps, for promotion, or for assignment to special tasks.

To assist in discovering soldiers of superior mental ability who should be selected for officers' training camps, for promotion, or for assignment to special tasks.

However, military selection takes different format and processes. The format may be categorized into three and they include:

Uniform format: This is where all candidate for military course take the same test batteries

Diversified format: This is where candidates for different military course are given different test battery according to the course.

Mixed format: This is where the entire candidates are firstly given a uniform test battery and then subsequently given specific test battery according to their chosen course.

The processes involved in the military selection include three steps and they are

- **Administrative Assessment:** This process involved the fact that every candidate for recruit or cadet course must meet certain legal criteria, including minimum and maximum age, educational qualification or level and

nationality. This is conducted by military administrative support personnel.

Medical and Physical Assessment: This includes the fact that candidates for recruits and cadets must meet criteria with respect to physical conditions like heights, absence of colour blindness and backbone problems etc. This is conducted by medical doctors, and physical trainers, but psychologist can also help during neuropsychological and psychopathological screening.

Psychological Assessment: According to Devriendt (2013) psychological selection is broad and involves assessing the candidate's abilities in relation to mental, intellectual, social and motivational criteria. It further assesses the candidate's ability to feel at ease within the organizational climate or military culture. As mentioned earlier, the psychological assessments are conducted using various tools such as psychological tests, interviews, case study, observation, specially designed apparatus and measurement procedures etc. The psychological assessment depending on the type may be conducted by trained selection officers, psychologists, military specialists who have experience in certain jobs and high-ranking officers who often decide on acceptance or rejection of applicants as members or chairperson of a selection board. Most psychological assessments are achieved using the mixed format (that is uniform and diversified).

TYPES OF PSYCHOLOGICAL ASSESSMENT TOOLS USED IN MILITARY SELECTION.

Among the many psychological assessment technique available for military selections which include observation (participant and non-participant); interview (structured,

unstructured and semi structured); biographical data; case study, the psychological tests are the most popular and reliable, because of its psychometric qualities. Types of psychological tests used in selection of military personnel include, aptitude test, achievement tests, and personality tests.

APTITUDE TESTS:

This is the most frequently used psychological testing tools. It is sometimes called cognitive or mental ability or intelligence test. It is usually administered in the uniform format and most psychological aptitude tests used for military selection purposes have time limit by implication, tests performance and speed are equally important. They also contain subtests which give indications about cognitive qualities such as numerical reasoning, verbal reasoning, spatial ability, mechanical comprehension and general reasoning.

Scores can be calculated separate per subtest or scores can be grouped in a total score. The performances in aptitude test batteries reflect a general level of cognitive functioning and ability to learn. Generally, for most aptitude test used in military higher scores indicate superior and better level of cognitive functioning and ability to learn. Examples of some standardized military aptitude tests for selection into the military include, Army Alpha test and Beta test and the Armed Services Vocational Aptitude Battery (ASVAB)

ARMY ALPHA TEST

This was created by a group of psychologists led by Robert Yerkes, to aid in segregating the mentally incompetent, classifying applicants according to their mental capacity and assisting in selecting competent applicants for responsible positions. The Army Alpha tests measured verbal ability, numerical ability, ability to follow directions and knowledge of information. The Army Alpha test is for literate applicants who can read and write. While Army beta test are for applicants who cannot speak or understand the language of the test and illiterate. Armed Services

Vocational Aptitude Battery (ASVAB) This is an improvement on Alpha test. It is a cognitive aptitude battery measuring domains such as General Science (GS), Arithmetic Reasoning (AR), Word Knowledge (WK), Paragraph Comprehension (PC) Numerical Operations (NO), Coding Speed (CS), Auto and shop Information (AS), Mathematics Knowledge (MK), Mechanical Comprehension (MC) and Electronics Information (EI). The most important aspect of the test involves summing composites $AR + WK + PC + NO$ = Armed Forces Qualification Test (AFQT).

ACHIEVEMENT TESTS

These can also be referred to as knowledge tests or skill test. They are traditional measures of cognitive ability and are better

suited to get an impression of scholastic knowledge. The most common type of achievement test is a standardized test developed to measure skills and knowledge learned in a given grade level, usually

through planned instruction, such as training or classroom instruction. Achievement test scores are often used in an educational system to determine what level of instruction for which a student is prepared. High achievement scores usually indicate a mastery of grade-level material, and the readiness for advanced instruction. Low

achievement scores can indicate the need for remediation or repeating a course grade. In the military most achievement test are diversified and are used for placement or promotion or selection for special mission. It is usually based on the supposed experiences on the job and can take different forms at different times.

PERSONALITY TESTS

The nature and personality structure of a candidate is very useful in view of the candidate's adaptation skills to military training, military jobs and the prospective behavior in operations. The main aim of personality selection is to make statements about the level of a candidate's fit into the military organization culture. It is also used in diagnostic assessment for psychopathology among serving personnel. Most personality tests are standardized and they have high psychometric validity. They

also have standardized norms or cutoff scores which are either high, borderline, or low and whether an individual possesses desired or undesired personality depends on the test interpretation. The most common personality tests include Objective and projective tests.

Minnesota multiphase Personality Inventory (MMPI) and Sixteen Personality Factor Questionnaire (or 16PF) are examples of objective test, while Rorschach Inkblot Test is an example of projective test.

MINNESOTA MULTIPHASE PERSONALITY INVENTORY (MMPI)

The most common objective personality test is the MMPI is the most common objective personality test; it contains 567 true/false test that is a good measure of dysfunction within personality. It is less useful as a measure of healthy or positive personality traits, because its design was based on helping a professional to find a psychiatric diagnostic label that best suited an individual. It was originally developed in the 1940s, and significantly revised in 1989 (and had another minor revision in 2001). MMPI measures personality traits such as Hypochondriasis (Hs), Depression (D),

Hysteria (Hy), Psychopathic Deviate (Pd), Masculinity/Femininity (Mf), Paranoia (Pa), Psych asthenia (Pt) Schizophrenia (Sc) Hypomania (Ma), Social Introversion (Si). It does this by connecting an individual's responses to dozens of questions scattered throughout the test that are positively or negatively correlated with a particular personality trait. Because the questions are not always obviously related to the trait to which they are correlated, it is difficult to "fake" this test. The MMPI is most often self-administered on a computer in a clinician's office.

SIXTEEN PERSONALITY FACTOR QUESTIONNAIRE (OR 16PF)

This is an ideal measure for people with healthy personalities on like MMPI which is used mainly for assessment of psychopathology. The 16PF measures 16 basic personality traits which contain 185 items and can help a person better understand where their personality falls amongst those traits: Warmth (Reserved vs. Warm; Factor A), Reasoning (Concrete vs. Abstract; Factor B) Emotional Stability (Reactive vs. Emotionally Stable; Factor C), Dominance (Deferential vs. Dominant; Factor E), Liveliness (Serious vs. Lively; Factor F), Rule-Consciousness (Expedient vs. Rule-Conscious; Factor G), Social Boldness (Shy vs. Socially Bold; Factor H),

Sensitivity (Utilitarian vs. Sensitive; Factor I), Vigilance (Trusting vs. Vigilant; Factor L), Abstractedness (Grounded vs. Abstracted; Factor M), Privateness (Forthright vs. Private; Factor N), Apprehension (Self-Assured vs. Apprehensive; Factor O), Openness to Change (Traditional vs. Open to Change; Factor Q1), Self-Reliance (Group-Oriented vs. Self-Reliant; Factor Q2), Perfectionism (Tolerates Disorder vs. Perfectionistic; Factor Q3), Tension (Relaxed vs. Tense; Factor Q4). This type of assessment might be administered so that a person can better understand themselves, and it can also help a professional better understand what type of approach or strategy to employ in treatment to best help the person.

RORSCHACH INKBLOT TEST

The test is composed 5 black and white inkblot cards and 5 colored inkblot cards that an individual is shown and then asked to tell the professional what they see. The most popular scoring system for the Rorschach is the Exner system, developed in the 1970s.

Responses are scored based the location described in the inkblot, and its determinants (that is the things in the blot that prompted the person's response).

PLACEMENT, TRAINING AND PROMOTION

As noted earlier, most psychological testing and assessment for the purposes of placement, or training or promotion are diversified form. They depend on whether the officers are in Army, Navy or Air force or the subsections of the units conducting the assessment. Specifically, each unit design or adapt a special psychological test to march the needs and skills required for such unit. The common types of tests used

for placement, on the job training and promotion are achievement test and personality test. However, other forms of psychological assessments such as interview and observation are also utilized. The tests used for this purpose also target cognitive, mental and intellectual abilities and achievements in relation to the work. Generally, psychological tests such as MMPI, Executive Function test, 16PF, Thematic Apperception Test and indeed many other cognitive, intelligent, personality

and special adapted achievement tests could

be utilized.

TESTING & ASSESSMENT IN VARIOUS OTHER SETTINGS:

The three main types of test to be considered here are tests of intelligence, tests of achievement, and tests of personality.

TESTS OF INTELLIGENCE

Modern intelligence tests can be classified into two categories. The first category contains what are called individual tests of intelligence. The best known of these are the Stanford Binet, now up to its fourth edition, and the Wechsler Adult Intelligence Scale (WAIS-III), the fourth version of which has just been released in Australia. Both of these takes anything up to 2 hours to administer, although the time taken is typically less. Test items cover a range of abilities employing both verbal and non-verbal item types. Test takers may be asked to give the meanings of words, to complete a series of numbers, to recall a list of numbers, and so on. Both tests yield overall estimates of intelligence (IQ) and estimates of a range of specific abilities. Because they are administered in an interview situation, both tests are capable of yielding a lot of clinical information as well. The Fourth Edition of the Stanford Binet can be used with both adults and children. The WAISIII is similar in format and style to the Stanford Binet but comes in different versions: a) an adult version (WAIS-III); b) a children's version (WISC-III); and c) a preschool version (WPPSI). Both the Stanford Binet and the

Wechsler tests are updated from time to time, with new revisions subjected to thorough validation procedures. There are other individual tests of intelligence, some that are arguably more reliable and more valid for certain purposes, but none that would match the Stanford Binet or the Wechsler tests in terms of popularity. The second category consists of the group tests of intelligence. These tests were first developed to handle the very large number of soldiers recruited into the US army during the First World War. Results helped with placement and classification decisions. Most group tests of intelligence now yield scores on a broad range of abilities, such as vocabulary, numerical ability, spatial ability, memory, reasoning, and many others. Testing has become a very commercialized enterprise and major test publishers offer a variety of group tests of intelligence. Examples of some ones include the Differential Aptitude Test (DAT: Bennett, Seashore, & Wesman, 1989) and the Australian Council of Educational Research (ACER) Advanced Tests AL-AQ and BL-BQ (ACER, 1982). Such tests can be administered to many people at the same time, scoring is generally easy, and norm tables can be compiled without any great difficulty. For this reason, there tends to be many more group tests than individual tests. Some group tests of intelligence have also been developed for special populations, such as those with language difficulties or hearing or sight impairment.

APPLICATIONS OF TESTS OF INTELLIGENCE

Intelligence tests, whether group or individual, have been in widespread use since the start of this century. They were developed initially for use in educational settings but quickly found their way into occupational and clinical settings.

TESTS OF INTELLIGENCE IN CLINICAL SETTINGS

Tests of intelligence are routinely used in clinical settings. Level of intellectual functioning provides insights into general level of health. The fact is that a lot of the problems that are referred to a psychologist or psychiatrist either have their origins in intellectual weaknesses or can be better understood following a diagnosis of the individual's intellectual strengths and weaknesses. Invariably, the tests used are individual tests, such as the Stanford Binet or one of the Wechsler tests. A major use of intelligence tests in clinical settings is in neuropsychological assessment, where the aim is to assess possible brain damage as a consequence of trauma, usually caused by a car accident.

TESTS OF INTELLIGENCE IN EDUCATIONAL SETTINGS

Intelligence tests were originally developed to measure learning potential, something that they still do very well. The author has for many years used measures of numerical

ability, verbal comprehension, vocabulary, and abstract reasoning to predict educational achievement. There is no doubt that tests such as these give a fairly accurate indication of success in various subject areas. That is, they have good predictive validity. Used in conjunction with measures of actual academic performance, intelligence tests can help guide people towards appropriate career choices. Although not as popular as they were in the 1950's and 1960's when most school children underwent IQ testing, intelligence tests are still very much part of the educational environment. They are used as the basis for awarding scholarships, for gaining entrance to some prestigious courses (especially in the United States), and they are widely used for diagnostic assessment where learning difficulties are suspected.

TESTS OF INTELLIGENCE IN OCCUPATIONAL SETTINGS

The role of intelligence testing in occupational psychology was summarized in a review by Hunter (1986) who pointed out that although intelligence testing has not been as successful in the occupational field as the educational field, it nevertheless predicts a reliable proportion of job performance, and it does so better than alternative measures, such as interviews or personality assessment. The relationship between intelligence and job performance, however, depends very much on the individual's familiarity with the job. In the early stages when there is a lot of learning occurring, tests of intelligence predict performance quite well, probably because performance is closely linked with the

ability to learn rapidly. Once the individual has settled into the job, however, the strength of the relationship between intelligence and job performance starts to decrease. In jobs which impose variable demands and where learning is constantly occurring, intelligence tests will prove more useful for predicting performance. Furthermore, there are many occupational settings where the tasks are quite complex and in these situations intelligence tests can be useful. Indeed, with the increasing complexity of modern day work situations, it is possible that the predictive validity of intelligence tests will increase in occupational settings. The introduction of automation is a familiar scene everywhere in the workplace. Tasks that were once performed by manual labour are now being

completed by a machine. As this happens, the job requirements are shifting from physical strength and motor coordination to cognitive dexterity. In Reich's (1991) terms, we are moving from a world of doers to a world of symbol analysts. The new technologies devalue experience and increase the value of the ability to learn (Hunt, 1995), precisely the sort of thing that is predicted by tests of intelligence. However, the trend is not completely in the direction of greater complexity. Some jobs that formerly required cognitive skills no longer do so because a machine (e.g., a calculator) now takes care of the cognitive work. Time will tell whether intelligence becomes more or less important in the workforce of the future. For a thorough analysis of this issue, see Hunt (1995).

TESTS OF ACHIEVEMENT

For most people, the most commonly experienced tests are the ones that we sit as students in educational institutions or as adults seeking professional or trade qualifications. These so-called achievement tests are designed to measure the effects of a specific programme of instruction or training (Anastasi & Urbina, 1997). They usually take the form of either free-response questions such as essays, or objective questions such as the popular multiple choice format. A problem with many achievement tests is that they are never standardized or validated in the manner suggested in this chapter. It is not hard to see why: most people who construct achievement tests have neither the time nor the expertise to undertake the necessary analyses. Free-response format tests, for

example, should really be checked for inter-rater reliability to make sure that different subject matter experts rate the answers in the same way. For reasons mentioned above, this rarely happens. Multiple choice tests are a different story: software is readily available to score these tests and at the same time give valuable feedback about questions that are unreliable and therefore decreasing the reliability of the whole test. If a test is unreliable, it cannot be valid. Unfortunately, it is probable that many constructors of achievement tests do not even use test specifications when selecting questions for inclusion. At the very least, the specifications should take account of the objectives, the content areas covered, and topic weightings. Having said this, there are excellent examples of achievement tests that are properly standardized and validated. The

Progressive Achievement Tests (PAT) published by ACER are widely used in Australia to measure a student's level of attainment in key academic areas such as vocabulary, comprehension, and mathematics. Test norms are available, so it is possible to see how a student compares

with other students throughout Australia. Tests like this can be extremely helpful for designing curricula and making decisions about which students can be directed to extension classes and which ones might benefit from supplementary work.

PERSONALITY TESTS

In terms of widespread usage, the assessment of personality ranks second only to intelligence and achievement testing. There are two basic forms of personality testing: self-report measures and projective techniques such as the Rorschach and the Thematic Apperception Test. The Rorschach, better known to most people as the inkblot test, is one of the earliest forms of personality assessment, having first made its appearance in 1921. The test presents a series of 10 stimulus cards to the test taker, who is required to state what he or she can see in the card. The theory upon which the test is based claims that the way a person perceives and interprets the test material reflects fundamental aspects of his or her psychological functioning, including personality. The Thematic Apperception Test (TAT) also makes use of pictures, but employs them in a different way. A series of 19 pictures and one blank card is shown to the test taker who is asked to make up a story about each picture. In the case of the blank card, the task is to imagine a picture on the card and then tell a story about it. The rationale underlying the use of the TAT is much the same as that for the Rorschach; there is an expectation that people will project much of themselves into the stories

they tell. However, although projective techniques are powerful tools for personality assessment they are not used by many practitioners. They take a lot of time to administer and a lot of training before reaching a reasonable degree of proficiency. Self-report methods of personality assessment have proven to be much more popular. As the term implies, self-report tests rely upon the test-taker responding to a set of standard statements by indicating whether they agree or disagree with the statements (if the answer is a simple yes-no) or choosing a number to indicate the extent of their agreement or disagreement with the item. There are so many self-report forms around these days that it is extremely unlikely that the reader has not encountered this form of test before. In the development of self-report personality inventories, several approaches have been followed in formulating, assembling, selecting, and grouping items. Among the major procedures in current use are those based on

a) content validation, b) empirical criterion keying, and c) factor analysis.

a) Content-related validation. These personality inventories are generally formed from lists of known problems which the individual can then tick as affecting them or not affecting them. This is the technique

used in the development of the Life stress Inventory described earlier in this chapter.

Empirical criterion keying. This method builds upon the previous method but takes a more statistical approach, looking for items that separate "normal" from "abnormal" response patterns. In a purely hypothetical example, if it became known that schizophrenics showed a fear of clocks, an item assessing attitude to clocks could be included in a test designed to detect schizophrenia. It is not important that we have no idea why clocks might inspire fear in this group. The important thing is that people with the disorder have the fear whilst others don't, so empirical criterion keying would suggest that such an item could be included. The best known example of a personality test developed through the use of empirical criterion keying is the Minnesota Multiphasic Personality Inventory (MMPI). The MMPI is a very large self-administered test, comprising numerous sub-scales. The sub-scales were developed empirically by criterion keying of items, the criterion being traditional psychiatric diagnosis. The latest revision of the MMPI has resulted in it being separated into two forms, the MMPI-2 and the MMPI-A (for use with adolescents). The California Psychological Inventory (CPI) is another very well-known instrument that was based on the MMPI. It consists of 434 items to be answered true or false. Half of these items came from the MMPI. The CPI has been widely used in industry as well as in clinical practice. **c) Factor analysis.** As mentioned earlier, factor analysis is a technique for detecting patterns of correlations among test scores that indicate underlying dimensions that are responsible

for scores on the test. Factor analysis can be used to help select items for inclusion in a personality test or to identify how many dimensions underlie tests developed by either of the first two methods. Cattell's Sixteen Personality Factor Questionnaire (16 PF) was developed using this method. The so-called Big Five Factor Model (Costa & McCrae, 1991), perhaps the dominant model of personality in occupational testing settings, was also based on factor analysis.

The big five personality factors are: 1. Neuroticism (N): indicates an individual's level of emotional stability, ranging from calm and even-tempered up to maladjustment and emotional distress. 2. Extraversion (E): indicates a person's degree of sociability and preference for interacting with people. 3. Openness (O): measures openness to experience, and is related to divergent thinking and creativity. Low scorers tend to be conventional and conservative. 4. Agreeableness (A): measures how a person views others. Low scorers tend to be competitive while high scorers favor cooperative interactions with others. 5. Conscientiousness (C): indicates a person's ability to control impulses and desires. High C is associated with strong will and high need for achievement, while low C is associated with a more lackadaisical approach to life.

APPLICATIONS OF PERSONALITY TESTING

The two traditional areas for the application of personality tests have been clinical settings and occupational settings. Recently, personality tests have become popular in the

new field of sport psychology, where they are used to gain insights into factors that affect performance.

PERSONALITY TESTING IN CLINICAL SETTINGS

Personality testing has a long history in clinical settings, where it has obvious relevance to the analysis of personality disorders. Perhaps the most common use of personality tests stems from the profile that can be obtained following their administration. A profile is a line linking an individual's scores on various parts of a test. Figure 15.1 shows a profile on an ability test. Similar profiles can be constructed for personality tests. The resulting pattern can be inspected for signs of abnormality. A single high or low score on its own may not indicate any problems but a combination of test scores may well be indicative of particular syndromes, such as schizophrenia. These forms of profile analysis have not lived up to expectations for two main reasons. Firstly, variations among subtest scores could arise from a variety of circumstances, only some of them pathological. Secondly, the diagnostic categories that provided the criteria for profile analysis are themselves subject to debate. For example, what does it mean to say that a particular pattern indicates schizophrenia when schizophrenia itself is not a clearly defined condition? Profile analysis may tell us something about group characteristics but it is prone to error when applied to individual cases. A more fruitful approach is to treat the pattern information as a source of hypotheses that can be tested

against the wealth of other data collected in individualized testing.

PERSONALITY TESTING IN OCCUPATIONAL SETTINGS

Personality testing also has a very strong tradition in job selection testing where the 16 PF and more recently the Five Factor Inventory, a measure of the big five personality factors (NEO-FFI: Costa & McCrae, 1991), have proved very popular. This popularity continues despite evidence that personality tests do not predict job performance very well, even for sales positions (Hunter, 1986). Robertson and Smith (1989) report a validity coefficient as low as .15 in personnel selection testing. In contrast, the coefficient for ability tests ranges between .25 and .45. The use of personality testing in occupational settings is not confined to selection testing, it has also proved very popular as an aid in training courses. The Myers Briggs Type Indicator (MBTI) is one of the best-known personality tests because it is used so often in workshops on career decision making, team building, conflict resolution, time management, relationship counselling, and a number of other applications. The MBTI is based on the theory of psychological types proposed by Carl Jung. Psychological types represent combinations of two or more traits or attributes that are stable and shape the way individuals think and behave. The MBTI classifies people into 16 types and one of the reasons for its success lies in the fact that all types are seen as being valuable with each having particular strengths and weaknesses. There are many clones of the MBTI that also seek to describe people in

terms of types. They are frequently used in management courses. Such type indicators can be extremely valuable in workshop settings where they serve as a basis for discussion of the different perspectives individuals may have on work situations, home life, and so on. For career selection purposes, there are as yet no data to support claims that knowledge of type (or personality) is a useful in predicting job performance.

PERSONALITY TESTING IN SPORT SETTINGS

The latest field of psychology to embrace personality testing in a big way is sport psychology. Much of the testing centers on what is now known as "sport psychology" - the study of personality characteristics as determinants of sporting success. As in occupational testing, the findings so far have not been very promising. When the personality profiles of elite athletes are compared with those of novice athletes, there are differences. Elite athletes are more aggressive, more focused, less anxious, and so on, but individual differences on these traits do not predict who is going to be an elite athlete. That is to say, the tests of personality have concurrent validity but poor predictive validity. Talent identification programmes have grappled unsuccessfully with this problem for years. Perhaps more situation-specific personality tests will help to improve the predictive validity of personality tests in both occupational and sports settings. There is no doubt that serious attempts are now being made to develop personality tests that are suited to sports situations. Ostrow (1990, p.

has included a graph which shows a quite steady increase in the number of sports-specific tests since 1975. The proportion is now close to 45%. As one might expect, given the nature of sport, these new tests are primarily in the areas of anxiety, motivation, mental skills, and specific sporting factors such as team cohesion (Fogarty, 1995).

TESTS OF VOCATIONAL AND CAREER INTERESTS

Another category of test that has proved to be very popular in educational and occupational settings is the career interest inventory. The best known of these are the tests based on Holland's model of career decision making. The Self-Directed Search (SDS) is the most popular of these tests and has an Australian version which is in widespread use in this country. Holland (1985) believed that the most productive approach to career decision making involved an investigation of the individual's personality type. He proposed a six-category typology: Realistic (R), Investigative (I), Artistic (A), Social (S), Enterprising (E), and Conventional (C). He believed that this six-category system could be used to not only describe the major types of people but also to describe the work environments they are likely to encounter in Western society. Holland's assumption was that people seek environments that allow them to express their interests, and by knowing something about their general orientation we are in a better position to judge where they will be happiest working. Holland's tests, or derivatives of them, are widely used in educational and occupational settings to assist with career decision making.

MISCELLANEOUS TESTS

There are many more test types than can be described here, some of them adapted to particular situations. Areas not covered in this chapter include stress (e.g., Osipow and Spokane, 1987), values (e.g., Schwartz & Bilsky, 1987), decision styles (e.g., Driver, Brousseau, & Hunsaker, 1990), learning styles (Entwistle, 1983), and perhaps it is better to stop here because the list could go on and on. The reader is referred to a text devoted exclusively to psychological testing,

COMPUTERIZED TESTING

It would be a mistake to conclude this chapter leaving the reader with the impression that tests are available only in paper-and-pencil format. Testing was one of the earliest areas within psychology to benefit from computer applications with standardized, objective-type personality tests being particularly well-suited to automation (Bartram & Bayliss, 1984). Initially, interest focused on automated scoring but later expanded to include the computerized administration of existing pencil-and-paper tests. Currently, almost every facet of personality testing has been computerized, from test design and development, through item generation and analysis, to test interpretation and report generation. In a typical computerized test presentation, individual questions or stimuli are presented on a video display unit (VDU) attached to the computer, a set of limited responses is offered, and test-takers record their selected response via a keyboard or some other interface. The advantages of this form of

such as Anastasi and Urbina (1997), for an overview of virtually the whole testing domain. For information on tests available here in Australia, the best place to approach is the Australian Council for Educational Research (ACER), which has a number of test catalogues containing descriptions of individual tests, including details of what qualifications you need to administer the tests and suitable areas of application.

administration over conventional administration are well-documented in several reviews (e.g., Bartram & Bayliss, 1984). The move from paper-and-pencil tests to computer-based formats, however, represents a major shift in the way tests are administered and it is important that research is conducted to check the equivalence of the two methods. Work has already started in this area, especially on the equivalence of paper-and-pencil versus computerized presentation. Reviews of these studies report conflicting findings, with many uncontrolled variables influencing the outcomes (e.g., Burke & Normand, 1987; Webster & Campeau, 1996). The author's own experience with this form of testing is that it does not appear to make a noticeable difference and that test manuals developed on the basis of paper-and-pencil tests are still applicable to computerized versions of tests (Fogarty, 1998). One of the major benefits of computerized assessment is undoubtedly the increased efficiency of administration made possible by software that adapts the presentation of items for each user. Thus, when assessing abilities there is

no need to present a whole lot of easy items to a very capable person. It is a waste of time. Similarly, there is no need to present a lot of difficult items to a person who has no chance of solving them. In a traditional paper-and-pencil test, everyone is given the same instructions, the same items, and the same time in which to complete the test. In an adaptive, computerized test situation, the

algorithms built into the software can quickly estimate a person's level of cognitive functioning on the ability being measured, rather than the person's total score. Such an estimation is possible because the difficulty level of each item is known beforehand and the test can draw upon a large bank of items covering all possible ability levels.

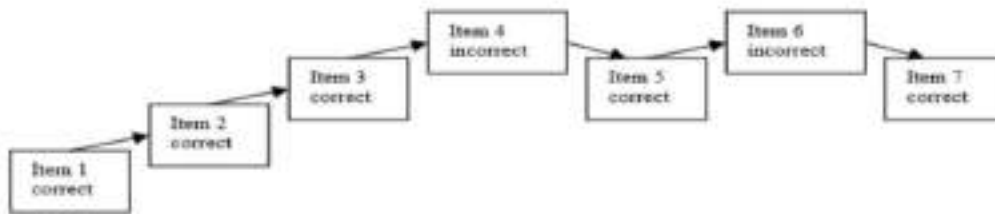


Figure 15.4. Illustration of possible sequence of items in an adaptive computerised test

If you can imagine that the higher items are more difficult you can see that each time the person is correct, a more difficult item is selected. When the item is incorrect, the computer selects an item from an easier level. In the oversimplified representation shown in Figure 15.4, the person's ability level is somewhere near to the levels assessed by items 3, 5, and 7. Above that level, the items are incorrect. Below that level and the items are correct. In actual practice, it is not quite this easy, but the principle is clear. At some point, performance will alternate between success and failure. The test is then ended and the level at which this occurred is reported as the ability estimate for that individual. Adaptive tests are not new in psychology and education, many individually administered tests are adaptive (e.g., the Stanford Binet), but computers allow whole groups of people to be tested

simultaneously. In some cases, testing time is halved because fewer items have to be presented. Adaptive computer testing can be used with other types of test (e.g., personality), but so far, their application has mostly been with intelligence and achievement tests. Apart from test administration, computers are increasingly being used to write test reports on the basis of scores collected during a computerized administration of the test or entered by the test administrator.

There is a real question mark hanging over the issue of the lack of flexibility of reports written by a computer. Such reports are based on test scores only and omit the large amount of data that can be collected in a face-to-face testing situation. When using tests of maximal performance, for example, a clinician can report the amount of effort put into the test, a computer cannot. In

personality testing, many signs of abnormality may be evident in the person's bearing and manner, things that are not available to the computer that generates the report. It may be some time before there is good research data on the validity of

computer-generated reports. Certainly, they should be supplemented by other reports from the training officer, psychologist, or whoever arranged the test.

COMPLETE

COMPLIMENTARY CONTENT – OUT OF SYLLABUS

ERRORS IN MEASUREMENT:

These are the discrepancies between the observed value of your measurement and the 'true' value. Every time a student takes a test there is a possibility that the raw score (observed score) obtained maybe less or more than the score the student should actually have received (true score). The difference between the observed score and the true score is called the error score.

There is a simple formula to illustrate this:

Observed score = True score + Error.

The true score of a person can be found by taking the mean score that the person would get on the same test if they had an infinite number of testing sessions. Because it is not possible to obtain an infinite number of test scores, T is a hypothetical, yet a central aspect. Hence, True value is the value that would be gained under ideal situations when using perfect measurement techniques. Based on the above formula, subtracting the true value from the measured value would produce the error component:

Measured value – true value = **Error**

Psychologists have divided the errors of measurement in two types:

Systematic bias

Random error

Therefore: Total measurement error = systematic bias + random error both of these types of errors occur in any measurement. Therefore, understanding how they occur and minimizing them as much as possible is necessary to reduce total measurement error

Systematic Error

Systematic bias refers to 'predictable errors in measurement' that occur in the same direction (under- or over-estimation) and have the same magnitude. As these errors are in a specific direction, they can produce quite misleading results. These errors are a result of polluted thoughts, personal biases, improper morals, and so on, such as-

Personal errors which arise from mistakes made by the tester during the test administration or lack of alertness on his part. These errors are due to subjective element of an individual. For example, errors made in reading the thermometer, stop watch or the test manual.

Such errors occur when in a haste, the tester starts or stops the stop watch early or when the start or stop of the stop watch is delayed due to some reason. These errors can be removed by comparing the results obtained by several testers/experimenters.

The second type of biased errors arise from carelessness. These errors are more evident when, for example, the experimenter instead of writing 0.1 writes 1 or instead of writing 0.1

writes 0.01, or instead of +ve (positive) he writes -ve (negative).

STATISTICAL ERRORS

Generally, statistical errors are divided into two types:

Errors of Descriptive Statistics:

Under these are included standard errors and probable errors.

Inferential Errors: Under these are Type I and Type II errors.

ERRORS OF DESCRIPTIVE STATISTICS

There are two ways of calculating errors of descriptive statistics. The first is the standard error and the second is the probable error. Standard errors get quite affected by extremes because standard errors are calculated using standard deviations.

Both standard error and probable error indicate the amount of error or expected error in the data obtained. After calculating these errors, we can also know whether these errors are due to chance factors or due to deviation in variables. If these are due to chance factors, the degree would be less. But if these are due to differences in variables, the degree of error would be high. The determination of error depends to a great extent on the nature of research and planning.

STANDARD ERRORS

The standard deviation of the distribution of random errors around the true score is called

the standard error of measurement. The lower it is, the more tightly packed around the true score the random errors will be.

Whenever the mean of a sample is calculated, the obvious argument is whether this mean is appropriate or not, that is, what may be the difference between the sample's mean and the mean of population from which the sample is drawn. The difference between the two means can be expressed through standard error.

Standard error is, therefore, that value which implies the amount of variation in the mean and the standard deviation of a sample, and the mean and the standard deviation of the population from which the sample is drawn.

PROBABLE ERRORS

If the difference in results of two observations on different occasions is less or if the observations of two persons on a problem are not very different, then it is not considered true or reliable, but a small deviation in two observations make it reliable. Thus, when we try to know the truth through measures of variance, the result is probable error.

INFERENTIAL ERRORS

The following are two types of the inferential errors:

- Type I error
- Type II error

TYPE I ERROR

This type of error is committed when a null hypothesis (H_0) is rejected and, in fact, it is true. This error is represented by (α). There exists an inverse relationship between α and the significance level. If the significance level is high, (α) will be less, and if the significance level is low, Type I error will be more. For example, if the significance level for any hypothesis is

kept at 0.01, then there would be more chances of committing Type I error, and if the significance level is kept at 0.05, then the chances of committing Type I error would be comparatively less.

TYPE II ERROR

Such an error is committed when H_0 is accepted and, in fact, it is false. This error is denoted by (β). Type II error has direct relation with the significance level. Higher the significance level, more are the chance (β) of being committed; for example, Type II error will be more when the significance level is 0.5 as compared to when it is 0.1. This is important in psychological experimentation.

Figure 2.1 Type I and Type II Errors

	Null Hypothesis True	Null Hypothesis False
Reject Null Hypothesis	Type I Error	Correct
Accept Null Hypothesis	Correct	Type II Error

Both the types of errors are related to each other. If we try to reduce Type I error, we increase the chance of Type II error being committed and, similarly, if we wish to reduce Type II error, we are taking a chance with Type I error.

The two types of errors can be written as:

- Type I error rejecting null hypothesis (H_0) when it is true,
- Type II error accepting null hypothesis (H_0) when it is false.

The most often used measures of 'test errors' are the four types of errors, namely:

- Error of measurement.
- Error of substitution.
- Error of estimating observed scores.
- Error of estimating true scores.

ERROR OF MEASUREMENT

The error of measurement is nothing but the error committed while substituting the observed/obtained score for the corresponding true score. Suppose we are interested in assigning every person a true score, but instead of true score, we assign the observed score. Whatever be the difference between these two scores, it is

nothing but the error of measurement. Thus, the error of measurement is the error made while substituting the observed score for the true score.

Based on this theory, error of measurement is:

$$E = X - t$$

Where, E is the error score

X is the observed score t

is the true score.

ERROR OF SUBSTITUTION

Error of substitution is the error made while substituting a score on one test for a score on a parallel test. In other words, error of substitution is the difference between two observed scores on parallel tests.

Mathematically, error of substitution is:

$$d = x_1 - x_2$$

Where, d is the difference

x_1 is the observed score on one test and

x_2 is the observed score on another test.

The above definition of the error of substitution would be applicable in case we are interested to find the differences between the results obtained by one researcher employing one psychological test and another researcher employing a parallel form of the same psychological test. [Standard Error of Estimate Conceptually, the standard error of estimate is related to regression analysis in that it typically provides an estimate of the dispersion of the prediction errors when you are trying to predict Y values from X values in a regression analysis.]

In order to understand the previous sentence, you will first need to understand three bits of jargon: prediction errors, Y values, and X values. Again, I would like to deal with

those terms in reverse order. In a regression analysis, X values are any values from which you want to predict, and Y values are any values to which you want to predict. Unfortunately, those predictions are never perfect because prediction errors occur. Such errors may be due to unreliable measurement in either the Y or X variable, or due to unsystematic differences between the two sets of numbers.

When you are trying to predict Y values from X values, it would be useful to know what the distribution of those prediction errors is so you can interpret your predictions wisely. An example of such a situation might be a case where you use regression analysis to predict TOEFL scores from the PERFECT test at your institution. You must first conduct a study based on a large number of students who took both tests. Then using regression analysis, you build a regression equation of the form $Y = a + bX$. Based on your analysis, you will know the values of a (the intercept) and b (the slope), and can then plug in the X value (or PERFECT test score) for a student who has never taken the TOEFL. Solving for Y will then give you that student's predicted Y (or predicted TOEFL score). All of this is beyond the scope of this explanation, but is necessary in order to get even a basic understanding of what the standard error of estimate is. We assume that any student's predicted Y score is our best estimate of that score, but we recognize that there are sampling errors around that estimate, just as there were for estimating the population mean and true scores. Those sampling errors are normally distributed and, in this case, have a standard deviation called the standard error of estimate.]

• ERROR OF ESTIMATING OBSERVED SCORE

Error of estimating or predicting observed score is the error made while a regression equation is used to estimate or predict the scores on one test from scores on a parallel test. In other words, it is the error made while predicating the score on one form of the test on the basis of the other form of the same test with the use of least square solutions, that is, regression equation.

ERROR IN ESTIMATING TRUE SCORE

Error in estimating true score is the error made in predicting the true score from the observed score while applying the best fitting regression equation or least square solutions. In other words, error of estimating the true score is the error difference between the true score and the predicated value of the true score.

By taking the common factor out, that is, $S_x = S_x \sqrt{(1-r_{xx})}$ from the four factors, we can write these four errors as,

$$\begin{aligned} S_e &= S_x \sqrt{(1-r_{xx})} \sqrt{1} \\ S_d &= S_x \sqrt{(1-r_{xx})} \sqrt{2} \\ S'_d &= S_x \sqrt{(1-r_{xx})} \sqrt{1+r_{xx}} \quad (\Theta 1-r_{xx}^2 = (1-r_{xx})(1+r_{xx})) \\ S'_e &= S_x \sqrt{(1-r_{xx})} \sqrt{r_{xx}} \end{aligned}$$

As we know that the reliability coefficient of the test rages from 0 to 1.0, hence,

$$2 > 1+r_{xx} > 1 > r_{xx}$$

SPEED vs. POWER TESTS

A psychological test is "an objective and standardized measure of a sample of behavior". Psychological Tests can be classified on various dimensions. One classification which is based on the rate of performance distinguishes between Speed Test and Power Test.

SPEED TESTS:

Speed Tests are the ones in which individual differences depend entirely on the speed of performance. Items are of uniform

difficulty, all of which are well within the ability level of the persons for whom the test is designed; but the time limit is such that no examinee can attempt all the items. Under such conditions, each person's score reflects only the speed with which he worked.

According to Gulliksen, a **pure Speed Test** is a test composed of items so easy that the subjects never give the wrong answer to any of them, that is, there would be no attempted item that would be incorrect and, consequently, the score for each person would equal the number of items attempted.

For example, a test in grade school where students are asked to complete as many multiplication facts as possible in two minutes is an example of a pure speeded test. Pure speeded tests are not useful unless the underlying construct being measured is one where speed is important (e.g., a typing task).

In speeded tests, the only way to get variance on the test scores for a group of test takers on a speeded test is to control the amount of time given to the test taker. Variability of test scores is a strong determinant of a test's reliability, so the time limit should be set to ensure maximum variance in the test scores.

Some speeded tests often take into account the individual difference of age. It is a well-established empirical finding that as individuals age, their response times to stimuli slow. Older individuals simply take longer to complete tasks than do younger individuals. When administering a speeded test, ensure that the test manual has addressed the issue of age. Usually this is done by adding a constant to scores of individuals within certain age bands. For example, individuals between 30–39 years of age taking the Wonderlic Personnel Test (1999) add one point to their raw score. Similar adjustments to raw scores are made for each decade up to 60 years of age and over.

POWER TESTS:

Power Tests, on the other hand, have time limits long enough to permit everyone to attempt all the items. But the difficulty of items increases gradually and the test includes certain items which are too difficult for anyone to solve, so that no one can get a

perfect score. Achievement examinations, for e.g., fall in this category.

In a **pure Power Test**, all the items are attempted so that the score on the test depends entirely upon the number of items that are answered and answered correctly. Therefore, in such tests, the items cannot be of trivial difficulty because that would not produce a desirable distribution of scores. Power tests assess individual differences without any effects of imposed time limits changing scores. Power tests often are made up of items that vary in their level of difficulty. The power test designer should ensure that the test is long enough to cover the content domain, but not so long that there is not enough time available for most people to complete the test.

DIFFERENCES

A speed test is one in which each item is trivially easy but is administered within such a strict time limit that it is impossible to answer all questions. For example, the examinee might be given 60 seconds for a 100-item test made up of questions. It is assumed that, given enough time, the person could answer every item correctly. However, the strict time limit makes it impossible to complete every item. Individual differences in test scores are thus a function of the speed with which the examinee responds to individual items; the faster the person works, the more items he or she will complete. The test-retest method is useful for assessing the reliability of speed tests. In a speed test, items are often trivial and fairly meaningless. It is precisely for this reason that the usual barrier to the use of the test-

retest method-subjects' tendency to remember their responses-tends not to be a problem with speed tests. When the test content is meaningful, subjects are likely to remember their responses to test items; when test items involve trivial perceptual or motor tasks, over time it becomes more difficult to remember or duplicate one's responses to a large number of test items. Thus, the test-retest method might be quite useful in assessing the reliability of a speed test, particularly in settings where it is difficult to obtain scores on separately timed half-tests.

In contrast, a power test is one in which speed in responding has no effect on the test score. For example, a 20-item introductory psychology exam administered with no time limit would probably be considered a power test. No matter how quickly (or slowly) the examinee works, the final score is determined by the person's knowledge of the material, together with his or her luck in guessing or having studied just the right material.

SCALE/ TEST CONSTRUCTION:

STEPS FOLLOWED IN SCALE CONSTRUCTION:

STEP 1: DEFINING THE TESTING UNIVERSE, AUDIENCE AND PURPOSE:

Testing universe refers to defining the behaviors that the test aims to measure or the body of knowledge that the test represents. The developer prepares an operational definition of the construct that the test will measure which mainly involves rigorous review of literature.

For a pure Power Test, the odd-even method to find split-half reliability should be employed. Whether a test is sufficiently close to being a pure Power Test can be indicated by a criterion which would make one sure that the odd-even reliability will not be spuriously high or low.

EFFECTS OF UNATTEMPTED QUESTION:

If a test is a Power Test, then it is possible to employ the split-half reliability to estimate the extent of error of measurement. For a Power Test, error of measurement would be **greater** if the subject is allowed to attempt all the items in a test.

If a test is a Speed Test, then a test-retest of an alternate form of reliability method should be used to calculate the reliability of the test.

In case a test is a speeded-up test, the retest or alternate form is to be used to calculate the reliability of the test. Such reliability will correctly represent the functioning reliability of the test.

Defining the audience: This involves making a list of characteristics of the people who will take the test. E.g.: Race, Gender, Socio economic Status.

Purpose of the test: It explains how the test users, can use the test scores

STEP 2: DEVELOPING A TEST PLAN:

It involves defining the construct i.e. defining the behavior that are related to the construct. It also involves deciding upon the length of the test. Choosing the test format is another important consideration in this step. It refers to the type of questions that the test

will contain. The format of the test is decided on the basis of the purpose of the test.

STEP 3: COMPOSING TEST ITEMS:

While composing test items it is preferred that two and a half times more items are developed as we go further with the steps of construction. In multiple choice questionnaire format, a multiple choice item consists of a problem, known as the stem, and a list of suggested solutions, known as alternatives. The alternatives consist of one correct or best alternative, which is the answer, and incorrect or inferior alternatives, known as distracters. The distracters shouldn't be too easy or too difficult they should be realistic more the options lessen the chance of guessing the right answers. In true and false that is dichotomous items format the effect of guessing is higher as only two options are available. Forced choice question format which are usually used in scales, chances of guessing is low.

Subjective answers like projective techniques, there are chances of response biases. Social desirability, there is tendency to choose responses which are socially desirable it can be reduced by using ipsative, a scale could also be used to see social desirability. Acquiescence that is the tendency to agree with the idea presented which can be controlled by using reverse scoring. Random responding or faking, various tests like lie scales and acquiescence scale are developed also usually questions are framed differently and randomly if the testee responds randomly to the similar questions measuring the same thing that means he/she is faking.

STEP 4: WRITING DOWN THE ADMINISTRATION INSTRUCTIONS:

The administration and instruction should include whether the test should be conducted on individual or in group setting, specific requirements should be mentioned (stationary, setting etc) the time limit to complete the test should be mentioned, there should be a proper script for administrator a good script encourages the testee to give accurate answers and also it should be mentioned if prior training is required. The instructions should also have an example for the testee; this will make him/her more clear and have a better understanding.

STEP 5: PILOT TESTING:

Pilot testing is done in two phases. The number of participants depends upon the size and complexity of target audience. The setting of the pilot test should mirror the planned test setting. The questions that should be asked after administration include, "Do you understand meaning of each item?", "Do you understand test instructions?", "Do you object to content of any item? Why?", "Do you think test assesses your skills properly?" Then revise or drop certain items and collect the data.

STEP 6: ITEM ANALYSIS:

Item analysis is analyzing property of each item.

Item difficulty is done only in test construction and not in scale construction because in scale items are not measuring success or failure in an item but the degree to which a trait is present.

$$D = R/N$$

Where, D is difficulty value; also denoted by P.

R is number of people answered correctly to an item.

N is total number of people answering an item.

Item discrimination is the ability to discriminate people who have high ability and low ability in test and high trait and low trait in scale.

$$V = D_U - D_L$$

Where, V is item discrimination

D_U is difficulty value of upper group
 D_L is difficulty value of lower group

Item reliability is inter-item correlation. Forced Nominal is found in scale and perfectly nominal in test. In scale, correlation is found by tetrachoric/ rank order/ Pearson moment.

Item validity is the correlation of an item to the test scores. It is found by bi-serial/ point bi-serial/ Pearson product moment formula.

STEP 7: REVISING THE TEST:

Test revision as a stage in developing a new test

After the new test is conceptualized, constructed, tried out and item analyzed (both quantitatively and qualitatively), what remains is to mould the test into its final form. On the basis of the huge amount of information that is generated at the item-analysis stage, some items from the original item pool will be eliminated and others will be re-written.

There are several ways to approaching test revision:

One is to characterize each item according to its strengths and weaknesses and balance them.

The advantage to making a large item pool becomes more apparent as test revision proceeds. Poor items can be eliminated in favor of those that were shown on the test tryout to be good items. Even when working with a large item pool, the revising test developer must be aware of the domain the test should sample.

Having balanced all these concerns, the test developer ends up coming with a better revised test. The next step includes administering this revised test under standardized conditions to a second appropriate sample of examinees.

When the item analysis of data derived from a test administration indicates that the test is not yet in its finished form, the steps of revision, tryout and item analysis are repeated until the test is satisfactory and standardization can occur.

Test Revision in the life cycle of an existing test:

There are no hard and fast rules of revising an existing test as such. Test are usually revised due to the following conditions -

The stimulus materials look dated and the current test takers cannot relate to them.

The verbal content of the test is not readily understood by current test takers.

The test norms are no longer adequate as a result of group membership changes in the population of potential test takers.

The steps to revise an existing test parallel those to create a brand new one. In test conceptualizing phase, the test developer must think through the objectives of the revision and how they can best be met. In the test construction phase, the proposed changes are made. Test tryout, item analysis and test revision follow.

CROSS -VALIDATION AND CO-VALIDATION:

Cross Validation: Revalidation of a test on a sample for test takers other than those on whom test performance was originally found to be a valid predictor of some criterion. **Co Validation:** Test validation process conducted on 2/more tests using the same sample of test takers.

IRT (Item Response Theory) can be used to build and revise tests as well.

STEP 8: ESTABLISHING RELIABILITY OF THE TEST:

Reliability refers to how consistently a test measures a characteristic. More the consistency, greater the reliability. In other words, a test is reliable if it yields the same scores on being administered on the same set of people for the second (or more) time. Factors such as test taker's psychological/physical state, environmental conditions, test form, etc are sources of chance or random measurement error in the assessment process. If there were no random errors of measurement, the individual would get the same test score, the individual's "true" score, each time. The degree to which test scores are unaffected by measurement errors is an indication of the reliability of the test. The reliability of a test is indicated by the reliability coefficient. It is denoted by the letter "r," and is expressed as a number ranging between 0 and 1.00, with $r = 0$ indicating no reliability, and $r = 1.00$ indicating perfect reliability. It is normal to not find a test with perfect reliability. The larger the reliability coefficient, the more reliable the test scores.

The following are the types of reliability estimates: -

Test retest reliability

Inter-rater reliability

Internal consistency reliability

- Alternate/Parallel form of reliability

STEP 9: ESTABLISHING VALIDITY:

Test validity may be defined as the degree to which a test measures what it is intended to measure. A test can be reliable without being valid, but not the other way round (Groth-Marnat 2003). For example, tests of processing speed that involve naming of alphabets or digits are usually highly reliable, but they may actually measure reading skills and not processing speed (Roivainen 2011). Most psychological constructs, such as processing speed, intelligence or personality are abstract by nature and cannot be directly observed. They must be studied by indirect means. However, psychological theories and the definition of psychological constructs change over time. Therefore, the first concern of a test constructor is to consider whether the test items they have selected are representative of the construct being measured. This aspect of validity is usually referred to as content validity. Content validity is subjective by nature, as it is based on the judgment of the test developers. Criterion validity refers to the correlation between a test and an outside measure. For example, IQ scores are highly correlated with school grades. If performance on a test item has zero correlation with education, then that item is probably a poor measure of intelligence. Predictive validity involves outside measurements that are performed sometimes after the psychological test, for example, the correlation of school grades with IQ scores from a test taken a year earlier (Blackmer & Endicott 2000, Morgan et al. 2001)

Validity is arguably the most important criteria for the quality of a test. The term validity refers to whether or not the test measures what it claims to measure. On a test with high validity the items will be

closely linked to the test's intended focus. For many certification and licensure tests this means that the items will be highly related to a specific job or occupation. If a test has poor validity, then it does not measure the job-related content and competencies it ought to. When this is the

case, there is no justification for using the test results for their intended purpose. There are several ways to estimate the validity of a test including content validity, concurrent validity, and predictive validity. The face validity of a test is sometimes also mentioned.

TYPES OF VALIDITY

CONTENT VALIDITY

While there are several types of validity, the most important type for most certification and licensure programs is probably that of content validity. Content validity is a logical process where connections between the test items and the job-related tasks are established. If a thorough test development process was followed, a job analysis was properly conducted, an appropriate set of test specifications were developed, and item writing guidelines were carefully followed, then the content validity of the test is likely to be very high. Content validity is typically estimated by gathering a group of subject matter experts (SMEs) together to review the test items. Specifically, these SMEs are given the list of content areas specified in the test blueprint, along with the test items intended to be based on each content area. The SMEs are then asked to indicate whether or not they agree that each item is appropriately matched to the content area indicated. Any items that the SMEs identify as being inadequately matched to the test blueprint or flawed in any other way, are either revised or dropped from the test.

CONCURRENT VALIDITY

Another important method for investigating the validity of a test is concurrent validity. Concurrent validity is a statistical method using correlation, rather than a logical method. Examinees who are known to be

either masters or non-masters on the content measured by the test are identified, and the test is administered to them under realistic exam conditions. Once the tests have been scored, the relationship is estimated between the Test Validity examinees' known status as either masters or non-masters and their classification as masters or non-masters (i.e., pass or fail) based on the test. This type of validity provides evidence that the test is classifying examinees correctly. The stronger the correlation is, the greater the concurrent validity of the test is.

PREDICTIVE VALIDITY

Another statistical approach to validity is predictive validity. This approach is similar to concurrent validity, in that it measures the relationship between examinees' performances on the test and their actual status as masters or non-masters. However, with predictive validity, it is the relationship of test scores to an examinee's future performance as a master or non-master that is estimated. In other words, predictive validity considers the question, "How well does the test predict examinees' future status as masters or non-masters?" For this type of validity, the correlation that is computed is between the examinees' classifications as master or non-master based on the test and their later performance, perhaps on the job. This type of validity is especially useful for test purposes such as selection or admissions.

FACE VALIDITY

One additional type of validity that you may hear mentioned is face validity. Like content validity, face validity is determined by a review of the items and not through the use of statistical analyses. Unlike content validity, face validity is not investigated through formal procedures and is not determined by subject matter experts. Instead, anyone who looks over the test, including examinees and other stakeholders, may develop an informal opinion as to whether or not the test is measuring what it is supposed to measure. While it is clearly of some value to have the test appear to be valid, face validity alone is insufficient for establishing that the test is measuring what it claims to measure. A well-developed exam program will include formal studies into other, more substantive types of validity. The validity of a test is critical because, without sufficient validity, test scores have no meaning. The evidence you collect and document about the validity of your test is also your best legal defense should the exam program ever be challenged in a court of law. While there are several ways to estimate validity, for many certification and licensure exam programs the most important type of validity to establish is content validity

TYPES OF NORMS

NATIONAL NORMS:

The most general and most commonly used type of norms is the **National Norms**, appropriate to the educational and age level (or levels) for which the test is constructed. One of the problems in defining the national norms group arises from the large number, variety, and communities, that are correlated with and relevant to the test scores. The variables that are associated with the characteristics of students include

STEP 10: DEVELOPING NORMS:

Valid norms are an essential aspect of test validity. Test results are usually interpreted by comparing the performance of an individual to norms based on a representative sample of the population. The more closely the standardization sample resembles the respondent in terms of age, education, sex and health, the more reliable the comparison is between test norms and the respondent's test score. Sir Francis Galton at the first time developed the logic for norm based testing in the 18th century. Norms refers to information regarding the group performance of a particular reference on a particular measure for which a person can be compared to. Norms mean standardized score. Scores on psychological test are most commonly interpreted by reference to norm that represents the test performance on standardization sample. Norms always represent the best performance.

Basically there are two purposes of norms:

Norms indicate the individual's relative standing in the normative sample and thus permit evaluation of his/her performance in refer to other persons.

Norms provide compared measures that permitted a direct comparison of the individual performance on difference test.

educational level, age, sex, race, present or intended field of study, socioeconomic level of parents (educational, occupational, and economic determinations), and sometimes, for achievement test norms.

Schrader has pointed out that national norms have the distinct advantage of being simple, definite and unique. National norms also have the advantage that, to the extent that publishers succeed in providing truly precise and unbiased national norms, it is possible to achieve score comparability across the tests of different publishers. On the other hand,

the availability of a single norms table tends to obscure the fact that a percentile rank is not unique but represents only one of many possible evaluations of a test score. Furthermore, national norms may frequently be too general to permit specific action. Clearly, the more specific and. Of course, the more relevant the norms group, the easier it is to make appropriate decisions based on test scores.

LOCAL NORMS:

Although it is unquestionably the responsibility of the test publisher to make available the kinds of norms that are appropriate for the uses he claims for his test, in many instances the most useful kind of norms are the **local norms** collected by the user itself and based on students enrolled in his own institution. These norms have the advantage of homogeneity, since the students included in the norms all come from the same educational and social milieu and constitute a group with which the test user has first-hand knowledge and familiarity.

Local norms are especially valuable when they are collected in a way that will permit the formation of particular decisions – for example, in the identification of students who would benefit from special instruction or in separating a total class into homogenous subgroups for whom the instructional pace can be more clearly defined. The data that help to make such decisions are those that relate to one set of scores to another, as in local studies of growth; or those that relate aptitude test scores to achievement; or those that relate test scores to individual or group characteristics, as in the preparation of differentiated local norms, for example by curriculum, by sex, or by intended major field of study.

AGE AND GRADE EQUIVALENTS:

In general, there are two kinds of normative or reference group, comparisons (Lindquist

Hieronymus, 1964). One kind of comparison makes use of a single reference group and describes the standing of an individual's scores in relation to the distribution of scores for that group. This type of reference is exemplified by the percentile distributions mentioned above and also by the normative scales. The second kind of comparison makes use of the mean scores on a series of reference groups and essentially identifies the group whose mean score is most nearly the score like the score under consideration. This approach is exemplified by the **grade equivalents**, which are appropriate for subject-matter achievement tests highly dependent on the curriculum and on the grade in which the subject is taught. It is also exemplified by the **age equivalents**, which appear to be more appropriate for such measure as general aptitude and intelligence which are less highly independent on the curriculum.

The imperfect correlation between age or grade and test performance leads to a number of anomalies, ambiguities and inconsistencies that impair the usefulness of the age or grade equivalents. This is unfortunate because except for these defects, “equivalents” appear other wise to have the ideal characteristics of interpretive data-clarity, definiteness, and direct meaning.

NORMS BY AGE AND GRADE:

A kind of normative data that makes use of the relationship between test performance and age (or grade) but avoids many of the problems associated with the equivalents, is the age or grade norms. These data are essentially nationally representative percentile rank distributions, differentiated by age or grade. Instead of age or grade

equivalents, they yield the usual kind of percentile rank that describes the person's relative standing in relation to other individuals who are of the same age or in the same grade.

It is important to note that the substitution of age and grade norms for age and grade equivalents effectively separates the function of norms from that of scales. Thus it is possible to develop a metric for a system of test forms and, entirely independently, to develop a collection of different kinds of norms – differentiated, for example, by grade, by region, by sex, by type of community, etc. – without committing the scale to any one of these sets of norms. The scale remains constant, so long as the test is appropriate and relevant to times. The norms on the other hand are free to develop and change as necessary to

provide the particular interpretive information required at the time.

STEP 11: DEVELOPING THE INSTRUCTIONAL MANUAL:

An ideal test manual is expected to include standard stimulus content, ways to administer the test, way to score objectively, specific, local normative data and standardized method of interpretation, with established reliabilities and validities.

Publishers may, however, present correlations of test scores with some other criterion (perhaps another well-known achievement test) as evidence of the validity or usefulness of the new test. The test publisher will probably collect the pertinent information and data into a manual for the test.

CONTENTS

What is Biopsychology? Divisions of Biopsychology.

Sensory systems: General and specific sensations, receptors and processes

Neurons: Structure, functions, types, neural impulse, synaptic transmission.
Neurotransmitters.

The Central and Peripheral Nervous Systems – Structure and functions. Neuroplasticity.

Methods of Physiological Psychology: Invasive methods – Anatomical methods, degeneration techniques, lesion techniques, chemical methods, microelectrode studies. Non-invasive methods – EEG, Scanning methods.

Muscular and Glandular system: Types and functions Biological basis of Motivation: Hunger, Thirst, Sleep and Sex. Biological basis of emotion: The Limbic system, Hormonal regulation of behavior. Genetics and behavior: Chromosomal anomalies; Nature-Nurture controversy [Twin studies and adoption studies]

UNIT 4

Biological Basis Of Behavior

NATURE AND SCOPE

Neuroscience is the scientific study of the nervous system may prove to be the brain's ultimate challenge. Neuroscience comprises several related disciplines. The primary discipline being biopsychology.

Biopsychology is a field devoted to our understanding of the physiology/biology of behavior. It is also known as **physiological psychology**, psychobiology, behavioral biology, or behavioral neuroscience; but preferable term is biopsychology because it denotes a biological approach to the study of psychology rather than a psychological approach to the study of biology. Psychology is the scientific study of behavior, the scientific study of all overt activities of the organism as well as all the internal processes that are presumed to underlie them (e.g., learning, memory, motivation, perception, and emotion).

The study of the biology of behavior has a long history, but biopsychology did not develop into a major neuroscientific discipline until the 20th century. Although it is not possible to specify the exact date of biopsychology's birth, the publication of *The Organization of Behavior* in 1949 by D. O. Hebb played a key role in its emergence. Hebb developed the first comprehensive theory of how complex psychological phenomena, such as

perceptions, emotions, thoughts, and memories, might be produced by brain activity. Hebb's theory did much to discredit the view that psychological functioning is too complex to have its roots in the physiology and chemistry of the brain. Hebb based his theory on experiments involving both humans and laboratory animals, on clinical case studies, and on logical arguments developed from his own insightful observations of daily life. This eclectic approach has become a hallmark of biopsychological inquiry.

Bio-psychologists draw together knowledge from the other neuroscientific disciplines and apply it to the study of behavior. The following are a few of the disciplines of neuroscience that are particularly relevant to biopsychology:

Neuroanatomical: The study of the structure of the nervous system.

Neurochemistry: The study of the chemical bases of neural activity.

Neuroendocrinology: The study of interactions between the nervous system and the endocrine system.

Neuropathology: The study of nervous system disorders.

Neuropharmacology: The study of the effects of drugs on neural activity.

Neurophysiology: The study of the functions and activities of the nervous system.

DIVISIONS OF BIOPSYCHOLOGY

Biopsychology has six major divisions:

Physiological Psychology
 Psychopharmacology
 Neuropsychology
 Psychophysiology
 Cognitive Neuroscience
 Comparative Psychology

For simplicity, they are presented as distinct approaches; but there is much overlap among them, and many bio psychologists regularly follow more than one approach.

TABLE 1.2 The Six Major Divisions of Biopsychology, with Examples of How They Have Approached the Study of Memory

<i>The Six Divisions of Biopsychology</i>	<i>Examples of How the Six Approaches Have Pursued the Study of Memory</i>
Physiological psychology: study of the neural mechanisms of behavior by manipulating the nervous systems of nonhuman animals in controlled experiments.	Physiological psychologists have studied the contributions of the hippocampus to memory by surgically removing the hippocampus in rats and assessing their ability to perform various memory tasks.
Psychopharmacology: study of the effects of drugs on the brain and behavior.	Psychopharmacologists have tried to improve the memory of Alzheimer's patients by administering drugs that increase the levels of the neurotransmitter acetylcholine.
Neuropsychology: study of the psychological effects of brain damage in human patients.	Neuropsychologists have shown that patients with alcohol-produced brain damage have particular difficulty in remembering recent events.
Psychophysiology: study of the relation between physiological activity and psychological processes in human subjects by noninvasive physiological recording.	Psychophysiolgists have shown that familiar faces elicit the usual changes in autonomic nervous system activity even when patients with brain damage report that they do not recognize a face.
Cognitive neuroscience: study of the neural mechanisms of human cognition, largely through the use of functional brain imaging.	Cognitive neuroscientists have used brain-imaging technology to observe the changes that occur in various parts of the brain while human volunteers perform memory tasks.
Comparative psychology: study of the evolution, genetics, and adaptiveness of behavior, largely through the use of the comparative method.	Comparative psychologists have shown that species of birds that cache their seeds tend to have big hippocampi, confirming that the hippocampus is involved in memory for location.

PHYSIOLOGICAL PSYCHOLOGY:

It is the division of biopsychology that studies the neural mechanisms of behavior through the direct manipulation of the brain in controlled experiments surgical and electrical methods of brain manipulation. The

subjects of physiological psychology research are almost always laboratory animals, because the focus on direct brain manipulation and controlled experiments precludes the use of human subjects in most instances. There is also a tradition of pure research in physiological psychology; the emphasis is usually on research that contributes to the development of

theories of the neural control of behavior rather than on research that is of immediate practical benefit.

PSYCHOPHARMACOLOGY:

Psychopharmacology is similar to physiological psychology, except that it focuses on the manipulation of neural activity and behavior with drugs. In fact, many of the early psychopharmacologists were simply physiological psychologists who moved into drug research, and many of today's bio psychologists identify closely with both approaches. However, the study of the effects of drugs on the brain and behavior has become so specialized that psychopharmacology is regarded as a separate discipline. A substantial portion of psychopharmacological research is applied. Although drugs are sometimes used by psychopharmacologists to study the basic principles of brain behavior interaction, the purpose of many psychopharmacological experiments is to develop therapeutic drugs or to reduce drug abuse. Psychopharmacologists study the effects of drugs on laboratory species and on humans, if the ethics of the situation permits it.

NEUROPSYCHOLOGY:

Neuropsychology is the study of the psychological effects of brain damage in human patients. Obviously, human subjects cannot ethically be exposed to experimental treatments that endanger normal brain function. Consequently, neuropsychology deals almost exclusively with case studies and quasi experimental studies of patients with brain damage resulting from disease, accident, or neurosurgery. The outer

layer of the cerebral hemispheres, the cerebral cortex is most likely to be damaged by accident or surgery; this is one reason why neuropsychology has focused on this important part of the human brain. Neuropsychology is the most applied of the biopsychological subdisciplines.

PSYCHOPHYSIOLOGY:

Psychophysiology is the division of biopsychology that studies the relation between physiological activity and psychological processes in human subjects. Because the subjects of psychophysiological research are human, psychophysiological recording procedures are typically noninvasive; that is, the physiological activity is recorded from the surface of the body. The usual measure of brain activity is the scalp electroencephalogram (EEG). Other common psychophysiological measures are muscle tension, eye movement, and several indicators of autonomic nervous system activity (e.g., heart rate, blood pressure, pupil dilation, and electrical conductance of the skin). The autonomic nervous system (ANS) is the division of the nervous system that regulates the body's inner environment. Most psychophysiological research focuses on understanding the physiology of psychological processes, such as attention, emotion, and information processing, but there have also been a number of interesting clinical applications of the psychophysiological method. For example, psychophysiological experiments have indicated that schizophrenics have difficulty smoothly tracking a moving object such as a pendulum (e.g., Chen et al., 2008).

COGNITIVE NEUROSCIENCE:

Cognitive neuroscience is the youngest division of biopsychology, but it is currently among the most active and exciting. Cognitive neuroscientists study the neural bases of cognition, a term that generally refers to higher intellectual processes such as thought, memory, attention, and complex perceptual processes. Because of its focus on cognition, most cognitive neuroscience research involves human subjects; and because of its focus on human subjects, its methods tend to be noninvasive, rather than involving penetration or direct manipulation of the brain. The major method of cognitive neuroscience is functional brain imaging (recording images of the activity of the living human brain) while a subject is engaged in a particular cognitive activity.

COMPARATIVE PSYCHOLOGY:

Comparative psychology is the division of biopsychology that deals generally with the biology of behavior, rather than specifically with the neural mechanisms of behavior. Comparative psychologists compare the behavior of different species in order to understand the evolution, genetics, and adaptiveness of behavior. Some comparative psychologists study behavior in the laboratory; others engage in ethological research i.e. the study of animal behavior in its natural environment. The two important areas of biopsychological research that also employ comparative analysis are often considered as part of comparative psychology. One of these is evolutionary psychology (a subfield that focuses on understanding behavior by considering its likely evolutionary origins. The other is behavioral

genetics (the study of genetic influences on behavior.

SENSORY SYSTEMS: GENERAL AND SPECIFIC SENSATIONS, RECEPTORS AND PROCESSES

Sensory receptors are specialized cells, usually neurons, which detect and respond to physical and chemical stimuli. Most are exquisitely sensitive to highly specific inputs, or sensory modalities, such as movement, odor chemicals or visible light photons. Detection of the physical stimulus commonly occurs in a morphologically and physiologically specialized cell region, while information about the stimulus is transmitted by action potentials along a sensory axon to the animal's central nervous system.

Thus, a major role of **sensory receptors is to help us learn about the environment around us, or about the state of our internal environment.** Stimuli from varying sources, and of different types, are received and changed into the electrochemical signals of the nervous system. This occurs when a stimulus changes the cell membrane potential of a sensory neuron. The stimulus causes the sensory cell to produce an action potential that is relayed into the central nervous system (CNS), where it is integrated with other sensory information—or sometimes higher cognitive functions—to become a conscious perception of that stimulus. The central integration may then lead to a motor response. Describing sensory function with the term sensation or perception is a deliberate distinction. **Sensation** is the activation of sensory receptor cells at the level of the stimulus. **Perception** is the central processing of sensory stimuli into a meaningful pattern. **Perception is dependent on sensation, but not all**

sensations are perceived. Receptors are the cells or structures that detect sensations. A receptor cell is changed directly by a stimulus. A trans membrane protein receptor is a protein in the cell membrane that mediates a physiological change in a neuron, most often through the opening of ion channels or changes in the cell signaling processes. Trans membrane receptors are activated by chemicals called ligands. For example, a molecule in food can serve as a ligand for taste receptors. Other trans membrane proteins, which are not accurately called receptors, are sensitive to mechanical or thermal changes. Physical changes in these proteins increase ion flow across the membrane, and can generate an action potential or a graded potential in the sensory neurons.

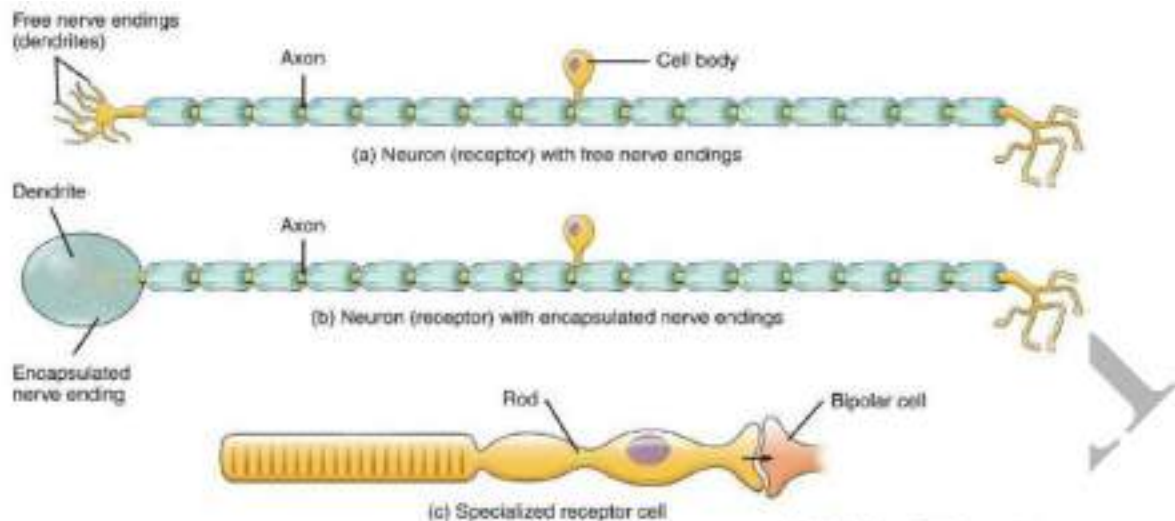
SENSORY RECEPTORS

Stimuli in the environment activate specialized receptor cells in the peripheral nervous system. Different types of stimuli are sensed by different types of receptor cells. Receptor cells can be classified into types on the basis of three different criteria: cell type, position, and function. Receptors can be classified structurally on the basis of cell type and their position in relation to stimuli they sense. They can also be classified functionally on the basis of the transduction of stimuli, or how the mechanical stimulus, light, or chemical changed the cell membrane potential.

STRUCTURAL RECEPTOR TYPES

The cells that interpret information about the environment can be either (1) a neuron that has a free nerve ending, with dendrites embedded in tissue that would receive a sensation; (2) a neuron that has an encapsulated ending in which the sensory nerve endings are encapsulated in connective tissue that enhances their sensitivity; or (3) a specialized receptor cell, which has distinct structural components that interpret a specific type of stimulus. The pain and temperature receptors in the dermis of the skin are examples of neurons that have free nerve endings. Also located in the dermis of the skin are lamellate corpuscles, neurons with encapsulated nerve endings that respond to pressure and touch. The cells in the retina that respond to light stimuli are an example of a specialized receptor, a photoreceptor.

Another way that receptors can be classified is based on their location relative to the stimuli. An **exteroceptor** is a receptor that is located near a stimulus in the external environment, such as the somatosensory receptors that are located in the skin. An **interoceptor** is one that interprets stimuli from internal organs and tissues, such as the receptors that sense the increase in blood pressure in the aorta or carotid sinus. Finally, a **proprioceptor** is a receptor located near a moving part of the body, such as a muscle, that interprets the positions of the tissues as they move.



FUNCTIONAL RECEPTOR TYPES

A third classification of receptors is by how the receptor transduces stimuli into membrane potential changes. Stimuli are of three general types. Some stimuli are ions and macromolecules that affect transmembrane receptor proteins when these chemicals diffuse across the cell membrane. Some stimuli are physical variations in the environment that affect receptor cell membrane potentials. Other stimuli include the electromagnetic radiation from visible light. For humans, the only electromagnetic energy that is perceived by our eyes is visible light. Some other organisms have receptors that humans lack, such as the heat sensors of snakes, the ultraviolet light sensors of bees, or magnetic receptors in migratory birds.

Receptor cells can be further categorized on the basis of the type of stimuli they transduce. Chemical stimuli can be interpreted by a **chemoreceptor** that interprets chemical stimuli, such as an object's taste or smell. **Osmoreceptors** respond to solute concentrations of body fluids. Additionally, pain is primarily a chemical sense that interprets the presence of chemicals from tissue damage, or similar intense stimuli, through a

nociceptor. Physical stimuli, such as pressure and vibration, as well as the sensation of sound and body position (balance), are interpreted through a **mechanoreceptor**. Another physical stimulus that has its own type of receptor is temperature, which is sensed through a **thermoreceptor** that is either sensitive to temperatures above (heat) or below (cold) normal body temperature.

SENSORY MODALITIES

Ask anyone what the senses are, and they are likely to list the five major senses—taste, smell, touch, hearing, and sight. However, these are not all of the senses. The most obvious omission from this list is balance. Also, what is referred to simply as touch can be further subdivided into pressure, vibration, stretch, and hair-follicle position, on the basis of the type of mechanoreceptors that perceive these touch sensations. Other overlooked senses include temperature perception by thermoreceptors and pain perception by nociceptors. Within the realm of physiology, senses can be classified as either **general or specific**.

A **GENERAL SENSE** is one that is distributed throughout the body and has

receptor cells within the structures of other organs. Mechanoreceptors in the skin, muscles, or the walls of blood vessels are examples of this type. General senses often contribute to the sense of touch, as described above, or to **proprioception** (body movement and detect changes that occur in muscles, tendons, ligaments and joint tissues) and **kinesthesia** (body movement), or to a **visceral sense**, which is most important to autonomic functions (i.e. detect changes that occur in internal organs).

A **SPECIAL SENSE** is one that has a specific organ devoted to it, namely the eye, inner ear, tongue, or nose. The senses of taste and smell (detected by chemoreceptors), hearing and equilibrium (detected by mechanoreceptors), and vision (detected by photoreceptors) are the (detected by photoreceptors) are the five special senses. They require specialized sensory receptors within large, complex sensory organs in the head region.

Each of the senses is referred to as a **sensory modality**. Modality refers to the way that information is encoded, which is similar to the idea of transduction. The main sensory modalities can be described on the basis of how each is transduced. The chemical senses are taste and smell. The general sense that is usually referred to as touch includes chemical sensation in the form of nociception, or pain. Pressure, vibration, muscle stretch, and the movement of hair by an external stimulus, are all sensed by mechanoreceptors. Hearing and balance are also sensed by mechanoreceptors. Finally, vision involves the activation of photoreceptors.

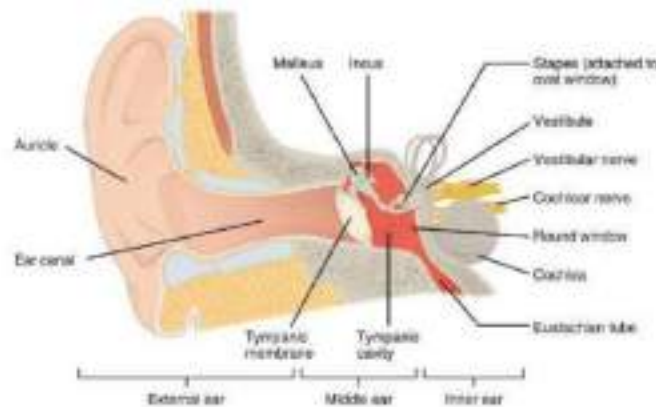
Listing all the different sensory modalities, which can number as many as 17, involves separating the five major senses into more specific categories, or **submodalities**, of the larger sense. An individual sensory

modality represents the sensation of a specific type of stimulus. For example, the general sense of touch, which is known as **somatosensation**, can be separated into light pressure, deep pressure, vibration, itch, pain, temperature, or hair movement.

(EXTRA FOR REFERENCE – IMPORTANT FOR NET)

AUDITION (HEARING)

Hearing, or audition, is the transduction of sound waves into a neural signal that is made possible by the structures of the ear (Figure 4). The large, fleshy structure on the lateral aspect of the head is known as the auricle. Some sources will also refer to this structure as the pinna, though that term is more appropriate for a structure that can be moved, such as the external ear of a cat. The C-shaped curves of the auricle direct sound waves toward the auditory canal. The canal enters the skull through the external auditory meatus of the temporal bone. At the end of the auditory canal is the tympanic membrane, or ear drum, which vibrates after it is struck by sound waves. The auricle, ear canal, and tympanic membrane are often referred to as the external ear. The middle ear consists of a space spanned by three small bones called the ossicles. The three ossicles are the malleus, incus, and stapes, which are Latin names that roughly translate to hammer, anvil, and stirrup. The malleus is attached to the tympanic membrane and articulates with the incus. The incus, in turn, articulates with the stapes. The stapes is then attached to the inner ear, where the sound waves will be transduced into a neural signal. The middle ear is connected to the pharynx through the Eustachian tube, which helps equilibrate air pressure across the tympanic membrane. The tube is normally closed but will pop open when the muscles of the pharynx contract during swallowing or yawning.



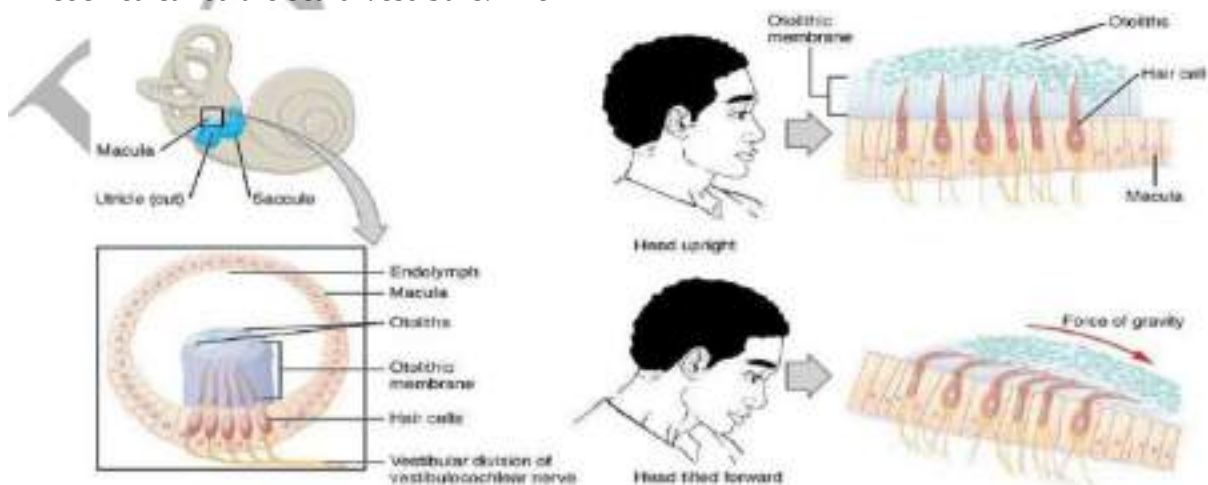
The inner ear is often described as a bony labyrinth, as it is composed of a series of canals embedded within the temporal bone. It has two separate regions, the **cochlea** and the **vestibule**, which are responsible for hearing and balance, respectively. The neural signals from these two regions are relayed to the brain stem through separate fiber bundles. However, these two distinct bundles travel together from the inner ear to the brain stem as the vestibulocochlear nerve. Sound is transduced into neural signals within the cochlear region of the inner ear, which contains the sensory neurons of the **spiral ganglia**. These ganglia are located within the spiral-shaped cochlea of the inner ear. The cochlea is attached to the stapes through the **oval window**.

The oval window is located at the beginning of a fluid-filled tube within the cochlea called the **scala vestibule**. The

scala vestibule extends from the oval window, travelling above the **cochlear duct**, which is the central cavity of the cochlea that contains the sound-transducing neurons.

EQUILIBRIUM (BALANCE)

Along with audition, the inner ear is responsible for encoding information about **equilibrium**, the sense of balance. A similar mechanoreceptor—a hair cell with stereocilia—senses head position, head movement, and whether our bodies are in motion. These cells are located within the vestibule of the inner ear. Head position is sensed by the **utricle** and **sacculus**, whereas head movement is sensed by the **semicircular canals**. The neural signals generated in the **vestibular ganglion** are transmitted through the vestibulocochlear nerve to the brain stem and cerebellum.



NEURONS: STRUCTURE, FUNCTIONS, TYPES, NEURAL IMPULSE, SYNAPTIC TRANSMISSION. NEUROTRANSMITTERS.

STRUCTURE AND FUNCTIONS OF NEURONS

NEURONS

The neuron or nerve cell is the information-processing and information-transmitting element of the nervous system. It is the basic unit of our nervous system and are specialized cells, which possess the unique property of converting various forms of stimuli into electrical impulses. They are also specialized for reception, conduction and transmission of information in the form of electrochemical signals. They receive information from sense organs or from other adjacent neurons, carry them to the central nervous

SOMA: The soma (cell body) contains the nucleus and much of the machinery that provides for the life processes of the cell. The genetic material of the neuron is stored inside the nucleus and it becomes

system (brain and the spinal cord), and bring motor information from the central nervous system to the motor organs (muscles and the glands).

BASIC STRUCTURE

Nearly 12 billion neurons are found in the human nervous system. They are of many types and vary in shapes, size, chemical composition, and function. Most neurons have, in one form or another, the following four structures or regions:

- Cell body (Soma)
- Dendrites
- Axons
- Terminal buttons.

actively engaged during the cell reproduction and protein synthesis. Its shape varies considerably in different kinds of neurons.

DENDRITES: Dendron is the Greek word for tree, and the dendrites of the neuron look very much like tree. Dendrites are branch-like specialized structures emanating from the soma.

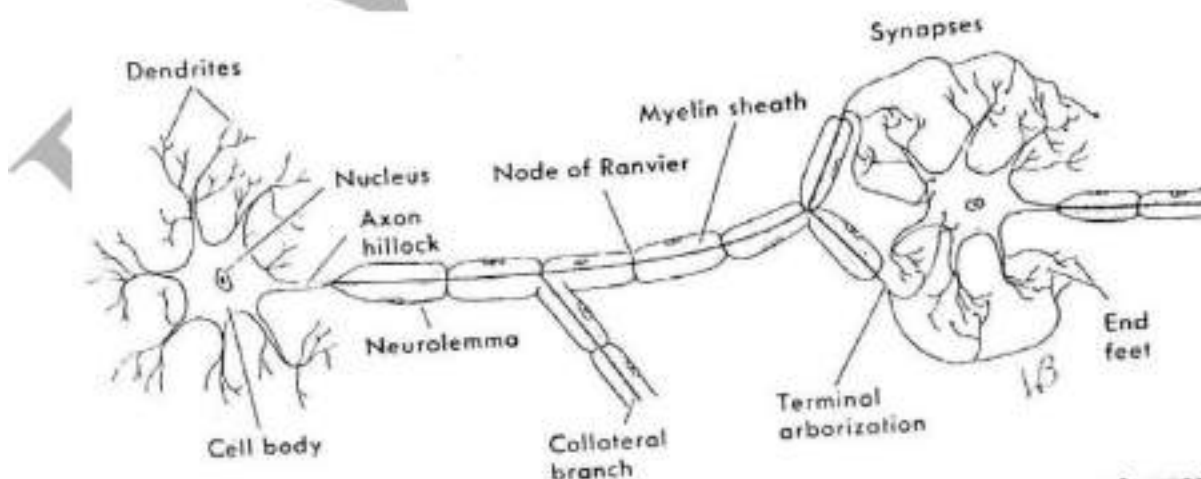


Fig. 4-1. Semidiagrammatic view of a "typical" neuron, showing its major parts and synapses with a second neuron.

They are the receiving ends of a neuron. Their function is to receive the incoming neural impulses from adjacent neurons or directly from the sense organs. On dendrites are found specialized receptors,

AXON: The axon is a long, slender tube, often covered by myelin sheath, which is a fatty covering interrupted at the **nodes of ranvier**. It carries information from the cell body to the terminal buttons. The transferred excitation is focused by the **axon hillock** on the axon and carried down the axon to its terminal arborization at the next set of synapses. The basic message it carries is called an action potential. When it reaches a point where the axon branches, it splits but does not diminish in size. Each branch receives a full-strength action potential.

which become active when a signal arrives in electrochemical or biochemical form. The received signals are passed on to soma and then to axon so that the information is relayed to another neuron.

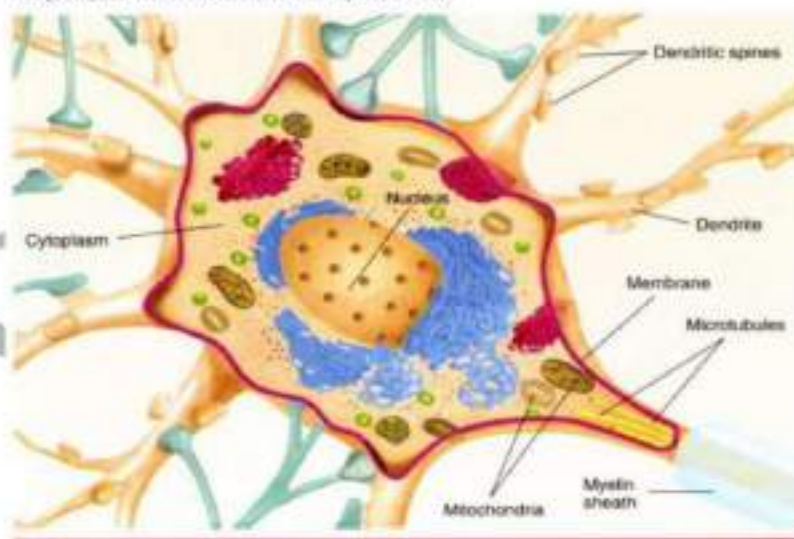
TERMINAL BUTTONS: The bud at the end of a branch of an axon, forms synapses with another neuron, sends information to that neuron. Terminal buttons have a very special function: when an action potential traveling down the axon reaches them; they secrete a chemical called a neurotransmitter. This chemical either excites or inhibits the receiving cell and thus helps to determine whether an action potential occurs in its axon. An individual neuron receives information from the terminal buttons of axons of other neurons, and the terminal buttons form synapses with other neurons.

INTERNAL STRUCTURE

The membrane defines the boundary of the cell. It consists of a double layer of lipid (fatlike) molecules. Embedded in the

membrane are a variety of protein molecules that have special

The principal internal structures of a multipolar neuron.



functions. Some proteins detect substances outside the cell (such as hormones) and pass information to the interior of the cell. Other proteins control access to the interior of the cell,

permitting some substances to enter but barring others and some proteins act as transporters, actively carrying certain molecules into or out of the cell.

The cell is filled with cytoplasm, a jellylike substance that contains small specialized structures such as **mitochondria**, which break down nutrients such as glucose and provide the cell with energy to perform its functions. Mitochondria produce a chemical called adenosine triphosphate (ATP), which can be used throughout the cell as an energy source.

Mitochondria contain their own genetic information and multiply independently of the cells in which they live. We inherit our mitochondria from our mothers. The nucleus is present deep inside the cell and contains the chromosomes. Chromosomes consist of long strands of deoxyribonucleic acid (DNA). The chromosomes contain the recipes for making proteins.

Proteins are important in cell functions. The **cytoskeleton** gives the neuron its shape. It is made up of various kinds of protein strands, linked to each other and forming a cohesive mass. Besides providing structure, proteins also serve as enzymes, causing particular molecules to join together or split apart. Proteins are also involved in transporting substances within the cell. For e.g., **axoplasmic transport**, is an active process that propels substances from one end of the axon to the other. This transport is accomplished by long protein strands called **microtubules**. Microtubules serve as railroad tracks, guiding the progress of the substances being transported. Movement from the soma to the terminal buttons is called **anterograde axoplasmic transport** (Antero- means “toward the front”). **Retrograde axoplasmic transport** carries substances from the terminal buttons back to the soma. (Retro- means “toward the back”). Energy for both forms of transport is supplied by ATP, produced by the mitochondria.

SUPPORTING CELLS

Neurons constitute only about half the volume of the CNS. The rest consists of a variety of supporting cells. Neurons have a very high rate of metabolism but they have no means of storing nutrients, they must constantly be supplied with nutrients and oxygen or they will quickly die. Thus, supporting cells support and protect neurons, which is very important to our existence.

GLIA or GLIAL CELLS

The most important supporting cells of the central nervous system are the neuroglia, or “nerve glue”.

Glial cells surround neurons and hold them in place, controlling their supply of nutrients and some of the chemicals they need to exchange messages with other neurons.

They insulate neurons from one another so that neural messages do not get scrambled.

They even act as housekeepers, destroying and removing the carcasses of neurons that are killed by disease or injury.

There are three most important types of glial cells:

1. **ASTROCYTE** (or astroglia): Astrocyte means “star-shaped cell”.

It plays the following functions:

Astrocytes provide physical support to neurons. They serve as the matrix that holds neurons in place. These cells also surround and isolate synapses, limiting the dispersion of neurotransmitters that are released by the terminal buttons.

PHAGOCYTOSIS (PHAGEIN, “TO EAT”; KUTOS, “CELL”): When neurons die, certain kinds of astrocytes take up the task of cleaning away the debris. When these astrocytes contact a piece of debris from a dead neuron,

they push themselves against it, finally engulfing and digesting it. This process is called phagocytosis. If there is a considerable amount of injured tissue to be cleaned up, astrocytes will divide and produce enough new cells to do the task. Once the dead tissue is broken down, a framework of astrocytes will be left to fill in the vacant area, and a specialized kind of astrocyte will form **scar tissue**, walling off the area.

They help to control the chemical composition of the fluid surrounding neurons by actively taking up or releasing substances whose concentrations must be kept within critical levels.

They produce some chemicals that neurons need to fulfill their functions.

Finally, astrocytes are involved in providing nourishment to neurons. Recent evidence suggests that astrocytes receive nutrients from the capillaries, store them, and release them to neurons when needed (Tsacopoulos and Magistretti, 1996; Magistretti et al., 1999).

2. **OLIGODENDROCYTES:** The principal function of oligodendrocytes is to provide support to axons and to produce the **myelin sheath**, which insulates most axons from one another. Myelin, 80 percent lipid and 20 percent protein, is produced by the oligodendrocytes in the form of a tube surrounding the axon. This tube does not form a continuous sheath rather it consists of a series of segment with a

small portion of uncoated axon between the segments. This bare portion of axon is called a **node of Ranvier**, after its discoverer. During the development of the CNS, oligodendrocytes form processes shaped something like canoe paddles. Each of these paddle-shaped processes then wraps itself many times around a segment of an axon and thus, produces layers of myelin. Each paddle thus becomes a segment of an axon's myelin sheath.

MICROGLIA: They are the smallest of the glial cells. Like some types of astrocytes, they act as phagocytes, engulfing and breaking down dead and dying neurons. They importantly serve as one of the representatives of the immune system in the brain, protecting the brain from invading microorganisms. They are primarily responsible for the inflammatory reaction in response to brain damage.

SCHWANN CELLS

Most axons in the PNS are myelinated. In the peripheral nervous system, the Schwann cells perform the same functions as the oligodendrocytes in the CNS of supporting axons and producing myelin. The myelin sheath occurs in segments, as it does in the CNS; each segment consists of a single Schwann cell, wrapped many times around the axon. In the PNS, a Schwann cell provides myelin for only one axon, and the entire Schwann cell and not merely a part of it surrounds the axon.

(Extra for References)**THE BLOOD-BRAIN BARRIER**

Blood-brain barrier is a semipermeable barrier between the blood and the brain produced by the cells in the walls of the brain's capillaries.

Over one hundred years ago, **Paul Ehrlich** discovered that if a blue dye is injected into an animal's bloodstream, all tissues except the brain and spinal will be tainted blue. However, if the same dye is injected into the fluid-filled ventricles of the brain, the blue color will spread throughout the CNS (Bradbury, 1979). This experiment demonstrates that a barrier exists between the blood and the fluid that surrounds the cells of the brain- the blood-brain barrier.

Some substances can cross the blood-brain barrier; others cannot. Thus, it is selectively permeable. In most of the body, the cells that line the capillaries do not fit together absolutely tightly. Small gaps are found between them that permit the free exchange of most substances between the blood plasma and the fluid outside the capillaries that surrounds the cells of the body. In the central nervous system, the capillaries lack these gaps; therefore, many substances cannot leave the blood. Thus, **the walls of the capillaries in the brain constitute the blood-brain barrier.** Other substances must be actively transported through the capillary walls by special proteins. For example, glucose transporters bring the brain its fuel, and other transporters rid the brain of toxic waste products.

SUMMARY

Neurons are the most important cells of the nervous system. The central nervous system (CNS) includes the brain and spinal cord; the peripheral nervous system (PNS) includes nerves and some sensory organs.

FUNCTION OF THE BLOOD-BRAIN BARRIER

Transmission of messages from place to place in the brain depends on a delicate balance between substances within neurons and in the extracellular fluid that surrounds them. If the composition of the extracellular fluid is changed even slightly, the transmission of these messages will be disrupted, which means that brain functions will be disrupted. The presence of the blood-brain barrier makes it easier **to regulate the composition of this fluid.** In addition, many of the foods that we eat contain chemicals that would interfere with the transmission of information between neurons. The blood-brain barrier prevents these chemicals from reaching the brain.

In several places the barrier is relatively permeable, allowing substances that are excluded elsewhere to cross freely. For example, the **area postrema** is a part of the brain that controls vomiting. The blood-brain barrier is much weaker there, permitting neurons in this region to detect the presence of toxic substances in the blood. A poison that enters the circulatory system from the stomach can thus stimulate this area to initiate vomiting. If the organism is lucky, the poison can be expelled from the stomach before it causes too much damage.

Neurons have four principal parts: soma (cell body), dendrites, axon, and terminal buttons. They communicate by means of synapses, located at the ends of the axons.

When an action potential travels down an axon, the terminal buttons secrete a chemical that has either an excitatory or an inhibitory effect on the neuron with which it communicates. The

interactions of circuits of neurons, with their excitatory and inhibitory synapses, are responsible for all of our perceptions, memories, thoughts, and behavior.

Neurons contain a quantity of clear cytoplasm, enclosed in a membrane. Embedded in the membrane are protein molecules that have special functions, such as the transport of particular substances into and out of the cell. The **nucleus** contains the genetic information- the recipes for the all the proteins that the body can make. **Microtubules** and other protein filaments compose the cytoskeleton and help to transport chemicals from place to place. **Mitochondria** serve as the location for most of the chemical reactions through which the cell extracts energy from nutrients.

Neurons in the central nervous system are supported by **glial cells**. Within the

CNS, astrocytes provide the primary support and also remove debris and form scar tissue in the event of tissue damage. Oligodendrocytes form myelin, the substance that insulates axons, and also support unmyelinated axons. Microglia are phagocytes that serve as the representatives of the immune system.

Within the PNS, support and myelin are provided by the Schwann cells.

In most organs, molecules freely move between the blood within the capillaries that serve them and the extracellular fluid that bathes their cells. The molecules pass through gaps between the cells that line the capillaries. The walls of the capillaries of the CNS lack these gaps and thus form a barrier between the blood and the brain. As a consequence, fewer substances can enter or leave the brain across the blood-brain barrier.

TYPES, NEURAL IMPULSE, SYNAPTIC TRANSMISSION. NEUROTRANSMITTERS.

TYPES OF NEURONS

Neurons vary in structure, function, and genetic makeup. Given the sheer number of neurons, there are thousands of different types, much like there are thousands of species of living organisms on Earth.

In terms of function, scientists classify neurons into three broad types: sensory, motor, and interneurons.

- SENSORY NEURONS

Sensory neurons help you: taste, smell, hear, see, and feel things around you.

Sensory neurons are triggered by physical and chemical inputs from your environment. Sound, touch, heat, and light are physical inputs. Smell and taste are chemical inputs.

For example, stepping on hot sand activates sensory neurons in the soles of your feet. Those neurons send a message to your brain, which makes you aware of the heat.

MOTOR NEURONS

Motor neurons play a role in movement, including voluntary and involuntary movements. These neurons allow the brain and spinal cord to communicate with muscles, organs, and glands all over the body.

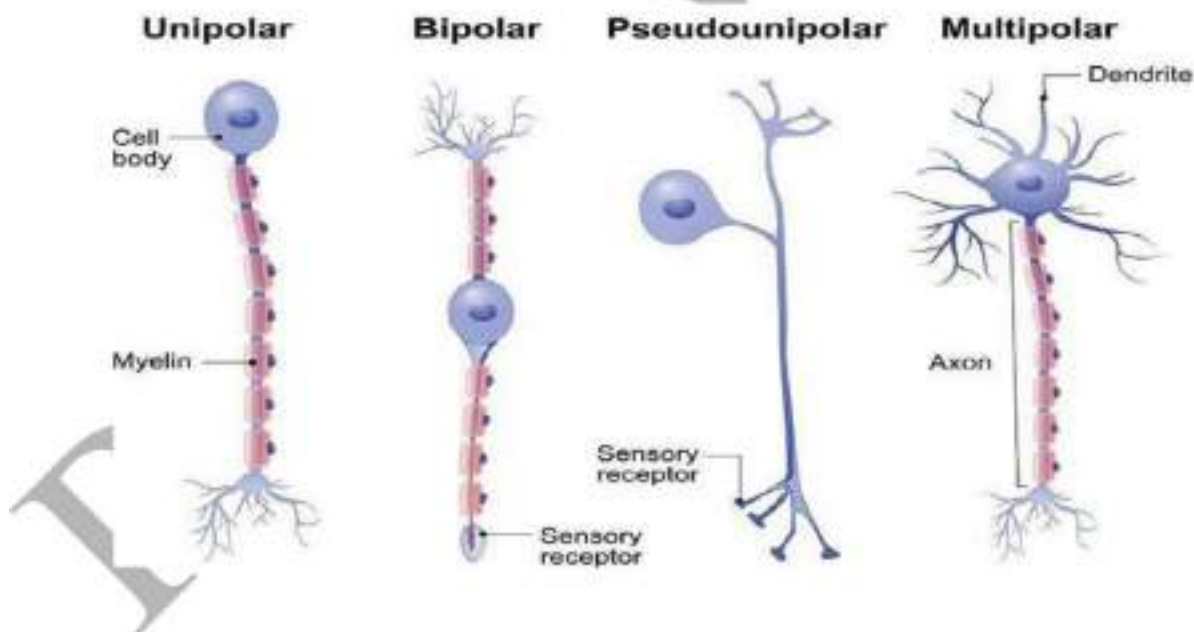
There are two types of motor neurons: lower and upper. Lower motor neurons carry signals from the spinal cord to the smooth muscles and the skeletal muscles. Upper motor neurons carry signals between your brain and spinal cord.

When you eat, for instance, lower motor neurons in your spinal cord send signals to the smooth muscles in your esophagus, stomach, and intestines. These muscles contract, which allows food to move through your digestive tract.

INTERNEURONS

Interneurons are neural intermediaries found in your brain and spinal cord. They're the most common type of neuron. They pass signals from sensory neurons and other interneurons to motor neurons and other interneurons. Often, they form complex circuits that help you to react to external stimuli.

For instance, when you touch something hot, sensory neurons in your fingertips send a signal to interneurons in your spinal cord. Some interneurons pass the signal on to motor neurons in your hand, which allows you to move your hand away. Other interneurons send a signal to the pain center in your brain, and you experience pain.



NEURAL IMPULSES

The central nervous system (CNS) goes through a three-step process when it functions: sensory input, neural processing, and motor output. The sensory input stage is when the neurons (or excitable nerve cells) of the sensory organs are excited electrically. Neural impulses from sensory receptors are sent to the brain and spinal cord for processing. After the brain has processed the information, neural impulses are then conducted from the brain and spinal cord to muscles and glands, which is the resulting motor output.

STAGES OF NEURAL IMPULSES

” Resting potential” is the name for the electrical state when a neuron is not actively being signaled. A neuron at resting potential has a membrane with established amounts of sodium (Na^+) and potassium (K^+) ions on either side, leaving the inside of the neuron negatively charged relative to the outside.

The action potential is a rapid change in polarity that moves along the nerve fiber

A neuron affects other neurons by releasing a neurotransmitter that binds to chemical receptors. The effect upon the postsynaptic (receiving) neuron is determined not by the presynaptic (sending) neuron or by the neurotransmitter itself, but by the *type of receptor* that is activated. A neurotransmitter can be thought of as a key, and a receptor as a lock: the key unlocks a certain response in the postsynaptic neuron, communicating a particular signal. However, in order for a presynaptic neuron to release a neurotransmitter to the next neuron in the chain, it must go through a series of changes in electric potential from neuron to neuron. In order for a neuron to move from resting potential to action potential—a short-term electrical change that allows an electrical signal to be passed from one neuron to another—the neuron must be stimulated by pressure, electricity, chemicals, or another form of stimuli. The level of stimulation that a neuron must receive to reach action potential is known as the threshold of excitation, and until it reaches that threshold, nothing will happen. Different neurons are sensitive to different stimuli, although most can register pain.

The action potential has several stages.

1. **DEPOLARIZATION:** A stimulus starts the depolarization of the membrane. Depolarization, also referred to as the “upswing,” is caused when positively charged sodium ions rush into a nerve cell. As these positive ions rush in, the membrane of the stimulated cell reverses its polarity so that the outside of the membrane is negative relative to the inside.

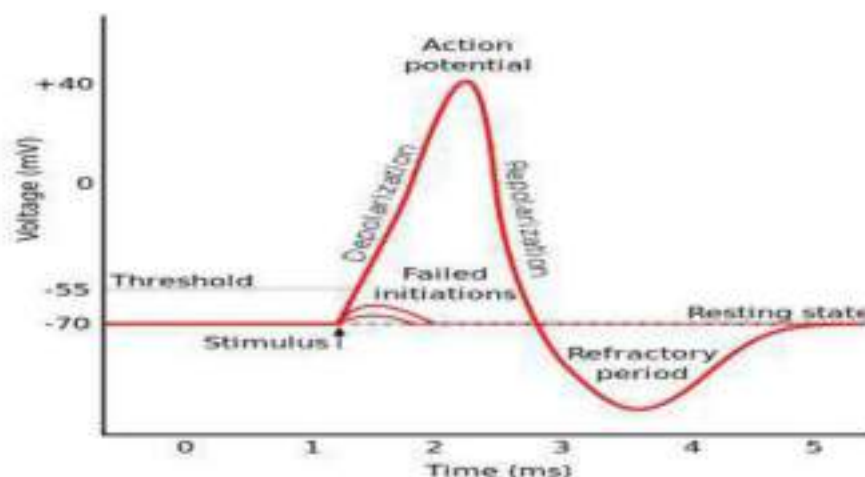
2. **REPOLARIZATION:** Once the electric gradient has reached the threshold of excitement, the “downswing” of repolarization begins. The channels that let the positive sodium ion channels through close up, while channels that allow positive potassium ions open, resulting in the release of positively charged potassium ions from the neuron. This expulsion acts to restore the localized negative membrane potential of the cell, bringing it back to its normal voltage.

3. REFRACTORY PHASE: The refractory phase takes place over a short period of time after the depolarization stage. Shortly after the sodium gates open, they close and go into an inactive conformation. The sodium gates cannot be opened again until the membrane is repolarized to its normal resting potential. The sodium-

This process of depolarization, repolarization, and recovery moves along a nerve fiber from neuron to neuron like a very fast wave. While an action potential is in progress, another cannot be generated under the same

potassium pump returns sodium ions to the outside and potassium ions to the inside. During the refractory phase this particular area of the nerve cell membrane cannot be depolarized. Therefore, the neuron cannot reach action potential during this “rest period.”

conditions. In unmyelinated axons (axons that are not covered by a myelin sheath), this happens in a continuous fashion because there are voltage-gated channels throughout the membrane. In myelinated axons (axons



Action potentials: A neuron must reach a certain threshold in order to begin the depolarization step of reaching the action potential.

covered by a myelin sheath), this process is described as saltatory because voltage-gated channels are only found at the nodes of Ranvier, and the electrical events seem to “jump” from one node to the next. Saltatory conduction is faster than continuous conduction. The diameter of the axon also makes a

difference, as ions diffusing within the cell have less resistance in a wider space. Damage to the myelin sheath from disease can cause severe impairment of nerve-cell function. In addition, some poisons and drugs interfere with nerve impulses by blocking sodium channels in nerves.

ALL-OR-NONE SIGNALS

The amplitude of an action potential is independent of the amount of current that produced it. In other words, larger currents do not create larger action potentials. Therefore, action potentials are said to be all-or-none signals, since either they occur fully or they do not occur at all. The frequency of action potentials is correlated with the intensity of a stimulus. This is in contrast to receptor potentials, whose amplitudes are dependent on the intensity of a stimulus.

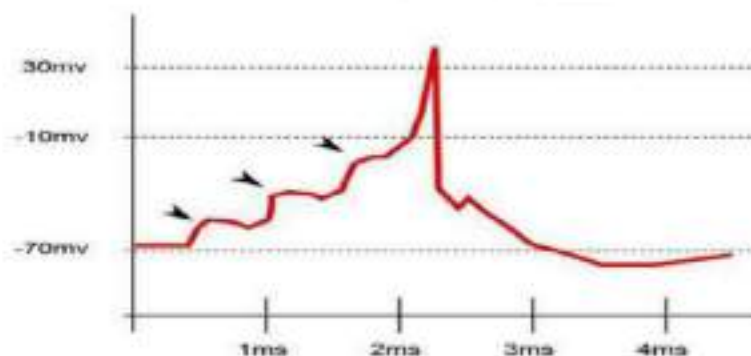
SYNAPSES

Synapse is the junction where neurons trade information. It is not a physical component of a cell but rather a name for the gap between two cells: the presynaptic cell (giving the signal) and the postsynaptic cell (receiving the signal).

REUPTAKE

Reuptake refers to the reabsorption of a neurotransmitter by a presynaptic (sending) neuron after it has performed its function of transmitting a neural impulse. Reuptake is necessary for normal synaptic physiology because it allows for the recycling of neurotransmitters and regulates the neurotransmitter level in the synapse, thereby controlling how long a signal resulting from neurotransmitter release lasts.

at the synapse—chemical or electrical. During a chemical reaction, a chemical called a neurotransmitter is released from one cell into another. In an electrical reaction, the electrical charge of one cell is



The electrical response of a neuron to multiple synaptic inputs: Synaptic responses summate in order to bring the postsynaptic neuron to the threshold of excitation, so it can fire an action potential (represented by the peak on the chart).

There are two types of possible reactions

influenced by the charge an adjacent cell.

All synapses have a few common characteristics:

- Presynaptic cell: a specialized area within the axon of the giving cell that transmits information to the dendrite of the receiving cell.
- Synaptic cleft: the small space at the synapse that receives neurotransmitters.
- G-protein coupled receptors: receptors that sense molecules outside the cell and thereby activate signals within it.
- Ligand-gated ion channels: receptors that are opened or closed in response to the binding of a chemical messenger.
- Postsynaptic cell: a specialized area within the dendrite of the receiving cell that contains receptors designed to process neurotransmitters.

THE ELECTRICAL SYNAPSE

The stages of an electrical reaction at a synapse are as follows:

1. **RESTING POTENTIAL.** The membrane of a neuron is normally at rest with established concentrations of sodium ions (Na^+) and potassium ions (K^+) on either side. The membrane potential (or, voltage across the membrane) at this state is -70 mV , with the inside being negative relative to the outside.

DEPOLARIZATION.

A stimulus begins the depolarization of the membrane. Depolarization, also referred to as the “upswing,” occurs when positively charged sodium ions

(Na^+) suddenly rush through open sodium gates into a nerve cell. If the membrane potential reaches -55 mV , it has reached the threshold of excitation. Additional sodium rushes in, and the membrane of the stimulated cell actually reverses its polarity so that the outside of the membrane is negative relative to the inside. The change in voltage stimulates the opening of additional sodium channels (called a voltage-gated ion channel), providing what is known as a positive feedback loop. Eventually, the cell potential reaches $+40\text{ mV}$, or the action potential.

3. **REPOLARIZATION.** The “downswing” of repolarization is caused by the closing of sodium ion channels and the opening of potassium ion channels, resulting in the release of positively charged potassium ions (K^+) from the nerve cell. This expulsion acts to restore the localized negative membrane potential of the cell.
4. **REFRACTORY PHASE.** The refractory phase is a short period of time after the repolarization stage. Shortly after the sodium gates open, they close and go into an inactive conformation where the cell’s membrane potential is actually even lower than its baseline -70 mV . The sodium gates cannot be opened again until the membrane has completely repolarized to its normal resting potential, -70 mV . The sodium-potassium pump returns sodium ions to the outside and potassium ions to the inside. During the refractory phase this particular area of the nerve cell membrane cannot be depolarized; the cell cannot be excited.

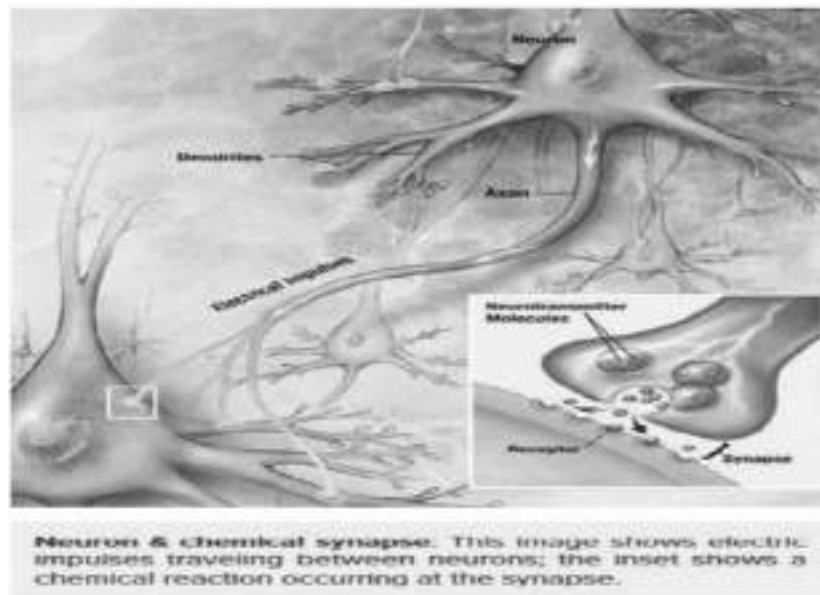
THE CHEMICAL SYNAPSE

The process of a chemical reaction at the synapse has some important differences

from an electrical reaction. Chemical synapses are much more complex than electrical synapses, which makes them slower, but also allows them to generate

different results. Like electrical reactions, chemical reactions involve electrical modifications at the postsynaptic

membrane, but chemical reactions also require chemical messengers, such as neurotransmitters, to operate.



A basic chemical reaction at the synapse undergoes a few additional steps:

The action potential (which occurs as described above) travels along the membrane of the presynaptic cell until it reaches the synapse. The electrical depolarization of the membrane at the synapse causes channels to open that are selectively permeable, meaning they specifically only allow the entry of positive sodium ions (Na^+).

The ions flow through the presynaptic membrane, rapidly increasing their concentration in the interior.

The high concentration activates a set of ion-sensitive proteins attached to vesicles, which are small membrane compartments that contain a neurotransmitter chemical.

These proteins change shape, causing the membranes of some “docked” vesicles to fuse with the membrane of the presynaptic cell. This opens the vesicles, which releases their

neurotransmitter contents into the synaptic cleft, the narrow space between the membranes of the pre- and postsynaptic cells.

The neurotransmitter diffuses within the cleft. Some of it escapes, but the rest of it binds to chemical receptor molecules located on the membrane of the postsynaptic cell.

The binding of neurotransmitter causes the receptor molecule to be activated in some way. Several types of activation are possible, depending on what kind of neurotransmitter was released. In any case, this is the key step by which the synaptic process affects the behavior of the postsynaptic cell.

Due to thermal shaking, neurotransmitter molecules eventually break loose from the receptors and drift away.

The neurotransmitter is either reabsorbed by the presynaptic cell or repackaged for future release, or else it is broken down metabolically.

DIFFERENCES BETWEEN ELECTRICAL AND CHEMICAL SYNAPSES

Electrical synapses are faster than chemical synapses because the receptors do not need to recognize chemical messengers. The synaptic delay for a chemical synapse is typically about 2 milliseconds, while the synaptic delay for an electrical synapse may be about 0.2 milliseconds.

Because electrical synapses do not involve neurotransmitters, electrical neurotransmission is less modifiable than chemical neurotransmission.

SYNAPTICS TRANSMISSION

Synaptic transmission is the process by which one neuron communicates with another. Information is passed down the axon of the neuron as an electrical impulse known as action potential. Once the action potential reaches the end of the axon it needs to be transferred to another neuron or tissue. It must cross over the synaptic gap between the presynaptic neuron and post-synaptic neuron. At the end of the neuron (in the axon terminal) are the synaptic vesicles, which contain chemical messengers, known as neurotransmitters.

I. SYNTHESIS AND STORAGE

Neurotransmitters are divided into two categories: (1) small-molecule neurotransmitters and (2) neuropeptides. Small-molecule neurotransmitters are processed within the axon terminal. An example of this is acetylcholine (ACh), which is particularly known as an excitatory neurotransmitter. On the other hand, neuropeptides are larger than small-molecule neurotransmitters and act as messengers. Because of their large size, they are made inside the neuronal cell body.

The response is always the same sign as the source. For example, depolarization of the presynaptic membrane will always induce a depolarization in the postsynaptic membrane, and vice versa for hyperpolarization.

The response in the postsynaptic neuron is generally smaller in amplitude than the source. The amount of attenuation of the signal is due to the membrane resistance of the presynaptic and postsynaptic neurons.

Long-term changes can be seen in electrical synapses. For example, changes in electrical synapses in the retina are seen during light and dark adaptations of the retina.

When the electrical impulse (action potential) reaches these synaptic vesicles, they release their contents of neurotransmitters. Neurotransmitters then carry the signal across the synaptic gap. They bind to receptor sites on the post-synaptic cell, thereby completing the process of synaptic transmission.

Chemical synapses enable the transmission of information (action potentials) from one neuron to another. These synapses require chemicals known as neurotransmitters.

The process of synaptic transmission involves four steps:

Unlike the small-molecule neurotransmitters, the synthesis of neuropeptides requires more effort and is likened to that of the synthesis of an ordinary secretory protein. The first step in neuropeptide synthesis is DNA transcription, followed by messenger RNA or mRNA construction and travel, and then translation. After the synthesis of neurotransmitters, they are stored in vesicles located at the axon terminal. While in storage, they await the arrival of an action potential, which is the triggering factor for their release.

II. NEUROTRANSMITTER RELEASE

The terminal of a neuron serves as the storage of vesicles containing neurotransmitter when at rest. These vesicles are strategically located in active

III. NEUROTRANSMITTER POSTSYNAPTIC RECEPTORS

Neurotransmitters released into the synaptic cleft go into an interaction at the postsynaptic cell together with receptor proteins. In order to interact, these

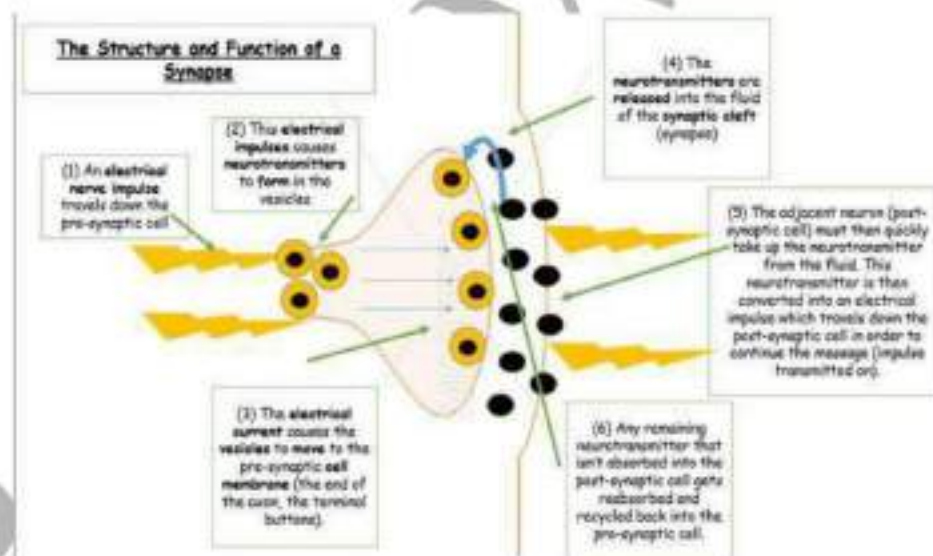
IV. INACTIVATION OF NEUROTRANSMITTERS

Some neurotransmitters are sent back to the synaptic cleft once they have been identified by the appropriate post-synaptic receptors. Special transporter proteins

zones, places found at the pre-synaptic membrane. Once an action potential arrives at the presynaptic neuron terminal, there would be a considerable influx of calcium ions, causing the neurotransmitter to be released from the vesicles.

neurotransmitters must be recognized by postsynaptic receptors first. As a result, membrane ionic channels open, and another action potential is initiated. In turn, this leads to depolarization.

transfer these neurotransmitters back to the pre-synaptic cells. Then, they undergo re-packaging and re-storage in a vesicle until it is needed once again for chemical signaling. Other neurotransmitters simply diffuse away



NEUROTRANSMITTERS

Neurotransmitters are chemicals that transmit signals from a neuron across a synapse to a target cell. Neurotransmitters are chemicals that transmit signals from a neuron to a target cell across a synapse. When called upon to deliver messages, they are released from their synaptic vesicles on the presynaptic (giving) side of the synapse, diffuse across the synaptic cleft, and bind to receptors in the membrane on the postsynaptic (receiving) side.

An action potential is necessary for neurotransmitters to be released, which means that neurons must reach a certain threshold of electric stimulation in order to complete the reaction. A neuron has a negative charge inside the cell membrane relative to the outside of the cell membrane; when stimulation occurs and the neuron reaches the threshold of excitement this polarity is reversed. This allows the signal to pass through the neuron. When the chemical message reaches the axon terminal, channels in the

postsynaptic cell membrane open up to receive neurotransmitters from vesicles in the presynaptic cell.

Inhibitory neurotransmitters cause hyperpolarization of the postsynaptic cell (that is, decreasing the voltage gradient of the cell, thus bringing it further away from an action potential), while excitatory neurotransmitters cause depolarization (bringing it closer to an action potential). Neurotransmitters match up with receptors like a key in a lock. A neurotransmitter binds to its receptor and will not bind to receptors for other neurotransmitters, making the binding a specific chemical event.

There are several systems of neurotransmitters found at various synapses in the nervous system. The following groups refer to the specific chemicals, and within the groups are specific systems, some of which block other chemicals from entering the cell and some of which permit the entrance of chemicals that were blocked before.

CHOLINERGIC SYSTEM

The cholinergic system is a neurotransmitter system of its own, and is based on the neurotransmitter acetylcholine (ACh). This system is found in the autonomic nervous system, as well as distributed throughout the brain.

The cholinergic system has two types of receptors: the nicotinic receptor and the

acetylcholine receptor, which is known as the muscarinic receptor. Both of these receptors are named for chemicals that interact with the receptor in addition to the neurotransmitter acetylcholine. Nicotine, the chemical in tobacco, binds to the nicotinic receptor and activates it similarly to acetylcholine. Muscarine, a chemical product of certain mushrooms, binds to the muscarinic receptor. However, they cannot bind to each other's receptors.

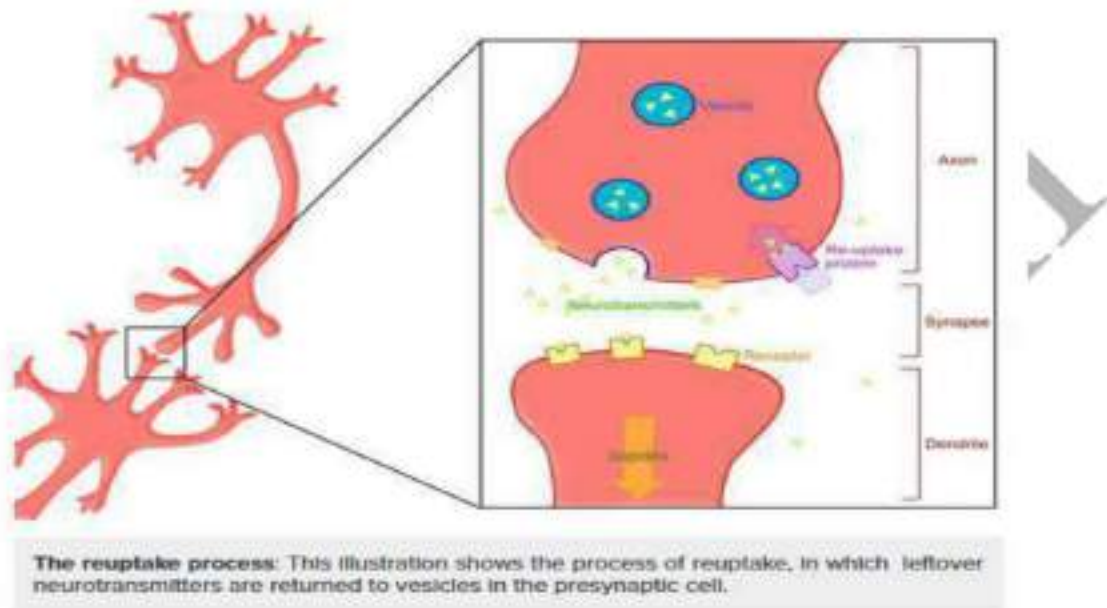
AMINO ACIDS

Another group of neurotransmitters are amino acids, including glutamate (Glu), GABA (gamma-aminobutyric acid, a derivative of glutamate), and glycine (Gly). These amino acids have an amino group and a carboxyl group in their

chemical structures. Glutamate is one of the 20 amino acids used to make proteins. Each amino acid neurotransmitter is its own system, namely the glutamatergic, GABAergic, and glycinergic systems. They each have their own receptors and do not interact with each other. Amino acid neurotransmitters are eliminated from the

synapse by reuptake. A pump in the cell membrane of the presynaptic element, or sometimes a neighboring glial cell, clears

the amino acid from the synaptic cleft so that it can be recycled, repackaged in vesicles, and released again.



BIOGENIC AMINES

Another class of neurotransmitter is the biogenic amine, a group of neurotransmitters made enzymatically from amino acids. They have amino

groups in them, but do not have carboxyl groups and are therefore no longer classified as amino acids.

NEUROPEPTIDES

A neuropeptide is a neurotransmitter molecule made up of chains of amino acids connected by peptide bonds, similar to proteins. However, proteins are long molecules while some neuropeptides are quite short. Neuropeptides are often released at synapses in combination with another neurotransmitter.

DOPAMINE

Dopamine is the best-known neurotransmitter of the catecholamine group. The brain includes several distinct dopamine systems, one of which plays a major role in reward-motivated behavior. Most types of reward increase the level of dopamine in the brain, and a variety of addictive drugs increase dopamine neuronal activity. Other brain dopamine systems are involved in motor control and in controlling the release of several other important hormones.

EFFECT ON THE SYNAPSE

The effect of a neurotransmitter on the postsynaptic element is entirely dependent on the receptor protein. If there is no receptor protein in the membrane of the postsynaptic element, then the neurotransmitter has no effect. The depolarizing (more likely to reach an action potential) or hyperpolarizing (less likely to reach an action potential) effect is also dependent on the receptor. When acetylcholine binds to the nicotinic receptor, the postsynaptic cell is depolarized. However, when acetylcholine binds to the muscarinic receptor, it might cause depolarization or hyperpolarization of the target cell.

The amino acid neurotransmitters (glutamate, glycine, and GABA) are almost exclusively associated with just one

effect. Glutamate is considered an excitatory amino acid because Glu receptors in the adult cause depolarization of the postsynaptic cell. Glycine and GABA are considered inhibitory amino acids, again because their receptors cause hyperpolarization, making the receiving cell less likely to reach an action potential.

THE RIGHT DOSE

Sometimes too little or too much of a neurotransmitter may affect an organism's behavior or health. The underlying cause of some neurodegenerative diseases, such as Parkinson's, appears to be related to overaccumulation of proteins, which under normal circumstances would be regulated by the presence of dopamine. On the other hand, when an excess of the neurotransmitter dopamine blocks glutamate receptors, disorders like schizophrenia can occur.

(EXTRA FOR REFERENCES)

KEY TERMINOLOGIES:

ACTION POTENTIAL: change in voltage of a cell membrane in response to a stimulus those results in transmission of an electrical signal; unique to neurons and muscle fibers.

CEREBRAL CORTEX: outermost layer of gray matter in the brain, where conscious perception takes place.

GRADED POTENTIAL: change in the membrane potential that varies in size, depending on the size of the stimulus that elicits it

LOWER MOTOR NEURON: second neuron in the motor command pathway that is directly connected to the skeletal muscle

NEUROTRANSMITTER: chemical signal that is released from the synaptic

end bulb of a neuron to cause a change in the target cell

PRECENTRAL GYRUS OF THE FRONTAL CORTEX: region of the cerebral cortex responsible for generating motor commands, where the upper motor neuron cell body is located

PROPAGATION: movement of an action potential along the length of an axon

THALAMUS: region of the central nervous system that acts as a relay for sensory pathways

THERMORECEPTOR: type of sensory receptor capable of transducing temperature stimuli into neural action potentials

THRESHOLD: membrane voltage at which an action potential is initiated

UPPER MOTOR NEURON: first neuron in the motor command pathway with its cell body in the cerebral cortex that

synapses on the lower motor neuron in the spinal cord

THE CENTRAL AND PERIPHERAL NERVOUS SYSTEMS – STRUCTURE AND FUNCTIONS.

The nervous system controls bodily function by gathering sensory input, integrating that information internally, and communicating proper motor output.

The nervous system can be divided into two major regions: the central and peripheral nervous systems. The **central nervous system (CNS)** is the brain and spinal cord, and the **peripheral nervous system (PNS)** is everything else. The brain is contained within the cranial cavity of the skull, and the spinal cord is contained within the vertebral cavity of the vertebral column. It is a bit of an oversimplification to say that the CNS is what is inside these two cavities and the peripheral nervous system is outside of them, but that is one way to start to think about it. In actuality, there are some elements of the peripheral nervous system that are within the cranial or vertebral cavities. The peripheral

nervous system is so named because it is on the periphery—meaning beyond the brain and spinal cord. Depending on different aspects of the nervous system, the dividing line between central and peripheral is not necessarily universal.

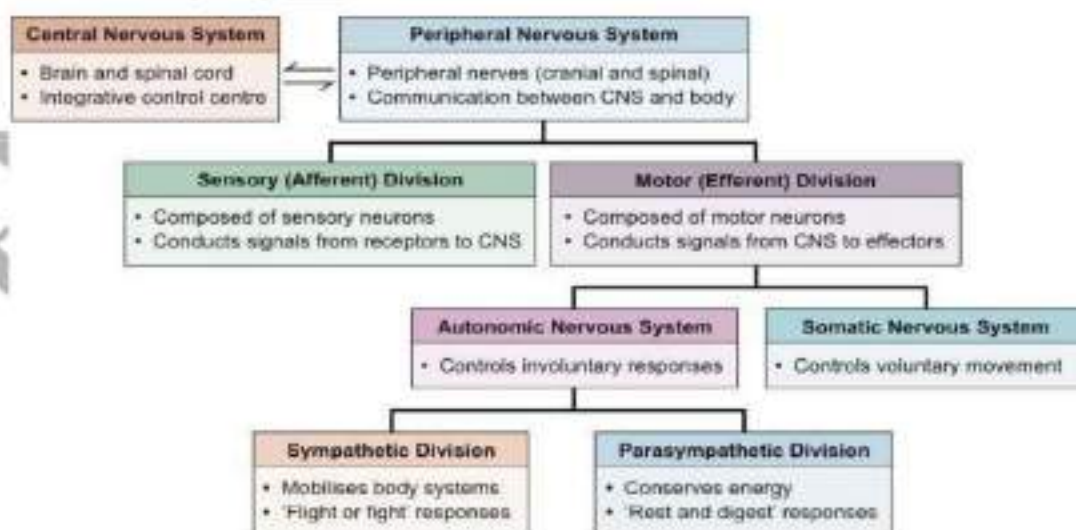
The nervous system is the body's main communication system; it gathers, synthesizes, and uses data from the environment.

The most basic unit of the nervous system is the neuron, which serves as both a sensor and communicator of internal and external stimuli.

The nervous system can be broken down into two major parts—the central nervous system and the peripheral nervous system.

The central nervous system, the main data center of the body, includes the brain and spinal cord.

The peripheral nervous system includes all of the neurons that sense and communicate data to the central nervous system.



THE CENTRAL NERVOUS SYSTEM (CNS)

The central nervous system is made up of the brain and spinal cord, which process sensory input and provide instructions to the body.

The CNS includes the brain and spinal cord, which together comprise the body's main control center. Together with the peripheral nervous system (PNS), the CNS performs fundamental functions that contribute to an organism's life and behavior.

STRUCTURE OF THE CENTRAL NERVOUS SYSTEM

The CNS is comprised of the **brain, brain stem, and spinal cord**.

- BRAIN

The brain is found in the cranial cavity and consists of the cerebrum and cerebellum. It houses the nerve centers responsible for coordinating sensory and motor systems in the body. The cerebrum, or the top portion for the brain, is the seat of higher-level thought. It is comprised of two hemispheres, each controlling the opposite side of the body.

The cerebellum is located underneath the backside of the cerebrum, and governs balance and fine motor movements. Its main function is maintaining coordination throughout the body.

(Discussed in detail ahead)

- BRAIN STEM

The brain stem is connected to the underside of the brain. It consists of the midbrain, pons, and medulla. The midbrain is found in between the hindbrain and the forebrain. It regulates motor function and allows motor and sensory information to pass from the brain to the

ACTIVITY OF THE CNS

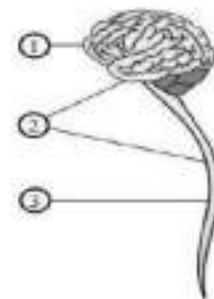
The nervous system has three main functions: gathering sensory information from external stimuli, synthesizing that information, and responding to those stimuli. The CNS is mainly devoted to the "information synthesizing" function. During this step in the process, the brain and spinal cord decide on appropriate motor output, which is computed based on the type of sensory input. The CNS regulates everything from organ function to high-level thought to purposeful body movement. Thus, the CNS is commonly thought of as the control center of the body.

rest of the body. The pons houses the control centers for respiration and inhibitory functions. The medulla also helps regulate respiration, as well as cardiovascular and digestive functioning.

- SPINAL CORD

The spinal cord is a tail-like structure embedded in the vertebral canal of the spine. The adult spinal cord is about 40 cm long and weighs approximately 30 g. The spinal cord is attached to the underside of the medulla oblongata, and is organized to serve four distinct tasks:

- to convey (mainly sensory) information to the brain;
- to carry information generated in the brain to peripheral targets like skeletal muscles;
- to control nearby organs via the autonomic nervous system;
- to enable sensorimotor functions to control posture and other fundamental movements.



The central nervous system. The three major components of the central nervous system: 1) the brain, 2) brain stem, and 3) spinal cord.

The spinal cord connects the brain and brain stem to all of the major nerves in the body. Spinal nerves originate from the spinal cord and control the functions of the rest of the body. Impulses are sent from receptors through the spinal cord to the

brain, where they are processed and synthesized into instructions for the rest of the body. This data is then sent back through the spinal cord to muscles and glands for motor output

STRUCTURE AND FUNCTION OF THE BRAIN

The human brain is one of the most complex systems on earth. Every component of the brain must work together in order to keep its body functioning. The brain and the spinal cord make up the central nervous system, which alongside the peripheral nervous system is responsible for regulating all bodily functions.

Psychology seeks to explain the mental processes and behavior of individuals by studying the interaction between mental

processes and behavior on a systemic level. Therefore, the field of psychology is tightly intertwined with the study of the brain.

THE STRUCTURE OF THE BRAIN

The developing brain goes through many stages. In the embryos of vertebrates, the predecessor to the brain and spinal cord is the neural tube. As the fetus develops, the grooves and folds in the neural tube deepen, giving rise to different layers of the brain. The human brain is split up into three major layers: the hindbrain, the midbrain, and the forebrain.

HINDBRAIN

The hindbrain is the well-protected central core of the brain. It includes the cerebellum, reticular formation, and brain stem, which are responsible for some of the most basic autonomic functions of life, such as breathing and movement. The brain stem contains the pons and medulla oblongata. Evolutionarily speaking, the hindbrain contains the oldest parts of the brain, which all vertebrates possess, though they may look different from species to species.

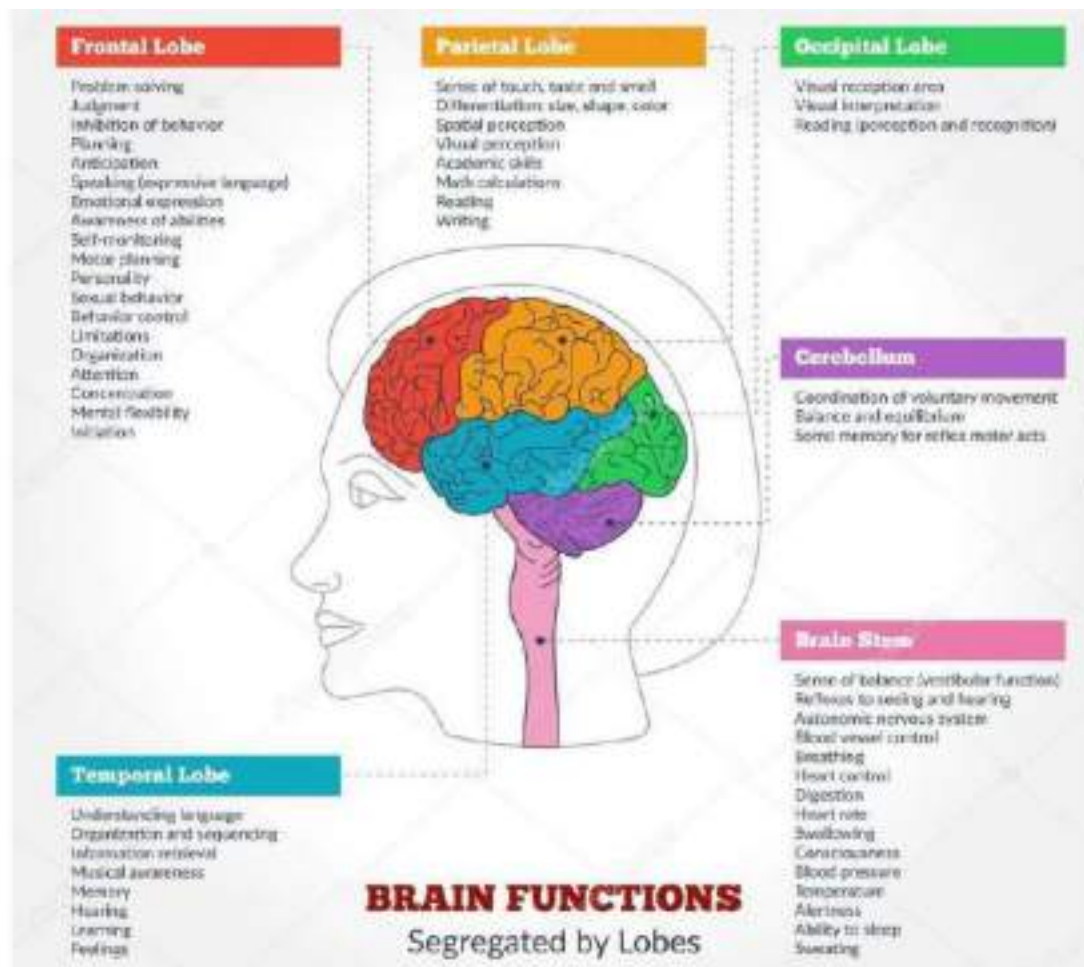
MIDBRAIN

The midbrain makes up part of the brain stem. It is located between the hindbrain and forebrain. All sensory and motor information that travels between the

forebrain and the spinal cord passes through the midbrain, making it a relay station for the central nervous system.

FOREBRAIN

The forebrain is the most anterior division of the developing vertebrate brain, containing the most complex networks in the central nervous system. The forebrain has two major divisions: the diencephalon and the telencephalon. The diencephalon is lower, containing the thalamus and hypothalamus (which together form the limbic system); the telencephalon is on top of the diencephalon and contains the cerebrum, the home of the highest-level cognitive processing in the brain. It is the large and complicated forebrain that distinguishes the human brain from other vertebrate brains.



LOWER-LEVEL STRUCTURES:

The brain's lower-level structures consist of the brain stem, the spinal cord, and the cerebellum.

The brain's lower-level structures consist of the brain stem and spinal cord, along with the cerebellum. With the exception of the spinal cord, these structures are largely located within the hindbrain, diencephalon (or interbrain), and midbrain. These lower dorsal structures are the oldest parts of the brain, having existed for much of its evolutionary history. As such they are geared more toward basic bodily processes necessary to survival. It is the more recent layers of the brain (the forebrain) which are responsible for the higher-level cognitive functioning (language, reasoning) not strictly necessary to keep a body alive.

THE HINDBRAIN

The hindbrain, which includes the medulla oblongata, the pons, and the cerebellum, is responsible some of the oldest and most primitive body functions. Each of these structures is described below.

MEDULLA OBLONGATA

The medulla oblongata sits at the transition zone between the brain and the spinal cord. It is the first region that formally belongs to the brain (rather than the spinal cord). It is the control center for respiratory, cardiovascular, and digestive functions.

PONS

The pons connects the medulla oblongata with the midbrain region, and also relays signals from the forebrain to the cerebellum. It houses the control centers for respiration and inhibitory functions. The cerebellum is attached to the dorsal side of the pons.

CEREBELLUM

The cerebellum is a separate region of the brain located behind the medulla oblongata and pons. It is attached to the rest of the brain by three stalks (called pedunculi), and coordinates skeletal muscles to produce smooth, graceful motions. The cerebellum receives information from our eyes, ears, muscles, and joints about the body's current positioning (referred to as proprioception). It also receives output from the cerebral cortex about where these body parts should be. After processing this information, the cerebellum sends motor impulses from the brain stem to the skeletal muscles so that they can move. The main function of the cerebellum is this

muscle coordination. However, it is also responsible for balance and posture, and it assists us when we are learning a new motor skill, such as playing a sport or musical instrument. Recent research shows that apart from motor functions the cerebellum also has some role in emotional sensitivity.

THE MIDBRAIN

The midbrain is located between the hindbrain and forebrain, but it is actually part of the brain stem. It displays the same basic functional composition found in the spinal cord and the hindbrain. Ventral areas control motor function and convey motor information from the cerebral cortex. Dorsal regions of the midbrain are involved in sensory information circuits. The substantia nigra, a part of the brain that plays a role in reward, addiction, and movement (due to its high levels of dopaminergic neurons) is located in the midbrain.

In Parkinson's disease, which is characterized by a deficit of dopamine, death of the substantia nigra is evident.

THE DIENCEPHALON ("INTERBRAIN")

The diencephalon is the region of the embryonic vertebrate neural tube that gives rise to posterior forebrain structures. In adults, the diencephalon appears at the upper end of the brain stem, situated between the cerebrum and the brain stem. It is home to the limbic system, which is considered the seat of emotion in the human brain. The diencephalon is made up of four distinct components: the thalamus, the subthalamus, the hypothalamus, and the epithalamus.

THALAMUS

The thalamus is part of the limbic system. It consists of two lobes of grey matter along the bottom of the cerebral cortex. Because nearly all sensory information passes through the thalamus it is considered the sensory "way station" of the brain, passing information on to the cerebral cortex (which is in the forebrain). Lesions of, or stimulation to, the thalamus are associated with changes in emotional reactivity. However, the importance of this structure on the regulation of emotional behavior is not due to the activity of the thalamus itself, but to the connections between the thalamus and other limbic-system structures.

HYPOTHALAMUS

The hypothalamus is a small part of the brain located just below the thalamus. Lesions of the hypothalamus interfere with motivated behaviors like sexuality, combativeness, and hunger. The hypothalamus also plays a role in emotion: parts of the hypothalamus seem to be involved in pleasure and rage, while the central part is linked to aversion, displeasure, and a tendency towards uncontrollable and loud laughing. When external stimuli are presented (for example, a dangerous stimuli), the hypothalamus sends signals to other limbic areas to trigger feeling states in response to the stimuli (in this case, fear).

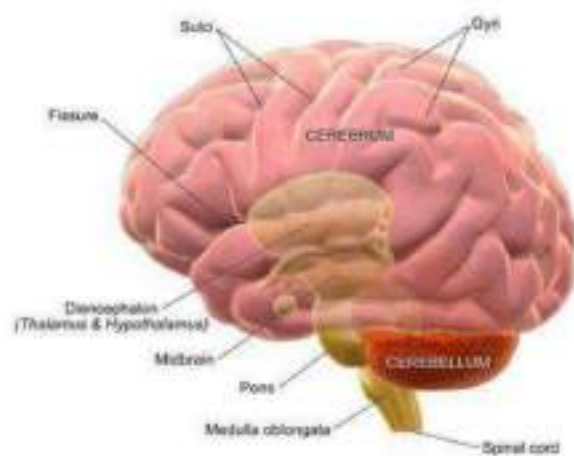
CEREBRAL CORTEX

The cerebral cortex is the outermost layered structure of the brain and controls higher brain functions such as information processing. The cerebral cortex, the largest part of the mammalian brain, is the wrinkly gray outer covering of the cerebrum. While the cortex is less than 1/4" thick, it is here that all sensation, perception, memory, association, thought, and voluntary. The "valleys" of the wrinkles are called *sulci* (or sometimes, fissures); the "peaks" between wrinkles are called *gyri*. While there are variations from person to person in their sulci and gyri, the brain has been studied enough to identify patterns. One notable sulcus is the central sulcus, or the wrinkle dividing the parietal lobe from the frontal lobe.

physical actions occur. The cerebral cortex is considered the ultimate control and information-processing center in the brain.

The cortex is made of layers of neurons with many inputs; these cortical neurons function like mini microprocessors or logic gates. It contains glial cells, which guide neural connections, provide nutrients and myelin to neurons, and absorb extra ions and neurotransmitters. The cortex is divided into four different lobes (the parietal, occipital, temporal, and frontal lobes), each with a different specific function.

The cortex is wrinkly in appearance. Evolutionary constraints on skull size brought about this development; it allowed for the cortex to become larger without our brains (and therefore craniums) becoming disadvantageously large. At times it has been theorized that brain size correlated positively with intelligence; it has also been suggested that surface area of cortex (basically, "wrinkles" of the brain) rather than brain size that correlates most directly with intelligence. Current research suggests that both of these may be at least partially true, but the degree to which they correlate is not clear.



Sulci and gyri: As depicted in this diagram of brain structures, sulci are the "valleys" and gyri are the "peaks" in the folds of the brain.

CEREBRUM

Beneath the cerebral cortex is the cerebrum, which serves as the main thought and control center of the brain. It is the seat of higher-level thought like emotions and decision making (as opposed to lower-level thought like balance, movement, and reflexes).

The cerebrum is composed of gray and white matter. Gray matter is the mass of all the cell bodies, dendrites, and synapses of neurons interlaced with one another, while white matter consists of the long, myelin-coated axons of those neurons connecting masses of gray matter to each other. (Discussed more in *referenced section*)

CEREBRAL HEMISPHERES AND LOBES OF THE BRAIN

The brain is divided into two hemispheres and four lobes, each of which specializes in a different function.

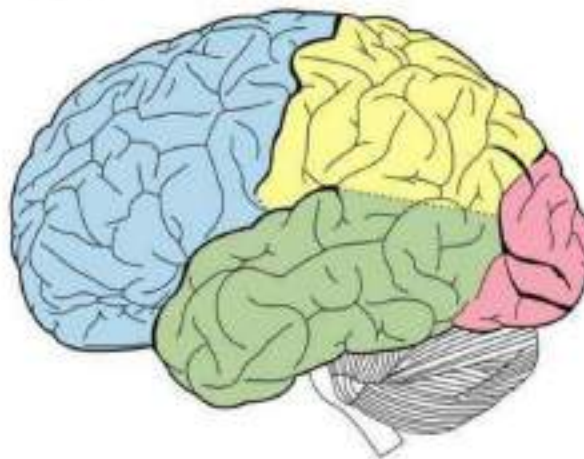
BRAIN LATERALIZATION

The brain is divided into two halves, called hemispheres. There is evidence that each brain hemisphere has its own distinct

to dominate the functions of speech, language processing and comprehension, and logical reasoning, while the right is more dominant in spatial tasks like vision-independent object recognition (such as identifying an object by touch or another nonvisual sense). However, it is easy to exaggerate the differences between the functions of the left and right hemispheres; both hemispheres are involved with most processes. Additionally, neuroplasticity (the ability of a brain to adapt to experience) enables the brain to compensate for damage to one hemisphere by taking on extra functions in the other half, especially in young brains.

CORPUS CALLOSUM

The two hemispheres communicate with one another through the corpus callosum. The corpus callosum is a wide, flat bundle of neural fibers beneath the cortex that connects the left and right cerebral hemispheres and facilitates interhemispheric communication. The corpus callosum is sometimes implicated in the cause of seizures; patients with epilepsy sometimes undergo a corpus



Lobes of the brain: A diagram of the brain identifying the different lobes by color. Counterclockwise from bottom: it contains the parietal lobe (green), the occipital lobe (red), the temporal lobe (yellow), and the frontal lobe (blue).

functions, a phenomenon referred to as lateralization. The left hemisphere appears

colostomy, or the removal of the corpus callosum.

THE LOBES OF THE BRAIN

The brain is separated into four lobes: the frontal, temporal, occipital, and parietal lobes.

THE FRONTAL LOBE

The frontal lobe is associated with executive functions and motor performance. Executive functions are some of the highest-order cognitive processes that humans have. Examples include:

- planning and engaging in goal-directed behavior;**
- recognizing future consequences of current actions;**
- choosing between good and bad actions;**
- overriding and suppressing socially unacceptable responses;**
- determining similarities and differences between objects or situations.**

The frontal lobe is considered to be the moral center of the brain because it is responsible for advanced decision-making processes. It also plays an important role in retaining emotional memories derived from the limbic system, and modifying those emotions to fit socially accepted norms.

THE TEMPORAL LOBE

The temporal lobe is associated with the retention of short- and long-term memories. It processes sensory input including auditory information, language comprehension, and naming. It also creates emotional responses and controls biological drives such as aggression and sexuality.

The temporal lobe contains the hippocampus, which is the memory center of the brain. The hippocampus plays a key

role in the formation of emotion-laden, long-term memories based on emotional input from the amygdala. The left temporal lobe holds the primary auditory cortex, which is important for processing the semantics of speech.

One specific portion of the temporal lobe, **Wernicke's area**, plays a key role in speech comprehension. Another portion, **Broca's area**, underlies the ability to produce (rather than understand) speech. Patients with damage to Wernicke's area can speak clearly but the words make no sense, while patients with damage to Broca's area will fail to form words properly and speech will be halting and slurred. These disorders are known as Wernicke's and Broca's aphasia respectively; aphasia is an inability to speak.

THE OCCIPITAL LOBE

The occipital lobe contains most of the visual cortex and is the visual processing center of the brain. Cells on the posterior side of the occipital lobe are arranged as a spatial map of the retinal field. The visual cortex receives raw sensory information through sensors in the retina of the eyes, which is then conveyed through the optic tracts to the visual cortex. Other areas of the occipital lobe are specialized for different visual tasks, such as visuospatial processing, color discrimination, and motion perception. Damage to the primary visual cortex (located on the surface of the posterior occipital lobe) can cause blindness, due to the holes in the visual map on the surface of the cortex caused by the lesions.

THE PARIETAL LOBE

The parietal lobe is associated with sensory skills. It integrates different types of sensory information and is particularly useful in spatial processing and navigation. The parietal lobe plays an important role in integrating sensory information from various parts of the body, understanding numbers and their relations, and manipulating objects. It also processes information related to the sense of touch.

The parietal lobe is comprised of the somatosensory cortex and part of the visual system. The somatosensory cortex consists of a “map” of the body that processes sensory information from specific areas of the body. Several portions of the parietal lobe are important to language and visuospatial processing; the left parietal lobe is involved in symbolic functions in language and mathematics, while the right parietal lobe is specialized to process images and interpretation of maps (i.e., spatial relationships).

THE PERIPHERAL NERVOUS SYSTEM (PNS)

The peripheral nervous system connects the central nervous system to environmental stimuli to gather sensory input and create motor output.

The peripheral nervous system (PNS) is one of the two major components of the body’s nervous system. In conjunction with the central nervous system (CNS), the PNS coordinates action and responses by sending signals from one part of the body

to another. The CNS includes the brain, brain stem, and spinal cord, while the PNS includes all other sensory neurons, clusters of neurons called ganglia, and connector neurons that attach to the CNS and other neurons.

The peripheral nervous system (PNS) provides the connection between internal or external stimuli and the central nervous system to allow the body to respond to its environment.

The PNS is made up of different kinds of neurons, or nerve cells, which communicate with each other through electric signaling and neurotransmitters.

The PNS can be broken down into two systems: the autonomic nervous system, which regulates involuntary actions such as breathing and digestion, and the somatic nervous system, which governs voluntary action and body reflexes.

The autonomic nervous system has two complementary parts: the sympathetic nervous system, which activates the “fight-or-flight-or-freeze” stress response, and the parasympathetic nervous system, which reacts with the “rest-and-digest” response after stress.

The somatic nervous system coordinates voluntary physical action. It is also responsible for our reflexes, which do not require brain input.

DIVISIONS OF THE PERIPHERAL NERVOUS SYSTEM

The PNS can also be divided into two separate systems: the autonomic nervous system and the somatic nervous system.

AUTONOMIC NERVOUS SYSTEM

The autonomic nervous system regulates involuntary and unconscious actions, such as internal-organ function, breathing, digestion, and heartbeat. This system consists of two complementary parts: the sympathetic and parasympathetic systems. Both divisions work without conscious effort and have similar nerve pathways, but they generally have opposite effects on target tissues.

The sympathetic nervous system activates the **“fight or flight”** response under sudden or stressful circumstances, such as taking an exam or seeing a bear. It increases physical arousal levels, raising the heart and breathing rates and dilating the pupils, as it prepares the body to run or confront danger. These are not the only two options; “fight or flight” is perhaps better phrased as “fight or flight or freeze,” where in the third option the body stiffens and action cannot be taken. This is an autonomic response that occurs in animals and humans; it is a survival mechanism thought to be related to playing dead when attacked by a predator. Post-traumatic stress disorder (PTSD) can result when a human experiences this “fight or flight or freeze” mode with great intensity or for large amounts of time.

The parasympathetic nervous system activates a “rest and digest” or “feed and breed” response after these stressful events, which conserves energy and replenishes the system. It reduces bodily arousal, slowing the heartbeat and

breathing rate. Together, these two systems maintain homeostasis within the body: one priming the body for action, and the other repairing the body afterward.

SOMATIC NERVOUS SYSTEM

The somatic nervous system keeps the body adept and coordinated, both through reflexes and voluntary action. The somatic nervous system controls systems in areas as diverse as the skin, bones, joints, and skeletal muscles. Afferent fibers, or nerves that receive information from external stimuli, carry sensory information through pathways that connect the skin and skeletal muscles to the CNS for processing. The information is then sent back via efferent nerves, or nerves that carry instructions from the CNS, back through the somatic system. These instructions go to neuromuscular junctions—the interfaces between neurons and muscles—for motor output.

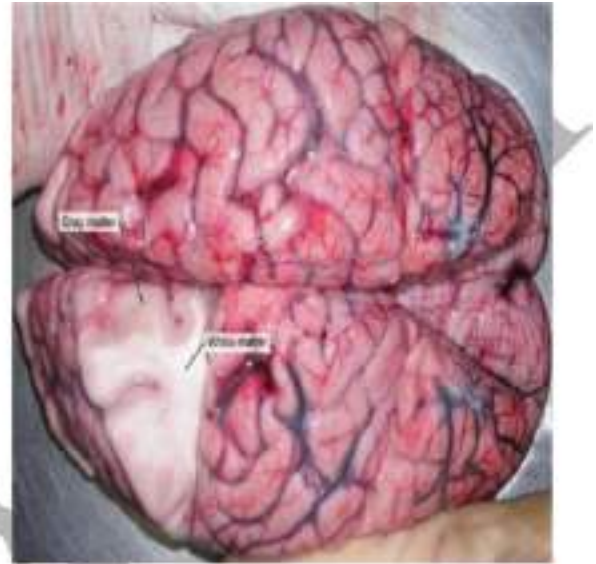
The somatic system also provides us with reflexes, which are automatic and do not require input or integration from the brain to perform. Reflexes can be categorized as either monosynaptic or polysynaptic based on the reflex arc used to perform the function. Monosynaptic reflex arcs, such as the knee-jerk reflex, have only a single synapse between the sensory neuron that receives the information and the motor neuron that responds. Polysynaptic reflex arcs, by contrast, have at least one interneuron between the sensory neuron and the motor neuron. An example of a polysynaptic reflex arc is seen when a person steps on a tack—in response, their body must pull that foot up while simultaneously transferring balance to the other leg.

For References (This will help you connect the dots and know the concept in detail)

Nervous tissue, present in both the CNS and PNS, contains two basic types of cells: neurons and glial cells. (Discussed above)

Looking at nervous tissue, there are regions that predominantly contain cell bodies and regions that are largely composed of just axons. These two regions within nervous system structures are often referred to as **gray matter** (the regions with many cell bodies and dendrites) or **white matter** (the regions with many axons). The colors ascribed to these regions are what would be seen in “fresh,” or unstained, nervous tissue. Gray matter is not necessarily gray. It can be pinkish because of blood content, or even slightly tan, depending on how long the tissue has been preserved. But white matter is white because axons are insulated by a lipid-rich substance called **myelin**. Lipids can appear

as white (“fatty”) material, much like the fat on a raw piece of chicken or beef. Actually, gray matter may have that color ascribed to it because next to the white matter, it is just darker—hence, gray.



FUNCTIONAL DIVISIONS OF THE NERVOUS SYSTEM

The nervous system can also be divided on the basis of its functions, but anatomical divisions and functional divisions are different. The CNS and the PNS both contribute to the same functions, but those functions can be attributed to different regions of the brain (such as the cerebral cortex or the hypothalamus) or to different ganglia in the periphery. The problem with trying to fit functional differences into anatomical divisions is that sometimes the same structure can be part of several functions. For example, the optic nerve carries signals from the retina that are either used for the conscious perception of

visual stimuli, which takes place in the cerebral cortex, or for the reflexive responses of smooth muscle tissue that are processed through the hypothalamus.

There are two ways to consider how the nervous system is divided functionally. First, the basic functions of the nervous system are sensation, integration, and response. Secondly, control of the body can be somatic or autonomic—divisions that are largely defined by the structures that are involved in the response. There is also a region of the peripheral nervous system that is called the enteric nervous system that is responsible for a specific set of the functions within the realm of autonomic control related to gastrointestinal functions.

BASIC FUNCTIONS

The nervous system is involved in receiving information about the environment around us (sensation) and generating responses to that information (motor responses). The nervous system can be divided into regions that are responsible for **sensation** (sensory functions) and for the **response** (motor functions). But there is a third function that needs to be included. Sensory input needs to be integrated with other sensations, as well as with memories, emotional state, or learning (cognition). Some regions of the nervous system are termed **integration** or association areas. The process of integration combines sensory perceptions and higher cognitive functions such as memories, learning, and emotion to produce a response.

SENSATION

The first major function of the nervous system is sensation—receiving information about the environment to gain input about what is happening outside the body (or, sometimes, within the body). The sensory functions of the nervous system register the presence of a change from homeostasis or a particular event in the environment, known as a **stimulus**. The senses we think of most are the “big five”: taste, smell, touch, sight, and hearing. The stimuli for taste and smell are both chemical substances (molecules, compounds, ions, etc.), touch is physical or mechanical stimuli that interact with the skin, sight is light stimuli, and hearing is the perception of sound, which is a physical stimulus similar to some aspects

INTEGRATION

Stimuli that are received by sensory structures are communicated to the nervous system where that information is

of touch. There are actually more senses than just those, but that list represents the major senses. Those five are all senses that receive stimuli from the outside world, and of which there is conscious perception. Additional sensory stimuli might be from the internal environment (inside the body), such as the stretch of an organ wall or the concentration of certain ions in the blood.

RESPONSE

The nervous system produces a response on the basis of the stimuli perceived by sensory structures. An obvious response would be the movement of muscles, such as withdrawing a hand from a hot stove, but there are broader uses of the term. The nervous system can cause the contraction of all three types of muscle tissue. For example, skeletal muscle contracts to move the skeleton, cardiac muscle is influenced as heart rate increases during exercise, and smooth muscle contracts as the digestive system moves food along the digestive tract. Responses also include the neural control of glands in the body as well, such as the production and secretion of sweat by the eccrine and merocrine sweat glands found in the skin to lower body temperature.

Responses can be divided into those that are voluntary or conscious (contraction of skeletal muscle) and those that are involuntary (contraction of smooth muscles, regulation of cardiac muscle, and activation of glands). Voluntary responses are governed by the somatic nervous system and involuntary responses are governed by the autonomic nervous system, which are discussed in the next section.

processed. This is called integration. Stimuli are compared with, or integrated with, other stimuli, memories of previous stimuli, or the state of a person at a particular time. This leads to the specific

response that will be generated. Seeing a baseball pitched to a batter will not automatically cause the batter to swing. The trajectory of the ball and its speed will need to be considered. Maybe the count is

three balls and one strike, and the batter wants to let this pitch go by in the hope of getting a walk to first base. Or maybe the batter's team is so far ahead; it would be fun to just swing away.

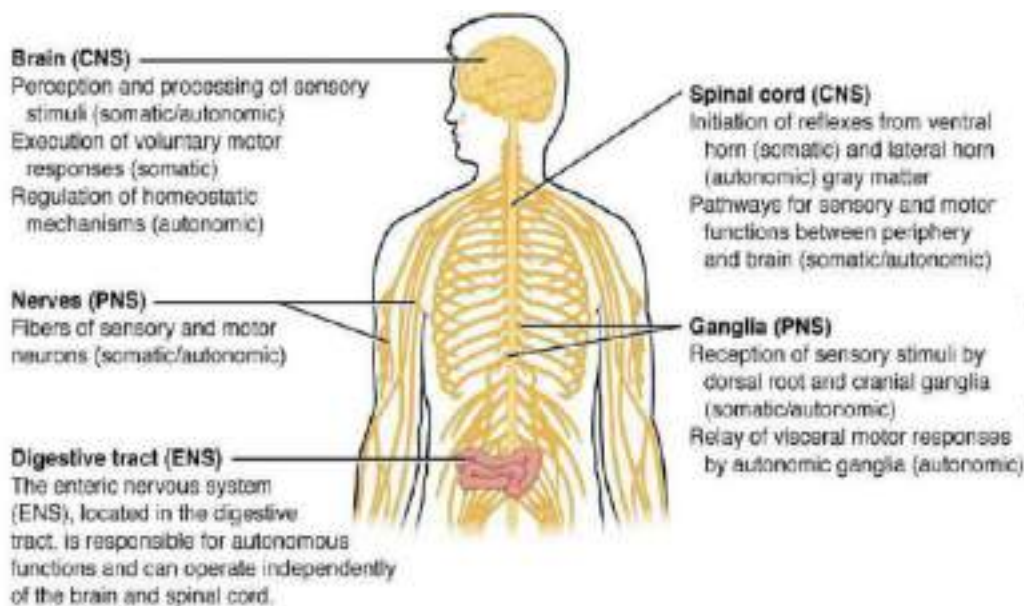
CONTROLLING THE BODY

The nervous system can be divided into two parts mostly on the basis of a functional difference in responses. The **somatic nervous system (SNS)** is responsible for conscious perception and voluntary motor responses. Voluntary motor response means the contraction of skeletal muscle, but those contractions are not always voluntary in the sense that you have to want to perform them. Some somatic motor responses are reflexes, and often happen without a conscious decision to perform them. If your friend jumps out from behind a corner and yells "Boo!" you will be startled and you might scream or leap back. You didn't decide to do that, and you may not have wanted to give your friend a reason to laugh at your expense, but it is a reflex involving skeletal muscle contractions. Other motor responses become automatic (in other words, unconscious) as a person learns motor skills (referred to as "habit learning" or "procedural memory").

The **autonomic nervous system (ANS)** is responsible for involuntary control of the body, usually for the sake of homeostasis (regulation of the internal environment). Sensory input for autonomic functions can be from sensory structures tuned to

external or internal environmental stimuli. The motor output extends to smooth and cardiac muscle as well as glandular tissue. The role of the autonomic system is to regulate the organ systems of the body, which usually means to control homeostasis. Sweat glands, for example, are controlled by the autonomic system. When you are hot, sweating helps cool your body down. That is a homeostatic mechanism. But when you are nervous, you might start sweating also. That is not homeostatic; it is the physiological response to an emotional state.

There is another division of the nervous system that describes functional responses. The **enteric nervous system (ENS)** is responsible for controlling the smooth muscle and glandular tissue in your digestive system. It is a large part of the PNS, and is not dependent on the CNS. It is sometimes valid, however, to consider the enteric system to be a part of the autonomic system because the neural structures that make up the enteric system are a component of the autonomic output that regulates digestion. There are some differences between the two, but for our purposes here there will be a good bit of overlap. See Figure 5 for examples of where these divisions of the nervous system can be found.



HOW MUCH OF YOUR BRAIN DO YOU USE?

Have you ever heard the claim that humans only use 10 percent of their brains? Maybe you have seen an advertisement on a website saying that there is a secret to unlocking the full potential of your mind—as if there were 90 percent of your brain sitting idle, just waiting for you to use it. If you see an ad like that, don't click. It isn't true.

An easy way to see how much of the brain a person uses is to take measurements of brain activity while performing a task. An example of this kind of measurement is functional magnetic resonance imaging (fMRI), which generates a map of the most active areas and can be generated and presented in three dimensions. This procedure is different from the standard MRI technique because it is measuring changes in the tissue in time with an experimental condition or event.

ENDOCRINE GLANDS

Endocrine Glands

There are eight major endocrine glands, each with a different function.

- Pituitary gland: the "brain" of the endocrine system; regulates all seven of the other glands and secretes growth hormone.
- Thyroid: regulates a person's metabolic rate, which is the amount of energy expended daily by a person at rest.
- Thymus: assists in the development of a person's immune system.
- Adrenal gland: regulates fluid and sodium balance within the body, and secretes epinephrine ("adrenaline") when the body is under stress, producing the fight-or-flight response.
- Ovaries (in females) and testes (in males): control the development of secondary sex characteristics.
- Pancreatic islets: regulate blood sugar.
- Pineal gland: regulates biorhythms and mood, and stimulates the onset of puberty.

- The endocrine system acts as a communication tool for the human body, working in tandem with the nervous system to communicate with the body's other internal systems.
- The endocrine system differs from the nervous system in that its chemical signals are slower-moving and longer-lasting.
- Hormones act as chemical messengers within the body, telling it to perform specific physical and mental functions.
- There are eight major endocrine glands, each performing a different function: the pituitary gland, the thyroid, the thymus gland, the adrenal gland, the ovaries (female) and testes (male), the pancreatic islets, and the pineal gland.
- The HPA axis is a complex set of direct influences and feedback interactions among three organs crucial to endocrine function: the hypothalamus, the pituitary gland, and the adrenal glands.

NEUROPLASTICITY

(NOTE: Topic is covered in detail, not required to go into this much detail. Details are given so as to give you a comprehensive idea of the concept and its practicality)

HEBBIAN THEORY

In 1949, neuroscientist Donald Hebb proposed that simultaneous activation of

cells leads to pronounced increase in synaptic strength between those cells, a theory that is widely accepted today. Cell assembly, or Hebbian theory, asserts that “cells that fire together wire together,” meaning neural networks can be created through associative experience and learning. Since Hebb’s discovery, neuroscientists have continued to find evidence of plasticity and modification within neural networks.

Key Points

- “Neuroplasticity” refers to changes in neural pathways and synapses that result from changes in behavior, environmental and neural processes, and changes resulting from bodily injury.
- Neuroplasticity has replaced the formerly held theory that the brain is a physiologically static organ, and explores how the brain changes throughout life.
- Neuroplasticity occurs on a variety of levels, ranging from minute cellular changes resulting from learning to large-scale cortical remapping in response to injury.
- Synaptic pruning, or apoptosis, is the programmed neuron cell death that takes place during early childhood and adolescence.
- Pruning strengthens important connections and eliminates weaker ones, creating more effective neural communication.

The brain is constantly adapting throughout a lifetime, though sometimes over critical, genetically determined periods of time. Neuroplasticity is the brain’s ability to create new neural pathways based on new experiences. It refers to changes in neural pathways and

synapses that result from changes in behavior, environmental and neural processes, and changes resulting from bodily injury. Neuroplasticity has replaced the formerly held theory that the brain is a physiologically static organ, and explores how the brain changes throughout life.

Neuroplasticity occurs on a variety of levels, ranging from minute cellular changes resulting from learning to large-scale cortical remapping in response to injury. The role of neuroplasticity is widely recognized in healthy development, learning, memory, and recovery from brain damage. During most of the 20th century, the consensus among neuroscientists was that brain structure is relatively immutable after a critical period during early childhood. It is true that the brain is especially “plastic” during childhood’s critical period, with new neural connections forming constantly. However, recent findings show that many aspects of the brain remain plastic even into adulthood.

Plasticity can be demonstrated over the course of virtually any form of learning. For one to remember an experience, the circuitry of the brain must change. Learning takes place when there is either a change in the internal structure of neurons or a heightened number of synapses between neurons. Studies conducted using

SYNAPTIC PRUNING

“Synaptic (or neuronal or axon) pruning” refers to neurological regulatory processes that facilitate changes in neural structure by reducing the overall number of neurons and synapses, leaving more efficient synaptic configurations. At birth, there are approximately 2,500 synapses in the cerebral cortex of a human baby. By three years old, the cerebral cortex has about 15,000 synapses. Since the infant brain has such a large capacity for growth, it must eventually be pruned down to remove unnecessary neuronal structures from the brain. This process of pruning is referred to as apoptosis, or programmed cell death. As the human brain develops, the need for more complex neuronal associations becomes much more pertinent, and simpler associations formed at childhood are

rats illustrate how the brain changes in response to experience: rats that lived in more enriched environments had larger neurons, more DNA and RNA, heavier cerebral cortices, and larger synapses compared to rats that lived in sparse environments.

A surprising consequence of neuroplasticity is that the brain activity associated with a given function can move to a different location; this can result from normal experience, and also occurs in the process of recovery from brain injury. In fact, neuroplasticity is the basis of goal-directed experiential therapeutic programs in rehabilitation after brain injury. For example, after a person is blinded in one eye, the part of the brain associated with processing input from that eye doesn’t simply sit idle; it takes on new functions, perhaps processing visual input from the remaining eye or doing something else entirely. This is because while certain parts of the brain have a typical function, the brain can be “rewired”—all because of plasticity.

replaced by more intricately interconnected structures.

Pruning removes axons from synaptic connections that are not functionally appropriate. This process strengthens important connections and eliminates weaker ones, creating more effective neural communication. Generally, the number of neurons in the cerebral cortex increases until adolescence. Apoptosis occurs during early childhood and adolescence, after which there is a decrease in the number of synapses. Approximately 50% of neurons present at birth do not survive until adulthood. The selection of the pruned neurons follows the “use it or lose it” principle, meaning that synapses that are frequently used have strong connections, while the rarely used synapses are eliminated.

EXTRA FOR REFERENCES (Read only if you find Neuroplasticity interesting and want to know more about it)

1. NEUROGENESIS IN ADULT MAMMALS:

Prior to the early 1980s, brain development after the early developmental period was seen as a downhill slope: Neurons continually die throughout a person's life, and it was assumed that the lost cells are never replaced by new ones. Although researchers began to chip away at this misconception in the early 1980s, it persisted until the turn of the century as one of the central principles of neurodevelopment.

The first serious challenge to the assumption that neurogenesis is restricted to early stages of development came with the discovery of the growth of new neurons in the brains of adult birds. Nottebohm and colleagues (e.g., Goldman Nottebohm, 1983) found that brain structures involved in singing begin to grow in songbirds just before each mating season and that this growth results from an increase in the number of neurons. This finding stimulated the re-examination of earlier unconfirmed claims that new neurons are created in the adult rat hippocampus. Then, in the 1990s, researchers, armed with newly developed immunohistochemical markers that have a selective affinity for recently created neurons, convincingly showed that adult neurogenesis does indeed occur in the rat hippocampus. And shortly thereafter, it was discovered that new neurons are also continually added to adult rat olfactory bulbs. Subsequently, it was reported that new neurons are added to the cortex, but apparently this report was erroneous.

In adult mammals, substantial neurogenesis seems to be restricted to the olfactory bulbs and hippocampus. At first, reports of adult neurogenesis were not

embraced by a generation of neuroscientists who had been trained to think of the adult brain as fixed, but acceptance grew as confirmatory reports accumulated. Particularly influential were reports that new neurons are added to the hippocampus of primates, including humans (Eriksson et al., 1998), and that the number of new neurons added to the adult hippocampus is substantial, an estimated 2,000 per hour.

New olfactory bulb neurons and new hippocampal neurons come from different places (Hagg, 2006). New olfactory bulb neurons are created from adult neural stem cells at certain sites in the subventricular zone of the lateral ventricles and then migrate to the olfactory bulbs. In contrast, new hippocampal cells are created near their final location in the dentate gyrus of the hippocampus.

Since the hippocampus is involved in some kinds of memory, perhaps exercise can be used as a treatment for memory problems. Indeed, aerobic exercise has been shown to improve cognitive function in older adults. It is now established that neurons generated during adulthood survive, become integrated into neural circuits, and begin to conduct neural signals. Adult-generated olfactory neurons become interneurons, and adult-generated hippocampal neurons become granule cells in the dentate gyrus. However, although there has been some progress in understanding the anatomy and physiological functions of adult-generated neurons, understanding of their behavioral functions has proven to be elusive.

EFFECTS OF EXPERIENCE ON THE REORGANIZATION OF THE ADULT CORTEX

The second line of research on adult neuroplasticity deals with the effects of experience on the reorganization of adult cortex. Experience in adulthood can lead to reorganization of sensory and motor cortical maps. For example, Mühlnickel and colleagues (1998) found that tinnitus (ringing in the ears) produces a major reorganization of primary auditory cortex; Elbert and colleagues (1995) showed that adult musicians who play stringed instruments that are fingered with the left hand (e.g., the violin) have an enlarged hand representation area in their right

Neuroplastic Responses to Nervous System Damage: Degeneration, Regeneration, Reorganization, and Recovery

There are four major neuroplastic responses of the brain to damage:

somatosensory cortex; Thus, once the brain has adapted to abnormal environmental conditions, it acquires the ability to adapt more effectively if it encounters the same conditions again. Although the cellular mechanisms underlying changes to adult sensory cortex are unknown, there are numerous possibilities. Experience has been shown to increase, decrease, or otherwise modify cortical synapses, buttons, and dendritic spines.

The discovery of adult neuroplasticity is changing the way that we humans think about ourselves. More importantly for those with brain damage, it has suggested some promising new treatment options.

Degeneration.
Regeneration.
Reorganization.
Recovery of function.

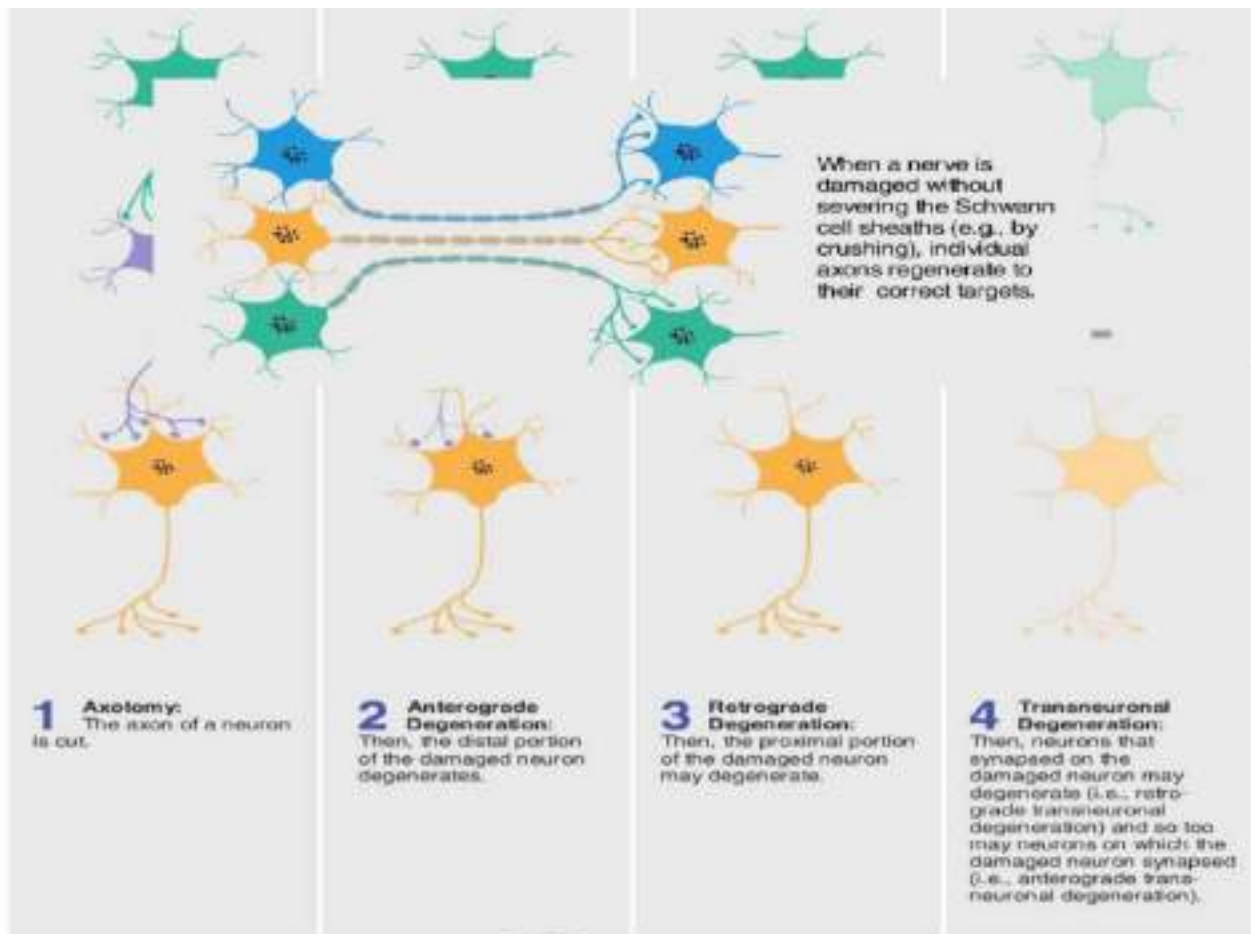
1. NEURAL DEGENERATION:

Neural degeneration (neural deterioration) is a common component of both brain development and disease. A widely used method for the controlled study of neural degeneration is to cut the axons of neurons. Two kinds of neural degeneration ensue: anterograde degeneration and retrograde degeneration. **Anterograde degeneration is the degeneration of the distal segment the segment of a cut axon between the cut and the synaptic terminals. Retrograde degeneration is the degeneration of the proximal segment the segment of a cut axon between the cut and the cell body.**

Anterograde degeneration occurs quickly following axotomy, because the cut

separates the distal segment of the axon from the cell body, which is the metabolic center of the neuron. The entire distal segment becomes badly swollen within a few hours, and it breaks into fragments within a few days.

The course of retrograde degeneration is different; it progresses gradually back from the cut to the cell body. In about 2 or 3 days, major changes become apparent in the cell bodies of most axotomized neurons. These early cell body changes are either degenerative or regenerative in nature. Early degenerative changes to the cell body (e.g., a decrease in size) suggest that the neuron will ultimately die usually by apoptosis but sometimes by necrosis or a combination of both.



Early regenerative changes (e.g., an increase in size) indicate that the cell body is involved in a massive synthesis of the proteins that will be used to replace the degenerated axon. But early regenerative changes in the cell body do not guarantee the long-term survival of the neuron; if the regenerating axon does not manage to make synaptic contact with an appropriate target, the neuron eventually dies. Sometimes, degeneration spreads from damaged neurons to neurons that are

linked to them by synapses; this is called **Trans neuronal degeneration**. In some cases, transneuronal degeneration spreads from damaged neurons to the neurons on which they synapse; this is called **anterograde transneuronal degeneration**. And in some cases, it spreads from damaged neurons to the neurons that synapse on them; this is called **retrograde transneuronal degeneration**.

2. NEURAL REGENERATION:

Neural regeneration- the regrowth of damaged neurons does not proceed as successfully in mammals and other higher vertebrates as it does in most invertebrates and lower vertebrates.

The capacity for accurate axonal growth, which higher vertebrates possess during their early development, is lost once they reach maturity. Regeneration is virtually nonexistent in the CNS of adult mammals, and is at best a hit-or-miss affair in the PNS. In the mammalian PNS, regrowth from the proximal stump of a damaged

nerve usually begins 2 or 3 days after

axonal damage.

There are three possibilities. First, if the original Schwann cell myelin sheaths remain intact, the regenerating peripheral axons grow through them to their original

PNS by producing both neurotrophic factors and cell-adhesion molecules (CAMs). The neurotrophic factors released by Schwann cells stimulate the growth of



targets at a rate of a few millimeters per day. Second, if the peripheral nerve is

new axons, and the cell-adhesion molecules on the cell membranes of



severed and the cut ends become separated by a few millimeters, regenerating axon tips often grow into incorrect sheaths and are guided by them to incorrect destinations; that is why it is often difficult to regain the coordinated use of a limb affected by nerve damage even if there has been substantial regeneration. And third, if the cut ends of a severed mammalian peripheral nerve become widely separated or if a lengthy section of the nerve is damaged, there may be no meaningful regeneration at all; regenerating axon tips grow in a tangled mass around the proximal stump, and the neurons ultimately die.

Schwann cells mark the paths along which regenerating PNS axons grow. In contrast, oligodendroglia, which myelinate CNS axons, do not clear debris or stimulate or guide regeneration; indeed, they release factors that actively block regeneration. Furthermore, oligodendroglia tends to survive for long periods of time (e.g., months) after nerve damage, thus chronically inhibiting regeneration of the axons.

If the factors that promote accurate regeneration in lower vertebrates can be identified and applied to the human brain, it might be possible to cure currently untreatable brain injuries. Remarkably, when invertebrates lose an entire limb, the regenerating axons release a factor that promotes regeneration of that limb.

Some CNS neurons are capable of regeneration if they are transplanted to the PNS, whereas some PNS neurons are not capable of regeneration if they are transplanted to the CNS. Clearly, there is something about the environment of the PNS that promotes regeneration and something about the environment of the CNS that does not. Schwann cells, which myelinate PNS axons, clear the debris resulting from neural degeneration and promote regeneration in the mammalian

3. NEURAL REORGANIZATION:

Adult mammalian brains have the ability to reorganize themselves in response to experience.

Most studies of neural reorganization following damage have focused on the sensory and motor cortex of laboratory animals. Sensory and motor cortex are ideally suited to the study of neural reorganization because of their topographic layout. The damage-induced reorganization of the primary sensory and motor cortex has been studied under two conditions: following damage to peripheral nerves and following damage to the cortical areas themselves.

The following studies have been particularly influential:

Kaas and colleagues (1990) assessed the effect of making a small lesion in one retina and removing the other. Several months after the retinal lesion was made, primary visual cortex neurons that originally had receptive fields in the lesioned area of the retina were found to have receptive fields in the area of the retina next to the lesion; remarkably, this change began within minutes of the lesion.

Sanes, Suner, and Donoghue (1990) transected the motor neurons that controlled the muscles of rat's vibrissae (whiskers). A few weeks later, stimulation of the area of motor cortex that had previously elicited vibrissae movement now activated other muscles of the face.

CORTICAL REORGANIZATION FOLLOWING DAMAGE IN HUMANS:

Demonstrations of cortical reorganization in controlled experiments on nonhumans provided an incentive to search for similar effects in human clinical populations. One such line of research has used brain-

imaging technology to study the cortices of blind individuals. The findings are consistent with the hypothesis that there is continuous competition for cortical space by functional circuits. Without visual input to the cortex, there is an expansion of auditory and somatosensory cortex, and auditory and somatosensory input is processed in formerly visual areas. There seems to be a functional consequence to this reorganization: Blind volunteers have demonstrated skills superior to those of sighted controls on a variety of auditory and somatosensory tasks.

MECHANISMS OF NEURAL REORGANIZATION:

Two kinds of mechanisms have been proposed to account for the reorganization of neural circuits: a strengthening of existing connections, possibly through release from inhibition, and the establishment of new connections by collateral sprouting.

Indirect support for the first mechanism comes from two observations: Reorganization often occurs too quickly to be explained by neural growth, and rapid reorganization never involves changes of more than 2 millimeters of cortical surface.

Indirect support for the second mechanism comes from the observation that the magnitude of long-term reorganization can be too great to be explained by changes in existing connections. Although sprouting and release from inhibition are considered to be the likely mechanisms of cortical reorganization following damage, these are not the only possibilities.

4. RECOVERY OF FUNCTION AFTER BRAIN DAMAGE:

Recovery of function in humans after nervous system damage is a poorly understood phenomenon. Nevertheless, there is a general consensus that recovery of function is most likely when lesions are small and the patient is young.

Neuroplastic phenomena are presumed to underlie recovery of function. However, it has proven difficult to provide strong evidence for this assumption. Recovery of function after nervous system damage is difficult to study because there are other compensatory changes that can easily be confused with it.

Cognitive reserve (roughly equivalent to education and intelligence) is thought to play a role in the improvements observed after brain damage that do not result from true recovery of brain function. Let me explain. Kapur (1997) conducted a

biographical study of doctors and neuroscientists with brain damage, and he observed a surprising degree of what appeared to be cognitive recovery. His results suggested, however, that the observed improvement did not occur because these patients had actually recovered lost brain function but because their cognitive reserve allowed them to accomplish tasks in alternative ways. Cognitive reserve has also been used to explain why educated people are less susceptible to the effects of aging-related brain deterioration.

Several recent experiments in laboratory animals have shown that various treatments are maximally effective only if initiated within a few days of the brain damage. This finding has a potentially vital implication: It suggests that rehabilitation of human patients with brain damage should begin as soon as possible after the damage occurs.

NEUROPLASTICITY AND THE TREATMENT OF NERVOUS SYSTEM DAMAGE:

The dream of recent discoveries about neuroplasticity can be applied to the treatment of brain damage in human .

patients. The following four subsections describe research on some major new treatment approaches. Most of this research has focused on animal models, but some of it has progressed to clinical trials with human patients

— REDUCING BRAIN DAMAGE BY BLOCKING NEURODEGENERATION:

Several studies in animals have shown that it may be possible to reduce brain damage by blocking neural degeneration in human patients. For example, in one study, Xu and colleagues (1999) induced cerebral ischemia in rats by limiting blood flow to the brain. This had two major effects on rats in the control group: It produced damage to the hippocampus, a structure that is particularly susceptible to ischemic damage, and it produced deficits in the rat's performance in the Morris water maze.

The hippocampuses of rats in the experimental group were treated with viruses genetically engineered to release apoptosis inhibitor protein. Amazingly, the apoptosis inhibitor protein reduced both the loss of hippocampal neurons and the deficits in Morris water maze performance. In addition to apoptosis inhibitor protein, several other neurochemicals have been shown to block the degeneration of damaged neurons. The most widely studied of these is nerve growth factor. Estrogens have a similar effect. Estrogens are a class of steroid hormones that are released in large amounts by the ovaries (the female

gonads). These hormones have several important effects on the maturation of the female body. They also have been shown to limit or delay neuron death in animal

PROMOTING RECOVERY FROM CNS DAMAGE BY PROMOTING REGENERATION:

Although regeneration does not normally occur in the mammalian CNS, several studies in laboratory animals have shown that it can be induced. The following two studies are particularly promising because they have shown that such regeneration can be associated with functional recovery. Cheng, Cao, and Olson (1996) transected the spinal cords of rats, thus rendering them paraplegic (paralyzed in the posterior portion of their bodies). The researchers

PROMOTING RECOVERY FROM CNS DAMAGE BY NEUROTRANSPLANTATION:

Efforts to treat CNS damage by neurotransplantation have taken two different approaches. The first is to transplant fetal tissue into the damaged area; the second is to transplant cultures of stem cells.

TRANSPLANTING FETAL TISSUE:

The first approach to neurotransplantation was to replace a damaged structure with fetal tissue that would develop into the same structure. This approach focused on Parkinson's disease. Early signs were positive. Bilateral transplantation of fetal substantia nigra cells was successful in treating the MPTP monkey model of Parkinson's disease (Bankiewicz et al., 1990; Sladek et al., 1987).

Soon after the favorable effects of neurotransplants in the MPTP monkey model were reported, neurotransplantation was offered as a treatment for Parkinson's disease at major research hospitals. The results of the first case studies were promising. More importantly, some of the

models. These neuroprotective effects of estrogens may explain why several brain disorders (e.g., Parkinson's disease) are more prevalent in males than in females.

then transplanted sections of myelinated peripheral nerve across the transection. As a result, spinal cord neurons regenerated through the implanted Schwann cell myelin sheaths, and the regeneration allowed the rats to regain use of their hindquarters.

patients improved. The results of these case studies triggered a large-scale double-blind evaluation study of patients suffering from advanced Parkinson's disease. The study was extremely thorough; it even included placebo controls patients who received surgery but no implants.

Unfortunately, however, about 15% of these patients started to display a variety of uncontrollable writhing and chewing movements about a year after the surgery. The incidence of adverse motor side effects is likely to stifle further attempts to develop neurotransplantation as a treatment for Parkinson's disease. However, many still believe that this is an extremely promising therapeutic approach, but that the large-scale clinical trial was premature. Researchers do not yet know how to maximize the survival and growth of neurotransplants and how to minimize their side effects.

Investigators are trying to develop procedures for repairing brain damage by injecting embryonic stem cells into the damaged site. Once injected, the stem cells could develop and replace the damaged

neurons or myelin, under guidance from surrounding tissue.

Many studies triggered a frenzy of research activity. Effective treatment for severe CNS damage appeared to be within reach. However, it quickly became apparent that much research still needs to

PROMOTING RECOVERY FROM CNS DAMAGE BY REHABILITATIVE TRAINING:

Several demonstrations of the important role of experience in the organization of the developing and adult brain kindled a renewed interest in the use of rehabilitative training to promote recovery from CNS damage.

One of the principles that have emerged from the study of neurodevelopment is that neurons seem to be in a competitive situation: They compete with other neurons for synaptic sites and neurotrophins, and the losers die. Weiller and Rijntjes (1999) designed a rehabilitative program based on this principle. Their procedure, called constraint-induced therapy, was to tie down the functioning arm for 2 weeks while the affected arm received intensive training. Performance with the affected arm improved markedly over the 2 weeks, and there was an increase in the area of motor cortex controlling that arm.

In one approach to treating spinal injuries, patients who were incapable of walking were supported by a harness over a moving treadmill. With most of their weight supported and the treadmill providing feedback, the patients gradually learned to make walking movements. Then, as they improved, the amount of support was gradually reduced.

BENEFITS OF COGNITIVE AND PHYSICAL EXERCISE

There are numerous studies linking cognitive and physical activity to

be done (Li et al., 2008). First, techniques for promoting the survival, maturation, and the establishment of correct connections with surviving cells need to be identified. Second, methods for encouraging functional recovery have to be developed.

beneficial neurological outcomes in human patients. Individuals who are cognitively and physically active are less likely to contract neurological disorders; and if they do, their symptoms tend to be less severe and their recovery better. However, in such correlational studies, there are always problems of causal interpretation.

Physical exercise has also been shown to have a variety of beneficial effects on the rodent brain. For example, Van Praag and colleagues found that wheel running can increase adult neurogenesis in the hippocampus (2002), reduce age-related declines in the number of neurons in the hippocampus (2005), and improve performance on tests of memory and navigation, two abilities that have been linked to the hippocampus.

PHANTOM LIMBS: NEUROPLASTIC PHENOMENA

Most amputees continue to experience the limbs that have been amputated a condition referred to as phantom limb. Phantom limbs are the product of neuroplasticity. The most striking feature of phantom limbs is their reality. Their existence is so compelling that a patient may try to jump out of bed onto a nonexistent leg or to lift a cup with a nonexistent hand. In most cases, the amputated limb behaves like a normal limb; for example, as an amputee walks, a phantom arm seems to swing back and forth in perfect coordination with the intact arm. About 50% of amputee's experience chronic severe pain in their phantom limbs. A typical complaint is that an amputated hand is clenched so tightly that

the fingernails are digging into the palm of the hand.

Based on the premise that phantom limb pain results from irritation at the stump, surgical efforts to control it have often

involved cutting off the stump or various parts of the neural pathway between the stump and the cortex. Unfortunately, these treatments haven't worked.

METHODS OF PHYSIOLOGICAL PSYCHOLOGY

AN OVERVIEW

EXPERIMENTAL ABLATION

Evaluating the Behavioral Effects
of Brain Damage.

Producing Brain Lesions.

Stereotaxic Surgery

Histological Methods

Tracing Neural Connections

NON-INVASIVE METHODS

Studying the Structure of the
Living Human Brain

EVALUATING THE BEHAVIORAL EFFECTS OF BRAIN DAMAGE:

A lesion is a wound or injury, and a researcher who destroys part of the brain usually refers to the damage as a brain lesion. Experiments in which part of the brain is damaged and the animal's behavior is subsequently observed are called lesion studies. The rationale for lesion studies is that the function of an area of the brain can be inferred from the behaviors that the animal can no longer perform after the area is damaged. For example, if, after part of the brain is destroyed, an animal can no longer perform tasks that require vision, we can conclude that the animal is blind- and that the damaged area plays some role in vision. The most difficult task in such studies is interpreting the effects of brain

An enormous array of research methods is available to the investigator. One of the most important research methods used to investigate brain functions is:

EXPERIMENTAL ABLATION: From the Latin word *ablatus*, a "carrying away". The removal or destruction of a portion of the brain of a laboratory animal; presumably, the functions that can no longer be performed are the ones the region previously controlled. It is the oldest method used in neuroscience, and it remains one of the most important ones today.

lesions. For example, how do we ascertain that the lesioned animal is blind? Does it bump into objects, or fail to run through a maze toward a light that signals the location of food, or no longer constrict its pupils to light? An animal could bump into objects because of deficits in motor coordination, it could have lost its appetite for food (and thus its motivation to run through the maze), or it could see quite well but could have lost its visual reflexes. Researchers can often be fooled.

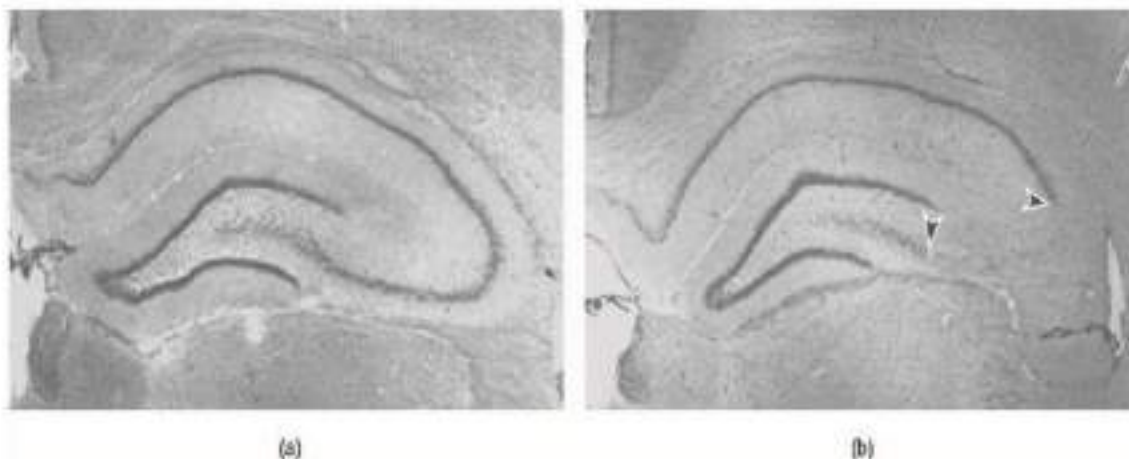
The distinction between brain function and behavior is an important one. Circuits within the brain perform functions, not behaviors. No one brain region or neural circuit is solely responsible for a behavior; each region performs a function (or set of functions) that contributes to performance of the behavior.

PRODUCING BRAIN LESIONS:

It is easy to destroy parts of the brain immediately beneath the skull; we anesthetize the animal, **cut open its scalp, remove part of its skull, and cut through the dura mater, bringing the cortex into view.** Then we can use a suction device to aspirate the brain tissue. To accomplish this tissue removal, we place a glass pipette on the surface of the brain and suck away brain tissue with a vacuum pump attached to the pipette. **Brain lesions of subcortical regions (regions located beneath the cortex) are usually produced by passing electrical current** through a stainless steel wire that is coated with an insulating varnish except for the very tip. We guide the wire stereotactically so that its end reaches the appropriate location. **(Stereotaxic surgery: Brain surgery using a stereotaxic apparatus to position an electrode or cannula in a specified position of the brain).** Then we turn on a lesion-making device, which produces radio frequency (RF) current-alternating current of a very high frequency. The passage of the current through the brain tissue produces heat that kills cells in the region surrounding the tip of the electrode. Lesions produced by these means destroy everything in the vicinity of the electrode tip, including neural cell bodies and the axons of neurons that pass through the region. A more selective method of producing

brain lesions employs an **excitatory amino acid such as kainic acid**, which kills neurons by stimulating them to death. Lesions produced this way are referred to as excitotoxic lesions. When an excitatory amino acid is injected through a cannula into a region of the brain, the chemical destroys neural cell bodies in the vicinity but spares axons that belong to different neurons that happen to pass nearby. This selectivity permits the investigator to determine whether the behavioral effects of destroying a particular brain structure are caused by the death of neurons located there or by the destruction of axons that pass nearby. When we produce subcortical lesions by passing RF current through an electrode or infusing a chemical through a cannula, we always cause additional damage to the brain. Therefore, we cannot simply compare the behavior of brain-lesioned animals with that of unoperated control animals; the incidental damage to the brain regions above the lesion may actually be responsible for some of the behavioral deficits we see. What we do is operate on a group of animals and **produce sham lesions-** a “placebo” procedure that duplicates all the steps of producing a brain lesion except turn on the lesion maker or start the infusion. This group of animals serves as a control group; if the behavior of the animals with brain lesions is different from that of the sham-operated control animals, we can conclude that the lesions caused the behavioral deficits.

Excitotoxic lesion. (a) Section through a normal hippocampus of a rat brain. (b) A lesion produced by infusion of an excitatory amino acid in a region of the hippocampus. Arrowheads mark the ends of the region in which neurons have been destroyed.



Most of the time, investigators produce permanent brain lesions, but sometimes it is advantageous to disrupt the activity of a particular region of the brain temporarily. The easiest way to do so is to inject a local anesthetic or a drug called **Muscimol** into the appropriate part of the brain. The anesthetic blocks action potentials in axons

STEREOTAXIC SURGERY

Stereotaxis literally means “solid arrangement”; more specifically, it refers to the ability to locate objects in space. A stereotaxic apparatus contains a holder that

STEREOTAXIC ATLAS

A collection of drawings of sections of the brain of a particular animal with measurements that provide coordinates for stereotaxic surgery.

No two brains of animals of a given species are completely identical, but there is enough similarity among individuals to predict the location of particular brain structures relative to external features of the head. For instance, a subcortical nucleus of a rat might be so many millimeters ventral, anterior, and lateral to a point formed by the junction of several bones of the skull.

The skull is composed of several bones that grow together and form sutures (seams). The heads of newborn babies contain a soft spot at the junction of the coronal and sagittal sutures called the fontanelle. Once this gap closes, the

STEREOTAXIC APPARATUS

A stereotaxic apparatus operates on simple principles. The device includes a head holder, which maintains the animal’s skull in the proper orientation, a holder for the electrode, and a calibrated mechanism that moves the electrode holder in measured

from entering or leaving that region, thus effectively producing a temporary lesion (usually called a reversible brain lesion).

Muscimol is a drug that stimulates GABA receptors, inactivates a region of the brain by inhibiting the neurons located there. (GABA is the most important inhibitory neurotransmitter in the brain.)

fixes the animal’s head in a standard position and a carrier that moves an electrode or a cannula through measured distances in all three axes of space. To perform stereotaxic surgery, one must first study a stereotaxic atlas.

junction is called **bregma**, from the Greek word meaning “front of head.” Bregma in a rat’s skull serves as a convenient reference point. If the animal’s skull is oriented in a specific way, a particular region of the brain is found in a fairly constant position, relative to bregma. **A stereotaxic atlas contains photographs or drawings that correspond to frontal sections taken at various distances rostral and caudal to bregma.**

Thus, by finding a neural structure (which we cannot see in our animal) on one of the pages of a stereotaxic atlas, we can determine the structure’s location relative to bregma (which we can see). Because of variations in different strains and ages of animals, the atlas gives only an approximate location. We always have to try out a new set of coordinates, slice and stain the animal’s brain, see the actual location of the lesion, correct the numbers, and try again.

distances along the three axes: anterior–posterior, dorsal–ventral, and lateral–medial. Once we obtain the coordinates from a stereotaxic atlas, we anesthetize the animal, place it in the apparatus, and cut the scalp open. We locate bregma, dial in

the appropriate numbers on the stereotaxic apparatus, drill a hole through the skull, and lower the device into the brain by the correct amount. In all cases, once surgery is complete, the wound is sewn together,

HISTOLOGICAL METHODS

After producing a brain lesion and observing its effects on an animal's behavior, we must slice and stain the brain so that we can observe it under the microscope and see the location of the lesion. **Brain lesions often miss the**

FIXATION AND SECTIONING

If we hope to study the tissue in the form it had at the time of the organism's death, we must destroy the **autolytic enzymes (autolytic means "self-dissolving")**, which will otherwise turn the tissue into mush. The tissue must also be preserved to prevent its decomposition by bacteria or molds. To achieve both of these objectives, we place the neural tissue in a **fixative**. The most commonly used fixative is **formalin**, an aqueous solution of formaldehyde, a gas. Formalin halts autolysis, hardens the very soft and fragile brain, and kills any microorganisms that might destroy it. Once the brain has been fixed, we must slice it into thin sections

STAINING

If we looked at an unstained section of brain tissue under a microscope, no fine details would be revealed. For this reason, the study of microscopic neuroanatomy requires special histological stains. Researchers have developed many different stains to identify specific substances within and outside of cells. For verifying the location of a brain lesion, we will use one of the simplest: **a cell-body stain**. In the late nineteenth century Franz Nissl, a German neurologist, discovered that a dye known as methylene blue would stain the cell bodies of brain tissue.

and the animal is taken out of the stereotaxic apparatus and allowed to recover from the anesthetic. Stereotaxic apparatuses can also be made for humans.

mark, so we have to verify the precise location of the brain damage after testing the animal behaviorally. To do so, we must fix, slice, stain, and examine the brain. Together, these procedures are referred to as **histological methods**. (The prefix histo- refers to body tissue.)

and stain various cellular structures to see anatomical details. **Slicing is done with a microtome (literally, "that which slices small")**. Slices of brain tissue are usually referred to as sections. After the tissue is cut, we attach the slices to glass microscope slides. We can then stain the tissue by putting the entire slide into various chemical solutions. Finally, we cover the stained sections with a small amount of a transparent liquid known as a **mounting medium and place a very thin glass coverslip over the sections**. The mounting medium keeps the coverslip in position.

Incidentally, the dyes were not developed specifically for histological purposes but were originally formulated for use in dyeing cloth. The discovery of cell-body stains made it possible to identify nuclear masses in the brain. **Cresyl violet is one such stain**. The stain is not selective for neural cell bodies; all cells are stained, neurons and glia alike. It is up to the investigator to determine which is which by size, shape, and location.

ELECTRON MICROSCOPY

The light microscope is limited in its ability to resolve extremely small details. A beam of electrons is passed through the tissue to be examined. A shadow of the tissue is then cast onto a sheet of

photographic film, which is exposed by the electrons. Electron photomicrographs produced in this way can provide information about structural details on the order of a few tens of nanometers.

TRACING NEURAL CONNECTIONS

The next step in a research program often requires the investigator to discover the afferent and efferent connections of the region of interest with the rest of the brain. Efferent connections (those that carry information from the region in question to other parts of the brain) are revealed with anterograde tracing methods, such as the one that uses PHA-L. Afferent connections (those that bring information to the region in question from other parts of the brain) are revealed with retrograde tracing methods, such as the one that uses fluorogold (A dye that serves as a retrograde label; taken up by terminal buttons and carried back to the cell bodies). To identify axons leaving a particular region and the terminal buttons of these axons, we use anterograde tracing method, such as PHA-L. In recent years, researchers have developed very precise methods that make specific neurons stand out from all of the others.

TRACING EFFERENT AXONS

To trace the efferent axons of any structure, we use an anterograde labeling method (**Anterograde means “moving forward”**). Anterograde labeling methods employ chemicals that are taken up by dendrites or cell bodies and are then transported through the axons toward the

terminal buttons. Over the years' neuroscientists have developed several different methods for tracing the pathways followed by efferent axons. Cell biologists have discovered that a family of proteins produced by plants bind with specific complex molecules present in cells of the immune system. These proteins, called lectins, have also found a use in tracing neural pathways. A particular lectin produced by the kidney bean, PHA-L is used to identify efferent axons. To discover the destination of the efferent axons of neurons located within the region, we inject a minute quantity of PHA-L into that nucleus by using a stereotaxic apparatus. The molecules of PHA-L are taken up by dendrites and are transported through the soma to the axon, where they travel by means of fast axoplasmic transport to the terminal buttons. Within a few days the cells are filled in their entirety with molecules of PHA-L: dendrites, soma, axons and all their branches, and terminal buttons. Then we kill the animal, slice the brain, and mount the sections on microscope slides. A special immunocytochemical method is used to make the molecules of PHA-L visible, and the slides are examined under a microscope.

Eventually, we will discover the relevant pathways from the region to the motor neurons whose activity is necessary for copulatory behavior.

TRACING AFFERENT AXONS

To find the afferent connections to the region, we employ a retrograde labeling method. **Retrograde means “moving backward”.** Retrograde labeling methods employ chemicals that are taken up by terminal buttons and carried back through the axons toward the cell bodies. The method for identifying the afferent inputs to a particular region of the brain is similar to the method used for identifying its efferent. First, we inject a small quantity of a chemical called fluorogold into the region. The chemical is taken up by terminal buttons and is transported back by means of retrograde axoplasmic transport to the cell bodies. A few days later we kill the animal, slice its brain, and examine the tissue under light of the appropriate wavelength. The anterograde and retrograde labeling methods (retrograde labeling method is a

histological method that labels cell bodies that give rise to the terminal buttons that form synapses with cells in a particular region) identify a single link in a chain of neurons—neurons whose axons enter or leave a particular brain region.

Transneuronal tracing methods identify a series of two, three, or more neurons that form serial synaptic connections with each other. The most effective transneuronal tracing method uses a pseudorabies virus—a weakened form of a pig herpes virus used for retrograde transneuronal tracing, which labels a series of neurons that are interconnected synaptically.

Together, anterograde and retrograde labeling methods—including transneuronal methods—enable us to discover circuits of interconnected neurons. Thus, these methods help to provide us with a “wiring diagram” of the brain.

RECORDING AND STIMULATING NEURAL ACTIVITY

Recording Neural Activity

Recording the Brain’s Metabolic and Synaptic Activity

Stimulating Neural Activity

INTERIM SUMMARY

When circuits of neurons participate in their normal functions, their electrical activity, metabolic activity, and chemical secretions increase. Thus, by observing these processes as an animal perceives various stimuli or engages in various behaviors, we can make some inferences about the functions performed by various regions of the brain. **Microelectrodes can be used to record the electrical activity of individual neurons.** Chronic recordings require that the electrode be attached to an electrical socket, which is fastened to the skull with a plastic adhesive. **Macroelectrodes record the**

activity of large groups of neurons. In rare cases macroelectrodes are placed in the depths of the human brain, but most often they are placed on the scalp and their activity is recorded on a polygraph.

Metabolic activity can be measured by giving an animal an injection of radioactive 2-DG, which accumulates in metabolically active neurons. The presence of the radioactivity is revealed through autoradiography: Slices of the brain are placed on microscope slides, and after a while developed like photographic negatives. When neurons are stimulated, they synthesize the nuclear protein **Fos (A protein produced in the nucleus of a neuron in response to synaptic stimulation).**

The presence of Fos, revealed by a special staining method, provides another way to discover active regions of the brain. The metabolic activity of various regions of the living human brain can be revealed by the 2-DG method, but a PET scanner is used to detect the active regions.

Another noninvasive method of measuring regional brain activity is provided by **functional MRI, which detects localized changes in oxygen level**. The secretions of neurotransmitters and neuromodulators can be measured by implanting the tip of a micro dialysis probe in a particular region of the brain. A PET scanner can be used to perform similar observations of the human brain. Researchers can stimulate various regions of the brain by implanting a macroelectrode and applying mild electrical stimulation. Alternatively, they can implant a guide cannula in the brain;

after the animal has recovered from the surgery, they insert a smaller cannula and inject a weak solution of an excitatory amino acid into the brain. The advantage of this procedure is that only **neurons whose cell bodies are located nearby will be stimulated; axons passing through the region will not be affected**. Transcranial magnetic stimulation induces electrical activity in the human cerebral cortex, which temporarily disrupts the functioning of neural circuits located there.

Recording and Stimulating neural activity involves studying the brain by recording or stimulating the activity of particular regions. Brain functions involve activity of circuits of neurons; thus, different perceptions and behavioral responses involve different patterns of activity in the brain. Researchers have devised methods to record these patterns of activity or to artificially produce them.

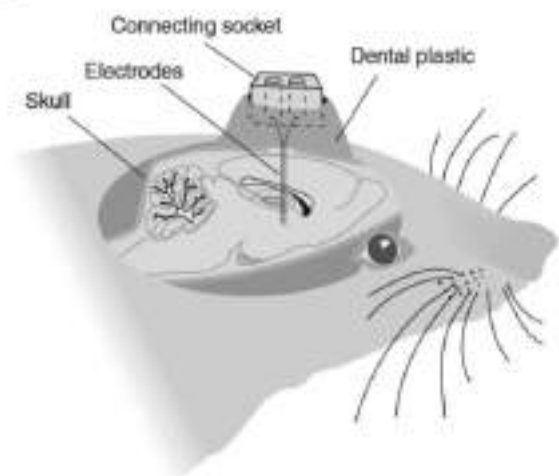


FIGURE 5.22 Implantation of Electrodes

The drawing shows a permanently attached set of electrodes, with a connecting socket cemented to the rat's skull.

Research Methods: Part II		
Goal of Method	Method	Remarks
Record electrical activity of single neurons	Glass or metal microelectrodes	Metal microelectrodes can be implanted permanently to record neural activity as animal moves
Record electrical activity of regions of brain	Metal macroelectrodes	In humans, usually attached to the scalp with a special paste
Record magnetic fields induced by neural activity	Magnetoencephalography; uses a neuro-magnetometer, which contains an array of SQUIDS	Can determine the location of a group of neurons firing synchronously
Record metabolic activity of regions of brain	2-DG autoradiography	Measures local glucose utilization
	Measurement of Fos protein	Identifies neurons that have recently been stimulated
	2-DG PET scan Functional MRI	Measures regional metabolic activity of human brain
Measure neurotransmitters and neuromodulators released by neurons	Microdialysis	A wide variety of substances can be analyzed
Measure neurochemicals in the living human brain	PET scan	Can localize any radioactive substance in the human brain
Stimulate neural activity	Electrical stimulation	Stimulates neurons near the tip of the electrode and axons passing through region
	Chemical stimulation with excitatory amino acid	Stimulates only neurons near the tip of the cannula, not axons passing through region
	Transcranial magnetic stimulation	Stimulate neurons in the human cerebral cortex with an electromagnet placed on the head

RECORDING NEURAL ACTIVITY

Axons produce action potentials, and terminal buttons elicit postsynaptic potentials in the membrane of the cells with which they form synapses. These electrical events can be recorded, and changes in the electrical activity of a particular region can be used to determine whether that region plays a role in various

behaviors. For example, recordings can be made during stimulus presentations, decision making, or motor activities. Recordings can be made chronically, over an extended period of time after the animal recovers from surgery, or acutely, for a relatively short period of time during which the animal is kept anesthetized.

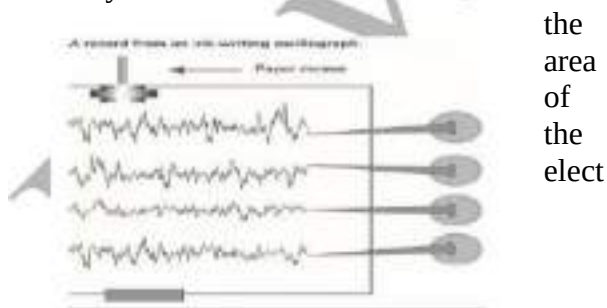
RECORDINGS WITH MICROELECTRODES

Microelectrodes have a very fine tip, small enough to record the electrical activity of individual neurons. This technique is usually called **single-unit recording (a unit refers to an individual neuron)**. Because we want to record the activity of single neurons over a long period of time

in **unanesthetized animals**, we want more durable electrodes so we can use arrays of very fine wires, gathered together in a bundle. The wires are insulated with a special varnish so that only their tips are bare. **The wires are flexible** enough that they can **follow movements of the brain tissue caused by movements of the animal's head**. As a result, they are less likely to damage the neurons whose

signals they are receiving. In addition, we can record the activity of several individual neurons in a particular region of the brain. We implant the electrodes in the brains of animals through stereotaxic surgery. Then, after recovery from surgery, the animal can be “plugged in” to the recording system. **Laboratory animals pay no heed to the electrical sockets on their skulls** and behave quite normally. Researchers often attach rather complex devices to the animals’ skulls when they implant microelectrodes so that they can record from several different parts of the brain during the course of their observations. **The electrical signals detected by microelectrodes are quite small and must be amplified** using amplifiers for converting the weak signals recorded at the brain into stronger ones. These signals can be stored in the memory of a computer for analysis at a later time.

Recordings with Macroelectrodes: At times we may need to record the activity of a region of the brain as a whole, not the activity of individual neurons located there. Hence, we use macroelectrodes. Macroelectrodes do not detect the activity of individual neurons; rather, the records that are obtained with these devices represent the postsynaptic potentials of many thousands—or millions—of cells in



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rode. These electrodes can consist of unsharpened wires inserted into the brain, screws attached to the skull, or even metal disks attached to the human scalp with a special paste that conducts electricity.

Occasionally, neurosurgeons implant macroelectrodes directly into the human brain. The reason for doing so is to detect the source of abnormal electrical activity that is giving rise to frequent seizures. Once the source is determined, the surgeon can open the skull and remove the source of the seizures—usually scar tissue caused by brain damage that occurred earlier in life. Most often, the electrical activity of a human brain is recorded through electrodes attached to the scalp and displayed on an ink-writing oscillograph, commonly called a polygraph. **A polygraph contains a mechanism that moves a very long strip of paper past a series of pens.** Such records are called

electroencephalograms (EEGs), or “writings of electricity from the head.” They can be used to diagnose epilepsy or brain tumors or to study the stages of sleep and wakefulness, which are associated with characteristic patterns of electrical activity. Another use of the EEG is to monitor the condition of the brain during procedures that could potentially damage it.

MAGNETOENCEPHALOGRAPHY

When electrical current flows through a conductor, it induces a magnetic field. This means that as action potentials pass down axons or as postsynaptic potentials pass down dendrites or sweep across the

somatic membrane of a neuron, magnetic fields are also produced. These fields are exceedingly small, but engineers have developed **superconducting detectors (called superconducting quantum interference devices, or SQUIDs)**, that can detect magnetic fields that are

approximately one-billionth of the size of the earth's magnetic field. Magnetoencephalography is performed with devices that contain an array of several SQUIDS, oriented so that a computer can examine their output and

RECORDING THE BRAIN'S METABOLIC AND SYNAPTIC ACTIVITY

Electrical signals are not the only signs of neural activity. **If the neural activity of a particular region of the brain increases, the metabolic rate of this region increases, too, largely as a result of increased operation of ion pumps in the membrane of the cells.** This increased metabolic rate can be measured. The experimenter injects **radioactive 2-deoxyglucose (2-DG)** into the animal's bloodstream. Because this chemical resembles glucose (the principal food for the brain), it is taken into cells. Thus, the most active cells, which use glucose at the highest rate, will take up the highest concentrations of radioactive 2-DG. **But unlike normal glucose, 2-DG cannot be metabolized,** so it stays in the cell. The experimenter then kills the animal, removes the brain, slices it, and prepares it for autoradiography. **Autoradiography can be translated roughly as “writing with one's own radiation.”** Sections of the brain are mounted on microscope slides. **The slides are then taken into a darkroom,** where they are **coated with a photographic emulsion** (the substance found on photographic film). Several weeks later, the slides, with their coatings of emulsion, are developed, just like photographic film. The molecules of radioactive 2-DG show themselves as spots of silver grains in the developed emulsion because the radioactivity exposes the emulsion, just as X-rays or light will do. The most active regions of the brain contain the most radioactivities, showing

calculate the source of particular signals in the brain. These devices can be used clinically—for example, to find the sources of seizures so that they can be removed surgically.

this radioactivity in the form of dark spots in the developed emulsion. Another method of identifying active regions of the brain capitalizes on the fact that when neurons are activated (for example, by the terminal buttons that form synapses with them), **particular genes in the nucleus called immediate early genes are turned on and particular proteins are produced.** These proteins then bind with the chromosomes in the nucleus. The presence of these nuclear proteins indicates that the neuron has just been activated. One of the nuclear proteins produced during neural activation is called FOS.

The metabolic activity of specific brain regions can be **measured in human brains,** too, using a method known as **positron emission tomography (PET).** First, the patient receives an injection of radioactive 2-DG. The person's head is placed in a machine similar to a CT scanner. When the radioactive molecules of 2-DG decay, they emit subatomic particles called positrons, which are detected by the scanner. The computer determines which regions of the brain have taken up the radioactive substance, and it produces a picture of a slice of the brain, showing the activity level of various regions in that slice. One of the **disadvantages of PET scanners is their operating cost.** The most recent development in brain imaging is **functional MRI (fMRI).** Engineers have devised modifications to existing MRI scanners that acquire images very rapidly and permit the measurement of regional metabolism by detecting levels

of oxygen in the brain's blood vessels. Functional MRI scans have a higher resolution than PET scans do, and they can

MEASURING THE BRAIN'S SECRETIONS:

Sometimes we are interested not in the general metabolic activity of particular regions of the brain, but in the secretion of specific neurotransmitters or neuromodulators in these regions. Say, **to measure the secretion of acetylcholine in a region of the medulla**, we use **microdialysis**. Dialysis is a process in which substances are separated by means of an artificial membrane that is permeable to some molecules but not others. A microdialysis probe consists of a small metal tube that introduces a solution into a

be acquired much faster. Thus, they reveal more detailed information about the activity of particular brain regions.

section of dialysis tubing. We use stereotaxic surgery to place a microdialysis probe in a rat's brain so that the tip of the probe is located in the region we are interested in. The microdialysis procedure has been applied to study of the human brain, but ethical reasons prevent us from doing so for research purposes. Fortunately, there is a noninvasive way to measure neurochemicals in the human brain. Although PET scanners are expensive machines, they are also versatile. They can be used to localize any radioactive substance that emits positrons.

BRAIN IMAGING TECHNIQUES

Recent advances in X-ray techniques and computers have led to the development of several methods for studying the anatomy of the living brain. These advances permit researchers to study the location and extent of brain damage while the patient is still living. All this is provided by the processes called MRI and CT.

Neuroimaging, or brain scanning, includes the use of various techniques to directly or indirectly image the structure, function, or pharmacology of the brain. It is a relatively new discipline within medicine, neuroscience, and psychology.

Physicians who specialize in the performance and interpretation of neuroimaging in the clinical setting are known as neuroradiologists.

Neuroimaging falls into two broad categories:

Structural imaging, which deals with the structure of the brain and the diagnosis of large-scale intracranial disease (such as a tumor), as well as injury.

Functional imaging, which is used to diagnose metabolic diseases and lesions on a finer scale (such as Alzheimer's disease), and also for neurological and cognitive-psychology research. Functional imaging allows the brain's information processing to be visualized directly, because activity in the involved area of the brain increases metabolism and "lights up" on the scan.

Four of the most common types of brain scans are **EEG**, **PET**, **MRI**, and **fMRI**. They scan the brain through a variety of methods and have varying degrees of specificity and invasiveness.

ELECTROENCEPHALOGRAPHY (EEG)

Electroencephalography (EEG) is used to show brain activity in certain psychological states, such as alertness or drowsiness. It is useful in the diagnosis of seizures and other medical problems that involve an overabundance or lack of activity in certain parts of the brain.

To prepare for an EEG, electrodes are placed on the face and scalp. After placing each electrode in the right position, the electrical potential of each electrode can be measured. According to a person's state (waking, sleeping, etc.), both the frequency and the form of the EEG signal differ. Patients who suffer from epilepsy show an increase of the amplitude of firing visible on the EEG record. The disadvantage of EEG is that the electric conductivity—and therefore the measured electrical potentials—may vary widely from person to person and also over time, due to the natural conductivities of other tissues such as brain matter, blood, and

MAGNETIC RESONANCE IMAGING (MRI)

Magnetic resonance imaging (MRI) and functional magnetic resonance imaging (fMRI) scans are the form of neural imaging most directly useful to the field of psychology. It is a technique whereby the interior of the body can be accurately imaged; involves the interaction between radio waves and a strong magnetic field. MRI passes an extremely strong magnetic field through the patient's head. The scanners can use this information to prepare pictures of slices of the brain.

An MRI uses strong magnetic fields to align spinning atomic nuclei (usually hydrogen protons) within body tissues, then disturbs the axis of rotation of these nuclei and observes the radio frequency

bones. Because of this, it is sometimes unclear exactly which region of the brain is emitting a signal.

POSITRON EMISSION TOMOGRAPHY (PET)

Positron emission tomography (PET) scans measure levels of the sugar glucose in the brain in order to illustrate where neural firing is taking place. This works because active neurons use glucose as fuel. As part of the scan, a tracer substance attached to radioactive isotopes is injected into the blood. When parts of the brain become active, blood (which contains the tracer) is sent to deliver oxygen. This creates visible spots, which are then picked up by detectors and used to create a video image of the brain while performing a particular task. However, with PET scans, we can only locate generalized areas of brain activity and not specific locations. In addition, PET scans are costly and invasive, making their use limited. However, they can be used in some forms of medical diagnosis, including for Alzheimer's.

signal generated as the nuclei return to their baseline status. Through this process, an MRI creates an image of the brain structure. MRI scans are noninvasive, pose little health risk, and can be used on infants and in utero, providing a consistent mode of imaging across the development spectrum. One disadvantage is that the patient has to hold still for long periods of time in a noisy, cramped space while the imaging is performed. The fMRI is a series of MRIs that measures both the structure and the functional activity of the brain through computer adaptation of multiple images. Specifically, the fMRI measures signal changes in the brain that are due to changing neural activity. In an fMRI, a patient can perform mental tasks and the area of action can be detected through blood flow from one part of the brain to

another by taking pictures less than a second apart and showing where the brain “lights up.” For example, when a person processes visual information, blood rushes to the back of the brain, which is where the occipital lobe is located. FMRIIs make it possible to show when things happen, how brain areas change with experience, and which brain areas work together. They

have been used to study a wide range of psychological phenomena, including (but by no means limited to) the neural activity of telling a lie, the differences between novices and experts when playing a musical instrument, and what happens inside our heads when we dream.

(For References)

Computerized Tomography (CT) is the use of a device that employs a computer to analyze data obtained by a scanning beam of X-rays to produce a two-dimensional picture of a “slice” through the body.

BIOLOGICAL BASIS OF MOTIVATION: HUNGER, THIRST, SLEEP AND SEX.

A **motive** is an impulse that causes a person to act. **Motivation** is an internal process that makes a person move toward a goal. Motivation, like intelligence, can’t

be directly observed. Instead, motivation can only be inferred by noting a person’s behavior. Researchers have proposed theories that try to explain human motivation. These theories include **drive reduction theories** and Maslow’s **hierarchy of needs theory**.

DRIVE REDUCTION THEORIES

Drive reduction theories of motivation suggest that people act in order to reduce needs and maintain a constant physiological state. For example, people eat in order to reduce their need for food. The idea of homeostasis is central to drive reduction theories. **Homeostasis** is the maintenance of a state of physiological equilibrium. Drive reduction theories fail to explain several aspects of motivation:

People sometimes aren’t motivated by internal needs.

Example: Some people fast for long periods for political causes, despite feeling extreme hunger. Sometimes, people continue being motivated even when they have satisfied internal needs.

Example: People sometimes eat even when they don’t feel hungry.

BIOLOGICAL BASIS OF HUNGER

Hunger is a complicated motivation; people don’t eat only because they need food. Many factors, both biological and environmental, influence hunger. These factors interact with one another in many ways.

There are a number of physiological mechanisms that serve as the basis for hunger. When our stomachs are empty, they contract, causing both hunger pangs and the secretion of chemical messages that travel to the brain to serve as a signal to initiate feeding behavior. When our blood glucose levels drop, the pancreas and liver generate a number of chemical signals that induce hunger (Konturek et al.,

2003; Novin, Robinson, Culbreth, & Tordoff, 1985) and thus initiate feeding

BIOLOGICAL FACTORS

Researchers believe certain genetic differences among individuals play a role

GENETIC DIFFERENCES AMONG INDIVIDUALS

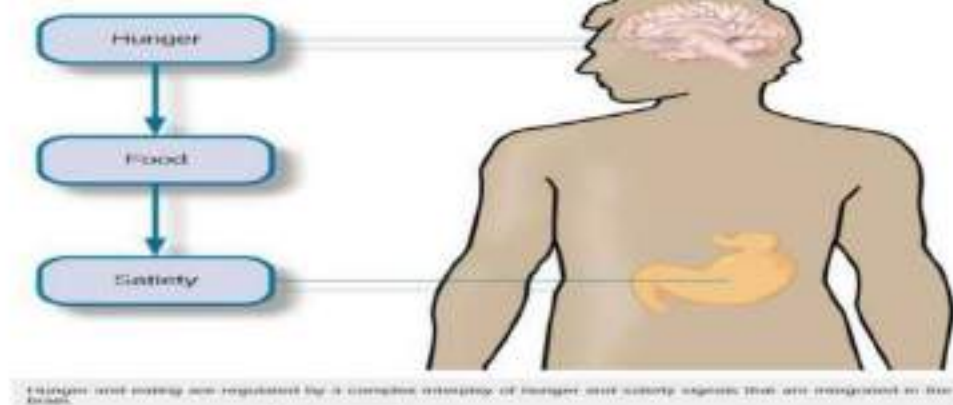
Researchers theorize that people have a genetically influenced **set point** for body weight. If a person's weight rises too far above his set point, his appetite decreases, or

behavior.

in hunger. The brain, the digestive system, and hormones are all involved in influencing hunger at the biological level.

Once again, he returns to his set point. The set point is maintained not only by food intake and energy expenditure but also by the body's basal metabolic rate, another genetically influenced variable.

Basal metabolic rate is the rate at which a person at complete rest uses energy. Some researchers disagree about set points and believe that



he uses up more energy. His weight then returns to its set point. If, on the other hand, his weight falls too far below his set point, his appetite increases, or he uses less energy.

people can reset their normal weight if they add or lose pounds slowly. They also point out that people usually gain weight when they have easy access to rich foods.

THE BRAIN

Researchers believe three areas in the hypothalamus play a key role in regulating hunger:

The lateral hypothalamus is involved in recognizing hunger. In rats, damage to the lateral hypothalamus results in loss of interest in eating.

The ventromedial nucleus of the hypothalamus is involved in recognizing satiety or fullness. In rats, damage to the ventromedial nucleus results in excessive eating and weight gain. The paraventricular nucleus of the hypothalamus is also involved in hunger regulation. When the paraventricular nucleus of a rat is

damaged, the rat will eat a very large quantity of food at each meal.

THE DIGESTIVE SYSTEM

The digestive system influences hunger in several ways. For instance, after a meal, the stomach and intestines send nerve impulses to the brain to help people recognize that they are full.

The body converts food to **glucose**, a simple sugar that acts as an energy source for cells. The level of glucose in the blood affects hunger. Low blood glucose increases hunger; high blood glucose decreases hunger.

ENVIRONMENTAL FACTORS

Many environmental factors influence hunger, including the availability of rich foods, taste preferences, habits, memory, stress, and cultural attitudes.

KINSEY'S STUDIES

One of the first researchers to give a modern account of human sexuality was **Alfred Kinsey**. In the 1940s, he and his colleagues interviewed more than 18,000 U.S. men and women about their sexual behavior and attitudes. In his comprehensive reports about human sexuality, Kinsey denounced the repressive social attitudes of his time, which he said

Critics of Kinsey's research maintained three arguments:

Kinsey's sample was not random. Instead, it consisted largely of well-educated, white city dwellers. Kinsey and

HORMONES

The hormone insulin also plays an important role in regulating hunger. Insulin allows cells to access glucose in the blood. When the pancreas secretes insulin, hunger increases. Diabetes is a condition caused by a deficiency of insulin. People who have diabetes take injections of insulin. Without these injections, their cells would be unable to use the glucose in their blood.

Another hormone involved in hunger regulation is leptin. Fat cells in the body secrete leptin and release it into the blood. When the leptin level in the blood is high, hunger decreases.

BIOLOGICAL BASIS OF SEX

Unlike hunger, sexual drive does not motivate people to fulfill a basic biological need. A lack of food leads to death; a lack of sex, on the other hand, does not. Both biological and psychological factors strongly influence sexual drive.

bore little relation to actual sexual practices. Kinsey provided statistics showing that sexual practices varied widely and that even in the 1940s there was a high prevalence of masturbation and premarital sex. These statistics shocked many people of his day.

his colleagues used questionable methods to gather their data, especially asking leading questions when interviewing subjects. Kinsey may have let his own beliefs influence his results

MASTERS AND JOHNSON'S STUDIES

Other pioneers of sexual research were **William Masters** and **Virginia Johnson**. In the 1960s, they studied several hundred male and female volunteers who agreed to either masturbate or have intercourse in a laboratory. Masters and Johnson hooked up the volunteers to instruments that measured various physiological indicators during sexual activity. Using the results of these studies, they described the sexual response cycle.

THE SEXUAL RESPONSE CYCLE

Masters and Johnson divided the human sexual response cycle into four phases:

1. **Excitement phase:** Physiological arousal increases quickly. Muscle tension, heart rate, blood pressure, and breathing rate increase. In men, the penis gets erect and the testes swell. In women, the clitoris hardens and swells, the vaginal lips open, and the vagina lubricates.

Plateau phase: Physiological arousal continues. In women, the clitoris retracts under the clitoral hood. Men

THE ROLE OF TESTOSTERONE

Sexual drive is related to testosterone level in both men and women, but the relationship is a complex one. Sexual activity increases testosterone levels, and testosterone levels increase sex drive.

Psychological Factors in Sexual Motivation: Hormones alone cannot cause sexual arousal. Psychological factors are also highly influential.

may secrete a small amount of fluid from the penis.

Orgasm phase: Physiological arousal peaks. Men ejaculate seminal fluid. Both men and women experience muscular contractions in the pelvic area, along with a sensation of pleasure.

Resolution phase: Physiological responses return to normal levels. Men then go through a refractory period that can vary in length, during which they are not responsive to stimulation. The refractory period tends to get longer as men age.

Critics of Masters and Johnson's research maintained two arguments:

Masters and Johnson studied a biased sample of people. The sample included only people who were both willing and able to perform sexual acts in a laboratory setting.

Masters and Johnson didn't pay attention to individual differences.

In reality, people's sexual responses vary according to factors such as age, amount of sexual experience, and cultural background.

EROTIC STIMULI: Both men and women can become sexually aroused by external and internal erotic stimuli. External erotic stimuli include sexually exciting material that is read, heard, or seen. Internal erotic stimuli include thoughts, fantasies, and memories of past sexual experiences. What is considered erotic varies according to the individual, historical period, and cultural context.

DESIRES: People have an infinite number of desires that influence the motivation for sex, including to procreate, to express love, to have physical enjoyment, to cope with difficult situations and emotions, to validate one's desirability, and to do what peers do.

CULTURAL CONTEXT: Having a strong influence on sexual behavior, cultures inform people about **sexual scripts**, or implicit rules that allow a person to judge the appropriate sexual behavior for a given situation. For example, people follow sexual scripts

when deciding whether they should initiate sexual activity or wait to receive a partner's advances.

A culture's social and economic structure determines the gender roles that men and women adopt. These gender roles in turn determine people's attitude toward sexual activity. In some cultures, for instance, women need marriage to get access to status and wealth. In such cultures, a woman is less likely to be interested in sex for its own sake, since casual sex can damage her reputation and reduce her chances of marriage.

SEX AND THE BRAIN

Psychological influences are clearly powerful motivators for sex, and the brain is highly involved in sexual arousal. People who have lost all sensation in their genitals because of spinal injuries, for example, are still capable of sexual desire.

GENDER DIFFERENCES IN SEXUAL BEHAVIOR AND PARTNER CHOICE

Many researchers have found that some differences exist between men and women in sexual behavior and partner choice, though all men and all women do not behave the same way or feel the same things.

Men	Women
More interested in sex; initiate and think about sex more often	Less interested in sex
Want sex with more partners	Not as interested in sex with many partners
Desire sex without emotional commitment	Desire sex with emotional commitment
Focus on youth and physical attractiveness when choosing a sex partner	Focus on social and economic status when choosing a sex partner
Feel more jealous when partner is physically unfaithful	Feel more jealous when partner is emotionally unfaithful

EVOLUTIONARY EXPLANATIONS

Some theorists use evolutionary theory to explain these gender differences. Their explanations are generally based on Robert Trivers's idea that men and women make different parental investments in order to produce offspring. From a biological

standpoint, men invest no more than the energy required for intercourse. Women, on the other hand, invest time and energy in pregnancy and breast feeding. Because of these biological differences, females can produce only a limited number of offspring, whereas males can potentially produce virtually unlimited offspring.

Males can increase their reproductive success by producing as many offspring as possible. Evolutionary theory predicts that men tend to choose attractive, youthful partners because these qualities imply good health and an ability to reproduce successfully. Females increase their reproductive success by being highly discriminating when choosing mates. They try to select males who have the most access to material resources, because such males can contribute the most to caring for offspring. Furthermore, men must contend with paternity uncertainty—they can never be certain that they are the fathers of their partners' offspring. Evolutionary theorists predict that men would therefore tend to have concerns about their partners' sexual infidelity. Women, on the other hand, *can* be certain that their offspring are their own, though they cannot be certain that their partners will provide for their offspring. Therefore, they are more likely to be concerned about the emotional fidelity of their partners.

POSSIBLE BIOLOGICAL FACTORS

Researchers have many ideas about the possible biological factors of homosexuality:

HORMONES: Some researchers have suggested that homosexuals and heterosexuals have different levels of various hormones in the blood. However, research in this area has failed consistently to find hormonal variations that could account for differences in sexual orientation.

PROBLEMS WITH EVOLUTIONARY EXPLANATIONS

Many people criticize the use of evolutionary explanations of gender differences in sexual behavior. Some critics argue that alternative explanations can account equally well for the observed gender differences. For example, women's history of social and economic subservience may have taught them to place a high value on their partners' access to material resources. Men's preferences and behaviors may likewise be a product of socialization. See page 43 for more information on problems with evolutionary explanations.

SEXUAL ORIENTATION

Sexual orientation is such a controversial subject that people cannot even agree about how the term *sexual orientation* should be defined. Some people argue over whether it refers to sexual behavior, sexual attraction, emotional attraction, or all three. Researchers define sexual orientation in a variety of ways, which means there is no clear idea about what proportion of the population is homosexual. Researchers also have many different opinions regarding how many biological and environmental factors contribute to sexual orientation.

GENES: Others have proposed that there is a genetic basis for predisposition to homosexuality. To investigate the possibility of a genetic basis, researchers have studied the sexual orientations of the identical, fraternal, and adoptive siblings of homosexual people. This research has shown that the identical twins of homosexuals are much more likely to be homosexual than the fraternal twins of homosexuals. In turn, the fraternal twins of homosexuals are more likely to be homosexual than the adoptive siblings of homosexuals.

- **PRENATAL FACTORS:** Some researchers have focused on prenatal environment. These researchers believe that the level of hormones present during a critical period in prenatal development can affect the organization of the brain, which in turn can influence sexual orientation. Research shows that women who were exposed to high prenatal levels of androgens are more likely to be homosexual. Critics point out that not all women who were exposed to

prenatal Androgens became homosexual and that many homosexual women were not exposed to androgens prenatally.

- **BRAIN DIFFERENCES:** One researcher, Simon LeVay, examined anatomical differences in the brains of homosexual and heterosexual men. He found that a specific area of the hypothalamus tended to be smaller in homosexual men and in heterosexual women than in heterosexual men.

ENVIRONMENTAL FACTORS

Many researchers believe biological factors alone can't explain the origin of homosexuality. For example, there is only about a 50 percent chance that the identical twins of homosexual men will also be homosexual. Therefore, some other factor must make the other 50 percent heterosexual. Although this other factor remains unknown, researchers have proposed a number of environmental situations that might influence sexual orientation:

- An ineffectual, distant father and an overly close, domineering mother
- Seduction in childhood by a homosexual adult
- Same-sex sexual play as children

- Many of these Proposals lack empirical support.

At this time, no one knows exactly what determines sexual orientation. Possibly, men and women develop homosexual orientations through various pathways. It is also possible that the cause of homosexual orientation differs from individual to individual.

SEXUAL ORIENTATION IN THE ANIMAL WORLD

Humans are not the only species to engage in homosexual or bisexual activity. Biologists have documented that animals belonging to hundreds of different species engage in homosexual or bisexual behavior. Many both male and female form exclusive, long-term homosexual pairs.

(EXTRA FOR REFERENCE – IMPORTANT FOR NET)

EATING DISORDERS

While nearly two out of three US adults struggle with issues related to being overweight, a smaller, but significant, portion of the population has eating disorders that typically result in being normal weight or underweight. Often, these individuals are fearful of gaining weight. Individuals who suffer from bulimia nervosa and anorexia nervosa face

many adverse health consequences (Mayo Clinic, 2012a, 2012b). People suffering from bulimia nervosa engage in binge eating behavior that is followed by an attempt to compensate for the large amount of food consumed. Purging the food by inducing vomiting or through the use of laxatives are two common compensatory behaviors. Some affected individuals engage in excessive

amounts of exercise to compensate for their binges. Bulimia is associated with many adverse health consequences that can include kidney failure, heart failure, and tooth decay. In addition, these individuals often suffer from anxiety and depression, and they are at an increased risk for substance abuse (Mayo Clinic, 2012b). The lifetime prevalence rate for bulimia nervosa is estimated at around 1% for women and less than 0.5% for men (Smink, van Hoeken, & Hoek, 2012).

As of the 2013 release of the *Diagnostic and Statistical Manual, fifth edition*, Binge eating disorder is a disorder recognized by the American Psychiatric Association (APA). Unlike with bulimia, eating binges are not followed by inappropriate behavior, such as purging, but they are followed by distress, including feelings of guilt and embarrassment. The resulting psychological distress distinguishes binge eating disorder from overeating (American Psychiatric Association [APA], 2013).

Anorexia nervosa is an eating disorder characterized by the maintenance of a

body weight well below average through starvation and/or excessive exercise. Individuals suffering from anorexia nervosa often have a distorted body image, referenced in literature as a type of body dysmorphia, meaning that they view themselves as overweight even though they are not. Like bulimia nervosa, anorexia nervosa is associated with a number of significant negative health outcomes: bone loss, heart failure, kidney failure, amenorrhea (cessation of the menstrual period), reduced function of the gonads, and in extreme cases, death. Furthermore, there is an increased risk for a number of psychological problems, which include anxiety disorders, mood disorders, and substance abuse (Mayo Clinic, 2012a). Estimates of the prevalence of anorexia nervosa vary from study to study but generally range from just under one percent to just over four percent in women. Generally, prevalence rates are considerably lower for men (Smink et al., 2012).

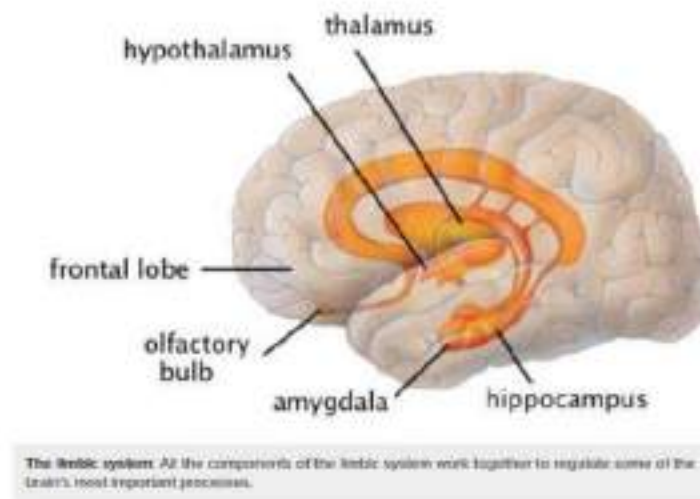
BIOLOGICAL BASIS OF EMOTION: THE LIMBIC SYSTEM, HORMONAL REGULATION OF BEHAVIOR.

THE LIMBIC SYSTEM

The limbic system combines higher mental functions and primitive emotion into one system.

The limbic system is a complex set of structures found on the central underside of the cerebrum, comprising inner sections of the temporal lobes and the bottom of the frontal lobe. It combines higher mental functions and primitive emotion into a single system often referred to as the emotional nervous system. It is not only responsible for our emotional lives but

also our higher mental functions, such as learning and formation of memories. The limbic system is the reason that some physical things such as eating seem so pleasurable to us, and the reason why some medical conditions, such as high blood pressure, are caused by mental stress. There are several important structures within the limbic system: the amygdala, hippocampus, thalamus, hypothalamus, basal ganglia, and cingulate gyrus.



THE AMYGDALA

The amygdala is a small almond-shaped structure; there is one located in each of the left and right temporal lobes. Known as the emotional center of the brain, the amygdala is involved in evaluating the emotional valence of situations (e.g., happy, sad, scary). It helps the brain recognize potential threats and helps prepare the body for fight-or-flight reactions by increasing heart and breathing rate. The amygdala is also responsible for learning on the basis of reward or punishment.

Due to its close proximity to the hippocampus, the amygdala is involved in the modulation of memory consolidation, particularly emotionally-laden memories. Emotional arousal following a learning event influences the strength of the subsequent memory of that event, so that greater emotional arousal following a learning event enhances a person's retention of that memory. In fact, experiments have shown that administering stress hormones to individuals immediately after they learn something enhances their retention when they are tested two weeks later.

THE HIPPOCAMPUS

The hippocampus is found deep in the temporal lobe, and is shaped like a

seahorse. It consists of two horns curving back from the amygdala. Psychologists and neuroscientists dispute the precise role of the hippocampus, but generally agree that it plays an essential role in the formation of new memories about past experiences. Some researchers consider the hippocampus to be responsible for general declarative memory (memories that can be explicitly verbalized, such as memory of facts and episodic memory).

Damage to the hippocampus usually results in profound difficulties in forming new memories (anterograde amnesia), and may also affect access to memories formed prior to the damage (retrograde amnesia). Although the retrograde effect normally extends some years prior to the brain damage, in some cases older memories remain intact; this leads to the idea that over time the hippocampus becomes less important in the storage of memory.

THE THALAMUS AND HYPOTHALAMUS

Both the thalamus and hypothalamus are associated with changes in emotional reactivity. The thalamus, which is a sensory “way-station” for the rest of the brain, is primarily important due to its connections with other limbic-system structures. The hypothalamus is a small part of the brain located just below the

thalamus on both sides of the third ventricle. Lesions of the hypothalamus interfere with several unconscious functions (such as respiration and metabolism) and some so-called motivated behaviors like sexuality, combativeness, and hunger. The lateral parts of the hypothalamus seem to be involved with pleasure and rage, while the medial part is linked to aversion, displeasure, and a tendency for uncontrollable and loud laughter.

THE BASAL GANGLIA

The basal ganglia are a group of nuclei lying deep in the subcortical white matter of the frontal lobes that organizes motor behavior. The caudate, putamen, and globus pallidus are major components of the basal ganglia. The basal ganglia appears to serve as a gating mechanism for physical movements, inhibiting potential movements until they are fully appropriate for the circumstances in which they are to be executed. The basal ganglia are also involved with:

THE CINGULATE GYRUS

The cingulate gyrus is located in the medial side of the brain next to the corpus callosum. There is still much to be learned about this gyrus, but it is known that its frontal part links smells and sights with pleasant memories of previous emotions. This region also participates in our emotional reaction to pain and in the regulation of aggressive behavior.

rule-based habit learning (e.g., initiating, stopping, monitoring, temporal sequencing, and maintaining the appropriate movement);
inhibiting undesired movements and permitting desired ones;
choosing from potential actions;
motor planning;
sequencing;
predictive control;
working memory;
attention.

Key Points

- The limbic system, located just beneath the cerebrum on both sides of the thalamus, is not only responsible for our emotional lives but also many higher mental functions, such as learning and formation of memories.
- The primary structures within the limbic system include the amygdala, hippocampus, thalamus, hypothalamus, basal ganglia, and cingulate gyrus.
- The amygdala is the emotion center of the brain, while the hippocampus plays an essential role in the formation of new memories about past experiences.
- The thalamus and hypothalamus are associated with changes in emotional reactivity.
- The cingulate gyrus coordinates smells and sights with pleasant memories, induces an emotional reaction to pain, and helps regulate aggressive behavior.
- The basal ganglia is a group of nuclei lying deep in the subcortical white matter of the frontal lobes; its functions include organizing motor behavior and coordinating rule-based, habit learning.

EARLY LANDMARKS IN THE BIOPSYCHOLOGICAL INVESTIGATION OF EMOTION:

<i>Event</i>	<i>Date</i>
Case of Phineas Gage	1848
Darwin's theory of the evolution of emotion	1872
James-Lange and Cannon-Bard theories	about 1900
Discovery of sham rage	1929
Discovery of Kluver-Bucy syndrome	1939
Limbic system theory of emotion	1952

There are six major early landmarks in the biopsychological investigation of emotion.

THE CASE OF PHINEAS GAGE (1848):

In 1994, Damasio and her colleagues brought the power of computerized reconstruction to bear on Gages classic case. They began by taking an X-ray of the

skull and measuring it precisely, paying particular attention to the position of the entry and exit holes. From these measurements, they reconstructed the accident and determined the likely region of Gages brain damage. It was apparent that **the damage to Gages brain affected both medial prefrontal lobes, which are involved in planning and emotion.**

CASE STUDY: PHINEAS GAGE

In 1848, Phineas Gage, a 25-year-old construction foreman for the Rutland and Burlington Railroad, was the victim of a tragic accident. In order to lay new tracks, the terrain had to be leveled, and Gage was in charge of the blasting. His task involved drilling holes in the rock, pouring some gun powder into each hole, covering it with sand, and tamping the material down with a large tamping iron before detonating it with a fuse. On the fateful day, the gunpowder exploded while Gage was tamping it, launching the 3-cm-thick, 90-cm-long tamping iron through his face, skull, and brain and out the other side. Amazingly, Gage survived his accident, but he survived it a changed man. Before the accident, Gage had been a responsible, intelligent, socially well-adapted person,

religious man who was well liked by his friends and fellow workers. Once recovered, he appeared to be as able-bodied and intellectually capable as before, but his personality and emotional life had totally changed. He became irreverent and impulsive. He became so unreliable and undependable that he soon lost his job, and was never again able to hold a responsible position. Gage became itinerant, roaming the country for a dozen years until his death in San Francisco.

Five years later, neurologist John Harlow was granted permission from Gages family to exhume the body and tamping iron to study them. Since then, Gages skull and the tamping iron have been on display in the Warren Anatomical Medical Museum at Harvard University.

DARWIN'S THEORY OF THE EVOLUTION OF EMOTION:

The first major event in the study of the biopsychology of emotion was the publication in 1872 of Darwin's book:

The Expression of Emotions in Man and Animals. In it, Darwin argued, largely on the basis of anecdotal evidence that particular emotional responses, such as human facial expressions, tend to accompany the same emotional states in all members of a species. Darwin believed that expressions of emotion, like other behaviors, are products of evolution; he therefore tried to understand them by comparing them in different species.

From such interspecies comparisons, Darwin developed a theory of the

evolution of emotional expression that was composed of three main ideas:

Expressions of emotion evolve from behaviors that indicate what an animal is likely to do next.

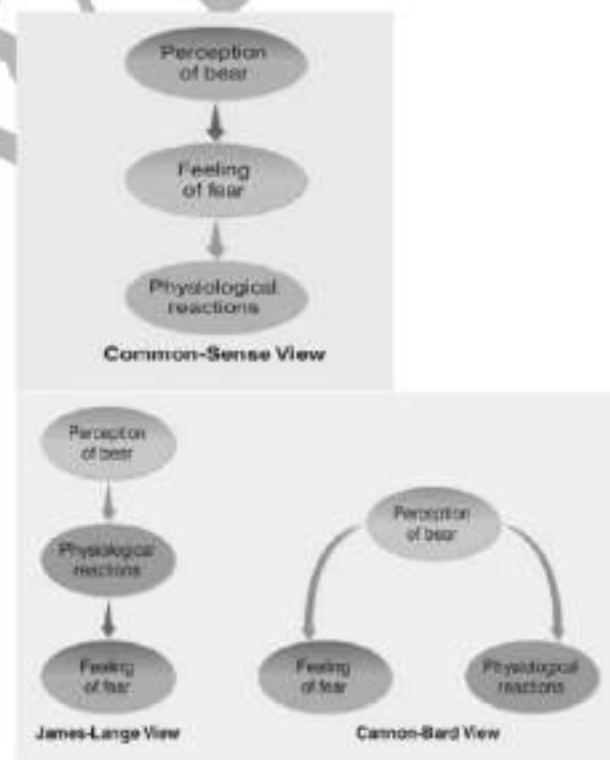
If the signals provided by such behaviors benefit the animal that displays them, they will evolve in ways that enhance their communicative function, and their original function may be lost.

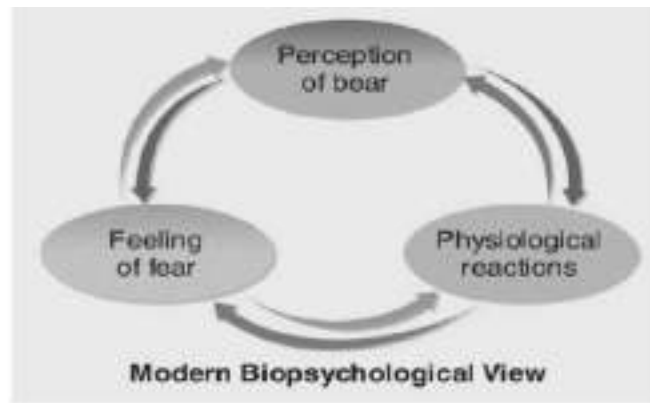
Opposite messages are often signaled by opposite movements and postures, an idea called the principle of antithesis.

JAMES-LANGE AND CANNON-BARD THEORIES:

The first physiological theory of emotion was proposed independently by James and Lange in 1884. According to the **James-Lange theory, emotion-inducing sensory stimuli are received and interpreted by the cortex, which triggers changes in the visceral organs via the autonomic nervous system and in the skeletal muscles via the somatic nervous system.** Then, the autonomic and somatic responses trigger the experience of emotion in the brain.

Four ways of thinking about the relations among the perception of emotion inducing stimuli, the autonomic and somatic responses to the stimuli, and the emotional experience.





James and Lange argued that the autonomic activity and behavior that are triggered by the emotional event (e.g., rapid heartbeat and running away) produce the feeling of emotion, not vice versa.

Around 1915, Cannon proposed an alternative to the James-Lange theory of emotion, and it was subsequently extended and promoted by Bard. **According to the Cannon-Bard theory, emotional stimuli have two independent excitatory effects: They excite both the feeling of emotion in the brain and the expression of emotion in the autonomic and somatic nervous systems.** That is, the Cannon-Bard theory, in contrast to the James-Lange theory, views emotional experience and emotional expression as parallel processes that have no direct causal relation. According to the James-Lange theory, emotional experience depends entirely on feedback from autonomic and somatic.

nervous system activity; according to the Cannon-Bard theory, emotional experience is totally independent of such feedback. Both extreme positions have proved to be incorrect. **On the one hand, it seems that the autonomic and somatic feedback is not necessary for the experience of emotion: Human patients whose autonomic and somatic feedback has been largely eliminated by a broken neck are capable of a full range of emotional experiences (e.g., Lowe & Carroll, 1985).**

Failure to find unqualified support for either the James-Lange or the Cannon-Bard theory led to the modern biopsychological view. According to this view, each of the three principal factors in an emotional response the perception of the emotion-inducing stimulus, the autonomic and somatic responses to the stimulus and the experience of the emotion can influence the other two.

4. SHAM RAGE:

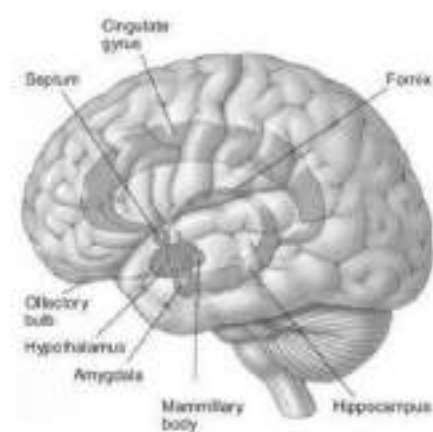
In the late 1920s, Bard (1929) discovered that decorticate cats whose cortex has been removed respond aggressively to the slightest provocation i.e., after a light touch, they arch their backs, erect their hair, growl, hiss, and expose their teeth. The aggressive responses of decorticate animals are abnormal in two respects: they are inappropriately severe, and they are not directed at particular targets. **Bard referred to the exaggerated, poorly directed aggressive responses of**

decorticate animals as sham rage. Sham rage can be elicited in cats, whose cerebral hemispheres have been removed down to, but not including, the hypothalamus; but it cannot be elicited if the hypothalamus is also removed. On the basis of this observation, **Bard concluded that the hypothalamus is critical for the expression of aggressive responses and that the function of the cortex is to inhibit and direct these responses.**

5. LIMBIC SYSTEM AND EMOTION:

In 1937, Papez (pronounced Payps) proposed that emotional expression is controlled by several interconnected nuclei and tracts that ring the thalamus.

The figure illustrates some of the key structures in this circuit, now known as the limbic system (limbic means border): the amygdala, mammillary body, hippocampus, and fornix, cortex of the cingulate gyrus, septum, olfactory bulb, and hypothalamus.



Papez proposed that emotional states are expressed through the action of the other structures of the circuit on the hypothalamus and that they are experienced through their action on the cortex. Papez's theory of emotion was revised and expanded by Paul MacLean in 1952 and became the influential limbic system theory of emotion.

6. KLUVER-BUCY SYNDROME:

In 1939, Kluver and Bucy observed a striking syndrome (**pattern of behavior**) **in monkeys whose anterior temporal lobes had been removed**. This syndrome, which is commonly referred to as the Kluver-Bucy syndrome, includes the following behaviors: **the consumption of almost anything that is edible, increased sexual activity often directed at inappropriate.**

EXTRA FOR REFERENCES (TRY TAKING OUT TIME TO ATLEAST GO THROUGH THIS SECTION)

EMOTIONS AND THE AUTONOMIC NERVOUS SYSTEM

Research on the role of the autonomic nervous system (ANS) in emotion has focused on two issues: the degree to which specific patterns of ANS activity are associated with specific emotions and the effectiveness of ANS measures in polygraphs (lie detection).

EMOTIONAL SPECIFICITY OF THE AUTONOMIC NERVOUS SYSTEM:

The James-Lange and Cannon-Bard theories differ in their views of the emotional specificity of the autonomic nervous system. The James-Lange theory says that different emotional stimuli induce different patterns of ANS activity and that these different patterns produce different emotional experiences.

There is ample evidence that not all emotions are associated with the same pattern of ANS activity. However, there is insufficient evidence to make a strong case for the view that each emotion is characterized by a different pattern of ANS activity.

POLYGRAPHY:

It is a method of interrogation that employs autonomic nervous system indexes of emotion to infer the truthfulness of the subject's responses. Polygraph tests administered by skilled examiners can be useful additions to normal interrogation procedures, but they are far from infallible. The main problem in evaluating the effectiveness of polygraphy is that it is rarely possible in real-life situations to know for certain whether a suspect is guilty or innocent.

The usual interrogation method is the control-question technique, in which the physiological response to the target question (e.g., did you steal that purse?) is compared with the physiological responses to control questions whose answers are known (e.g., Have you ever been in jail before?).

The assumption is that lying will be associated with greater sympathetic activation. The average success rate in various mock-crime studies using the control-question technique is about 80%.

Despite being commonly referred to as lie detection, polygraphy detects emotions, not lies. Consequently, it is less likely to successfully identify lies in real life than in experiments. **In real-life situations, questions such as did you steal that purse? Are likely to elicit a reaction from all suspects, regardless of their guilt or innocence, making it difficult to detect deception.**

The guilty-knowledge technique circumvents this problem. In order to use this technique, the polygrapher must have a piece of information concerning the

crime that would be known only to the guilty person. Rather than attempting to catch the suspect in a lie, the polygrapher simply assesses the suspects reaction to a list of actual and contrived details of the crime. Innocent suspects, because they have no knowledge of the crime, react to all such details in the same way; the guilty react differentially.

EMOTIONS AND FACIAL EXPRESSION:

Ekman and his colleagues have been preeminent in the study of facial expression. They began in the 1960s by analyzing hundreds of films and photographs of people experiencing various real emotions. From these, they compiled an atlas of the facial expressions that are normally associated with different emotions (Ekman & Friesen, 1975). For example, to produce the facial expression for surprise, models were instructed to pull their brows upward so as to wrinkle their forehead, to open their eyes wide so as to reveal white above the iris, to slacken the muscles around their mouth, and to drop their jaw.

Universality of Facial Expression:

Several studies have found that people of different cultures make similar facial expressions in similar situations and that they can correctly identify the emotional significance of facial expressions displayed by people from cultures other than their own.

Remarkably, human facial expressions are similar in many respects to those of our primate relatives. **Ekman and Friesen concluded that the facial expressions of the following six emotions are primary: surprise, anger, sadness, disgust, fear, and happiness.** They further concluded that all other facial expressions of genuine emotion are composed of predictable mixtures of these six primaries.

FACIAL FEEDBACK HYPOTHESIS:

Is there any truth to the old idea that putting on a happy face can make you feel better? Research suggests that there is.

The hypothesis that our facial expressions influence our emotional experience is called the facial feedback hypothesis. In a test of the facial feedback hypothesis, Rutledge and Hupka (1985) instructed subjects to assume one of two patterns of facial contractions while they viewed a series of slides; the patterns corresponded to happy or angry faces, although the subjects were unaware of that. The subjects reported that the slides made them feel more happy and less angry when they were making happy faces, and less happy and more angry when they were making angry faces.

VOLUNTARY CONTROL OF FACIAL EXPRESSION:

Because we can exert voluntary control over our facial muscles, it is possible to inhibit true facial expressions and to substitute false ones. There are many reasons for choosing to put on a false facial expression. Some of them are positive (e.g., putting on a false smile to reassure a worried friend), and some are negative (e.g., putting on a false smile to disguise a lie). In either case, it is difficult to fool an expert.

There are two ways of distinguishing true expressions from false ones (Ekman, 1985).

First, **micro-expressions (brief facial expressions)** of the real emotion often break through the false one. Such micro-expressions last only about 0.05 second, but with practice they can be detected without the aid of slow-motion photography.

Second, there are often subtle differences between genuine facial expressions and false ones that can be detected by skilled observers.

The most widely studied difference between a genuine and a false facial

expression was first described by the French anatomist Duchenne in 1862. Duchenne said that the smile of enjoyment could be distinguished from deliberately produced smiles by consideration of the two facial muscles that are contracted during genuine smiles: orbicularis oculi, which encircles the eye and pulls the skin from the cheeks and forehead toward the eyeball, and zygomaticus major, which pulls the lip corners up.



FIGURE 17.7 A fake smile. The orbicularis oculi and the zygomaticus major are two muscles that contract during genuine (Duchenne) smiles. Because the lateral portion of the orbicularis oculi is difficult for most people to contract voluntarily, fake smiles usually lack this component. This young man is faking a smile for the camera. Look at his eyes.

According to Duchenne, the zygomaticus major can be controlled voluntarily, whereas the orbicularis oculi is normally contracted only by genuine pleasure. Thus, inertia of the orbicularis oculi in smiling unmasks a false friend a fact you would do well to remember. Ekman named the genuine smile the Duchenne smile (see Ekman & Davidson, 1993).

Facial Expressions: Current Perspectives

Ekman's work on facial expression began before video recording became commonplace. Now, video recordings provide almost unlimited access to natural facial expressions made in response to real-life situations. As a result, it is now clear that Ekman's six primary facial

expressions of emotion rarely occur in pure form they are ideals with many subtle variations. Also, the existence of other primary emotions has been recognized. For example, Ekman (1992) agrees that there is evidence for adding contempt and embarrassment to his original six. Have you noticed that only one of the eight primary emotions, happiness, has a positive emotional valence? (Emotional valence refers to the general positive or negative character of an emotion).

This imbalance has led some to hypothesize that all positive emotions may

share the same facial expression. The research on pride by Tracy and Robins (2004) argues against this view. The expression of pride is readily identified by individuals of various cultures, cannot be created from a mixture of other primary expressions, and involves postural as well as facial components. Pride is expressed through a small smile, with the head tilted back slightly and the hands on the hips, raised above the head, or clenched in fists with the arms crossed on the chest.

BRAIN MECHANISMS OF HUMAN EMOTION

COGNITIVE NEUROSCIENCE OF EMOTION:

Cognitive Neuroscience is currently the dominant approach being used to study the brain mechanisms of human emotion. There have been many functional brain imaging studies of people experiencing or imagining emotions or watching others experiencing them. These studies have established three points that have advanced our understanding of the brain mechanisms of emotion in fundamental ways. Brain activity associated with each human emotion is diffuse there is not a center for each emotion. Think mosaic, not center, for locations of brain mechanisms of emotion. There is virtually always activity in motor and sensory cortices when a person experiences an emotion or empathizes with a person experiencing an emotion. Very similar patterns of brain activity tend to be recorded when a person experiences an emotion, imagines that emotion, or sees somebody else experience that emotion.

These three fundamental findings are influencing how researchers are thinking about the neural mechanisms of emotion. For example, the activity observed in sensory and motor cortex during the experience of human emotions is now believed to be an important part of the mechanism by which the emotions are experienced. **The re-experiencing of related patterns of motor, autonomic, and sensory neural activity during emotional experiences is generally referred to as the embodiment of emotions.** These three fundamental findings may also help explain the remarkable ability of humans to grasp the emotional states of others.

Mirror neurons, which have been identified in nonhuman primates, are neurons that fire when a specific response is performed by a subject or the subject watches the response being performed. The discovery that certain patterns of brain activity are observed on fMRI scans when individuals experience an emotion or watch somebody else experience the same emotion suggests that a **mirror-like system** might be the basis for human empathy.

AMYGDALA AND HUMAN EMOTION:

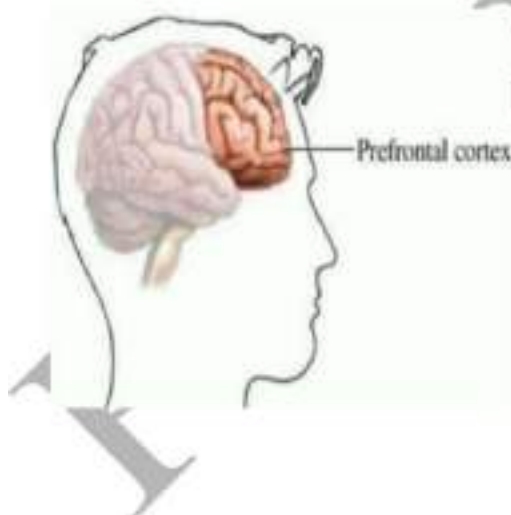
Amygdala plays an important role in fear conditioning in rats. Numerous functional brain-imaging studies have found the amygdala to be involved in human emotions particularly in fear and other negative emotions. Furthermore, the amygdala appears to be involved in only some aspects of human fear. It seems to be more involved in the perception of fear in others than in its experience.

Although recent research has focused on the role of the amygdala in the recognition of negative facial expressions, patients with Urbach-Wiethe disease sometimes have difficulty recognizing the negative expressions and other complex visual stimuli.

MEDIAL PREFRONTAL LOBES AND HUMAN EMOTION:

Emotion and cognition are often studied independently, but it is now believed that they are better studied as components of

the same system. The medial portions of the prefrontal lobes (including the medial portions of the orbitofrontal cortex and cingulate cortex) are the sites of emotion-cognition interaction that have received the most attention. Functional brain-imaging studies have found evidence of activity in the medial prefrontal lobes when emotional reactions are being cognitively suppressed or re-evaluated. Most of the studies of medial prefrontal lobe activity employ **suppression paradigms or reappraisal paradigms**. In studies that use suppression paradigms, participants are directed to inhibit their emotional reactions to unpleasant films or pictures; in studies that use reappraisal paradigms, participants are instructed to reinterpret a picture to change their emotional reaction to it. The medial prefrontal lobes are active when both of these paradigms are used, and they seem to exert their cognitive control of emotion by interacting with the amygdala. The medial prefrontal lobes are large and complex, and they likely perform many functions.



LATERALIZATION OF EMOTION:

There is considerable evidence that emotional functions are lateralized, that is, that the left and right cerebral hemispheres are specialized to perform different emotional functions. This evidence has led to several theories of the cerebral lateralization of emotion; the following are the two most prominent:

The right-hemisphere model of the cerebral lateralization of emotion holds that the right hemisphere is specialized for all aspects of emotional processing: perception, expression, and experience of emotion.

The valence model proposes that the right hemisphere is specialized for processing negative emotion and the left hemisphere is specialized for processing positive emotion.

Most studies of the cerebral lateralization of emotion have employed functional brain-imaging methods, and the results have been complex and variable. Overall comparisons made in many studies between left and right hemispheres revealed no interhemispheric differences in either the amount of emotional processing or the valence of the emotions being processed. However, when the comparisons were conducted on a structure-by-structure basis, they revealed substantial evidence of lateralization of emotional processing. Some kinds of emotional processing were lateralized to the left hemisphere in certain structures and to the right in others.

Clearly, neither the right-hemisphere model nor the valence model of the lateralization of emotion is supported by the evidence. The models are too general. Another approach to studying the lateralization of emotions is based on observing the asymmetry of facial expressions. **In most people, each facial expression begins on the left side of the face and, when fully expressed, is more pronounced there which implies right-hemisphere dominance for facial expressions.**

INDIVIDUAL DIFFERENCES IN THE NEURAL MECHANISMS OF EMOTION:

In general, more complex brain functions tend to show more individual differences in cerebral localization. The cortical localization of language processes varies substantially from person to person. Nevertheless, few studies of the neural mechanisms of emotion have focused on individual differences.

Let's consider this study by Canli and colleagues (2002) that used functional MRIs to compare the reactions of healthy

participants who scored high on extraversion with those of healthy participants who scored high on neuroticism. These personality dimensions were selected because of their relation to emotion people high on the extraversion scale have a tendency toward positive emotional reaction; people high on the neuroticism scale have a tendency toward negative emotional reaction. Although all the participants displayed increased activity in the amygdala when viewing fearful faces, only the extraverts displayed increased amygdalae activity when viewing happy faces. The following case

study reveals that the brain mechanisms of emotion differ from person to person.

GENETICS AND BEHAVIOR: CHROMOSOMAL ANOMALIES; NATURE-NURTURE CONTROVERSY [TWIN STUDIES AND ADOPTION STUDIES]

CHROMOSOMES AND GENES

Key Points

- Chromosomes are structures in the nucleus of a cell containing DNA coiled around histone proteins.
- All animals have some number of chromosomes, which transmit genetic material. Human beings have 46 chromosomes (23 pairs).
- Humans have two types of chromosomes: autosomes and sex chromosomes.
- Chromosomal abnormalities can result in genetic conditions such as Down syndrome.

Key Terms

- **chromosome:** A structure in the cell nucleus that contains DNA, histone protein, and other structural proteins.
- **gene:** A unit of heredity; a segment of DNA or RNA transmitted from one generation to the next, carrying genetic information such as the sequence of amino acids for a protein.
- **autosome:** Any chromosome that is not a sex chromosome.

Chromosomes contain genetic material that can determine a person's characteristics.

Chromosomes are structures in the nucleus of a cell containing DNA, histone protein, and other structural proteins. Chromosomes also contain genes, most of which are made up of DNA and RNA.

DNA, or deoxyribonucleic acid, determines whether our eyes are blue or brown, how tall we will be, and even our preference for certain types of behavior. Known as our "genetic code," it is shaped like a double helix, made of sequences of nucleic acids attached to a sugar phosphate backbone. Genes are subsections of DNA molecules linked together that create a particular characteristic.

Each chromosome is made up of a single DNA molecule coiled around histone

proteins. Research dating back to the 1800s shows that every living creature has a specific set of chromosomes in the nucleus of each of its cells.

Human chromosomes are divided into two types—autosomes and sex chromosomes. Some genetic traits are linked to a person's sex and therefore passed on by the sex chromosomes. The autosomes contain the remainder of a person's genetic information. All human beings have 23 pairs of chromosomes by which genetic material is developed and characteristically demonstrated; 22 of these are autosomes, while the remaining pair (either XX, female, or XY, male) represents a person's sex chromosomes. These 23 pairs of chromosomes work together to create the person we ultimately become.

Chromosomal abnormalities can occur during fetal development if something goes wrong during the replication of the cells. Common abnormalities include Down syndrome (caused by an extra chromosome #21), Klinefelter syndrome (caused by an extra X chromosome), and Turner syndrome (caused by a missing X chromosome). Genetic counseling is available for families in order to determine if any abnormalities exist that may be passed along to offspring. Many chromosomal abnormalities are of psychological importance, with substantial

impacts on mental processes; for example, Down syndrome can cause mild to moderate intellectual disabilities.

As science advances, the ability to manipulate chromosomes is becoming more realized. Cloning is an example of taking chromosomal and genetic material and creating a new animal, and was first successfully achieved in the famous example of Dolly the sheep. There is much controversy surrounding the manipulation of chromosomes in human beings, with many people believing it to be unethical.



Dolly the sheep: The first successfully cloned animal. Chromosomal and genetic manipulation are controversial topics.

NATURE VS NURTURE

Genetic expression can be influenced by various social factors, as well as environmental factors, from light and temperature to exposure to chemicals.

- Today it is generally accepted that nature and nurture work in tandem to create the people we ultimately become.
- Adoption and twin studies show that both nature and nurture are factors in human development.
- The environment in which a person is raised can trigger expressions of behavior for which that person is genetically predisposed; genetically identical people raised in different environments may exhibit different behavior.
- Three types of gene-environment correlations (rGE) exist: passive (ambiguous correlation), evocative (one factor invokes a response in the other), and active (one factor influences a preference for another).

Key Terms

- **gene-environment correlation:** A relationship in which exposure to environmental conditions correlates with an individual's genotype.
- **phenotype:** The observable expression of a gene.

Our genetic destiny is not necessarily written in stone; it can be influenced by several factors, such as social factors, as well as environmental influences among which we live, including anything from light and temperature to exposure to chemicals. The environment in which a person is raised can trigger the expression of behavior for which a person is genetically predisposed, while the same person raised in a different environment may exhibit different behavior.

Long-standing debates have taken place over the idea of which factor is more important, genes or environment. Is a person destined to have a particular outcome in life because of his or her genetic makeup, or can the environment (and the people in it) work to change what might be considered “bad” genes? Today, it is generally agreed upon that neither

genes nor environment work alone; rather, the two work in tandem to create the people we ultimately become.

Environmental elements like light and temperature have been shown to induce certain changes in genetic expression; additionally, exposure to drugs and chemicals can significantly affect how genes are expressed. People often inherit sensitivity to the effects of various environmental risk factors, and different individuals may be differently affected by exposure to the same environment in medically significant ways. For example, sunlight exposure has a much stronger influence on skin cancer risk in fair-skinned humans than in individuals with an inherited tendency for darker skin. The color of a person's skin is largely genetic, but the influence of the environment will affect these genes in different ways.

(EXTRA FOR REFERENCES)

Gene-environment correlations, can be explained in 3 particular ways—passive, evocative, or active.

PASSIVE GENE-ENVIRONMENT CORRELATIONS

In passive gene-environment correlation, an association exists between a person's genetic makeup and the environment in which he or she is raised. In other words,

the person's environment, particularly in the case of children, is largely determined by the parent's genetic characteristics. Parents create a home environment that is influenced by their own heritable characteristics. When the children's own genotype influences their behavior or cognitive outcomes, the result can be a misleading relationship between environment and outcome. For example,

an intelligent parent is likely to create a home environment rich in educational materials and experience. Since intelligence is moderately heritable, it can be argued that intelligence in the child is inherited rather than a factor of the home environment created by the parents. It is relatively unclear whether the genetic or environmental factors had more to do with the child's development.

EVOCATIVE GENE-ENVIRONMENT CORRELATIONS

Evocative gene-environment correlation happens when an individual's (heritable) behavior evokes an environmental response. For example, the association between marital conflict and depression may reflect the tensions that arise when engaging with a depressed spouse rather than a causal effect of marital conflict on risk for depression.

ACTIVE GENE-ENVIRONMENT CORRELATIONS

In active gene-environment correlation, the person's genetic makeup may lead them to select particular environments. For example, a shy person is likely to choose quiet activities and less boisterous environments than an extroverted individual. He or she may be more likely to spend time at the library than at a dance club.

TWIN STUDIES:

A powerful method for estimating the influence of heredity on a particular trait is to compare the concordance rate for this trait in pairs of monozygotic and dizygotic twins. Monozygotic twins (identical twins) have identical genotypes—that is, their chromosomes, and the genes they contain, are identical. In contrast, the genetic similarity between dizygotic twins (fraternal twins) is, on the average, 50 percent. Investigators study records to identify pairs of twins in which at least one member has the trait—for example, a

ADOPTION AND TWIN STUDIES IN THE NATURE VS. NURTURE DEBATE

Adoption and twin studies can help make sense of the influence of genes and the environment. Studies of adult twins are used to investigate which traits are heritable. Identical twins share the same genotype, meaning their genetic makeup is the same. Twins raised apart tend to be similar in intelligence and, in some cases, life events and circumstance, when studied years later, even when raised separately.

However, researchers have discovered that the phenotype (or the observable expression of a gene) of identical twins grows apart as they age. In adoption studies, identical twins raised by different families can give insight into the nature-versus-nurture debate. Since the child is being raised by parents who are genetically different from his or her biological parents, the influence of the environment shows in how similar the child is to his or her adoptive parents or siblings. Adoption studies make a strong case for the influence of environment, whereas twin studies make a strong case for genetic influence.

diagnosis of a particular mental disorder. If both twins have been diagnosed with this disorder, they are said to be concordant.

If only one has received this diagnosis, the twins are said to be discordant. Thus, if a disorder has a genetic basis, the percentage of monozygotic twins who are concordant for the diagnosis will be higher than the percentage of dizygotic twins. For e.g., the concordance rate for schizophrenia in twins is at least four times higher for monozygotic twins than for dizygotic twins, a finding that provides strong evidence that schizophrenia is a heritable trait. Twin studies have found that many

individual characteristics, including personality traits, prevalence of obesity, incidence of alcoholism, and a wide variety of mental disorders, are influenced by genetic factors.

ADOPTION STUDIES:

Another method for estimating the heritability of a particular behavioral trait is to compare people who were adopted early in life with their biological and adoptive parents. All behavioral traits are affected to some degree by hereditary factors, environmental factors, and an interaction between hereditary and environmental factors. Environmental factors are both social and biological in nature. For example, the mother's health, nutrition, and drug-taking behavior during pregnancy are prenatal environmental factors, and the child's diet, medical care, and social environment (both inside and outside the home) are postnatal environmental factors. If a child is adopted soon after birth, most of the postnatal

environmental factors will be associated with the adoptive parents, the genetic factors will be associated with the biological parents, and the prenatal environmental factors will be associated with the biological mother.

Adoption studies require that the investigator know the identity of the parents of the people being studied and be able to measure the behavioral trait in the biological and adoptive parents. If the people being studied strongly resemble their biological parents, we conclude that the trait is probably influenced by genetic factors. To be certain, we will have to rule out possible differences in the prenatal environment of the adopted children. If, instead, the people resemble their adoptive parents, we conclude that the trait is influenced by environmental factors. Of course, it is possible that both hereditary and environmental factors play a role, in which case the people being studied will resemble both their biological and adoptive parents.

(EXTRA FOR REFERENCE)

TARGETED MUTATIONS:

A recently developed method has put a powerful tool in the hands of neuroscientists. Targeted mutations are mutated genes produced in the laboratory and inserted into the chromosomes of mice. These mutated genes (also called **knockout genes**) are defective—they fail to produce a functional protein. In many cases, the target of the mutation is an enzyme that controls a particular chemical

reaction. For example, the lack of a particular enzyme interferes with learning. This result suggests that the enzyme is partly responsible for changes in the structure of synapses required for learning to occur. In other cases, the target of the mutation is a protein that itself serves useful functions in the cell. For example, a particular type of opiate receptor is involved in the reinforcing and analgesic effects of opiates.

EXTRA FOR REFERENCES (IMPORTANT FOR NET-JRF)

SPEECH PRODUCTION

Both perceptions of current events and memories of events that occurred in the past involve brain mechanisms in the posterior part of the cerebral hemispheres (the occipital, temporal, and parietal lobes). Thus, this region is largely responsible for our having something to say. Damage to a region of the inferior left frontal lobe (Broca's area) disrupts the ability to speak:

It causes **Broca's aphasia**. This disorder is characterized by slow, laborious, and nonfluent speech. When trying to talk with patients who have Broca's aphasia, most people find it hard to resist supplying the words the patients are obviously groping for. People with Broca's aphasia find it easier to say some types of words than others. They have great difficulty saying the little words with grammatical meaning, such as *a*, *the*, *some*, *in*, or *about*. These words are called function words, because they have important grammatical functions. The words that they do manage to say are almost entirely content words—words that convey meaning, including nouns, verbs, adjectives, and adverbs, such as *apple*, *house*, *throw*, or *heavy*.

One of the usual symptoms of Broca's aphasia is agrammatism; a difficulty in comprehending or properly employing grammatical devices, such as verb endings and word order. Functional-imaging

studies by Opitz and Friederici (2003, 2007) found that Broca's area was activated when people were taught an artificial grammar, which supports the conclusion that this region is involved in learning grammatical rules—especially complex ones.

Patients mispronounce words, often altering the sequence of sounds. For example, lipstick might be pronounced “likstip.” People with Broca's aphasia recognize that their pronunciation is erroneous, and they usually try to correct it.

Prosody includes changes in intonation, rhythm, and stress that add meaning, especially emotional meaning, to the sentences that we speak. The neural mechanisms that control the prosodic elements of speech appear to be in the right hemisphere.

Stuttering is not a result of abnormalities in the neural circuits that contain the motor programs for speech. For example, stuttering is reduced or eliminated when a person reads aloud with another speaker, sings, or reads in cadence with a rhythmic stimulus. The problem appears to lie more in the neural mechanisms that are involved in planning and initiation of speech. Functional imaging indicates deficient auditory feedback produced by the stutterer's own voice. Delayed auditory feedback, which impairs the speech of most fluent speakers, often facilitates the speech of stutterers.

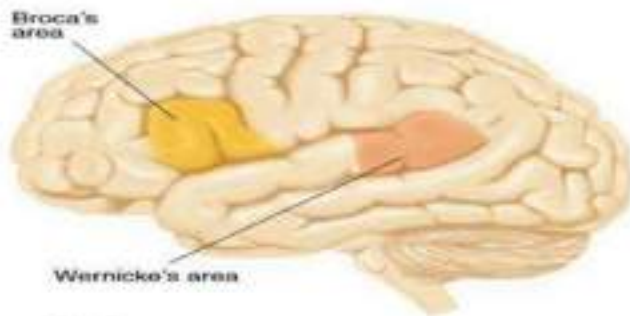


FIGURE 14.2 Speech Areas

The figure shows the location of the primary speech areas of the brain. (Wernicke's area will be described later.)

SPEECH COMPREHENSION:

Comprehension of speech obviously begins in the auditory system, which detects and analyzes sounds. But recognizing words is one thing; comprehending them—understanding their meaning—is another. Recognizing a spoken word is a complex perceptual task that relies on memories of sequences of sounds.

Wernicke's area is a region on the left temporal lobe of humans, which is

important in the comprehension of words and the production of meaningful speech.

WERNICKE'S APHASIA:

The primary characteristics of Wernicke's aphasia are poor speech comprehension and production of meaningless speech. Unlike Broca's aphasia, Wernicke's aphasia is fluent and unlabored; the person does not strain to articulate words and does not appear to be searching for them. The patient maintains a melodic line, with the voice rising and falling normally.

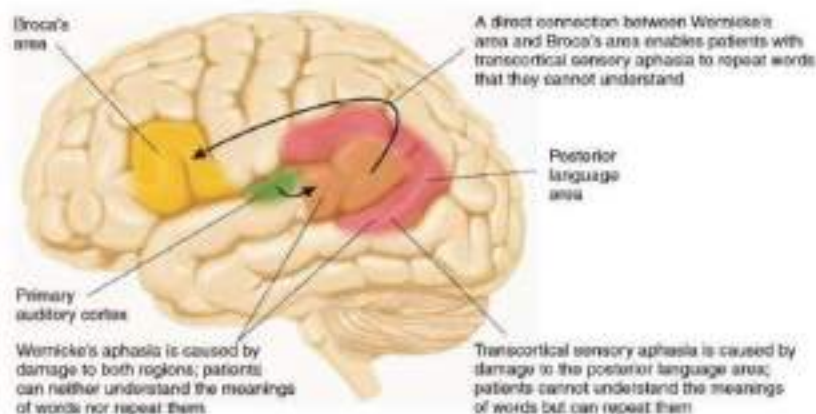


FIGURE 14.10 Transcortical Sensory Aphasia and Wernicke's Aphasia

This schematic shows the location and interconnections of the posterior language area and an explanation of its role in transcortical sensory aphasia and Wernicke's aphasia.

When you listen to the speech of a person with Wernicke's aphasia, it appears to be grammatical.

The abilities that are disrupted include recognition of spoken words, comprehension of the meaning of words, and the ability to convert thoughts into words. Let us consider each of these abilities in turn.

However, the person uses few content words, and the words that he or she strings together just do not make sense. In the extreme, speech deteriorates into a meaningless jumble, illustrated by the following quotation:

Examiner: What kind of work did you do before you came into the hospital?

Patient: Never, now mista oyge I wanna tell you this happened when happened when he rent. His—his kell come down here and is—he got ren something. It happened. In these ropier were with him for hi—is friend—like was. And it just happened so I don't know; he did not bring around anything.

A remarkable fact about people with Wernicke's aphasia is that they often seem

unaware of their deficit. That is, they do not appear to recognize that their speech is faulty, nor do they recognize that they cannot understand the speech of others.

Damage restricted to Wernicke's area causes pure word deafness—loss of the ability to understand speech but intact speech production, reading, and writing. The left hemisphere is involved in analysis of rapid changes in sounds, which is consistent with its role in the analysis of speech sounds, which are characterized by such changes. Wernicke's aphasia, caused by damage to Wernicke's area and the posterior language area, consists of poor speech comprehension, poor repetition, and production of fluent, meaningless speech. Transcortical sensory aphasia, caused by damage to the posterior speech area, consists of poor speech comprehension and production, but the patients can repeat what they hear. Thus, the symptoms of Wernicke's aphasia consist of those of transcortical sensory aphasia plus those of pure word deafness. (WA = TSA + PWD.) Feedback from mirror neurons that are activated when people hear the speech of other people may facilitate speech recognition

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UNIT 5

ATTENTION, PERCEPTION, LEARNING, MEMORY And FORGETTING

ATTENTION

Attention is defined as the mental process of concentrating effort on a stimulus or mental event the limited mental energy or resource that powers the mental system.

Attention. Attention has also been referred to as the allocation of processing resources. Therefore, **the process through which certain stimuli are selected from a group of others is generally referred to as attention.**

It can also be defined as the cognitive process of selectively concentrating on one aspect of the environment while ignoring other things. Attention is the term used or given to the perceptual processes that select certain inputs for inclusion in our conscious experience, or awareness at any given time. It is the process involving the act of listening, and concentrating on a topic, object or event for the attainment of desired ends.

Attention has a focus as well as a fringe. When the field of awareness is centered on a particular object or event, it is called focus or the focal point of attention. On the contrary, when the objects or events are away from the centre of awareness and one

is only vaguely aware of them, they are said to be at the fringe of attention. Let's understand it with an example that include

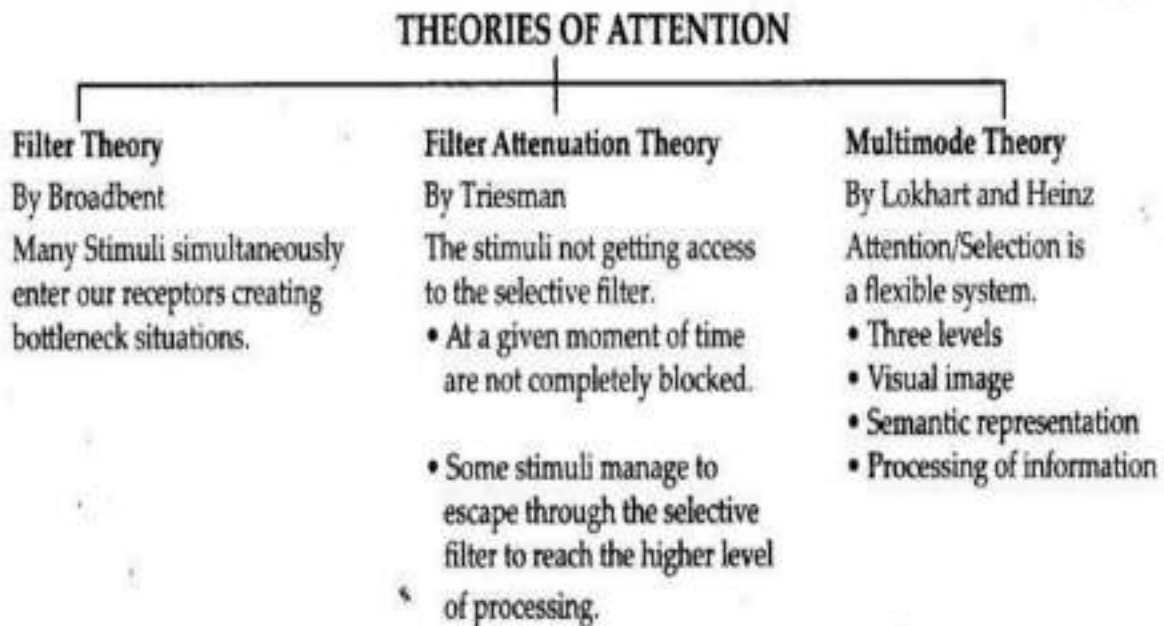
carefully listening to what someone is saying while ignoring other conversations in a room (the cocktail effect) or listening to a cell phone conversation while driving a car. Attention has been classified in a number of ways. A process-oriented view divides it into two types, namely selective and sustained- Selective attention is concerned mainly with the selection of a limited number of stimuli or objects from a large number of stimuli. While selective attention is mainly concerned with the selection of stimuli, sustained attention is concerned with concentration. It refers to our ability to maintain attention on an object or event for longer durations. It is also known as "vigilance".

Sometimes we can also attend to two different things at the same time. When this happens, it is called divided attention.

MODELS OF ATTENTION There

are many theories of attention

Among them are among them is the Attenuation theory, Filter theory, Capacity model



, Deutsch and Deutsch model,

Multimode theory and the Schema theories of attention. But here, we would only

BROADBENT'S FILTER MODEL

Broadbent (1958) proposed that physical characteristics of messages are used to select one message for further processing and that all others are lost. Information from all of the stimuli presented at any given time enters an unlimited capacity sensory buffer. One of the inputs is then selected on the basis of its physical characteristics for further processing by being allowed to pass through a filter.

TREISMAN'S ATTENUATION MODEL

Filter-attenuation theory was developed by Treisman (1962) by modifying early bottleneck filter.

focus on few of the models that are relevant to the exams.

Because we have only a limited capacity to process information, this filter is designed to prevent the information-processing system from becoming overloaded.

The inputs not initially selected by the filter remain briefly in the sensory buffer store, and if they are not processed, they decay rapidly. Broadbent assumed that the filter rejected the unattended message at an early stage of processing.

Broadbent's theory. Treisman (1964) agrees with Broadbent's theory of an

However, the difference is that Treisman's filter attenuates rather than eliminates the

This theory proposes that the stimuli not getting access to the selective filter at a given moment of time are not completely blocked. The filter only attenuates (weakens) their strength. Thus, some stimuli manage to escape through the selective filter to reach higher levels of

In her experiments, Treisman demonstrated that participants were still able to identify the contents of an

unattended material.

processing. It is indicated that personally relevant stimuli (e.g., one's name in a collective dinner) can be noticed even at a very low level of sound. Such stimuli, even though fairly weak, may also generate response occasionally by slipping through the selective filter unattended message, indicating that they were able to process the meaning of both the attended and unattended messages.

MULTIMODE THEORY:

Multimode theory was developed by Johnston and Heinz (1978). This theory believes that attention is a flexible system that allows selection of a stimulus over others at three stages. At stage one the sensory representations (e.g., visual images) of stimuli are constructed; at stage two the semantic representations (e.g., names of objects) are constructed; and at

stage three the sensory and semantic representations enter the consciousness. It is also suggested that more processing requires more mental effort. When the messages are selected on the basis of stage one processing (early selection), less mental effort is required than when the selection is based on stage three processing (late selection).

PERCEPTION

Sensation is the stimulus detection process by which our sense organs respond to and translate environmental stimuli into nerve impulses that are sent to the brain.

The scientific area of psychophysics, studies relations between the physical characteristics of stimuli and sensory capabilities.

Perception—making “sense” of what our senses tell us—is the active process of organizing this stimulus input and giving it meaning (Mather, 2006; May, 2007). Perception is influenced by perceptual sets: a readiness to perceive stimuli in a particular way.

Recognizing a stimulus implies that we have a perceptual schema—a mental representation or image containing the critical and distinctive features of a person, object, event, or other perceptual phenomenon. Schemas provide mental templates that allow us to classify and identify sensory input in a top down fashion.

To create our perceptions, the brain carries out two different kinds of processing functions: In bottom-up processing, the system takes in individual elements of the stimulus and then combines them into a unified perception.

In top-down processing, sensory information is interpreted in light of existing knowledge, concepts, ideas, and expectations.

GESTALT APPROACHES

The Gestalt psychologists indicate that our cerebral processes are always oriented towards the perception of a good figure or *pragnanz*. That is the reason why we perceive everything in an organised form. The most primitive organisation takes place in the form of figure- ground segregation. The Gestalt theorists emphasized the importance of figure-ground relations, our tendency to organize stimuli into a central or foreground figure and a background.

In vision, the central figure is usually in front of or on top of what we perceive as background. It has a distinct shape and is

more striking in our perceptions and memory than the background. We perceive borders or contours wherever there is a distinct change in the color or brightness of a visual scene, but we interpret these contours as part of the figure rather than background. Likewise, we tend to hear instrumental music as a melody (figure) surrounded by other chords or harmonies (ground).

Four Gestalt laws of perceptual organization namely: **similarity, proximity, closure, and continuity** have been suggested according to which people group and interpret stimuli.

GESTALT LAW OF SIMILARITY: When parts of a configuration are perceived similar, they will be perceived as belonging together.



THE LAW OF PROXIMITY: Elements that are near each other are likely to be perceived as part of the Same configuration.



THE LAW OF CLOSURE:

People tend to close the open so that their identification of (in this case, a circle) is more complete than what is actually

edges of a figure or fill gaps in an incomplete figure, the form there.



THE LAW OF CONTINUITY holds that people link individual elements together so they form

a continuous line or pattern that makes sense.

PERCEPTUAL CONSTANCIES:

They allow us to recognize familiar stimuli under varying conditions. Without these, we would have to literally rediscover what something is each time it appeared under different conditions. The various types of perceptual constancies are:

BRIGHTNESS CONSTANCY:

The relative brightness of objects remains the same under different conditions of illumination, such as full sunlight and shade. Brightness constancy occurs because the ratio of light intensity between an object and its surroundings is constant. The actual brightness of the light that illuminates an object does not matter, as long as the same light intensity illuminates both the object and its surroundings.

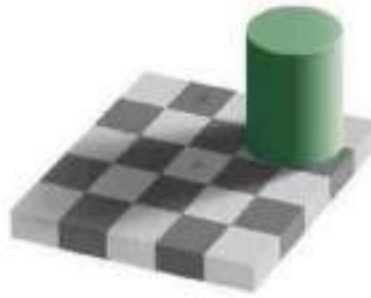
SHAPE CONSTANCY: It

allows us to recognize people and other objects from many different angles.

SIZE CONSTANCY: It is the perception that the size of objects remains relatively constant even though images on our retina change in size with variations in distance.



Thus, a man who is judged to be 6 feet tall when standing 5 feet away is not perceived to be 3 feet tall at a distance of 10 feet, even though the size of his image on the retina is reduced to half its original size.



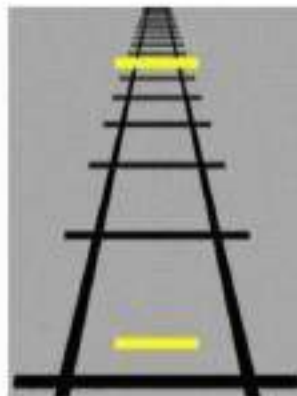
Checker-shadow illusion

Color constancy tricks our brains into seeing squares A and B as two different colors; however, they are the exact same shade of gray.



Shape constancy

This form of perceptual constancy allows us to perceive that the door is made of the same shape despite different images being delivered to our retinas.



The Ponzo Illusion

This famous optical illusion uses size constancy to trick us into thinking the top yellow line is longer than the bottom; they are actually the exact same length.

PERCEPTION OF DEPTH AND DISTANCE:

A very intriguing aspect of visual perception is our ability to perceive depth.

The retina receives information in only two dimensions (length and

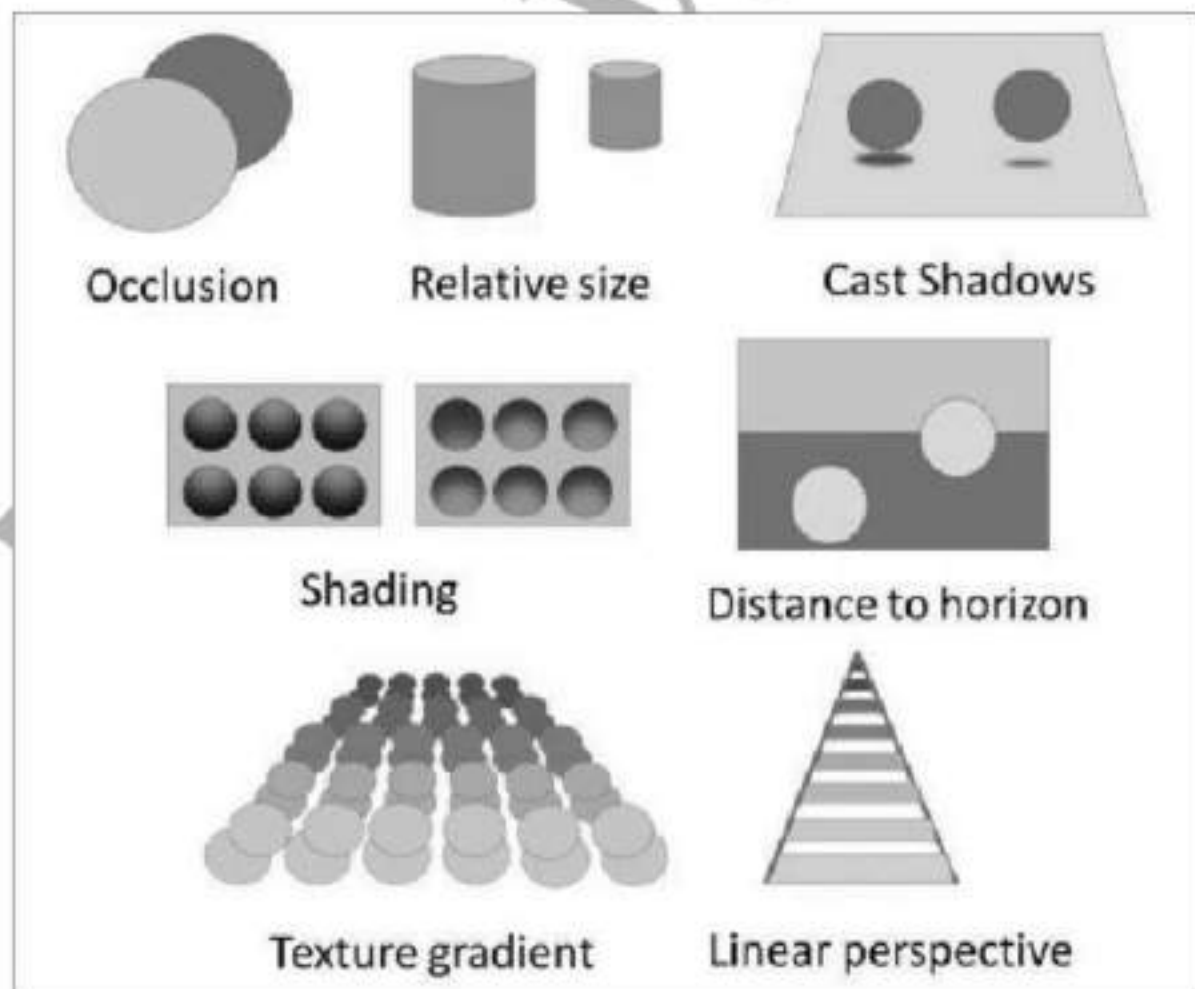
MONOCULAR DEPTH CUES: 8 Monocular cues have been recognized.

One such cue is patterns of light and shadow.

Another cue, **linear perspective**, refers to the perception that parallel lines converge, or angle toward one another, as they recede into the distance. Thus, if we look down railroad tracks, they appear

width), but the brain translates these cues into three-dimensional perceptions. It does this by using both **monocular depth cues**, which require only one eye, and **binocular depth cues**, which require both eyes.

to angle toward one another with increased distance, and we use this as a depth cue. The same occurs with the edges of a highway. Another cue for perception of depth and distance is **Interposition**, in which objects closer to us may cut off part of our view of more distant objects



An object's height in the horizontal plane provides another source of information. For example, a ship 5 miles offshore appears in a higher plane and closer to the horizon than does one that is only 1 mile from shore.

Texture is a fifth cue, because the texture or grain of an object appears finer as distance increases.

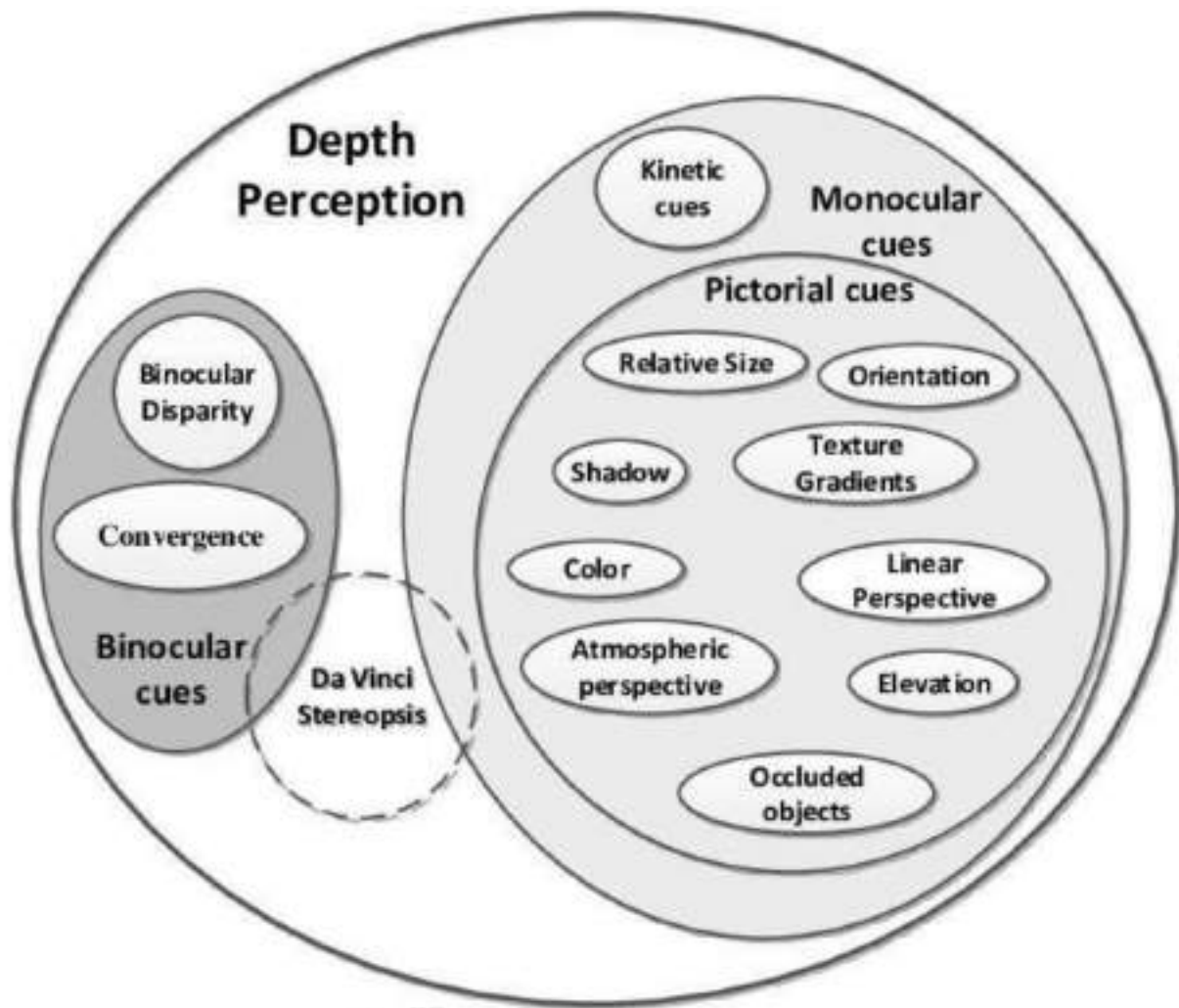
Likewise, clarity can be an important cue for judging distance; we can see nearby hills more clearly than ones that are far away, especially on hazy days.

Relative size is yet another basis for distance judgments. If we see

two objects that we know to be of similar size, then the one that looks smaller will be judged to be farther away.

It is important to remember that none of these monocular cues mentioned above involve movement of the object(s).

However, a final monocular cue, motion parallax, tells us that if we are moving, nearby objects appear to move faster in the opposite direction than do faraway ones.



BINOCULAR DEPTH CUES:

These require the use of both eyes.

The principle of **binocular disparity**, in which each eye sees a slightly different image is a binocular depth cue. Within the brain, the visual input from the two eyes is analyzed by feature detectors that are attuned to depth (Howard, 2002; Livingstone & Hubel, 1994). Some of the feature detectors respond only to stimuli that are either in front of or behind the point on which we are fixing our gaze. The responses of these

depth-sensitive neurons are integrated to produce our perception of depth (Goldstein, 2002).

A second binocular distance cue, **convergence**, is produced by feedback from the muscles that turn your eyes inward to view a close object. An example of this can be by holding a finger about 1 foot in front of your face and then moving it slowly toward you. Messages sent to our brain by the eye muscles provide it with a depth cue.

It has been recognized that the primary cue for perceiving motion is the movement of the stimulus across the retina. Gestalt psychologist **Max Wertheimer** (1912) demonstrated this in his studies of **stroboscopic movement** (termed the “**phi phenomenon**” by Wertheimer), an illusory movement produced when a light is briefly flashed in darkness and then, a few milliseconds later, another light is flashed nearby. If the timing is just right, the first light seems to move from one place to the other in a manner indistinguishable from real movement.

STROBOSCOPIC-

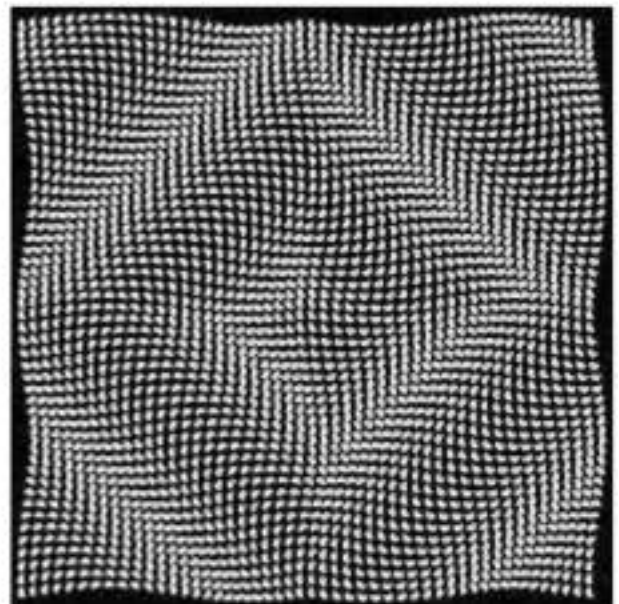
MOVEMENT is also the principle behind motion pictures, which consist of a series of still photographs, or frames, that are projected on a screen in rapid succession with dark intervals in between. The rate at which the frames are projected is critical to our perception of smooth movement. Early movies, such as the silent films of the 1920s, projected the stills at only 16 frames per second, and the movements appeared fast and jerky. Today the usual speed is 24 frames per second, which more perfectly produces an illusion of smooth movement. Television presents at 30 images per second.

Monocular depth cues	Binocular depth cues	
Monocular static depth	Monocular dynamic depth cues	
Relative size Interposition Elevation Lighting and shading Texture gradient Aerial perspective	Motion parallax Deletion Accretion	Binocular parallax Binocular convergence

ILLUSIONS

Illusions are compelling but incorrect perceptions. Most visual illusions can be attributed to perceptual constancies that ordinarily help us perceive more accurately (Frisby, 1980).

Sometime we fail to interpret the sensory information correctly. This results in a mismatch between the physical stimuli and its perception.



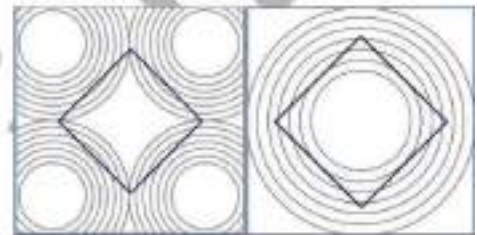
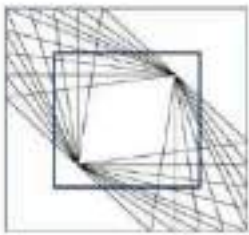
These misperceptions resulting from misinterpretation of information

These are experienced more or less by all of us. They result from an external stimulus situation and generate the same kind of experience in each individual. That is **why illusions are also called “primitive organisations**. Some perceptual illusions are universal and found in all

received by our sensory organs are generally known as illusions. individuals. For example, the rail tracks appear to be converging to all of us. These illusions are called universal illusions or permanent illusions as they do not change with experience or practice. Some other illusions seem to vary from individual to individual; these are called personal illusions.

HERING'S OPTICAL ILLUSIONS:

The slanted lines cause the illusion that the blue lines (straight lines) are not parallel. In fact, they are!

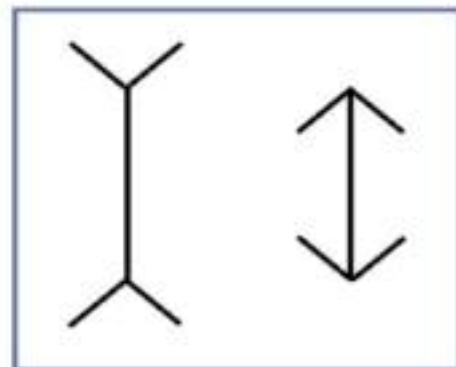


EHRENSTEIN'S OPTICAL ILLUSIONS:

Slanted lines make the blue square seem distorted.

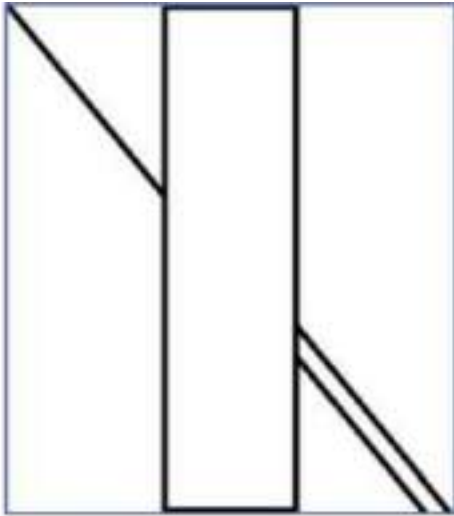
MÜLLER-LYER'S OPTICAL ILLUSIONS: Probably the most famous and most studied illusion was created by German psychiatrist Franz Müller-Lyerin

1889. Although your eyes tell you that the left horizontal line is longer than the right one, they are equal in length.



POGGENDORF'S OPTICAL ILLUSIONS: The single line if continued

joins with the lower of the pair, not the top.

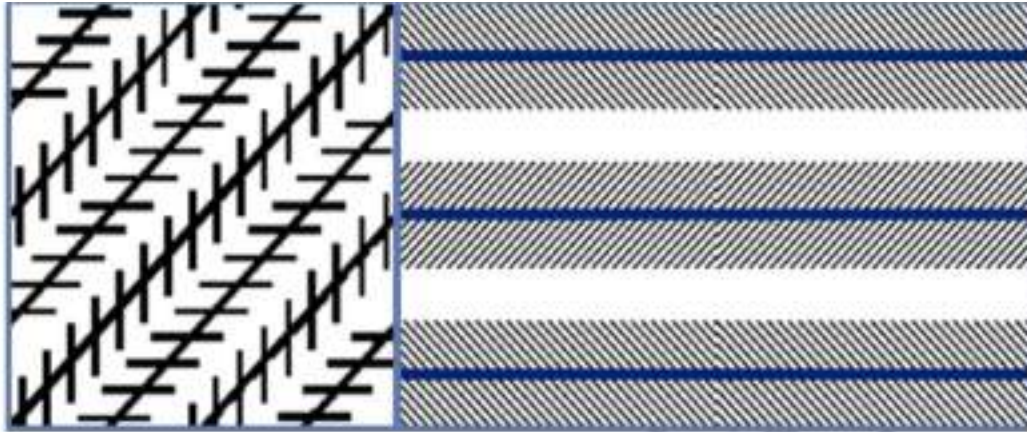


MEYER'S OPTICAL ILLUSIONS:
The center shape is in fact a circle.



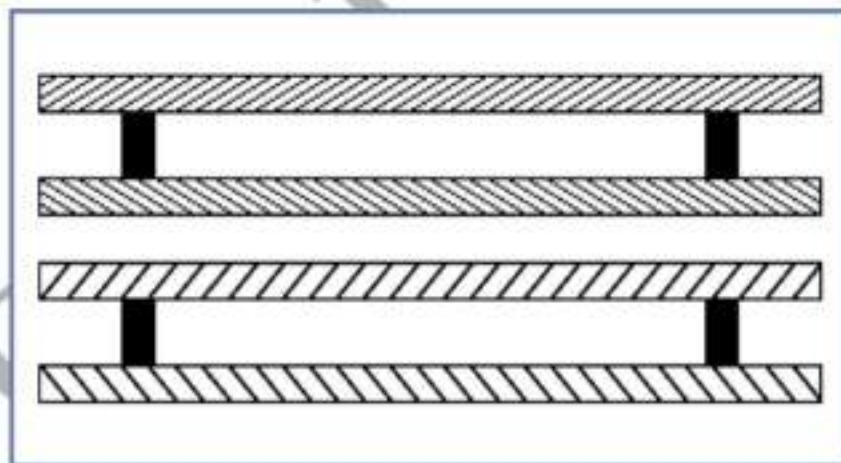
ZÖLLNER'S OPTICAL ILLUSIONS:
Parallel lines intersected by a pattern of short diagonal lines appear to diverge.

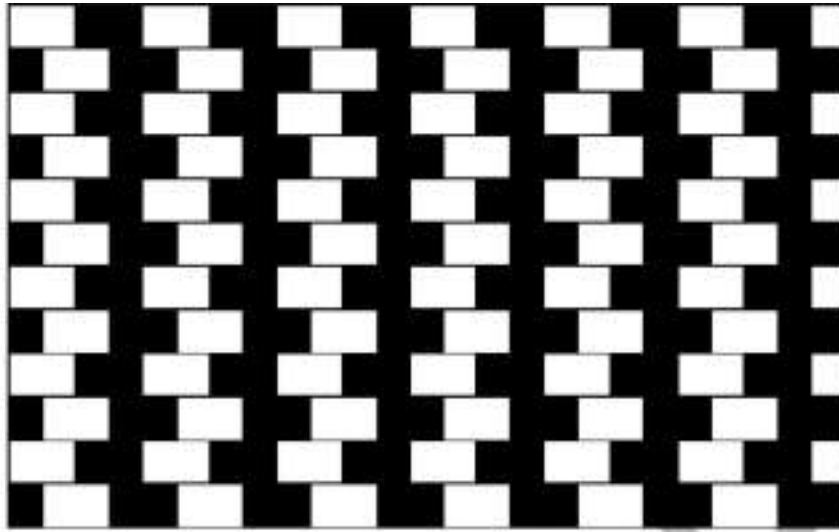
EXPLANATION: the brain is attempting to interpret this image as if it were part of a three- dimensional scenes.



Now let's see a variant of Zöllner's optical illusion based on parallel bars. Note that the strength of the

illusion depends on the pattern, especially on the slope of the diagonal pattern lines.





MÜNSTERBERG ILLUSION

PONZO'S OPTICAL ILLUSIONS: The two slanted lines arranged like an inverted V evoke the impression of perspective. Thus, the upper objects (line, circle, barrel), which are

framed by the diagonally converging line segments, are perceived to differ in length from the twin objects beneath.



SIGNAL DETECTION THEORY

Signal detection theory is concerned with the factors that influence sensory judgments. It is based on **3 assumptions**:

Neurons are constantly sending information to the brain, even

when no external stimuli are present. This is called internal neural 'noise.' The level of neural noise fluctuates constantly.

When a faint (external) stimulus called a 'signal' occurs, it creates a neural response which adds to internal noise.

The brain must decide whether the

neural activity it is getting reflects noise alone, or whether there was

SENSORY RECEPTORS-These are of two types:

EXTEROCEPTORS: sensory receptors that respond to light, sound, smell, touch, pain, etc., to create conscious sensation.

PROPRIOCEPTORS: sensory receptors that respond to joint movement (kinesthesia) and joint position (joint position sense), but do not typically contribute to conscious sensation.

The theory involves treating detection of the stimulus as a decision-making process, part of which is determined by the nature of the stimulus, by how sensitive a person is to the stimulus, and by cognitive factors. In other words, a

also a signal.

person will be able to detect more intense sounds or lights more easily than less intense stimuli. Further, a more sensitive person requires less stimulus intensity than a less sensitive person would. Finally, when a person is quite uncertain as to whether the stimulus was present, the individual will decide based on what kind of mistake in judgment is worse: to say that no stimulus was present when there actually was one or to say that there was a stimulus when, in reality, there was none. Signal-detection research shows us that perception is, in part, a decision. **Thus, decision criterion, a standard of how certain individuals must be that a stimulus is present before they will say they detect it becomes significant.**

Signal detection theory predicts when an individual will decide if a weak stimulus is present or not based on psychological factors as opposed to the biological theories of absolute threshold. Psychologists use signal detection theory to evaluate how accurately participants are able to recognize faint stimuli under a variety of different conditions. In a typical experiment the participants are asked to indicate if the stimulus was either present or absent, and the researcher assesses if the decision was correct. For example, Mark is waiting for an important text message from his girlfriend letting him know when to meet at the movie theater. According to the signal detection theory, various psychological factors, including Mark's level of attention, motivation, and expectations, will impact whether or not he detects the incoming text message or not. There are four possible outcomes for Mark in this situation based on the signal detection theory described in the Table below.

	Present (signal occurred)	Absent (signal did NOT occur)
Yes (participant noticed a signal)	HIT	FALSE ALARM
No (participant did NOT notice a signal)	MISS	CORRECT REJECTION

Directions: Read the following scenarios and determine which of the four potential outcomes for the signal detection theory is being presented.

1. _____ A brother and sister are playing hide and seek and the brother has hid under a pillow on the sofa. His sister comes into the room and looks at the sofa, but seeing nothing, she leaves.
2. _____ The sister then goes into her brother's room and sees a lump under the sheets, she sneaks up to the lump and screams, "I found you", but soon she finds out this is her brother's stuffed up teddy bear.
3. _____ An administrative assistant is expecting an important e-mail from her boss, she must temporarily leave her desk to make some copies, as she makes copies she thinks she hears the "ding" of her computer indicating she has an incoming message, it turns out that she did not receive any incoming messages.

CULTURE AND PERCEPTION

This concept suggests that the habits of perception are learnt differently in different cultural settings. Different experiences and learning opportunities available to people in different cultural settings also influence their perception. People coming from a picture less environment fail to recognize objects in pictures.

Example: Hudson studied the perception of pictures by African subjects, and noted several difficulties. Many of them were unable to identify objects depicted in pictures (e.g., antelope, spear). They also failed to perceive distance in pictures, and interpreted pictures incorrectly.

Another example is that of Eskimos who have been found to make fine distinction among a variety of snow that we may be unable to notice.

Some of the perceptual styles proposed by researchers are:

ACTIVITY: People with the Activity Perceptual Style jump into life with both feet. They fully engage with the confidence that the details will sort themselves out. Direction, ideas, and pursuits emerge as the result of constant action and involvement with others and their surroundings. They engage until some new possibility or interest emerges to capture their attention. They cultivate extensive networks of friends and associates.

ADJUSTMENTS: People with the

PERCEPTUAL STYLES

Perceptual Style acts as a filter between sensation and understanding. It is at the core of who we are, and it impacts our values, our beliefs and our feelings. They are also known as cognitive style. Cognitive style refers to a consistent way of dealing with our environment. It significantly affects the way we perceive the environment. There are several cognitive styles that people use in perceiving their environment. One most extensively used in studies is the “field dependent and field independent” cognitive style.

Field dependent people perceive the external world in its totality, i.e. in a global or holistic manner. On the other hand, **field independent people** perceive the external world by breaking it into smaller units, i.e. in an analytic or differentiated manner.

Adjustments Perceptual Style see the world as an objective reality that can be known if they take the time to gather complete information about its intricacies and complexities. They pursue the acquisition and application of knowledge as the basis for their life experience. They enjoy sharing their knowledge with others and gathering new information from research or conversation. They have a strong sense of diplomacy and project a calm certainty.

FLOW: People with the Flow Perceptual Style are instinctive

advocates for the natural rhythms of life. They see the complex connectivity among seemingly unrelated people, environments, and situations. They intuitively integrate and harmonize their actions within a broadly defined community that provides them and others with a sense of belonging. They honor the continuity between past, present, and future.

GOALS: People with the Goals Perceptual Style stride through life focused on the accomplishment of specific results and well-defined objectives. They experience a sense of urgency and clarity of purpose. They believe achievement is primary and method or process secondary – the end justifies the means. They evaluate all activities based on possible contribution towards the achievement of the results they expect. They thrive on competition and believe that life is a constant competition with winners and losers.

METHODS: People with the Methods Perceptual Style approach life in a practical, matter-of-fact manner. They focus on how things need to be done. They believe that ordered processes, properly followed, will produce the desired results. They will discern the best process or technique to apply to any specific situation in order to produce reliable, repeatable outcomes. They impose order and they believe that everyone prefers to use well known and proven methods.

VISION: People with the Vision Perceptual Style approach life as a

singular experience, a journey toward the future. They face the realities of a situation with serious intent, an optimistic perspective that a solution will be found, and confidence that if one is not, there are always other alternatives to explore. They intuitively see new directions, and actions are taken or dropped opportunistically based on a sense of future possibilities and potential. They are highly persuasive and easily convince others to follow their vision.

All six Perceptual Styles provide distinctly different experiences of the world. These differences result in a profound psychological and perceptual diversity that is the most important diversity there is because it helps explain the differences between people.

PATTERN RECOGNITION:

Human pattern recognition can be considered as a typical perception process which depends on knowledge and experience people already have. Generally, **pattern recognition refers to a process of inputting stimulating (pattern) information and matching with the information in long-term memory, then recognizing the category which the stimulation belongs to.** Therefore, pattern recognition depends on people's knowledge and experience. Without involving individual's knowledge and experience, people cannot understand the meanings of the stimulating information pattern inputted, then neither possible to recognize the patterns, which means to recognize the objects. **The process by which a person distinguishes a pattern he**

percepts with others and identifies what it is refers to pattern recognition.

ECOLOGICAL PERSPECTIVES ON PERCEPTION

Has been proposed by **Gibson**.

According to him, the concept of the world can be analyzed at different levels. Gibson is mainly concerned with the world at the level of **ecology**, in which animal and environment form an integrated system of mutual constraint. To draw this distinction, Gibson offers a **structural analysis** of the environment. This amounts to an explanatory scheme based on principles of self-organization. This analysis leads Gibson to formulate the concepts of the meaningful environment and the **perceiver-environment coupling** he calls **reciprocity**.

LEARNING:

Learning refers to the process by which experience produces a relatively enduring change

in an organism's behavior or capabilities.

There have been three major theories proposed to explain the concept of learning. These are:

THORNDIKE'S LEARNING THEORY:

Thorndike's theory of learning is also known as connectionism. He acknowledged the existence of thought, which he called mental units. A mental unit was anything sensed or perceived or the sensing, perceiving bit of consciousness. A physical unit was a stimulus or response (observable behaviour).

Thorndike proposed that learning was a matter of making **four kinds of connections**:

mental and physical units,

physical units with mental,

mental units with other mental units, and

physical units with other physical units.

His experiments looked for those things that strengthened these connections. Based on the experiments he conducted, Thorndike came up with three laws of learning which form a prominent part of his theory:

LAWS OF LEARNING

Based on his experiments, Thorndike came up with three laws of learning.

LAW OF EFFECT

Law of effect states that in a given situation, a response followed by a satisfying consequence will become more likely to occur and a response followed by an annoying consequence will become less likely to occur.

The law of effect states that an action followed by a pleasurable consequence is more likely to be repeated and one followed by an annoying or painful consequence is less likely to be repeated. This basically implies that the strength of a connection is influenced by consequences of a response. Put simply, actions that are rewarded tend to be strengthened and repeated, those that are punished tend to be weakened and not repeated.

An important finding by Thorndike was that pleasure was more potent for stamping out response than pain. For instance, if someone wants a negative behaviour to go away, it is more effective to reward a conflicting positive behaviour than to simply punish the negative behaviour such as providing rewards to students for doing positive things that would make it impossible for them to do the negative things.

LAW OF EXERCISE

The law of exercise states that the connections between a stimulus and response becomes strengthened with

practice and weakened when practice is discontinued.

The more a stimulus-induced response is repeated the longer it will be retained. The more often the cat is put in the puzzle box to make the connection between lever and gate opening, the longer this behaviour will be retained. However, if the cat was only put in the puzzle box once every other week, the learning it had gained would quickly recede. That is, the number of tries and the amount of time it took to press the level would increase.

LAW OF READINESS

Law of readiness states that in situations when an individual is ready to act, it is reinforcing for them to do so and annoying for them not to do so. On the other hand, when a human is not ready to act, forcing them to do so is annoying.

HULL'S DRIVE REDUCTION THEORY OF MOTIVATION:

In his theory, Hull tried to explain behaviour and learning through the means of drive reduction. He proposed that learned connections existed directly between the neural impulses from activated sensory receptors and certain muscular reactions- the term Hull used was '**receptor-effector connections.**'

Hull adopted the term 'reinforcement' in his theory for an event which strengthened stimulus- response connections, and thus Thorndike's Law of Effect became the **Law of Primary Reinforcement.**

Hull based his theory on the concept of homeostasis, the idea that the body

actively works to maintain a certain state of balance or equilibrium. In his theory, Hull used the term **drive** to refer to the state of tension or arousal caused by biological or physiological needs. **Thirst, hunger, and the need for warmth** are all examples of drives. A drive creates an unpleasant state, a tension that needs to be reduced.

In order to reduce this state of tension, humans and animals seek out ways to fulfill these biological needs. We get a drink when we are thirsty. We eat when we are hungry. We turn up the thermostat when we are cold. He suggested that humans and animals will then repeat any behaviour that reduces these drives.

The **reduction of the drive** acts as a **reinforcement** for that behaviour. This reinforcement increases the likelihood that the same behaviour will occur again in the future when the same need arises. In order to survive in its environment, an organism must behave in ways that meet these survival needs. In a stimulus-response (S-relationship, when the stimulus and response are followed by a reduction in the need, it increases the likelihood of that behaviour occurring again.

GUTHRIE'S CONTIGUITY THEORY:

Guthrie suggested that contiguity (i.e. the sequential occurrence or proximity of stimulus and response, causing their association in the mind) alone, without reward, was sufficient for learning. For instance, hanging up the hat and coat must be associated with coming in the door (contiguous in time and space).

GUTHRIE'S VIEW OF REWARD:

Reward didn't strengthen the association but it was important. In Guthrie's view, reward changes the stimulus context. A context can be internal or external. When the stimulus is changed, further conditioning to another response is prevented. Reward must be immediate.

GUTHRIE'S VIEW OF PUNISHMENT: Punishment only works when the response it elicits is different than the response being punished. For instance, the act of punishing a child by spanking him for hitting others may not work according to Guthrie. The reason is that punishment elicits aggression which is compatible with hitting – so hitting will not be eliminated that way.

CLASSICAL CONDITIONING:

In this, an organism learns to associate two stimuli (e.g., a song and a pleasant event), such that one stimulus (the song) comes to elicit a response (feeling happy) that originally was elicited only by the other stimulus (the pleasant event). It was proposed by Pavlov.

FAMOUS EXPERIMENT OF PAVLOV: Ivan Pavlov while doing a study on dogs noticed that they tend to salivate to food. Gradually upon repeatedly pairing food with a neutral stimulus (one that did not lead to salivation) the presence of the neutral stimulus itself led to the dogs salivating (even when food was not presented with it).

BEFORE CONDITIONING:

Unconditioned Stimulus(UCS)(Food) -----
Unconditioned Response(UCR)
(Salivation)

Neutral Stimulus(Bell) -----

Orientation but no salivation

During conditioning:

UCS (Food) paired with
Neutral stimulus(bell) -----
UCR (Salivation)

AFTER CONDITIONING:

Conditioned Stimulus(CS) (Bell) -----
Conditioned Response (CR)(salivation)

PRINCIPLES OF CLASSICAL CONDITIONING:

ACQUISITION: Occurs when the Conditioned stimulus (bell) after repeated pairings with the unconditioned stimulus (food) comes to evoke the conditioned response (saliva).

EXTINCTION; Occurs when after conditioning, CS ceases to produce CR.

SPONTANEOUS RECOVERY: Refers to the reappearance of the CR after a rest period following extinction.

STIMULUS

GENERALISATION: Refers to the tendency to respond not only to the original CS but also to stimuli that are similar to it in the same manner.

STIMULUS

DISCRIMINATION: Refers to the ability to differentiate between a

conditioned stimulus and other similar stimuli that have not been paired with the UCS. So, the animal responds to the CS but not to similar stimuli.

APPLICATIONS:

TREATMENT OF DISORDERS:

Classical Conditioning techniques are useful in the treatment of phobias or anxiety problems.

COUNTER-CONDITIONING:

Involves pairing the stimulus that elicits fear (CS) with a stimulus (UCS) that elicits positive emotion (UCR).

AVERSION THERAPY: This helps to remove an undesired behaviour by associating it with unpleasant feelings.

SYSTEMATIC

DESENSITIZATION: It is a learning-based treatment for anxiety disorders. Has been developed by Joseph Wolpe. This therapy aims to remove the fear response of a phobia, and substitute a relaxation response to the conditional stimulus gradually using counter conditioning. There are three phases to the treatment:

First, the individual is taught a deep muscle relaxation technique and breathing exercises. Then the individual creates a fear hierarchy starting at stimuli that create the least anxiety (fear) and building up in stages to the most fear provoking images. The list is crucial as it provides a structure for the therapy. **Third**, the individual works their way up the

fear hierarchy, starting at the least unpleasant stimuli and practising their relaxation technique as they go. When they feel comfortable with this (they are no longer afraid) they move on to the next stage in the hierarchy. If the client becomes upset, they can return to an earlier stage and regain their relaxed state. The client repeatedly imagines (or is confronted by) this situation until it fails to evoke any anxiety at all, indicating that the therapy has been successful. This process is repeated while working through all of the situations in the anxiety hierarchy until the most anxiety-provoking.

FLOODING: In this, the individual is exposed to the phobic object until the fear fades away. This technique is based on the principle of extinction.

COGNITION IN CLASSICAL CONDITIONING:

Cognitive learning theorists believe that classical conditioning forms a CS-UCS link. In cognitive terminology, the link is an **expectancy** that the CS will be followed by the UCS. This expectancy model states that the most important factor in classical conditioning is not how often the CS and the UCS are paired, but how well the CS predicts (i.e., signals) the appearance of the UCS (Lovibond, 2006; Rescorla & Wagner, 1972).

OPERANT CONDITIONING:

Operant conditioning is a

type of learning in which behaviour is influenced by the consequences that follow it (Skinner, 1938, 1953). It was proposed by **B.F. Skinner**.

Skinner designed what has come to be known as a **Skinner box**, a special chamber used to study operant conditioning experimentally. A lever on one wall is positioned above a small cup. When the lever is depressed, a food pellet automatically drops into the cup. A hungry rat is put into the chamber and, as it moves about, it accidentally presses the lever. A food pellet clinks into the cup, and the rat eats it. The rat's behaviour is recorded on a cumulative recorder, which shows that the rat presses the bar more frequently over time.

The important principles of operant conditioning:

REINFORCEMENT- A response is strengthened by an outcome that follows it.

PUNISHMENT- Occurs when a response is weakened by outcomes that follow it.

Discriminative stimulus, a signal that

a particular response will now produce certain consequences. Discriminative stimuli set the occasion for operant responses. The sight of the teacher raising chalk to the blackboard was—in operant conditioning terms—a discriminative stimulus signalling it was time for the students to put their fingers in their ears. Discriminative stimuli guide much of our everyday behaviour. Food on our plate, classroom bells, the words people speak to us, and the sight of a friend's face are all discriminative stimuli that set the occasion for us to make certain responses.

Shaping (also called the method of successive approximations), involves reinforcing successive approximations toward a final response. Using a shaping procedure similar to the one just described, researchers took little time to get Mark playing on the monkey bars

Chaining, is used to develop a sequence (chain) of responses by reinforcing each response with the opportunity to perform the next response.

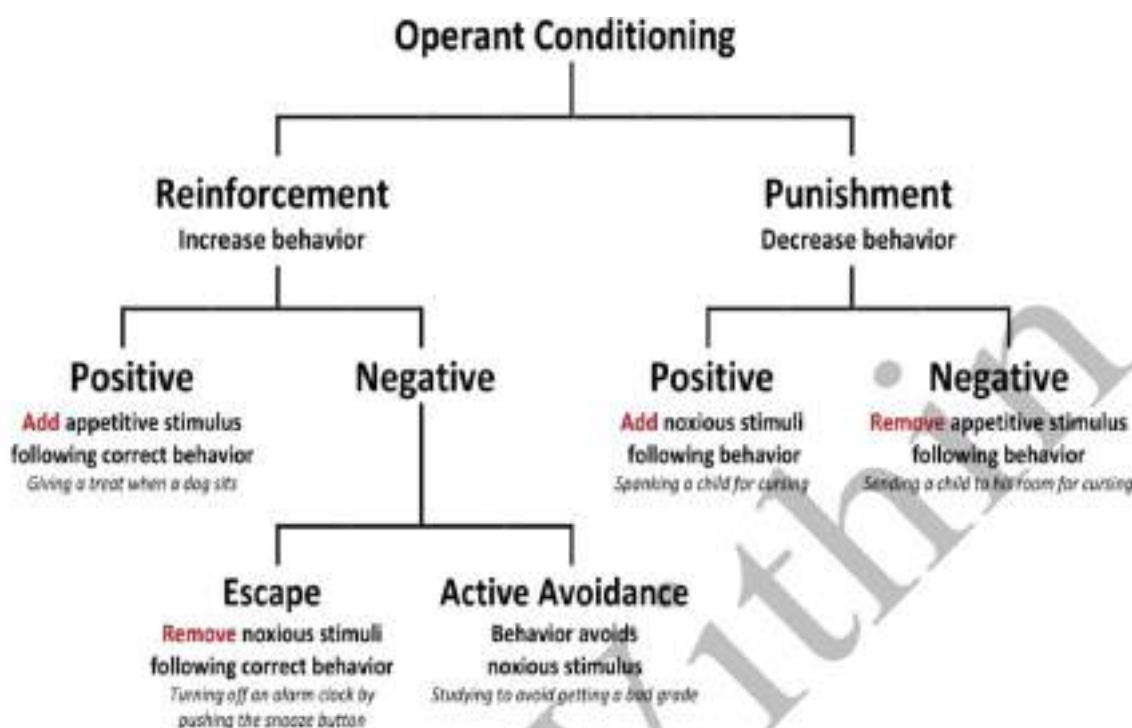


Fig: Components of Operant Conditioning

SCHEDULES OF REINFORCEMENT:

Schedules of reinforcement patterns of stimulus- response behaviours which have strong and predictable effects on learning, extinction, and performance (Ferster & Skinner, 1957; Soto et al., 2006). The most basic distinction is **between continuous and partial reinforcement.**

With **continuous reinforcement**, every response of a particular type is reinforced. Every press of the lever results in food pellets.

With **partial (intermittent) reinforcement**, only a portion of

the responses of a particular type are reinforced.

On a **fixed schedule**, reinforcement always occurs after a fixed number of responses or after a fixed time interval. On a **variable schedule**, the required number of responses or the time interval between them varies at random around an average.

The schedules of reinforcement have primarily been classified into four types:

On a **fixed-ratio (FR) schedule**, reinforcement is given after a fixed number of responses. For example, FR-3 means that reinforcement occurs after every third response, regardless of how long it takes for those responses to occur.

On a **variable-ratio (VR) schedule**, reinforcement is given after a variable number of responses, all centered around an average. A VR-3 schedule means that, on average, 3 responses are required for reinforcement. For example, for the first 12 responses, reinforcement might occur after responses 2, 3, 6, and 11.

On a **fixed-interval (FI) schedule**, the first response that occurs after a fixed time interval is reinforced.

On a **variable-interval (VI) schedule**, reinforcement is given for the first response that occurs after a variable time interval, centred around an average.

APPLICATIONS OF OPERANT CONDITIONING:

EDUCATION AND THE WORKPLACE: **Token economies**, in which desirable behaviours are quickly reinforced with tokens (e.g., points, gold

stars) that are later turned in for tangible rewards (e.g., prizes, recreation time), have been used to enhance academic and job performance.

- **MODIFYING PROBLEM BEHAVIOURS:** Operant conditioning research gave rise to a field called applied behaviour analysis, which combines a behavioural approach with the scientific method to solve individual and societal problems (Johnston et al., 2006; Kazdin, 1975).

Analysts design and implement a program to change behaviour, and they measure its effectiveness objectively by gathering data before and after the program is in place. The procedures (e.g., reinforcement, shaping) that are used to change behaviour are collectively known as behaviour modification. Applied behaviour analysis has reduced an incredible range of behaviour problems, from chronic hair pulling to drivers' failure to use seat belts (Byrd et al., 2002). It has improved students' academic performance and social skills.

	Reinforcement/ punishment is predictable	Reinforcement/ punishment is unpredictable
A given behaviour produces reinforcement (or punishment) every time it occurs.	<i>Continuous schedule</i>	
A given behaviour produces reinforcement (or punishment) only after a certain number of responses have been performed.	<i>Fixed ratio schedules</i>	<i>Variable ratio schedules</i>
A given behaviour produces reinforcement (or punishment) only after a certain amount of time has passed.	<i>Fixed interval schedules</i>	<i>Variable interval schedules</i>

COGNITION IN OPERANT CONDITIONING:

LEARNED HELPLESSNESS:

Has been proposed by Seligman in 1975. Refers to the tendency to fail to act to escape from an aversive situation because of the history of repeated failures in the past even where escape is possible. The organism learns to be helpless i.e. become passive.

COGNITIVE APPROACHES TO LEARNING

These approaches demonstrate the importance of cognition in learning. Some learning theorists challenged the S-R model, arguing that in between stimulus and response there is something else: the organism's (O) cognitive representation of the world. This came to be known as the S-O-R, or cognitive model of learning.

INSIGHT LEARNING

Köhler exposed chimpanzees to novel learning tasks and concluded that they were able to learn by insight, the sudden perception of a useful relationship that helps to solve a problem. One of his apes solved the problem of how to retrieve bananas that were dangling beyond reach. Köhler emphasized that the apes often spent time staring at the bananas and available tools, as if they were contemplating the problem, after which the solution suddenly appeared. Behaviourists however, have argued that insight actually represents the combining of previously learned responses (**Epstein et al., 1984**).

LATENT LEARNING

Another cognitive pioneer, American learning theorist Edward Tolman, studied spatial learning in rats. He developed a maze at the end of which was a box containing

food. He found that rat runs to an open circular table, continues across, and follows the only path available to a goal box containing food. After 12 trials, the rat easily negotiates the maze. Next, the maze is changed. The rat runs its usual route and reaches a deadend.

Tolman found that when confronted with this situation, rats returned to the table, briefly explored most of the 18 new paths for just a few inches, and then chose one. By far, the largest number—36 percent—chose the fourth path to the right of their original route, which took them closest to where the goal box had been. In short, the rats behaved as though they had a map.

Tolman (1948) argued that reinforcement theory could not explain this behaviour but that he could: The rats had developed a **cognitive map**, a mental representation of the spatial layout. The concept of cognitive maps supported Tolman's belief that learning does not merely "stamp in" stimulus response connections. Rather, learning provides knowledge, and based on their knowledge, organisms develop an expectancy, a cognitive representation, of "what leads to what."

OBSERVATIONAL LEARNING

Observational learning is that which occurs by observing the behaviour of a model. It was proposed by Albert Bandura keeping his famous **Bobo Doll**

experiment as the basis.

Bandura's social-cognitive theory, also known by its former name social-learning theory, emphasizes that people learn by observing the behaviour of models and acquiring the belief that they can produce behaviours to influence events in their lives (Bandura, 1969, 2006b).

The Modeling Process and Self-Efficacy

Bandura views modelling as a four-step process that includes several cognitive factors:

ATTENTION: First, we must pay attention to the model's behaviour.

RETENTION: Second, we must retain that information in memory so that it can be recalled when needed.

REPRODUCTION: Third, we must be physically capable of reproducing the model's behaviour or something similar to it.

MOTIVATION: Fourth, we must be motivated to display the behaviour.

According to Bandura, **self-efficacy**, which represents people's belief that they have the capability to perform behaviours that will produce a desired outcome, is a key motivational factor in observational learning.

VERBAL LEARNING

Verbal learning refers to the **acquisition and retention of**

verbal information. It is typically associated with the memorization and retention of lists of words, in order to describe basic elements of associative learning.

It is the process of actively memorizing new material using mental pictures, associations, and other activities.

Verbal learning was first studied by **Hermann Ebbinghaus**, who used lists of nonsense syllables to test recall. Later, Frederic **Bartlett** refuted Ebbinghaus' beliefs that nonsense syllables had an advantage over using words when testing verbal learning because the use of words adds the element of making a meaningful connection between words.

Ebbinghaus's book, *On Memory* (1885), showed how experimental methods could be used to measure the learning and retention of verbal items arranged in lists.

TYPES OF VERBAL LEARNING METHODS

SERIAL LEARNING:

Serial learning is memorizing a list of words in a particular order. Serial Learning involves having subjects learn a list of items according to the order in which the items appear in the list. Each item that appears provides feedback on the previous response and acts as a cue for saying the next response. In this, starting and ending points may possess some type of distinctiveness to set them apart from the rest of the list. Middle items have

more of a likelihood of being interfered with from earlier and later items, while the initial and terminal items do not have to face as much interference.

An important component of serial learning is the concept of **Serial Position Effect**-- lower recall error rates for the first (primacy effect) and last few (recency effect) items on the list, with higher recall error rates for items appearing in the middle of the list.

PAIRED-ASSOCIATE LEARNING

This refers learning to make a verbal response when a specific stimulus is presented (for example, learning foreign language vocabulary: stimulus = foreign word, response = English word). Paired Associate (PA) learning involves having 2 items (a Stimulus and Response item) paired as stimuli (e.g., BOAT-CHAIR).

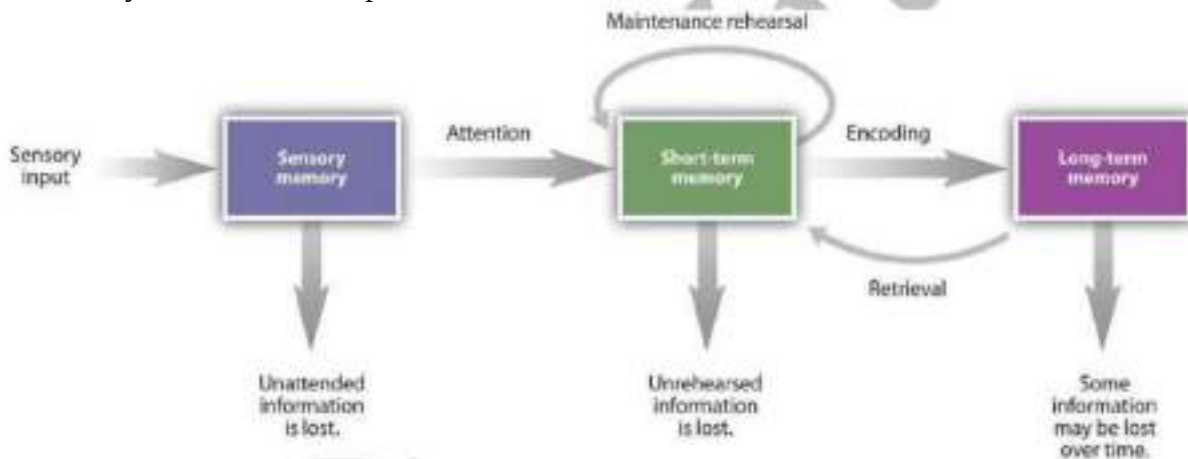
Each pair of items is presented in a 2-step procedure. In the first step, the stimulus is presented alone, and the subject tries to state the response that goes with it. In the second step, the stimulus and response items appear together. This gives the subject feedback and another opportunity to learn the association.

There are two kinds of trials. On a **study trial**, the subject sees each complete pair of items. Then there is a **test trial** in which the subject is presented the first item of each pair and is asked to state the item that goes with it. When the items are

committed to memory, the presentation of the first word (the stimulus word) should evoke the second word (the response word). So presenting BOAT should elicit response of CHAIR. Pre-existing associations between the stimulus and response items can either help or hinder the association process.

FREE RECALL:

Free Recall is a fairly unstructured method; one can recall words in any order they would like. This procedure



resembles serial learning. A person is given a list of words and immediately afterwards they try to recall them. But it can be in any order. It doesn't have to be the original order like in serial learning. It's like remembering a shopping list: **cereal, apples, milk, bread, cookies**. When we get to the store, it doesn't matter what order we remember them in as long as we remember them all.

Verbal learning is usually intentional but a person may learn some features of the words unintentionally or incidentally. In

this kind of learning, participants notice features such as whether two or more words rhyme, start with identical letters, have same vowels, etc. Thus, verbal learning is both intentional as well as incidental.

NEUROPHYSIOLOGY OF LEARNING

INFORMATION PROCESSING SYSTEM

The **brain's reticular activating system** filters incoming

information to focus on what's important, excluding anything trivial which leads to meaningful perception (Wolfe, 2001 as cited by Schunk, 2012). This process is adaptive and perceived importance is based on factors such as novelty, intensity, and movement. In terms of its application in the classroom, these aspects can be used to maintain student's attention and educators can find ways to build these factors into their lesson and student activities. Stangor, C. (2010). Memory duration

specifically, sensory inputs are processed in the **sensory memories** area of the brain and the ones that are retained move to the working memory where it primarily resides in the prefrontal cortex of the frontal lobe. The cortex and medial temporal lobe are involved in memory and information processing. The **LTM** is located in the frontal and temporal cortex. The parts of the brain responsible for long term memory differ based on the type of information whether it is declarative memory (facts, definitions, events), or procedural memory (procedures, strategies).

According to **Bandura** (1986), **cognitive neuroscience** supports the idea that learning occurs through observation. **Nonmotor procedures** (i.e. decoding words) involve the use of the visual cortex. Yet, repetition can change the neural structure of the **visual cortex**. These changes allow us to quickly recognize visual stimuli (i.e. words, numbers) without consciously having to process their meaning. Conversely, conscious processing is involved when it requires extended activity (i.e.

pausing to understand a passage).

NEURAL CONNECTIONS IN MEMORY NETWORKS

From a cognitive neuroscience perspective, learning involves forming and strengthening neural connections and networks. **Hebb's theory** elaborates on this by explaining the role of two cortical structures: **cell assemblies** and **phase sequences**. Cell assemblies are structures with cells in the cortex and subcortical centers. It is the neural counterpart of a simple association that is developed through repeated stimulations.

Phase sequence is a series of cell assemblies which form an organized pattern. Hebb believed that when a cell assembly was activated that it would in turn activate the neural and motor responses. For example, if we repeatedly see a friend's face then multiple cell assemblies are activated enabling us to meaningfully perceive who it is. Thus, Hebb's theory can be summarized best by this statement

"cells that fire together, wire together" as it describes associative

Table: Illustrates how long-term memory works in the brain

Type of information	How LTM occurs in the brain
1. Declarative	Sensory registers in the cerebral cortex (visual, auditory) receives input and transfers it to the hippocampus and nearby medial temporal lobe. Inputs are registered in the same format as they appear (auditory, visual stimulus). The hippocampus acts as a processor and conveyor of inputs. Stronger neural connections develop based on multiple activations of memories in the neural networks.
2. Procedural	Becomes automated as certain procedures can be done without conscious awareness (i.e. typing). Initial procedural learning occurs from the prefrontal cortex, the parietal lobe, and cerebellum which ensure that we consciously attend to the steps in the correct sequence. Practice helps these areas to become less involved and other areas such as the motor cortex become more involved in their place.

learning. Hebb's theory is still relevant today as it is linked with recent research on how learning occurs and memories are formed. For instance, several studies have shown that enriched environments are key to leading to improved learning outcomes (van Praag et al.,2000).

Generally, people are born with a large quantity of neural connections and our experiences work on this system so that our connections are selected or ignored, strengthened or lost. According to the National Research Council (2000), the process of forming and strengthening synaptic connections, essentially learning, changes the physical structure of the brain and alters its organization. Although we tend to think that the brain determines learning, it is a reciprocal

relationship since The **"neuroplasticity"** of the brain or its capacity to change its structure and function is a result of An individual's experience.

Recent research indicates Tha memory is not formed at the time initial learning occurs. Memory forms as a continuous process called **consolidation** where stabilizing and strengthening of neural connections occur in the **hippocampus**. The brain An plays an active role in storing d retrieving information. **Instruction** also plays an important role In helping impose a desired structure on learning.

Table : Illustrates instructional applications to help improve memory

Factors that improve consolidation(memory)	Instructional applications
Organization	▪ Creating timelines, labelling, use of visual organizers
Rehearsal	Practicing for a play, repeating vocabulary words, using lists, essays
Elaboration	Surveying students to gauge their interests, relating concepts, encouraging students to elaborate on their questions/answers

LANGUAGE LEARNING

The brain's **cerebral cortex** is responsible for reading and the posterior cortical association areas of the left hemisphere are important for understanding language and normal reading. There are several brain structures involved in the following

aspects of reading. For educators it is important to know how the brain develops since developmental changes need to be considered in planning instruction to help students learn best. The table illustrates the brain structures associated with reading

Brain Structures	Aspects of Reading
Primary visual area	Orthographic- refers to letters or characters processing
Superior temporal lobe	Phonological- phonemes or syllables
Broca's area in the frontal lobe and medial temporal lobe	Semantic- meanings
▪ Broca's area in the frontal lobe	▪ Syntactic- sentence structure processing

BRAIN DEVELOPMENT

Educational implications for teaching and learning vary depending on the level of brain development. Factors influencing brain development are illustrated below.

The table below illustrates how factors influence brain development

Factors influencing brain development	How it occurs
▪ Genetics	Human brains have a similar genetic structure but they differ in size and structure. Genetic instructions determine the size, structure, and neural connectivity of the brain.
Environmental stimulation	Brain development requires experiences from the environment. These experiences develop neural circuitry that can receive and process stimuli and experiences.
▪ Nutrition	Lack of good nutrition can have major effects on brain development. Crucial period is between the 4 th and 7 th month of gestation when most brain cells are produced.
▪ Steroids	Hormones can affect brain development. Excess stress hormones can lead to the death of neurons.
Teratogens	Foreign substances (i.e. alcohol, viruses) can cause abnormalities in the developing embryo or fetus. Teratogens can have effects on the development and interconnections of neurons and glial cells and can cause birth defects.

CRITICAL DEVELOPMENTAL PHASES

At birth, the human brain has over a million connections which represent about 60% of the peak number of synapses that will develop over a lifetime. **Brain connections** that are not used or needed disappear.

KEY HIGHLIGHTS INCLUDE:

2 years old- a child will have as many synapses as an adult

3 years old- a child will have billions more than an adult

5 years old- a child's brain has acquired a language and developed sensory motor skills and other competencies.

Teenage years- major structural changes occur in the brain. The frontal lobes which handle

reasoning and problem solving mature and the parietal lobes increase in size. The prefrontal cortex which controls judgments and impulses matures slowly. There are also changes in neurotransmitters especially dopamine that can leave the brain more sensitive to the effects of drugs and alcohol. There is a thickening of the brain cells and massive reorganization of synapses which makes it a key time for learning.

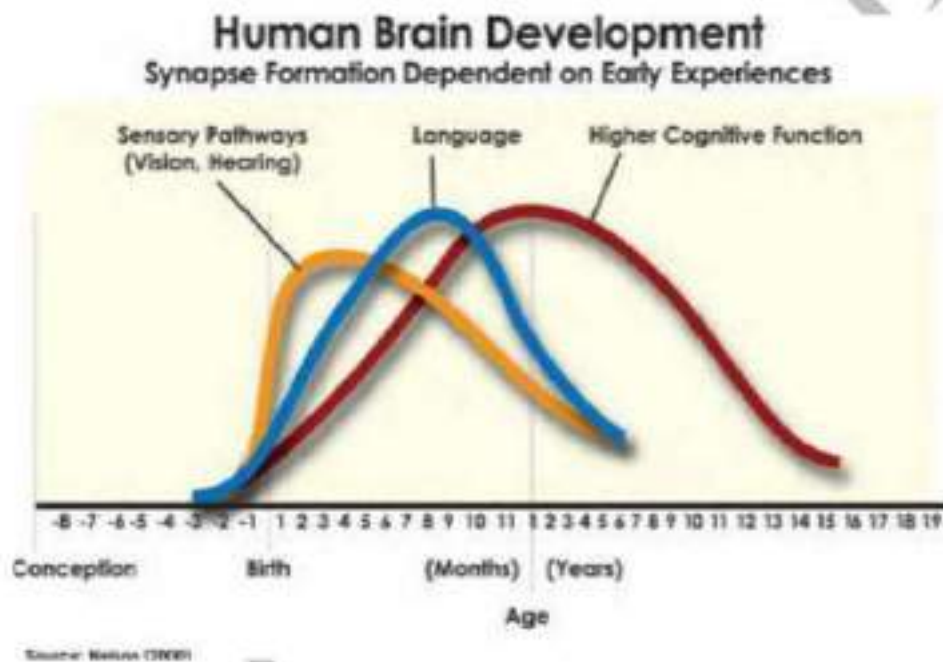


Table: Four aspects of crucial brain development

Type of Brain Development	Why it is important
Sensory motor development	Systems associated with vision, hearing and motor movements develop extensively through experiences during the first 2 years of life.
Auditory development	By age 6 months children can distinguish between most sounds in their environment. In the first two years children's auditory systems mature in terms of range of sounds heard and ability to discriminate among sounds. Problems in auditory development can lead to problems Learning language as it is dependent on hearing the speech of others.

Vision	Develops in the first year of life especially after 4 months. Proper visual development requires a visually rich environment where infants can explore objects and movements.
Language	Children who are normally developing show extensive bilateral and anterior cortical activation and left-sided activation in language and speech areas. Reading development is based on anterior activation on both sides of the brain. A critical period of language development is from birth to age 5. During this time, a child's brain develops most of their language capabilities. There is a rapid increase in vocabulary between age 19-31 months. Development of language is enhanced when children are in language-rich environments where parents and others talk with children.

INSTRUCTIONAL APPLICATIONS

Instruction can help facilitate language development by stressing perceptual, motor, and language functions. Teachers can work with students of all ages to help develop their language skills by coordinating the components of language- seeing, hearing, thinking and speaking. Some examples include:

Helping students learn phonemes using cards and practicing writing/reading it in sentences

Using picture and word cards for names and spellings

Using magnetic number lines students put the magnetic bar on the appropriate number line. Next converting the numbers to a common denominator to place in the correct order.

For learning historical events, have students engage in role-playing activities where they read documents as historical characters

Using case studies

MEMORY

Memory refers to the processes that allow us to record, store, and later retrieve experiences and information.

Encoding refers to getting information into the system by translating it into a neural code that

your brain processes. This processing can be of two types:

Effortful processing, encoding that is initiated intentionally and requires conscious attention (Adam et al., 2005; Hasher & Zacks, 1979). When we rehearse information, make lists, and take notes, we are engaging in effortful processing.

Automatic processing on the other hand, refers to encoding that occurs without intention and requires minimal attention. Information about the frequency, spatial location, and sequence of events is often encoded automatically (Jimenez

& Mendez, 2001). According to the concept of levels of processing, the more deeply we process information, the better we will remember it (Craik & Lockhart, 1972).

REHEARSAL AND EXPOSURE:

Rehearsal goes beyond mere exposure. When we rehearse information, we are thinking about it. This rehearsal can be of varied types: One approach, called **maintenance rehearsal**, involves simple, rote repetition, and some students rely on it to learn their course material. Maintenance rehearsal keeps information active in working memory, as when someone tells us a phone number and we repeat it to ourselves as we place the call.

The techniques actors use are examples of **elaborative rehearsal**, which involves focusing on the meaning of information or expanding (i.e., elaborating) on it in some way. According to **Craik and Lockhart** (1972), elaborative rehearsal involves deeper processing than maintenance rehearsal, and experiments show that it is more effective in transferring information into long-term memory (Benjamin & Bjork, 2000).

DUAL CODING THEORY:

Allan Paivio (1969, 2006) has proposed that information is stored in long-term memory in two forms: verbal codes and visual codes. According to his dual coding theory, encoding information using both verbal and visual codes enhances memory because the odds improve

that at least one of the codes will be available later to support recall. Memory experts recommend using imagery to dual-code information. The ancient Greeks developed the **method of loci** (loci are Latin for “places”), a memory aid that associates information with mental images of physical locations.

Mnemonic Devices

The term **mnemonics** refers to the art of improving memory, and a mnemonic device is a memory aid. Mnemonic devices reorganize information into more meaningful units and provide extra cues to help retrieve information from long-term memory. A few examples of

mnemonic devices are chunking, method of loci and soon.

ASSOCIATIVE NETWORKS

One group of theories proposes that long-term memory can be represented as an **associative network**, a massive network of associated ideas and concepts (Collins & Loftus, 1975, Hekkanen & McEvoy, 2005). The term **priming**

STORAGE involves retaining information over time. Memory is enhanced by forming associations between new information and other items already in memory.

refers to the activation of one concept (or one unit of information) by another. For instance, listening to the word “fire engine” primes the node for “red,” making it more likely that our memory for this colour will be accessed (Chwilla & Kolk, 2002).

In **neural network (connectionist) models**, each memory is represented by a unique pattern of interconnected and simultaneously activated nodes.

As we look across the entire neural network, various nodes distributed throughout the network fire in parallel at each instant and simultaneously spread their activation to

other nodes. In this manner, certain nodes prime other nodes, and concepts and information are retrieved. For this reason, neural network (connectionist) models are often called parallel distributed processing (PDP) models (Rumelhart et al., 1986).

RETRIEVAL refers to processes that access stored information from the long term memory. A **retrieval cue** is a stimulus, whether internal or external, that activates information stored in long-term memory. Distinctive events stand a greater chance of etching long-term memories that seem vivid and clear.

EMOTION AND AROUSAL:

Researchers have found that emotional arousal enhances **autobiographical memories**, recollections of personally experienced events

that make up the stories of our lives (Schaefer & Philippot, 2005). When people are asked to record their unique daily experiences in a diary and rate the emotional pleasantness and intensity of each event (e.g., say, on a 7-point scale ranging from “extremely unpleasant” to “extremely pleasant”), it typically is the more intense events that they recall most vividly when tested days, months, or years later.

THREE STAGE MODEL OF MEMORY

Developed by **Richard Atkinson and Richard Shiffrin** (1968) and subsequently modified, it depicts memory as having three major components: **sensory memory, working (short-term) memory, and long-term memory**.

SENSORY MEMORY

Sensory memory briefly holds incoming sensory information. It comprises different subsystems, called sensory registers, which are the initial information processors. Our **visual sensory register** is called the **iconic store**, and in 1960 George Sperling conducted a classic

experiment to assess how long it holds information.

SHORT-TERM MEMORY

It is a memory store that temporarily holds a limited amount of information. Once information leaves sensory memory, it must be represented by some type of code if it is to be retained in short-term memory. **Memory codes** are mental representations of some type of information or stimulus, and they can take various forms. We may try to form mental images (visual codes), code something by sound (phonological codes), or focus on the meaning of a stimulus (semantic codes). For physical actions, such as learning sports or playing musical instruments, we code patterns of movement (motor codes).

Short-term memory can hold only a limited amount of information at a time. The limit on short-term storage capacity concerns the number of meaningful units that can be recalled, and the 20 letters have been combined into 6 meaningful units (words). Combining individual items into larger units of meaning is called **chunking**, which aids recall.

Working memory is a limited-capacity system that temporarily stores and processes information (Baddeley, 2007). Has primarily three components:

One component, the **phonological loop**, briefly stores mental representations of sounds. The phonological loop is active when you listen to a spoken word or when you sound out a word to yourself as you read. A second component, the **visuospatial sketchpad**, briefly stores visual and spatial information, as occurs when you form a mental image of someone's face or of the spatial layout of your bedroom. A third component, called the **episodic buffer** provides a temporary storage space where information from long-term memory and from the phonological and/or visuospatial

subsystems can be integrated, manipulated, and made available for conscious awareness. The episodic buffer also comes into play when you chunk information.

The **fourth component** of working memory, called the **central executive**, directs the overall action.

When solving arithmetic problems, for example, the central executive doesn't store the numbers or rules of addition. Instead, it plans and controls the sequence of actions that need to be performed, divides and allocates attention to the other subsystems, and integrates information within the episodic buffer. It also may monitor the progress as interim steps are completed (DeStefano & LeFevre, 2004).

LONG-TERM MEMORY: Long-term memory is our vast library of more durable stored memories. It can store limitless information and it becomes easier to add new information - encoding based on meaning. Forming and using long-term memories is an active, creative, highly personal process

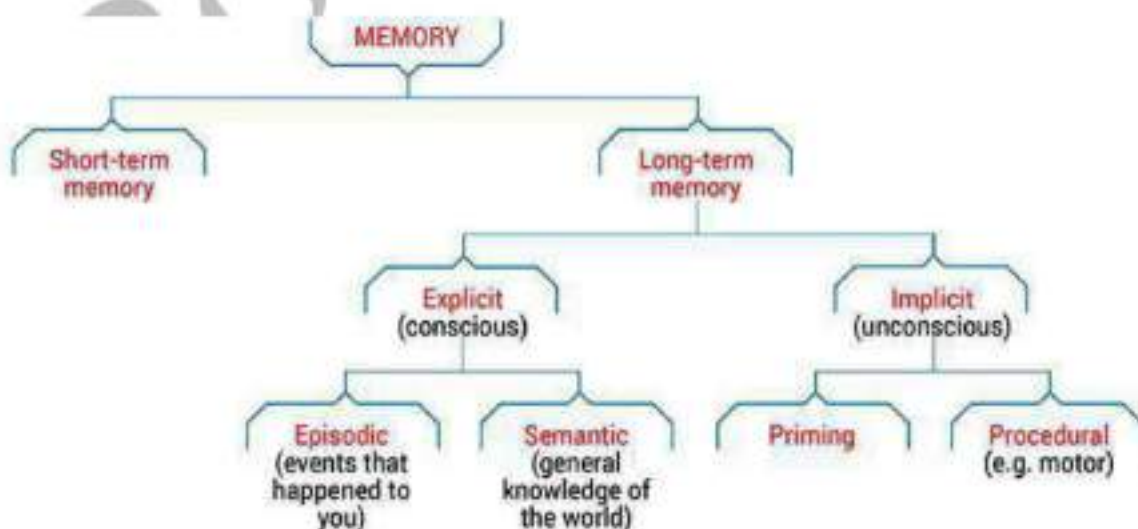


Fig: Types of Long- Term Memory (Source: Queensland Brain Institute)

TYPES OF LONG-TERM MEMORY:

DECLARATIVE AND PROCEDURAL MEMORY

Declarative memory involves factual knowledge Includes two sub-categories.

Episodic memory is our store of knowledge concerning personal experiences: when, where, and what happened in the episodes of our lives. Your recollections of childhood friends, a favourite movie, and what you ate this morning represent episodic memories.

Semantic memory represents general factual knowledge about the world and language, including memory for words and concepts. You know that Mount Everest is the world's tallest peak and that $e=mc^2$. Episodic and

semantic memories are called declarative because to demonstrate our knowledge, we typically have to declare it: We tell other people what we know.

Procedural (non-declarative) memory is reflected in skills and actions (Cohen et al., 2005). One component of procedural memory consists of skills that are expressed by doing things in particular situations, such as typing or riding a bicycle.

Explicit memory involves conscious or intentional memory retrieval, as when you consciously recognize or recall something. Recognition requires us to decide whether a stimulus is familiar,

Implicit memory occurs when memory influences our behavior without conscious awareness.

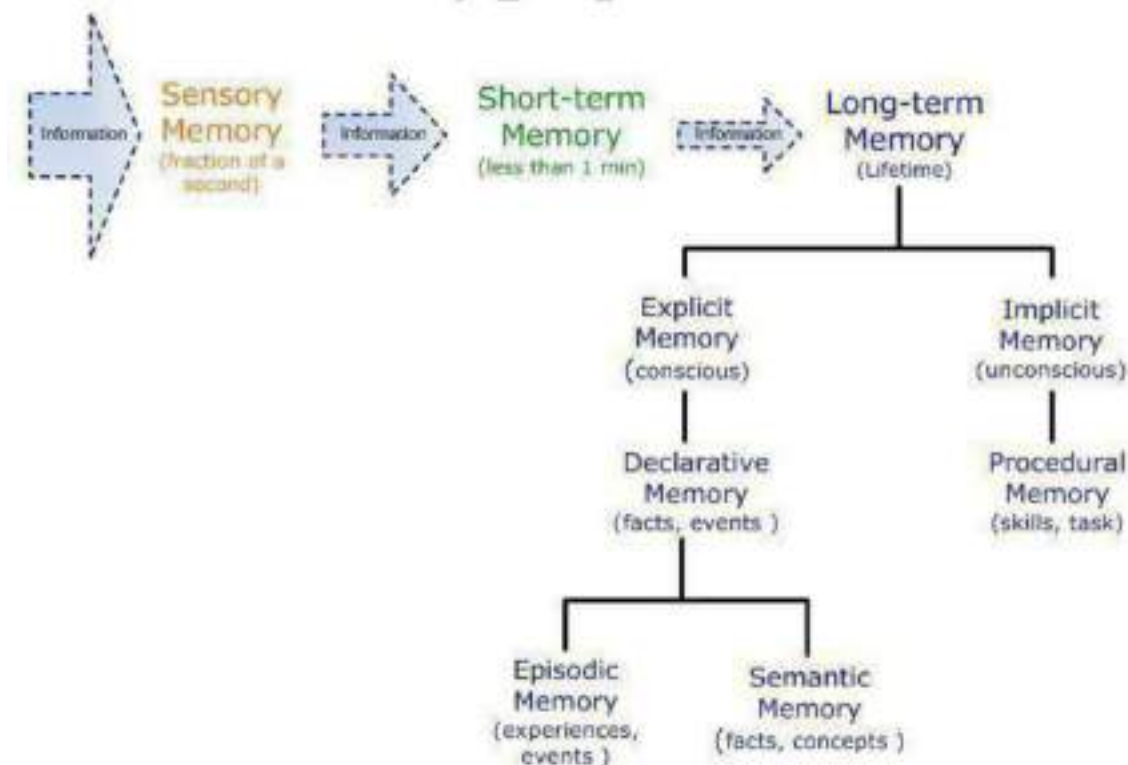


Fig: Various types of memory (Source: Peak performance center)

MEASUREMENT OF MEMORY

TIP OF TONGUE: Feeling that memory is available but not retrievable

RECALL: To supply or reproduce memorized information with a minimum of external cues.

RECOGNITION: An ability to correctly identify previously learned information

RELEARNING: Learning again something that was previously learned. Used to measure memory of prior learning – measured by saving score (amount of time saved in relearning).

PRIMING: Facilitating the retrieval of an implicit memory by using cues to activate hidden memories

CONSOLIDATION: Process by which relatively permanent memories consolidate (damaged for new memories) in brain development. Hippocampus acts as switching station (inability b/w STM to & LTM).

FORGETTING:

THEORIES OF FORGETTING:

ENCODING FAILURE

Many memory failures result not from forgetting information that we once knew, but from failing to encode the information into long-term memory in the first place. Even when we notice information, we may fail to encode it deeply because we turn our attention to something else.

DECAY OF THE MEMORY TRACE

Decay theory proposed that with time and disuse the long-term physical memory trace in the nervous system fades away. Decay theory soon fell into disfavor because scientists could not locate physical memory traces nor measure physical decay.

Unfortunately, decay theory's prediction—the longer the time interval of disuse between learning and recall, the less should be recalled—has been found to be problematic. Some professional actors display perfect memory for words they had last spoken on stage 2 years earlier—this despite having moved on to new acting roles and scripts (Noice & Noice, 2002). Moreover, when research participants learn a list of words or a set of visual patterns and are retested at two different times, they sometimes recall material during the second testing that they could not remember during the first. This phenomenon, called reminiscence, seems inconsistent with the concept that a memory trace decays over time (Greene, 1992).

THEORY OF INTERFERENCE

According to interference theory, we forget information because other items in long-term memory impair our ability to retrieve it (Postman & Underwood, 1973; Feredoes et al., 2006)

– **Proactive interference** occurs when material learned in the past interferes with recall of

newer material.

- **Retroactive interference** occurs when newly acquired information interferes with the ability to recall information learned at an earlier time.

Proposed **reasons for interference** to occur: It takes time for the brain to convert short-term memories into long-term memories, and some researchers propose that when new information is entered into the system, it can disrupt (i.e., retroactively interfere with) the conversion of older information into long-term memories (Wixted, 2005). Others believe that on long-term memories are formed, retroactive and proactive interference are caused by **competition among retrieval cues** (Anderson & Neely, 1996).

RETRIEVAL FAILURE THEORY:

Retrieval failure occurs when the information is present in the long-term memory but cannot be accessed due to the absence of adequate retrieval cues.

Retrieval cues can be:

External: context dependent/ in the environment

Internal: state dependent/ inside of us

Context-dependent memory proposes that it is typically easier to remember something in

the same environment in which it was originally encoded.

State-dependent memory proposes that our ability to retrieve information is greater when our internal state at the time of retrieval matches our original state during learning

MOTIVATED FORGETTING:

Psychodynamic and other psychologists propose that, at times, people are consciously or unconsciously motivated to forget. They have proposed the concept of **repression** which refers to a motivational process that protects us by blocking the conscious recall of anxiety-arousing memories.

The concept of repression, however, is controversial. Some evidence supports it, and other evidence does not (Karon, 2002). People certainly do forget unpleasant events— even traumatic ones—yet they also forget very pleasant events. Overall, it has been difficult to demonstrate experimentally that a special process akin to repression is the cause of memory loss for anxiety-arousing events (Holmes, 1990).

FORGETTING TO DO THINGS:

Prospective Memory concerns remembering to perform an activity in the future.

Amnesia: The term amnesia commonly refers to memory loss due to special conditions, such as brain injury, illness, or psychological trauma. Amnesia takes several forms.

Retrograde amnesia represents memory loss for events that took place sometime in life before the onset of amnesia.

Anterograde amnesia refers to memory loss for events that occur after the initial onset of amnesia.

Dementia: Dementia refers to impaired memory and other cognitive deficits that accompany brain degeneration and interfere with normal functioning.

Alzheimer's Disease: Alzheimer's disease (AD) is a progressive brain disorder that is the most common cause of dementia among adults over the age of 65.

CONTENTS

Theoretical perspectives on thought processes: Associationism, Gestalt, Information processing, Feature integration model

Concept formation: Rules, Types, and Strategies; Role of concepts in thinking Types of

Reasoning Language and thought

Problem solving: Type, Strategies, and Obstacles

Decision-making: Types and models

Metacognition: Metacognitive knowledge and Metacognitive regulation

Intelligence: Spearman; Thurstone; Jensen; Cattell; Gardner; Stenberg; Goleman; Das, Kar & Parrila

Creativity: Torrance, Getzels & Jackson, Guilford, Wallach

& Kogan Relationship between Intelligence and Creativity

UNIT 6

Unit 6: Thinking, Intelligence and Creativity

THEORETICAL PERSPECTIVES

ASSOCIATIONISM

Our mental processing is based on our senses, conceptions and perceptions. These elements are stored as varied associations that help us relate and understand new things generally through what we have already experienced, or gained knowledge about, through other mediums. Aristotle (350 B.C.E) the Greek scientist, who studied in Plato's academy in Athens, came up with four laws of associationism

Law of Similarity: When we are reminded of, or recall one thing from the past experiences, similar instances will also be recalled.

Law of Contrast: When we are reminded of, or recall things from the past experiences, instances exactly opposite to them, will also be recalled.

Law of Contiguity: When we are reminded of, or recall things from the past experiences, we also recall actions or other associations with that object that were originally experienced along with that entity.

Law of Frequency: If a series of events keep occurring frequently, the probability of recalling similar associated instances will increase.

GESTALT

The term “Gestalt,” comes from a German word that roughly means **pattern or form**. The main tenet of the Gestalt theory is that the whole is greater than the sum of its parts; learning is more than just invoking mechanical responses from learners. As with other learning theories, the Gestalt theory has **laws of organization** by which it must function. These organizational laws already exist in the make-up of the human mind and how perceptions are structured. Gestalt theorists propose that the experiences and perceptions of learners have a significant impact on the way that they learn. One aspect of Gestalt is **phenomenology**, which is the study of how people organize learning by looking at their lived experiences and consciousness. Learning happens best when the instruction is related to their real life experiences. The human brain has the ability to make a map of the stimuli caused by these life experiences. This process of mapping is called **“isomorphism.”**

Whenever the brain sees only part of a picture, the brain automatically attempts to create a complete picture. This is the first organizational law, called the

“factor of closure,” and it does not only apply to images, but it also applies to thoughts, feelings and sounds.

Based upon Gestalt theory, the human brain maps elements of learning that are presented close to each other as a whole, instead of separate parts. This organizational law is called the **“factor of proximity,”** and is usually seen in learning areas such as reading and music, where letters and words or musical notes make no sense when standing alone, but become a whole story or song when mapped together by the human brain.

INFORMATION PROCESSING

The Information Processing Model is a framework used by cognitive psychologists to explain and describe mental processes. The model likens the thinking process to how a **computer** works.

Just like a computer, the human mind takes in information, organizes and stores it to be retrieved at a later time.

Just as the computer has an input device, a processing unit, a storage unit, and an output device, so does the human mind have equivalent structures.

In a computer, information is entered by means of input devices like a keyboard or scanner. In the human mind, the input device is called the **Sensory Register**, composed of sensory organs like the eyes and the ears through which we receive information about our surroundings.

As information is received by a computer, it is processed in the Central Processing Unit, which is equivalent to the Working Memory or **Short-Term Memory**. In the human mind, this is where information is temporarily held so that it may be used, discarded, or transferred into **long-term memory**.

In a computer, information is stored in a hard disk, which is equivalent to the **long-term memory**. This is where we keep information that is not currently being used. Information stored in the Long-Term Memory may be kept for an indefinite period of time.

When a computer processes information, it displays the results by means of an output device like a computer screen or a printout. In humans, the result of information processing is exhibited through **behaviour or actions** - a facial expression, a reply to a question, or body movement.

FEATURE INTEGRATION THEORY

Feature Integration Theory is a perceptual and attentional theory that explains how an individual combines pieces of observable information about an object in order to form a complete perception of the object. This theory was developed by **Gelade and Treisman** and focuses on the visual search component of stimuli perception. During a visual search a person uses cues like colors, shape, and size to distinguish objects from one another. There are two stages that comprise this theory.

The first is the **pre-attention stage** in which the individual focuses on one distinguishing attribute of the object. The pre-attention phase is an automatic process which happens unconsciously. The second stage is focused attention in which an individual take all of the observed features and combines them to make a complete perception.

This **second stage process** occurs if the object doesn't stand out immediately. For example, if you were looking for a penny in a handful of quarters it would be easy to spot and only require pre-attention focus. If you were looking for a nickel in a handful of quarters it would be more difficult to spot because of the same colors and similar size- focused

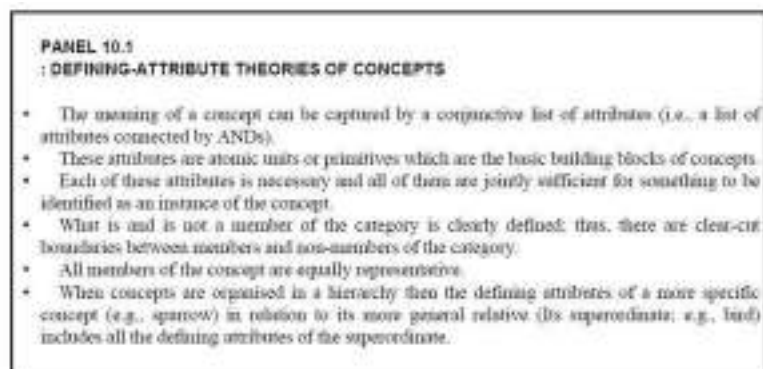
attention would be required to spot the nickel.

CONCEPT

A concept is a **mental representation** that is used for a variety of cognitive functions, including memory, reasoning, and using and understanding language (Solomon et al., 1999). Thus, when you think about cats, you are drawing on your concept, or mental representation, of cats, which includes information about what cats are, what they usually look like, how they behave, and so on. Concepts are **mental groupings** of similar objects, events, or people. Concepts enable us to organize complex phenomena into cognitive categories that are easier to understand and remember. By far the most commonly studied function of concepts is **categorization**, which is the process by which things are placed into groups called **categories**. For example, when you see vehicles in the street you can place them into categories such as cars, SUVs, Chevrolets, Fords, American cars, and foreign cars.

THE DEFINING-ATTRIBUTE VIEW

This view, elaborated by the logician Gottlob Frege (1952), maintains that a concept can be characterised by a set of *defining attributes*.



THE PROTOTYPE VIEW

In some theories, the prototype is a set of characteristic attributes; there are no defining attributes but rather only characteristic attributes of differential importance within the concept. An object is a member of the concept if there is a good match between its attributes and those of the prototype. In other prototype theories, the prototype is captured by a specific instance of the category, the best example of the concept (e.g., Rosch, 1978).

PANEL 10.3:

PROTOTYPE THEORY OF CONCEPTS

- Concepts have a prototype structure; the prototype is either a collection of characteristic attributes or the best example (or examples) of the concept.
- There is no delimiting set of necessary and sufficient attributes for determining category membership; there may be necessary attributes, but they are not jointly sufficient; indeed membership often depends on the object possessing some set of characteristic, non-necessary attributes that are considered more typical or representative of the category than others.
- Category boundaries are fuzzy or unclear; what is and is not a member of the category is ill-defined; so some members of the category may slip into other categories (e.g., tomatoes as fruit or vegetables).
- Instances of a concept can be ranged in terms of their typicality; that is, there is a typicality gradient which characterises the differential typicality of examples of the concept.
- Category membership is determined by the similarity of an object's attributes to the category's prototype.

THE EXEMPLAR-BASED VIEW**PANEL 10.4:****THE EXEMPLAR-BASED VIEW OF CONCEPTS**

- Categories are made up of a collection of instances or exemplars rather than any abstract description of these instances (e.g., a prototype summary description).
- instances are grouped relative to one another by some similarity metric.
- Categorisation and other phenomena are explained by a mechanism that retrieves instances from memory given a particular cue.
- When exact matches are not found in memory the nearest neighbour to the cue is usually retrieved.

LEARNING NEW CONCEPTS

There is an experiment containing cards, each of which contains one of three shapes, one of three colors, and one of three borders. Participants were given an instances (such as “black circles”), without the concept itself. They then had to pick additional cards which they thought might be instances of the same concept. They received feedback on their guesses.

Strategy for learning new concepts

LEARNING STRATEGIES

They might use a conscious, strategic, or unconscious strategy. When the task is simple (few objects and categories), conscious learning is the way to go. Learning a language is more complex, so an unconscious approach would be better in that situation.

Simultaneous scanning: testing multiple hypotheses at the same time (such as “white circles”). This has heavy demands on working memory.

Successive scanning: testing one hypothesis at a time (such as “black figures”). This is inefficient but has low demands on working memory.

Conservative focusing: choosing cards that vary in only one respect from a positive instance (known as the “focus card”). This is both efficient and easy.

REASONING

Reasoning is the process of **drawing conclusions from principles and from information**. In reasoning, we move from what is already known to infer a new conclusion or to evaluate a proposed conclusion. Reasoning is often

divided into two types: deductive and inductive reasoning

DEDUCTIVE REASONING

Deductive reasoning is the process of reasoning from one or more general statements regarding what is known to reach a logically certain conclusion. It often involves reasoning from one or more general statements regarding what is known to a **specific application** of the general statement. Deductive reasoning is based on logical propositions. A **proposition** is basically an assertion,

Examples are “Cognitive psychology

students are brilliant,”

CONDITIONAL REASONING

One of the primary types of deductive reasoning is conditional reasoning, in which the reasoner must draw a conclusion based on an **if-then proposition**. The conditional if-then proposition states that if antecedent condition p is met, then consequent event q follows. For example, “If students study hard, then they score high on their exams.”

Laird, 2000). The key feature distinguishing inductive from deductive reasoning is that, in inductive reasoning,

Table 12.2 Conditional Reasoning: Deductively Valid Inferences and Deductive Fallacies

Two kinds of conditional propositions lead to valid deductions, and two others lead to deductive fallacies; p is called the antecedent; q is called the consequent. $\rightarrow$ stands for then; and $\therefore$ stands for therefore.

Type of Argument		Conditional Proposition	Existing Condition	Inference
Deductively valid inferences	Modus ponens—affirming the antecedent	$p \rightarrow q$ If you are a mother, then you have a child.	p You are a mother.	$\therefore q$ Therefore, you have a child.
	Modus tollens—denying the consequent	$p \rightarrow q$ If you are a mother, then you have a child.	$\neg q$ You do not have a child.	$\therefore \neg p$ Therefore, you are not a mother.
Deductive fallacies	Denying the antecedent	$p \rightarrow q$ If you are a mother, then you have a child.	$\neg p$ You are not a mother.	$\therefore \neg q$ Therefore, you do not have a child.
	Affirming the consequent	$p \rightarrow q$ If you are a mother, then you have a child.	q You have a child.	$\therefore p$ Therefore, you are a mother.

SYLLOGISTIC REASONING: CATEGORICAL SYLLOGISMS

In addition to conditional reasoning, the other key type of deductive reasoning is syllogistic reasoning, which is based on the use of syllogisms. Syllogisms are deductive arguments that involve **drawing conclusions from two premises** (Maxwell, 2005; Rips, 1994, 1999). All syllogisms comprise a major premise, a minor premise, and a conclusion. In the case of the categorical syllogism, the premises state something about the category memberships of the terms.

An example of a categorical syllogism would be as follows:

- P1: All cognitive psychologists are pianists.
- P2: All pianists are athletes.
- C: Therefore, all cognitive psychologists are athletes.

INDUCTIVE REASONING

Inductive reasoning is the process of reasoning from specific facts or observations to reach a likely conclusion that may explain the facts. The inductive reasoner then may use that **probable conclusion to attempt to predict future specific instances** (Johnson-

we never can reach a logically certain conclusion. We only can reach a particularly well-founded or probable conclusion.

LANGUAGE AND THOUGHT

JEAN PIAGET'S THEORY OF RELATIONSHIP BETWEEN LANGUAGE AND THOUGHT

Jean Piaget's theory claims that "Thought determines Language".

He says that because a child has some thought, they learn the language. Imitating other's is a proof that a child is thinking. Thinking is the basic need and necessity to learn a language.

BENJAMIN LEE WHORF'S THEORY OF RELATIONSHIP BETWEEN LANGUAGE AND THOUGHT

Benjamin Lee Whorf's theory claims that "Language determines Thought".

In this theory, he claims that language determines the contents of thought which is also known

as **linguistic relativity hypothesis**.

For example: In India, we have names for every relationship such as kaka, chachaa, mausi etc but in western languages, they do not have the same. They mostly have uncle and aunt which does not determine any specific relationship. Due to this difference, Indian children may have better clarity in relationships than western children. In short, he wants to say that the words we use to express ourselves in our mother tongue or any language determine how we think.

LEV VYGOTSKY'S THEORY OF RELATIONSHIP BETWEEN LANGUAGE AND THOUGHT

Lev Vygotsky's theory claims that "Language and Thought are Independent processes until the child is 2 years."

He argued that thoughts and language are interdependent and develop in a child separately until about two years of age. Crying when uncomfortable is simply an automatic reflex than thought or language based. Only around 2 years of age a child is able to express thought verbally and her/his speech reflects rationality.

NOAM CHOMSKY'S THEORY OF RELATIONSHIP BETWEEN LANGUAGE AND THOUGHT

Noam Chomsky's theory claims that "Language and Thought are Innate Proposition."

Also known as Nativist Theory. Noam Chomsky claims that we actually do have a device in our brain. The device is **Language Acquisition Device (LAD)**. In the argument of Language Vs Thought, Chomsky suggests that

Language in an innate proposition. We are all born with the LAD which has a set of language learning tools such as

Universal Grammar rules based on which a child learns a language. This is the Natives theory. He mentions that, there is a critical period in every child's life when learning must occur successfully to learn a language. If it doesn't then after the **critical period** it becomes very difficult for the child to learn a language. That is why adults find more difficulty learning a new language than a child.

WHAT IS METACOGNITION?

In the late 1970's, John Flavell originally coined the word "metacognition." He defined the word as "**cognition about cognitive phenomenon**", or basically **thinking about thinking**. Subsequent studies on metacognition described the term comparative to Flavell's meaning. Cross and Paris (1988) defined it as "**the knowledge and control children have over their own thinking and learning activities**." For Hennessey (1999), it is the "**awareness of one's own thinking**, awareness of the content of one's conception, as active monitoring of one's cognitive processes, an attempt to regulate one's cognitive.

COMPONENTS OF METACOGNITION

Metacognition is classified as having three components –

Metacognitive knowledge,

Metacognitive regulation, and

Metacognitive experiences.

Metacognitive knowledge refers to the awareness individuals possess about themselves and other people as

cognitive processors. Metacognitive regulation, on the other hand, has to do with people's control over cognition and learning experiences through a set of methods that help people regulate their learning while metacognitive experiences involve cognitive efforts that are currently taking place.

TYPES OF METACOGNITIVE KNOWLEDGE

Aside from these three components, metacognition also has three different types of metacognitive knowledge –

(1) **Declarative knowledge:** Declarative knowledge refers to the factual information that one knows, and can both be spoken or written. This is also the knowledge about oneself as a learner and about what factors can influence one's performance

(2) **Procedural knowledge:** Procedural knowledge comprises information on how to do something or how to perform the procedural steps that make up a task. A high degree of procedural knowledge allows individuals to perform tasks more

automatically through a variety of strategies.

(3) **Conditional knowledge:** Conditional knowledge, on the other hand, refers to the knowledge about when to use a procedure, skill, or strategy or when not to, information as to why a procedure works and under what conditions it works, in addition to why one procedure is better than another. Such knowledge affords the individual a chance to assign resources when using strategies.

SKILLS IN METACOGNITIVE REGULATION

There are three important skills in metacognitive regulation –

(1) **Planning:** Planning involves suitable selection of strategies and the right assignment of resources.

(2) **Monitoring:** Monitoring includes awareness of understanding and task performance

(3) **Evaluating:** Evaluating refers to the assessment of the final result of a task and the efficiency carried out during task performance.



PROBLEM SOLVING

Syllabus: Problem solving: Type, Strategies, and Obstacles **Contents**

Problem solving cycle

Types of problems

Well defined

Solving well defined problems (heuristics)

Ill-defined

Solving ill defined problems(insights)

Obstacles

Mental sets

Positive and negative transfer

Analogy

Incubation

Knowledge experts and novices

Automaticity



Figure 11.2 The Problem-Solving Cycle.

The steps of the problem-solving cycle include problem identification, problem definition, strategy formulation, organization of information, allocation of resources, monitoring, and evaluation.

TYPES OF PROBLEMS

Problems can be categorized according to whether they have clear paths to a solution (Davidson & Sternberg, 1984).

Well-structured problems have clear paths to solutions. These problems also are termed **well-defined problems**. An example would be, “How do you find the area of a parallelogram?”

Ill-structured problems lack clear paths to solutions (Shin et al., 2003). These problems are also termed ill-defined problems. An example is shown in Figure 11.1: “How do you tie together two suspended strings, when neither string is long enough to allow you to reach the other string while holding either of the strings?”

People seem to make three main kinds of **ERRORS** when trying to solve well-

structured problems (Greeno, 1974; Simon & Reed, 1976; Thomas, 1974). These errors are:

(1) Inadvertently **moving backward**:

They revert to a state that is further from the end goal, for instance, moving all of the “orcs” and “hobbits” back to the first side of the river.

(2) Making **illegal moves**: They make an illegal move—that is, a move that is not permitted according to the terms of the problem. For example, a move that resulted in having more than two individuals in the boat would be illegal.

(3) Not realizing the nature of the next legal move: They **become “stuck”**—they do not know what to do next, given the current stage of the problem. An example would be realizing that you must bring one “orc” or “hobbit” back across the river to its starting point before you can move any of the remaining characters.

A **problem space** is the universe of all possible actions that can be applied to

solving a problem, given any constraints that apply to the solution of the problem.

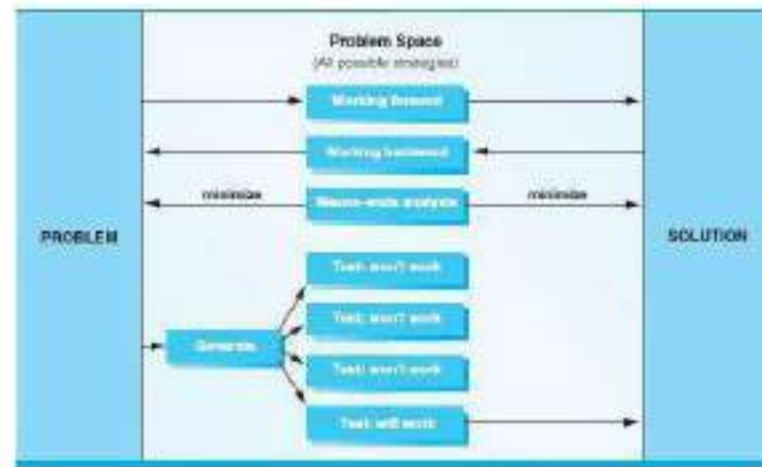


Figure 11.4 Problem Space.

A problem space contains all the possible strategies leading from the initial problem state to the solution (the goal state). This problem space, for example, shows four of the heuristics that might be used in solving the move problem illustrated in Figure 11.3.

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Algorithms are sequences of operations (in a problem space) that may be repeated over and over again and that, in theory, guarantee the solution to a problem (Hunt, 1975; Sternberg, 2000). Generally, an algorithm continues until it satisfies a condition determined by a program.

forward, working backward, and generate and test.

ISOMORPHIC PROBLEMS

Sometimes, two problems are isomorphic; that is, their formal structure is the same, and only their content differs.

Newell and Simon recognized these limits and observed that humans must use **mental shortcuts** for solving problems. These mental shortcuts are termed **heuristics**—informal, intuitive, speculative strategies that sometimes lead to an effective solution and sometimes do not when problem solvers are confronted with a problem for which they cannot immediately see an answer, effective problem solvers use the heuristic of **means–ends analysis**. In this strategy, the problem solver continually compares the current state and the goal state and takes steps to minimize the differences between the two states. Various other problem-solving heuristics include **working**

Table 11.1 Four Heuristics

These four heuristics may be used in solving the move problem illustrated in Figure 11.3.

Heuristic	Definition of Heuristic	Example of Heuristic Applied to the Move Problem (Greeno & Simon, 1988)	Example of Heuristic Applied to an Everyday Problem: How to Travel by Air from Your Home to Another Location Using the Most Direct Route Possible (Hunt, 1994)
Means-ends analysis	The problem solver analyzes the problem by viewing the end—the goal being sought—and then tries to decrease the distance between the current position in the problem space and the end goal in that space.	Try to get as many individuals on the far bank and as few people on the near bank as possible.	Try to minimize the distance between home and the destination.
Working forward	The problem solver starts at the beginning and tries to solve the problem from the start to the finish.	Evaluate the situation carefully with the six people on one bank and then try to move them step by step to the opposite bank.	Find the possible air routes leading from home toward the destination, and take the routes that seem most directly to lead to the destination.
Working backward	The problem solver starts at the end and tries to work backward from there.	Start with the final state—having all hobbits and all orcs on the far bank—and try to work back to the beginning state.	Find the possible air routes that reach the destination, and work backward to trace which of these routes can be most directly traced to originate at home.
Generate and test	The problem solver simply generates alternative courses of action, not necessarily in a systematic way, and then notices in turn whether each course of action will work.	This method works fairly well for the move problem because at most steps in the process, there is only one allowable forward move, and there are never more than two possibilities, both of which eventually will lead to the solution.	Find the various possible alternative routes leading from home, then see which of these routes might be used to end up at the destination. Choose the most direct route. Unfortunately, given the number of possible combinations of routes for air travel, this heuristic may not be very helpful.

ILL-STRUCTURED PROBLEMS AND THE ROLE OF INSIGHT

The following problems illustrate some of the difficulties created by the representation of ill-structured problems (after Sternberg, 1986). Be sure to try all three problems before you read about their solutions.

Haughty Harry has been asked to build a hat rack with a few given materials (see Figure 11.8). Can you help him construct the hat rack?

A woman who lived in a small town married 20 different men in that same town. All of them are still living, and she never divorced any of them. Yet she broke no laws. How could she do this?

You have loose black and brown socks in a drawer, mixed in a ratio of five black socks for every brown one. How many socks do you have to take out of that drawer to be assured of having a pair of the same colour?

Both the two-string problem and each of the three preceding problems are ill-structured problems. There are no clear, readily available paths to solution. **By definition, ill-structured problems do not have well-defined problem spaces.**

Insight is a distinctive and sometimes seemingly sudden understanding of a problem or of a strategy that aids in solving the problem. Often, an insight involves reconceptualizing a problem or a strategy in a totally new way. Insight often involves detecting and combining relevant old and new information to gain a **novel view of the problem or of its solution**. Although insights may feel as though they are sudden, they are often the result of much prior thought and hard work.

EARLY GESTALTIST VIEWS

Gestalt psychologist Max Wertheimer (1945/1959) wrote about **productive thinking**, which involves insights that go beyond the bounds of existing associations. He distinguished it from **reproductive thinking**, which is based on existing associations involving what is already known. According to Wertheimer, insightful (productive) thinking differs fundamentally from reproductive thinking. It involves thinking that differs from normal, linear information processing.

THE NEO-GESTALTIST VIEW

Some researchers have found that insightful problem solving can be distinguished from non-insightful problem solving in two ways (Metcalf, 1986; Metcalf & Wiebe, 1987). For one thing, when given routine problems to solve, problem solvers show remarkable accuracy in their ability to **predict their own success** in solving a problem prior to any attempt to solve it. In contrast, when given insight problems, problem solvers show poor ability to predict their own success prior to trying to solve the problems. Not only were successful problem solvers **pessimistic** about their ability to solve insight problems, but unsuccessful problem solvers were often optimistic about their ability to solve them.

OBSTACLES AND AIDS TO PROBLEM SOLVING

Several factors can hinder or enhance problem solving. Among them are **mental sets** as well as **positive and negative transfer**. **Incubation** plays a role in problem solving as well. In the next sections, we will explore these factors in more detail. One factor that can hinder problem solving is **mental set**—a frame of mind involving an existing model for representing a problem, a problem

context, or a procedure for problem solving. Another term for mental set is **entrenchment**. When problem solvers have an entrenched mental set, they fixate on a **strategy** that normally works well in solving many problems but that does not work well in solving this particular problem.

Another type of mental set involves fixation on a **particular use (function)** for an object. Specifically, **functional fixedness** is the inability to realize that something known to have a particular use may also be used for performing other functions (German & Barrett, 2005; Rakoczy et al., 2009). Functional fixedness prevents us from solving new problems by using old tools in novel ways

Another type of mental set is considered an aspect of social cognition. **Stereotypes** are beliefs that members of a social group tend more or less uniformly to have particular types of characteristics. We observe a particular instance or set of instances of some pattern. We then may overgeneralize from those limited observations. We may assume that all future instances similarly will demonstrate that pattern.

NEGATIVE AND POSITIVE TRANSFER

Transfer is **any carryover of knowledge or skills** from one problem situation to another (Detterman & Sternberg, 1993; Gentile, 2000). Transfer can be either negative or positive.

Negative transfer occurs when solving an earlier problem makes it harder to solve a later one. Sometimes an early problem gets an individual on a wrong track.

Positive transfer occurs when the solution of an earlier problem makes it easier to solve a new problem.

That is, sometimes the transfer of a mental set can be an aid to problem solving.

TRANSFER OF ANALOGIES

They found that when the domains or the contexts for the two problems were more similar, participants were more likely to see and apply the analogy (see Holyoak, 1990). Perhaps the most crucial aspect of these studies is that ***people have trouble noticing analogies unless they explicitly are told to look for them.*** Positive transfer from solved examples to unsolved problems was more likely among students who specifically tried to understand why particular examples were solved as they were, as compared with students who sought only to understand how particular problems were solved as they were (Chi et al., 1989). Based on these findings, we generally need to be looking for analogies to find them. The opposite phenomenon is **transparency**, in which people see analogies where they do not exist because of similarity of content. In making analogies, we need to be sure we are focusing on the relationships between the two terms being compared, not just their surface content attributes.

INCUBATION

Incubation—putting the problem aside for a while without consciously thinking about it—offers one way in which to minimize negative transfer. It involves **taking a pause** from the stages of problem solving.

Knowledge Experts and novices differ in how they classify various problems, describe the essential nature of problems, and how they determine and describe solutions (Chi, Glaser, & Rees, 1982; Larkin et al., 1980). Another difference between experts and novices can be observed by asking

problem solvers to report aloud what they are thinking as they are attempting to solve various problems (Bilalic, 2008; Dew et al., 2009). Statements made by problem solvers are called **verbal protocols**. An interesting effect of verbal protocols is that they can lead to increased problem-solving ability.

In another study, problem-solving ability was enhanced when participants wrote a description of their problem-solving strategy as compared with when they spoke about their strategy (Pugalee, 2004). Thus, it seems that, **for novice problem solvers, communicating problem-solving strategies improves performance.**

Another difference between expert and novice problem solvers is the **time spent** on various aspects of problems, and the relationship between problem-solving strategies and the solutions reached.

Experts appear to spend proportionately more time determining how to represent a problem than do novices (Lesgold, 1988; Lesgold et al., 1988), but they spend much less time than do novices actually implementing the strategy for solution.

Experts also involve more **procedural knowledge** about strategies relevant to that domain. Perhaps because of their better grasp of the strategies required, experts more accurately predict the difficulty of solving problems than do novices. Experts also monitor their problem-solving strategies more carefully than do novices (Schoenfeld, 1981).

Novices seem to spend relatively little time trying to represent the problem. Instead, they choose to work backward from the unknown information to the given information. That is, they go from asking what they need to find out to asking what information is offered and what strategies do they know that can help them find the missing information.

Often, novices use means–ends analysis (see Hunt, 1994). Thus, novices often consider more possible strategies than experts consider (see Holyoak, 1990).

AUTOMATIC EXPERT PROCESSES

Through practice in applying strategies, experts may automatize various operations. They can retrieve and execute these operations easily while working forward (see VanLehn, 1989). They use two important processes: One is **schematization**, which involves developing rich, highly organized schemas; the other is **automatization**, which involves consolidating sequences of steps into unified routines that require little or no conscious control.

Automaticity can be seen in mathematics, for example, where low-level skills, such as counting and adding, become automatic (Tronsky, 2005). These skills reduce the working-memory load and allow for higher-level mathematical procedures to be complete.

Table 11.3 What Characterizes Expertise?

	Experts	Novices
Schemas	Have large, rich schemas containing a great deal of declarative knowledge about domain Schemas contain a great deal of procedural knowledge about problem-solving strategies relevant to a given domain	Have relatively impoverished schemas containing relatively less declarative knowledge about domain Schemas contain relatively little procedural knowledge about problem strategies relevant to the given domain
Organization	Have well-organized, highly interconnected units of knowledge in schemas	Have poorly organized, loosely interconnected, scattered units of knowledge
Use of time	Spend proportionately more time determining how to represent a problem than in searching for and executing a problem strategy	Spend proportionately more time searching for and executing a problem strategy than in determining how to represent a problem
Representation of problems	Develop sophisticated representation of problems based on structural similarities among problems	Develop relatively poor and naive representation of problems based on superficial similarities among problems
Work direction	Work forward from given information to implement strategies for finding unknown information	Work backward from focusing on the unknown to finding problem strategies that make use of given information
Strategy	Generally choose a strategy based on elaborate schema of problem strategies; use means-ends analysis only as a backup strategy for handling unusual, atypical problems	Frequently use means-ends analysis as a strategy for handling most problems; sometimes choose a strategy based on knowledge of problem strategies
Automatization	Have automatized many sequences of steps within problem strategies	Show little or no automatization of any sequences of steps within problem strategies
Efficiency	Show highly efficient problem solving; when time constraints are imposed, solve problems more quickly than novices	Show relatively inefficient problem solving; solve problems less quickly than experts
Prediction of difficulty	Accurately predict the difficulty of solving particular problems	Do not accurately predict the difficulty of solving particular problems
Monitoring	Carefully monitor own problem-solving strategies and processes	Show poor monitoring of own problem-solving strategies and processes
Accuracy of solution	Show high accuracy in reaching appropriate solutions	Show much less accuracy than experts in reaching appropriate solutions
Confronting unusual problems	When confronting highly unusual problems with atypical structural features, take relatively more time than novices both to represent the problem and to retrieve appropriate problem strategies	When confronting highly unusual problems with atypical structural features, novices take relatively less time than experts both to represent the problem and to retrieve problem strategies

DECISION MAKING

SYLLABUS: DECISION-MAKING: TYPES AND MODELS

Classical Decision theory

The Model of Economic Man and Woman o Subjective Expected Utility Theory

Naturalistic decision making

Group decision making

Heuristics

Biases

Fallacy

CLASSICAL DECISION THEORY

The earliest models of how people make decisions are referred to as classical decision theory. Most of these models were devised by economists, statisticians, and philosophers, not by psychologists. Hence, they reflect the strengths of an economic perspective.

THE MODEL OF ECONOMIC MAN AND WOMAN

Among the early models of decision making crafted in the 20th century was that of economic man and woman. This model assumed three things:

Decision makers are fully informed regarding all possible options for their decisions and of all possible outcomes of their decision options. They are infinitely sensitive to the subtle distinctions among decision options. They are fully rational in regard to their choice of challenges of options (Edwards, 1954; see also Slovic, 1990).

The **assumption of infinite sensitivity** means that people can evaluate the difference between two outcomes, no matter how subtle the distinctions among options may be. **The assumption of rationality** means that people make their choices to maximize something of value, whatever that something may be.

SUBJECTIVE EXPECTED UTILITY THEORY

An alternative model makes greater allowance for the psychological makeup of each individual decision maker. According to subjective expected utility theory, the goal of human action is to seek pleasure and avoid pain. According to this theory, in making decisions, people will seek to **maximize pleasure** (referred to as positive utility) and to **minimize pain** (referred to as negative utility). In doing so, however, each of us uses calculations of two things. One is **subjective utility**, which is a calculation based on the individual's judged weightings of utility (value),

rather than on objective criteria. The second is **subjective probability**, which is a calculation based on the individual's estimates of likelihood, rather than on objective statistical computations. The difference between this model and the former one is that here the ratings and weights are subjective, whereas in the former model they are supposedly objective.

NATURALISTIC DECISION MAKING

Many researchers contend that decision making is a complex process that cannot be reproduced adequately in the laboratory because real decisions are frequently made in situations where there are high stakes. These situations share a number of features, including

- ill-structured problems, changing situations, high risk, time pressure, and sometimes, a team environment (Orasanu & Connolly, 1993).

A number of models are used to explain performance in these high-stakes situations. These models allow for the consideration of **cognitive, emotional, and situational factors** of skilled decision makers; they also provide a framework for advising future decision makers

(Klein, 1997; Lipshitz et al., 2001). For instance, Orasanu (2005)

developed recommendations for training

astronauts to be successful decision makers by evaluating what makes current astronauts successful, such as developing team cohesion and managing stress. Naturalistic decision making can be applied to a broad range of behaviors and environments. These applications can include individuals as diverse as badminton players, railroad controllers, and NASA astronauts.

GROUP DECISION MAKING

Groups form decisions differently than individuals. Often, there are benefits to

making decisions in groups. However, a phenomenon called “groupthink” can occur that seriously impairs the quality of decisions made. By forming decision-making teams, the group benefits from the **expertise** of each of the members. There is also an **increase in resources and ideas** (Salas, Burke, & Cannon-Bowers, 2000). Another benefit of group decision making is improved **group memory over individual memory** (Hinsz, 1990).

Groups that are successful in decision making exhibit a number of similar characteristics, including the following:

the group is small;

HEURISTICS

The mental shortcuts of heuristics and biases lighten the cognitive load of making decisions, but they also allow for a much greater chance of error.

SATISFICING

We humans are not necessarily irrational. Rather, we show bounded rationality—we are rational, but within limits (Simon, 1957). In satisficing, we consider options one by one, and then we select an option as soon as we find one that is satisfactory or just good enough to meet our minimum level of acceptability. When there are limited working-memory resources available, the use of satisficing for making decisions may be increased (Chen &

going to college);

form a minimum
criterion for that aspect
(tuition must be under
\$20,000 per year);

eliminate all options that do not meet that criterion (e.g., Stanford University is more than \$30,000 and would be eliminated);

for the remaining options, select a second aspect for which we set a minimum criterion by which to eliminate additional options (the college must be on the West Coast); and

it has open communication;
members share a common mind-set;
members identify with the group; and
members agree on acceptable group behaviour (Shelton, 2006).

Groupthink is a phenomenon characterized by premature decision making that is generally the result of group members attempting to avoid conflict (Janis, 1971). Groupthink frequently results in suboptimal decision making that avoids non-traditional ideas (Esser, 1998).

Sun, 2003). Satisficing is also used in industrial contexts in which too much information can impair the quality of decisions, as in the selection of suppliers in electronic marketplaces (Chamodrakas, et al., 2010).

ELIMINATION BY ASPECTS

in which we eliminate alternatives by focusing on aspects of each alternative, one at a time. If you are trying to decide which college to attend, the process of elimination by aspects might look like this:

focus on one aspect (attribute) of the various options (the cost of

continue using a sequential process of elimination of options by considering a series of aspects until a single option remains (Dawes, 2000).

One of the key ways in which we use mental shortcuts centres on our estimations of probability. Consider some of the strategies used by statisticians when **calculating probability**. Another kind of probability is conditional probability, which is the likelihood of one event, given another.

Table 12.1 Rules of Probability

Hypothetical Example	Calculation of Probability
Lee is one of 10 highly qualified candidates applying for one scholarship. What are Lee's chances of getting the scholarship?	Lee has a 0.1 chance of getting the scholarship.
If Lee is one of 10 highly qualified scholarship students applying for one scholarship, what are Lee's chances of not getting the scholarship?	$1 - 0.1 = 0.9$ Lee has a 0.9 chance of not getting the scholarship.
Lee's roommate and Lee are among 10 highly qualified scholarship students applying for one scholarship. What are the chances that one of the two will get the scholarship?	$0.1 + 0.1 = 0.2$ There is a 0.2 chance that one of the two roommates will get the scholarship.

REPRESENTATIVENESS HEURISTIC

In representativeness, we judge the probability of an uncertain event according to:

how obviously it is similar to or representative of the population from which it is derived; and

the degree to which it reflects the salient features of the process by which it is generated (such as randomness) (see also Fischhoff, 1999; Johnson-Laird, 2000, 2004).

For example, suppose we have not heard a weather report prior to stepping outside. We informally judge the probability that it will rain. We base our judgment on how well the characteristics of this day (e.g., the month of the year, the area in which we live, and the presence or absence of clouds in the sky) represent the characteristics of days on which it rains.

AVAILABILITY HEURISTIC

Most of us at least occasionally use the availability heuristic, in which we make judgments on the basis of how easily we can call to mind what we perceive as relevant instances of a phenomenon (Tversky & Kahneman, 1973; see also Fischhoff, 1999; Sternberg, 2000). For example, consider the letter R. Are there more words in the English language that begin with the letter R or that have R as their third letter? Most respondents say that there are more words beginning with the letter R (Tversky & Kahneman, 1973). Why? Because generating words beginning with the letter R is easier than generating words having R as the third letter. In fact, there are more English-language words with R as their third letter.

Anchoring A heuristic related to availability is the **anchoring-and-adjustment heuristic**,

by which people adjust their evaluations of things by means of certain reference points called end-anchors. For example, when the price of a TV set is given as \$3,000, people adjust their estimate of its production costs more than when the

price is given as \$2,991 (Janiszewski & Uy, 2008).

FRAMING

Another consideration in decision theory is the influence of framing effects, in which the way that the options are presented influences the selection of an option (Tversky & Kahneman, 1981). For instance, we tend to choose options that demonstrate risk aversion when we are faced with an option involving potential gains.

Biases

In the next section, we discuss several biases that frequently occur when people make decisions: illusory correlation, overconfidence, and hindsight bias.

ILLUSORY CORRELATION

We are predisposed to see particular events or attributes and categories as going together, even when they do not. This phenomenon is called illusory correlation (Hamilton & Lickel, 2000). In the case of events, we may see spurious cause-effect relationships. In the case of attributes, we may use personal prejudices to form and use stereotypes (perhaps as a result of using the representativeness heuristic). For example, suppose we expect people of a given political party to show particular intellectual or moral characteristics. The instances in which people show those characteristics are more likely to be available in memory and recalled more easily

OVERCONFIDENCE

Another common error is overconfidence—an individual's overvaluation of her or his own skills, knowledge, or judgment. For example, people answered 200 two-alternative statements, such as "Absinthe is (a) a liqueur, (b) a precious stone." (Absinthe is a licorice-flavored liqueur.) People were asked to choose the correct answer and to state the probability that their answer was correct (Fischhoff, Slovic, &

Lichtenstein, 1977). People were overconfident. For example, when people were 100% confident in their answers, they were right only 80% of the time.

HINDSIGHT BIAS

Finally, a bias that can affect all of us is hindsight bias—when we look at a situation retrospectively, we believe we easily can see all the signs and events leading up to a particular outcome (Fischhoff, 1982; Wasserman, Lempert,

Hastie, 1991). For example, suppose people are asked to predict the outcomes of psychological experiments in advance of the experiments. People rarely are able to predict the outcomes at better-than-chance levels. However, when people are told of the outcomes of psychological experiments, they frequently comment that these outcomes were obvious and could easily have been predicted in advance. The outcomes of psychological experiments, they frequently comment that these outcomes were obvious and could easily have been predicted in advance

GAMBLER’S FALLACY AND THE HOT HAND

Gambler’s fallacy is a mistaken belief that the probability of a given random event, such as winning or losing at a game of chance, is influenced by previous random events. For example, a gambler who loses five successive bets may believe that a win is therefore more likely the sixth time. A tendency opposite to that of gambler’s fallacy is called the **“hot hand” effect**.

It refers to a belief that a certain course of events will continue. Apparently, both professional and amateur basketball players, as well as their fans, believe that a player’s chances of making a basket are greater after making a previous shot than after missing one.

SUNK-COST FALLACY

An error in judgment that is quite common in people’s thinking is the sunk-cost fallacy (Dupuy, 1998, 1999; Strough et al., 2008). This fallacy represents the decision to continue to invest in something simply because one has invested in it before and one hopes to recover one’s investment. For example, suppose you have bought a car. It is a lemon. You already have invested thousands of dollars in getting it fixed. Now you have another major repair on it confronting you. You have no reason to believe that this additional repair really will be the last in the string of repairs. You think about how much money you have spent on repairs and reason that you need to do the additional repair to justify past amounts already spent. So you do the repair rather than buy a new car. You have just committed the sunk-cost fallacy.

THE CONJUNCTION FALLACY

This intuitive feeling is an example of a fallacy—erroneous reasoning—in judgment and reasoning. The ‘Conjunction Fallacy’ is a fallacy or error in decision making where people judge that a conjunction of two possible events is more likely than one or both of the conjuncts.



INTELLIGENCE

SYLLABUS : Intelligence: Spearman; Thurstone; Jensen; Cattell; Gardner; Stenberg; Goleman; Das, Kar & Parrila

THEORIES

Galton- father of individual differences -

Binet – father of iq testing -

William stern – IQ formula -

Terman – Stanford binet revision -

Yerkes – group testing -

Spearman – 2 factor theory

Weschler – wis, wisc, wais

Thurstone – group factor theory, primary mental abilities

Cattell – fluid & crystallised intelligence

Guilford – si model

Vernon & carroll – hierarchical approach

Vernon –
levels o Burt
model

o Carroll – 3 stratum model

Gardner – multiple intelligences

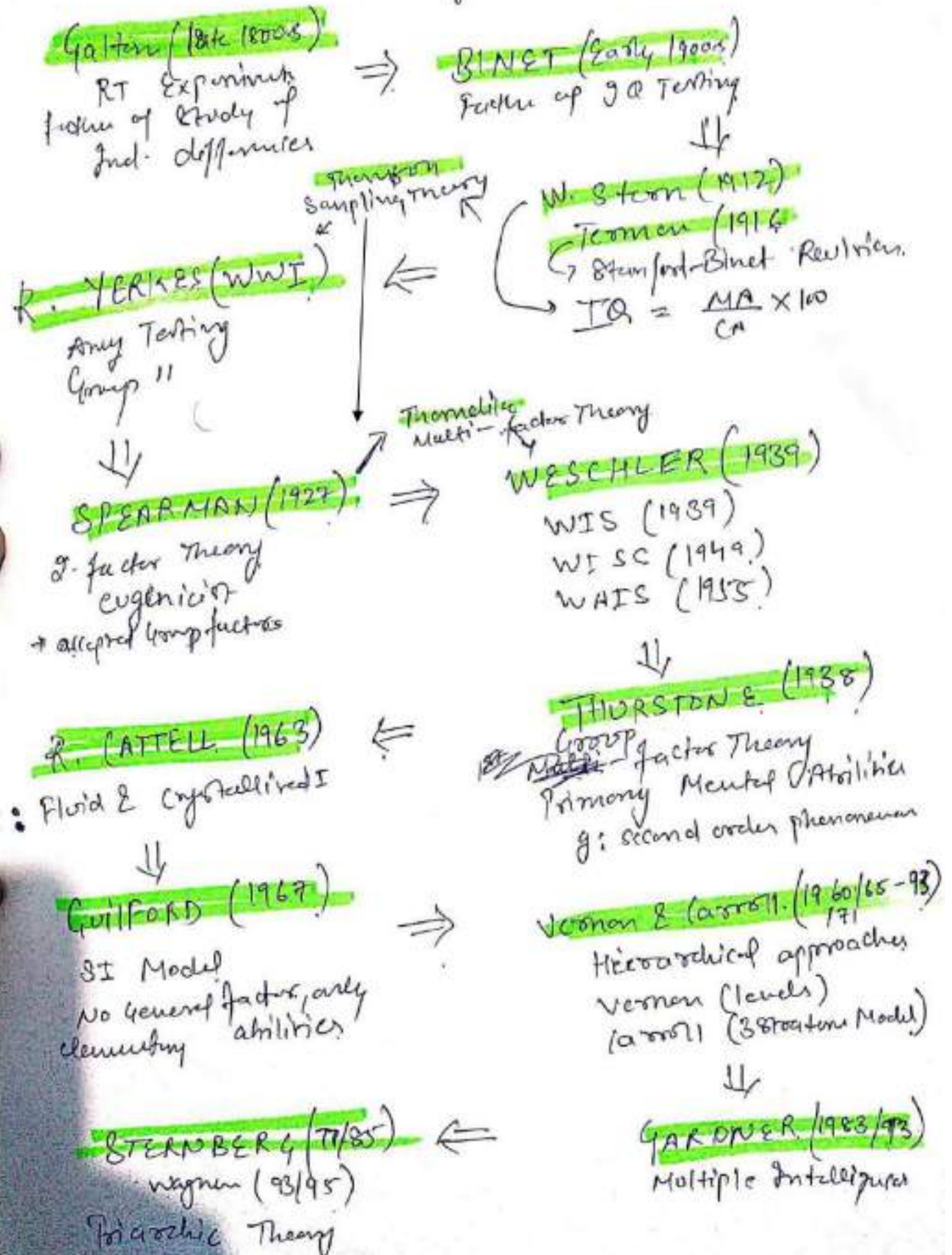
Sternberg – triarchic theory

Hebb's theory

Zajonc's theory of confluence -

Thomson's model – sampling theory

Intelligence Timeline



WHAT IS INTELLIGENCE? – DEFINING INTELLIGENCE

“Individuals differ from one another in their ability to understand complex ideas, to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by taking thought.”

American Psychological Association

“It seems to us that in intelligencethereisa fundamental faculty, the alteration or the lack of which, is of the utmost importance for practical life. This faculty is judgement, otherwise called good sense, practical sense, initiative, the faculty of adapting ones self to circumstances.” **A. Binet**

“... the ability to plan and structure one’s behavior with an end in view.” **J. P. Das**

“Intelligence A: the biological substrate of mental ability, the brains’ neuroanatomy and physiology; Intelligence B: the manifestation of intelligence A, and everything that influences its expression in real life behavior; Intelligence C: the level of performance on psychometric tests of cognitive ability.” **H. J. Eysenck.**

“An intelligence is the ability to solve problems, or to create products, that are valued within one or more cultural settings.”

Gardner

“... performing an operation on a specific type of content to produce a particular product.”

P. Guilford

“Intelligence is a general factor that runs through all types of performance.” **A. Jensen**

“... I prefer to refer to it as ‘successful intelligence.’ And the reason is that the emphasis is on the use of your intelligence to

achieve success in your life. So I define it as your skill in achieving whatever it is you want to attain in your life within your sociocultural context — meaning that

5 people have different goals for themselves, and for some it’s to get very good grades in school and to do well on tests, and for others it might be to become a very good basketball player or actress or musician.” **R. J. Sternberg**

“Intelligence, considered as a mental trait, is the capacity to make impulses focal at their early, unfinished stage of formation. Intelligence is therefore the capacity for abstraction, which is an inhibitory process.” **L. L. Thurstone**

“A global concept that involves an individual’s ability to act purposefully, think rationally, and deal effectively with the environment.” **D. Wechsler**

Sir Francis Galton is a key figure in modern intelligence testing. As the first cousin of Charles Darwin, he attempted to apply Darwin’s evolutionary theory to the study of human abilities. He postulate that intelligence was quantifiable and normally distributed. In other words, he believed that we could assign a score to intelligence **where the majority of people fall in the average range** and the percentage of the population decreases the farther from the middle their score gets.

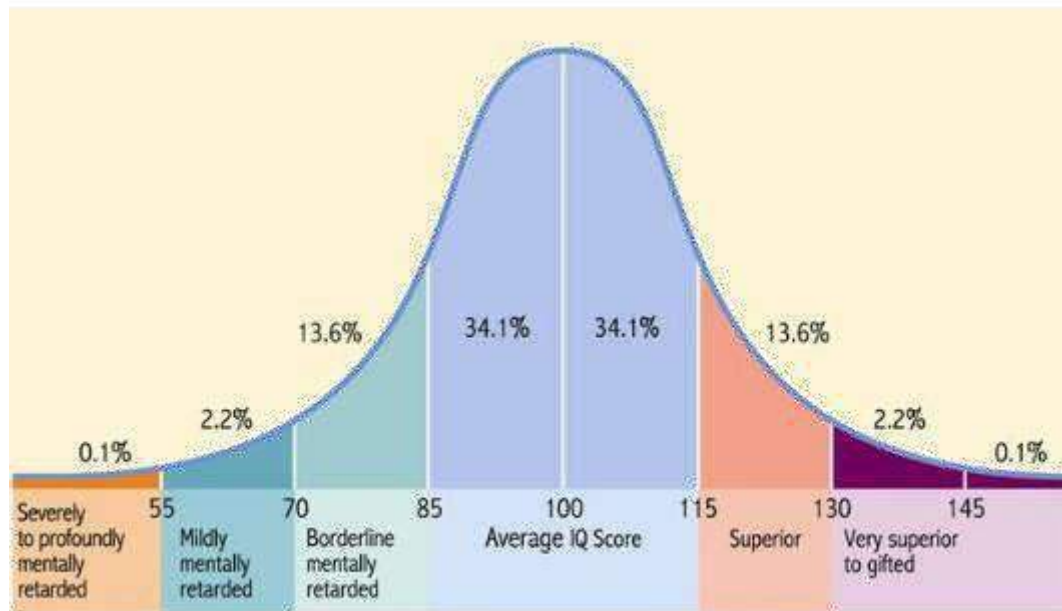
The first workable intelligence test was developed by French psychologist **Alfred Binet**. He and his partner, **Theodore Simon**, were commissioned by the French government to improve the teaching methods for developmentally disabled children. They believed that intelligence was the key to

effective teaching, and developed a strategy whereby a mental age (MA) was determined and divided by the child's chronological age (CA). This formula, stated as "MA/CA X 100."

German psychologist **William Stern** formulated the basic definition of IQ in 1912 when he defined intelligence quotient as a ratio of an estimated "mental age" and "actual chronological age":

$$100 \times \frac{\text{mental age}}{\text{chronological age}}$$

Frequency Distribution of IQ Scores



In 1915, **Yerkes** helped to create the Yerkes-Bridges Point Scale of Intelligence. In 1917, he served as president of the American Psychological Association (APA), which under his leadership, began several programs devoted to the war effort in World War I. As chairman of the Committee on the Psychological Examination of Recruits, **Yerkes developed the Army's Alpha and Beta Intelligence Tests**, given to over 1 million United States soldiers during the war.

to this theory, mind is made up of different faculties like reasoning, memory, discrimination, imagination, etc. These faculties are independent of each other and can be developed by vigorous training. Faculty Theory had been under criticism by experimental psychologists who disproved the existence of independent faculties in the brain.

One factor theory	Two factor theory
32	33

THEORIES OF INTELLIGENCE

1. FACULTY THEORY

It is the oldest theory regarding the nature of intelligence and flourished during 18th and 19th century. According

2. STERN : UNI-FACTOR THEORY

Around 1911, Wilhelm Stern introduced his uni-factor theory of intelligence which has also been known as the general capacity theory.



Fig. 1 : Stern's uni-factor theory of intelligence, showing intelligence to consist of only a general capacity, G. (Bischof, p. 6)

Stern considered that all persons are endowed with G, though in unequal amounts. It is applied to any problem solving situation, the efficiency of which depends solely upon the environment. Thus, Stern's uni-factor theory is the simplest one.

3. SPEARMAN TWO-FACTOR THEORY

It was developed in 1904 by an English Psychologist, Charles Spearman, who proposed that intellectual abilities were comprised of two factors: one general ability or common ability known as 'G' factor and the other a group of specific abilities known as 'S' factor. 'G' factor is universal inborn ability. Greater 'G' in an individual lead to greater success in life. 'S' factor is acquired from the environment. It varies from activity to activity in the same individual.

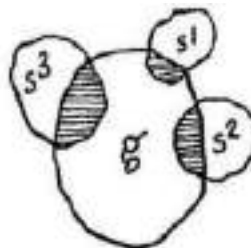


Fig. 2 : Spearman's two-factor theory, with a general capacity (g) and three specific kinds of

4. THOMSON SAMPLING THEORY

In 1916 G. H. Thomson first proposed his sampling theory which he refined in 1935 and still later in 1948. By the throwing of dice and the random selection of cards, that is, by the study of purely chance selection, Thomson indicated that any given mental test may sample a number of independent, abilities.

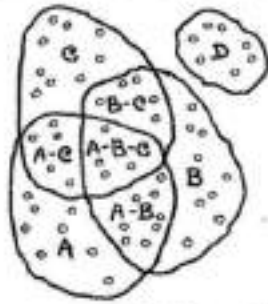


Fig. 4 : Godfrey Thomson's sampling theory of intelligence, with small circles to represent independent mental abilities and circles A, B, C and D to indicate tests (Bischof, p. 10)

Notes: the controversy between two - factor theory and sampling theory has been reduced as Spearman later on accepted the existence of group factors.

5. THORNDIKE MULTI-FACTOR THEORY

Thorndike believed that there was nothing like General Ability. Each mental activity requires an aggregate-different set of abilities. He distinguished the following four attributes of intelligence:

(a) Level—refers to the level of difficulty of a task that can be solved.

means the total number of situations at each level to which the individual is able to respond.

Speed—is the rapidity with which we can respond to the items.

THURSTONE MULTIPLE-FACTOR THEORY(GROUP FACTOR THEORY)

This theory states that Intelligent Activities are not an expression of innumerable highly specific factors, as Thorndike claimed. Nor is it the expression primarily of a general factor that pervades all mental activities. It is the essence of intelligence, as Spearman held. Instead, the analysis

GUILFORD'S MODEL OF STRUCTURE OF INTELLECT

interpretation of Spearman and others led them to the conclusion that 'certain' mental operations have in common a 'primary' factor that gives them psychological and functional unity and that differentiates them from other mental operations. These mental operations then constitute a group. A second **group** of mental operation has its own unifying primary factor, and so on. In other words, there are a number of groups of mental abilities, each of which has its own **primary factor**, giving the group a functional unity and cohesiveness. Each of these primary factors is said to be relatively Independent of the others.

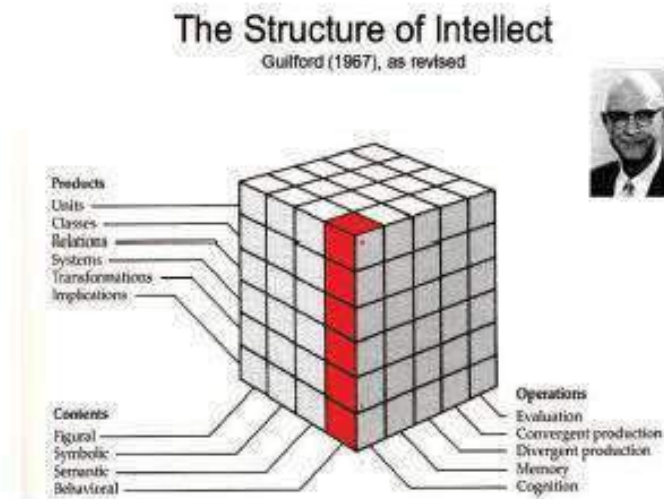
Guilford (1967, 1985, 1988)

Based on these factors Thurstone constructed a new test of intelligence known as "**Test of Primary Mental Abilities (PMA)**." proposed a three-dimensional structure of intellect model. According to Guilford every intellectual task can be classified according to it's

content, the mental operation involved and the product resulting from the operation.

He further classified content into five categories, namely, Visual, Auditory, Symbolic, Semantic and Behavioural. He classified operations into five

categories, namely, Cognition, Memory retention, Memory recording, Divergent production, Convergent production and evaluation. He classified products into six categories, namely, Units, Classes, Relations, Systems, Transformations and Implications.



Vernon's description of different levels of intelligence may fill the gaps between two extreme theories, the two-factor theory of Spearman, which did not allow for the existence of group factors, and the multiple-factor theory of Thurstone, which did not allow a "g" factor. Intelligence can be described as comprising abilities at varying levels of

VERNON'S HIERARCHICAL THEORY

1. The highest level : "g" (general intelligence) factor with the largest source of variance between individuals. (Spearman)
2. The next level : major group factors such as verbal-numerical-educational (v.ed) and practical-mechanical-spatial-physical (k.m.) ability.
3. The next level : minor group factors are divided from major group factors.
4. The bottom level : "s" (specific) factor. (Spearman)

Beginning in 1969, Vernon became increasingly involved in studying the contributions of environmental and genetic factors to intellectual development. Vernon continued to analyze the effects of genes and the environment on both individual and group difference in intelligence. He concludes that individual difference in intelligence are approximately 60 percent attributable to genetic factors, and that there is some evidence implicating genes in racial group differences in average levels of mental ability.

BURT'S HIERARCHICAL GROUP FACTOR THEORY

According to this theory, abilities are classified into three types:

General Ability entering into every test belonging to a certain broad genesis.

Special Abilities, each limited to certain group, and

3. Individual Abilities, peculiar to a particular test.

Burt's says that "The set of factors can be arranged in a hierarchical order as shown in the schemes. While

recognizing group and specific factor, he does not deny the 'g' factor"

CATTELL'S FLUID AND CRYSTALLIZED THEORY

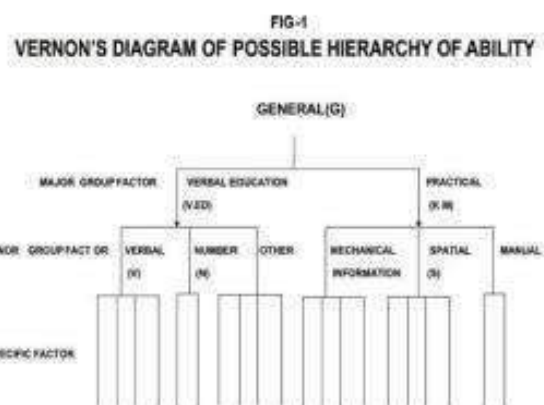
The fluid aspect of this theory says that intelligence is a basic capacity due to genetic potentiality. While this is affected by the past and new experiences, the crystallized theory is a capacity resultant of experiences, learning and environment.

GARDENER'S THEORY OF MULTIPLE INTELLIGENCE :

The multiple intelligence theory is that people possess eight types of intelligence : linguistic, logical, spatial, musical, motor ability, interpersonal, intrapersonal and naturalistic intelligence.

STERNBERG'S TRIARCHIC THEORY

Psychologist Robert Sternberg (1985) has constructed a three—pronged, or triarchic theory of intelligence. The Three types are :



Analytical Intelligence—is what we generally think of as academic ability. It enables us to solve problems and to acquire new knowledge. Problem—solving skill include encoding information, combining and comparing pieces of information and generating a solution.

Creative Intelligence—is defined by the abilities to cope with novel situations and to profit from experience. The ability to quickly relate novel situations to familiar situations (that is, to perceive similarities and differences) fosters adaptation. Moreover, as a result of experience, we also become able to solve problems more rapidly.

Practical Intelligence—or “street smarts”, enable people to adapt to the demands of their environment. For example, keeping a job by adapting one’s behavior to the employer’s requirements is adaptive. But if the employer is making unreasonable demands, reshaping the environment (by changing the employer’s attitudes) or selecting an alternate environment (by finding a more suitable job) is also adaptive.

ANDERSON’S THEORY : COGNITIVE DEVELOPMENT

Anderson proposes that human cognitive architectures will have adapted optimally to the problems posed in their environment. Therefore, discovering the optimal solution to the problem posed by the environment, independent of the architecture, is equivalent to discovering the mechanism used by the architecture. A ‘**Rational Analysis**’, as it is called, takes into account the available information in the environment, the goals of the agent, some basic

assumptions about computational cost (in terms of a ‘general’ architecture mechanism), and produces the optimal behavioral function. This function then of course can be tested empirically and assumptions modified if it proves inaccurate. A contrasting point of view to this is espoused by Simon, and is centered around the claim that, in a rational analysis, the assumptions about the architecture actually do most of the work.

EYSENCK’S STRUCTURAL THEORY

Eysenck discovered the neurological correlates of intelligence. He identified three correlates of intelligence i.e. reaction time, inspection time and average evoked potential. First two are observed behavior. Third behavior, is description of mental waves. Brighter individual progressively takes less time in responding. They show less variability in reaction time. Their inspection time is also less as compared to less intelligent. Average evoked potential is often measured by the wavelength in electroencephalogram and complexities of waveform. He found that the waves of intelligent individuals are complex.

CECI’S BIOLOGICAL THEORY

Ceci (1990) proposes that there are multiple cognitive potentials. These multiple intelligence’s are biologically based and place limits on mental processes. These are closely linked to the challenges and opportunities in the individual’s environment. In his view, context is essential to the demonstration of cognitive abilities. By context, he means domain of knowledge and other factors such as personalities, motivation and education. Context can be mental, social or physical.

HEBB’S THEORY

“Intelligence A is the basic potentiality of the organism, whether animal or human, to learn and to adapt to its

environment...Intelligence A is determined by the genes but is mediated mainly by the complexity and plasticity of the central nervous system...Intelligence B is the level of ability that a person actually shows in behaviour—cleverness, the efficiency and complexity of perceptions, learning, thinking, and problem solving. This is *not* genetic...Rather, it is the product of the interplay between genetic potentiality and environmental stimulation...I have suggested that we should add a third usage to Hebb's Intelligence A and B, namely Intelligence C, which stands for the score or IQ obtained from a particular test"

CATTELL'S CRYSTALIZED AND FLUID INTELLIGENCE

Cattell introduced the concepts of fluid and crystallized intelligence, how they differed in children and adults and how they were influenced by brain injury. His description of these concepts is as follows:

Adult mental capacity is of two kinds, the chief characteristics of which may be best connoted by the use of the terms "fluid" and "crystallized."

Fluid ability has the character of a purely general ability to discriminate and perceive relations between any fundamentals, new or old. It increases until adolescence and then slowly declines. It is associated with the action of the whole cortex. It is responsible for the intercorrelations, or general factor, found among children's tests and among the speeded or adaptation-requiring tests of adults.

Crystallized ability consists of discriminatory habits long established in a particular

field, originally through the operation of fluid ability, but not [sic] longer requiring insightful perception for their successful operation.

Intelligence tests test at all ages the combined resultants of fluid and crystallized ability, but in childhood the first is predominant whereas in adult life, owing to the recession of fluid ability, the peaks of performance are determined by the crystallized abilities (Cattell, 1943, p. 178).

18. JENSEN THEORY

Jensen (1969) proposed two level-theories of mental abilities. In other words intelligence consists of two levels of abilities. The Level I ability is associative learning, which consists of Short-Term Memory (STM), rote learning, attention and simple associative skills. The Level

is called cognitive learning, which consists of abstract thinking, symbolic thought, conceptual learning, and use of language in problem solving.

He argued that Level I ability, i.e. associative learning is equally distributed across all racial and national groups but on the contrary, Level II, i.e., cognitive learning is concentrated more in the middle class Anglo-American populations than in the lower class black populations.

According to him, genetic differences in intelligence exist among people coming from different races, nationalities, and social classes. Jensen conducted studies of intelligence comparing identical fraternal twins. Identical twins have identical genetic dispositions for that their IQs are assumed to be similar.

Fraternal twins like ordinary siblings genetically less similar, which results in the assumption that their IQs less similar. Jensen found support for his views in these findings. Studies with identical twins produced an average correlation of .86, whereas studies of ordinary siblings produced an average correlation of .47.

Jensen on basis of his studies believed that genetic factors are more important than environmental factors for one's intelligence.

GOLEMAN'S EMOTIONAL INTELLIGENCE

Daniel Goleman's "Emotional Intelligence"

Element	Definition	Characteristics
Self-Awareness	•The ability to recognize and understand your moods, emotions, and drives, as well as their effect on others	•Self-confidence •Realistic self-assessment •Self-deprecating sense of humor
Self-Regulation	•The ability to control or redirect disruptive impulses and moods •The propensity to suspend judgment – to think before acting	•Trustworthiness and integrity •Comfort with ambiguity •Openness to change
Motivation	•A passion to work for reasons that go beyond money or status •A propensity to pursue goals with energy and persistence	•Strong drive to succeed •Optimism, even in the face of failure •Organizational commitment
Empathy	•The ability to understand the emotional makeup of other people •Skill in treating people according to their emotional reactions	•Expertise in building and retaining talent •Cross-cultural sensitivity •Service to clients and customers
Social Skill	•Proficiency in managing relationships and building networks •An ability to find common ground and build rapport	•Effectiveness in leading change •Persuasiveness •Expertise in building and leading teams

20. PASS MODEL

The PASS (Planning, Attention-arousal, and Simultaneous-successive)THEORY of intelligence has been developed by J.P Das, Jack Naglieri, and Kirby (1994). They proposed that three functional units of brain determine the intellectual activity of an individual. These three units are responsible for

planning, arousal/attention and simultaneous/Successive processing. These PASS processes are interactive in nature yet each has its own distinctive functions.

Attention-Arousal: This process is basic to any behavior and it is processed by 1st functional unit of brain that involves the ability to selectively attend to stimuli while ignoring other distractions. Arousal keeps people awake and alert. The arousal functions are generally associated with the brain stem and thalamus. Individuals with Attention Deficit Disorder (ADD) have impairments in this area. An optimal level of arousal focuses our attention to the relevant portion of a problem.

Simultaneous Processing: This involves the ability to integrate separate stimuli/information to our knowledge system as an interrelated whole. The occipital and parietal lobes are thought to be important for these functions. For example, in Raven's Progressive Matrices (RPM) Test, a design is given and one of its parts has been removed. We are required to choose one of the six options which completes the design. Simultaneous processing helps us in finding relationships between the given abstract figures. Simultaneous processing is broadly with occipital and parietal lobes.

Successive Processing: This involves the ability to integrate stimuli/information into a sequential order. Learning of digits, alphabets, multiplication tables, etc. are examples of successive processing. This type of processing is related to the temporal lobe.

Planning: This is the ability of an individual to make decisions about how to solve problems and how to carry out the task. It involves setting goals, courses of action to reach the goal and anticipating their consequences. Planning is associated with the frontal lobes of the brain.

THE CONFLUENCE MODEL

Firstborn IQ advantage in terms of the ever-changing intellectual environment within the family. It uses a simple mathematical formula to compute the relative advantages and disadvantages of these factors:

The Confluence Model proposed by R.B. Zajonc & Markus (1975) and Zajonc (1976, 2001) explains the

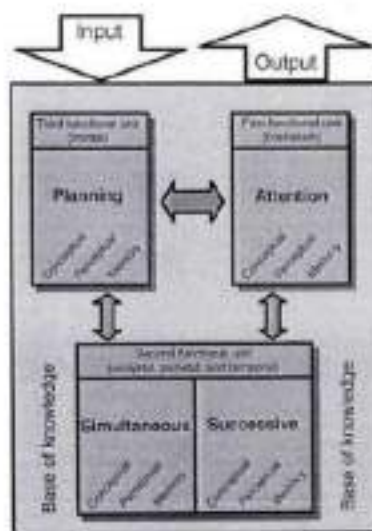


FIGURE 7.1. PASS theory.

Firstborns do not have to share their parents' attention, so they benefit from their parents' complete absorption in the new responsibility. Laterborn children never experience this advantage. Moreover, additional siblings automatically limit the amount of attention any of the siblings get-and this includes the firstborn. This would explain the Belmont and Marolla (1973) finding that firstborns from smaller families have higher IQs than firstborns from larger families.

Firstborn children are exposed to more adult language. Laterborns are exposed to the less mature speech of their siblings. This may affect their performance on the verbal scales of intelligence tests. Moreover, the linguistic environment becomes increasingly less mature as more children enter the family. This

also gels with the finding that children in larger families have lower IQ scores.

As more children enter the family, the general intellectual environment becomes less mature. This would explain why firstborns and older children from large families have lower IQs than firstborns and older children from smaller families.

Firstborns (and older siblings in general) often have to answer questions and explain things to their younger siblings. It is believed that the act of tutoring helps the older children to cognitively process information. Further, teaching others may improve their verbal abilities. Except in very rare cases, youngest siblings do not get the opportunity to tutor their brothers and sisters. This is the reason why only children do not tend to have higher IQs than firstborns.



CREATIVITY

Syllabus : Creativity: Torrance, Getzels & Jackson, Guilford, Wallach &

Kogan Relationship between Intelligence and Creativity

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Associative theory

DEFINING CREATIVITY

Creativity is defined as the tendency to generate or recognize ideas, alternatives or possibilities that may be useful in solving problems, communicating with others and entertaining ourselves as well as others.

BY AUTHORS

Thurston (1955) “Any action is to be creative if it has immediate solution to the kind of thinking which has always been innovative.” Baron (1961) “Creativity is the essence of making pre-existing objects and elements as new one.” E.P. Torrance (1965) “Creative thinking is the process of understanding errors un-received, and rare elements, making concepts in their relationship imagining notions and making tests ,carrying results to other persons and making improvement by reviewing the concepts.” E.P Torrance-“A process of becoming sensitive to problems, deficiencies, gaps in knowledge, missing elements, disharmonies and so on, identifying the difficulty, searching for solution, making guesses, or formulating hypotheses about the deficiencies, testing and 57 retesting these hypotheses and possibly modifying and retesting them, and finally communicating the results.” James Drever - “Creativity is the ability to create new objects. Creativity in the broad sense refers to the imagination of the sum of new ideas and talents. (If the self -motivated, do not follow others) and synthesis of ideas and where mental functions are not only the sum of the other's views. Dr. Passi - “Creative power is a multi-faceted trait and power. The verbal and nonverbal traits and power are distributed in different proportion in different human beings. The main factors responsible for it are eloquence, flexibility or morbidity, curiosity or

proneness to invention and toughness and good trend.”

Bertlet- “Creativity is bold thinking. Bold thinking means the ability of thinking beyond the mainstream, receiving new experience and adding the present relationship with that of the future. Bold thinking or creativity is to work or think with new attitude by going beyond mutual thinking.” Rollo May- “Creativity is the process of bringing something new in human being. Creativity requires passion and commitment. It brings to our awareness what was previously hidden and points to new life. The experience is one of heightened consciousness ecstasy.”

The four P's of creativity

Creativity may also be considered in terms of which aspect or facet of creativity they emphasize (Rhodes, 1961; Runco, 2004b). Traditionally, these aspects have been referred to as the "four p's of creativity": **process**, **product**, **person** (or personality), and **place** (or press). More recent versions of this framework (e.g., Runco, 2007a) have extended it to six P's, adding **persuasion** (Simonton, 1990) and **potential** (Runco, 2003)'

Theories that focus on the **creative process** aim to understand the nature of the mental mechanisms that occur when a person is engaged in creative thinking or creative activity. Process theories typically specify different stages of processing (e.g., Mace & Ward, 2002; Simonton, 1984; Wallas, 1926; Ward, Smith, & Finke, 1999) or particular mechanisms as the components of creative thought (e.g., Mumford, Baughman, Maher, Costanza, & Supinski, 1997; Mumford, Mobley, Uhhnan, ReiterPalmon, onking involves the same basic cognitive mechanisms as non-creative thinking, the relative 44

roles of conscious versus unconscious processes, the relative contributions of chance or stochastic processes versus more controlled and guided processes, and the nature and reliability of evaluative processes during the process of creation.

Probably the most objective approach to creativity focuses on **products**: works of art} inventions, publications, musical compositions, and so on. Products can usually be counted, thus permitting considerable quantitative objectivity, and they are often available for viewing or judging, so interrater reliability can be readily determined - two substantial advantages. A down side is that when studying a product, little can be directly said about the process leading to it or the creator's personality; inferences are thus necessary to inform the creative process or person. Moreover, unambiguously creative products are constructed by unambiguously creative persons. Thus, studies of products tell us about highly creative individuals but not about persons with as-yet unfulfilled creative potential (Runco, 1996).

Another longstanding perspective on creativity has focused on the **creative person** (or personality). Much early research compared mathematicians, architects, writers, and other groups in terms of the traits that may be indicative or contraindicative of creative potential. Several traits cut across domains; these include intrinsic motivation} wide interests, openness to experience, and autonomy (Barron, '995; Helson, 1972). A number of personality traits also appear to be more pervasive either among persons in artistic domains or in scientific domains (Feist, '998, 1999)' Personality is now usually viewed as one influence on creative behavior, rather than a complete explanation (Feist & Barron, 2003).

The expression of personality often depends on the **setting or climate** in which an individual resides. The research on places or "**press**" **factors** (press from pressures) is especially useful in defining such interactions between persons and environments. There are individual differences in terms of preferred environments, but again also general tendencies: Creativity tends to flourish when there are opportunities for exploration and independent work, and when originality is supported and valued (Amabile, 1990; Witt & Boerkem, 1989)'

Simonton (1990) offered another perspective following the alliterative scheme by describing creativity as **persuasion**: Creative people change the way others think, so they must then be persuasive to be recognized as creative. The notion of creativity as persuasion shares assumptions with the social perspective (Amabile, 1990), the attributional theory of creativity (Kasof, 1995), and Csikszentmihalyi's (1988a) systems model. In the last of these, persuasive individuals are the ones who are likely to influence the direction taken by a domain. The emphasis on persuasion implies that everyday originality (Runco & Richards, 1998) may not be deemed creative, since it is often largely personal.

Runco (2008) recently suggested that this might be further organized into a hierarchy that starts with theories of **creative performances versus creative potentials**. The former is divided into products and persuasion theories, and any other perspective that focuses on manifest, unambiguously creative behavior; the latter is divided into creative personality and places, and any other perspective that appreciates yet-unfulfilled possibilities and subjective processes. This hierarchical framework captures the earlier alliterative scheme

but allows research on everyday creativity and the creative potentials of children and others who may have most

of what it takes but require educational opportunities or other support before they can perform in a creative fashion.

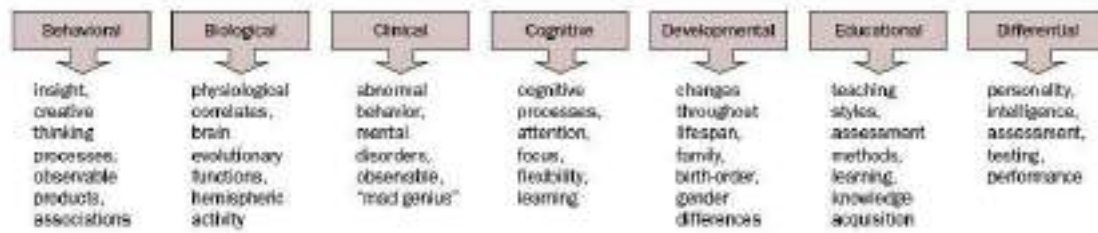
Table 2.1: Summary of Theories of Creativity

Category	Primary Assertion	Key Concepts	Str.P's Focus	Levels of Magnitude	Major Studies and Examples
Developmental	Creativity develops over time (from potential to achievement); mediated by an interaction of person and environment.	Place and family structures Role of play Support during transitions Longitudinal process Multivariate influences	Person, Place, Potential, & Product	Mini-c to Pro-c	Helson (1999) Subotnik & Arnold (1996) Albert & Runco (1985)
Psychometric	Creativity can be measured reliably and validly; differentiating it from related constructs (IQ) and highlighting its domain-specific nature.	Reliable and valid measurement Discriminant validity Thresholds Domain specificity	Primarily Product	Little-c to Big-C	Grefford (1997) Wallace & Kogan (1976)
Economic	Creative ideation and behavior is influenced by "market forces" and cost-benefit analyses.	Influence of market-level factors Psychosociometric perspectives Markets of creativity Investment decisions	Person, Place, Product, & Persuasion	Little-c to Big-C	Rubenstein & Runco (1992, 1995) Florida (2002) Sternberg & Lubart (1995, 1996)
Stage & Componential Process	Creative expression proceeds through a series of stages or components; the process can have linear and recursive elements.	Preparation stages Incubation and insight Verification and evaluation Component mechanisms	Primarily Process	Mini-c to Big-C	Wallas (1926) Runco & Chand (1995) Amabile (1995)
Cognitive	Ideational thought processes are foundational to creative persons and accomplishments	Remote association Divergent/convergent thinking Conceptual combination, expansion Metaphorical thinking, imagery Metacognitive processes	Person & Process	Little-c to Big-C	Mednick (1969) Guilford (1968) Finke, Ward, & Smith (1992)

(continued)

Table 2.1 (continued)

Category	Primary Assertion	Key Concepts	Str.P's Focus	Levels of Magnitude	Major Studies and Examples
Problem Solving & Expertise-Based	Creative solutions to ill-defined problems result from a rational process, which relies on general cognitive processes and domain expertise.	Ill-defined problems Cognitive, computational approach Expertise-based approaches Problem representations & heuristics	Person, Process, & Product	Little-c to Big-C	Ericsson (1993) Simon (1973, 1989) Weisberg (1999, 2006)
Problem Finding	Creative people proactively engage in a subjective and exploratory process of identifying problems to be solved.	Subjective creative processes Exploratory behaviors On-line discovery	Process, Person, & Potential	Primarily Mini-c	Geroski & Calamandrei (1995) Runco (1998)
Evolutionary (Darwinian)	Emergent creativity results from the evolutionary-like processes of blind generation and selective retention.	Chance-configuration Blind generation of ideas Selective retention of ideas Equal-odds rule Social judgment and choice	Person, Process, Place, & Product	Primarily Big-C	Campbell (1960) Sternberg (1978, 1992)
Typological	Creativity varies along key individual differences, which are related to both macro- and micro-level factors and can be classified via typologies.	Individual differences Comparison of creators Societal versus individual Integrating multiple levels of analysis	Primarily Person; but also Process, Product, & Place	Little-c to Big-C	Galenson (1991, 2006) Rothstein (2003)
Systems	Creativity results from a complex system of interacting and interrelated factors.	Evolving system Network of interconnects Domain and field Development Collaborative Creativity Chaos and Complexity	Varying emphasis across all Ps.	Little-c to Big-C	Gruber (1992a) Calamandrei (1995) Runco (2006)



STAGES OF CREATIVITY

The history of research on stages of creativity began with Graham Wallas (1926) who suggested that creative thinking follows four successive steps:

Stage of preparation: The subject begins to gather information about the problem to be solved and attempts some solutions. This stage is characterised by a state of trial -and-error in learning. Therefore, the subject is advised to learn as much as possible about the problem area. In preparation the thinker begins recalling personal experiences and investigating in all different directions to gather information about the problem to be solved. The object of defining the focus question of interest is to list all concepts associated with the focus question. Since the goal from this procedure is to generate the largest possible list, the thinker should not worry about redundancy, relative importance, or relationships at this point.

Stage of incubation: In the second stage the solution exists but is not clear. The subject must not intentionally work on the problem. Instead it is allowed to sink into the unconscious. In this stage the solution exists but is not clear. Therefore, the thinker must not intentionally work on the problem. Instead, he/she should be allowed to sink into the unconscious and the thinker is advised to relax and reflect on his/her focus question which might lead him/her to modification of the focus question.

Stage of illumination: In the third stage the subject suddenly experiences insight into the problem when a new solution,

idea, or relationship emerges. In other words, the subject attempts to reformulate his/her ideas or to formulate new ones. The subject is more active and more conscious work is needed in this stage. In the stage of illumination the thinker experiences insight into the problem when a new solution, idea, or relationship emerges. Thus, he/she attempts to reformulate his/her ideas or to formulate new ones.

Stage of verification: Finally, the subject tries and checks the solution. In this stage some modification may also occur to ideas reached in the previous stages. In the stage of verification the thinker tests, tries and checks the solution he/she created. Since this stage is the final one, the thinker may well make some modification to his/her ideas which he/she reached in the previous stages. In this stage thinker should rework the structure of his/her map to represent his/her collective understanding of the interrelationships and connections among groupings, which may include adding, subtracting, or changing super-ordinate concepts, thus, he/she may need to review his/her concept map as he/she gains new knowledge or new insights.

In some situations, the above stages may appear in a different order, or combined into two or three stages. They also do not occur regularly. For example, sometimes the subject's knowledge of the problem area allows him/her to pass over the first stage (preparation) and move on to the next stage (incubation) or even to the third stage.

THEORIES OF CREATIVITY

Torrance Tests of Creative

Thinking (TTCT)

Torrance chose to define creativity as a process because he thought if we understood the creative process, we could predict what kinds of person could master the process, what kind of climate made it grow and what products would be involved (Torrance, 1995). Building on J.P. Guilford's work, the Torrance Tests of Creative Thinking (TTCT) originally involved simple tests of divergent thinking and other problem-solving skills, which were scored on four scales:

Fluency.

Flexibility.

Originality.

Elaboration.

In longitudinal studies (1958-2000), Torrance found that students identified as creatively gifted but not intellectually gifted (IQ of 130+), out achieved the intellectually in adulthood. He found that characteristics of the creative thinking abilities differ from those of the abilities involved in intelligence and logical reasoning. Torrance's research has demonstrated that a variety of techniques for training in creative problem solving produce significant creative growth without **Manifesto for Children**

By: E. Paul Torrance

1. Don't be afraid to fall in love with something pursue it with intensity.
2. Know, understand, take pride in, practice, develop, exploit, and enjoy your greatest strengths.

interfering with traditional kinds of educational achievement. Creative growth seems to be the greatest and most predictable when deliberate, direct teaching of creative thinking skills are involved. Torrance believed that each person is unique and has particular strengths that are of value and must be respected; therefore, education must be built upon strengths rather than weaknesses. It takes courage to be creative. Just as soon as you have a new idea, you are a minority of one. Torrance found that learning and thinking creatively takes place in the process of sensing difficulties, problem, and gaps in information; making guesses or formulating hypotheses about these deficiencies; in testing these guesses and possibility, revising and retesting them; and finally in communicating the results. Vital human needs are involved in each of these four stages. If we sense that something is missing or wrong our tension is aroused and we become uncomfortable. To relieve our tension we try to make guesses in order to fill gaps and make connections.

3. Learn to free yourself from the expectations of others and to walk and away from the games they impose on you. Free yourself to play your own game.
4. Find a great teacher or mentor who will help you.
5. Don't waste energy trying to be well-rounded.
6. Do what you love and can do well.
7. Learn the skills of interdependence

MANIFESTO FOR ADULTS

Being a Beyonder means doing your very best, going beyond where you have been before, and going beyond where others have gone.

- They are tolerant of mistakes by themselves and others.
- The beyonders take delight in deep thinking.
- They are able to feel comfortable as a minority of one.
- They love the work that they do and do it well.
- They have a sense of mission and have the courage to be creative.

-They do not waste needless energy trying to be well-rounded.

GETZEL AND JACKSON THEORY OF INTELLIGENCE

Perhaps the best known research to produce systematic evidence that creativity and intelligence might be independent traits was that of Getzels and Jackson (1962). They studied a group of children and adolescents in a private school in Chicago, whose ages ranged quite widely and whose average IQ was over (In an unselected group less than three per cent would reach this level). Within this sample, they formed two contrasting sub-groups, one highly 'creative', as indicated by aggregate score on a battery of creativity tests, including some originally devised by Guilford and by Cattell, but relatively less 'intelligent'; the other group consisted of lower scorers on the creativity tests but was exceptionally 'intelligent' even by the standards of the whole sample. Getzels and Jackson then compared the two groups on numerous other characteristics, including school

degree of conformity to accepted values, sense of humour, and so on. They found low correlation between creativity and intelligence and thus maintained that intelligence and creativity are different in their existence.

Interesting and suggestive differences

were found in many of these respects; in

particular, the 'highly creative' children

appeared to have a more lively sense of

humour, to come from less 'academic'

families, to hold less conformist values, and to 'over-achieve' in school work

both because of the atypicality of the group studied and because within this relatively to their IQ. But the generality of these findings is open to some doubt, attainment, popularity with teachers, home background, career aspirations,

special group only contrasting and extreme minorities were studied in detail, and the majority of the children whose performance was not so fully described were neither exceptionally 'creative' nor exceptionally 'intelligent but scored more or less equally on both sets of tests.

GUILFORD'S MODEL OF CREATIVITY

Guilford (1986) considered creative thinking as involving divergent thinking, which emphasises fluency, flexibility, originality, and elaboration. Guilford, however, noted that creative thinking is not the same as divergent thinking, because creativity requires sensitivity to problems as well as redefinition abilities, which include transformations of thought, reinterpretations, and freedom from functional fixedness in driving unique solutions. In order to develop Torrance Tests of Creative Thinking (TTCT) and in its further revisions, Torrance

(1966, 1974) has explained six components of creativity. He has described these aspects of creativity in terms of their mode of measurement. These aspects of creativity are:

Fluency: The number of relevant ideas; shows an ability to produce a number of figural images.

Flexibility: Flexibility is the individual's ability to produce not only a large number of responses, ideas or solutions to a problem, but also a variety of responses, ideas or solutions to a problem.

Originality: The number of statistically infrequent ideas; shows an ability to produce uncommon or unique responses.

Elaboration: The number of added ideas; demonstrates the subject's ability to develop and elaborate on ideas.

Abstractness of Titles: The degree beyond labeling; based on the idea that creativity requires an abstraction of thought. It measures the degree a title moves beyond concrete labeling of the pictures drawn.

Resistance to Premature Closure: The degree of psychological openness; based on the belief that creative behaviour requires a person to consider a variety of information when processing information and to keep an "open mind."

(1) Fluency

It refers to a rapid flow of ideas and tendencies to change directions and modify information*. It is the quantitative representation of the units of products*. It emphasises the rate of production of all the units within all classes. Fluency is of four types :

Ideational Fluency - It is the generation or production of ideas where free expression is encouraged and where quality is not evaluated.

Expressional Fluency - It refers to the production of new

ideas to fit a system or logical theories. This facilitates construction of sentences.

Associational Fluency- It indicates production of ideas or words from a restricted area with equal relationships.

Word Fluency - It is the generating of words of specifically required epithets. It is concerned only with words. It has been drawn by divergent production process. Using semantic contents to give a product of units in a table. Various tests to measure word fluency, use prefix, suffix or first or last letters of words.

(2) Flexibility

It is the readiness to change behaviour to meet changing circumstances. It represents a number of classes of objects or trains ideas produced, It indicates in how many distinct different ways an individual can respond to a stimulus. Flexibility is of two types

Spontaneous - It is the production of diversity of ideas in a relatively unrestricted situation.

Adaptive Flexibility - It is some divergent transformation quality which involves changes.

(3) Originality

It refers to the unusual applications of partial uncommonness or newness in the content.

(4). Elaboration

It refers to the expand of higher thought. It shows pro variety of implications and uses quantitatively measured.

(5) Creative Production

It refers to possessing both creativeness and yield. The semantic contents thinking give units and figural content transformations.

(6) Ingenious Solution to Problems

It is inventive, many alternative options. It is the right option from many.

(7) Sensitivity to Problems

It indicates the Creativity for problems, creator sees defects,, needs, deficiencies, unusual ties and sees what must be done. Whether the problem is simple or complex he attacks it from various angles.

(8) Re-definition

It is closely, related to flexibility and originality that arises from transformation, specially of convergent Thinking productions. It is ability to rearrange ideas, concepts, people and things to shift the function of object and use them in new ways. It can be applied to different type of contents in the same way to figural symbolic, semantic etc., and they can be named with their names as figural redefinition, symbolic redefinition etc., thus it is evident that the concept of creativity component emerged from Guilford divergent thinking technology. The components of

A paradigmatic model that regards creativity as the precondition for intelligence is Sternberg

and Lubart's (1995, 1996) , named after

the idea that creative individuals have an

extraordinary ability **investment** to invest

theory in ideas, “buying low and selling

high” (Sternberg & O'Hara, 2000). The

authors also posit that creativity is an

important determinant of intelligent

creativity remain unique in their content-production Their predominant mental operation is of the divergent type.

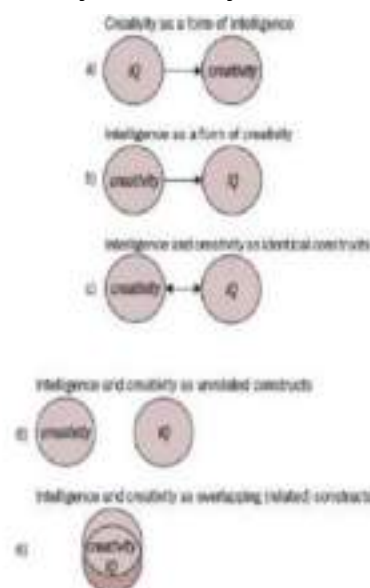
WALLACH AND KOGAN STUDY

Wallach and Kogan[72] highlighted that the creativity measures were not only weakly related to one another (to the extent that they were no more related to one another than they were with IQ), but they seemed to also draw upon non-creative skills. McNemar[101] noted that there were major measurement issues, in that the IQ scores were a mixture from 3 different IQ tests.

Wallach and Kogan[72] administered 5 measures of creativity, each of which resulted in a score for originality and fluency; and 10 measures of general intelligence to 151 5th grade children. These tests were untimed, and given in a game-like manner (aiming to facilitate creativity). Inter-correlations between creativity tests were on average $r = .41$. Inter-correlations between intelligence measures were on average $r = .51$ with each other. Creativity tests and intelligence measures correlated $r = .09$.

CREATIVITY AND INTELLIGENCE

Thinking and intelligent behavior because it enables individuals to “think differently” and “defy the crowd.”



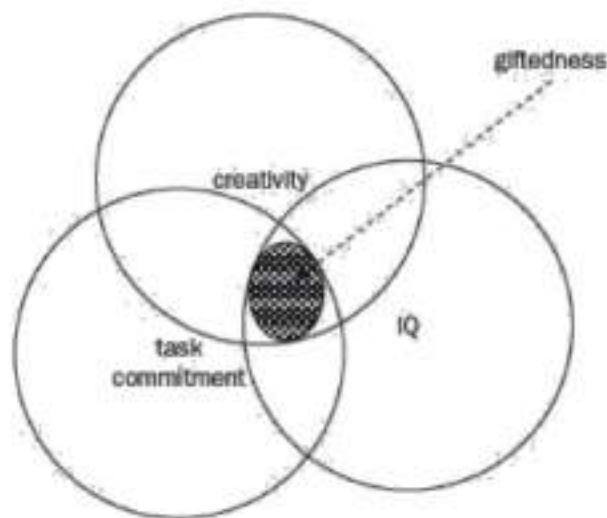


Figure 19.7 Renzulli's (1986) "three-ring" model of giftedness.

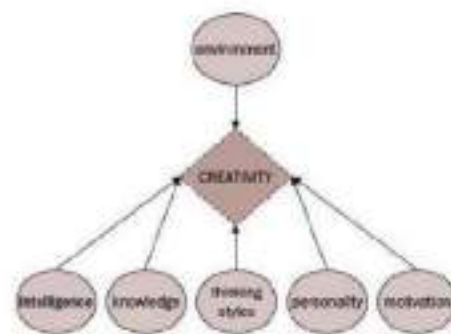


Figure 19.8 Sternberg and Lubart's (1995) model: creativity as a product of intelligence.

This idea, often referred to as the **threshold** **theory of creativity** significantly lower in creativity, but those with higher intelligence scores were neither significantly higher nor lower in creativity. Thus intelligence is necessary but not sufficient for creative thinking (Schubert, 1973).

Hayes (1989) proposed an alternative **'certification hypothesis'**, which doubted intrinsic links between creativity and intelligence. Instead, it stated that most possibilities to display a recognisable level of creativity, like occupations in architecture or science, simply require a high level of formal education. Since academic performance is correlated with IQ, society simply denies creative individuals of low IQ the chance to express their talent adequately.

ASSOCIATIVE THEORY

Associative theory was developed in 1962 by Sarnoff Mednick and falls into the cognitive group of theories as the theory primarily focuses on three types of creativity: serendipitous, similarity and mediation, which refer to cognitive mechanisms we use to generate new ideas. Mednick defined creativity as:

‘...the forming of associative elements into new combinations which either meet specified requirements or are in some way useful [...] the more mutually remote the elements of the new combination, the more creative the process or solution’. (Mednick, 1962).

An important aspect to understand from Mednick’s definition is that creativity arises from the **combination of remote elements**. An example therefore would be the combination of computer networks and libraries to create Google, a library of sorts housed in a world wide network accessible through personal computers.

As mentioned above Mednick proposed three distinct types of creativity into which any idea would likely fall. **Serendipitous creativity** occurs when a combination appears by chance (Example: Larry Page was eating his lunch next to a server and BANG he comes up with the idea for Google). **Similarity** refers to the combination of concepts which may appear remote but do actually share some similarities which prompt us to try combining them (Example: Larry likes libraries and notices that the internet is one big unorganised collection of knowledge and so invents Google to organise the internet, just like a library). Finally, **Mediation** occurs when we make combinations of completely remote elements using a mediatory concept that has links to both of the initial ideas (Example: Larry is inspired to combine the concept of ‘the internet’ and libraries because his second name is Page and both libraries and the internet have pages in them).

CONTENTS

Determinants of personality: Biological and socio-cultural

Approaches to the study of personality: Psychoanalytical, Neo-Freudian, Social learning, Trait and Type, Cognitive, Humanistic, Existential, Transpersonal psychology.

Other theories: Rotter's Locus of Control, Seligman's Explanatory styles, Kohlberg's theory of Moral development.

Basic motivational concepts: Instincts, Needs, Drives, Arousal, Incentives, Motivational Cycle.

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Exploratory behavior and curiosity

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Chapter 7

PERSONALITY, MOTIVATION, EMOTION, STRESS AND COPING

PERSONALITY

Personality is a concept to be used to recognize *stability and consistency of behaviour across different situations, uniqueness of the person and individual differences. There are controversies among psychologists to define personality.* Among the numerous definitions of personality, the most commonly accepted definition is given by Allport (1937). The word personality has been derived from the Latin word '**Persona.**' At first this word was used for the mask worn by the actors (Roman and Greek) in ancient times, - 8 - to indicate to the audiences whether they played the villain's or the hero's role in a drama. Thus the mask gave the actor his characteristic features.

DEFINITIONS

He defines personality as; "Personality is the dynamic organization within the individual of those psychophysical system that determine his unique adjustment to the environment." (Allport-1947) Later on, **Allport** (1965) revised his definition of personality. The revised definition of personality is 'personality is a dynamic organization within the individual of those psychophysical systems that determine his characteristic behaviour and thought.' (Allport-1965)

Personality is : ...a person's unique pattern of traits (**Guilford**, 1959, p.5) ...the most adequate conceptualization of a person's behavior in all its detail." (**McClelland**, 1951, p. 69). - 10 -

"Personality consists of the distinctive patterns of behavior including thoughts and emotions that characterize each individual's adaption to the situations of his or her life" - **Walter Mischel** (1976)

"Personality is the sum of activities that can be discovered by actual observations over a long enough period of time to give reliable information." - **Watson**

"Personality refers to deeply ingrained patterns of behavior, which include the way one relates to, perceives and thinks about the environment and one self." - **American Psychiatric Association-1987**

"An individual's pattern of psychological processes arising from motives, feelings, thoughts, and other major psychological function. Personality is expressed through its influences on the body, in conscious mental life, through the individual's social behavior." - **Mayer**, 2005

"Personality refers to individuals' characteristic patterns of thought, emotion and behavior together with the psychological mechanisms hidden or not-behind those patterns this definition means that among their colleagues in other subfields of psychology, those psychologists who study personality have a unique mandate: to explain whole persons." - **Funder**, D. C., 1997

“The more or less stable and enduring organization of a person’s character, temperament, Intellect, and physique that determines his unique adjustment to his environment.” - H.

Eysenck

“That which permits a prediction of what a person will do in a given situation.” - **R. B. Cattell**

DETERMINANTS OF PERSONALITY

Factors Influencing Personality The factors affecting personality can be divided into - two classes - biological and environmental.

BIOLOGICAL FACTORS

The biological factors affecting the development of personality are **heredity, endocrine glands, physique and nervous system**.

Heredity: The principal raw materials of personality - physique, intelligence and temperament are the result of heredity. How they will develop will depend on environmental influences. Many aspects of human behavior and development ranging from physical characteristics such as height, weight, eye and skin color, hair, the complex patterns of social and intellectual behavior, are influenced by a person’s genetic endowment.

Endocrine Glands

The secretions of endocrine glands affect physical growth, emotional growth and mental growth.

have an impact on the total personality of an individual.

The **thyroid gland** secretes a hormone called thyroxin, and the main function of this hormone is regulation of body metabolism. If the thyroid gland is under active, the result is usually mental dullness, inactivity, depression, fatigue and poor appetite. Hyper secretion of these glands leads to extreme over activity.

The **parathyroid gland** regulates calcium metabolism excitability of the nervous system is directly dependent on the amount of calcium in the blood. Deficient working of this gland leads to the development of an irritable, quick reactive, distracted, nervous and a tense person. Similarly other glands like pituitary, the adrenal and the gonads have their tremendous impact on various personality traits.

Nervous System : Entire behavior is effectively managed and controlled by the coordination and functioning of the nervous system. How we will behave in a particular situation depends upon the judgment of our brain. The sense impressions, which are received through sense organs, do not bear any significance unless they are given a meaning by the nervous system.

ENVIRONMENTAL FACTORS

Family : Among environmental factors, the most important is the family environment. The reaction of the family environment towards an individual, and the role of parents, are very important in the molding of personality, parents serve as a model whom the child imitates, and their influence is considerable on the child. Parents influence the development of child’s personality in a wide variety of ways.

Children learn the moral values, code of conduct, social norms and methods of interacting with others from parents.

On the whole friendly and tolerant fathers help their children to have greater emotional stability, self-esteem And self-confidence. **Domineering and rigid fathers will only foster the development of submissive and frightened, dependent children.**

Over protective mothers will influence their children in the direction of dependence and a total disregard for others. Nagging mothers will cause their children to be shy, submissive and emotionally unstable.

Besides the role of the parents, the atmosphere in the family is greatly influencing. A peaceful and loving atmosphere results in children being orderly, peace-loving and very affectionate. Without undue strain they develop nature and pleasant personalities.

In a family where there is tension, anxiety, constant quarrels and incompatibility among parents, the child is likely to develop strong feelings of insecurity and inferiority.

Birth Order : This is another familial factor that can have an important influence on the personality development. All children has a unique position in the family, such as the eldest, youngest, second or third. This position has a definite influence on personality.

The eldest child is very often overburdened with responsibility, hence, he or she grows up to be very independent, while the youngest being the baby of the family is petted and spoiled. The common view of an only child would be that he or she will be pampered and spoiled.

School : The children spend much of their time in the schools and hence it can play a very significant part in the formation of the personality of the child.

Teacher : A teacher is the most important person in the school who can help in modifying the children's personalities. He or she is the most powerful source of stimulation for the child. If he/she possess desirable personal and - 15 - social modes or reactions he/she will inculcate them among his/her student or the other hand, **effects of prejudicial treatment on the part of teachers can make the child lose self-confidence and develop low self-esteem.**

Peer Group : From their peers children learn many forms of behavior, some socially appropriate and others socially undesirable. For example, **by striving to be accepted and like by their peers, they gain new insights into the meaning of friendship. Through give and take with friends, they learn the importance of sharing, reciprocity and cooperation.** By trying to get peers to understand their thoughts and feelings, they learn to communicate more effectively. Within the friend's group, child also learn sex-role norms. In general, **boys become rougher, boisterous, more compulsive,** and from larger groups, while girls tend to form more intimate and exclusive groups. Feelings of

masculine superiority, sex bias and other attitudes and behaviors develop with gender identification.

Sibling Relationships : The number of sibling as well as their sex and age has a considerable influence on the development of both favorable and unfavorable personality traits like cooperativeness, sharing, aggressiveness, jealousy, etc. Although sibling rivalry is common, older sibling invariably teach the infant a great deal and they can even function as a source of security. On the other hand, unhealthy comparisons can also develop, for instance, **an athletic child who is favored by an athletic father over a less active sibling, may suffer from an inferiority complex or develop low self esteem & self confidence.**

Mass Media : Mass media includes films, television, radio, printed literature, etc. Mass media has a considerable impact on attitudes, values, beliefs and behavior patterns. **Baron and Bryne (1986) have shown that individuals, especially child, imitate specific aggressive acts of models. They have proposed that human personality formation is a result of modeling** and imitating the behavior of significant others. Many abnormal forms of behavior can be learned by imitating models from the mass media.

Culture : Culture influences personality because every culture has a set of ethical and moral values, beliefs and norms which considerably shapes behavior. Cross-cultural studies have pointed out the importance of cultural

environment in shaping our personality.

Individuals of certain cultures are more generous, open-hearted and warm where as individuals of some other cultures are suspicious, introverted and self-centered. It has also been found that certain cultural communities are more prone to develop certain abnormal behaviors as compared to others, probably due to the influence of geographical, dietary, hormonal or genetic influences within the community.

APPROACHES TO PERSONALITY

PSYCHOANALYTIC THEORY

Introduction

It revolutionized the concept of treatment for psychological disturbances by proposing a method of psychological treatment based on psychological concepts such as understanding of the unconscious, instincts, free association, dream interpretation, and so on. The history of modern psychotherapy in fact begins with the work of Sigmund Freud. Third, psychoanalysis is both a theory of personality as well as a system of psychotherapy. As a theory of personality, it aims at providing explanations of human behavior and experience by revealing the underlying mental forces. As a system of psychotherapy its basic goal is to bring to consciousness the unconscious impulses causing neurotic conflict and thereby help the person gain greater self-knowledge and self-control.

Theory of personality

The psychoanalytic theory of personality can be considered from three points of view: physiological, social, and psychological.

Physiological view

Physiological structures present in an organism are referred to as instincts, drives, instinctual drives, sexuality, psychosexuality or libido in the psychoanalytic theory. According to Freud, the physiological drives are more important in childhood than in later life as they exert a significant influence in the formative years. However, Freud was aware of the dichotomy present between drive and control of the instincts – expressed in the form of defenses against anxiety, growing from the pleasure principle to the reality principle, and as the conflict between ego and id.

Social view

The impact of social influences has received due importance in psychoanalysis. Society is seen as a collection of individuals, in which the family is the core of the social structure. Every culture has specific socialization procedures which give rise to conflicts between parents and children over appropriate behavior. These interactions may either be gratifying or frustrating depending on the love or hate within the family structure.

Psychological view

Inner psychological states begin to develop at a very early stage. These inner psychological states manifest themselves in behavior, fantasy and life. A person may become psychologically

sick because of frustrating early experiences which become part of the inner states. These experiences are internalized by the individual and are difficult to dislodge, making the person inaccessible to others' influences. Covert anxiety, which can only be inferred from a total picture of the personality, has been given enormous importance, and studied extensively by psychoanalytic theory.

Concepts of Psychoanalysis

The psychoanalytic theory of personality explains any human event or experience with reference to a number of different points of view. These views are: (1) topographic, (2) genetic, (3) dynamic, (4) economic, (5) structural, (6) interpersonal, (7) cultural, and (8) adaptive.

The **topographic** point of view explains the concept of the **unconscious**. According to this view, a mental event can be unconscious, preconscious, or conscious. The unconscious consists of wishes which are repressed because of anxiety. The presence of the unconscious is felt only when these repressed wishes are expressed in the form of fantasies and dreams, neurotic symptoms, or overt active behavior.

The **genetic-developmental** view is one of the fundamental bases for the understanding of the individual. The development process is described in Freud's psychosexual stages of development. Sexuality is divided into three main periods: from birth to the ages of three to five, from age five to puberty, and puberty, when sexuality reaches its adult form.

The **dynamic** point of view understands the human psyche as an interplay between dynamic emotional forces while the **economic** point of view stresses upon the quantitative factor in mental functioning. It was originally hypothesized that mental energy could be quantified.

The **structural** or **tripartite** view sees the personality as divided into three parts – the **id**, **ego** and **superego**. The **id** is the source of all drives, and is the reservoir of instincts. The **ego** is the executive of the personality, and is the mediator between the demands of the **id** and the external environment. The **superego** consists of moral and social values, and is the internal representative of parents. It acts as the mediator between the individual and the environment.

The **interpersonal** point of view stresses on the interpersonal context of all human activities. In the early years of a person's life, the interpersonal context consists almost entirely of the family, from which all later relationships develop. The essential traits of the personality become fixed by the time the child begins to go to school, by which time the **superego** has formed.

The **cultural** point of view has stressed upon the role that the broader culture plays in the formation and maintenance of the personality structure. Freud revealed that the same psychological mechanisms can be found in all cultures. Thus, it can be deduced that the same needs exist in human beings in all cultures, but are moulded in different ways in different cultures.

The **adaptive** point of view focuses on the need of the human being to adapt himself to his environment. However, psychoanalytic thought is more concerned with the individual's ability

to adapt to other people – the interpersonal and cultural aspect of adaptation is considered the real problem.

View of the individual

A major tenet of psychoanalysis, which Freud developed, is the principle of **psychic determinism**. According to this principle, the human being is seen as an organism driven by unconscious, innate forces. Behavior is determined by previous behavior and by the biological drives. Freud believed that nothing happens by chance or at random and all behavior is meaningful if the person has sufficient insight to understand it. According to Freud, the basic personality pattern is established by the age of five.

Topography of the mind

Freud evolved the concept of the **unconscious** and defined three constructs of the mind: the conscious, unconscious, and preconscious. The **conscious** is all that we are aware of in ourselves and in our environment. The **unconscious** is the bigger area consisting of events, wishes, desires, and impulses that have been repressed or censored from the conscious mind. These mental events are not accessible to the conscious mind. The **preconscious** is the censor that acts on the memories in the unconscious. After a memory is scrutinized, it is allowed to pass into consciousness; if not, it is repressed into the unconscious.

Structure of personality

According to Freud, the personality is made up of three major systems: the *id*, the *ego* and the *superego*. Each of these structures has its own functions, properties, and mechanisms.

Behavior is almost always the product of an interaction between these three systems.

The **id** consists of all aspects that are physiological in nature and inherited – aspects that are present at birth. It is the reservoir of psychic energy and functions to reduce tension. Its sole purpose is to obtain pleasure, and operates on the *pleasure principle*. The id functions by two processes: the **primary process**, present in the unconscious, which seeks to alleviate tension immediately and the **pleasure principle** that makes the organism seek immediate satisfaction of instinctual needs.

The **ego** evolves out of the id as the child develops. The ego operates on the *reality principle* through the **secondary process** that develops at the conscious level of thinking. The reality principle ensures that pleasure is obtained in accordance with the demands of reality. The ego is said to be the **executive** of the personality as it decides what needs should be satisfied and to what extent. The ego also functions to protect the self by employing appropriate Défense mechanisms.

The **superego** is the moral arm of the personality – it represents the *ideal* rather than the real and strives for perfection rather than pleasure. The superego is our *conscience* and develops as a result of the rewards and punishments given by parents. Whatever is taught as improper is incorporated into the conscience and what is approved of is incorporated into the *ego ideal* through *introjection*. This is the process by which parental and societal values are internalized by the individual. The superego functions unconsciously to a large extent, and with its formation,

Self-control takes over from parental control. The main functions of the superego can be summed up as:

- ❑ to inhibit sexual or aggressive impulses of the id, as the expression of these impulses is condemned by society,
- ❑ to persuade the ego to substitute moralistic goals for realistic ones, and, to strive for
- ❑ perfection.

the ego, and function as a whole. The id may be thought of as the **biological** component of personality, the ego as the **psychological** and the superego as the **social** component.



ANXIETY

Anxiety is a state of tension. The external world plays a part in shaping the personality as it can threaten as well as satisfy. When threatened, the ego becomes anxious. The function of anxiety is to warn the person of impending danger and signal the ego that appropriate measures are to be taken. Freud identified three types of anxieties: **reality anxiety**, **neurotic anxiety**, and **moral anxiety**. Reality anxiety is the fear of real dangers in the external world. Neurotic anxiety is the fear that instincts will get out of control and that the individual will be punished for it. Moral anxiety is fear of the conscience; people with well-developed superegos often feel moral anxiety. When the ego cannot cope with anxiety by rational methods it has to fall back upon unrealistic ones called the *Défense* mechanisms.

EGO DEFENSE MECHANISMS

When the ego is undergoing extreme anxiety, it is forced to take extreme measures called *defense mechanisms* to relieve the pressure. These defenses are used to keep the unacceptable impulses of the id out of consciousness and prevent their open expression. All defense mechanisms have two common characteristics:

they deny, falsify, or distort reality, and they operate unconsciously so that the person is not aware of what is taking place. Some of the most common defense mechanisms are described below.

❑ **Repression.** Repression is the ego's refusal to allow a forbidden id impulse or accompanying memories, emotions, desires, or wish-fulfilling fantasies to become conscious. It is the "forgetting" of unacceptable thoughts or impulses - unconscious forgetting. The person who is

repressing a memory is usually not aware that something is forgotten.

❑ **Reaction Formation.** It is the blocking of an impulse that the ego labels dangerous and whose presence causes anxiety. By strengthening and emphasizing the opposite impulse, reaction formation keeps the forbidden impulse out of awareness. This process is also unconscious. For example, hate could appear as a reaction formation against love, if love was viewed as a threat.

❑ **Isolation.** It is the recollection of memories of the past without the accompanying feeling or emotion. In other words, the affect is isolated from the memory. The affect or the emotion is repressed, but the memory of the wish or incident remains conscious. In this way, we block frightening or painful memories of emotion from the consciousness, thereby reducing the threat.

❑ **Undoing.** This is the attempt to "undo" an act or impulse from the id that the ego considers dangerous, such as a hostile or sexual act. For example, a four-year-old hits his younger sister and then hugs her to "undo" the harm.

❑ **Denial.** When certain unpleasant or unwanted aspects of the outside world's reality are blocked from the consciousness, denial is in effect. Thus, a person who believes that he is a bright, gifted student denies the external reality of his poor school grades.

❑ **Projection.** This is the act of not being aware of one's own wish or attitude and instead, attributing it to some other person or external object. The child who is jealous of a classmate might explain to his friends how that particular classmate is jealous of *him* because of his toys, or his school grades, etc.

❑ **Regression.** It occurs when the ego is faced with severe conflicts with the id impulses. The ego may regress to a previous stage where the sense of equilibrium is maximized and sense of anxiety is minimized. The points to which a person regresses are called *points of fixation*.

❑ **Sublimation.** This is the unconscious procedure of accepting a forbidden impulse through the secondary process and turning it into a related, yet socially acceptable activity that gratifies the basic impulse. It is commonly believed that many of our aggressive drives are sublimated into acceptable sporting activities. For example, a very aggressive person might sublimate his aggression and become a boxer. Sexual drives are sublimated into activities such as working, seeking power, influence and money.

❑ **Rationalization.** It is the process of organizing facts, attitudes, and beliefs into an explanation for an individual's behavior that is believed to be far more acceptable both socially and personally. The facts are usually distorted and organised to support the individual's behavior or beliefs. A student, for instance, who has failed may rationalize his failure by his

belief that the teachers were not fair in evaluation.

❑ **Displacement.** It is the purposeful and unconscious shifting from one object to another in order to solve a conflict. Although the object is changed, the impulse and its aim remain unchanged. For example, the individual who has been reprimanded by his boss in the office may shout at his wife once he reaches home, displacing his anger toward his boss onto his wife.

❑ **Intellectualization.** This is a systematic manner of thinking where the affect is removed from the event or situation in order to defend against anxiety caused by these unacceptable impulses. By merely thinking about them, instead of experiencing them, the person tries to avoid the negative associations of the impulses.

Defenses have further been classified into four types according to the level of adaptation and use. **Narcissistic** defenses are used by children and psychotics. **Immature** defenses are used by adolescents and are seen in depression, obsessions, and compulsions. **Neurotic** defenses are seen in adults under stress and can be observed in obsessive-compulsive and hysteric persons. **Mature** defenses are normal adult adaptive mechanisms.

PSYCHOSEXUAL STAGES OF DEVELOPMENT

Psychosexual stages of development are innately determined stages of sexual development through which all individuals pass. According to Freud, the first few years of life are decisive and extremely important for the formation of the personality. Freud introduced the concept of **infantile sexuality** when he described the five stages of development: the *oral, anal, phallic,*

latency and *genital* stages. These stages are not discrete; as development occurs one stage merges with the other, producing a smooth transition.

Oral Stage

The oral stage occurs at birth and lasts about one and a half years. The mouth is center of gratification. The primary sexual organ is the mouth as it seeks to suck, bite, and put objects in the mouth as its source of pleasure. When gratification is thwarted, the person may become fixated at the oral stage and may develop into either of two types of personality: **oral aggressiveness** is shown by sarcasm and argumentativeness, and biting, chewing and destroying; and **oral eroticism** where in the person will swallow almost anything, and acquires pleasure in amassing knowledge or possessions.

ANAL STAGE

This stage extends from 1-3 years approximately, and the anus becomes the central area of sexual tension and pleasure. The child has to learn to control the pleasure that follows from relieving anal tension. If the parents are very strict and repressive in their methods, the child may hold back its feces and become constipated and will develop into an **obstinate** and **stingy** personality. If the parents plead and cajole the child, and praise it on expulsion of faeces, the child understands the importance of bowel movement and may develop into a **creative** and **productive** adult personality.

PHALLIC STAGE

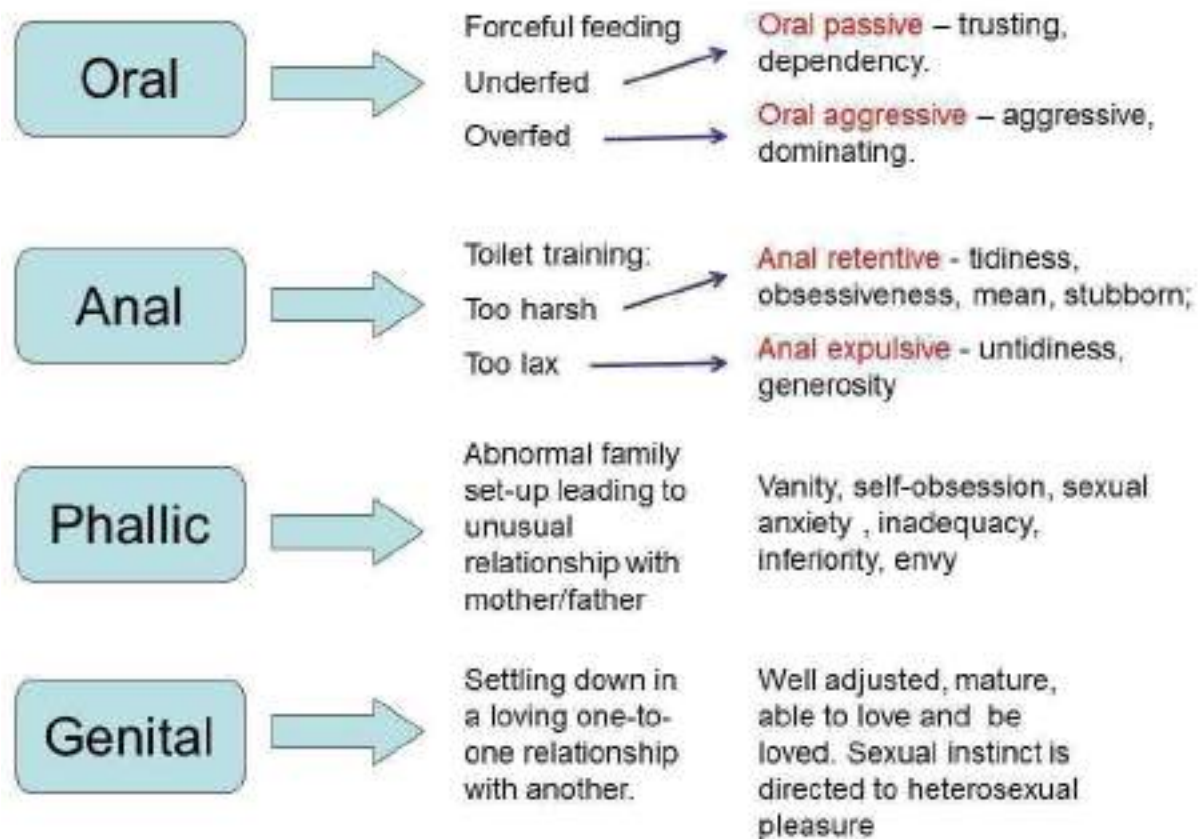
In this stage of personality development, sexual interests, stimulation and excitement of the genital area become primary. The **Oedipus complex** appears in this stage. The Oedipus complex is a sexual attraction for the parent of the opposite sex and hostile feelings for the parent of the same sex. Boys experience **castration anxiety** – the fear of the removal of his genital organs – and girls experience **castration complex** or penis envy – she does not have something which the boy has. The repression of the Oedipus complex causes the superego to undergo the final development.

LATENCY STAGE

The above mentioned three stages are known as the pregenital stages. The child then goes into a prolonged latency period, or the quiet years, as there is a repression of sexual urges. In this period, the child is more involved in same sex friendships, learning and play activities, exploring the environment, and learning how to deal with the adult world. This is a crucial period for developing essential skills

GENITAL STAGE

The genital stage extends from the onset of puberty till the person reaches young adulthood. The adolescent begins to love others for altruistic purposes and sexual attraction, socialization, group activities, vocational planning, and preparations for marrying and raising a family begin to appear. The person becomes a reality-oriented and socialized adult. The principal biological function of the genital stage is that of reproduction. The final organization of personality represents contributions from all stages of development.



THEORY OF INSTINCTS

Freud classified instincts as *life* instincts (**Eros**) and *death* instincts (**Thanatos**). An *instinct* is an inborn psychological representation of an internal bodily arousal. The psychological representation is called a *wish* and the physiological arousal arising from it is called *need*. According to Freud, these instincts represent the forces that underlie the sexual and aggressive drives.

turned outward onto substitute objects. A person fights with other people and is destructive because the death wish is blocked by the forces of the life instincts and other obstacles in the personality.

LIFE INSTINCTS

Hunger, thirst and sex are life instincts as they serve the purpose of individual survival. The energy of life instincts is called *libido*. Life instinct refers to the tendency of particles to unite, as in sexual reproduction.

DEATH INSTINCTS

Freud stated that “the goal of all life is death”, and defined Thanatos as the tendency of organisms and their cells to return to an inanimate state. An example of the death instinct is the *aggressive drive*. Aggressiveness is self-destruction

JUNG'S ANALYTIC THEORY

INTRODUCTION

Carl Jung's psychoanalytic school, known as **analytical psychology**, includes basic ideas related to Freud's theories. He expanded on Freud's concept of the unconscious by describing the collective unconscious as consisting of all humankind's common and shared mythological and symbolic past. Jung felt that modern humans have been shaped and moulded into their present form by the cumulative experiences of past generations. An individual's Centre for Distance personality is a resultant of inner forces acting upon outer forces and outer forces acting on inner forces.

STRUCTURE OF PERSONALITY

Jung described the total personality or psyche as consisting of a number of different but interacting systems. The main systems are the *ego*, the *personal unconscious*, and its *complexes*, the *collective unconscious* and its *archetypes*, the *persona*, the *anima* and *animus* and the *shadow*.

THE EGO

The ego is the conscious mind. It is made up of conscious perceptions, memories, thoughts, and feelings. The ego is responsible for one's feeling of identity and continuity, and from the individual's viewpoint, is considered as being at the centre of consciousness.

THE PERSONAL UNCONSCIOUS

The personal unconscious consists of experiences that were conscious at one time, but have been repressed, suppressed, forgotten or ignored. It also consists of the experiences that were too weak to make an impact

on the conscious person. The contents of the personal unconscious are accessible to consciousness, and there is a lot of interaction between the ego and the personal unconscious. **Complexes.** A complex is an organized group of feelings, thoughts, perceptions and memories that exist in the personal unconscious. The *mother complex*, for example, consists of ideas, feelings, and memories related to the mother. Thus, an individual whose personality is dominated by the mother is said to have a strong mother complex.

THE COLLECTIVE UNCONSCIOUS

It is the most powerful and influential system of the psyche (personality) and has the ability to overshadow the ego and the personal unconscious. The collective unconscious is made up of latent memory traces inherited from one's ancestral past. This unconscious is the remainder of human evolutionary development that has accumulated over many generations. It is universal in nature and is detached from anything personal. All human beings have more or less the same collective unconscious.

ARCHETYPES. The structural component of the collective unconscious, an archetype is a universal idea that contains a large element of emotion. Archetypes are the images that predispose us to perceive the external world in certain ways. For example, the archetype of mother produces an image of a mother figure that is then identified with the actual mother. The baby's experience is the joint product of an inner predisposition to perceive the world in a

certain manner and the actual nature of the world. Some of the archetypes we have in our collective unconscious are birth, power, death, magic, hero, child, God, etc. Some of the important archetypes are:

The Persona. This is a mask adopted by the person in response to the demands of social convention and tradition and to his or her own inner needs. It is the role assigned by society and often conceals the real nature of the person. The persona is the public personality which the world sees, while the private personality exists behind the mask.

The Anima and Animus. Animus is the masculine side of females and anima is the feminine side of males. These archetypes are the products of racial experiences of man living with woman and vice-versa.

The Shadow. This archetype consists of the animal instincts that humans inherited in their evolution from lower forms of life. The shadow represents the animal side of human nature and is responsible for the unpleasant and socially unacceptable thoughts, feelings, and actions in our consciousness and behavior.

The self. It is the mid-point of personality and all systems are built around it. The self provides the personality with unity, equilibrium and stability. The self is the goal that people constantly strive for, but cannot reach. It motivates human behavior and becomes evident only when a person has reached middle age.

INTROVERSION-EXTRAVERSION

Jung also suggested that we are all born with innate tendencies to be concerned mostly with ourselves or with the outside world. The extravert is oriented toward the external, objective world and is open, confident and takes part in many activities. The introverted person is oriented toward the inner, subjective world and is hesitant, cautious and prefers to observe the world than get involved.

DEVELOPMENT OF PERSONALITY

Jung emphasized on the forward going character of personality development – humans are constantly progressing or attempting to progress from a less complete stage of development to a more complete one. What is the goal of development? In Jung's theory, the ultimate goal is *self-realization* or *individuation*. Self-realization is the blending of all aspects of a human's total personality. It means that the psyche has evolved into the *self* in place of the *ego*.

CAUSALITY VS. TELEOLOGY

The idea that a goal guides and directs human destiny is the basis of the teleological viewpoint. According to this viewpoint, human personality can be explained in terms of where it is going, not where it has been. It explains the present in terms of the future. On the other hand, the present may be explained by the past. This is the viewpoint of causality which holds that present events are the consequences or effects of antecedent conditions or causes. A look into the person's past will account for his or her present behavior. Jung believed that both standpoints are necessary for a complete understanding of personality. The present is not only determined by the past (causality) but it is also determined by the future (teleology). When the two views are combined, we get a complete picture of the person.

PROGRESSION AND REGRESSION

Development may follow either a progressive forward movement or a regressive, backward movement. By progression, Jung meant the satisfactory adjustment of the conscious ego to the demands of both external environment and the needs of the unconscious. When this progression is thwarted by a frustrating circumstance, the libido is unable to handle the environment, as a result of which, it regresses into the unconscious and indulges in introverted values. The ego loses its objectivity and becomes subjective. However, regression does not necessarily have a permanently bad effect on adjustment. It may help the ego to find a way around the obstacle and move forward again. For example, a young adult, living on his own may face a problem which he is unable to handle by himself. Though he has left his parents, he will still turn to them for help and advice, or rather the parental images in his unconscious.

THE INDIVIDUATION PROCESS

The central feature of Jung's psychology is that personality has a tendency to develop in the direction of a stable unity. The ultimate goal of development is the realization of selfhood. In order to realize this aim it is necessary for all the various systems of personality to be fully developed. A neglected part will offer resistance and sap energy from other developed systems. If too many resistances develop, the person becomes neurotic. This usually happens when archetypes are not allowed to express themselves through the conscious ego, as the persona has smothered the personality. To have a healthy integrated personality, every system must be allowed to reach the fullest degree of

differentiation, development and expression. This is known as the *individuation process*.

ADLER'S INDIVIDUAL PSYCHOLOGY

Alfred Adler expanded on Freud's theories and formed *Individual Psychology*. Opposing both Freud's assumption of inborn instincts, and Jung's inborn archetypes as the factors motivating behavior, Adler suggested that humans are motivated primarily by **social urges**. According to him humans are inherently social beings.

His second major contribution was his concept of the **creative self**. This self is a highly personalized, subjective system that interprets and makes the experiences of the organism meaningful.

Thirdly, Adler considered the personality to be **unique**. Each person is a unique configuration of motives, traits, interests, and values; and each act of the individual has a distinctive style.

Lastly, Adler considered consciousness to be the centre of the personality, and considered that humans are conscious and aware of their behavior.

Adler's theory of personality consists of the following concepts: **striving for superiority, inferiority feelings and compensation, social interest, style of life and the creative self**.

STRIVING FOR SUPERIORITY

Superiority denotes a striving for perfect completion. This striving is innate, and is part of life. This striving for superiority carries the person from one stage of development to the other right from birth to death. Each person has his or her own way of achieving or trying to achieve perfection. Adler identified power with masculinity and weakness with femininity. In line with this suggestion, he spoke of “masculine protest”, a form of overcompensation that both men and women indulge in when they feel inadequate and inferior. Superiority in this context does not mean social distinction, leadership, or an eminent position in society. What causes these various modes of striving to come into the individual?

INFERIORITY FEELINGS AND COMPENSATION

According to Adler, the causes of the striving for superiority can be traced to inferiority feelings and compensation. Adler observed that a person with a defective organ tries to compensate for the weakness by strengthening it through intensive training. He explained that feeling of inferiority arise from a sense of incompleteness or imperfection in any sphere of life. Adler was of the opinion that inferiority feelings are not a sign of abnormality, but are the cause of all improvement in mankind. In other words, human beings are pushed by the need to overcome their inferiority and pulled by the desire to be superior. Perfection, not pleasure, was the goal of life.

Social Interest

Adler believed that social interest is inborn; that humans are social creatures by nature, not by habit. However, social interest can be brought to a culmination

only with training and guidance. By definition, social interest consists of the individual helping society to attain the goal of a perfect society. The person is embedded in a social context from the first day of life, and is continuously involved in a network of interpersonal relations that shape the personality and provide outlets for striving for superiority. Striving for superiority becomes socialized, and by working for the common good, humans compensate for their individual weakness.

STYLE OF LIFE

Style of life is the principle by which the individual personality functions; it is the principle that explains the uniqueness of the person. Everyone has a style of life, but no two people develop the same style. Style of life is formed very early in childhood, and experiences are taken in according to this unique style of life. What determines an individual's style of life? Style of life is a compensation for a particular inferiority; the dull child will strive for intellectual superiority. For example, Napoleon's conquering style of life was determined by his slight physical build. Adler however, was not satisfied with this concept, and came up with the concept of creative self.

THE CREATIVE SELF

The theory of creative self asserts that humans make their own personalities. They construct them out of the raw material of heredity and experience. The creative self is the catalyst which transforms raw data into a personality that is subjective, dynamic, unified, personal, and uniquely stylized. The creative self gives meaning to life and is the goal as well as the means to the goal. It is the active principle of human life, and is unlike the soul.

ERIKSON'S CONTEMPORARY PSYCHOANALYTIC THEORY

INTRODUCTION

Erik Erikson's most significant contribution is his formulation of the *psychosocial theory of development* from which he expanded the conception of the ego. Erikson's view of psychosocial development brings together a number of important ideas for the understanding of personality growth. Building on Freud's psychosexual stages, Erikson added ego development (role of sensorimotor and cognitive capacities), and interpersonal interactions. Thus, he combined the concepts of dynamic motivation, ego functioning and social behavior into a single model of personality development.

THE PSYCHOSOCIAL THEORY OF DEVELOPMENT

Development proceeds in eight stages according to Erikson. The first four stages occur during infancy and childhood, the fifth stage during adolescence, and the last three stages during the adult years up to and including old age. Erikson places particular emphasis on the adolescent period as it is the transition stage between childhood and adulthood.

Erikson felt that each child has its **own timetable**, and thus, it is very difficult to frame a strict chronological schedule. Further, each stage is not passed through, and left behind. Each stage contributes to the formation of the total personality. At each stage, the maturing person faces new and important encounters with his world

(developmental tasks) in which his growing abilities are tested. The resolution of each task provides a base for further growth; unsolved developmental crises block further development and may lead to neurotic residuals in the later character structure. Erikson also describes **ritualizations** that are peculiar to each stage. By this he means a playful, yet culturally determined way of doing or experiencing something in the daily interplay of individuals. The basic purpose of these ritualizations is to turn the maturing individual into an effective and familiar member of the community.

KAREN HORNEY: BASIC ANXIETY

Horney emphasized on the importance of social relationships in personality development. **Basic anxiety** refers to the feeling of a child of being isolated and helpless in a potentially hostile world.

ERICH FROMM

Erich Fromm was a neo-Freudian psychoanalyst who suggested a theory of personality based on two primary needs: **the need for freedom and the need for belonging**. He suggested that people develop certain personality styles or strategies in order to deal with the anxiety created by feelings of isolation. Of these character types, he suggested that four of them are unproductive orientations, while one is a productive orientation. Fromm believed that character is something that stems both from our genetic inheritance and from our learning experiences.

Stages of Erikson Development Theory

Approximate Age	Virtues	Psycho-social crisis	Significant relationship	Existential question	Examples
Infancy 0-2 years	Hope	Basic trust vs. mistrust	Mother	Can I trust the world?	Feeding, abandonment
Early childhood 2-4 years	Will	Autonomy vs. shame and doubt	Parents	Is it okay to be me?	Toilet training, clothing themselves
Preschool age 4-5 years	Purpose	Initiative vs. guilt	Family	Is it okay for me to do, move and act?	Exploring, using tools or making art
School age 5-12 years	Competence	Industry vs. inferiority	Neighbours, school	Can I make it in the world of people and things?	School, sports
Adolescence 13-19 years	Fidelity	Identity vs. role confusion	Peers, role model	Who am I? Who can I be?	Social relationships
Early adulthood	Love	Intimacy vs. isolation	Friends, partners	Can I love?	Romantic relationships
Adulthood 40-64 years	Care	Generativity vs. stagnation	Household, workmates	Can I make my life count?	Work, parenthood
Maturity 65-death	Wisdom	Ego integrity vs. despair	Mankind, my kind	Is to okay to have been me?	Reflection on life

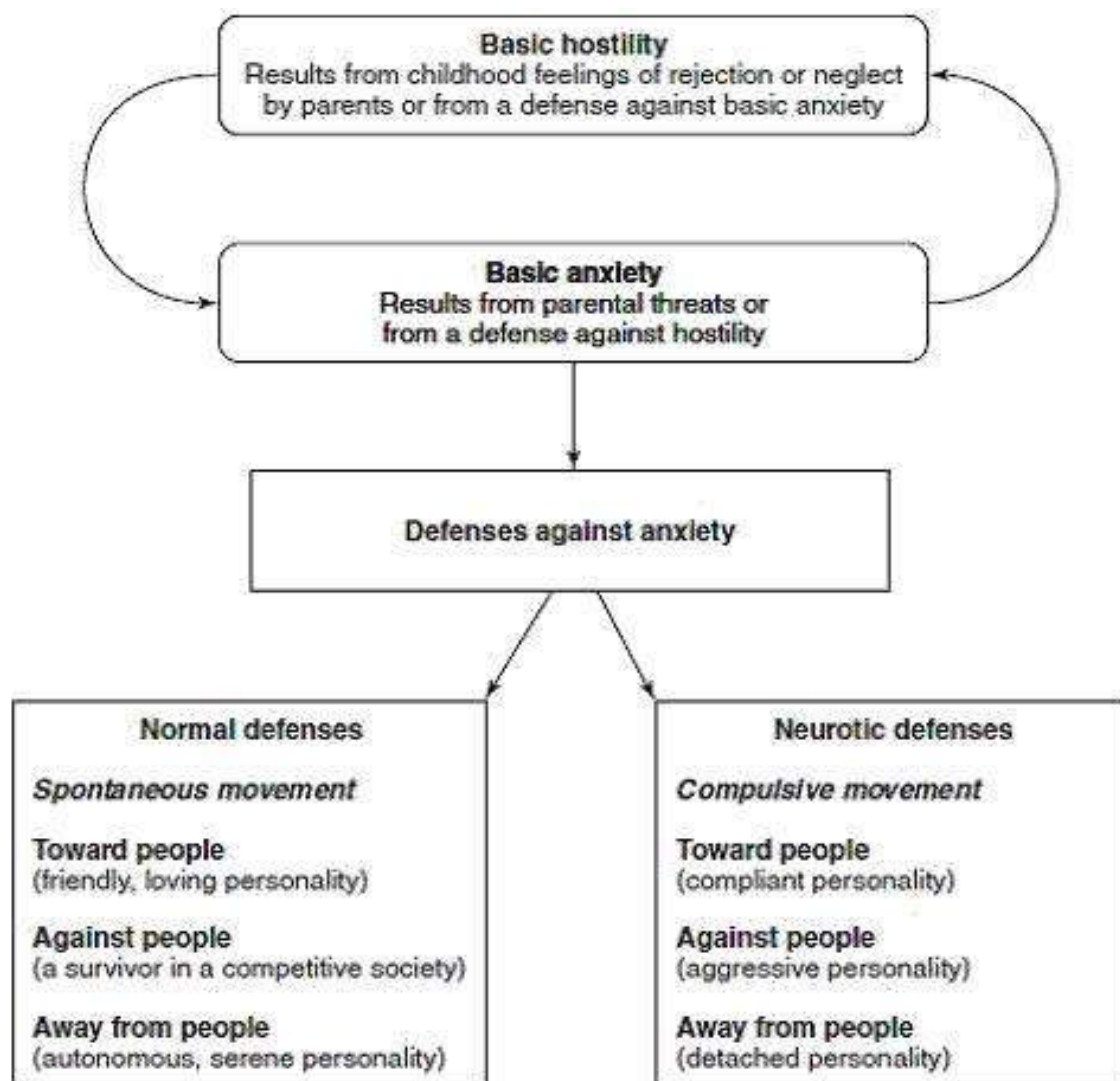


FIGURE 6.1 *The Interaction of Basic Hostility and Basic Anxiety with the Defenses against Anxiety.*

TYPE THEORIES

SHELDON'S SOMATOTYPE THEORY

The type approach has mainly

evolved out of the medical sciences.

It makes the following assumptions.

- a. People can be classified into a few categories or types depending on their behaviour patterns.
- b. These types or categories are qualitatively ^{characterized} different from each other.
- c. The behavioural variations among the different types are stable, describable and even measurable.
- d. It is possible to relate these behavioural types to constitutional and body characteristics.

Sheldon set out with the intention of establishing a relationship between body type and psychological or temperamental types. The human body

consists of three layers, the **ectoderm**, the **mesoderm** and the **endoderm**.

According to him an ectomorphic individual tends to be **cerebrotonic** in temperament. Cerebrotonics are

characterized by greater nervous and cerebral activities and

are given to activities like thinking, reading, etc, The endomorphic physique was associated with a **viscerotonic**

temperament likely to be more

interested in visceral activities like eating and drinking. Lastly, the

mesomorphic physique is associated with the **somatotonic** temperament, being more given to muscular activity.

Sheldon's Somatotype	Character	Shape	Sample Picture
Endomorph [viscerotonic]	relaxed, sociable, tolerant, comfort-loving, peaceful	plump, buxom, developed visceral structure	
Mesomorph [somatotonic]	active, assertive, vigorous, combative	muscular	
Ectomorph [cerebrotonic]	quiet, fragile, restrained, non-assertive, sensitive	lean, delicate, poor muscles	

HIPPOCRATE'S TYPOLOGY

In 400 B.C. Hippocrates attempted to explain personality in terms of body fluid or humours. He postulated that our body has **four types of fluid; yellow bile, black bile, blood and phlegm**. Every person is characterised by the prominence of one type of fluid which determines the temperament of the person concerned. Thus, he classified people into four types which are given below:

Choleric – people with predominance of yellow bile are irritable, restless and hot blooded.

Melancholic – people with high black bile are sad, depressed and devoid of hope in life

Sanguinary – When blood content is high the person remains cheerful, active and he is optimistic in life.

Phlegmatic – predominance of phlegm makes a person calm and quite and usually their behaviour is marked by inactiveness.

JUNG'S TYPOLOGY

Jung postulated personality theory based on psychological characteristics. He divided people into two broad types i.e. extroverts and introverts.

Extroverts – Such people are socially oriented. They like to mix up with people, are fun loving, optimistic. They are realistic in their approach towards life. Often such people exhibit leadership qualities.

Introverts – They are the opposite of extroverts. They do not like to mingle with people. They have very few friends. They are self-centred and conservative. Such people are dogmatic in the sense that they follow traditions and customs of the society without ever giving thought to their justifiability.

Jung's classification has been criticised on the ground that it is not possible to divide people into two watertight compartments because a significant number of people do not fall in either of the categories. They exhibit characteristics of both the extrovert and introvert type. To compensate for this psychologist placed such persons into another category called **ambiverts**.

TRAIT THEORIES

According to trait theory personality is made up of different traits. Traits are the building blocks and human behaviour can be described in terms of these traits. **A trait** generally is a description of behaviour, for example, friendliness, social, assertive are words that describe human behaviour.

Now the question is do these words really describe behaviour? No, because to be called a trait it must manifest consistency. For example, if a person shows assertiveness in all situations he is said to show consistency but if he does not show assertiveness when the situation demands it he is not showing consistency. Besides consistency of a trait should also show the characteristics of stability and relative permanency. In other words, the trait should remain manifest itself in the person's behaviour for long periods of time. For example, if a person shows honesty in his behaviour across all situations for a month or so but after that honesty is not reflected in his behaviour, then the trait of honesty does not have stability.

According to Atkinson, Atkinson and Hilgard **"A trait refers to any characteristic that differs from person to person in a relatively permanent and consistent way."**

ALLPORT'S TRAIT THEORY

Allport mentioned two types of traits ; common traits and personal traits.

Common Traits – are the traits found in the majority of persons living in a society or culture. Thus people of a society or culture can be compared on that trait. Common traits thus are those which are reflected in the behaviour of most of the persons in a society or a community or culture.

Personal Traits – This refers to the unique characteristics of a person and not shared by other members of the society or community or culture. Such a personal trait is not comparable with those of others in that culture. These traits are inculcated by a person more in the process of socialisation and thus many of do's and don'ts of the parents or caregivers become part of the personality and these traits are unique to this individual. Another important aspect is that, these traits are highly consistent and can be seen in almost all behaviours of this individual irrespective of the situation concerned. To give an example, trait of parsimony, is something which an individual will show in almost every aspect of his behaviour whether he is at home or office or school or anywhere. He will for example put off the lights to economise on electricity consumption whether at office or at home. Similarly he would be very careful not to waste paper and will use for rough work one side pages in the office as well as at home.

Allport further divided personal traits into three subcategories: a) cardinal

dispositions,
b) central dispositions, and
secondary dispositions.

Cardinal dispositions: such traits have overriding and overwhelming influence on the behaviour of a person in that they manifest themselves in all that a person does and guides the behaviour of that person. For example, Mahatma Gandhi had firm belief and conviction in peace and non-violence. Message of peace and nonviolence were explicitly seen in whatever Mahatma Gandhi did in his life, whether at home or abroad.

Central dispositions: This is found in all persons and one can have 5 to 10 central dispositions. These are not equivalent to cardinal traits but one can assess the personality of an individual in terms of these traits. These traits actually define the personality of a person. Let us take an example of a person who has the traits of honesty, punctuality, parsimony, cleanliness and generosity. Such a person will be always on time to the office, and keep the scheduled meetings on time and never will waste anyone's time, will be always straight forward and deal directly with his employer

and employees, and would ensure that nothing is wasted and will make sure others do not waste anything and when ever someone comes for help would be generous enough to offer help and solve the problem.

c) Secondary dispositions:

These traits of a person are less consistent, less explicit and less meaningful for the person and hence are called secondary traits. These traits are of not much help in explaining the personality. For example, hair style, dressing sense, eating pattern or preferences etc.

CATTELL'S TRAIT THEORY

After Allport, major contribution to trait theory was made by R.B. Cattell. He divided traits into two categories, viz., surface traits, and source traits.

Surface Traits – As the name suggests these are found on the periphery of personality i.e. these are reflected in the day to day interactions of the person. Their expression is so explicit that it leaves no doubt about their existence in the personality.

Source Traits – These represent the structure of personality. They are present in less number than surface traits. These traits are not observable in day to day interactions of the person. Source traits come to notice when some of the surface traits are joined together. For example, sociability, unselfishness and humor are surface traits which when joined together create a source trait known as friendliness.

Cattell mentioned two types of source traits i.e. **environmental mould traits** and **constitutional traits**. In the development of some source traits environmental factors play more important role than genetic factors, therefore, such traits are called environmental mould traits. Source traits determined by genetic factors are called constitutional traits.

Cattell also divided traits **according to the behaviour** they are related to and these include the dynamic traits and ability traits.

Dynamic Traits are those which direct the behaviour of the person in a particular direction. Attitude and sentiments are examples of dynamic traits. For example, suppose a person feels strongly for girls education / women's education, then he may join an NGO or an organisation which is concerned with spreading women's education and offering education to girls and women. This trait will be reflected in all his activities in terms of writing articles in the newspapers and various other magazines advancing the cause of women and girls education.

Ability Traits – traits that are instrumental in reaching to a goal are called ability traits. For example, musical ability is a must for becoming a musician.

Temperamental Traits – These develop out of a person's efforts to reach a goal and relate to emotional state and energy of the person.

EYSENCK'S TRAIT THEORY

H.J. Eysenck proposed that personality could be reduced into two broad dimensions. These dimensions are **Neuroticism, and Extraversion-Introversion dimensions**. According to Eysenck, these are biologically and genetically based and each dimension subsumes under it a number of specific traits.

Neuroticism: Neuroticism is the name

Eysenck gave to a dimension that had persons who tend to be quite “nervous.”

These people tend to suffer more frequently from a variety of “nervous disorders”, hence the name of the dimension. But it does not mean that people who score high on the neuroticism scale are necessarily suffering from neurotic disorders, but it only shows that such persons are relatively more susceptible to develop neurotic problems as compared to the normal persons.

Extraversion-introversion: His second dimension is extraversion-introversion. By this he means something very similar to what Jung meant by the same terms, and something very similar to our

common sense understanding of them,

that is Shy, quiet people “versus” out-

going, loud people. This dimension, too,

is found in everyone, but the

physiological explanation is a bit more complex. According to Eysenck, extraversion-introversion is a matter of the balance of “**inhibition**” and “**excitation**” in the brain itself.

Excitation is the brain waking itself up, getting into an alert, learning state.

Inhibition is the brain calming itself down,

either in the usual sense of relaxing and

going to sleep, or in the sense of protecting itself in the case of

overwhelming stimulation. **Thus someone who is extraverted**

, he hypothesized, has good, strong inhibition: When confronted by traumatic stimulation

such as a car crash — the extravert’s brain inhibits itself, which means that it becomes “numb,” you might say, to the trauma, and therefore will remember very little of what happened. After the car crash, the extravert might feel as if he had “blanked out” during the event, and may ask others to fill them in on what happened. Because they don’t feel the full mental impact of the crash, they may be ready to go back to driving the very next day. **The introvert, on the other hand,**

has poor or weak inhibition: When

trauma, such as the car crash, hits them,

their brains don’t protect them fast

enough, don’t in any way shut down.

Instead, they are highly alert and learn

well, and so remember everything that

happened. They might even report that

they saw the whole crash “in slow

motion!” They are very unlikely to want to

drive anytime soon after the crash, and

may even stop driving altogether.

Neuroticism and extraversion-introversion

Another thing Eysenck looked into was

the interaction of the two dimensions and

what that might mean in regard to various

psychological problems. He found, for

example, that people with phobias and

obsessive-compulsive disorder tended to

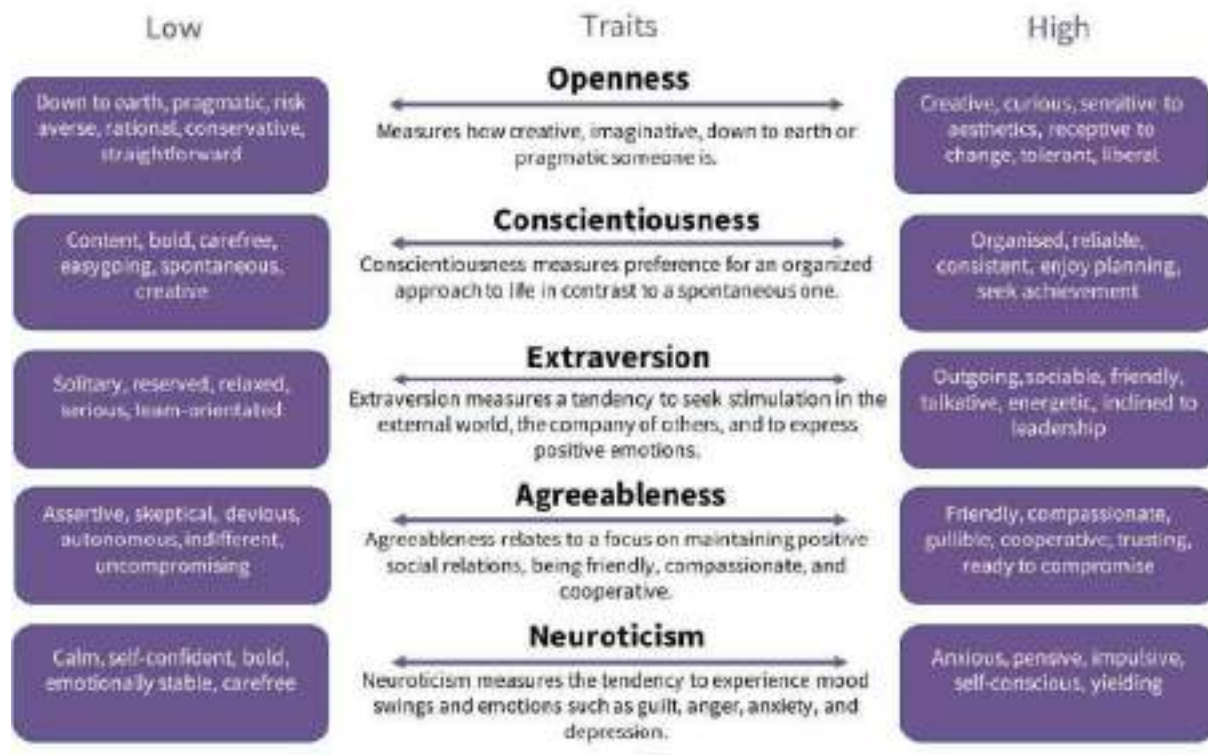
be quite introverted, whereas people with

conversion disorders (e.g. hysterical

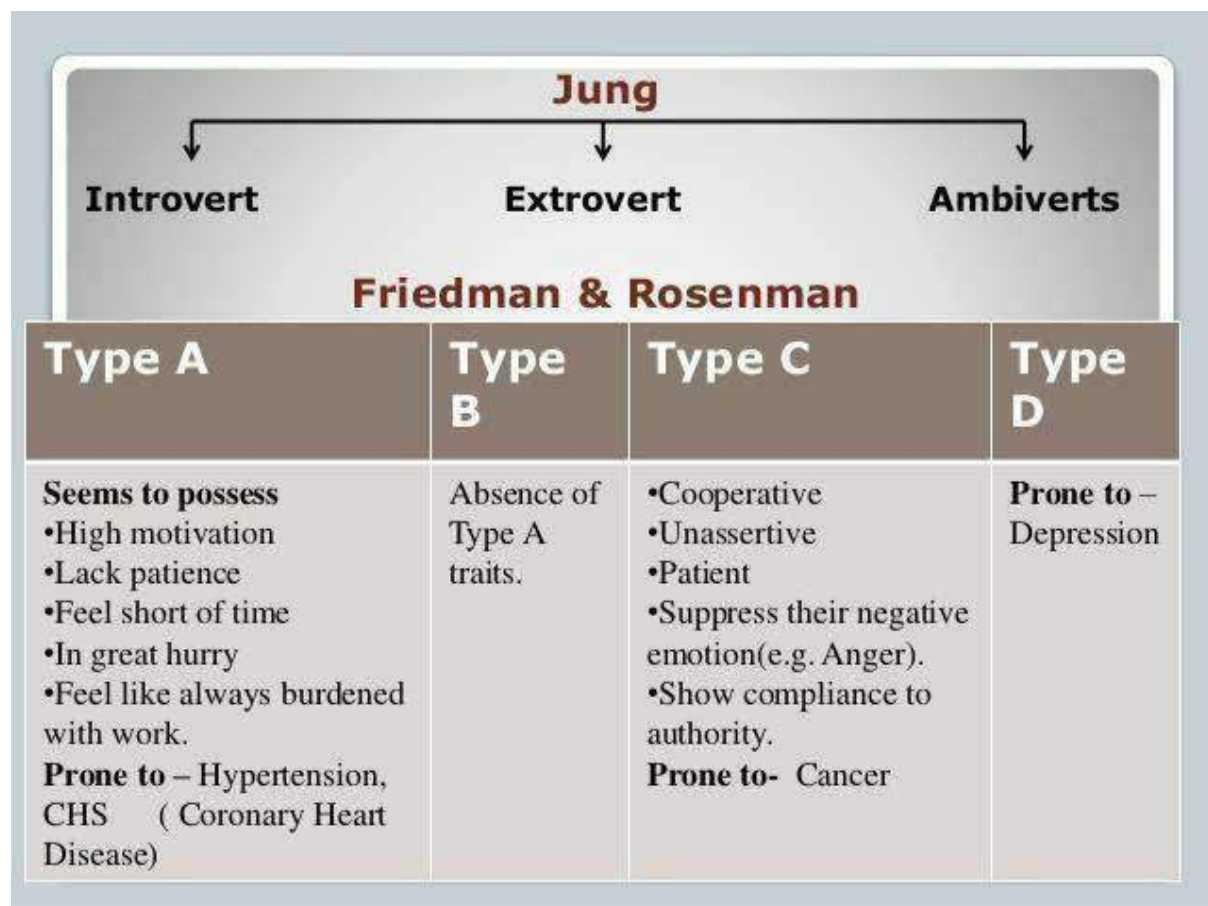
paralysis) or dissociative disorders (e.g.

amnesia) tended to be more extraverted.

OCEAN MODEL BY COSTA AND MC CRAE



PERSONALITY TYPES BY FRIEDMAN AND ROSENMAN



SOCIAL COGNITIVE THEORY

Social cognitive theory is basically a social learning theory based on the ideas that people learn by watching what **others** do and that **human thought processes**. People's interaction with the **environment, behaviour of others, and one's own cognition** etc. act as chief factors in influencing the development of personality of an individual.

Main Tenets of Social Cognitive Theory
People learn by observing others, a process known as vicarious learning. Although learning can modify behaviour, people do not always apply what they have learned.

Individual's choice is based on perceived or actual consequences of the concerned behaviour.

People are more likely to follow the behaviours modeled by someone with whom they can identify. The more perceived commonalities and/or emotional attachments between the observer and the model, the more likely the observer will learn from the model.

Also, the degree of self-efficacy that a learner possesses directly affects his or her ability to learn. Self-efficacy is a fundamental belief in one's ability to achieve a goal. If a person believes that he or she can learn new

behaviours, that would make the person much more successful in doing so.

ALBERT BANDURA'S SOCIAL LEARNING THEORY

"In social learning theory, **a self-system** is not a psychic agent that controls behaviour. Rather it refers to cognitive structures that provide reference mechanisms to set of functions for perception, evaluation and regulation of behaviour". According to Bandura the three component processes involved in self regulation of behaviour through the activation of self-prescribed contingencies. **The three components involved in self-system** are self-observation, judgmental processes and self-response.

Self-observation: We look at ourselves, our behaviour, and keep tabs on it.

Judgment: We compare what we see with a standard.

Self-response: If you did well in comparison with your standard, you give yourself rewarding self-responses. If you did poorly, you give yourself punishing self-responses.

Self efficacy: In other words, self-efficacy is a person's belief in his or her ability to succeed in a particular situation

People with a strong sense of self-efficacy have the following characteristics:

They view challenging problems as tasks to be mastered.

They develop deeper interest in the activities in which they participate.

They form a stronger sense of commitment to their interests and activities.

They recover quickly from setbacks and disappointments. On the other hand, people with a weak sense of self-efficacy show the following characteristics:

They avoid challenging tasks.

They believe that difficult tasks and situations are beyond their capabilities.

They focus on personal failings and negative outcomes.

They quickly lose confidence in personal abilities (Bandura, 1994).

According to Bandura, there are **four major sources of self-efficacy**.

Mastery Experiences: The most effective way of developing a strong sense of efficacy is through mastery experiences. (Bandura 1994).

Performing a task successfully strengthens our sense of self-efficacy.

However, failing to adequately deal with a task or challenge can undermine and weaken self-efficacy.

Social Modeling: Witnessing other people successfully completing a task is another important source of self-efficacy. According to Bandura, when a person sees another person or persons similar to oneself succeeding by one's sustained efforts, makes the person raise the belief that they too possess the capabilities that could

help them master comparable activities and succeed.

Social Persuasion: Bandura also asserted that people could be persuaded to believe that they have the skills and capabilities to succeed. Consider a time when someone said something positive and encouraging that helped you achieve a goal.

Getting verbal encouragement from others helps people overcome self-doubt and instead focus on giving their best effort to the task at hand.

Psychological Responses: Our own responses and emotional reactions to situations also play an important role in self-efficacy. Moods, emotional states, physical reactions, and stress levels can all impact how a person feels about their personal abilities in a particular situation. A person who becomes extremely nervous before speaking in public may develop a weak sense of self-efficacy in these situations.

Principles of Observational Learning

A Study of Aggression. In this experiment Bandura exposed a group of children to a video, featuring violent and aggressive actions. For the experiment Bandura made of film of one of his students, a young woman, essentially beating up a bobo doll. Bobo doll is an inflatable, egg-shaped balloon creature with a weight in the bottom that makes it bob back up when you knock him down. The woman punched the clown, shouting "sockeroo!" She kicked it, sat on it, hit with a little hammer, and so on, shouting various aggressive phrases.

Bandura showed this film to groups of kindergartners who, as you might predict, liked it a lot. They then were let out to play. In the play room, of course, were several observers with pens and clipboards in hand, a brand new bobo doll, and a few little hammers.

All these variations allowed Bandura to establish that there were certain **steps involved in the modeling** process:

Attentional Processes: In order to learn, you need to be paying attention. Anything that detracts your attention is going to have a negative effect on observational learning

Retentional Processes: The ability to store information is also an important part of the learning process. Retention can be affected by a number of factors, but the ability to pull up information later and act on it is vital to observational learning

Reproduction Processes: Once you have paid attention to the model and retained the information, it is time to actually perform the behaviour you observed.

In the theory of observational learning Bandura mentions a number of **motives**, which are:

a) **past reinforcement**

b) **ala traditional behaviourism.**

C. Reinforcement Value Reinforcement value is a person's preference for any particular reinforcement over other reinforcements if all are equally likely to

D. Psychological Situation The psychological situation is that part of the external and internal world to which a person is responding. Behavior is a function of the interaction of people with their meaningful environment.

promised reinforcements (incentives) that we can imagine. c) **vicarious reinforcement** — seeing and recalling the model being reinforced.

ROTTER'S SOCIAL LEARNING THEORY

Rotter's interactionist position holds that human behavior is based largely on the interaction of people with their meaningful environments. Human behavior is most accurately predicted by an understanding of four variables: behavior potential, expectancy, reinforcement value, and the psychological situation.

A. Behavior Potential Behavior potential is the possibility that a particular response will occur at a given time and place in relation to its likely reinforcement.

B. Expectancy People's expectancy in any given situation is their confidence that a particular reinforcement will follow a specific behavior in a specific situation or situations. Expectancies can be either general or specific, and the overall likelihood of success is a function of both generalized and specific expectancies.

occur. Internal reinforcement is the individual's perception of an event, whereas external reinforcement refers to society's evaluation of an event.

E. Basic Prediction Formula Hypothetically, in any specific situation, behavior can be predicted by the basic prediction formula, which states that the potential for a behavior to occur in a

particular situation in relation to a given reinforcement is a function of people's expectancy that the behavior will be followed by that reinforcement in that situation.

KELLY'S COGNITIVE THEORY: PERSONAL CONSTRUCTS THEORY

Kelly's theory of personal constructs can be seen as a meta-theory, or a theory in relation to the theories. It holds that **people expect events through the meanings or interpretations that they place on those events**. Kelly described these interpretations personal constructs. His philosophical position, described **constructive alternativism**, assumes that alternative interpretations are always accessible to people.

Psychotherapy: Kelly insisted that clients **should set their own goals** for therapy and that they should be active participants in the therapeutic process. He sometimes used a procedure described **fixedrole therapy** in which clients act out a predetermined role for many weeks. Through playing the part of a psychologically healthy person, clients may discover previously hidden characteristics of themselves. **The Rep Test:** The purpose of the Rep test is to discover methods in which clients construe important people in their lives. Clients place names of people they know on a repertory grid in order to identify both similarities and differences among these people.

INTRODUCTION TO HUMANISTIC THEORIES

Accordingly, humanistic psychology focuses on **subjective experiences** of persons as opposed to forced, definitive factors that determine behaviour. Abraham Maslow and Carl Rogers were proponents of this view, which is based on the "phenomenal field" theory of Combs and Snygg. Maslow and Rogers emphasised a view of the **person as an active, creative, experiencing human being who lives in the present** and subjectively responds to current perceptions, relationships, and encounters. They disagree with the dark, pessimistic outlook of those in the Freudian psychoanalysis ranks, but rather view humanistic theories as positive and optimistic proposals which stress the tendency of the human personality **toward growth and self-actualisation**.

THEORY OF ABRAHAM MASLOW

"Self Actualization is the intrinsic growth of what is already in the organism, or more accurately, of what the organism is."

Characteristics of Self Actualizing People

Realistic Realistically oriented, SA persons have a more efficient perception of reality, they have comfortable relations with it. This is extended to all areas of life. SA persons are unthreatened, unfrightened by the unknown. they have a superior ability to reason, to see the truth. They are logical and efficient.

Acceptance Accept themselves, others and the natural world the way they are. Sees human nature as is, have a lack of crippling guilt or shame, enjoy themselves without regret or apology, they have no unnecessary inhibitions.

Spontaneity, Simplicity, Naturalness Spontaneous in their inner life, thoughts and impulses, they are unhampered by convention. Their ethics is autonomous, they are individuals, and are motivated to continual growth.

Problem Centering Focus on problems outside themselves, other centered. They have a mission in life requiring much energy, their mission is their reason for existence. They are serene, characterized by a lack of worry, and are devoted to duty.

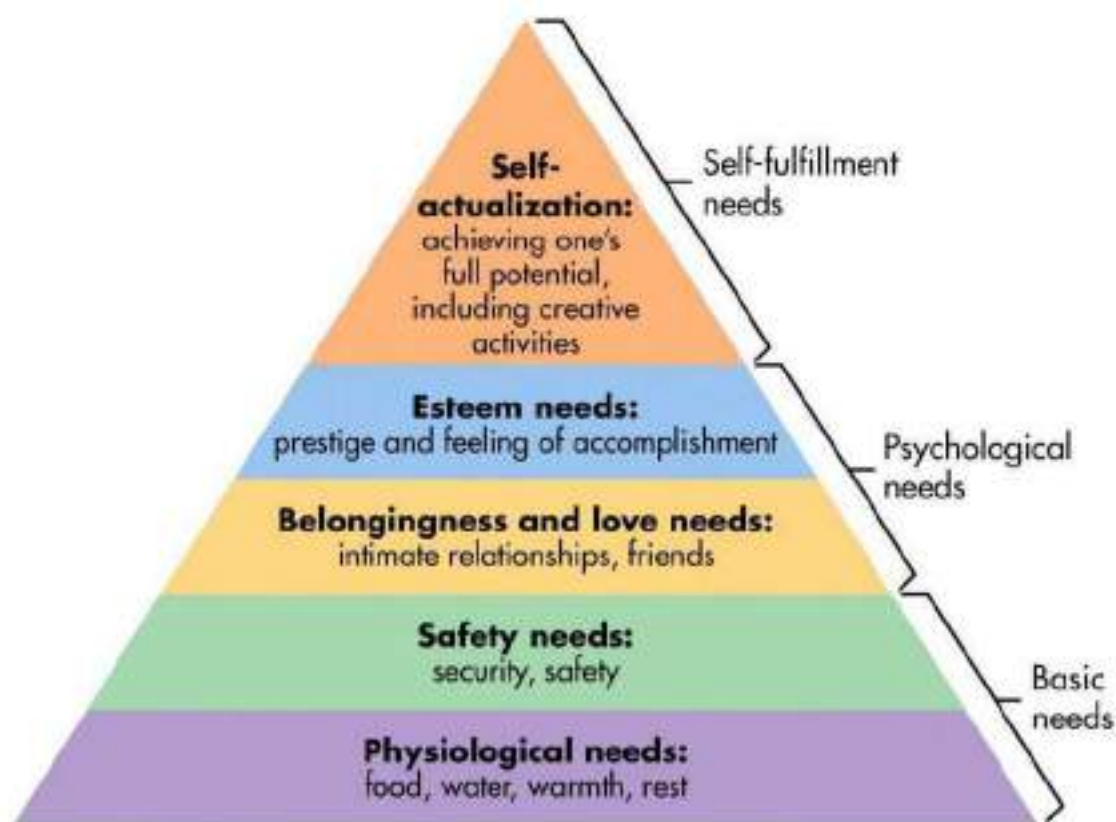
Detachment: The Need for Privacy Alone but not lonely, unflappable, retain dignity amid confusion and personal misfortunes,

objective. They are self starters, responsible for themselves, own their behavior.

Autonomy: Independent of Culture and Environment SA's rely on inner self for satisfaction. Stable in the face of hard knocks, they are self contained, independent from love and respect.

Continued Freshness of Appreciation Have a fresh rather than stereotyped appreciation of people and things. Appreciation of the basic good in life, moment to moment living is thrilling, transcending and spiritual. They live the present moment to the fullest.

Peak experiences "Feelings of limitless horizons opening up to the vision, the feeling of being simultaneously more powerful and also more helpless than one ever was before, the feeling of ecstasy and wonder and awe, the loss of placement in time and space with, finally, the conviction that something extremely important and valuable had happened, so that the subject was to some extent transformed and strengthened even in his daily life by such experiences.



Homeostasis Maslow also talks about these levels in terms of homeostasis. Homeostasis is the principle by which your furnace thermostat operates: When it gets too cold, it switches the heat on, and when it gets too hot, it switches the heat off. Maslow sees all these needs as essentially survival needs. Even love and esteem are needed for the maintenance of health. He says we all have these needs built in to us genetically, like instincts. In fact, he calls them instinctoid, that is, instinct like needs. Under stressful conditions, or when survival is threatened, we can “regress” to a lower need level. When a person’s good career is in jeopardy, the person might seek out a little attention

THEORY OF CARL ROGERS

A person’s identity is formed through a series of personal experiences, which reflect how the individual is perceived by both him or herself and the outside world – the **phenomenological field**.

Rogerian personality theory distinguishes between two personalities. The **real self** is created through the actualising tendency, it is the self that one can become. The demands of society, however, do not always support the actualising tendency and we are forced to live under conditions that are out of step with our tendencies. The **ideal self** is the ideal created through the demands of society.

Rogers tells us that organisms know what is good for them. Evolution has provided us with the senses, the tastes, the discriminations we need: When we are hungry, we find food not just any food, but food that tastes good. Food

that tastes bad is likely to be spoiled, rotten and unhealthy. This is called **organismic valuing**. Among the many things that we instinctively value is **positive regard**, Rogers umbrella term for things like love, affection, attention, nurturance, and so on. It is clear that babies need love and attention.

Another thing that we value is positive **self regard**, that is, self esteem, self worth, and a positive self image. We achieve this positive self regard by experiencing the positive regard others show us over our years of growing up. Getting positive regard on “on condition” Rogers calls **conditional positive regard**. Because we do indeed need positive regard, these conditions are very powerful, and we bend ourselves into a shape determined, not by our organismic valuing or our actualising tendency, but by a society that may or may not truly have our best interests at heart

Incongruity: The gap between the real self and the ideal self, the “I am” and the “I should be” is called incongruity. The greater the gap, the more is the incongruity. The more the incongruity, the more the individual suffers.

Defenses: When you are in a situation where there is an incongruity between your image of yourself and your immediate experience of yourself (i.e. between the ideal and the real self), you are in a threatening situation. Rogers also has a partial explanation for psychosis: **Psychosis** occurs when a person’s defense are overwhelmed, and their sense of self becomes “shattered” into little disconnected pieces.

The Fully Functioning Person

Rogers, like Maslow, is just as interested in describing the healthy person. His term is “fully-functioning,” and involves the following qualities:

Openness to experience. This is the opposite of defensiveness. It is the accurate perception of one’s experiences in the world, including one’s feelings. It also means being able to accept reality, again including one’s feelings.

Existential living. This is living in the here-and-now. Rogers, as a part of getting in touch with reality, insists that we not live in the past or the future — the one is gone, and the other isn’t anything at all, yet! The present is the only reality we have.

Organismic trusting. We should allow ourselves to be guided by the organismic valuing process. We should trust ourselves, do what feels right, what comes naturally. Rogers meant trust your real self, and you can only know what your real self has to say if you are open to experience and living existentially.

Experiential freedom: It means that we feel free when choices are available to us. Rogers says that the fully-functioning person acknowledges that feeling of freedom, and takes responsibility for his choices.

Creativity: A fully-functioning person, in touch with actualisation, will feel obliged by their nature to contribute to the actualisation of others, even life itself. This can be through creativity in

the arts or sciences, through social concern and parental love, or simply by doing one’s best at one’s job. Creativity as Rogers uses it is very close to Erikson’s generativity.

Therapy: Carl Rogers is best known for his contributions to therapy. He originally called his therapy to be **non-directive**, because he felt that the therapist should not lead the client, but rather be there for the client while the client directs the progress of the therapy. Rogers felt that **a therapist**, in order to be effective, must have three very special qualities:

Congruence — genuineness, honesty with the client.

Empathy — the ability to feel what the client feels.

Respect — acceptance, unconditional positive regard towards the client.

EXISTENTIAL THERAPY

Existential therapy focuses on free will, self-determination, and the search for meaning—often centering on you rather than on the symptom. The approach emphasizes your capacity to make rational choices and to develop to your maximum potential.

The existential approach stresses that:

All people have the capacity for self awareness.

Each person has a unique identity that can be known only through relationships with others.

People must continually re-create themselves because life's meaning constantly changes.

Anxiety is part of the human condition

ROLLO MAY THEORY

Rollo May is the best known American existential psychologist. Much of his thinking can be understood by reading about existentialism in general, and the overlap between his ideas and the ideas of Ludwig Binswanger is great.

Stages of development

Innocence -- the pre-egoic, pre-self-conscious stage of the infant. The innocent is pre-moral, i.e. is neither bad nor good. Like a wild animal who kills to eat, the innocent is only doing what he or she must do. But an innocent does have a degree of will in the sense of a drive to fulfil their needs!

Rebellion -- the childhood and adolescent stage of developing one's ego or self-consciousness by means of contrast with adults, from the "no" of the two-year-old to the "no way" of the teenager. The rebellious person wants freedom, but has as yet no full understanding of the responsibility that goes with it. The teenager may want to spend

their allowance in any way they choose - yet they still expect the parent to provide the money, and will complain about unfairness if they don't get it!

Ordinary -- the normal adult ego, conventional and a little boring, perhaps. They have learned responsibility, but find it too demanding, and so seek refuge in conformity and traditional values.

Creative -- the authentic adult, the existential stage, beyond ego and self-actualizing. This is the person who, accepting destiny, faces anxiety with courage!

These are not stages in the traditional sense. A child may certainly be innocent, ordinary or creative at times; An adult may be rebellious.

Love and Will

His basic motivational construct is **the daimonic**. The daimonic is the entire system of motives, different for each individual. It is composed of a collection of specific motives called **daimons**. The word daimon is from the Greek, and means little god. But originally, a daimon could be **bad or good**. Daimons include **lower needs**, such as food and sex, as well as **higher needs**, such as love.

For May, one of the most important daimons is **eros**. Eros is love (not sex), and in Greek mythology was a minor god pictured as a young man.

Another important concept for May is **will**: The ability to organize oneself in order to achieve one's goals. This makes will roughly synonymous with ego and reality-testing, but with its own store of energy, as in ego psychology

TYPES:

There is the type he refers to as "**neo-Puritan**," who is all will, but no love. They have amazing self-discipline, and can "make things happen"... but they have no wishes to act upon. So they become "anal" and perfectionistic, but empty and "dried-up." The archetypal example is Ebenezer Scrooge.

The second type he refers to as "**infantile**." They are all wishes but no will. Filled with dreams and desires, they don't have the self-discipline to make anything of their dreams and desires, and so become dependent and conformist. They love, but their love means little. Perhaps Homer Simpson is the clearest example!

The last type is the "**creative**" type. May recommends, wisely, that we should cultivate a balance of these two aspects of our personalities. He said "Man's task is to unite love and

will." This idea is, in fact, an old one that we find among quite a few theorists

VICTOR FRANKL THEORY

Understanding Logotherapy

Frankl believed that humans are motivated by something called a "will to meaning

"Everything can be taken from a man but one thing: the last of the human freedoms—to choose one's attitude in any given set of circumstances."

Fundamentals of Logotherapy

"Logos" is the Greek word for meaning, and logotherapy involves helping a patient find personal meaning in life.

Core Properties

Frankl believed in three core properties on which his theory and therapy were based:

Each person has a healthy core.

One's primary focus is to enlighten others to their own internal resources and provide them tools to use their inner core.

Life offers purpose and meaning but does not promise fulfillment or happiness.

BASIC ASSUMPTIONS

Logotherapy consists of six basic assumptions that overlap with the fundamental constructs and ways of seeking meaning listed above:

1. Body, Mind, and Spirit

The human being is an entity that consists of a body (*soma*), mind (*psyche*), and spirit (*noos*).

2. Life Has Meaning in All Circumstances

Frankl believed that life has meaning in all circumstances, even the most miserable ones. This means that even when situations seem objectively terrible, there is a higher level of order that involves meaning.

3. Humans Have a Will to Meaning

Logotherapy proposes that humans have a will to meaning, which means that meaning is our primary motivation for living and acting, and allows us to endure pain and suffering.

4. Freedom to Find Meaning

Frankl argues that in all circumstances, individuals have

the freedom to access that will to find meaning.

5. Meaning of the Moment

The fifth assumption argues that for decisions to be meaningful, individuals must respond to the demands of daily life in ways that match the values of society or their own conscience.

6. Individuals Are Unique

Frankl believed that every individual is unique and irreplaceable.

LOGOTHERAPY IN PRACTICE

Frankl believed that it was possible to turn suffering into achievement and accomplishment. He viewed **guilt** as an opportunity to change oneself for the better, and life transitions as the chance to take responsible action.

Three techniques used in logotherapy include dereflection, paradoxical intention, and Socratic dialogue.

Dereflection: Dereflection is aimed at helping someone focus away from themselves and toward other people so that they can become whole and spend less time being self-absorbed about a problem or how to reach a goal.

. **2) Paradoxical intention:**

Paradoxical intention is A technique that has the patient wish for the thing that is feared most. This was suggested for use in the case of anxiety or phobias, in which humour and ridicule can be used when fear is paralyzing. For example, A person with a fear of looking foolish might be encouraged to try to look foolish on purpose. Paradoxically, the fear would be removed when the Intention involved the thing that was feared most.

- . **3) Socratic dialogue:** Socratic dialogue would be used in logotherapy as a tool to help a patient through the process of self-discovery through his or her own words. In this way, the therapist would point out patterns of words and help the client to see the meaning in them. This process is believed to help the client realize an answer that is waiting to be discovered.



Concepts

BASIC MOTIVATIONAL CONCEPTS:

Instincts,
Needs,
Drives,
Arousal,
Incentives,
Motivational Cycle.

APPROACHES TO THE STUDY OF MOTIVATION:

Psychoanalytical,
Ethological,
S-R Cognitive,
Humanistic

MOTIVATION

A motive is a reason for doing something. Motivation is concerned with the strength and direction of behaviour and the factors that influence people to behave in certain ways. The term ‘motivation’ can refer variously to the goals individuals have, the ways in which individuals choose their goals and the ways in which others try to change their behaviour.

Types of motivation

The two types of motivation are **intrinsic motivation** and **extrinsic motivation**.

Intrinsic motivation

Intrinsic motivation can arise from the self-generated factors that influence people’s behaviour. It is not created by external incentives. It can take the form of motivation by the work itself when individuals feel that their work is important, interesting and challenging and provides them with a reasonable degree of autonomy (freedom to act), opportunities to achieve and advance, and scope to use and develop their skills and abilities. **Deci and Ryan (1985)** suggested that intrinsic motivation is based on the needs to be competent and self-determining (that is, to have a choice). Intrinsic motivation can be enhanced by job or role design. According to an early writer on the significance of the motivational impact of job design (**Katz, 1964**): ‘The job itself must provide sufficient variety, sufficient complexity, sufficient challenge and sufficient skill to engage the abilities of the worker.’ In their job

characteristics model, **Hackman and Oldham (1974)** emphasized the importance of the core job dimensions as motivators, namely skill variety, task identity, task significance, autonomy and feedback.

Extrinsic motivation

Extrinsic motivation occurs when things are done to or for people to motivate them. These include rewards, such as **incentives, increased pay, praise, or promotion; and punishments, such as disciplinary action, withholding pay, or criticism**. Extrinsic motivators can have an immediate and powerful effect, but will not necessarily last long.

The intrinsic motivators, which are concerned with the ‘quality of working life’ (a phrase and movement that emerged from this concept), are likely to have a deeper and longer-term effect because they are inherent in individuals and their work and not imposed from outside in such forms as incentive pay.

CONCEPTS RELATED TO MOTIVATION

Drive

Need leads to drive, which is the second step towards achieving goal. Drive can be defined as the state of **tension or arousal produced by need**. The drive can also be considered as the original source of energy that activates an organism. For instance, when an organism is hungry

and/or thirsty, the organism seeks to reduce this drive by eating and/or drinking. Drive acts as a strong persistent stimulus to push an organism towards its goal. It is the state of heightened tension leading to restless activity and preparatory behavior.

NEED AND MOTIVES

A need is a **condition of lack or deficit of something** required by the organism. In order to maintain **homeostasis** or balance the organism finds it necessary to satisfy the needs. The needs are of different types. The need for food or water is a **physiological need**, which arises out of lack or deficit of food or water in the organism. The needs for excretion and urination are also physiological needs. They are due to the organism's necessity to eliminate waste matter from the body. The need for contact with other persons is a **social need**. The other social needs include need for prestige, status, affection, self-esteem, and so on. A person becomes more aware of his needs when they are not fulfilled. In other words, when you are hungry, you need food, and, when you are thirsty you need water

Needs for food, water, sex, sleep and rest, and elimination are **primary needs**. Needs for achievement, affiliation, power are examples of social needs. The term '**motive**' refers to goal directed behaviour and energising conditions within the organism that drive behaviour. It is generally used to refer to certain conditions which, besides arousing, predispose a person to respond, or behave in a way appropriate to that motive. **Motives direct the activity** of the individual towards person's goals.

Incentives

Incentives refers to the **goal objects which satisfy the needs**. Incentives vary in quality and quantity which make them

less or more satisfying and attractive. Thus, one can put in greater amount of effort to attain a more attractive incentive. As a matter of fact, many incentives assume considerable significance in the lives of people and they do everything possible to attain those incentives.

Instincts

Instinct is an old concept in the field of motivation. It is defined as an **innate biological force that predisposes the organism to act in a certain way**. At one time all behaviours were supposed to be results of certain instincts. Some of the instincts identified by early psychologists are **fight, repulsion, curiosity, self abasement, acquisition** etc. It was thought that instincts were inherited and compelling sources of conduct, but can be modified by learning and experience. This term is no more used in relation to human behaviour. Animal behaviour is sometimes explained using this term. In current usage 'instinct' is reserved for innate response tendencies found among animals

Values

Values work as important motivators. They are considered as **desirable and cheris-able goals** that serve as guiding principles in people's lives. Values help to make choices. Values **prioritize** needs. It is only because of values that people take purposeful long range actions. **Moral values** differentiate between good and bad.

Power: This includes **social status and prestige, control and dominance** over people and resources

Achievement: This includes personal success by demonstrating competence according to social standards.

Self-direction: This includes independent thought and action, choosing, creating, and exploring.

Universalism : This includes understanding, appreciation, tolerance and protection for the welfare of all people.

Benevolence: This includes preservation and enhancement of the welfare of people with whom one is in frequent personal contact.

Tradition : This includes respect, commitment and acceptance of the customs and ideas that are given importance in the traditional cultures or religions.

Conformity: This includes restraint of action, inclination, and impulses likely to upset or harm others and violate social expectations or norms.

Security: This includes safety, harmony and stability of society, of relationships and of self.

In the Indian context the framework of Dharma provides a set of values which are considered central to the sustenance of

life. They include truth (Satya), non stealing (asteya), keeping tolerance (driti), intellect (dhi), knowledge (vidya), non-anger (akrodh), forgiveness (kshama), purity (saucha), control of sense organs (indriya nigraha) and self control (dam). These values provide basis for maintaining and promoting life at the individual and the social levels. It maintains a the view that holds entire universe into account.

Arousal's Contribution to Motivation

Arousal is the **energy source** for behaviour; it is the primary key in the **initiation, strength and persistence** of all motivational behaviours (Pfaff, 2006, p. 2). You can be aroused without being motivated, yet **you cannot be motivated without being aroused**. Motivational arousal can increase and decrease with its level of intensity being guided by how an individual perceives a task. If a task is expected to be difficult the need for motivational arousal will increase, whereas if the task is expected to be easy motivational arousal will decrease (Miron, Parkinson & Brehm, 2007).

Motivation cycle



APPROACHES TO STUDY OF MOTIVATION

PSYCHOANALYTICAL

This theory which has been explained by Sigmund Freud, deals with **unconscious motivation**.

According to Freud, the inborn tendencies called **instincts** influence our behaviour.

Life instincts (Eros): these instincts have the life energy called Libido-which motivates the individual towards constructive activities like love, sympathy/helping others, etc.

Death instincts (Thanatos)-motivate the individual for destructive activities like murder, suicides, aggression, attack, etc. According to him, our **unconscious mannerisms, slips of tongue and pen, phobias** are the result of these hidden motives.

ETHOLOGICAL APPROACH

Ethology is a study of behavior based on two core principles:

behavior changes to achieve survival
(sometimes referred to as an adaptive trait)
behavioural traits are inherited

The work of Charles Darwin laid the foundation for ethology. Darwin believed biological traits could adapt (or

evolve) over time to promote survival. Ethological theories are in large part about how **behavior adapts to better ensure survival** and is passed down to the next generations

One of the pioneers of instinct theories of motivation is the English-born social psychologist, William **McDougall**, who formed the Hormic Psychology, with 'hormic' meaning animal impulse or urge. **Hormic Psychology** is based on determined and goal-oriented behaviors that are supposed to be motivated by instincts, which are spontaneous, persistent, variable, and repetitive.

Humanistic approach

This theory believes in striving tendency of the individual for realizing his **potentialities**, especially **creative** ones, strengthening self-confidence and attaining the **ideal self**. There are two important persons related to this theory— Abraham Maslow and Carl Rogers.

Maslow has explained that every individual struggle to fulfil basic needs first, and then followed by safety, love, esteem and finally **actualisation needs**.

Self-actualisation means becoming everything one is capable of, or becoming what he can, that is, fulfilment of his basic potentialities. Maslow explains that the self-actualised people experience, what he calls the ‘peak experiences’, when they fulfil the need for self-actualisation.

Carl Rogers, as a humanist believes in the strength and potentialities of human beings. Every individual will strive to realise his potentialities and to grow to become a **fully functioning person**. Hence in the view of Rogers, the motivation for self-growth and becoming a fully functioning person are important concepts.

S-R COGNITIVE APPROACH

Cognitive theories of motivation seek to explain human behavior as a product of the careful study and active **processing and interpretation of information** received. Such a perspective runs counter to rationalizing human behavior as a result of automatic responses governed by pre-programmed rules or innate mechanisms involving drives, needs and reactions. The actions of humans, in addition to what motivates them to engage in particular actions, are therefore, the **product of deliberate thought processes such as beliefs, expectations, knowledge** about things and past experiences.

Cognitive motivation is said to be rooted on two basic factors. The first involves **information available** to the individual. Initially, an individual will process a situation based on whatever input is

immediately available to his senses. The second factor involves the **individual's past experience**, which the person refers to when trying to make sense of information presently available and in determining how to respond or relate to the current situation.

THEORIES of MOTIVATION

Content (needs) theory

The theory focuses on the content of motivation in the shape of needs. Its basis is the belief that an **unsatisfied need creates tension and a state of disequilibrium**. To restore the balance a goal is identified that will satisfy the need, and a behaviour pathway is selected that will lead to the achievement of the goal and the satisfaction of the need. All behaviour is therefore motivated by unsatisfied needs.

Maslow's hierarchy of needs

The most famous classification of needs is the one formulated by Maslow (1954). He suggested that there are five major need categories that apply to people in general, starting from the fundamental physiological needs and leading through a hierarchy of safety, social and esteem needs to the need for self-fulfilment, the highest need of all. When a lower need

is satisfied the next highest becomes dominant and the individual's attention is turned to satisfying this higher need. The need for self-fulfilment, however, can never be satisfied. Psychological development takes place as people move up the hierarchy of needs, but this is not necessarily a straightforward progression. The lower needs still exist, even if temporarily dormant as motivators, and individuals constantly return to previously satisfied needs.

ERG theory (Alderfer)

Alderfer (1972) devised a theory of human needs that postulated three primary categories:

Existence needs such as hunger and thirst – pay, fringe benefits and working conditions are other types of existence needs.

Relatedness needs, which acknowledge that people are not self-contained units but must engage in transactions with their human environment—acceptance, understanding, confirmation and influence are elements of the relatedness process.

Growth needs, which involve people in finding the opportunities 'to be what they are most fully and to become what they can'.

McClelland's achievement–affiliation–power needs

An alternative way of classifying needs was developed by McClelland (1961), who based it mainly on studies of managers. He identified three needs as being most important:

The need for achievement, defined as the need for competitive success measured against a personal standard of excellence.

The need for affiliation, defined as the need for warm, friendly, compassionate relationships with others.

The need for power, defined as the need to control or influence others.

Process theory

In process theory, the emphasis is on the psychological processes or forces that affect motivation, as well as on basic needs. It is also known as ‘**cognitive theory**’ because it is concerned with people’s perceptions of their working environment and the ways in which they interpret and understand it. According to Guest (1992), process theory provides a much more relevant approach to motivation that replaces the theories of Maslow and Herzberg which, he claims, have been shown by extensive research to be wrong.

Expectancy theory

Expectancy theory states that motivation will be high when people know what they have to do to get a reward, expect that they will be able to get the reward and expect that the reward will be worthwhile. The concept of expectancy was originally contained in the **valency–instrumentality–expectancy (VIE) theory formulated by Vroom (1964)**.

Valency stands for value, instrumentality is the belief that if we do one thing it will lead to another, and expectancy is the probability that action or effort will lead to an outcome. The strength of expectations may be based on past experiences (reinforcement), but individuals are frequently presented with new situations – a change in job, payment system, or working conditions imposed by management – where past experience is an inadequate guide to the implications of the change. In these circumstances, motivation may be reduced. Motivation is only likely when

a clearly perceived and usable relationship exists between performance and outcome, and the outcome is seen as a means of satisfying needs.

This theory was developed by **Porter and Lawler (1968) into a model** that follows Vroom’s ideas by suggesting that there are two factors determining the effort people put into their jobs:

first the value of the rewards to individuals in so far as they satisfy their needs for security, social esteem, autonomy, and self-actualization, and second the probability that rewards depend on effort, as perceived by individuals – in other words, their expectations about the relationships between effort and reward.

Thus, the greater the value of a set of awards and the higher the probability that receiving each of these rewards depends upon effort, the greater the effort that will be expended in a given situation.

The two variables additional to effort that affect task achievement are

ability – individual characteristics such as intelligence, knowledge, skills, and

role perceptions – what individuals want to do or think they are required to do.

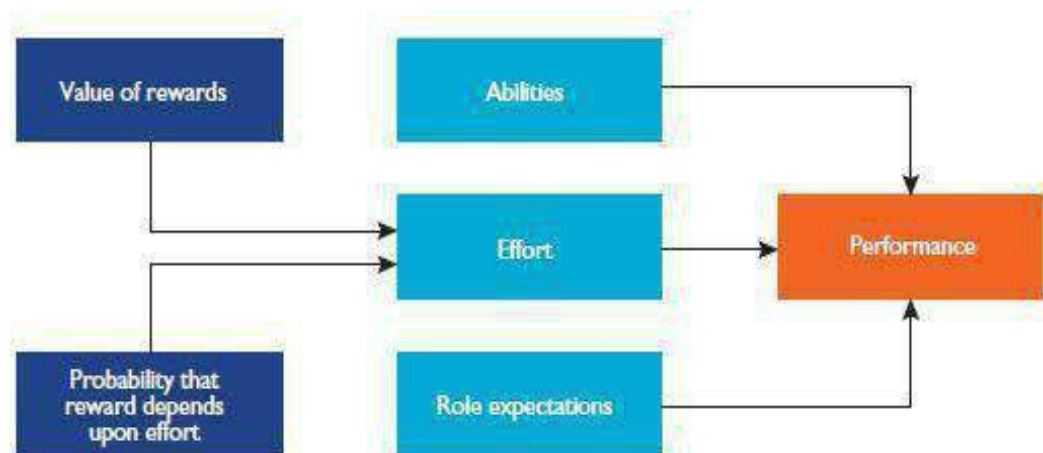


Figure 19.2 Motivation model (Porter and Lawler)

Goal theory

Goal theory as developed by **Latham and Locke** (1979) states that motivation and performance are higher when individuals are set **specific goals**, when **goals are difficult but accepted**, and when there is **feedback on performance**. Participation in goal setting is important as a means of getting agreement to the setting of higher goals. Difficult goals must be agreed and their achievement reinforced by guidance and advice. Finally, feedback is vital in maintaining motivation, particularly towards the achievement of even higher goals.

Social learning theory

Social learning theory as developed by Bandura (1977) combines aspects of both behavioural and expectancy theory. It recognizes the significance of the basic behavioural concept of **reinforcement** as a determinant of future behaviour but also emphasizes the importance of internal psychological factors, especially **expectancies about**

the value of goals and the individual's ability to reach them. The term '**reciprocal determinism**' is used to denote the concept that while the situation will affect individual behaviour, individuals will simultaneously influence the situation.

Equity theory

Equity theory (Adams, 1965) is concerned with the perceptions people have about how they are being treated as compared with others. To be dealt with equitably is to be treated fairly in comparison with another group of people (a reference group) or a relevant other person. **Equity involves feelings and perceptions and it is always a comparative process**. It is not

synonymous with equality, which means treating everyone the same, since this would be inequitable if they deserve to be treated differently.

Equity theory states, in effect, that people will be better motivated if they are treated equitably and demotivated if they are treated inequitably. It explains only one aspect of the processes of motivation and job satisfaction, although it may be significant in terms of morale.

There are two forms of equity:

distributive equity, which is concerned with the fairness with which people feel they are rewarded in accordance with their contribution and in comparison with others; and **procedural equity**, which is concerned with the perceptions employees have about the fairness with which company procedures in such areas as performance appraisal, promotion and discipline are being operated.

Herzberg's two-factor model

The two-factor model of satisfiers and dissatisfiers was developed by Herzberg *et al* (1957) following an investigation into the sources of job satisfaction and dissatisfaction of accountants and engineers. The wants of employees divide into **two groups**. One group revolves around the need to develop in one's occupation as a source of **personal growth**. The second group operates as an essential base to the first and is associated with **fair treatment in compensation, supervision, working conditions and administrative practices**. The fulfilment of the needs of the second group does not motivate the individual to high levels of job satisfaction and to extra performance on the job. All we can expect from satisfying this second group of needs is the prevention of dissatisfaction and poor job performance. The second group forms the **hygiene factors** in the medical use of the term, meaning preventive and environmental. Herzberg pointed out that while financial incentives may

motivate in the short term, the effect quickly wears off. According to Herzberg's theory, only challenging jobs that have the opportunities for achievement, recognition, responsibility, advancement and growth will motivate personnel.

McGregor's Theory X and Y

Douglas McGregor (1960) produced his analysis of the different views about people and how they should be motivated. **Theory X** is the traditional view that the average human dislikes work and wishes to avoid responsibility and that, therefore, 'most people must be coerced, controlled, directed, threatened with punishment to get them to put forward adequate effort towards organizational objectives'. In contrast, **theory Y** emphasizes that people will exercise self-direction in the service of objectives to which they are committed and that commitment to objectives is a function of the rewards associated with their achievement.

Achievement Motivation

David McClelland and his associates (Notably John Atkinson), focused on needs similar to the higher order (social and esteem) needs identified by Maslow. McClelland's needs theory, also called Achievement motivation theory is concerned with how individual needs and environmental factors combine to form three basic human motives.

Need for Achievement (n Ach):

People who want to take responsibility for finding solutions to problems, who seek challenge, who is willing to work hard and who has the mental vigour to reach the ultimate goal are considered to have a high

need for achievement. Achievement motivated people tend to get more pay raises and are promoted faster because they are constantly trying to think of better ways of doing things. They have a desire to do something better or more efficiently than it has been done before. This drive is the achievement need.

Need For Power (n POW): Need for power is the desire to have impact to be influential, and to control others. A high need for power means that an individual seeks to influence or control others. Individuals with this need are concerned with acquiring, exercising and retaining power or influence over others. They prefer to be placed into competitive and status oriented situations. They generally tend to seek positions of leadership. Top level managers, politicians etc., have a high need for power.

Need For Affiliation (n Aff): Need for affiliation is related to the desire for affection and establishing friendly relationships. People who have a high need for affiliation view the organization as a chance to form new and satisfying relationships. They are motivated by the jobs that provide frequent interaction with colleagues. They derive pleasure from being loved by the group.

Each of the McClelland's three motives evokes a different type of feeling of satisfaction. Achievement motive tends to evoke a sense of accomplishment, power motive tends to evoke a sense of authority and affiliation motive tends to evoke love and affection. Tools like **Thematic Apperception Test (TAT)** are used to measure and determine the strength of these needs.

Comparison of needs in four theories of motivations

MASLOW	ALDERFER	MCCLELLAND	HERZBERG
Self-actualization	Growth	Need for Power	Motivators
Esteem or Ego		Need for Achievement	
Belongingness and love	Relatedness	Need for Affiliation	Hygiene
Safety and Security	Existence		
Physiological			

Source: Gordon R. Judith et al, *Management and Organizational Behavior* , Allyn and Bacon, 1990, p.428.

IMMATURITY - MATURITY THEORY

Argyris postulated a descriptive and multidimensional developmental process

along which individuals in an organization naturally grow towards maturity.

2.6.(i) Immaturity - Maturity Continuum

Passive	-	Active
Dependence	-	Independence
Behave in a few ways	-	Capable of behaving in many ways
Shallow abilities and interests	-	Developing stronger and deeper Interests and abilities
Short time perspective	-	Long time perspective
Subordinate Position	-	Equal or super ordinate position
Lack of self-awareness	-	Self-control and self-awareness.

Thus Argyris proposes that the existing bureaucratic - pyramidal organization structure (the organizational counterpart to Theory X assumptions about people) should give way to humanistic - democratic value system (the organizational counterpart to Theory Y assumptions about people'2).

REINFORCEMENT THEORY

Reinforcement theory applies the behaviourist **learning theories** to motivation. Also called **Organizational Behaviour Modification Theory** or O.B. Model developed by B.F Skinner is quite different from the cognitive theories of Motivation. Reinforcement

theory reveals that human behaviour can be explained in terms of the **previous positive or negative outcomes of that behaviour**. This is truly a behaviouristic approach where one can see that reinforcement conditions behaviour. The repeating behaviours that people have learned will produce pleasant outcomes. Those behaviors, which are rewarded, tend to be repeated and those behaviors, which are either not rewarded or punished, tend to disappear. It should be applied in conjunction with the principles of social learning. Rewards or reinforcements must meet an employee's specific needs and must be applied equitably.

CONCEPTS

Exploratory behavior and curiosity

Achievement, Affiliation and Power

Motivational Competence

Self-regulation

Flow



CURIOSITY

A second factor influencing individual motivation is *curiosity*. This is an *individual* factor because a person's curiosity can be aroused without involving other people. Curiosity is stimulated when something in the physical environment **attracts our attention** or when there is an optimal level of **discrepancy between present knowledge or skills and what these could be** if the learner engaged in some activity. **Novelty and interest** are good synonyms for the motivational use of curiosity.

There are two types of curiosity that can stimulate intrinsic motivation:

Sensory curiosity occurs when physical factors such as changes in tone of voice, light, or sound attract the attention of learners.

Cognitive curiosity, on the other hand, is evoked when learners believe that it may be useful to modify existing cognitive structures.

The feeling of **loss of control** is one of the most powerful **anti-motivating** factors in education. Drive theories differ on whether they view curiosity as a primary or secondary drive. Some research has shown that unsatisfied curiosity tends to intensify over some interval as do other drives such as hunger and thirst.

CURIOSITY AND EXPLORATION

What exactly are curiosity and exploration? Loewenstein (1994) points out **four central issues of curiosity**: definition and dimensionality, cause, voluntary exposure to curiosity, and situational determinants.

He adds a fifth issue of superficiality and intensity since he states that curiosity can arise, change focus or end abruptly.

Loewenstein believed that despite its transience, curiosity can be a powerful motivational force. **Langevin** (1971) conducted research in the area of curiosity and classified measures of curiosity into two categories.

Curiosity is viewed as a **motivational state** and measured with behavioural indices.
Curiosity is viewed as a **personality trait** that is assessed by personality measures.

It has been suggested that curiosity is not a unitary construct. At the conceptual level there are numerous definitions of curiosity which tend to encompass a broad range of characteristics. Fowler (1965) stated that **boredom is generally a prerequisite for curiosity** (exploration). Curiosity and exploration are difficult to define independently when looking at them from a psychological perspective. This is so because whenever we consider exploration and curiosity, the concepts of motivation and drive come into play and become intertwined.

KINDS OF CURIOSITY

William James (1890) pointed out to two kinds of curiosity.

i) He emphasised the biological function of curiosity as a mechanism of **instinct driven behaviour** that serves in approaching new objects. Approach and exploration are described as being characteristic forms of behaviour.

The second kind of curiosity pointed out by James is “**scientific curiosity**” and “metaphysical wonder” with which “the practical instinctive root has probably nothing to do” rather “the philosophical brain responds to an inconsistency or a gap in its knowledge”.

In the psychoanalytical literature **Freud** views curiosity as a derivative of the **sex drive**. The partial impulse of looking motivates the child’s great interest in all things and all events that have to do with sexuality. Whereas the looking impulse and curiosity are primarily sexual in origin, the child’s exploratory interest and desire for knowledge can be considered to be a by product of cognitive development. Due to social pressure, sexual exploration is later abandoned.

Blarer proposed curiosity to be intrinsic to the individuals perceptions and world experiences and thus Blarer’s view is the basis for the **intrinsic motivation viewpoint in curiosity theory**.

ZUCKERMAN’S SENSATION SEEKING

novel and highly stimulating

Sensation seeking is a personality trait defined by the degree to which an individual seeks **activities** and experiences. People who are high in sensation seeking are attracted to the unknown and as a result consistently seek the new, varied, and unpredictable. Examples of such behaviors are varied, but sensation seekers may be attracted to extreme sports, frequent travel, diverse foods and music, new sexual partners and experiences, and challenging existing viewpoints. Specifically, Zuckerman’s basic proposition is that sensation seeking is based on individual differences in the **optimal level**

of sensation caused by biological nervous-system differences. People who are **high in sensation seeking** are individuals who have relatively **low-level nervous system activation** and therefore seek arousal from their external environment by looking for novel stimuli and engaging in varied experiences. In contrast, individuals who are **low in sensation seeking** have a naturally **higher level of internal activation** and thus do not tend to seek sensation from external sources.

Zuckerman posits that sensation seeking is **genetically influenced** because it is evolutionary adaptive. Across the animal kingdom, engaging in a certain degree of risky behaviors will increase the likelihood of survival and reproductive success (e.g., seeking new territories for food and new potential mates).

Zuckerman first created the **Sensation Seeking Scale** in 1964 to measure an individual's overall level of susceptibility to excitement or boredom in the context of sensory deprivation experiments. Current versions of the self-report measure include four subscales:

Thrill and Adventure Seeking—the extent to which individuals engage in or are interested in participating in risky activities such as parachuting or skiing;

Experience Seeking—the degree to which one seeks excitement through the mind, such as from music, art, and travel;

Disinhibition—seeking sensations through social stimulation and disinhibitory behaviors such as drinking and sex; and

Boredom Susceptibility—avoiding monotonous, repetitive, and boring situations, people, and activities.

MOTIVATIONAL COMPETENCE

A theory of achievement motivation based on a person's feelings of personal

competence. According to the theory competence **motivation increases when a person successfully masters a task**. This encourages the person to master more tasks.

Harter's Model

Harter's competence motivation theory (1978) can be used to explain motivation to participate and drop out of physical activity. This theory represents an interactionist view of behavior, incorporating both individual and situational factors that impact one's motivation. The original model came from White (1959). Harter (1978) expanded on the premise that **people are intrinsically motivated to master specific domains**. As a consequence, they gain and display competence and control over their environment, which in turn increases pleasure (positive affect or enjoyment). If people succeed, it increases motivation to adhere to physical activity. However, if people do not believe they are competent in that domain, they will have negative affect, and this will decrease their motivation. In addition, her model incorporates the impact of significant others on one's perceptions of competence, affect, and motivation (Harter, 1978).

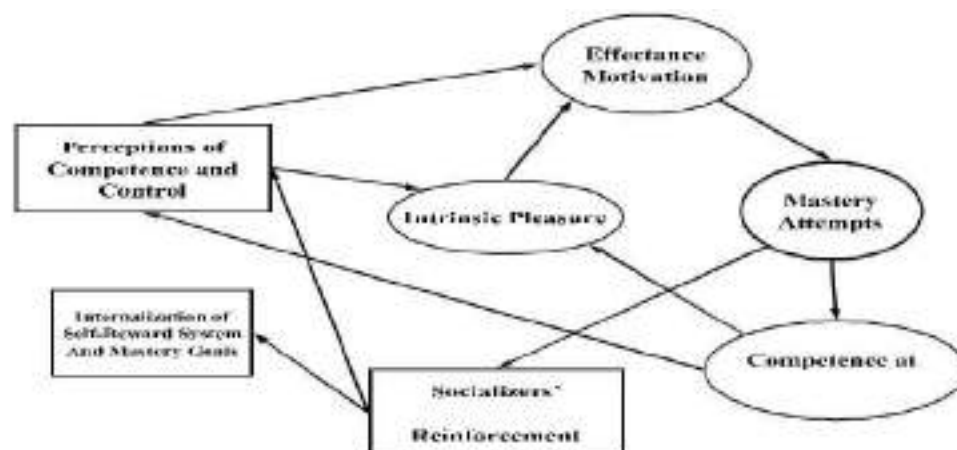


Figure 1. Harter's (1978) competence motivation theory.

SELF-REGULATION

Self-regulation models are concerned with what individuals choose to do and how they go about trying to accomplish their goals. In more formal terms, we can distinguish three components of the self-regulation process:

goal selection, (2) preparation for action, and (3) a cybernetic cycle of behavior (made up of several component processes) (Markus & Wurf, 1987).

Goal Selection- The first stage in the self-regulation process is the goal-selection stage. Before they can effectively regulate their behavior, people must select a goal; they must decide what they intend to do.

Preparation for action- Having adopted a goal, people prepare to attain it. This is the second stage in the self-regulation process. Here, people gather information, construct scenarios regarding possible outcomes, and engage in behavioral practice (rehearsal). In short, they design and prepare to implement a plan to achieve their goal. Of course, not all behavior fits this model.

Cybernetic Cycle of Behavior The third stage in the self-regulation process has been conceptualized as a cybernetic cycle of action. Cybernetics is the study of how entities use information to regulate their actions (Wiener, 1948). It is also called **control theory**, as it emphasizes negative feedback control as the means by which machines (e.g., thermostats, guided missiles, cruise control settings in automobiles) as well as animals adjust their behavior to match some standard.

Formally, this process is known by the acronym **TOTE**, as it involves four stages:

A test phase, in which a present value is compared against some relevant standard (the current temperature in the room is

compared with the desired temperature);

an operate stage, in which an action is undertaken to bring the present value in line with the standard (the heat comes on if the room temperature is below the standard);

another test phase, in which the new value is compared with the standard (the new room temperature is compared with the desired temperature); and

an exit, or quit, stage, which occurs when the desired goal is reached (the furnace shuts off when the room reaches the selected temperature).

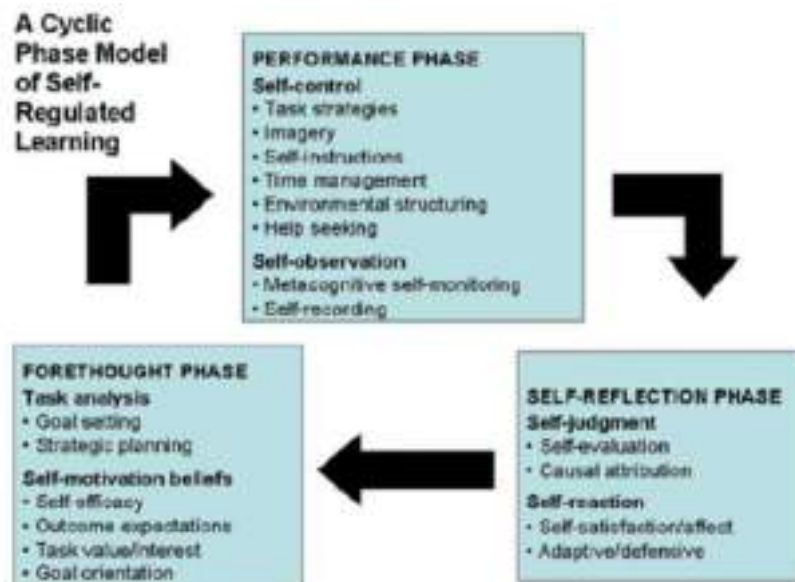
THREE SELF-RELEVANT PHENOMENA

Self-Efficacy-Beliefs People's beliefs about their ability to succeed exert a strong influence on the self regulation process. Bandura (1986, 1989) refers to such beliefs as self-efficacy beliefs. People with high self-efficacy beliefs think they have the ability to succeed at a task, to overcome obstacles, and to reach their goals. People with low self-efficacy beliefs doubt their ability to succeed and do not believe they have what it takes to reach their goals.

Possible Selves- People's ideas about what they may be like in the future also influence motivated behavior. Markus and her colleagues (Markus & Nurius, 1986; Markus & Ruvolo, 1989) coined the term possible selves to refer to these beliefs. To illustrate, an aspiring gymnast might have a clearly articulated "Me winning an Olympic gold medal" possible self. This person is able to vividly imagine herself on the victory stand with the national anthem playing in the background and the crowd cheering while she receives her medal.

Self-Awareness- A third variable of interest to a motivational analysis of behavior is self-awareness. As discussed throughout this text, the self has a reflexive quality: People are capable of taking themselves as the object of their own attention. But our attention is not always focused inward.

Much of the time (perhaps most of the time) our attention is focused outward on the environment. This means that attentional focus is variable and that self-awareness is a transient state. Sometimes we are aware of ourselves; other times we are not.



Bandura's self-regulation proposal

Self-Observation	Self-Judgment	Self-Reaction
Regularity	Types of standards	Evaluative motivators
Proximity	Goal properties	Tangible motivators
Self-recording	Goal importance	
	Attributions	

FLOW

Viewed through the experiential lens of flow, *a good life is one that is characterized by complete absorption in what one does.*

Studying the creative process in the 1960s (Getzels & Csikszentmihalyi, 1976), **Csikszentmihalyi** was struck by the fact that when work on a painting was going well, the artist **persisted single-mindedly, disregarding hunger, fatigue, and discomfort**—yet rapidly lost interest in the artistic creation once it had been completed.

The **conditions of flow** include:

- Perceived challenges, or opportunities for action, that stretch (neither overmatching nor underutilizing) existing skills; a sense that one is engaging challenges at a level appropriate to one's capacities clear proximal goals and immediate feedback about the progress that is being made.

Under these conditions, experience seamlessly unfolds from moment to moment, and **one enters a subjective state** with the following characteristics:

- Intense and focused concentration on what one is doing in the present moment
- Merging of action and awareness
- Loss of reflective self-consciousness (i.e., loss of awareness of oneself as a social actor)
- A sense that one can control one's actions; that is, a sense that one can in principle deal with the situation

because one knows how to respond to whatever happens next Distortion of temporal experience (typically, a sense that time has passed faster than normal). Experience of the activity as intrinsically rewarding, such that often the end goal is just an excuse for the process. Because the direction of the unfolding flow experience is shaped by both person and environment, we speak of **emergent motivation** in an open system (Csikszentmihalyi, 1985): what happens at any moment is responsive to what happened immediately before within the interaction, rather than being dictated by a pre-existing intentional structure located within either the person (e.g., a drive) or the environment (e.g., a tradition or script). Here, motivation is emergent in the sense that **proximal goals arise out of the interaction.**

THE AUTOTELIC PERSONALITY

From the beginning, Csikszentmihalyi (1975/2000) recognized the possibility of an autotelic personality, a person who tends to enjoy life or “**generally does things for their**

own sake, rather than in order to achieve some later external goal” (Csikszentmihalyi, 1997, p. 117). This kind of personality is distinguished by several **meta-skills** or competencies that enable the individual to enter flow and stay in it. These meta-skills include a **general curiosity and interest in life, persistence, and low self-centeredness**, which result in the ability to be motivated by intrinsic rewards.

The **Flow Questionnaire** presents respondents with several passages describing the flow state and asks whether they have had the experience, how often, and

in what activity contexts (Csikszentmihalyi & Csikszentmihalyi, 1988).

The quotations used were drawn from the original interviews about flow activities (Csikszentmihalyi, 1975/2000), one each from a dancer, a rock climber, and a composer.



PHYSIOLOGICAL CORRELATES OF EMOTION

Any stimulus that arouses emotion alters the activity of the autonomic nervous system, *the section of the nervous system that controls the organs* such as the heart and intestines. The word *autonomic* means “independent” (autonomous). Biologists once believed that the autonomic nervous system was independent of the brain and spinal cord. We now know that the brain and spinal cord regulate the autonomic nervous system, but the term **autonomic** remains.

The autonomic nervous system consists of the sympathetic and the parasympathetic nervous systems. *Chains of neuron clusters just to the left and right of the spinal cord* comprise the **sympathetic nervous system**, *which arouses the body for vigorous action. It is often called the “fight-or-flight” system because it increases your heart rate, breathing rate, sweating, and flow of epinephrine (EP-i-NEF-rin; also known as adrenaline), thereby preparing you for vigorous activity.* Different situations activate different parts of the sympathetic nervous system to facilitate different kinds of activity.

The **parasympathetic nervous system** consists of *neurons whose axons extend from the medulla (see Figure 12.1) and the lower part of the spinal cord to neuron clusters near the organs.* The *parasympathetic nervous system decreases the heart rate and promotes digestion and other nonemergency functions.* Both the sympathetic and parasympathetic systems send axons to the heart, the digestive system, and most other organs. A few organs, such as the adrenal gland, receive only sympathetic input.

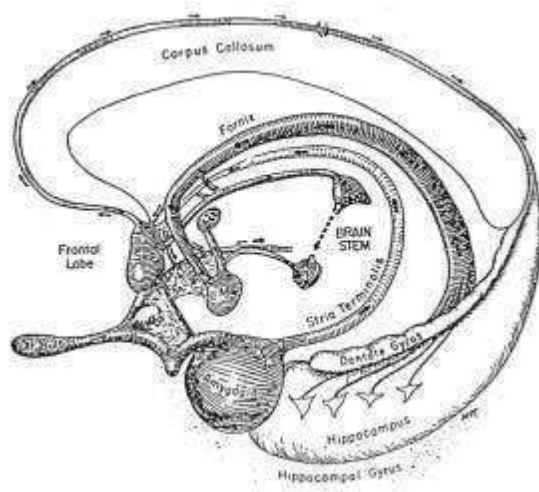
Both systems are constantly active, although one system can temporarily dominate. If you spot danger at a distance (in either time or space), you pay attention to it with mainly parasympathetic activity. If the danger is close enough to require action, you shift to vigorous sympathetic activity (Löw, Lang, Smith, & Bradley, 2008). Many situations activate parts of both systems (Berntson, Cacioppo, & Quigley, 1993). Some emergency situations increase your heart rate and sweating (sympathetic responses) and also promote bowel and bladder evacuation (parasympathetic responses). Have you ever been so frightened that you thought you might lose your bladder control?

To measure emotion, researchers measure sympathetic nervous system arousal as indicated by **heart rate, breathing rate, or momentary changes in the electrical conductivity across the skin.** However, remember that the sympathetic nervous system is the fight-or-flight system, so its responses could indicate anger, fear, or any other intense emotion. Physiological measurements do not tell us *which* emotion someone is feeling.

Limbic system

The limbic system was originally proposed to consist of interconnected subcortical structures with pathways to the hypothalamus. The limbic system was proposed to modulate the emotional quality of stimuli and support autonomic effector mechanisms associated with emotional states. A key limbic structure that has a critical role in **emotional** expression is the **amygdala**. The amygdala has an important role in evaluating the emotional valence of stimuli. Support for this view arises from extensive work done with lesions of the amygdala. For

example, animals with **amygdala lesions** have difficulty learning associations between environmental stimuli and emotional states. They may fail to learn that a stimulus predicts reward or danger, they may fall in social rank, or show decreased affiliative behavior. Damage to other limbic structures can also produce changes in emotional behavior.



It is important to note that an **interaction exists between cortical brain regions and the limbic system**. There are massive connections between cortical regions, particularly from the frontal and temporal lobes, to subcortical limbic structures (Fig. 2). The implication of these connections is that complex sensory information processing occurring in the cortex can directly influence the limbic system. Conversely, limbic processing can strongly influence higher-level cognitive integration occurring in the cortex. Disconnection in

the transmission of information between the cortical and subcortical limbic structures can have dire consequences. For example, **patients with frontal lobe lesions show inappropriate emotional and social behavior in the absence of intellectual deficits. These patients might cry or laugh inappropriately, urinate in public, or use profanity.**

Monoaminergic systems (serotonin, norepinephrine, dopamine)

The neural circuits and brain structures involved in emotions are modulated by a myriad of chemical **neurotransmitters**. The ascending monoamine systems have received considerable attention over the past several decades. These include the **serotonin, norepinephrine, and dopamine** systems. Prior to the discovery of neurotransmitters, researchers believed that a major ascending neural system was responsible for arousal of forebrain (epithalamus, thalamus, subthalamus) and telencephalon (cerebral cortex, basal ganglia and associated structures like the nucleus basalis of Meynert and the nucleus accumbens). This neural system used to be called the **ascending reticular activating system**, before the monoamines were characterized. It is believed that a balance among these systems (as well as other neurotransmitters) is necessary for normal emotional states and arousal. Over the last three decades, the neurochemical basis of this ascending system was described and receptors identified.

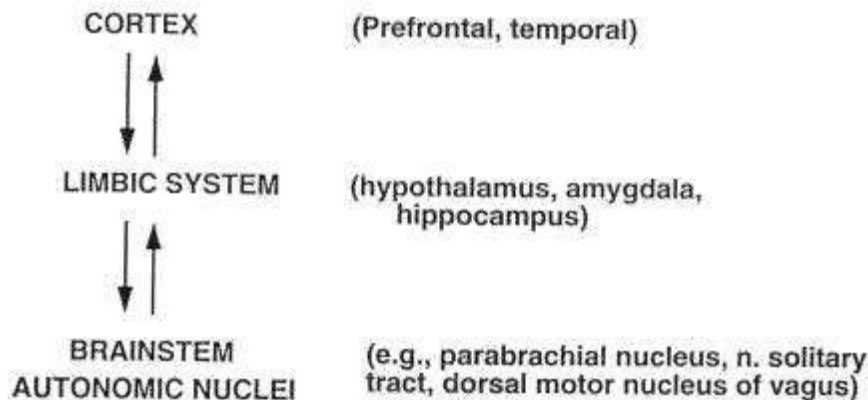


Fig. 2. Schematic diagram emphasizing the interactions between structures in the limbic system

THE NEURAL SUBSTRATES OF FEAR AND ANXIETY

This class of emotion is elicited by threatening situations and it functions as an internal signal to alert the organism to potential danger. In response to fear, individuals engage in defensive or protective acts that serve to promote survival. These behaviors include **fleeing** or **withdrawing** from a situation, freezing to remain inconspicuous, or fighting.

The amygdala: A key structure mediating fear. In the 1930's, Kluver and Bucy noted that large lesions of the temporal lobe made monkeys tame in the presence of previously fearful stimuli, such as humans and snakes. The absence of an appropriate behavioral response to fear-eliciting stimuli was termed "**psychic blindness**" because it was presumed that the **cognitive** processing of emotional stimuli was altered. This taming effect could be produced by lesions restricted

to the amygdala. In contrast to the taming effects of amygdala lesions, electrical stimulation of the amygdala elicits defensive and flight reactions in cats and feelings of fear and anxiety in humans. Increased autonomic activity, i.e., heart rate, blood pressure also occurs after electrical stimulation of the amygdala.

The amygdala may also be involved in what is termed "**emotional memory**." We seem to be able to better recall events surrounding a strong, negative emotional experience than events not linked to any particular experience. Many people in the U.S. of a certain generation can remember exactly where they were and what they were doing when informed that President Kennedy was shot. Younger people may clearly remember the day the space shuttle Challenger blew up.

A human case study illustrates the importance of the amygdala in the recognition of emotional **stimuli**. A patient suffered from **Urbach-Wiethe disease**, a rare genetic disorder, that resulted in bilateral calcification and atrophy of her amygdala. When asked to rate the intensity of various facial expressions, the patient judged faces showing fear expressions to be considerably less intense than ratings made by normal control subjects. Other facial expressions (smiling for instance) were also judged by the patient to be less intense than those reported by controls, but not to the degree made when viewing **fear** expressions. **Bilateral** amygdala damage appears to produce a profound insensitivity to the intensity of

fear shown by faces. The patient, however, had no difficulty recognizing people by their faces and could rapidly learn the identity of new faces. Role of corticotrophin-releasing hormone (CRH) systems in fear and anxiety

In animals, administration of CRH into the cerebral ventricles (so as to eventually reach receptors on amygdala and LC cells) effectively **induces**

anxiety responses, including hypervigilance, enhancement of the freezing posture, and decreased exploration in unfamiliar situations. Furthermore, in anxiety-provoking situations that typically elicit these behavioural responses, administration of a CRH antagonist produces a reduction in the occurrence of these reactions. In rats, infusion of a CRH antagonist into the central nucleus reduces expression of fear behavior (“freezing” in an environment where the animal had been previously shocked) suggesting that blockade of CRH receptors in the central nucleus has an antianxiety effect.

SADNESS AND NEGATIVE

AFFECT An important clinical observation was made in the 1950's when the antihypertensive agent reserpine was prominently used. Clinicians noted that some individuals became markedly depressed after taking this drug, which produces a long-lasting depletion of monoamines (norepinephrine, serotonin and dopamine). Other work demonstrated that drugs that increased the level of monoamines were effective in the treatment of depression. Together, these observations led to the **monoamine hypothesis of depression.** According to this hypothesis, depression results from a deficit in brain norepinephrine or serotonin, or both.

Lowered brain serotonin is associated with Suicide Monoamine oxidase inhibitors (MAOI), tricyclic antidepressants, and selective serotonin reuptake inhibitors (SSRIs) are effective antidepressants that share The pharmacological Property of

increasing the level of biogenic amines (DA, NE, epineph serotonin, histamine) but in different ways. **MAOIs** are a class of drugs that block MAO, the major enzyme responsible for the oxidation of monoamines. The **tricyclic drugs** work by blocking the reuptake (keep it around longer) of NE and serotonin into the presynaptic terminal resulting in a net increase in neurotransmitter availability. Consequently, there is an increase in postsynaptic receptor activity. **SSRIs** work by selectively blocking the reuptake of serotonin. SSRIs are as effective as the tricyclic compounds but without some of the sedating and cardiovascular side effects of tricyclic antidepressants. As a result, SSRIs (e.g., Prozac, Zoloft) are now used widely and underscore the importance of serotonergic systems in regulating mood.

It is presently unclear why reduced brain **serotonin function predisposes individuals to commit suicide.** One hypothesis is that low brain 5-HT values produce an **increase** in impulsive behavior. **Impulsivity** refers to a propensity to act without considering alternative options in a decision-making process. Although impulsivity is not synonymous with acting rapidly, impulsive individuals tend to act without time for reflection. In people with personality disorders characterized by chronic problems with impulsive behavior, the rate of completed suicide can be as high as 25%. The underlying cause of reduced brain serotonin remains unknown but could be related to heritable factors and/or neurological insults during development.

THEORIES OF EMOTION

JAMES LANGE THEORY

Proposed independently by psychologist William James and physiologist Carl Lange, the James-Lange theory of emotion proposes that emotions occur as a result of physiological reactions to events. According to this theory, you see an external stimulus that leads to a physiological reaction. Your emotional reaction is dependent upon how you interpret those physical reactions. For example, suppose you are walking in the woods and you come across face to face with a bear. You begin to tremble and your heart begins to race. The James-Lange theory proposes that you will interpret your physical reactions and conclude that you are frightened (“I am trembling, therefore I am afraid.”)

William James explained the same thing differently. He said that the bodily changes follow directly the perception of the exciting fact, and that our feeling of the same changes as the facts occur and these feelings are called emotions.

The **James-Lange theory** states:

**Environmental influence (event) —>
Physiological change —>
Psychological experience**

Problems:

Later studies separated the internal organs that James said caused arousal from the CNS, but this did not eliminate emotional responding. So, perceptions of bodily changes could not be the only factor involved in emotions. James Lange theory offers a more surprising

view of emotion. It suggests that the subjective emotional experiences are actually the result of physiological changes within our bodies. In other words you feel frightened when making your speech because you notice that your heart is racing, your mouth is dry, and so on. We feel sorry because we cry, angry because we strike, and afraid because we tremble

CANNON – BARD THEORY OF EMOTION

This theory suggests that various emotion provoking events induce simultaneously the subjective experiences which are labeled by humans as emotions. All these emotions are accompanied by physiological reactions and thus in the situation given as an example of a student going to the stage to give a lecture, the sight of the audience and of his professor, whose pen is poised to evaluate the student's performance, causes the student to experience a racing heart, a dry mouth and other signs of physiological arousal and at the same time, also to experience subjective feelings which is labeled as fear. In other words it may be stated that this situation stimulates various portions of the nervous system so that both arousal, mediated by the autonomic nervous system and subjective feelings, mediate in part by the individual's cerebral cortex .

The Cannon-Bard theory is a physiological explanation of emotion developed by Walter Cannon and Philip Bard, and states that **we feel the emotions and experience the**

physiological reactions such as sweating, trembling and muscle tension simultaneously. More specifically, it is suggested that emotions result when the thalamus sends a message to the brain in response to a stimulus, resulting in a physiological reaction.

For example: I see a snake → I am afraid → I begin to tremble.

According to the Cannon-Bard theory of emotion, we react to a stimulus and experience the associated emotion at the same time. The Cannon Bard theory of emotion differs from other theories of emotion such as the James Lange theory of emotion, which argues that physiological responses occur first and this in turn cause the emotions. The questions in regard to the emotions that arise are for instance

Where do our emotions come from?
What is it about the human body and mind that has the capacity to form emotions and reactions to such a variety of situations?
Why do we form these emotions?

According to Cannon-Bard, emotions do not come as a response to physiological conditions. However, at the same time, neither do emotions come and then the body creates a physical reaction to the said emotions. When we consider this philosophy we realise that according to Cannon-Bard **“emotions and physiological responses occur at exactly the same time.”** When both these theories (James Lange and Cannon Bard’s) are considered, the issue arises is that which of these theories is closer to the truth. Until recent decades, most of the psychologists believed that Cannon-Bard theory was more accurate. They reached this conclusion on the

basis of several forms of evidence. More recently though the pendulum of scientific opinion has begun to swing the other way and certain aspects of James-Lange approach have gained increasing acceptance. Additional support for the James-Lange theory of emotion is provided by studies of the **facial feedback hypothesis** according to which changes in our facial expression sometimes produce shifts in our emotional experiences rather than merely mirroring them. In other words, as James would suggest, we feel happier while we smile, sadder when we frown, and so on. While there are many complexities in examining the hypothesis, the results of several studies offer support for accuracy

SCHACHTER-SINGER’S TWO-FACTOR THEORY OF EMOTIONS

This theory is similar to Bem’s explanation for the cause of behaviour in general. Schachter’s theory looks specifically at how we decide what emotions we are experiencing especially when we are experiencing one. When trying to understand what kind of person we are, we first watch what we do and feel and then deduce our nature from this.

This means that the first step in the experience of motions is to experience physiological arousal. You are physiologically up or down compared to normal. We then try to find a label to explain our feelings usually by looking at what we are doing (behaviour) and what else is happening at the time of arousal (environment) Thus we do not just feel angry, happy or what ever. We experience general feeling and then decide what the experience mean, a specific emotion.

Physiological arousal + Environment circumstances = Attributed emotions

- e.g. 1. I am tense and sweating + a gun is being pointed at me = I am afraid.
e.g.2. I am tense and sweating + Sheela is looking at me and smiling = I am in love.

In both the examples, the state of arousal is the same. What changes is the environment.

Two factor theory argues that the **cues** in the environment are what determine the emotions that we believe we are experiencing. **Change the environment the emotions will also change.** Like the James Lange theory of emotion, Schachter and Singer felt that **physical arousal** plays a primary in emotions. However, they suggested that this arousal was the same for a wide variety of emotions, so physical arousal alone could not be responsible for emotional responses. The two-factor theory of emotion focuses on the **interaction between physical arousal and how we cognitively label** that arousal.

SCHACHTER AND SINGER'S EXPERIMENT

In a 1962 experiment, Schachter and Singer put their theory to the test. A group of 184 male participants were injected with epinephrine, a hormone that produces arousal including increased heartbeat, trembling and rapid breathing. All of the participants were told that they were being injected with a new drug to test their eyesight. However, one group of participants was informed of the symptoms the injection might cause, while other participants were not. Participants were then placed in a room with another participant who was actually a confederate in the experiment. The confederate either acted in one of two ways: euphoric or angry. Participants who had not been informed

about the effects of the injection were more likely to feel either happier or angrier than those who had been informed. Schachter and Singer developed the two-factor theory of emotion. The two-factor theory suggests that emotion comes from a combination of a state of arousal and a cognition that makes best sense of the situation the person is in. For example, the two-factor theory of emotion argues that when people become aroused they look for cues as to why they feel the way they do. If a person experiences a state of arousal for which they have no immediate explanation, they will describe their emotions in terms of the cognitions available to them at the time. If a person experiences a state of arousal for which they have an appropriate explanation e.g. 'I feel this way because I have just received an injection of adrenalin', then they will be unlikely to describe their emotion

Criticism of Two-Factor Theory

While Schachter and Singer's research spawned a great deal of further research, their theory has also been subject to criticism. Other researchers have only partially supported the findings of the original study, and have also shown contradictory results. Other criticisms of the two-factor theory:

Sometimes emotions are experienced *before* we think about them.

There are actual physiological differences between emotions.

To sum up this theory, strong emotions are a common part of daily life, but how do we tell them apart? How do we know that we are angry rather than frightened, sad rather than surprised? One potential answer is provided by a third theory of emotion, that is the two-factor theory of emotions by Schachter and Singer.

According to this view, emotion provoking events produce increased arousal. In response to the feelings of arousal, we search the external environment in order to identify the cause of such arousal feelings. These causes that we select play a key role in determining the importance we place on our arousal. If we feel aroused after a near-miss in traffic, we will probably label our emotion as fear or perhaps anger. If instead we feel aroused in the presence of an attractive person, we may label our arousal as attraction or love. In short, we perceive ourselves to be experiencing the emotion based on the external cues, as well as our processing them, and then suggest what we should be feeling. The theory is described as a two factor theory because it considers both arousal and cognitive appraisal that we perform in our efforts to identify the cause of such arousal

LAZARUS'S COGNITIVE THEORY

The importance of cognitive interpretation of circumstances in determining the arousal levels is central to Lazarus Theory. In this theory he suggests that some degree of **cognitive processing** is essential before an emotional reaction, either overt or internal, can occur.

s in terms of the alternative cognitions available. If a person is put in a situation, which in the past could have made them feel an emotion, they will react emotionally or experience emotions only if they are in a state of physiological arousal.

Experiment

The participants were 184 male college students, taking classes in introductory psychology at Minnesota University. As soon as the participant arrived, he was

taken to a private room by the experimenter and told that the aim of the experiment was 'to look at the effects of vitamin injections on visual skills', and was asked if he would mind having an injection of 'Suproxin' (made up name). 184 out of 195 participants agreed to the injection. They were given an injection (by a doctor) of either adrenalin (epinephrine) or a placebo, which was actually a saline solution, which has no side effects at all. The effects of the adrenalin are an increase in blood pressure, heart rate, blood sugar level, respiration rate, and blood flow to the muscles and brain, with an accompanying decrease in blood flow to the skin. This is often experienced as palpitations, tremors, flushing and faster breathing. The effects begin after three minutes and last from ten minutes to an hour.

The participants were then put in one of four experimental conditions:

Adrenalin Ignorant - participants were given no idea of the injection adrenalin

Adrenalin informed - participants were informed what adrenalin will do

Adrenalin misinformed - participants were given wrong information about adrenalin effect 4) Control group.

The researchers then made observational measures of emotional response through a one-way mirror, and also took self-report measures from the participants. In the euphoria condition the misinformed participants were feeling happier than all the others. The second happiest group was the ignorant group. This demonstrates that these participants were more susceptible to the stooge because they had no explanation of why their bodies were reacting differently.

Lazarus's Cognitive Theory

Lazarus et al.(1980) proposed a theory suggesting that **emotion is a cognitive function**, arising as result of appraisal of a situation. A situation may be appraised as nonthreatening leading to positive emotional states. The actual emotion experienced depends upon other characteristics or circumstances. A stimulus appraised as threatening leads to direct action, such as attack, retreat or freezing, together with physiological responses such as these which accompany negative states, including fear, anger or depression. If direct action is impossible, coping strategies may be employed to reappraise the situation benignly, in order to live with the threat. These coping strategies may be simply to redefine the threat as 'not' as bad as it was first perceived. For example, why does a wife continue to live with her husband who is violent towards her? The husband who is the stimulus is appraised as threatening, but for various reasons the wife cannot take direct action. Probably she cannot fight back, as her husband is stronger, she cannot run away, as she has no one and nowhere to go. In such cases, the situation is to be reappraised and a different solution is to be found such as staying on with the husband and losing the extreme fear and doing things that may prevent violence etc. A different thinking pattern may also perhaps help. In other words, the woman may start thinking that the husband may not perhaps hit her again, and that he is very considerate the next day and so on. Living with this reappraised threat is less problematic than fear of the outside of that situation.

This theory is primarily descriptive, but the underlying mechanisms involved in emotion such as the relationship between cognitions, feelings and expressions of emotion are undefined to a great extent. Thus to sum up the cognitive theory of emotions, it may be stated that as follows: Lazarus' cognitive-mediational theory: This is an extension of Schacter-Singer

theory. **Cognitive appraisal** of a situation is of primary importance in emotional states. In contrast, Zajonc (pronounced "ZI-ANTS," rhymes with "science") holds that we feel first and think later. In other words, the emotional state strongly influences the cognitive appraisal. More recently, cognitive scientists have proposed that there may be different components to our response to stimuli, but it is not productive to segregate them into cognitive vs. emotional categories

LINDSLEYS ACTIVATION THEORY

The American psychologist and neuroscientist Donald Benjamin Lindsley's (1907-2003) germinal work on activation in the general area of emotion shows the importance of the **brain-stem portion** of the brain called the **reticular substance**. Lindsley's activation theory is based, in particular, on research involving the electroencephalogram (**EEG**) and its relevance toward understanding the interaction of the cerebral cortex and the subcortical structures. The activation theory was advanced not only as an explanatory concept for emotional behavior, but it was related also to the phenomenon of sleep-wakefulness, to EEG recordings of cortical activity, and to different types of abnormal behavior involving various psychiatric symptomatology.

The activation theory states: the EEG in emotion shows an activation pattern with **reduction in alpha** (synchronized) rhythms and induction of low-amplitude, fast activity;

the EEG activation pattern is reproducible by electrical stimulation of the **brain-stem reticular formation** (BSRF);

destruction of the rostral end of the BSRF abolishes EEG activation and allows restoration of rhythmic discharges in the thalamus/cortex;

the **behavior associated with destruction** of the rostral end of the BSRF is the opposite of emotional excitement, namely, apathy, somnolence, lethargy, and catalepsy;

the combined mechanism of the **basal diencephalon and lower BSRF** is identical with, or overlaps, the EEG activating mechanism, and this mechanism causes the objective features of emotional expressiveness to appear.

Lindsley (1951) concludes that it is not legitimate on the basis of the existing experimental evidence to attempt to account for all the varieties of emotional expression, and further research is advised on the influences of learning, habituation, and memory on emotional expression. The activation theory is able to account for the extremes of emotional behavior but is not able to explain completely the intermediate and mixed states of emotional expressiveness.

DEFINING EMOTION REGULATION

Emotion regulation refers to the processes by which individuals influence which emotions they have, when they have them, and how they experience and express these emotions. Emotion regulatory processes may be automatic or controlled, conscious or unconscious, and may have their effects at one or more points in the emotion generative process (which I describe in a later section). Because emotions are multi-componential processes that unfold over time, emotion regulation involves changes in "emotion dynamics" (Thompson, 1990), or the latency, rise

time, magnitude, duration, and offset of responses in behavioral, experiential, or physiological domains. Emotion regulation also involves changes in how response components are interrelated as the emotion unfolds, such as when large increases in physiological responding occur in the absence of overt behavior.

Focus on five aspects of this definition of emotion regulation.

First, individuals increase, maintain, and decrease negative and positive emotions (Parrott, 1993).

Second, neural emotion circuits do not appear to overlap completely (LeDoux, 1994; Panksepp, 1982; 1998). This suggests that circuits involved in regulating these emotions also may not overlap completely, and that there may be important differences in emotion regulatory processes across emotions.

Third, this definition of emotion regulation emphasizes regulation in self. Other definitions include attempts to influence others' emotions (e.g., Gross & Levenson, 1993; Masters, 1991; Thompson, 1994).

Fourth, prototypic examples of emotion regulation are conscious, such as deciding to change an upsetting conversational topic or squelching laughter at a child's inappropriate antics. One can imagine, however, emotion regulatory activity that occurs without conscious awareness, such as hiding one's disappointment at an unattractive present (Cole, 1986) or turning one's attention away from potentially upsetting material (Boden & Baumeister, 1997).

Fifth, I make no a priori assumptions as to whether emotion regulation is good or bad (Thompson & Calkins, 1996). This circumvents the confusion that was created in the stress and coping literature by

predefining defenses as maladaptive and coping as adaptive (Parker & Endler,

1996). Thus, cognitive strategies that dampen negative emotions may permit medical professionals to operate successfully (Lief & Fox, 1963; Smith & Kleinman, 1989). The same strategies, however, may be used to dehumanize an enemy and neutralize empathic distress that could interfere with state-sanctioned killing (Bandura, 1977).

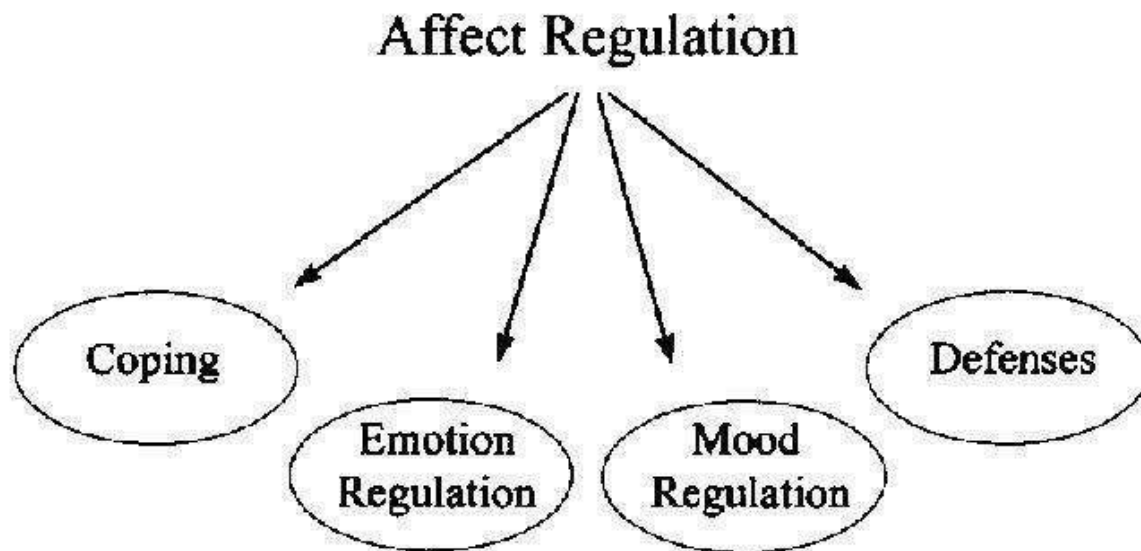


Figure 2. A hierarchical conception of affect regulation.

TABLE 1
Target by function classification of emotion-regulation strategies

<i>Emotion-generating system</i>	<i>Psychological function</i>		
	<i>Need-oriented</i>	<i>Goal-oriented</i>	<i>Person-oriented</i>
Attention	Thinking pleasurable or relaxing thoughts (Langens & Mörth, 2003); Attentional avoidance (Derakshan et al., 2007)	Effortful distraction (Van Dillen & Koole, 2007); Thought suppression (Wenzlaff & Wegner, 2000)	Attentional counter-regulation (Rothermund et al., 2008); Meditation (Cahn & Polich, 2006); Mindfulness training (Brown et al., 2007)
Knowledge	Cognitive dissonance reduction (Harmon-Jones & Mills, 1999); Motivated reasoning (Kunda, 1990); Self-defence (Tesser, 2000)	Cognitive reappraisal (Gross, 1998b; Ochsner & Gross, 2008)	Expressive writing (Pennebaker, 1997); Specification of emotional experience (Neumann & Philippot, 2007); Activating stored networks of emotion knowledge (Barrett et al., 2001)
Body	Stress-induced eating (Greeno & Wing, 1994); Stress-induced affiliation (Taylor et al., 2000)	Expressive suppression (Gross, 1998a); Response exaggeration (Schmeichel et al., 2006) Venting (Bushman et al., 2001)	Controlled breathing (Philippot et al., 2002); Progressive muscle relaxation (Esch et al., 2003)

Note: Cited articles refer to relevant empirical demonstrations or literature reviews.

CONFLICT

Conflict is any situation in which there are incompatible goals, cognitions, or emotions within or between individuals or groups that lead to opposition or antagonistic interaction. It is the struggle between incompatible and opposing needs, wishes, ideas, interests, or people. Conflict is a form of interaction among parties who differ in interests, perceptions, goals, values, or approaches to problems. Conflict arises when we begin to feel that the other person is interfering with our ability to attain a certain objective. It begins when we believe the other party is interfering or standing in the way of an action we want to take, an idea we want to pursue, or a belief we hold. Conflicts may involve individual or group disagreements, struggles, disputes, quarrels, or even physical fighting and wars. Because human beings are unique—possessing a variety of physical, intellectual, emotional, economic, and social difference—conflict is inevitable.

Conflict can be either positive or negative. The outcomes of conflict depend on how the conflict is managed or resolved. **Positive conflict** is functional and supports or benefits the organization or person's main objectives.⁵ Conflict is constructive when it leads to better decisions, creativity, and innovative solutions to long-standing problems. Conflict is viewed as positive when it results in:

- *Increased Involvement*—Organizational members have the opportunity to develop goals, share ideas, and voice opinions, gaining greater insight into others and situations.

Increased Cohesion—Members build strong bonds from learning how to resolve differences; “if we can survive this, we must have a true relationship”

embodies this benefit of conflict. In some cases conflict initially reduces cohesion that can in turn reduce the likelihood of “group think” occurring. In this case conflict is positive.

- *Increased Innovation and Creativity*—Members are encouraged to “put their ideas on the table”;⁷ this can lead to more discoveries, improvements, and creative solutions. “Two heads are truly better than one” when conflict brings about synergy instead of chaos.

Positive Personal Growth and Change—Individuals learn their strengths and weaknesses; conflict of ideas challenges individuals to learn and grow by expressing their ideas and thoughts through self-disclosure and sharing of important concepts with others.

Clarification of Key Issues—Through discussion, members reduce ambiguity and focus energy on the real sources of conflict, then work together to target remaining issues that need to be addressed.

Values Clarification—Members clarify who they are and what they stand for, understand who the other party is and what his or her values are, and learn when to sublimate personal interests to the larger needs of the group or organization.

Negative conflict is dysfunctional and hinders the organization's or the person's performance or ability to attain goals or objectives. Conflict is destructive when it leads to stress and anxiety, inability to take action, and loss of esteem or purpose. Conflict is viewed as negative when it results in:

Unresolved Anger—Members leave the interaction believing they have legitimate concerns that have not been addressed appropriately or goals that cannot be achieved; companies can be slowly poisoned by anger and hostility.

Personality Clashes—Members lack understanding of their style differences and how to work cooperatively and are more tied to their own interests than those of others.

Low Self-esteem or Self-confidence—Members have a diminished sense of selfworth or identity as a result of the conflict. Often this results from impulsive things said or done in the heat of the conflict.

Unclear or Opposing Views on Who Is or Should Be Responsible for What—Members have different expectations of each other and their roles; the conflict was unresolved, unproductive, or ended too soon, leaving ambiguity in its wake.

Problems of Efficiency—Members decide they are unwilling or unable to work together, resulting in redundancies and poor use of existing resources.

“Unfinished Business”—

Members are still unclear about the issue or have remaining concerns that will get in the way of being able to move forward.

SOURCES OF INTERPERSONAL CONFLICT

LIMITED RESOURCES

Despite clear differences between units within an organization, one commonality remains. In general, all are vying for the same resource pool. This pool is usually limited, causing the various units within an organization to compete against each other for finite resources. No matter how prosperous an organization might appear from its facilities, salary levels, or private jets and limousines, few if any organizations have infinite resources. This usually results in competition among business units for the restricted resources available through the parent organization. People in organizations compete for what they consider to be their fair share of resources such as money, time, senior management attention, technology, supplies, equipment, and human talent. This inevitably results in conflict.

DIFFERENCES IN GOALS/OBJECTIVES

A common source of conflict within organizations is differences in personal and/or professional goals and objectives. If we are working on a project with someone whose objective is different from ours, tension or conflict is likely to occur. For example, perhaps one team member wants to “coast” or do as little work as possible toward the team's expected

output or deliverable. If this person is on a team of individuals who are committed to a high-quality output, there will be differences between them on a host of items, such as approach to the work, ways to get the work done, and standards of work quality and quantity. This tension can be from **intragroup conflict**, differences between members of one group, or from **intergroup conflict**, differences between competing subgroups of an organization. For example, the marketing department might have a different goal than the finance department. Marketing folks might push to increase spending on advertising and promotion in order to improve sales, while finance folks push for increased cost-cutting efforts.

MISCOMMUNICATION

Many times, personal and professional conflicts arise due to poor communication. Seldom is miscommunication intentional. More often than not, it's the result of our not taking time to clarify our understanding of something, or gender or cultural differences, or errors in semantics. Often we say one thing and mean another. Or in our haste, we will speak quickly and cryptically in hopes that others know what we want. Or perhaps we speak clearly but our nonverbal communication contradicts the verbal message. In any case, misunderstanding is likely to occur. These communication issues are further compounded by the jargon shared by and understood within specific groups of people, such as engineers and military personnel. The processes and principles of communication may also differ between work groups. For example, one group might have a division newsletter, through which employees are kept informed of important organizational changes, while another group might rely on word of mouth to spread key bits of information. This results in each group having a very different understanding of what's going on in the organization. Interaction between these groups could lead to

numerous miscommunications, each one a potential source of conflict.

DIFFERING ATTITUDES, VALUES, AND PERCEPTIONS

Many conflicts are the result of differences in attitudes, values, and perceptions. Sometimes, without even realizing it, we bring feelings or concerns into an interaction that predisposes us to react in a certain way. For example, if you are afraid of dogs and encounter a neighbor with a dog while out walking one morning, you may react with fear or even hostility. Upon reflection, you realize this fear is due to a fear of animals you've had since you were a child. But the neighbor, without knowing this background, might misinterpret your strong reaction and conclude you dislike the neighbor rather than fear the dog. Without a chance to communicate—for the neighbor to share his or her perception with you and for you to explain the background behind your reaction—it is likely that you will each emerge from the interaction with a vastly different understanding of what just occurred, and with different, possibly negative, opinions of each other. Conflicting values are a common and difficult-to-resolve source of conflict between people. Differences in religious beliefs, attitudes towards diverse others, clashes in family values, or in work ethic might result in interpersonal differences that surface in the work environment.

PERSONALITY CLASHES

Another common source of conflict is differences in personal style or personality. An obvious example of this is the predictable tension between two roommates who are on different “body clocks.” The early riser who gets up at dawn and the night owl who sleeps until noon are almost certain to get into conflict with each other. Conflicts likely occur between the

“slob” and the “neat freak.” The manager who is task-oriented and the employee who is a perpetual socializer are likely to encounter much tension and conflict in their boss–subordinate relationship. Personality conflicts can result in unproductive behaviors at work including gossip, jealousy, insults, taking sides or playing favorites, slowing of work speed, forming of cliques, and even looking for another job!

Conflict is classified into the following FOUR TYPES:

- **Interpersonal conflict** refers to a conflict between two individuals. This occurs typically due to how people are different from one another. We have varied personalities which usually results to incompatible choices and opinions. Apparently, it is a natural occurrence which can eventually help in personal growth or developing your relationships with others. In addition, coming up with adjustments is necessary for managing this type of conflict. However, when interpersonal conflict gets too destructive, calling in a mediator would help so as to have it resolved.
- **Intrapersonal conflict** occurs within an individual. The experience takes place in the person’s mind. Hence, it is a type of conflict that is psychological involving the individual’s thoughts, values, principles and emotions. Interpersonal conflict may come in different scales, from the simpler mundane ones like deciding whether or not to go organic for lunch to ones that can affect major decisions such as choosing a career path. Furthermore, this type of conflict can be quite difficult to handle if you find it hard to decipher your inner struggles. It leads to restlessness and uneasiness, or can
- Even cause depression. In such occasions, it would be best to seek a way to let go of the anxiety through communicating with other people. Eventually, when you find yourself out of the situation, you can become more empowered as a person. Thus, the experience evoked a positive change Which will help you in your own personal growth.
- **Intragroup conflict** is a type of conflict that happens among individuals within a team. The incompatibilities and misunderstandings among these individuals lead to an intragroup conflict. It arises from interpersonal disagreements (e.g. team members have different personalities which may lead to tension) or differences in views and ideas (e.g. in a presentation, members of the team might find the notions presented by the one presiding to be erroneous due to their differences in opinion). Within a team, conflict can be helpful in coming up with decisions which will eventually allow them to reach their objectives as a team. However, if the degree of conflict disrupts harmony among the members, then some serious guidance from a different party will be needed for it to be settled.
- **Intergroup conflict** takes place when a misunderstanding arises among different teams within an organization. For instance, the sales department of an organization can come in conflict with the customer support department. This is due to the varied sets of goals and interests of these different groups. In addition, competition also contributes for intergroup conflict to arise. There are other factors which fuel this type of conflict. Some of these factors may

include a rivalry in resources or the boundaries set by a group to others which establishes their own identity as a team.

STRESS

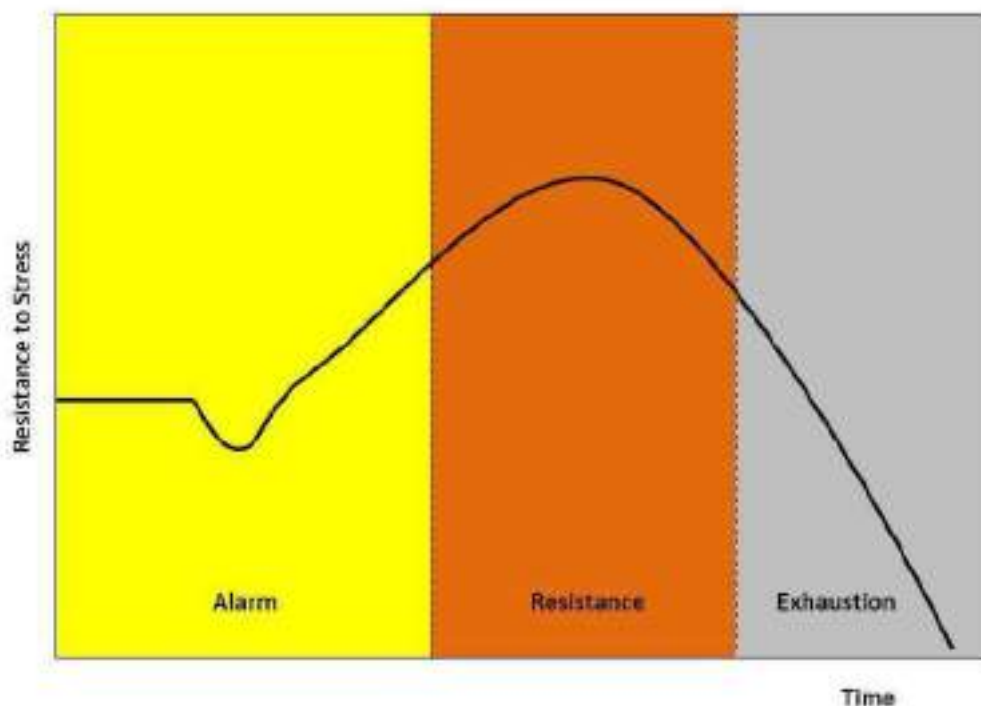
Stress has been viewed as a *response*, a *stimulus*, and a *transaction*. How an individual conceptualizes stress determines his or her response, adaptation, or coping strategies.

Stress As a Response

Stress as a **response** model, initially introduced by Hans Selye (1956), describes stress as a *physiological response pattern* and was captured within his **general adaptation syndrome (GAS)** model (Figure 16.3). This *model describes stress as a dependent variable and includes three concepts*:

Stress is a defensive mechanism. Stress follows the three stages of *alarm*, *resistance*, and *exhaustion*. If the stress is prolonged or severe, it could result in diseases of adaptation or even death.

Later, in *The Stress Concept: Past, Present and Future* (1983), Selye introduced the idea that the stress response could result in positive or negative outcomes based on cognitive interpretations of the physical symptoms or physiological experience (Figure 16.3, “The General Adaptation to Stress Model”). In this way, stress could be experienced as **eustress (positive)** or **dystress (negative)**. However, Selye always considered stress to be a physiologically based construct or response. Gradually, other researchers expanded the thinking on stress to include and involve psychological concepts earlier in the stress model.



The response model of stress incorporates coping within the model itself. The idea of adaptation or coping is inherent to the GAS model at both the alarm and resistance stages. When confronted with a negative stimulus, the alarm response initiates the sympathetic nervous system to combat or avoid the stressor (i.e., increased heart rate, temperature, adrenaline, and glucose levels). The resistance response then initiates physiological systems with a **fight or flight** reaction to the stressor, returning the system to homeostasis, reducing harm, or more generally accommodating the stressor, which can lead to adaptive diseases such as sleep deprivation, mental illness, hypertension, or heart disease. Thus, along with the early conceptualization of stress as a physiological response, early research on coping was also born. As early as 1932, Walter Cannon described the notion of self-regulation in his work *The Wisdom of the Body*.

Stress As a Stimulus

The theory of stress as a **stimulus** was introduced in the 1960s, and viewed stress as *a significant life event or change that demands response, adjustment, or adaptation*. Holmes and Rahe (1967) created the **Social Readjustment Rating Scale (SRRS)** consisting of 42 life events scored according to the estimated degree of adjustment they would each demand of the person experiencing them (e.g., marriage, divorce, relocation, change or loss of job, loss of loved one). Holmes and Rahe theorized that stress was an independent variable in the **health-stress-coping equation** — the cause of an experience rather than the experience itself. While some correlations emerged between SRRS scores and illness (Rahe, Mahan, & Arthur, 1970; Johnson & Sarason, 1979), there were problems with the

stress as stimulus theory. The stress as stimulus theory assumes:

Change is inherently stressful.

Life events demand the same levels of adjustment across the population.

There is a common threshold of adjustment beyond which illness will result.

Rahe and Holmes initially viewed the human subject as a passive recipient of stress, one who played no role in determining the degree, intensity, or valence of the stressor. Later, Rahe introduced the concept of interpretation into his research (Rahe Arthur, 1978), suggesting that a change or life event could be interpreted as a positive or negative experience based on cognitive and emotional factors. However, the stress as stimulus model still ignored important variables such as prior learning, environment, support networks, personality, and life experience.

Stress As a Transaction

In attempting to explain stress as more of a dynamic process, Richard Lazarus developed the **transactional theory of stress and coping (TTSC)** (Lazarus, 1966; Lazarus & Folkman, 1984), which presents *stress as a product of a transaction between a person (including multiple systems: cognitive, physiological, affective, psychological, neurological) and his or her complex environment*. Stress as a transaction was introduced with the most impact when Dr. Susan Kobasa first used the concept of hardiness (Kobasa, 1979). Hardiness refers to a pattern

of personality characteristics that distinguishes people who remain healthy under life stress compared with those who develop health problems. In the late 1970s, the concept of hardiness was further developed by Salvatore Maddi, Kobasa, and their graduate students at the University of Chicago (Kobasa, 1982; Kobasa & Maddi, 1981; Kobasa, Maddi, & Kahn, 1982; Kobasa, Maddi, Puccetti, & Zola, 1985; Maddi & Kobasa, 1984). Hardiness has some notable similarities with other personality constructs in psychology, including locus of control (Rotter, 1966), sense of coherence (Antonovsky, 1987), self-efficacy (Bandura, 1997), and dispositional optimism (Scheier &

Carver, 1985), all of which will be discussed in the next section.

Researchers introduced multiple variables to the stress-as-transaction model, expanding and categorizing various factors to account for the complex systems involved in experiencing a stressor (Werner, 1993). The nature of stress was described in multiple ways: acute, episodic or intermittent, and chronic. Different types of stressors emerged, such as event, situation, cue, and condition, which then fell into categories based on locus of control, predictability, tone, impact, and duration. Figure 16.4 illustrates theories of stress as a response, stimulus, and transaction.

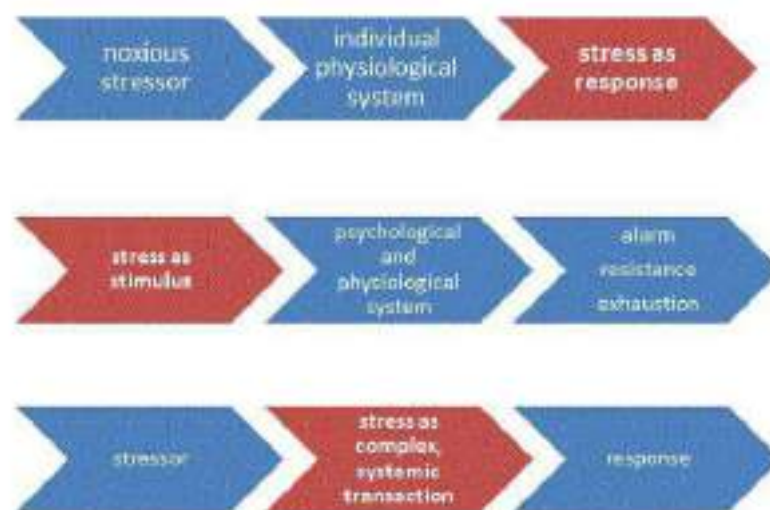


Figure 16.4 Theories of stress as response, stimulus, and transaction.

In his book *Psychological Stress and the Coping Process* (1966), Lazarus presented an elegant integration of previous research on stress, health, and coping that placed a person's appraisal of a stressor at the centre of the stress experience. How an individual appraises a stressor determines how he or she copes with or responds to the stressor. Whether or not a stressor is experienced as discomforting is influenced by a variety of personal and contextual factors including capacities, skills and abilities, constraints, resources, and norms (Mechanic, 1978). Lazarus and Folkman (1984) unpacked the concept of interpretation further in their model of stress appraisal, which includes primary,

secondary, and reappraisal components (see Figure

16.5, "The Transactional Theory of Stress and Coping"). **Primary appraisal** involves *determining whether the stressor poses a threat*. **Secondary appraisal** involves *the individual's evaluation of the resources or coping strategies at his or her disposal for addressing any perceived threats*. The process of **reappraisal** is *ongoing and involves continually reappraising both the nature of the stressor and the resources available for responding to the stressor*.

Figure 16.5 The Transactional Theory of Stress and Coping, by J. Walinga.



STRESS COPING, as described by researchers such as Lazarus and Folkman, implies a more specific process of cognitive appraisal to determine whether an individual believes he or she has the resources to respond effectively to the challenges of a stressor or change (Folkman & Lazarus, 1988; Lazarus & Folkman, 1987). The appraisal literature explains the response or coping process in terms of problem-focused coping or emotion-focused

coping (Folkman & Lazarus, 1980;

as **active and passive coping** styles

well, **approach and avoidance-style** measures of withdrawal (Anshel, 1996; Anshel & Weinberg, 1999; Roth & Cohen, 1986). When faced with a challenge, an individual primarily appraises the challenge as either threatening or non-threatening, and secondarily in terms of whether he or she has the resources to respond to or cope with the challenge effectively. If the individual does not believe he or she has the capacity to respond to the challenge or feels a lack of control, he or she is most likely to turn to an emotion-focused coping response such as wishful thinking (e.g., I wish that I could change what is happening or how I feel), distancing (e.g., I'll try to forget the whole thing), or emphasizing the positive (e.g., I'll just look for the silver lining) (Lazarus & Folkman, 1987). If the person has the resources to manage the challenge, he or she will usually develop a problem-focused coping response such as analysis (e.g., I try to analyze the problem in order to understand it better; I'm making a plan of action and following it). It is theorized and empirically demonstrated that a person's secondary appraisal then determines coping strategies (Lazarus & Folkman, 1987). Coping strategies vary from positive thinking to denial (see Figure 16.7, "COPE Inventory") and are measured and tested using a variety of instruments and scales such as the **COPE inventory** (Carver, Scheier, & Weintraub, 1989).

BIOFEEDBACK

Biofeedback is a form of self-regulation in which individuals learn to control physiological responses by providing them with an information signal, as sensory feedback, about biological conditions of which they may not be ordinarily aware. Feedback responses include muscle tension, skin surface temperature, brain wave activity, electrodermal response, blood pressure and heart rate. In combination with

Lazarus & Folkman (1984), also referred to as the **stress management** approach, is a form of biofeedback that involves the use of physiological signals to help individuals learn to control their physiological responses (Jex, Bliese, & Primeau, 2001). As to become active participants in the rehabilitation or health maintenance process.¹ Biofeedback involves the application of operant conditioning to gain control of visceral, somatomotor, or central nervous system activities.² Biofeedback emerged as a treatment method from pioneering learning research during the 1950s of experimental psychologists, such as Neal Miller. It became a clinical modality in the late 1960s.

HOW DOES BIOFEEDBACK WORK?

As a Stress Modifier Biofeedback operates on the notion that we have the innate ability and potential to influence the autonomic functions of our bodies through the exertion of will and mind. Biofeedback shows the physiological effects of stress and our 'behavior' directly as they occur and in the hands of a trained clinician or therapist, can help the patient to go through a learning process. In this process the patient becomes aware of the effects of stress on the body and learns how to make the required 'behavioral' or 'lifestyle' changes. Biofeedback is often aimed at changing habitual reactions to stress that can cause pain or disease. Most patients who benefit from biofeedback are

trained to relax and modify their behavior. Relaxation is a key component in biofeedback treatment of many disorders, particularly those brought on or made worse by stress. In itself, relaxation is characterized by a decreased metabolism, heart rate, blood pressure, and rate of breathing as well as an increase in skin temperature. Feedback of physical responses such as skin temperature and muscle tension provides information to help patients recognize a relaxed state. The feedback signal may also act as a kind of reward for reducing tension. Constitutive nitric oxide as a result of the relaxation approach may crucially contribute to potentially beneficial outcomes and effects in diverse pathologies, exerting a global healing effect.

Music therapy can be effective in helping the individual to learn to recognize feelings and behaviours associated with stress and to express these emotions in positive, appropriate ways. Stress and stress-related disorders can be terribly disruptive to normal functioning and interrelating. The music therapy session can heighten an individual's awareness of the emotional toll stress is taking on his or her life. If stress-related feelings and emotions are not identified and dealt with, they may simply exacerbate the situation. For the individual who has so successfully internalized these feelings, musical experience, being both personal and internal, may provide an important cathartic release (Brown, et al., 1989).

Lienhard (as cited in Peters, 1987), also discusses the integral part that music therapy can play as an adjunct to group therapy in stress treatment programmes. According to Lienhard, instrumental improvisation, music listening, their reactions. Song lyrics, for example, can be used as a catalyst for discussion, helping clients to identify areas of stress, identify and express feelings associated with this stress, discuss emotions, and share constructively with the group and receive valuable feedback.

Provide alternative coping mechanisms for dealing more successfully with stress

While it is critically important for persons with stress-related disorders to increase or enhance their ability to relax, it is just as important that they learn alternate ways of dealing with the stress in their lives. Music therapy can be effective in this role. Wolfe (1978) successfully employed music therapy discussion groups designed to encourage group members to talk to one another as well as to stimulate constructive discussion concerning ways of improving unpleasant and stressful situations. Learning to play instruments while in music therapy has the added benefit of providing the client with a leisure skill that serves as a means of tension release.

Further, in using the music setting as a model for the environment in which stress-evoking interactions and/or events may occur, problems may be identified and solved within the confines of the therapy session (Hanser, 1985). Stress-related behaviours and/or patterns can be observed within the music therapy context and dealt with or changed within this controlled environment. Also, beyond simply learning relaxation techniques, Saperston

(1989), for example, uses the music therapy setting to help individuals learn to recognize when they would benefit from performing their relaxation induction, providing another effective means of coping with stress.

PROGRESSIVE MUSCLE RELAXATION

Progressive muscle relaxation is an exercise that relaxes your mind and body by progressively tensing and relaxation muscle groups throughout your entire body. You will tense each muscle group vigorously, but without straining, and then suddenly release the tension and feel the muscle relax. You will tense each muscle for about 5 seconds. If you have any pain or discomfort at any of the targeted muscle groups feel free to omit that step. Throughout this exercise you may visualize the muscles tensing and a wave of relaxation flowing over them as you release that tension. It is important that you keep breathing throughout the exercise. Now let's begin.

GUIDED IMAGERY

Guided imagery is a **stress management technique**, where you use your imagination to picture a person, place, or time that makes you feel relaxed, peaceful and happy. Imagery is slightly different from other stress management techniques, in that it relies on the use of all of your senses. For instance, in your imagination you hear the sound of birds chirping, you see the drops of dew on the grass, you feel the breeze on your skin, you smell the wildflowers, and you taste the cold drink. In imagery, using all of your senses is what creates such a

powerfully relaxing experience, and this is why it's so useful in managing stress and coping with difficult situations. There are several other ways that you can use imagery to help you relax. For example, you could create mental pictures of stress flowing out of your body, or of your problems, your distractions, and your everyday concerns being folded away and stashed in a padlocked chest.

MINDFULNESS

Mindfulness is a state of intentional, nonjudgmental focus on the present moment. Mindfulness techniques are often incorporated into other practices, such as yoga or meditation. It is often thought to include the following elements:

Awareness – Being tuned in to what is happening in the present moment, including sights, sounds, smells, or physical sensations you might usually ignore.

Focus – Paying attention to the present moment, without thinking about past or future events.

Acceptance – Accepting whatever you might be feeling or thinking, without judging it or trying to change it or your reactions.

Observation – Recognizing unpleasant sensations, thoughts, and feelings as temporary and fleeting, observing them objectively without reaction or judgment.

What Are the Benefits of Mindfulness?
Mindfulness has been studied for many years. According to recent research, it has several benefits for physical and mental health. Some of the strongest health benefits include:

Improves mood – Mindfulness training may reduce depression and anxiety. One study found that mindfulness training was as effective as antidepressant medications in preventing a depression relapse.

Reduces stress and its consequences – Mindfulness can lead to less intense stress responses. This has many health benefits, such as lowering your blood pressure and strengthening your immune system.

Improves coping with pain – People with chronic pain who practice mindfulness meditation report less severe pain and pain-related distress. They are also more active in spite of their pain.

Improves brain functions – Practicing mindfulness helps build your ability to pay attention and focus. Over time, this training can sharpen memory and improve mental performance.

Helps with weight management – Some mindfulness techniques have been shown to reduce overeating and obesity.

Meditation can give you a sense of calm, peace and balance that can benefit both your emotional well-being and your overall health.

And these benefits don't end when your meditation session ends. Meditation can help carry you more calmly through your day and may help you manage

Symptoms of certain medical conditions.

TYPES OF MEDITATION

Meditation is an umbrella term for the many ways to a relaxed state of being. There are many types of meditation and relaxation techniques that have meditation components. All share the same goal of achieving inner peace.

Ways to meditate can include:

Guided meditation. Sometimes called guided imagery or visualization, with this method of meditation you form mental images of places or situations you find relaxing.

You try to use as many senses as possible, such as smells, sights, sounds and textures. You may be led through this process by a guide or teacher.

Mantra meditation. In this type of meditation, you silently repeat a calming word, thought or phrase to prevent distracting thoughts.

Mindfulness meditation. This type of meditation is based on being mindful, or having an increased awareness and acceptance of living in the present moment.

In mindfulness meditation, you broaden your conscious awareness. You focus on what you experience during meditation, such as the flow of your breath. You can observe your thoughts and emotions, but let them pass without judgment.

Qi gong. This practice generally combines meditation, relaxation, physical movement and breathing exercises to restore and maintain balance. Qi gong (CHEE-gung) is part of traditional Chinese medicine.

Tai chi. This is a form of gentle Chinese martial arts. In tai chi (TIE-CHEE), you perform a self-paced series of postures or movements in a slow, graceful manner while practicing deep breathing.

Transcendental Meditation®. Transcendental Meditation is a simple, natural technique. In Transcendental Meditation, you silently repeat a personally assigned mantra, such as a word, sound or phrase, in a specific way.

This form of meditation may allow your body to settle into a state of profound rest and relaxation and your mind to achieve a state of inner peace, without needing to use concentration or effort.

Yoga. You perform a series of postures and controlled breathing exercises to promote a more flexible body and a calm mind. As you move through poses that require balance and concentration, you're encouraged to focus less on your busy day and more on the moment.

STRESS INOCULATION TRAINING (SIT) is a multifaceted type of cognitive-behavioral therapy designed to help individuals cope with

stress. It was initially developed by psychologist Donald Meichenbaum in the 1980s and has been employed to mitigate the sequelae of stress in a variety of situations. The essence of SIT is that by exposing people to increasing levels of perceived stress, they practice employing different coping skills and eventually develop increased tolerance³ or immunity to a particular stimulus.

The training consists of three phases that are designed to empower individuals and enhance their repertoire of coping skills^{4,5,6}:

Conceptualization – This phase is designed to achieve two specific goals. First, it builds a relationship between the therapist, coach, or trainer. Second, it educates an individual, increasing their understanding and awareness of his or her stress response and existing coping skills. According to Meichenbaum, is it preferable for this phase to be less didactic and more Socratic in nature. He suggests using “curious questions” to promote an individual's processing and discovery.

Skills Acquisition and Consolidation – This phase is all about developing and practicing individual psychological tools and cognitive restructuring techniques (breathing techniques, relaxation techniques, negative thought stopping, etc.). The goal is build the coping techniques to they can be applied in the next phase to regulate negative emotions and increase control over physiological responses. They are discovered, nurtured, and strengthened in collaborative manner with a trainer or therapist.

Application and Follow-through

- This phase is, in essence, the phase of inoculation. On a graduated basis, an individual is exposed to increasing levels of a particular stressor and practices applying the skills they have developed to mitigate his or her stress response.

MODIFICATIONS OF STRESS INOCULATION TRAINING: STRESS EXPOSURE TRAINING

It is important to recognize that Meichenbaum's intention was really to help individuals cope with physical pain, anxiety, anger, and fear as a result of traumatic experiences. This was a clinical intervention designed to treat pathological psychiatric conditions. However, various organizations and individuals saw the potential benefit of instituting this technique prophylactically. Organizations, like the military and NASA, began to adapt the traditional structure of SIT in demonstrated substantial improvements^{7,8,9,10,11,12} in performance.

One adaptation of SIT is Stress Exposure Training (SET). SET, originally proposed by Driskell and Johnston, takes a slightly different¹¹ approach. Its general structure is similar to Meichenbaum's cognitive

behavioral approach. It is divided into three analogous phases:

Information provision – This phase provides information on the human stress response, conditions participants should expect to encounter, and other preparatory information

Skills acquisition – This phase is designed to develop and refine behavioral, technical, and cognitive skills

Application and practice – This phase includes practicing skills under conditions that approximate the operational environment and that gradually attain the level of stress expected

SET is different in that, quite simply, it takes a proactive approach. It provides prophylaxis. It prepares individuals without psychiatric pathology for potential stressors and situations that are likely to encounter.

I'll make a point to emphasize that whether we call it Stress Inoculation or Stress Exposure, makes little difference. Truth be told, I prefer the term "inoculation." However, the important thing is that we maintain the salient training objectives and strategies as they have been developed for SET.

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UNIT 8

UNIT 8: SOCIAL PSYCHOLOGY

WHAT IS SOCIAL PSYCHOLOGY?

Social psychology is a science that studies the influences of our situations, with special attention to how we view and affect one another. More precisely, it is the scientific study of how people think about, influence, and relate to one another. Social psychology lies at psychology's boundary with sociology. Compared with sociology (the study of people in groups and societies), social psychology focuses more on individuals and uses more experimentation. Compared with personality psychology, social psychology

focuses less on individuals' differences and more on how individuals, in general, view and affect one another. Social psychology is still a young science. The first social psychology experiments were reported barely more than a century ago (1898), and the first social psychology texts did not appear until just before and after 1900 (Smith, 2005). Not until the 1930s did social psychology assume its current form. And not until World War II did it begin to emerge as the vibrant field it is today.

WHAT IS SOCIAL IN SOCIAL PSYCHOLOGY?

Every science has its substantive field, a core by which it is identified. The field of social psychology is usually defined as that branch of science that deals with human interaction, i.e., the interaction between man and man, and man and society. It aims to search out general laws of social behavior. Social psychologists are trained in using the tools of conceptual

analysis and scientific methodology in the explaining relationship between person and society. Obviously, there could be several social psychologies, depending on the meaning of the term 'social'. **Staat (1983)** posited that the concept of social refers to both - social environment and social behavior. Social environment, in a sense, extends the analogy of physical environment to social setting. It refers to social groups, organizations, structures, norms, obligations, support, etc., which

provide the context within which an individual performs. Social behavior refers to affects, attitudes, activities, and motivations in response to any social environment. The study of such social behavior is mostly at the individual level; more precisely, the study of individual in a group. This emphasis on individual behavior got accentuated with the rise of American influence on social psychology, individual level processes to the society.

Thus, if psychology is defined as a science of behavior (than that of mind), social psychology can be defined as a science of social behavior. The same methodology which is used at the individual level is employed to study societies.

NATURE, SCOPE AND HISTORY OF SOCIAL PSYCHOLOGY

SCIENTIFIC NATURE OF SOCIAL PSYCHOLOGY:

The term science does not refer to a special group of highly advanced fields. Rather, it refers to two things: (1) a set of values and (2) several methods that can be used to study a wide range of topics. In deciding whether a given field is or is not scientific, therefore, the critical question is, does it adopt these values and methods? To the extent it does, it is scientific in nature. To the extent it does not, it falls outside the realm of science. Here we focus on the core values that all fields must adopt to be considered scientific in nature.

Four of these are most important:

Accuracy: A commitment to gathering and evaluating information about the world (including social behavior and

particularly with popularity of the experimental approach. **Graumann (1986)** called it the individualization of 'social'; the reduction of social variables to the level of the individual. Another way of looking at individualization of the social is to treat society as individual and imputing the extent they have been verified over and over again. thought) in as careful, precise, and error-free a manner as possible.

- **Objectivity:** A commitment to obtaining and evaluating such information in a manner that is as free from bias as humanly possible.
- **Skepticism:** A commitment to accepting findings as accurate only to.
- Open-mindedness:** A commitment to changing one's views—even views that are strongly held—if existing evidence suggests that these views are inaccurate.

Social Psychology, as a field, is deeply committed to these values and applies them in its efforts to understand the nature of social behavior and social thought. For this reason, it makes sense to describe it as scientific in orientation. In contrast, fields that are not scientific make assertions about the world, and about people, that are not put to the careful test and analysis required by the values listed above.

• SOCIAL PSYCHOLOGY FOCUSES ON THE BEHAVIOR OF INDIVIDUALS

Societies differ greatly in terms of their views concerning courtship and marriage, yet it is still individuals who fall in love. Similarly, societies vary greatly in terms of

their overall levels of violence, yet it is still individuals who perform aggressive actions or refrain from doing so. The same argument applies to virtually all other aspects of social behavior, from prejudice to helping: the actions are performed by, and the thoughts occur in, the minds of individuals, although they may, of course, be strongly influenced by other people. Because of this basic fact, the focus in social psychology is strongly on individuals. Social psychologists realize, of course, that we do not exist in isolation from social and cultural influences—far from it.

FOCUS ON UNDERSTANDING THOUGHTS AND BEHAVIOR.

Social psychologists are primarily interested in understanding the many factors and conditions that shape the social behavior and thought of individuals—their actions, feelings, beliefs, memories, and inferences concerning other people. Obviously, a huge number of variables play a role in this regard. Most, though, fall under the four major headings described below.

HISTORY OF SOCIAL PSYCHOLOGY

The science of social psychology began when scientists first started to systematically and formally measure the thoughts, feelings, and behaviors of human beings (Kruglanski & Stroebe, 2011). The earliest social psychology experiments on group behavior were conducted before 1900 (Triplet, 1898), and the first social psychology textbooks were published in 1908 (McDougall, 1908/2003; Ross, 1908/1974). During the 1940s and 1950s, the social psychologists **Kurt Lewin** and **Leon Festinger** refined the

experimental approach to studying behavior, creating social psychology as a rigorous scientific discipline. Lewin is sometimes known as “the father of social psychology” because he initially developed many of the important ideas of the discipline, including a focus on the dynamic interactions among people. In 1954, Festinger edited an influential book called *Research Methods in the Behavioral Sciences*, in which he and other social psychologists stressed the need to measure variables and to use laboratory experiments to systematically test research hypotheses about social behavior. He also noted that it might be necessary in these experiments to deceive the participants about the true nature of the research.

Social psychology was energized by researchers who attempted to understand how the German dictator Adolf Hitler could have produced such extreme obedience and horrendous behaviors in his followers during the World War II. The studies on conformity conducted by **Muzafir Sherif** (1936) and Solomon Asch (1952), as well as those on obedience by **Stanley Milgram** (1974), showed the importance of conformity pressures in social groups and how people in authority could create obedience, even to the extent of leading people to cause severe harm to others. **Philip Zimbardo**, in his well-known “prison study” (Haney, Banks, & Zimbardo, 1973), found that the interactions of male college students who were recruited to play the roles of guards and prisoners in a simulated prison became so violent that the study had to be terminated early.

Social psychology quickly expanded to study other topics. **John Darley** and **Bibb Latané** (1968) developed a model that

helped explain when people do and do not help others in need, and **Leonard Berkowitz** (1974) pioneered the study of human aggression. Meanwhile, other social psychologists, including Irving Janis (1972), focused on group behavior, studying why intelligent people sometimes made decisions that led to disastrous results when they worked together. Still other social psychologists, including **Gordon Allport** and **Muzafir Sherif**, focused on intergroup relations, with the goal of understanding and potentially reducing the occurrence of stereotyping, prejudice, and discrimination. Social psychologists gave their opinions in the 1954 *Brown v. Board of Education* U.S. Supreme Court case that helped end racial segregation in American public schools, and social psychologists still frequently serve as expert witnesses on these and other topics (Fiske, Bersoff, Borgida, Deaux, & Heilman, 1991). In recent years insights from social psychology have even been used to design anti-violence programs in societies that have experienced genocide (Staub, Pearlman, & Bilali, 2010).

The latter part of the 20th century saw an expansion of social psychology into the field of attitudes, with a particular emphasis on cognitive processes. During this time, social psychologists developed the first formal models of persuasion, with the goal of understanding how advertisers and other people could present their messages to make them most effective (Eagly & Chaiken, 1993; Hovland, Janis, & Kelley, 1963). These approaches to attitudes focused on the cognitive processes that people use when evaluating messages and on the relationship between attitudes and behavior. Leon

Festinger's important cognitive dissonance theory was developed during this time and became a model for later research (Festinger, 1957).

In the 1970s and 1980s, social psychology became even more cognitive in orientation as social psychologists used advances in cognitive psychology, which were themselves based largely on advances in computer technology, to inform the field (Fiske & Taylor, 2008). The focus of these researchers, including Alice Eagly, Susan Fiske, E. Tory Higgins, Richard Nisbett, Lee Ross, Shelley Taylor, and many others, was on social cognition—an understanding of how our knowledge about our social worlds develops through experience and the influence of these knowledge structures on memory, information processing, attitudes, and judgment. Furthermore, the extent to which humans' decision making could be flawed due to both cognitive and motivational processes was documented (Kahneman, Slovic, & Tversky, 1982).

In the 21st century, the field of social psychology has been expanding into still other areas. Examples that we consider in this book include an interest in how social situations influence our health and happiness, the important roles of evolutionary experiences and cultures on our behavior, and the field of social neuroscience—the study of how our social behavior both influences and is influenced by the activities of our brain (Lieberman, 2010). Social psychologists continue to seek new ways to measure and understand social behavior, and the field continues to evolve. We cannot predict where social psychology will be directed in the future, but we have no doubt that it will still be alive and vibrant.

TRADITIONAL THEORETICAL PERSPECTIVES:

FIELD THEORY

Kurt Lewin (Luh-veen) was considered by some as the father of modern social psychology due to his act of breaking new ground in employing scientific methods and experimentation in the study of social behavior.

Lewin not only adapted Gestalt principles but further applied them to a theory of personality and development into what is now known as the **Psychological Field Theory**. He translated Gestalt philosophy into social experience involving people who should be considered as wholes instead of being composed of discrete parts. A person is presented as a whole system consisting of subsystems that are somewhat separate yet are still capable of interacting and combining with each other.

He was one of the first psychologists to propose that the development of an individual was the product of the interaction between inborn predispositions (nature) and life experiences (nurture). This conception was presented by Lewin in the form of a mathematical equation known as Lewin's Equation for behavior, stating that behavior is the function of the person interacting within his environment or $B = f(P, E)$.

As such, Lewin accounted for human behavior by emphasizing forces and tensions that influence it. He asserted that the behavior of an individual is always

geared toward some goal or objective and it is precisely this intention that matters most in the performance of behavior. These intentions supposedly follow field principles and are influenced by psychological forces such as how the individual perceives a situation.

According to Levin, behavior exists in a totality of interacting facts which comprise a dynamic field. The circumstances or conditions in any part of the field are influenced by and depend on every other part of the field. This psychological field is otherwise known as the life space which comprises the individual and his psychological or behavioral environment also known as facts that affect the behavior or thoughts of the individual at a certain point in time.

Life space is most frequently determined by the physical and social environment that the individual finds himself in. It may include places where he goes, events that occur, feelings about places and people encountered, what he sees on TV or reads in books, his imagined thoughts and goals. Encompassed by a child's life space are forces which the child may be aware of or not, in addition to forces which are accepted by the child as true though they may not be so.

COGNITIVE DISSONANCE

Leon Festinger's (1957) cognitive dissonance theory suggests that we have an inner drive to hold all our attitudes and behavior in harmony and avoid

disharmony (or dissonance). This is known as the principle of cognitive consistency.

When there is an inconsistency between attitudes or behaviors (dissonance), something must change to eliminate the dissonance.

Cognitive dissonance refers to a situation involving conflicting attitudes, beliefs or behaviors. This produces a feeling of mental discomfort leading to an alteration in one of the attitudes, beliefs or behaviors to reduce the discomfort and restore balance.

PSYCHODYNAMIC APPROACHES

Psychodynamic approach includes all the theories in psychology that see human functioning based upon the interaction of drives and forces within the person, particularly unconscious, and between the different structures of the personality. Freud's psychoanalysis was the original psychodynamic theory, but the psychodynamic approach as a whole includes all theories that were based on his ideas, e.g., Jung (1964), Adler (1927) and Erikson (1950).

Sigmund Freud (writing between the 1890s and the 1930s) developed a collection of theories which have formed the basis of the psychodynamic approach to psychology.

His theories are clinically derived - i.e., based on what his patients told him during therapy. The psychodynamic therapist would usually be treating the patient for depression or anxiety related disorders.

SOCIO-BIOLOGY

Sociobiology, also called behavioral ecology, is the study of the evolution of social behavior in all

organisms, including human beings. The highly complex behaviors of individual animals become even more intricate when interactions among groups of animals are considered. Animal behavior within groups is known as social behavior. Sociobiology asks about the evolutionary advantages contributed by social behavior and describes a biological basis for such behavior. It is theory that uses biology and genetics to explain why people (and animals) behave the way they do.

Sociobiology is a relatively new science. In the 1970s, Edward O. Wilson, now a distinguished professor of biology at Harvard University, pioneered the subject. In his ground-breaking and controversial book, *Sociobiology: The New Synthesis*, Dr. Wilson introduced for the first time the idea that behaviour is likely the product of an interaction between an individual's genetic makeup and the environment (or culture in the case of human beings). Wilson's new ideas rekindled the debate of "Nature vs. Nurture," wherein nature refers to genes and nurture refers to environment.

Socio-biology is often subdivided into three categories: narrow, broad, and pop socio-biology. Narrow socio-biology studies the function of specific behaviours, primarily in non-human animals. Broad socio-biology examines the biological basis and evolution of general social behavior. Pop socio-biology is concerned specifically with the evolution of human social behaviour.

Sociobiologists focus on reproductive behaviors because reproduction is the mechanism by which genes are passed on to future generations. It is believed that behavior, physically grounded in an

individual's genome (or genes), can be acted upon by natural selection. Natural selection exerts its influence based upon the fitness of an organism. Individuals that are fit are better suited (genetically) to their environment and therefore reproduce more successfully. An organism that is fit has more offspring than an individual that is unfit. Also, fitness requires that the resulting offspring must survive long enough to themselves reproduce. Because sociobiologists believe that social behavior is genetically based, they also believe that behavior is heritable and can therefore contribute to (or detract from) an individual's fitness. Examples of the kinds of reproductive interactions in which sociobiologists are interested include courtship, mating systems like monogamy (staying with one mate), polygamy (maintaining more than one female mate), and polyandry (maintaining more than one male mate), and the ability to attract a mate (called sexual selection.)

Sociobiology also examines behavior that indirectly contributes to reproduction. An example is the theory of optimal foraging which explains how animals use the least amount of energy to get the maximum amount of food. Another example is altruistic behavior (altruism means selfless). Dominance hierarchies, territoriality, ritualistic (or symbolic) behavior, communication (transmitting information to others through displays), and instinct versus learning are also topics interpreted by sociobiology.

Sociobiology applied to human behavior involves the idea that the human brain evolved to encourage social behaviors that increase reproductive fitness. For example, the capacity for learning in human beings is a powerful

characteristic. It allows people to teach their relatives (or others) important life skills that are passed-down from generation to generation. However, the ability to learn is also a variable trait. That is, not every person learns as quickly or as well as every other person. A

sociobiologist would explain that individuals who learn faster and more easily have increased fitness. Another example is smiling. The act of smiling in response to pleasurable experiences is a universal social behavior among people. Smiling is observed in every culture. Furthermore, smiling is an example of an instinct that is modified by experience. Therefore, because the behavior is instinctual, it has a genetic and inheritable basis. Because it is altered by experience, the behavior is socially relevant. Sociobiologists might speculate, then, that since smiling is a visual cue to other individuals that you are pleased, people who tend to smile more easily are more likely to attract a suitable mate, and are therefore more fit.

SOCIAL COGNITION

Social cognition – the way in which we encode, process, remember and use information in social contexts in order to make sense of others behaviors.

DEFINITIONS OF SCHEMAS

Baron and Byrne define schemes figuratively as cognitive scaffolds in our minds. Also, as cognitive categories or frameworks into which we mentally sort out our cumulative information on given topics.

Taylor and Crocker define schemes as cognitive structures containing our

knowledge about some specific domain and its attributes allowing us to categorize and interpret new information related to that domain.

Fiske and Linville define social schemes as richly interconnected networks of information relevant to various concepts that is to various aspects of our social world

Information in **social schemas** belongs to 3 categories of groups:

Physical appearance categories

Personality traits

Characteristic behaviors

TYPES/CATEGORIES OF SCHEMAS

Schemas are classified into the following categories

Role schemes

Person schemes

Event schemes

Self Schemes

ROLE SCHEMA

Role schema is a cognitive category or a cognitive structure concerning a particular group and its members **especially the roles and behaviors they are expected to display or engage in**. Thus role schemes can be formed for different occupational groups, religious groups, nationalities, gender groups etc. Role schema is very similar to the concept of stereotype.

PERSON SCHEMA

A person schema is a cognitive framework concerning a specific individual that includes information about his or her physical features, personality traits, and characteristic behavior. In developing a person schema we often use or apply

schemas that we might hold about certain personality types to one specific case.

EVENT SCHEMA

Event schemes are often referred to as individuals internal scripts. Thus an event schema is a cognitive structure concerning the usual normal expected sequence of events that should occur in various types of situations. For example at a wedding, an interview, a funeral etc. Such event schemes lead us to expect how the actors in that situation should behave. If there is a violation of the expectations, we often experience a host of negative emotions ranging from surprise or shock to disillusionment.

SELF SCHEMA

Self schema is a cognitive framework that each person holds about himself or herself. It is equivalent to what is known as the self concept. The relationship between the self concept/self schema on one hand and the actual or real self on the other hand has been a topic of great interest. McDavid and Harrari have diagrammatically represented this relationship through what they call the **cheeseburger self**. The real self or actual self is represented by the circle i.e. the totality of the self or the holistic self. The self concept is represented by the cheese that covers most of the burger bun. However, there are parts of the self concept that extend beyond the real self called **areas of exaggeration**. These are most often positive traits and behaviors that are socially desirable that a person attributes to himself or herself though the evidence for these is meager or non existent in the real self. They result from a mechanism known as perceptual vigilance. We are specially or extraattentive to instances of positive

behaviors and traits even though they might be very few.

AREAS OF DENIAL

They are most often negative i.e. socially undesirable behavior that are part of our real self but are denied entry into our self-concept. They result from a mechanism known as perceptual defense to prevent the ego from damage and devaluation by ignoring certain information about oneself especially anxiety arousing information. That the burger and cheese overlap to a large extent suggests that for most humans the self-concept and the real self are congruent with each other. This is the basis for a well adjusted personality.

Effects of schemas on cognitive processes:

Schemas tend to act like filters i.e. they allow some information to penetrate them enabling us therefore to perceive and remember information that is congruent to our schemas. On the other hand, information incongruent with our schemas tends to be ignored, misperceived, distorted repressed and or reconstructed.

ATTENTION

Stimuli that fit in with our schemas are more likely to be noticed and attended to and thus become the focus of our attention, whereas those that do not fit in with our schemas tend to remain at the periphery of our attentional field.

PERCEPTION

Like in the case of attention, schemas also lead to selective perception. We are more likely to

accurately interpret and perceive information consistent with our schemas.

We are more likely to

distort, misinterpret and misperceive information that is inconsistent with our schemas. If however,

we recognize that inconsistent information incongruent with our schemas has some truth, we

explain it away as being an exception to the rule. The relationship between schemas and perception

was studied by Sagar and Schofield 1980 who got supportive evidence.

MEMORY

Selective memory also results from the operation of our schemas in that we tend to remember

information that is congruent with our schemas. We are more likely to selectively forget or

reconstruct information incongruent with our schemas. Research in this area began with the work of Frederick Bartlett. Unlike Ebbinghaus he was more interested in the qualitative changes in memory over time.

He identified 3 main qualitative changes in memory

Leveling refers to the fact that unimportant irrelevant details are excluded from the person's memory of the story. The story becomes crisper and more concise over time.

Sharpening refers to the qualitative change where certain elements that are different,

salient, unusual to the story are exaggerated, highlighted and begin to dominate ones memory of the story.

Assimilation refers to the fact that new information in the story is distorted or changed to make it fit better with ones existing schemas, ideas, values, beliefs etc.

SOCIAL PERCEPTION [COMMUNICATION, ATTRIBUTIONS]

Social perception—the process through which we seek to know and understand other people.

Nonverbal communication

Basic channels through which such communication takes place. Research findings indicate that five of these channels exist: **facial expressions, eye contact, body movements, posture, and touching.**

FACIAL EXPRESSIONS AS CLUES TO OTHERS' EMOTIONS

Cicero stated: “The face is the image of the soul.” By this he meant that human feelings and emotions are often reflected in the face and can be read there in specific expressions. Modern research suggests that Cicero was correct: It is possible to learn much about others’ current moods and feelings from their facial expressions. In fact, it appears that five different basic emotions are represented clearly, and from a very early age, on the human face: anger, fear, happiness, sadness, and disgust (Izard, 1991; Rozin, Lowery, & Ebert, 1994). (Surprise, has also been suggested as a basic emotion reflected clearly in

facial expressions, but recent evidence concerning this suggestion is mixed, so it may not be as basic or as clearly represented in facial expressions as other emotions; Reisenzein, Bordgen, Holtbernd, & Matz, 2006).

Early research seemed to suggest that facial expressions are universal in both respects (e.g., Ekman & Friesen, 1975) and with few exceptions, these results have been confirmed in more recent research (Effenbin&Ambady, 2002). In fact, it has been found that certain facial expressions—smiles, frowns, and other signs of sadness) occur, and are recognized as representing basic underlying emotions (e.g., happiness, anger, sadness) in many different cultures (e.g., Shaver, Murdaya,

Fraley, 2001). It seems reasonable to conclude that some facial expressions provide clear signals of underlying emotional states, and are recognized as doing so all over the world. Cultural differences certainly do exist with respect to the precise meaning of facial expressions, but unlike spoken languages, they do not seem to require much in the way of translation.

GAZES AND STARES: EYE CONTACT AS A NONVERBAL CUE

Taking note of the importance of cues provided by others’ eyes, ancient poets often described the eyes as “windows to the soul.” In one important sense, they were correct: We do often learn much about others’ feelings from their eyes. For example, we interpret a high level of gazing from another as a sign of liking or friendliness (Kleinke, 1986). In contrast, if others avoid eye contact with us, we may conclude that they are unfriendly, don’t like us, or are simply shy. If another

person gazes at us continuously and maintains such contact regardless of what we do, he or she can be said to be staring. A stare is often interpreted as a sign of anger or hostility—as in cold stare—and most people find this particular nonverbal cue disturbing (Ellsworth & Carlsmith, 1973). This is one reason why experts on “road rage”—highly aggressive driving by motorists, sometimes followed by actual assaults—recommend that drivers avoid eye contact with people who are disobeying traffic laws and rules of the road (e.g., Bushman, 1998). Apparently, such people, who are already in a highly excitable state, interpret anything approaching a stare from another driver as an aggressive act, and react accordingly.

BODY LANGUAGE: GESTURES, POSTURE, AND MOVEMENTS

our current moods or emotions are often reflected in the position, posture, and movement of our bodies. Together, such nonverbal behaviors are termed body language, and they, too, can provide useful information about others. First, body language often reveals others’ emotional states. Large numbers of movements—especially ones in which one part of the body does something to another part (touching, rubbing, scratching)—suggest emotional arousal. The greater the frequency of such behavior, the higher the level of arousal or nervousness.

More specific information about others’ feelings is often provided by **gestures**. These fall into several categories, but perhaps the most important are emblems—body movements carrying specific meanings in a given culture.

TOUCHING: WHAT DOES IT CONVEY?

Suppose that during a brief conversation with another person, he or she touched you briefly. How would you react? What information would this behavior convey? The answer to both questions is, it depends. And what it depends on is several factors relating to who does the touching (a friend, a stranger, a member of your own or the other gender); the nature of this physical contact (brief or prolonged, gentle or rough, what part of the body is touched); and the context in which the touching takes place (a business or social setting, a doctor’s office). Depending on such factors, touch can suggest affection, sexual interest, dominance, caring, or even aggression. Despite such complexities, existing evidence indicates that when touching is considered appropriate, it often produces positive reactions in the person being touched.

One acceptable way in which people in many different cultures touch strangers is through handshaking. “Pop psychology” and even books on etiquette (e.g., Vanderbilt, 1957) suggest that handshakes reveal much about other people—for instance, their personalities—and that a firm handshake is a good way to make a favorable first impression on others. Research findings (e.g., Chaplin, Phillips, Brown, Clanton, & Stein, 2000) suggest that it is. **The firmer, longer, and more vigorous others’ handshakes are, the higher we tend to rate them in terms of extraversion and openness to experience, and the more favorable our first impressions of them tend to be.**

Levav and Argo (2010) found that a light, comforting pat on the arm can induce feelings of security among both women

and men—but only if the touching is performed by a woman. Such feelings of security, in turn, influence actual behavior: individuals touched on the shoulder by a female experimenter actually showed greater risk taking in an investment task than those not touched, or ones who were touched only through handshakes.

Paralinguistic cues—changes in the tone or inflection of others’ voices (quite apart from the meaning of their words). And recent research indicates that even subtle cues relating to others’ body chemistry can be revealing. For instance, research by **Miller and Maner (2010)** indicates that changes in women’s internal chemistry occurring during the menstrual cycle can be transmitted to others (especially, perhaps, men) through subtle olfactory cues—changes in the aromas emitted by their bodies. The men couldn’t report detecting differences in the scents of the shirts worn during ovulation and after it was over, but their testosterone levels still differed. Overall, these findings indicate that shifts in body chemistry, too, can provide nonverbal cues about other people—at least in the case of women and their menstrual cycle.

FACIAL EXPRESSIONS. WHY IS IT IMPORTANT?

In a sense, this is not surprising because we direct lots of attention to others’ faces as we interact with them. In support of this basic fact, several different research findings combine to suggest that **facial expressions** are indeed a uniquely crucial source of information about others. First, it is almost impossible to ignore such information. Even after viewing them once, they still grip our attention the next time they are presented (e.g., Blagrove & Watson, 2010). Moreover, this is

especially true for negative facial expressions. Even if such expressions are seen on one occasion, they are still easier to notice than other stimuli on later occasions. For example, individuals can spot an angry face in an array of faces more quickly than neutral or smiling faces.

The **facial feedback hypothesis (Laird, 1984)** suggests that there is a close link between the facial expressions we show and our internal feelings, and that this relationship works both ways: yes, the expressions we show reflect our internal feelings or emotions, but in addition, these expressions also feed back into our brains and influence our subjective experiences of emotion. In short, we don’t only show what we feel inside on our faces—we also sometimes feel, inside, what we show.

McCanne and Anderson (1987) asked female participants to imagine positive and negative events (e.g., “You inherit a million dollars,” “You lose a really close friendship”). While imagining these events, they were told to either enhance or suppress tension in certain facial muscles. One of these muscles is active when we smile or view happy scenes. The other is active when we frown or view unhappy scenes. Measurements of electrical activity of both muscles indicated that after a few practice trials, most people could carry out this task quite successfully. They could enhance or suppress muscle tension when told to do so, and could do this without any visible change in their facial expressions. After imagining each scene, participants rated their emotional experiences in terms of enjoyment or distress. **If the facial feedback hypothesis is correct, these ratings should be affected by participants’ efforts to enhance or suppress muscle tension. If**

they enhanced activity in muscles associated with smiling, they would report more enjoyment of the positive events. If they suppressed such activity, they would report less enjoyment. Results offered clear support for these predictions.

DECEPTION: RECOGNIZING IT THROUGH NONVERBAL CUES, AND ITS EFFECTS ON SOCIAL RELATIONS

Be honest: how often do you tell lies? This includes very small “white lies” designed to avoid hurting others’ feelings or accomplish other positive social purposes to ones designed to get us out of trouble or further our own goals (“I’m sorry, Professor—I missed the exam because of an unexpected death in my family . . .”). In fact, research findings indicate that most people tell at least one lie every day (DePaulo&Kashy, 1998) and use deception in almost 20 percent of their social interaction.

Given the fact that nearly everyone engages in deception at least occasionally, how can we recognize such actions? The answer seems to involve careful attention to both nonverbal and verbal cues that can reveal the fact that others are trying to deceive us. With respect to nonverbal cues, the following information has been found to be very helpful (e.g., DePaulo et al., 2003):

MICRO EXPRESSIONS: These are fleeting facial expressions lasting only a few tenths of a second. Such reactions appear on the face very quickly after an emotion-provoking event and are

difficult to suppress. As a result, they can be very revealing about others’ true feelings or emotions.

INTER-

CHANNELDISCREPANCIES: A second nonverbal cue revealing of deception is known as interchannel discrepancies. (The term channel refers to type of nonverbal cues; for instance, facial expressions are one channel, body movements are another.) These are inconsistencies between nonverbal cues from different basic channels. These result from the fact that people who are lying often find it difficult to control all these channels at once. For instance, they may manage their facial expressions well, but may have difficulty looking you in the eye as they tell their lie.

EYE CONTACT: Efforts at deception are often revealed by certain aspects of eye contact. People who are lying often blink more often and show pupils that are more dilated than people who are telling the truth. They may also show an unusually low level of eye contact or—surprisingly—an unusually high one as they attempt to fake being honest by looking others right in the eye.

EXAGGERATEDFACIALEXPRESSIONS: Finally, people who are lying sometimes show exaggerated facial expressions. They may smile more—or more broadly—than usual or may show greater sorrow than is typical in a given situation. A prime example: someone says no to a request you’ve made and then shows

exaggerated regret. This is a good sign that the reasons the person has supplied for saying “no” may not be true.

In addition to these nonverbal cues, other signs of deception are sometimes present in nonverbal aspects of what people actually say, or in the words they choose. When people are lying, the pitch of their voices often rises—especially when they are highly motivated to lie. Similarly, they often take longer to begin—to respond to a question or describe events. And they may show a greater tendency to start sentences, stop them, and begin again. In other words, certain aspects of people’s **linguistic style** can be revealing of deception.

THEORIES OF ATTRIBUTION: FRAMEWORKS FOR UNDERSTANDING HOW WE MAKE SENSE OF THE SOCIAL WORLD

Because attribution is complex, many theories have been proposed to explain its operation.

FROM ACTS TO DISPOSITIONS: USING OTHERS’ BEHAVIOR AS A GUIDE TO THEIR LASTING TRAITS

The first of these theories—**Jones and Davis’s (1965) theory of correspondent inference**—asks how we use information about others’ behavior as a basis for inferring their traits. In other words, the theory is concerned with how we decide, on the basis of others’ overt actions,

whether they possess specific traits or dispositions likely to remain fairly stable over time.

Others’ behavior provides us with a rich source on which to draw, so if we observe it carefully, we should be able to learn a lot about them. Up to a point, this is true. The task is complicated, however, by the following fact: Often, individuals act in certain ways not because doing so reflects their own preferences or traits, but rather because external factors leave them little choice. For example, suppose you go to a restaurant and the young woman who greets you at the “Please Wait to Be Seated” sign smiles and acts in a friendly manner. Does this mean that she is a friendly person who simply “likes people”? It’s possible, but According to **Jones and Davis’s theory (Jones & Davis, 1965; Jones & McGillis, 1976)**, we accomplish this task by focusing our attention on certain types of actions—those most likely to prove informative. First, we consider only behavior that seems to have been freely chosen, while largely ignoring ones that were somehow forced on the person in question. Second, we pay careful attention to actions that show what Jones and Davis term **noncommoneffects**—effects that can be caused by one specific factor, but not by others. Perhaps she is acting in this way because that is what her job requires; she has no choice.

According to the theory proposed by Jones and Davis, we are most likely to conclude that others’ behavior reflects their stable traits (i.e., we are likely to reach correspondent inferences about them), when that behavior **(1) is freely chosen; (2) yields distinctive, non-common**

effects; and (3) is low in social desirability.

KELLEY'S THEORY OF CAUSAL ATTRIBUTIONS: HOW WE ANSWER THE QUESTION "WHY?"

According to Kelley, in our attempts to answer the why question about others' behavior, we focus on three major types of information. First, we consider **consensus—the extent to which other people react to a given stimulus or event in the same manner as the person we are considering. The higher the proportion of people who react in the same way, the higher the consensus.**

Second, we consider **consistency—the extent to which the person in question reacts to the stimulus or event in the same way on other occasions, over time.**

And third, we examine **distinctiveness—the extent to which this person reacts in the same manner to other, different stimuli or events.**

According to Kelley's theory, we are most likely to attribute another's behavior to **internal causes** under conditions in which consensus and distinctiveness are **low** but consistency is **high**. In contrast, we are most likely to attribute another's behavior to **external causes** when consensus, consistency, and distinctiveness are **all high**. Finally, we usually attribute another's behavior to a **combination of internal and external factors** when consensus is **low** but consistency and distinctiveness are **high**.

OTHER DIMENSIONS OF CAUSAL ATTRIBUTION While we are often very interested in knowing whether others'

behavior stemmed mainly from internal or external causes, this is not the entire story. In addition, we are also concerned with two other questions: (1) Are the causal factors that influenced their behavior likely to be stable over time or likely to change?

Are these factors controllable—can the individual change or influence them if he or she wishes to do so (Weiner, 1993, 1995)

ACTION IDENTIFICATION AND THE ATTRIBUTION PROCESS

When we see other people perform some action, and try to understand it—why they are doing it, what they want to accomplish—we have a wide range of interpretations open to us. For instance, suppose you saw someone putting loose change into a jar. You could conclude: "She wants to avoid losing the change so she puts it into the jar." Alternatively, you could conclude: "She is trying to save so that she can contribute to her own education." The **first** is a **low level interpretation** that focuses on the **action itself** and involves little in the way of planning or long-range goals to the person involved; the **second**, in contrast, **attributes such plans, intentions, and goals to this person**. The action is the same (putting changes into a jar) but our interpretation of it—and of why it occurs—is very different. The level of interpretation we use is known as **action identification**.

THE CORRESPONDENCE BIAS:

overestimating the role of dispositional causes **Correspondence bias**—the tendency to explain others' actions as stemming from (corresponding to) dispositions even in the presence of clear

situational causes (e.g., Gilbert & Malone, 1995). This bias seems to be so general in scope that many social psychologists refer to it as the fundamental attribution error. In short, we tend to perceive others as acting as they do because they are “that kind of person,” rather than because of the many external factors that may influence their behavior. This tendency occurs in a wide range of contexts but appears to be strongest in situations where both consensus and distinctiveness are low, as predicted by Kelley’s theory, and when we are trying to predict others’ behavior in the far-off future rather than the immediate future.

WHY DOES THE FUNDAMENTAL ATTRIBUTION ERROR OCCUR?

When we observe another person’s behavior, we tend to focus on his or her actions and the context in which the person behaves; hence potential situational causes of his or her behavior often fade into the background. As a result, dispositional causes (internal causes) are easier to notice (they are more **salient**) than situational ones. In other words, from our perspective, the person we are observing is high in perceptual salience and is the focus of our attention, whereas situational factors that might also have influenced this person’s behavior are **less salient** and so seem less important to us. Another explanation is that we notice such situational causes but give them insufficient weight in our attributions. Still another explanation is when we focus on others’ behavior, we tend to begin by assuming that their actions reflect their underlying characteristics. Then, we attempt to correct for any possible effects of the external world—the current situation—by taking

these into account the mental shortcut known as **anchoring** and **adjustment**.

THE ACTOR–OBSERVER EFFECT: “YOU FELL; I WAS PUSHED”

The actor–observer effect (Jones & Nisbett, 1971), the tendency to attribute our own behavior to situational (external) causes but that of others to dispositional (internal) ones. Thus, when we see another person trip and fall, we tend to attribute this event to his or her clumsiness. If we trip, however, we are more likely to attribute this event to situational causes, such as ice on the sidewalk.

THE SELF-SERVING BIAS: “I’M GOOD; YOU ARE LUCKY”

The self-serving bias: This tendency to attribute our own positive outcomes to internal causes but negative ones to external factors, and it appears to be both general in scope and powerful in its effects (Brown & Rogers, 1991; Miller & Ross, 1975).

Why does this tilt in our attributions occur? Several possibilities have been suggested, but most of these fall into two categories: cognitive and motivational explanations. The cognitive model suggests that the self-serving bias stems mainly from certain tendencies in the way we process social information (Ross, 1977; see also Chapter 2). Specifically, it suggests that we attribute positive outcomes to internal causes, but negative ones to external causes because we expect to succeed and have a tendency to attribute expected outcomes to internal causes more than to external causes. In contrast, the motivational explanation suggests that the self-serving bias stems from our need to

protect and enhance our self-esteem or the related desire to look good to others (Greenberg, Pyszczynski, & Solomon, 1986). While both cognitive and motivational factors may well play a role in this kind of attributional error, research evidence seems to offer more support for the motivational view (e.g., Brown & Rogers, 1991).

ATTRIBUTION AND DEPRESSION

Depression is the most common psychological disorder. Although many factors play a role in depression, one that has received increasing attention is what might be termed **a self-defeating pattern of attributions**. In contrast to most people, who show the self-serving bias described above, depressed individuals tend to adopt an opposite pattern. They attribute negative outcomes to lasting, internal causes such as their own traits or lack of ability, but attribute positive outcomes to temporary, external causes such as good luck or special favors from others. As a result, such people perceive that they have little or no control over what happens to them—they are simply being blown about by the winds of unpredictable fate. Little wonder that they become depressed and tend to give up on life! And once they are depressed, the tendency to engage in this self-defeating pattern is strengthened, and a vicious cycle is often initiated.

ATTITUDE AND ITS CHANGE WITHIN CULTURAL CONTEXT

Social psychologists use the term **attitude** to refer to people's evaluation of almost any aspect of the world. People can have favorable or unfavorable reactions to

issues, ideas, objects, actions (do you like white water rafting), a specific person (such as Barack Obama) or entire social groups (Muslims). Some attitudes are quite stable and resistant to change, whereas others may be unstable and show considerable variability depending on the situation (Schwarz & Bohner, 2001). We may hold some attitudes with great certainty, while our attitudes toward other objects or issues may be relatively unclear or uncertain (Tormala & Rucker, 2007).

Attitudes can influence our thoughts, even if they are not always reflected in our overt behavior. Moreover, while many of our attitudes are **explicit attitudes**—conscious and reportable—other attitudes may be **implicit attitudes**—uncontrollable and perhaps not consciously accessible to us.

ATTITUDES FORMATION

CLASSICAL CONDITIONING: It is a basic principle of psychology that when a stimulus that is capable of evoking a response—the unconditioned stimulus—regularly precedes another neutral stimulus, the one that occurs first can become a signal for the second—the conditioned stimulus. Advertisers and other persuasion agents have considerable expertise in using this principle to create positive attitudes toward their products.

Attitudes can be influenced by **subliminal conditioning**—classical conditioning that occurs in the absence of conscious awareness of the stimuli involved.

MEREEXPOSURE—having seen an object before, but too rapidly to remember having seen it—can result in attitude formation (Bornstein & D’Agostino, 1992).

ILLUSION OF TRUTH EFFECT- The mere repetition of information creates a sense of familiarity and more positive attitudes.

INSTRUMENTALCONDITIONING- A basic form of learning in which responses that lead to positive outcomes or which permit avoidance of negative outcomes are strengthened.

OBSERVATIONALLEARNING-A third means by which attitudes are formed can operate even when direct rewards for acquiring or expressing those attitudes are absent. It occurs when individuals acquire attitudes or behaviors simply by observing others (Bandura, 1997).

SOCIALCOMPARISONthe process through which we compare ourselves to others to determine whether our view of social reality is, or is not, correct.

REFERENCE GROUPS-People often adjust their attitudes so as to hold views closer to those of others who they value and identify with.

PROSOCIAL BEHAVIOR

Staub (1979) defined pro-social behavior as voluntary behavior intended to benefit another person. “Voluntary” emphasises the spontaneous initiative by the Pro-social behavior is defined as “...any act performed with the goal of benefiting another person” (Aronson, Wilson, & Akert, 2004 p. 382). Pro-social are voluntary made with the intention of

benefiting others (Eisenberg & Fabes, 1998). This definition focuses on the potential benefits to the person performing the pro-social behavior.

PRO-SOCIAL BEHAVIOR AND ALTRUISM

Pro-social behavior is often accompanied by psychological and social rewards for its performer. In the long run, individuals can benefit from living in a society where prosociality is common. It has been difficult for researchers to identify purely altruistic behaviors, benefiting only the recipient and not the performer. Altruism is generally defined as any form of voluntary act intended to favour another without expectation of reward.

Perhaps the first person to utilise the term altruism was the French sociologist Auguste Comte, who declared that humans have inborn drives to behave sympathetically toward others (Lee, Lee and Kang, 2003). It is a specific kind of motivation to benefit another without consciously considering for one’s own self interest (Hall, 1999). In other words, altruism refers to a kind of selfless help, which is based on pure desire to help others (Aronson, Wilson, Akert, & Fehr, 2004). Nevertheless, altruism is not a synonym for pro-social behavior. Prosocial behavior refers to helping behavior of favouring another person with the goal that may involve benefits to self (Smith & Mackie, 2000; Aronson et. al., 2004). For instance, people donating money to Tsunami relief fund may not always be selfless. In the case that donation is for the sake of tax exemption, its motive would not be regarded as altruistic. The major difference between altruism and pro-social

behavior is that altruism does not involve the element of self interest (Myers, 1996).

FACTORS AFFECTING HELPING BEHAVIOR PHYSICAL

ATTRACTIVENESS: Attractiveness is defined as physical attractiveness or the attractiveness of a person's personality or behavior (DeVito, 1976). Researchers believe physical attractiveness can be defined for any one individual situationally (DeVito, 1976). Physically attractive people are more likely to receive help than unattractive people (Harrell, 1978). The explanation lies in the fact, that as a society, we consciously or subconsciously tend to treat attractive individuals differently, expecting better lives for them (Berscheid, Walster, Bohrnstedt, 1973). Adams and Cohen (1976) feel physical attractiveness is a major factor in the development of prosocial behavior in a child.

SIMILARITY AND KINSHIP Finally, individuals are more likely to behave prosocially towards similar or likable others (Penner et al., 2005), and towards others considered to be close, especially kin (Graziano et al., 2007). Genetic relatedness aside, pro-social behavior towards family members probably involves a sense of duty, reciprocity, and affective relationships. Individuals care more for victims who belong to their in-group rather than to their out-group (Dovidio et al. 1997; Flippen et al. 1996; Levine et al. 2002). Park and Schaller (2005) found that attitude similarity serves as a heuristic cue signaling kinship, which may motivate kin-recognition responses

(e.g., prosocial behavior) even to unrelated individuals.

RELIGIOSITY Although several studies have examined the impact of donor characteristics across various domains, the findings are not as robust as those about victim characteristics. One consistent finding is that humanitarian values and religiosity are correlated with giving (Burnett 1981; Pessemier, Bemmaor, and Hanssens 1977).

VICTIM'S PERSPECTIVE Batson and colleagues have shown consistently greater empathy and altruistic behavior by individuals who are primed to take the victim's perspective (Batson, Early, and Salvarani 1997; Batson et al. 2003).

PERSONAL EXPERIENCE A vast literature examines the impact of personal experience on self-protective behavior (Weinstein, 1989, for a critical review). Although the majority of studies examine effects on victims themselves, a few assess the impact of knowing a victim as a form of personal experience (Manheimer, Mellinger & Crossley 1966 and Schiff 1977). Barnett et al. (1986) found that participants who had been raped reported greater empathy when watching a video about a rape victim than did those who had never been raped. Batson et al. (1996) found that for females but not males, the expectation of oneself receiving a shock affected self-reported empathy when one observed a same-sex peer receiving a shock. Christy and Voigt (1994) found that those who reported being abused as a child indicated that they would be more likely than those who had never been abused to intervene if they saw a child being abused.

IDENTIFIABLE VICTIM EFFECT Previous research has shown that people give more to identifiable victims

than to unidentifiable or statistical victims (Kogut and Ritov 2005a, b; Small, Loewenstein, and Slovic 2006). This effect has even been demonstrated when no meaningful information is provided about the identified victim (Small and Loewenstein 2003). Other identifying factors, such as showing a victim's face or being in the presence of a victim, also increase pro-social behavior (Bohnet and Frey 1999). Charities do often describe or show images of specific victims to potential donors in their advertising campaigns, but such attempts seem designed to benefit from the identifiable victim effect (Kogut and Ritov, 2005a, b; Small et al. 2006), rather than to create "friendship" between donors and victims.

ATTRIBUTIONS CONCERNING VICTIM'S RESPONSIBILITY

People also give more to victims who are perceived as "deserving," in other words, whose needs arise from external rather than internal causes (Weiner 1980). Thus, disabled children are deemed deserving; healthy unemployed men are not (Schmidt and Weiner 1988). Finally, the effect of deservingness on prosocial behavior is mediated by sympathy, suggesting that giving decisions are not based on cold mental calculations (Weiner, 1980). A study carried out on the New York subway showed that people were more likely to help 'blind' rather than 'drunk' confederates who had collapsed (Piliavin, 1969).

Positive FRIEND INFLUENCE Barry and Wentzel (2006) supported the notion that friends in particular can be important socialisers of pro-social behavior. Children are similar to their friends in the degree to which they display pro-social behavior and are motivated to behave this way

(Wentzel, Barry, & Caldwell, 2004; Wentzel & Caldwell, 1997). Adolescents who have friends are more likely to be pro-social than those without friends (McGuire & Weisz, 1982).

GENDER Females engage in prosocial behaviors more frequently than males (Fabes, Carlo, Kupanoff, & Laible, 1999), which is consistent across ratings from parents, teachers, and peers (Holmgren, Eisenberg, & Fabes, 1998). Additionally, observational studies have indicated that females are more likely than males to share and cooperate when interacting (Burford, Foley, Rollins, & Rosario, 1996). Beutel and Johnson (2004) reported that in a study of 12 through 17 year-olds, females placed more importance on prosocial values than males at younger ages, and the gender gap in prosocial values was larger at older ages. Eagly and Crowley (1986) did a meta-analysis and found that men are more likely to help in chivalrous, heroic ways, and women are more likely to help in nurturant ways involving long-term commitment.

AGE Older adolescent males placed less importance on prosocial values than younger adolescent males (Beutel & Johnson, 2004). Further, in a study of adolescent soccer players' behaviors, recruited from age groups of under 13, under 15, and under 17, significant differences among the age groups indicated that the oldest group displayed more frequent antisocial behaviors and less frequent prosocial behaviors compared to the younger groups (Kavussanu, Seal, & Phillips, 2006). However, there appears to be an increase in the use of some prosocial behaviors after a certain point in adolescence, as Eisenberg et al. (2005) found that prosocial moral reasoning and

perspective-taking abilities showed increases with age from late adolescence to early adulthood, whereas helping and displaying sympathy did not increase with age.

PERSONALITY Research following children from early childhood to adulthood supports the existence of the long-debated altruistic or prosocial personality (Eisenberg et al., 1999). Individual differences in prosociality are linked to sociability, low shyness, extroversion, and agreeableness, although specific prosocial behaviors may require a combination of additional traits, such as perceived self-efficacy in the case of helping (Penner et al., 2005). Personality and contextual variables are likely to interact in determining prosocial behavior. For example, agreeable individuals were more likely to help an outgroup member than low-agreeableness individuals, but agreeableness was not associated with helping an ingroup member (Graziano et al., 2007). While, Hartshorne and May (1929) found only a .23 correlation between different kinds of helping behaviors in children, and several studies have found that those who scored high on a personality test of altruism were not much more likely to help than those who scored low. People's personality is clearly not the only determinant of helping. Instead, it seems to be that different kinds of people are likely to help in different situations.

EFFECTS OF POSITIVE MOODS: Feel Good, Do Good People who are in a good mood are more likely to help. For example, Isen and Levin (1972) did a study in a shopping mall where subjects either found or did not find a dime in a phone booth. As the person emerged from

the booth, a confederate walked by and dropped a sheaf of papers; 84% of those who found the dime helped compared with 4% of those who did not find the dime. North, Tarrang, & Hargreaves (2004) found that people are more likely to help others when in a good mood for a number of other reasons, including doing well on a test, receiving a gift, thinking happy thoughts, and listening to pleasant music. Good moods can increase helping for three reasons: (1) good moods make us interpret events in a sympathetic way; (2) helping another prolongs the good mood, whereas not helping deflates it; (3) good moods increase self-attention, and this in turn leads us to be more likely to behave according to our values and beliefs (which tend to favor altruism).

THEORETICAL PERSPECTIVES

SOCIAL LEARNING THEORY

Social learning theory suggests that pro-social behavior is learned (Bandura, 1977; Bandura & McDonald, 1963; Batson, 1998). Observing role models who are loved or respected, such as parents or authorities, engaged in pro-social behavior, demonstrates how people can and should behave prosocially. Rewards reinforce helping behavior; punishments reduce unhelpful or hurtful behavior. Within a group context, social recognition, not just private reward, increases pro-social behavior (Fisher & Ackerman, 1998). Observational modeling processes with reinforcement will result in learning over time (Compeau & Higgins, 1995; Lim et al., 1997).

MOTIVATION PERSPECTIVE

Theorists differentiate altruistic prosocial behavior from egoistic prosocial behavior depending upon the motivation of the helper (Batson, 1991; Nelson, 1999; Piliavin & Charng, 1990). Altruistic prosocial behavior is motivated purely by the desire to increase another person's welfare; egoistic prosocial behavior is motivated by the desire to increase one's own welfare or that of one's group or cause through helping others (Batson, 1998; MacIntyre, 1967).

SOCIAL IDENTITY THEORY

Social identity theory and self-categorisation theory (Tajfel & Turner, 1986; Turner et al., 1987) are helpful in understanding why some people exhibit substantial prosocial behavior over time. Social identity theory is based on the premise that people identify with particular groups in order to enhance their self-esteem. Identification leads to selective social comparisons that emphasise intergroup differences along dimensions. This leads to favouring the ingroup and confer positive distinctiveness on the ingroup when compared to the salient outgroup (Hogg & Abrams, 1988). Categorising the self and others in terms of groups accentuates the similarities between group members with respect to their fit with the relevant group prototype or 'cognitive representation of features that describe and prescribe attributes of the group' (Hogg & Terry, 2000). The prototype guides the participants' understanding of the group and its expected behaviors and attitudes. People identified with a group will thus be more likely to exhibit behaviors that are consistent with shared group norms and will cooperate with the group and its members.

BIOLOGICAL PERSPECTIVE

Empathy, altruism and prosocial behavior are considered vital for the good functioning of society. Although psychological theories emphasise the importance of cognition and socialisation, genes also have a role to play. Monozygotic (MZ) twin pairs share 100% of their genes, whereas dizygotic (DZ) twin pairs share 50%; thus the comparison of MZ and DZ twin similarities and differences allows for estimates to be made of genetic influences (Plomin et al. 2001). Several studies have found that by adulthood, approximately 50% of the variance in altruism, empathy and social responsibility is due to genes and 50% to nongenetic factors (Rushton et al. 1986; Rushton 2004).

NEGATIVE-STATE HYPOTHESIS

RELIEF

Negative State Relief Model, views empathic concern as being accompanied by feelings of sadness that the helper tries to relieve through helping someone in need (Smith, Keating, & Stotland, 1989; Cialdini, et al., 1987; Schroeder, Dovidio, Sibicky, Matthews, & Allen, 1988). Here, the motivation for prosocial behavior is based on increasing the welfare of both the helper and helpee. Three prominent features of the Negative State Relief Model are that: (1) helpers experience empathic concern; (2) such concern is accompanied by feelings of sadness and helpers attempt to relieve such feelings by helping others. Cialdini's (1987) experiments involved participants taking the place of people receiving electric shocks. However, high empathy participants were less inclined to help if they had been praised by the researchers. It is thought that this praise helped to lift

their mood so that it was not necessary to help the person receiving the shocks. When people feel guilty, they are more likely to help. For example, Harris et al. (1975) found that churchgoers were more likely to donate money after confession

EMPATHY – ALTRUISM HYPOTHESIS

Batson (1987, 1991) introduced the empathy-altruism hypothesis, which states refers to “the claim that feeling empathic emotion for someone in need evokes altruistic motivation to relieve that need has been called the empathy-altruism hypothesis. According to this hypothesis, the greater the empathetic emotion, the greater the altruistic motivation.” (Batson et. al., 2002). According to the Empathy-Altruism Hypothesis, empathic concern motivates helpers to enhance the welfare of those in need rather than avoid the situation instead (Smith, Keating, & Stotland, 1989; Batson, 1987).

EMPATHIC-JOY HYPOTHESIS

Smith, Keating, and Stotland's (1989) hypothesis proposes that empathic concern is based on a helper's overarching sensitivity to a victim's emotional state and a subsequent heightened sense of vicarious happiness and relief upon the fulfillment of the recipient's needs. The authors propose that empathic witnesses to someone in need may regard empathic joy as being more achievable and rewarding than would be a self-focused witness, and thus have greater motivation to help. The three prominent features of the Empathic-Joy Hypothesis are that:

- helpers experience empathic concern;
- this concern is a function of their sensitivity to another's needs; and

the awareness of relief for another's distress promotes subsequent relief of the helper's empathic concern as well as a sense of joy.

SELF-EFFICACY HYPOTHESIS

This hypothesis reflects a combination of proposals from authors regarding correlates to helping behaviors. According to Midlarsky (1968) individuals' level of competence with a given skill can influence helping behavior, especially in times of need. Such competence may increase the likelihood of helping through increased certainty over what to do, along with the decreased fear of making a mistake and decreased stress over the situation (Withey, 1962; Janis 1962, Midlarsky, 1968; Staub, 1971).

GROUP AND SOCIAL INFLUENCE [SOCIAL FACILITATION, SOCIAL LOAFING]

WHAT IS A GROUP?

Group dynamics expert **Marvin Shaw** (1981) argued that all groups have one thing in common: Their members interact. Therefore, he defines a group as two or more people who interact and influence one another. Moreover, notes Australian National University social psychologist **John Turner** (1987), groups perceive themselves as “us” in contrast to “them.” A pair of jogging companions, then, would indeed constitute a group. Different groups help us meet different human needs—to affiliate (to belong to and connect with others), to achieve, and to gain a social identity (Johnson & others, 2006).

By Shaw's definition, students working individually in a computer room would not be a group. Although physically together, they are more a collection of individuals than an interacting group (though each may be part of a group with dispersed others in an online chat room). The distinction between collections of unrelated individuals in a computer lab and the more influential group behavior among interacting individuals sometimes blurs. People who are merely in one another's presence do sometimes influence one another. At a football game, they may perceive themselves as "us" fans in contrast with "them" who root for the other team.

SOCIAL FACILITATION

HOW ARE WE AFFECTED BY THE PRESENCE OF OTHERS?

Social psychology's most elementary issue concerns the mere presence of others. Some early experiments on this question found that performance improved with observers or co-actors present. Others found that the presence of others can hurt performance. **Robert Zajonc** reconciled those findings by applying a well-known principle from experimental psychology: **Arousal** facilitates dominant responses. Because the presence of others is arousing, the presence of observers or co-actors boosts performance on easy tasks (for which the correct response is dominant) and hinders performance on difficult tasks (for which incorrect responses are dominant).

Being in a crowd, or in crowded conditions, is similarly arousing and facilitates dominant responses.

But why are we aroused by others' presence? Experiments suggest that the arousal stems partly from evaluation apprehension and partly from distraction—a conflict between paying attention to others and concentrating on the task. Other experiments, including some with animals, suggest that the presence of others can be arousing even when we are not evaluated or distracted.

SOCIAL LOAFING

DO INDIVIDUALS EXERT LESS EFFORT IN A GROUP?

In a team tug-of-war, will eight people on a side exert as much force as the sum of their best efforts in individual tugs-of-war? If not, why not? What level of individual effort can we expect from members of work groups?

Nearly a century ago, French engineer **Max Ringelmann** (reported by Kravitz & Martin, 1986) found that the collective effort of tug-of-war teams was but half the sum of the individual efforts. Contrary to the presumption that "in unity there is strength," this suggested that group members may actually be less motivated when performing additive tasks.

Researchers **Bibb Latané, Kipling Williams, and Stephen Harkins (1979; Harkins & others, 1980)** kept their ears open for other ways to investigate this phenomenon, which they labeled **social loafing**. They observed that the noise produced by six people shouting or clapping "as loud as you can" was less than three times that produced by one person alone. Like the tug-of-war task, however, noisemaking is vulnerable to group inefficiency. So Latané and his associates followed Ingham's example by

leading their Ohio State University participants to believe others were shouting or clapping with them, when in fact they were doing so alone.

**SOCIAL INFLUENCE
[CONFORMITY, PEER PRESSURE,
PERSUASION, COMPLIANCE,
OBEDIENCE, SOCIAL POWER,
REACTANCE].**

SOCIAL INFLUENCE

WHAT IS CONFORMITY?

Let us take this question first. Is conformity good or bad? That question has no scientific answer. Assuming the values most of us share, we can say that conformity is at times **bad** (when it leads someone to drive drunk or to join in racist behavior), at times **good** (when it inhibits people from cutting into a theatre line), and at times **inconsequential** (when it disposes tennis players to wear white). In Western individualistic cultures, where submitting to peer pressure is not admired, the word “conformity” tends to carry a negative value judgment. How would you feel if you overheard someone describing you as a “real conformist”? I suspect you would feel hurt. North American and European social psychologists, reflecting their individualistic cultures, give social influence negative labels (conformity, submission, compliance) rather than positive ones (communal sensitivity, responsiveness, cooperative team play).

In Japan, going along with others is a sign not of weakness but of tolerance, self-control, and maturity (Markus & Kitayama, 1994). The moral: We choose labels to suit our values and judgments. Labels both describe and evaluate, and they are

inescapable. So let us be clear on the meanings of the following labels: conformity, compliance, obedience, and acceptance. Conformity is not just acting as other people act; it is also being affected by how they act. It is acting or thinking differently from the way you would act and think if you were alone. Thus,

conformity is a change in behavior or belief to accord with others. When, as part of a crowd, you rise to cheer a game-winning goal, are you conforming? When, along with millions of others, you drink milk or coffee, are you conforming? When you and everyone else agree that women look better with longer hair than with crewcuts, are you conforming? Maybe, maybe not. The key is whether your behavior and beliefs would be the same apart from the group. Would you rise to cheer the goal if you were the only fan in the stands? There are several varieties of conformity (Nail & others, 2000). Consider three: compliance, obedience, and acceptance. Sometimes we conform to an expectation or a request without really believing in what we are doing. We put on the necktie or the dress, though we dislike doing so. This insincere, outward conformity is **compliance**. We comply primarily to reap a reward or avoid a punishment. If our compliance is to an explicit command, we call it **obedience**.

Sometimes we genuinely believe in what the group has persuaded us to do. We may join millions of others in exercising because we all have been told that exercise is healthy and we accept that as true. This sincere, inward conformity is called

acceptance. Acceptance sometimes follows compliance; we may come to inwardly believe something we initially questioned.

Three classic sets of experiments illustrate how researchers have studied conformity.

Muzafer Sherif observed that others' judgments influenced people's estimates of the movement of a point of light that actually did not move. Norms for "proper" answers emerged and survived both over long periods of time and through succeeding generations of research participants.

Solomon Asch had people listen to others' judgments of which of three comparison lines was equal to a standard line and then make the same judgment themselves. When the others unanimously gave a wrong answer, the participants conformed 37 percent of the time.

Stanley Milgram's obedience experiments elicited an extreme form of compliance. Under optimum conditions—a legitimate, close-at-hand commander, a remote victim, and no one else to exemplify disobedience—65 percent of his adult male participants fully obeyed instructions to deliver what were supposedly traumatizing electric shocks to a screaming, innocent victim in an adjacent room.

These classic experiments expose the potency of several phenomena. Behavior and attitudes are mutually reinforcing, enabling a small act of evil to foster the attitude that leads to a bigger evil act. The power of the situation is seen when good people, faced with dire circumstances, commit reprehensible acts (although dire situations may produce heroism in others).

WHAT PREDICTS CONFORMITY?

Using conformity testing procedures, experimenters have explored the circumstances that produce conformity. Certain situations appear to be especially powerful. For example, conformity is affected by the characteristics of the group: People conform most when three or more people, or groups, model the behavior or belief.

Conformity is reduced if the modelled behavior or belief is not unanimous.

Conformity is enhanced by group cohesion.

The higher the status of those modeling the behavior or belief, the greater likelihood of conformity.

People also conform most when their responses are public (in the presence of the group).

A prior commitment to a certain behavior or belief increases the likelihood that a person will stick with that commitment rather than conform.

WHY CONFIRM?

Experiments reveal two reasons people conform. **Normative influence** results from a **person's desire for acceptance**: We want to be liked. The tendency to conform more when responding publicly reflects normative influence.

Informational influence results from others' **providing evidence about reality**. The tendency to conform more on difficult decision-making tasks reflects informational influence: We want to be right.

WHO CONFIRMS?

The question “Who conforms?” has produced few definitive answers. Personality scores are poor predictors of specific acts of conformity but better predictors of average conformity. **Traiteffects** are **strongest** in “**weak**” situations where social forces do not overwhelm individual differences.

Although conformity and obedience are universal, different cultures socialize people to be more or less socially responsive.

Social roles involve a certain degree of conformity, and conforming to expectations is an important task when stepping into a new social role.

PEER PRESSURE

Peer pressure (or **social pressure**) is the direct influence on people by peers, or the effect on an individual who gets encouraged to follow their peers by changing

their attitudes, values or behaviors to conform to those of the influencing group or individual. This can result in either a positive or negative effect, or both. Social groups affected include both *membership groups*, in which individuals are “formally” members (such as political parties and trade unions), and cliques, in which membership is not clearly defined. However, a person does not need to be a member or be seeking membership of a group to be affected by peer pressure. Peer pressure can decrease one's confidence. It can affect the lives of the students drastically.

There has been considerable study regarding peer pressure's effects on children and adolescents, and in popular discourse the term is mostly used in the contexts of those age groups. For children, the common themes for study regard their abilities for independent decision making; for adolescents, peer pressure's relationship with sexual intercourse and substance abuse have been significantly researched. Peer pressure can affect individuals of all ethnicity, genders and ages, however. Peer pressure has moved from strictly face-to-face interaction to digital interaction as well. Social media offers opportunities for adolescents and adults alike to instil and/or experience pressure every day.

COMPLIANCE

THE UNDERLYING PRINCIPLES

Some years ago, **Robert Cialdini**, a well-known social psychologist, decided that **the best way to find out about compliance was to study what he termed compliance professionals**—people whose success (financial or otherwise) depends on their ability to get others to say yes. Who are such people? They include salespeople, advertisers, political lobbyists, fund-raisers, politicians, con artists, professional negotiators, and many others. Cialdini's technique for learning from these people was simple: He temporarily concealed his true identity and took jobs in various settings where gaining compliance is a way of life.

FOLLOWING ARE THE PRINCIPLES OF COMPLIANCE:

FRIENDSHIP/LIKING: In general, we are more willing to

comply with requests from friends or from people we like than with requests from strangers or people we don't like.

The term **ingratiation** refers to behaviors that a person illicitly enacts to make others like him or her or think well of his or her qualities as a person. There are many ways in which people can ingratiate themselves. One that is frequently used is to show interest in another person; ask questions, pay attention, and single out the person so that you make him or her feel special. A second strategy is do favors or to help or assist a person. For instance, you can bring your colleague a cup of coffee or help an attractive stranger with car trouble. Third, you may show support and loyalty, for instance during a meeting, when you express agreement with your supervisor. A fourth way to make people like you is simply to smile and be friendly, cheerful, and positive. Fifth, you can directly express admiration by flattering people and telling them what you like or admire about them.

Another means is **incidental similarity—calling attention to small and slightly surprising similarities between them and ourselves**. In several recent studies, Burger, Messian, Patel, del Pardo, and Anderson (2004) found that research participants were more likely to agree to a small request (make a donation to charity) from a stranger when this person appeared to have the same first name or birthday as they did than when the requester was not similar to them in these ways. Apparently, these trivial forms of similarity enhance liking or a feeling of affiliation with the requester and so increase the tendency to comply with this person's requests.

COMMITMENT/CONSISTENCY:

Once we have committed ourselves to a position or action, we are more willing to comply with requests for behaviors that are consistent with this position or action than with requests that are inconsistent with it.

This is the basic idea behind an approach for gaining compliance known as **the foot-in-the-door technique**. Basically, this involves inducing target people **to agree to a small initial request (“Accept this free sample”)** and **then making a larger request**—the one desired all along. The results of many studies indicate that this tactic works—it succeeds in inducing increased compliance. Once we have said yes to the small request, we are more likely to say yes to subsequent and larger ones, too, because refusing these would be inconsistent with our previous behavior.

Another is the **lowball procedure**. In this technique, which is often used by automobile salespersons, a very good deal is offered to a customer. After the customer accepts, however, something happens that makes it necessary for the salesperson to change the deal and make it less advantageous for the customer—for example, the sales manager rejects the deal. The totally rational response for customers, of course, is to walk away. Yet, often they agree to the changes and accept the less desirable arrangement.

Clear evidence for the importance of an initial commitment in the success of the lowball-technique is provided by research conducted by **Burger and Cornelius (2003)**. These researchers phoned students living in dorms and asked them if they would contribute \$5.00 to a scholarship fund for underprivileged students. **In the**

lowball condition, she indicated that people who contributed would receive a coupon for a free smoothie at a local juice bar. Then, if the participant agreed to make a donation, she told them that she had just run out of coupons and couldn't offer them this incentive. She then asked if they would still contribute.

In another condition (the interrupt condition), she made the initial request but before the participants could answer yes or no, interrupted them and indicated that there were no more coupons for people who donated. In other words, this was just like the “lowball condition”, except that participants had no opportunity to make an initial commitment to donating to the fund. Finally, in a **third (control) condition**, participants were asked to donate \$5.00 with no mention of any coupons for a free drink. **Results indicated that more people in the lowball condition agreed to make a donation than in either of the other two conditions.**

SCARCITY: In general, we value, and try to secure, outcomes or objects that are scarce or decreasing in availability. As a result, we are more likely to comply with requests that focus on scarcity than ones that make no reference to this issue.

We are often willing to expend more effort or go to greater expense to obtain items or outcomes that are scarce than to obtain ones that are in large supply. This principle serves as the foundation for several techniques for gaining compliance. One of the **most common of these is playing hard to get—a tactic often used in the area of romance. What it involves is actions by a person using this**

technique suggesting that they have very little interest in the target person—the one toward whom playing hard to get is directed. For instance, a person playing hard to get might drop hints to the effect that a potential partner (the target person) has a lot of competition—many rivals. When it works, this tactic can fan the flames of passion in the people who are on the receiving end (e.g., Walster, Walster, Piliavin, & Schmidt, 1973).

A related procedure also based on the **“what’s-scarce-is-valuable” principle** is one frequently used by department stores. Ads using this **deadline technique state that a special sale will end on a certain date, implying that after that, the prices will go up. In many cases, the time limit is false: the prices won’t go up after the indicated date and may, in fact, continue to drop if the merchandise remains unsold.** Yet many people reading such ads believe them and hurry down to the store to avoid missing out on a great opportunity. So when you encounter an offer suggesting that “the clock is ticking” and may soon run out, be cautious: this may simply be a technique for boosting sales.

RECIPROCITY: We are generally more willing to comply with a request from someone who has previously provided a favor or concession to us than to someone who has not. In other words, we feel obligated to pay people back in some way for what they have done for us.

Tactics Based on Reciprocity: The Door-in-the Face and the “That’s-Not-All” Approach Reciprocity is a basic rule of social life: we usually “do unto others as they have done unto.” If they have done a favor for us, therefore, we feel that we

should be willing to do one for them in return. While this is viewed by most people as being fair and just, the principle of reciprocity also serves as the basis for several techniques for gaining compliance.

The door in-the-face technique(because the first refusal seems to slam the door in the face of the requester), and several studies indicate that it can be quite effective. For example, in one well-known experiment, Cialdini and his colleagues (1975) stopped college students on the street and presented a huge request: Would the students serve as unpaid counselors for juvenile delinquents 2 hours a week for the next 2 years! As you can guess, no one agreed. When the experimenters then scaled down their request to a much smaller one—would the same students take a group of delinquents on a 2-hour trip to the zoo—fully 50 percent agreed. In contrast, less than 17 percent of those in a control group agreed to this smaller request when it was presented cold rather than after the larger request.

A related procedure for gaining compliance is known as the **“that’s-not-all technique”**. Here, **an initial request is followed, before the target person can say yes or no, by something that sweetens the deal—a small extra incentive from the people using this tactic** (e.g., a reduction in price, “throwing in” something additional for the same price). For example, television commercials for various products frequently offer something extra to induce viewers to pick up the phone and place an order—for instance a “free” knife or a “free “cookbook.

SOCIAL VALIDATION: We are generally more willing to comply

with a request for some action if this action is consistent with what we believe people similar to ourselves are doing (or thinking). We want to be correct, and one way to do so is to act and think like others.

AUTHORITY: In general, we are more willing to comply with requests from someone who holds legitimate authority—or simply appears to do so.

SOCIAL POWER

Keltner, Gruenfeld, and Andeson (2003) have noted that the restrictions that often influence the thought, expression, and behavior of most people don’t seem to apply to the powerful. And in fact, there are several reasons why this might be so. Powerful people are less dependent on others for obtaining social resources. As a result, they may not pay much attention to threats from others or efforts to constrain their actions in some way. They may be less likely to take the perspective of other people and so be less influenced by them. Instead, their thoughts and actions are more directly shaped by their own internal states; in other words, there is a closer correspondence between their traits and preferences and what they think or do than is true for most people. Overall, then, situational information might have less influence on their attitudes, intentions, actions, and creative expressions.

Research conducted by **Galinsky et al.** (2008) indicates that it is. **In a series of related studies, they found that people who possessed power, or were merely primed to think about it, were in fact less likely to show conformity to the actions or judgments of others than**

people lower in power. In one study, for instance, participants were asked to think either about a situation in which they had power over someone (high power) or a situation in which someone else had power over them (low power). In a third condition they did not think about power one way or the other. Following these conditions, they performed a tedious word construction task—one that most people do not find interesting or enjoyable. Then, they were asked to rate this task.

Before doing so, however, they learned that 10 other students rated it very high on both dimensions. (In a control, baseline condition, they did not receive this information.) It was predicted that the people primed to think about times when they had power over others would rate the task less favorably than those who thought about times when others had power over them—in other words, their feelings of power would affect the extent to which they were influenced by the judgments of other people. In contrast, those not asked to think about power would be influenced by others' opinions and therefore rate the task more favorably.

REACTANCE

Individuals value their sense of freedom and self-efficacy. When blatant social pressure threatens their sense of freedom, they often rebel. The theory of psychological **reactance**—that people act to protect their sense of freedom—is supported by experiments showing that attempts to restrict a person's freedom often produce an **anti-conformity “boomerang effect”** (Brehm & Brehm, 1981; Nail & others, 2000). In one field experiment, many nongeeky students stopped wearing a “Livestrong” wristband when nearby geeky academic students

started wearing the band (Berger & Heath, 2008). Likewise, rich Brits dissociated themselves from a dissimilar group when they stopped wearing Burberry caps after they caught on among soccer hooligans (Clevstrom&Passariello, 2006). Reactance may contribute to underage drinking. A survey of 18- to 24-year-olds by the Canadian Centre on Substance Abuse (1997) revealed that 69 percent of those over the legal drinking age (21) had been drunk in the last year, as had 77 percent of those under 21. In the United States, a survey of students on 56 campuses revealed a 25 percent rate of alcohol abstinence among students of legal drinking age (21) but only a 19 percent abstinence rate among students under 21 (Engs& Hanson, 1989).

AGGRESSION

There are three key issues with defining human aggression. **First**, it is hard to interpret research findings and theories about aggression without a clear definition. Historically, however, many different definitions have been used. As a result, many studies of aggressive behavior are hard to meaningfully compare. More recently, definitions of aggression among social psychologists have converged around the notion that **aggression is any behavior enacted with the intention to harm another person who is motivated to avoid that harm** (e.g., Anderson and Bushman, 2002; Bushman and Huesmann, 2010). Such a definition is wide enough to capture the full range of aggressive behaviors, and to make allowance for activities that can ‘hurt’ a target person but to which the target of the hurt willingly consents (such as undergoing surgery or engaging in sadomasochistic sex).

The second issue is that many laypersons and misinformed professionals use the term aggression interchangeably with related but conceptually distinct phenomena such as anger, hostility and competitiveness. There is no question that in the field of psychology, aggression refers only to a behavior, and not to a mindset or an emotional state. Feelings such as anger, attitudes such as wishing the worst for another, and motivations such as the desire to win or control one's environment may contribute to a person behaving aggressively but are not aggression per se. To study aggression effectively, such factors need to be clearly differentiated from aggression and from each other. A third definitional issue involves the common practice of using the term 'violence' interchangeably with the term 'aggression.' Treating these as synonymous creates miscommunications and confusion among researchers, public policy-makers, and the general public. Among most social psychologists, violence is a subtype of aggression. More precisely, 'violence' is aggression that is intended to cause harm extreme enough to require medical attention or to cause death. Many social psychologists extend this definition to include causing severe emotional harm. Thus, all violent behavior is aggression, but most aggression is not violence. Note that this definition of violence is not synonymous with 'violent crime,' which is a legal term, not a scientific one.

TYPES AND CHARACTERISTICS OF AGGRESSION

Different forms of aggression include physically harming another (i.e., **physical aggression** such as

hitting, biting, kicking, clubbing, stabbing, shooting), hurting another with spoken words (i.e., **verbal aggression** such as yelling, screaming, swearing, name calling), or hurting another's reputation or friendships through what is said to others verbally or digitally (i.e., **relational aggression**).

Aggression may also be **direct** (with the victim physically present) or **indirect** (enacted in the absence of the victim; for example, smashing someone's property or spreading rumors about them).

Aggression **also differs by function**. It may involve a relatively pure intent to punish/hurt the target person, as in reacting aggressively to provocation (i.e., reactive, affective, hostile, hot, impulsive, **or retaliatory aggression**) or it may involve a considered and deliberate plan to harm another to gain a desired outcome (i.e., **instrumental**, proactive, planned, or cold aggression). Aggression may be an automatic response driven by hard-wired self-protection mechanisms (e.g., fight or flight) or involve a script for aggressive behavior that is so commonly enacted that the response is no longer thought-through.

Of course, such distinctions can be problematic. What about a person whose rage drives them to carefully plan the death of another? Such instances do not fit any of these traditional categorical or dichotomous distinctions. A viable alternative approach to understanding the function of aggression is to locate aggressive acts on three dimensions –

- the degree to which the goal is to harm the victim versus benefit the perpetrator;
- the level of hostile or agitated emotion that is present; and

the degree to which the aggressive act was thought-through.

THEORIES OF AGGRESSION

Early Social Psychological Theories For more than 70 years, social psychology has provided a variety of frameworks from which hypotheses about the causes and consequences of aggression could be derived and tested. These theories, although distinct, have also tended to overlap as new knowledge has extended an existing framework of aggressive behavior. The earliest influential theory from social psychology was the frustration-aggression hypothesis.

1. THE FRUSTRATION-AGGRESSION HYPOTHESIS

During 1939, partially in response to World War II and partially in response to the spreading influence of psychodynamic theories in the US, Dollard et al. (1939) proposed the first systematic theory of aggression. Using assumptions from psychoanalytic theory, they focused on the frustration caused

when a goal is blocked, and suggested that “the occurrence of aggressive behavior always presupposes the existence of frustration,” and that “the existence of frustration always leads to some form of aggression”(p. 1). Although this theory enjoyed some empirical support, it quickly became obvious that

frustration does not always lead to aggression, and that not every act of aggression can be traced back to frustration. Frustration-aggression theory was revised to incorporate the possibility that frustrations can elicit responses other than aggression (e.g., to escape or to find another way to achieve a goal), and that

the inclination which will be acted upon is the one that best reduces frustration. In this revised formulation, people learn through experience to respond to frustrations with aggressive or nonaggressive responses.

2. LEARNING THEORIES

The earliest theory of learning in modern psychology explains behavior in terms of classical conditioning –learning to associate one thing with another.

Pioneered by Pavlov, this approach suggests that once people mentally pair things together, they become ‘conditioned’ to expect those things to always occur together. This theory was later supplemented with theories of operant conditioning developed by Thorndike and Skinner, which suggest that people are more likely to repeat a behavior that has been rewarded and less likely to repeat a behavior that has been punished. In aggression research it has been shown that children can be taught to behave aggressively through rewarding aggressive behavior (positive reinforcement) or removing a painful consequence after aggression (negative reinforcement). In addition, children learn to discriminate between situations where aggression has a desirable consequence and when it does not, and to generalize this knowledge to new situations. Although such research demonstrates that aggression can be learned through conditioning (e.g., Eron et al., 1971), it was clear by the 1960s that such processes could not explain the acquisition of all learned aggression. Bandura proposed that social behaviors, including aggression, could be learned through observing and imitating others (i.e., via observational learning). In his classic experiments, children observed a

film of an actor hitting a ‘Bobo Doll’ in several novel ways. The children later imitated the behavior in the absence of any classical or operant conditioning. Bandura also developed the concept of vicarious learning of aggression, and showed that children were especially likely to imitate models that had been rewarded for behaving aggressively. **In social learning theory (later called social cognitive theory)**, Bandura hypothesized that the way people mentally construct their experiences is crucial. People may see one person hit another, but will also decide how competent they feel to do the same, and will make assumptions about what constitutes a normal way to respond when someone provokes you. In this way,

making inferences about observed aggression not only increases the likelihood of imitating it, but also expands the range of situations to which that response might be generalized (see Bandura, 1986). There is considerable research support for social cognitive explanations of aggression. People sometimes imitate aggressive models, especially if the aggressive behavior is rewarded or carried out by a person who is heroic, admired, of high status, attractive, or similar.

3. AROUSAL: COGNITIVE LABELING AND EXCITATION TRANSFER

The emergence of cognitive psychology inspired a plethora of new approaches to aggression by social psychologists. Early in this period, researchers explored the way people make meaning of physiological arousal, a known precursor to aggression. Researchers such as Schacter found that when people

to thesecond event will be misattributed as being relevant to all of thearousal experienced, thus producing an inappropriately strongresponse (e.g., becoming angry to a level far greater than mightbe expected for a minor provocation). Because the cognitivelabel (or attribution) is crucial in determining

	<i>Aggression conceptualized as . . .</i>	<i>Data base</i>
Biological approaches		
Ethology	. . . internal energy released by external cues; steam-boiler model	Animal studies
Behaviour genetics	. . . transmitted as part of genetic make-up	Twin and adoption studies
Hormonal explanations	. . . influenced by male sex hormones and cholesterol	Developmental studies
Psychological approaches		
Frustration-aggression hypothesis	. . . as a likely response to frustration, likelihood enhanced by aggressive cues	Experimental studies
Cognitive neo-associationist model and excitation transfer	. . . as a result of affect elicited by aversive stimulation that is interpreted as anger	Experimental studies
Learning theory	. . . as a result of reinforcement, either direct or indirect (observed)	Experimental + observational studies
Social cognitive approaches	. . . as a result of social information processing, enactment of learned scripts	Experimental + longitudinal studies

Are aroused, they look for cues in the environment to help themattribute the cause of their arousal. For example, Schacter andcolleagues found that if aroused people were exposed toanother person who was angry, they tended to cognitively labeltheir arousal as being angry themselves. Zillmann (1979)extended this concept with

excitation-transfer theory (ETT). Physiological arousal, however produced, dissipates slowly.ETT posits that if two arousing events are separated by a shortamount of time, arousal from the first event will add to arousalfrom the second. However, the cognitive label given

behavior, stronganger related to excitation transfer may persist long after thearousal itself has dissipated.

MAINSTREAM COGNITIVE THEORIES

INFORMATION PROCESSING AND SCRIPT THEORIES

The confluence of computer availability and the growing dominance of cognitive approaches to psychology in the 1980s heralded a major change of direction in social psychological

aggression research. For the first time, researchers started to conceptualize the acquisition of social behavior in terms of computerlike processes –inputs, outputs, and the processing of information. Two key theories of aggression emerged –the Social Information Processing (SIP) theory of Dodge (1980)

and Script theory from Huesmann (1982). SIP theory emphasized the way people perceive the behavior of others and make attributions about their motives. A key construct in SIP theory is the hostile attributional bias –a tendency to interpret ambiguous events (such as being bumped in a corridor) as

being motivated by hostile intent. This bias has been extensively studied and has been found to reliably predict aggressive behavior. Script theory emphasizes the acquisition of scripts for behavior (much like an actor’s script) through either direct experience or observational learning. Once encoded in semantic memory, scripts define particular situations and provide a guide for how to behave in them. In script theory, a person faced with a particular situation first considers a script relevant to that situation, assumes a role in the script, assesses the appropriateness or likely outcome of enacting the script,

and if judged appropriate, then behaves according to the script. If a person habitually responds to conflict by using scripts that include behaving aggressively, these scripts may become more easily brought to mind (i.e., chronically accessible), become automatic, and generalize to other situations, increasing the likelihood of aggression in a growing number of spheres of life.

COGNITIVE NEOASSOCIATION THEORY

Cognitive Neoassociation Theory (CNA) reformulated the frustration-aggression hypothesis within the framework of emerging knowledge about neural connectivity. Assuming that concepts, emotions, memories, and action tendencies are interconnected within the brain’s associative neural network,

Berkowitz (1989) posited that aversive events such as frustrations, provocations, or unpleasant physical environments produce negative affect, which is neurally linked to various thoughts, feelings, and behavioral tendencies that are themselves linked to both fight and flight tendencies. Depending on the characteristics of the person and the situation, one response set will eventually dominate, with dominant ‘fight’ responses linked with anger and being more likely to elicit aggression. Importantly, higher-order processes such as making attributions about another’s motives or thinking through the consequences of an aggressive response may cause a person to moderate an aggressive impulse in this model.

CURRENT THEORIES

THE GENERAL AGGRESSION MODEL

The General Aggression Model (GAM; Anderson and Bushman, 2002) is the most recent and broadest theory of aggression processes to date. It is a biosocial-cognitive model designed to account for both short- and long-term (developmental) effects of an extensive range of variables on aggression. GAM can explain the widest range of aggressive behaviors, including those not based around aversive events or

negative affect. In addition, it is arguably the model that has the most empirical support. GAM unifies previous major models of aggression from the field of social psychology into a single framework, but also incorporates knowledge from other disciplines in psychology. The model itself is deceptively parsimonious. Every instance of aggression involves a person, with all their characteristics (e.g., biology, genes, personality, attitudes, beliefs, behavioral scripts), responding to an environmental trigger such as a provocation, an aversive event, or an aggression-related cue. These person and situation variables influence the person's present internal state—cognitions, affects, and physiological arousal. Depending on the nature of activated knowledge structures (which include affect), and on how aroused the person is, the person's immediate response may be an impulse to aggress. The person may act on this impulse, but if they have the time and cognitive resources to do so, and if the immediate response is undesirable, a period of appraisal and reappraisal will follow. Consequences are then thought-through, alternate responses considered, and a considered response made. The resulting behavioral action may or may not be aggressive, but in any case all actions feed back into the immediate situation and also influence the person's psychological make-up (i.e., their personality). Underlying the GAM are detailed

assumptions that take into account a myriad of within-person factors, a range of possible triggers for aggression, known internal psychological processes, and the means by which behavior is reinforced and learned. In terms of the latter, knowledge structures such as schemas (a grouping of knowledge, feelings, memories, perceptions and notions about typical behavior that is centered around a particular theme) and scripts (knowledge about how people typically behave in a given situation such as during conflict) are person factors that can not only impel a person to be aggressive in the moment, but also change to reflect our experiences (upper portion of Figure 1). Thus, experience leads to changes in the type, content, and accessibility of knowledge structures, which are seen as the basis of personality. Together, these features of GAM can be used to explain short- and long-term aggression across a range of forms and functions, including the three key dimensions already noted: degree of hostile/agitated affect; degree of automaticity versus conscious thought; and degree to which the goal is to harm the victim versus benefit the perpetrator. Phenomena as different as sexual and nonsexual aggression against women (e.g., Anderson and Anderson, 2008), personality effects on violent crime (Hosie et al., 2014), and dozens more are well explained by GAM.

taken into account in current social

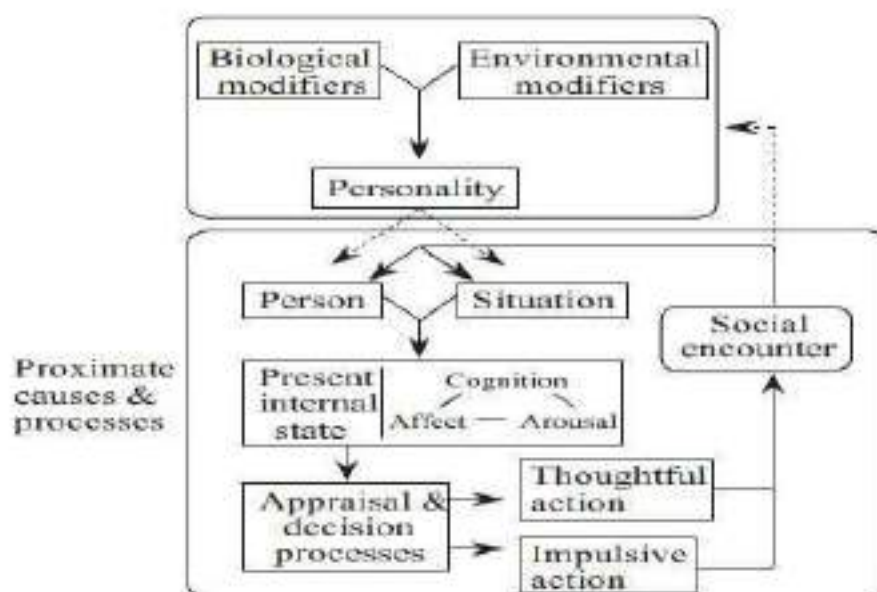


Figure 1 General Aggression Model. From Anderson, C.A., Anderson, K.B., 2008. Men who target women: specificity of target, generality of aggressive behavior. *Aggressive Behavior* 34, 605–622. Reprinted by permission.

RESEARCH FINDINGS: DETERMINANTS OF AGGRESSION

1. DEVELOPMENT AND STABILITY OVER TIME

Scholars studying social development have shown that the frequency of physical aggression typically peaks in the toddler years and then decreases across the life span. Importantly, the degree to which one person is aggressive relative to others of the same age is fairly stable across the life span. Aggressive children tend to become adolescents and adults who are more aggressive than their peers (Bushman and Huesmann, 2010).

2. PERSON FACTORS

Numerous factors in a person's make-up have been shown to increase the likelihood of aggressive behavior. Not all are studied directly in social psychology, but all are

psychological models of aggression.

3. GENDER DIFFERENCES IN AGGRESSION

Overall, males are generally more aggressive than females, and this applies from early in childhood through the life span. This is especially true for physical aggression and violent behavior, although women are as physically aggressive as men when strongly provoked. Men are more likely than women to use direct forms of aggression, but the reverse is true for women, who are more likely to use forms of indirect aggression, including relational aggression. Within intimate relationships, however, women are somewhat more likely to use physical aggression than men, though for different purposes and with different results. For example, men are much more likely to strike with a fist (women with an open slap), which is one reason why intimate

partner violence yields many more women requiring medical attention than men.

4. TRAIT ANGER

Trait anger reliably predicts an aggressive predisposition. It is characterized by extreme sensitivity to provocation and a considerably increased inclination to respond with aggression once provoked.

5. CALLOUS UNEMOTIONAL PERSONALITY TRAITS

There are three personality styles under this umbrella – psychopathy, Machiavellianism, and narcissism. All three are linked with high levels of aggression, lack of empathy, and curtailed emotional responding. Individuals of all three types routinely use aggression instrumentally to obtain desired goals, but narcissists and psychopaths are also prone to reactive aggression. Narcissists often respond aggressively when they feel threatened (particularly by insults, humiliations, or other threats to their inflated ego), or when they fear that their flaws may be exposed. Psychopaths, particularly those with secondary psychopathy characteristics, are often impulsive, fearless, and unconcerned about negative consequences to themselves or victims – a potent mix for a person already predisposed to aggression. Machiavellians most typically use instrumental aggression to achieve their goals and feel little or any remorse for harmful consequences to others. They do, however, consider potential consequences to themselves, and are thus more likely to aggress indirectly so that there is little likelihood of being held responsible for their actions.

6. IMPULSIVITY, EXECUTIVE CONTROL AND SELF-CONTROL

Impulsivity is a temperament variable often noticeable from early infancy, and is a reliable predictor of aggression, presumably because impulsive people have difficulty curbing aggressive impulses. In contrast, people are less aggressive if they have greater control over their emotions, greater self-control, and a stronger capacity to inhibit their impulses (Moffitt et al., 2011).

7. INTELLIGENCE

There is not a great deal of research on IQ and aggression, but some studies have found links between low IQ and higher levels of aggression in children, particularly in children with poor verbal intelligence and/or with low self-control.

8. PERSONALITY TRAITS – THE ‘BIG FIVE’

Research on the ‘Big Five’ personality traits and aggression has generally found that people low in agreeableness and high in neuroticism are more aggressive and violent. Furthermore, both of these dimensions are associated with aggressive emotions, and low agreeableness is also associated with greater aggressive thinking (Barlett and Anderson, 2012).

9. HORMONES

The hormone most consistently linked with aggression is testosterone. Males have around 10 times as much testosterone as females, and levels are much higher in older teenagers and young adults than in

older men. Interestingly, when people dominate others, their testosterone levels typically increase, along with their levels of aggression. There also is evidence that testosterone's effect on aggression is a by-product of its effect on dominance. There also may be links between low levels of estrogen and progesterone and aggression, although results are mixed. Finally, emerging evidence suggests that low levels of oxytocin may be linked with increased aggression.

GENETIC PREDISPOSITIONS

Although aggressive behavior has a considerable learned component, studies show that inherited characteristics account for perhaps a quarter to a third of an aggressive predisposition (Tuvblad et al., 2009). More than a dozen genetic markers have been linked with aggressive and antisocial behavior, although links are rarely direct. Typically, genetic predispositions more directly relate to temperament variables such as impulsivity, which are themselves linked with greater aggression. The two most widely studied genetic markers of aggression are a polymorphism in the promoter of the monoamine oxidase A gene (MAOA) and a variation in the 5-HT serotonin transporter gene. Crucially, in line with the emerging field of epigenetics, the MAOA gene polymorphism seems to interact with a child's early environment, so that aggression and antisocial behavior are most likely in those who have this genetic trait and also experience childhood maltreatment (Kim-Cohen et al., 2006).

FACTORS FROM THE ENVIRONMENT AND CUES FOR AGGRESSION

1. PROVOCATION

Perhaps the single greatest trigger for aggression is provocation by another person (Bettencourt et al., 2006). However, provocation does not need to be direct. People can be provoked to aggression by social exclusion, having rumors spread about them and a range of other 'indirect' provocations.

2. WEAPONS

Weapons are one stimulus that almost all people conceptually link with aggressive behavior. Research consistently shows a 'weapons effect' whereby people who view a real or virtual weapon tend to have aggression-related cognitions primed in semantic memory, and become more likely to behave aggressively. Interestingly, this effect varies by type of weapon and hunting experience (see Figure 2).

VIOLENT ENVIRONMENT

According to social cognitive models, people who are exposed to a lot of violence, virtual or real, will have an associative neural network with a lot of aggression-related knowledge structures, including aggressive behavioral scripts. This is borne out by research demonstrating that people from violent environments, whether homes, neighborhoods, or war-torn countries, have a greater predisposition to be aggressive (e.g., Aguilar et al., 2000).

VIOLENT MEDIA

The same principle applies to exposure to violent media. It is one of the most studied

phenomena by social psychologists, and several hundred studies converge across all major research methodologies in finding that violent media exposure increases the likelihood of aggressive behavior and causes desensitization to violence in both the short- and long-term (Warburton, 2014). In addition, greater exposure to media violence has been linked to hostile biases in thinking, increases in aggressive thoughts and feelings, and decreases in empathy and prosocial behavior (see Anderson et al., 2003; Krahe et al., 2012, for reviews).

5. ENVIRONMENTAL STRESSORS

A variety of environmental stressors can increase the tendency to aggress. The most notable are physical pain, bad-smelling odors, loud or aversive noises, and hot temperatures. Importantly, it seems that aggression is most likely when the individual has no control over those environmental stressors.

6. ANONYMITY

Anonymity in some circumstances increases the likelihood of aggressive behavior. It is much easier to hurt another if an individual believes there will be no consequences, and anonymity allows a person to experience ‘deindividuation’—a lessening of the restraints on antisocial behavior normally accorded to people perceived as being ‘individuals.’

7. SOCIAL REJECTION

Humans have a fundamental need to feel socially included and to have supportive and enduring relationships. When this need is thwarted through social exclusion or rejection, people sometimes behave

more prosocially to facilitate reinclusion. However, the dominant response to such rejection is to aggress, especially when the person can do so without significant social reprisals (e.g., Warburton et al., 2006).

8. SUBSTANCES

Alcohol intoxication consistently causes both men and women to behave more aggressively inside and outside the laboratory, and is linked with a substantial proportion of murders, assaults, rapes, and incidents of intimate partner violence. Importantly, this increase is due to the aggressor experiencing a diminished ability to inhibit their aggressive impulses. Thus people who are predisposed to behave aggressively are most affected (Giancola, 2000). Aggression has also been linked with other substances that cause disinhibition and/or an increase in physiological arousal, such as stimulants, amphetamines, and methamphetamines.

RESEARCH FINDINGS: FACTORS THAT MEDIATE AGGRESSION

The previous section examined factors within the person and in the environment that can trigger or increase the likelihood of aggression. This section deals with the three key types of internal processes noted in GAM that can increase or decrease the likelihood of aggression.

1. EMOTION/AFFECT

Early models of aggression placed considerable emphasis on the role of negative emotions in causing aggressive behavior, and research has linked several emotions to an increased likelihood of aggression, most notably anger, shame,

jealousy, and frustration. Of these, anger is the most researched. Although anger can precede aggression, the pathway is far from a simple cause and effect. Anger increases aggression primarily through reducing inhibitions, narrowing attentional focus to cues for aggression, and alerting people to cues for potential threats (see Anderson and Bushman, 2002). Shame has also been linked to increases in aggression, primarily when the shamed person feels their personal flaws have been exposed. Jealousy has also been linked with aggression and in particular with intimate partner violence. Recent research suggests that the anticipation of how one will feel in the future can be as important as how one currently feels in determining whether a person will be aggressive. It should be noted that some emotions can be a protective factor for aggression. For example, empathy (taking another person's perspective and having concern for them) is consistently related to lower aggression.

2. COGNITION

Recent models of aggression have focused increasingly on the cognitions that may underlie aggression. These include attitudes, beliefs, expectations, perceptions, ideas, and concepts as well as aggregated cognitions such as schemas and scripts. It is clear that a variety of external triggers can increase the accessibility of aggressive cognitions in semantic memory. These cognitions may be activated but below the threshold of awareness, or activated to the point of conscious awareness. In either case they can elicit an aggressive behavioral tendency through the priming of aggression-related action-tendencies, the activation of aggressive scripts for behavior, or through the

influence of hostile biases on the interpretation of cues from the environment.

3. AROUSAL

Physiological arousal and emotional arousal are both linked with increased aggression, and this is true regardless of what caused the arousal in the first instance. This may be due to one or more of several factors. First, arousal increases the likelihood that a person will act on an aggressive action tendency or impulse rather than think through the consequences of an aggressive action. Second, excessive levels of

arousal feel unpleasant, and can elicit aggression in the same way as other unpleasant experiences. Third, arousal may be part of a fight or flight response system that bypasses rational thought and impels aggressive action. Fourth, arousal may be cognitively labeled as resulting from anger, thus causing the

person to feel and act angry. Excitation transfer may compound this effect, leading to a disproportionately aggressive response. Finally, low levels of arousal may facilitate aggression if people lack the energy and motivation to inhibit aggressive impulses.

PERSUASION

WHAT PATHS LEAD TO PERSUASION?

Sometimes persuasion occurs as people focus on arguments and respond with favorable thoughts.

Such systematic, or "central route," persuasion occurs when people are naturally analytical or involved in the issue.

- When issues don't engage systematic thinking, persuasion may occur through a faster, "peripheral route," as people use heuristics or incidental cues to make snap judgments.
- Central route persuasion, being more thoughtful and less superficial, is more durable and more likely to influence behavior.

WHAT ARE THE ELEMENTS OF PERSUASION?

Among the ingredients of persuasion explored by social psychologists are these four:

**the communicator,
the message,
how the message is communicated,
and
the audience.**

In other words, who says what, by what method, to whom? How do these factors affect the likelihood that we will take either the central or the peripheral route to persuasion?

WHAT MAKES PERSUASION EFFECTIVE?

Researchers have explored four factors: the communicator (who says it), the message (what is said), the channel (how it is said), and the audience (to whom it is said).

Credible communicators have the best success in persuading. People who speak unhesitatingly, who talk fast, and who look listeners straight in the eye seem more credible. So are people who argue against their own self-interest. An attractive communicator also is effective on matters of taste and personal values.

- The message itself persuades; associating it with good feelings makes it more convincing. People often make quicker, less reflective judgments while in good moods. Fear-arousing messages can also be effective, especially if the recipients feel vulnerable but can take protective action.

How discrepant a message should be from an audience's existing opinions depends on the communicator's credibility. And whether a one- or two-sided message is more persuasive depends on whether the audience already agrees with the message, is unaware of opposing arguments, and is unlikely later to consider the opposition.

When two sides of an issue are included, the primacy effect often makes the first message more persuasive. If a time gap separates the presentations, the more likely result will be a recency effect in which the second message prevails.

Another important consideration is how the message is communicated. Usually, face-to-face appeals work best. Print media can be effective for complex messages. And the mass media can be effective when the issue is minor or unfamiliar, and when the media reach opinion leaders.

Finally, it matters who receives the message. The age of the audience makes a difference; young people's attitudes are more subject to change. What does the audience think while receiving a message? Do they think favourable thoughts? Do they counter argue? Were they forewarned?

HOW DO PEOPLE RESIST PERSUASION?

A prior public commitment to one's own position, stimulated perhaps by a mild attack on the position, breeds resistance to later persuasion.

A mild attack can also serve as an **inoculation**, stimulating one to develop counterarguments that will then be available if and when a strong attack comes.

This implies, paradoxically, that one way to strengthen existing attitudes is to challenge them, though the challenge must not be so strong as to overwhelm them.

LEADERSHIP STYLE AND EFFECTIVENESS.

THE ROLE OF LEADERSHIP IN GROUP SETTINGS

LEADERSHIP—the very word conjures up images of heroic figures leading their followers toward something better: victory, prosperity, or social justice.

Leadership involves influence—influencing others in a group by establishing a direction for collective effort and then encouraging the activities needed to move in that direction. Consistent with that definition, being a leader involves exerting influence—changing the behavior and thoughts of other members of the group so that they work together to attain the group's common goals. We consider three key aspects of the findings of research on this topic in terms of (1) why some individuals, but not others, become leaders; (2) when non-traditional leaders are most likely to emerge; and (3) how leaders influence group members' satisfaction with their performance.

To explain which individual from within those categories becomes a leader, early researchers formulated the great person theory of leadership—the view that great leaders possess certain traits that set them apart from other human beings—traits that differentiate them from those who are merely followers.

AUTOCRATIC LEADERSHIP

Autocratic leadership style is centered on the boss. In this leadership the leader holds all authority and responsibility. In this leadership, leaders make decisions on their own without consulting subordinates. They reach decisions, communicate them to subordinates and expect prompt implementation. Autocratic work environment does normally have little or no flexibility. In this kind of leadership, guidelines, procedures and policies are all-natural additions of an autocratic leader. Statistically, there are very few situations that can actually support autocratic leadership. Some of the leaders that support this kind of leadership include: Albert J Dunlap (Sunbeam Corporation) and Donald Trump (Trump Organization) among others.

DEMOCRATIC LEADERSHIP

In this leadership style, subordinates are involved in making decisions. Unlike autocratic, this leadership is centered on subordinates' contributions. The democratic leader holds final responsibility, but he or she is known to delegate authority to other people, who determine work projects. The most unique feature of this leadership is that communication is active upward and downward. With respect to statistics, democratic leadership is one of the most preferred leadership, and it entails the

following: fairness, competence, creativity, courage, intelligence and honesty.

■ **STRATEGIC LEADERSHIP STYLE**

Strategic leadership is one that involves a leader who is essentially the head of an organization. The strategic leader is not limited to those at the top of the organization. It is geared to a wider audience at all levels who want to create a high performance life, team or organization. The strategic leader fills the gap between the need for new possibility and the need for practicality by providing a prescriptive set of habits. An effective strategic leadership delivers the goods in terms of what an organization naturally expects from its leadership in times of change. 55% of this leadership normally involves strategic thinking.

TRANSFORMATIONAL LEADERSHIP

Unlike other leadership styles, transformational leadership is all about initiating change in organizations, groups, oneself and others. Transformational leaders motivate others to do more than they originally intended and often even more than they thought possible. They set more challenging expectations and typically achieve higher performance. Statistically, transformational leadership tends to have more committed and satisfied followers. This is mainly so because transformational leaders empower followers.

TEAM LEADERSHIP

Team leadership involves the creation of a vivid picture of its future, where it is heading and what it will stand for. The

vision inspires and provides a strong sense of purpose and direction. Team leadership is about working with the hearts and minds of all those involved. It also recognizes that teamwork may not always involve trusting cooperative relationships. The most challenging aspect of this leadership is whether or not it will succeed. According to Harvard Business Review, team leadership may fail because of poor leadership qualities.

CROSS-CULTURAL LEADERSHIP

This form of leadership normally exists where there are various cultures in the society. This leadership has also industrialized as a way to recognize front runners who work in the contemporary globalized market. Organizations, particularly international ones require leaders who can effectively adjust their leadership to work in different environs. Most of the leaderships observed in the United States are cross-cultural because of the different cultures that live and work there.

FACILITATIVE LEADERSHIP

Facilitative leadership is too dependent on measurements and outcomes – not a skill, although it takes much skill to master. The effectiveness of a group is directly related to the efficacy of its process. If the group is high functioning, the facilitative leader uses a light hand on the process. On the other hand, if the group is low functioning, the facilitative leader will be more directives in helping the group run its process. An effective facilitative leadership involves monitoring of group dynamics, offering process suggestions and interventions to help the group stay on track.

LAISSEZ-FAIRE LEADERSHIP

Laissez-faire leadership gives authority to employees. According to azcentral, departments or subordinates are allowed to work as they choose with minimal or no interference. According to research, this kind of leadership has been consistently found to be the least satisfying and least effective management style.

TRANSACTIONAL LEADERSHIP

This is a leadership that maintains or continues the status quo. It is also the leadership that involves an exchange process, whereby followers get immediate, tangible rewards for carrying out the leader's orders. Transactional leadership can sound rather basic, with its focus on exchange. Being clear, focusing on expectations, giving feedback are all important leadership skills. According to Boundless.com, transactional leadership behaviors can include: clarifying what is expected of followers' performance; explaining how to meet such expectations; and allocating rewards that are contingent on meeting objectives.

COACHING LEADERSHIP

Coaching leadership involves teaching and supervising followers. A coaching leader is highly operational in setting where results/performance require improvement. Basically, in this kind of leadership, followers are helped to improve their skills. Coaching leadership does the following: motivates followers, inspires followers and encourages followers.

CHARISMATIC LEADERSHIP

In this leadership, the charismatic leader manifests his or her revolutionary power. Charisma does not mean sheer behavioural change. It actually involves a

transformation of followers' values and beliefs. Therefore, this distinguishes a charismatic leader from a simply populist leader who may affect attitudes towards specific objects, but who is not prepared as the charismatic leader is, to transform the underlying normative orientation that structures specific attitudes.

VISIONARY LEADERSHIP

This form of leadership involves leaders who recognize that the methods, steps and processes of leadership are all obtained with and through people. Most great and successful leaders have the aspects of vision in them. However, those who are highly visionary are the ones considered to be exhibiting visionary leadership. Outstanding leaders will always transform their visions into realities.

THEORIES OF INTERGROUP RELATIONS [SOCIAL IDENTITY THEORY, RELATIVE DEPRIVATION THEORY, REALISTIC CONFLICT THEORY, BALANCE THEORIES, EQUITY THEORY, SOCIAL EXCHANGE THEORY]

SOCIAL IDENTITY THEORY

Henri Tajfel's greatest contribution to psychology was social identity theory.

Social identity is a person's sense of who they are based on their group memberships. Tajfel (1979) proposed that the groups (e.g. social class, family, football team etc.) which people belonged to were an important source of pride and self-esteem. Groups give us a sense of social identity: a sense of belonging to the

social world. In order to increase our self-image, we enhance the status of the group to which we belong. For example, England is the best country in the world! We can also increase our self-image by discriminating and holding prejudice views against the out group (the group we don't belong to). For example, the Americans, French etc. are a bunch of losers!

Therefore, we divided the world into “them” and “us” based through a process of social categorization (i.e. we put people into social groups).

This is known as in-group (us) and out-group (them). Social identity theory states that the in-group will discriminate against the out-group to enhance their self-image.

Henri Tajfel proposed that stereotyping (i.e. putting people into groups and categories) is based on a normal cognitive process: the tendency to group things together. In doing so we tend to exaggerate:

The differences between groups

The similarities of things in the same group.

We categorize people in the same way. We see the group to which we belong (the in-group) as being different from the others (the out-group), and members of the same group as being more similar than they are. Social categorization is one explanation for prejudice attitudes (i.e. “them” and “us” mentality) which leads to in-groups and out-groups.

THE OUTLINE

Tajfel and Turner (1979) proposed that there are three mental processes involved in evaluating others as “us” or “them” (i.e.

“in-group” and “out-group”. These take place in a particular order.

The first is **categorization**. We categorize objects in order to understand them and identify them. In a very similar way we categorize people (including ourselves) in order to understand the social environment. We use social categories like black, white, Australian, Christian, Muslim, student, and bus driver because they are useful.

If we can assign people to a category then that tells us things about those people, and as we saw with the bus driver example, we couldn't function in a normal manner without using these categories; i.e. in the context of the bus.

Similarly, we find out things about ourselves by knowing what categories we belong to. We define appropriate behavior by reference to the norms of groups we belong to, but you can only do this if you can tell who belongs to your group. An individual can belong to many different groups.

In the second stage, **social identification**, we adopt the identity of the group we have categorized ourselves as belonging to. If for example you have categorized yourself as a student, the chances are you will adopt the identity of a student and begin to act in the ways you believe students act (and conform to the norms of the group). There will be an emotional significance to your identification with a group, and your self-esteem will become bound up with group membership. The final stage is **social comparison**. Once we have categorized ourselves as part of a group and have identified with that group, we then tend to compare that group with other groups. If our self-esteem is to be maintained our

group needs to compare favorably with other groups.

This is critical to understanding prejudice, because once two groups identify themselves as rivals, they are forced to compete in order for the members to maintain their self-esteem. Competition and hostility between groups is thus not only a matter of competing for resources (like in Sherif's Robbers Cave) like jobs but also the result of competing identities.

METHODOLOGY OF THE MINIMAL GROUP PARADIGM

The **minimal group paradigm** is the typical design used in experiments that inspired and support SIT. The basic idea is that participants (adults and children have been used in studies) are randomly divided into groups. They are then asked to award rewards, prizes or even money to other participants in specially designed booklets. The recipients are anonymous, except for a number and which group they are in (e.g. Member #28, Group X; Member #3, Group Y).

Originally, the Tajfel and Turner hypothesized that they would have to gradually increase the similarities between group members before they would observe in-group bias (e.g. positive distinctiveness). They were surprised to learn that even when groups were formed using complete arbitrary criteria, such as flipping a coin, they demonstrated in-group bias. Even when they were directly informed that the groups were meaningless, they still were biased to their in-group. This initial discovery is what led to further development and elaboration of the SIT; they concluded that the mere existence of an out-group was

enough for social comparison and in-group bias to occur.

The findings, from numerous studies, show that the in-group will act favorably towards members of their own in-group. Moreover, they will even sacrifice rewards for themselves to increase the difference in rewards given between the in-group members and the out-group members.

RELATIVE DEPRIVATION THEORY

In sociology, relative deprivation theory is a view of social change and movements, according to which people take action for social change in order to acquire something (for example, opportunities, status, or wealth) that others possess and which they believe they should have, too. Some sociologists believe relative deprivation theory explains why people join social movements or advocate social change. For example, in this view, gay people join the movement for gay marriage in order to acquire something (the right to marry) they believe others already possess; relative to these people, such advocates of gay marriage believe they are deprived. Critics claim that relative deprivation theory does not explain why some people join movements that apparently do not benefit them directly (animal rights movements, say)

REALISTIC GROUP CONFLICT THEORY CLASSIC STUDY

Muzafer Sherif's Robbers Cave experiment is a demonstration of this theory. Sherif is credited as one of the most important social psychologists of his time. With his colleagues, he set up a 2-week experiment involving White, middle-class, 12-year-old boys at a summer camp.

At first, the boys interacted only with their own group members because Sherif wanted them to develop a sense of group identity. The boys did develop a group identity and called themselves the Eagles or the Rattlers. In the second phase of the study, the boys were introduced to the other group and were required to engage in a series of competitive activities. Rewards and prizes were handed out to the winning team. Sherif and his colleagues purposely set up these games and rewards so that the boys would have reason to compete intensely. During these fierce competitions, both groups became suspicious of and hostile toward one another. As tensions increased, the boys demonstrated allegiance to their group by discouraging one another from establishing friendships across group lines. No one wanted to be seen as a traitor, so the boys stuck to their own groups. Hostility increased to the point that physical fights and acts of vandalism broke out. Despite direct interventions by adults, the two groups could not seem to reconcile.

Unity was restored only when Sherif and colleagues created situations requiring both groups of boys to depend on each other to achieve important goals equally valued by both groups. In other words, harmony was restored when both groups were equally invested in achieving a goal that required everyone's help and cooperation. For example, Sherif set up a situation in which a truck carrying their food supply broke down and the help of all the boys was needed to bring the food to camp. After completing a series of such tasks requiring interaction and everyone's involvement, positive behavior toward the other group members increased. The boys

began to behave more like individuals rather than group members and formed friendships across group lines. Psychologically, they began as two distinct groups, but when the perception of threat was replaced by cooperation and interdependence, the groups re-established themselves as one large group. Therefore, the group distinctions made between Eagles and Rattlers disappeared and everyone felt as if they belonged to the same group.

BALANCE THEORY

It was proposed independently by **Newcomb (1961)** and by **Heider (1958)**.

This framework suggests that people naturally organize their likes and dislikes in a symmetrical way (Hummert, Crockett, & Kemper, 1990). When two people like each other and discover that they are similar in some specific respect, this constitutes a state of balance, and balance is emotionally pleasant. When two people like each other and find out that they are dissimilar in some specific respect, the result is imbalance. Imbalance is emotionally unpleasant, causing the individuals to strive to restore balance by inducing one of them to change and thus create similarity, by misperceiving the dissimilarity, or simply by deciding to dislike one another. Whenever two people dislike one another, their relationship involves non-balance. This is not especially pleasant or unpleasant because each individual is indifferent about the other person's similarities or dissimilarities.

EQUITY THEORY

It focuses on determining whether the distribution of resources is fair to both relational partners. Equity is measured by

comparing the ratio of contributions (or costs) and benefits (or rewards) for each person. Considered one of the justice theories, equity theory was first developed in the 1960s by J. Stacy Adams, a workplace and behavioral psychologist, who asserted that employees seek to maintain equity between the inputs that they bring to a job and the outcomes that they receive from it against the perceived inputs and outcomes of others (Adams, 1963). According to Equity Theory, in order to maximize individuals' rewards, we tend to create systems where resources can be fairly divided amongst members of a group. Inequalities in relationships will cause those within it to be unhappy to a degree proportional to the amount of inequality. The belief is that people value fair treatment which causes them to be motivated to keep the fairness maintained within the relationships of their co-workers and the organization. The structure of equity in the workplace is based on the ratio of inputs to outcomes. Inputs are the contributions made by the employee for the organization.

SOCIAL EXCHANGE THEORY

Social exchange theory is based on the idea that people seek to maximize rewards and minimize costs in any given social relationship. Rewards can consist of anything tangible or intangible that an individual considers valuable. For instance, business relationships may provide several concrete benefits, such as income or material goods, in addition to several more abstract benefits, such as prestige and a sense of security. Costs include anything that an individual considers to be unrewarding or sees as

requiring a significant amount of time or effort. For example, romantic relationships may involve costs such as shared housework and spending vacations with one's in-laws (which, for some people, can be extremely unpleasant). Of course, the evaluation of rewards and costs is highly subjective because that which is rewarding for one individual might not be quite as rewarding for another person. Similarly, that which is considered rewarding in one relationship might not be perceived as rewarding in a different social involvement.

People's evaluations of perceived rewards and costs influence how satisfied they are with their relationships and the relative stability of those relationships. Satisfaction with a relationship is determined by considering one's outcome comparison level (i.e., the standard by which one judges his or her current relationship's outcomes). For instance, a person may compare his or her current outcomes with those he or she has received in a past relationship of a similar type. So, you might compare how things are going now with your current boyfriend or girlfriend with how things went with past romantic partners. To the extent that a person's current outcomes exceed his or her previous outcomes, the person is satisfied with a relationship and desires it to continue. However, if a person's current outcomes don't compare favorably to his or her previous outcomes, the person becomes dissatisfied and is less likely to work at furthering the relationship. People compare their current outcomes not only to past outcomes but also to those that they could be receiving now in other potential relationships (referred to as the comparison level for alternatives). To the

extent that the outcomes people perceive as possible within an alternative relationship are better than those that they are receiving in their current relationship, they are less likely to continue in the current relationship.

Reward-to-cost ratios and comparison levels are subject to change over time, as individuals continually take stock of what they have gained and lost in their relationships. This implies that relationships that a person found satisfying at one point in time may become dissatisfying later because of changes in perceived rewards and costs. This may occur because certain factors may become less rewarding or costlier over time. For instance, sex may be extremely rewarding for members of a newly married couple but may become less so as passion and spontaneity decrease over the years.

Finally, people's perceptions of their relationships also depend on whether the exchanges that occur are viewed as equitable. Equitable or fair exchanges are necessary to avoid conflict between relationship partners. For instance, assume that there is favorable exchange for all parties involved in an ongoing relationship, but one party is receiving substantially greater benefits than the other. Such a scenario may be perceived as unfair because distributive justice is not present (i.e., outcomes are being distributed unequally). In this case, individuals with worse outcomes may feel exploited and have negative feelings about their exchange partner, which may ultimately affect how committed they are to continuing the relationship.

APPLIED SOCIAL PSYCHOLOGY: ENVIRONMENT AND LAW; PERSONAL SPACE, CROWDING, AND TERRITORIALITY.

ENVIRONMENT

The first topic is **resource dilemmas**, which are sometimes called *commons dilemmas*. These are situations in which individuals must choose between self-interest (taking or using unsustainable amounts of a natural resource such as water or fish) and the interests of the community or environment (taking a sustainable share, or less, of the resource). Which social factors do you think might come into play as individuals make these decisions? Given that no one person is likely to be given control of an entire water supply, fishing grounds, or the climate, how might you conduct some research to understand what causes greed or cooperation in these situations?

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likely to be given control of an entire water supply, fishing grounds, or the climate, how might you conduct some research to understand what causes greed or cooperation in these situations?

Defensible space represents a way of fighting crime through careful arrangement of the physical aspects of communities, retail buildings, and residences. The way in which a building or community is designed can encourage or discourage burglars, robbers, and vandals. What could those design factors be? How could social psychology be a part of this kind of research?

Social dilemmas are a group of situations in which individuals face important choices. Sometimes individuals do not realize how important their choices are—or even that they are making choices—but that is a separate problem. In social dilemmas, the rewards to the individual for noncooperation are greater than the rewards for cooperation no matter what others do; however, if most individuals involved fail to cooperate, then everyone receives lower rewards (Dawes, 1980).

Public goods problems involve dilemmas about whether to contribute (e.g., time, effort, money) to a project that would benefit everyone when such a contribution is voluntary. For example, one may decide to help (or not help) build a neighborhood children's playground. The dilemma is that contributing costs something (in this case, one's money or time), but if not enough others contribute, the playground project will not be successful.

In a limited common consisting of some desirable resource, individuals acting in

self-interest might lead to a process called the **tragedy of the commons**, which occurs when “each [person] is locked into a system that compels him to increase his [harvesting] without limit—in a world that is limited. Ruin is the destination toward which all [persons] rush, each pursuing his own best interest” (Hardin, 1968, p. 1244). **Hardin's** (1968) article in the journal *Science* on the ultimate consequences of resource dilemmas has been very influential. He concluded that commons dilemmas probably would be fatal to the entire planet eventually. In terms of one enormous environmental-social problem, climate change, many tendencies to *not* take appropriate action have been identified. These “**dragons of inaction**” (Gifford, 2011) include over 30 “species” in seven “genera” – limited cognition, certain ideologies, social norms and comparison, discredence (mistrust of experts), perceived risks (of changing one's behavior), sunk costs (e.g., investments in resource extraction), and limited behavior (e.g., “I recycle, so I have done enough”).

Grounded theory analysis is a sophisticated way of learning how people think about particular issues by asking them what they are doing, and why, as they are considering something such as making a decision about using a resource. It uses the **think-aloud procedure**, in which individuals explain their decisions moment by moment as they make them. This procedure enables the researcher to track online cognitive processing as it naturally occurs. As noted earlier, in most resource dilemma studies, researchers simply impose a set of conditions on the harvesters and then observe their resultant decisions. This tells us little about exactly

how the decision makers arrived at their decisions because it ignores what goes on in the “black boxes” of the people’s minds.

Getting people to reduce household energy usage. A third major environmental issue is household energy use, which is a major contributor to the steady increases in greenhouse gas emissions. In 2008, households accounted for about 21% of U.S. carbon dioxide emissions (U.S. Department of Energy, 2009). Therefore, effective interventions aimed at lowering household energy usage can help to ameliorate the negative impact of households on the environment. The Internet has been used as an intervention tool to encourage consumers to use less energy. It was used by Abrahamse, Steg, Vlek, and Rothengatter (2007) in the Netherlands to encourage households to reduce their use of direct energy (gas, electricity, and fuel) and indirect energy (that which is used to produce, transport, and dispose of consumer goods). A combination of three intervention strategies was employed: (a) a list of energy-saving measures with potential savings tailored to each household, (b) a goal-setting request to reduce energy consumption by 5% over five months, and

customized feedback about changes in energy use and amount of money saved. The 5% goal was achieved. Households consumed 5.1% less energy, whereas a control group had a slight increase in energy consumption.

Schultz and colleagues reasoned that the occurrence of unintended **boomerang effects** may account for some of the disappointing results. They suggested that the normative information provided might act as a “magnet” for people who engage

in more of the undesirable behavior than the norm *and* also for those who engage in less of the undesirable behavior than the norm. Thus, in shifting their behavior toward greater conformity with the norm, the former individuals will show reductions in the undesirable behavior, whereas, contrary to the objectives of the intervention, the latter individuals will show increases in the undesirable behavior (i.e., the boomerang effect).

Schultz and colleagues suggested the possible importance of a second type of norm, an **injunctive norm**, that is, a norm that communicates/defines what is the culturally appropriate and approved behavior. The researchers hypothesized that introducing an injunctive normative message along with a descriptive normative message will prevent the occurrence of a boomerang effect. That is, individuals whose levels of undesirable behavior are already relatively low will be made aware of this fact (by the descriptive norm) but will be less tempted to shift toward the norm because at the same time they will receive a message that approves of their current behavior.

Have you ever had to study, live, or work in a school, home, or workplace that just did not work well and foster the types of behavior the situation called for? Certainly, some parts of the built environment need much improvement. One well-known example is a large apartment complex in St. Louis, Missouri, that was completed in 1954. The Pruitt–Igoe project was designed with the admirable intention of replacing deteriorating inner-city housing. The design for this complex, which contained 43 eleven-story buildings to house 12,000

people, was praised in an architectural journal for having vandal-resistant features, individualistic design, and no wasted space (“Slum Surgery in St. Louis,” 1951). **The Pruitt–Igoe design** saved space in part by having elevators stop only at every third floor, so that most residents would walk up or down one flight of stairs to their apartments. Pruitt–Igoe cost much less per unit than did comparable buildings. The design changes were considered so admirable that the architect even applied for a patent on the design.

But problems appeared soon after Pruitt–Igoe opened. The failure to carefully examine its design in relation to human social behavior contributed to high rates of fear, vandalism, serious crime, and vacancy. A particular problem was crime in the stairwells that residents were forced to use caused by the “innovative” elevator savings plan. The situation was so bad that, after only 18 years, the city began to demolish the entire complex. In this example, insufficient consideration of how the physical structure would influence social behavior led to the ultimate failure of the project. Whether the architect ever received his patent is unknown.

Pruitt–Igoe is the most dramatic example of building design failure, but many other buildings also pose problems for their users. Take a look at Figure 13.3 for a different example of architecture that fails to suit human needs. *Hard architecture* is aimed at preventing vandalism but goes so far toward that goal that it is uncomfortable and, therefore, rarely used.

However, when architects and social designers do collaborate, they begin to think of architecture as **placemaking**, that

is, real people imagined in real spaces (Schneekloth&Shibley, 1993). To “make a place,” architects and social designers work together to create an “envelope for behavior,” meaning that they think mainly about what people actually do in a building rather than think of the building mainly as a sculptural object without much regard for the people who will be using it.

Social design is not *always* needed in the design process. It is not required, for example, in times, places, and cultures where buildings are constructed by small communities in which everyone works together in accordance with a time-tested architectural tradition. These traditions, called **preindustrial vernacular** (Rapoport, 1969), evolved an architecture that already quite well fits community and cultural norms, individual interests, local climate, geography, and building materials. When community members are both builders and occupants, the design process does not need separate financiers, architects, boards of directors, and construction firms (for an example of vernacular architecture, see Figure 13.4).

SIX GOALS OF SOCIAL DESIGN. Social design researchers and practitioners have six main goals, with some being broader than others and some overlapping with others (Steele, 1973):

- Create physical settings that match the needs and activities of their occupants. This goal is probably the most important one of all.

- Satisfy building users. Occupant satisfaction is important because occupants must spend significant parts of their lives working, residing, or relaxing in the setting.

Change behavior. Such changes might include increasing office worker productivity, enhancing social ties among institutionalized elderly people, reducing aggression in a prison, or increasing communication among managers in an administrative office. As we will see, the behavior change goal can be both difficult to attain and controversial.

Enhance the building users' personal control (Holahan, 1983). The more building users are able to alter the setting to make it suit their needs, the less stressful that setting will be.

Facilitate social support (Holahan, 1983). Designs that encourage cooperation, assistance, and support are desirable primarily for building occupants who are disadvantaged in one way or another but also for active and successful individuals.

Employ "imageability." This refers to the ability of the building to help occupants, and (especially) visitors and newcomers, to find their way around without getting lost or confused.

MATCHING.How well the occupants' activities and needs are met by the setting is called **matching**. An example of poor matching might be a gymnasium when it is used for a concert. It is done, but gyms are not very well suited to that task. Ideally, of course, buildings should match their occupants' needs and behaviors perfectly. However, whether the degree of match is high or low sometimes depends in part on whose viewpoint is considered (Michelson, 1976). For some, the gym

might seem to be a fine place to stage a concert, but for concert-goers and bands, the acoustics and general aesthetics will often be quite inappropriate for a great musical experience.

The early personality theorist Henry Murray and his collaborators distinguished between two forms of **press**, which refers to properties or characteristics of environmental features that shape behavior (Murray, 1938). **Alpha press** refers to actual reality that can be assessed through objective inquiry. **Beta press** refers to people's interpretation of external reality. For example, a person may act toward a conversation partner in an objectively neutral fashion (alpha press) but be perceived by the partner as aggressive (beta press).

Alpha matching, or **congruence**, refers to how well the setting fits the person from an objective point of view. For example, there is a good (objective) height for kitchen counters for persons of different heights. Beta matching, or **habitability**, is "environmental quality as perceived by occupants of buildings or facilities" (Preiser & Taylor, 1983, p. 6). Some kitchen workers might not think that a certain counter height is good for them, even if experts claim that the existing counter height is correct.

SOCIAL SUPPORT. Personal control is an individual phenomenon, whereas social support is a group phenomenon. **Social support** is a process in which a person receives caring, kind words, and helpfulness from those around him or her. Many social problems would be eased if more and better social support were available. Common psychological

problems, such as depression and anxiety, have been shown to increase when social support is absent or inadequate. Social support may be seen as an antistress process (Moos, 1981).

What can social design do to facilitate social support? On a small scale, furniture can be arranged in a sociopetal fashion instead of a sociofugal fashion. **Sociopetal arrangements** are those that encourage social interaction (e.g., when people sit facing each other), whereas **sociofugal arrangements** discourage social interaction (e.g., when people sit in rows or even facing away from one another; Mehrabian & Diamond, 1971). At the building level, open-space areas may be arranged to facilitate social interaction (Holahan, 1972). Of course, if the personal control goal, as well as the social support goal, is to be met, the increased social interaction must be controllable; occupants should be able to find social interaction when and if they want it but should not be faced with unwanted social encounters.

In office buildings, social support may be fostered through the provision of high-quality lounge space for employees. The mere existence of such space does not guarantee that valuable social support will be available, but with inadequate space for employees to share coffee and conversation, the likelihood of supportive social networks declines.

Finally, in some cases, social support may result from a design that provides optimal privacy (being able to filter one's interactions). Consider shelters for victims of domestic violence. A study of alternative designs for such shelters showed that designs characterized by

anonymity and safety were most preferred (Refuerzo&Verderber, 1990). Sometimes social support is maximized when a person simultaneously can be near a helper and far from an abuser. The difference in helpfulness and caring is especially large when the contrast is between a residence that is full of hostility and violence and one that is dominated by caring and understanding.

OUTDOOR SPACES.

Many of the same social design ideas apply to outdoor public areas such as plazas, parks, and streets. In one of the most widely used changes wrought by environmental psychology principles, the very fabric of many cities has been changed by a concept called *density bemusing*, which can be traced to the pioneering work of William Whyte (1980). Recognizing the need for some open space in the city core in 1961, the City of New York offered developers a deal: For every square foot of plaza they included in a new project, their new building could exceed normal zoning restrictions by 10 square feet. Developers liked the idea, and this deal certainly increased New York City's supply of open space downtown

IMAGEABILITY.

Buildings should be **imageable** (i.e., clearly understandable or legible) to the people who use them (Hunt, 1985). When you walk into a building, you should immediately be able to find your way around or, in more technical terms, be capable of *purposeful mobility*. In simple terms, you should not be confused.

Too often, a person enters a building that is unfamiliar and is unable to figure out where to go next. Unless we realize that buildings should be imageable, there is a

tendency to blame ourselves (e.g., “I never did have a good sense of direction”). Sometimes observation reveals that you are not the first to have problems. Perhaps you have seen handmade signs that occupants have made to be helpful and/or to save themselves from answering the same question about where such-and-such is “for the hundredth time.” Such signs represent a failure to make the building imageable, either through good signage or through good and legible design of the building itself.

DEFENSIBLE SPACE

As noted previously, in Baltimore, the use of speed bumps and barriers has helped to promote feelings of safety among residents of neighborhoods. How might the physical setting influence the actual likelihood of crime? Most evidence bearing on this question has emerged from the observations and ideas of Jane Jacobs and Oscar Newman that led to **defensible space theory, which** deals with both crime and the fear of crime (Jacobs, 1961; Newman, 1972). This theory proposes that certain design features will increase residents’ sense of security and decrease crime in the territory. Some of the features include the use of real or symbolic barriers to separate public territory from private territory and the provision of opportunities for territory owners to observe suspicious activity in their spaces (surveillance).

Quite a number of field studies have tested defensible space theory, and most of them provide support for it (Schneekloth&Shibley, 1993). For example, one would expect more crime in areas that offer fewer opportunities for surveillance and do not appear to be controlled by anyone. A study of crime in university residence halls showed that

halls with defensible space features (e.g., more areas over which residents could feel some control and exercise more “surveillability”) suffered less crime than did halls on the same campus without such features (Sommer, 1987). A survey of 16 well-conducted studies in which multiple design changes were made in accordance with defensible space theory found reductions in robberies of 30% to 84% (Casteel & Peek-Asa, 2000). In what follows, we consider the notion of defensible space in several settings.

PERSONAL CONTROL.

Good social design will provide building occupants with real options to control their proximate environment. What does this mean in specific terms? Consider, for example, publicly funded residential space for students (dormitories) and poor people (housing projects). Some buildings, high-rises in particular, seem designed to overload residents with social stimulation. Too few elevators and long, narrow hallways, for example, result in the sense that people are everywhere and inescapable. Residents may develop the feeling that they cannot control the number of social contacts—especially unwanted social contacts—they must face daily. This loss of control can negatively affect feelings of security and self-esteem.

Two other common examples of low-control settings are crowded retail stores and traffic jams.

CROWDING

It refers to the subjective sense that too many people are around; it may be distinguished from **population density**, which is an objective measure of persons per unit area. High density does not always

lead to crowding, and crowding is not always the result of high density. Crowding is caused, in part, by social overload and informational overload, which in turn lead to the sense that one has lost control. Designing *against* crowding is, in part, designing *for* personal control. Again, simple design changes can be effective. By merely adding a few entrances to a mental health center, clients' sense of freedom (and thus control) was increased. Furthermore, the various treatment units within the center experienced a greater sense of identity because therapists felt as though they had their "own" entrances (Gutkowski, Ginath, & Guttman, 1992).

Stress is often related to lack of personal control over physical and social input.

Noise, unwanted social contact, congestion, and a lack of places of refuge are examples of primary sources of stress (Evans & McCoy, 1998). Good social design can anticipate and attempt to overcome such sources or at least buffer the user from them.

CULTURAL DIFFERENCES IN PERSONAL SPACE

Environmental psychologists are acutely aware that human behavior varies considerably around the world. The ways in which people celebrate birth, teach their children, dress, get married, work, and are treated at death are like a colorful tapestry of swirling colors. Yet in another way, and at another level, people are the same everywhere. They celebrate births, teach their children, dress, get married, work, and recognize death in some kind of ceremony. Personal space is like that; the distance across which individuals interact

with one another varies from culture to culture. Yet in every culture, there are rules that govern the choice of those interactional distances.

Personal space has been described as hidden, silent, and invisible, yet everyone possesses and uses personal space every day. Personal space stretches and shrinks with circumstances. It is interpersonal, so it depends on with whom people are interacting. It refers to the distance people choose to stay from others, but social interaction, involving angle of orientation and eye contact, is also part of personal space. Finally, personal space can be invaded, although such invasions are a matter of degree (Patterson, 1975). In sum, **personal space** is the geographic component of interpersonal relations, that is, the distance and angle of orientation (e.g., side by side, face to face) between individuals as they interact (Gifford, 2014).

Beyond these within-culture variations, personal space is used differently around the world. In one study, for example, groups of four male students came to the laboratory and were told that they would be observed but were given no other instructions (Watson & Graves, 1966). Half of the groups were composed of Arabs, and half were composed of Americans. The average interpersonal distance chosen by Arabs was about the length of an extended arm, whereas the average interpersonal distance chosen by Americans was noticeably farther. The Arabs touched one another much more often, and their orientation was much more direct. In general, the Arabs were much more "immediate" (close) with one another than were the Americans.

Such findings might lead to overly simplistic generalizations or stereotypes about cultural differences, for example, that some cultures are “close” and others are “distant.” However, two studies (Forston & Larson, 1968; Mazur, 1977) revealed that students from supposedly close cultures (Latin America, Spain, and Morocco) chose seating positions that were farther apart from one another than did students from a supposedly distant culture (United States). Furthermore, not all Latin Americans use the same amount of space (Shuter, 1976). Costa Ricans, for example, choose smaller interpersonal distances on average than Panamanians and Colombians.

Despite some oversimplifications, personal space does vary with culture. In one study, for example, Japanese people used more distance in conversations than did Americans, who in turn used more than did Venezuelans. But when the same Japanese and Venezuelans spoke English instead of their first languages, their conversational distance moved toward that of the Americans (Sussman & Rosenfeld, 1982). Language, an important part of culture, can modify one’s cultural tendencies to use more or less interpersonal distance.

The study of personal space is not merely academic; it also has important implications for cultural understanding and conflict. For example, a researcher taught some English students how to act more like Arabs in their nonverbal behavior (Collett, 1971). Arabs who interacted with the trained students liked them more than they did students who had not received

such training. Consider the implications for diplomats or even ordinary tourists.

APPLICATION OF SOCIAL PSYCHOLOGY TO DIFFERENT LEVELS OF JUSTICE

KIRK BLOODSWORTH:

Is the first American sentenced to death row to be exonerated by DNA, although his death sentence had already been commuted to two consecutive life sentences by the time his exoneration based upon DNA evidence was in the works. Bloodsworth was released as a result of DNA testing, a new science, revealed that the evidence left on

Dawn’s clothing could not have come from him.

The question is how could the legal system have incarcerated an innocent man? Bloodsworth *not fit* the description given by eye witness accounts of a tall, 6 ft all man with curly blond hair.

It seems as if the jurors convicted solely on the eyewitness testimony provided by two young boys and other neighbours.

But could their identifications have been influenced by the procedures the police used to collect this eyewitness evidence?

Could the jurors have been affected by the process used to select jurors for cases in which the death penalty is a sentencing option?

Could the overwhelming media attention to the crime have prejudiced the jury against KB? Long before reaching adulthood, most people

understand that laws are developed to help regulate human behavior.

However, lawmakers, law enforcement personnel, judges, and jurors are social and cultural beings whose decisions about which behaviors need regulating and how to respond to people who break the law are influenced by their assumptions about human behavior. Thus our criminal justice system is based on many psychological assumptions about behavior: Highly confident witnesses to a crime must have

accurate memories. Are highly confident witnesses necessarily accurate? Can jurors disregard information when judges instruct them to do so? Social Psychologists are in a unique position to test whether these psychological assumptions are warranted.

EYEWITNESS MEMORY:

Eyewitness testimony is a legal term. It refers to an account given by people of an event they have witnessed. For example, they may be required to give a description at a trial of a robbery or a road accident someone has seen. This includes identification of perpetrators, details of the



Three different kitchens, with similar features that might be a part of a 'kitchen schema'

crime scene etc.

One important application of research into memory is eyewitness testimony (EWT). EWT is used as evidence in criminal trials

in countries all over the world. Juries tend to pay extra attention to eyewitness testimony and generally see it as very trustworthy and convincing. However, a great deal of research in cognitive psychology tells us that, in general, people's memories are fairly fallible. This section examines some of the psychological factors that can affect the accuracy of EWT.

It is split into two main sections:
Reconstructive Errors & Leading Questions
Weapon Focus and Violence Distraction

RECONSTRUCTIVE ERRORS

Many people believe that memory works something like a videotape. Storing information is like recording and remembering is like playing back what was recorded, with information being retrieved in much the same form as it was encoded. However, memory does not work in this way. It is a feature of human memory that we do not store information exactly as it is presented to us. Rather, people extract from information the gist, or

underlying meaning. In other words, people store information in the way that makes the most sense to them. We make sense of information by trying to fit it into schemas, which are a way of organising

information. Schemas are general purpose 'packets' of knowledge that correspond to frequently encountered people, objects or situations. They allow us to make sense of what we encounter in order that we can predict what is going to happen and what we should do in any given situation. Schemas are a very effective way of processing information. Besides making the world more predictable, they remove the need to store similar information more than once. For example, if you think about a kitchen, you will probably find that your idea of kitchens includes features like a cooker, a fridge, cupboards, work surfaces

So, whilst you might encode a few details about it, for example, the layout, or the colour of the walls and so on, you would not need to store any more than this. For this reason, schema driven processing increases the efficiency or cognitive economy with which memory operates. However, schema driven processing has an important consequence for the way we store information. By forcing new situations to fit into our schemas, we may distort them in some way. So the information encoded in memory will not correspond exactly to what we actually



The stimulus Bartlett instructed his participants to remember

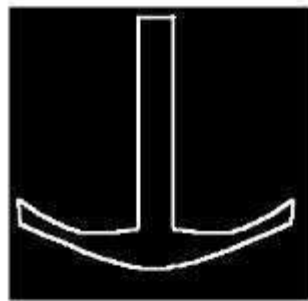
and so on. Your schema for 'kitchen' includes these features, because you have discovered through your experiences that most kitchens have them in some form. Now, suppose you visit someone's house for the first time, and they ask you to get something from the kitchen. You may not know where the kitchen is, but you would be able to recognise it when you found it because it would contain all or most of the things that feature in your 'kitchen schema'. Additionally, when you got there, it would not be necessary for you to store information about its contents, because you would already know most of what was in there due to your having a schema for that particular type of room.

encounter. When we later recall the information, these distortions will have been incorporated into our recall that hence may not be entirely accurate.

THE WORK OF FREDERICK BARTLETT

These general ideas were first formulated by Frederick Bartlett in the 1930s. Bartlett carried out a large number of studies in which he showed that the ways in which participants make sense of something (i.e. the schema they apply to it) affects the way they recall it later. In one study, participants were shown unfamiliar line drawings like the one below and instructed to memorise them.

Bartlett asked his participants to talk aloud whilst trying to memorise the figures. He found that different participants would make sense of the figures in different ways. For the figure above, for example, one participant called the drawing a pickaxe, whilst another called it a turf cutter. When the participants were later asked to draw the figures as they remembered them, Bartlett found that the labels (or schemas) they had applied during initial encoding influenced their recall:



'Pickaxe' schema



'Turf cutter' schema

These differences in recall show that the schema applied by the participants affected their recall of the stimulus. Another study by Bartlett illustrated further the role that schemas can play in the distortion of information in memory. Participants were asked to memorise a short story called 'the war of the ghosts'. The important thing about this story is that it comes from a native American tradition, whilst the participants were British. Native American storytelling uses different conventions to European storytelling, and different assumptions are made about the knowledge that the listener is likely to have. So the schemas needed to understand 'the war of the ghosts' would not necessarily be possessed by a European

participant. Bartlett hypothesised that his participants would find the story difficult to understand and memorise and, hence, that when they recalled it they would distort it in a number of ways. As he predicted, the participants' retellings of the story differed from the original in several characteristic ways:

The story became significantly shorter.

Much of the detail was lost.

Some details were changed e.g. 'seal hunting' became 'fishing'.

The structure altered to become more 'Westernised'.

The participants attempted to fit the story into their western schemas and, as a result, distorted it during recall. This showed that they were not recalling the information exactly as it had been presented to them, but were making a 'best guess' at the story, based on their own understanding. Bartlett called this the tendency to make 'efforts after meaning'. He concluded that we always try to recall things in a way that is consistent with our schemas and, hence, that memory is 'the imaginative reconstruction of experience'. The Work of Elizabeth Loftus You may be wondering what kitchens, pickaxes and folk tales have to do with eyewitness testimony. The link between reconstructive memory and witness testimony was made by Elizabeth

Loftus in the 1970s. Following on from Bartlett's work, she reasoned that witnesses to crimes, rather than recalling events exactly as they happened, are creating reconstructions of the crime based on their own schematic understanding of the world. However, Loftus Contributed by Aidan Sammons took this idea one step further. She suggested that any new information about the crime that the witness took in had the potential to distort their recall of events. Where might this new information come from? There are a number of possible sources. For example, witnesses might confer with each other about what they saw, thereby contaminating each other's accounts. But the most problematic source of new information is the leading questions that may be asked by police and lawyers. A leading question is a question that contains information previously unknown to the witness. For example, a police officer that asks 'how many times did Joe Bloggs hit the victim?' is not only requesting information about an assault but incidentally conveying the idea that it was Joe Bloggs that hit the victim and not Joe Blow. This information has the potential to affect the witness' understanding of the event so that, when they are later asked who carried out the assault, the understanding that it was Joe Bloggs has been implanted in their recollection of what happened. This is a fairly trite example, and it is unlikely that a police officer or lawyer would ask such a blatantly leading question. However, the information suggested by real leading questions can be subtle and go unnoticed by the witness. Loftus carried out a number of studies in which participants were shown films of car crashes. Different groups of participants were asked slightly

different questions about what they had seen. Loftus and her colleagues found that the different questions caused the participants to recall the events differently. For example, Loftus & Zanni (1975) found that asking the participants if they had seen 'the' broken headlight rather than 'a' broken headlight increased the likelihood that they would report seeing one, even though there was no broken headlight in the film. Similarly, Loftus & Palmer (1974) found that participants who were asked how fast the cars were going when they 'smashed into' each other gave higher estimates of speed than participants who were asked how fast the cars were going when they 'hit' each other. In a further study, Loftus, Miller & Burns (1978) found that they were able to get some participants to recall having seen a sign by the road, even though there had been no sign shown. This study also showed that most participants were unaware that they had been misled, which helps to reduce the possibility that results like these are simply the result of demand characteristics. Loftus' basic position on eyewitness testimony is that, frequently, witness' accounts of what they have seen are little better than guesses. This finding was highly influential in the writing of The Devlin Report, which concluded that, unless the circumstances were highly unusual, eyewitness testimony should not result in a conviction in an English court in the absence of other corroborating evidence. However, not all researchers agree with Loftus' point of view. Some research has shown that, if the circumstances are right, witness recall can be extremely accurate. For example, Yuille

Cutshall (1986) examined the recall of witnesses to a shooting in a town in Canada. A man had attempted to rob a gun shop. During the course of the robbery the

shop owner was shot. He returned fire, killing the would-be robber. The incident occurred in broad daylight, in front of a large number of witnesses. Some months after the event, Yuille and Cutshall tracked down the witnesses and asked to interview them. Fifteen of them agreed to take part in the study. Examining their accounts, Yuille and Cutshall made several important findings:

The witnesses were able to recall the incident in a great deal of detail.

There was a very high level of agreement between the accounts given by the different witnesses.

The witnesses' accounts did not alter in response to leading questions.

These findings are contrary to those that Loftus might lead us to expect. The reconstructive memory approach would suggest first, that the witnesses' memory of the event would fade with time, second that the witnesses' accounts would differ according to their different interpretations of the incident and, third, that the witnesses would be susceptible to leading questions. Such findings, which are obtained from real-world witnesses and hence are high in ecological validity cast doubt on the validity of Loftus' conclusions. However, it would be unwise to dismiss Loftus' research purely on the basis of Yuille and Cutshall's findings. The incident witnessed by their participants was shocking, highly unusual and was observed in ideal viewing conditions. The same is not true of most witness incidents. In general, the available research supports Loftus' view that witness testimony can be affected by leading questions, but only under certain circumstances:

The witness believes the questioner knows what really happened.

The witness is unaware they may be misled.

The misleading information concerns peripheral details of the incident, rather than central ones.

The misleading information is not blatantly incorrect.

WEAPON FOCUS

One of the main assumptions of the cognitive approach is that people have a limited capacity for processing information. This can be a problem because there is a huge amount of information in the environment. It would be impossible for a person to take in and process all this information. Therefore, people are selective about what they take in. The cognitive system is equipped with a sort of filtering system (called attention) that is used to select some information and reject other information. Under normal circumstances, people are quite good at choosing which information will be processed. However, there appear to be some sorts of information that draw attention to themselves. Weapons and other threats are one example. When a person witnesses a crime in which a weapon was used, their attention tends to focus on the weapon. This is hardly surprising, as it is the weapon that represents the most obvious threat. However, whilst the person's attention is focused on the weapon, it is not focused elsewhere, so the person is not taking in much information about the person holding it, for example. In other words, weapon focus is the tendency for witnesses

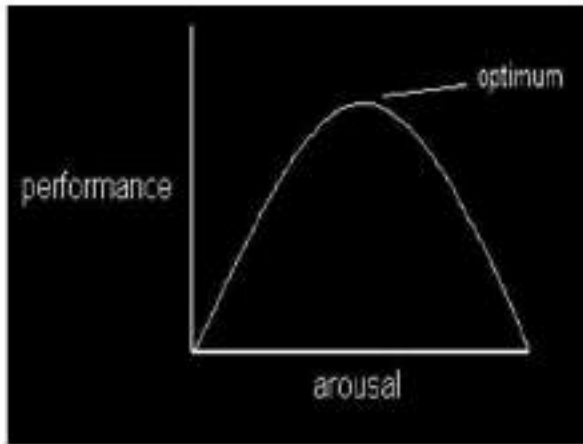
to violent crimes to focus their attention on the weapon used. Weapon focus usually results in poor quality testimony, as the witness is unable to describe much that is useful about other aspects of the incident. This is why bank tellers (whom you might expect to be able to describe an armed robber quite well, if they have faced one) can typically describe the weapon used in great detail, but not the person who was pointing it at them. Studies of Weapon Focus Loftus et al (1987) showed participants one of two films. In one, a customer in a restaurant was holding a cheque, in the other he was holding a gun. It was found that participants had a higher recall for the 'cheque' condition. Maass and Kohnen (1989) carried out a field experiment in which participants were approached by a woman holding either a pen or a syringe. Participants in the 'pen' condition were able to supply more accurate descriptions of the woman. Studies like these give support to the suggestion that witnesses' attention tends to be drawn towards a weapon, thereby preventing them from taking in other details about the situation. However, this theory rests on the assumption that the weapon represents a threat. What about situations in which weapons do not pose a threat? It might be suggested that in these situations weapon focus does not occur. The research appears to support this

hypothesis. Weapon focus appears most likely to occur when the weapon is incongruous (for example, a nun holding a gun will be more poorly recalled than a soldier holding a gun) and when a weapon is not appropriate to the situation (e.g. a person holding a gun is poorly recalled if they are depicted in a restaurant but reasonably well recalled if they are depicted on a firing range). Arousal, Anxiety and Violence Distraction Some researchers have questioned whether witnesses' poor recall of violent incidents is solely due to attentional focus. For example, Clifford and Scott (1978) found that witnesses to violent incidents generally recall less than witnesses to non-violent incidents, regardless of whether a weapon was used or not. They showed participants one of two films involving the same people. In one film there was a physical assault. Participants who viewed this film were less likely to identify the people involved than those who had viewed the non-violent film. Clifford and Hollin (1981) examined the relationship between the level of violence and recall. They found that the higher the level of violence depicted, the poorer participants' recall of an assault.

There are two possible reasons why this might be:

Violent incidents, like weapons, draw the focus of the witnesses' attention,

improves performance but only up to a point. Once arousal has passed a



A graph illustrating the Yerkes-Dodson law. As arousal increases, performance improves, but only until the optimum point is reached. Thereafter, as arousal continues to increase, performance goes into decline.

leading to little or no processing of other information

Violent incidents increase anxiety and autonomic arousal, which has a detrimental effect on memory generally. According to the Yerkes-Dodson Law, an increase in arousal

critical point called the optimum, performance tends to decline.

A possible interpretation of the research on violence distraction is that witnessing violence raises witnesses' arousal level past optimum, leading to poorer memory performance.

Power Within

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Successful aging.

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Applications of theories of motivation and learning in School

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UNIT

9

HUMAN DEVELOPMENT and INTERVENTIONS

WHAT IS LIFE SPAN DEVELOPMENT?

Lifespan development, also known as human **development**, is a field of **study** that is devoted to understanding constancy and change throughout a person's **lifespan** (Berk, 2010). **Life span development** begins with infancy and looks at several points in a person's life in which significant change takes place. The **study** of children did not begin until the late nineteenth and early twentieth centuries (Berk, 2010). Throughout the

years, there was much speculation about how people grow and change and when combined.

The study of developmental psychology is essential to understanding how humans learn, mature and adapt. Throughout their lives, humans go through various stages of development. Developmental psychologists study how people grow, develop and adapt at different life stages.

DEVELOPMENT: Pattern of change that begins at conception and continues throughout the life span (Santrock, 1998)

DEVELOPMENTAL PSYCHOLOGY:

Branch of psychology devoted to the understanding of all changes that human being experience throughout the life span.

CHILD DEVELOPMENT: all changes/ aspects of human growth from conception through adolescence . Developmental psychology, which is the study of how humans grow and change, was confined to the years from birth through adolescence. By the time people reach adulthood, psychologists believed, they are through developing:

- **The traditional approach** emphasizes extensive Change from birth to

adolescence, little or no change in adulthood, and decline in late old age.

Infancy is especially thought to be a time of considerable change, in the traditional approach.

- **The life-span approach** emphasizes developmental change during adulthood as well as childhood (Salthouse, 2000). Baltes saw the way that people changed throughout the entirety of their lives. To challenge the traditional view of development, he established the **life span perspective of development**, which views growth and change as occurring at all points in a person's life, as well as in many different directions at once.No age

period dominates development (so don't give importance to childhood only). Whether a child is learning to feed himself or a new parent is learning how to make decisions based on more than their own selfish interests, people are always growing and changing.

more applied settings like health care facilities or clinics help to assess, evaluate may also work in assisted living homes for the elderly, hospitals, mental health clinics and centres for the homeless.

Developmental psychologist's study human growth and development over the lifespan, including physical, cognitive, social, intellectual, perceptual, personality and emotional growth. Developmental psychologists working in colleges and universities tend to focus primarily on research or teaching. Others working in

and treat people living with developmental disabilities. Developmental psychologists

TABLE: COMPARISON OF GROWTH AND DEVELOPMENT

GROWTH	DEVELOPMENT
Growth refers to physiological changes.	Development refers to overall changes in the individual. It involves changes in an orderly and coherent type towards the goal of maturity.
Changes in the quantitative respect is termed as growth.	Development changes in the quality along with quantitative aspect.
Growth does not continue throughout life.	Development continues throughout life.
Growth stops after maturation.	Development is progressive.
Growth occurs due to the multiplication of cells.	Development occurs due to both maturation and interaction with the environment.
Growth is cellular.	Development is organizational.
Growth is one of the parts of the developmental process.	Development is a wider and comprehensive term.
Growth may be referred to describe the changes in particular aspects of the body and behavior of the organism.	Development describes the changes in the organism as a whole.
The changes produced by growth are subjects of measurements. They may be quantified and observable in nature.	Development brings qualitative changes which are difficult to measure directly. They are assessed through keen observation of behavior in different situations.
Growth may or may not bring development.	Development is possible without growth.

Tenets of a life span perspective/Characteristics of development (Baltes, 1987)

DEVELOPMENT IS LIFELONG: Lifespan theorists believe that development is life-long, and change is apparent across the lifespan. No single age period is more crucial, characterizes, or dominates human development. Consequently, the term lifespan development will be used throughout the textbook.

2. DEVELOPMENT IS

MULTIDIRECTIONAL: Humans change in many directions. We may show gains in some areas of development, while showing losses in other areas. Every change, whether it is finishing high school, getting married, or becoming a parent, entails both growth and loss.

3. DEVELOPMENT IS AN INTERACTIVE PROCESS: Intersection between the environment and the individual

DEVELOPMENT IS MULTIDIMENSIONAL: We change across three general domains/dimensions; physical, cognitive, and psychosocial. The **physical domain** includes changes in height and weight, sensory capabilities, the nervous system, as well as the propensity for disease and illness. The **cognitive domain** encompasses the changes in intelligence, wisdom, perception, problem-solving, memory, and language. The **psychosocial domain** focuses on changes in emotion, self-perception and interpersonal relationships with families, peers, and friends. All three domains influence each other. It is also important to note that a change in one domain may cascade and prompt changes in the other domains. For instance, an infant who has started to crawl or walk will encounter more objects and people, thus fostering developmental change in the child's understanding of the physical and social world.

DEVELOPMENT IS MULTIDISCIPLINARY: As mentioned at the start of the chapter, human development is such a vast topic of study that it requires the theories, research methods, and knowledge base of many academic disciplines.

DEVELOPMENT IS CHARACTERIZED BY PLASTICITY: *Plasticity is all about our ability to change and that many of our characteristics are malleable.* For instance, plasticity is illustrated in the brain's ability to learn from experience and how it can recover from injury.

DEVELOPMENT IS MULTI-CONTEXTUAL: Development occurs in many contexts.

Baltes (1987) identified three specific contextual influences.

NORMATIVE AGE-GRADED INFLUENCES: An **age-grade** is a specific age group, such as toddler, adolescent, or senior. Humans in a

specific age-grade share particular experiences and developmental changes.

2. **NORMATIVE HISTORY-GRADED INFLUENCES:** The time period in which you are born (see Table 1.1) shapes your experiences. A **cohort** is a group of people who are born at roughly the same period in a particular society. These people travel through life often experiencing similar circumstances.
3. **NON-NORMATIVE LIFE INFLUENCES:** Despite sharing an age and history with our peers, each of us also has unique experiences that may shape our development. A child who loses his/her parent at a young age has experienced a life event that is not typical of the age group.

FACTORS AFFECTING HUMAN DEVELOPMENT:

1. **HEREDITARY FACTORS:** Heredity exerts an influence on human development. The child carries genetic endowments from his/her parents. It is genetically transmitted characteristics from one generation to the next. The physical characteristics like height, weight, eye color etc. and psychological characteristics such as intelligence, personality, creativity and soon are innately determined and hereditary. The genetic code provides the base on which brain and body grow and manifest in observable appearance and behavior.

ENVIRONMENTAL FACTORS:

Another important factor of human development is the environment where an individual lives. The child lives and grows in his environment. Environment consists of a wide range of stimuli and it provides the necessary input and experiential base for development of the child. Enrichment or impoverishment of the environment would produce differences in his abilities. For example, a child may have inherited

music talent from his parents through transmission of genes, but he may not excel in music field if he does not get the proper environment and support to develop his innate ability.

HOME ENVIRONMENT: Home environment exerts tremendous influence on child's understanding of the external world. It builds self-concept and prepares him to face the external world. The child begins to acquire knowledge through interaction with parents and other family members. During his early years of development, the behavior of the child is modulated by the home environment. The environment of the family can be supportive or stressful for the child. If it is supportive, warm and harmonious environment, the child develops normally. In unsupportive and stressful home environment, broken families or uncaring parents in the family, children may develop as mal-adjusted persons.

CULTURAL FACTORS: Culture is the totality of our shared language, knowledge, material objects, and behavior. Culture teaches us how to live in a society and allows us to advance because each new generation can benefit from the solutions found and passed down from previous generations. Culture is learned from parents, schools, houses of worship, media, friends and others throughout a lifetime. The kinds of traditions and values that evolve in a particular culture serve to help members function and value their own society. We tend to believe that our own culture's practices and expectations are the right ones. *This belief that our own culture is superior is called ethnocentrism* and is a normal by-product of growing up in a culture. It becomes a roadblock, however, when it inhibits understanding of cultural practices from other societies. **Cultural relativity** is an

appreciation for cultural differences and the understanding that cultural practices are best understood from the standpoint of that particular culture. The socialization processes of the child take place according to the culture, customs and traditions of the society. For example, greeting someone is a familiar experience but behavioral experiences are different in different cultures. In Indian culture, people greet others by saying namaskar, folding hands or lying down near the feet but in Western culture, people greet by handshake or kissing or saying hello etc.

SOCIOECONOMIC STATUS

(SES): Socioeconomic Status plays a pivotal role in human development. The index of socioeconomic status is determined by parental education, occupation and income. The children of low socioeconomic status may develop as mal-nourished, suffer from lack of knowledge in many aspects and the normal development may get hampered. The parenting in high socioeconomic status families would be different from low socioeconomic status families. Children from the high socioeconomic groups of the society get better social opportunities, are nurtured with better nutrition, good medical treatment and are exposed to more intellectual stimulation than low socioeconomic group.

NORMATIVE INFLUENCES:

Normative influences occur in a similar way for majority of people in a particular group. These influences may be biological or environmental. For example, biological events like sexual maturity or deterioration in old age. Environmental events, like entering the school at about 6 yrs. of age, parenthood etc. have the same influences on individuals. Most of the people of the same age, at same place and time and generation have common biological and environmental influences such as

floods, famines and other natural disasters. Non-normative influences include the unusual life events in an individual's life. For example, death of a parent when a child is young or birth defect etc.

EDUCATION AND TRAINING:

Each child is equipped with certain abilities which need to be nurtured through proper education and training. Therefore, the first and foremost step is to identify and recognize the ability of the child and the next step is to provide adequate opportunities to develop the same. If proper identification of the ability is not possible and adequate facilities are not available to the child, then his innate ability may not be developed. Thus, adequate education and training have influence on human development.

CONCEPTIONS OF AGE

1. **CHRONOLOGICAL AGE:** The number of years that have elapsed since a person's birth.

BIOLOGICAL AGE: Another way developmental researcher can think about the concept of age is to examine *how quickly the body is aging, this is your biological age*. Several factors determine the rate at which our body ages. Our nutrition, level of physical activity, sleeping habits, smoking, alcohol consumption, how we mentally handle stress, and the genetic history of our ancestors, to name but a few.

PSYCHOLOGICAL AGE: *Our psychologically adaptive capacity compared to others of our chronological age is our psychological age*. This includes our cognitive capacity along with our emotional beliefs about how old we are. An individual who has cognitive impairments might be 20 years of age, yet has the mental capacity of an 8-

year-old. A 70-year-old might be travelling to new countries, taking courses at college, or starting a new business. Compared to others of our age group, we may be more or less adaptive and excited to meet new challenges. Remember you are as *young* or *old* as you feel.

SOCIAL AGE: Our **social age** is *based on the social norms of our culture and the expectations our culture has for people of our age group*. Our culture often reminds us whether we are "on target" or "off target" for reaching certain social milestones, such as completing our education, moving away from home, having children, or retiring from work. However, there have been arguments that social age is becoming less relevant in the 21st century (Neugarten, 1979; 1996). If you look around at your fellow students in your courses at college you might notice more people who are older than the more traditional aged college students, those 18 to 25. Similarly, the age at which people are moving away from the home of their parents, starting their careers, getting married or having children, or even whether they get married or have children at all, is changing.

ISSUES IN LIFESPAN DEVELOPMENT

NATURE AND NURTURE: Why are you the way you are? As you consider some of your features (height, weight, personality, being diabetic, etc.), ask yourself whether these features are a result of heredity or environmental factors, or both. Chances are, you can see the ways in which both heredity and environmental factors (such as lifestyle, diet, and so on) have contributed to these features. For decades, scholars have carried on the "nature/nurture" debate. For any particular feature, those on the side of **nature** would argue that heredity plays the most

important role in bringing about that feature. Those on the side of **nurture** would argue that one's environment is most significant in shaping the way we are. This debate continues in all aspects of human development, and most scholars agree that there is a constant interplay between the two forces. It is difficult to isolate the root of any single behavior as a result solely of nature or nurture.

CONTINUITY

VERSUS

DISCONTINUITY: Is human development best characterized as a slow, gradual process, or is it best viewed as one of more abrupt change? The answer to that question often depends on which developmental theorist you ask and what topic is being studied. The theories of Freud, Erikson, Piaget, and Kohlberg are called stage theories. **Stage theories or discontinuous development** assume that developmental change often occurs in distinct stages that are qualitatively different from each other, and in a set, universal sequence. At each stage of development, children and adults have different qualities and characteristics. Thus, stage theorists assume development is more discontinuous. Others, such as the behaviorists, Vygotsky, and information processing theorists, assume development is a more slow and gradual process known as **continuous development**. For instance, they would see the adult as not possessing new skills, but more advanced skills that were already present in some form in the child. Brain development and environmental experiences contribute to the acquisition of more developed skills.

developmental path? Are you at the whim of your genetic inheritance or the environment that surrounds you? Some theorists see humans as playing a much more active role in their own development. Piaget, for instance believed that children actively explore their world and construct new ways of thinking to explain the things they experience. In contrast, many behaviorists view humans as being more passive in the developmental process.

STABILITY VERSUS CHANGE: How similar are you to how you were as a child? Were you always as out-going or reserved as you are now? Some theorists argue that the personality traits of adults are rooted in the behavioral and emotional tendencies of the infant and young child. Others disagree, and believe that these initial tendencies are modified by social and cultural forces over

ACTIVE VERSUS PASSIVE: How much do you play a role in your own

HISTORICAL FOUNDATIONS

1.	Original Sin	• In the original sin view: especially advocated during the Middle Ages, children were perceived as being basically bad, born into the world as evil beings. • The goal of child rearing was salvation, which was believed to remove sin from the child's life
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2.	Tabula Rasa	• Toward the end of the seventeenth century, the tabula rasa view was proposed by English philosopher John Locke. • He argued that children are not innately bad. • children are like a “blank tablet/slate”, • They acquire their characteristics through experience. • Locke believed that childhood experiences are important in determining adult characteristics. • He advised parents to spend time with their children and help them become contributing members of society.
3.	Innate Goodness: children are inherently good	• In the eighteenth century, the innate goodness view was presented by Swiss-born French philosopher Jean-Jacques Rousseau. • He stressed that children are inherently good. Rousseau said that because children are basically good, they should be permitted to grow naturally with little parental monitoring or constraint.

RESEARCH DESIGNS FOR DEVELOPMENTAL STUDIES

1. THE LONGITUDINAL DESIGN.

PARTICIPANTS are studied repeatedly, and changes are noted as they get older.

The time spanned may be relatively short (a few months to several years) or very long (a decade or even a lifetime).

2 major strengths:

It tracks the performance of each person over time, researchers can identify common patterns as well as individual differences in development.

Longitudinal studies permit investigators to examine relationships between early and later events and behaviors.

The granddaddy of longitudinal studies, a classic study, is a study of gifted children begun by Lewis Terman about 80 years ago. In the study—which has yet to be concluded—a group of 1,500 children with high IQs were tested about every 5 years. Now in their 80s, the participants—who call themselves “Termites”—have provided information on everything from intellectual accomplishment to personality and longevity (Feldhusen, 2003; McCullough, Tsang, & Brion, 2003; Subotnik, 2006).

Longitudinal research has also provided great insight into language development.



2. THE CROSS-SECTIONAL DESIGN

The length of time it takes for many behaviors to change
In the **cross-sectional design**, groups



LIMITATIONS

For instance, by tracing how children's vocabularies increase on a day-by-day basis, researchers have been able to understand the processes that underlie the human ability to become competent in using language (Gershkoff-Stowe & Hahn, 2007; Oliver & Plomin, 2007; Childers, 2009; Fagan, 2009).

LIMITATIONS

Participants moving away or Dropping out

Changes the original sample: questions of generalizability

Becoming test wise (practice effect)/ becoming aware of one's own T/F/A due to repeated testing/questioning (age related change not being reflected)

Cohort effect: Individuals born in the same time period are influenced by a particular set of historical and cultural conditions. Results based on one cohort may not apply to people developing in other times.

of people differing in age are studied at the same point in time.

An **efficient strategy** for describing age-related trends.

Participants are **measured only once**. Hence no concern about as participant dropout or practice effects.

Does not provide evidence about development at the level at which it actually occurs: the individual (Kraemer et al., 2000).
Comparisons are limited to age-group

averages. **Within individual changes cannot be analysed.**

Cannot tell if important individual differences exist.

Indeed, **longitudinal findings reveal that adolescents vary considerably** in the changing quality of their sibling relationships. Although many become more distant, others become more supportive and intimate, and still others more rivalrous and antagonistic (Branje et al., 2004; Dunn, Slomkowski, & Beardsall, 1994).

Cross-sectional studies—especially those that cover a wide age can be **threatened by cohort effects**.

For example, comparisons of 10-year-old cohorts, 20-year-old cohorts, and 30-year-old cohorts—groups born and reared in different years—may not really represent age-related changes. Instead, they may reflect unique experiences associated with the historical period in which the age groups were growing up.

3. SEQUENTIAL DESIGNS

Utilizing the strengths of both the designs

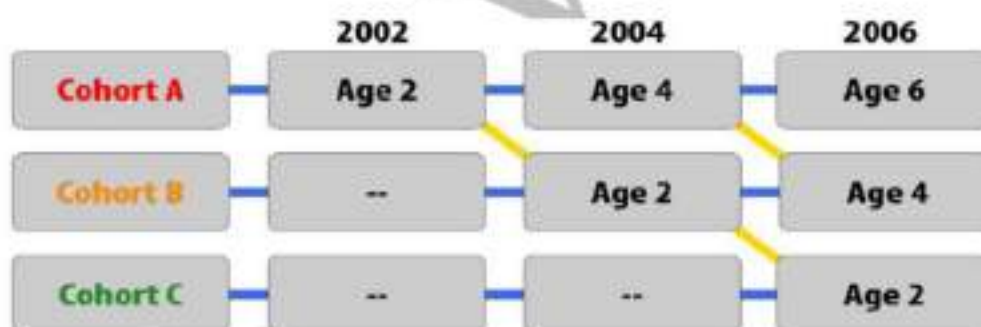
Examining a number of different age groups at several points in time.

Conducting several similar cross-sectional or longitudinal studies (called *sequences*) at varying times.

ADVANTAGES:

We can **find out whether cohort effects are operating** by comparing participants of the same age who were born in different years. E.g., we can compare the three longitudinal samples at ages 20, 30, and 40. **If they do not differ, we can rule out cohort effects.**

We can make longitudinal and cross-sectional comparisons. If outcomes are similar in both, then we can be especially confident about our findings.



Issues of research in Life Span Development: Methodological & Ethical

PERCEPTION OF CHILDREN

Until relatively recently, research was ***fundamentally on children, rather than with children or for children*** (Mayall, 2000; O’Kane, 2000; Darbyshire et al., 2005).

Historically, **children were seen as objects to be studied**, being regarded as incompetent, unreliable and incomplete (e.g. Barker and Weller, 2003).

With the emergence of the ‘new

social studies of childhood’ (James, Jenks, and Prout, 1998) and the **children’s right discourse** (the United Nations Conventions on the Rights of the Child, 1989 (1); the Children’s Act, 2004), children are now viewed as social actors who are ‘experts’ on their own lives (e.g. Mauthner, 1997; Kellett and Ding, 2004).

METHODOLOGICAL SHIFT

Emergence of **new ‘participatory’ research methodologies**, the adaptation of more traditional methods, such as observation and questionnaires (Punch, 2002), and the development of **multi-method**

approaches, such as the ‘mosaic approach’, developed by Clark and Moss (2001).

In recent years, **children have started to become involved in the various stages of the research process**, such as formulating the research questions, planning the methodology, collecting and/or analysing data, drafting recommendations and disseminating findings (Coad and Evans, 2008). This has involved differing levels of control-sharing and of participation in the research process (Brownlie et al., 2006; Alderson, 2000; McNeish, 1999).

POWER RELATIONS

Fraser (2004) argues for the importance of ‘research with...’ rather than ‘research on...’ Children and young people **can be involved is as active participants**, like children and young people actually carrying out the research (for example interviewing other children or young people. Our beliefs about the competence of children and young people and how these affect research, questions of power relations, and who should provide consent

Robinson and Kellett (2004), Jones (2004) point out, when children and young people become co-researchers in a project this can be genuinely empowering. Involving children and young people as co-researchers raises additional issues, such as ensuring that the research activities and arrangements are appropriate for the child researchers and considering the consequences for the children when they cease being researchers.

EVER CHANGING CONTEXT AND RELEVANCE OF RESEARCH FINDINGS:

Fashions and trends shift and change in research and while we may believe that we have identified and solved a problem once

and for all it is a fact of research, as it is of life, that many of the problems we identify and views which we hold today will be challenged by future generations and surpassed as different questions and approaches to research emerge.

DIVERSITY OF CHILDREN PARTICIPANTS

Pattman and Kehily remind us that it is essential that researchers think **carefully about the suitability of their methodological approach, given the diversity of participants.**

GAINING ACCESS AND SEEKING CONSENT:

When undertaking research with children, researchers must gain the **co-operation of a range of different ‘gatekeepers’**, such as school staff and parents (Cree et al., 2002).

Informed consent should be freely given (without coercion, threat or persuasion) by children who can make an appropriately informed decision.

Competent minors less than 16 years old of age can give consent, with competence being defined as having enough knowledge to understand what is proposed and enough discretion to be able to make a wise decision in light of one’s own interests (Alderson and Morrow, 2004).

It has often been assumed that ‘children are not competent enough to give their informed consent, that this needs to be gained from a ‘more competent adult’ and the simpler level of ‘assent’ (agreement to participate) is sufficient from the child’ (Kellett and Ding, 2004).

Some researchers have questioned this assumption and claim that children are fully capable of giving their informed consent.

Cocks (2006) argues that the notion of consent might exclude some children, such as disabled or refugee children, since it might not always be possible to

obtain in those particular contexts.

She endorses the concept of ‘assent’ (their like or dislike in participating)

as a sensitive and appropriate option to include all children in research on issues that affect them, and argues that the notion of ‘assent’ ‘removes the reliance on the child demonstrating adult-centric attributes such as maturity, competence and completeness’

CONFIDENTIALITY AND CHILD PROTECTION ISSUES

Parents’ curiosity and concern for their child as well as child protection regulations.

Children might reveal that they are seriously harmed or ill-treated/abused, or the researcher, when interviewing a child, might identify a medical condition or learning difficulty which the parents could take action about.

Children should be informed about the limitations of confidentiality before participating in the research in order to enable them to give fully informed consent (Williamson et al, 2005).

This could ‘be expressed as the difference between what can be ‘just between you and me’ and what may need to be told to others ‘to stop someone from getting hurt’

(Thompson and Rudolph, 2000: 35)

DATA COLLECTION: DEBRIEFING AND REWARDS

DEBRIEFING

After the interview, it might be appropriate to have some debriefing (Clark, 2005), and **suitable support might be needed in cases where the participants’ feelings may become overwhelming** (Kay et al., 2003).

Giving payments (whether cash or vouchers) to participants is a somewhat controversial issue.

Payments can pressurise people into taking part in the research and into saying what they think researchers want to hear. An EU Directive

(L121/34) from 2001 advised that paying children to take part in research should be illegal (Cree et al., 2002)

REWARDS

Payments should be given as an acknowledgement of the participants’ time and contribution. (Children in Scotland, 2001)

Cree and others (2002) decided not to pay or give vouchers to the participants in their research, but to offer them a pack of paper and pens as a sign of their gratitude. In contrast, Bushin

(2007) decided to give a ‘thank you’ voucher to children, but only informed them about the voucher during the interviews with them, rather than prior to participating in the research, since that might have acted as an incentive.

USE OF DECEPTION:

When is deception acceptable? Never, if there is another way to elicit and study the behaviour.

If deception is a critical element of the research, however, the importance of the research is paramount in determining if deception is acceptable.

Debriefing is necessary to fulfil our responsibility to the rights of subjects. It is intended to **remove any unwarranted negative effects of participation**, especially when stress and deception are involved.

Even when there is no stress and virtually no possibility of harm to subjects, debriefing is included as a standard part of experimental procedure.

TECHNIQUES, METHODS AND TOOLS WHEN RESEARCHING CHILDREN

Various methods and techniques have been used when conducting research with children. However, it has been argued that researchers need to critically reflect on the

methods and techniques they use and the ways they use them (Barker and Weller, 2003; Sanders and Munford, 2005). The methodology chosen needs to match the research questions of the project, respect limitations of time and resources, be sensitive and ethical, and consider the particular characteristics and needs of the participants, as well as the cultural and physical setting where it takes place (e.g. Christensen and Prout, 2002; Punch, 2002a

THEORIES OF HUMAN DEVELOPMENT

FREUDIAN MODEL OF PSYCHOSEXUAL STAGES

Sigmund Freud (1953, 1964a, 1964b) proposed that people are born with biological drives that must be redirected so as to live in society. The personality of the child is formed in childhood, as children deal with unconscious conflicts between these inborn urges and the requirements of civilized life. These conflicts occur in an unvarying sequence of five maturational based stages of psychosexual development, in different stages of development, sexual pleasure shifts from one body zone to another, i.e. from the mouth to the anus and then to the genitals.

According to Sigmund Freud there are five stages of development. Freud believed that personality is formed in first three stages which include first few years of life, as children deal with conflicts between their biological, sexually related urges and the pressures of society. At each stage there is a change in the main source of gratification. Each stage is characterized by a typical pleasure gaining behavior as well as a conflict due to the social pressures. These stages are named according to the

source of pleasure at the particular stage. They are:

ORAL STAGE (BIRTH TO 18 MONTHS): It begins at birth and extends till 12 to 18 months. In this stage the mouth is the primary organ of giving pleasure. The experience of the child and mother relationship has an impact on the unconscious mind of the child. This stage relates to the activities like feeding, crying, teething, biting and thumb sucking, etc.

ANAL STAGE (18 MONTHS TO 3 YEARS): The child gets sensual gratification from withholding and expelling feces during this stage. Anus is the chief source of pleasure. The child's real experiences during this stage have an impact upon the unconscious and behaviors. Pleasure derived from body parts are the centre of the world during this stage.

PHALLIC STAGE (3 TO 6 YEARS): During this period the child derives pleasure from genital region. Many of the normal sexual behaviour of human personality develop in this period. The child's sexual longing is intensified at this time. This is the stage in which conflicts related to reproductive issues are resolved.

LATENCY STAGE (6 TO PUBERTY): This stage is a stage of learning skills rather than a psychosexual stage. There is no sexual development taking place during this period. This is the time of relative calm between more turbulent stages. Previous experiences, fears and conditioning have shaped many of the child's feelings and attitudes.

Genital Stage (Puberty onwards):

This the time of mature adult sexuality. In this stage many hormonal and physical changes take place. Children are engaged in sex-

Erik Erikson (1902–1994) was a stage theorist who took Freud's controversial theory of psychosexual development and modified it as a psychosocial theory. Erikson emphasized that the ego makes

Erikson's Psychosocial Crisis Stages (Syntonic v Dystonic)	Freudian Psycho-sexual Stages	Life Stage / Relationships / Issues	Basic Virtue & Second Named Strength (Potential Positive Outcomes from Each Crisis)	Maladaptation / Malignancy (Potential Negative Outcome from Each Crisis)
Trust v Mistrust	Oral	infant / mother / feeding and being comforted, teething, sleeping	Hope and Drive	Sensory Distortion / Withdrawal
Autonomy v Shame & Doubt	Anal	toddler / parents / bodily functions, toilet training, muscular control, walking	Willpower and Self-Control	Impulsivity / Compulsion
Initiative v Guilt	Phallic	preschool / family / exploration and discovery, adventure and play	Purpose and Direction	Ruthlessness / Inhibition
Industry v Inferiority	Latency	schoolchild / school, teachers, friends, neighborhood / achievement and accomplishment	Competence and Method	Narrow Virtuosity / Inertia
Identity v Role Confusion	Puberty and Genitality	adolescent / peers, groups, influences / resolving identity and direction, becoming a grown-up	Fidelity and Devotion	Fanaticism / Repudiation
Intimacy v Isolation	Genitality	young adult / lovers, friends, work connections / intimate relationships, work and social life	Love and Affiliation	Promiscuity / Exclusivity
Generativity v Stagnation	-	mid-adult / children, community / 'giving back', helping, contributing	Care and Production	Overextension / Rejection
Integrity v Despair	-	late adult / society, the world, life / meaning and purpose, life achievements	Wisdom and Renunciation	Presumption / Disdain

related thoughts and feelings. Narcissism takes place during this stage.

ERIKSON'S THEORY OF PSYCHOSOCIAL DEVELOPMENT

positive contributions to development by mastering **attitudes**, ideas, and skills at each stage of development. This **mastery** helps children grow into successful, contributing members of society. During each of Erikson's eight stages, there is a psychological conflict

that must be successfully overcome in order for a child to develop into a healthy, well-adjusted adult.

According to Erikson, a person passes through eight developmental stages that build on each other. At each stage we face a crisis. By resolving the crisis, we develop psychological strengths or character traits that help us become confident and healthy people. One of the main elements of Erikson's psychosocial stage theory is the development of ego identity. 1 Ego identity is the conscious sense of self that we develop through social interaction. According to Erikson, our ego identity is constantly changing due to new experience and information we acquire in our daily interactions with others. In addition to ego identity, Erikson also believed that a sense of competence

1. TRUST VS. MISTRUST

From birth to 12 months of age, infants must learn that adults can be trusted. This occurs when adults meet a child's basic needs for survival. Infants are dependent upon their caregivers, so caregivers who are responsive and sensitive to their infant's needs help their baby to develop a sense of trust; their baby will see the world as a safe, predictable place. Unresponsive caregivers who do not meet their baby's needs can engender feelings of **anxiety**, fear, and mistrust; their baby may see the world as unpredictable. If infants are treated cruelly or their needs are not met appropriately, they will likely grow up with a sense of mistrust for people in the world.

AUTONOMY VS. SHAME/DOUBT

As toddlers (ages 1–3 years) begin to explore their world, they learn that they can control their actions and act on their environment to get results. They begin to show clear preferences for certain elements of the environment, such as food, toys, and clothing. A toddler's main task is to resolve the issue of *autonomy vs. shame*

also motivates behaviors and actions. Each stage in Erikson's theory is concerned with becoming competent in an area of life. If the stage is handled well, the person will feel a sense of mastery, which he sometimes referred to as ego strength or ego quality. 2 If the stage is managed poorly, the person will emerge with a sense of inadequacy. In each stage, Erikson believed people experience a conflict that serves as a turning point in development. In Erikson's view, these conflicts are centered on either developing a psychological quality or failing to develop that quality. During these times, the potential for personal growth is high, but so is the potential for failure.

and doubt by working to establish independence. This is the "me do it" stage. For example, we might observe a budding sense of autonomy in a 2-year-old child who wants to choose her clothes and dress herself. Although her outfits might not be appropriate for the situation, her input in such basic decisions has an effect on her sense of independence. If denied the opportunity to act on her environment, she may begin to doubt her abilities, which could lead to low **self-esteem** and feelings of shame.

3. INITIATIVE VS. GUILT

Once children reach the preschool stage (ages 3–6 years), they are capable of initiating activities and asserting control over their world through social interactions and play. According to Erikson, preschool children must resolve the task of *initiative vs. guilt*. By learning to plan and achieve goals while interacting with others, preschool children can master this task. Initiative, a sense of ambition and responsibility, occurs when parents allow a child to explore within limits and then support the child's choice. These children will develop self-confidence and feel a sense of purpose. Those who are

unsuccessful at this stage—with their initiative misfiring or stifled by over-controlling parents—may develop feelings of guilt.

4. INDUSTRY VS. INFERIORITY

During the elementary school stage (ages 6–12), children face the task of *industry vs. inferiority*. Children begin to compare themselves with their peers to see how they measure up. They either develop a sense of pride and accomplishment in their schoolwork, sports, social activities, and family life, or they feel inferior and inadequate because they feel that they don't measure up. If children do not learn to get along with others or have negative experiences at home or with peers, an inferiority complex might develop into **adolescence** and adulthood.

5. IDENTITY VS. ROLE CONFUSION

In adolescence (ages 12–18), children face the task of *identity vs. role confusion*. According to Erikson, an adolescent's main task is developing a sense of self. Adolescents struggle with questions such as “Who am I?” and “What do I want to do with my life?” Along the way, most adolescents try on many different selves to see which ones fit; they explore various roles and ideas, **set** goals, and attempt to discover their “adult” selves. Adolescents who are successful at this stage have a strong sense of identity and are able to remain true to their beliefs and values in the face of **problems** and other people's perspectives. When adolescents are apathetic, do not make a conscious search for identity, or are pressured to conform to their parents' ideas for the future, they may develop a weak sense of self and experience role confusion. They will be unsure of their identity and confused about the future. Teenagers who struggle to adopt a positive role will likely struggle to “find” themselves as adults.

6. INTIMACY VS. ISOLATION

People in early adulthood (20s through early 40s) are concerned with *intimacy vs. isolation*. After we have developed a sense of self in adolescence, we are ready to share our life with others. However, if other stages have not been successfully resolved, young adults may have trouble developing and maintaining successful relationships with others. Erikson said that we must have a strong sense of self before we can develop successful intimate relationships. Adults who do not develop a positive **self-concept** in adolescence may experience feelings of loneliness and emotional isolation.

7. GENERATIVITY VS. STAGNATION

When people reach their 40s, they enter the time known as middle adulthood, which extends to the mid-60s. The social task of middle adulthood is *generativity vs. stagnation*. Generativity involves finding your life's work and contributing to the development of others through activities such as volunteering, mentoring, and raising children. During this stage, middle-aged adults begin contributing to the next generation, often through childbirth and caring for others; they also engage in meaningful and productive work which contributes positively to society. Those who do not master this task may experience stagnation and feel as though they are not leaving a mark on the world in a meaningful way; they may have little connection with others and little interest in productivity and self-improvement.

8. INTEGRITY VS. DESPAIR

From the mid-60s to the end of life, we are in the period of development known as late adulthood. Erikson's task at this stage is called *integrity vs. despair*. He said that people in late adulthood reflect on their lives and feel either a sense of satisfaction or a sense of failure. People who feel proud of their accomplishments feel a sense of integrity, and they can look back

on their lives with few regrets. However, people who are not successful at this stage may feel as if their life has been wasted. They focus on what “would have,” “should have,” and “could have” been. They face the end of their lives with feelings of bitterness, depression, and despair.

BEHAVIORISM THEORY

Whereas, Erikson introduced the importance of social context to development, the Behavioral Theory, at least in its classical form, placed nearly sole emphasis on the impact of environment, experience, and learning about the unfolding development of the human condition. This orientation can best be illustrated by a quote offered by John B. Watson (1878–1958), deemed the father of American Behaviorism. Watson’s emphasis on the role of environment in the shaping of human development is concretized in his statement “Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I will guarantee to take anyone at random and train him to become any type of specialist I might select... doctor, lawyer, artist, merchant-chief... and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors” (Watson, 1958, p.82). That is quite a guarantee and clearly highlights the valuing and focus on this behaviorist placed on the role of environment, experience, and learning in the creation of the human condition.

Two main themes or forms of behavioral theory have been presented in explaining how the environment has such a formational impact. One proffered by B. F. Skinner (1904–1990) suggested that behavior was formed, or shaped, as a result of the consequences experienced. His operant conditioning model argued that behavior followed by a rewarding stimulus would be more likely to recur and endure than that followed by a punishing consequence (Cohen, 1987; Skinner, 1974). Thus, an infant who experiences the comfort of being picked up and cradled

following his or her crying is more likely to employ crying behavior in the future. Or, an individual who has experienced ridicule following his or her initiation of a social contact may soon employ withdrawal and isolation as a developmental coping style. A second behavioral approach to the explanation of the influence of environment on development is that initially presented by Albert Bandura (1997, 2008) as social learning theory. Bandura’s model expanded the classic behavioral theory to include cognitive elements. Bandura’s work emphasized the importance of observational learning (also called imitation or modeling). For example, consider the situation of a child who was raised in an environment where there was much domestic arguing and physical violence and the employment of alcohol as a stress-reducing strategy. Raised in this setting and observing these social exchanges and coping styles, the child, according to social learning theory, would be very likely to model the observed behavior and engage in similar domestic violence and alcohol use behaviors.

According to Bandura, Barbaranelli, Caprara, and Pastorelli (2001), social learning is connected to perceptions and interpretations of the individual’s experience. Self-efficacy, the belief that personal achievement depends on one’s actions, teaches people to have high aspirations and to strive for notable accomplishments when they see others solve problems successfully. This premise is contrary to Behavioral Theory, which holds that behavior depends on associations between one stimulus and another and also assumes that all behaviors result from a chain of learned responses. In contrast, social learning maintains that behaviors come from people acting on the stimulation of the environment.

COGNITIVE DEVELOPMENT THEORY (JEAN PIAGET)

Jean Piaget, a Swiss psychologist was particularly concerned with the way

thinking develops in children from birth till they become young adults. To understand the nature of this development, Piaget carefully observed the behavior of his own three kids. He used to present problems to them, observe responses slightly after the situations and again observe their responses. Piaget called this method of exploring development clinical interview. Piaget believed that humans also adapt to their physical and social environments in which they live.

THERE ARE THREE BASIC COMPONENTS TO PIAGET'S COGNITIVE THEORY:

1. Schemas- (building blocks of knowledge).
 2. Adaptation processes that enable the transition from one stage to another (equilibrium, assimilation, and accommodation).
- Stages of Cognitive Development-

sensorimotor,
preoperational,
concrete operational,
formal operational.

SCHEMAS

Schemas are the basic building blocks of such cognitive models, and enable us to form a mental representation of the world. Piaget (1952, p. 7) defined a schema as: "a cohesive, repeatable action sequence possessing component actions that are tightly interconnected and governed by a core meaning."

When Piaget talked about the development of a person's mental processes, he was referring to increases in the number and complexity of the schemata that a person had learned. When a child's existing schemas are capable of explaining what it can perceive around it, it is said to be in a state of equilibrium, i.e., a state of cognitive (i.e., mental) balance.

Piaget emphasized the importance of schemas in cognitive development and described how they were developed or acquired. A schema can be defined as a set of linked mental representations of the

world, which we use both to understand and to respond to situations. The assumption is that we store these mental representations and apply them when needed.

Piaget believed that newborn babies have a small number of innate schemas - even before they have had many opportunities to experience the world. These neonatal schemas are the cognitive structures underlying innate reflexes. These reflexes are genetically programmed into us. For example, babies have a sucking reflex, which is triggered by something touching the baby's lips. A baby will suck a nipple, a comforter (dummy), or a person's finger. Piaget, therefore, assumed that the baby has a 'sucking schema.'

Similarly, the grasping reflex which is elicited when something touches the palm of a baby's hand, or the rooting reflex, in which a baby will turn its head towards something which touches its cheek, are innate schemas. Shaking a rattle would be the combination of two schemas, grasping and shaking.

ASSIMILATION AND ACCOMMODATION

Jean Piaget (1952; see also Wadsworth, 2004) viewed intellectual growth as a process of **adaptation** (adjustment) to the world. This happens through:

ASSIMILATION: Which is using an existing schema to deal with a new object or situation. Example: A 2-year-old child sees a man who is bald on top of his head and has long frizzy hair on the sides. To his father's horror, the toddler shouts "Clown, clown" (Siegler et al., 2003).

ACCOMMODATION: This happens when the existing schema (knowledge) does not work, and needs to be changed to deal with a new object or situation. Example: In the "clown" incident, the boy's father explained to his son that the man was not a clown and that even though his hair was like a clown's, he wasn't wearing a funny costume and wasn't doing silly things

to make people laugh. With this new knowledge, the boy was able to change his schema of “clown” and make this idea fit better to a standard concept of “clown”.

EQUILIBRATION- This is the force which moves development along. Piaget believed that cognitive development did not progress at a steady rate, but rather in leaps and bounds. It occurs when a child's schemas can deal with most new information through assimilation. However, an unpleasant state of disequilibrium occurs when new information cannot be fitted into existing schemas (assimilation). It is the force which drives the learning process as we do not like to be frustrated and will seek to restore balance by mastering the new challenge (accommodation). Once the new information is acquired the process of assimilation with the new schema will continue until the next time we need to make an adjustment to it.

PIAGET'S 4 STAGES OF COGNITIVE DEVELOPMENT

Jean Piaget's theory of cognitive development suggests that children move through four different stages of intellectual development which reflect the increasing sophistication of children's thought

His theory focuses on understanding how children acquire knowledge regarding fundamental concepts such as object permanence, number, categorization, quantity, causality, and justice.

Piaget's four stages of cognitive development include:

Sensorimotor: Birth to ages 18-24 months.

Preoperational: Toddlerhood (18-24 months) through early childhood (age 7).

Concrete operational: Ages 7 to 11 years.

Formal operational: Adolescence to adulthood. Each child goes through the stages in the same order, and child development is determined by biological maturation and interaction with the environment.

Stage	Age Range	Description
Sensorimotor	0-2 years	Coordination of senses with motor response, sensory curiosity about the world. Language used for demands and cataloguing. Object permanence developed
Preoperational	2-7 years	Symbolic thinking, use of proper syntax and grammar to express full concepts. Imagination and intuition are strong, but complex abstract thought still difficult. Conservation developed.
Concrete Operational	7-11 years	Concepts attached to concrete situations. Time, space, and quantity are understood and can be applied, but not as independent concepts
Formal Operations	11+	Theoretical, hypothetical, and counterfactual thinking. Abstract logic and reasoning. Strategy and planning become possible. Concepts learned in one context can be applied to another.

1.SENSORIMOTOR STAGE- This stage begins at birth and lasts till the child is

about 2 years old. It is called Sensorimotor Stage, because children's thinking involves seeing, hearing, moving, touching, testing and so on. This stage marks a transitional stage for a person from a biological to a psychological being. In the first few weeks of life the baby's behavior consists simply of reflex responses, such as sucking, stepping and grasping. Later the reflex disappears and the baby chooses what and when to grasp. During this period the infants attain the concept of **object permanence**. This refers to the understanding that objects and events continue to exist even when they cannot directly be seen, heard or touched. Till this kind of understanding is achieved, an object that is out of sight remains out of mind and therefore, becomes non-existent.

A second major accomplishment in the Sensorimotor period is learning to reverse actions. E.g., we give a toy to a child that has ten detachable parts. We detach all parts. Through trial and error, the child gradually learns to attach all the parts of the toy.

PRE-OPERATIONAL STAGE (2 TO 7 YEARS)-The preoperational period is divided into two stages:

The **Symbolic Function Substage** occurs between 2 and 4 years of age and is characterized by the child being able to mentally represent an object that is not present and a dependence on perception in problem solving.

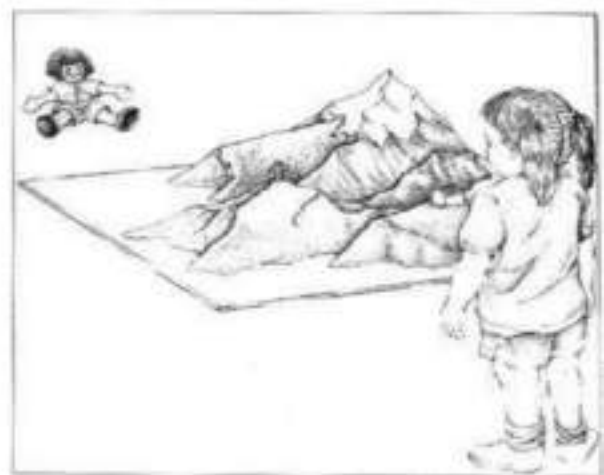
The **Intuitive Thought Sub-stage**, lasting from 4 to 7 years is marked by greater dependence on intuitive thinking rather than just perception (Thomas, 1979). At this stage, children ask many questions as they attempt to understand the world around them using immature reasoning.

Let's examine some of Piaget's assertions about children's cognitive abilities at this age.

Pretend Play: Pretending is a favorite activity at this time. A toy has qualities beyond the way it was designed to function and can now be used to stand for a character or object unlike anything originally intended. A teddy bear, for example, can be a baby or the queen of a faraway land. Piaget believed that children's pretend play helped children solidify new schemata they were developing cognitively. This play, then, reflected changes in their conceptions or thoughts. However, children also learn as they pretend and experiment. Their play does not simply represent what they have learned (Berk, 2007).



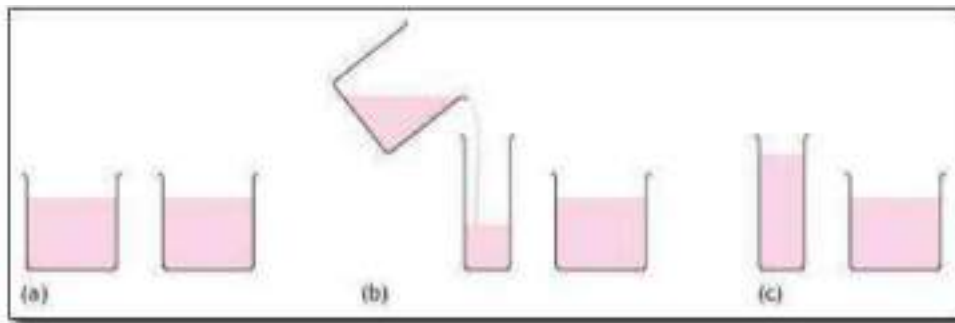
Egocentrism: *Egocentrism in early childhood refers to the tendency of young children not to be able to take the perspective of others, and instead the child thinks that everyone sees, thinks, and feels just as they do.* An egocentric child is not able to infer the perspective of other people and instead attributes his own perspective to situations. For example, ten-year-old Keiko's birthday is coming up, so her mom takes 3-year-old Kenny to the toy store to choose a present for his sister. He selects an Iron Man action figure for her, thinking that if he likes the toy, his sister will too.



Piaget's classic experiment on egocentrism involved showing children a three-dimensional model of a mountain and asking them to describe what a doll that is

looking at the mountain from a different angle might see. Children tend to choose a picture that represents their own, rather than the doll's view. By age 7 children are less self-centered.

However, even younger children when speaking to others tend to use different sentence structures and vocabulary when addressing a younger child or an older adult. This indicates some awareness of the views of others.



Conservation of Liquid. Does pouring liquid in a tall, narrow container make it have more?

Conservation

Errors: Conservation refers to the ability to recognize that moving or rearranging matter does not change the quantity. Let's look at Kenny and Keiko again. Dad gave a slice of pizza to 10-year-old Keiko and another slice to 3-year-old Kenny. Kenny's pizza slice was cut into five pieces, so Kenny told his sister that he got more pizza than she did. Kenny did not understand that cutting the pizza into smaller pieces did not increase the overall amount exhibited.

Centration, or focused on only one characteristic of an object to the exclusion of others. Kenny focused on the five pieces of pizza to his sister's one piece even though the total amount was the same. Keiko was able to consider several characteristics of an object than just one. Because children have not developed this understanding of conservation, they cannot perform mental operations.

The classic Piagetian experiment associated with conservation involves liquid (Crain, 2005). As seen in Figure 4.10, the child is shown two glasses (as shown in a) which are filled to the same

level and asked if they have the same amount. Usually the child agrees they have the same amount.

The experimenter then pours the liquid in one glass to a taller and thinner glass (as shown in b). The child is again asked if the two glasses have the same amount of liquid. The preoperational child will typically say the taller glass now has more liquid because it is taller (as shown in c). The child has concentrated on the height of the glass and fails to conserve.

Classification Errors: Preoperational children have difficulty understanding that an object can be classified in more than one way. For example, if shown three white buttons and four black buttons and asked whether there are more black buttons or buttons, the child is likely to respond that there are more black buttons. They do not consider the general class of buttons. Because children lack these general classes, their reasoning is typically.

Transductive, that is, *making faulty inferences from one specific example to another*. For example, Piaget's daughter Lucienne stated she had not had her nap, therefore it was not afternoon. She did not understand that afternoons are a time period and her nap was just one of many events that occurred in the afternoon (Crain,

2005). As the child's vocabulary improves and more schemata are developed, the ability to classify objects improves.



Animism: Animism refers to attributing life-like qualities to objects.

The cup is alive, the chair that falls down and hits the child's ankle is mean, and the toys need to stay home because they are tired. Cartoons frequently show objects that appear alive and take on lifelike qualities. Young children do seem to think that objects that move may be alive, but after age three, they seldom refer to objects as being alive (Berk, 2007).

CONCRETE OPERATIONAL STAGE (7 TO 11 YEARS).

At this stage a child is concerned with the integration of stability of his cognitive systems. He learns to add, subtract, multiply and divide. He is in a position to classify concrete objects. In short, children develop the abilities of rational thinking but their thinking is tied to concrete objects. At the concrete operational stage, children are able to use inductive logic – the type of reasoning that starts from a specific idea and leads to a generalization. They can also distinguish facts from fantasies, as well as formulate judgements about cause and effect.

Another important milestone at this stage is the idea of reversibility – children understand that some objects can be altered and then shaped back to their original shape. For example, a deflated balloon can be filled with air again to become an inflated balloon.

FORMAL OPERATIONAL STAGE (11 & ABOVE). This type is characterized by the emergence of logical thinking and reasoning. Other important cognitive attainments during this period are: the ability to think about the hypothetical possibilities and to solve problems through logical deductions and in a systematic manner.

EDUCATIONAL IMPLICATIONS:

Piaget's concept of development process of understanding working of the child's mind can be helpful to those who are involved in teaching and other educational practices. He says that children pass through number of stages before the age of 14 years and a lot of care should be taken in child's training and development. Most of the teachers are now in agreement with him that it is waste of time to take those things to children which cannot be experienced through sense organs. When the children form many direct experiences then only they are in a position to understand the abstract ideas and concept. Piaget does not like to looking at education, therefore, the teacher must always make an effort to orient education around the child. The most important function of school is to provide good stimulating environment within the school for the proper development of their mental abilities. The school should provide good library opportunities for free discussions and community services. The needs of adolescents should be given proper place on the school curriculum.

VYGOTSKY'S SOCIOCULTURAL THEORY OF COGNITIVE DEVELOPMENT

Vygotsky has developed a sociocultural approach to cognitive development. He developed his theories at around the same time as Jean Piaget was starting to develop his ideas (1920's and 30's), but he died at the age of 38, and so his theories are incomplete - although some of his writings are still being translated from Russian. No single principle (such as Piaget's equilibration) can account for development. Individual development cannot be understood without reference to the social and cultural context within which it is embedded. Higher mental processes in the individual have their origin in social processes. Vygotsky's theory differs from that of Piaget in a number of important ways:

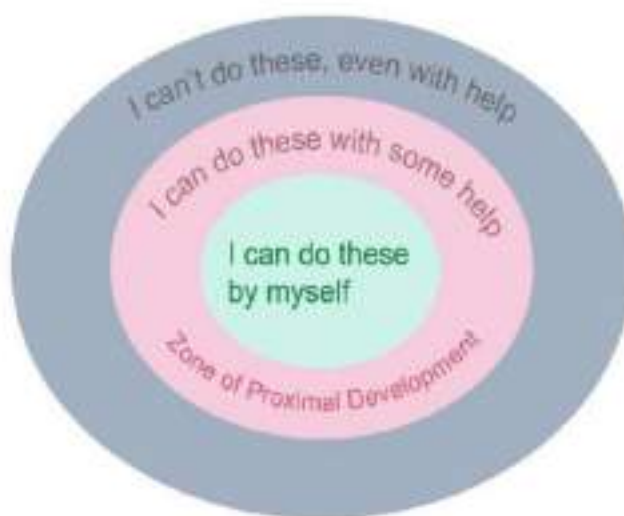
	Piaget	Vygotsky
Sociocultural context	Little emphasis	Strong emphasis
Constructivism	Cognitive constructivist	Social constructivist
Stages	Strong emphasis on stages of development	No general stages of development proposed
Key processes in development & learning	Equilibration; schema; adaptation; assimilation; accommodation	Zone of proximal development; scaffolding; language/dialogue; tools of the culture
Role of language	Minimal – Language provides labels for children's experiences (egocentric speech)	Major – Language plays a powerful role in shaping thought
Teaching implications	Support children to explore their world and discover knowledge	Establish opportunities for children to learn with the teacher and more skilled peers

The main assertion of the Vygotsky theory is that the cognitive development of children is advanced through social interaction with other people, particularly those who are more skilled. In other

words, Vygotsky believed that social learning comes before cognitive development, and that children construct knowledge actively.

VYGOTSKY'S CONCEPT OF ZONE OF PROXIMAL DEVELOPMENT

Vygotsky's Sociocultural Theory of Cognitive Development



Vygotsky is most recognized for his concept of Zone of Proximal Development (ZPD) pertaining to the learning process of children. According to the Vygotsky theory of cognitive development, children who are in the zone of proximal development for a particular task can almost perform the task independently, but not quite there yet. They need some help in order to perform the task successfully.

VYGOTSKY'S CONCEPT OF MORE KNOWLEDGEABLE OTHER

Vygotsky's sociocultural theory emphasizes that children learn through social interaction that include collaborative and cooperative dialogue with someone who is more skilled in tasks they're trying to learn. Vygotsky called these people with

higher skill level the More Knowledgeable Other (MKO). They could be teachers, parents, tutors and even peers.

VYGOTSKY'S CONCEPT OF SCAFFOLDING

Vygotsky's concept of scaffolding is closely related to the concept of the Zone of Proximal Development. Scaffolding refers to the temporary support given to a

child by a More Knowledgeable Other that enables the child to perform a task until such time that the child can perform this task independently.

According to Vygotsky's theory, scaffolding entails changing the quality and quantity of support provided to a child in the course of a teaching session. The MKO adjusts the level of guidance in order to fit the student's current level of performance.

Vygotsky's Sociocultural Theory of Cognitive Development



For novel tasks, the MKO may utilize direct instruction. As the child gains more familiarity with the task and becomes more skilled at it, the MKO may then provide less guidance.

EVIDENCE FOR VYGOTSKY AND THE ZPD-

Freund (1990) conducted a study in which children had to decide which items of furniture should be placed in particular areas of a dolls house. Some children were allowed to play with their mother in a similar situation before they attempted it alone (zone of proximal development) while others were allowed to work on this by themselves (Piaget's discovery learning). Freund found that those who had previously worked with their mother

(ZPD) showed the greatest improvement compared with their first attempt at the task. The conclusion being that guided learning within the ZPD led to greater understanding/performance than working alone (discovery learning).

VYGOTSKY AND LANGUAGE:

Vygotsky believed that language develops from social interactions, for communication purposes. Vygotsky viewed language as man's greatest tool, a means for communicating with the outside world. According to Vygotsky (1962) language plays two critical roles in cognitive development:

It is the main means by which adults transmit information to children.

Language itself becomes a very powerful tool of intellectual adaptation.

VYGOTSKY (1987) DIFFERENTIATES BETWEEN THREE FORMS OF LANGUAGE:

Social speech which is external communication used to talk to others (typical from the age of two); private speech (typical from the age of three) which is directed to the self and serves an intellectual function; and finally, private speech goes underground, diminishing in audibility as it takes on a self-regulating function and is transformed into silent inner speech (typical from the age of seven). For Vygotsky, thought and language are initially separate systems from the beginning of life, merging at around three years of age. At this point speech and thought become interdependent: thought becomes verbal, speech becomes representational. When this happens, children's monologues internalized to become inner speech. The internalization of language is important as it drives cognitive development. 'Inner speech is not the interior aspect of external speech - it is a function in itself. It still remains speech, i.e., thought connected with words. But while in external speech thought is embodied in words, in inner speech words die as they bring forth thought. Inner speech is to a large extent thinking in pure meanings.' (Vygotsky, 1962: p. 149).

Vygotsky (1987) was the first psychologist to document the importance of private speech. He considered private speech as the transition point between social and inner speech, the moment in development where language and thought unite to constitute verbal thinking. Thus, private speech, in Vygotsky's view, was the earliest manifestation of inner speech. Indeed, private speech is more similar (in its form and function) to inner speech than social speech. Private speech is 'typically defined, in contrast to social speech, as speech addressed to the self (not to others)

for the purpose of self-regulation (rather than communication).' (Diaz, 1992, p.62)

Unlike inner speech which is covert (i.e., hidden), private speech is overt. In contrast to Piaget's (1959) notion of private speech representing a developmental dead-end, Vygotsky (1934, 1987) viewed private speech as: 'A revolution in development which is triggered when preverbal thought and pre-intellectual language come together to create fundamentally new forms of mental functioning.' (Ferryhough & Fradley, 2005: p. 1)

Through private speech, children begin to collaborate with themselves in the same way a more knowledgeable other (e.g., adults) collaborate with them in the achievement of a given function. Vygotsky (1987) notes that private speech does not merely accompany a child's activity but acts as a tool used by the developing child to facilitate cognitive processes, such as overcoming task obstacles, enhancing imagination, thinking, and conscious awareness. Vygotsky (1987) proposed that private speech is a product of an individual's social environment. This hypothesis is supported by the fact that there exist high positive correlations between rates of social interaction and private speech in children. Children raised in cognitively and linguistically stimulating environments (situations more frequently observed in higher socioeconomic status families) start using and internalizing private speech faster than children from less privileged backgrounds. Indeed, children raised in environments characterized by low verbal and social exchanges exhibit delays in private speech development.

CLASSROOM APPLICATIONS

A contemporary educational application of Vygotsky's theories is "reciprocal teaching," used to improve students' ability to learn from text. In this method, teachers and students collaborate in learning and practicing four key skills: summarizing, questioning, clarifying, and predicting.

The teacher's role in the process is reduced over time. Also, Vygotsky is relevant to instructional concepts such as "scaffolding" and "apprenticeship," in which a teacher or more advanced peer helps to structure or arrange a task so that a novice can work on it successfully. Vygotsky's theories also feed into the current interest in collaborative learning, suggesting that group members should have different levels of ability so more advanced peers can help less advanced members operate within their ZPD.

LANGUAGE

Language is a system of communication using gestures, sounds, or written symbols that have significance for those who use the language and follow its rules. In speech, **phonemes** are the smallest units of sound in a language, and although they individually have no meaning, they acquire it when combined. For example, the phonemes *k* and *r* alone convey no meaning (other than that they are letters), but together they sound like *car*, which is a meaningful sound in the English language.

SEMANTICS. The term **semantics** refers to the study of meaning in a language. The smallest unit of meaning in spoken language is called a **morpheme**, which in many instances is itself a word. The word **overcoat** is composed of two morphemes, *over* and *coat*. Language rules govern the combination of morphemes to create meaning; *overcoat*, for example, means something different than does *coatover*.

SENTENCES. Language rules also dictate **syntax**, how morphemes are put

together to form **sentences**, groups of words that make meaningful statements. **Pragmatics** is the study of language as used in particular situations, which may affect its meaning. Consider the statement, "What a wonderful day!" and its various meanings if the sun is shining, if the rain is pouring down, and if the speaker has just received a traffic ticket.

ACQUISITION OF LANGUAGE. B. F. Skinner believed that language acquisition, an important development in childhood, occurs because of reinforcement, that is, because children's parents or other caregivers reward them when their initially random sounds most resemble speech. Linguist Noam Chomsky contested Skinner's approach and proposed the well-known, but controversial, theory that children have an innate neural mechanism called a **language acquisition device (LAD)** (not yet discovered), which allows them to master language.

Developmental psychologists have subsequently documented the general process of language acquisition, which is usually thought to progress through the stages shown in Table 1.

TABLE 1 Language Development

Approximate Age	Language Response
4 months	Babbling—a variety of sounds, often repeated
10 months	Babbling that includes language used in the household
12 to 18 months	Single words (sometimes one word has a variety of meanings— holophrastic speech)
18 to 24 months	Two-word statements (often a noun and a verb— telegraphic speech)
30 + months	More complex sentences, greater vocabulary, rapid language expansion

KOHLBERG'S MORAL DEVELOPMENT THEORY

Lawrence Kohlberg was, for many years, a professor at Harvard University. He became famous for his work there beginning in the early 1970s. He started as a developmental psychologist and then moved to the field of moral education. He was particularly well-known for his theory of moral development which he popularized through research studies conducted at Harvard's Center for Moral Education.

His theory of moral development was dependent on the thinking of the Swiss psychologist Jean Piaget and the American philosopher John Dewey. He was also inspired by James Mark Baldwin. These men had emphasized that human beings develop philosophically and psychologically in a progressive fashion. He used Piaget's storytelling technique to tell people stories involving moral dilemmas. In each case, he presented a choice to be considered, for example, between the rights of some authority and the needs of some deserving individual who is being unfairly treated.

One of the best known of Kohlberg's (1958) stories concerns a man called Heinz who lived somewhere in Europe: (Link

here)<https://www.youtube.com/watch?v=5czp9S4u26M>

Kohlberg believed and was able to demonstrate through studies...that people progressed in their moral reasoning (i.e., in their bases for ethical behavior) through a series of stages. He believed that there were six identifiable stages which could be more generally classified into three levels.

The first level of moral thinking is that generally found at the elementary school level. In the first stage of this level, people behave according to socially acceptable norms because they are told to do so by some authority figure (e.g., parent or teacher). This obedience is compelled by the threat or application of punishment. The second stage of this level is characterized by a view that right behavior means acting in one's own best interests.

The second level of moral thinking is that generally found in society, hence the name "conventional." The first stage of this level (stage 3) is characterized by an attitude which seeks to do what will gain the approval of others. The second stage is one oriented to abiding by the law and responding to the obligations of duty.

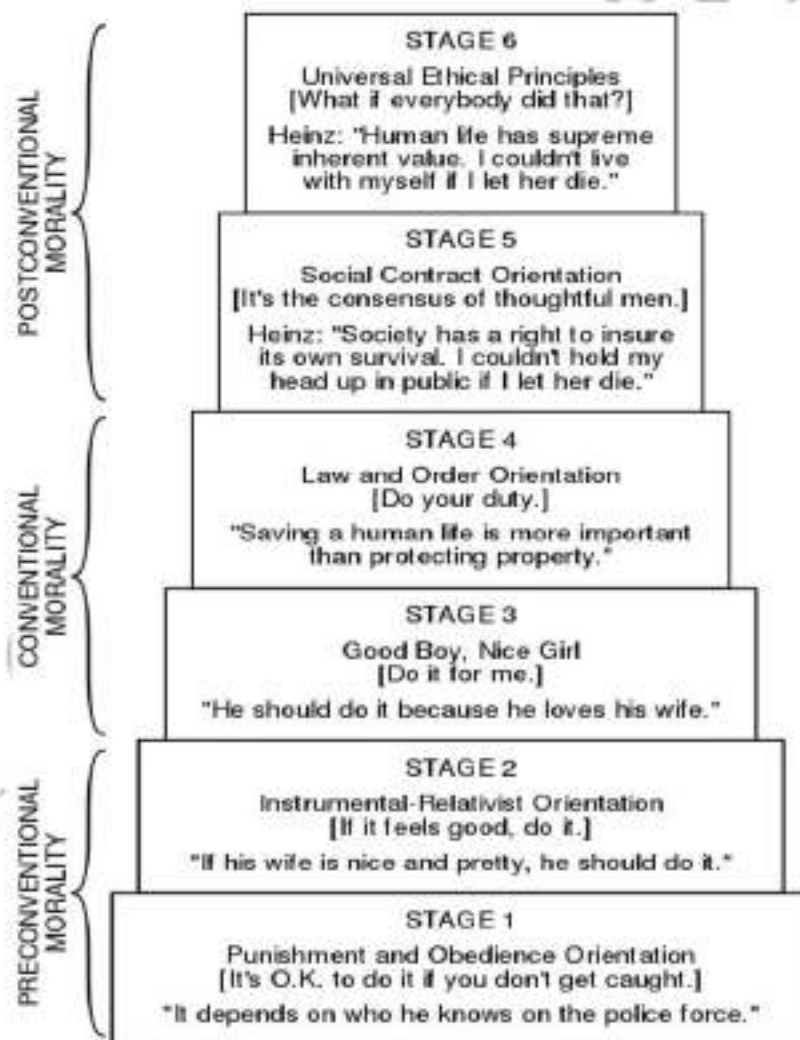
The third level of moral thinking is one that Kohlberg felt is not reached by the majority of adults. Its first stage (stage 5) is an understanding of social mutuality and a genuine interest in the welfare of others.

The last stage (stage 6) is based on respect for universal principle and the demands of individual conscience. While Kohlberg always believed in the existence of Stage 6 and had some nominees for it, he could never get enough subjects to define it, much less observe their longitudinal movement to it.

Kohlberg believed that individuals could only progress through these stages one stage at a time. That is, they could not "jump" stages. They could not, for example, move from an orientation of selfishness to the law and order stage without passing through the good boy/girl stage. They could only come to a comprehension of a moral rationale one stage above their own. Thus, according to

Kohlberg, it was important to present them with moral dilemmas for discussion which would help them to see the reasonableness of a "higher stage" morality and encourage their development in that direction. The last comment refers to Kohlberg's moral discussion approach. He saw this as one of the ways in which moral development can be promoted through formal education. Note that Kohlberg believed, as did Piaget, that most moral development occurs through social interaction. The discussion approach is based on the insight that individuals develop as a result of cognitive conflicts at their current stage.

Kohlberg's classification can be outlined in the following manner:



CRITICISM:

Kohlberg's work has been criticized by a lot of researchers. A prominent one among them is Carol Gilligan who argues that

Kohlberg's theory reflects a gender bias. In contrast with Kohlberg's justice perspective, Gilligan argues for a care perspective, which is a moral perspective that views people in terms of their connectedness with others and emphasizes interpersonal communication, relationships with others and concern for others. According to Gilligan, Kohlberg greatly underplayed the care perspective, perhaps because he was a male, because most of his research was with males rather than females, and because he used male responses as a model for his theory.

In Gilligan's conception of morality, which is different from psychologists like Freud, Piaget or Kohlberg, the moral problem arises from conflicting responsibilities rather than from competing rights and requires for its resolution a mode of thinking that is contextual and narrative rather than formal and abstract. This conception concerned with the activity of care centers moral development around the understanding of responsibility and relationships, just as the conception of morality as fairness ties moral development to the understanding of rights and rules. The morality of rights differs from the morality of responsibility in its emphasis on separation rather than connection and in its consideration of the individual rather than the relationship as primary.

This dissimilar construction of the moral problem by women has been cited as the critical reason for their failure to develop within the constraints of Kohlberg's system.

The differences in morality in men and women may as well be understood by differences in their formation of gender identity as pointed out by Gilligan (1979) in her paper 'Woman's Place in Man's Life Cycle.'

For boys and men, separation and individuation are critically tied to gender identity since separation from the mother is essential for the development of masculinity. On the other hand, for girls and women, issues of femininity or feminine identity do not depend on the

achievement of separation from the mother or on the progress of individuation. Since masculinity is defined through separation while femininity is defined through attachment, male gender identity is threatened by intimacy while female gender identity is threatened by separation. Thus, males tend to have difficulty with relationships, while females tend to have problems with individuation.

Even though Gilligan's critique is an important one, questions have been raised about Gilligan's gender conclusions (Walker and Frimer, 2009). For example, a meta-analysis casts doubt on Gilligan's claim of substantial gender differences in moral judgements (Jaffee and Hyde, 2000). Also, a research review concluded that girls' moral orientations are "somewhat more likely to focus on care for others than on abstract principles of justice, but they can use both moral orientations when needed (as can boys...)" (Blakemore, Brenbaum and Liben, 2009).

Some other criticisms of Kohlberg's theory include:

Moral thought and moral behavior:

Kohlberg's theory has been criticized for placing too much emphasis on moral thought and not enough emphasis on moral behavior (Walker, 2004).

Culture and Moral Reasoning:

Kohlberg emphasized that his stages of moral reasoning are universal, but some critics claim his theory is culturally biased (Gibbs, 2010; Miller, 2007).

Families and Moral Development:

Kohlberg argued that family processes are essentially unimportant in children's moral development. However, most experts on children's moral development conclude that parent's moral values and actions influence children's development of moral

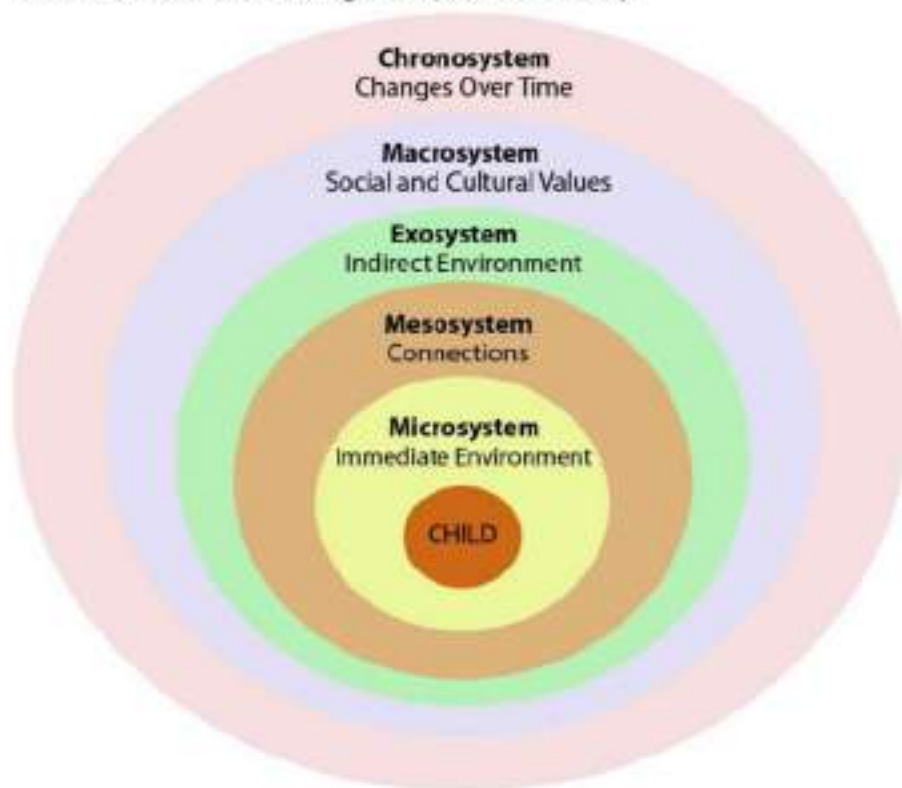
reasoning (Laible and Thompson, 2007).

BRONFENBRENNER'S ECOLOGICAL SYSTEMS THEORY

Ecological systems theory: This theory looks at a child's development within the context of the system of relationships that form his or her environment. Bronfenbrenner's theory defines complex "layers" of environment, each having an effect on a child's development. This theory has recently been renamed

"bioecological systems theory" to emphasize that a child's own biology is a primary environment fueling her development. The interaction between factors in the child's maturing biology, his immediate family/community environment, and the societal landscape fuels and steers his development. Changes or conflict in any one layer will ripple throughout other layers. To study a child's development then, we must look not only at the child and her immediate environment, but also at the interaction of the larger environment as well.

Bronfenbrenner's Ecological Systems Theory



THE MICROSYSTEM– this is the layer closest to the child and contains the structures with which the child has direct contact. The microsystem encompasses the relationships and interactions a child has with her immediate surroundings (Berk, 2000). Structures in the microsystem include family, school, neighborhood, or childcare environments. At this level, relationships have impact in two directions both away from the child and toward the child. For example, a child's parents may affect his beliefs and behavior; however,

the child also affects the behavior and beliefs of the parent. Bronfenbrenner calls these bi-directional influences, and he shows how they occur among all levels of environment. The interaction of structures within a layer and interactions of structures between layers is key to this theory. At the microsystem level, bi-directional influences are strongest and have the greatest impact on the child. However, interactions at outer levels can still impact the inner structures. The mesosystem – this layer provides the connection between the structures of the child's microsystem (Berk, 2000).

Examples: the connection between the child's teacher and his parents, between his church and his neighborhood, etc.

THE EXO-SYSTEM – this layer defines the larger social system in which the child does not function directly. The structures in this layer impact the child's development by interacting with some structure in her microsystem (Berk, 2000). Parent workplace schedules or community-based family resources are examples. The child may not be directly involved at this level, but he does feel the positive or negative force involved with the interaction with his own system.

THE MACROSYSTEM– this layer may be considered the outermost layer in the child's environment. While not being a specific framework, this layer is comprised of cultural values, customs, and laws (Berk, 2000). The effects of larger principles defined by the macrosystem have a cascading influence throughout the interactions of all other layers. For example, if it is the belief of the culture that parents should be solely responsible for raising their children, that culture is less likely to provide resources to help parents. This, in turn, affects the structures in which the parent's function. The parents' ability or inability to carry out that responsibility toward their child within the context of the child's microsystem is likewise affected.

THE CHRONOSYSTEM– this system encompasses the dimension of time as it relates to a child's environments. Elements within this system can be either external, such as the timing of a parent's death, or internal, such as the physiological changes that occur with the aging of a child. As children get older, they may react differently to environmental changes and may be more able to determine more how that change will influence them.

DEVELOPMENTAL STAGES

PRENATAL

DEVELOPMENT:Development happens

quickly during the **Prenatal Period**, which is the time between conception and birth.

Fetal development occurs in **cephalocaudal** order, beginning with the head and ending with the lower body and extremities. This sequence of development results in the head of a typical fetus being disproportionately larger than the rest of its body. This period is generally divided into three stages: the germinal stage, the embryonic stage, and the fetal stage.

STAGE 1: THE GERMINAL STAGE

The two-week period after conception is called the **Germinal Stage**. Conception occurs when a sperm cell combines with an egg cell to form a **Zygote**. About thirty-six hours after conception, the zygote begins to divide quickly. The resulting ball of cells moves along the mother's fallopian tube to the uterus.

Around seven days after conception, the ball of cells starts to become embedded in the wall of the uterus. This process is called **Implantation** and takes about a week to complete. If implantation fails, as is quite common, the pregnancy terminates. One key feature of the germinal stage is the formation of a tissue called the **Placenta**. The placenta has two important functions:

- Passing oxygen and nutrients from the mother's blood into the embryo or fetus

- Removing waste materials from the embryo or fetus

STAGE 2: THE EMBRYONIC STAGE

The **Embryonic Stage** lasts from the end of the germinal stage to two months after conception. The developing ball of cells is now called an **Embryo**. In this stage, all the major organs form, and the embryo becomes very fragile. The biggest dangers are **teratogens**, which are agents such as viruses, drugs, or radiation that can cause deformities in an embryo or fetus. At the end of the embryonic period, the embryo is only about an inch long.

STAGE 3: THE FETAL STAGE

The last stage of prenatal development is the **Fetal Stage**, which lasts from two months after conception until birth. About one month into this stage, the sex organs of the fetus begin to form. The fetus quickly grows as bones and muscles form, and it begins to move inside the uterus. Organ systems develop further and start to function. During the last three months, the brain increases rapidly in size, an insulating layer of fat forms under the skin, and the respiratory and digestive systems start to work independently.

FETAL VIABILITY

Around twenty-two to twenty-six weeks after conception, the fetus reaches the age of viability, after which it has some chance of surviving out-side the womb if it is born prematurely. The chances of a premature baby's survival increase significantly with each additional week it remains in the mother's uterus.

ADVERSE FACTORS AFFECTING FETAL DEVELOPMENT

Although the womb provides protection, the fetus remains indirectly connected to the outside world through its mother. Several factors that are linked to the mother can harm the fetus:

- Poor nutrition
- Use of alcohol
- Smoking
- Use of certain prescription or over-the-counter drugs
- Use of recreational drugs such as cocaine, sedatives, and narcotics
- X-rays and other kinds of radiation
- Ingested toxins, such as lead
- Illnesses such as AIDS, German measles, syphilis, cholera, smallpox, mumps, or severe flu

FETAL ALCOHOL SYNDROME

Mothers who drink heavily during pregnancy may have babies with fetal alcohol syndrome. Babies with this

syndrome may have problems such as small head size, heart defects, irritability, hyperactivity, mental retardation, or slowed motor development. Fetal alcohol syndrome is incurable.

INFANCY AND TODDLER-HOOD STAGE:

Developmental psychologists use the term **infancy** to denote the period of development that generally is from birth to two years of age. The word **infant** means "without language". Infancy includes development in the areas of cognition, perception, motor activity, emotion, sociability and language. In the beginning of infancy period infants can recognize human faces and after that they can differentiate between known and unknown faces and react differently. Development on different areas take place throughout infancy to toddlerhood, i.e., the first three years of life.

a. PHYSICAL DEVELOPMENT:

It refers to the changes in the body. This

development is rapid during infancy. Infants increase

their body weight almost triple and increase in height by about one-third during the first year alone. Not only body size and weight of the infant increase but also brain size expands rapidly during the first 18 months and brain weight of the infant reaches more than half of the adult brain due to rapid growth of dendrites and axons within the brain and glia cells. Physical development also includes development in vision, hearing, perceptual development etc. Infants motor development takes place in a sequential order and this type of development proceeds from head towards the limbs. Infant first controls his head and trunk, then lift his chest, sit-up-right, crawling, creeping, stand with help, stand alone with holding some

objects, and soon walking.

This development occurs due to improvement of skills and control of other body parts like legs, arms, etc. Gradually children develop their eye, head and hand coordination and are able to pick up things.

In the early life, infant's vision, hearing and perception are not clear and focused. The neonate prefers to perceive brighter colors and which are nearer to him. These senses develop as the child matures from infancy to toddlerhood. The new born is well equipped with sensory functioning for life. The child is capable to turn his head toward a loud noise or clapping. Gradually he tries to locate the source of noise and he can differentiate between the voice of mother and the father and also recognizes the voices of other people. Another capacity of the infant is his ability to learn from experience. For example, the child learns to suck faster when sucking is pleasant.

Piaget noted that the sensory motor stage of cognitive development occurs during infancy. During infancy period there are development in vision, control of muscles and nervous system, start to eat and sleep on regular intervals, sit on their own and to hold objects themselves.

REFLEXES AND MOTOR SKILLS

Because infants cannot endure on their own, newborns have specific built-in or prewired abilities for survival and adaptive purposes. **Reflexes** are automatic reactions to stimulation that enable infants to respond to the environment before any learning has taken place. For instance, babies automatically suck when presented with a nipple, turn their heads when a parent speaks, grasp at a finger that is pressed into their hand, and startle when exposed to loud noises. Some reflexes, such as blinking, are permanent. Others, such as grasping, disappear after several months and eventually become voluntary responses.

COMMON INFANT MOTOR REFLEXES APPEAR IN TABLE 1.

(OVER LEAF)

TABLE 1 Common Infant Motor Reflexes

<i>Reflex</i>	<i>Stimulus/Action</i>
Blinking	In response to a puff of air, the infant closes both eyes.
Babinski	In response to stroking the side of its foot, the infant twists its foot inward and fans out its toes.
Grasping	In response to an object pressed against its palm, the infant attempts to grasp the object.
Moro	In response to a shock or loud noise, the infant arches its back and throws its arms outward.
Rooting	In response to stroking its cheek, the infant turns its head toward the touch and attempts to suck.
Stepping	In response to holding the infant so that its feet barely touch a surface, the infant "walks."
Sucking	In response to inserting a finger or nipple into its mouth, the infant begins rhythmically sucking.
Babkin	In response to stroking its forehead, the infant turns its head and opens its mouth.
Plantar	In response to touching the ball of the foot, the infant curls its toes under.

Motor skills, or behavioral abilities, develop in conjunction with physical growth. In other words, infants must learn to engage in motor activities within the context of their changing bodies. At about 1 month, infants may lift their chins while lying flat on their stomachs. Within another month, infants may raise their chests from the same position. By the fourth month, infants may grasp rattles, as well as sit with support. By the fifth month, infants may roll over, and by the eighth month, infants may be able to sit without assistance. At about 10 months, toddlers may stand while holding onto an object for support. At about 14 months, toddlers may stand alone and perhaps even walk. Of course, these ages for each motor-skill milestone are averages; the rates of physical and motor developments differ among children depending on a variety of factors, including heredity, the amount of activity the child participates in, and the amount of attention the child receives.

Motor development follows **cephalocaudal** (center and upper

body) and **proximodistal** (extremities and lower body) patterns, so that motor skills become refined first from the center and upper body and later from the extremities and lower body. For example, swallowing is refined before walking, and arm movements are refined before hand movements.

TEMPERAMENT: Some babies have fussy personalities, while others have chirpy or quiet natures. These differences result from **Temperament**, the kind of personality features babies are born with, individual differences in behavioural styles, emotions, and characteristic ways of responding. Temperament has an emotional basis; but while emotions such as fear, excitement, and boredom come and go, temperament is relatively consistent and enduring.

Temperament—sometimes defined as a person's characteristic, biologically based way of approaching and reacting to people and situations—has been described as the *how* of behaviour: not *what* people do, but how they go about doing it (Thomas & Chess, 1977).

Temperament appears to be largely inborn and probably hereditary. But not fully

formed at birth. It can change during the early months in response to parental attitudes and treatment.

Alexander Thomas and Stella Chess, two researchers who study temperament, described three basic types of temperament: easy, slow to warm up, and difficult. In their research, 40 percent of the children were easy, 15 percent were slow to warm up, and 10 percent were difficult. The remaining 35 percent of the children displayed a mixture of these temperaments:

Easy children tend to be happy and adapt easily to change. They have regular sleeping and eating patterns and don't upset easily.

- **Slow-To-Warm-Up** children tend to be less cheerful and less adaptable than easy children. They are cautious about new experiences. Their sleeping and eating patterns are less regular than those of easy children.

Difficult children tend to be glum and irritable, and they dislike change. Their eating and sleeping patterns are irregular.

Table 8-2 Three Temperamental Patterns (according to the New York Longitudinal Study)

"Easy" Child	"Difficult" Child	"Slow-to-Warm-Up" Child
Has moods of mild to moderate intensity; usually positive	Displays intense and frequently negative moods; cries often and loudly; also laughs loudly	Has mildly intense reactions, both positive and negative
Responds well to novelty and change	Responds poorly to novelty and change	Responds slowly to novelty and change
Quickly develops regular sleep and feeding schedules	Sleeps and eats irregularly	Sleeps and eats more regularly than difficult child, less regularly than easy child
Takes to new foods easily	Accepts new foods slowly	Shows mildly negative initial response to new stimuli (a first encounter with a new person, place, or situation)
Smiles at strangers	Is suspicious of strangers	
Adapts easily to new situations	Adapts slowly to new situations	
Accepts most frustrations with little fuss	Reacts to frustration with tantrums	
Adapts quickly to new routines and rules of new games	Adjusts slowly to new routines	Gradually develops liking for new stimuli after repeated, unpressured exposures

Source: Adapted from A. Thomas & Chess, 1964.

ATTACHMENT

Attachment is a close emotional bond between two people. It is the strong affectionate tie we have with special people in our lives that leads us to experience pleasure and joy when we interact with them and to be comforted by their nearness during times of stress. By the second half of the first year, infants become attached to familiar people who have responded to their needs.

Various psychologists have provided insight on the development of attachment.

Freud: He emphasised that infants become attached to the person or object that provides oral satisfaction. For most infants, this is the mother, since she is most likely to feed the infant.

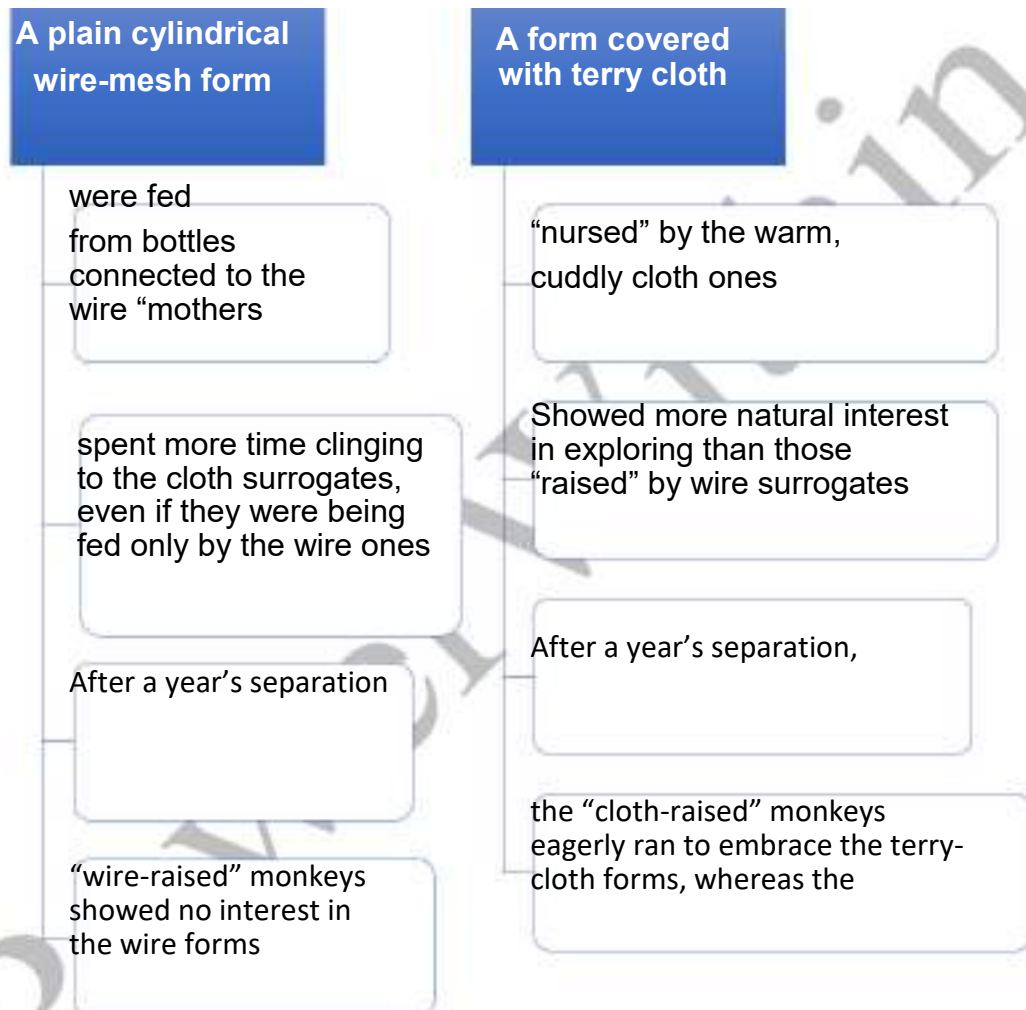
Erik Erikson: Physical comfort plays a crucial role in his view of infant's development. Physical comfort and sensitive care, according to Erikson (1968), are key to establishing a basic trust in infants.

Behaviourism: Behaviourism emphasizes the importance of feeding. According to a

well-known behaviourist account, as the mother satisfies the baby's hunger, the infants learn to prefer her soft caresses, warm smiles, and tender words of comfort because these events have been paired with tension relief.

Harlow: He emphasised the importance of contact comfort in the attachment process. A classic study done by him in 1958 revealed this. Harlow removed infant

Rhesus monkeys from their mothers at birth; for six months they were reared by surrogate mothers. One surrogate mother was made of wire, the other of cloth. Regardless of which mother fed them, the infant monkeys spent more time with the cloth mother. Whether the mother provided comfort seemed to determine whether the monkeys associated the mother with security.



Bowlby: The ethological perspective given by John Bowlby(1969,1989) also stresses the importance of attachment in the first year of life and the responsiveness of the caregiver.

BOWLBY'S ETHOLOGICAL THEORY OF ATTACHMENT

The ethological theory of attachment formulated by Bowlby views the infant's emotional tie to the familiar caregiver as

an evolved response that promotes survival through ensuring both safety and competence. In his theory, he retained the psychoanalytic idea that quality of attachment to the caregiver has profound implications for the child's feelings of security and capacity to form trusting relationships. At the same time, Bowlby was inspired by Konrad Lorenz's studies of imprinting.

DEVELOPMENTAL COURSE OF ATTACHMENT:

According to Bowlby, attachment develops in four phases:

Pre-attachment (birth to 6 weeks):

Built-in signals, such as crying and cooing, grasping, smiling and gazing into the adult's eyes- bring a newborn baby into close proximity with their caregiver. Babies of this age recognize their own mothers smell and voice. However, they are not yet attached to her, so the baby is still comfortable being left with an unfamiliar person.

Attachment-in-the-making (6 weeks to 8 months): During this phase, infants respond differently to familiar caregivers than they do to strangers. As infants interact with their parents and experience relief from distress, they learn that their own actions affect the behaviour of those around them. They begin to develop a sense of trust- the expectation that the caregiver will respond when signalled. Though they recognize the parents babies still do not protest when separated from her. Parents continue to build attachment by meeting the baby's basic needs for food, shelter, and comfort.

Clear-cut attachment (8 months to 18 months): In this phase, attachment to the familiar caregiver is evident. Babies display separation anxiety- they become upset when the adult on whom they have come to rely, leaves. Separation anxiety does not always occur; it depends on infant temperament, context and the adult's behaviour. During this phase, infants use their mother as a secure base from which to explore.

Formation of reciprocal attachment (18 months to 2 years): By the end of the second year, Rapid growth in representation and language permits toddlers to understand some of the factors that influence parent's coming and going and to predict her return. As a result, separation protest declines. Now children start to negotiate with the caregiver using requests and persuasion to alter her goals rather than clinging to her.

Bowlby argued that infants develop an internal working model of attachment, a supplemental model of the caregiver, their relationship and the self as deserving of nurturant care.

INDIVIDUAL DIFFERENCES IN ATTACHMENT PATTERNS

Mary Ainsworth (1979) proposed that the quality of babies' attachment experiences vary. She created the Strange Situation, an observational measure of infant attachment that takes about 20 minutes in which the infant experiences a series of introductions, separations and reunions with the caregiver and an adult stranger in a prescribed order.

Observing the responses of infants to these episodes, researchers have identified a secure attachment pattern and three patterns of insecurity; a few babies cannot be classified (Ainsworth et al., 1978; Barnett and Vondra, 1999; Main and Solomon, 1990).

Attachment pattern	Description
Secure attachment	➤ These infants use the parent as a secure base. ➤ When separated they may or may not cry, but if they do, it is due to the parent's absence, Since they show A strong preference for her over the stranger.

	➤ When the parent returns they actively seek contact, and their crying is reduced immediately.
Avoidant attachment	➤ These infants seem unresponsive to the parent when she is present. ➤ When she leaves, they usually are not distressed, and they react to the stranger in much the same way as to the parent. ➤ During reunion they avoid or are slow to greet the parent, and when picked up, they often fail to cling.
Resistant attachment	➤ Before separation, these infants often seek closeness to the parent and fail to explore. ➤ When she returns, they display angry, resistive behavior, sometimes hitting and pushing. ➤ In addition, many continue to cry and cling after being picked up and cannot be comforted easily.

Disorganized - disoriented attachment

- This pattern reflects the greatest insecurity.
- At reunion, these infants show confused, Contradictory behaviors.
- They might look away while being held by the parent or approach her with flat, depressed emotion.
- Most Communicate their disorientation with a dazed expression on the face.
- A few cries out after having calmed down or display odd, frozen postures.

CAREGIVING STYLES AND ATTACHMENT:

Dozens of studies report that sensitive caregiving-responding promptly, consistently and appropriately to infants and holding them tenderly and carefully-is moderately related to attachment security in diverse cultures.

Caregivers of avoidant babies tend to be unavailable or rejecting (Posada and Kaloustian, 2010). They often don't respond to their babies' signals and have little physical contact with them. When they do interact with their babies, they may behave in an angry or irritable way.

Resistant infants often experience inconsistent care. Their mothers are unresponsive to infant signals. Yet when the baby begins to explore, these mothers interfere, shifting the infant's attention back to themselves. In general, the caregivers tend not to be very affectionate

with their babies and show little synchrony when interacting with them.

Caregivers of disorganized babies often neglect or physically abuse them (Bernard and others, 2012; Bohlin and others, 2012; Groh and others, 2012).

Across several studies, a special form of communication called interactional synchrony separated the experiences of secure and insecure babies. It is best described as a sensitively tuned “emotional dance” in which the caregiver responds to infant signals in a well-timed, rhythmic, appropriate fashion.

THE CHILDHOOD STAGE:

Ages 2 through 6 are the **early childhood** years, or preschool years. This stage includes two sub-stages: one is early childhood and another is late childhood. This period covers the period between the age of 3 to 11 years. Sometimes the girls at 13 years and boys at 14 years of age are considered as adolescents and till then the child is considered being in the childhood stage. During this period significant physical and psychological changes take place.

a. PHYSICAL CHANGES

Children begin to lose their baby fat, or chubbiness, around age 3. Toddlers soon acquire the leaner, more athletic look associated with childhood. The child's trunk and limbs grow longer, and the abdominal muscles form, tightening the appearance of the stomach. Even at this early stage of life, boys tend to have more muscle mass than girls. The preschoolers' physical proportions also continue to change, with their heads still being disproportionately large, but less so than in toddlerhood.

Three-year-old preschoolers may grow to be about 38 inches tall and weigh about 32 pounds. For the next 3 years, healthy preschoolers grow an additional 2 to 3 inches and gain from 4 to 6 pounds per year. By age 6, children reach a height of

about 46 inches and weigh about 46 pounds. Of course, these figures are averages and differ from child to child, depending on socioeconomic status, nourishment, health, and heredity factors.

b. BRAIN DEVELOPMENT

Brain and nervous system developments during early childhood also continue to be dramatic. The better developed the brain and nervous systems are, the more complex behavioral and cognitive abilities children are capable of.

The brain is comprised of two halves, the right and left **cerebral hemispheres**. **Lateralization** refers to the localization of assorted functions, competencies, and skills in either or both hemispheres. Specifically, language, writing, logic, and mathematical skills seem to be located in the left hemisphere, while creativity, fantasy, artistic, and musical skills seem to be located in the right hemisphere. Although the hemispheres may have separate functions, these brain masses almost always coordinate their functions and work together.

The two cerebral hemispheres develop at different rates, with the left hemisphere developing more fully in early childhood (ages 2 to 6), and the right hemisphere developing more fully in middle childhood (ages 7 to 11). The left hemisphere predominates earlier and longer, which may explain why children acquire language so early and quickly.

Another aspect of brain development is **handedness**, or preference for using one hand over the other. Handedness appears to be strongly established by middle childhood. About 90 percent of the general population is right-handed, while the rest of the population is left-handed and/or **ambidextrous**. A person is ambidextrous if he or she shows no preference for one hand over the other. Typically, right-handedness is associated with left-cerebral dominance and left-handedness with right-cerebral dominance.

The nervous system undergoes changes in early childhood, too. The majority of a child's **neurons**, or cells that make up nerves, form prenatally. However, the **glial cells**, (nervous system support cells surrounding neurons) that nourish, insulate, and remove waste from the neurons without actually transmitting information themselves, develop most rapidly during infancy, toddlerhood, and early childhood. The **myelin sheaths** that surround, insulate, and increase the efficiency of neurons (by speeding up the action potential along the axon) also form rapidly during the first few years of life. The postnatal developments of glial cells and myelin sheaths help to explain why older children may perform behaviors that younger children are not capable of.

c. MOTOR SKILLS

Motor skills are physical abilities or capacities. **Gross motor skills**, which include running, jumping, hopping, turning, skipping, throwing, balancing, and dancing, involve the use of large bodily movements. **Fine motor skills**, which include drawing, writing, and tying shoelaces, involve the use of small bodily movements. Both gross and fine motor skills develop and are refined during early childhood; however, fine motor skills develop more slowly in preschoolers. If you compare the running abilities of a 2-year-old and a 6-year-old, for example, you may notice the limited running skills of the 2-year-old. But the differences are even more striking when comparing a 2-year-old and 6-year-old who are tying shoelaces. The 2-year-old has difficulty grasping the concept before ever attempting or completing the task.

Albert Bandura's theory of **observational learning** is applicable to preschoolers' learning gross and fine motor skills. Bandura states that once children are biologically capable of learning certain behaviors, children must do the following in order to develop new skills:

Observe the behavior in others.

Form a mental image of the behavior.

Imitate the behavior.

Practice the behavior.

Be motivated to repeat the behavior.

In other words, children must be ready, have adequate opportunities, and be interested in developing motor skills to become competent at those skills.

d. FAMILIAL RELATIONSHIPS

DIANA BAUMRIND'S (1966, 67) PROTOTYPICAL DESCRIPTIONS OF 3 PARENTING STYLES

The **permissive** parent attempts to behave in a nonpunitive, acceptant and affirmative manner towards the child's impulses, desires, and actions. She [the parent] consults with him [the child] about policy decisions and gives explanations for family rules. She makes few demands for household responsibility and orderly behavior. She presents herself to the child as a resource for him to use as he wishes, not as an ideal for him to emulate, nor as an active agent responsible for shaping or altering his ongoing or future behavior. She allows the child to regulate his own activities as much as possible, avoids the exercise of control, and does not encourage him to obey externally defined standards. She attempts to use reason and manipulation, but not overt power to accomplish her ends (p. 889).

The **authoritarian** parent attempts to shape, control, and evaluate the behavior and attitudes of the child in accordance with a set standard of conduct, usually an absolute standard, theologically motivated and formulated by a higher authority. She [the parent] values obedience as a virtue and favors punitive, forceful measures to curb self-will at points where the child's actions or beliefs conflict with what she thinks is right conduct. She believes in keeping the child in his place, , in restricting his autonomy, and in assigning household responsibilities in order to inculcate respect for work. She regards the

preservation of order and traditional structure as a highly valued end in itself. She does not encourage verbal give and take, believing that the child should accept her word for what is right (p. 890).

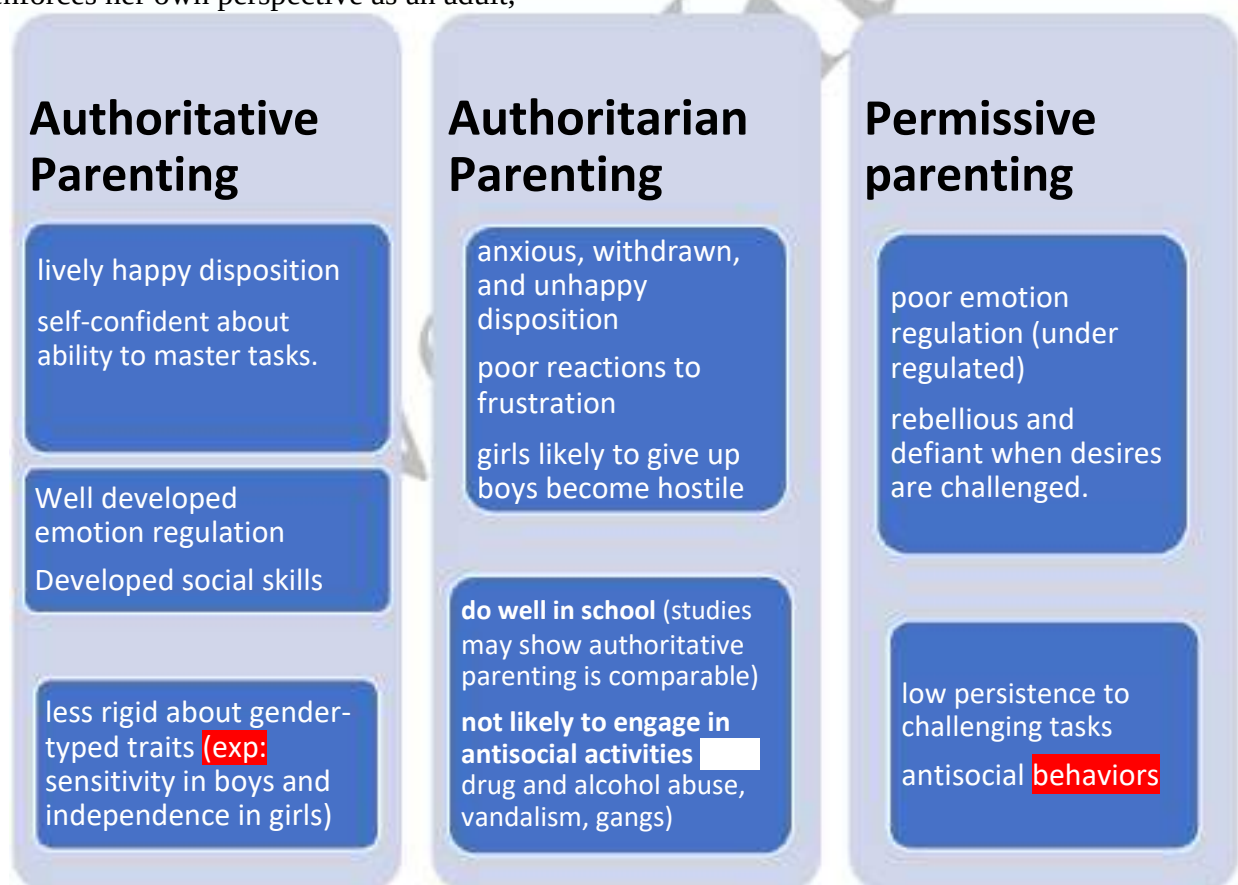
The **authoritative** parent attempts to direct the child's activities but in a rational, issue-oriented manner. She [the parent] encourages verbal give and take, shares with the child the reasoning behind her policy, and solicits his objections when he refuses to conform. Both autonomous self-will and disciplined conformity are valued. [She values both expressive and instrumental attributes, both autonomous self-will and disciplined conformity] ...

Therefore, she exerts firm control at points of parent-child divergence, but does not hem the child in with restrictions. She enforces her own perspective as an adult,

but recognizes the child's individual interests and special ways. The authoritative parent affirms the child's present qualities, but also sets standards for future conduct. She uses reason, power, and shaping by regime and reinforcement to achieve her objectives, and does not base her decisions on group consensus or the individual child's desires. [... but also, does not regard herself as infallible, or divinely inspired.]

[Note that portions in brackets are significant additions to the prototype in Baumrind (1967).]

CONSEQUENCES OF PARENTING STYLES



WHY DOES AUTHORITATIVE PARENTING WORK?

Control that appears fair and reasonable (i.e. not arbitrary) to the child is far more likely to

be complied with and internalized. Nurturing parents who are secure in the standards they hold for their children provide models of caring concern as well as confident, self-controlled behavior. A child's modeling of these parents provides emotion regulation skills, emotional understanding, and social understanding. Parents who combine warmth and rational and reasonable control are likely

to be more effective reinforcing agents. They praise children for striving to meet their expectations and making good use of disapproval, which works best when applied by an adult who has been warm and caring.

Authoritative parents make demands that fit with children's ability to take responsibility for their own behavior. Children subsequently learn that they are competent individuals who can do things successfully for themselves. This fosters high self-esteem, cognitive development, and emotional maturity.

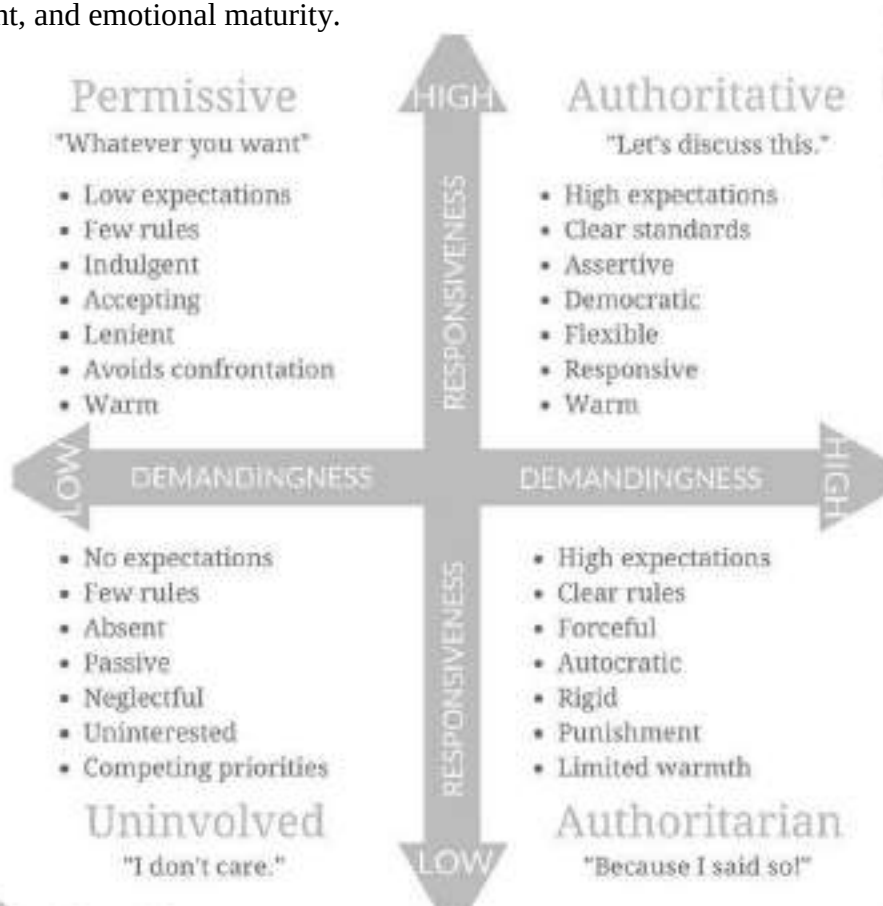
NEGLECTFUL OR UNINVOLVED PARENTING

4th style

Added by Maccoby & Martin (1983)

Parents, because of stress or depression focus on their own needs rather than on their child.

Related with a number of behavioral disorders in childhood & adolescence (Baumrind, 1991)



PEERS- PLAY:

The significance of peers is made evident very early on. Play is a voluntary, intrinsically motivated, usually non-literal activity which helps children develop new skills, mastery over their own body, cognitive skills, physical health and

PIAGET (1951) PARTEN (1932)

inculcates initiative taking and adherence to social rules.

Piaget (1951) and Parten (1932) divided playing behaviour on the basis of content and social dimension:



Stages of Play

Unoccupied play	The child is not playing, just observing
Solitary play	The child plays alone, focussed on their own activity, unaware and uninterested in what others are doing. Most common in children aged under three.
Onlooker play	The child watches others at play but does not engage in it. They may talk about the play, but they do not join in. Most common in children aged 2-3 years
Parallel play	The child plays separately but close to others often mimicking their actions. Often seen as the beginning of more complex social play. Most common in children aged 2 1/2-4 years.
Associative play	The child is interested in playing with others and interacts with others during play, but the activity is not organised or co-ordinated. Most common in children aged 3-4 years
Cooperative play	The child is interested both in the people playing and in the activity. The activity is organised, and participants have assigned roles. This is the beginning of 'team work'. Most common in children aged 4 1/2 - 6+ years.

Adapted from Mildred Parten's Stages of Play theory
http://en.wikipedia.org/wiki/Parten's_stages_of_play

ADOLESCENCE STAGE

The term adolescence is derived from the Latin word *adolescere*, which means to sprout into maturity. **Adolescence** is a transitional period in the human life span, linking childhood and adulthood. Understanding the meaning of adolescence is important because adolescents are the future of any society. Most researchers currently involved in adolescent research agree that development includes a combination of socioemotional, cognitive, and biological processes. However, the balance of how

these processes are associated with adolescent development depends on how and when adolescents grow.

Adolescence typically describes the years between ages 13 and 19 and can be considered the transitional stage from childhood to adulthood. However, the physical and psychological changes that occur in adolescence can start earlier, during the preteen or "tween" years (ages 9 through 12). As this is the transitional phase of life adolescents' rebel against their parents and society most often. Therefore, this period is labelled as storm and stress period.

Adolescence can be a time of both disorientation and discovery. This transitional period can bring up issues of independence and self-identity; many adolescents and their peers face tough choices regarding schoolwork, sexuality, drugs, alcohol, and social life. Peer groups, romantic interests, and appearance tend to naturally increase in importance for some time during a teen's journey toward adulthood.

PHYSICAL CHANGES

Pubescence refers to the two years before puberty. The adolescent growth spurt actually begins during pubescence, at about age eleven in girls and about age thirteen in boys. At this time, children get taller and heavier and develop secondary sex characteristics. **Secondary Sex Characteristics** are sex-specific physical characteristics that are not essential for reproduction. Girls develop breasts, widened pelvic bones, and wider hips. Boys develop facial hair, broader shoulders, and deeper voices. After pubescence and at the beginning of adolescence, **Puberty** occurs. Puberty is the point at which sexual organs mature. Sexual organs include the ovaries in girls

and the penis and testes in boys. **Menarche**, or the first menstrual period, marks the onset of puberty in girls. The average age of menarche for American girls is about twelve and a half. The beginning of **Nocturnal Emissions**, so-called wet dreams, marks the onset of puberty in boys. American boys typically begin to produce sperm by fourteen years of age. Girls reach full sexual maturation around age sixteen, and boys reach sexual maturity at around eighteen.

IDENTITY

**David Elkind [1967, 1976, 1978]:
IMMATURE ASPECTS OF
ADOLESCENT THOUGHT:**

Imaginary audience is the belief that others are as pre-occupied with the adolescent's behavior as he or she is. Attention-seeking behavior, so common in early adolescence, may reflect this interest in an, imaginary audience, that is the desire to be noticed, visible, and on stage. Particularly during early adolescence, individuals see themselves as constantly on stage, believing they are the main actors and all others are the audience.

The construction or representation **personal fable** on the other hand refers to the adolescent's sense of personal **uniqueness and indestructibility**. Their sense of personal uniqueness makes them feel that no one can understand how they really feel - their perception that they are special. Another aspect of the personal fable involves the belief that one is indestructible. This results in feelings of invulnerability or insusceptibility and therefore risk-taking behaviors such as alcoholism, drug and substance abuse as well as sexual promiscuity. For reasons likely tied to an emerging interest in idealism and the ability to think in more

abstract and hypothetical ways, young adolescents often get caught up in a mental world far removed from reality. This may entail the belief that things just can't or won't happen to them and that they are omnipotent and indestructible.

ERIKSON AND IDENTITY

5th stage: Identity vs. Role confusion

Developing a coherent sense of self and role one has to take up

Modifying & synthesizing earlier identifications into a new

PSYCHOLOGICAL STRUCTURE

greater than the sum of its parts (Kroger, 1993)

Organizing abilities, needs, interests, desires that can be expressed in a social context.

Danger of this stage is: identity or role confusion

- Confusion takes 2 forms:
 - isolation or lost identity in the crowd

Some degree of identity confusion always present: accounting for **the chaos associated with this stage**.

MARCIA'S THEORY

Probably the most famous person who studied the formation of identity is a man named James Marcia. Marcia decided to interview lots of different people to ask how they developed their beliefs, preferences, and so on. He believed that identity is mostly based on two things. The first is what he called **occupation**, which refers to the main role one has in society. Occupation might refer to a job, like a doctor or bus driver, but it could also apply to roles such as father, student, or celebrity. The second base for identity, according to Marcia, is **ideology**, which refers to a person's fundamental beliefs

about topics like religion or politics. Are you religious? Are you involved in politics? Marcia tried to identify different people's occupations and ideologies in order to identify their current identities.

Based on his interviews, Marcia proposed that there are four major **identity statuses**,

or degrees to which a person has achieved a set identity. As we go through them, think about your own identity.

IDENTITY STATUS

		COMMITMENT	
		PRESENT	ABSENT
Exploration	Present	Identity Achievement Description: Has committed to an identity after exploring multiple possibilities Characteristics: Active problem-solving style, high self-esteem, feelings of control, high moral reasoning, and positive views of work and school	Identity Moratorium Description: Has not committed to an identity but is exploring alternatives Characteristics: Information-seeking, active problem-solving style, open to experience, anxiety, experimentation with alcohol or substance use
	Absent	Identity Foreclosure Description: Has committed to an identity without having explored multiple possibilities Characteristics: Avoid reflecting on their identity choice, not open to new information, especially if contracts their position, rigid and inflexible	Identity Diffusion Description: Has neither committed to an identity nor explored alternatives Characteristics: Avoidance; tend to not solve personal problems in favor of letting issues decide themselves, academic difficulties, apathy, and alcohol and substance use

ADOLESCENT-PEER RELATIONS

The phase of adolescence is dominated by peer influence and the formation of more intense and intimate bonds between individuals.

PEER PRESSURE AND CONFORMITY: Adolescents conform to peer standards much more than children do. Students of approximately 9th grade peak at adolescent peer conformity, especially to antisocial standards (Brendt, 1979). This influence can be both positive and negative.

CLIQUEs AND CROWDS: A clique is a group of several people, but it is small

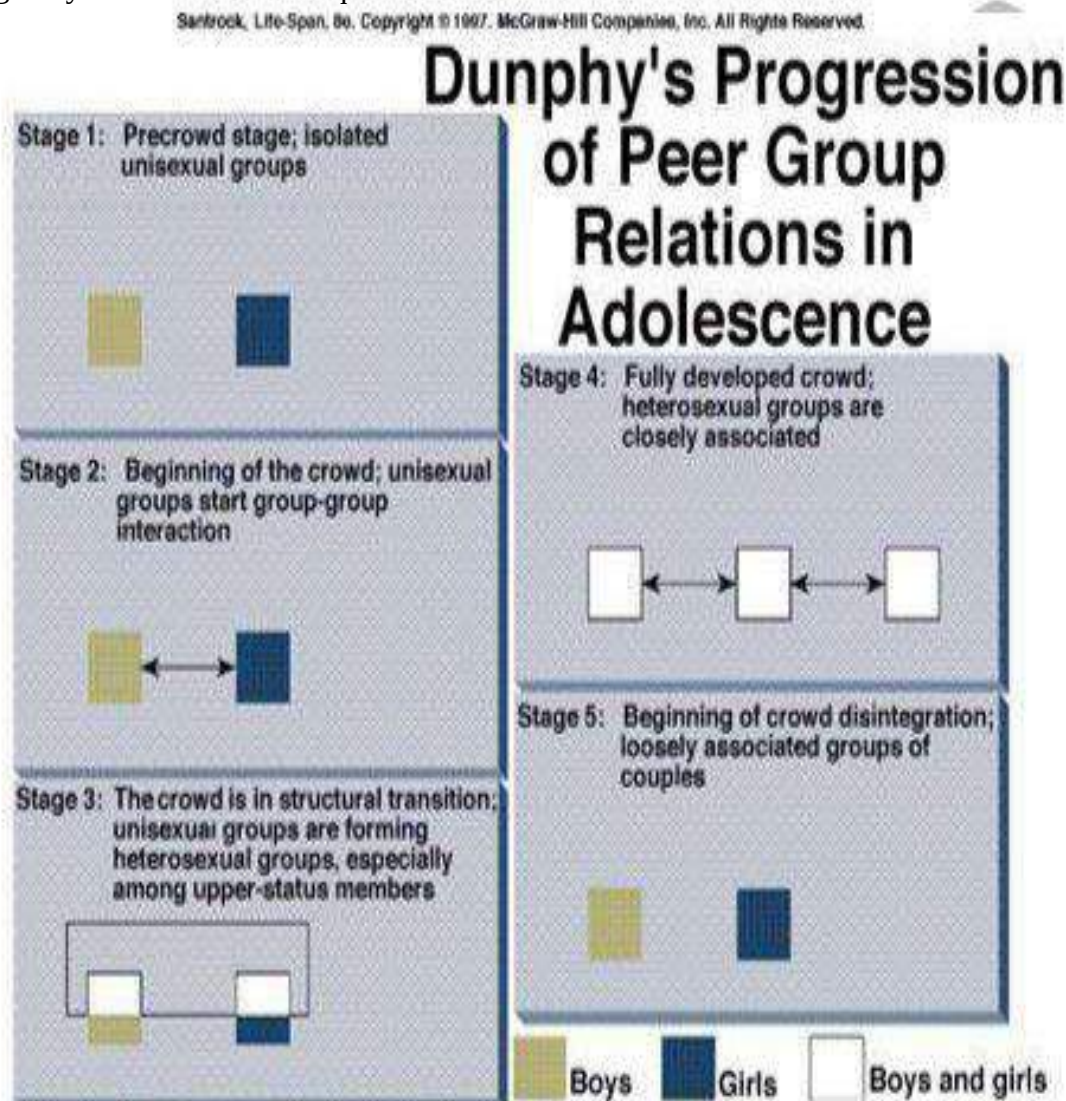
enough that all members interact. Often cliques are made up of same sex and same age individuals with similar interests. The choice of membership in certain cliques is evidence of a child's increasing control of their choice in their peer interactions.

Crowds are more reputation based. The individuals in these crowds may not necessarily be friends. They are often groups that have been labelled and stereotyped. There are three ways in which crowds influence adolescent social life: help them learn about alternative social identities that are available to them, they strongly influence whom they are likely to meet and spend time with and end up

shaping their members interpersonal relations.

ADOLESCENT GROUPS V/S CHILD GROUPS: The number of cross-sex relations increase in adolescent groups. These adolescent groups include a broader array of members and have more defined rules. Dexter Dunphy's (1963) supports the notion that while late childhood sees the development of same-sex cliques, during early adulthood these cliques start

interacting and this leads to the formation of unisex cliques (especially comprising of high-status members of the former cliques). These unisex cliques are closely associated with other such cliques, making a cohesive crowd. As adolescence progresses, these crowds disintegrate and become loosely associated groups of couples.



BULLYING

Bullying is defined as verbal or physical behavior that is intended to disturb someone less powerful (Nansel & others, 2001). A national US survey found that nearly one in three students have had occasional or frequent involvement as either victim, or perpetrator of bullying

(Nansel and others, 2001). The same study found that girls were bullied more for their looks, speech and are subject to rumors and sexual comments or gestures. Boys were bullied more for their religion, race and were hit, slapped and pushed. Children who were bullied reported more loneliness and difficulty in making friends. Research has also shown that anxious, socially withdrawn (since they won't retaliate) and aggressive children (since

they irritate the perpetrators) are more likely to get bullied (Hanish and Guerra, 2004). A 2004 Netherlands study found that victims of bullying are prone to having from headaches, sleeping problems, abdominal pain, lethargy and depression (Fekkes et. al., 2004). Perpetrators were more likely to have low grades and smoke and drink alcohol. Children who've been both bullies and victims display the highest level of conduct, school and relationship problems and bullies enjoyed the highest standing among classmates (Juvonen, Graham and Schuster, 2003).

The following methods can be adopted to curb bullying:

➤ Get older peers to monitor and intervene	➤ Policy level changes in institutions
➤ Provide peer support groups for those who're bullied	➤ advertising of anti-bullying messages at a community level
➤ Involving parents to inculcate positive behaviour	➤ skills training help bullies in early stages to modify behaviour
➤ Take help from mental health professionals	➤ Reinforce positive behaviour and model positive behaviour

GENDER AND PEER RELATIONS

Gender influences the children's group composition, size, and interactions, It has been seen that gender typed behaviors increase in children if their playmates are of the same sex.

Group Composition: Even from the tender age of 3 years, preference for same-sex playmates is evident and increasingly prominent. This preference continues to increase from age 4-10.

Group size: From age 5 onwards, boys tend to having larger groups of playmates as compared to girls.

Interactions in same-sex groups- Boys are more likely to participate in organized group games and rough and tumble play. They engage in competition, conflict, ego display, risk taking, dominance seeking, and group goal oriented behaviours. Girls are more likely to engage in "collaborative discourse".

ADULTHOOD STAGE

An adult is someone who is responsible, mature, self-supporting and well-integrated into society. Also, people do not develop these attributes and characteristics at the same time and with same skills. This adult stage has three sub-stages of development. First stage is early adulthood, second stage is middle adulthood, and third stage is late adulthood.

In early childhood period adults are at the peak of physical health, strength and energy. Sensory and motor systems are at their highest functioning. Adults gain weight and their height increases. They develop intimate relationships, they marry and begin their families. At this period of life adults take many important decisions of life like choice of career, type of friends, residence, etc. independently and accept responsibility and consequences for their own decisions. Every adult tries for recognition, job security and to excel in his job.

Middle adulthood period is explained in terms of a gradual decline in one's physical abilities, physical health, stamina etc., but the decline is gradual in nature. Both men and women feel tired easily. In this period people experience stress due to the double responsibility of caring for the aged parents and the growing children. Women usually experience hormonal changes which results in the loss of ability to reproduce, a process called menopause. Many psychological and

intellectual changes occur in middle age.

During this period, women express more assertiveness and men are more nurturant. The term “late adulthood” is roughly equivalent to old age. This is the final stage of physical change.

Robert Havighurst lists major tasks in the middle years:



Accepting and adjusting to physiological changes, such as menopause.

Reaching and maintaining satisfaction in one's occupation.

Adjusting to and possibly caring for aging parents.

Helping teenage children to become responsible adults.

Achieving adult, social and civic responsibility.

Relating to one's spouse as a person.

Developing leisure time activities.

Some people suffer from depression due to middle age crisis by thinking that some important

years of their lives are over while some others try hard to achieve their goals by changing their

lifestyles. Some people engage themselves in creative activities to get self-

satisfaction. Other people get worried about the future of their children and some start worrying about their post retirement activities.

PHYSICAL DEVELOPMENT

Physical growth and development is at its maximum during this period. Physical strength usually is more than previous years due to mature physical structures. strength, speed, coordination and endurance for activity is greatest during this period. A number of sensory

and neural functions are optimal levels during this period. Full brain weight and mature brain wave patterns are observed at this stage of development. Changing life style pattern has an impact on growth and development.

In the middle adulthood period (during 20s) there is a decline in our physical stamina and health. At that time physical functioning are generally perfect, but physical attractiveness declines and biological changes in the sexual life occurs. The physical

changes like loss of elasticity of the skin, resulting in facial lines

as well as looseness in other parts of the body are easily noticeable through external

physical appearance. There is a marked change in visual activity. People suffer

from presbyopia – the far sightedness associated with aging. The

auditory problem is also associated with increasing age, i.e. presbycusis. It means

progressive loss of hearing. In this period loss of taste, smell and

sensitivity to touch also occurs. Some physiological changes like enlargement

of prostate gland, less enzyme secretion in the

gastrointestinal tract, diminished ability to pump blood, reduced kidney functioning, etc.

take place.

PSYCHO-SOCIAL DEVELOPMENT

During early years of adulthood, people face the problem of choosing,

preparing for and entering into careers brings a lot

of social changes in the adult. They have cordial relationships with their siblings.

They spend very few times with friends because their energies are consumed

for family and work. Middle adulthood people have children of adolescent age.

It is very difficult to handle their problems. Conflicts occur between

parents and children regarding “giving” and “getting” independence. Parents

have to help their children in their search for identity. Middle aged adult

also has important responsibilities towards their

parents. In this period, they also try to be satisfied at workplace. People are worried about

their jobs and pay packages because they have to meet the daily needs of

themselves and of the family.

COGNITIVE DEVELOPMENT

Intellectual ability and cognitive skills are high in early adulthood. Middle adulthood people cannot learn new skills. Recent research suggests that intellectual development continues into late adulthood. Intellectual development continued and that are influenced by the accumulation of the experiences of life, i.e. verbal skills, social knowledge and moral judgments. During this period people show changes in logic and morality. It is observed that creativity peaks in the early adulthood but forms of creativity that require experience, revision and interpretation either remain unchanged or increase in middle age. People utilize their cognitive skills and creativity particularly at the workplace in order to get recognition. Studies revealed that intelligence declines with age, but there is no certainty that intelligence and age are related with each other.

LATE ADULTHOOD STAGE

The term late adulthood is rightly equivalent to old age. This is the final stage of physical change. It is difficult to consider the exact age of a person for being old. The onset of old age may vary from country to country. Although the retirement age is considered as one of the criteria of old age, it varies from job to job. For example, in India the retirement age for government organization is 58 years while for Universities it is 62 years and in some other institutions like IIT and IIM it is 65 years. Now-a-days people beyond 65 years are considered as old. It is the period of decline where the person thinks that he has done what he wanted to do and most of his life span is over.

Today researchers used **two categories to describe old age people.**

The young Old: This category of older people irrespective of their

actual age, are vital, vigorous and active.

The Old Old: Those older individuals who are infirm, inactive and suffer from physical problems include in this group.

As most of older persons suffer from physical problems, they became less active. There are changes in body like wrinkles in face, rough skin, gray hairs, problems in hearing and eyesight, bones and joints problems, change in voice, etc. They may suffer from various diseases. Old people also suffer from irreversible mental problem, which is organic in nature and called Alzheimer's disease.

During late adulthood period significant changes take place which are mentioned below.

a. PHYSICAL DEVELOPMENT

The experiences of old age are not same for everybody. Some people are worried about old age particularly those who are single or there is nobody in the family to support them economically or have continued physical ailments. It is a period in one's life span that is characterized by physical decline. There are two types of physical changes during old age. They are (a)

Primary ageing and (b) Secondary ageing. Primary ageing is that in which bodily deterioration occurs by the passage of time or by genetic factors which is gradual and inevitable. Secondary ageing results from disease, disuse or abuse of body which is preventable. Many structural and functional changes occur as people reach old age. Common physical changes like decrement in sensory abilities of vision, hearing, taste, smell and other senses, skin becomes inelastic, decrease in fat and muscle tissue, loss of teeth, deterioration of bone tissue, decline in

height due to compression of spinal disc and postural drop etc. occur due to primary ageing. Physical changes occur due to secondary ageing are also different for different people. For example, physical changes of diabetic people are different from people who are suffering from arthritis. Old people are also inefficient in sensory-motor coordination. They cannot respond to the physical demands of stressful situations as quickly or efficiently as possible. They are prone to accidents due to decline of reserve capacity of physical energy. Thus, ageing affects appearance, sensation and motor abilities of a person.

However, despite the problems of old age some people lead a happy life due to their positive thinking towards life and they are physically fit by doing yoga and exercises. They are developing and enjoying their hobbies and their perception of old age for relaxation help them free from different ailments.

PSYCHO-SOCIAL DEVELOPMENT

Many psycho-social changes happen to people during old age. The statement that old age is a period for waiting death can prove to be a myth. Commonly people believe that old age is a curse so far as physical and psychological aspects of life are concerned.

According to Erikson during late adulthood the outcome of a well-integrated person is wisdom. Wisdom means accepting life as it is, accepting imperfections in self, parents and in life and having no regrets. Old people remain in a state of inward depression and dissatisfaction. These internal feelings are manifested in reaction formations for which they become over sensitive and demanding. Some old people depend on their children to a large extent.

This undue dependence makes them sad. But those people who are independent and less expectations from their children and other support system, they are happy in their old age and they engage themselves in various activities. They think that old age is the golden age because they are free from family responsibilities and burdens of job. There are many social and cultural factors

Influences the process of successful ageing. Financial security and close relationship with children or other relatives or good friends make people healthy in old age.

Ageing is a process of growing old. The most important theory that explains the psychosocial aspects of ageing are:

(I) THE ACTIVITY THEORY, AND (II) THE DISENGAGEMENT THEORY.

The Activity Theory: This theory states the relationship between keeping active and ageing well. According to this theory, persons who age successfully keep themselves busy in many activities and they find substitutes for activities which they have lost due to retirement and also, they perform many roles. As role decreases, ageing increases.

The Disengagement Theory: This theory reveals that successful ageing by mutual withdrawal between society and the older person. Generally, older people voluntarily minimize their activities by retiring and also society encourages this by making individuals retire.

COGNITIVE DEVELOPMENT

People commonly believe that with old age intellectual decline occurs. Intellectual decline is associated with respect to certain functions such as; speed, perceptual integration ability, memory and inductive reasoning. People show variability in intellectual functioning in late adulthood period. Researchers classified intelligence into two types. They are: (1) Fluid intelligence and (2) Crystallized intelligence. Fluid intelligence declines as age increases. Fluid intelligence involves ability to perceive relations, form concepts and reason abstractly. Crystallized intelligence involves the ability to

remember and use learned information. Fluid intelligence includes reasoning, memory and information, processing capabilities, whereas crystallized intelligence is the acquired knowledge of the person. During old age people forget that may lead to dementia or Alzheimer's disease which is possible due to Fluid intelligence, because it relates to neurological functioning. On the other hand, crystallized intelligence helps in storing information. Therefore, in old age people should keep their minds active by mental exercises such as problem-solving analysis and other operations etc.

	Domain of Development				
	Age Range	Age Period	Physical Development	Cognitive Development	Emotional & Social Development
P e r i o d o f D e v e l o p m e n t	conception to birth	Prenatal	Physical growth is the most rapid of the lifespan.	Research indicates that recognition of such items as a story may begin.	Recognition of significant voices, such as mother, is evident.
	birth to age 2	Infancy & Toddlerhood	Brain reaches more than 2/3 of its adult size.	Comprehension and use of language develop rapidly.	Attachment to parents and others form.
	2 to 6 years	Early Childhood	Appearance becomes more slender and proportions become more adult like.	Immaturity of thought leads to some illogical ideas about the world.	Gender identity develops.
	6 to 11 years	Middle Childhood	Growth slows.	Children begin to think logically but concretely.	Peers become increasingly important.
	11 to 18 years	Adolescence	Reproductive maturity arrives.	Immature thinking persists in some attitudes and behaviors.	Search for identity becomes central.
	18-40 years	Early Adulthood (including Emerging Adulthood)	Athletic skills peak then decline.	Narrows vocational options and settles into specific career.	May cohabitate, marry and bear children.
	40-65 years	Middle Adulthood	Vision often declines.	Decreases in processing speed and working memory; increases in knowledge and problem solving.	Desire to give to and guide the next generation intensifies.
	65-years to death	Late Adulthood	Overall decline in physical abilities.	Overall decline in memory abilities; increased use of external aids.	Comes to terms with life (and death).

PSYCHOPATHOLOGY

The term "psychopathology" was first used in 1913 when this scientific discipline was introduced by Karl Jaspers, a German/Swiss philosopher and psychiatrist. This new framework for understanding the mental experience of individuals followed a long history of varied attempts at making meaning out of the "abnormal experiences."

Just like *pathology* is the study of the nature of disease (including causes, development, and outcomes), *psychopathology* is the study of the same concepts within the realm of mental health (or illness).

What we might consider normal behavior is difficult to define. Equally difficult is understanding what abnormal behavior. The American Psychiatric Association, in its publication, the Diagnostic and

Statistical Manual of Mental Disorders, 5th edition (DSM-5 for short), states that though “no definition can capture all aspects of all disorders in the range contained in the DSM-5” certain aspects are required. These include:

- **DYSFUNCTION**— includes “clinically significant disturbance in an individual’s cognition, emotion regulation, or behavior that reflects a dysfunction in the psychological, biological, or developmental processes underlying mental functioning”. Abnormal behavior, therefore, has the capacity to make our well-being difficult to obtain and can be assessed by looking at an individual’s current performance and comparing it to what is expected in general or how the person has performed in the past. As such, a good employee who suddenly demonstrates poor performance may be experiencing an environmental demand leading to stress and ineffective coping mechanisms. Once the demand resolves itself the person’s performance should return to normal according to this principle.
- **DISTRESS**— When the person experiences a disabling condition “in social, occupational, or other important activities”. Distress can take the form of psychological or physical pain, or both concurrently. Alone though, distress is not sufficient enough to describe behavior as abnormal. Why is that? The loss of a loved one would cause even the most “normally” functioning individual pain. An athlete who experiences a career ending injury would display distress as well. Suffering is part of life and cannot be avoided. And some people who display abnormal behavior are generally positive while doing so.

DEVIANCE— Closer examination of the word abnormal shows that it

indicates a move away from what is normal, or the mean (i.e. what would be considered average and in this case in relation to behavior), and so is behavior that occurs infrequently (sort of an outlier in our data). Our **culture**, or the totality of socially transmitted behaviors, customs, values, technology, attitudes, beliefs, art, and other products that are particular to a group, determines what is normal and so a person is said to be deviant when he or she fails to follow the stated and unstated rules of society, called **social norms**. What is considered “normal” by society can change over time due to shifts in accepted values and expectations. For instance, homosexuality was considered taboo in the U.S. just a few decades ago but today, it is generally accepted.

The APA (2000) defines a psychological disorder as “a clinically significant behavioral or psychological syndrome or pattern that occurs in an individual and that is associated with present distress (e.g., a painful symptom) or disability (i.e., impairment in one or more important areas of functioning) or with a significantly increased risk of suffering death, pain, disability, or an important loss of freedom”

ATTEMPTS TO DEFINE ABNORMAL BEHAVIOR ARE:

1. VIOLATION OF SOCIAL NORMS

What is considered socially acceptable changes overtime (e.g. homosexuality) and differs between cultures. Also, a lot of great leaders violate social norms (Martin Luther King)

2. STATISTICAL DEVIATION

Many behaviors are normally distributed (bell shaped curve).

Not all statistically infrequent qualities considered bad e.g. Superior intelligence

3. MALADAPTIVE BEHAVIOR

- Behavior that causes impaired functioning for people
- Not necessarily the behavior which is the problem, it's the outcome

CRITICISM-

Definition doesn't consider personal context/distress i.e. John's gambling may be causing him a lot of distress, but it isn't affecting his work, relationship or health, is it still normal? Another criticism that it is difficult to identify what may be considered a maladaptive behavior with reference to psychological processes (i.e. thought/cognition)

4. PERSONAL DISTRESS

Cause a person to be upset or uncomfortable

- Assumes it is normal not to feel upset
- Emphasises subjective experience and on the reaction to a situation as opposed to the outcome
- Criticism- person could feel no distress (lack of insight) but behaviour could still be considered abnormal (i.e. psychopath) Also, distress is not necessarily abnormal

NORMALITY - IS EVEN MORE DIFFICULT TO DEFINE, BUT MOST WOULD AGREE ON:

Efficient perception of reality
Self-knowledge
Voluntary control of behavior
Self-esteem & acceptance
Ability to form affectionate relationships
Productivity

ABNORMAL PSYCHOLOGY: The scientific study of abnormal behavior, with the intent to be able to reliably predict, explain, diagnose, identify the causes of, and treat maladaptive behavior is what we refer to as **abnormal psychology**.

Abnormal behavior can become pathological in nature and so leads to the scientific study of psychological disorders, or **psychopathology**. **Mental disorders** are characterized by psychological dysfunction which causes physical and/or psychological distress or impaired functioning and is not an expected behavior according to societal or cultural standards.

Epidemiology is the scientific study of the frequency and causes of diseases and other health-related states in specific populations such as a school, neighborhood, a city, country, and the world.

Psychiatric or mental health epidemiology refers to the occurrence of mental disorders in a population. In mental health facilities, we say that a patient presents with a specific problem, or the **presenting problem**, and we give a **clinical description** of it which includes information about the thoughts, feelings, and behaviors that constitute that mental disorder. We also seek to gain information about the occurrence of the disorder, its cause, course, and treatment possibilities.

Occurrence can be investigated in several ways. First, **prevalence** is the percentage of people in a population that has a mental disorder or can be viewed as the number of cases per some number of people. For instance, if 20 people out of 100 have bipolar disorder, then the prevalence rate is 20%. Prevalence can be measured in several ways:

✓ **Point prevalence** indicates the proportion of a population that has the characteristic at a specific point in time. In other words, it is the number of active cases.

Period prevalence indicates the proportion of a population that has the characteristic at any point during a

given period of time, typically the past year.

Lifetime prevalence indicates the proportion of a population that has had the characteristic at any time during their lives.

Incidence indicates the number of new cases in a population over a specific period of time. This measure is usually lower since it does not include existing cases as prevalence does.

Finally, **comorbidity** describes when two or more mental disorders are occurring at the same time and in the same person.

The **etiology** is the cause of the disorder. There may be social, biological, or psychological explanations for the disorders beginning which need to be understood to identify the appropriate treatment. Likewise, the effectiveness of a treatment may give some hint at the cause of the mental disorder.

The **course** of the disorder is its particular pattern. A disorder may be *acute* meaning that it lasts a short period of time, or *chronic*, meaning it lasts a long period of time. It can also be classified as *time-limited*, meaning that recovery will occur in a short period of time regardless of whether any treatment occurs.

Prognosis is the anticipated course the mental disorder will take. A key factor in determining the course is age, with some disorders presenting differently in childhood than adulthood.

Treatment is any procedure intended to modify abnormal behavior into normal behavior. The person suffering from the mental disorder seeks the assistance of a trained professional to provide some degree of relief over a series of therapy sessions. The trained mental health professional may prescribe medication or utilize psychotherapy to bring about this change.

MENTAL STATUS EXAMINATION:

The mental status examination (MSE) is a is based on your observations of the client. It is not related to the facts of the client's situation, but to the way the person acts, how the person talks, and how the person looks while in your presence. The purpose is to evaluate, quantitatively and qualitatively, a range of mental functions and behaviors at a specific point in time. The MSE provides important information for diagnosis and for assessment of the disorder's course and response to treatment. Observations noted throughout the interview become part of the MSE, which begins when the clinician first meets the patient. Information is gathered about the patient's behaviors, thinking, and mood. At an appropriate point in the evaluation the formal MSE is undertaken to compile specific data about the patient's cognitive functioning. Earlier informal observations about mental state are woven together with the results of specific testing. For example, the interviewer will have considerable information about attention span, memory, and organization of thought from the process of the interview. Specific questions during the formal exam clarify more precisely the degree of attention or memory dysfunction.

To assess properly the patient's mental status, it is important to have some understanding of the patient's social, cultural, and educational background. What may be abnormal for someone with more intellectual ability may be normal for someone with less intellectual ability. Patients for whom English is a second language may have difficulty understanding various components of the MSE, such as the proverbs. Age may be a factor. In general, patients over the age of 60 years tend to do less well on the cognitive elements of the MSE. Often this is related to less education rather than to aging alone.

MAIN COMPONENTS OF MSE (MENOMONIC- ASEPTIC)

APPEARANCE AND BEHAVIOUR

Age, sex, race, body build, posture, eyecontact, dress, grooming, manner, attentiveness to examiner, distinguishing features, prominent physical abnormalities, emotional facial expression, alertness

SPEECH

Rate, rhythm, volume, amount, articulation, spontaneity

EMOTION (MOOD AND AFFECT)

Stability, range, appropriateness, intensity, affect, mood

PERCEPTION

Hallucinations, illusions, depersonalization, derealization

THOUGHT CONTENT AND PROCESS

Suicidal ideation, death wishes, homicidal ideation, depressive cognitions, obsessions, ruminations, phobias, ideas of reference, paranoid ideation, magical ideation, delusions, overvalued ideas.

Thought content- Associations, coherence, logic, stream, clang associations, perseveration, neologism, blocking, attention.

INSIGHT AND JUDGEMENT

Insight	Assuming the patient has difficulties and/or an illness, does the patient understand this? Good insight: Patient understands they are ill and need treatment (similar to being in action phase)) Partial insight may indicate that the patient acknowledges a problem, but is not willing
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	to seek appropriate help or treatment (similar to being contemplative) Poor insight means that the patient does not see that they are ill nor does the patient need any help or treatment (similar to being pre-contemplative)
Judgment	Is patient able to use facts and make reasonable decisions? May be good, fair, impaired

7. COGNITION

Level of consciousness	Alert, confused, lethargic, stuporous
Orientation in 3 spheres	Name: What is your name? Place: Where are you right now? Time: What year, month, day is it?
Attention/Concentration	How well does the patient seem to be able to focus? (Good, poor)
Memory	How well can the patient remember? Short-term: Can the patient recall recent things that have happened? Long-term:

	Can the patient recall distant events?
Intelligence (globally and intellectual functions)	Based on your observations and patient's use of speech, does the patient's overall intelligence and cognition appear to be 1) below average, 2) average, or 3) above average?

Several shortened forms of the MSE have been developed as screening instruments. All are composed of a combination of measures to detect cognitive impairments more accurately. Although helpful, such exams must be combined with clinical history. The diagnosis of dementia and delirium also requires the demonstration of a decline in cognitive functioning from a higher baseline. All screening exams have difficulty in identifying patients with mild cognitive impairment and patients with focal neurologic lesions, such as subdural hematomas or meningiomas. The key point is that *MSEs should not be used as the sole criteria for diagnosing delirium or dementia.*

OTHER SCREENING EXAMS-

The Mini-Mental State Examination (MMSE)

is probably the best known.

The MMSE tests orientation, immediate and short-term memory, concentration, arithmetic ability, language, and praxis. It takes about 10 minutes to administer.

The Cognitive Capacity Screening Examination (CCSE) tests orientation, serial subtraction, memory, and similarities. It is less sensitive to delirium or dementia in the elderly.

Neurobehavioral Cognitive Status Examination (NCSE) is especially good for medically ill patients; it focuses on consciousness, orientation, attention, language, construction, memory, calculations, and reasoning. It tends to be more sensitive in detecting impairment because it is more detailed.

CLASSIFICATION-

The term classification refers to any effort to construct groups or categories and to assign objects or people to these categories or groups on the basis of their attributes, characteristics or relations.

Emil Kraepelin (late 19th century): The "father of psychiatry" (Weckowicz, 1984) gave the first truly comprehensive system medical classification scheme. In it he defines two major groups of mental disorders: the manic-depressive psychoses, and dementia praecox (i.e. schizophrenia) with further subdivisions. Kraepelin's system is important because it defined the discipline of psychiatry as distinct from neurology, and as a part of the general field of medicine.

Considering the difficulty in distinguishing normal from abnormal, categorizing & diagnosing different types of abnormalities can be difficult.

TERMINOLOGY In the past decade or so, psychologists have begun to change the terminology used in referring to people with psychological disorders.

We say "*a person with autism*" instead of an "*autistic person*" for very good reasons. People are not their disorders, & much is happening in this child's life that has nothing to do with autism.

Similarly, we say a *person with schizophrenia* rather than a *schizophrenic*.

CLASSIFICATION SYSTEMS

There are two classification systems in use currently in the world: DSM and ICD.

DSM: The Diagnostic and Statistical Manual of Mental Disorders (DSM) given by American Psychiatric Association (APA). This is different from American psychological association also referred to as APA).

ICD: International Statistical Classification of Diseases (ICD) developed by WHO

DSM (DIAGNOSTIC AND STATISTICAL MANUAL OF MENTAL DISORDERS)

The Diagnostic and Statistical Manual of Mental Disorders is the handbook widely used by clinicians and psychiatrists in the United States to diagnose psychiatric illnesses. Published by the American Psychiatric Association (APA), the DSM covers all categories of mental health disorders for both adults and children. It contains descriptions, symptoms, and other criteria necessary for diagnosing mental health disorders. It also contains statistics concerning which gender is most affected by the illness, the typical age of onset, the effects of treatment, and common treatment approaches.

DSM HISTORY

The *Diagnostic and Statistical Manual* has been updated seven times since it was first published in 1952.

Version	Year	Number of	diagnoses	Version	Year	Number of diagnoses
DSM-I	1952	106	DSM-II	1968		182

Version	Year	Number of diagnoses	disorder and histrionic personality disorder. Personality disorders cause significant problems in how a person relates to the world, while mental retardation is characterized by intellectual impairment and deficits in other areas such as self-care and interpersonal skills.
DSM-III	1980	265	
DSM-III-R	1987	292	
DSM-IV	1994	297	
DSM-IV-TR	2000	365	

The newest version of the DSM, the DSM-5, was published in May of 2013. This latest revision was met with considerable discussion and some controversy.

DSM-IV-TR MULTIAXIAL SYSTEM

The DSM-IV was originally published in 1994 and listed more than 250 mental disorders. An updated version, called the DSM-IV-TR, was published in 2000. This version utilized a multi-axial or multidimensional approach for diagnosing mental disorders.

The multi-axial approach was intended to help clinicians and psychiatrists make comprehensive evaluations of a client's level of functioning because mental illnesses often impact many different life areas.

It described disorders using five DSM "axes" or dimensions to ensure that all factors—psychological, biological, and environmental—were considered when making a mental health diagnosis.

AXIS I – CLINICAL SYNDROMES

Axis I consisted of mental health and substance use disorders that cause significant impairment. Disorders were grouped into different categories such as mood disorders, anxiety disorders, or eating disorders.

AXIS II – PERSONALITY DISORDERS AND MENTAL RETARDATION

Axis II was reserved for mental retardation and personality disorders, such as antisocial personality

AXIS III – GENERAL MEDICAL CONDITIONS

Axis III was used for coding medical conditions that influence or worsen Axis I and Axis II disorders. Some examples include HIV/AIDS and brain injuries.

AXIS IV – PSYCHOSOCIAL AND ENVIRONMENTAL PROBLEMS

Any social or environmental problems that may impact Axis I or Axis II disorders were accounted for in this axis. These include such things as unemployment, relocation, divorce, or the death of a loved one.

AXIS V – GLOBAL ASSESSMENT OF FUNCTIONING

Axis V is where the clinician gives their impression of the client's overall level of functioning. Based on this assessment, clinicians could better understand how the other four axes interacted and the effect on the individual's life.

CHANGES IN THE DSM-5

The DSM-5 contains a number of significant changes from the earlier DSM-IV. The most immediately obvious change is the shift from using Roman numerals to Arabic numbers.

Perhaps most notably, the DSM-5 eliminated the multi-axial system. Instead the DSM-5 lists categories of disorders along with a number of different related disorders. Example categories in the DSM-5 include anxiety disorders, bipolar and related disorders, depressive disorders, feeding and eating

disorders, obsessive-compulsive and related disorders, and personality disorders.

A few other changes in the DSM-5:

Asperger's disorder was removed and incorporated under the category of autism spectrum disorders.

Disruptive mood dysregulation disorder was added, in part to decrease over-diagnosis of childhood bipolar disorders.

Several diagnoses were officially added to the manual including binge eating disorder, hoarding disorder, and premenstrual dysphoric disorder

While the DSM is an important tool, only those who have received specialized training and possess sufficient experience are qualified to diagnose and treat mental illnesses.

ICD 10 (INTERNATIONAL CLASSIFICATION OF DISEASES)

The International Classification of Diseases (ICD) is published by the WHO in Geneva, Switzerland. The latest ICD is ICD-10 (1992). This classification system includes diagnoses for all the systems in the human body. Of the 21 chapters in ICD 10, Chapter V is devoted to mental health disorders. Chapter V is divided into 10 groups with every main group has the identification letter "F". The ICD (for clinical practice - CDDG) and the DSM are now broadly similar.

Table

ICD-10 coding structure and rules for psychiatric disorders

F00-F09 *Neurocognitive Disorders and Mental Disorders Associated with a General Medical Condition*

- F00 Alzheimer dementia
- F01 Vascular dementia
- F05 Delirium, etc.

F10-F19 *Substance Use Related Disorders*

A third digit denotes the substance

F20-F29 *Schizophrenia and Related Psychotic Disorders*

- F20 Schizophrenia
- F22 Delusional disorder
- F25 Schizoaffective disorder

F30-39 *Major Mood Disorders*

- F30 Initial manic episode
- F31 Bipolar disorder
- F32 Initial depressive episode
- F33 Recurrent depressive disorder

F40-F49 *Anxiety, Stress-Related, Dissociative, and Somatoform Disorders*

- F40 Phobic anxiety
- F41 Other anxiety disorders
- F42 Obsessive-compulsive disorder
- F43 PTSD and adjustment disorders
- F44 Dissociative disorders
- F45 Somatoform disorders

F50-F59 *Eating and Sleep Disorders, and Sexual Dysfunction*

- F50 Eating disorders
- F51 Sleep disorders
- F52 Sexual dysfunction

F60-F69 *Personality Disorders; Impulse-Control and "Habit" Disorders; Gender Identity and Sexual Preference Disorders*

- F60.x Specific personality disorders with "x" denoting the type

F70-F79 *Intellectual Disability ("mental retardation")*

- F70-F73 Mild, moderate, severe, and profound (respectively)

F80-F89 *Specific Learning Disabilities and Autism Spectrum Disorders*

- F84 Autism spectrum disorder

F90-F99 *ADHD, Conduct Disorders, Childhood Anxiety Disorders, and Tic Disorders*

ADHD: attention-deficit/hyperactivity disorder; ICD-10: International Classification of Diseases, 10th revision; PTSD: posttraumatic stress disorder

character if it is necessary to subdivide further.

All of the diagnostic codes start with the letter F and, like the other chapters, it has 10 major divisions, each of which can be divided into 10 subdivisions, and so on. For example, F20 schizophrenia can be followed by a further number for the category within the group (e.g. F20.1, disorganized schizophrenia), and a fourth

DIFFERENCE BETWEEN DSM AND ICD

	ICD-10	DSM-IV
Origin	International (World Health Organization)	American Psychiatric Association
Presentation	Different versions for clinical work, research, and use in primary care	A single document
Languages	Available in all widely spoken languages	English version only
Structure	Part of overall ICD Single axis in Chapter V; separate multi-axial systems available	Multi-axial
Content	Guidelines and criteria do not include social consequences of disorders	Diagnostic criteria usually include significant impairment in social functions

MAJOR CATEGORIES OF DISORDERS IN DSM-IV-TR

Anxiety disorders
 Somatoform disorders
 Dissociative disorders
 Mood disorders
 Schizophrenia
 Personality disorders
 Sexual disorders
 Substance-related disorders
 Delirium, dementia, amnesia, and other cognitive disorders.

ANXIETY DISORDERS

Disorders in which anxiety becomes an impediment in a person's routine functioning. Anxiety is a reaction to real or imagined threat that may hamper the daily functioning and results in uneasiness, worry, and apprehension. In anxiety disorders, anxiety occurs without an

obvious external cause, to an extent that it affects routine functioning of the person. Stress is the part of daily routine in a person's life but the reactions to stress vary from individual to individual.

Anxiety is one of the various reactions to stress. Whether or not one will develop anxiety, and to what extent, will depend on the nature of stress faced, family history, and fatigue or over work, and the person's coping strategies.

MAJOR SYMPTOMS OF STRESS INCLUDE:

Sleeplessness, Headaches, Twitching and trembling, Dry mouth, Memory problems, Nightmares, Irritability, Fatigue, Sweating, Muscle tension, Insomnia

SUBCATEGORIES OF ANXIETY DISORDERS:

Panic Disorder consists of recurrent unexpected Panic Attacks. A panic attack is generally accompanied by physical symptoms, such as a pounding heart, sweating, and rapid breathing.

Person suffers from *panic attacks* (brief attacks of intense terror & apprehension, often marked by trembling, shaking, confusion, dizziness, nausea, and/or difficulty breathing).

≈40% of young adults have occasional attacks.

In addition to panic attacks, this diagnosis requires that the attacks have chronic consequences, either:

- worry over the attacks'
- potential implications
- persistent fear of future attacks
- significant changes in behavior related to the attacks

33-50% develop agoraphobia.

A phobia is an irrational fear that produces a conscious avoidance of the feared object. The affected person usually recognizes that the reaction is excessive. Specific phobias (formerly known as “simple phobias”) involve fear of specific objects; social phobias – involves fear of social situations and agoraphobias (fear of situations where escape is difficult).

AGORAPHOBIA (GREEK - “FEAR OF MARKETPLACE”)

- ◆ Fear anyplace where might be trapped or unable to receive help in an emergency.
- ◆ Often accompanies panic disorder.
- ◆ Are usually very dependent people.
- ◆ Is the most common & the hardest to treat.

SOCIAL PHOBIAS

- ◆ Are insecure in social situations.

Have a fear of embarrassing themselves.

SIMPLE PHOBIAS

Is a fear of an animal, object or situation.

Over 300 have been named.

Example:

Animals

Zoophobia

Being touched

Aphephobia

Blood

Hemophobia

Confinement

Claustrophobia

Darkness

Nyctophobia

Death

Thanatophobia

Dirt

Mysophobia

Dog

Cynophobia

Fire

Pyrophobia

God

Theophobia

Heights

Acrophobia

Marriage

Gamophobia

Money

Chrematophobia

	Naked body
Gymnophobia	
	Robbers
Harpaxophobia	
	Sex
Genophobia	
	Sin
Harmartophobia	
	Sleep
Hypnophobia	
	Spiders
Arachnophobia	
	Strangers
Xenophobia	
	Thunder
Brontophobia	
	Travel
Hodophobia	
	Women
Gynophobia	
	Work
Ponophobia	

✓

GENERALIZED ANXIETY DISORDER is defined by excessive worry about a number of events or activities, accompanied by restlessness, fatigue, or other physical symptoms.

Characterized by long-lasting anxiety that is not focused on any one object or situation. Those suffering from generalized anxiety disorder experience non-specific persistent fear and worry, & become overly concerned with everyday matters.

Diagnosis of GAD is made when a person has been excessively worried about an everyday problem for >6 months.

Anxiety here is *free floating*. It is the most common anxiety disorder in older adults.

OBSESSIVE-COMPULSIVE DISORDER is marked by the presence of either an obsession or a compulsion that interferes with their life, either by taking up too much time or impairing their ability to function adequately.

Obsession - an idea you cannot get out of your head.

Compulsion - a behavior you cannot stop performing.

Washing, cleaning, & checking are the most common.

POSTTRAUMATIC STRESS DISORDER (PTSD) often occurs after the person has been exposed to a traumatic effect. It involves the persistent reexperiencing of a traumatic event (flashbacks, nightmares), avoidance of stimuli associated with the event, and symptoms of increased arousal lasting at least 1 month. Acute Stress Disorder involves similar features lasting less than 1 month.

Results from a traumatic experience. Post-traumatic stress can result from an extreme situation, such as combat, natural disaster, rape, hostage situations, child abuse, bullying or even a serious accident. It can also result from chronic exposure to a severe stressor.

Symptoms - avoidance & numbing, intrusive memories, anxiety & emotions. Became widely accepted as a diagnostic category because of

difficulties experienced by Vietnam War veterans.

SOMATOFORM DISORDERS

A disorder in which psychological problems take the physical (somatic) form without any apparent physical cause; a state where there are physical symptoms present but no medical cause.

Somatization Disorder involves a history of multiple physical complaints over a period of several years without any organic pathology, pain in at least four different sites on the body (head, abdomen, back, joints etc.) two gastrointestinal problems (e.g. diarrhea, vomiting, bloating) one sexual symptom (impotence, sexual indifference, irregular menses), and one pseudo-neurological symptom similar to those seen in Conversion disorder (fainting, blindness). They have no organ pathology to account for their multiple symptoms.

Conversion Disorder is distinguished by the development of a symptom suggestive of a neurological disorder affecting senses (e.g. blindness, double vision, deafness, loss of touch or pain sensation), or voluntary motor function (e.g. aphonia, impaired coordination or balance, paralysis or localized weakness, difficulty swallowing, urinary retention, seizures). However, the underlying cause is not neurological but severe psychological conflict.

Pain Disorder is characterized by pain occurring in association with psychological factors.

Hypochondriasis is the fear of, or the idea that one has, a serious physical illness.

Body Dysmorphic Disorder is a preoccupation with an imagined defect in appearance

DISSOCIATIVE DISORDER

Dissociative Disorders: The essential feature of Dissociative Disorders is a disturbance in the integrative functions of identity, memory, and consciousness. All of the disorders are trauma-based (such as rape, wartime trauma or severe accident.) *(Dissociation is the disconnection from full awareness of self – don't know who you are, time – don't know what time it is, and/or external circumstances – don't know where you are)*

TYPES OF DISSOCIATIVE DISORDER

1- DISSOCIATIVE AMNESIA

A state when a selective loss of memory occurs. The person is unable to recall specific events often as a result of extreme stress. Significant memory loss occurs about personal information that is not due to an organic cause. This disorder vanishes abruptly as it begins and rarely re-occurs.

2- DISSOCIATIVE FUGUE

Fugue means, "Flight" Fugue is a form of amnesia. The sufferer takes sudden impulsive trips, at times assuming a new identity. Dissociative fugue includes forgetting as well as fleeing from one's home for days and weeks, also being unable to remember one's identity. Unconscious wandering in which the person has limited social contacts.

In some instances, person may take over another personality that is more sociable than the previous one.

3- DISSOCIATIVE IDENTITY DISORDER/MULTIPLE PERSONALITY

Rare disorder in which the person may take over two or more personalities that are entirely different from one another. The first one is usually restrained, restricted and dull but the other one is entirely different from the previous one; one's mannerisms, vocal, movements are entirely different from one another.

MOOD DISORDERS

Disturbances of **mood** are the primary manifestation of Mood Disorders. The primary Mood Disorders are divided into *unipolar* and *bipolar* disorders.



In *unipolar disorders*, patients experience emotional extremes at just one end of the continuum i.e. depression. The unipolar Depressive Disorders include Major Depressive Disorder (severe depression) and Dysthymic Disorder (mild depression that lasts for at least two years)

Bipolar disorder may involve both depressive and manic periods or only manic period. Even though the patient may be exhibiting only manic symptoms, it is assumed that a depressive episode will eventually occur and so no “unipolar” mania exists. These are divided into Bipolar - I disorder (most severe form of mania), Bipolar - II disorder (mild form of mania) and Cyclothymic disorder (mildest form of mania). Bipolar - I is characterized by episodes of both severe depression (usually but not always) and severe mania; BP II by severe depression with occasional episodes of “mild” mania, called hypomania and cyclothymic disorder alternates between hypomania and mild depression. (BP-I; pronounced “type one bipolar disorder”).

SCHIZOPHRENIA

Schizophrenia is a category of mental disorders marked by severe distortion of reality. Schizophrenia is a disorder with a minimum of 6 months' duration and is characterized by an active psychotic phase for at least 1 month: delusions, hallucinations, disorganized speech, grossly disorganized behavior, and negative symptoms (e.g. alogia or little speech, or avolition or lack of motivation).

There is a deep division between the real world and the schizophrenic's world. Have difficulty sorting out the real from unreal, in keeping track of their thoughts, & responding to the everyday events in life.

Involves personality disintegration & a loss of contact with reality.

This group occupies about half the beds in mental hospitals.

Occurs equally in men & women, but tends to occur at an earlier age in men than women.

Typical age of onset is 15 - 35.

Major Symptoms

Disordered Thinking

Autistic Thinking -

Absorption in fantasy.

Prelogical Thinking - Thought processes are primitive & incomplete.

Delusions - False beliefs.

Several types: Persecution, Grandeur, Control, & Identity.

Disturbances of Perception

Attention & Filtering -

Seem to have trouble focusing attention & filtering out irrelevant stimuli.

Louis Wain (1860-1939) - A famous animal artist. His drawings of a cat show his progressive deterioration & some disturbing distortions of perception.

Hallucinations - False perceptions. Are usually auditory (hear voices), but may also be visual or olfactory.

Disturbances of Emotion

Flattened emotions (blunted affect).

Inappropriate emotions. Communication Difficulties

Echolalia - Repeating the last word or phrase spoken by another.

Neologisms - Made up words.

Word Salad - Words haphazardly thrown together.

Verbal Exhibitionism - A grandiose manner of speech.

Bizarre Motor Behavior

Unusual Motions - May grimace or gesture in peculiar ways.

Catalepsy - Holding a particular posture for a long time.

Waxy Flexibility - Posture can be molded.

Catataplexy - Loss of muscle tone.

TYPES OF SCHIZOPHRENIA

Disorganized or Hebephrenic Type	Marked by inappropriate emotion: inappropriate giggling, laughter, silliness, incoherent speech, infantile behavior, and strange and at times obscene behavior.
Paranoid Schizophrenia	The patient experiences delusions and hallucinations of his own greatness. Behavior is unpredictable, and erratic. Sense of judgment is lost.
Catatonic Schizophrenia	Catatonic schizophrenia is marked by disturbances in the motor activity and muscular control. Major disturbances occur in movement. At times all motion stops and the patient just freezes in one position. This frozen posture may last for hours and even days. In some phases the patient exhibits wild, free floating, and even violent movement.
Undifferentiated Schizophrenia	This variety of schizophrenia involves a combination of the major symptoms found in other varieties. This diagnosis is used when patients do not fit into any one of the major categories of schizophrenia.
Residual Schizophrenia	Residual schizophrenia consists of minor signs of schizophrenia after a major, more serious, episode.

Taken together, these data suggest Genetics a very strong genetic component to the disorder.

The more closely related a person is to a patient with schizophrenia, the more likely that person is to develop schizophrenia (Gottesman, 1991).

Adopted children with schizophrenia are the most likely to have symptomatic biological relatives.

PERSONALITY DISORDERS

as “an enduring pattern of inner experience and behavior that deviates markedly from the expectations of the individual's culture.” This pattern is manifested in two (or more) of the following areas-

Cognition (i.e., ways of perceiving and interpreting self, other people, and events):
e.g. suspicious, self-critical.

Affectivity (i.e., the range, intensity, lability or abnormal sudden rapid shifts in mood, and appropriateness of emotional response) e.g. indifferent, flattened affect

Interpersonal functioning: Does not have healthy and satisfying relationships with friends and family members.

Impulse control: This is characterized by an inability to resist the impulse to perform an action that is harmful to one's self or others e.g. aggressive outbursts, sudden changes of jobs, residences, or relationships.

CLUSTERS OF PERSONALITY DISORDERS

Cluster A (odd or eccentric): Paranoid | Schizoid | Schizotypal (treatment resistant category)

Cluster B (dramatic, emotional, or erratic): Histrionic/ Antisocial | Borderline//Narcissistic (mixed results)

Cluster C (anxious or fearful): Avoidant | Dependent | Obsessive-Compulsive (treatment responsive category)

Odd or eccentric cluster	Schizotypal -
	behavior odd which leads to social isolation Schizoid - social isolation by choice (loner) Paranoid - suspicious, mistrusting, jealous, and cold
Dramatic, emotional or erratic cluster	Histrionic -
	Dramatic and Attention seeking, Shallow and Superficial in relationships Narcissistic - overcompensate for low self-esteem by building self-up/self-preoccupation Borderline - attention seeking, manipulative, mood swings, struggles with identity Antisocial - no conscience/remorse, regularly violate the rights of others, superficial
Anxious/fearful cluster	Avoidant -
	fear rejection therefore socially isolated though desire intimacy Obsessive compulsive - preoccupied with

	organization and perfection
	Dependent - inability to take responsibility for self, need others to make decisions

CAUSES OF PSYCHOLOGICAL DISORDERS

Knowledge of the causes of psychological disorders is important for two main reasons. First, in everyday clinical work it helps the psychologist to understand possible causes of an individual patient's psychological disorder. Second, it adds to the general understanding of mental disorders, which may contribute to advances in diagnosis, treatment, or prognosis.

A necessary cause is a condition that must be there for a disorder to occur. For example, general paresis, a degenerative brain disorder cannot develop unless a person had syphilis. A necessary cause is not always sufficient to cause a disorder and therefore other factors may also be required.

A sufficient cause is a condition that guarantees the occurrence of a disorder. For example, one theory says that hopelessness is a sufficient cause of depression (Abramson et al 1995). According to this theory if you are hopeless enough about your future then you will become depressed. A contributory cause is one that increases the probability of developing a disorder. For example, parental rejection could increase the probability that the child may have problems in handling close relationships later. When there are multiple causes it is useful to group them into predisposing, precipitating, and perpetuating factors.

Predisposing factors determine the vulnerability to other causes that act close to the time of the illness. Many predisposing factors act early in life, for example, genetic endowment, the

environment in utero, trauma at birth, and social and psychological factors in infancy and childhood. Precipitating factors are events that occur shortly before the onset of a disorder and appear to have induced it. They may be physical, psychological, or social. Physical precipitating factors include diseases such as cerebral tumor, traumatic brain injury that is caused by accidents, and the effects of drugs taken for treatment or used illegally.

As for psychological causes, there are many as for example a psychological cause can be bereavement, losing a very near and dear person. This causes depression. While moving home is a social cause, some causes may act in more than one way; for example, a head injury may induce a psychological disorder through physical changes in the brain and through psychological effects.

Perpetuating factors (or maintaining factors) prolong a disorder after it has begun. Sometimes a feature of a disorder makes itself perpetuating (e.g. some ways of thinking commonly prolong anxiety disorders). Social factors are also important (e.g. overprotective attitudes of parents or care givers or relatives). Awareness of perpetuating factors is particularly important in planning treatment because they may be modifiable even when little can be done about predisposing and precipitating factors.

BIOLOGICAL FACTORS

According to this view, mental disorders are seen as disorders of the central nervous system and thus are sometimes inherited or caused by some medical factors like injuries or physical diseases. These are given below:

Neurotransmitter and Hormonal imbalances in the brain,

Genetics,

Constitutional liabilities, Brain

structure,

Physical deprivation or
disruption.

Neurotransmitter and Hormonal Imbalances in the Brain

Factors affecting neurotransmitter imbalance include:

Excessive production and release of the neurotransmitter substance into the synapses, causing an excess in levels of that neurotransmitter.

Dysfunctions in the normal

processes by which neurotransmitters are deactivated after they are released into the synapse. This deactivation is done in two ways. They are either deactivated by enzymes present in the synapse or reabsorbed or sucked back into the presynaptic axon button, a process called re-uptake.

Dysfunctions can occur when the enzymes in the synapse are deficient or there is slowing of the process of re-uptake.

Problems in the receptors in the postsynaptic neuron, which may either be abnormally sensitive or insensitive.

HORMONAL IMBALANCE

Hormones are chemical messengers secreted by the endocrine glands (e.g., pituitary). They play a role in the functioning of the nervous system and in the regulation of behavior

(e.g., during adolescence, changes in the hypothalamic-pituitary-adrenal axis are involved in the increase in cortisol, a stress-related hormone). Malfunction of this system has been said to be responsible for various forms of psychopathology.

Hormonal influences are also responsible for the differences in behaviour between men and women.

GENETICS Genes play an important role in determining risks for both psychotic and non-psychotic disorders. For example, the lifetime risk of schizophrenia is 1%, but

for the offspring of an affected person it becomes 10% and in bipolar disorder is 20%.

For many years, twin studies served as the most direct way of determining whether or not a disorder has a genetic basis. In the classic twin study design, the similarity of monozygotic ('identical') twins and dizygotic ('fraternal') twins are compared. Because monozygotic twins share all of their genes and dizygotic twins share only half their genes, greater similarity among monozygotic twins than among dizygotic twins implies a genetic component for most major mental disorders (e.g., schizophrenia, the major mood disorders, anxiety disorders, alcohol-related disorders). Twin and/or adoption studies have demonstrated significant genetic effects. However, these studies also demonstrate equally, if not more, important environmental effects. However, psychologists often assume that 'environment' means the family or psychosocial environment and this is not necessarily the case. Factors such as the prenatal environment and viral infections are all part of the 'environment' in genetic terms.

Using several different techniques to analyze DNA (the molecular basis of genes), researchers are now in a position to correlate the presence of specific genes with specific mental disorders. The task is complicated because existing research suggests that most mental disorders are caused by multiple genes (polygenic), making it very difficult to discover each individual gene that is associated with a disorder.

CONSTITUTIONAL LIABILITIES

The term 'constitutional' is used to describe any characteristic that is either innate or acquired early in life often at prenatal stage and in such strength that it is functionally similar to a genetic characteristic. Physical handicaps and temperament are some of the traits included in this category. Embryonic

abnormalities or environmental conditions operating before or after birth may result in physical defects. The most common birth difficulty associated with learning disabilities and behavioural and emotional disorders is low birth weight. Prenatal conditions that can lead to premature birth and low birth weights include nutritional deficiencies, disease, and exposure to radiation, drugs, severe emotional stress or mother's excessive use of alcohol or tobacco. Socio-economic status is also related to fetal and birth difficulties. The temperament of an infant or young child has profound effects on a variety of important developmental processes (Rothbart & Ahadi, 1994).

BRAIN STRUCTURE

Knowledge about brain structure has increased with the advances in computed tomography (CT) scanning and magnetic resonance imaging (MRI). This has led to many notable observations. For instance, neuroimaging in some patients with schizophrenia shows dilated cerebral ventricles and reduced frontal lobe density. This evidence indicates that schizophrenia may be neurodevelopmental in origin. Exposure to adverse conditions which can affect brain development (in utero or in early life) may lead to changes in the frontal lobes that increase the risk of schizophrenia. Neuroimaging also helps us to distinguish between different types of dementia. Also, some older people experiencing severe depression for the first time might have underlying cerebro-vascular disease.

PHYSICAL DEPRIVATION OR DISRUPTION The most basic human requirements are those of food, water, oxygen, sleep and elimination of wastes. Insufficient rest, inadequate diet or working too hard when ill, can all interfere with a person's ability to cope and might predispose him or her to a variety of problems. Experimental studies of volunteers who have gone without sleep for 72 to 98 hours show psychological problems like disorientation for time and place and

feelings of depersonalization. Prolonged food deprivation also affects psychological health. Severe malnutrition in children not only impairs physical development and lowers resistance to disease but it also stunts brain growth, results in lowered intelligence and increases risk for disorders like attention-deficit disorder.

PSYCHOLOGICAL FACTORS

Personality 'traits' are usually defined as those aspects of how one person relates to others, reacts to interpersonal stimuli and evaluates themselves over time. For all this, a person may laugh, cry, be angry, be thoughtful and act unkindly at different times. This does not mean that their 'personality' is constantly changing—it is the overall balance of thinking, feeling and behaviour that is important, not how they react in any specific situation.

PSYCHODYNAMICS AND THE PARENT-CHILD RELATIONSHIP

Freud emphasized the role of the early parent-child relationship in the development of mental illness. According to Freud, to the extent that the child did not successfully negotiate the psychosexual stages, mental illness would develop.

Contemporary psychodynamic models (e.g., Kohut, 1977; Kernberg, 1976; Mitchell, 1988) also suggest that the early parent child relationship is the original source of mental illness, and that what goes on in the mind of the child (and the adult) is important. The attachment model of psychopathology, developed by Bowlby (1969; 1973; 1980) resembles the contemporary psychodynamic models in that it also emphasizes the early parent child relationship and how the resulting

models of self and others guide development

LEARNED BEHAVIOUR

Behavioral models suggest that all behaviours, abnormal included, is a product of learning, that is mainly learning by association. For example, according to the classical conditioning model of learning (e.g. Pavlov, 1928), if a person experiences chest pain which results in anxiety while shopping in a departmental store, he may develop a fear of departmental stores and begin to avoid them because he associates them with anxiety. There is nothing inherently frightening about departmental stores, but this person fears them because of the association that he has formed with his earlier anxiety about having a possible heart attack.

DISTORTED THINKING Cognitive models of abnormal behaviour focus on the way people think about themselves, others and the world (e.g. Beck et al., 1979). Distorted cognitive processes—such as selectively attending to some information and ignoring other information, exaggerating negative feelings, expecting the worst, or making inaccurate attributions about events have been shown to play an important role in various types of psychological disorders.

FAMILY THEORIES

Fromm-Reichmann (1948) coined the term schizophrenogenic mother, to describe a parent who was cold, domineering and manipulative, and who had a marked tendency to induce conflict in others around her.

Bateson et al. (1956) argued that schizophrenia develops as a result of repeated exposure to a process called the double-bind. This term means communications that are inherently contradictory or conflicting, and which put the child in a 'no win' situation.

Lidz et al. (1958) suggested that abnormalities in the relationships between the parents of mentally ill patients were primarily responsible for their developing psychological problems. They coined the terms skew and schism to describe such relationships. Skew describes a marriage where one partner is dominant and the other submissive, while schism describes the relationship in which parents are emotionally distant from one another.

SOCIO-CULTURAL FACTORS

One, called social causation, suggests that lower economic class people are more prone to mental disorder because they are more likely to experience social stress (e.g. unemployment, separation), to suffer from psychic frailty, infectious diseases, neurological impairments, and to lack good medical treatment, coping ability and social support. Through an accumulation of these problems, and the stresses that result, low social status becomes a cause of mental illness. The other explanation emphasizes social selection or drift. This suggests that mentally ill people from higher social classes often drift downward into the lower-class areas, (due to job loss, unemployment) helping to increase the rate of mental illness in such neighborhoods. This explanation suggests that being lower economic class is a result of mental illness among formerly higher status individuals. Both explanations may be true to some extent.

GENDER

The female role is relatively restrictive and oppressive, likely to confine the woman to her inner self, such that she tends to keep her frustration and anger to herself rather than aggressively pour it out on others. Hence women are more likely to fall victim to depression and anxiety. Men, on the other hand, have a more liberated role, and they are encouraged

to be bold, assertive and aggressive in social relations. If frustrated and angry, they are more likely to take it out on others — behaving as antisocial and paranoid individuals.

AGE

Another social factor that has been associated with mental disorder is age. Studies conducted before the 1980's suggested that older persons were more likely to suffer from mental disorders. This was attributed to societal neglect of the elderly eventually resulting in institutionalization, where the neglect can continue. Yet, more recent studies in the 1980's and 1990s show that the elderly are the least likely among all age groups to become mentally ill.

RACE AND ETHNICITY

A third social factor in mental disorders is race and ethnicity. Like gender, these have not been consistently found to be related to mental illness in general. While many studies have shown higher rates of psychological stress among minorities, the standard explanation has been that these groups experience more social stresses stemming from discrimination, poverty and cultural conflict.

THE DIATHESIS-STRESS MODEL OF PSYCHOLOGICAL DISORDERS

Despite advances in understanding the biological basis of psychological disorders, the psychosocial perspective is still very important. This perspective emphasizes the importance of learning, stress, faulty and self-defeating thinking patterns, and environmental factors. Perhaps the best way to think about psychological disorders, then, is to view them as originating from a combination of biological and psychological processes. Many develop not from a single cause, but from a delicate fusion between partly biological and partly psychosocial factors.

The diathesis-stress model (Zuckerman, 1999) integrates biological and

psychosocial factors to predict the likelihood of a disorder. This diathesis-stress model suggests that people with an underlying predisposition for a disorder (i.e., a diathesis) are more likely than others to develop a disorder when faced with adverse environmental or psychological events (i.e., stress), such as childhood maltreatment, negative life events, trauma, and so on. A diathesis is not always a biological vulnerability to an illness; some diatheses may be psychological (e.g., a tendency to think about life events in a pessimistic, self-defeating way).

The key assumption of the diathesis-stress model is that both factors, diathesis and stress, are necessary in the development of a disorder. Different models explore the relationship between the two factors: the level of stress needed to produce the disorder is inversely proportional to the level of diathesis.

TREATMENT OF DISORDERS

There are different approaches to therapy but they can be basically divided into two groups:

1) pharmacological and 2) psychotherapy

DRUGS:

Antipsychotic medications are used in psychotic disorders such as schizophrenia, and bipolar disorder. They reduce or eliminate symptoms such as delusions and hallucinations. Patients begin to feel better within six weeks of starting antipsychotic medication.

ANTIPSYCHOTIC

MEDICATIONS: There are two types of antipsychotic medications. These are known as:

TYPICAL / 1ST GENERATION ANTIPSYCHOTICS (E.G.

CHLORPROMAZINE (THORAZINE)

–this refers to the older type of medications that first appeared in the mid-1950s that work by blocking the action of dopamine. These medications were likely to cause side effects called ‘extrapyramidal symptoms’ or EPS, such as: Rigidity (muscle stiffness) and ‘Tardive dyskinesia’ – movement of the mouth, tongue and sometimes other parts of the body over which the person has no control.

ATYPICAL / 2ND GENERATION ANTIPSYCHOTICS (E.G. CLOZAPINE (CLOZARIL) AND RISPERIDONE (RISPERDAL)).

This works on both dopamine and serotonin and also have fewer side effects. Clozapine is considered a very effective but it can cause ‘agranulocytosis’, a loss of the white blood cells and hence blood cell need to be monitored every week. Common side-effects of atypical antipsychotic include diabetes and weight gain.

INJECTIONS – DEPOT

MEDICATION: Depot medication is given by injection, which releases the drug slowly over some weeks. Some people prefer injections as they find remembering to take tablets difficult.

ANTIDEPRESSANT DRUGS: These drugs are used for treating depression, anxiety, eating disorders and certain personality disorders such as borderline. There are several types of antidepressants available to treat depression, such as:

NEWER/SECOND GENERATION TREATMENT; They have less severe side effects than the older drugs. Common side effects of newer antidepressants include: Feeling sick, dry mouth, insomnia and weight gain

- Selective serotonin reuptake inhibitors (SSRIs): They act by increasing the amount of serotonin in the brain. The SSRIs include drugs such as Fluoxetine (Prozac) and Sertraline (Zoloft).

Atypical/Serotonin-norepinephrine reuptake inhibitors (SNRIs): These target serotonin and norepinephrine.

Serotonin, noradrenaline, and dopamine reuptake inhibitor (or SNDRI). Drugs like WellbutrinTM and EffexorTM block the reabsorption of the three neurotransmitters – serotonin, dopamine and norepinephrine.

OLDER/CLASSICAL ANTIDEPRESSANTS

Tricyclic antidepressants: Tricyclics are among the older antidepressants. They work by inhibiting the brain's reuptake of dopamine, serotonin and norepinephrine. Because the tricyclics have such a broad mechanism of action, they tend to cause more side effects than the other classes of antidepressants.

Monoamine oxidase inhibitors (MAOIs): MAOIs are not typically chosen as a first-line depression treatment because of their side effects.

ANTI-ANXIETY/ANXIOLYTIC DRUGS: These reduce anxiety and are also prescribed as sleeping pills and muscle relaxants.

Benzodiazepines: they enhance the activity of GABA which then promotes relaxation by slowing the

CNS. They include:
Valium(diazepam), Xanax
(alprazolam).

- Beta blockers. Trade names: *Inderal*, *Tenormin*. These cardiac drugs counteract the effect of adrenaline and alleviate certain anxiety symptoms such as shaking, palpitations and sweating.

buspirone (Buspar®): Buspar, or buspirone hydrochloride, is in a class of its own. It stimulates both serotonin and dopamine.

LITHIUM AND OTHER MOOD STABILIZING DRUGS (E.G.,

DEPAKOTE): Lithium is used to treat the mania. Lithium affects the flow of sodium through nerve and muscle cells in the body.

BRAIN STIMULATION TECHNIQUES:

Brain stimulation therapies involve activating the brain directly with electricity (ECT), magnets (TMS), or implants (VNS) to treat depression and other disorders like bipolar, schizophrenia and OCD. These are usually considered only after a patient's illness has not improved after other treatment options.

ECT: With ECT, electrodes are put on the patient's scalp and a finely controlled electric current is applied causing a brief seizure. In Unilateral ECT current flow is applied to only one side of the brain (nondominant side) while in bilateral ECT (which is considered more effective), electrical current is applied to both sides. ECT is much safer today; it still causes some side effects such as temporary memory loss and confusion.

- **TRANSCRANIAL MAGNETIC STIMULATION (TMS):** Transcranial magnetic stimulation (TMS) creates a magnetic field to induce a much smaller electric current without causing seizure or loss of consciousness.

VAGUS NERVE STIMULATION (VNS): Vagus nerve stimulation (VNS) works by sending electrical pulses through the left vagus nerve. The vagus nerves carry messages from the brain to the body's major organs like the heart, lungs and intestines, and to areas of the brain that control mood, sleep, and other functions.

NEUROSURGERY:

During neurosurgery a small piece of brain is destroyed. In lobotomy fiber tracts connected to the frontal lobe are destroyed. It was believed that this could help could reduce problems, such as depression and anxiety. Unfortunately, the technique left some patients with severe side effects, including seizures, and changes in personality. By the 1980s, the neurosurgery field moved away from destruction to deep-brain stimulation where electrodes are placed into the neural tissue and current is passed to activate or inactivate patches of the brain.

Evaluation: Many pharmacological treatments are effective (especially for major depression and bipolar disorder) but side effects need to be controlled. Moreover, they need to be combined with psychotherapy for long term results.

PSYCHOTHERAPY-

Psychotherapy is a form of treatment for problems of an emotional nature in which a trained person deliberately establishes a professional relationship with a patient for

the purpose of removing, modifying, or retarding existing symptoms, of mediating disturbed patterns of behavior, and of promoting positive personality growth and development (Wolberg, 1967).

Psychologists generally draw on one or more theories of psychotherapy.

A theory of psychotherapy acts as a roadmap for psychologists: It guides them through the process of understanding clients and their problems and developing solutions.

Psychoanalysis and psychodynamic therapies.

This approach focuses on changing problematic behaviors, feelings, and thoughts by discovering their unconscious meanings and motivations.

Psychoanalytically oriented therapies are characterized by a close working partnership between therapist and patient. Patients learn about themselves by exploring their interactions in the therapeutic relationship. While psychoanalysis is closely identified with **Sigmund Freud**, it has been extended and modified since his early formulations.

The goal of psychoanalytic therapy is to help patients better understand the unconscious forces that can play a role in current behaviors, thoughts, and emotions. This type of therapy is based upon the theories and work of Sigmund Freud, who founded the school of thought known as psychoanalysis. Freud described the unconscious as the reservoir of desires, thoughts, and memories that are below the surface of conscious awareness. He believed that it was these unconscious influences that could often lead to psychological distress and disturbances.

THE HISTORY OF PSYCHOANALYTIC THERAPY

Psychoanalytic theory grew out of the work of the famous psychoanalyst Sigmund Freud who began developing his therapeutic techniques in the late 1800s. In 1885, Freud began to study and work with Jean-Martin Charcot at the Salpêtrière in Paris. Charcot used hypnosis to treat women suffering from what was then known as hysteria. Symptoms of the illness included partial paralysis, hallucinations, and nervousness.

Freud continued to research hypnotism in treatment, but his work and friendship with colleague Josef Breuer led to the development of his most famous therapeutic technique. Breuer described his treatment of a young woman, known in the case history as Anna O., whose symptoms of hysteria were relieved by talking about her traumatic experiences. Freud and Breuer collaborated on a book called *Studies on Hysteria* and Freud continued to develop his use of this "talk therapy." This approach proposed that simply talking about problems could help relieve psychological distress.

THE PROCESS

Psychoanalytic therapy is insight driven, and therefore looks to foster change by helping you understand your past and how events from your early life could be affecting you now. Sessions will vary according to why you are seeking therapy, and where you are in the therapy journey, but much of the time will be spent talking freely to your therapist in a safe, non-judgmental environment.

The therapist will listen to your concerns and look out for patterns or certain events that may hold significance. In this type of therapy, it is believed that our unconscious feelings and certain childhood events play a key role in mental distress.

As well as listening to you talk about your experiences and your concerns, the therapist may use other techniques to help you understand and identify potential causes for your concerns, such as free

association, therapeutic transference and interpretation.

FREE ASSOCIATION

Free association involves you talking about whatever comes into your mind without censoring or editing the flow of memories or ideas. Your therapist will encourage you to speak freely to help you return to an earlier emotional state, so they can better understand any recurrent patterns of conflict you may be experiencing.

THERAPEUTIC TRANSFERENCE

Transference relates to the way you may be transferring thoughts or feelings connected to influential figures in your life (for example your parents or siblings) onto your therapist. While this may not happen in every case, if it does, your therapist should discuss transference with you to help you gain further insight into the way you deal with people in your daily life.

INTERPRETATION

A key element of psychoanalytic therapy is interpreting and 'reading between the lines'. While your therapist is likely to stay relatively quiet and allow you to talk freely, they will occasionally interject with thoughts or interpretations of the topics you discuss. They may also ask you about your dreams; Freud wrote a lot on the subject of dream analysis and believed that dreams were important resources for understanding the unconscious.

APPLICATIONS OF PSYCHOANALYTIC THERAPY

Psychoanalytic therapy can be used by those with a specific emotional concern, as well as those who simply want to explore themselves. Understanding why we are the way we are, often brings with it a sense of well-being and a strong sense of self. As psychoanalytic therapy is considered one of the more long-term therapy types, it is perhaps less useful for those seeking

quick, solution-focused therapies. Psychoanalytic therapy is a gradual process that takes time, yet the results are said to be life changing.

Some believe that due to the nature of therapy, psychoanalytic work is better suited to more general concerns such as anxiety, relationship difficulties, sexual issues or low self-esteem. Phobias, social shyness and difficulties sleeping are further examples of areas that may be effectively managed with the help of psychoanalytic therapy.

Psychoanalytic therapy can also be applied in a group setting. This is called group analysis. This form of therapy brings together psychoanalytic techniques with interpersonal functions.

BEHAVIOR THERAPY.

This approach focuses on learning's role in developing both normal and abnormal behaviors.

Ivan Pavlov made important contributions to behavior therapy by discovering **classical conditioning**, or associative learning. Pavlov's famous dogs, for example, began drooling when they heard their dinner bell, because they associated the sound with food.

"Desensitizing" is classical conditioning in action: A therapist might help a client with a phobia through repeated exposure to whatever it is that causes anxiety.

Another important thinker was **E.L. Thorndike**, who discovered **operant conditioning**. This type of learning relies on rewards and punishments to shape people's behavior.

Several variations have developed since behavior therapy's emergence in the

1950s. One variation is **cognitive-behavioral therapy**, which focuses on both thoughts and behaviors.

In behavioral therapy, the goal is to reinforce desirable behaviors and eliminate unwanted or maladaptive ones. Behavioral therapy is rooted in the principles of behaviorism, a school of thought focused on the idea that we learn from our environment. The techniques used in this type of treatment are based on the theories of classical conditioning and operant conditioning.

One important thing to note about the various behavioral therapies is that unlike some other types of therapy that are rooted in insight (such as psychoanalytic and humanistic therapies), behavioral therapy is action-based. Behavioral therapists are focused on using the same learning strategies that led to the formation of unwanted behaviors.

Because of this, behavioral therapy tends to be highly focused. The behavior itself is the problem and the goal is to teach clients new behaviors to minimize or eliminate the issue. *Old learning* led to the development of a problem, and so the idea is that *new learning* can fix it.

There are three major areas that also draw on the strategies of behavioral therapy:

Cognitive-behavioral

therapy relies on behavioral techniques but adds a cognitive element, focusing on the problematic thoughts that lie behind behaviors.

Applied behavior analysis uses operant conditioning to shape and modify problematic behaviors.

Social learning theory centers on how people learn through observation. Observing others being rewarded or punished for their actions can lead to learning and behavior change.

Edward Thorndike was one of the first to refer to the idea of modifying behavior. Other early pioneers of behavior therapy included psychologists Joseph Wolpe and Hans Eysenck. Behaviorist B.F. Skinner's work had a major influence on the development of behavior therapy and his work introduced many of the concepts and techniques that are still in use today.

Later on, psychologists such as Aaron Beck and Albert Ellis began adding a cognitive element to behavioral strategies to form a treatment approach known as cognitive-behavioral therapy (CBT).

THE FOUNDATION OF BEHAVIORAL THERAPY

In order to understand how behavioral therapy works, let's start by exploring the two basic principles that contribute to behavioral therapy: Classical and operant conditioning.

- **Classical conditioning** involves forming associations between stimuli. Previously neutral stimuli are paired with a stimulus that naturally and automatically evokes a response. After repeated pairings, an association is formed and the previously neutral stimulus will come to evoke the response on its own.

Operant conditioning focuses on how reinforcement and punishment can be utilized to either increase or decrease the frequency of a behavior. Behaviors followed by desirable consequences are more likely to occur again in the future, while those followed by negative consequences become less likely to occur.

BEHAVIOR THERAPY BASED ON CLASSICAL CONDITIONING

Classical conditioning is one way to alter behavior, and a number of techniques exist that can produce such change. Originally known as behavior modification, this type of therapy is often referred to today as

applied behavior analysis. There are several different techniques and strategies used in this approach to therapy.

FLOODING

This process involves exposing people to fear-invoking objects or situations intensely and rapidly. It is often used to treat phobias. During the process, the individual is prevented from escaping or avoiding the situation.

For example, flooding might be used to help a person who is suffering from an intense fear of dogs. At first, the client might be exposed to a small, friendly dog for an extended period of time during which they cannot leave. After repeated exposures to the dog during which nothing bad happens, the fear response begins to fade.

SYSTEMATIC DESENSITIZATION

In this technique, a client makes a list of fears and then learns to relax while concentrating on these fears. The use of this process began with psychologist John

Watson and his famous Little Albert experiment in which he conditioned a young child to fear a white rat. Later, Mary Cover Jones replicated Watson's results and utilized counterconditioning techniques to desensitize and eliminate the fear response.

Systematic desensitization is often used to treat phobias and other anxiety disorders. The process follows three basic steps.

First, the therapist teaches the client relaxation techniques.

Next, the individual creates a ranked list of fear-invoking situations.

Starting with the least fear-inducing item and working their way up to the most fear-inducing item, the client confronts these fears under the guidance of the therapist while maintaining a relaxed state.

For example, an individual with a fear of the dark might start by looking at an image of a dark room, before moving on to thinking about being in a dark room, and then actually confronting his fear by sitting in a dark room. By pairing the old fear-producing stimulus with the newly learned relaxation behavior, the phobic response can be reduced or even eliminated.

AVERSION THERAPY

This process involves pairing an undesirable behavior with an aversive stimulus in the hope that the unwanted behavior will eventually be reduced. For example, someone suffering from alcoholism might take disulfiram, a drug which causes severe symptoms (such as headaches, nausea, anxiety, and vomiting) when combined with alcohol.

BEHAVIOR THERAPY BASED ON OPERANT CONDITIONING

Many behavior techniques rely on the principles of operant conditioning, which means that they use reinforcement, punishment, shaping, modeling and related techniques to alter behavior. These methods have the benefit of being highly focused, which means that they can produce fast and effective results.

TOKEN ECONOMIES

This type of behavioral strategy relies on reinforcement to modify behavior. Clients are allowed to earn tokens that can be exchanged for special privileges or desired items. Parents and teachers often use token economies to reinforce good behavior.

Kids earn tokens for engaging in preferred behaviors and may lose tokens for displaying undesirable behaviors. These tokens can then be traded for rewards such as candy, toys, or extra time playing with a favorite toy.

CONTINGENCY MANAGEMENT

This approach uses a formal written contract between a client and a therapist (or parent or teacher) that outlines behavior-change goals, reinforcements, rewards, and penalties. Contingency contracts can be very effective in producing behavior changes since the rules are spelled out clearly, preventing both parties from backing down on their promises.

MODELING

This technique involves learning through observation and modeling the behavior of others. The process is based on Albert Bandura's social learning theory, which emphasizes the social components of the learning process.

Rather than relying simply on reinforcement or punishment, modeling allows individuals to learn new skills or acceptable behaviors by watching someone else perform those desired skills. In some cases, the therapist might model the desired behavior. In other instances, watching peers engage in sought-after behaviors can also be helpful.

EXTINCTION

Another way to produce behavior change is to stop reinforcing a behavior in order to eliminate the response. Time-outs are a perfect example of the extinction process.

During a time-out, a person is removed from a situation that provides reinforcement. For example, a child who starts yelling or striking other children would be removed from the group and required to sit quietly in a place where there are no opportunities for attention and reinforcement. By taking away the attention that the child found rewarding, the unwanted behavior is eventually extinguished.

COGNITIVE THERAPY.

Cognitive therapy emphasizes what people think rather than what they do. Cognitive therapists believe that it's dysfunctional thinking that leads to dysfunctional emotions or behaviors. By changing their thoughts, people can change how they feel and what they do. Major figures in cognitive therapy include **Albert Ellis** and **Aaron Beck**.

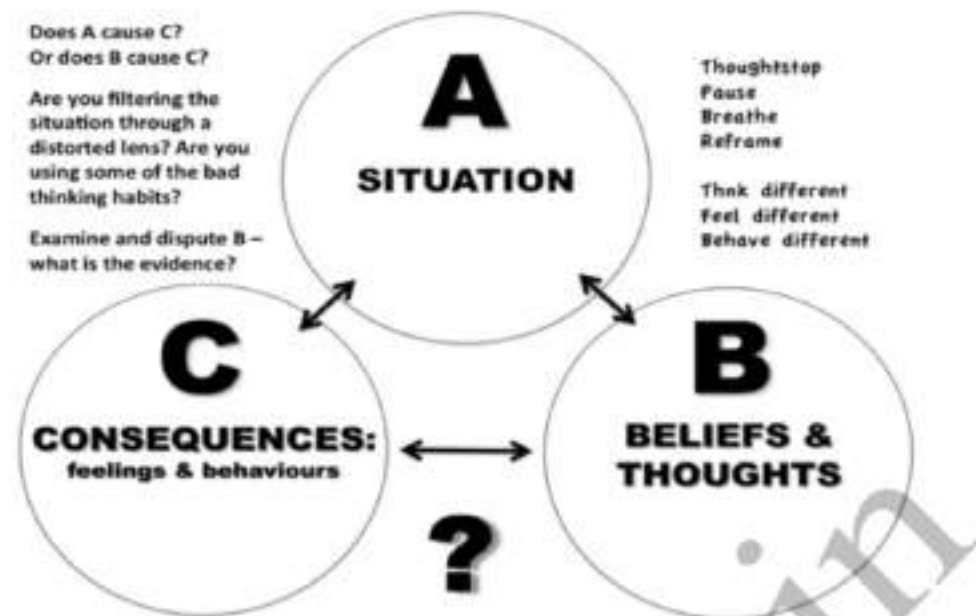
AN OVERVIEW AND SUMMARY OF CBT

Cognitive Behavioral Therapy is utilized in treating multiple types of psychiatric problems. The treatment is typically done between 3 and 6 months, depending on the problem. The following is a list of psychological problems where CBT has been utilized.

- depression
- anxiety
- eating disorders
- anger management
- marital crisis
- obsessive-compulsive disorder
- schizophrenia
- post-traumatic stress disorder
- chronic pain

There are 5 five areas that are believed to be interconnected and affecting one another. For instance, how one feels about a certain situation can cause physical and emotional feelings, resulting in varying behaviors in response.

- situations
- thoughts
- emotions
- physical feelings
- behaviors



understanding that they will be utilized in their future.

CBT breaks problems down into smaller pieces to give detailed attention to each part. The techniques aid patients in **disrupting** negative, automatic thoughts, and replacing them with more helpful ones. The overall goal is to teach the skill of breaking down negative thought patterns and changing them into a more **helpful approach** to handling daily life.

KEY CONCEPTS AND PRINCIPLES OF THE APPROACH

The patient's active participation in therapy is a key principle in CBT. Without it, this goal-oriented and problem focused approach would not be effective. Sessions in CBT are well structured and the client's better understanding of the role of cognition in correcting behavioral dysfunctions is paramount to their success. This educative approach allows the client-therapist relationship to deepen, which is also an important principle in this therapy.

CBT is a time limited approach, and work outside of the therapy office is vital to success. While this approach is initially present focused, an emphasis on adaptive thinking allows for relapse prevention. It allows the patient to be taught **techniques** to change their thinking, mood, and behavior with the

In cognitive behavior therapy, psychological problems are believed to develop through the use of **cognitive distortions**. Aaron Beck's work suggests that by correcting these distortions, a more accurate experience of events is created. Through this work, a patient is better able to develop skills to properly process exposure to life events.

Here is a list of Cognitive Distortions.

Personalization means the attribution of the negative feelings of others and the world around them. For instance, a gymnastics coach is cross, so a gymnast automatically feels it is her fault.

Fallacy of change is assuming that other people will change to suit them if pressured enough. This is a common distortion found in relationships. For instance, a woman feeling that if her partner improved himself, she would be happier.

Emotional reasoning is the distortion that occurs when feelings are considered as facts. For example, saying, "I feel that way, it must be true."

Fallacy of Fairness is a distortion that measures all things by an imaginary ruler of fairness. A person may feel resentful because they feel that they have a clear definition of fairness, but that others might not agree with them.

Polarization or “Black and White” Thinking is a distortion that occurs when things are all or nothing. Someone might believe they have to be perfect, or they are a failure.

Overgeneralization means coming to broad negative conclusions based on a single, insignificant event. An example would be telling yourself you’re a terrible baker based on failing at one recipe attempt.

Control fallacies are distortions in which a person would feel that everything that happens to you is a result of external actions or of their own behavior. For instance, believing that your work is not good because you’re dealing with disruptive coworkers.

Jumping to Conclusions is a distortion that occurs when assumptions about another’s feelings or beliefs are made. An example might be a child feeling that they know how someone else feels about them. It can also be described as perceived mind reading.

Catastrophizing is a distortion that assumes the expectation that the worst is going to happen. An example is believing that a small error at work is going to result in you being fired.

Always Being Right is a distortion that occurs when the person is always putting others on trial to prove their opinions are absolutely correct.

Filtering is a distortion that includes both *Minimization and Magnification*. Minimization means underplaying the significance of an event. For instance, you are praised for your job performance, but you see it as trivial.

Magnification means exaggerating the importance of an undesirable event. For instance, allowing getting cut off by another driver as the impetus to ruin your entire day. Filtering also includes selective abstraction. This is focusing on a single aspect of a situation and ignoring others.

Blaming occurs when a person holds someone else responsible for their emotional pain, or hold themselves responsible for every problem.

Global labeling is a distortion that occurs when a person generalizes single qualities into a global judgment. For instance, “I failed a test, therefore I’m stupid.”

Heaven’s Reward Fallacy is the distortion that self-sacrifice will eventually pay off.

Shoulds are distortions that occur when a person has hard set rules about how every person should behave. An example would be saying, “I should work out. I shouldn’t be so lazy.” The resulting emotional consequence is guilt.

Albert Ellis developed the ABC technique that is still utilized in CBT today. The ABC Technique of Irrational Beliefs analyzes the first three steps in which someone might develop an irrational belief: A) Activating event B) Belief C) Consequences.

Activating Event. This is an event that would lead someone to a type of high emotional response, and/ or negative dysfunctional thinking.

Beliefs. The client would write down the negative thoughts that occurred to them around the activating event.

Consequences. These are the negative feelings and behaviors that occurred as a result. The beliefs are to be viewed as a bridge to the negative feelings and behaviors that occurred as a result of the activating event.

Ellis believed that it was not the activating event (A) that causes the negative beliefs and consequences (C), but rather how the patient interprets or misinterprets the meaning of the event (B) that helps cause the consequences (C).

Helping a patient reinterpret their irrational belief system helps to forge new ways for them to interpret their beliefs resulting in alternative behaviors. A person can utilize this technique, even in the absence of a therapist.

Journaling for the awareness of cognitive distortions is a powerful way to better understand personal cognition. A person keeps track of their **automatic thoughts** and an analysis of the presence of various distortions is detected.

Once better understood, a person can utilize different methods to reevaluate these automatic thoughts with evidence. Well trained practitioners in CBT can aid someone who has difficulty in unraveling these distortions.

Rescripting is a technique used to help patients suffering from nightmares. When the emotion that is brought to the surface from the nightmare is exposed, a therapist can help the patient to redefine the emotion desired and to develop a new image to elicit that emotion.

Exposure therapy is used in **OCD** and anxiety phobias. Exposing yourself to the trigger reduces the response to the trigger. Many therapists recommend mild exposure 3 times daily. While this may be uncomfortable during the first exposures, the increase in exposure reduces phobic reactions.

The Worst Case/ Best Case/Most Likely Case Scenario technique is used to help people overcome fear or anxiety. Allowing the brain to ruminate to the point of ridiculousness allows the person to “play out” the fear to an unrealistic end. Then the person is brought to the best case and again allowed to let their thoughts “play out” to the ridiculous. Then, a most likely

scenario is explored with actionable steps attached, so that control over behavior is realized.

A recent, popular technique being utilized in CBT is called *Acceptance and Commitment Therapy*. It differs from traditional CBT in that it is not trying to teach people to better control their thoughts around their activating events; instead this approach is teaching people to “just notice,” accept and embrace the feelings around the activating events. This approach utilizes techniques from CBT as well, but ACT focuses on freeing the patient from the grip of the event itself.

Mindfulness techniques like deep breathing and *Progressive Muscle Relaxation (PMR)* play a big role in CBT. These techniques allow the person to be present in the moment and **calmly soothe** the unfocused mind. With the relaxation comes the stronger ability to tune in and alter automatic negative thoughts.

Cognitive restructuring is a CBT **technique** that helps people examine their unhelpful thinking. It helps them to redevelop ways to react in situations that have in the past proven problematic. Keeping a daily record of the automatic negative thoughts creates a way to find the patterns in these thoughts. With an identified pattern, alternative reactions and adaptive thoughts can be forged.

Treating thoughts as guesses is a technique that helps to gather evidence to combat automatic negative thoughts. When a person takes their thoughts to “court” proof of truth must be found for the thought to be held as accurate. If any proof against the thought is found, it must be tossed and replaced with a more accurate thought.

A *cognitive pie chart* is a fun way for **kids to utilize CBT**. The first step is identifying the automatic negative thoughts. For example, “I’m dumb because I failed a test.” Step two is devising a list of alternative explanations to those ANTs.

Finding as many alternatives as possible is helpful. The third step is giving each explanation a percentage in the contribution to the outcome of failing the test. Step four is placing these explanations in a pie chart.

Activity Scheduling is a powerful technique in CBT. It helps people engage in activities that they are not normally used to doing. It presents as a way to slowly reintroduce rewarding behavior that has been excluded from people's routines. The technique is helpful in increasing positive emotion when performed incrementally.

Graded exposure is a technique used to help expose anxiety sufferers to contact with what is feared. The underlying theory is that people who avoid situations that induce fear or anxiety will increase the anxiety. The slowly increased exposure aids to decrease that fear.

Many deficits in *social skills* can be improved through CBT techniques. *Modeling, role playing, and instruction* can be used to increase social skills like communication and assertiveness. **Communication** skills, or rather the lack of, are a massive obstacle for many. Improving these skills bolsters confidence and abilities to interact with others, dramatically reducing levels of anxiety inducing situations.

RATIONAL EMOTIVE BEHAVIOR THERAPY

Rational Emotive Behavior Therapy (REBT) is a short-term form of psychotherapy that helps you identify self-defeating thoughts and feelings, challenge the rationality of those feelings, and replace them with healthier, more productive beliefs. REBT focuses mostly on the present time to help you understand how unhealthy thoughts and beliefs create emotional distress which, in turn, leads to unhealthy actions and behaviors that interfere with your current life goals. Once identified and understood, negative thoughts and actions can be changed and replaced with more positive and

productive behavior, allowing you to develop more successful personal and professional relationships.

Ellis theorized that many of our emotional and behavioral problems spring from basic irrational assumptions or assumptions that are not totally grounded in reality and influence people to act in ways that are inappropriate, unhelpful, or even destructive (McLeod, 2015).

Based on this idea, Ellis developed a model to help explain, describe, and treat emotional and behavioral disturbances.

THE ABCDE MODEL OF EMOTIONAL DISTURBANCE

Ellis hypothesized that irrational beliefs are the result of a person's goals or desires being inhibited or blocked. When we don't get or accomplish what we wanted to, we may develop irrational beliefs about ourselves or the world that help explain what happened.

For example, imagine you are dead set on getting a job you applied for. You study up on the company, practice your interview answers, and make sure you're looking extra sharp the day of the interview. Although you prepared extensively, the hiring manager decided to go with another candidate.

You may accept that this just wasn't meant to be, or that you just weren't the right fit for the job. However, you may also be heavily impacted by the decision and develop an irrational belief about why you didn't get the job.

You might think, "I didn't get this job because they can see that I'm a loser. I'm not good at anything and I never will be."

Or, you might think, "The only reason I didn't get this job is because the hiring manager had it out for me. It's like the universe has it out for me!"

Both of these are thoughts that can help you explain why you didn't get the job, but

they are irrational and can lead to negative emotions and behavior down the road.

Using this scenario as an example, this is how the **ABCDE model** can explain the development (and the solution) of such problems:

A – ACTIVATING EVENT / ADVERSITY

An activating event or adversity is something that triggers you to form an irrational belief, such as being turned down for the position. It is the first step in developing an irrational thought because the irrational thought is formed to **help you deal** with the event.

B – IRRATIONAL BELIEF

The “B” stands for the irrational belief that is formed in response to the activating event. This is a belief that you use to cope with the event, such as “I’m a loser, I’m useless, and I wouldn’t be able to do the job anyway.” While this is, of course, an incredibly hurtful thought, it can still be more comforting than having no idea why you didn’t get the job. Irrational beliefs are surprisingly easy to develop.

C – EMOTIONAL AND BEHAVIORAL CONSEQUENCES

The third component is the consequences of this irrational belief. Irrational beliefs always have consequences, sometimes emotional, sometimes behavior, and sometimes both. In this case, the

consequences may be that you lose your **self-confidence** or frequently feel sad (emotional) and stop applying to any jobs (behavioral).

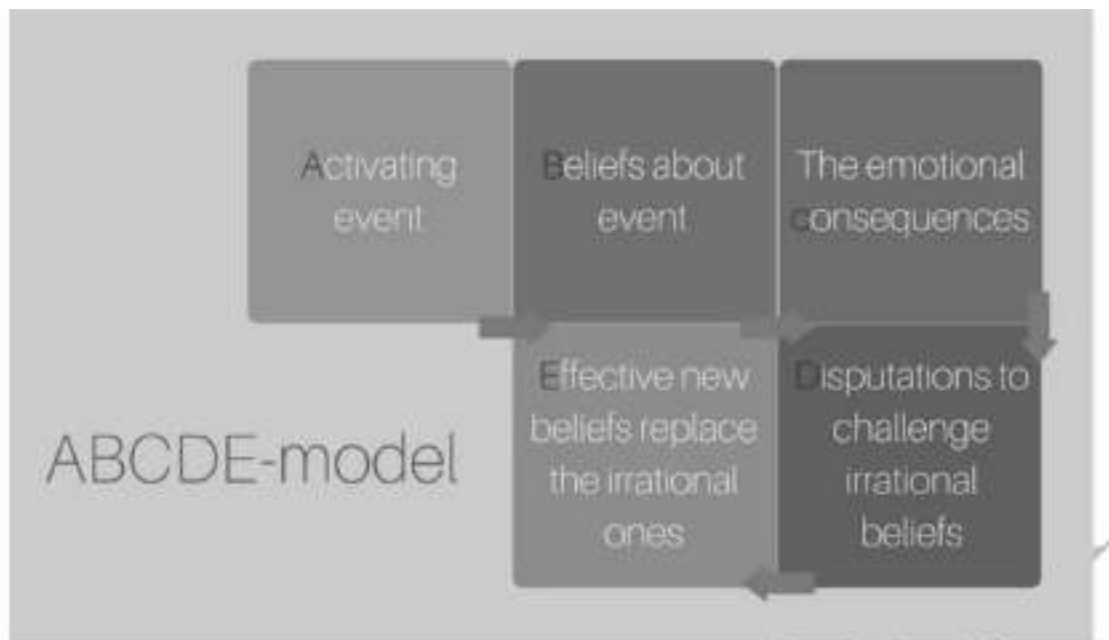
D – DISPUTES OR ARGUMENTS

At some point, you may realize that you have an irrational belief that is causing you problems. You notice your loss of self-confidence and negative thoughts about yourself and begin to argue against your irrational belief. If you’re working with a therapist, the therapist may help guide you in developing arguments against the belief and help you come up with evidence to the contrary, such as “I have an amazing spouse. My spouse wouldn’t be with a ‘loser’ so I must not be a loser.”

E – NEW EFFECT

When you have successfully countered the irrational belief, you will notice new (hopefully more positive!) consequences or effects. In our scenario, these effects might be increased confidence, applying to more jobs, and feeling good about your abilities. These effects are the positive outcomes of holding more rational thoughts, like “I just wasn’t a good fit for that job, but I’ll find another” or “Maybe the hiring manager really didn’t like me, but that’s her loss” (McLeod, 2015).

The ABCDE model can be extremely helpful in tracing the development of irrational thought and providing a high-level outline of how to challenge and replace it.



ourselves and others only leads to disappointment, recrimination, regret, and anxiety.

THE BASIC STEPS OF REBT

In order to better understand how REBT looks, it is important to take a closer look at the therapeutic process itself.

IDENTIFY IRRATIONAL THOUGHT PATTERNS AND BELIEFS

The very first step in the process is to identify the underlying, irrational thoughts, feelings, and beliefs that lead to psychological distress. In many cases, these irrational beliefs are reflected as absolutes, as in "I must," "I should," or "I cannot." According to Ellis, some of the most common irrational beliefs include:

- Feeling excessively upset over other people's mistakes or misconduct
- Believing that you must be 100% competent and successful in everything to be valued and worthwhile
- Believing that you will be happier if you avoid life's difficulties or challenges
- Feeling that you have no control over your own happiness, that your contentment and joy are dependent upon external forces

Holding such unyielding beliefs makes it almost impossible to respond to activating situations in a psychologically healthy way. Possessing such rigid expectations of

CHALLENGE IRRATIONAL BELIEFS

Once these underlying feelings have been identified, the next step is to challenge the mistaken beliefs. In order to do this, the therapist disputes these beliefs using very direct and even confrontational methods. Ellis suggested that rather than simply being warm and supportive, therapists need to be blunt, honest, and logical in order to push people toward changing their thoughts and behaviors.

GAIN INSIGHT AND CHANGE BEHAVIOR

As you might imagine, REBT can be a daunting process for the client. Facing irrational thought patterns can be difficult, especially because accepting these beliefs as unhealthy is far from easy. Once the client has identified the problematic beliefs, the process of actually changing these thoughts can be even more challenging.

While it is perfectly normal to feel upset when making a mistake, the goal of REBT is to help people respond rationally to such situations. When faced with this type of situation in the future, the emotionally

healthy response would be to realize that it is not realistic to expect success in every endeavor. You made a mistake, but that's okay. Everyone makes mistakes. All you can do is learn from the situation and move on.

While REBT uses cognitive strategies, it focuses on emotions and behaviors as well.² In addition to identifying and disputing irrational beliefs, therapists and clients also work together to target the emotional responses that accompany problematic thoughts. Clients are encouraged to change unwanted behaviors using strategies such as meditation, journaling, and guided imagery.

HUMANISTIC THERAPY.

This approach emphasizes people's capacity to make rational choices and develop to their maximum potential. Concern and respect for others are also important themes. Humanistic philosophers like **Jean-Paul Sartre**, **Martin Buber** and **Søren Kierkegaard** influenced this type of therapy.

- a. Three types of humanistic therapy are especially influential. **Client-centered therapy** rejects the idea of therapists as authorities on their clients' inner experiences. Instead, therapists help clients change by emphasizing their concern, care and interest.

Gestalt therapy emphasizes what it calls "organismic holism," the importance of being aware of the here and now and accepting responsibility for yourself.

Existential therapy focuses on free will, self-determination and the search for meaning.

Person-centred therapy, also known as person-centred or client-centred counselling, is a humanistic approach that deals with the ways in which individuals

perceive themselves consciously, rather than how a counsellor can interpret their unconscious thoughts or ideas.

Created in the 1950s by psychologist Carl Rogers, the person-centred approach ultimately sees human beings as having an innate tendency to develop towards their full potential. However, this ability can become blocked or distorted by certain life experiences, particularly those the experiences which affect our sense of value.

The therapist in this approach works to understand an individual's experience from their perspective. The therapist must positively value the client as a person in all aspects of their humanity, while aiming to be open and genuine. This is vital in helping the client feel accepted, and better able to understand their own feelings. The approach can help the client to reconnect with their inner values and sense of self-worth, thus enabling them to find their own way to move forward and progress.

The core purpose of person-centred therapy is to facilitate our ability to self-actualise - the belief that all of us will grow and fulfil our potential. This approach facilitates the personal growth and relationships of a client by allowing them to explore and utilise their own strengths and personal identity. The counsellor aids this process, providing vital support to the client and they make their way through this journey.

The person-centred counsellor is not an expert; rather the client is seen as an expert on themselves and the person-centred counsellor encourages the client to explore and understand themselves and their troubles.

There are three conditions believed to help achieve this environment, particularly in the therapy room.

Congruence - the counsellor must be completely genuine.

Empathy - the counsellor must strive to understand the client's experience.

Unconditional positive regard - the counsellor must be non-judgmental and valuing.

THE IMPORTANCE OF SELF-CONCEPT

Self-concept also plays an important role in person-centered therapy. Rogers defined self-concept as an organized set of beliefs and ideas about the self. The self-concept play an important role in determining not only how people see themselves, but also how they view and interact with the world around them.

Sometimes self-concept lines up well with reality, which Rogers referred to as congruence. In other cases, self-perceptions are sometimes unrealistic or not in tune with what exists in the real world. Rogers believed that all people distort reality to some degree, but when self-concept is in conflict with reality, incongruence can result. For example, a young boy might perceive himself as a strong athlete, despite the fact that his actual performance on the field reveals that he is not particularly skilled and could use extra practice.

Through the process of person-centered therapy, Rogers believed that people could learn to adjust their self-concept in order to achieve congruence and a more realistic view of themselves and the world. For example, imagine a young woman who views herself as uninteresting and a poor conversationalist despite the fact that other people find her fascinating and quite engaging. Because her self-perceptions are not congruent with reality, she may experience poor self-esteem as a result. The client-centered approach focuses on providing unconditional positive regard, empathy, and genuine support in order to help the client reach a more congruent view of herself.

B. GESTALT THERAPY

*"I do my thing and
you do your thing.
I am not in this
world to live up to
your expectations
And you are not in
this world to live up
to mine.
You are you and I
am I,
And if by chance we
find each other, it's
beautiful.
If not, it cannot be
helped."*

*(Fritz Perls, 1969,
in Gladding, 2000)*

Gestalt Therapy was developed in the 1940's by Fritz and Laura Perls and further influenced by the likes of Kurt Lewin and Kurt Goldstein (Corsini & Wedding, 2000). It was developed as a revision to psychoanalysis and focuses on an experiential and humanistic approach rather than analysis of the unconscious which was one of the main therapeutic tools at the time Gestalt Therapy was employed.

Gestalt Therapy rejects the dualities of mind and body, body and soul, thinking and feeling, and feeling and action. According to Perls, people are not made up of separate components, this is, mind, body and soul, rather human beings function as a whole. In doing so, one defines who one is (sense of self) by choice of responses to environmental interactions (boundaries). The word "Gestalt" (of German origin) refers to a "whole, configuration, integration, pattern or form" (Patterson, 1986). The form of Gestalt Therapy practiced today utilizes ideas, data and interventions from multiple sources, as well as some of the original techniques known to be 'Gestalt Therapy techniques. It is noted that Gestalt Therapy has a history of being an approach which creates or borrows specific techniques that are focused on

assisting the client to take the next step in their personal growth and development.

KEY CONCEPTS

Several key concepts underlie Gestalt Therapy, many of which are similar to that of person-centred and existential therapy. However, what does differentiate Gestalt Therapy from these therapies are some of the ideas added by Perls and associates as well as distinctive therapeutic techniques that will be covered further down (Seligman, 2006). The following are the key concepts of Gestalt Therapy:

Wholeness and Integration: Wholeness refers to the whole person or the individual's mind and body as a unit rather than as separate parts (Seligman, 2006). Integration refers to how these parts fit together and how the individual integrates into the environment. Often people who come to therapy do not have these parts fitting together in their environment, Gestalt Therapy is about facilitating clients to integrate themselves as whole persons and help restore balance in their environment.

Awareness: Awareness is one of the most important elements in Gestalt Therapy as it is seen as a "hallmark of the healthy person and a goal of treatment" (Seligman, 2006). When individuals are "aware", they are able to self-regulate in their environment.

THERE ARE TWO MAIN CAUSES LACKING AWARENESS:

1. Preoccupation with one's past, fantasies, flaws and strengths that the individual becomes unaware of the whole picture.
2. Low self-esteem.

There are three ways people may achieve awareness through therapy:

Contact with the environment: This is through looking, listening, touching, talking, moving, smelling, and tasting.

This enables the individual to grow in his or her environment through reacting to the environment and changing.

Here and now: This is the individual living in and being conscious at the present moment rather than worrying about the past or the future.

Responsibility: This refers to the individual taking responsibility for his or her own life rather than blaming others.

Energy and blocks to energy: Gestalt Therapists often focus on where energy is in the body, how it is used, and how it may be causing a blockage (Corey, 2005). Blocked energy is a form of resistance, for example, tension in a part of the body, not breathing deeply, or avoiding eye contact. Gestalt Therapy is about finding and releasing the blockages that may be inhibiting awareness.

Growth Disorders: Growth disorders refer to emotional problems that are caused by people who lack awareness and do not interact with their environment completely. In doing so, people are unable to cope with the changes in their lives successfully and, instead deal with the problems in a defensive manner (Seligman, 2006).

Unfinished business: Unfinished business refers to people who do not finish things in their lives and is often related to people with a "growth disorder" (Seligman, 2006). People with unfinished business often resent the past and because of this are unable to focus on the here and now. One of the major goals of Gestalt Therapy is to help people work through their unfinished business and bring about closure.

GENERAL IDEAS ABOUT PERSONALITY DEVELOPMENT

Gestalt Therapy deems that people cannot be considered as separate from their environment or from interpersonal relations. The individual is seen as being

self-regulating and is able to motivate oneself to solve problems. Individuals are able to work towards growth and develop as their environments allow.

A psychologically healthy person is someone who is self-regulating through the changes in life and has developed a sense of “wholeness” between mind and body (Corsini & Wedding, (2000).

THERAPEUTIC TECHNIQUES & METHODS OF WORKING

The most important goal of Gestalt Therapy is that Gestalt Therapists do not aim to change their clients. The therapist's role is to assist clients in developing their own self-awareness of how they are in the present moment. This will therefore allow them to rectify issues affecting his or her life.

“The therapist's job is to invite clients into an active partnership where they can learn about themselves by adopting an experiential attitude toward life in which they try out new behaviours and notice what happens” (Perls, Hefferline and Goodman, 1954, in Corey, 2005).

A focus of developing awareness is that of clients' awareness of their own realities. In order to do this, clients must first accept responsibility for choosing their present situations. Language plays a big part in accepting responsibility. The client may attempt to use avoidance responses or project individual traits onto other people or external causes, for example “She makes me so angry”; “It's his fault”. Both avoidance responses and projection of traits attempt to displace ownership and responsibility onto an external cause.

Another goal of Gestalt Therapy is that therapists should work to create an “I-thou” relationship with clients in which both the therapist and client are present in the here-and-now rather than focusing on the past or future (Seligman, 2006).

Also, an understanding of the whole of the client's experience is required by the therapist. This involves considering the client's — verbal — and — non-verbal — communication. In fact, the nonverbal

communication is seen to provide more information about the real essence of the person.

Thus, an important function of the Gestalt Therapist is paying attention to the client's body language such as the client's posture, movements, gestures, voice, and hesitations as the body language is considered to be reflective of what the client is going through at that point in time.

Experiments: Gestalt Therapists use the technique of experiments or learning experiences with their clients. The experiments are designed for the individual and take the form of an enactment, role play, homework, or other activity which promotes the individual's self-awareness (Seligman, 2006).

An example of this technique is with a man who feels insecure in social situations. He has a work function to go to in two weeks' time so the therapist gives him the experiment of starting a conversation at the function with someone he does not normally speak to. Spending time thinking about what he might say promotes self-awareness and the experiment itself gives him more confidence in social situations.

Use of Language: Gestalt Therapists choose language that will encourage change in the client. The following are ways that this can be accomplished (Seligman, 2006):

Emphasis on statements rather than questions to highlight a collaborative client-therapist relationship.

“What” and “How” questions (when questions are used) to keep the client in the present and promote integration. “I” statements are used to promote client's ownership of feelings rather than placing blame on others.

The present tense is used so the focus is on the present rather than the past. Encouraging responsibility for clients of their words, emotions, thoughts, and

behaviors so they recognize and accept what they are feeling.

Empty Chair: The empty chair technique is a “method of facilitating the role-taking dialogue between the patient and others or between parts of the patient’s personality. It is generally used in a group situation” (Patterson, 1986). Two chairs are placed facing each other: one represents the patient or one aspect of the patient’s personality, and the other represents another person or the opposing part of the personality. As the patient alternates the role, he or she sits in one or the other chair.

The therapist may simply observe as the dialogue progresses or may instruct the patient when to change chairs, suggest sentences to say, call the patient’s attention to what has been said, or ask the patient to repeat or exaggerate words or actions.

In the process, emotions and conflicts are evoked, impasses may be brought about and resolved, and awareness and integration of polarities may develop – polarities or splits within the patient, between the patient and other persons, or between the patient’s wants and the social norms (Patterson, 1986).

Topdog – Underdog: A commonly utilized Gestalt technique is that of the top dog-underdog dialogue. This technique is used when the therapist notices two opposing opinions/attitudes within the client. The therapist encourages the client to distinguish between these two parts and play the role of each in a dialogue between them (Patterson, 1986).

The tyrannical ‘top dog’ demands that things be a particular way whilst the ‘underdog’ plays the role of disobedient child. The individual becomes split between the two sides struggling for control.

Dreams: Dreams are used to bring about integration by the client. The focus of a client’s dream is not on the

unconscious, rather on projections or aspects of the dreamer (Seligman, 2006). The therapist would get clients to talk about their dream/s in terms of the significance of each role in the dream and this allows clients to take responsibility for the dreams and increase awareness of their thoughts and emotions.

Fantasy: Fantasy is used in Gestalt Therapy to increase clients’ self-awareness of their thoughts and emotions and to bring about closure to unfinished business (Seligman, 2006). Therapists use guided imagery techniques (fantasy) to encourage clients to imagine situations such as what they would do in a certain situation or by projecting themselves into different roles.

The Body as a Vehicle of Communication: Gestalt Therapy sees that not only are thoughts and emotions important to creating a feeling of “wholeness” for the client, the physical sensations are also important. Seligman (2006) has identified three strategies to help with focusing attention on the physical sensations:

Identification: Gestalt Therapists should be able to recognize physical signs of their clients. For example, a client might be tapping their feet on the ground. The therapist may say “Become your leg and give it a voice?” This creates awareness of the client’s physical sensations and emotions.

Locating emotions in the body: Gestalt Therapists may ask clients where they are experiencing the emotion in their body. For example, a client may say they are feeling nervous about something. The therapist may ask where this is coming from in the body and the response from the client may be that the feeling is butterflies in the stomach. This helps the client to bring about more awareness into sensations and their emotions.

Repetition and exaggeration: If there is repetition such as the example of the client tapping their feet on the ground, the therapist would get them to exaggerate the movement and talk about feelings that come up. This in turn focuses on the emotion and should help to release the blocked awareness.

Confusion: The technique of dealing with confusion of the client is about drawing attention to the client's hesitation in talking about something unpleasant. The hesitation can be shown through avoidance, blanking out, verbalism and fantasy (Patterson, 1986). By drawing attention to the hesitation, it creates self-awareness for the client and allows the client to work through the issue.

Confrontation: In Gestalt Therapy, confrontation means 'to challenge or frustrate the client'. The client is challenged with sensitivity and empathy on the part of the therapist to face the issues important to them. It is an invaluable tool for bringing clients into clear awareness of their realities, when used appropriately. However, confrontation is not a technique that can be used with all clients.

APPLICATIONS

Originally Gestalt Therapy was predominantly used to treat individuals who were anxious and/or depressed and who were not showing serious pathological symptoms. Although still used in the treatment of anxiety and depression, Gestalt Therapy has been effective in treating clients with personality disorders such as borderline personality disorder.

Gestalt Therapy is also effective in counselling groups, couples, and families (Corsini& Wedding, 2000).

STRENGTHS

There is empirical research to support Gestalt Therapy and its techniques (Corsini& Wedding, 2000).

Gestalt Therapy is equal to or greater than other therapies in treating various disorders, Gestalt Therapy has a beneficial impact with personality disorders, and the effects of therapy are stable.

Works with the past by making it relevant to the present (Corey, 2005).

Versatile and flexible in its approach to therapy. It has many techniques and may be applied to different therapeutic issues.

WEAKNESSES

For Gestalt Therapy to be effective, the therapist must have a high level of personal development (Corey, 2005).

Effectiveness of the confronting and theatrical techniques of Gestalt Therapy is limited and has not been well established.

It has been considered to be a self-centered approach which is concerned with just individual development.

Potential danger for therapists to abuse the power they have with clients (Corey, 2005).

Lacks a strong theoretical base.

Deals only with the here and now.

Does not deal with diagnosis and testing.

CONCLUSION

Gestalt Therapy focuses on the integration between the "whole" person and his or her environment. This therapy sees a healthy individual as being someone who has awareness in his or her life and lives in the here and now rather than focusing on the past or future. Gestalt Therapy has a number of successful techniques that are applicable in therapy today and may be utilized across a broad spectrum of emotional issues.

EXISTENTIAL THERAPY

Existential therapy is a unique form of psychotherapy that looks to explore difficulties from a philosophical perspective. Focusing on the human condition as a whole, existential therapy highlights our capacities and encourages us to take responsibility for our successes.

Emotional and psychological difficulties are viewed as inner conflict caused by an individual's confrontation with the givens of existence. Rather than delve into the past, the existential approach looks at the here and now, exploring the human condition as a whole and what it means for an individual.

On this page, we will look at the background of existential therapy, including the philosophers who influenced it, the associated theories and how it could benefit you.

EXISTENTIAL THERAPY: BACKGROUND

The roots of existential psychotherapy lie in philosophy from the 1800s, and more importantly with philosophers whose work dealt with human existence. The philosophers most commonly associated with existential therapy are Søren Kierkegaard and Friedrich Nietzsche. While the two influential thinkers were in conflict regarding the ideologies of their time, they were committed to the exploration of reality and how it was experienced.

Kierkegaard theorized that human discontent could only be overcome via internal wisdom, while Nietzsche introduced the idea of free will and personal responsibility. By the 1900s, philosophers like Sartre and Heidegger had begun exploring the role interpretation and investigation had in the healing process. Over the next few decades, other contemporaries began to acknowledge the importance of 'experiencing' in terms of achieving psychological well-being.

THEORIES OF THE EXISTENTIAL APPROACH

A key element of existential counselling is that it does not place emphasis on past events like some other therapy types. The approach does take the past into consideration and together, the therapist and individual can understand the implications of past events.

Instead of putting blame on events from the past, however, existential counselling uses them as insight, becoming a tool to promote freedom and assertiveness. Coming to the realization that you are not defined by your history and that you are not destined to have a certain future is often a breakthrough that offers liberation.

Practitioners of existential therapy say that its role is to help facilitate an individual's own encounter with themselves and to work alongside them as they explore values, assumptions and ideals. An existential therapist will avoid any form of judgement and instead help the individual speak from their own perspective.

The therapist should enter sessions with an open mind and be ready to question their own biases and assumptions. The goal of the therapist is to understand the individual's assumptions with a clarity that the individual themselves may not be able to muster.

A belief that lies at the heart of existential counselling is that even though humans are essentially alone in the world, they long to be connected with others. This belief can help to explain why certain concerns appear and may help the individual understand why they feel the way they do sometimes.

Another interesting theory is that inner conflict stems from an individual's confrontation with the givens of existence. These givens were noted by influential psychotherapist Irvin D. Yalom, and include:

freedom and responsibility
 existential isolation
 meaninglessness
 death

These four givens (also known as 'ultimate concerns') are the cornerstones of existential psychotherapy. Once the issue has been conceptualized by the therapist, a method of treatment can be developed. Due to the all-encompassing nature of existential therapy, it is near impossible to identify one single cause for psychological upset. The therapy instead treats each person as an individual, exploring their experience and relating it to the experience of all humankind.

THE FOUR REALMS

Within existential psychotherapy, there is a description of four different levels of experience and existence with which people are inevitably confronted. These can often help individuals understand the context of their concerns. It is believed that a person's orientation towards the world and the four realms define their reality. There are various names for the four realms within existential therapy, however, the following are perhaps the best-known:

The physical realm : This world or realm is centred around physicality. It is the world we share with animals, the world of bodily needs. It is the world that stores desire, relief, sleep/wake cycles and nature. Birth, death and physical feelings/symptoms are also part of this realm.

2. The social realm: Within the social realm lies everything to do with relationships. Culture, society and language are here as well as work, attitudes towards authority, race and family. Emotions, friendships and romantic relationships are also part of the social world.

The personal realm: The personal realm is concerned with issues of the self. This includes intimacy (with self and

others), identity, personal characteristics and overall sense of self. Personal strengths and weaknesses are also important as well as the question of being authentic.

The making realm: The final realm is considered our 'ideal' world. Included within it are religion, values, beliefs and transformation. This is the dimension where we make sense of our lives and is considered the realm of transcendence.

One of the primary aims of existential therapy is to help people face anxieties of life and to embrace the freedom of choice humans have, taking full responsibility for these choices as they do so. Existential therapists look to help individuals live more authentically and to be less concerned with superficiality. They also encourage clients to take ownership of their lives, to find meaning and to live fully in the present.

Individuals who are interested in self-examination and who view their concerns as issues of living rather than symptoms of a psychiatric illness are more likely to benefit from this approach. Existential therapy is also well suited to those facing issues of existence, for example, those with a terminal illness, those contemplating suicide, or even those going through a transition in their life.

INTEGRATIVE OR HOLISTIC THERAPY OR ECLECTIC APPROACH

Many therapists don't tie themselves to any one approach. Instead, they blend elements from different approaches and tailor their treatment according to each client's needs.

MBCT (MINDFULNESS BASED COGNITIVE THERAPY)

MBCT is based on **Jon Kabat Zinn's** Stress Reduction program at the University of Massachusetts Medical Center, which was developed to help people suffering with chronic physical pain and disease. It includes simple

meditation techniques to help participants become more aware of their experience in the present moment, by tuning into moment-to-moment changes in the mind and the body. Participants learn the practice of mindfulness meditation through a course of eight weekly classes (the atmosphere is that of a class, rather than a therapy group), and through daily practice of meditation skills while listening to tapes at home. MBCT also includes basic education about depression (and, in our new study, suicidality), and a number of exercises derived from cognitive therapy that demonstrate the links between thinking and feeling and how best participants can care for themselves when they notice their mood changing or a crisis threatens to overwhelm them.

PLAY THERAPY

Play therapy is a therapeutic modality wherein play is the principle therapeutic factor. Because play is a child's natural medium of communication, play therapy is an ideal modality in which to allow children to express their feelings and deal with their emotional problems. In essence, play therapy is for children what 'talk therapy' or counselling is for adults. Play therapy provides the child with a consistent and predictable therapeutic relationship and environment in which to explore their fears, difficulties, struggles and pain, as well as hopes, dreams and fantasies. The underpinning assumption of play therapy is that, given this relationship and environment, the child has the inner resources to bring about growth and change in their own lives.

POSITIVE PSYCHOTHERAPY (PPT)

PPT is a therapeutic approach based on a premise, articulated and empirically explored by **Chris Peterson**, who emphasized that psychology ought to be concerned with strength as with weakness; as interested in building the best things in life as in repairing the worst; and as concerned with making lives of normal people fulfilling as with healing pathology

(Peterson, 2006a). Strongly influenced by Peterson's seminal work **Classification of Virtues and Strengths** (CVS; Peterson & Seligman, 2004), PPT which integrates symptoms with strengths, resources with risks, weaknesses with values, and hopes with regrets in order to understand inherent complexities of human experiences in a balanced way. Clients seeking therapy are neither mere conglomerate of symptoms nor embodiments of strengths. PPT systematically amplifies their positive resources; specifically, positive emotions, character strengths, meaning, positive relationships, and intrinsically motivated accomplishments.

TRANSACTIONAL ANALYSIS (TA) (IMPORTANT TOPIC)

The development of counseling and psychotherapy over last 100 years has drawn a significant number of influential psychological theories on mental illness from different perspectives. Transactional analytic (TA) therapy has its root in psychodynamic therapy, shares its philosophy with humanistic approach, and its characteristics with the behavioral approaches (Stewart & Tilney, 2000). The fundamental teaching of this process, as

determined by Berne (1961) is that there are historical mal-adaptations embedded in what is known as the childhood script, namely the process of repeating childhood behaviors and ideas throughout life. These need to be addressed through problem solving as opposed to avoidance or passivity in relation to new challenges in the person's life (Berne, 1961).

Transactional analysis is a type of psychotherapy developed by **Canadian psychiatrist Eric Berne** in the 1940s. Berne believed that communication with others comes from three distinct parts of self which he called **ego states**.

WHAT IS TA?

TA is a theory of personality and human relations offering systematic methods for personal growth and professional development.

TA gives us a picture of how people are structured psychologically.

TA also provides a theory of communication; how people communicate and why communication breakdowns occur.

TA offers a theory of child development- the concept of life script explains how our present life patterns originated in childhood- how decisions made in childhood may continue to influence behavior later in life despite painful consequences.

PHILOSOPHY OF TA:

We all are born OK, as princes and princesses (I'm OK, You're OK)
All of us have the capacity to think except the severely brain damaged
People decide their own destiny
and these decisions can be changed.

CONTRACTUAL METHODS:

A contract is an Adult commitment to one's self and/or to someone else to make a change. Contracts can be established to

change feelings, behavior, or psychosomatic problems.

According to **Claude Steiner**, a valid contract will fulfil the following four criteria:

Mutual Consent: both parties (client and counselor) agree and commit themselves to the terms of the contract

Valid consideration: all parties involved need to have clear and specific benefits

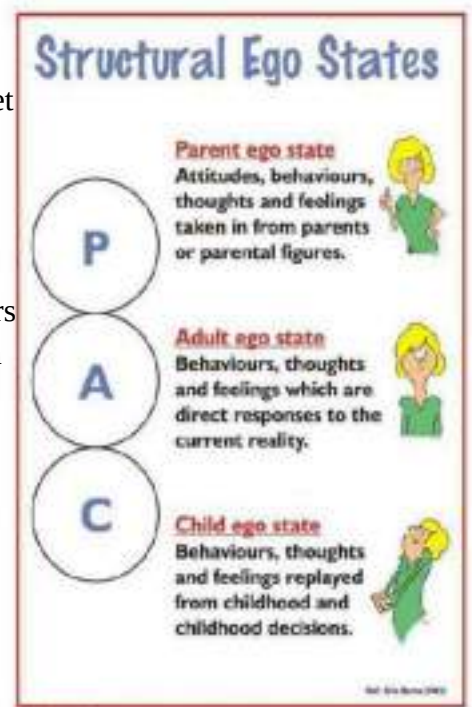
Competence: all parties involved need to be competent to enter into and fulfil the terms of the contract

Lawful object: all actions bound by the contract need to be aligned with the law of the land, and accepted ethical norms of the individual, group, society or the organization.

STRUCTURAL ANALYSIS OF EGO STATES

An **Ego State** is a consistent pattern of feeling and experience directly related to a corresponding consistent pattern of behavior.

Parent ego state is the set of feeling s, attitude s, values and prejudi ces and behavi ors taken in from parents and significant parental figures.



ego state is the Blind part (concept from Johari Window) or the unconscious part which is uncovered by asking from others or coming in contact with others (religion, society, parents, etc.).

Adult ego state is those feelings, attitudes, behaviors which are direct response to current here-and-now reality.

Child ego state is the archaic feelings, emotions, attitudes, and behaviors, which are remnants of the person's past, significantly from childhood and childhood decisions. It is the hidden part (concept from Johari Window) that we do not show to the world.

The goal of TA is to help clients strengthen their **Adult Ego State** and enhance their communications with other people.

THE FUNCTIONAL MODEL OF EGO STATES

Functionally ego states manifest as Critical Parent (CP), Nurturing Parent (NP), Adult (A), Adapted Child (AC) and Free or

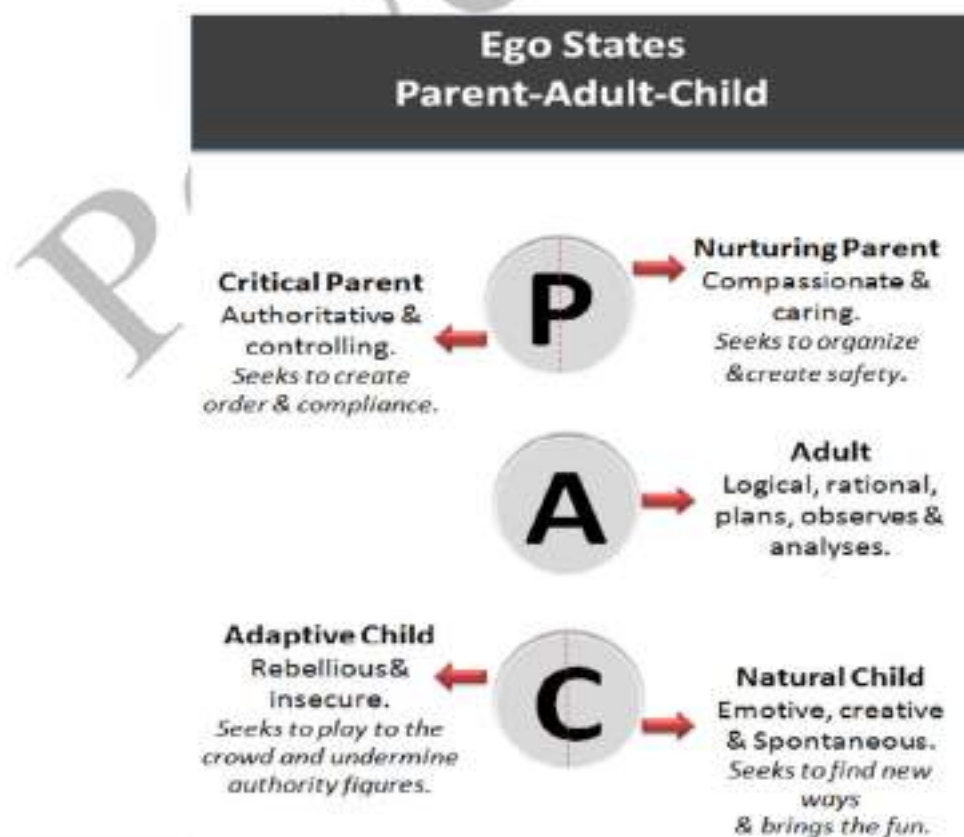
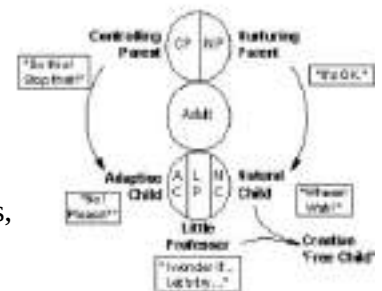
Natural Child (FC or NC).

While in **Critical Parent** people manifest themselves as disappointed, aggrieved, feeling 'always' right, patronizing, moralistic, judgmental, authoritarian, controlling, critical, putting down others. As **Nurturing Parent**, people act loving, reassuring, caring, concerned, encouraging, supportive, understanding, etc.

From **adult** we are open-minded, interested, confident, reality-based, process data, organize information, estimate probabilities, make logical statements, and provide non-judgmental feedback.

Rebellious Child is defiant and/or complaining, disobeys, throws temper tantrums, feels bored and distracts.

Adapted Child acts compliant, passive, loyal, withdrawn, pleases others, feels hurt, sulks and 'always' feels one down.



(Sometimes RC and AC are seen as one and the same form that manifests rebellion and insecurity).

When we are in our **Free Child (Natural Child)**, we are curious, energetic, fun-loving, spontaneous, excited and enthusiastic, and we express our anger, sadness and fear freely without any inhibition.

TRANSACTIONAL ANALYSIS: THE METHOD

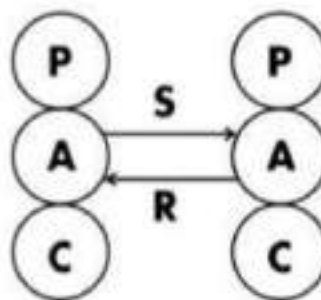
Definition of Transaction: A transaction is defined as a transactional stimulus plus a transactional response. It is the basic unit of social discourse or communication. The stimulus and response could be either verbal or non-verbal or both

TYPES OF TRANSACTIONS:

COMPLEMENTARY

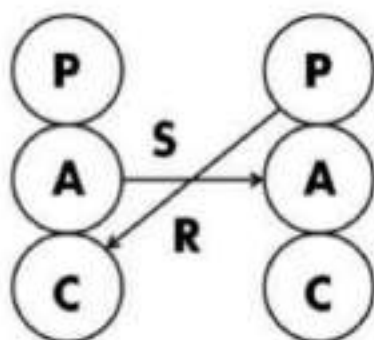
TRANSACTIONS: In a complementary transaction the transactional vectors are parallel and the ego state addressed is the

one which responds. It can be from any ego state to any other ego state.

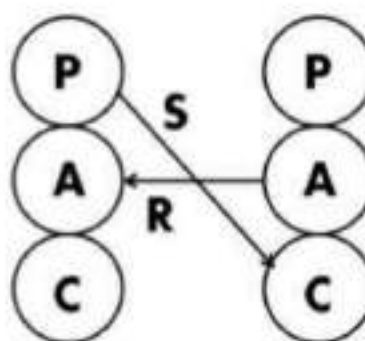


Adult - Adult complementary transaction

CROSSED TRANSACTIONS: When the transactional vectors are not parallel, or in which the ego state addressed is not the one which responds, it is called a crossed transaction. Here the response comes from an unexpected ego state.



A to A, P to C, crossed transaction

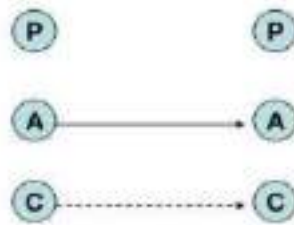


P to C, A to A crossed transaction

ULTERIOR TRANSACTIONS: They happen when a person appears to be sending one message overtly (social level) but secretly sending the other (psychological level).

Uterior 'Transact

- It is most complex because communication
- When an ulterior message is sent, it is often in an acceptable way.
- On the surface level, it has a clear adult message but a hidden message on the psychological level.
- Interactions, responses, actions which are explicitly stated.



THE KEY CONCEPTS AND PRINCIPLES OF TRANSACTIONAL ANALYSIS

Transference and Counter-transference. How memories from past relationships can impact on your current relationships.

Phenomenology. How our individual perception can impact on how we conduct relationships (this idea is also the basis of Gestalt)

Ego States -Structural and Functional Models in TA.

Structural – The basic idea of ego states (Parent, Child and Adult)

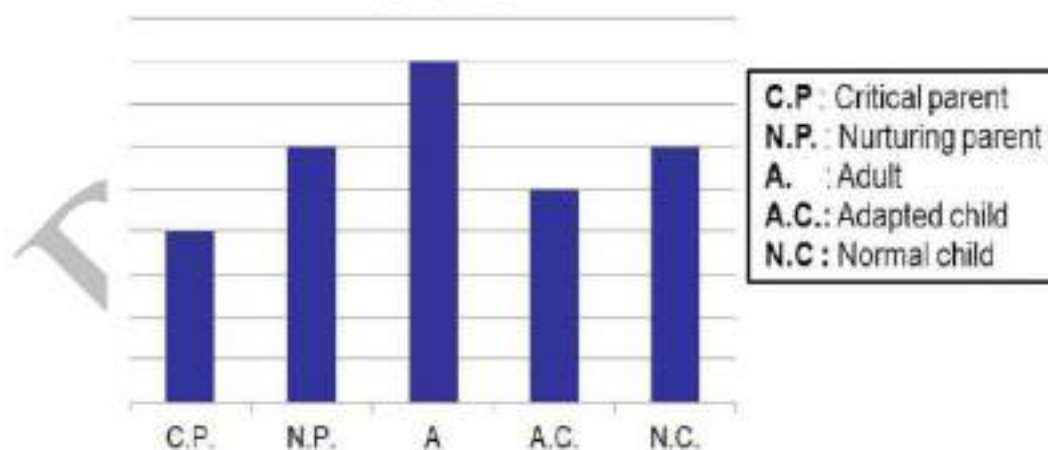
Functional – How the theory of ego states is applied (CP, NP, A, FC, AC/RC)

Games. How a series of interactions between two or more people follow predictable patterns and outcomes.

Racket feelings. Feelings that the child learns are acceptable and/or desired, which are used to replace the real feelings which a parent may find unacceptable.

KEY TERMINOLOGIES:

Egogram: The egogram is a relation diagram, depicting the amount of energy a person uses externally, or actively, as one relates to others. It is a bar chart representing the person's entire personality. It is also drawn as a way of providing feedback to someone regarding how others experience him or her.



CONSTANCY

HYPOTHESIS: Egograms are based on “constancy hypothesis”, which states that the amount of psychic energy within a person remains constant. For example, if a person starts to increase the energy in his Natural Child (Free Child), there will be

lesser energy available for his other ego states.

STROKES: A stroke is defined as a “unit of recognition”. They fulfil our hunger for recognition. They can be verbal or nonverbal, positive or negative, and conditional or unconditional.

For e.g., a positive stroke invites us to feel OK about others and ourselves whereas a negative stroke invites us to feel now OK about others and ourselves.

SCRIPT:

“All the world is a stage and all men and women the actors” said Shakespeare. From the TA perspective also, one’s life is a long drama, right from childhood to death – it begins, progresses and ends.

DEFINITION OF SCRIPT:

Life scripts consists of a set of decisions that are made by the child in response to the parental messages about self, others and the world.

Script is a life plan made in early childhood, reinforced by the

parents, justified by subsequent events, and culminating in a chosen alternative.

It is an on-going programme developed in the early childhood under parental influences, which directs the individual’s behavior in the most important aspects of his life.

LIFE POSITION:

According to Eric Berne, Life positions are psychological senses regarding self, others and life, which the person takes. These positions determine the person’s attitudes and perceptions. Depending on the experience and message the child encounters, each child takes one of the four life positions under stressful situations. This is called the Basic Life Position.

		You are okay with me			
I am not okay with me	I am not OK You are OK <i>The One-down position</i> <i>"I wish I could do that as well as you do."</i>	I am OK You are OK <i>The Healthy position</i> <i>"Hey, we're making good progress now."</i>		I am okay with me	
	I am not OK You are not OK <i>The Hopeless position</i> <i>"Oh this is terrible – we'll never make it."</i>	I am OK You are not OK <i>The One-up position</i> <i>"You're not doing that right – let me show you."</i>			
		You are not okay with me			

- ◆ The **I’m OK—You’re OK** position is known as the healthy position and is generally game-free. It is the belief that people have basic value, worth, and dignity as human beings. That people are OK is a statement of

their essence, not necessarily their behavior. This position is characterized by an attitude of trust and openness, a willingness to give and take, and an acceptance of others as they are. People are close to themselves and to others. There are no losers, only winners.

The **I'm OK—You're not OK** is the position of people who project their problems onto others and blame them, put them down, and criticize them. The games that reinforce this position involve a self-styled superior or one-up (the "I'm OK") who projects anger, disgust, and scorn onto a designated inferior, or scapegoat (the "You're not OK"). This position is that of the person who needs an underdog to maintain his or her sense of "OKness."

The **I'm not OK—You're OK** is known as the depressive or one-down position and is characterized by feeling powerless in comparison with others. Typically, such people serve others' needs instead of their own and generally feel victimized. Games supporting this position include "Kick me" and "Martyr"—games that support the power of others and deny one's own.

The **I'm not OK—You're not OK** is known as the position of hopelessness, futility and frustration. Operating from this place, people have lost interest in life and may see life as totally without promise. This self-destructive stance is characteristic of people who are unable to cope in the real world, and it may lead to extreme withdrawal, a return to infantile behavior, or violent behavior resulting in injury or death of themselves or others.

TRANSACTIONAL ANALYSIS: MANY NAMES ONE MASTER

The therapist analyses your past and how it affects the here and now, which is why it is known as an **Active Directive form of therapy**.

TA therapists believe that past life events (things that happened maybe in childhood) hold the key to understanding what our present difficulties are. This is known in TA therapy as the **presenting past**.

TYPES OF TRANSACTIONAL ANALYSIS

Since the death of Eric Berne in 1970, Transactional Analysis has developed along differing theoretical paths, sometimes referred to as 'Schools'.

The Classical School - sometimes referred to as **transactional analysis proper**, includes ego state analysis, life scripts and game analysis, is the original concept based on the work of Eric Berne, Claude Steiner and several close associates.

The Re-decision School - developed by Bob and Mary Goulding who integrated TA with Gestalt to make a powerful, vibrant means of personal change.

The Cathexis School - developed initially by Jacquie Schiff who worked with clients who had been diagnosed with severe mental health problems. Her approach was very controversial and was based on the theory of radical reparenting.

Modern Schools of TA are referred to as 'second wave' and have significantly influenced TA practice in recent years. Practitioners combine these new ideas with the Classical School to produce a contemporary, research-based approach to client work.

Relational school - outlined by Charlotte Sills and Helena Hargaden in their book "Transactional Analysis a Relational Perspective". This school draws strongly on the humanistic idea that the therapeutic relationship and empathy are central to the healing process.

Integrative psychotherapy -

developed by Richard Erskin. Like the relational school, this approach emphasizes the empathic nature of the relationship in helping clients regain the parts of self, which become lost or fragmented due to traumatic experiences.

SUMMARY OF TA

Transactional therapy is developed on the philosophical assumptions that human interact with others by subconsciously utilizing three ego-states (Adult, Parent, and Child) based on the life-script once written to meet survival needs in childhood but may no longer valid, and everyone has the capacity to consciously make decisions and change through therapy (Stewart & Tilney, 2000). In the Theory of TA, it's believed that individuals have three separate, functional ego states: Child, Adult, and Parent. Child Ego State: childlike behaviors and feeling, including Free Child & Adaptive Child. Free Child: spontaneous, creative, impulsive, feeling oriented, and self – centered. Adaptive Child: the compliant self, follows the rules learned in childhood to get needs met. Parent Ego State is originated in early childhood interactions, are carried through the life span, including Nurturing Parent (comforts, praises) and Critical Parent (disapproves, finds faults). Adult Ego State approaches life events by gathering the facts from external and internal ego status, processing information, and maintaining balance among Parent and Child ego states. In TA therapy, both counselors and clients must understand which ego states are functioning in their specific problem areas.

One of the limitations of TA may be its emphasis of cognitive aspect of human relationship and personality, and leave very limited attention for exploration of emotions. If this approach can be utilized with Gestalt approach, both cognitive and emotional dimension can be explored and integrated. Another limitation of TA is that it seems to have plenty of labels and jargon in this therapy, some clients may

get ‘lost in translation’ especially for those who are not familiar with western psychology. This therapy stresses achieving autonomy through a progressive process including social control, symptomatic relief, transference relief, and script cure, hence, TA may seem confrontational on some points, the therapist has to make efforts to respect others’ cultural beliefs and values while encouraging them break free from old “script” and live their life in a new and more resourceful way.

DIALECTICAL BEHAVIOUR THERAPY (DBT)

It is a type of talking therapy which was originally developed by an American psychologist named Marsha Linehan. It is based on cognitive behaviour therapy (CBT), but has been adapted to meet the particular needs of people who experience emotions very intensely. In a nutshell, ‘dialectics’ means trying to balance seemingly contradictory positions. For example, in DBT you will work with your therapist to find a good balance between: • Acceptance – accepting yourself as you are. • Change – making positive changes in your life. You might eventually come to feel that these goals are not as conflicting as they seem at first.

Art therapy involves the use of creative techniques such as drawing, painting, collage, coloring, or sculpting to help people express themselves artistically and examine the psychological and emotional undertones in their art. With the guidance of a credentialed art therapist, clients can "decode" thenon-verbal messages, symbols, and metaphors often found in these art forms, which should lead to a better understanding of their feelings and behavior so they can move on to resolve deeper issues.

Performing art generally includes dance, music, songs, drama in organized fashion. Besides, painting, sculpturing, Yoga, farming etc. are the performing arts. Performing art therapy is the therapeutic process in which performing art therapist tends to explore one's needs, values, crisis in the life cycle; improvise performing art in such a fashion that client can understand own needs and can develop coping strategy to overcome the crisis. In performing art, client is not merely receptive but productive and creative. Therefore, the objective of therapist will be to stimulate the client to understand self and socially accepted coping strategies. It develops adaptability within individual.

THEORIES OF MOTIVATION IN EDUCATION

Motivation itself has a vast scope to cater for, and several **motivational theories** are relevant to the learning domain. The following theories contribute to the essential outcomes of the learning process without being dependent on any other theories in the education domain:

- intrinsic and extrinsic motivation theory
- self-determination theory (SDT)
- the ARCS model
- social cognitive theory
- expectancy theory

Self-determination theory (SDT) and the ARCS model are widely utilized in the motivation domain for learning discipline. The implementation level of theories such as social cognitive theory and expectancy theory is still in initial stages but can significantly contribute to understanding motivation in learning as well as other aspects of life where motivation is crucial.

INTRINSIC AND EXTRINSIC MOTIVATION THEORY

According to Ryan and Deci, **intrinsic motivation** defines an activity done for its own sake without the anticipation of external rewards and out of a sense of the sheer satisfaction it provides (2000).

The right level of challenge, coupled with adequate skills, sense of control, curiosity, and fantasy, are some key factors that can trigger intrinsic motivation. And when combined with will power and positive attitude, these elements can help sustain motivation over time.

Some studies show that intrinsic motivation and academic achievement share significant and positive correlates (Pérez-López & Contero, 2013). Intrinsic motivation can direct students to participate in academic activities to experience the fun, the challenge, and the novelty away from any external pressure or compulsion and without expectations of rewards (Ryan, & Deci, 2000).

In contrast, extrinsic motivation describes activities students engage in while anticipating rewards, be it in the form of good grades or recognition, or out of compulsion and fear of punishment (Tohidi, & Jabbari, 2021).

Motivation can be cultivated extrinsically at the initial stage, particularly when it comes to activities that are not inherently interesting, as long as the ultimate goal is to transform it into intrinsic motivation as the learning process unfolds. The rationale for this has to do with a short shelf life and a potential dependence on rewards.

Although extrinsic motivation can initially spark a high level of will power and engagement, it does not encourage perseverance and is challenging to sustain over time due to hedonic adaptation. Finally, external rewards or compliments undermine the possibility that students will engage in the educational activities for their own sake or to master skills or knowledge.

Nevertheless, both types of motivation have their place in the process of learning.

While intrinsic motivation can lead to greater levels of **self-motivation**, extrinsic motivation often offers that initial boost that engages students in the activity and can help sustain motivation throughout the process of learning over time (Ti, & Lynch, 2016).

It is no easy undertaking to guide students to learn **how to be highly motivated**, to face the challenges, understand the process, and to be able to apply their new-found knowledge in real-life circumstances.

2. SELF-DETERMINATION THEORY

Self-determination Theory addresses intrinsic and extrinsic motivation further. It explains it in terms of self-regulation, where extrinsic motivation reflects external control of behavior, and inherent motivation relates to true self-regulation (Ryan, & Deci, 2000).

SDT tells us that intrinsic motivation is closely related to the satisfaction of basic psychological needs of autonomy, competence, and relatedness, and illustrates how these natural human tendencies related to several key features in the learning process.

Here autonomy is related to volition and independence, and competence is associated with the feeling of effectiveness and self-confidence in pursuing and accomplish academic tasks. Relatedness provides the feeling of safety and connectedness to the learning environment where such enables and enhances students' academic performance and motivation (Ulstad, et al., 2016).

Self-determination theory evolved out of five other sub-theories that further support its claims. First, the Cognitive Evaluation Theory (CET), which explains the effects

of external consequences on internal motivation, draws our attention to the critical role autonomy and competence plays in fostering intrinsic motivation by showing how it is vital in education, arts, sports, and many other domains.

Second, Organismic Integration Theory (OIT) and Causality Orientations Theory (COT) further explain motivation as occurring along a spectrum from an amotivational stage toward motivational states where the focus is on competence.

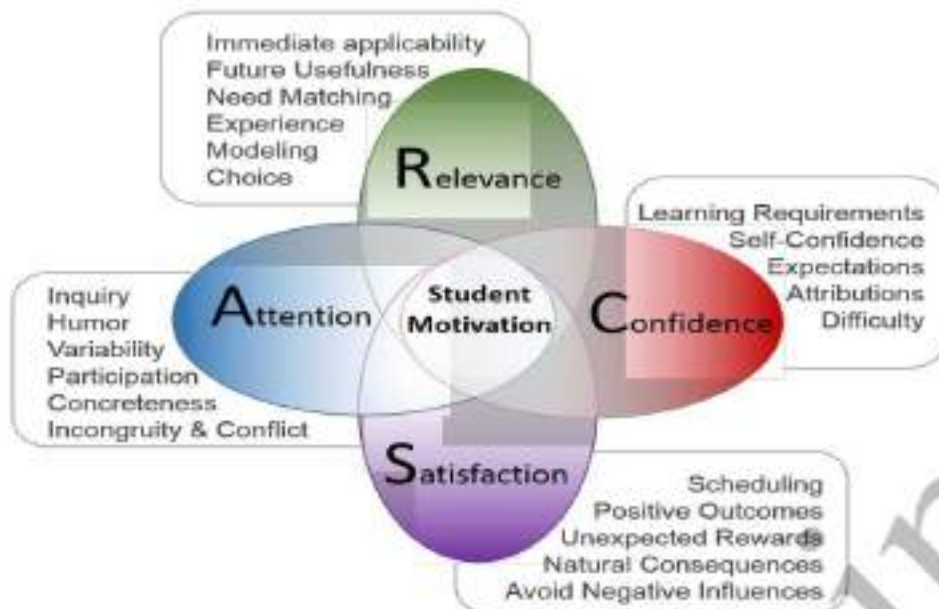
Next, Basic Psychological Needs Theory (BPNT), which classifies human needs into three primary psychological needs for autonomy, competence, and relatedness, shows how the satisfaction of those needs is crucial for engagement, motivation, healthy progress and well-being among students (Gagne, & Deci, 2014).

Finally, Goal Contents Theory (GCT) shows the relationship between the fundamental needs satisfaction and **well-being** based on intrinsic and extrinsic goal motivation, where intrinsic goals lead to greater achievement and better academic performance, especially within the social setting of the educational environment (Ryan, & Deci, 2000).

3. ARCS MODEL

ARCS is an abbreviation for Attention, Relevance, Confidence, and Satisfaction. The ARCS model is an approach to instructional design that focuses on the motivational aspects of the learning environment by addressing four components of motivation:

arousing interest
creating relevance
developing an expectancy of success,
and
increasing satisfaction through
intrinsic and extrinsic rewards
(Keller, 1983).



The ARCS model stresses capturing students' attention as critical to gaining and sustaining their engagement in learning and shows how this can be accomplished through the use of attractive and stimulating medium or learning

material that is relevant to their experiences and needs.

It recognizes how confidence is related to the students' anticipation of success and how positive feelings about the learning process lead to greater satisfaction from the acquisition of knowledge (Keller, 2008).

Major Categories and Definitions		Process Questions
Attention	Capturing the interest of learners; stimulating the curiosity to learn	How can I make this learning experience stimulating and interesting?
Relevance	Meeting the personal needs/ goals of the learner to effect a positive attitude	In what ways will this learning experience be valuable for my students?
Confidence	Helping the learners believe/ feel that they will succeed and control their success	How can I via instruction help the students succeed and allow them to control their success?
Satisfaction	Reinforcing accomplishment with rewards (internal and external)	What can I do to help the students feel good about their experience and desire to continue learning?

4. SOCIAL COGNITIVE THEORY

Social Cognitive Theory (SCT), implemented today in various domains from education and communication to psychology, refers to the acquisition of knowledge by direct observation, interaction, experiences, and outside media influence (Bryant, & Oliver, 2002).

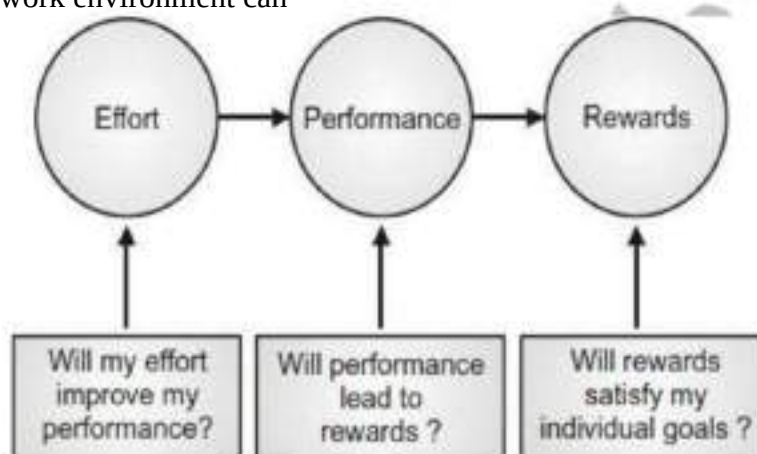
It rests on the assumption that we construct meaning and acquire knowledge though

social influence from daily communication to the use of the internet and explains the relationships between behavior, social and physical environment, and personal factors.

SCT illustrates how people gain and maintain several behavioral patterns and provides basic intervention strategies like interactive learning, which allows students to gain confidence through practice (Bandura, 1997).

5. EXPECTANCY THEORY

Expectancy theory, originally developed to explain how the work environment can



In the educational context this would translate into the student perception that their effort will lead to good or better performance (expectancy) followed by student's belief that their performance will lead to achieving the desired goal and rewards (instrumentality) and lastly, the value of the rewards is satisfactory and support the goals of the student (valence) (Bauer et al., 2016).

FACTORS INFLUENCING THE ACADEMIC PERFORMANCE OF THE STUDENTS:

The factors that influence the academic performance of the students have been stated as follows:

motivate employees, strives to show the relationship between the expectations of success and anticipation of rewards and the amount of effort expended on a task and how it relates to the overall performance (HemaMalini, & Washington, 2014). Simply put, it explains motivation as a choice based on the expectation of the results of selected behavior.

The expectancy theory explains motivation in terms of reasons we engage in specific behaviors where we expect that effort will lead to better performance, which in turn will lead to valued rewards.

ATTITUDE OF THE STUDENTS–

In secondary schools, the students are between 12 to 18 years of age. They possess the abilities to differentiate between what is appropriate and what is inappropriate. Goal-oriented students usually possess positive feelings regarding their school experiences, they possess the traits of discipline, diligence, and resourcefulness, are avid readers and tend to devote less time towards recreation and leisure activities. It is vital for the students to possess positive thinking in terms of their schools, teachers and academic subjects. With positive attitude, they will be able to dedicate themselves wholeheartedly towards learning and generate the desired academic outcomes (Maina, 2010).

SCHOOL RESOURCES– Within schools, it is vital to make provision of

resources that can be utilized to enhance the academic performance of students. The textbooks, notes, learning materials, hand-outs, technology, library facilities and laboratory facilities, especially in science subjects should include the essential materials. When students will be provided the necessary tools and equipment, they will be able to acquire a better understanding regarding academic concepts and how to perform the experiments. In some cases, especially the students belonging to deprived, marginalized and socio-economically backward sections of the society, cannot afford the books and materials required for learning, hence, they are dependent upon the library facilities and fellow students to obtain the books and other materials (Maina, 2010).

LEADERSHIP ASPECTS— The principal, teachers and the administrative staff members of the school are vested with the authority to implement the managerial functions of planning, organizing, controlling, and directing the activities. The principal primarily has the right to make decisions. The decisions can be made either at one's own discretion or they may consult others and seek ideas and suggestions from teachers and other staff members of the school. The major role of leadership aspects in influencing the academic outcomes of the students is based upon the administration and management of the school. When there are proper rules, policies and management is put into practice in an appropriate manner, then there would be improvement in academic performance of the students (Maina, 2010).

SKILLS AND ABILITIES OF THE TEACHERS— Teachers have an imperative role in influencing the academic performance of the students. They are bestowed with the authority to direct all the classroom activities and administer learning

It is vital for the teachers to possess the traits of professionalism and conscientiousness. They need to possess an approachable nature, listen and provide solutions to the problems experienced by the students. They should possess adequate knowledge and information regarding the subjects that they are teaching, usage of technology, modern and innovative methods in the teaching and learning processes, managing discipline and directing all of the classroom as well as school activities and functions in a well-organized manner. The teachers in some cases are strict, but strictness should be maintained within limits. The main objective of the teachers should only be to enhance the academic performance of the students and lead to their effective development (Maina, 2010).

CLASSROOM ENVIRONMENT— The academic concepts are made known to the students by the teachers within classroom. Teachers have the main job duty of completing the subject syllabus. Therefore, it is vital that classroom environment should be disciplined, and well-ordered (Kudari, 2016). Within the classroom, it is vital for the teachers and the students to implement the traits of morality and ethics. It is vital to promote mutual understanding, amiability and co-operation among the teachers and students as well as among the fellow students. The efficiency in the management of the classroom, introduces a well-organized and efficient management of the lesson plans, instructional strategies, teaching-learning processes and so forth. When there is discipline and effective communication among the individuals, then it would help the students learn better and improve their academic performance.

ROLE OF PARENTS— Home is referred to as the place from where the foundation of learning and education takes place. In order to produce good

ood academic outcomes, it is vital for the parents, children and other family members to encourage a learning atmosphere within homes. For instance, when students experience problems in certain subjects, then parents are responsible for providing help. This help may be in the form of private tuitions or they themselves may teach their children. They make provision of technology and other learning materials at home to enhance the academic performance of their children. Parents play an important role in leading to operative growth and development of their children (Kudari, 2016). In schools, whatever problems that children go through regarding academics and other areas, they normally communicate to their parents. Parents are sources of security, encouragement and help their children in providing solutions to their problems.

SOCIAL CIRCLE– Children get enrolled in schools not only to learn academic concepts, but they also learn, how to interact and socialize with others. Students usually form friendly terms and relationships with the fellow students. Forming a social circle and friendships have a positive effect upon the academic outcomes of the students. As when one has to work on a project or prepare for a test, then group study is in most cases beneficial. It also causes social satisfaction and happiness in one's student life (Kudari, 2016). Forming a social circle proves to be beneficial to the individuals in number of ways, such as, solving academic problems, getting involved in leisure activities, sharing one's joys and sorrows, and so forth.

HEALTH RELATED FACTORS– Learning is not an easy task for the students. It is essential for them to possess diligence, resourcefulness and conscientiousness to enhance their learning. In order to generate positive academic outcomes, it is essential for the students to maintain their

psychological and physical health. When a student is healthy, then he will be able to contribute an active role towards learning. On the other hand, factors such as, stress, anxiety, fear, trauma, depression or physical health problems prove to be impediments within the course of their academic achievement. It is vital for the students to take pleasure and look towards their school and classroom activities from a positive viewpoint. Consuming a healthy and nutritious diet, getting engaged in extra-curricular activities, staying calm and taking pleasure in studying are some of the aspects that help in maintaining good psychological and physical health (Srinivas, & Venkatkrishnan, 2016).

MOTIVATING AND ENCOURAGING STUDENTS

– In academic learning, some of the concepts are difficult to learn and understand. When problems and difficulties are experienced by the students, then they need to obtain assistance from others. When students are unable to achieve the desired grades, then instead of getting angry on them, the teachers and parents need to make provision of help and assistance. They should motivate the students and encourage them to do well in future. They need to understand their weaknesses and help them. When students find certain areas difficult to learn, then teachers should repeat the concepts, provide them class and homework assignments, so that they are able to acquire complete understanding

of the concepts (Srinivas, & Venkatkrishnan, 2016).

VISUAL AND HEARING IMPAIRMENTS

– Visual and hearing impairments are stated as the disease related factors. There are students, who experience these problems. But when they are achievement-oriented, they do not let these problems become barriers

within the course of achievement of their academic goals. When these problems are severe and the students do not have the necessary resources to facilitate learning, then it is likely that their academic performance may get affected in a negative manner. Visual impairment usually imposes problems in handwriting and observing the blackboards. Hearing impairment imposes problems in listening to the instructions and explanations given by the teachers within classrooms and it is associated with poor language development. It impedes the communication abilities of the students and hence their academic performance gets influenced in a negative manner (Srinivas, & Venkatkrishnan, 2016).

COUNSELLING AND GUIDANCE SERVICES–

In secondary schools, there are numerous problems and difficulties that students experience. Some may even get involved into violent and criminal acts, drug abuse, HIV, other sexually transmitted infections, teenage pregnancies, induced abortions and unemployment. These factors have a direct impact upon not only their academic performance, but also their lives. These may hamper their psychological approach and they may even experience problems of depression, trauma or stress to a major extent. Counselling and guidance services need to be established in schools to help the students to provide solutions to their problems, focus upon their studies, and become responsible members of the community (Maganga, 2016). It is vital for the individuals, who work as counsellors to specialize in a particular area and possess adequate knowledge, expertise and skills. Counselling is related to various areas, such as career, academics, depression, stress, health, family issues and so forth.

DEVELOPMENT OF STUDY SKILLS–

In order to enhance one's academic performance, it is vital for the individuals to develop study skills within themselves. The students themselves need to generate awareness regarding study skills, so that they are able to produce desired academic outcomes. Some of the study skills include, memorizing from the textbooks or other materials, making notes, practicing writing essays and articles, especially in languages, practicing calculations in mathematics and so forth. One of the important areas is, when one is studying, it is vital to completely concentrate towards one's studies. Inability to completely concentrate is one of the factors that leads to undesired academic outcomes. Memorizing is regarded as one of the rare techniques, hence, the teachers encourage students to acquire understanding of the concepts instead of memorizing.

TIME MANAGEMENT–

Students in secondary school have a busy schedule, hence it is vital for them to generate awareness in terms of effective time management. Research has indicated that the normal schedule of the secondary school students comprises of school hours, then they need to spend some time in completion of home-work assignments. They also get involved in some kinds of extra-curricular activities and sports. Playing and getting engaged in creative activities, not only help them concentrate better, but they are able to stimulate their mind-sets. For the secondary school students, it is important to get engaged in extracurricular activities and for this purpose, they need to implement proper time management skills. It is essential for the students to create a balance between all the tasks and activities. The activities that are more important should be given more time and lesser amount of time can be spent on the activities that are less important.

HOME ENVIRONMENT– The home environment should be amiable and pleasant in order to generate appropriate academic outcomes. Within home, among the family members, it is vital to initiate measures to form effective relationships. They should communicate with each other in an appropriate manner and minimize the occurrence of conflicts and disputes. Research has indicated that conflicts may take place with respect to the Availability of Materials and amiable environmental conditions within the home are essential for the students to focus upon studies and generate the desired academic outcomes.

TEACHING-LEARNING

METHODS– The teaching-learning methods and strategies should be appropriate and encouraging to the students. The teachers in school are the ones that contribute an imperative part in promoting learning among the students. It is essential for them to ensure that the teaching methods used should prove to be beneficial to the students. For instance, if the students are able to learn better through dictation of notes, then teachers should provide notes. If the students are able to learn better through verbal explanation, then they should promote verbal explanation and so forth. Within home, if students are taking private tuition of certain academic subjects such as, mathematics or science, or their parents teach them, then too, it is vital for the parents and tutors to make sure that effective teaching learning methods are implemented, which may encourage learning among the students and help them understand better.

APPROACHABILITY AND PROFESSIONALISM

– The teachers in school are required to be approachable and professional in their conduct. They are the ones, whom students' approach, in case they have any problems and difficulties. When the teachers are friendly and generous, then the students feel comfortable in not only approaching them, but also in clarifying their doubts. The professionalism and approachable attitude on the part of the teachers is of utmost significance in influencing the academic performance of the students in a positive manner. On the other hand, at home, parents or tutors are the ones, who supervise their studies, hence, it is vital for them to be professional in their conduct. Teaching should be implemented in a calm and pleasant manner. Any kind of harsh attitude should be avoided, as it may demotivate the students.

TEACHER'S EFFECTIVENESS

There are various feedback devices to be used to modify the teacher behavior. The following are few commonly used such as Simulated Social Skill Training, Micro-Teaching, Programmed Instruction, Team Teaching, Interaction Analysis and T. Group Training. These are briefly explained as below:

Simulated Social Skill Training: The simulation technique is to induce certain behavior in an artificial situation. Pupil teacher has to play several roles as a teacher, as a student and as a supervisor. It's a feedback mechanism. It is a socio-drama related to practice and gives control over teaching variables. Important is pupil teacher is teaching in non-stressful conditions.

Micro-Teaching: Micro-Teaching provides teachers with a practice setting or instruction in which the normal

complexities of class room are reduced and in which the teacher gets feedback on performance.

Programmed Instruction: The method in which students are individualized in instruction in which students are active and proceed at his own pace and provided with immediate knowledge of result. The programmed learning is a strategy in which various kinds of intellectual, emotional and motor experiences are provided to learner in a controlled situation through a variety of devices like book, teaching machines, teacher, radio, television, etc.

Team Teaching: It is an instructional situation where two or more teachers possessing complementary teaching skills cooperatively plan and implement the instruction for a single group of students using flexible scheduling and grouping techniques to meet the particular instruction.

Interaction Analysis: It is a technique for analyzing and observing the classroom behavior. It provides the structure, component and flow of behavior of classroom activities. It is a feedback device.

T-Group Training: It is also a feedback device. It is a leaderless group of trainees numbering eight to twelve, discuss their own problems of teaching without any agenda and suggest some solutions on basis of their experiences.

SUGGESTIONS TO IMPROVE TEACHER'S EFFECTIVENESS:

More easily access high-quality curricula aligned to the common, college and career-ready, internationally-benchmarked standards: Teachers should increasingly be easily able to go online and find a myriad of resources linked to the specific standards they are teaching, with high-quality resources "rising to the top" because they are downloaded

and cited often and rated highly by other teachers. This includes a growing repository of open education resources – online open source instructional materials that can be modified and customized by end users.

Collaborate with and learn from one another across geographic boundaries: The web should enable teachers to find excellent lesson plans created by others, view video soft-top-notch delivery, and tap the wisdom of other teachers in meeting their toughest challenges via online discussion boards and blogs, and via direct communication with other teachers.

Use online and software-based tools to assess students regularly, diagnose learning challenges, and select the best responses to the barrier's students face.

Benefit from the trove of data that should be generated if more instruction happens online. The more students receive instruction and do their work online, the more possible it should be to capture data from their experiences about the most effective way to convey content, motivate students, and address students' challenges.

ROLE OF A STATE FOR IMPROVING TEACHER EFFECTIVENESS

States are in a strong position to play several roles in this process:

- **Committing to reaching increasing portions of students with top-quintile instructors and instruction** through various reach extension methods (both in-person and remotely with technology);

Accelerating progress by creating demand for outstanding remote instruction where it can help most (e.g., by requiring or encouraging districts to

offer it if they cannot fill teaching slots with effective instructors, and by requiring that remotely offered instruction meet a top-tier learning progress standard);

- **Reducing state-level policy barriers** to the use of these mechanisms (e.g., rigid seat-time requirements, upper-grade class size maximums, teacher certification requirements that would block high-quality out-of-state instructors from teaching remotely); and,

Directing the benefits to students who need them the most by providing funding or other inducements for hard-to-staff schools in particular to make use of the emerging opportunities. Otherwise, the benefits of new technologies should tend to flow more naturally to advantaged schools and students first.

GUIDANCE IN SCHOOLS: NEEDS, ORGANIZATIONAL SET UP AND TECHNIQUES

In India various education commissions have recommended importance of these services in the education system and thus two distinct points of view emerged. According to the first view point guidance and counselling services should be an integral part of the educational process and the educational functionaries (namely subject teachers) should render these services during teaching in the classroom. The second view point emphasizes a separate guidance programme organized by fully professionally trained guidance personnel. On the basis of these two viewpoints guidance and counselling programmes can be categorized as unstructured or structured.

We have briefly described here the unstructured and structured guidance and counselling programme.

NEED FOR GUIDANCE AND COUNSELLING PROGRAMME

Guidance and Counselling programme aims to fulfill the following needs of students in school system.

Educational Guidance Need: In educational guidance the emphasis is on providing assistance to students to perform satisfactorily in their academic work. Other needs include selection of appropriate course of study, overcome learning difficulties, foster creativity, and improve levels of motivation and soon.

Vocational Guidance Need: Students also need guidance in selection of a vocation and preparation for the same. Vocational guidance enables students to acquire information about career opportunities, career growth and educational/ training facilities.

Personal, Social and Emotional Guidance Need: Guidance concerned with personal need of individuals enable them to adjust themselves to their environment so that they become productive and efficient human being. The main purpose of social guidance is to make individual's effective and responsible citizen by enabling them to contribute to the society, assume leadership, conform to social norms, develop healthy and positive attitude towards different sections of the society, etc. Guidance also helps individuals in attaining emotional maturity. In addition, guidance services also assist individuals in maintaining health, both mental and physical and spending leisure time more meaningfully by developing good hobbies.

ROLE OF GUIDANCE AND COUNSELLING FUNCTIONARIES

Guidance and counselling services are rendered by trained professionals. They play the following roles to fulfill the guidance and counseling needs of individuals. These roles are related to guidance needs of an individual such as educational, vocational and personal social and emotional.

EDUCATIONAL ROLES: It includes

- interpreting the educational programmes to the parents and the community.
- helping parents understand the abilities and interests of their children and respective educational opportunities available.
- providing assistance to children in the choice of elective subjects – academic and vocational courses.
- maintenance of record of the students' progress in the form of cumulative records card (CRC).
- assisting students in exploring their potentials through hobby clubs, students' house, etc.
- orienting new students about curricular and co-curricular activities of the institutions.
- assisting students and parents in appraising students' academic progress.
- diagnosing learning difficulties in students.
- organizing remedial teaching based upon the diagnosis.
- identifying students (gifted and disabled) with special needs.
- assisting students in adjustment to school subjects.
- conducting follow-up of school pass-outs.

VOCATIONAL ROLES: It includes

- organizing group guidance activities such as talks (orientation/ career / class), exhibition, field visits, etc. to help students in exploring world of work.

providing occupational information to students related to general and vocational subjects.

administering psychological tests such as achievement, general and special ability, interest, personality, etc. to identify personal characteristics.

assisting students in changing their academic options, if necessary.

helping students in developing acquaintance with occupations during educational visits.

orienting students about vocational subjects and the career opportunities available through individual counselling and group guidance activities.

PERSONAL SOCIAL ROLE: It deals with the following:

counselling students with adjustment and behavioural problems.

providing counselling to students and their parents who are on the verge of dropping out.

keeping personal record of students about their personal characteristics and family background.

ensuring involvement of other teachers in all round development of students.

To cater to these guidance functions, a counsellor/career teacher can organize various guidance services in a school setting. The planning and organisation of these services depend upon the needs of the students and availability of resources (time, budget and support). Here we are briefly describing these guidance services.

ORIENTATION

SERVICE: Under the school guidance programme, orientation service assists the new students to know about the institution through orientation talk, tour of the school, students' handbooks and assemblies.

INDIVIDUAL INVENTORY

SERVICE: Individual analysis is the activity of the school guidance programme which follows systematic

assessment procedures to identify the characteristics and potentials of every student. Standardized test results, school records and observation reports such as anecdotal records, rating scales, self-reporting techniques, (questionnaires and autobiography) are used for collecting data about students' family background, abilities, aptitudes, interests, achievements and other psychological variables. A counsellor/ guidance worker takes help of teacher, peers and parents to prepare a cumulative record (CRC) of students.

CAREERS INFORMATION

SERVICE: Careers information includes information about all types of occupations and industries, educational and training facilities, apprenticeship facilities, scholarships and stipends, local and national employment trends and opportunities and occupational structure of the country. For providing such information to students, group guidance activities such as field trips, career conference-cum-exhibition, career talks, etc. are organized. A Career Information Corner may be set up to display career information materials such as booklets, monographs, pamphlets and posters and charts, newspaper cuttings on occupational and educational opportunities.

COUNSELLING

SERVICE: Counselling is generally a one-to-one helping relationship which focuses upon the individuals' growth and adjustment, and problem solving and decision-making needs. The aim of counselling is self-understanding, self-acceptance and self-realization. Counselling requires a high level of professional training and skill therefore; it should be done by trained counsellors who have the necessary qualifications.

PLACEMENT

SERVICE: Placement, as understood in the guidance field, usually indicates assistance offered to individuals in taking the next step, whether toward further training, a job situation or a different course of study. At present job placement in selected areas is done by the employment exchanges, however, many private placement agencies are also working to help people in their proper placement. A counsellor/ career teacher can provide information to students about such agencies.

FOLLOW-

UP SERVICE: Through this service, a record of school leavers (drop-outs and pass-outs) is kept. Before leaving the school, the students are told about follow-up and its purpose and through questionnaires, interviews, letters and telephone calls, data is collected. Professional institutions use email for getting information from the pass outs.

REFERRAL SERVICE: Students with special needs and with specific problems (emotional, behavioural or educational) may require special referral to special institutions or professionals for proper care and treatment. The counsellor or career teacher keeps addresses of such institutions/ agencies/ professionals who can be approached at the time of any need.

RESEARCH AND EVALUATION IN GUIDANCE AND COUNSELLING

SERVICE: Research is necessary for the advancement of the profession of counselling, especially to judge the effectiveness of various techniques and methods being used with particular groups. Evaluation is a means of assessing the programme effectiveness and to

reduce improvements in the programme. The evaluation could be done by obtaining feedback or follow-up of recipients of guidance activities. The findings of evaluation programmes can be used to establish the validity of the guidance programme. A counsellor/career teacher should plan this aspect while planning guidance programme in school.

CHARACTERISTICS OF A COMPREHENSIVE GUIDANCE AND COUNSELLING PROGRAMME

A comprehensive guidance and counselling programme should be characterized with the following:

- Careful and consistent development programme
- It should be developmental, preventative, remedial and corrective. It should address the needs of the individuals. It should be goal-oriented and accountable. It should be an integral part of the school curriculum and should be complementary to other school activities. It should be balanced, encompassing the four fundamental fields/areas of guidance e.g. personal, educational, social, and vocational/career. It should determine the services to be offered such as orientation, information, counselling, etc. It should involve staff members as per the requirement of the programme. It should create and demonstrate an atmosphere of teamwork. It must be flexible. Should take into account age, location/environment, cultural background, sex, economic status of the target group. Should benefit all students rather than just a few. Should be printed and displayed for

information of the beneficiaries.

COMPONENTS OF A COMPREHENSIVE GUIDANCE AND COUNSELLING PROGRAMME

Gysbers and Handerson (1994) are of the opinion that School Guidance Programme should be comprehensive. According to them an educational and vocational guidance programme comprises following four components:

SCHOOL GUIDANCE AND COUNSELLING CURRICULUM:

School guidance and counselling curriculum is designed systematically according to grade level so that it serves all students at the classroom or group level. The main emphasis of this curriculum is on decision-making, self-knowledge, career exploration and career development. The curriculum is organized around three major areas:

Learning to Live (Understanding and appreciating the self

, others, home and family; developing a sense of

community; making decisions and setting goals; understanding safety and survival);

Learning to Learn (Making decisions, setting goals, and taking action; understanding interaction between home, family, school and community; and understanding factors which affect school achievement); and

Learning to Work (Learning the relationship between personal qualities and work; exploring careers; learning how to use leisure time; learning the relationship between education and work; learning

to work together; and learn in how the community relates to work).

INDIVIDUAL

PLANNING: Individual planning includes the counseling activities meant to assist all students to plan, monitor and manage their own personal, educational, social and career development. Activities of this component of the programme are offered to all students to assist them in the development and implementation of their personal, educational, social and career components, in accordance with the skills and information they gather. This component generally offers students the opportunity for self-appraisal, and to plan for the world of work. One way would be to help students to understand themselves through individual or group activities.

RESPONSIVE

SERVICES: Above two components are for all students. This component provides special help to those students who are facing problems that interfere with their educational, or career or healthy personal social development. This component includes activities such as individual and small group counselling, consultation with staff and parents and referral of students and family members to other specialist or programmes, special education, social worker, or medical specialists, etc. Services like Counselling service, Referral service, Consultation service, Placement and Follow-up service, Research and Evaluation service are generally used for responsive service.

4) **SYSTEM SUPPORT:** This component has following two parts:

i) Activities necessary to support

guidance curriculum, individual planning and responsive service. The support from the system required for these activities include orientation of school staff, identification and utilization of community resources, budget, infrastructure and policy support.

ii) Activities implemented by counsellor/teacher counsellor support educational programmes. The counsellor / teacher counsellor provides support to the educational programme includes the assistance through individual planning activities such as selection of courses/ subjects and co-curricular activities by student; linkage with special education programme and preparation of students' personal information including curricular /co-curricular achievements record with the help of school staff.

COUNSELLING

Guidance and Counseling is defined as a planned and organized work aimed at assisting the trainee to understand himself/herself and his/her abilities and develop his/her potentialities in order to solve problems and achieve psychological, social, educational and professional compatibility, and also to achieve objectives within the framework of teachings. Guidance is a kind of advice or help given to the individual especially students, on matters like choosing course of study or career, work or preparing for vocation, from a person who is superior in the respective field or an expert. It is the process of guiding, supervising or directing a person for a particular course of action. Counseling is not just giving advice or making a judgment, but helping the client to see clearly the root of problems and identify the potential solutions to the issues. The counselor also changes the viewpoint of the client, to help him/her take the right decision or choose a course of action. It will also help the

client to remain intuitive and positive in the future.

Counseling takes place in the context of a helping relationship in which the counselor and the client work together to resolve a problem, change behavior or foster personal growth and awareness. Although clients may have a number of helping relationships with friends or family, the counseling relationship is different in a number of ways –

The counseling relationship is not reciprocal. The counselor's job is to focus on the clients' concerns and offer them support and encouragement. This is the clients' time to focus on themselves.

The counselor is a trained professional who has spent several years learning about different ways to help clients, resolve their particular problem.

The counseling relationship is confidential. Whereas with friends or family, clients might hope that they will respect their privacy, a counselor is ethically and legally bound by confidentiality. Unless the client is an imminently dangerous to themselves or others, their conversations with a counselor will be private.

Clients can depend on the counselor to meet them at their set appointment times

STAGE ONE: RELATIONSHIP BUILDING - INITIAL DISCLOSURE The first step involves building a relationship and focuses on engaging clients to explore issue that directly affect them. The first interview is important because the client is reading the verbal and nonverbal

messages and makes inferences about the counselor and the counseling situation.

STAGE TWO: IN DEPTH EXPLORATION - PROBLEM ASSESSMENT

While the counselor and the client are in the process of establishing a relationship, a second process is taking place, i.e. problem assessment. This step involves the collection and classification of information about the client's life situation and reasons for seeking counseling. Assessment refers to anything counselors do to gather information and draw conclusions about the concerns of clients. Assessment takes place at the beginning of the counseling process. But, some degree of assessment takes place throughout the counseling process with purpose of finding missing pieces of the puzzle, i.e. the client. According to Seligman (1996), assessment should attempt to recognize the importance and uniqueness of the client. In a way, saying to the person, "You are special and I want to get to know you and understand why you are the way you are". The counselor needs such knowledge about the client so that nothing is left out. For example, you might have left out an important piece of information about your client which may affect the counseling or therapeutic process.

STAGE THREE: GOAL SETTING - COMMITMENT TO ACTION

Setting goals is very important to the success of counseling. It involves making a commitment to a set of conditions, to a course of action or an outcome. Goals are the results or outcomes that client wants to achieve at the end of counseling. Goals help the counselor and client determine what can and what

cannot be accomplished through counseling. In goal setting, the client identifies with the help of the counselor, specific ways in which they want to resolve the issues and what course of action should be taken to resolve the problem.

STAGE FOUR: COUNSELING INTERVENTION

There are different points of view concerning what a good counselor should do with clients depending on the theoretical position that the counselor subscribes to. For example, the person-centered approach suggests that the counselor gets involved rather than intervenes by placing emphasis on the relationship. The behavioral approach attempts to initiate activities that help clients alter their behavior. After completion of the counseling intervention –

- Client acting upon plans;
- Client managing and coping with difficulty functioning;
- Existence of a support system and supports being accessed; and
- Identification of strategies for maintenance of change.

STAGE FIVE: EVALUATION, TERMINATION OR REFERRAL

Terminating the counseling process will have to be conducted with sensitivity with the client knowing that it will have to end. Note that each of these stages continues even though the counselor and the client move to the next step. For example, after building a relationship, the counselor moves to Stage Two which involves assessing the problem encountered by the client. In the meantime, the counselor continues to strengthen the relationship that has been built.

In other words, Stage One or the building of a relationship does not stop but is ongoing until Stage Five which involves termination or perhaps the client is referred to for further action. In this stage – Closure discussed and planned; Appointment intervals lengthened; Available resources and referrals identified and accessed; Assurance provided to the client of the option to return to counseling if necessary.

THE TOP TEN BASIC COUNSELING SKILLS

Research is increasingly finding that the type of therapy used is not as important to outcomes as are specific counselor behaviors such as (1) Enthusiasm, (2) Confidence, (3) Belief in the patient's ability to change.

Although there is nothing which will ensure change, it would appear that clients are more likely to achieve their goals when a good and positive relationship exists between them and their therapist. In essence the counselor's interactions with the client are a powerful tool in the helping relationship.

LISTENING

a. **Attending** - orienting oneself physically to the patient (pt) to indicate one is aware of the patient, and, in fact, that the client has your full, undivided attention and that you care. Methods include eye contact; not moving around, not being distracted, eye contact, encouraging verbalizations; mirroring body postures and language; leaning forward, etc. Researchers

estimate that about 80 percent of communication takes place non-verbally.

- b. **Listening/observing** - capturing and understanding the **verbal** and **nonverbal** information communicated by that pt.

TWO PRIMARY SOURCES OF INFORMATION:

- **CONTENT** - what is specifically said. Listen carefully for, not only what a person says, but also the words, expressions and patterns the person is using, which may give you a deeper insight. Counselors should develop their ability to remember what was said, as well as to clarify what was said or finding out what was not said.

PROCESS - all nonverbal phenomena, including how content is conveyed, themes, body language, interactions, etc.

EMPATHY

The ability to perceive another's experience and then to communicate that perception back to the individual to clarify and amplify their own experiencing and meaning. **It is not identifying with them or sharing similar experiences-- not "I know how you feel"!**

PRIMARY SKILLS ASSOCIATED WITH THE COMMUNICATION OF EMPATHY INCLUDE:

- nonverbal and verbal attending
- paraphrasing content of client communications
- reflecting patient feelings

and implicit messages

ATTENDING – involves our behaviors which reflect our paying full attention, in an accepting and supportive way, to the client.

PARAPHRASING - Selective focusing on the cognitive part of the message – with the client's key words and ideas being communicated back to the patient in a rephrased, and shortened form.

There are four steps in effective paraphrasing:

Listen and recall. The entire client message to ensure you recalled it in its entirety and do not omit any significant parts.

Identify the content part of the message by deciding what event, situation, idea, or person the client is talking about.

Rephrase, in as concise a manner as possible, the key words and ideas the client has used to communicate their concerns in a fresh or different perspective.

Perception check is usually in the form of a brief question, e.g., "It sounds like...", "Let me see if I understand this," which allows the client to agree or disagree with the accuracy of your paraphrasing.

Effectively reflecting on patient's feelings-

Affective reflection in an open-ended, respectful manner of what the client is communicating verbally and nonverbally, both directly through words and nonverbal behaviors as well as reasonable inferences about what the client might be experiencing emotionally. It is important for the helper to think carefully about which words he/she chooses to communicate these feelings back to the client. The skill lies in choosing words which use different words that convey the same or similar. For example, if a poorly skilled helper reflected to the client that he/she

was “very angry and depressed,” when the client had only said they were irritated by a certain event, and had felt very sad over the death of a family pet, the result could be counterproductive to the process of change.

Genuineness

Ability of counselor to be freely themselves. Includes congruence between outer words or behaviors and inner feelings; non-defensiveness; non-role-playing; and being unpretentious. For example, if the helper claims that they are comfortable helping a client explore a drug or sexual issue, but their behavior (verbally and nonverbally) shows signs of discomfort with the topic this will become an obstacle to progress and often lead to client confusion about and mistrust of the helper.

UNCONDITIONAL POSITIVE REGARD

An expression of caring and nurturance as well as acceptance.

Includes conveying warmth
Also conveying acceptance by responding to the patient's messages (verbal and nonverbal) with nonjudgmental or noncritical verbal & nonverbal reactions. Respect-ability to communicate to the counselor's sincere belief that every person possesses the inherent strength and capacity to make it in life, and that each person has the right to choose his own alternatives and make his own decisions.

CONCRETENESS

Keeping communications specific -- focused on facts and feelings of relevant concerns, while avoiding tangents, generalizations, abstract discussions, or talking about counselor rather than the client.

Includes the following functions:

- a. Assisting client to identify and work on a specific problem from the various ones presented.
- b. Reminding the client of the task and re-describing intent and structure of the session.
- c. Using questions and suggestions to help the client clarify facts, terms, feelings, and goals.
- d. Use a here-and-now focus to emphasize process and content occurring in current session, which may help to elucidate the problem being worked on or improving the problem-solving process.

OPEN QUESTIONS

A questioning process to assist the client in clarifying or exploring thoughts or feelings. Counselor is not requesting specific information and not purposively limiting the nature of the response to only a yes or no, or very brief answer.

- a. Goal is to facilitate exploration – not needed if the client is already doing this.
- b. Have an intention or therapeutic purpose for every question you ask.
- c. Avoid asking too many questions, or assuming an interrogatory role.
- d. Best approach is to follow a response to an open-ended question with a paraphrase or reflection which encourages the client to share more and avoids repetitive patterns of question/answer/question/answer, etc.

8. COUNSELOR'S SELF-DISCLOSURE

The counselor shares personal feelings, experiences, or reactions

to the client. Should include relevant content intended to help them. As a rule, it is better to not self-disclose unless there is a pressing clinical need which cannot be met in any other way. Remember empathy is not sharing similar experiences but conveying in a caring and understanding manner what the client is feeling and thinking.

collaboration with the client identify possible solutions and alternatives.

UNIT 9 IS COMPLETED

9. INTERPRETATION

Any statement to the client which goes beyond what they have said or are aware of. In interpretation the counselor is providing new meaning, reason, or explanation for behaviors, thoughts, or feelings so that patient can see problems in a new way. Interpretations can help the client make connections between seemingly isolated statements of events, can point out themes or patterns, or can offer a new framework for understanding. An interpretation may be used to help patient focus on a specific aspect of their problem, or provide a goal.

- a. Keep interpretations short, concrete (see concreteness), and deliver them tentatively and with empathy.
- b. Use interpretations sparingly and do not assume a patient's rejection of your insight means they are resistant or that you are right.

10. INFORMATION GIVING AND REMOVING OBSTACLES TO CHANGE

Supplying data, opinions, facts, resources or answers to questions. Explore with client possible problems which may delay or prevent their change process. In

CONTENTS

SOCIAL ISSUES & ROLE OF PSYCHOLOGY.

ISSUES OF GENDER, POVERTY, DISABILITY, AND MIGRATION:Cultural bias and discrimination. Stigma, Marginalization, and Social Suffering; Child Abuse and Domestic violence.

PEACE PSYCHOLOGY:Violence, non-violence, conflict resolution at macro level, role of media in conflict resolution.

WELLBEING AND SELF-GROWTH:Types of wellbeing [Hedonic and Eudemonic], Character strengths, Resilience and Post-Traumatic Growth.

HEALTH: Health promoting and health compromising behaviors, Life style and Chronic diseases [Diabetes, Hypertension, Coronary Heart Disease], Psychoneuroimmunology [Cancer, HIV/AIDS]

PSYCHOLOGY AND TECHNOLOGY INTERFACE:Digital learning; Digital etiquette: Cyber bullying; Cyber pornography: Consumption, implications; Parental mediation of Digital Usage.

UNIT

10

EMERGING AREAS

SOCIAL ISSUES

A social problem is any condition or behavior that has negative consequences (i.e. preventing society from functioning at an optimal level) for large numbers of people and that is generally recognized as a condition or behavior that needs to be addressed. This definition has both an *objective* component and a *subjective* component.

It is important to understand that not all things that occur in society are raised to the level of social problems. **Four factors** have been outlined that seem to

Social issues are distinguished from economic issues. Some issues have both social and economic aspects, such as

characterize a social issue or problem. These include:

The public must recognize the situation as a problem.

The situation is against the general values accepted by the society.

A large segment of the population recognizes the problem as a valid concern.

The problem can be rectified or alleviated through the joint action of citizens and/or community resources.

immigration. There are also issues that don't fall into either category, such as wars.

ROLE OF PSYCHOLOGY

Psychology plays a major role in understanding diverse social issues. This includes using social-psychological theories and concepts for understanding and predicting human behavior in the context of intergroup relations and social unrest. The role of Social psychology is,

An example in this case would be the PATH model.

Following a general overview of the different stages in doing applied social psychology, from problem definition to problem solution, we start by illustrating the disadvantages of a mono-theory approach over a multi-theory approach in explaining and predicting real-life human behavior. A systematic working tool, called PATH (a method for accessing scientific literature to answer practical questions with empirical data and theory) developed by Buunk & Van Vugt in 2008, is provided that can help in applying social psychology theory to practical cases to better understand human behavior and thus identify possible target points for intervention programmes. Next, we make the switch from behavior explanation to behavior change. Behavior change is a planned activity.

TYPES OF SOCIAL ISSUES

Issues around Gender (majorly Gender Discrimination)

however, not only a basic social science that studies the nature and determinants of human social behavior; it is also an applied discipline of relevance for all kinds of societal problems and issues. Applying social-psychology theory helps in making sense of everyday human behavior and provides tools to change behavior in preferred directions.

Poverty
Disability
Migration: Cultural bias and discrimination
Stigma
Marginalization
Social Suffering
Child Abuse
Domestic violence.

SUSTAINABLE DEVELOPMENT GOALS (SDGS): INDIA

SDG 1: No Poverty
SDG 2: Zero Hunger
SDG 3: Good Health and Well-Being
SDG 4: Quality Education
SDG 5: Gender Equality
SDG 7: Affordable and Clean Energy
SDG 8: Decent Work and Economic Growth
SDG 9: Industry, Innovation and Infrastructure

SDG 11: Sustainable Cities and Communities

SDG 12: Responsible Consumption and Production

SDG 17: Partnerships for the Goals

ISSUES OF GENDER

There are two Divisions of APA that directly deal with these issues:

Division 35: Society for the Psychology of Women, established 1973.

Division 51: Society for the Psychological Study of Men and Masculinity, established 1995

Let's first understand some key terms:

WHAT IS GENDER?

The perception of being a male or a female.

DIFFERENCE BETWEEN “GENDER” AND “SEX”?

Gender refers to the psychological aspect whereas sex refers to the physiological/physical aspect of the same phenomenon. “Gender” refers to culturally constructed categorization of “maleness” and “femaleness” or masculinity and femininity. The biologically based distinctions between males and females are referred to as “Sex” of the person.

WHO MAKES US BEHAVE AS MALES OR FEMALES?

Human-beings are born as male or female babies, BUT they become masculine or feminine as a result of child rearing practices and socio-cultural variables.

WHAT IS REFERRED TO AS GENDER ROLES?

Expectations of appropriate behavior for males and females set by a society. Gender role expectations are very similar across societies.

WHAT ARE GENDER DIFFERENCES?

Actual differences between males and females in terms of average capacity, ability, or typical behavior.

WHAT ARE GENDER STEREOTYPES?

“Widely held beliefs about females’ and male’s abilities, personality traits, and social behavior” (Weiten, 2001).

WHAT IS ANDOCENTRICITY?

Most gender research, especially the initial one, had been androcentric i.e., looking at issues from the male perspective. For example: for many years the researchers studying occupations concentrated upon men’s formal work and ignored female’s domestic work and childcare. The other extreme may be gynocentricity (female centered) i.e., evaluating things from women’s point of view alone.

Before going deep into the concept of social role and how it is shaped, **let's ponder over some questions...**

- Why are women more conscious about body weight than men?
 - Why don't men wear makeup as women do?
 - Can men become good "mothers"?
 - Why are women not hired as guards?
 - Why no men sew clothes at home but most professional tailors are men?
 - Why most chefs are men but very few men cook at home?
 - Why are women considered talkative whereas on average men talk more?
- The answers to these questions pertain to societal role allocation and shaping.

FEMINISM

Feminism is a collection of political movements, social movements and ideologies that defend the political, the economic, the personal and the social rights of women. Feminist movements aim at achieving and establishing equality between women and men.

The modern feminist movement emerged from the west. Men and Women both contributed to the cause of improving women's status in the society, in the political, economic, and social spheres.

All societies were patriarchal, male dominated, where women were generally oppressed and home bound. Women were

denied the right to franchise, to go for education, and profession. Woman was treated as a second-rate citizen with limited or no ability, little or lower grade intelligence and non-trustable skill. These were the major concerns that led to the beginning of the Feminist Movement to establish equality between men and woman.

FORMS OF FEMINISM

Liberal Feminism: (1960s- 1970s) Liberal feminism had roots in classic liberal thought; women should have the same rights as men.

Radical Feminism: They believed that the cultural notion of gender needs to be eliminated. In their view, men are the oppressors and women are the ones who are being oppressed. A similar Pattern of oppression was also seen in Racial or social classes.

Socialist Feminism: Rather than seeking liberation at individual level, men and women should be together in this pursuit.

Cultural Feminism: They insist that social change is a must and that societies need to accept and appreciate traditionally feminine values.

FEMINIST:

A person who believes in the social, political and economic equality of the sexes. They act, speak, write and advocate on behalf of women's issues and rights and identify injustice to females in the social status quo.

MYTHS AROUND FEMINISM:

Feminism is the belief that women are superior
 Feminism is hating men (misandry)
 Feminism is male oppression

THE FEMINIST MOVEMENT:

A movement or way of thinking that proposes, advocates, supports, and struggles for social equality of the two sexes. The difference between men and women should be looked into from a “gender” perspective rather than a “sexist” perspective. The difference between the sexes is learned and attained, not ascribed or inherent. Society is the seed bed of our gender roles, gender Identity, and gender stratification leading to social status and ranking of the genders. Feminist movement had its roots in struggle for enlightenment, women’s rights, and legal and political rights.

The feminists advocate:

- The significance of change.
- Reintegration of humanity: all human traits are found in all humans (French, 1985).
- Equal rights and elimination of gender stratification.
- Curbing and bringing an end to all forms of violence against women.
- Promoting sexual autonomy.

THE THREE WAVES OF FEMINISM:

The history of the feminist movements is divided into three "waves".

The **first wave** refers to the movement of the 19th through

early 20th centuries, which dealt mainly with suffrage, working conditions and educational rights for women and girls.



First-wave feminism involved a period of feminist activity during the 19th and early 20th centuries, especially in Europe and in the United States.

Key concerns:

women's suffrage (the right to vote)

the right to education

better working conditions

marriage and property laws

reproductive rights

The **second wave** (1960s-1980s) dealt with the inequality of laws, as well as cultural inequalities and the role of women in society. By this time, women in most countries had been granted the right to franchise and to contest election for public office. These feminists were active after 1960. They were working for equality, both economic and social. They fought for the right to contraception, birth control and sexual liberation. Sexual liberation became a much-debated issue and was criticized by many feminists.

Second-wave feminism is a period of feminist activity and thought that first began in the early 1960s in USA and spread all over the western world and beyond.

Key concerns:

raising consciousness
about
sexism and
patriarchy

raising
consciousness about
gender-based
violence, domestic
abuse and marital
rape

inequalities in the
workplace

legalizing abortion
and birth control

sexual liberation of
women

social
identities, as
race, gender,
sexuality,
and class,
contributes
to the
specific type
of systemic
oppression
and
discrimination
experienced
by an
individual)

The **third wave** of feminism (1990s-2000s) is seen as both a continuation of the second wave and a response to the perceived failures.

The third wave of feminism (1990s-2000s) arose partially as a response to the perceived failures of second-wave feminism.

As the earlier feminists failed to see and deal with variations within women. Thus, the third wave feminists try to see how different groups of women may be having different needs, and circumstances, and requiring different solutions. Womanism, Queer theory, and Post-Modern Feminism are a part of this wave.

Key concerns:

Intersectionality (the theory that the overlap of various

- The diversity of "women" is recognized and emphasis is placed on identity, gender, race, nation, social order and sexual preference
- Changes on stereotypes, media portrayals and language used to define women.
- Sexual identities

SOME IMPORTANT HISTORICAL FACTS AND EVENTS:

Christine de Pizan: Generally considered the first feminist writer. Wrote in the medieval period.

In the 1700's women like Lady

Mary Wortley Montagu, Marquis de Condorcet propagated the significance of women's education

1785: The first scientific Society for Women established in Middleburg, Dutch, Republic.

1791: "Declaration of the Rights of Women & the Female Citizen"
The "Declaration of the Rights of

Man and of the Citizen (1789) was paraphrased. It was the French revolution's central document.

1792: Mary Wollstonecraft wrote

her famous book: "A Vindication of the Rights of Woman".

1848: New York: The first

Women's Rights Convention was held.

J.S. Mill wrote "The Subjection of Women" in the mid-1800s. **He was influenced by his wife Harriet Taylor.**

Emmeline Pankhurst formed

WSPU/Women's Social and Political Union.

Emma Goldman, Elizabeth Cady

Stanton, Dame Ethel Mary Smyth, & Margaret Sanger were among the 19th century feminists.

In the East **Qasim Amin** an

Egyptian jurist is known as the Father of Egyptian feminist movement. In 1899 he wrote "Women's Liberation" and proved to be a very influential writer (Tahrir u Mar'a).

FEMINIST MOVEMENT IN INDIA

1947: The freedom struggle saw active participation by women.

1946-52: Telengana agitation in which women were trained as guerrillas.

1972: Shahada and anti-price rise agitation in Maharashtra with women in the forefront

1972: Anti-alcohol agitation in parts of western India. Self-

Employed Women's Association (SEWA) set up by Ela Bhatt.

1973: Anti-price rise agitations in Gujarat.

1974: NavNirman agitation.

1978: First national conference of socialist feminists in Bombay.

1979: StriSangharsh formed in Delhi.

1980-83: Campaigns against dowry.

1987: Campaigns against domestic violence and rape.

Early '80s: Establishment of

Centre for Women's Development Studies.

1985: Agitations in solidarity with Shah Bano.

1986: Movements against Muslim

Women's Bill.

1987: Agitation against sati

Late '80s: Struggle for a safe

environment, demonstrations against

early '90s: Union Carbide after the Bhopal gas tragedy.

DIFFERENT PHASES OF WOMEN'S MOVEMENTS IN INDIA

If we go through the whole concept of women movements and their struggle, we will notice that no social change of much importance had been brought about among them most of the women in villages are still illiterate and backward they do not participate in the political, any social and

economic life of the nation. Rural women remaining backward and orthodox, due to tradition, illiteracy, ignorance, social evils and many other factors. Hence, women emancipation in rural India is still need some improvement and it is an essential notion for social progress of the nation. Let's take a peek at some of the important movements in India.

SOCIAL REFORM

MOVEMENT: Social reform movement in India was a part of struggle that were made by different social reformers. During this time, on one side, India was suffering from stagnating traditional culture and society, while on the other hand, India possessed a still traditional society in the creative excitement of modernizing and of emerging as a new nation. The nineteenth century initiated this process of transformation in the religious, social, economic, political and cultural spheres. The impact of the British empire influenced administration, legislation, trade, network of communication, industrialization and urbanization in India, affecting not only society as a whole but also the traditional pattern of life. British scholars, educators and missionaries also impacted the cultural field. The reformers consciously reacted to the new situation and advocated deliberate changes in social, religious and customs. This social and religious reform movement arose among all communities of the Indian people. Reformers, worked for abolition of castes and untouchability, purdah system, sati, child marriage, women education and social inequalities.

Raja Rammohan Roy, abolished Sati in 1929. His campaign against sati aroused the opposition of the orthodox Hindus

who bitterly attacked women. For instance, the position of the nineteenth century high –caste women need to be understood within the context of the patriarchal social system. The caste hierarchy and gender hierarchy are the organizing principles of the Brahmanical social order and are closely interconnected. The Brahmanical patriarchy in early India reveals that the structure of social relations which shaped gender was by achieving the compliance of women. There was the case of Pandita Ramabai, whose father was an unconventional social reformer who began with social transformation in his own house by educating his wife Lakshmi Bai even at the cost of being exiled by his own community for this. Lakshmi Bai taught her daughter as is recorded in, Pandita's book (1886) *The high caste Hindu women*. The book argues on women oppression and the treatment given to Hindu women throughout their life. She argues about the evil in child marriage and the taboo associated with widowhood. She also talks about the money and commercial nature of arrange marriages. She raised the issue of inhuman expectation from the young girl. Marriage was like a central theme as for a woman. According to her, marriage is like hypocrisy of old times, women were not allowed to say a single word in the decision making or in choice for her husband, and Women were not considered as capable of thinking about their future or any chance to give her opinion. Also, in early nineteenth century women were not encouraged to get an education. Some people believed that if women were well educated it would ruin their marriage prospects and to be harmful to their mind. Women's career was very limited so people didn't think they needed the

education in which men did. A lot of the focus was to teach women how to run a household. The live of Indian women began to change significantly in late 19th century when the colonial government critical of the treatment of both Hindu and Muslims women, allies among Indian reformers these men agreed that women should be educated and play some role in public life.

Pundit Ishwarchandra vidyasagar consolidated the way for the remarriage for socially forsaken widows through the widow's remarriage act no XV of 1856. He empathetically lamented the deep distress of Indian womanhood. The air of reformation held high the urgency for casting the enlightenment of education on women. The British government eager to prove their liberal, ethical and pro-modernity attitude resorted to the women question. This is the fundamental feminist question concerned with the rights and progress of women. British denounced the exciting insignificance of Indian womanhood and tried to initiate some feminist welfare activities to show their social-cultural advancement. They took help of the indigenous modern minds like Raja Rammohan Roy, Vidyasagar and other prominent Indians. As mention above the early Vedic society, allowed girls to acquire knowledge, but in the later society, the orthodox Brahmins introduced blind superstition, rituals and rigid customs which forbade girls from learning. We recognize a severe disappointment of the western educational entrepreneurs with the oppression of social norms. In Bengal the overriding tension was that 'knowledge of letter' would teach female to oppress the meaningless coercion. The terrible

situation made women absolutely dependent on father in their childhood, then on husband in marital life and then on the sons during their old age. As per the customs, her life gets devoted in religious practices wishing for the long and safe life of the male members of her family.

Rassundaridevi, the author of the autobiographical, "Amar Jiban" she poignantly describes her in domitable struggle for reaching the gateways of education. She records her childhood marriage, the daily task of household she explained the unspoken agony of a young girl. She has desire to read, she stole pages from a book and kept them in kitchen at a hidden place. She teaches herself domestically as she writes, "Is this my destiny that I am a woman"? Just because of I am a woman does it necessarily mean that trying to educate myself is a big crime in a patriarchal society. Both Pandita Ramabai and Rassundaridevi raised the issue related to women education and child marriage. The Calcutta school society was built in 1816 to improve the status of female education. The church missionary society tasted greater success in south India. The importance of female education in India cannot be over rated. Women education started spreading its wings. The outcome was the evolution of nineteenth century generation of the "new woman". The later half of this century started seeing the raise of Indian womanhood to freedom and assertion.

Kamini Roy, Pandita Ramabai, Rassundari Devi were the celebrated examples of the "new woman" of this time.

The period between the 1930s- and 1947- marks Indian women's entry in to the

nationalist struggle for freedom. The nationalist cause was gathering strength and women were called upon to serve the nation with their abilities of caring and nurturing. Women have her own unique contribution to make for the purpose of enriching life so that it may fulfill its own divine mission to perfection. So in the struggle for freedom they are destined to play their legitimate part and bear their share of the responsibility in breaking the chain that weight heavily on people. Mahatma Gandhi had also discussed the importance of women in social revolution, in reconstruction and in national struggle.

Women have worked and participated in different way in the movements and contributed their efforts individually.

- Participation in the Nationalist movement
- Resurgence in women's political activity
- Women's empowerment
- Role in Freedom struggle

PritilataWaddedar, the most celebrated female martyr of the freedom movement asked an important question related to dominant gender ideology of nationalism:

“I wonder why there should be any distinction between males and females in a fight cause of the country's freedom. If our brothers can join a fight for the cause of the motherland then why can't the sisters'? The pages of history are replete with high admiration for the historic exploits of distinguished ladies. Then why should we, the modern Indian women, be deprived of joining this noble fight to redeem our country from foreign domination?”

JUDICIAL APPROACHES TO EQUALITY RIGHTS IN INDIA

The following section elaborates the equality rights as per articles of the Indian constitutions:

Article 14-It guarantees equality before the law and equal protection under the law. It allows the states to make classifications on reasonable ground. The article embodies a guarantee against arbitrariness.

Article 15- I t prohibits discrimination on the ground of religion, race, caste, sex and place of birth. As per 15(3), which allows states to make special provisions for women.

Article16- It guarantees equality of opportunity and prohibits discriminations in matters of employment.

JUDICIAL APPROACHES TO SEX DISCRIMINATION

The sex discrimination case law remains overly determined by a formal model of equality. There are three major approaches in relevance of gender differences protectionist, sameness and corrective. As per protectionist approach women are constituted as weak and subordinate and are thus in need of protection. In this approach the courts understanding of women's differences is asserted as justification for differential treatment. This approach tends to essentialist the difference i.e. to take the existence of the differences as natural.

As per sameness approach women are constructed as same as men and treated similarly in law.

As per corrective approach women are seen to require special treatment as a result of past discrimination. Within this approach gender differences are seen as relevant and requiring recognition in law.

In the post-independent India, we had series of laws passed for the upliftment of women status. This legislation has been brought in orders to give equal rights and privileges with men, to eliminate discriminations against women, remove inequality between sexes and remove external barriers coming in the ways of their development. The important act passed for the upliftment of women is:

The Hindu marriage Act of 1955

The Hindu adoption and maintenance Act of 1956

The Hindu succession Act of 1956: That woman has got equal rights in the inheritance of family property.

The Dowry prohibition Act of 1961

The equal remuneration Act of 1976: This act does not permit wage discrimination between male and female workers.

EXTRA FOR REFERENCES (All part in *Italics* is out of syllabus and for reference only)

HISTORICAL BACKGROUND

Two main viewpoints exist in psychology regarding gender differences:

Gender variation is biological and evolution-based in nature... hence fixed and un-alterable. Biological differences are there, BUT it is the society that shapes and nurtures gender roles. The biological approach has roots in evolutionary psychology and has an essentialist nature.

The bio-social approach has a broader perspective and benefits from knowledge and research in Sociology,

Anthropology, Ethnology, and Biology.

The main issues are tackled in these theories: Who made us what we are? Were we born like that or are we a product of the environment and the society that we exist in?

BIOLOGICAL DIFFERENCES BETWEEN GENDERS

Biological differences between males and females are found in three ways:

In chromosomes

In hormones

In structure of the nervous system

Hormones and chromosomes are the:

- Bases of masculinity and femininity

Structural units and carriers of heredity Play important role in various human behaviors Endocrine system is a network of glands that produce and secrete chemical messengers (hormones), directly into the blood stream. This activity is controlled by hypothalamus: the important relay station and significant brain structure.

HORMONES AND CHROMOSOMES:

Hormones and chromosomes are not only bases of masculinity and femininity and structural units of heredity, they play important role in various human behaviors. Endocrine System is a network of glands that produce and secrete chemical messengers that are called hormones, directly into the blood stream. These glands are controlled by hypothalamus that is important relay station and significant brain structure. Pituitary glands (also

called master glands) stimulate or inhibit secretion of hormones from other glands. Sex and reproduction hormones are called, Steroid hormones. Reproductive organs ovaries (in females) and testes (in males) are called gonads.

SEXUAL DIFFERENTIATION AND ROLE OF CHROMOSOMES:

Sexual dimorphism is existence of two sexes (male & female) in a species, including differences in genetics, gonads, hormones, internal genitalia and external genitalia. The process of sexual differentiation starts when conception takes place. Chromosomes are threadlike structures that are structural carriers of heredity, play very important role in sex differentiation. In human beings there are 23 pairs of chromosomes. At the time of conception half number of chromosomes is received from mother side and half from father side. 23rd pair of chromosomes is called sex chromosome in females it is XX and in males it is XY. Genes contain genetic information carried on chromosomes. Researchers claim that XY chromosomal makeup may not be as stable as XX and therefore play a role in higher mortality rates for men. XX chromosomal makeup and the hormone estrogen seem to make women less vulnerable to physical problems so women live longer lives.

SEX CHROMOSOMAL ABNORMALITIES: Sudden structural changes in the genes are called mutations. These mutations lead to negative consequences in individual's development. Sometimes deleted or duplicated number of chromosomes also causes abnormalities in an individual. Researchers have found

that disorders in sex chromosomes lead to problems in gender role development, intelligence, growth and hormone production.

FEMALE SEX CHROMOSOME ABNORMALITIES:

- **TURNER'S SYNDROME:** This syndrome occurs when individual inherit only one X chromosome and their sex chromosomal structure is XO. Such cases are females in sex with abnormal growth patterns. They are short in stature, averaging 4 foot 7 inches as adults, and often have distinctive webbed necks (i.e., extra folds of skin), small jaws, and high arched palates. They generally lack prominent female reproduction ability. They have exceptionally small, widely spaced breasts, broad shield-shaped chests, and turned-out elbows. Their ovaries do not develop normally and they do not ovulate. They also have a higher than average incidence of thyroid disease. In some individuals, there is slight mental retardation. Turner's syndrome is rare. Current estimates of its frequency range from 1 in 3,000 female infants to 1 in 5,000.

If diagnosed in early childhood, regular injections of human growth hormones can increase their stature by a few inches. Beginning around the normal age of puberty, estrogen replacement therapy can result in some breast development and menstruation. These treatments allow Turner's syndrome women to appear relatively normal.

METAFEMALES OR TRIPLE-X FEMALES:

Such females inherit three X chromosomes. Their chromosomal structure is XXX. As adults, these "super-females" are usually an inch or so taller than average with unusually long body structure. They have normal development of sexual characteristics and are fertile. They may have slight learning difficulties and are usually in the low range of normal intelligence. They tend to be emotionally immature for their size during childhood. This type of chromosomal abnormality is less rare than Turner syndrome.

MALE SEX CHROMOSOME ABNORMALITIES:

KLINEFELTER SYNDROME:

Individual with this syndrome inherit on extra X chromosome and their chromosomal structure is XXY. They characteristically have relatively high-pitched voices, feminine body contours as well as breast enlargement, and comparatively little facial and body hair. They are sterile or nearly so, and their testes and prostate glands are small. As a result, they produce relatively small amounts of testosterone. The feminizing effects of this hormonal imbalance can be significantly diminished if Klinefelter syndrome boys are regularly given testosterone from the age of puberty on. Klinefelter syndrome men are an inch or so above average height. They also are likely to be overweight. They usually have learning difficulties as children, especially with language and short-term memory. If not given extra help in early childhood, this often leads to poor school grades and a subsequent

low self-esteem. Klinefelter syndrome is not diagnosed until they are tested for infertility. Klinefelter syndrome males with more than two X chromosomes usually have extreme symptoms and are often mentally retarded.

XYY SYNDROME: The individual with this abnormality receive extra Y chromosome and their chromosomal structure is XYY. As adults, these "super-males" are usually tall (above 6 feet) and generally appear and act normal. However, they produce high levels of testosterone. During adolescence, they often are slim, have severe facial acne, and are poorly coordinated. They are usually fertile and lead ordinary lives as adults. Majority remain unaware that they have a chromosomal abnormality. Early studies of XYY syndrome done in European prisons initially led to the erroneous conclusion that these men were genetically predisposed to antisocial, aggressive behavior, below average intelligence, and homosexuality. Contributing to the early view that XYY syndrome men have serious personality disorders. However, some researchers suggest that the high testosterone levels of XYY men can make them somewhat more prone to violence and that this may cause higher rates of wife beating.

SEX DIFFERENTIATION IN REPRODUCTIVE SYSTEM

X and Y chromosomes determine the sex of child being male or female. Both sexes have two sex related internal systems:

- **WOLFFIAN SYSTEM:**The system with a capacity to develop into the male internal reproductive system.
- **MULLERIAN SYSTEM:**The system with a capacity to develop into the female internal reproductive system

THEORIES OF GENDER:

The gradual process of development of gender schema:

Stage	Age	Description
Gender identification	2-3.5 years	Children believe it is possible to change sex by wearing clothes of opposite sex. Can identify household gender stereotypes: (mechanical tools used by father and the mother works in kitchen). Gender labeling is learned.
Gender stability	3.5-4.5 years	Sex is stable over time, not situations. Older peers and siblings are models for social learning of gender roles. Boys may play with the toys socially labeled as girl's toys but do not own them ultimately. Child applies labels to self and others, but they may be inconsistent
Gender consistency	4.5-7 years	Sex is stable. Children value and imitate same sex behaviors. This leads to development of gender appropriate attributes

THE COGNITIVE APPROACH TO GENDER ROLES

The theory proposes the interaction of mental schema and social experience in directing gender role behavior. The cognitive approach focuses upon the child's "understanding". A child's understanding refers to the way he/she perceives and tackles a phenomenon. Information about gender is organized into sets of beliefs about the sexes i.e. gender schema. Gender schema (plural schemata or schemas) is a mental framework that organizes and guides a child's understanding of information relevant to gender. Example: information about which toys are for girls and which toys are for boys forms a schema that guides behavior. Example: If a child has seen women being respected in his family, he will perceive women as a respectable being; and if he has seen women being battered and maltreated he will perceive them as some low-grade creature.

THE PSYCHODYNAMIC APPROACH

Freudian Perspective: Psychosexual stages of development take place with possibility of particular conflicts at different stages. Gender roles develop as a result of the resolution process of conflict at the phallic stage. Feelings of rivalry and hatred

develop against the father at this stage. The father is seen as stronger and unconquerable; this leads to a conflict. The defense mechanism of identification is used for resolving the conflict. This gender identification leads to sex-typed behavior and development of gender roles. Absence of a parent, particularly, the same-sex parent affects the normal process of gender development. Stevenson & Black (1988): boys with absent fathers around the oedipal stage show less sex-typed behavior.

POVERTY

Let's take a quick peek into the data first.

POVERTY DATA: INDIA

In India, 21.9% of the population lives below the national poverty line in 2011.

In India, the proportion of employed population below \$1.90 purchasing power parity a day in 2011 is 21.2%.

For every 1,000 babies born in India in 2017, 39 die before their 5th birthday.



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The global economic crisis, natural disasters, high migration rates and unstable political situations in different parts of the world have caused a decrease in financial wealth among the global population and an increase in poverty incidence rates. According to statistics by the World Bank, 702.1 million people worldwide are living in extreme poverty [World Bank, 2015].

Poverty is a social condition that is characterized by the lack of resources necessary for basic survival or necessary to meet a certain minimum level of living standards expected for the place where one lives. The income level that determines poverty is different from place to place, so social scientists believe that it is best defined by conditions of existence, like lack of access to food, clothing, and shelter. People in poverty typically experience persistent hunger or starvation, inadequate or absent education and health care, and are usually alienated from mainstream society. Poverty is a consequence of the uneven distribution of

material resources and wealth on a global scale and within nations. Sociologists see it as a social condition of societies with an unequal and inequitable distribution of income and wealth, of the de-industrialization of Western societies, and the exploitative effects of global capitalism. Poverty is not an equal opportunity social condition. Around the world and within the U.S., women, children, and people of color are far more likely to experience poverty than are white men. While this description offers a general understanding of poverty, sociologists recognize a few different types of it.

Types of Poverty Defined: Absolute poverty is what most people probably think of when they think of poverty, especially if they think about it at the global level. It is defined as the total lack of resources and means required to meet the most basic standards of living. It is characterized by lack of access to food, clothing, and shelter. The characteristics of this type of poverty are the same from place to place. Relative poverty is defined differently from place to place because it depends on the social and economic contexts in which one lives. Relative poverty exists when one lacks the means and resources required to meet a minimum level of living standards that are

considered normal in the society or community where one lives. In many parts of the world, for example, indoor plumbing is regarded as a sign of affluence, but in industrial societies, it is taken for granted and its absence in a household is taken as a sign of poverty. Income poverty is the type of poverty measured by the federal government in the U.S. and documented by the U.S. Census. It exists when a household does not meet a set national minimum income considered necessary for the members of that household to achieve basic standards of living. The figure used to define poverty on a global scale is living on less than \$2 per day. In the U.S., income poverty is determined by the size of household and number of children in the household, so there is no fixed income level that defines poverty for all. According to the U.S. Census, the poverty threshold for a single person living alone was \$12,331 per year. For two adults living together it was \$15,871, and for two adults with a child, it was \$16,337. Cyclical poverty is a condition in which poverty is widespread but limited in its duration. This type of poverty is typically linked to specific events that disrupt a society, like war, an economic crash or recession, or natural phenomena or disasters that disrupt the distribution of food and other resources. For example, the poverty rate within the

U.S. climbed throughout the Great Recession that began in 2008, and since 2010 has declined. This is a case in which an economic event caused a cycle of more intensive poverty that was fixed in duration (about three years). Collective poverty is a lack of basic resources that is so widespread that it afflicts an entire society or subgroup of people within that society. This form of poverty persists over periods of time stretching across generations. It is common in formerly colonized places, frequently war-torn places, and places that have been heavily exploited by or excluded from participation in global commerce, including parts of Asia, the Middle East, much of Africa, and parts of Central and South America. Concentrated collective poverty occurs when the kind of collective poverty described above is suffered by specific subgroups within a society, or localized in particular communities or regions that are devoid of industry, good paying jobs, and that lack access to fresh and healthy food. For example, within the U.S., poverty within metropolitan regions is concentrated within the principal cities of those regions, and often also within specific neighborhoods within cities. Case poverty occurs when a person or family is unable to secure resources required to meet their basic needs despite the fact that resources are not scarce and those around

them are generally living well. Case poverty might be produced by sudden loss of employment, inability to work, or injury or illness. While it might at first glance seem like an individual condition, it is actually a social one, because it is unlikely to occur in societies that provide economic safety nets to their populations. Asset poverty is more common and widespread than income poverty and other forms. It exists when a person or household does not have enough wealth assets (in the form of property, investments, or money saved) to survive for three months if necessary. In fact, many people living in the U.S. today live in asset poverty. They may not be impoverished so long as they are employed, but could be thrown immediately into poverty if their pay were to stop.

THE PSYCHOLOGY OF POVERTY

(Taken from the 2010–2011 IRP New Perspectives in Social Policy Seminar given by SendhilMullainathan, Professor of Economics at Harvard University on September 21, 2010.)

Theories about poverty often fall into two general categories: that the behaviors of poor people reflect the best choices they can make in unfavorable circumstances, and, alternatively, that **these behaviors**

Theories about poverty often fall into two **are a result of a unique “culture of poverty” based on deviant values.**

The first view presumes that people are highly rational, hold coherent and well-informed beliefs, and pursue their goals effectively with no need for help. The second view attributes to the poor a variety of shortcomings that make them misguided, uninformed, impulsive, and in need of paternalistic guidance in order to make reasonable choices. While there is no doubt that people—the poor included—are at times methodical and calculating, and at other times fallible or misguided, a third, alternate theory takes a different tack and is informed by recent behavioral research. According to this view, scarcity experienced as a result of economic instability and poverty reduces already limited cognitive resources, resulting in detrimental behaviors and ineffective decision-making. Current policies designed to improve outcomes for poor people may be effective when successfully implemented, but program administrators

may find it difficult to get people in the door, and then to carry through with the program. Similarly, early childhood programs that rely on parental participation and complementary parental behaviors may not succeed if poor parents do not follow through. **A variety of costly behaviors by the poor such as debt traps, failure to take available and necessary medications, or obesity, may further inhibit economic mobility.** These behaviors, as suggested by the author, are a result of increased psychological stress caused by poverty, and that it may be possible to design antipoverty programs that make it easier for poor people to succeed if this reduced functioning is considered.

PSYCHOLOGY OF SCARCITY

Mullainathan argues that cognitive resources, such as attention and self-control, are limited. Using both laboratory and field research, he demonstrates that scarcity further reduces those already limited resources, hampering the ability of poor people to follow through on tasks or to make effective decisions.

ATTENTION

Attention is a scarce resource; people can only focus on a limited number of things at

one time. One must choose what to focus on, although this choice is not always conscious. Numerous laboratory experiments have demonstrated that people have limited attention and that they have the capacity to allocate this attention. In a case when subjects must choose between two things to pay attention to, unattended things are generally not remembered at all. In the real world, this means that parents may not be able to attend fully to their jobs if they are also worrying about problems at home, while inattention at home could result in early symptoms of a child's illness going unnoticed, or medication for a chronic condition not being taken. Mullainathan and his colleagues did a field study in three countries that demonstrated that simply directing someone's attention to something they have stated they wanted to do, but might otherwise forget, can have significant results. In this case, people voluntarily agreed to participate in a savings program. Those who received a single text message reminder if they failed to meet their savings goal in a given month had a 6 percent higher savings rate than those who did not receive a reminder. Similar kinds of studies on medication-taking by HIV patients have found very high compliance rates achieved with simple reminders such as a pill bottle that lights up and beeps if it is not opened each

day. In both cases, people have the intention to do something, but on their own may fail to allocate some of their limited attention to achieving that goal. Simply focusing people's attention on the stated goal may be sufficient to get them to achieve it.

SELF-CONTROL

Another limited cognitive resource with similar consequences is self-control. The Stanford marshmallow experiment, conducted in 1972 by Walter Mischel, is a classic illustration of self-control. Children were led into a room, given a choice of treats, then left alone in the room with the instruction that they could eat the treat, but if they could wait for 15 minutes without eating it, they would be rewarded with a second treat. Video of the children shows that this was a difficult task. Another study conducted by Baba Shiv and Alexander Fedorikhin demonstrated how depletion limits self-control. Subjects asked to remember either a 2-digit or 7-digit number for 10 minutes waited in a room with a choice of cake or fruit salad. Those working to remember the longer number were considerably more likely to choose cake than those asked to remember the shorter number. Mullainathan and his colleagues conducted a field study to investigate the real-world implications of these psychological findings. In this case,

the researchers confirmed their theory that workers may face self-control problems that limit their productivity, and that making a binding commitment to work harder can actually produce that result. In a data-entry firm, where workers are compensated on a piece-rate basis, employees were given the opportunity to improve their production (and thus increase their pay) by setting a threshold for themselves. If they met the threshold, they would be paid the regular rate, but if they fell short, they would only earn half the rate. Economically, choosing to agree to a threshold is clearly a bad deal, since they do not earn anything extra if they meet it, and earn half what they would have otherwise if they do not meet it. Still, given this opportunity, workers chose to set the threshold about a third of the time. Workers randomly assigned to receive the threshold offer on a given day (whether or not they chose it) had production and earnings that were higher than those not given the offer. This illustrates that lack of self-control did limit productivity, and that it is possible for an individual to increase his or her self-control.

HOW DOES SCARCITY TAX THESE RESOURCES?

Having established that cognitive resources such as attention and self-control

are both important and limited, the next issue is to demonstrate how having very little of them taxes these resources. This section begins with a conceptual argument, which is followed by experimental evidence.

THE PACKING PROBLEM: A SUITCASE METAPHOR

The conceptual argument derives from the “knapsack problem” in complexity and computational theory. Imagine packing for a trip, using either a small or large suitcase. If you have a large suitcase, it is an easy task to pack everything important with room to spare. You may even choose not to completely fill the suitcase. With a small suitcase, however, the task becomes much more complex. If not, all important items will fit, you must consider trade-offs, such as what to take out if one more item is added. The suitcase can represent any resource, such as money. In that case, someone with ample resources can easily purchase all needed items with money left over. They may consider the wisdom and value of a particular small purchase, but are not likely to explicitly consider what another item must be given up in its place. In contrast, someone with limited funds must spend a lot of time and mental energy thinking about what to purchase, as each item chosen means some other item or

items is foregone. In other words, having fewer resources makes decision-making much more complex. Complex problems draw on limited cognitive resources, which in turn means that there are fewer resources available for self-control. Evidence on the effects of scarcity Mullainathan and his colleagues have experimental evidence that people with less money are much more likely to know how much things cost than those with more money, even for items that poorer people were less likely to spend money on. In a study conducted in Boston, people at a train station were asked what the initial fare was when you first get into a taxi. People with a low socioeconomic status (SES) were considerably more likely to correctly identify this amount than people with a high SES, even though they were much less likely to actually take a taxi. People with less money were paying better attention to the price, because prices matter more to them. Similar studies have been done with people leaving a supermarket, who are asked the price of specific items and the total amount they spent. Again, people with a low SES are much more likely to be able to answer these questions correctly than people with a high SES.

Mullainathan and his colleagues were also able to show that poorer people have a clear and absolute understanding of the value of a dollar, while more wealthy people may infer the value of a dollar based on the context. People were asked to imagine that a friend went to buy an appliance priced at \$100, \$500, or \$1,000. The friend was informed that a store 45 minutes away offered the same item on sale for \$50 less. Subjects were asked if they would advise their friend to travel to the other store to save \$50. The response to this question varied greatly depending on the income level of the respondents. In a high-income area, subjects were much less likely to advise traveling to save \$50 as the initial price of the item rose. In contrast, in

low-income area, subjects were much more likely to advise travel, and the initial price of the item made little difference to their recommendation. An experiment conducted in a New Jersey mall showed that asking poor people to think about money depleted their cognitive resources. Participants were asked to consider either an easy or hard financial problem or an easy or hard non-financial mathematics problem. While they were considering the problem, they were asked to complete a test of cognitive control that required concentration. For those with below-median income, there was a significant drop in the cognitive control test score,

while those with above-median income showed little change. A final piece of evidence that scarcity is depleting comes from a real-world example, harvest of sugar cane in India. Sugar cane is a crop that is harvested once a year, but the harvests are staggered over some months, so the same calendar month could be a pre-harvest month for some farmers and a post-harvest month for others. Since sugar cane farmers receive all of their annual income at once, they will be poor immediately before the harvest, and rich after. This creates panel data that allowed the researchers to compare pre- and post-harvest spending while controlling for month effects such as festival spending and seasonality. Farmers were not very good at smoothing spending across the year; while expenditures on food were similar in pre- and post-harvest months, post-harvest spending on other items was dramatically higher. Study outcomes included the Stroop test, a psychological test of attention, as well as allostatic load, a physiological measure of stress.

Mullainathan and his colleagues found that farmers scored significantly better on the Stroop test in the month after harvest than they did in the month before harvest. They also found that farmers had significantly lower stress levels after the harvest.

These results support the researchers' argument that scarcity taxes cognitive resources.

WHAT ARE THE CONSEQUENCES OF REDUCED COGNITIVE RESOURCES?

The evidence presented above shows that scarcity can be distracting, since managing tight resources requires more attention and self-control. One final piece of evidence from a laboratory experiment suggests some of the implications of reduced cognitive resources on the lives of the poor, including facilitating self-destructive credit practices. This experiment uses a "Family Feud" game to create a condition of scarcity. Subjects played a game in which they had to guess popular answers to a question. They earned points for correct answers, and received a monetary reward based on their total number of points over 20 rounds. All participants had the opportunity to complete practice rounds before playing for money. Half of the subjects were in a "rich" group, and had ample time (50 seconds) to complete each round of the task, while the other half were in a "poor" group, and had quite limited time available for each round (20 seconds). Within each group, one-third were permitted to "borrow" seconds from a future round, one-third could borrow at a

100 percent interest rate (one additional second in the current round "cost" two seconds from a future round), and the remaining third could not borrow. The intention of the borrowing condition was to allow participants to allocate their time according to how familiar they were with each question, so that, ideally, they could have more time to answer questions with which they were less familiar. Those in the poor group were more likely to borrow than those in the rich group. Those in the poor group were also less sensitive to the interest rate; they were nearly as likely to borrow at the 1:2 rate as at the 1:1 rate, while those in the rich group were much less likely to borrow seconds when they cost more. As one would expect, those in the poor group earned fewer points than those in the rich group. The more interesting result, however, comes from comparing the different borrowing conditions within each group. Those in the rich group gained no particular benefit from being able to borrow, but since they did not borrow often, it did not particularly hurt them either; there is no significant difference between points earned in the no-borrow and 1:2 interest rate conditions. For those in the poor group, however, borrowing consistently lowers their point total, and higher interest rate borrowing lowers their points more. The data show that this is because of over-

borrowing; those in the poor group tended to borrow seconds a lot in the early rounds, thus leaving themselves little time for later rounds.

This experiment illustrates two important points. First, attention was required; participants needed to choose how much time to focus on the current problem, as compared to future problems. Those in the “poor” group tended to focus on the current problem to the exclusion of future problems, and to their ultimate detriment. In this situation, credit turns out to be a bad thing; it may help with the current problem, but hurt with future problems, and thus overall. Second, this experiment was done with Princeton undergraduate students, so one can assume that outside factors such as financial literacy, upbringing, and early childhood development have no effect in this particular case. Nor could they have any effect in the sugar cane experiment, since the same people were being compared at two different times. This supports Mullainathan’s contention that detrimental decision-making by the poor is attributable to the condition of having very little, not to shortcomings that are unchangeable characteristics of poor people.

POLICY IMPLICATIONS

In his seminar, Mullainathan concluded that these results have important public policy implications, particularly related to take-up and retention in programs designed to help the poor. As currently designed, many of these programs actually create cognitive burdens, thus adding unnecessary challenges for those they are intended to assist. One example of this can be seen in programs designed to increase college attendance for low-income teens, which tends to be much lower than for their higher-income peers. Prevailing wisdom would say that reducing the cost would increase attendance, but simply making Pell Grants available did not significantly increase low-income attendance. Another explanation is that the student aid application was too complicated, and required more attention than people had. Having an administrator complete the form, rather than just providing information about it, resulted in significantly higher college enrollment.

Simplification works because instability makes dealing with complexity particularly challenging; forms are tough for all of us, but toughest when attention is most depleted. Forward-looking actions require attention and self-control. Instability taxes both of these, and thus makes economic mobility harder.

POLICY RESPONSES

Mullainathan identified two broad categories of policy responses—creating stability, and creating mobility programs that are resistant to the effects of instability. An example of the first type of policy response would be supplementing Unemployment Insurance with wage or hours worked insurance to help maintain a consistent salary for people facing a cutback yet retaining their job. Another approach would be a crisis-triggered social safety net card that people could have in-hand in the event of a sudden drop in income. Newly available financial products such as a debit card that includes a saving mechanism could also help in this area. As currently constructed, many mobility programs rely on stability as a condition of success. How could these programs be structured differently in order to remove this condition? An illustration of this is training classes. Most of these are designed so that each class builds on the last, making them particularly prone to instability. If you miss one class, it is much harder to get yourself to the next class, and this only gets more difficult the more you miss. An “instability proof” alternative would be to have rotating training class opportunities, where one could, for example, attend three of the next 10 classes

over the next four months in order to receive a training certificate. This approach would present some challenges, since new curricula would be required, but the effort could pay off in creating a program that was much easier for participants to follow through on. Poverty and economic instability reduce cognitive resources such as attention and self-control. These conditions make it much harder for the poor to behave in a way that will improve their economic fortunes, and much easier for them to make decisions that impede their mobility. Public policies should be designed to offset this scarcity phenomenon.

(For References – Just go through once, don’t dwell into detail. Know the word Easterlin Paradox)

THE EFFECT OF POVERTY ON AFFECT AND STRESS: CORRELATIONS BETWEEN POVERTY, AFFECT, AND STRESS

For several decades, the prevalent view on the relationship between income and psychological well-being was what became known as the **Easterlin Paradox**, according to which income, self-reported happiness, and life satisfaction are correlated within but not across countries and are uncorrelated above income levels

required to meet basic needs. In addition, higher incomes were thought to be uncorrelated with increased happiness and satisfaction over time. However, larger and newer data sets now suggest that higher incomes are associated with more happiness and life satisfaction both within and across countries, that no saturation point exists (although there are decreasing happiness returns to income), and that as countries grow richer, they also grow happier.

In addition to happiness and life satisfaction, poverty is also more broadly related to mental health. According to the 2003 World Health Report, the poorest population quintiles in rich countries exhibit a depression and anxiety disorder prevalence that is 1.5 to 2 times as high as that of the richest quintiles. A recent comprehensive review of 115 studies on the relationship between mental health and poverty in low- and middle-income countries finds a negative association between poverty indicators and good mental health outcomes in 79% of studies. Finally, income and socioeconomic status are also correlated with levels of the stress hormone cortisol. Several studies have shown elevated cortisol levels in persons with lower income and education and lower lifetime economic position as measured by occupational status. Similar results have

been obtained in infants and children. Together, these findings show that poverty correlates with unhappiness, depression, anxiety, and cortisol levels. But is this relationship causal?

CAUSAL EFFECT OF POVERTY ON AFFECT AND STRESS

The effect of reductions in poverty on affect and stress is usually studied in the context of randomized field experiments or natural experiments such as lottery wins. One such study examined the effects of an unconditional cash transfer program in Kenya on psychological well-being. Households were randomly chosen to receive unconditional transfers of either \$0, \$400, or \$1500. Psychological well-being was measured with the happiness and life satisfaction questions from the World Values Survey, and stress and depression were measured using the Center for Epidemiologic Studies Depression Scale, Cohen's Perceived Stress Scale, and levels of the stress hormone cortisol in saliva. The study finds substantial improvements in all of these variables when households receive positive transfers, but the stress hormone cortisol was only reduced in those who received large transfers. Similarly, several other studies reports results

from randomized controlled trials that show that cash transfers reduce distress and depression scores. Similarly, using natural experiments such as the introduction of guaranteed incomes, lottery payouts, access to a pension scheme, and payouts to Native Americans from a casino opening, several studies find that the resulting increases in income lead to a reduction in hospitalization for mental health problems, lower consumption of anxiolytics, and increases in self-reported mental health. Less direct alleviations of poverty have also shown effects; several randomized controlled trials report increases in psychological well-being when participants receive health insurance, improved housing, and access to water. Conversely, the effect of increases in poverty on well-being is usually studied using unexpected shocks such as spells of bad weather for farmers. One such study examined whether random negative income shocks to farmers in Kenya, generated by periods of drought, lead to increases in cortisol levels. The study finds that farmers have higher levels of cortisol and self-reported stress during drought periods when crops are likely to fail. This relationship does not hold for nonfarmers and is more pronounced among farmers who depend solely on agriculture for their income than among those who also have other sources of earnings. In addition,

it is robust to controlling for physical activity, suggesting that changes in labor supply are not the driving factor; the plausibility of this alternative account is further reduced by the fact that the increase in cortisol levels is mirrored by an increase in self-reported stress. Another study measured cortisol levels in a sample of 354 Swedish blue-collar workers before and after a subset of these workers lost their jobs. Cortisol levels were significantly higher in those workers who lost their jobs. Importantly, the layoffs were due to a plant closure, arguing against the possibility that job loss might be a consequence rather than a cause of high cortisol levels in individual workers. However, the fact that only one plant was studied and attrition among participants over the course of the study was non-negligible weakens the finding. A further study (50) uses declining industries as an exogenous source of variation for job loss and finds an effect of job loss on family mental health using this approach. These findings thus suggest causal links between poverty, psychological well-being, and stress levels. Altogether, we identified 25 studies that report the effect on psychological well-being of an increase or decrease in poverty, induced either in randomized controlled trials or natural experiments [see the supplementary material (51)]. Of these, 18

studies show a positive effect of poverty alleviation on psychological well-being or stress, 5 studies show effects on some psychological variables related to well-being or stress (e.g., certain mental disorders), but not others, and 2 show no results. The mixed or inconsistent findings in these studies may reflect deficiencies or noise of some of the measures used, heterogeneity in the interventions tested, or heterogeneity in the effect of changes in poverty on particular psychological constructs; future studies need to assess these different explanations. Thus, the large majority of the findings suggest that increases in poverty often lead to negative affect and stress, and decreases in poverty have the opposite effect. We now ask whether negative affect and stress influence risk-taking and time discounting and could therefore be among the channels through which poverty affects economic behavior.

DISABILITY

(Referenced from a well-versed article by Prashant Srivastava & Pradeep Kumar, published in the Delhi Psychiatric Journal)

OVERVIEW

In India, the numbers of disabled are so large, their problems are complex, available resources also scarce, social stigma still attached and people attitudes so damaging. Attitudinal barriers engrained as part of India's historical response to disability must be changed through education programs for both teachers and the general populace. These programs require financial and collaborative commitment from key national and state education stakeholders, and partnership with universities to support research-based initiatives. It is only legislation which can eventually bring about a substantial change in a uniform manner. Although legislation cannot alone radically change the fabric of a society in a short span of time, it can nevertheless, increase accessibility of the disabled to education and employment, to public buildings and shopping centres, to means of transport and communication. Therefore, in country like India mainstreaming of these people is challenging issue. For achieving this task it's necessary to change public attitudes, remove social stigma, provide barrier free

environment, needs reformation in the area of policy and institutional level.

Throughout centuries, the disabled have been oppressed marginalized and stigmatized in almost all societies. They constitute a section of the population, which is most backward least served and grossly neglected. Person with disability are the poorest of the poor and weakest of the weak, who have been socially, educationally and economically disadvantaged; thus, having customarily denied their right to self-assertion, identity and development. Now where is this victimization more glaring than in matters of education, employment and physical access. Disability is not all alone sometimes impairment and handicap were used interchangeably, but these terms has different meanings and describedifferent concepts. To promote appropriate use ofthese terms, in 1980 the World Health Organizationestablished the international classification ofimpairment, disability and handicap, which definethese concepts: Impairment – refers to the loss orabnormality of psychological, physical, or anatomicalstructure or function at the system or organ levelthat may or may not be permanent and that may ormay not result in disability.

DEFINING DISABILITY

Disability – refers to anindividual limitation or restriction of an activity asthe result of impairment. Handicap – refers to thedisadvantage to the individual resulting from animpairment or disability that presents a barrier tofulfilling a role or reaching a goal.

Disability is arelative term in so far as different cultures definetheir norms of being and doing differently.Conceptions of disability are therefore highlycontextual and subjective.As per 2001 census, 21.9 million or 21,906,769people are disabled in India, who constitutes 2.13

per cent of the total population. Out of the 21,906,769people with disabilities, 12,605,635 are males and9,301,134 are females.This includes persons withvisual, hearing, speech, locomotor and mentaldisabilities. Seventy-five per cent of persons withdisabilities live in rural areas, 49 per cent of disabledpopulation is literate and only 34 per cent areemployed.

The earlier emphasis on medicalrehabilitation has now been replaced by an emphasison social rehabilitation. The experience of disabledpeople in rural India suggests that this medical modeldoes not adequately capture the phenomena ofdisability. What

prevents disabled peoples in villages from living a life like anyone else, going to school, participating in family celebrations and working in fields is not individual's impairment, but how society interpreted and reacts to it. Life is made difficult not so much by the individual's medical condition, but mainly by a hostile physical and social environment which excludes disabled people from all spheres of social life. It is not the medical impairment, but the way in which society reacts to it, that exclude disabled from taking part in celebration, political decision-making or religious worship.

CAUSES OF DISABILITY

Analysis of the causes of disability from a medical or bio-centric standpoint tends to emphasise disease, hereditary and birth defects over systemic and environmental factors. Genetic factors and lack of access to basic services can also lead to a person becoming disabled, for example, before birth (poor nutrition, improper medication, taking drugs, smoking cigarettes, mother exposed to disease, mental or physical trauma), During birth (premature delivery, complicated delivery), After birth (Malnutrition, Lack of vaccination, Infections like meningitis polio, accident, trauma, toxic substance).

FACTORS OF DISABILITY

There are some factors which are responsible for disability, such as : Poverty and Disability -There is a high correlation between disability and poverty but very few studies have investigated how poverty and disability influence each other and with their combination create new forms of barriers. In general, people with disabilities are estimated to make up to 15 to 20% of the poor in developing countries. Inequitable economic and social policies have contributed to large numbers of people living in extreme poverty. Poor families often do not have sufficient income to meet their basic needs. Inadequate shelter, unhygienic living conditions, lack of sanitation and clean drinking water combined with poor access to health facilities lead to disability.

MALNUTRITION AND DISABILITY-

Malnutrition in its various forms is a cause of disability as well as a contributory factor in other ailments that increase susceptibility to disabling conditions. Common micro-nutrient deficiencies that affect disability include:

Vitamin A deficiency – blindness, Vitamin complex deficiency – beri-beri (inflammation or degeneration of the

nerves, digestive system and heart), pellagra (central nervous system and gastro-intestinal disorders, skin inflammation) and anaemia, Vitamin D deficiency – rickets (soft and deformed bones), Iodine deficiency – slow growth, learning difficulties, intellectual disabilities and goitre, Iron deficiency – anaemia, which impedes learning and activity, and is a significant cause of maternal mortality, Calcium deficiency – osteoporosis (fragile bones).

At the present rate, by the year 2010 there could still be some 680 million chronically undernourished people whose disabilities are likely to have roots in micro-nutrient deficiencies.

OCCUPATIONAL HAZARDS-

Around 90% of the workforce in India is in the unorganised sector, which is characterised by low levels of technology, low standards of safety and hazardous working conditions. Occupation-related health problems of workers employed in stone quarrying, leather industry, glasswork, weaving, diamond cutting, hand embroidery, and children employed in carpet, cracker and match industry have been recognised but have not received appropriate and sustained attention by those responsible for regulating work

standards. For example, 44,000 people lost their limbs in industrial accidents during the period of Vietnam War in which 17,000 American soldiers became disabled.⁴ Wars and Disability- War has been the single largest factor responsible for causing permanent disablement not only to combatants in the battlefield but also to civilians who are forced to bear the hazards of lethal, chemical and nuclear weapons. Based on figures from a study carried out in 206 communities, including Afghanistan and Cambodia, landmine triggered disability rate among survivors is about 0.9%. About 6% of households in Afghanistan are affected by landmine accidents alone. Surveys of four countries in 1995 found that between 12% & 60% landmine victims and to sell assets to meet their medical bills.

CRIME AND DISABILITY –

Violent crimes underline shortcomings in the social, political and economic arrangements of a society. Many children and women are abducted to be used in prostitution, slavery and beggary. The Bhagalpur blinding case in India is a well-known and documented illustration of this menace. Traffic Hazards- Unplanned cities with narrow roads, rapid growth in number of vehicles and disregard of traffic regulations have been responsible for

increasing the number of road accidents in India. If current trends continue, road accidents may become the leading cause of death and disability in the country. An expert in the field, Dr Leslie G Norman, estimates that for every road accident death there are 30-40 light injuries and 10-15 serious injuries, which may lead to disability. It is estimated that by 2020, road traffic accidents will be ranked as the third leading cause of disability in the Asian and Pacific region. Quadriplegia, paraplegia, brain damage and behavioral disorders are some common disabilities among survivors of traffic accidents.

CONSTITUTIONAL FRAMEWORK IN INDIA

The Constitution of India applies uniformly to every legal citizen of India, whether they are healthy or disabled in any way (physically or mentally) and guarantees a right of justice, liberty of thought, expression, belief, faith and worship and equality of status and of opportunity and for the promotion of fraternity. To safeguard the interests of the disadvantaged sections of the Society, the Constitution of India guarantees that no person will be denied 'equality' before the law (Article 14 of the Indian Constitution). Relevant Articles in Indian Constitution providing constitutional guarantees to all

including disabled are: Article 15(1): It enjoins on the Government not to discriminate against any citizen of India (Including disabled) on the ground of religion, race, caste, sex or place of birth. Article 15(2): It states that no citizen (including the disabled) shall be subjected to any disability, liability, restriction or condition on any of the above grounds in the matter of their access to shops, public restaurants, hotels and places of public entertainment or in the use of wells, tanks, bathing places (ghats), roads and places of public resort maintained wholly or partly out of government funds or dedicated to the use of the general public. Article 17: No person including the disabled irrespective of his belonging can be treated as an untouchable. It would be an offence punishable in accordance with law. Article

Every person including the disabled has his life and liberty guaranteed. Article

There can be no traffic in human beings (including the disabled), and beggar and other forms of forced labour is prohibited and the same is made punishable in accordance with law. Article 29(2): The right to education is available to all citizens including the disabled. No citizen shall be denied admission into any educational institution maintained by the State or receiving aid out of State funds. Article 32: Every disabled person can move the Supreme Court of India to

enforce his fundamental rights and the rights to move the Supreme Court.⁷

ACTS FOR DISABILITY

The legislative framework for the protection of the rights of disabled people is covered by given acts: The Rehabilitation Council of India Act 1992: Act sets out to regulate the training of professionals in rehabilitation and sets out a framework for a Central Rehabilitation Register. In order to give statutory powers to the Council for carrying out its duties effectively the Rehabilitation Council of India Act was passed by the Parliament which came into force with effect from 1993. The amendment in the Act in 2000 gave the additional responsibility of promoting research to the Council. The major functions of the council include the recognition of qualifications granted by Universities in India for Rehabilitation Professionals and also the recognition of qualification by Institutions outside India. The Persons with Disabilities (Equal Opportunities, protection of Rights and full Participation) Act 1995: This act provides 3% reservations for disabled people (blind or low vision, hearing impairment and locomotor disability or cerebral palsy in poverty alleviation programs, government posts, and in state

educational facilities, as well as other rights and entitlement). The specific objectives of the Act are: Prevention and Early Detection of Disabilities, Education – all Government educational institutions reserves more than 3% seats for disabled. Employment – with 3% reservation the disability vacancies not filled up to be carried forward for next three years and after that the vacancy will be filled by a non-disabled person. Schemes for ensuring employment of person with disabilities are: Training and welfare, Relaxation of upper age limit, Regulation the employment, Health and safety measures. Affirmative Action – Preferential allotment of land for certain purpose – Government or local authorities for: House; Setting up business; Setting up special recreation centers; Establishment of special school; Establishment of research centers; Establishment of factories by entrepreneurs with disability.

CRITICISM OF PERSON WITH DISABILITIES ACT 1995:

The Persons with Disabilities Act (PWD), 1995 has been landmark legislation for the disabled in India. This Act that is related to mental illness (MI) and provides recommendations aimed at making it an official instrument for equal opportunities, participation and protection

of rights Act in its true sense. But there are some serious flaws in the Act that have to be set right to ensure equitable distribution of benefits to all. The Act defines a disabled person as one who is “suffering from 40% or more disability”.

However, as far as MI is concerned, this quantification is a mystification because such a tool is unavailable. The PWD Act unfortunately, turns out to be an instrument of injustice and discrimination albeit unwittingly. According to the chairperson of the Amendments Committee, MI missed this opportunity for employment rights because of the absence of a well-informed advocacy platform that coalesced into a lobby.⁹ The National Trust for Welfare of Persons with Autism, Cerebral Palsy, Mental Retardation and Multiple Disabilities Act 1999: This Act provides for the constitution of a national body for the Welfare of Persons with Autism, Cerebral Palsy, Mental Retardation and Multiple Disabilities. The main objectives are: To enable and empower persons with disability to live as independently and as fully as possible within and as close to the community to which they belong; To strengthen facilities to provide support to persons with disability to live within their own families; To extend support to registered organization to provide need

based services during the period of crises in the family of persons with disability; To deal with problems of persons with disability who do not have family support.¹⁰ National Policy for Persons with Disabilities Act 2005: The National Policy, released in February 2006 recognizes that Persons with Disabilities are valuable human resource for the country and seeks to create an environment that provides them equal opportunities, protection of their rights and full participation in society. Its aim is to ensure better coordination between various wings of the State and Central Governments. The focus of the policy is on the following: Prevention of Disabilities, Rehabilitation Measures, Physical Rehabilitation Strategies, Early Detection and Intervention, Counselling & Medical Rehabilitation. In addition to the legal framework, extensive infrastructure has been developed in India for disabled persons under this Act and includes the establishment of the following institutions: Institute for the Physically Handicapped, New Delhi, National Institute of Visually Handicapped, Dehradun, National Institute for Orthopedically Handicapped, Kolkata, National Institute for Mentally Handicapped, Secunderabad, National Institute for Hearing Handicapped, Mumbai, National Institute of Rehabilitation Training & Research,

Cuttack, National Institute for Empowerment of Persons with Multiple Disabilities, Chennai. A 2004 survey in India revealed the following: Limited information on the definition of disability; Limited information on access to education to disabled people; No mention on disability in the Indian constitution; No standardized sign language.

Ministry of Social Justice and Empowerment, in pursuance of the provisions of Section 32 of the Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995 (1 of 1996), had constituted an Expert Committee on the 30th December, 2010 under the Chairmanship of Additional Secretary, Ministry of Social Justice and Empowerment. The Expert Committee, with the help of three sub-committees, made an in-depth study of various jobs performed in Government of India Ministries/Departments, public sector undertakings and autonomous bodies including Universities. The Sub-committees also reviewed jobs notified in 2007 and prepared a detailed list of Groups A, B, C and D posts which were identified suitable for persons with disabilities. The Expert Committee submitted its report on 24th January, 2012.

The Central Government accepted the report and the list of posts identified for Persons with Disabilities has been published vide Notification No. 16-15/2010-DD-III dated 29th July 2013. 11 Programmes and concessions for the disabled Government of India has taken the responsibility of providing optimal environment to ensure full participation of the persons with disabilities. In this context, it has introduced a number of welfare programs, schemes, concessions and facilities for educational, vocational and economic rehabilitation.

EDUCATION:

Children's educational allowance - The reimbursement of tuition fee in respect of physically handicapped and mentally retarded children of the Central Government employee has been enhanced from Rs.50/- to Rs.100/- per month. Scheme of integrated education for the disabled children - Under the scheme, handicapped children are sought to be integrated in the normal school system. 100% assistance is provided to the States/UTs for education of children suffering from certain mild handicaps in common schools with the help of necessary aids, incentives and specially trained teachers. Financial assistance to persons with disabilities: - The National

Handicapped Finance and Development Corporation NHFDC, a subsidiary under the Ministry of Social Justice and Empowerment, Government of India, promotes infrastructure development schemes directly leading to income generation. This corporation provides self-employment opportunities to the persons with disabilities by providing loans at very low rates of interest. Travel concession: By Rail- Orthopedically handicapped persons / mentally retarded persons with an escort, on production of a medical certificate, are eligible to 75% concession. By Air- Locomotors disabled persons with 80% disability are allowed 50% concession in Indian Airlines. Employment of handicapped: Assistance to disabled persons in getting gainful employment either through special cells in National Employment Exchanges or through Special Employment Exchanges for physically handicapped. Up to 100% financial assistance is provided in the case of special cells and 80% in the case of Special Employment Exchanges to state governments / union territory administrations. Communication: Educated unemployed persons are eligible for allotment of STD, PCOs. The educational qualifications for the applicants are 8th or middle class passed from rural areas, at least matriculation or high

school for urban areas. Conveyance allowance: Orthopedically handicapped with disability of lower extremities will be paid transport allowance at double the normal rates. Handicapped employees who have been provided with government accommodation within a distance of 1 km from the place of work or within the campus, the allowance shall be admissible at normal rates.

MISCELLANEOUS PROGRAMMES:

Family pension to disabled children - Handicapped children shall be eligible for the benefit of family pension even if they have been born after the retirement of the Government Servant from a marriage solemnized after retirement. Ad-hoc allotment of central pool residential accommodation to the physically handicapped employees- Government employees suffering from tuberculosis, cancer and physically handicapped persons may get ad-hoc allotment of general pool residential accommodation on request after recommendation of the special recommendation committee and on the approval of the urban development ministry.

(ONLY FOR REFERENCE)

CERTIFICATION IN DISABILITY

The medical boards have been constituted at the State level, District level and Taluk level to assess the percentage of disability / the level of disability and then to issue disability certificates whose disability is more than 40% and above. The composition of the Medical Board is as follows: State Level Medical Board, District Level Medical Board, Taluk Level Medical Board. Because of the UN

proclamation in 1981, subsequent declaration of Decade for Disabled and the Biwako Millennium Framework of Actions in 2003, extended further from 2003-2012, to which India is a signatory, it is binding on the member countries to protect the rights, provide equal opportunities and empower persons with disability. The guidelines notified, are for assessment of disability in the respective area/body part (function) and to quantify in terms of percentage of disability, to avail facilities & concessions viz. Reservation in job, Travel concession, soft loan for entrepreneurship development, Scholarship, Income Tax / Custom rebate, Age relaxation in employment etc. As per the Act, authorities to give a disability certificate will be a medical board duly constituted by the central and state government. According to the PWD Rules, 1996, the Committee recommended that certification of disability for the purposes of the Act had to be carried out by

a medical board comprising of the following members:

- (a) The Medical Superintendent/Principal/Director/Head of the institution or his nominee Chairperson,
Psychiatrist—Member,
Physician — Member.

At least two of the members, including Chairperson of the board must be present and sign the disability certificate. The certificate would be valid for a period of five years for those whose disability is 'temporary' and are below the age 18 years. For those who acquire permanent disability, the validity can be shown as 'permanent' in the certificate. When there are no chances of variation in the degree of disability, a permanent disability certificate is given. The PWD Act (1995) did not prescribe guidelines for evaluation and assessment of mental illness and the specific procedures for certification. With regard to assessment of disability related to mental illness it was agreed that the Indian Disability Evaluation and Assessment Scale (IDEAS, 2002) developed by the Rehabilitation Committee of the Indian Psychiatric Society (IPS) through a Task Force should be used with modifications for the purposes of the Act.

MEASUREMENT OF DISABILITY FOR MENTAL ILLNESS:

The assessment tools have already been existed for the visually impaired, hearing impaired and orthopaedic handicap and persons with mental retardation. These people are certified by the

authentic body and become eligible by having disability certificates to avail the benefits under the PWD Act 1995. But there were no assessment tools for the certification of mentally ill people and yet these people are not availed any benefits even as disabled. Looking that perspective and to justify these people

rehabilitation committee of Indian Psychiatric Society has developed the assessment tool for disability certification in 2002. This tool is known as Indian Disability Evaluation and Assessment Scale in short IDEAS. This IDEA has opened new horizon for mentally ill people. This committee has developed clear guideline to make use of it very easy. General Guidelines: IDEAS are suited best for the purpose of measuring and certifying Disability, It is therefore a brief and simple instrument, which can be used, even in busy clinical settings. Some training is required in the use of IDEAS. This is to be used only on out patients and those living in the community. Not

appropriate for in-patients. Rating should be done only based on interviews of the Primary Care Givers. Case records and patients' interviews can be used to supplement information. Only in rare instances when no primary care giver is available should the rating be based only on patient interview. This should then be documented. The gender specification "he" has been used for convenience and refers to both genders. Patients with only the following diagnosis as per ICD or DSM criteria are eligible for disability benefits: Schizophrenia, Bipolar Disorder, Dementia, and Obsessive-Compulsive Disorder.

DURATION OF ILLNESS:

The total duration of illness should be least two years. For the purpose of scoring, the number of months the patients were symptomatic in the last two years (MI 2Y – months of illness in the last two years) should be determined. Only the Psychiatrist can do diagnosis and certification. Trained social workers, psychologist, or occupational therapists can do administration of IDEAS. Psychiatric Disability will be reassessed every two years and re-certified. The feasibility of doing this in the rural areas will however have to be examined. Items in IDEAS: Self-care: Include taking care of body

hygiene, grooming, and health including bathing, toileting, eating and taking care of one's health. Interpersonal Activities (Social Relationship) : Includes initiating and maintaining interactions with others in a contextual and socially appropriate manner. Communication and Understanding: Include communication and conversation with others by producing and comprehending spoken/ written/ nonverbal messages. Work: Three areas are Employment/ House work/ Education measures any one aspect. 1- Performing in Work/Job: Performing in work / employment (paid) employment / self-employment family concern or otherwise. Measures ability to perform tasks at employment completely and efficiently and in proper time. Includes seeking employment.

PERFORMING IN HOUSEWORK:

Maintaining household including cooking, caring for other people at home, taking care of belongings etc. Measures ability to take responsibility for and perform household tasks completely and efficiently and in proper time. Performing in school / college: measures performance in education related tasks.

Scoring:

0 – No Disability,

1 – Mild Disability,

2 - Moderate Disability,

3 – Severe Disability,

4 – Profound Disability.

Total Score (range 0-20).

PERCENTAGE:

For the purpose of welfare benefits, 40% will be cut off point. The scores above 40% have been categorized as Moderate, Severe, and profound based on the Global disability score. This grading will be used to measure change over time. 0-No disability
0% , 1-7 – Mild Disability = < 40%,
8 and above = > 40%, (8-13 moderate disability; 14-19 Severe Disability; 20 Profound Disability).¹⁴

HURDLES IN ACCESSING THE BENEFITS & SERVICES

Apart from that there are many problems facing by People with Disabilities to access the services easily, such as: Employment - There seems to be a lack of coordination between different government services so that people with disabilities are unclear about which Commonwealth or State agencies can provide them with assistance. Information about the financial and health impacts of entering the

openworkplace is hard to find, there is limited or delayedaccess to open employment services.

Financial -Additional transport, support and other costsassociated with work, many of which are not subsidized foregoing the Disability Support Pensionand associated benefits, such as the PensionersHealth Care Card. Inflexible workingenvironment

Difficulties with physical access tothe workplace, and getting to and from work,inadequateadjustmentsand adaptations toworkplace equipment, inflexible working hours.

More limited opportunities for people withdisabilities - Poor links between State-administeredschool and post-school programs and Common-wealth-administered disability employment servicesunder-representation in the vocational, education andtraining systems, limited scope and variety of jobsoffered to people with disabilities, lower possibilitiesfor promotion, lower paying jobs & lower retentionrates.Issues related to Psychiatric DisabilityImpact of disability on Psychiatric Disability isbased on the major four functional aspects of personwhich are:

PSYCHOLOGICAL ISSUES OF PSYCHIATRIC DISABILITY-

Individuals with psychiatric disabilityexperiences a wide range of symptoms that affectpsychological and cognitive function, and their needsare multifaceted and complex. Although the benefitsof medication in the treatment of psychiatric disabilityare substantial, medication usually does not cure thecondition but rather controls the symptoms.Individuals often have residual symptoms, deficitsand impairments as a result of their condition andmany are subjected to periodic relapses withrecurrence of symptoms. Individuals with psychiatricdisability may be particularly vulnerable to stressand may lack the ability to withstand the pressureor to cope with the normal stressors of everydaylife. They may have limited problem-solving abilityor find it difficult to engage in self-directed activity.Others experience subjective distress, such as aninner sense of weakness, jealousy, or anxiety,although function in most of their life is minimallydisturbed. Some psychiatric disabilities arecharacterized by disorganization of mentalcapacities, which can be affect individual's ability tofunction in an unstructured environment.

(II) LIFESTYLE ISSUES OF PSYCHIATRIC DISABILITY –

Thedegree to which psychiatric disability affects individual's lifestyle depends to a

great extent on the nature of the condition. Some psychiatric disabilities so severely impair individual's ability to carry on the activities of daily living that constant supervision or hospitalization is necessary. In other instances, individuals are able to carry on these activities, but in an altered manner. At times the treatment itself requires lifestyle changes. Individuals may need to rearrange their schedules so that they can attend therapy sessions. Some medications used in the treatment of psychiatric disability may require special lifestyle considerations. For example, the use of MAO inhibitors in the treatment of depression requires careful monitoring of diet. Other medications have side effects, such as drowsiness and sedation that also affects daily functioning. In addition, subjective manifestations of lowered self-esteem and self-confidence may make it more difficult for individuals to form intimate relationship.

(III) SOCIAL ISSUES OF PSYCHIATRIC DISABILITY-

The impact of a psychiatric disability on social function also depends on the nature of the condition. Individuals who experience mania as part of their disability may enjoy the euphoria and feel that it contributes to their social well-being. Even though attitudes of society become

more accepting of individuals with mental illness, family members may continue to be resistant to recognizing the problem and pursuing the appropriate treatment. If, however, individuals manifest bizarre, abusive, or socially offensive behavior, family members or others within a social group may avoid the individual altogether, leaving him or her socially isolated. Other psychiatric disabilities may lead to social withdrawal. Families of individuals with psychiatric disability may experience a variety of stresses engendered by the condition. In some instances, the demand of caregiving may require family members to curtail their social activities or alter their relationship with friends and acquaintances. The time commitments of caregiving may lead to neglect of other family member's needs, further disrupting the family as a unit. Social barriers are frequently erected against individuals with mental disorders and against their families. Social stigma may be the result of fear of individual's behaviour, ignorance about psychiatric disability, or feelings of inadequacy interacting with those who have psychiatric disability. Regardless of the cause, the result can be a source of continuing stress for individuals and their families, as well as a barrier to social activity and interaction.

(IV) VOCATIONAL

Issues of Psychiatric Disability-Individuals with psychiatric disability have a condition that limits their capacity to perform certain tasks and functions and their ability to perform certain roles. The ability to work depends on the type of disability, the type of work in which they are involved, and the attitude of those within the work setting. Although work is important to increase self-esteem for those with a number of disabilities, it can be an especially strong therapeutic tool for those with a psychiatric disability. Job restriction may be related to job pressure or ability to work with others, regardless of the individual's level of skill or physical and cognitive ability to perform work related tasks. Some medications used in treatment may produce side effects such as drowsiness or sedation that could adversely affect work performances. Individual's reactions to the work environment, including noise and distractions, should be considered as should their level of personal responsibility and ability for self-direction and decision making. Some individuals may need more structured work environment; in some instances, a workshop environment may be preferable. The unemployment rate for individuals with psychiatric disabilities continues to be high. Although

it is believed that people with psychiatric disability have been significantly discriminated against in the workplace, the extent to which discrimination exists is difficult to determine because of lack of relevant data.

PSYCHOSOCIAL ISSUES OF MENTAL RETARDATION

The opinions and expectations most people have about themselves are influenced to a great degree by the behaviour of those around them. When minimum expectation or lack of belief in individual's ability to achieve is communicated, the chances for individuals to progress in attaining goals are diminished. Because a number of inaccurate and stereotypical ideas about individuals with mental retardation still exist, barriers to reaching optimal function and independence continue to be present. Although societal and employer attitudes are changing slowly, there is continued need for education and integration of individuals with MR into society and into the workplace. Although all individuals with MR can experience stresses due to societal stereotypes and attitudes, individuals with mild MR may confront specific stresses because they may appear normal to others and consequently limitations may not be recognised as a disability. Lack of

acceptance and devaluation can result in low self-esteem and isolation, which in turn can lead to deviant behaviours or acting out. In more severe cases, a psychiatric disorder may be developed as a means of coping. Vocational issues of Mental Retardation - The level of occupational functioning for individual with MR depends to some extent on the degree of disability. Because mental retardation is often accompanied by other medical conditions, the physical limitations associated with any medical condition must also be considered. Individuals with MR usually perform better in a structured environment. Many individuals may need to be taught how to function independently and may need accompanying social skill training. Although there has been heightened effort and interest towards increasing integrated employment opportunities for individuals with mental retardation, rehabilitation outcomes, especially for individuals of racial and ethnic underrepresented groups, have been less than ideal.

MIGRATION: CULTURAL BIAS AND DISCRIMINATION

(Taken from Migration Psychology: New Behavioral Models by James T. Fawcett)

BRIEF ABOUT MIGRATION:

Just 20 years ago the geographer Julian Wolpert (1965) published his influential conceptual paper, "Behavioral Aspects of the Decision to Migrate" which led to a landmark change in the introduction of Migration as a topic of focus in Psychological researches.

Migration is the movement of a person from one location to another. It can be understood to take many forms, including a relatively permanent change of residence from one city or rural area to another, a move from one neighborhood to another within the same city (residential mobility), temporary moves for purposes such as seasonal employment or to attend school (circulation), and voluntary or involuntary movements across national boundaries (immigration, international labor migration, refugee movements). Migration is one of the main factors in population redistribution and thus plays a dynamic role in such areas of critical concern as rapid urbanization of the world's

population, the balance between population and resources, and human impacts on the natural environment. Most studies of migration have been conducted from a macroeconomic or social-demographic perspective, looking at migration as an aggregate phenomenon.

A complementary perspective is taken in migration psychology, where the conceptual focus is at the microlevel. Migration psychology is the study of individual behavior with respect to movement across space. Studies in this field often deal with the subjective correlates of migration behavior, such as underlying attitudes, values, perceptions and migration intentions. Two broad questions define the scope of migration psychology: What factors are responsible for a person's voluntary change of location? And, what are the consequences for the person of the new environmental factors associated with a change of location? The first question points toward the application of motivational constructs and decision models in migration psychology. The second question relates to adjustment, acculturation, or adaptation models and brings in issues of environmental stress, mental health, and community intervention. We will focus here only on the motivational and decisional approaches to migration behavior. (For psychologically-oriented reviews on the consequences of migration, see Brody, 1969; Olmedo, 1979). Psychological studies of the determinants and consequences of migration would seem to be a natural focus for the American Psychological Association's

Division of Population and Environmental Psychology.

MIGRATION IN CONTEXT

Although often identified with massive refugee movements, migration is usually undertaken to increase economic or socio-cultural opportunities for the migrant. Migrations are local, national or international. They involve small groups of varying size as well as mass displacements of entire populations.

The history of migration to Britain highlights some of the reasons why people migrate. Significant migration to Britain started in the nineteenth century. Irish immigration has been marked by periods of influx and efflux to and from Britain, as people have come to either settle permanently or work temporarily with ultimate return to Ireland as a goal. Eastern European Jews came at the latter part of the nineteenth century to escape both religious persecution and poverty, with additional numbers arriving in Britain both before and after World War II. In the 1960s, employers, especially in urban areas, recruited people from the West Indies to fill low paying jobs which were less attractive to the local population. People from the Indian subcontinent migrated to Britain for educational and

economic reasons, the peak of which occurred about the same time as the West Indian migration. Asian people expelled by Idi Amin's government came from Uganda in the late 1970s. The 1980s saw a change in the immigration laws limiting the numbers of people allowed to relocate to Britain. Today, people from around the globe choose to migrate to the UK as well as other developed countries, both legally and illegally, for better educational and employment opportunities, to escape persecution, to relocate after catastrophic events, including terrorism, disasters and war, and/or to join relatives who migrated at an earlier time.

Most migrations are voluntary; however, forced resettlement is as ancient as the Babylonian captivity of the Jews and as recent as the vast population transfers following the separation of India and Pakistan. In the United States, the policies of Andrew Jackson led to the expulsion of the "civilized tribes" from the Southwest to Oklahoma, the Trail of Tears still anguishing the Choctaw, Cherokee and other Indians. Perhaps the greatest forced migration has been the African diaspora resulting from 1,000 years of Arab and 350 years of European slaving. All migrations, even slaving, lead to permanent transfers of populations and produce groups which return to areas of

origin. This kind of circular travel characterizes traditional nomadism, episodic wage work in mining (Africa) or agriculture (United States), pilgrimages (Mecca, Hindu sacred places) and tourism.

Although there exists no systematic history of migration, this phenomenon invariably occurs in response to high regional or international imbalances of populations and resources on the one hand; on the other, to new perceptions of opportunity, changes in transportation, and politico-military events. One important concomitant of migration has been epidemic disease, such as the influenza pandemic of 1919-21 and the great African cholera epidemic of 1974.

Notable migrations include those resulting from Alexander's conquests and Greek city building from Egypt to Central Asia. From about 100 B.C. to 600 A.D., Eurasia from the Pacific to the Atlantic witnessed very large population movements, such as the migration of the Huns from eastern Mongolia to Europe and India. But these earlier migrations have been far overshadowed by the upwellings of Europeans and Chinese since about 1850. The former populated the Americas, Australia and New Zealand, and Siberia; the latter, Manchuria and Southeast Asia. Today, the great migratory movement,

especially in the Third World, is urbanization. Even cities as grim as Calcutta, where thousands are permanently homeless, attract villagers seeking a better, freer, more exciting life. Delhi today is serviced by entire villages of commuters, who retain a part-time village life.

Migration can be classified in a number of ways; e.g., by the reasons for the migration, the social class and education of the migrating people, the duration of relocation and the geographic distribution of the resettlement. Further, a distinction can be made in the classification of migrants according to whether their contact with the 'majority' or 'dominant' culture is deemed voluntary. Migrants can be classified as immigrants and sojourners when the change in their location results in contact voluntarily, whereas refugees are deemed to change their location involuntarily. For example, immigrants choose to migrate, and thus be in increased and regular contact with the 'majority' culture in preparation for migration, for potential economic and/or educational advancement, whereas refugees are forced to migrate, and thus be in contact with the 'majority' population involuntarily, to escape persecution. Additionally, rural-urban migration has been associated with economic and educational reasons for relocation, whereas migration across

nations has been associated with economic, educational, social and political reasons.

The process of migration has been described as occurring in broadly three stages. The first stage is pre-migration, involving the decision and preparation to move. The second stage, migration, is the physical relocation of individuals from one place to another. The third stage, postmigration, is defined as the "absorption of the immigrant within the social and cultural framework of the new society". Social and cultural rules and new roles may be learnt at this stage. The initial stage of migration may have comparatively lower rates of mental illness and health problems than the latter stages, due to the younger age at the initial stage of migration and the problems with acculturation and the potential discrepancy between attainment of goals and actual achievement in the latter stages. It is worth noting that the stages are often not discrete and merge into one another.

CULTURAL BIASES IN MIGRATION

Culture is not new to the study of migration. It is interesting to see how culture manifests itself in the migration process for three groups of actors: the migrants, those remaining in the sending areas, and people already living in the

recipient locations. Actors behave differently, within a group there are economically important shared beliefs (customs, values, attitudes, etc.), which we commonly refer to as culture. Culture and identity play a central role in our understanding of migration as an economic phenomenon and also in sorting the determinants of identity and the determinants of culture (prices and incomes, broadly defined).

‘Culture matters’ has become a general wisdom in the economics literature. Indeed, recent literature shows that norms and values, such as trust, which is an important component of culture, determine economic activity and eventually growth. Specifically, cultural biases hinder economic exchange across location. Since culture only changes slowly over time, it is also a main source of the legacy of history emphasized by economic historians.

Conceptually, the decision to migrate, and where exactly to go, is determined by comparing the costs and benefits of moving to the costs and benefits of alternatives (Todaro 1969; Harris and Todaro 1970). Benefits and costs can be monetary or non-monetary; also including the non-monetary psychic migration costs of moving from a familiar to an unfamiliar surrounding (Sjaastad 1962). Examples for

these costs are that migrants might have to leave family and friends (combined with the non-existence of a social network at the new destination; Munshi 2003), have to cope with different cultural traits and habits or with higher prices or even unavailability of origin-specific local goods in the new destination. A complementary interpretation of non-monetary psychic migration costs are that they represent labour market frictions, preventing the efficient allocation of labour. Labor market search imply that individuals accept wage offers only when they are compensated for migration cost. That gross migration is larger between culturally more similar counties implies that searching for a new job is not random, but directed toward more similar regions. This leads to inefficient search behavior in terms of wage income as individuals miss better paid job opportunities elsewhere. The individual, however, is compensated by a higher familiarity with the destination region, but one can wonder whether society as a whole does not suffer from such labor market frictions.

PSYCHOLOGICAL IMPACT OF MIGRATION

Migration has contributed to the richness in diversity of cultures, ethnicities and

racess in developed countries. Individuals who migrate experience multiple stresses that can impact their mental wellbeing, including the loss of cultural norms, religious customs, and social support systems, adjustment to a new culture and changes in identity and concept of self. Indeed, the rates of mental illness are increased in some migrant groups. Mental health practitioners need to be attuned to the unique stresses and cultural aspects that affect immigrants and refugees in order to best address the needs of this increasing and vulnerable population.

If the individual feels isolated from his or her culture, unaccepted by the 'majority culture' and has a lack of social support, a consequent sense of rejection, alienation and poor self-esteem may occur. During the stages of migration, there may be factors that predispose individuals to mental disorders. Pre-migration factors include the personality structure of an individual, forced migration, and persecution, among others. Migration factors include bereavement, culture shock, a discrepancy between expectations and achievement, and acceptance by the new nation are potential post-migration factors. These factors can be thought of as vulnerability factors along with biological, social and psychological variables. For example, personality structure can be

thought of as a biological factor as well as in cultural terms. Personality is influenced by cultural factors and influences patterns of child rearing, responding to stress and accepting social support. National character and personality factors are interlinked.

CULTURAL BEREAVEMENT

The loss of one's social structure and culture can cause a grief reaction, as has been described by Eisenbruch. Migration involves the loss of the familiar, including language (especially colloquial and dialect), attitudes, values, social structures and support networks. Grieving for this loss can be viewed as a healthy reaction and a natural consequence of migration; however, if the symptoms cause significant distress or impairment and last for a specified period of time, psychiatric intervention may be warranted. Eisenbruch has defined cultural bereavement as "the experience of the uprooted person - or group - resulting from loss of social structures, cultural values and self-identity: the person - or group - continues to live in the past, is visited by supernatural forces from the past while asleep or awake, suffers feelings of guilt over abandoning culture and homeland, feels pain if memories of the past begin to fade, but finds constant images of the past

(including traumatic images) intruding into daily life, yearns to complete obligations to the dead, and feels stricken by anxieties, morbid thoughts, and anger that mar the ability to get on with daily life".

CULTURAL IDENTITY

It is important to define basic sociologic terms of identity to understand cultural identity. Culture is learned and passed through generations and includes the beliefs and value system of a society. Culture has been described as features that are shared and bind people together into a community. Identity is the totality of one's perception of self, or how we as individuals view ourselves as unique from others. Bhugra notes that racial, cultural and ethnic identities form part of one's identity, and identity will change with development at a personal as well as at a social level along with migration and acculturation. Social identity can be thought of as the culturally defined personality characteristics, which are ascribed to social roles, such as the role of being a father, mother, friend, employer, employee, etc. Ethnicity is a source of social identity. Ethnic groups are composed of people who may or may not share the same race but do share common cultural characteristics, including history, beliefs, values, food and entertainment preferences, religion and language.

Ethnicity typically incorporates both race and culture. Race is based on biologic constructs, such as sharing certain physical attributes; it may or not be also a social and political construct. For example, people from the West Indies, Africa and parts of North and South America may share the same race but have different beliefs, value systems, social norms and idioms of distress.

CULTURAL CONGRUITY

Migrating people come from diverse cultural backgrounds, with already formed cultural identities. As noted above, cultural identity is influenced by various factors both during and after the migration process, and cultural bereavement is a potential inherent consequence in people who have migrated. Cultural identities interact, as people who have migrated come into contact not only with people of the majority culture but also with immigrants of both similar and disparate cultures. Resultant feelings of a sense of belonging and comfort or a sense of alienation and distress may occur. Bhugra and Jones proposed that various personal and relational factors during the migration process impact the mental wellbeing of migrating people. During the post-migration phase, personal factors of importance in coping with adversity

include cultural identity, social support networks, self-esteem, and self-concept. Achievement, racism, ethnic density, social isolation and unemployment are among the relational factors of importance in migrants during the post-migration phase.

Ethnic density, the size of a particular ethnic group in proportion to the total population in a specified area, may be a factor that influences the rates of mental illness in ethnic minorities. Additionally, a sense of alienation may occur if the cultural and social characteristics of an individual differ from those of the surrounding population, whereas a sense of belonging tends to occur if the individual and surrounding population have similar cultural and social characteristics.

(Let's define Discrimination: It is the phenomenon of treating a person differently from other persons based on group membership and an individual's possession of certain characteristics such as age, class, gender, race, religion, and sexuality. Discriminatory behavior can take various forms from relatively mild behavior, such as social avoidance, to acts of violence, including hate crimes and genocide.)

STIGMA, MARGINALIZATION, AND SOCIAL SUFFERING

(Taken some parts from Prof. S.N. Chaudhary's article on 'Social Problems and Marginalized Groups')

The issue of social exclusion has become the centrality of current development paradigm in India. Social exclusion is one of the negative effects of economic restructure and transformation. It is a phenomenon wherein individuals, social groups, or geographical areas tend to experience political, economic and/or social disadvantages, lack of confidence, a sense of powerlessness or a degree of social alienation. In other words, it is used to describe a situation in which an individual, group or class is unable to participate in the basic political, economic and social functions of society (Silver 1994; Haan 1998; Byrne 1999). Social exclusion is, therefore, not merely a problem of economic disadvantages such as low income or unemployment, but it is also social disintegration, alienation/isolation from society, powerlessness and inability to take part in, and contribute to, mainstream development. Social exclusion is a multi-dimensional concept. It refers to exclusion (deprivation) in economic, social and political spheres. It goes beyond the analysis of resource all

ocation mechanisms, and includes power relations, agency, culture and social identity. Social exclusion can refer to a state or situation, but it often refers to a process, to the mechanism by which people are excluded.

Combating social exclusion of certain groups or classes from main stream development has become an essential feature of the long-term development paradigm in most developed and developing economies of the 21st century. The term “social exclusion” was originally coined by Rene Lenoir in 1974, the French Social Action Secretary of State in the Chirac government (published a document entitled *Les Exclus: Un Français sur Dix*) referring to various social categories of people, such as the mentally and physically handicapped, single parents, substance users and other groups unprotected by social insurance.

THE INDIAN CONTEXT

No society is free from marginalised groups and weaker sections. However, the core bases of marginalisation differ from society to society. While in most of the European and Western societies, skin colour is one of the important bases of marginalisation of blacks, in Indian society, division of population on the bases of birth as per the preambles of Varna system became core base of marginalisation of

significant segment of population who were placed outside for fold Varna system, who are today constitutionally known as Scheduled Castes. Irrespective of their achieved background, SCs are perceived as marginalised groups. STs are outside the Varna system. They are animist with distinct cultural heritage. It is claimed that they are the victim of geographical segregation and maternal exploitation by outsiders. It is claimed that with the passage of time, due to multiple reasons, their subjugation has become intense. They are another category of marginalised groups. Women as a category are perceived as marginalised because in the patriarchal framework, historically they have suffered denial and subjugation.

After becoming independent and adopting democratic form of governance, India made effort to enlarge space meant for marginalised groups. Today its boundary is more spacious and in broad comparison to situation few decades back. There are many groups and social categories which have been termed as marginalised or who are making effort to be declared as marginalised groups. Important among these groups are widows, war victims, persons with disability, destitute, lesbians, trans-genders, AIDS victims and so on. It is the moral responsibility of the state to identify, protect and promote concerns of

all these segments of population. Many policies and schemes have been formulated in favour of such population.

Social suffering is eventual resultant caused by cumulative interactions of disadvantages such as income, health, education, access to services, housing, debt, quality of life, dignity and autonomy (Cheetam and Fuller 1998; Barry 1998). The consequences of social exclusion are manifested through social disintegration, rupture of social bonds and loss of social cohesion, resulting in state dependency, family breakdown, homelessness, rising cases of criminality and substance abuse (Bynner 1998) and also institutional problems in education, health, nourishment, crime violence, social division, racism and xenophobia (Klasen 1998).

Nevertheless, there is lack of clarity about the concept of social exclusion and underestimation of its consequences affecting the human being.

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THORAT'S CONCEPT OF SOCIAL EXCLUSION

According to Thorat (2007), social exclusion and discrimination are closely associated and reinforcing factors. In India, the groups such as SCs, STs, OBC and minority have been excluded from main stream development. Exclusion of these social groups has to be seen within the purview of the forceful obstruction to exercise of their rights and intrusions against the vector of freedom (Thorat, 2007). These processes of elimination are more critical than the outcome facing the marginalized social groups. Such forms of exclusion are not just worrisome, but they simultaneously, culminate in similar typologies of anti-welfare customs. Untouchability, for instance, is one of the outcomes of this process of exclusion. In India, exclusion is structured around the societal processes that exclude, discriminate, isolate and deprive some communities on the basis of their social group characteristics, for instance, caste and ethnicity. A large section of the Indian population constitutes

ch groups, which include SCs, STs, OBCs and the Muslims and others. Even after several decades of independence, there is still visible inequality among marginalized sections and (Scheduled Caste and Tribes) in human development and socio-economic variables. Over a period of time, the disparity between these two sections of society was found to be decreased but at a slower pace.

STIGMA

Stigma is an attribute or characteristic that marks a person as different from others and that extensively discredits his or her identity. Ancient Greeks coined the term stigma to describe a mark cut or burned into the body that designated the bearer as someone who was morally defective, such as a slave, criminal, or traitor. Sociologist Erving Goffman resurrected the term, defining stigma as an attribute that spoils a person's identity, reducing him or her in others' minds "from a whole and usual person to a tainted, discounted one." Stigmatizing marks are associated with negative evaluations and devaluing stereotypes. These negative evaluations and stereotypes are generally well known among members of a culture and become a basis for excluding, avoiding, and discriminating against those who possess (or are believed to possess) the stigmatizing mark. People who are closely

associated with bearers of stigma may also experience some of the negative effects of stigma, a phenomenon known as stigma by association.

Stigma does not reside in a person but in a social context. For example, within the United States, gays and lesbians are stigmatized across a range of situations, but not in a gay bar. African Americans are stigmatized in school but not on the basketball court. This contextual aspect of stigma means that even attributes that are not typically thought of as being stigmatizing may nonetheless lead to social devaluation in some social contexts (e.g., being heterosexual at a gay pride rally). Some marks, however, are so pervasively devalued in society that they cause bearers of those marks to experience stigmatization across a wide range of situations and relationships. The consequences of stigmatization are far more severe for these individuals than for those who experience stigmatization only in very limited contexts.

STIGMA TYPES AND DIMENSIONS

Goffman categorized stigmatizing marks into three major types: tribal stigma, abominations of the body, and blemishes of character. Tribal stigmas are passed from generation to generation and include membership in devalued racial, ethnic, or

religious groups. Abominations of the body are uninherited physical characteristics that are devalued, such as obesity or physical deformity. Blemishes of character are individual personality or behavioral characteristics that are devalued, such as being a child abuser or rapist.

Stigmas also differ on important dimensions, such as the extent to which they are concealable, controllable, and believed to be dangerous. These differences have important implications for how the stigmatized are treated by others, and how stigma is experienced by those who have a stigmatizing condition.

Some marks (e.g., obesity) are visible or cannot be easily concealed from others, whereas others (e.g., being a convicted felon) are not visible or can more easily be concealed. Individuals whose stigma is visible must contend with different issues than do those whose stigma is invisible. The visibly stigmatized are more likely to encounter avoidance and rejection from others than those whose stigmas are concealed. Consequently, the former may be more likely to interpret others' behavior in terms of their stigma and be more concerned with managing others' treatment of them. People whose stigmas are concealable, in contrast, have a

different set of concerns. Although they may be able to "pass" or hide their stigma from others, they may be preoccupied with figuring out the attitudes of others toward their (hidden) stigma and with managing how and when to disclose their stigma to others. They must live with the fear of others finding out about their stigma, and of being discredited. They may also have a harder time finding others like themselves to interact with, which may lead to social isolation and lowered self-esteem.

FUNCTIONS OF STIGMA

Most scholars regard stigma as socially constructed, meaning that the particular attributes or characteristics that are stigmatized are determined by society. This view is supported by evidence of variability across cultures in the attributes that are stigmatized. For example, obesity is severely stigmatized in the United States, far less so in Mexico, and is prized in some cultures. Even within the same culture, the degree to which a particular attribute is stigmatizing can change over time. For example, in the United States, being divorced was much more stigmatizing in earlier than it is today. Some commonalities exist across cultures, however, in what attributes are stigmatized.

Social stigma occurs in every society. This universality suggests that stigmatization may serve some functional value for individuals, groups, or societies. At the individual level, putting someone else down may make one feel better about oneself as an individual. At the group level, devaluing other groups may help people feel better about their own groups by comparison. At the societal level, negatively stereotyping and devaluing people who are low in social status may make their lower status seem fair and deserved, thereby legitimizing social inequalities in society. Stigmatization may also serve a fourth function. Evolutionary psychologists propose that it may have evolved among humans to avoid the dangers that accompany living with other people. Specifically, they posit that humans have developed cognitive adaptations that cause them to exclude (stigmatize) people who possess (or who are believed to possess) attributes that (a) signal they might carry parasites or other infectious diseases (such as a having a physical deformity or AIDS), (b) signal that they are a poor partner for social exchange (such as a having a criminal record), or (c) signal they are a member of an outgroup that can be exploited for one's own group's gain.

CONSEQUENCES OF STIGMA

Stigmatization has profound and wide-ranging negative effects on those who bear (or who are thought to bear) stigmatizing marks. Stigmatization has been linked to lower social status, poverty, impaired cognitive and social functioning, poorer physical health, and poorer mental health. These negative effects can occur through several pathways.

DIRECT EFFECTS

Stigma has direct negative effects on bearers by increasing their likelihood of experiencing social rejection, exclusion, prejudice, and discrimination. Research has established that the stigmatized are vulnerable to a variety of types of social rejection, such as slurs, slights, derision, avoidance, and violence. People who are stigmatized also receive poorer treatment in the workplace, educational settings, healthcare system, housing market, and criminal justice system. Stigma even has negative effects on family relationships. For example, parents are less likely to pay for the college education of their daughters who are heavy than of daughters who are thin. Discrimination can be interpersonal (e.g., when a woman is rejected by a man because of her weight) or institutional (e.g., when a woman is denied a job as a flight attendant because of institutionalized height and weight requirements).

Stigma also can have direct, negative effects on the stigmatized through the operation of expectancy confirmation processes. When people hold negative beliefs about a person because of the person's stigma (e.g., believe that someone who has been hospitalized for mental illness is dangerous), their beliefs (incorrect or correct) can lead them to behave in certain ways toward the stigmatized that are consistent with their beliefs (e.g., avoid the stigmatized, watch them suspiciously, refuse to hire them). These behaviors can cause the stigmatized to respond in ways that confirm the initial evaluation or stereotype (e.g., they get angry, hostile). This can happen without the stigmatized person even being aware that the other person (perceiver) holds negative stereotypes, and even when the perceiver is not conscious of holding negative stereotypes.

INDIRECT EFFECTS

Stigma also has indirect effects on the stigmatized by influencing how they perceive and interpret their social worlds. Virtually all members of a culture, including bearers of stigma, are aware of cultural stereotypes associated with stigma, even if they do not personally endorse them. People who are stigmatized are aware that they are devalued in the

eyes of others, know the dominant cultural stereotypes associated with their stigma, and recognize that they could be victims of discrimination. These beliefs are collective representations, in that they are typically shared by others who bear the same stigma. These collective representations influence how bearers of stigma approach and interpret situations in which they are at risk of being devalued, negatively stereotyped, or targets of discrimination. For some, their stigma may become a lens through which they interpret their social world. They may become vigilant for signs of devaluation and anticipate rejection in their social interactions.

Collective representations can have negative effects on the stigmatized by increasing their concerns that they will be negatively evaluated because of their stigma, a psychological state termed identity threat. Identity threat is not chronic, but situational; it occurs only in situations in which people are at risk of devaluation because of their stigma. When experienced, identity threat can interfere with working memory, performance, and social relationships and can increase anxiety and physiological stress responses. One form of identity threat is stereotype threat, concern that one's behavior will be interpreted in light of or confirm negative stereotypes associated with one's stigma.

Stereotype threat occurs in situations in which negative group stereotypes are relevant and may be applied to the self and can impair performance in those domains.

Collective representations can also lead bearers to experience attributional ambiguity in situations in which their stigma is relevant. Attributional ambiguity stems from bearers' awareness that they may be targets of prejudice and discrimination. As a consequence of this awareness, bearers of stigma (particularly those whose stigma is visible) who are treated negatively may be unsure whether it was caused by something about themselves (such as their performance or lack of qualifications) or was caused by prejudice and discrimination based on their stigma. Positive outcomes can also be attributionally ambiguous. As noted earlier, bearers of stigma are often exposed to inconsistent treatment and are aware of discrepancies between how the non-stigmatized feel and how they behave toward the stigmatized. As a consequence, bearers of stigma may mistrust the validity, sincerity, and diagnosticity of positive as well as negative feedback. This, in turn, can negatively affect their social relationships as well as interfere with their abilities to make accurate self-assessments.

Collective representations associated with stigma influence how bearers of stigma perceive, interpret, and interact with their social world. Through this process, stigma can have negative effects on bearers in the absence of any obvious forms of discriminatory behavior on the part of others, even if a stigmatizing mark is unknown to others, and even when no other person is present in the immediate situation.

COPING STRATEGIES

Some psychological theories describe bearers of stigma as passive victims who cannot help but devalue themselves because they are devalued by society. In fact, research shows that not all bearers of stigma are depressed, have low self-esteem, or perform poorly. Indeed, members of some stigmatized groups have higher self-esteem on average than do members of non-stigmatized groups. How bearers of stigma respond to their predicament varies tremendously. An important determinant of their response is how they cope with the threats to their identity that their stigma poses.

Bearers cope with stigmatization in a variety of ways. Some coping efforts are problem focused. For example, the stigmatized may attempt to eliminate the mark that is the source of stigmatization,

such as when an obese person goes on a diet or a stutterer enrolls in speech therapy. This strategy, of course, is not available to bearers whose stigma cannot be eliminated. Bearers may also cope by trying to avoid stigmatization, such as when a person with a concealable stigma “passes” as a member of more valued group, or an overweight person avoids going to the gym or the beach. This coping strategy may severely constrain the everyday lives of the stigmatized. The stigmatized may also cope by attempting to overcome stigma by compensating, or striving even harder in domains where they are negatively stereotyped or devalued. For example, one study showed that overweight women who believed that an interaction partner could see them (and hence believed their weight might negatively affect the interaction) compensated by behaving even more sociably compared with overweight women who thought their interaction partner could not see them. Although this strategy can be effective, it can also be exhausting, especially in the face of enormous obstacles.

Other coping strategies focus on managing the negative emotions or threats to self-esteem that stigmatization may cause. For example, the stigmatized may cope with threats to their identity by disengaging

their self-esteem from domains in which they are negatively stereotyped or fear being a target of discrimination and investing themselves more in domains in which they are less at risk. When they encounter negative treatment, another coping strategy they may use is to (often correctly) shift the blame from stable aspects of themselves (“I am stupid,” “I am unlikable”) to the prejudice of others. This strategy may protect their self-esteem from negative outcomes, especially when prejudice is blatant. Bearers of stigma may also cope by identifying or bonding with others who share their stigma. Similarly, stigmatized others can provide social support, a sense of belonging, and protect against feelings of rejection and isolation. Furthermore, bonding with others who are similarly stigmatized may also enable bearers to enact social changes that benefit their stigmatized group, as demonstrated by the success of the civil rights movement and the gay pride movement. In sum, through various coping strategies, bearers of stigma may demonstrate resilience even in the face of social devaluation.

CHILD ABUSE AND DOMESTIC VIOLENCE.

(Referred from APA and an article Child Sexual Abuse in India: Current Issues

and Research by David K. Carson & Nishi Tripathi)

CHILD ABUSE

Child sexual abuse (CSA) is a serious and widespread problem in India as it is in many parts of the world today. The trauma associated with sexual abuse can contribute to arrested development, as well as a host of psychological and emotional disorders, that some children and adolescents may never overcome. When sexual abuse goes unreported and children are not given the protective and therapeutic assistance they need, they are left to suffer in silence.

Child maltreatment in India is a pervasive problem that often results in immediate negative effects on children, followed by the potential for numerous problems throughout the lifespan. Research has documented that child sexual abuse (CSA) may hinder proper growth and development (Cicchetti and Toth 2006; Foster 2011; Goodman et al. 2010) and place children at risk for a host of mental health disorders, including but not limited to: anxiety, depression, anger, cognitive distortions, posttraumatic stress, dissociation, identity disturbance, affect dysregulation, interpersonal problems,

substance abuse, self-mutilation, bulimia, unsafe or dysfunctional sexual behavior, somatization, aggression, suicidality, and personality disorders. Hence, the experience of CSA can have a profound influence on a child's functioning.

Common sequelae for adult survivors of CSA include: mental health problems (e.g., depression, anxiety, substance abuse, posttraumatic stress), relational challenges (e.g., sexual health, intimacy, and increased risk for sexual assault and domestic violence), and spiritual concerns (e.g., shattered assumptions about life, people, and self, as well as changing belief systems, following the trauma) (Chawla 2004; Davidson et al. 2009; Deb and Sen 2005). However, some adult survivors of CSA demonstrate resiliency and posttraumatic growth (Wright et al. 2007). Hence, healing and change are possible.

KEY TERMINOLOGIES:

Child sexual abuse (CSA) is defined as the misuse of power and authority, combined with force or coercion, which leads to the exploitation of children in situations where adults, or children sufficiently older than the victim to have greater strength and power, seek sexual gratification through those who are developmentally immature, and where, as

a result, consent from the victim is a non-concept. Such gratification can involve explicit sexual acts, or may involve invasive and inappropriate actions not directly involving contact (Miller et al. 2007).

Child sexual exploitation can involve the following: possession, manufacture and distribution of child pornography; online enticement of children for sexual acts; child prostitution; child sex tourism; and child sexual molestation.

Trauma is defined as “the realization of one’s worst fears, the experiences that every human being would never want to have”(Klempner 2000,p.77).

Grooming is defined as methods used by perpetrators to earn trust and keep children involved in sexual acts. Common strategies for such manipulation include giving the victim gifts or special treatment or privileges, which is often a confusing experience for the child victim (Lanktree and Briere 2008). Abusers gain access to their child victims and attain their trust through the giving of special attention and time. Perpetrators often trick or deceive the child and others in order to ensure that the abuse is kept secret.

Re-victimization is that which places a person who was sexually abused as a minor at greater risk for further abuse in adulthood. Re-victimization may occur in

the form of unwanted sexual contact, physical abuse, and psychological maltreatment.

Sexual Assault is a class of sexual conduct prohibited by the law that includes forcible sex offenses such as rape and sodomy of a perpetrator toward or upon a victim. The victim may be a minor or an adult.

OVERVIEW OF CHILD SEXUAL ABUSE IN INDIA

Estimated Incidence Rates of Child Sexual Abuse India has a large child population that is vulnerable to all types of abuse, neglect and exploitation (Chawla 2004;Deb2005, 2009; Priyabadini 2007). According to Deb (2002,2009), Deb and Mukherjee (2009), and Iravani (2011), child sexual abuse (CSA) in India has been an age-old and deep-rooted social problem, and child trafficking for commercial sexual abuse has become a serious issue for policy makers. Of the total population in modern day India, about 44.4 % are under 18 years of age (children and adolescents), and one in every two children is deprived in terms of not receiving primary education, adequate nutrition and medical care (National Family Health Survey 2005–2006). Presently there is a dearth of information about the extent

of CSA in India with the exception of a few recent studies. However, there appears to be a gross under-reporting of crimes against children in India (as in the United States and other countries), including various types of child sexual abuse. Thus, there is also a general consensus that the problem of child abuse is much more prevalent than what is commonly understood or acknowledged (Chawla 2004; Deb and Mukherjee 2009).

Researchers in India estimate that between 18 % and 50 % of their country's population may have experienced some type of sexual abuse in their life time (Chatterjee et al. 2006; Chawla

2004; Deb 2006, 2009; Deb and Mukherjee 2009; Deb and Walsh 2012). These statistics may not account for the number of children (1 in 5) who are sexually solicited while using the internet, and the high number of victims who never disclose their sexual abuse from in and outside the family.

Children who fail to disclose may be between 30 % and 87 % (Deb 2005, 2009; Deb and Mukherjee 2009, 2011). There is additional empirical evidence which supports the assertion that incidences of CSA in India are high. In a

recent study, Deb and Walsh (2012) found, for example, that of 160 boys and 160 girls who were randomly selected from Grades 8 and 9 in school in the state of Tripura an average of 18 % of the children had experienced sexual abuse in the home environment. Girls reported higher incidences of sexual abuse than boys, whereas boys were more likely to have experienced physical and psychological abuse in the home. Overall social adjustment scores for girls were significantly lower than those for boys.

According to Iravani's (2011) examination of studies of CSA in India based on lengthy interviews with adults, approximately 30 % of men and 40 % of women remember having been sexually molested during childhood, with "molestation" defined as actual genital contact and not just exposure. This researcher noted that about half of these incidences were directly incestuous with family members (although with the knowledge or complicity of other caretakers in at least 80 % of the cases) and the other half occurred with perpetrators outside the immediate or extended family. Other studies examined in this article support these high incidences of CSA.

Iravani (2011) concluded that:

“These experiences of seduction are not just pieced together from fragmentary memories, but are remembered in detail, are usually for an extended period of time and have been confirmed by follow-up reliability studies in 83 % of the cases, so they are unlikely to have been fantasies. These seductions occurred at much earlier ages than had been previously assumed, with 81 % occurring before puberty and an astonishing 42 % under age 7 (p. 151).”

SOCIO-CULTURAL AND FAMILY FACTORS

Involved in Child Sexual Abuse in India. The most significant challenges to addressing all types of child abuse and neglect (CAN) in India include overpopulation that involves poor service delivery for children and families, poverty, illiteracy, abandonment of children, underreporting of CAN, and cultural beliefs and practices pertaining to parental rights and styles. These include parents believing that children are their personal property, and that the rights and choices of children solely belong to the parents (Deb 2009; Deb and Mukherjee 2009). Deb (2005) and Deb and Mukherjee (2009) also note that parents and/or close relatives are the most common perpetrators of CAN, which includes child sexual abuse (Virani 2000). Girl children, who occupy a

lower status in the family and society, are particularly vulnerable to CAN, including sexually abusive acts (Chawla 2004; Deb and Mukherjee 2009). Further, girls in India, especially in rural areas, are discriminated against in terms of education, nutrition, and medical care, are more likely to experience infanticide, and are often treated as more of a burden to the family (Deb 2006; India Country Report on Violence against Children 2005). In addition, boy children are typically valued and preferred in Indian families, and boy children often reap the better fruits of what parents have to offer. All of these factors put girl children especially at greater risk for child sexual abuse and exploitation.

Another socio-cultural factor in child sexual abuse is family secrecy. In India the business of the family stays in the family, especially with regard to any actions that are considered inappropriate or taboo (Choudhury 2006). This is because in India there are cultural elements of blame and shame (including in family systems), and families will go to great lengths to protect the reputation of the family in the community (Baradha 2006; Choudhury 2006). It is also not unusual for children to be blamed for their own abuse because the rights and statements of adults tend to trump those of children (Baradha 2006; Priyabadini

2007). Moreover, since the child's identity is rooted in the family's identity and standing in the community, anything that would embarrass the family or tarnish their good name is kept private –in some cases even from other immediate or extended family members (Patnaik 2007; Priyabadini 2007). This practice of secrecy only serves to protect the sexual perpetrator and allows the cycle of abuse to continue (Baradha 2006; Patnaik 2007). In addition, the parents or caregiver's refusal to believe the child victim about the sexual abuse or cover it up further exacerbates the child's distress (i.e., betrayal trauma) and prevents her or him from getting therapeutic help when needed (Priyabadini 2007).

According to Kacker and Kumar (2008), traditionally the care and protection of children in India has been the responsibility of families and communities. They may be correct in their observation that a strongly knit patriarchal family system has seldom held the belief that children are individuals with their own rights. These authors note that even though the Constitution of India guarantees many fundamental rights to children, these rights are more needs based than rights based, and the government has the challenging task of implementing constitutional and statutory provisions for children. Hence,

“..... with an increasing incidence of child abuse, India needs both legislation and large scale interventions to address this problem”(2008, p. 98). Widespread public education about child sexual abuse and exploitation is also sorely needed, especially in Indian schools and families (Deb and Mukherjee 2009; Priyabadini 2007). The protection of children against all forms of child abuse and exploitation needs to be a chief priority at the local, state and national level, and current laws need to be enforced when children and adolescents become victims of a perpetrator's acts, including perpetrators being prosecuted to the full extent of the law.

THE CHILD ABUSE AND PREVENTION TREATMENT ACT

The Child Abuse and Prevention Treatment Act defines child abuse and neglect or child maltreatment as:

Any recent act or failure to act on the part of a parent or caretaker, which results in death, serious physical or emotional harm, sexual abuse or exploitation, or an act or failure to act which presents an imminent risk of serious harm.

Neglect is a failure to meet the child's basic needs, e.g., not providing enough food, shelter or basic supervision, necessary medical or mental health treatment, adequate education or emotional comfort.

Physical abuse refers to the injury of a child on purpose, e.g., striking, kicking, beating, biting or any action that leads to physical injury.

Sexual abuse is the use, persuasion or forcing of a child to engage in sexual acts or imitation of such acts.

Abuse and neglect of children occurs in families from all walks of life, of all incomes, religions and ethnicities. There is no single cause of child maltreatment; rather, it occurs as a result of many forces working together to impact the family.

PARENT OR CAREGIVER RISK FACTORS

Low self-esteem, poor impulse control, depression, anxiety or antisocial behavior.

Experiencing or witnessing violence as a child, which teaches violent behavior or justifies it as proper behavior.

Substance abuse, which interferes with mental functioning, judgment, self-control, ability to be protective of one's child and making the child's needs a priority.

Lack of knowledge about normal child development and unrealistic expectations, frustration and/or inappropriate methods of discipline.

FAMILY RISK FACTORS

Children living with single parents are more likely to live in poverty with fewer social supports, which may contribute to stress and increase risks of maltreatment.

Children in violent homes may witness intimate partner violence, may be victims of physical abuse themselves and may be neglected by parents or caregivers who are focused on their partners or unresponsive to their children due to their own fears.

Stressful life events, parenting stress and emotional distress (e.g., losing a job, physical illness, marital problems or the death of a family member) may worsen hostility, anxiety or depression among family members and increase the level of family conflict and maltreatment.

Maltreating parents or caregivers are less supportive, affectionate, playful and responsive with their children and are more likely to use harsh discipline and verbal aggression than positive parenting strategies (e.g., using time

outs, reasoning, and recognizing and encouraging the child's successes).

CHILD RISK FACTORS

Infants and young children, because they are small and need constant care, are more likely to experience certain forms of maltreatment such as being shaken by parents or caregivers frustrated or overwhelmed by persistent crying. Teenagers, on the other hand, are at greater risk for sexual abuse.

Children with physical, cognitive and emotional disabilities or chronic illnesses may be at greater risk of maltreatment. Parents or caregivers of children with disabilities are more likely to experience high levels of stress, depression and anger. Children with disabilities may not understand that abusive behaviors are inappropriate and are unable to defend themselves.

Aggression, attention deficits, difficult temperaments and behavior problems in children have been associated with increased risk for maltreatment, especially when parents have poor coping skills, are unable to empathize with the child or have difficulty controlling emotions. Maltreatment often exacerbates the problem. A

physically abused child may develop aggressive behaviors that lead to recurring maltreatment.

ENVIRONMENTAL RISK FACTORS

The vast majority of parents or caregivers who live in these types of environments are not abusive. However, these stresses can increase the risk of abuse for some:

- Poverty and unemployment can increase the likelihood of maltreatment, especially in combination with family stress, depression, substance abuse and social isolation.
- Parents with less material and emotional support and who do not have positive parenting role models feel less pressure to conform to conventional standards of parenting behaviors.
- Children living in dangerous neighborhoods are at higher risk than children from safer neighborhoods for severe neglect, physical abuse and sexual abuse. It is possible that violence may seem an acceptable response or behavior to individuals who witness it more frequently.

Child abuse and neglect can result in physical and psychological developmental delays. A neglectful mother may not feed

her baby properly, which can slow brain development, or an emotionally abusive father may damage his child's ability to form trusting relationships. Abused or neglected children can see the world as an unstable, frightening and dangerous place, which can undermine their sense of self-worth and their ability to cope with and adapt to their environments as they grow up. If unaddressed, maltreatment may contribute to later problems, such as alcoholism/substance abuse, depression, domestic violence, multiple sexual partners and exposure to sexually transmitted diseases, suicidal thoughts and attempts.

The impact of abuse can vary depending on:

- Age and developmental status of the child when the abuse or neglect occurred,
- Type of abuse (physical abuse, neglect, sexual abuse, etc.),
- Frequency, duration and severity of abuse,
- Relationship between the victim and his or her abuser.

WHAT FACTORS PROTECT A CHILD FROM RISK OF ABUSE OR NEGLECT?

Children's optimism, high self-esteem, intelligence, creativity, humor and independence, which enhance their coping skills in the face of adversity.

The acceptance of peers and positive influences such as teachers, mentors and role models

The family's access to social supports, neighborhood stability and access to safe schools and adequate health care

The child's experience of love, acceptance, positive guidance and protection from a caring adult, which encourages trust that their parents or caregivers will provide what they need to thrive

Parent's or caregiver's respectful communication and listening, consistent rules and expectations, and safe opportunities that promote independence

Parents or caregivers who can cope with the stresses of everyday life and have the inner strength to bounce back when things are not going well.

Parents or caregivers with a social network of emotionally supportive friends, family and neighbors

Families who can meet their own basic needs for food, clothing, housing and transportation and know how to access essential services such as childcare, health care and mental health services

WHAT PREVENTIONS ARE THERE FOR CHILD MALTREATMENT?

Psychologists are heavily involved in the development and implementation of prevention programs for child abuse and neglect. Primary prevention programs raise awareness among the public, service providers and policymakers about the scope of issues involved in child maltreatment. Secondary prevention programs target populations with one or more risk factors for child maltreatment.

Tertiary prevention programs target families where maltreatment has already taken place and aim to reduce the impact and prevent it from reoccurring.

CHALLENGES FOR COUNSELORS IN THE TREATMENT PROCESS

There are two major goals in working with child victims of sexual abuse (Anderson and Hiersteiner 2008; Briere and Scott 2006). First is to help the victim express and work through her/his emotions regarding the abuse, including about and toward the perpetrator, in the here and now. This is a long-term process for many victims of CSA. The second goal is to help the child or adolescent move from a sense of victim to survivor to victor; i.e., the Resilient Self –characteristics that include: independence, connectedness, creativity, insight, play and humor, morality, self-regulation, initiative, and spirituality.

For victims of CSA, processing and working through their sexual abuse is often an extremely difficult task (Oz 2005). Common challenges that counselors need to prepare for when the counseling process begins are: (a) increased symptomatology for some period of time in treatment, (b) non-linear, slowed or halted progress, and (c) drop out. Counselors are encouraged to confront these challenges through providing adequate levels of challenge and support for the child and the child's family. Moreover, exposure to traumatic memories and content before the child is ready can be damaging to the child's well-being and therapeutic relationship. The concept of the therapeutic window helps avoid both re-traumatization and failure to move towards recovery (Briere and Scott 2006). Therefore, when implementing any therapy with sexually abused children, therapists must determine whether the child is ready for trauma treatment or initially needs to be stabilized and made to feel safe following the abuse (and establish a safe place they can learn to go to in their thoughts and feelings). It is important to remember that children often experience some ambivalence between having a desire to protect the secret of CSA as well as unburden their story to a safe and caring person (Crenshaw and Hardy 2007). In order to begin the unburdening process,

children must feel secure, supported, and believed about the abuse. Therapy begins with establishing a therapeutic relationship between the child and counselor. Developing a relationship with children who have experienced extreme trauma, including sexual abuse, has been described as “a harrowing feat” (Crenshaw and Hardy 2007) and the counselor’s role as one of an “empathetic witness of injustice” (Kaminer 2006, p. 488). Due to the nature of CSA, trust is a central issue. Many abused children tend to respond to others with either blind trust (that does not distinguish between safe and unsafe people) or an inability to trust anyone in any circumstance. Other victims of CSA fear that the counselor will betray their trust and they will be harmed again. An important component to the healing process is for children to learn how to trust others again, a process which begins in the counseling relationship and continues over time (Kaminer 2006). Failure to establish a safe, trusting relationship often leads to the failure of any method or technique employed since the efficacy of counseling is directly related to the therapeutic relationship (Gil 2006).

During the trauma narrative process (the child telling or sharing her/his abuse story), the counselor works closely with the child to help them recall, write about, and

process their experience (Foster 2011). Children may initially fear recalling their trauma, believing that the remembering will lead to an unbearable reliving of the events. It is important to help the child distinguish between a memory (past events that are gone and not operating in the present) and the here-and-now and to make sure that the child understands this difference. When fears are expressed, it is helpful for the counselor to explain the rationale of the trauma narrative and what the counselor will do if symptoms arise. It is also important to assure the child that they will work at his/her own pace. At this stage, it is vital for the counselor to be an empathetic, nonjudgmental listener as the child becomes ready to tell his or her story (Kaminer 2006). When children share trauma in the form of the narrative, they are actively involved in the process of moving towards closure.

Closure is defined as the survivor becoming free from habitually thinking about the trauma in such a way that causes distress (Klempner 2000). During this process, children seek to understand their trauma and its impact, which may involve addressing why the trauma happened to them and understand that in no way is anything their fault. It also involves exploration of the ways in which the experience has changed their view of

self, others, and the world. Children (when they are developmentally capable) can explore and discover personal meanings within the traumatic experience. The act of making meaning out of one's trauma (e.g., "that which does not kill me makes me stronger") often helps children attain some level of closure (Briere and Lanktree 2008). Integrating the traumatic experience into one's life is the last portion of trauma recovery. For children, the ability to adapt and move forward often lies in their courage to face their pain and process the emotional impact of the abuse on them, while at the same time learning new ways of coping with life. It is important to remember that children need ample time to successfully complete treatment.

Finally, involvement of supportive parents or caregivers in treatment is recommended for children who have experienced sexual abuse (Lanktree and Briere 2008). This improves treatment outcomes for children (Cohen and Mannarino 2008) and helps promote positive family relationships (McPherson et al. 2012). One of the major goals is to increase parents' and caregivers' ability to talk openly about the trauma with their child (Cohen and Mannarino 2008). Many adults have difficulty talking about sexual abuse, which often leaves children feeling isolated

and alone. Adults may also fear that openly talking about the abuse will re-traumatize the child, and therefore they avoid the topic altogether (Ogawa 2004). Children are aware of whether or not the abuse can be talked about openly with their parents, and they too may avoid the topic out of fear that it will make their parents sad or angry. However, bringing the trauma out in the open and helping the child express her/his thoughts and deep inner feelings helps "demystify" the experience of CSA for them and emotionally work through the trauma. Children also need to know what to expect in sessions, what their role is and the role of the counselor. Parents and caregivers should also be informed of the potential increases in symptoms and decline in functioning for some period during treatment with some children and how to respond to their child during these sensitive periods.

DOMESTIC VIOLENCE

(Referred Article: Domestic Violence, Psychological Trauma and Mental Health of Women: A View from India an article by L. N. Suman)

The recognition of domestic violence as a serious social problem is an achievement of second wave feminism originating in the late 1960s. Feminist activists provided

safety and shelter for women and children escaping violence, and located the roots of domestic violence in gender inequality in social relationships.

With more than half a billion-female population, India has the second largest population of females in the world. This immense resource needs to be provided safety, security and nurturance for society, and thereby the nation, to benefit in multitude ways. It goes without saying that at first the girl child has to be given a chance to be born and subsequently, not only accepted, but cherished. This requires an attitudinal shift in the way the girl child is perceived and treated in the patriarchal Indian society. Several social ills that plague Indian society target the girl child and women. These include female feticide, neglect of the girl child, sexual abuse of the girl child, trafficking of girls and women, dowry harassment and domestic violence. Such violence against girls and women impairs their physical and psychological health which necessitates health professionals to be sensitive to such issues. Health professionals, including mental health professionals, need to be made aware of the underlying psychological trauma in order to ensure adequate and appropriate interventions. The government of India has passed legislations to punish abuse and violence

against girls and women by treating them as crimes. Domestic violence against women in India is especially significant in terms of prevalence and consequences. According to the Domestic Violence Act of India, domestic violence includes physical violence, emotional abuse and economic abuse. The National Family Health Survey of India for the period 2005-2006 examined the prevalence of domestic violence among 83,703 women in the age range of 15 to 49 years. Results revealed that 34% of the women had experienced domestic violence and that domestic violence was higher among women who had low educational and economic status. Physical violence was more prevalent than sexual and emotional abuse with one third of the victimized women reporting cuts, bruises, burns, dislocations, deep wounds, broken teeth and bones. However, 75% of them did not seek help to end the violence and the data also suggested that neither education nor wealth implied a greater likelihood that women would seek help against violence. Domestic violence, especially physical violence, has been found to be strongly associated with poor mental health among women and it has also been implicated as an important factor in women seeking safety in shelter homes. In a review of studies on domestic violence in India, reported that sexual violence by

husbands led to higher odds of women reporting gynecologic symptoms like blood after intercourse, vaginal discharge, burning micturition, pain during intercourse, and symptoms suggestive of sexually transmitted infections. Further, the risk for infant mortality was 36% higher among mothers who experienced domestic violence compared with mothers who did not. In this context, it should be noted that alcohol abuse has been found to be a significant factor among men who indulge in domestic violence. Interpersonal victimization of girls and women significantly impacts their ability to cope and can result in symptoms of psychological trauma such as dissociation, somatization, intrusive thoughts and images of the traumatic events, self-harm, and social withdrawal. Further, re-traumatization and multiple traumatization contribute significantly to the risk of Post-Traumatic Stress Disorder (PTSD). However, Cook and Newman reported that mental health practitioners lack evidence-based knowledge, assessment and psychotherapy skills to effectively treat trauma survivors. They recommended training for psychologists in ‘Trauma Psychology’ to obtain competencies in psychosocial trauma focused assessment, trauma focused psychosocial interventions and trauma informed professionalism apart from obtaining scientific knowledge and

cross cutting competencies to understand trauma reactions. A similar situation exists in India with little awareness about psychological trauma and its manifestations among counselors. Hence, it is important to train counsellors and other mental health professionals in trauma informed care. It was with this aim that a ‘Trauma Recovery Clinic’ was started in 2014 at NIMHANS urban well-being centre. The objectives of the clinic are to enable survivors to come to terms with the trauma and reclaim their lives by reducing emotional symptoms of trauma, enhancing self-esteem and resilience, improving social support and preventing re-traumatization. The clinic also conducts awareness and capacity building programmes for psychologists and counsellors. However, several such clinics are required in India to bring about a noticeable difference in the quality of life of girls and women in the country and ensure a life of freedom, empowerment and dignity for them.

DOMESTIC VIOLENCE VS INTIMATE PARTNER VIOLENCE

Intimate partner violence (IPV) is defined as any behavior within an intimate relationship (married, unmarried, and live-in) that causes physical, psychological, or sexual harm to those in that relationship.

This definition encompasses physical, sexual, and psychological aggression/abuse or controlling behavior of any kind. IPV differs conceptually from domestic violence. Domestic violence is defined as the physical, sexual, and emotional maltreatment of one family member by another. It typically includes all types of family violence such as elder abuse, child abuse, and marital rape; however, IPV is limited to acts of aggression between intimate partners. As females are more likely to be hurt in cases of IPV, male-to-female partner violence has been studied in much greater detail, though both male-to-female and female-to-male partner violence exist; and the rates of female-to-male partner violence equal or exceed male-to-female partner violence; in general household population. Almost 50% of IPV are bidirectional, and the rest divided between male-to-female only and female-to-male only partner violence.

One in three ever-partnered women worldwide has experienced physical and/or sexual violence by an intimate partner. Of these women, 42% sustained immediate physical injuries, and 13% were fatally injured. Nearly 13%–61% of women have experienced physical violence by a partner sometime in their lifetime. About 4%–49% have experienced severe physical violence, and 6%–59%

have experienced sexual violence. The incidence across South Asia ranges between 8% and 50%, and almost 31% of Indian women have experienced IPV at some point in their marital life.

PEACE PSYCHOLOGY: VIOLENCE, NON-VIOLENCE, CONFLICT RESOLUTION AT MACRO LEVEL, ROLE OF MEDIA IN CONFLICT RESOLUTION.

PEACE PSYCHOLOGY:

Psychologists have been interested in war and peace since the beginning of modern psychology. Though philosophizing about peace has ancient roots, the discipline of peace psychology essentially began soon after the establishment of modern scientific psychology with the 1910 essay by William James, “The Moral Equivalent of War.”

William James, the first peace psychologist, was a most distinguished scholar and also an insistent public voice on issues of war and peace. He, in a speech at Stanford University in 1906, coined the phrase “the moral equivalent of war” (James, 1910/1995, p. 22). James argued that war provides human beings with opportunities to express their spiritual inclinations toward self-sacrifice and

personal honor; consequently, to end war, societies must find alternative “moral equivalents” for the expression of these profoundly important human values.

Peace can be defined as a positive societal state in which violence, whether direct or structural, is not a likely occurrence, and in which all humans, animals, and ecology are treated with fairness, dignity, and respect. Peace psychology is the study of mental processes and behavior that lead to violence, prevent violence, and facilitate nonviolence as well as promoting fairness, respect, and dignity for all, for the purpose of making violence a less likely occurrence and helping to heal its psychological effects.

The Division of Peace Psychology (Division 48), which was established in 1991, shares with the American Psychological Association (APA) a commitment to promoting human well-being. The goals of peace psychology are to “increase and apply psychological knowledge in the pursuit of peace . . . [including] both the absence of destructive conflict and the creation of positive social conditions which minimize destructiveness and promote human well-being”

By the close of World War II, 13 well-known American psychologists (including

Gordon Allport, Edna Heidbreder, Ernest Hilgard, Otto Klineberg, Rensis Likert, and Edward Tolman) circulated a “Psychologists’ Manifesto: Human Nature and the Peace: A Statement by Psychologists” (reprinted in Jacobs, 1989, and in Murphy, 1945), which was signed by almost 4,000 psychologists (M. B. Smith, 1999). The Manifesto argued that “war can be avoided: War is built, not born” and urged lawmakers to work toward peace with attention to “the root desires of the common people of all lands”

Direct violence comes from individuals who hurt others in acute and discrete incidents. In a war, there will be a multitude of such incidents. Structural violence is inflicted on people through the structures of society and involves chronic forms of harm such as poverty, environmental damage, misallocated resources, or dangerous working conditions.

Structural violence is embedded in the values, social norms, laws, social structures, and procedures within a society or community which systematically disadvantage certain individuals and groups so that they are poorer, sicker, less educated, and more harmed than those who are not disadvantaged.

‘Positive’ peace denotes the simultaneous presence of many desirable states of mind

people relate those perspectives to each other within some coherent framework.

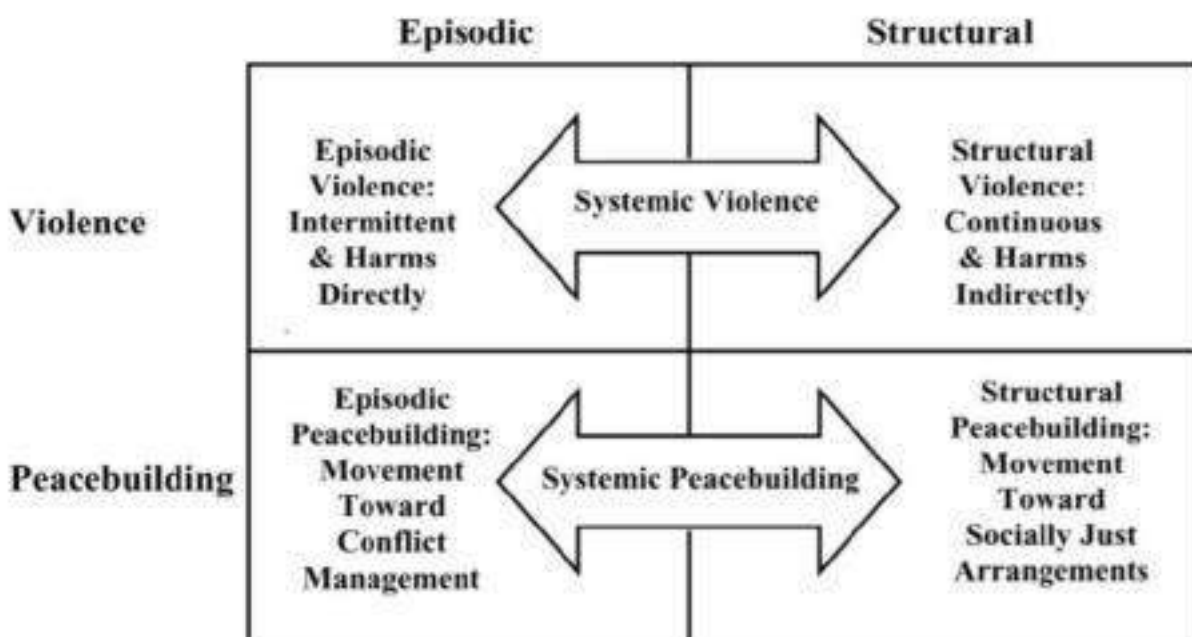


Fig. 1. Systems perspective on violence and peacebuilding.

and society, such as harmony, justice, equity, etc. ‘Negative’ peace has historically denoted the ‘absence of war’ and other forms of wide-scale violent human conflict.

IMPORTANT TERMINOLOGIES

INTEGRATIVE COMPLEXITY:

One concept for understanding the decision-making process for wars is integrative complexity. This construct has two features: differentiation or the degree to which people see differences among perspectives on a particular problem; and integration, which is the degree to which

Leaders who take an oversimplified, inflexible approach are more likely to escalate conflicts. More flexible leaders, able to understand the other side’s perspectives, are less likely to get into a war. Content analysis of public speeches and similar documents before various wars and conflicts shows that a fall in integrative complexity scores is a good predictor of whether war will be the outcome (Conway, Suedfeld, & Tetlock, 2001).

- **MECHANISMS OF MORAL DISENGAGEMENT:** Albert Bandura and colleagues argue that the most inhumane behavior occurs when original ideas of moral conduct are disengaged. People

disengage by changing how they think about the act, comparing it to worse conduct, scapegoating, deferring to authority, discounting the effects, or discounting the victims.

COMPARTMENTALIZING:It is related—putting different parts of life into different compartments, sealed off from one another. The officers ordering a massacre may go to church on Sunday and profess beliefs contrary to their actions. They are not thinking of their actions when they make those statements.

INTELLECTUALIZING:It refers to the focus on reasoning that allows for violence, avoiding the accompanying negative emotions. Discounting the victim is described in detail by Brennan (1995) who gives various categories of “linguistic warfare” used to facilitate violence: describing people as deficient humans, nonhumans, nonpersons, animals, parasites, diseases, inanimate objects, or waste products. He offers an array of quotations from throughout history to depict these attitudes aimed at vulnerable groups.

THEORIES:

A range of theories has focused on emotions and drives as bases for violent behavior.

Frustration–Aggression Hypothesis:

One of the earliest proposed explanations of aggression was the Frustration–Aggression Hypothesis, which holds that when people are thwarted from attaining an expected goal, aggression follows (Dollard, Doob, Miller, Mowrer, & Sears, 1939). Many but not all laboratory experiments showed frustration leading to greater aggression.

Catharsis: Freud believed that if anger builds up, it can explode into poorly timed, inappropriate action. If anger were aimed at an inanimate or symbolic target instead, it dissipates. The buildup subsides. He termed this process “catharsis.” This hypothesis has been tested extensively for decades.

Cognitive Dissonance and Effort

Justification: The theory of cognitive dissonance is that people find it stressful to have two different ideas in conflict with each other. To reduce this tension, they might take actions or make arguments that seem illogical. Suggestions of a contradiction are then met with belligerency. For wars, this can include effort justification—the belief that more effort is required to protect and justify the effort already made. If one must otherwise admit all the effort was wasted, then the effort must be continued. The continuation

of the American war in Vietnam for several years is one of the most cited examples.

Realistic Group Conflict Theory: It posited that hostility is likely to occur when groups are in competition for scarce resources (Campbell, 1965; Sherif&Sherif, 1953). Realistic group conflict is likely to increase as resources grow scarcer and environmental problems become more pressing (Winter, 2000; Winter & Koger, 2004). Depletion of natural resources such as clean water, arable land, and precious minerals, combined with population growth and displacement, will increase dramatically in the next decades. During such conflicts, attributional biases can cascade into scapegoating, antagonistic ideologies, and blaming.

RELATIVE DEPRIVATION THEORY:

As defined by social theorists and political scientists, Relative Deprivation Theory suggests that people who feel they are being deprived of almost anything considered essential in their society—whether money, rights, political voice or status—will organize or join social movements dedicated to obtaining the things of which they feel deprived. For example, relative deprivation has been cited as one of the causes of the U.S. Civil

Rights Movement of the 1960s, the struggle of Blacks to gain social and legal equality with whites. Similarly, many gay people join the same-sex marriage movement in order to acquire the same legal recognition of their marriages enjoyed by straight people.

In some cases, relative deprivation has been cited as a factor driving incidents of social disorder like rioting, looting, terrorism, and civil wars. In this nature, social movements and their associated disorderly acts can often be attributed to the grievances of people who feel they are being denied resources to which they are entitled.

Development of the concept of relative deprivation is often attributed to American sociologist Robert K. Merton, whose study of American soldiers during World War II revealed that soldiers in the Military Police were far less satisfied with their opportunities for promotion than regular GIs.

- ABSOLUTE DEPRIVATION THEORY:

Relative and absolute deprivation are measures of poverty in a given country. Absolute deprivation describes a condition at which household income falls below a level needed to maintain the basic

necessities of life like food and shelter. Relative deprivation describes a level of poverty at which household income drops to a certain percentage below the country's median income. For example, a country's level of relative poverty could be set at 50 percent of its median income. While absolute poverty can threaten one's very survival, relative poverty is more likely to limit one's ability to participate fully in their society.

In 2015, the World Bank Group set the worldwide absolute poverty level at \$1.90 a day per person based on purchasing power parities (PPP) rates.

CONFLICT RESOLUTION AT MACRO LEVEL:

Conflict management broadly to refer to efforts that prevent violent episodes by containing differences in views (conflict management) or by reaching an agreement (conflict resolution).

One of the earliest and most important insights about conflict resolution is that it often requires finding "integrative solutions," that is, outcomes that satisfy the needs of all parties (Follett, 1924). However, conflicts often arise out of negative interdependence between parties that have mixed motives with both competitive and cooperative dimensions

(Deutsch, 1973, 1985). Competitive relations are fraught with impaired communication, suspicion, criticism, disagreement, power plays, coercion, and the belief that solutions benefit one party but not all parties. However, competitive features of a relationship can be transformed by emphasizing cooperation, which is characterized by effective communication, friendliness, helpfulness, coordination of effort, shared values and beliefs, willingness to enhance the other's power, and the viewpoint that conflict is a mutual problem to be solved.

Various conflict resolution techniques used to reduce tension and conflict include negotiation, mediation, arbitration, diplomacy, interactive problem solving, cooperation on superordinate goals, and unilateral initiatives.

Approaches to conflict resolution tend to fall within two broad categories:

1. INTEREST BASED

The interest-based approach was pioneered by Fisher and Ury (1981) in their best-selling and highly influential book on negotiation strategies, *Getting to Yes*.

This technique has helped negotiators by encouraging intergroup empathy and mutual understanding, separating thinking

about players' personalities from the problem, avoiding criticism of the parties by each other while critically analyzing the issue, inventing options that yield mutual gains, and using objective criteria to judge whether proposed agreements satisfy everyone's interests. This "principled negotiation" tends to yield creative options, positional flexibility, mutually satisfying (win– win) solutions, and improved relationships through mutual learning and problem solving.

2. NEEDS BASED

needs-based approach called interactive problem solving (IPS) has been applied in a large variety of settings marked by violent episodes. Pioneered by Kelman and Fisher, IPS uses unofficial representatives of groups or states, who come together for several days to engage in dialogue and problem solving. The facilitators of these workshops are typically academics or other well-regarded citizens. IPS is designed to promote greater mutual understanding between parties, stronger intergroup relationships, new perspectives on old problems, and a loosening of entrenched and polarized positions. Agreements are nonbinding because participants do not officially represent their governments; but because participants are widely respected,

workshops not only induce change in the individuals participating but serve as a catalyst for change in wider political communities. IPS has been applied in many conflicts, including those in Northern Ireland (Hall, 1999), Cyprus (Fisher, 2001b), Israel/Palestine (Kelman, 1995; Rouhana&Kelman, 1994), Argentina (Mitchell, 2000), Sri Lanka (Hicks & Weisberg, 2001), and the Horn of Africa (Beyna, Lund, Stacks, Tuthill, &Vondal, 2001).

- Role of media in conflict resolution:

The successive evolution in communication technology has significantly altered the conduct of conflicts, warfare and conflict resolution in the world today. Compared to people of earlier ages, people around the world today know much more and much sooner about major developments in international relations. Similarly, there is millions of global news networks that broadcast live from all corners of the world, and via the internet, providing immediate access to unfolding events and under certain conditions, influencing the way those events develop and end. As a result, this has prompted a general feeling that media coverage, especially television, has had an increased influence on conflict management.

Triggered by rapid technological change in the media, information no longer simply describes on-scene developments, but increasingly shapes the dynamics by influencing perceptions, opinions and loyalties (Jakobsen, 2000; Galtung, 2004). This has made information to be an important means of warfare used by governments, military and leaders in all ages everywhere for propaganda or disinformation purposes or for great potential to promote peace building.

- MEDIA INFLUENCE

History has shown that media can incite people towards violence. Hitler used the media to create hatred for Jews (Vladimir & Schirch, 2007). Joseph Goebbels, Hitler's Minister of Propaganda, masterminded the most effective effort of mass persuasion, heavily relying on propaganda messages in motion pictures and radio broadcasting (Jowett & O'Donnell, 1999).

Apart from the negative roles of media, several cases have also linked media to the promotion of peace and reconciliation. For example, in Northern Uganda where the Lord Resistance Army (LRA) has camped for decades, media has been used to create the common good for the people to promote peace. Struges (2007) found that

Mega FM has promoted peace in Northern Uganda with positive effects since 2002.

Conflict is an inevitable part of the society today, the ability to manage it well is of importance as far as development is concerned. Conflict in it right is not evil but its management if not addressed with the keenest of attention can destroy developmental process. The media has a role to play in conflict and should be guided by the core ethics of its functions to handle conflict situations in the better of interests.

IMPORTANT TERMS

Hate speech - Is a form of speech that degrades others and promotes hatred and encourages violence against a group on the basis of a criteria including religion, race, color or ethnicity

Media - Refers to the several media or channels used in an organized manner to communicate information to groups of people, as a service to the public. It is divided into print media and electronic media.

Peacekeeping - A technique designed to preserve the peace, however fragile, where fighting has been halted, and to assist in implementing agreements achieved by the peacemakers.

Peace building - Involves a range of measures targeted to reduce the risk of lapsing or relapsing into conflict by strengthening national capacities at all levels for conflict management, and to lay the foundation for sustainable peace and development.

Peacemaking - Include measures to address conflicts in the progress and involves diplomatic action to bring hostile parties to a negotiated agreement.

Peace Enforcement - Application of military force or the threat of its use, normally pursuant to compel compliance with resolutions or sanctions designed to maintain or restore peace and order.

Peace Journalism- Peace Journalism is defined “when editors and reporters make choices - of what to report, and how to report it - that create opportunities for society at large to consider and value non-violent responses to conflict”.

This is journalism with peace as the main aim i.e. a normative mode of responsible and conscientious media coverage of conflict that aims at contributing to peacemaking, peacekeeping, and changing the attitudes of media owners, advertisers, professionals, and audiences towards war and peace. It shows backgrounds and contexts of conflicts; hears from all sides;

explores hidden agendas; highlights peace ideas and initiatives from anywhere at any time.

THE ROLE OF MEDIA IN CONFLICT MANAGEMENT

There are millions of global news networks that broadcast live from all corners of the world, and via the internet, providing immediate access to unfolding events and under certain conditions, influencing the way those events develop and end. As a result, this has prompted a general feeling that media coverage, especially television, has had an increased influence on conflict management since the collapse of the Soviet Union (Jakosben, 2000). Over the past few decades, awareness has increased with regard to the role local civilian population play in armed conflict. They are no longer considered to be a passive factor, but instead their needs and perceptions have explicit influence upon the success of a mission. Triggered by rapid technological change in the media, information no longer simply describes on-scene developments, but increasingly shapes the dynamics by influencing perceptions, opinions and loyalties (Jakosben, 2000; Galtung, 2004). This has made information to be an important means of warfare used by governments, military and leaders in all

ages everywhere for propaganda or disinformation purposes or for great potential to promote peace building.

The policies of the military and the attitudes of the press and the public towards media coverage of wars were influenced enormously by the experiences of the Vietnam War. The power of media influence in conflict was witnessed in the Vietnam War during the US involvement in the war due to the significant growth in information technology posed by discovery of television (Praeger, 1994). As remarked by Colonel Harry G. Summers – a battalion and corps operations officer in the Vietnam War; “....*there is a tendency to blame our problems with public support on the media it was this horror, not the reporting, that so influenced the American people,*” (Kull, 1995). Thus, the military realized after the Vietnam experience that to be able to wage war it would have to control the media, particularly the television media, so that the horrible images of war would not make it back to the home-front.

In order to objectively understand the nature and the role of media in peace and conflict management, it is important to understand the various ways through which media influence conflict and conflict management.

HISTORY OF MEDIA INFLUENCE

History has shown that media can incite people towards violence. Hitler used the media to create hatred for Jews (Vladimir & Schirch, 2007). Joseph Goebbels, Hitler's Minister of Propaganda, masterminded the most effective effort of mass persuasion, heavily relying on propaganda messages in motion pictures and radio broadcasting (Jowett & O'Donnell, 1999). In the 1992-5 Balkan conflict, electronic and print media helped promote ethnic conflict and hatred leading to violence. While the explicit broadcast of hate messages was rare, the cumulative impact of biased coverage fuelled the hatred over a long period polarizing local communities to the point where violence became an acceptable tool for addressing grievances (Buric, 2000; Sadkovich, 1998). In the past four years, the African continent experienced several political uprisings in Arab countries. Some of these uprisings were claimed to be engineered and subsequently escalated by media. Using the Libyan civil war as a case study, Viggo (2011) explored the media's role in modern conflicts. Emphasizing on the media's shortcomings in its reporting and analysis of the conflict, he highlighted the double standards that the media gave by drawing parallels between the wars in Iraq and Libya in an effort to illustrate the

errors repeatedly carried out by mass media. Eventually, Viggo concluded with reference to Muammar Qaddafi's capture and subsequent killing on October 20, 2011, "... the power of media in conflict cannot be assumed.... speed and technology with which images of the event recorded on cell phones were transmitted around the world via social media platforms, and widely disseminated even before Qaddafi's death was confirmed...."

During the 2010 general elections in Côte d'Ivoire, the media landscape was rife with partisan polemic and misinformation from opposing sides and incitements to violence, while members of the media themselves were frequent targets of violence and intimidation (Media Foundation for West Africa, 2011). During the post-election crisis that followed after the announcement of the presidential results, the print media and public television were used by both presidential candidates Laurent Gbagbo and Alassane Ouattara as a tool for propaganda and a means for mobilizing their electorate against their opponents (Electoral Reform International Services, 2011). The media occupied a central role in the post-election crisis and the descent into civil war that saw the displacement of more than 1 million people, the effective termination of many government services in parts of the

country and further economic decline (ibid). Apart from the negative roles of media, several cases have also linked media to the promotion of peace and reconciliation. For example, in Northern Uganda where the Lord Resistance Army (LRA) has camped for decades, media has been used to create the common good for the people to promote peace. Struges (2007) found that Mega FM has promoted peace in Northern Uganda with positive effects since 2002.

The station has encouraged LRA members to come out of the bush, joined radio phone-in talk shows and hold discussions with government and civil society representatives, a good step in peace building. Likewise, Open Broadcast Network (OBN), a media network established in Bosnia two decades ago, is one of the most ambitious and earliest intentional media attempts to reduce violent conflict. To this day, it remains the only television network established to promote peace and reconciliation. This followed the initial assessments of the Bosnian conflict that propaganda, through ethnic television stations, was instrumental in spreading messages of hate that incited and fuelled the conflict (Sadkovich, 1998). Search for Common Ground (SFCG), a US based NGO dealing with conflict resolution, launched Studio Ijambo in

1995 at the height of ethnic and political violence in Burundi and neighbouring Rwanda. The studio was established with the intention of promoting reconciliation, understanding and foster nonviolent conflict resolution (SFCG, 2004). Since its inception, Studio Ijambo has had a significant positive impact in mitigating conflict through high-quality radio programs that promoted reconciliation, dialogue, and collaboration, in addition to its credible, unbiased programming which included news, special features, round-table discussions, telephone call-ins, music, and highly popular soap-opera series (Slachmuislder&Nkurunziza, 1972).

WELLBEING AND SELF-GROWTH: TYPES OF WELLBEING [HEDONIC AND EUDEMONIC], CHARACTER STRENGTHS, RESILIENCE AND POST-TRAUMATIC GROWTH.

Psychological well-being refers to inter- and intraindividual levels of positive functioning that can include one's relatedness with others and self-referent attitudes that include one's sense of mastery and personal growth. Subjective

well-being reflects dimensions of affect judgments of life satisfaction.

Psychological well-being is simultaneously the absence of the crippling elements of the human experience—depression, anxiety, anger, fear—and the presence of enabling ones—positive emotions, meaning, healthy relationships, environmental mastery, engagement, self-actualization.

Psychological well-being is above and beyond the absence of psychological ill-being and it considers a broader spectrum of constructs than what is traditionally conceived of as happiness (Seligman and Csikszentmihalyi, 2000; Seligman, 2011).

The concept of well-being has received considerable interest in recent years, both in the scientific and lay literature. As such, well-being is frequently cited as a national priority for government policy around the world (Beddington et al. 2008). However, depending upon one's professional and personal perspective, the notion of well-being can have quite different connotations. For instance, economists may interpret well-being in terms of economic capacity and prosperity, the growth in both individual and national economic wealth.

Psychological wellbeing includes the absence of disorders, such as major depression or schizophrenia. An individual

suffering from mental disorders can hardly experience psychological wellbeing. However, absence of those disorders does not guaranty psychological flourishing. Since society traditionally supports mental illness within its healthcare system, it belongs in the health domain and it will not be a focus of this chapter. Only some of the interventions described below can both alleviate mental illness and improve positive psychological functioning. Most of the recommendations in this chapter are aimed at amplifying psychological health assets. Subjective well-being (SWB)- good mental states, including all of the various evaluations, positive and negative, that people make of their lives and the affective reactions of people to their experiences, is part of psychological well-being. It is often conceptualized as a set of two interrelated elements:

Life evaluation—a reflective assessment on a person's life or some specific aspect of it.

Affect —a person's feelings or emotional states, typically measured with reference to a particular point in time.

The notion of subjective wellbeing (SWB) is the currently dominant conception of happiness in psychological literature. There are several empirically informed models which aim to determine the structure of SWB. Bradburn (1969) found

SWB to be a function of the independent dimensions of general positive and negative affectivity.

TYPES OF WELLBEING (HEDONIC AND EUDEMONIC)

(Referred from an article by Henarath)

According to the scientific literature, there are two relatively distinct perspectives for empirical inquiries into happiness. The first perspective is hedonism that reflects maximizing pleasure and minimizing pain which was articulated thousands of years ago by the Geek philosopher, Arsttuppus (Watson, 1895).

His concept of hedonism has been further elaborated by many others. Accordingly, Epicurus (342 – 270 BCE) introduced ethical hedonism which holds our primary moral obligation to maximize experience of pleasure (Brunschwin and Nussbaum, 1993). Later, David Hume (1711 – 1776) and Jeremy Bentham (1748 – 1832) used the philosophy of hedonism to develop a foundation of utilitarianism (Peterson et al., 2005) the doctrine, based on the fact that utility promotes happiness. Bentham defined happiness, based on psychological experience as the cumulative value of pleasure and pain and he further argued that moral quality of human action should be judged by its consequence on human happiness (Veenhoven, 2010). Hedonism is being promoted even now, under the name of hedonic psychology (Kahneman et al., 1999). Preconceived understanding among hedonic psychologists is that, happiness or well-being comprises of subjective happiness and all good or bad judgments of life, based on pleasure versus displeasure (Ryan and Deci, 2001). Further, they stated that hedonic viewpoint

focuses on Subjective Well-Being (SWB) equated with positive affect and greater pleasure. Although, there is a continuous philosophical debate on validity and desirability of hedonism, most of the researchers who appreciate hedonism are still using SWB measurements to assess happiness, irrespective of availability of many tools, to evaluate pleasure and pain continuum in human lives. As an example, Diener (2000) highlighted that people's own evaluation of their lives must figure prominently in assessing the success of the society in terms of democratic notion. Further, Peterson (2005) pointed out that pursuit of pleasure is widely endorsed as a mean of happiness, especially by the western world.

The second perspective is eudemonism, which stands in contrast to hedonism reflecting the need of realizing one's true nature, in order to earn happiness. Many philosophers, religious masters and visionaries from both Western and Eastern worlds, challenged the hedonic view of happiness (Ryan and Deci, 2001). Eudemonic theories hold the stance that all desires and all outcomes that a person appreciates may not yield happiness or quality of life, (Ryan and Deci, 2001). Aristotle (384 – 322 BCE) clearly rejected Aristaeus's view on happiness and emphasized that true happiness entails only after identifying one's virtues, cultivating them and living according to them (Aristotle, 2000). Freud saw hedonic happiness as a short-lived orgasmic experience that arrives due to release of primitive urges (Veenhoven, 2010).

Similar position was further advanced by John Stuart Mill (1806 – 1873) and stressed that how self-fulfillment facilitates sense of happiness. Eudaimonism is an ethical doctrine which facilitates to recognize one's unique potentials and live according to virtues (Waterman, 1993). Further, he emphasizes that eudemonic happiness arrives when an individual's life is governed according to virtues and is fully engaged. In line with eudemonic principles, Ryff and Singer (1996) elaborated concept of Psychological Well-Being (PWB), which emphasizes life-time developmental perspectives from six distinct aspects of life: autonomy, personal growth, self-acceptance, life purpose, mastery and positive relatedness. Ryff and Singer (1998) argued that model of SWB is narrow as a measure of well-being and not reflects key aspects of a healthy life. Frey and Stutzer (2002) explain happiness from two perspectives: objective happiness related to physiological measures and subjective happiness related to psychological measures. Most importantly, Ryan and Deci (2000) presented the Self-Determination Theory (SDT), which posits the fact that satisfaction of basic psychological needs fosters both SWB and PWB. SDT is based on three basic psychological needs: autonomy, competence, and relatedness and theorizes

that the achievement of such needs is essential for both SWB as well as eudemonic well-being (Ryan and Deci, 2001). Further, they stated that eudemonic viewpoint focuses on PWB, which defines in terms of fully functioning of a person.

CHARACTER STRENGTHS

The research and practice of character strengths is an emerging science that has already elucidated important discoveries yet is marked by significant potential. Researchers in the science of character are drawing a number of fascinating connections between the use of character strengths and positive outcomes and the manifestation of optimal states. Novel synergies of character strengths education, research, and practice are being conceptualized (e.g., Niemiec & Wedding, 2008). Practitioners are finding quick and effective strategies for helping their clients get closer in touch with their best qualities, face suffering and challenges with strength, and pursue life goals by capitalizing on their strengths. In the end, this leads clients to be better informed to answer the question on identity that we opened with, and taken a step further, helps the clients use their positive core toward flourishing.

According to Peterson and Seligman (2004) character strengths are ubiquitous traits which are valued in their own right and not necessarily tied to tangible outcomes. Character strengths do not diminish others; rather they elevate those who witness the strength expression, producing admiration not jealousy.

Let's understand it by digging into the concept a little more.

Human beings have the unique ability to think about themselves: Who are we? We can recall our past and make predictions about our future. Our ability to think about ourselves, about our past and the future, however, is marked by negativity. We are hard-wired to focus on weaknesses than on strengths, and we are more risk averse than gain sensitive (Kahneman & Tversky, 1984).

Evolution has made most of us more adept at worrying than at appreciating. When we encounter challenges, we are more likely to recall shortcomings, failures, and setbacks - our own and those of others - and if we do this persistently, we develop psychological distress. Riding on this default wave of negativity, psychology has enumerated far more than two hundred different varieties of psychological distress

and numerous efficacious therapies to remediate this distress. Nonetheless, it has also largely made psychology a remedial or correctional discipline. Until recently, there wasn't a single coherent system which could reliably classify, define, and measure characteristics and conditions which make life worth living.

In the past decade, positive psychology has made concerted empirical efforts to advance the science behind what is strongest and best in people. Most prominent in these efforts is the field of character strengths. Whereas symptoms, in their varying combinations and severity, help us to understand nuanced levels of stress, sadness, and anxiety, character strengths help us to understand ways one can experience flourishing, positive relationships, engagement, meaning, and well-being.

Character strengths are capacities of cognition, affect, volition, and behavior. They are the basic psychological ingredients that enable us to act in ways that contribute to our well-being. Fearing that character strengths carry prescriptive and value-laden tone, psychology relegated them to philosophy and made the troubled aspects of human life the focus of its empirical investigation. Christopher Peterson and Martin Seligman (2004) spearheaded the first systematic effort to

classify core human strengths, acknowledging that character strengths are morally desired traits of human existence but at the same time they present them as descriptive traits open to empirical examination. Much like psychology has shown that individuals who experience negative emotions such as anger, hostility, vengeance, or narcissistic traits are more likely to develop a host of psychological ills, individuals who experience gratitude, forgiveness, humility, love, and kindness are more likely to report being happier and satisfied with life. Hence, character strengths describe various shades and dimensions of human flourishing and well-being.

According to Peterson and Seligman (2004) character strengths are ubiquitous traits which are valued in their own right and not necessarily tied to tangible outcomes. Character strengths do not diminish others; rather they elevate those who witness the strength expression, producing admiration not jealousy. Some individuals strikingly embody strength while others may have what appears to be no perceivable level of a given strength. Societal institutions, through rituals, attempt to cultivate character strengths. Peterson (2006) distinguishes character strengths (e.g., kindness, teamwork, zest) from talents and abilities. Athletic

prowess, photographic memory, perfect pitch, fine dexterity, physical agility, and such constitute talents and abilities.

Character strengths have moral flavor whereas talents and abilities do not. Also, talents, by and large are not buildable. Clearly, self-regulation and optimism can improve one's swimming performance but these increments tend to be relatively small. To swim like Michael Phelps and run like Usain Bolt, one needs certain physical abilities. Character strengths can be built on even frail foundations, and with enough practice, persistence, and good mentoring can take root to make life worth living. Also, one can squander a talent, but a character strength is usually not squandered.

The field of psychology has placed a predominant emphasis on weaknesses which has led mental health professionals and consumers to think of psychological disorders primarily as the presence of symptoms. Taking a similar categorical approach, disorders, can also be conceptualized as an overuse and/or underuse of multiple character strengths. However, we believe a dimensional approach makes more sense in attempting to understand the complexities of the character strength-psychopathology relationship. Such a dimensional approach would view strengths in terms of their

overuse and underuse, and their expression would exist in degrees. Character strength use varies by context so there is no perfect mean other than to reflect what Aristotle (2000) expressed as the golden mean of the expression of the right combination of strengths to the right degree in the right situation. From this angle, depression might be viewed as an underuse of the character strengths of hope/optimism, humor/playfulness, and zest; simultaneously, it may often be characterized as the overuse of the character strengths of judgment/critical thinking as well as of perseverance as reflected in thought rumination. Likewise, anxiety nearly always has an element involving the underuse of bravery/courage and the narrowing of attention reflects an underuse of the character strength of perspective. Knowing and using something positive that one already has increases one's self-efficacy to deal with challenges confidently and adaptively. What follows are some arguments for why character strengths are important, particularly in therapy, counseling, coaching, and teaching. While there is emerging research to support these points, further research is necessary before final conclusions can be drawn.

Repairing or fixing weakness doesn't necessarily make you stronger: The assumption that fixing all weaknesses

will make us happy or happier is a misconception. In writing or editing a paper one may correct all spelling and grammatical errors but this does not necessarily make the paper excellent in quality. Writing an excellent paper entails expressing creative craftsmanship.

Using strengths helps to reinterpret and reframe problems adaptively:

Using strengths increases one's self-efficacy and confidence in ways focusing on weakness cannot. Because, being aware of our character strengths, in addition to weaknesses, facilitates us to reinterpret and reframe problems from a strength's perspective rather than from a weakness perspective.

Fixing weakness doesn't necessarily

cultivate happiness: A popular working principle many practitioners take is that fixing weaknesses is of utmost importance for happiness and strengths will simply take care of themselves. A paralleled conception is that working on something one is already good at is wasting time and energy which could better be spent on correcting weaknesses. Much like weaknesses require fixing, strengths require nurturance. Fixing weakness yields remediation while strengths nurturance produces growth and most likely, greater happiness.

Be wary of trite proclamations:

There are many popular statements that are more likely to be misconceptions than conventional wisdom, e.g., "you can do anything, if you work hard at it" and "the sky is the limit." Instead of chanting these mantras, consider using your character strengths. A strengths-based approach helps frame specific, practical yet realistic goals geared toward alleviation of depression and creating enduring life satisfaction.

Strength awareness builds a cumulative advantage: Evidence shows that people who are aware of their strengths are able to build self-

confidence at a young age and tend to reap a "cumulative advantage" that continue to grow over a lifetime (Judge & Hurst, 2008). This is not dissimilar to Fredrickson's (2001) broaden-and-build theory of positive emotions, however, applied to character strengths would state that strengths broaden a repertoire of action potentials in the present and build resources in the future.

Using strengths to promote resilience: Adversities, traumas, and losses undermine physical and emotional health and pose a significant risk for thwarting growth. These challenges can strike anytime and are often unexpected. Knowing and using strengths, in good times helps you to learn strategies which, you can use in turn, during tough times. Being aware of and using strengths not only promotes resilience but also prepares individuals to encounter challenges adaptively.

Using of strengths to find balance in daily interactions and manage relational challenges:

If daily interactions between two partners, friends, or colleagues focused more on each other's weakness and deficits then unease, tension, and resentment is likely going to mark these interactions. A balanced approach is going to address conflict as well as cooperation, grudge as well as gratitude, hubris as well as humility, and self-centeredness as well as empathy. This will likely lessen tension and create more positive opportunities in these interactions.

CHARACTER STRENGTHS AND OPTIMAL EXPERIENCES

Strengths can be used to build optimal states such as flow. Mihaly Csikszentmihalyi (1990) articulated the phenomenology of flow as an engrossing and enjoyable state that is worth doing for its own sake. During flow, time passes, and attention is razor sharp on the present moment activity. The individual becomes one with the activity, thus both conscious experience of emotion and the self fades, but only in the immediacy of the experience because the aftermath of flow is invigorating. Flow is maintained through the balance between skill and challenge in which the activity is not so easy that the person finds it boring, nor is it too difficult that one is frustrated. Flow is intrinsically motivating and highly enjoyable. Seligman (2002) has proposed that one way to build flow is to identify the salient character strengths of individuals and then help them to find opportunities to use these strengths more often. Mindfulness is another optimal state that involves observing one's ever-changing present moment experience. Researchers operationally defined mindfulness as two

core character strengths stating it involves the self-regulation of attention while employing an attitude of curiosity, openness, and acceptance with what one places their attention on (Bishop et al., 2004). Thus, the use of these character strengths and others may help facilitate greater mindfulness (Niemiec, Rashid, & Spinella, in press), and in turn, the practice of mindfulness has been shown to nurture a variety of character strengths (e.g., kindness, forgiveness, zest, spirituality).

SIGNATURE STRENGTHS

Signature strengths are those highest strengths of character that an individual self-consciously owns and celebrates, which he or she feels a sense of ownership and authenticity ("this is the real me"); the individual feels excited while displaying these signature strengths, learns quickly as they are practiced, feels more invigorated than exhausted when using them, and creates and pursues projects that revolve around them. One particular intervention used widely by practitioners around the world is the "use signature strengths in new ways" intervention. This involves the client first taking the VIA Inventory of Strengths (www.viasurvey.org) to receive their rank ordering of character strengths per the VIA Classification, and second using one of their highest strengths in a

way that is new and unique for that client (Linley et al., 2010; Madden et al., 2011; Mitchell et al., 2009; Mongrain&Anselmo, 2009; Rust, Diessner, & Reade, 2009; Peterson & Peterson, 2008; Seligman et al., 2005). This intervention has been found to boost happiness and decrease depression for up to six months. But, the benefits of using character strengths is mounting beyond increases to happiness and decreases to depression.

Character strengths use has been linked to increased work satisfaction, work engagement, meaning, self-efficacy, self-esteem, goal achievement, positive affect, vitality, and lower perceived stress (Govindji& Linley, 2007; Linley et al., 2010; Littman-Ovadia&Davidovitch, 2010; Littman-Ovadia& Steger, 2010; Park, Peterson, & Seligman, 2004; Proctor, Maltby, & Linley, 2009; Wood et al., 2011).

Additional lines of research (Sheldon & Houser-Marko, 2001; Linley et al., 2010) have provided empirical evidence that employing signature strengths enhances people's well-being by facilitating a motivational sequence of goal selection (e.g., you select a goal that is meaningful for you), goal pursuit (in pursuing the goal you use your best internal resources), and goal attainment (you successfully complete a project which you find enjoyable, engaging, and meaningful). The whole process of goal pursuit creates an upward spiral of well-being (Fredrickson & Joiner, 2000). This spiral is not only therapeutic but also helps individuals to flourish.

	Character Strengths	Description
1	Appreciation of beauty & excellence	Being moved deeply by beauty in nature, in art (painting, music, theatre, etc) or in excellence in any field of life
2	Honesty	Authenticity; not pretending to be someone one is not; coming across as a genuine and honest person
3	Bravery	Valor; overcoming fears to do what needs to be done; not give up in face of a hardship or challenge
4	Creativity	Originality; thinking of new and better ways of doing things; not being content with doing things in conventional ways
5	Curiosity	Interest; openness to experience; being driven to explore things; asking questions; not tolerating ambiguity easily; being open to different experiences and activities
6	Fairness	Equity; standing up for others when they are treated unfairly, bullied or ridiculed; day-to-day actions show a sense of fairness
7	Forgiveness	Mercy; forgiving easily those who offend; not holding grudges
8	Gratitude	Expressing thankfulness for good things through words and actions; not take things for granted
9	Hope	Optimism; future-mindedness; hoping and believing that more good things will happen than bad ones; recovering quickly from setbacks and taking concrete steps to overcome them
10	Humor	Being playful, funny and uses humor to connect with others
11	Kindness	Generosity; compassion; doing kind deeds for others, often without asking; helping others regularly; being known as a kind person
12	Leadership	Organizing activities that include others; being someone others like to follow; being often chosen to lead by peers
13	Love	Having warm and caring relationships with family and friends; showing genuine love and affection through actions regularly
14	Love of learning	Loving to learn many things, concepts, ideas, facts in school or on one's own
15	Humility	Modesty; not liking to be the center of attention; not acting as being special; admitting shortcomings readily; knowing what one can and cannot do
16	Judgment	Critical thinking; thinking through and examining all sides before deciding; consulting with trusted others; being flexible to change one's mind when necessary
17	Perseverance	Diligence; industry; finishing most things; being able to refocus when distracted and completing the task without complaining; overcoming challenges to complete the task
18	Perspective	Putting things together to understand underlying meaning; settling disputes among friends; learning from mistakes
19	Prudence	Being careful and cautious; avoid taking undue risks; not easily yielding to external pressures
20	Spirituality	Believing in God or higher power; liking to participate in religious or spiritual practices e.g., prayer, meditation, etc.
21	Self-regulation	Self-control; managing feelings and behaviour well most of the time; following gladly rules and routines
22	Social intelligence	Easily understanding others' feelings; managing oneself well in social situations; displaying excellent interpersonal skills
23	Teamwork	Citizenship; loyalty; relating well with teammates or group members; contributing to the success of the group
24	Zest	Enthusiasm; energetic; cheerful and full of life

RESILIENCE

Human beings typically encounter a variety of difficulties and challenges during the course of their lives, ranging from daily hassles to major life events. A significant chunk of Indian youth is contributing tremendously to the global pool of education, business and information technology. At the same time, they have counterparts who cannot deal with competitiveness and the stress to perform. The young adults, therefore are able to bounce back from setbacks and deal with daily hassles and stressors through the process called Resilience.

The term resilience stems from Latin word 'resiliens' to refer to the pliant or elastic quality of a substance (Joseph, 1994). Although resilience remains a familiar word in everyday English language, the term resilience carries different meanings across different contexts.

Resilience is also a process of adapting well in the face of adversity, trauma, tragedy, threats or significant sources of stress such as family and relationship problems, serious health problems and financial stressors. It means "bouncing back" from difficult experiences. Moreover, Luthar and Cicchetti (2000) defined resilience as a "positive adaptation...is considered in a

demonstration of manifested behavior on social competence or success at meeting any particular tasks at a specific life stage"

Resilience is that ineffable quality that allows some people to be knocked down by life and come back stronger than ever. Rather than letting failure overcome them and drain their resolve, they find a way to rise from the ashes. Psychologists have identified some of the factors that make someone resilient, among them a positive attitude, optimism, the ability to regulate emotions, and the ability to see failure as a form of helpful feedback. Even after misfortune, resilient people are blessed with such an outlook that they are able to change course and soldier on.

CLOSELY RELATED TERMS FOR PSYCHOLOGICAL RESILIENCE

Relevant psychological literature on resilience hasn't always used the term 'resilience' or 'psychological resilience'. Consider searching for material using synonyms or closely related keywords, such as (major terms in bold):

Adaptive Coping
Adversity Quotient
Emotional Intelligence (Daniel Goleman)
Hardiness
Learned Optimism (Martin Seligman)
Learned Resourcefulness (Rosenbaum)
Life Orientation
Resourcefulness
Self-esteem, Self-concept, Self-confidence, Self-efficacy

10. Self-healing personality (Howard Friedman)
11. Sense of Coherence (Aaron Antonovsky)
Sense of Meaning (Viktor Frankl)
Thriving (Al Siebert - Thrivenet)

CHARACTERISTIC OF RESILIENCE

Ability to "bounce back" and "recover from almost anything".

Have a "where there's a will, there's a way" attitude.

Tendency to see problems as opportunities.

Ability to "hang tough" which things are difficult.

Capacity for seeing small windows of opportunity and making the most of them.

Have deep-rooted faith in a system of meaning.

Have a healthy social support network.

Has the wherewithal to competently handle most different kinds of situations.

Has a wide comfort zone.

Able to recover from experiences in the panic zone or of a traumatic nature.

ESTABLISHED MODELS AND THEORIES OF RESILIENCE

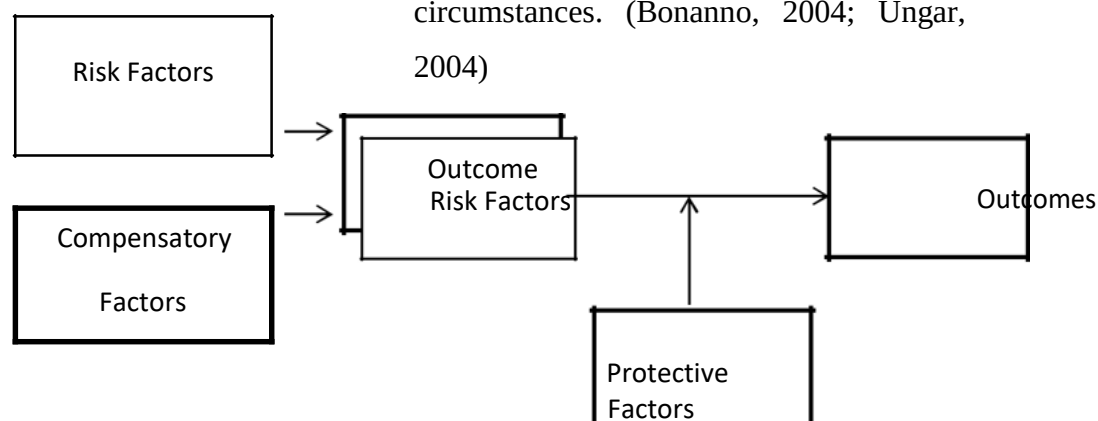
The concept of Resilience can be explained with the help of certain models. The three models mentioned below are based on the Resiliency Theory which works on a strength-based approach. It highlights the promotive factors of resilience that are known to operate in opposition to the risk factors. These promotive factors are the positive contextual, social and individual variables that help in overcoming the negative effects of risk exposure. Resiliency theory

allows to study and understand as to why some youth grow up to be healthy adults in spite of risks exposure (Garmezy, 1991). Further, while each of these models depicts different pathways of resilience, it seems that a model that can explain the combination of protective, challenge and compensatory effects of promotive and risk factors on the individual would be beneficial.

• COMPENSATORY MODEL (GARMEZY ET AL.1984)

This Model is an important one in the domain of Resiliency Theory, and examines how different variables (called compensatory factors) may directly and independently reduce negative outcomes associated with risk factors. Compensatory factors are variables that neutralize exposure to risk or operate in a counteractive fashion against the potential negative consequences introduced by a risk. Both the risk and compensatory factors contribute additively in the prediction of the outcome. Parental support, for example, was found to compensate for risks associated with fighting and being around violent adults (Zimmerman, et al., 1998). Here, parental support is a positive compensatory factor working in opposition to the risk factor of violence which predicts less violent behavior in adolescents.

exposure. These protective factors are known to foster positive outcomes and healthy personality characteristics despite unfavourable or aversive life circumstances. (Bonanno, 2004; Ungar, 2004)



RISK-PROTECTIVE MODEL OF RESILIENCE (GARMEZY ET AL. 1984; RUTTER, 1985)

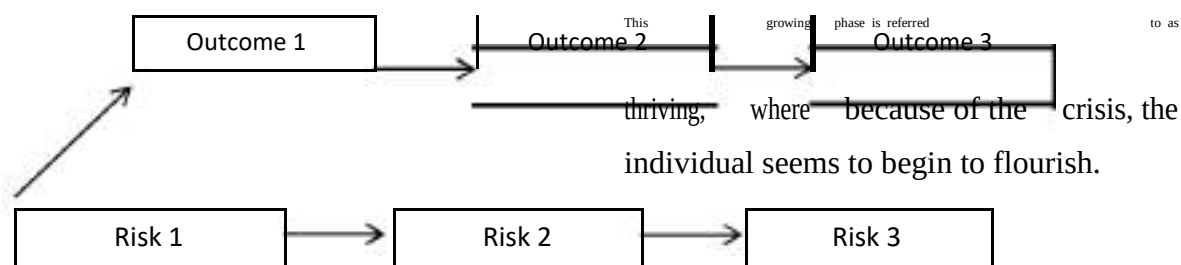
As per this model, by means of an interaction between the protective and risk factors, the probability of negative outcomes is reduced. The model examines whether protective factors modify the effects of risks in an interactive fashion. Although they may have direct effects on the outcomes, protective factors mainly act as moderators. Brook et al (1990) proposed that effects may be associated with a ‘risk- protective mechanism’ or a ‘protective- protective mechanism’. A risk protective variable functions to lessen the negative effect of a risk factor, whereas protective- protective variables enhance the positive effects of promotive factors found to decrease negative outcomes. Thus, resilience here can be said to moderate the negative influences of risk

• CHALLENGE MODEL OF RESILIENCE (RUTTER, 1987)

This model is based on the idea that exposure to moderate level of risks, helps in preparing individuals to overcome subsequent exposures. It is vital that the initial exposure is challenging enough to help develop the coping mechanisms to overcome its effects. This suggests that exposure to low levels and high levels of a risk factor are associated with negative outcomes, but moderate levels of the risk are related to less negative (or positive

outcomes). The underlying idea is that individuals exposed to moderate levels of risks develop skills to overcome it, whereas very high levels of risk may overwhelm their capacity to adapt. Moderate levels of risk may therefore be beneficial as long as the risk exposure is challenging enough to elicit a coping response, enabling the individual to learn from the process of overcoming the risk. An example of this is provided in the form of resolution of interpersonal conflicts in a friendly way, which later can help in solving a more heated social disagreement, which might lead to a violent response if not resolved appropriately.

The authors postulated that a resilience cycle may be used by people facing adversity. The researchers suggest that even the most resilient individuals experience a 'Rollercoaster Effect' as they work through a traumatic experience. A four-cycle phase to resilience was defined by the researchers, and includes: a deteriorating phase, an adapting phase, a recovery phase, and a growing phase. Resilience capacity, for the most part, largely determines where in the cycle the individual finds himself. Thus, if a person is unable to adapt to his challenging experience, he will most likely sink into a dysfunctional level and will be unable to cope or survive the adversity. However, some individuals may adapt but not fully recover, thus reaching survival level. Some may reach the recovery phase and return to the status quo. However, a small minority of individuals, those who are 'thrivers', are likely to reach the growing phase and achieve a strengthened resilience level.



RESILIENCE CYCLE (PATTERSON AND KELLEHER, 2005)

REVIEW OF LITERATURE:

The literature review seeks to better understand the construct of resilience and provide a context for how it can further be studied in graduate and postgraduate students. More specifically, the literature review is organized around 2 central questions:

How is resilience defined;

Is resilience an innate quality or a dynamic process;

RESILIENCE DEFINED

Nearly fifty years of research in resiliency has brought forth various perspectives and voices. Despite the vast body of research on resilience, there is little agreement on a single definition of resilience among scholars. In fact, scholars define the construct of resilience in a multitude of ways (Carle & Chassin, 2004).

Richardson and his colleagues (1990) contended that resiliency is “the process of coping with disruptive, stressful, or challenging life events in a way that provides the individual with additional protective and coping skills than prior to the disruption that results from the event” (p.34). Similarly, Higgins (1994) described resiliency as the “process of self-righting or growth” (p. 1), while Wolins (1993) defined resiliency as the “capacity to bounce back, to withstand hardship, and

to repair yourself” (p. 5). Other notable definitions are:

A dynamic process encompassing positive adaptation within the context of significant adversity (Luthar et al., 2000).

Resiliency is also defined as a “positive adaptation...is considered in a demonstration of manifested behavior on social competence or success at meeting any particular tasks at a specific life stage” (Luthar & Cicchetti, 2000, p. 110).

All of the theorists’ definitions of resilience contain two core ideas – that a person has experienced serious risk and has demonstrated positive functioning in some way. Thus, both these aspects were considered in the operational definition that was constructed.

RESILIENCE: AN INNATE QUALITY OR DYNAMIC PROCESS?

Despite the debate on resilience as a process or an ability, majority of the members of the group were largely convinced that resilience is more of a process and thus this aspect was included in the operational definition. Some of the studies that support this view are

During early waves of resilience research, researchers tended to regard and label individuals who transcended their adverse circumstances as “hardy,” “invulnerable,”

or “invincible” (Werner & Smith, 1982). Such labels implied that these individuals were in possession of a rare and remarkable set of qualities that enabled them to rebound from whatever adversity came their way – almost as if these fortunate individuals possessed a sort of magical force field that protected them from all harm. Increasingly, however, researchers have arrived at the consensus that resilience is not some remarkable, innate quality but rather a developmental process that incorporates the normative self-righting tendencies of individuals (Masten, 2001).

In her book, *Fostering Resiliency in Children*, Bonnie Benard (1995) claimed:

We are all born with an innate capacity for resilience, by which we are able to develop social competence, problem-solving skills, a critical consciousness, autonomy, and a sense of purpose (p. 17).

Researchers increasingly view resilience not as a fixed attribute but as an alterable set of processes that can be fostered and cultivated (Masten, 2001; Pardon, Waxman & Huang, 1999).

Garmezy’s (1991) triadic model of resilience provided a widely accepted ecological framework for understanding the resilience process. The triadic model described the dynamic interactions among risk and protective factors on three levels

(individual, family, and environmental). The model also emphasized that resilience is a process that empowers individuals to shape their environment and to be shaped by it in turn. Garmezy defined resilience as, “not necessarily impervious to stress. Rather, resilience is designed to reflect the capacity for recovery and maintained adaptive behavior that may follow initial retreat or incapacity upon initiating a stressful event” (Garmezy, 1991a).

Hence, implicit in the concept of resilience as a dynamic process is the understanding that resilience can grow or decline over time depending on the interactions taking place between an individual and their environment and between risk and protective factors in an individual’s life. Therefore, an individual may be resilient at certain times - and not at others - depending upon the circumstances and relative strength of protective factors compared to risk factors at the given moment (Winfield, 1991).

RESILIENCE PROMOTING WELL-BEING AND COMPETENCE

A large body of research also showed the positive effects of resilience on overall well-being and competence. These aspects were also considered important by the group members and thus were included in

the operation definition. Some of the research include:

Resilience shows a positive correlation with psychological well-being (Picardi et al., 2012; Sagone & De Caroli, 2013). Souri and Hasanirad (2011) found that resilience is also a positive and significant predictor for psychological well-being and this relationship is mediated by optimism. Furthermore, positive emotions predicted increase in both resilience and psychological well-being (Cohn, Fredrickson, Brown, Mikels & Conway, 2009). Due to these empirical findings and changes in laws and regulations surrounding health care and the way health care is funded, promoting mental health through cultivating resilience has become common practice in regard to treatment and prevention in counselling, mental health services, school environments and community programs (AIESG, 2010; Ager, 2013). Promoting mental health by enhancing resilience fits current financial cut backs because it locates resources within the individual and the community rather than in government programs and initiatives (Ager, 2013). Therefore, the demand for interventions that promote resilience and well-being is growing.

Resilient adolescents show such characteristics as social competence, problem-solving skills, mastery, autonomy

and a sense of purpose and future. Children and adolescents actually acquire much of their belief about their own competence from their families and home environments. A strong sense of competence is most likely to develop in a home environment that offers some challenges, encourages, sets high but realistic aspirations, contains positive role models, provides and supports mastery experiences, and teaches how to deal with difficulties. The ability to apply problem-solving skills to interpersonal problems or conflicts, to show empathy for the feelings of others, and to voluntarily help others, are hallmarks of positive development. Adolescents with good intimacy skills—that is, those who are able to be emotionally close to another individual—are also more likely to be resilient. (Hair et al., 2002).

POST-TRAUMATIC GROWTH

As Kay Wilson struggled to make her way through a Jerusalem forest after being repeatedly stabbed by a Palestinian terrorist, she distracted herself from her agony by playing the song "Somewhere Over the Rainbow" in her mind, composing a new piano arrangement while she fought for breath and forced herself to put one bare foot in front of the other.

Wilson, then 46, had been working as a tour guide when, on Dec. 18, 2010, she and a friend were ambushed by terrorists. Wilson witnessed her friend's murder and was herself viciously stabbed with a machete, ultimately playing dead as her attacker plunged his knife into her chest a final time.

She eventually recovered from her severe physical wounds and is healing from her psychological trauma. She now speaks to global audiences about her survival, hoping to "dispel hatred, whether toward Arabs or Jews."

The work "helps me make meaning out of something so senseless," says Wilson, who is also writing a book about her experiences.

After the attack, Wilson had flashbacks and deep survivor's guilt. But like many people who have survived trauma, she has found positive change as well—a new appreciation for life, a newfound sense of personal strength and a new focus on helping others.

Post-traumatic growth (PTG) is a theory that explains this kind of transformation following trauma. It was developed by psychologists Richard Tedeschi, PhD, and Lawrence Calhoun, PhD, in the mid-1990s, and holds that people who endure psychological struggle following adversity can often see positive growth afterward.

"People develop new understandings of themselves, the world they live in, how to relate to other people, the kind of future they might have and a better understanding of how to live life," says Tedeschi.

How can clinicians use PTG theory to help patients? How has new research helped refine understanding of it? Here's a look at developments in the field.

SIGNS OF POST-TRAUMATIC GROWTH

PTG can be confused with resilience, but the two are different constructs (see "The post-traumatic growth inventory" below).

"PTG is sometimes considered synonymous with resilience because becoming more resilient as a result of struggle with trauma can be an example of PTG—but PTG is different from resilience, says KanakoTaku, PhD, associate professor of psychology at Oakland University, who has both researched PTG and experienced it as a survivor of the 1995 Kobe earthquake in Japan.

"Resiliency is the personal attribute or ability to bounce back," says Taku. PTG, on the other hand, refers to what can happen when someone who has difficulty bouncing back experiences a traumatic event that challenges his or her core beliefs, endures psychological struggle (even a mental illness such as post-traumatic stress disorder), and then ultimately finds a sense of personal growth. It's a process that "takes a lot of time, energy and struggle," Taku says.

Someone who is already resilient when trauma occurs won't experience PTG because a resilient person isn't rocked to the core by an event and doesn't have to seek a new belief system, explains Tedeschi. Less resilient people, on the other hand, may go through distress and confusion as they try to understand why

this terrible thing happened to them and what it means for their world view.

To evaluate whether and to what extent someone has achieved growth after a trauma, psychologists use a variety of self-report scales. One that was developed by Tedeschi and Calhoun is the Post-Traumatic Growth Inventory (PTGI) (*Journal of Traumatic Stress*, 1996). It looks for positive responses in five areas:

Appreciation of life.

Relationships with others.

New possibilities in life.

Personal strength.

Spiritual change.

The scale is being revised to add new items that will expand the "spiritual change" domain, says Tedeschi. This is being done "to incorporate more existential themes that should resonate with those who are more secular" as well as reflect cross-cultural differences in perceptions of spirituality.

A PREDISPOSITION FOR GROWTH?

How many people experience PTG? Tedeschi prefers not to put a hard number on it.

"It all depends on the trauma, the circumstances, the timing of the measurement ... [and] on how you define growth using the PTGI, looking at total score, means, factors or individual items," he says. However, he estimates that about one-half to two-thirds of people show PTG.

Some PTG researchers have tried to corroborate self-reported growth by questioning friends and family members about whether growth "sticks."

"We are getting more studies that show that PTG is generally stable over time, with a few people showing increases and a few showing decreases," Tedeschi says. "It is now up to us to learn what is going on with those who change over time, but the evidence is for stability in general, and also corroboration by others."

There appear to be two traits that make some more likely to experience PTG, says Tedeschi: openness to experience and extraversion. That's because people who are more open are more likely to reconsider their belief systems, says Tedeschi, and extroverts are more likely to be more active in response to trauma and seek out connections with others.

Women also tend to report more growth than men, says Tedeschi, but the difference is relatively small.

Age also can be a factor, with children under 8 less likely to have the cognitive capacity to experience PTG, while those in late adolescence and early adulthood—who may already be trying to determine their world view—are more open to the type of change that such growth reflects, says Tedeschi.

There also may be genetic underpinnings for PTG, but researchers are just beginning to tease this out. In a 2014 study in the *Journal of Affective Disorders*, for example, Harvard social and psychiatric epidemiologist Erin Dunn, ScD, and a team of researchers examined data previously collected from over 200 Hurricane Katrina survivors and found that variants in the gene RGS2 significantly interacted with levels of exposure to the hurricane to predict PTG. RGS2 is linked to fear-related disorders, such as post-traumatic stress disorder, panic disorder and anxiety.

Dunn calls the results "very interesting" but notes that "we have to be somewhat cautious in interpreting it because we were unable to find a similar sample to replicate that finding."

Sarah Lowe, PhD, of Montclair State University, who worked with Dunn on the research, says one difficulty with gene studies for PTG is the concept's complexity. "If you look at what predicts PTG, it is often psychological stress and dysfunction—but also more positive personality traits like optimism and future orientation, which you'd expect would have a very different genetic basis," she says.

THE POST-TRAUMATIC GROWTH INVENTORY

To evaluate whether and to what extent someone has achieved growth after a trauma, psychologists look for positive responses in five areas.

Appreciation of life

Relationships with others

New possibilities in life

Personal strength

Spiritual change

**HEALTH: HEALTH
PROMOTING AND HEALTH
COMPROMISING
BEHAVIORS, LIFE STYLE AND
CHRONIC DISEASES
[DIABETES, HYPERTENSION,
CORONARY HEART DISEASE],
PSYCHONEUROIMMUNOLOG
Y [CANCER, HIV/AIDS]**

**HEALTH PROMOTING AND
HEALTH COMPROMISING
BEHAVIORS**

(Referred from an article on Health-Promoting and Health-Compromising Behaviors Among Minority Adolescents by Clark, R.)

A huge proportion of the world's population - more than 1.75 billion is young, aged between 10 and 24 years. Adolescents (aged 10 to 19 years) have specific health and development needs, and many face challenges that hinder their wellbeing, including poverty, a lack of access to health information and services, and unsafe environments. Interventions that address their needs can save lives and foster a new generation of productive adults who can help their communities progress. Many boys and girls in developing countries enter adolescence undernourished, making them more vulnerable to disease and early death. Conversely, overweight and obesity another form of malnutrition with serious health consequences - is increasing among other young people in both low- and high-income countries. Adequate nutrition and healthy eating and physical exercise habits at this age are foundations for good health in adulthood. (WHO Facts on Adolescent Health 2008). Viner R. et.al., 2005 in a clinical review on health promotion in adolescence has suggested that health promotion strategies and interventions can

be delivered at various levels including individual, family, school, community and national levels. The interventions at school level can be single issue, curriculum-based education and "whole school" organisation and behavior management interventions.

In India, health promotion is now receiving an increasing attention regarding the prominent role it plays in health. Health promoting lifestyle is one factor that positively contributes to quality of life. When a person engages in health promoting lifestyle, he/she has a greater potential to remain healthy and possibly live longer without the burden of the disease (Raj S, Senjam et al., 2012).

Promoting health through schools is a "life-course" approach to promote healthy behaviour among children. Many of today's and tomorrow's leading causes of death, disease and disability can be significantly reduced by preventing behaviour that is initiated during youth, through health education, understanding and motivation; and fostered by social and political policies and conditions. Almost all children attend school and spend 6-7 hours of their time every day in that learning environment. Incorporating health into the school curriculum can have substantial influence on health promoting behaviors (HRIDAY PHFI report 2010).

HEALTH BEHAVIORS COMPROMISING AMONG ADOLESCENTS

The major causes of adolescent mortality are not diseases, but are primarily related to preventable social, environmental and behavioral factors (Irwin and Millstein, 1986; Millstein, 1989). The three primary causes of mortality during adolescence are injuries, homicide and suicide; together they are responsible for 75% of all adolescent deaths (Millstein et al, 1993). Major sources of morbidity include injury and disability associated with the use of motor or recreational vehicles, pregnancy complications, sexually transmitted diseases and consequences of substance abuse (Millstein et al.,1993). Among adolescent females, eating disorders are another significant source of morbidity, and the use of unhealthy weight loss methods may have numerous psychological and physical health consequences (Nylander, 1971; Pugliese et al.,1983; French and Jeffery, 1994; Neumark-Sztainer,1995). To improve adolescent health, it is essential to reduce the frequency, delay the onset and aim towards the prevention of behaviors associated with morbidity and mortality among youth. Following the identification

of health-compromising behaviors to be targeted for intervention, the next step in building effective prevention programs is to understand the factors associated with these behaviors among adolescents at different stages of development. Numerous studies on health-compromising behaviors among adolescents indicate that these D. Neumark-Sztainer et al. behaviors tend to co-occur, leading to a health-compromising life-style (Donovan and Jessor,1985; Donovan et al., 1988, 1991; Osgood, 1991; Neumark-Sztainer et al., 1996). Research on Problem Behavior Theory suggests that alcohol and marijuana use, delinquent behaviors, and sexual intercourse may constitute a 'syndrome' of problem behavior in adolescence. However, strengths of associations between health-compromising behaviors differ, suggesting that there may be greater shared etiologic factors for certain sets of behaviors than for others (Osgood, 1991; Elliot, 1993). Studies on covariations of health-compromising behaviors provide information for hypothesis generation about common underlying factors leading to the clustering of behaviors.

The identification of etiologic factors common to an array of health-compromising behaviors may foster the design of more effective and cost-efficient

prevention programs. Social Cognitive Theory (SCT) discusses the importance of personal, socio-environmental and behavioral factors on behavior and the reciprocal relations between all of these factors (Bandura, 1977, 1986). SCT has particular relevance for explaining involvement in health-compromising behaviors among youth. Jessor has developed a model specifically aimed at explaining adolescent risk behavior/lifestyle which incorporates a number of principles similar to those in SCT (Jessor, 1991, 1992, 1993). Personal, socio-environmental and behavioral factors are shown to influence adolescent risk behavior with reciprocal relationships among all of the factors. However, personal factors are separated into biological and personality domains while the social environmental factors are divided into actual and perceived domains. Jessor indicates that both risk factors and protective factors may exist within each domain. Both of these models were used

as a guide in developing a theoretical framework for the present study and in selecting variables whose associations with health-compromising behaviors were to be examined (see Figure 1). Personal factors include those from the psychological domain: self-esteem, emotional well-being and risk-taking disposition. Socio-environmental factors include both actual and perceived components such as family structure and family connectedness, school connectedness, and stressful life experiences such as past physical and sexual abuse. Behavioral factors include school achievement, involvement in extracurricular activities and attendance at religious services. It should be noted that the placement of these variables into these larger categories is not clear-cut as most of these variables have personal socio-environmental and behavioral components (e.g. school connectedness and achievement, religiosity and physical and sexual abuse).

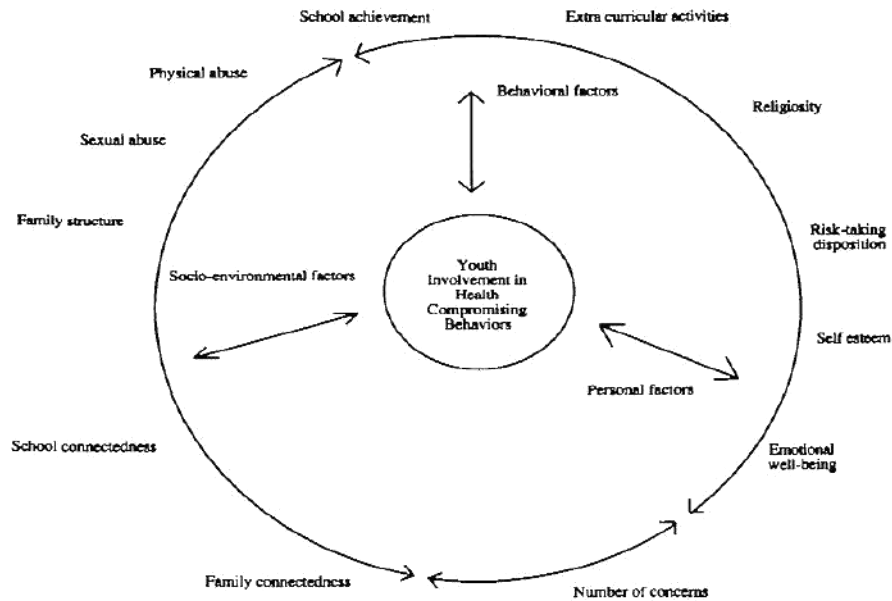


Fig. 1. A framework for understanding factors influencing youth involvement in health-compromising behaviors

LIFE STYLE AND CHRONIC DISEASES [DIABETES, HYPERTENSION, CORONARY HEART DISEASE], PSYCHONEUROIMMUNOLOGY [CANCER, HIV/AIDS]

Modern science through improved sanitation, vaccination, antibiotics and medical attention has eliminated the threat of death from most infectious diseases. But now, too many people are dying relatively young from noncommunicable diseases like heart diseases and cancer. The main contributing factor for this epidemiological

transition is change in our lifestyle toward the unhealthy continuum, e.g., tobacco use, excessive alcohol consumption, unhealthy dietary habits and physical inactivity. This means that death from lifestyle diseases like hypertension, diabetes and cancer are now the primary causes of death.

Lifestyle is the way a person lives. This includes patterns of social relations, consumption, entertainment and dress. The term lifestyle also reflects an individual's attitudes, beliefs and, essentially, the way the person is perceived by himself/herself and, at times, also how he/she is perceived by others. This rapidly growing epidemic of noncommunicable diseases is responsible for 60% of the world's deaths. At least

50% of the deaths in the US each year are due to unhealthy lifestyle. In India also, the situation of lifestyle diseases is quite alarming. The disease profile is changing rapidly. The World Health Organization (WHO) has identified India as one of the nations that is going to have most of the lifestyle disorders in the near future. Nowadays, not only are lifestyle disorders becoming more common but they are also affecting the younger population. Hence, the population at risk shifts from 40+ to maybe 30+ or even younger. Already considered the diabetes capital of the world, India now appears headed toward gaining another dubious distinction of becoming the lifestyle-related disease capital as well. A study conducted jointly by the All India Institute of Medical Sciences and the Max Hospital shows that the incidence of hypertension, obesity and heart disease is increasing at an alarming rate, especially in the young, urban population.

In most of these lifestyle diseases, the onset is insidious and is usually after the age of 30 years. By the time interventions are planned and implemented, the damage to health has already occurred.

Adolescents are a unique population with specific health concerns and needs. Adolescence is the peak age of onset for

serious mental illness like depression and psychosis. Physical, emotional, social and sexual change makes adolescents overloaded with stress, which can result in anxiety, withdrawal, aggression, poor coping skills and actual physical illness. Adolescents constitute about 22.8% of the population of India.

It is well documented that behaviors developed during this period influence health in adulthood. Several health-compromising behaviors (e.g., smoking, alcohol) as well as health-enhancing behaviors (e.g., physical exercise, nutrition) are adopted in adolescence, and they often persist into adulthood. The WHO estimates that 70% of premature deaths among adults are due to behavior (smoking, illicit drug use, reckless driving) initiated during adolescence. Therefore, helping adolescents establish healthy lifestyles and avoid developing health risk behaviors is crucial and should be started before these behaviors are firmly established. The demands on young people are new and unprecedented; their parents could not have predicted many of the pressures they face. How we help adolescents meet these demands and equip them with the kind of education, skills and outlook they will need in a changing environment will depend on how well we understand their world.

There is a definite need to investigate the health behavior of adolescents rather than focus on adults since it will be far more difficult for adults to change their unhealthy habits adopted in their youth. Many studies have demonstrated a positive relationship between health-promoting lifestyle and quality of life. McDaniel found a positive correlation between health-promoting lifestyle and quality of life in 91 older adults, aged 60-92 years. It was also demonstrated that there exists a relationship between positive health outcomes and such behaviors as regular exercise, not smoking, maintaining ideal weight, eating a well-balanced diet and managing stress.

DIABETES, HYPERTENSION, CORONARY HEART DISEASE

DIABETES & HYPERTENSION:

Hypertension and type 2 diabetes are common comorbidities. Hypertension is twice as frequent in patients with diabetes compared with those who do not have diabetes. Moreover, patients with hypertension often exhibit insulin resistance and are at greater risk of diabetes developing than are normotensive individuals. The major cause of morbidity and mortality in diabetes is cardiovascular disease, which is exacerbated by

hypertension. Accordingly, diabetes and hypertension are closely interlinked because of similar risk factors, such as endothelial dysfunction, vascular inflammation, arterial remodeling, atherosclerosis, dyslipidemia, and obesity. There is also substantial overlap in the cardiovascular complications of diabetes and hypertension related primarily to microvascular and macrovascular disease. Common mechanisms, such as upregulation of the renin-angiotensin-aldosterone system, oxidative stress, inflammation, and activation of the immune system likely contribute to the close relationship between diabetes and hypertension.

INTRODUCTION:

TYPE 2 DIABETES MELLITUS AND HYPERTENSION

The prevalence of obesity and type 2 diabetes (T2D) continues to rise worldwide as lifestyles associated with low energy expenditure and high caloric intake are increasingly adopted, particularly in lower-income and developing countries. It is predicted that the number of cases of T2D will rise from 415 million to 642 million by 2040. Hypertension is even more common, rising in prevalence in the same countries,

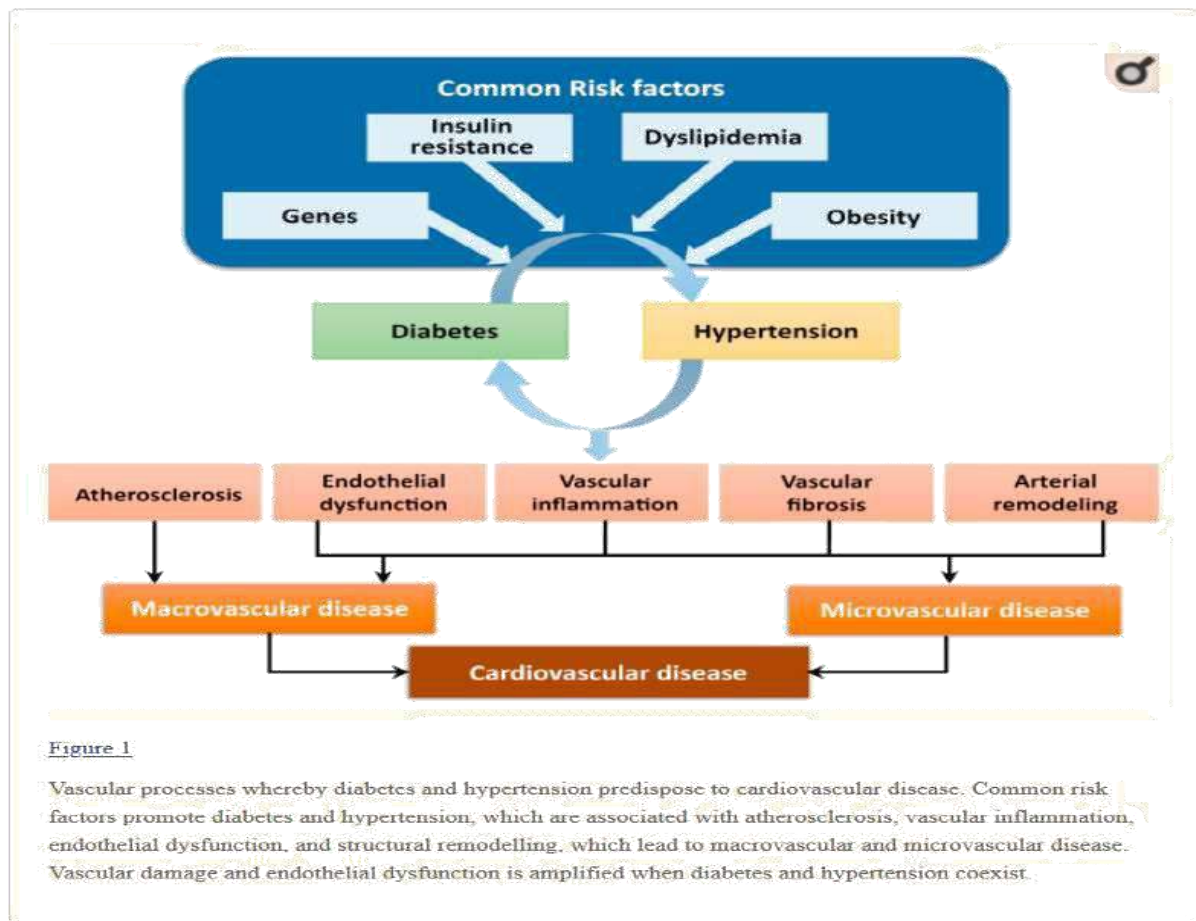
with a recent worldwide estimate of 1.39 billion cases.

Although T2D and hypertension can be simply diagnosed at the bedside, they are each complex and heterogeneous phenotype associated with an elevated risk of life-threatening cardiovascular disease (CVD). Their frequent coexistence in the same individual is not a coincidence, because aspects of the pathophysiology are shared by both conditions, particularly those related to obesity and insulin resistance. For example, in the San Antonio Heart Study, 85% of those with T2D had hypertension by the fifth decade of life, whereas 50% of those with hypertension experienced impaired glucose tolerance or T2D.

In health, insulin maintains glucose homeostasis by integrated actions on carbohydrate, protein, and lipid metabolism. Loss of sensitivity to aspects of insulin action (insulin resistance) principally affects the liver, muscle, and adipose tissues and is selective for glucose and lipid metabolism, e.g., sparing insulin's action to retain sodium in the distal tubule. Reduction in insulin-mediated glucose disposal leads to compensatory

hypersecretion of insulin to maintain homeostasis: Glucose intolerance ensues if this endocrine pancreas response is inadequate, although some obese individuals avoid T2D by virtue of a supranormal B-cell response. Recently, the role of adipose tissue in these associations has been increasingly appreciated.

Diabetes is associated with both macrovascular (involving large arteries such as conduit vessels) and microvascular (involving small arteries and capillaries) disease. Chronic hyperglycemia and insulin resistance play an important role in the initiation of vascular complications of diabetes and involve a number of mechanisms including (1) increased formation of advanced glycation end products (AGEs) and activation of the receptor for advanced glycation end products (RAGE) AGE-RAGE axis, (2) oxidative stress, and (3) inflammation. In addition, emerging evidence suggests a role for microRNAs (miRNAs) in the vasculopathy of diabetes (see further on). Hypertension is an important risk factor for diabetes-associated vascular complications, because hypertension itself is characterized by vascular dysfunction and injury (Fig. 1).



MACROVASCULAR DISEASE

CLINICAL FEATURES

Macrovascular (or cardiovascular) disease of larger conduit arteries is a complex inflammatory process leading to myocardial infarction, stroke, and peripheral artery disease. The primary pathologic process associated with macrovascular disease is atherosclerosis, which in diabetes is accelerated with extensive distribution of vascular lesions. T2D confers an approximate 2-fold elevation in CVD risk, equivalent to that

of a previous myocardial infarction. Moreover, patients with T2D have poorer outcomes after an acute coronary syndrome and higher rates of reinfarction and heart failure.

Elevation of CVD risk begins at the stage of prediabetes in association with insulin resistance and impaired glucose tolerance. As well as being the diagnostic hallmark of T2D, hyperglycemia is the principal determinant of microvascular complications of T2D and plays an important role in the pathogenesis of CVD. However, in established T2D, it is a relatively weak modifiable risk factor compared with hypertension, dyslipidemia,

and (unfortunately in many populations) cigarette smoking.

PATHOPHYSIOLOGICAL FEATURES

Insulin resistance is detectable for several years before the onset of T2D. It is associated with obesity, particularly central obesity, but may be present in lean individuals with hypertension. During calorie excess, adipocytes in obese humans—whether in subcutaneous or visceral areas—undergo hypertrophy. Visceral adipocytes are more susceptible to cellular death as they begin to enlarge and their stromal vascular fraction becomes infiltrated with macrophages.

These macrophages around dead adipocytes form “crown-like structures,” a histologic appearance that is associated with expression of cytokines (including tumor necrosis factor- α [TNF- α], interleukin-6 [IL-6]), and inducible nitric oxide synthase. These changes have been shown to coincide with the onset of insulin resistance and provide a pathophysiological link between metabolic and vascular disease.

In addition to these proinflammatory changes, adipocyte hypertrophy is associated with larger triglyceride stores, a higher lipolytic rate, and an atherogenic

lipid profile: elevated concentrations of small dense low-density lipoprotein cholesterol, high concentrations of triglycerides, triglyceride-rich remnants, very low-density lipoprotein cholesterol, and apolipoprotein B, usually in combination with low levels of high-density lipoprotein cholesterol. This profile is associated with increased production of leptin, decreased production of adiponectin, higher circulating levels of nonesterified fatty acids (NEFAs), and activation of mitochondrial oxidative stress pathways in vascular endothelial cells.⁷

These proinflammatory and metabolic consequences of obesity and insulin resistance result in endothelial dysfunction, a key antecedent and modulator of atherosclerosis that has been demonstrated not only in hypertension but also in prediabetes, first-degree relatives of individuals with T2D, and even insulin-resistant healthy individuals. It is characterized by disruption of the intricate physiological balance between vasoconstrictors (endothelin, angiotensin II) and vasodilators (nitric oxide, prostacyclin), growth promoting and inhibitory factors, proatherogenic and antiatherogenic factors, and procoagulant and anticoagulant factors. A substantial body of evidence suggests that impaired

endothelium-dependent vasodilation may in turn contribute to or exacerbate insulin resistance by limiting the delivery of substrate (glucose) to key target tissues.

In addition to these functional changes, an associated low-grade inflammation in endothelial and smooth muscle cells of the vascular wall causes cell proliferation, hypertrophy, remodelling, and apoptosis.²⁷

This accelerates disruption of the balance between the arterial wall scaffolding proteins elastin and collagen that determine vascular compliance, a form of “vascular aging,” which is a characteristic

phenotype in hypertension. Vascular stiffening leads to widening of arterial pulse pressure and increased pulsatile shear, exacerbating endothelial dysfunction and vascular disease.

INFLAMMATION AND THE IMMUNE SYSTEM

Links between inflammation and the immune system with metabolic dysfunction, hypertension, and cardiovascular morbidity are supported by extensive experimental data. This encompasses a number of immune metabolic aspects, including the key role of the tricarboxylic cycle or sphingosine-1-phosphate in the regulation of vascular inflammation. Clinical studies have shown that patients with T2D have increased total

leukocyte counts, particularly neutrophils and lymphocytes, that correlate with insulin sensitivity, which is in part mediated by inflammatory changes of adipose tissue. Inflammatory biomarkers are also useful in developing targeted cardiovascular therapies in the context of metabolic dysfunction. The link between inflammation and T2D is further supported by genetic studies and clinical trials showing protective effects of immune-targeted therapies and anti-inflammatory actions of classic antidiabetes drugs. Circulating and locally produced effector cytokines such as $\text{TNF-}\alpha$, interferon- γ , $\text{IL-1}\beta$, and IL-12 may influence insulin sensitivity of peripheral tissues and can modulate insulin release in the pancreatic islets. Increased glucotoxicity and lipotoxicity have been associated with immune cell infiltration of target tissues, thereby affecting diabetes-associated target organ damage and cardiovascular complications, including the development of metabolic cardiomyopathy. Inflammation is a key modulator of metabolic and diabetic CVD.

GENETIC EVIDENCE

Although genome-wide association studies (GWAS) for insulin resistance or T2D have not shown strong associations with immune-related genes, numerous

metabolic traits are linked to immune-related loci. Studies integrating metabochip approaches with GWAS have shown that classic immunometabolic genes including JNK signalling pathways (such as *MAP3K1*), nuclear factor kappa B (NF- κ B) regulators (*MACROD1*), inflammasome activators (NRF3), and interferon- γ receptor genes associate with T2D. This also corresponds to results of recent large T2D GWAS that identified genes related to macrophage function and antigen presentation (*MAEA*, *ST6GAL1*), and T-cell signalling (*CMIP* or *PTPRJ*). While trying to interpret these important studies, it should be appreciated that GWAS approaches have limitations, because only a small component of heritability of complex traits is directly explainable by single-gene variability.

CLINICAL EVIDENCE

Increasing clinical evidence indicates an immune component in T2D and its cardiovascular complications. Immune-targeted therapies currently available for the treatment of rheumatoid arthritis and autoimmune disorders, including anti-TNF therapies, may prevent insulin resistance as well as cardiovascular risk. A recent meta-analysis of studies with anti-TNF agents supports an overall protective effect of anti-TNF therapies on lifetime risk of

diabetes as well as insulin sensitivity and obesity.

A recent large proof of concept trial of anti-inflammatory therapy in patients after myocardial infarction (A Randomized, Double-blind, Placebo-controlled, Event-driven Trial of Quarterly Subcutaneous Canakinumab in the Prevention of Recurrent Cardiovascular Events Among Stable Post-Myocardial Infarction Patients With Elevated hsCRP [CANTOS]; canakinumab targeting IL-1 β) showed a clear reduction in the rate of cardiovascular events, albeit with an associated increase in the rate of severe infections. These results were particularly evident in high-risk patients, although effects on metabolic profile remain unclear. However, evidence that IL-1 β targeting may have significant metabolic benefits has been well established, as evidenced by improved profile insulin sensitivity in response to IL-1 β blockade. The potential beneficial effects of anti-inflammatory and immune-modulating agents in T2D and its complications may relate to direct vasoprotective effects. These studies have led to the rapid development of the concept of immunometabolism, clearly linking metabolic changes in the tissues to the regulation of inflammation as well as metabolic status of immune cells to their

activation. The latter can be characterized by a switch between oxidative phosphorylation and anaerobic glycolysis, which is observed in macrophages and T cells. This also emphasizes the importance of the interplay between vascular oxidative stress and the development of inflammation in adipose tissue and the vasculature.

ANTI-INFLAMMATORY PROPERTIES OF ANTIDIABETIC THERAPIES

Classic approaches improving metabolic health, such as weight reduction and the use of metformin, statin drugs, pioglitazone, and insulin have been shown to have anti-inflammatory effects. Metformin reduces C-reactive protein levels by 13%. More recently, a novel anti-inflammatory mechanism of metformin affecting M1/M2 polarization of macrophages has been shown to reduce obesity-associated low-grade inflammation, possibly because of adenosine monophosphate-activated protein kinase (AMPK) activation. These effects were modulated by AMPK and the AMPK analogue 5-aminoimidazole-4-carboxamide ribonucleotide, effects that appear stronger than those of metformin. Recent studies have shown that salicylates have anti-inflammatory effects that

involve inhibition of NF- κ B and that they also prevent diabetes and improve insulin resistance in experimental models and humans. Drugs such as glicazide and troglitazone, as well as N-acetylcysteine, decrease inflammatory markers in patients with diabetic nephropathy and diabetic retinopathy.

Epigenetics is another mechanism that may influence inflammation and immunometabolism in diabetes. Histone deacetylase (HDAC) inhibitors cause NF- κ B inhibition through acetylation of the p65 subunit. Givinostat (formerly ITF2357), an orally active HDAC inhibitor, has been shown to prevent the development of diabetes. Similarly, activation of sirtuin1, which is involved in inflammation, metabolism, and aging, has been shown to have anti-inflammatory properties in diabetes.

DIABETES, VASOPROTECTION, AND POTENTIAL NEW THERAPIES

Data from landmark clinical trials in T2D including UKPDS, ADVANCE, and Action to Control Cardiovascular Risk in Diabetes (ACCORD) demonstrate that treating comorbidities including hypertension and hypercholesterolemia is a more effective strategy for reducing cardiovascular complications than targeting blood glucose levels with

conventional agents. Antihypertensive drugs such as angiotensin-converting enzyme inhibitors, angiotensin-receptor blockers, mineralocorticoid-receptor blockers, and calcium-channel blockers may have direct vasoprotective effects, and their use may contribute, at least in part, to reduced vascular complications in patients with diabetes and concomitant hypertension. Tight control of BP has been shown to reduce cardiovascular risk in T2D: most recent US and Canadian guidelines recommend a target of < 130/80

Hg. Statin drugs and clopidogrel are also vasoprotective and may have extra benefit in patients with diabetes. Some of the beneficial effects of these drugs have been attributed to their antioxidant and anti-inflammatory properties.

New therapeutic approaches targeting oxidative stress, inflammation, and fibrosis are currently being developed to treat diabetes-associated cardiovascular complications. In particular, drugs that increase Nrf-2 activity, such as bardoxolone methyl, and strategies to inhibit the pyrin domain containing 3 (NLRP3) inflammasome, may have therapeutic potential. A novel bardoxolone methyl derivative, dh404, has been shown to attenuate endothelial dysfunction, reduce Nox1 expression, decrease oxidative stress, and inhibit inflammation

in diabetic mice, suggesting that upregulation of Nrf2 may have therapeutic potential to limit diabetes-associated vascular damage. Another example includes inhibition of dipeptidyl peptidase-4 by linagliptin, which reduces obesity-related insulin resistance and inflammation by regulating M1/M2 macrophage status. Other therapies on the horizon for the treatment of cardiovascular complications of diabetes include pentoxifylline (methylxanthine derivative and nonspecific phosphodiesterase inhibitor with anti-inflammatory and antifibrotic effects), ruboxistaurin (selective protein kinase C- β inhibitor), pirfenidone (TGF- β inhibitor), bindarit (MCP-1/CCL2 inhibitor), sulodexide (an oral formulation composed of 2 glycosaminoglycans), AKB-9778 (Tie2 activator), baricitinib (JAK/STAT inhibitor), and Nox inhibitors. The clinical benefit of these compounds awaits further confirmation, and novel nanotherapeutic approaches are being developed to target inflammation.

CARDIOVASCULAR DISEASES (CVDS)

(Sourced from WHO)

Cardiovascular diseases (CVDs) are a group of disorders of the heart and blood vessels and they include:

Coronary heart disease – disease of the blood vessels supplying the heart muscle;

Cerebrovascular disease – disease of the blood vessels supplying the brain;

Peripheral arterial disease – disease of blood vessels supplying the arms and legs;

Rheumatic heart disease – damage to the heart muscle and heart valves from rheumatic fever, caused by streptococcal bacteria;

Congenital heart disease – malformations of heart structure existing at birth;

Deep vein thrombosis and pulmonary embolism – blood clots in the leg veins, which can dislodge and move to the heart and lungs.

Heart attacks and strokes are usually acute events and are mainly caused by a blockage that prevents blood from flowing to the heart or brain. The most common reason for this is a build-up of fatty deposits on the inner walls of the blood vessels that supply the heart or brain. Strokes can also be caused by bleeding from a blood vessel in the brain or from blood clots. The cause of heart attacks and strokes are usually the presence of a combination of risk factors, such as tobacco use, unhealthy diet and obesity, physical inactivity and harmful use of

alcohol, hypertension, diabetes and hyperlipidaemia.

CORONARY HEART DISEASE:

Coronary heart disease happens when your coronary arteries get narrower and reduce the blood flow to the heart. It is the usual underlying cause of a heart attack

Coronary heart disease is associated with age and is a lifelong condition that affects many people. Coronary arteries are like small pipes that supply blood to your heart. They get narrower when fatty material gradually builds up inside and clogs them.

The fatty material is called plaque. The process of plaque building up is called atherosclerosis. This can start when you are young and may be well advanced by middle age. Stable plaque is generally not harmful but if it narrows the arteries too much, they may sometimes not deliver enough blood to your heart. The first effects of this are pain and discomfort called angina. Angina needs to be treated.

Unstable plaque has more fat, a thin cap and is inflamed. It may not severely narrow the artery, but it can develop a crack on the surface, exposing the contents of the plaque to the blood. Blood cells try to seal the gap in the surface with a blood

clot. The blood clot partially or completely blocks the artery.

A heart attack occurs when a blood clot completely blocks the flow of blood and seriously reduces blood flow to the heart muscle.

If you have had a heart attack, you have coronary heart disease.

There are risk factors that increase your chance of getting coronary heart disease. Know your risks

Coronary heart disease is sometimes referred to as CHD.

SYMPTOMS

Many people don't know they have heart disease until they have chest pain (angina or a heart attack).

If you are worried about your risk of heart disease, see your doctor or health practitioner to discuss your risks.

DIAGNOSIS

Your doctor will get some tests done to check your heart health. They may include:

electrocardiogram (ECG)

coronary angiogram or coronary computed tomography angiogram (CCTA)

echocardiogram (heart ultrasound)

blood tests

myocardial perfusion study.

TREATMENT

There's no cure for coronary heart disease. But there are treatments and lifestyle changes that can help you avoid a heart attack.

LIFESTYLE CHANGES

You may need to make some changes to stop your heart disease getting worse. Read more about looking after yourself.

PROCEDURES AND DEVICES

If your coronary heart disease is causing serious heart problems your doctor may want you to have a procedure done, like angioplasty and stent implantation, or coronary artery bypass graft surgery (CABG).

CARDIAC REHABILITATION

If you've had a heart attack or a procedure, you should be given information about a cardiac rehabilitation program where you can get help with living with coronary heart disease to reduce the risk of more problems.

IF YOU WANT TO KNOW MORE ABOUT CVDS (NOT FOR SYLLABUS):

detection and management using counselling and medicines, as appropriate.

DATA ON CVDS:

CVDs are the number 1 cause of death globally: more people die annually from CVDs than from any other cause. An estimated 17.9 million people died from CVDs in 2016, representing 31% of all global deaths. Of these deaths, 85% are due to heart attack and stroke. Over three quarters of CVD deaths take place in low- and middle-income countries.

Out of the 17 million premature deaths (under the age of 70) due to noncommunicable diseases in 2015, 82% are in low- and middle-income countries, and 37% are caused by CVDs.

Most cardiovascular diseases can be prevented by addressing behavioural risk factors such as tobacco use, unhealthy diet and obesity, physical inactivity and harmful use of alcohol using population-wide strategies.

People with cardiovascular disease or who are at high cardiovascular risk (due to the presence of one or more risk factors such as hypertension, diabetes, hyperlipidaemia or already established disease) need early

RISK FACTORS FOR CARDIOVASCULAR DISEASE

The most important behavioural risk factors of heart disease and stroke are unhealthy diet, physical inactivity, tobacco use and harmful use of alcohol. The effects of behavioural risk factors may show up in individuals as raised blood pressure, raised blood glucose, raised blood lipids, and overweight and obesity. These “intermediate risks factors” can be measured in primary care facilities and indicate an increased risk of developing a heart attack, stroke, heart failure and other complications.

Cessation of tobacco use, reduction of salt in the diet, consuming fruits and vegetables, regular physical activity and avoiding harmful use of alcohol have been shown to reduce the risk of cardiovascular disease. In addition, drug treatment of diabetes, hypertension and high blood lipids may be necessary to reduce cardiovascular risk and prevent heart attacks and strokes. Health policies that create conducive environments for making healthy choices affordable and available are essential for motivating people to adopt and sustain healthy behaviour.

There are also a number of underlying determinants of CVDs or "the causes of the causes". These reflect the major forces driving social, economic and cultural change – globalization, urbanization and population ageing. Other determinants of CVDs include poverty, stress and hereditary factors.

COMMON SYMPTOMS OF CARDIOVASCULAR DISEASES

Symptoms of heart attacks and strokes

Often, there are no symptoms of the underlying disease of the blood vessels. A heart attack or stroke may be the first warning of underlying disease.

Symptoms of a heart attack include:

- pain or discomfort in the centre of the chest;
- pain or discomfort in the arms, the left shoulder, elbows, jaw, or back.

In addition, the person may experience difficulty in breathing or shortness of breath; feeling sick or vomiting; feeling light-headed or faint; breaking into a cold sweat; and becoming pale. Women are more likely to have shortness of breath, nausea, vomiting, and back or jaw pain.

The most common symptom of a stroke is sudden weakness of the face, arm, or leg,

most often on one side of the body. Other symptoms include sudden onset of:

- numbness of the face, arm, or leg, especially on one side of the body;
- confusion, difficulty speaking or understanding speech;
- difficulty seeing with one or both eyes;
- difficulty walking, dizziness, loss of balance or coordination;
- severe headache with no known cause;
- and
- fainting or unconsciousness.

People experiencing these symptoms should seek medical care immediately.

(THE ITALICS PART IS NOT IN SYLLABUS BUT ONLY FOR REFERENCE AND MEANING MAKING OF THE ABOVE CONTENT)

RHEUMATIC HEART DISEASE

Rheumatic heart disease is caused by damage to the heart valves and heart muscle from the inflammation and scarring caused by rheumatic fever. Rheumatic fever is caused by an abnormal response of the body to infection with streptococcal bacteria, which usually begins as a sore throat or tonsillitis in children.

Rheumatic fever mostly affects children in developing countries, especially where

poverty is widespread. Globally, about 2% of deaths from cardiovascular diseases is related to rheumatic heart disease.

SYMPTOMS OF RHEUMATIC HEART DISEASE

Symptoms of rheumatic heart disease include: shortness of breath, fatigue, irregular heartbeats, chest pain and fainting.

Symptoms of rheumatic fever include: fever, pain and swelling of the joints, nausea, stomach cramps and vomiting.

WHY ARE CARDIOVASCULAR DISEASES A DEVELOPMENT ISSUE IN LOW- AND MIDDLE-INCOME COUNTRIES?

At least three quarters of the world's deaths from CVDs occur in low- and middle-income countries.

People in low- and middle-income countries often do not have the benefit of integrated primary health care programmes for early detection and treatment of people with risk factors compared to people in high-income countries.

People in low- and middle-income countries who suffer from CVDs and other noncommunicable diseases have less access to effective and equitable

health care services which respond to their needs. As a result, many people in low- and middle-income countries are detected late in the course of the disease and die younger from CVDs and other noncommunicable diseases, often in their most productive years.

The poorest people in low- and middle-income countries are affected most. At the household level, sufficient evidence is emerging to prove that CVDs and other noncommunicable diseases contribute to poverty due to catastrophic health spending and high out-of-pocket expenditure.

At macro-economic level, CVDs place a heavy burden on the economies of low- and middle-income countries.

HOW CAN THE BURDEN OF CARDIOVASCULAR DISEASES BE REDUCED?

“Best buys” or very cost-effective interventions that are feasible to be implemented even in low-resource settings have been identified by WHO for prevention and control of cardiovascular diseases. They include two types of interventions: population-wide and individual, which are recommended to be used in combination to reduce the greatest cardiovascular disease burden.

Examples of population-wide interventions that can be implemented to reduce CVDs include:

- comprehensive tobacco control policies
- taxation to reduce the intake of foods that are high in fat, sugar and salt
- building walking and cycle paths to increase physical activity
- strategies to reduce harmful use of alcohol
- providing healthy school meals to children.

At the individual level, for prevention of first heart attacks and strokes, individual health-care interventions need to be targeted to those at high total cardiovascular risk or those with single risk factor levels above traditional thresholds, such as hypertension and hypercholesterolemia. The former approach is more cost-effective than the latter and has the potential to substantially reduce cardiovascular events. This approach is feasible in primary care in low-resource settings, including by non-physician health workers.

For secondary prevention of cardiovascular disease in those with established disease, including diabetes,

treatment with the following medications are necessary:

- aspirin
- beta-blockers
- angiotensin-converting enzyme inhibitors
- statins.

The benefits of these interventions are largely independent, but when used together with smoking cessation, nearly 75% of recurrent vascular events may be prevented. Currently there are major gaps in the implementation of these interventions particularly at the primary health care level.

In addition, costly surgical operations are sometimes required to treat CVDs. They include:

- coronary artery bypass
- balloon angioplasty (where a small balloon-like device is threaded through an artery to open the blockage)
- valve repair and replacement
- heart transplantation
- artificial heart operations

Medical devices are required to treat some CVDs. Such devices include pacemakers, prosthetic valves, and patches for closing holes in the heart.

PSYCHONEUROIMMUNOLOGY [CANCER, HIV/AIDS]

(This is a vast topic filled with complex and big terminologies. Don't go very deep into the topic. Simply grasp the idea and certain name of the terms used here. This topic has been referred from an article by healthline)

Psychoneuroimmunology is the study of interactions between behavior, the brain, and the immune system. It addresses how psychological factors influence the immune system and physical health through neural and endocrinological pathways. These relationships are especially relevant to immunologically mediated health problems, including infectious disease, cancer, autoimmunity, allergy, and wound healing.

This new field of psychoneuroimmunology (PNI) has developed over the past 10 to 15 years.

(NOT PART OF THE SYLLABUS BUT IMPORTANT TO UNDERSTAND PSYCHONEUROIMMUNOLOGY)

BACKGROUND

The immune system is an integral part of the body involved in protection of body from various diseases. Its presence and role in protection against disease was known since ancient times. Its anatomy and physiology have become clearer only in the last fifty years or so. Broadly, its components can be divided into the cellular and humoral components. The cellular component i.e. the cell-mediated immunity is constituted by the T-lymphocytes which are derived from the thymus gland. The T-lymphocytes constitute about 75% of all lymphocytes. The humoral component is constituted by the bone marrow derived B-lymphocytes which make about 10% of all lymphocytes. The rest 15% of cells belong to a heterogeneous group which primarily include natural killer cells. These cells are histologically and physiologically different and can be recognized by the presence of unique cluster determinant antigens (CD) present on their cell membrane. T-lymphocytes are further differentiated into various categories depending on their functions and presence

of CD markers. The T-helper (Th) cells and the T-suppressor (Ts) cells are the main two categories of T lymphocytes which act to increase and decrease the immune response, therefore they work in coordination with each other to maintain homeostasis. They also play an important role in cancer immunity and various autoimmune diseases. The B-lymphocytes act by synthesis of antibodies – a specialized form of protein molecules active against the particles like bacteria, viruses, etc. which are pathogenic and/or foreign to the organism. The recognition of self and non-self molecules is again determined by a complex system of molecules in which the Major Histocompatibility Complex (MHC) is the most important. Both cell-mediated and humoral-mediated immunity also work in a coordinated way through secretion of cytokines and interleukins from lymphocytes which guide their interaction. Although the immune system is a perfectly specialized system of the body involved in defense against foreign and pathogenic particles, it also interacts bilaterally with the other body systems including the CNS. This interaction attracted the researchers all over so much that it acquired a separate meaning and status as **psychoneuroimmunology**. The immune molecules particularly the MHC also

regulate the development of nervous system. Like other body responses, the response of immune system can also be conditioned. The neurons have receptors for interleukins and the immune cells respond to the neurotransmitters through these receptors. Under the stressful situation whether physical or psychological the CNS releases stress biochemicals such as cortisol through HPA activation; activates ANS to prepare the body for fight and flight response. However, such activation paradoxically lowers the immune functions and makes the organism susceptible to infections and other pathologic processes as the requirement at such times is a “strong defense”. The evidence to this conclusion has come from various animal and human studies. In the depressive and anxiety disorders the indices of immune functioning have been consistently seen to be lower than the normal. Under social isolation and ambiguous situations, the immune response is consistently seen inadequate in various studies. The CNS-Immune System interaction is quite intricate one. Despite plethora of studies the exact mechanism of this interaction is still beyond a reasonable level of understanding. Clearly more studies are required to elucidate this complex mechanism.

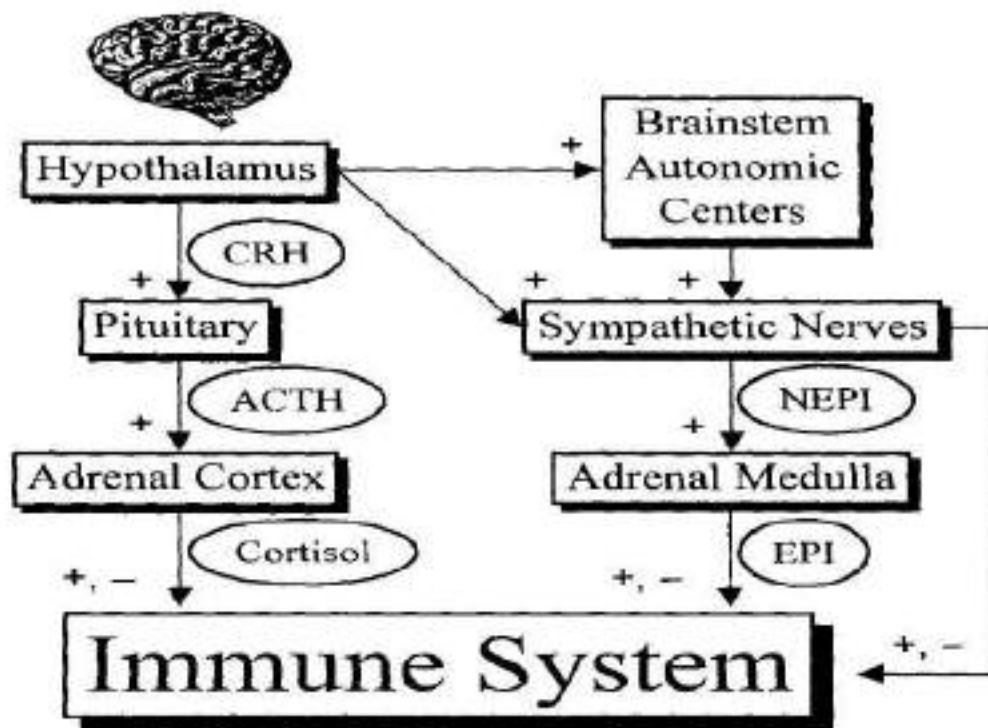


Figure 4.1 Neural and endocrine pathways that may modulate the immune system. The hypothalamic-pituitary-adrenal (HPA) and sympathoadrenomedullary (SAM) axes are represented, which influence the immune system in multiple ways.

Abbreviations:

CRH—corticotropin-releasing hormone,

ACTH—adrenocorticotropin hormone,

NEPI—norepinephrine,

EPI—epinephrine, (+) stimulation, (–) inhibition.

LET'S UNDERSTAND THIS IN SIMPLER TERMS WITH A PINCH OF DETAIL.

WHAT IS PSYCHONEUROIMMUNOLOGY?

Psychoneuroimmunology (PNI) is a relatively new field of study that looks at the interactions between your central nervous system (CNS) and your immune system. Researchers know that our CNS and immune system can communicate

with each other, but they only recently started to understand *how* they do it and what it means for our health.

The nerves in your brain and spinal cord make up your CNS, while your immune system is made up of organs and cells that defend your body against infection. Both systems produce small molecules and proteins that can act as messengers between the two systems. In your CNS,

these messengers include hormones and neurotransmitters. Your immune system, on the other hand, uses proteins called cytokines to communicate with your CNS.

WHAT THE RESEARCH SAYS

There's plenty of existing research about the effects of stress on the immune system. Many of these studies focus on the release of cytokines in response to both physical and psychological stress.

A cytokine is a small protein that's released by cells, especially those in your immune system. There are many types of cytokines, but the ones that are generally stimulated by stress are called pro-inflammatory cytokines.

Under normal circumstances, your body releases pro-inflammatory cytokines in response to an infection or injury to help destroy germs or repair tissue. When you're physically or emotionally stressed, your body also releases certain hormones, including epinephrine (adrenaline). These hormones can bind to specific receptors that signal for the production of pro-inflammatory cytokines.

Here's a look at some of the recent research and discussions around PNI in the medical community:

A 2016 review of existing studies found that stressful experiences during childhood can increase the release of cytokines by your immune system. This is associated with an increased risk of mental illness in adulthood. Researchers believe that this early release of cytokines may cause changes in the brain that increase a person's risk of developing a mental illness later in life.

A 2015 article noted that rats produced different types of cytokines depending on the type of stress they experienced. For example, an injury produced one type of pro-inflammatory cytokine. Meanwhile, exposure to a social stressor, such as separation from a close family member, released a different type of pro-inflammatory cytokine.

Another 2016 review found that both sleep disturbances and sleeping too much seemed to trigger the release of pro-inflammatory cytokines.

A 2011 review exploring the link between stress and the immune system found that stress may play a role in conditions that affect the immune

system, such as cancer, HIV, and inflammatory bowel disease.

WHAT ARE SOME EXAMPLES OF PNI?

What does all of this new knowledge mean for our health? Keep reading to learn more about the role that PNI plays in several common conditions.

PSORIASIS

Psoriasis is a great example of how your immune system, CNS, mental health, and stress levels are all intertwined. It's a chronic condition that causes your skin cells to grow too quickly. Your body usually sheds extra skin cells, but if you have psoriasis, these extra cells build up on your skin's surface. This can lead to intense itching and pain.

The overgrowth of skin cells in psoriasis is due to the release of cytokines from your immune system. We know that psychological stress may worsen or trigger episodes of psoriasis. Indeed, people with psoriasis tend to have increased levels of cortisol, a stress hormone.

Your hypothalamus, which is part of your CNS, is responsible for cortisol production. When it senses stressors, it signals your nearby pituitary gland, which

signals for cortisol production. This, in turn, can trigger the release of pro-inflammatory cytokines by your immune system. These cytokines then trigger an overgrowth of skin cells.

In addition, people with psoriasis often report having psychological conditions, such as depression, increased stress, and suicidal thoughts. Previous research has linked an increase in cytokine levels with major depression.

There's currently no cure for psoriasis, but new developments in the field of PNI could change this in the future. In the meantime, here's how to manage it at home.

CANCER

A 2013 review of many studies exploring the relationship between PNI and cancer found evidence to suggest that:

Women with genetic risk factors for developing cancer showed immune system abnormalities in response to stress.

There appears to be a link in people with breast cancer between depression, the quality of social support they have, and immune cell activity.

People with breast, cervical, or ovarian cancer who reported feeling stressed or lonely had abnormalities in their immune systems.

Communication between the immune system and brain may impact symptoms that are related to cancer treatment, including fatigue, depression, and difficulty sleeping. Stressful experiences and depression may be associated with a poorer survival rate for several types of cancer.

CORONARY ARTERY DISEASE

A review from 2010 looking at the relationship between stress, immune function, and coronary artery disease echoed other studies suggesting that psychological stress increases the production of pro-inflammatory cytokines.

This increase in pro-inflammatory cytokines is associated with an increase in heart rate and blood pressure. In addition, the production of cytokines by your immune system promotes feelings of sickness or fatigue. According to this review, this reaction isn't immediately harmful. However, long-term stress and cytokine production may contribute to the development of cardiac disease.

PSYCHOLOGY AND TECHNOLOGY INTERFACE: DIGITAL LEARNING; DIGITAL ETIQUETTE: CYBER BULLYING; CYBER PORNOGRAPHY: CONSUMPTION, IMPLICATIONS; PARENTAL MEDIATION OF DIGITAL USAGE.

Technology is everywhere ... at work, home or at leisure time. Obviously, this is not new, since cars, computers, mobile phones (especially, smartphones) and the various gadgets which had occupy our existence for quite a while. In the twentieth century, the exponential growth in various technical fields and the emergence of modern technologies (consecutive, especially to the progress of computer science) caused the appearance of the psychological ambivalence, because, modern technologies, generate, in the same measure, comfort and disasters.

Between the two extreme positions formed, represent by *technophilia* and *technophobia*, arose a variety of issues on the psychological and social impact of modern technology, which has fueled and

will further fuel a intense debate on the advantages and simultaneously carrying dangers involves by the development of techniques (and technologies!), but also on how its rules. At the psycho-dynamic level, *technophilia* (attraction to technology) generates its psychological opposite, namely, *technophobia* (rejection of technology). Technophilia and technophobia are the two extremes of the relationship between technology and the human being, but especially, between technology and society.

TECHNOPHILIA AND TECHNOPHOBIA – TERMINOLOGY

The person attracted to technology, the "technophile", takes the most or all technologies in a positive manner, enthusiastically adopting new forms of technology and view this as a way to improve his living conditions and combat social problems (Amichai-Hambrurger, 2009). However, it was found that, with the continued proliferation of modern technologies in almost every aspect of our existence, the number of people who manifest fear of them is increasing. Fear can go from avoiding technology to organic symptoms such sweating and palpitations, even if they only think about using

such technology ... The phenomenon would affect about one third of all population. The avoidance of the new technologies by some people, has led to the hypothesis of "technophobia" or "computerophobia" (these terms are used interchangeably). When factors as anxiety and attitude, or more specifically, the computer anxiety and the attitude toward computer are beginning to combine, the first condition is a requirement for the second, heving as result the appearance of irrational fears and anxieties expressed by avoiding behavior, paradoxical, sometimes absurd. Basically, the technophobia beginning to take shape. Mental resistance to new technology, manifested in the form of avoidance of computers was well presented in the literature, by the term "technophobe" or "computerophobe", used to describe people who refuse to use computers when they has this opportunity or are required to do. Although we can't speak about a phobia in the classic sense of the term (as in agoraphobia, for example), but there are many similarities at etiology and "treatment", that justify the term of "technophobia".

Technophobia not mean fears about giving up, change the job or concerns about radiation emitted from the screen, but rather an emotional response and negative attitudes relative to technology, that the

technophoberecognizes to be irrational. The prejudgement that technophobia is a phenomenon that only affects the elderly population, has been disproved long time ago. Current research shows that things are far from being improved.

TECHNOPHILIA

Technophilia (from the Greek τέχνη - *technē*, "art/ artifact, skill and understanding" and φίλος - *philos*, "love"), refers generally to the enthusiasm generated by the use of technology, particularly new technologies, such as: personal computers, Internet, mobile phones and even the technologies of "home cinema". The term emerged in the 1960s, is mainly used in sociology, when is examined the interaction between individuals and society. Technophilia is defined as attraction, enthusiasm of the human individual determined by the activities which involve the use of advanced technologies. It is expressed by easily adaptation to the social changes brought by technological innovations. The term of technophilia is used to highlight how technology can evoke strong futuristic positive feelings. However, reverent attitude towards technology that determines technophilia can sometimes prevent a realistic assessment of

environmental and social impact of technology on society. The technophiles has no fear about the effects of technological development on society, as is the case for technophobes. Technophilia refers to "technological determinism", theory emphasize that human society has not the power to resist towards the influences of technology. A number of technologies are used as an expression of personal narcissism. The technophiles enjoy using technology and focus on its egocentric benefits.

The concept of addiction is often associated negatively with technophilia, while targeting only those technophiles who become excessive and obsessive bound to the forms of technology they possess. So far as, in the eighteenth century, industrialized societies (notably the UK and France) have relied on their development and expansion of the multiplication and improvement techniques in order to obtain *effective /efficient* to their producers and *confort* to consumers it can be said that *such societies are by their nature technophiles*. Undoubtedly, *technophilia tend to be the norm* in most contemporary societies. As technophilia is associated with the phenomenon of psychological and social "normality", we will not insist on this theoretical

approach. We remember only that in the extreme forms, such as Internet Addiction that we expose briefly, in the following lines -, technophilia can acquire a pathological aspect.

INTERNET ADDICTION

Internet Addiction or *Internet Addiction Disorder* (IAD) is now known mainly as the *compulsive internet use* (CIU).

The tendency is to avoid the term addiction, reflecting the long-term dependence and is not limited to a single cause. Online activities (e.g., shopping on the net) are considered problematic if it has a compulsive nature. Other activities, such as reading or playing video games become problematic only when excessively interferes with daily life. In 1949, Otto Fenichel was the one who would speak first of "addiction without drugs", which is the direct expression of Internet addiction. The Anglophone term *Internet Addiction* was first used by the American psychologist Kimberly Young, at a colloquium of the American Psychological Association (APA), in Toronto (1996) and was subsequently taken over in specialized terminology. However, the term IAD was originally proposed, also by an American

psychologist, named Ivan Goldberg in 1995 (but he used it in the pejorative sense). Currently, IAD is an anachronism present in textbooks of DSM (*Diagnostic and Statistical Manual of Mental Disorders*), from DSM-IV. Although in the DSM-IV, Internet Addiction has emerged as an independent entity, *American Medical Association* (AMA) and the *American Society of Addiction Medicine* (ASAM) are against the inclusion of Internet Addiction as a

formal diagnosis in DSM and recommends the study of addictions, especially relative to video games. (in Price, 2011). In the opinion of I. Goldberg, Internet Addiction is more a symptom and not a disorder itself. To describe hypothetical Internet Addiction, Goldberg draws an analogy with gambling. Goldberg notes that "Internet addiction can cause the denial or the avoidance of other current issues of life." The addictive behaviour expresses the socioemotional immaturity in the individual's inability to build a true and solid psychosocial identity. Cyberdependence is characterized by "all mental disorders related to the use of computer equipment, harmful to the human individual". This latest disorder can be diagnosed with tests which highlighted the obsessional behavior. The results of these tests should be interpreted, but with caution. Associated to the Internet

Addiction appear the *Communication Addiction Disorder* (CAD or *compulsivetalking*), behavioral disorder linked to the need to constantly communicate with others, even when there is no real justification for such communication. CAD is a "theoretical" disorder in which users become to be addicted to social networks (or "social media elements" of the Internet!), such as Facebook or YouTube. Sometimes it happens that these activities generate intrapsychic conflict and guilt. (Bucy, & Newhagen, 2004).

TECHNOPHOBIA

Technophobia (from the Greek τέχνη - *technē* and φόβος - *phobos*, "fear") is fear, dislike or discomfort by using modern technologies and complex technical devices (especially computers). The term is related to *cyberphobia*. Technophobia is defined as an irrational fear or anxiety caused by side effects of advanced technologies. Definition involves two components: first the fear for side effects of technological development on society and the environment; and second, the fear of using technological devices such as computers and advanced technology. A number of authors consider that technophobia always has a pathological character, since it refers

to an exaggerated and unjustified fear. Although the accent is on pathological, abnormal aspects, some of these fears maybe rightly justified (e.g., radiation exposure). Technical progress can sometimes be detrimental to ecosystem health.

The main reasons for the opposition in terms of technical development are not only ecological (it is considered that technology destroys the environment), and ethics (biometrics and video surveillance, for example, are regarded as indicating serious damage of individual freedom, creating a progressively social control likely to degenerate into a new form of totalitarianism).

Some examples of technophobe ideas can be found in various art forms: from literary works such as *Frankenstein*, movies like *Metropolis* or famous Charlie Chaplin's, *Modern Times*. Many of these works portray the dark side of technology as perceived by technophobes.

DIGITAL LEARNING

Online Collaborative Learning (OCL)

proposes a modern learning theory with such a framework that considers the universality of the Internet and the societal shift toward collaborative learning. It features the intellectual growth rather than the conventional teaching approach.

Proposal is an Online Collaborative Learning (OCL) which renders a model of learning in which students are encouraged and supported to work together toward knowledge. To invent, to explore ways to innovate, and, by so doing, seek to understand, therefore,

stimulate a sharp solution-finding rather than presenting literature statements. This learning method is collaborative so that students are expected to work collectively through it, alternatively to the approach that merges individual and closed decisions to accomplish an outcome. A constructivist framework approach points to the value of educators as key drivers for the OCL instructional process. Aforementioned encourages an inherent disposition from student's awareness through real-case scenarios and encourages curiosity and discussions that support their autonomy. Likewise, the term participatory pedagogy represents a shift in educational paradigm that combines instruction with a net-worked and connected education. The results mentioned above ends in a promotion of the student's engagement and their enthusiasm to focus on enhancing theoretical transmission of knowledge.

DIGITAL LEARNING SUCCESS

To what regards the use of Information and Communication Technology (ICT) as a digital learning resource for OCL, educational stakeholder's attitudes and perceptions of ICT's significantly impact their student's digital literacy. The student's use of ICT is related to teacher's confidence level of their digital competencies. Evidence shows a significant relationship between the efficient use of ICT as a learning methodology and the "respondents' age", as well as, high relationship between teachers "technological anxiety" and age. Nevertheless, educators may require continuous professional development and training regarding their ICT capabilities. Skillful teachers in a poor ICT learning environment can perform better. Moreover, a survey reported that 20 to 25 percent of European students are taught by digitally competent teachers. ICT supports the learning process by the use of text, images and audio and by that, each person can create their story and engage in an in-depth learning. Digital Storytelling stands as one technology application that is well positioned to take advantage of user-directed content and to help instructors use technology productively in their classes. Nonetheless, the use of innovative ICTs alone does not guarantee effective learning processes and outcomes in OCL

environments. There is evidence to favor that digital storytelling (DST-OCL, here and further on) is more effective than a lecture-based (G-OCL, here and further on) instructional strategy. Adopting a DST-OCL strategy promotes student's cooperation in learning activities, such as script review, image selection and storyboarding as well as, "online communication", "interactivity" and "privacy". There is evidence to conclude also that using a DST-OCL student are relatively more motivated to have a discussion between group members, in a constructivist point of view. To what regards digital problem-based learning (PBL) research, concludes that students that are provided with autonomy support have a greater sense of autonomy and those who received structural directives, clear expectations, guidance and procedures contribute experience greater sense of competence. The study in matter also concludes that both autonomy support and structure in a OCL promotes a positive effect on intrinsic motivation. Even with a single aforementioned dimension, students fostered their intrinsic motivation, therefore, they are mutually supportive. However, if both are absent, autonomy support and structure, evidence shows a low intrinsic motivational usage of ICTs. Nonetheless, this study also confirmed that structure OCL strategy has a positive

learning outcome but autonomy support doesn't. Yet, if both are combined, autonomy support is able to stimulate greater outcomes.

ADVANTAGES:

1. Extendibility, Accessibility, and Suitability - Users can proceed through a training program at their own pace and at their own place. They can also access the training at any time, receiving only as much as they need. In other words, "just in time and just enough."

Quicker (and cheaper) turnaround of finished product.

Collaborative and exploratory learning environments.

Easy and affordable training delivery - Chances are (and you'll want to access this before embarking on WBT) your audience has access to a browser. Validate what browser(s) your audience has and what version(s) and you can capitalize on the delivery vehicle.

Cross Platform - WBT can be accessed by web browsing software on any platform: windows, MAC, Unix, etc. Basically, you can deliver your training course to any machine over the Internet or company intranet without having to develop a different course for each unique platform.

Inexpensive worldwide distribution - No separate or distinct distribution mechanism is needed (i.e., distributing CD-ROMs for CBT training). WBT can be accessed from any computer anywhere in the world while at the same time keeping delivery costs down.

Reduced technical support - Web-based training courses decrease some

of themore “potent and deadly” technical support issues that often enshroud technology-based training.

Ease of content update - The changes you make to any of your content are immediately available to your learning audience across the world.

Installation options on private networks for security or greater bandwidth. If you opt for intranet delivery, you have more control over plug-ins and bandwidth, giving you more options for inclusion in your WBT.

Travel cost and time savings - Learning is delivered directly to the learner instead of the other way around.

Web browsers and Internet connections are widely available - Most computer users have access to a browser, such as IE4 or Netscape and are connected to a company's intranet, and/or have access to the Internet.

WBT-based development is easier to learn and pick up than CBT-based development. You should be able to draw from a larger pool of WBT developers than is available for creating traditional CBT.

Vast, untapped market for training - The untapped WBT market is still large. If you're looking for commercial markets, the Internet also offers a huge audience hungry for material via the Net.

Access is controllable - You can direct and monitor who receives web training - when, how many times, and in what sequence.

Billing options - You can bill—and collect on that bill—through Net distribution, billing by user ID, number of accesses, date/time of access, or any other means by which you want to assess usage.

Direct access to many other training resources - The Internet gives access to the largest library in the world. Capitalize on the offerings that have already been created, and use them to enhance the learning you are distributing.

DISADVANTAGES:

Limited formatting of content in current browsers - The WBT you create will not resemble the CBT you might be familiar with because of Net bandwidth constraints. So if your content relies on a lot of media “bells and whistles,” or particular formatting, the Net might not be the best delivery medium.

Bandwidth/browser limitations may restrict instructional methodologies - Again, you are constrained by the technology. If your content relies on a lot of video, audio, or intense graphics, and your audience isn't on a T1 line, Net delivery will only frustrate your learners.

Limited bandwidth means slower performance for sound, video, and large graphics. Know the bandwidth available to your audience—and what's reasonable “wait” time for them—before you commit to the Net. On average, most mobile end-users are still running on a 28.8 kbps connection.

Someone must provide web server access, control usage, and bill users (if applicable) - The Net doesn't magically solve all of your resource issues. Plan on having someone oversee some of these issues.

Time required for downloading applications - Again, be very, very

aware of download times—and we still recommend looking at delivery options considering a 28.8 kbps modem if you are delivering training to an external audience or over the Internet. Your training might be great, but if your audience isn't patient enough to wait for it to download, it doesn't matter.

Student assessment and feedback is limited - The Internet provides a wonderful means to get all kinds of information back and forth to your audience, but it also makes it harder to assess some types of student feedback and information.

Many, if not most, of today's web-based training programs are too static, with little if any interactivity - This is probably due to the bandwidth limitation, but if we deliver poor, page-turning training, we can't expect stellar results from our learners.

Cannot design and develop robust multimedia courses - The bandwidth limitations of the Net constrain what can be delivered effectively.

Are computers replacing human contact? - The Net is not right for all training.

Newness- It may take longer designing and developing WBT courses, the first time around. Because of its recent emergence to the training arena, new technologies always require time, experience, and money in order to take full advantage of its capabilities. The great thing, however, is you'll learn new skills and knowledge with each new course.

Web-based training has high-fixed costs- Some people assume that since you don't need a CD-ROM drive, you have no additional costs. Not so. It's important that you consider doing a pilot before proceeding further into the

WBT game. Validate what works well, when, and at what it cost.

DIGITAL ETIQUETTE: CYBER BULLYING

(This section is also important for paper

I. This section has been referred from an article by Jennifer Preece)

Digital etiquette is a term that describes the positive behavioral or attitude of digital technology users. The scope of this term is wide, encompassing not only internet usage but also other technologies such as mobile phones, tablets etc. Some also refer to the term as Digital citizenship in view of the virtual world created by the usage of various technologies. Hence technology users are called digital citizens (Ribble, Bailey & Ross, 2004). In a focused scope of internet, this etiquette is called netiquette while the Internet users are often referred to as online citizens or netizens (Preece, 2004). In another perspective, this virtual world is also called the cyber-world. The power of the world-wide web is now bigger than its first initial introduction. This technology made the impossible possible (i.e. people all over the world could read one piece of content from the same source at any given time and location).

While we were cheering at the success of the technology, the setback was just around the corner. The different backgrounds of internet users determine the interpretation of content. As highlighted by Preece (2004), “Internet users come from many cultures and walks of life. They arrive with a mix of expectations using a variety of technologies, which they access in different ways.” The technology has also been misused by users in many ways. It has acted as a gateway for criminals to freely conduct their activities in anonymity. This is referred to as cyber-crimes. Despite the various setbacks, the diversity of internet usage alleviates its value with an increase in the demographic background of users accessing it especially at different ages range for different purposes. The technology has now reached our children as early as four months old. Without proper guidance, these children could become potential victims to various crimes in the virtual world.

Knowledge management is interpreted as the ability to ‘selectively capture, archive and access the best practices of work-related knowledge...for both employees and managers’ (Bergeron, 2003, pg. 6). In order to achieve this, a supportive working culture is essential within an organization. By cultivating positive digital activities from an early age, such working culture is feasible as a developing mind is easily molded and influenced (Siegel, 2012). The importance of digital etiquette from a knowledge management (KM) perspective is that it indirectly allows the children to be aware of acceptable online interactions at an early age. Such indirect training strengthens a child’s positive social norms, which would be brought into their adulthood and working environment. This training would provide a platform for developing an effective knowledge management structure, which depicts the best practices for an organization. In recent years, there is a significant increase in the number of internet usage among children (Madden et al., 2013) where not all children will use the Internet wisely according to the cyber ethics. “Cyber-ethics is the discipline of using appropriate and ethical behaviors and acknowledging moral duties and obligations pertaining to online environments and digital media”. Children like to use internet for socializing purposes

at home and they may have been exposed only to the basic online awareness via their parents. Although Internet use provides numerous valuable benefits but misusing it can cause serious risks. Due to the lack of awareness and knowledge about digital etiquettes, it can lead to undesired negative impacts on children. Therefore, it is important to expose and make them understand about unethical online activities (Freestone & Mitchell, 2010) such as plagiarism, software piracy and privacy violation, (Jung, 2009) not to mention the existing online threats (Lazarinis, 2010) such as viruses.

WHY IS ETIQUETTE ONLINE IMPORTANT?

As Internet settlers form cyber-communities, the importance of etiquette grows. Lack of etiquette is weakening sociability and even destroying online communities. Etiquette online is not just *nice* to have, it is *necessary*. Like the pioneers of the wild west, early Internet adopters were a rough and tumble gang. An occasional sarcastic comment, expletive, or confrontational challenge was part of

the fun. But times have changed. Today's settlers flock online in the millions. A single word aptly summarizes these settlers – '*diverse*'. These Internet users come from many cultures and walks of life. They arrive with a mix of expectations using a variety of technologies, which they access in different ways. The new settlers include children and adults, healthy and infirm, eager and reluctant. English-speakers dominate but other languages and cultures are gaining prominence. The number of women matches the number of men. Even, low-income families and those with limited education are beginning to appear (Pew, 2003). This diversity can have its charm, but also can lead to unpredictable encounters, misunderstandings, and frustrated expectations. One person's clever joke is another person's offensive insult. Connecting with others tops the 'to do' lists for new settlers. Internet shopping, gaming, and searching for information, particularly health information, are also near the top. The new settlers are a social bunch compared with the early pioneers, who focused on programming and information-oriented tasks. Different types of technology require different forms of etiquette (Marx, 1994). Texting via a mobile phone is different from instant messaging and worlds apart from the asynchronous experience of email. A short abrupt comment that is acceptable

in instantmessaging may not be in email where some people expect to be addressed by name. Emotional affordances, syntax and semantics vary across technology too.

Furthermore, new technologies may challenge previously accepted norms. Who would have guessed that having a stranger edit one's fastidiously composed prose without first asking permission would be acceptable? Yet this is exactly what happens in Wikis, which are designed for developing collaborative web pages. Therefore, rules of etiquette are needed that preserve this spirit of flexibility while supporting reasonable behavior and good will. With such a wide range of communications software now available to user's etiquette is challenged when users move from one type to another. It's particularly easy to forget more subtle

differences between the technologies. Access conditions may also cause poor etiquette. Hundreds of miles and many time zones separate some participants; most cannot rely on face-to-face meetings to learn about each other's norms. In addition, access to and experience with technology differs. Unwanted messages and large attachments that slightly annoy high-bandwidth users can be distressing for users with unreliable, expensive, dial-up facilities in remote locations. The complexity of this rich

mélange of users, goals, technology, and access conditions present new challenges to etiquette online, particularly for the growing number of support communities where kindness, help, and empathy is anticipated. Good approaches for fostering etiquette online are therefore needed.

HOW DOES ETIQUETTE DEVELOP?

Widely accepted ways of behaving reflect the attitudes and values of a community or society at large - they are its norms (Morton, 2003). Social norms are people's beliefs

about behaviors that are normal, acceptable, or even expected in a particular social context (Postmes, T., Spears, R. and Lea, M., 2000).

Norms, including rules of etiquette, are learned through experience in a community. For

example, children observe how adults and other children behave, absorb these norms, and

learn their community's etiquette at an early age. This role-modeling process continues

throughout life. Other community members correct those who do not conform to expectations. Problems arise when people go into other cultures with

different norms, particularly when the differences are subtle.

Gift giving provides a good example of an etiquette norm. If I receive a gift from American friends, I open it, thank them and comment enthusiastically to show my pleasure. In Japan, I would instead thank the person and carefully put my gift aside to be opened later; opening the gift there and then would not be polite. My behavior would contravene the norm in Japanese society and be interpreted as poor etiquette. In each culture, norms preserve or enforce comfort and empathy in the community. Consequently, when norms of etiquette are broken, discomfort, confusion, annoyance, embarrassment and even fear may ensue.

WHAT PROBLEMS OCCUR ONLINE?

To understand how to create and support etiquette norms online, first we must understand what challenges them. Ask Internet users what offends them most online and you will get a slew of comments. For example, two friends offered the following:

“Receiving notes with inappropriate automated signatures or not addressing me by

my name and ending without a farewell greeting and the sender’s name – that’s rude and unfriendly.”

Another said:

“One-word answers and comments that don’t refer to our conversation. How am I supposed to remember what ‘Yes’ refers to when I get 70-100 emails a day?”

“Attaching large files that take a long time to download is thoughtless.”

Surveys also provide examples of annoying behavior. A recent Internet survey used

convenience sampling to collect opinions from 4155

participants (istudio.vantagenet.com/cgi-bin/pollresults/002, accessed 10/03). The survey asked:

“Which netiquette issues aggravate you most?”

The following problems were identified and they are listed in order of most to least mentioned: sending spam; forwarding bogus virus warnings; sending dumb jokes; typing in all caps or all lower case; lack of basic grammar and

punctuation; including my email address in the CC. or TO. with a list of other email addresses; not editing emails; including no hello or thank you; and poor use of formatting styles. These responses suggest different reasons for the underlying

behavior: *unintentionally annoying behavior* that is due either to poorly developed skills (e.g., not mixing caps and lower case, not editing emails, poor grammar) or failing to appreciate what others care about (e.g., wasting time by sending dumb jokes); *potentially malicious behavior* (e.g., sending spam); and *absence of courtesy* (e.g., including hello and thank you). Surprisingly, aggressive comments – known as ‘flames’ – are not mentioned. Flames are

comparatively infrequent but when they do occur, they can be devastating to an individual or community.

WHAT ARE THE SOLUTIONS?

Many researchers have reported breakdowns in etiquette online over the years (e.g., Shapiro and Anderson, 1985; Sproull and Kiesler, 1992). We also know much about the causes of poor etiquette in textual communication; for example, absence of non-verbal feedback and reduced sense of responsibility between people who may never have to address each other face to face (e.g., Katz and

Rice, 2002). This knowledge has given rise to new interfaces that support identity and social interaction online (e.g., Preece, 2000). Two well-known approaches that specifically address etiquette solutions are setting rules (often called netiquette) and moderating discussions. These can be effective but often prove inadequate. New approaches are called for that combine human judgment with technical efficiency. The following discussions review current practice and suggest some other approaches.

HUMAN-ORIENTED PROCESSES

NETIQUETTE

The usual approach is to develop written lists of rules for online behavior. Typing ‘netiquette’ in Google produces a stream of links to rules for online etiquette. Recent examples like Nokia’s “Don’t be a txt msg abuser” offer the following guidance for text messaging with cell phones (www.thefeature.com accessed 10/03):

COMMON COURTESY STILL RULES.

Contrary to popular belief, composing an SMS while you're in a face-to-face conversation is just about as rude as ...

LEAVE THE SLANG TO THE KIDS.

Don't expect your stodgy superiors at work to be hip to the lingo of the SMS streets. And don't expect to win points with your kids by trying to be cool.

REMEMBER THAT YOUR PHONE HAS AN OFF BUTTON!

These basic, common sense rules can be effective, but they are often read and forgotten. When upheld by moderators, community leaders and participants they are more likely to be successful; particularly for setting standards in the early days of a community's life.

MODERATORS

Moderating is a well-known practice for preventing impolite behavior online. However, moderating can be demanding and time-consuming, particularly in active communities. Knowing when to control groups, when to let go, and how to bring in non-participants is not as obvious as it might seem.

ROLE MODELS

It seems obvious that role models could be used more online. After all, it's the way children learn etiquette (Morton, 2003). Online moderators and early adopters tend to be role models for those that follow. By greeting, acknowledging and praising participants they encourage a climate of appreciation and respect that fosters etiquette. Watching what others do is also a common strategy for newcomers to an online community. It enables them to judge the tone of the community before launching in, and so avoid causing offense, being ridiculed or put-down (Preece, Nonnecke and Andrews, 2004). Even established community participants adopt this wise strategy when joining new communities because each community has its own standards and ways of behaving. What is acceptable in one may not be in another.

Evidence of constructive discussions, information exchanges and empathic support are the trademarks of most successful communities. But there are other models for community. Some, like the saloons of Wild West, are rough and tumble places that thrive on sarcasm, bawdiness and punchy comments. Problems only arise when expectations are not met, especially in communities that are normally helpful and supportive.

MENTORS

Mentoring could be helpful in technical, specialist and cross-cultural communities. This approach has been used successfully in education environments. Based on principles advocated by the well-known psychotherapist Carl Rogers students were given templates to teach them polite critiquing skills (Zimmer and Alexander, 1996). These templates promote addressing a person politely by name and being careful to check that you understand what she is trying to say before jumping in with questions, suggestions or criticism; for example:

“[Name] What I think you mean is ... My own experience differs in this way ...”

“[Name] Please tell me what you want to do here, so that I can understand your point of view”

This approach encourages empathy and shared understanding, which is important online where non-verbal cues are reduced or missing.

CITIZEN REGULATION

Communities in which home-grown etiquette norms develop with little outside influence tend to be particularly successful. These citizen-regulated communities

moderate their own behavior. If someone steps out of line, community members remind the offender about what is expected gently or sternly, as the situation warrants. This behavior is particularly noticeable in patient support communities. For example, in a health support community one participant told another who was miserable and in pain not to be a wimp and to “*tough it out*” (Maloney-Krichmar and Preece, 2004). Quickly three people came to the rescue with comments like:

“Nonsense! A macho approach ... cannot avoid all problems, you were just lucky.”

How does a community become self-regulating? Participants who take on specific roles in the community help to establish the community's norms and express its values (Postmes et al., 2000). Good community management and skillful moderating can help to make this happen by ensuring that people treat each other politely and with respect. Even communities with a high turnover can develop strong etiquette norms and become self-regulating. Mary, a member of a vibrant patient support community that has existed for eight years, describes how this happens:

“Folks with different values eventually weed themselves out or are invited to

leave. There is definitely respect granted to members of the board. ... I am constantly amazed at how well members of the board get on on a day to day basis."

TECHNICALLY ORIENTED PROCESSES

Some tools exist to support etiquette online but more are needed.

FILTERS

Obvious obscenities and unwanted spam messages can be detected and eliminated using filters. Email systems enable users to set their own filters. Spam detecting software like SpamKiller and SpamAssassin offer increasingly sophisticated filtering capabilities. The software comes with a large list of pre-set filters that check senders' email addresses, subject lines, message bodies and embedded URLs for unwanted material. These lists are updated regularly with revised ones which can be downloaded free of charge. In addition, users can set their own word lists, email addresses and URLs for filtering. For example, ePrism (http://www.stbernard.com/products/eprism/products_eprism-spam.asp Accessed December, 2003) provides a checkbox interface so that users can

specify which filters they want in operation (Figure 1). Some text filters also substitute URLs, obscenities and other unwanted words with a message such as "Hi there!", "Cool it!" or nonsense words like "Hubbubb!"

COMMUNITY TOOLS

Many community-building software applications provide tools for moderators to identify, approve, reject, delete, and edit messages, or to request the sender to edit it herself. Moderators can also delete whole threads or lock a topic so that no further discussion can occur and send autoreplies. Some systems also make it easy for participants to request help directly from a moderator.

SEARCH AND VISUALIZATION TOOLS

Tools are available for identifying, rating and rewarding contributors to online discussions. Some offer data mining and visualization, such as Microsoft's Netscan (netscan.research.microsoft.com). Using Netscan researchers can search for the most active UseNet Newsgroups, or the most active contributors within a group or groups, and the most valued members (Smith and Fiore, 2001). Results for different types of searches can be displayed

in ranked tabular form (Figure 2), graphically as a treemap for showing hierarchical relationships (Figure or in clusters, as appropriate. A similar approach could be adopted for etiquette if dictionaries of good and bad etiquette were added.

RATING AND REWARD SCHEMES

Led by Amazon and e-Bay, several e-commerce sites have rating schemes for customers to evaluate their products and services. Often a five-point scale is used in which “1” represents “poor” and “5” represents “excellent”. Individual and average scores are then displayed for future customers to see. Slashdot, a large, technical discussion community that receives several thousand messages per day, has a more sophisticated scheme known as “karma” for evaluating participants’ contributions. Moderators award karma points to each contributor for the messages and stories they submit to the board. Karma can be rated as: “Terrible”, “Bad”, “Neutral”, “Positive”, “Good”, and “Excellent”. Karma ratings are also influenced by meta-moderating in which one moderator evaluates the ratings of another moderator ensures that moderation is done fairly and can adjust karma points if it is not. This clever scheme therefore also checks moderator performance and

enables moderation to be an open process that involves community members. Schemes like this provide a basis for self-government in which communities determine and maintain their own etiquette standards. They encourage commitment and community service. Becoming a moderator is an honor.

WHAT IS THE WAY FORWARD?

Just as today’s Internet settlers have different needs from the early pioneers, tomorrow’s settlers will be different too. More people from different cultures will come online. Ensuring good etiquette online will challenge moderators’ and community leaders’ unless better processes and tools for supporting good etiquette are developed. This should be a welcome challenge for researchers and developers. It is an opportunity to build bridges between people from different cultures, religions, genders, ages and educational achievements. The way forward is to develop processes that bring together the best human-oriented approaches with good technical support. New processes need to include rich social structures as well as being labor-saving and scalable. Processes that encourage communities to develop self-governing etiquette standards are promising. Creative newsocio-technical processes are needed that take

account of knowledge about: cross-cultural communication, counseling, group facilitation strategies, conflict management, community development, personal and group identity online, as well as strong technical know-how. In addition to preventing obvious breaches of etiquette, processes are needed for dealing with subtle etiquette problems such as clever pranks designed to incite reactions and inadvertent impoliteness due to cultural misunderstandings. A deeper knowledge of semiotics is needed to build these kinds of applications (de Souza, 2004). As the Internet population continues to grow and diversify etiquette will become increasingly important. Strong etiquette online is no longer just *nice* to have it is *necessary*.

CYBERBULLYING

(Important for paper I too. Referred from an article by R. Donegan)

The word “bully” can be traced back as far as the 1530s. (Harper, 2008). In its most basic sense bullying involves two people, a bully or intimidator and a victim. The bully abuses the victim through physical, verbal, or other means in order to gain a sense of superiority and power. These actions may be direct (i.e. hit-ting, verbally

assaulting face-to-face, etc.) or indirect (i.e. rumors, gossip, etc.).

TRADITIONAL BULLYING VS. CYBERBULLYING

Technology’s progression is often equated with the advancement of human societies. Pivotal innovations, such as the Internet, have forever changed how people interact. Though these developments have allowed the human race to make great strides in many fields, they have also allowed forms of transgression to become more rampant and widespread. This is evident when considering how traditional bullying has evolved into an issue today known as cyberbullying. While bullying and cyberbullying are often similar in terms of form and technique they also have many differences. Unlike traditional bullying, cyberbullying allows the offender to mask his or her identity behind a computer. This anonymity makes it easier for the offender to strike blows against a victim without having to see the victim’s physical response. The distancing effect that technological devices have on today’s youth often leads them to say and do crueler things compared to what is typical in a traditional face-to-face bullying situation.

TECHNOLOGICAL EVOLUTION

As technology has evolved, bullying has proliferated. With the advent of the Internet, chat rooms soon followed. Online forums provided a communal breeding ground for youth to assault one another (Subrahman-yam & Greenfield, 2008). Chat rooms were supplemented by AOL Instant Messenger (AIM), an online communication program that allowed teens to spend hours talking to one another in private, one-on-one conversations or in public chat rooms. The program further allowed youth to create group-specific chat rooms. This exclusive forum allowed for youth to get together with select groups of friends and talk about the latest gossip. Online innovations have continued due to telecommunication advances. The advent of cell phones in the late 1960s and early 1970s changed the way people communicated (Shiels, 2003). However, these portable communication devices did not become widespread, or make it into a majority of youth's hands, until the appearance of the second generation of digital network phones in the 1990s. After that, they spread like wildfire. According to a study conducted by the Pew Research Center, 75% of 12-17-year-olds own cell phones, which increased from 45% in 2004 and one-in-three teens sends 3,000 text messages per month (Lenhart, 2010). Though many parents believe that they are

purchasing a cell phone for their child for protective reasons, the opposite may be true as many youths admit to utilizing their phones as an instrument for cyberbullying. Further progress on the Internet brought about more and more websites and with this came the advent of social media. The site MySpace is often considered the pioneer of social media. MySpace allows individual users to create their own unique profiles and interact in cyberspace with friends and foes alike.

Online publication of personal information is dangerous because it allows many people to see a side of a person more often kept private in a face-to-face interaction. This vulnerability puts many teens in a position as either the victim or active offender partaking in cyberbullying actions. Another aspect of social media that can be misleading and hazardous is the ability to create alias profiles. The ability for teens to mask their identities provides them with an opportunity to say anything to another individual without the worry of any repercussions. Social media sites, such as Facebook and Google+, are prone to abuses like cyberbullying. Anonymous blogging is another technological advancement that has fostered cyberbullying activity and fueled ethical debate. On sites, such as College ACB and Juicy Campus, which

have both recently faced tightened regulations due to their verbally abusive nature, youth (typically of college age) were able to login and comment anonymously in an open forum. The forum included harsh topics ranging from “Most Attractive” to “Worst Hookup.” The sites even included certain topic headings that were simply a person’s name under which people could post insulting comments. These blogging sites are illustrative of the most dramatic forms of cyberbullying thus far.

STATUS OF CYBERBULLYING

Studies have indicated that the number of youth reporting cyberbullying instances varies greatly depending on the definition of the term and the age of those surveyed. In the following study, Sameer Hinduja and Justin W. Patchin, of the Cyberbullying Research Center, sampled 4,441 teens, ranging in age from 11 to 18, from a large school district in the southern U.S. In this study, the researchers defined cyberbullying as “when someone repeatedly makes fun of another person online or repeatedly picks on another person through email or text message or when someone posts something online about another person that they don’t like” (Hinduja&Patchin, 2010c, p. 1).

VICTIMIZATION

According to their results, cyberbullying victimization rates have varied in the past few years, ranging between 18.8 percent in May 2007 and 28.7 percent in Nov. 2009 with a mean of 27.32 percent based on 7 different studies from May 2007-Feb. 2010. Cyberbullying offending rates have varied in a broader spectrum than victimization rates, ranging between 20.1 percent in June 2004 and 11.5 percent in Nov. 2009 with a mean of 16.76 percent based on 7 different studies from June 2004 to February 2010 (Hinduja&Patchin, 2010g, p. 1). Several specific types of victimization and cyberbullying were discovered through a survey taken in 2010. The survey discovered that the highest concentration of victimizations and cyberbullying offenses occurred in the following areas respectively: mean or hurtful comments posted online (14.3%, 8.8%), rumors online (13.3%, 6.8%), threats through a cell phone text message (8.4%, 5.4%) (Hinduja&Patchin, 2010e, p. 1). Another important factor that Hinduja and Patchin brought to light was what type of technology a teen primarily uses. According to a 2010 study, which asked teens what role technology played in their daily lives, cell phones were used the most (83%), followed by the Internet for school work (50.8%), and then Facebook (50.1%)

(Hinduja&Patchin, 2010h, p. 1). This points to cell phones and the Internet as the two primary mediums used for cyberbullying. Cyberbullying proved to vary by gender as well. Based on a 2010 study involving a random sample of 2,212 teen males and 2,162 teen females, the male to female ratio varied the most in the following three areas: victimization within a person's lifetime (16.6% for males vs. 25.1% for females), admitted to a cyberbullying offense within a person's lifetime (17.5% for males vs. 21.3% for females), and had a hurtful comment posted about oneself online (10.5% for males vs. 18.2% for females) (Hinduja&Patchin, 2010b, p. 1). This information may be biased due to the reticence among males about admitting a past bullying experience. Nevertheless, it is interesting that females reported a higher percentage in all categories.

CLINICAL

PERSPECTIVE/REPERCUSSIONS

At first, one may believe that the effects of bullying are limited to initial responses that tend to fade within a few days or a week, at most. However, research indicates that the harm inflicted by bullying, whether physical or psychological, has many implications and

can result in a snowball effect of lasting painful emotions and negative impacts.

GENDER AND BULLYING

Though many students tend to deny the emotional harm caused by bullying tactics such as name-calling, rumor spreading, and teasing, research suggests the opposite. In a study that utilized a sample of over 3,000 students, researchers found that "38 percent of bully victims felt vengeful, 37 percent were angry and 24 percent felt helpless." Furthermore, in a study conducted by the Cyberbullying Research Center involving a sample size of 468 students revealed that females are typically more emotionally affected by cyberbullying than males. The females in the study reported being frustrated (39.6%), angry (36%), and sad (25.2%) more often than males who reported lower percentages in each category (27.5%, 24.3%, 17.9% respectively). This is not surprising due to the fact, as mentioned earlier, that males have a reluctance to admit weaknesses especially from an emotional standpoint. In reality, one would expect males to be at least equal if not higher in emotional response concerning anger and frustration. (Hinduja&Patchin, 2009c, p.1).

AGE AND BULLYING

In another study conducted by the Cyberbullying Research Center, the emotional repercussions of cyberbullying across age groups were observed. The study discovered that anger and frustration remain the dominant responses among senior and junior high students, but students at the elementary level are more likely to feel sad as a result of being bullied (Hinduja&Patchin, 2009c, p.1). This is likely because at a younger age child are not battling with the same kind of competitive social hierarchy typically found within upper level schools. So, rather than feeling the need to prove themselves among their peers, students at the elementary level tend to well-up within the initial emotional responses to bullying. This points to the idea that younger children may keep their initial emotional responses to themselves rather than acting out. Unfortunately, regardless of the initial emotional reaction to bullying, these emotions have the ability to continue to develop, with serious clinical implications. A 2003 intensive survey study, which focused on the clinical effects of cyberbullying, reported an increase in emotional distress specifically related to cyberbullying. The study involving 512 professionals coming from psychology, psychiatry and social work backgrounds reported that for “one-third (34%) of these youth, the Internet problem played a

primary role in the client’s treatment” (Mitchell, Finelhot& Becker-Blease, 2007, p. 48). This evidence proves that cyberbullying is having noticeable clinical effects on today’s youth. In his “General Strain Theory,” sociologist Robert Agnew hypothesized that the strain and stress exerted on an individual as a result of bullying “can manifest itself in problematic emotions that lead to deviant behavior,” possibly leading to delinquency (Agnew, 2006, pp. 659-660). This theory stresses the vicious cycle that many teens may go through while being victimized. The cyclical repercussions of this process are particularly alarming if it leads a victim to antisocial behaviors when they try to find an outlet for their emotions. In 2001, the Office of Juvenile Justice and Delinquency Prevention reported that “60 percent of males who were bullies in grades 6 through 9 were convicted of at least one crime as adults, compared with 23 percent who did not bully; 35 to 40 percent of these former bullies had three or more convictions by the age of 24, compared with 10 percent of those who did not bully” (Ericson, 2001). Offenders are likely to utilize bullying tactics as an outlet for other insecurities or problems in their lives. This utilization of bullying as a coping mechanism contributes to the cyclical nature that the process evidently has on victims and offender alike. Not

only do these flawed coping mechanisms fail to resolve the emotional distress caused by bullying, they also expand the overall problem of deviant behavior. This inability for bullied victims and offenders to find adequate relief for emotional wounds, coupled with the fact that youth are unlikely to seek relief through a mentor, explains why some youth begin to feel helpless. Feeling lost in emotional distress with seemingly no way for relief allows suicidal or even thoughts of violent response to creep in to a youth's consciousness.

The extreme consequences of bullying are suicidal thoughts or thoughts of violent revenge. According to a National Vital Statistics Report, suicide is the 3rd leading cause of death among youth ranging in age from 15 to 24 (Anderson & Smith, 2003). Furthermore, the U.S. Department of Health and Human Services reported "substantial increases in both homicide and suicide rates among males from 2000 to 2003". This statistic becomes chillingly relevant as more information is uncovered concerning the link between cyberbullying and suicidal ideation or action. According to Rigby and Slee, "Youth who are bullied or who bully others, are at an elevated risk for suicidal thoughts, attempts, and completed suicides" (Rigby & Slee, 1999, p. 119). Statistically both victims of

cyberbullying as well as offenders proved to be much more likely to have attempted "bullycide," the act of committing suicide due to the effects of bullying, than youth who had not been affected (High, 2007).

PREVENTION

As cyberbullying draws more attention, a universal definition has begun to take shape within the law. Though not acknowledged across all states, a common definition in congruence with a wider recognition of the problem makes addressing the elusive issue a bit easier. The problem is that technology will inevitably continue to advance. So, as technology progresses, local and national anti-bullying policies and laws must continue to evolve at a parallel rate. Many of the issues faced by federal and state governments concerning cyberbullying are avoidable for schools and communities at the local level through the implementation of procedures to limit the effects of cyberbullying.

RECOGNITION OF THE PROBLEM

One of the most notable issues that need to be addressed is recognition of the problem itself. Many people, whether parents, teachers, or even law enforcement officers, do not know what their specific state laws

are in regards to cyberbullying. In a formal survey of approximately 1,000 officers, “over 85 percent . . . said that cyberbullying was a serious concern that warrants the response of law enforcement. 90 percent of the school’s resource officers had dealt with a cyberbullying case ‘sometimes’ or ‘often.’” Despite this obvious concern, “25 percent of the school resource officers and over 40 percent of the traditional law enforcement officers did not know if their state had a law specific to cyberbullying” (Patchin, 2011). These statistics are alarming considering that the number of states without some kind of bullying or harassment specific law can be counted on one hand (HI,MI,MT,SD) and the number of states with cyberbullying specific laws are increasing. Every person that deals with children (i.e. education, parenting, law enforcement, etc.) should know the bullying or harassment law specific to his or her state, the physical and emotional signs of bullying, and his or her community or school prevention plan, including how to deal with and report a problem.

Along with recognizing the problem and being able to identify solutions, it is vital that cyberbullying be addressed in a consistent way. To effectively put a harness on the problem will require “a

concerted and coordinated effort – a partnership if you will – among our families, schools, youth organizations, and communities” (Morino, 1997). If American communities and schools address the issue with a clear preventative pro-gram that keeps each level of prohibition on the same page, children will in turn receive a consistent message from a young age, which will presumably resonate effectively. This message should cause children to feel comfortable with confronting and reporting the problem by portraying any form of bullying as unacceptable. Furthermore, this consistency across a given program will change the overall environment rather than just focusing on individual cases. According to Dorothy Espelage and Susan Swearer’s book *Bullying in American Schools*, “A comprehensive program . . . is generally more powerful in reducing bullying and increasing school safety than concentrating on individual students”.

POTENTIAL SOLUTIONS

Credited with initiating the first systematic bullying research in the early 1970s, Dr. Dan Olweus is primarily known for his bullying prevention programs. The Olweus

Bullying Prevention Program, enacted by Norway's Ministry of Education, develops methods of dealing with bullying on a variety of levels including school-level components, individual-level components, classroom-level components, and community-level components. This all-encompassing structure creates a cohesive plan in which each level reinforces the next. Since its creation, the Olweus Bullying Prevention Program has been adopted in schools across the world. According to a series of evaluations involving 40,000 students from 42 schools over a two-and-a-half-year period, the program proved to be successful. The studies verified the program's success by reporting "reductions by 20 to 70 percent in student reports of being bullied and bullying others," "reductions in student reports of antisocial behavior, such as vandalism, fighting, theft, and truancy" and "clear improvements in the class-room social climate" (Olweus, 2005, pp. 389-402). A mixed-methods research design carried out in Nevada, in which 118 middle school principals were surveyed about cyberbullying, identified components of a successful preventative program. Of those surveyed, 66 responded and 10 agreed to participate in a one-on-one interview. From these ten, three were selected and questioned intensively about cyberbullying and preventative strategies

within their school as well as about their thoughts on the issue in general. Based on the interview results, the following six components were identified as essential elements to a preventative cyberbullying program: "the importance of a reporting procedure; curriculum integration; student-centered productions through mediums such as Broadcast Journalism; a focus on prevention rather than solely on punishment; the importance of punishment as a part of an effective policy; and keeping up with changes in technology" (Wiseman, 2011).

CYBER PORNOGRAPHY: CONSUMPTION, IMPLICATIONS

It is important to note that, Indian law nowhere defines the term 'Pornography'. Even the term is not statutorily defined in the United States of America. It is also felt that as there is no uniform standard of moral culture and ethics there cannot be any fixed/uniform definition of this term. Pornography is often described as hard to define but easy to recognise.

The Internet is a vast matrix of computers that are interconnected with each other and which facilitates the individual to communicate and to collect information in very short time, and at cheaper cost. Its impact and influence on our daily lives is

immeasurable. It has changed the way individual communicate with another person, anywhere in the world, and provides information on any topic imaginable. Irrespective of caste, race, economic background, gender, and age, Internet provides everyone a platform through which they can advance their ideas, without facing the bias or prejudice of others. The way people do online trading, is just one simple example of why the Internet will continue to shape the way we communicate and transact businesses worldwide. The world has truly become a lot smaller. Unfortunately, the Internet is not all flowers and sunshine. There are elements within it that have grown to become sources of concern within our society. Among them is the persistent availability of Pornography or more commonly known as Cyberporn.

Cyberporn has become the most controversial topic arising from the use of Internet in recent years. Its availability on the Internet has caused fear and a 'moral panic' among the government, law enforcement bodies. There is no settled definition of pornography in a multinational environment such as the Internet. Cultural, moral and legal variations all around the world make it difficult to define 'pornographic content' in a global society. It is because of a reason that has caused many citizens worldwide to put pressure on their law enforcement mechanism to monitor the Internet. Many nations have framed legislation that would strictly regul

ate the Internet and still others are trying to legislate.

Individuals who choose to view Cyberporn have challenged these legislations. Pornography has been an issue of great controversy around the world for years. It has its history, in continental Europe since 1660's. During those days, pornography was associated with various and wide-ranging ideas, spanning from political message to obscenity. When pornography was first labeled as such in France in the early eighteenth century it was primarily used in political cartoon to help organizers stance against the French aristocracy. This type of pornography, referred to as "political pornography", was strikingly different from the pornography that we know today. The monarchy was often the subject of much of the pornography written between 1500 and 1800. The king's body and the body of the nation were frequently used by cartoonists to dramatize the political concerns of the citizenry. In addition, pornography was also used to criticize the excess of the Catholic Church. The Church was repeatedly referred to as the "Whore of Rome" by pornographic pamphlets printed during this time, and pornographic writers would use a rich and obscene woman to depict the ceremonies associated with the Roman Church. Towards the end of the period between 1500 and 1800, pornography lost its political role and instead, began to be produced only for individual sexual gratification. An industry was created that produced mass quantities of publications that were not intended to be politi

cal or cultural critiques. Once political pornography evolved (or devolved) to this phase, it ceased being political and began to be profit making. The political history of pornography has shaped the post-modern definition that we understand today. This change is what leads to the development of this type of pornography that we are most familiar with today. To understand today's pornography, it is necessary to understand and incorporate the political history that pornography has and the influence that its history contributed to our understanding of pornography in this post-modern period. It was not until the early 1800's when pornography became associated with the humanist and material movements that it was considered obscene and regulated by the State. Both of these movements were linked to the spread of democracy and corresponding rejection of the monarchy and the hierarchy of the Catholic Church in Europe. The emphasis that both humanism and materialism placed on the individual was threatening the political order of the period. This resulted in the creation of pornography as a category of publications, allowing the State to regulate and control the publication of pornographic works. Since that time, various developments in technology have changed the way pornography is disseminated. However, it was not until the advancement in Internet technology that access to pornography was virtually unlimited.

Before the Internet, one wishing to purchase or view pornographic material had to go to a place which provided such material. This allowed the government to regulate children's access to pornography. It also provided the government with an enforceable method of ensuring that all legal pornographic material was not sold. By reviewing the material sold in the establishments selling pornography, and by requiring publishers of the pornographic material to provide proof of the age of the subjects in the photographs, illegal pornographic material such as child pornography and bestiality could be monitored. Now, with the development of the Internet, the regulation of pornographic material has become much more difficult. Children can access pornographic materials from their homes or even from public libraries, and illegal materials can easily be disseminated. Anyone with a scanner and a modem can post obscene photographs (which are available to millions of people). Indeed, each day additional sites dedicated to pornography are created. The Internet was originally designed by the United States Department of Defense as a means of providing decentralized computer controls. The theory was that if part of the country were destroyed by a nuclear attack, the computer systems containing war planning would still be functional. Soon the opportunities of the Internet became clear to the private sector. Universities and businesses created connections to what has been termed "the informati-

On super high way" and eventually led to what is today known as the WorldWideWeb.

Technology serves to facilitate commercial activity and pornographers have never lagged behind from it. Huge amount of pornographic material

can be reproduced more quickly and cheaply on the new media like harddisks, floppydisks and CD-

ROMs. The new technology is not merely an extension of the existing forms like text, photographs and images. Apart from still pictures and images, full motion video clips with sound are also available. In fact, the latest trend seems to be towards interactive 'live sex' where people perform

and according to requests by subscribers to their services. The end user can view all of this on his/her monitor, save and can transmit it. In past, when technology was not so advanced government was able to keep some check. But, then the real nature of the medium also that plays the vital role. Any search engine can be used to generate a list of a vast number of sites with pornographic content. Many of these are free or have free trial offers and this defeats the very rationale of having an age limit under which pornography may not be sold to minors. This shows the serious need for regulation of pornographic materials, or at least regulation of where pornographic materials can be viewed.

PARENTAL MEDIATION OF DIGITAL USAGE

The increasing accessibility of consumer digital devices has seen a rise in digital media use among children. The prevalence of digital technology has not only changed the way children learn and absorb knowledge, but also transformed the way they communicate and interact with each other. While it is debatable whether these new technologies have an overall positive or negative impact on children, scholars suggest that the impact of media on children largely depends on the proper management and mediation of its use (Clark, 2011).

However, the changing media environment poses challenges for parents. Today's children, known as "digital natives," are tech-savvy and confident about their digital skills, and their knowledge and understanding of digital technology often exceeds that of their parents. In addition, many digital devices are portable and mobile, making it harder for parents to closely monitor children's use of them.

THE ROLE OF PARENTS IN CHILDREN'S MEDIA USE

Parents influence children's behavior from the moment they are born. Parent and child are "part of a biologically intertwined system that strongly supports the [child's] socialization process" and parents are often motivated by this strong relationship to handle and mediate the child's behavior in

the face of social demands and new situations. Parents, being “the most influential people in the development and socialization of children” also take primary responsibility for guiding their children’s media behavior. Proper mediation of children’s media use by parents is known to result in positive outcomes such as decreased exposure to privacy risks, decreased exposure to child-inappropriate media content, and heightened understanding of media content.

Three general approaches to the mediation of children’s media use have been identified. *Active mediation* refers to parents explaining and discussing media with their child. *Restrictive mediation* happens when parents set rules regarding the amount of time and the kind of content with which their child is allowed to engage. Lastly, *co-use* refers to shared usage of media between parents and children without parental instruction or discussion. Research suggests that different forms of the parental mediation result in different outcomes. One of the consistent findings in regard to the outcomes of different forms of parental mediation is that active mediation is more effective than the other two forms of mediation primarily because active mediation based on parent-child conversation and discussion is more likely to cultivate critical thinking skills in children and to lead children to be more responsive to parents’ initiatives. Research also suggests that too much restriction can cause unintended consequences such as a boomerang effect or a forbidden fruit syndrome, especially when it is imposed on older children who pursue more autonomy from parents. The effects of co-

use have not been compellingly verified as prior studies have demonstrated that co-use can also be viewed by children as parental endorsement to the media content rather than deliberate monitoring efforts, especially in television viewing contexts.

While the above three strategies have been identified in research on television viewing (Clark, 2011), a number of studies examining parental mediation of children’s use of the internet and video games have applied the three forms of parental mediation. However, the advent of the digital age has resulted in a generation of children that are experts at multitasking with digital technology, and the complexity of new digital technology, characterized by its focus on multiple activities, multi-media, active content production, and social interactions, have posed new challenges to parents guiding their children through all of these activities. Addressing the changes in the media environment, some studies have identified additional types of parental mediation strategies. For instance, Livingstone and Helsper (2008) identified a strategy known as *monitoring*, which refers to parents’ checking of children’s email messages or the sites that children have visited. This can be conducted overtly or covertly after the child’s internet use. A similar concept called *supervision* was identified by Nikken and Jansz (2014), which refers to parents’ keeping an

eye on the child and the device and allowing children to surf the web only when parents are present. While monitoring involves examining the sites and messages that a child is involved with and can happen with or without the child being around, supervision happens when the parent is physically together with the child.

Livingstone and Helsper (2008) surveyed teenagers aged 12–18 and their parents in the UK and found that parents often adopt strategies used in television mediation to deal with the new challenges posed by the internet. Nikken and Jansz (2014) surveyed parents of children aged 2–12 in the Netherlands and found that supervision was the most common parental mediation strategy parents used to mediate children's internet use. Sonck et al. (2013) found that active safety mediation was the most common strategy employed by Dutch parents of children aged 9–16 using the internet. Khurana, Bleakley, Jordan, and Romer (2015) surveyed parents of American adolescents and found that parental monitoring was practiced twice as often as restrictive strategies.

Overall, there seems to be a lack of consensus as to which mediation strategy is most popular among parents with regard to children's use of digital technology. In addition, each of the prior studies focused on a different age group. Thus, it is unclear whether particular parental mediation strategies are more or less prevalent among parents of a certain age group, such as primary school aged children. Finally, most existing studies have focused on children's internet use, mostly in Western societies with an individualistic orientation. It is unclear how parents in Singapore, a non-Western society with largely collectivistic values (Smith, Dugan, & Trompenaars, 1996), manage children's *overall* digital technology use and screen time. Hence, the following research question is proposed:

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